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Data Interpretation and Logical Reasoning for CAT

Trishna Knowledge Systems

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ISBN 978-93-530-6306-1
eISBN: 9789353066017

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Preface

Congratulations on arming yourself with Trishna's *Data Interpretation and Logical Reasoning for CAT* and thus taking an important step towards preparation for a career in management, one of the most challenging careers today!

Of course, success will come only after cracking one of the toughest competitive examinations—CAT, which means your preparation should be nothing short of top class, where each mark will decide your future. You must aim not only to maximize your percentile but also cross the cut-off in each section of the examination. Meeting such stringent criteria calls for a champion-like preparation.

The Data Interpretation and Logical Reasoning section accounts for one-third of the total marks in CAT and an equal percentage in other management entrance examinations as well. The section tests your ability to understand the data given and interpret it.

In previous years, merely common sense could help crack this section. Yet, in the last few years, the nature of questions in Logic and Data Interpretation has undergone a major shift. From calculation and observation-based to now being pure logic-based or a fine blend of logic and basic calculation.

The level of difficulty has also increased manifold. Cracking such questions requires the application of many skill sets. To succeed, you must prepare for all types of scenarios expected in CAT and other management entrance tests.

Over the last few years, there has been an increased focus on the reasoning/logic-based sets. The key to solving such sets is to understand the logic behind them. If one cracks the logic behind the set, most of the questions in the set can be solved and this is one big advantage of this section compared to other areas where each question can have different logic. Special care has been taken to ensure that plenty of such logic-based sets are given. One of the hallmarks of this book is that each set is unique, innovative and creative. Understanding the logic behind the sets and practicing similar sets helps during the actual examination. Most of the questions one encounters in the actual examination would be very similar or involve a logic which is like the questions one has seen in the book. So, one should go through each and every set that is covered in the book.

The book comprises a sufficient number of chapters, each of which begins with a concise presentation of the required fundamental concepts of that topic. Following the basics and solved examples in every chapter, there are three exercises. The first exercise has easy questions and is meant as the starting point for all the students who are preparing for the competitive examinations. The second and third exercises have questions that are of moderate to high level of difficulty and they facilitate students to upgrade their ability and get ready for the tough battle ahead when they face the toughest of the examinations—CAT.

This book is your ideal preparation resource with a wide range of questions, including models of problems, that appeared over the last few years in many competitive examinations.

In addition to these, the book comes with access to three free AIMCATs which can be accessed by following the instructions given on the last page of the book.

This compendium of class-tested content, extensive practice resources, time-tested strategies, and practical guidelines is the result of the collective effort of a team of well-qualified faculty members. Our content team has extensive experience of teaching and developing high-quality study materials to aid preparations for various competitive examinations. They have guided more than 21 lakh students in the last 26 years, helping them gain admission in some of the top management institutions in India and worldwide. The extensive experience of the dedicated team at our institute allows us to say that nobody understands the needs of students and the nature of entrance examinations better than we do.

Although this book focuses on helping you prepare for the CAT and other major MBA entrance examinations (OMETs), it will also guide you to build the right foundation to develop and hone your strategies and skills necessary for career advancement in business.

The Editorial Team
Trishna Knowledge Systems

CAT Pattern Analysis 2015–2017

CAT Journey So Far

We have come a long way from the long-drawn-out 40 slot-25 day window to a 2 slots—a one-day affair.

The CAT paper now comprises three sections, namely

- Verbal Ability and Reading Comprehension (VARC),
- Data Interpretation and Logical Reasoning (DILR), and
- Quantitative Ability/Aptitude (QA), each section has a sectional time limit of 60 minutes with 34, 32, and 34 questions, respectively.

The surprise element in 2015 was the introduction of Non-Multiple Choice Questions (Non-MCQs) across the three sections. It was noted that as many as one-third of the questions were non-MCQs (i.e. 33 questions out of 100 questions), which contributed to increasing the 'difficulty level' of the paper and applied brakes on all those who took chances and marked answers based on random guesses.

In terms of the test interface, there was a slight departure from what used to be shown in the sample test.

- Within the VARC section, VA and RC questions were now grouped separately and given under two separate tabs.
- Similarly, in the DILR section, the DI and LR questions appeared under two separate tabs. This helps students easily access the type of questions they would like to answer.
- Students were also able to look at their performance in the previous sections at any time during the test by clicking on the respective tab for that section.
- The number of questions attempted, left out, and marked for review were also displayed.
- Furthermore, towards the end of the test, a similar snapshot was provided for all the 3 sections.

An overall pattern analysis is provided to help students understand the changes that occurred over the last three years:

Section	Subject	2015			2016			2017		
		MCQ	Non-MCQ	Level of difficulty	MCQ	Non-MCQ	Level of difficulty	MCQ	Non-MCQ	Level of difficulty
I	Verbal Ability and Reading Comprehension (VA & RC)	24	10	Moderate-difficult	24	10	Moderate-difficult	27	7	Moderate-difficult
II	Data Interpretation and Logical Reasoning (DI & LR) (one set = 4 questions)	24	8	Very difficult	24	8	Difficult	24	8	Very difficult
III	Quantitative Aptitude (QA)	19	15	Moderate	27	7	Moderate-difficult	23	11	Easy
	Total	**67**	**33**		**75**	**25**		**74**	**26**	

❐ PATTERN ANALYSIS

❐ 2015

The difficulty of the paper, across sections, was largely similar in both the slots. It was observed that the QA section was relatively much easier compared to the other two, followed by VARC.

With 24 RC questions and only 10 VA questions, the VARC section was moderate to difficult. All the 10 questions on VA were non-MCQs, and this moved the difficulty level of the section up by a significant level.

The RC passages were between 350–550 words long and were not too tough to read. However, with 5 passages (3 passages with 6 questions each and 2 passages with 3 questions each), students found it difficult to attempt all of them.

DILR was the section that troubled many with an unexpectedly high level of difficulty across multiple sets. This was true for both the slots. However, there was a marginal respite because the calculator was available.

This led to two things—As the section was difficult overall, the number of attempts dropped for all the test takers; and with CAT announcing that scores will be normalized across sections, the impact of the section on the overall score was expected to be moderated.

Another significant observation on the DILR section was that while separate tabs were provided for the DI and LR areas, there was no clear-cut segregation. There were both reasoning-based DI sets in the DI area and quantitative-based LR sets in the LR area.

The third section, QA, was moderate in difficulty. The presence of 15 non-MCQs contributed to the increased difficulty level of this section, which otherwise had many direct questions, albeit tricky ones.

Section description	No. of questions	No. of MCQs	No. of non-MCQs	Difficulty level	No. of attempts for 95%ile	No. of attempts to cross 99%ile
VARC	34	24	10	Moderate-difficult	22–24	26–28
DILR	32	24	8	Very difficult	11–12	14–15
QA	34	19	15	Moderate	21–23	25–27
Total	**100**	**67**	**33**	**Difficult**	**54–59**	**65–70**

Note: We are assuming an accuracy of 80% for the above estimates.

❐ 2016

In line with the expectations that students had about the CAT exam throwing surprises at them, the CAT 2016 did amaze many. There were quite a few doable questions across sections, as there were last year. However, the number of tough questions went up significantly. The order of questions and options for the questions was different for different students. The presence of easy questions made some students feel that the section was not very tough. However, many felt it to be an arduous task to push their overall attempts beyond a certain level. This was because any further move beyond an easy question was blocked by difficult ones that were present aplenty across the sections.

One significant observation that was gathered from our expert analysis was that the level of difficulty of each of the three sections was very close across the two slots. This was unlike CAT 2015, where there was observable difference in difficulty level in two of the sections (DILR and VARC).

Let's look at the test pattern.

Section	No. of questions	No. of non-MCQs	Difficulty level
Verbal Ability and Reading Comprehension	34	10	Moderate-difficult
Data Interpretation and Logical Reasoning	32	8	Difficult
Quantitative Ability	34	7	Moderate-difficult
Total	**100**	**25**	**Moderate-difficult**

❐ 2017

In line with the expectations that students have about the CAT exam, CAT 2017 did not surprise them much as far as the difficulty level of the paper was concerned.

There were quite a few doable questions across VARC and QA. However, in DILR section, the sets were quite tough to crack, with only a few doable questions. Hence, many felt it to be really difficult to push their attempts in DILR and subsequently the overall attempts beyond a certain level. The order of questions and options for the questions was different for different students.

The level of difficulty was broadly similar across both the slots. However, a relatively higher number of students from the 2nd slot felt positive about their DILR performance than those in the 1st slot. This positive feeling did not exactly translate into better scores/percentile because of the process of equating and scaling that the IIMs have been following over the past few years.

Before we get into the detailed analysis, let us quickly look at the test pattern.

Section	No. of questions	No. of non-MCQs	Difficulty level
Verbal Ability and Reading Comprehension	34	7	Moderate-difficult
Data Interpretation and Logical Reasoning	32	8	Very difficult
Quantitative Ability	34	11	Easy
Total	**100**	**26**	**Moderate-difficult**

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Data Interpretation and Logical Reasoning Analysis 2015–2017

Analysis—2015

The Data Interpretation and Logical Reasoning (DI&LR) section could be classified as the toughest of all. This section mainly consisted of 4 sets of DI and 4 sets of LR accompanied with 8 questions of the non-MCQ form. The sole purpose of this method was to push the students to derive the complexity out of the given sets where they could answer only 2-3 complete sets. As a matter of fact, the students could perhaps dabble in other sets by managing to crack at least one odd question in them. There were one LR and one DI set which were considered straightforward. However, the remaining sets were considered to be more difficult. Notably, majority of the students scored low in this section and their cut-offs were below average.

Analysis—2016

By this time, CAT 2015 has raised the bars high in terms of difficulty for the DILR section. The aspirants who had put much effort after this period benefitted the most as CAT 2016 set a new benchmark too. The difficulty level in Data Interpretation & Logical Reasoning section certainly went up one notch in CAT 2016. The DI sets were not difficult in terms of interpretation but these type of questions were asked in an alarming number. Overall, the questions were tricky and it was not easy to solve more than 2-3 questions in each set. However, students who had put serious effort through the AIMCATs were able to keep their balance and found this section less intimidating.

Area	Topic	No. of Qs	Good Attempts
Data Interpretation	Veg/Non Veg (Set on Veg/Non veg topics)	4	2
	Exam Pass Percentage	4	2
	Movies	4	2
	Train	4	1-2
Logical Reasoning	T-Shirts	4	1-2
	Venn Diagram	4	2
	Marks	4	2
	Folders	4	2

Afternoon Slot: The afternoon slot was tougher with one factor that the students could take solace from was that the difficult sets were clearly unsolvable right from the outset, helping them drop out of those fairly soon.

Area	Topic	No. of Qs	Good Attempts
Data Interpretation	Technical/Non-technical	4	2
	Bank model	4	2
	Venn diagram	4	2
	Experts, products and features	4	1-2
Logical Reasoning	Water supply	4	1-2
	Restaurant ratings	4	1-2
	Coding	4	2
	Conference rooms	4	2

Analysis—2017

CAT 2016 had set a new benchmark in terms of raising the difficulty for the DILR section and yet it took just one year to break this record. The difficulty level of the Data Interpretation & Logical Reasoning section certainly rose up again when compared to 2016. The entire section had more difficult, lengthier and trickier questions. There was barely one full set in the section that could be called outright workable. In fact, in some of the sets, it was not easy to solve more than 1-2 questions, while the remaining were almost untouchable. While the first slot did not have any set in the DILR section that could be called familiar, the students from the second slot did report seeing at least one standard type of set in the section.

Unit 1

Introduction

1 Introduction to Data Interpretation

CHAPTER

❒ INTRODUCTION

The study of figures and statistics has been an integral part of any medium of course. The study and manipulation of such data leads us to an important area called data interpretation. Data can be organized in a number of ways so that larger volume of data can be presented in a more compact and precise form. Data thus presented has to be deciphered correctly by the user of the data. This process of deciphering the data from its compactly presented form is called data interpretation.

Unorganized and haphazard data does not make any sense to the top management for whom time is a very valuable and rare commodity. Hence, any data, be it daily production figures, daily sales figures, financial performance or productivity, such data will have to be presented in a concise manner. However, at the same time being precise is very significant so that the top management can study it with ease and thus it also facilitates faster decision making.

In this section, we will cover data interpretation questions which is almost certainly asked in every MBA entrance exams either as a part of Mathematics or as a separate section.

Over the last few years, the people who formulate question papers for competitive exams have developed enormous liking for this area. Consequently, the variety of questions asked and the degree of difficulty have increased over a period of time.

❒ METHODS OF PRESENTING DATA

Numerical data can be presented in one or more of the following ways.

1. Data tables
2. Pie charts
3. Two-dimensional graphs
4. Bar charts
5. Three-dimensional graphs
6. Venn diagrams
7. Geometrical diagrams
8. Pert charts
9. Others

The 'Others' category covers miscellaneous forms like descriptive case format, which is customized for the situation. Data can also be presented by using a combination of two or more of the above forms.

While some data can be presented in many different forms, some other data may be amenable to be presented only in a few ways. In real life situations, the style of data presentation is based on the end-objective. In certain situations, the data has to be presented as a combination of two or more forms of data presentation.

Let us understand each of the above forms of data presentation with an example.

❒ DATA TABLE

Here data is presented in the form of simple table. While any type of data can be presented in table form, that too in a very accurate manner, interpreting the data in table format is very difficult and time consuming than the other modes, all of which are basically pictorial or graphical in presentation.

Data tables can be of a number of types. They can be of a single-table variety or combination of tables. Some examples of tables are given below.

Table 1.1 Movement of Goods by Different Modes of Transport (in 000's of metric-ton-kms)

Year	Road	Rail	Air	Water	Total
1985	1000	1500	120	20	2640
1986	1600	2000	129	24	3753
1987	2907	3090	139	28	6164
1988	4625	5200	152	27	10004
1989	6346	7540	174	33	14093
1990	7920	10250	212	40	18422
1991	9540	13780	266	50	23636

Note: All figures are fictitious.

From the table, we can deduce the following:

- Rate of growth by each mode of transport in successive years as well as cumulative annual growth.
- Rate of growth of total haulage by all modes of transport together in any year.
- Contribution by each mode of transport to the total haulage in any given year.
- Trends of growth over time for various modes of transport.
- Given the cost of transportation for each mode, we can calculate the total annual cost of transportation over the years for various modes of transport as well as make a cost comparison.
- Finding out the mode of transportation in any given year that forms the largest percentage of total haulage.
- For a given mode of transport, finding out the year in which the percentage increases in haulage over the previous year was the highest.

Table 1.2 Railway Time Table – Coromandel Express

Place	Cumulative mileage	Arrival Time (in hours)	Departure Time (in hours)
Madras	0	—	08.00
Nellore	200	11.20	11.30
Vijayawada	525	15.30	16.00
Rajahmundry	700	19.20	19.30
Visakhapatnam	1100	01.10	01.30
Bhubaneswar	1450	03.45	04.00
Kharagpur	1600	07.25	07.30
Calcutta	1925	09.30	—

From the above time table, we can obtain the following:

- Distance between various stations.
- Total idle time as a proportion of total travel time.
- Average speed between stations as well as over the entire journey.
- Minimum and maximum speeds of the train between two stations.

❒ PIE-CHARTS

This is probably the simplest of all pictorial forms of data presentation. Here, the total quantity to be shown is distributed over one complete circle or 360 degrees. In pie charts, data is essentially presented with respect to only one parameter (unlike in two and 3-dimensional graphs described later). This form essentially presents the shares of various elements as proportion or percentage of the total quantity. Each element or group in the pie chart is represented in terms of quantity (or value, as the case may be) or as the angle made by the sector representing the elements or as a proportion of the total or as a percentage of the total.

Figure 1.1 depicts the distribution of the population in different geographical zones.

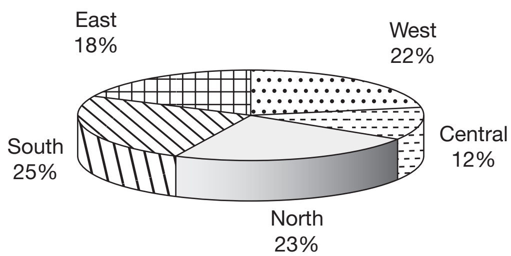


Fig. 1.1 Distribution of population in geographical zones

From the above pie chart, we can calculate the following:

- Population in any zone given the total population
- Population of any zone as a percentage of that of another zone.
- Percentage increase in the total population given the percentage increase in the population of one or more zones.

Pie charts are also very frequently used in combination with other forms of data or along with other pie charts.

❐ TWO-DIMENSIONAL GRAPHS

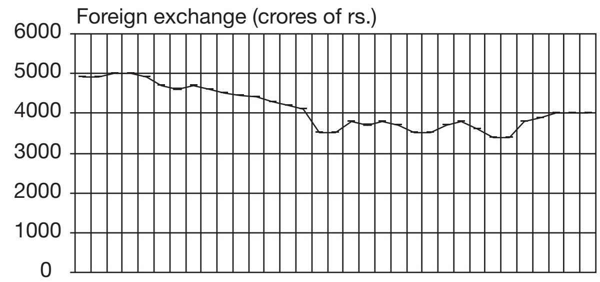


Fig. 1.2 Foreign exchange reserves of India

This is essentially used for continuous data but can also be used for depicting discrete data provided that we understand the limitation. The representation of this data is also known as Cartesian graphs, they represent the variation of one parameter with respect to another parameter each shown on a different axis. These types of graphs are useful in studying the rate of change or understanding the trends through extrapolations.

These graphs can be of various types and a few of them are shown below (Figure 1.2 to 1.4):

The graph in Figure 1.2 shows the changes in the foreign exchange reserves of our country during a period of time. One can find out the trends and the growth rates of foreign exchange reserves.

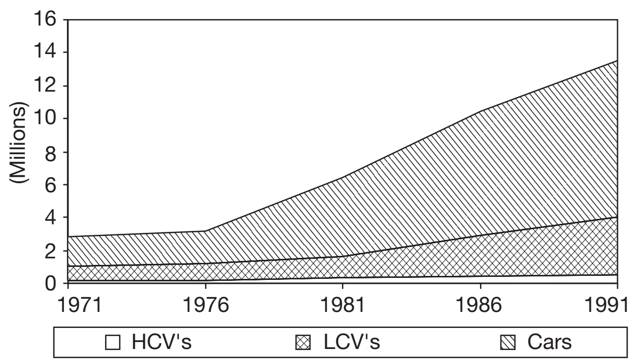


Fig. 1.3 Automobiles in India

Figure 1.3 shows a cumulative type of graph (stacked graph). This chart provides more information than the previous graph that you studied in Figure 1.2.

From the graph given in Figure 1.3, the relative proportion of different varieties of vehicles which constitute the total can be obtained along with the trends and growth rates, percentage variation, actual variations and trends for any period of time can be ascertained.

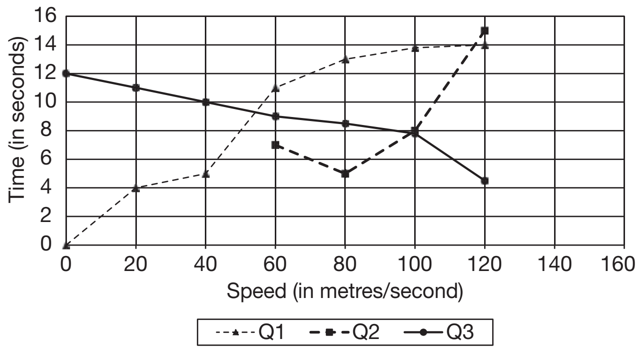


Fig. 1.4 Motion graph of Q1, Q2 and Q3

Figure 1.4 presents another type of 2-dimensional graph which is mostly used to depict scientific data like speed, velocity, vectors, etc. In the graph, the speed trends of three bodies Q1, Q2, Q3 is given.

❐ BAR CHARTS

This is a type of graph which is widely used to depict data in a discrete way. They are accurate and the comparison of variables is very convenient.

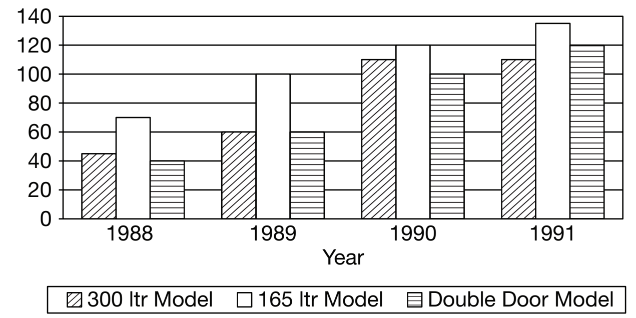


Fig. 1.5 Refrigerator sales of company abc (000's of units)

Figure 1.5 shows the model wise sales of refrigerators during four years. From this graph we can obtain the following:

- Percentage contribution of each model to the company's total sales for four years.
- Relative increase or decrease in the share of each model.
- Sales trend of various models.

Using this bar chart one can carry out a detailed performance evaluation of the company with respect to the sales of the four year period from 1988 to 1991 for any given model. These bar charts can also be depicted horizontally. Another variation could be showing each product at one place (rather than each year at one place).

❐ THREE-DIMENSIONAL GRAPH

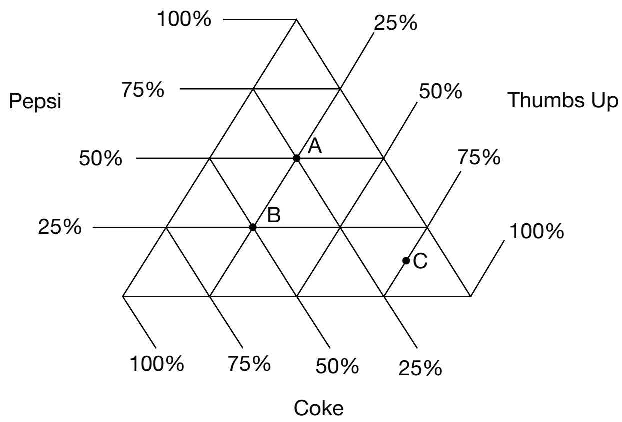


Fig. 1.6

The data (parameters) in a triangular graph are given on each side of the triangle. Each point represents a particular parameter in terms of the percentage, the same represents.

This graph represents the percentage of students who like the three colas, such as Pepsi, Thumbs Up and Coke in three colleges A, B and C.

❐ VENN-DIAGRAMS

You must be familiar with the concept of sets. Data is represented in the form of Venn diagrams when operations have to be carried out on different distinct sets of elements each following a different functional rule. All the elements in a set follow the same functional rule. By set union and intersection operations, you can establish new sets from the existing sets.

For example (Figure 1.7), consider three of the courses, such as Physics, Chemistry and Maths offered to B.Sc. students from various groups.

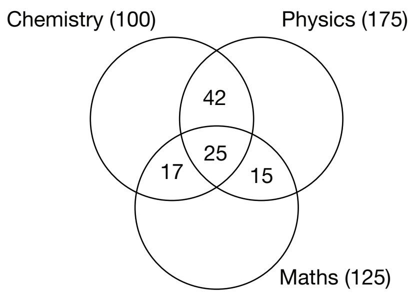


Fig. 1.7

From the chart, you can arrive at the number of students who are studying only one out of the above three subjects.

❐ PERT CHARTS

The term PERT stands for 'Project Evaluation and Review Techniques'. The progress of any project is monitored and the execution of various activities is scheduled keeping track of the resource constraints (like labour) and time constraints. For the purpose of data interpretation questions, the data may be given in the form of a table or a chart.

Let us create a table and draw a PERT chart from the table.

Table 1.3 Interior Decoration of an Office Room

The interior decoration work of an office is taken up and the activities involved along with the time taken by each activity are given below.

Activity	Duration (in weeks)	Other activities to be completed before this activity can be taken up
False roofing	2	—
Making furniture	1	—
Fixing furniture	1	False roofing, partition systems.
Fixing venetian blinds	1	Painting of doors and windows.
Fixing air-conditioner	1	—
Painting walls	1	False roofing
Partition systems	2	False roofing, laying the carpet.
Laying of the carpet	1	False roofing, painting of doors and windows, painting of walls.
Painting of doors and windows	1	False roofing

From the table, you can arrive at the minimum time after which a particular activity can be taken up or the whole task can be completed.

2 Speed Maths

❒ IMPORTANCE OF CALCULATION SPEED

Calculation speed plays a very important role in almost all the competitive exams—more so in MBA entrance exams. Some people have the natural ability to do calculations fast but, those of us who do not have good calculation speeds need not envy such people for their inborn talent. It is very easy to develop good calculation speed in a relatively short period of time. All it requires is taking care of one basic factor—that is spending a certain amount of time regularly practising calculations.

How does one improve calculation speed?

Spend just about 15 minutes a day over a three to six month period on calculation practice and you will find the difference in your calculation speed. The practice involves basic additions, subtractions, multiplications, percentage calculations, comparing fractions and calculating squares.

This practice does not need any material in the form of printed exercises or test papers. Take any figures that you can think of and work out the calculations (additions, subtractions, multiplications, etc.) mentally. What you should certainly try to ensure is that you are doing the calculations mentally wherever possible. Put away your calculators and avoid doing your calculations on paper to the extent possible.

What does this chapter consist of?

While you can always take figures at random for the purpose of practicing calculations mentally, in this chapter, we have put together a number of exercises which you can use for calculation practice.

How to gain from this chapter:

Before you move to the next unit, make sure that you revise the techniques discussed in this chapter. You should also make sure that you are thorough with the following:

Multiplication Tables (up to 20×10)
Squares (up to 25)
Cubes (up to 12)
Powers of 2 (up to 12)
Powers of 3 (up to 6)
Reciprocals of numbers (up to 12)
Complements of 100 (i.e., the difference between 100 and the given two-digit number).

While taking each exercise/test paper, please follow the instructions given below:

1. Check the starting time and keep in mind the time that has been allotted for that particular exercise.
2. Do not use a calculator.
3. Write as little as possible on paper. You should try doing as much of the calculation as possible mentally.
4. If you have to do rough work, do it in the book on the same page as the question that you are answering and not at any other place in the chapter.
5. Some questions require precise calculations whereas some other questions require only approximate calculations. Please remember that the level of accuracy to which you should work out the calculations will depend on the answer choices given in the question paper. So, do not spend more time than is necessary on each question.

6. Stop the exercise/test as soon as the prescribed time is over.
7. After you complete each exercise, spend time working out the questions that you could not complete in the given time. Then, check for the correctness of your answers. Rework all the questions in the test to see whether the method that you adopted was the best/shortest.
8. Even after you use up all the exercises given in this booklet, you should continue similar calculation practice on a regular basis to ensure that your calculation speed does not drop.

For any of the MBA entrance and similar other exams you will be appearing for, there are three areas that you have to take care of:

1. **Knowledge:** It is essential to have a certain level of knowledge in every area. It is not that a very high level of knowledge is required. A tenth or twelfth standard student should be able to answer these papers very comfortably, but nevertheless, some minimum level of knowledge is required.
2. **Speed:** One very important factor which determines success in MBA entrance exams is speed. The number of questions one can attempt correctly makes all the difference between the one who gets selected and the one who does not get selected. Speed in all areas of these exams is very important.
3. **Approach:** Knowledge alone is not sufficient to do well in these exams. For example, you cannot afford to leave out 20 questions out of 30 in a section and still hope to get selected. A person who does not take care of all the areas may not get through. This is where what we refer to as 'approach' is important in tackling the test papers. When you take comprehensive test papers, we will discuss this issue of 'approach' to test-taking.

Here, we will discuss the second of the three aspects mentioned above—speed. We will concentrate on certain speed methods of calculations which will be of great use to you in most of these exams.

As far as calculations are concerned, these exams do not allow the use of calculators or any other calculating aids. The ability to perform calculations faster is an advantage and you will solve more questions than the others in the given time. Even in your day-to-day work where you need to perform calculations, try not to use a calculator. This is a habit that you have to cultivate. If you continue using calculating aids like calculators, it is difficult to improve your calculation speed. However, please remember that any of the methods discussed in this chapter are useful only if you practice these methods regularly as well as consciously use such methods in calculations in your day-to-day work also.

❒ ADDITIONS, SUBTRACTIONS AND MULTIPLICATIONS

In this chapter, we will show you a number of calculations and take you through the different steps involved in each of the calculations. These steps are put down on paper here for the purpose of explanation but, when you are performing the calculations, you should do all these steps *mentally*.

❒ Some Ways of Simplifying Calculations

1. For multiplication by 5, you should multiply the figure given by 10 and then divide it by 2.
   E.g., $6493 \times 5 = 64930/2 = 32465$. This is a very simple method. You may feel that adopting this method will only save 5 seconds and wonder how you will benefit by it. If you adopt such methods at a number of places in the full paper and you can save even 4 to 5 minutes it will help you attempt at least 4/5 more questions. This itself may make all the difference to your chances of selection.
2. For multiplication by 25, you should multiply the figure given by 100 and divide it by 4. E.g., $6493 \times 25 = 649300/4 = 162325$.
3. For multiplication by 125, you should multiply the figure given by 1000 and divide by 8, e.g., $6493 \times 125 = 6493000/8 = 811625$.
   Alternatively, you can treat 125 as $100 + 25$. So, multiplication by 125 can be treated as multiplication by 100 and add to this figure one-fourth of itself (because 25 is one-fourth of 100).
4. For multiplication by 11, the rule is 'for each digit add the right hand digit and write the result as the corresponding figure in the product'. For the purpose of applying the rule, it will be easier if

you assume that there is one 'zero' on either side of the given number. E.g., 7469 × 11 → 0|7469|0 → 82159.

5. For multiplication by 12, the rule is 'double each digit and add the right hand digit and write the result as the corresponding digit of the product' e.g., 0|7469|0 × 12 = 89628.

   The carry forward digit has to be added to the subsequent step for multiplication by 11 or 12.

6. For multiplication by 13, the rule is 'three times each digit added to the right hand digit gives the corresponding digit in the product'. E.g., 0|92856|0 × 13 = 1207128.

7. Multiplication by 19, can be treated as multiplication by (20 – 1); e.g., 92856 × 19 = 92856 × 20 – 92856 = 1764264

   The important point to note here is that all the above calculations, after one or two examples each, should be done orally, and hence, the students also should practise accordingly. Only when large numbers are dealt with should the student put part of the figures on paper.

❐ Multiplying Two Numbers Both of Which are Close to the Same Power of 10

Suppose we want to multiply 97 with 92. The power of 10 to which these two numbers are close is 100. We call 100 as the base. Write the two numbers with the difference from the base, i.e., 100 (including the sign) as shown below.

97 → –3 (because 97 is obtained as 100 – 3)

92 → –8 (because 92 is obtained as 100 – 8)

Then, take the sum of the two numbers (including their signs) along either one of the two diagonals (it will be the same in both cases). In this example, the diagonal sum is 97 – 8 = 92 – 3 = 89. This will form the first part of the answer.

The second part of the answer is the product (taken along with the sign) of the difference from the power of 10 written for the two numbers – in this example, it is the product of –3 and –8 which is 24.

Hence, putting these two parts 89 and 24 together one next to the other, the answer is 8924, i.e., the product of 97 and 92 is 8924.

NOTE

The product of the two deviations should have as many digits as the number of zeros in the base. For example, in this case, the product of –8 and –3 has 2 digits which is the same as the number of zeroes in 100.

Two-digit/three-digit multiplication method: The usual process of multiplying two digit and three digit numbers is time consuming. For example, consider the multiplication 234 × 186 = 43524

```
   2 3 4
   1 8 6
 -------
 1 4 0 4
1 8 7 2
2 3 4
---------
4 3 5 2 4
```

In the above method, we observe that in order to find the product of 234 and 186, which is 43524, we wrote three steps (1404, 1872, 234) that are not required. By avoiding these steps we could have saved some amount of time. The amount of time saved may be only 10 seconds per calculation. However, as there will be a large number of such calculations in the exam, you will end up saving a significant amount of time by using this method.

❐ FRACTIONS AND PERCENTAGES

While solving questions on simplification, sometimes, we may come across simplification of fractions. Simplification of fractions may involve addition, subtraction, multiplication and division. In Addition as well as Subtraction of fractions, we may come across fractions with different denominators. In such cases, the denominators are to be made equal by converting the denominators to their L.C.M.

❐ RECIPROCALS AND ITS MULTIPLES

We come across a number of calculations of percentages in data interpretation and in some parts of quant. To do the calculations faster, if we can remember the reciprocals and its multiples, then we can do the calculations at a faster rate. For example, if we want to calculate 37.5% of 896, we can do it faster if we remember 37.5% (as 3/8) = 3/8 × 896 = 3 × 112 = 336.

The important reciprocals are from 1/2 to 1/12 and their multiples. Once we memorize these, upto 12, remembering its multiples is not that difficult. For example,

$1/8 = 12.5\%$; $2/8 = 2 \times 1/8 = 2 \times 12.5\% = 25\%$; $3/8 \Rightarrow 3 \times 1/8 = 3 \times 12.5 = 37.5\%$; $4/8 \Rightarrow 4 \times 1/8 = 4 \times 12.5 = 50\%$ or $4/8 = 1/2 = 50\%$; $5/8 \Rightarrow 5 \times 1/8 = 5 \times 12.5 = 62.5\%$; $6/8 \Rightarrow 3/4 = 75\%$; $7/8 \Rightarrow 7 \times 1/8 = 7 \times 12.5 = 87.5\%$.

Similarly, we can remember all the multiples of reciprocals upto 12.

Conversion of fractions to percentages

1/2 = 50%,	1/3 = 33.33%,	1/4 = 25%,
	2/3 = 66.66%,	3/4 = 75%,
1/5 = 20%,	1/6 = 16.66%,	1/7 = 14.28%
2/5 = 40%,	5/6 = 83.33%,	2/7 = 28.57%,
3/5 = 60%,		3/7 = 42.85%,
4/5 = 80%,		4/7 = 57.13%,
		5/7 = 71.42%,
		6/7 = 85.72%,
1/8 = 12.5%,	1/9 = 11.11%,	1/11 = 9.09%,
3/8 = 37.5%,	2/9 = 22.22%,	2/11 = 18.18%,
5/8 = 62.5%,	4/9 = 44.44%,	3/11 = 27.27%,
7/8 = 87.5%,	5/9 = 55.55%,	4/11 = 36.36%,
	7/9 = 77.77%,	5/11 = 45.45%,
	8/9 = 88.88%,	6/11 = 54.54%,
		7/11 = 63.63%,
		8/11 = 72.72%,
		9/11 = 81.81%,
		10/11 = 90.9%,
1/12 = 8.33%,		
5/12 = 41.66%,		
7/12 = 58.33%,		
10/12 = 83.33%,		
11/12 = 91.66%		

It will be very useful to memorize all the above values as it will help us to do the calculations very fast.

❐ Percentage Calculations

In calculating the percentage value of a number, we usually go for multiplication. But that does not give the answer easily and quickly in most cases. Hence, an easier method called 10% concept, is suggested. In this approach, we take 10% of the denominator. To get close to the answer take further values like 1% and 0.1%.

For example: $23\% = 10\% \times 2 + 1\% \times 3$

$43.2\% = 10\% \times 4 + 1\% \times 3 + 0.1\% \times 2$.

The following is the illustration of the same.

How to calculate the value of 36% of 1325?

Here, explain the concept of 10% and 1%. Therefore, for any value, say 1264, 10% of the value is obtained by simply shifting the decimal point by one place (or digit) to the left. Note that 1264 = 1264.0

10% of 1264.0 = 126.40 (i.e., the decimal point moves to the left by one place (or digit)). Similarly, 1% of 1264.0 will be obtained by shifting the decimal point by two places to the left. Hence, 1% of 1264.0 = 12.640.

Hence, 36% of 1325 = (40% – 4%) of 1325 = $(4 \times 10\% - 4 \times 1\%)$ of 1325 = $(4 \times 132.5 - 4 \times 13.25) = 530 - 53 = 477$.

Similarly, consider another example, say, 18% of 3250 = (20% – 2%) of 3250

$= (2 \times 10\% - 2 \times 1\%)$ of $3250 = (2 \times 325 - 2 \times 32.5) = 585$.

If there is a 10% increase, then the new value will become 1.1 times the old value and in general if there is an increase of $p\%$, the new value will become $\left(1 + \frac{p}{100}\right)$ times the old value. But sometimes converting the percentage into fraction maybe easier than this if there is an increase of 33.33%, then the new value will be 4/3 times the old value. Calculating in this way converting $33^1/_3$ into a fraction and simplifying is faster.

Whenever percentage increase cannot easily be converted into a convenient fraction, then the approximate percentage increase p, in integer form, must be found and then $1.p$ has to be used.

❐ COMPARISON OF FRACTIONS

Comparison of fractions will be required in a number of problems in Data Interpretation and Quantitative Ability. Let us study some of the common methods of identifying out the largest or smallest of a given set of fractions.

Type 1: When two or more fractions have the same numerators and different denominators, the fraction with the largest denominator is the smallest.

Type 2: When the numerators are different and the denominators are same, the fraction with the largest numerator is the largest.

Type 3: The fraction with the largest numerator and the smallest denominator is the largest.

Type 4: When the numerators of two fractions are unequal, we try and equate them by suitably cancelling factors or by suitably multiplying the numerators. Thereafter we compare the denominators as in Type 1.

Type 5: *A*: *For a fraction less than* 1.

If the difference between the numerator and the denominator is same, then the fraction with the larger values of numerator and denominator will be the largest.

B: *For a fraction greater than* 1.

If the difference between the numerator and denominator is same, then the fraction with the smaller values will be the largest.

Type 6: Another method of comparing fractions is by comparing the percentage changes in denominators and numerators. The important points to remember are that when two fractions are compared, if the percentage increase in the numerator is more than the percentage increase in the denominator (where the first fraction is taken as reference), then the second fraction is greater than the first fraction. Instead, if the percentage increase in the denominator is greater than that in the numerator, then the second fraction is smaller than the first.

❐ APPROXIMATIONS

'Approximate calculation' is one of the approaches in solving a problem / arriving at the answer to a question at a faster rate. With the help of approximate calculations, one can save a lot of time and this can be utilized in other areas.

In most cases in various exams, the approach towards a question depends on the answer choices. From the answer choices, one should decide which method to follow—actual calculations or approximate calculations.

In most exams, for solving questions based on simplifications, data interpretation, ratios, percentages, etc., the use of approximations is very handy for solving the question at a faster pace. Approximation in any calculation depends on the degree of accuracy required. The closer the given answer choices, the greater the need for closer approximation.

Example: For which of the following values is the increase the highest?

(a) 3164 to 4072

(b) 2422 to 3218

(c) 4234 to 5866

(d) 1876 to 2761

In order to solve the above question, if we calculate accurately, we will get 4072 – 3164 = 908; 3218 – 2422 = 796; 5866 – 5234 = 632; 2761 – 1876 = 885.

If we try approximate calculations,

Rounded off to 4072 4100

Rounded off to 3164 3200

Here, the subtraction is very simple.

4100 – 3200 = 900.

During the process of rounding off, if the last two digits are 50 or greater than 50, then the figure must be rounded off to the next highest hundred. Otherwise, it should be rounded off to the next lowest hundred. Thus, 3164 would be rounded off to 3200, while 4072 would be rounded off to 4100. In fact, in such calculations, even the hundreds (i.e., the last two zeroes) need not be considered since the two zeroes are present in every case. This means that the above calculation would be further simplified by mentally treating it as 41 – 32 = 9. Other calculations can be done in a similar manner.

In case of multiplications like 389 × 1456, suppose the answer choices are given as below:

(A) 564322 (B) 565400

(C) 566384 (D) 572356

We can go for 390 × 1450 which is 565500 whereas the actual answer here is 566384. This is far from the actual answer. But as none of the answer choices lie between these values, this is the required answer. In the above calculation, only one answer ends with 4; hence, without actually calculating, we can say that Choice (C) is the answer.

Suppose the answer choices are closer, like

(A) 565424 (B) 566644

(C) 566384 (D) 572354

then the above approximation will not be useful. Then, the approach should be 1455 × 389 which gives us 565995. Hence, the answer should be very much close to 565995 but should be more than that which is 566384.

Suppose we have to calculate 37.22% of 1384.

The actual calculation takes around 45 seconds to 75 seconds, depending on the speed of the person.

Suppose the answer choices for the above calculation are mentioned as

(A) 564 (B) 515 (C) 529 (D) 542

As the answer choices are not very close, calculating for 40% and reducing that by 3% may be sufficient for answering the question.

10% of 1384 = 138.4. Four times that is ~554 and 3% is approximately equivalent to 42. Hence, the answer is ~554 – 42, i.e., ~512.

Therefore, the closest answer is 510.

Suppose the answer choices for the above calculation are mentioned as:

(A) 510.264 (B) 515.124
(C) 519.316 (D) 522.356

It is clear that the answers are very close. But if you approximate 37.22% equivalent to 37.5%, you can simply convert the calculation into 3/8 of 1384. As 1384 goes 173 times, the answer is 519. Hence, 519.316 cannot be the answer, as it should be less than 519 but very close to 519, which is 515.124.

If the answer choices are even closer than the above example, we go for subtraction of 28%, which is approximately $1/400^{th}$ part of 1384. Hence, by using approximate calculations we can answer questions at a faster rate.

Now, let us consider another question.

Find the value of $\dfrac{5843}{31200} \times 100$

(A) 17.56 (B) 18.38 (C) 18.72 (D) 16.96

This ratio can be calculated faster by two very useful methods than by conventional division.

The first approach can be called the '*ten percent method*'. In this method, 10% of the denominator is first obtained by simply shifting the decimal point in the denominator it by one place to the left. Similarly, 1% of the denominator, 0.1%, etc., can also be successively obtained by shifting the decimal point to the left by one more place in each successive step. Then, the numerator is expressed as the nearest possible multiple of 10% of the denominator along with some excess or shortfall.

For example,

10% of 31200 = 3120

and

$5843 = 1 \times 3120 + 2723$ (excess)

Again, the excess of 2723 can be expressed as a multiple of say,

5% of denominator plus some excess.

Half of 10% of 31200 = 5% of 31200 = 1560

Hence, 2723 = 5% of 31200 + 1163

Further, 1163 is slightly less than 4 times (1% of 31200), i.e., 1248.

Thus,

$5843 \cong (10\% + 5\% + 4\%)$ of $31200 \cong 19\%$ of 31200.

The correct figure must be slightly less than 19% of 31200. Hence, from the choices, the answer can be Choice (C).

The second approach to quickly calculate the ratio $\dfrac{5843}{31200}$ is by using of the decimal equivalent values of the reciprocals of the first few natural numbers. In this approach, the numerator and denominator are first approximated as $\dfrac{5800}{31200}$, which is further approximated to $\dfrac{5.8}{31.2}$ which is close to $\dfrac{5.8}{3 \times 10.4}$

(i.e., 4% less than 19.3),

i.e., 18.54. But since in the first approximation we had taken 5800 instead of 5843, the answer has to be slightly more than 18.54. Thus, Choice (C) is the answer.

Find the value of $\dfrac{6164}{26879} \times 100$

(A) 21.68 (B) 22.16 (C) 22.93 (D) 23.37

By observing the given choices, we understand that the answer should be close to 22.22% (i.e., 2/9) (Here, one should remember reciprocals and their multiples). The calculation is:

$\dfrac{2}{9} \times 26879 = 5973$. As 5973 is about 190 less than 6164, we need to add about 190 to 5973. But as we are interested in percentages, 190 forms slightly more than 0.5% but less than 1%. The answer should be more than 22.7% but less than 23.22%. From the choices, only choice (C) is satisfied.

Find the value of $(2911 / (3784 \times 4)) \times 100$.

(A) 17.86 (B) 18.15 (C) 21. 76 (D) 19.23

$$\text{Required value} = \frac{\left(\dfrac{6695 - 3784}{3784}\right) \times 100}{4}$$

This is approximated as:

$$\frac{6700 - 3800}{3800 \times 4} \times 100$$

$$= \frac{2900}{38 \times 4} = \frac{2900}{152}$$

$\dfrac{2900}{152}$ is slightly less than $\dfrac{3000}{150} = 20$

Thus, 19.23% is close to 20%. Therefore Choice (D) is correct.

Approximations for divisions can be done in two ways. The first one is cross multiplication. The examples show how to solve an approximation problem using cross multiplication.

Therefore, we understand that approximations are very useful in additions, subtractions, multiplications, divisions, percentage calculations, etc.

❐ BODMAS—HIERARCHY OF ARITHMETIC OPERATIONS

To simplify arithmetic expressions, which involve various operations like brackets, multiplication, addition, etc. a particular sequence of the operations has to be followed. For example, $2 + 3 \times 4$ has to be calculated by multiplying 3 with 4 and the result 12 added to 2 to give the final result of 14 (you should not add 2 to 3 first to take the result 5 and multiply this 5 by 4 to give the final result as 20). This is because in arithmetic operations, multiplication should be done first before addition is taken up.

The hierarchy of arithmetic operations are given by a rule called BODMAS rule. The operations have to be carried out in the order in which they appear in the word BODMAS, where different letters of the word BODMAS stand for the following operations:

(V Vinculum)
B Brackets
O Of
D Division
M Multiplication
A Addition
S Subtraction

There are four types of brackets:

1. **Vinculum:** This is represented by a bar on the top of the numbers. For example,

   $2 + 3 - \overline{4 + 3}$; Here, the figures under the vinculum have to be calculated as 4 + 3 first and the 'minus' sign before 4 is applicable to 7. Thus the given expression is equal to 2 + 3 – 7 which is equal to –2.
2. **Simple Brackets:** These are represented by ( )
3. **Curly Brackets:** These are represented by { }
4. **Square Brackets:** These are represented by [ ]

The brackets in an expression have to be opened in the order of vinculum, simple brackets, curly brackets and square brackets, i.e., [{ (¯) }] to be opened from inside outwards.

After brackets is *O* in the BODMAS rule standing for 'of' which means multiplication. For example, 1/2 of 4 will be equal to 1/2 *x* 4 which is equal to 2.

After *O*, the next operation is *D* standing for division. This is followed by *M* standing for multiplication. After Multiplication, *A* standing for addition will be performed. Then, *S* standing for subtraction is performed.

❐ Squares and Cubes

In competitive examinations, there can be questions on direct application of squares, cubes, square-roots and cube-roots. For example, there can be a question which asks you to find the tens-digit of a four-digit perfect square. Also, an understanding of squares and cubes of useful while performing calculations.

Remembering squares (upto first 25 natural numbers), cubes (upto first 12 natural numbers) is very important in calculations. By remembering these (squares upto 25), one can calculate squares of any natural number from 26 to 125 in no time, which in turn will help in solving some other questions too. Similarly, by remembering cubes (upto 12) one can calculate cubes of any two-digit number with greater speed. Given below are some methods for finding squares and cubes of numbers.

How to find the square of a number ending in 5:

Getting the square of a number ending in 5 is very simple. If the last digit of the number is 5, the last two digits of the square will be 25. Consider the earlier part of the number and multiply it with one more than itself and that product will be the first part of the answer. (The second part of the answer will be 25 itself.)

$35^2 = 1225$ (Here, $3 \times 4 = 12$, so, the answer is 1225)
$45^2 = 2025$
$55^2 = 3025$
$75^2 = 5625$
$95^2 = 9025$
$125^2 = 15625$
$175^2 = 30625$
$195^2 = 38025$
$235^2 = 55225$
$245^2 = 60025$

So, now we know the squares of numbers 35, 45, 55, 75, etc. If we want to find the square of any other number ending in 5, we can find it using these squares which we already know.

To find the square of a number which is one more than the number whose square we already know:

For 26^2, we will go from 25^2; for 31^2 we go from 30^2 and so on.

One way is by writing $26^2 = (25+1)^2$. But we need not even calculate $(a + b)^2$ by adopting the following method;

$26^2 = 25^2 + 26^{th}$ odd number, i.e., $625 + 51 = 676$
$(a + b)^2 = a^2 + 2ab + b^2$

$26^2 = (25 + 1)^2$

$(25)^2 + 2(25 \times 1) + (1)^2$

$625 + 50 + 1 = 625 + 51 = 676$

But we will look at a different method which will enable the student perform the calculations for squares mentally.

$1^2 = 1 = 1$

$2^2 = 4 = 1 + 3$

$3^2 = 9 = 1 + 3 + 5$

$4^2 = 16 = 1 + 3 + 5 + 7$

$5^2 = 25 = 1 + 3 + 5 + 7 + 9$

i.e., to get n^2, we add up the first n odd numbers. If we want 13^2, it will be the sum of the FIRST 13 odd numbers.

n^{th} odd number is equal to $(2n - 1)$.

Suppose we want to find out 6^2, knowing what 5^2 is, we can move from 5^2 to 6^2.

6^2 will be the sum of 1^{st} 6 odd numbers. But the sum of the first 6 odd numbers can be written as 'sum of the first 5 odd numbers' + 'sixth odd number'. Since we already know that the sum of the first 5 odd numbers is 5^2, i.e., 25, we need to add the sixth odd number, i.e., $(2 \times 6 - 1 =)$ 11 to 25 to give us $6^2 = 36$.

Similarly

$31^2 = 900 + 31\text{st odd number} = 900 + 61 = 961$

$36^2 = 1225 + 36^{th}\text{ odd number} = 1225 + 71$

$= 1296$ (Since $35^2 = 1225$)

$41^2 = 1600 + 81 = 1681$

$46^2 = 2025 + 91 = 2116$

$126^2 = 15625 + 251 = 15876$

$196^2 = 38025 + 391 = 38416$

$216^2 = 46225 + 431 = 46656$

We have now seen how to find the squares of numbers which are one more than those numbers whose squares we already know (e.g., 25, 30, 35, etc.)

To find the square of a number which is one less than the number whose squares we already know

Similarly, we can find the squares of numbers which are one less than the numbers whose squares are known. For example,

$29^2 = 30^2 - 30^{th}\text{ odd number}$

$= 900 - 59 = 841$

$39^2 = 40^2 - 40^{th}\text{ odd number} = 1600 - 79 = 1521$

$34^2 = 1225 - 69 = 1156$

$54^2 = 3025 - 109 = 2916$

$74^2 = 5625 - 149 = 5476$

$94^2 = 9025 - 189 = 8836$

$214^2 = 46225 - 429 = 45796$

Thus, we have seen how to arrive at the squares of numbers which are one more or one less than the numbers whose squares we already know (i.e., 25, 30, 35, 40, 45, 50, 55, etc.).

To find the square of a number which is 2 more than the number whose squares we already know:

Now, we will see how to get the squares of numbers which are 2 more (or less) than the numbers whose squares we already know.

$27^2 = 26^2 + 27^{th}$ odd number $= 25^2 + 26^{th}$ odd number + 27^{th} odd number.

The sum of the 26^{th} odd number and 27^{th} odd number is the same as 4 times 26. Hence,

$27^2 = 25^2 + 4 \times 26 = 625 + 104 = 729$

$57^2 = 3025 + 224$ (4 times 56) $= 3249$

$77^2 = 5625 + 304$ (4 times 76) $= 5929$

$97^2 = 9025 + 384$ (4 times 96) $= 9409$

To find the square of a number which is 2 less than the number whose squares we already know

Similarly, we can find out the squares of numbers

which are 2 less than the numbers whose squares we know.

$28^2 = (30^2 - 4 \text{ times } 29) = 900 - 116 = 784$

$53^2 = (55^2 - 4 \text{ times } 54) = 3025 - 216 = 2809$

$93^2 = 9025 - 376 = 8649$

$243^2 = 60025 - 976 = 59049$

$143^2 = (145^2 - 4 \times 144) = 21025 - 576 = 20449$

To find the square of a number from 26 to 50

The squares of numbers from 26 to 50 can be calculated by writing down and adding two parts as explained below:

The first part is as many times 100 as the number is more than 25, for example in finding 31^2, as 31 is 6 more than 25, the first part is $100 \times 6 = 600$.

The second part is the square of the number that is as much less than 25 as the number is more than 25, i.e., in finding 31^2, the second part is the square of 6 less than 25, i.e., $(25 - 6)^2 = 19^2 = 361$.

Hence, 31^2 = First part + Second part = 600 + 361 = 961.

The above method can be summarized as

1. Finding 31^2
   (i) $31 = 25 + 6$
   (ii) $25 - 6 \rightarrow 19^2 \rightarrow 361$
   (iii) $31^2 = 6 \times 100 + 361 = 961$
2. Finding 33^2
   (i) $33 = 25 + 8$
   (ii) $25 - 8 \rightarrow 17^2 \rightarrow 289$
   (iii) $31^2 = 8 \times 100 + 289 = 1089$

To find the square of a number from 51 to 75:

The squares of numbers from 51 to 75 can be calculated by writing down two parts, each of which is a two-digit number, adjacent to each other as explained below:

The second part is the two-digit number formed by the two digits which are to the extreme right of the square of the number by which the given number is more than 50.

For example, in finding 63^2, as 63 is 13 more than 50, the second part will be the two digits to the extreme right of 13^2 (= 169), i.e., 69. Since there are more than two digits in 13^2, the digit to the extreme left, i.e., 1, is taken as carry forward and is to be added to the first part.

The first part is the sum of (i) the carry forward, if any, from the second part and (ii) the sum of 25 (for this range (i.e., 51 to 75) 25 is taken as the base) and the number by which the given number is more than 50.

As 63 is 13 more than 50, the first part will be 25 + 13 + 1 (carry forward from the second part) = 39

Therefore $63^2 = 39\ 69$

The above can be summarized as

First Part Second Part

1. $63^2 = (25 + 13)\ /\ 13^2 = 38\ /\ 69 = 38 + 1\ /\ 69 = 39\ 69$
C.F

First Part Second Part

2. $61^2 = (25 + 11)\ /\ 11^2 = 36\ /\ 21 = 37\ 21$
C.F

First Part Second Part

3. $56^2 = (25 + 6)\ /\ 6^2 = 31/36 = 31\ 36$

NOTE

This process is faster for squares in the range of 50 to 60, as there is no carry forward.

To find the square of a number from 76 to 100:

The squares of numbers from 76 to 100 can be calculated by writing down two parts, each of which is a two-digit number, adjacent to each other as explained below:

The second part is the two-digit number formed by the two digits to the extreme right of the square of the number by which the given number is less than 100.

For example in 88^2, as 88 is 12 less than 100, the second part will be the two digits to the extreme right of 12^2(= 144), i.e., 44. Since there are more than two digits in 12^2, the digit to the extreme left, i.e., 1, is taken as carry forward and is to be added to the first part.

The first part is the sum of (i) the carry forward, if any, from the second part and (ii) the difference between the given number and the number by which the given number is less than 100. As 88 is 12 less than 100, the first part will be 88 – 12 + 1(carry forward from the second part) = 77.

Therefore, $88^2 = 77\ 44$

The above can be summarized as

First Part Second Part

1. $88^2 = (88 - 12)\ /\ 12^2 = 76\ /\ 44 = (76 + 1)\ /\ 44 = 77\ 44$
C.F

First Part Second Part

2. $89^2 = (89 - 11)\ /\ 11^2 = 78\ /\ 21 = 79\ 21$
C.F

First Part Second Part

3. $96^2 = (96 - 4)\ /\ 4^2 = 92/16 = 92\ 16$

NOTE

This process is faster for squares in the range of 90 to 100, as there is no carry forward.

To find the square of a number from 101 to 125:

The squares of numbers from 101 to 125 can be calculated by writing down two parts, each of which is a two-digit number, adjacent to each other as explained below:

The second part is the two-digit number formed by the two digits to the extreme right of the square of the number by which the given number is more than 100.

For example, finding 112^2, as 112 is 12 more than 100, the second part will be the two digits of 12^2(= 144), i.e., 44. Since there are more than two digits in 12^2, the digit to the extreme left, i.e., 1, is taken as carry forward and is to be added to the first part.

The first part is the sum of (i) the carry forward, if any from the second part and (ii) the sum of the given number and the number by which the given number is more than 100. As 112 is 12 more than 100, the first part will be 112 + 12 + 1 (carry forward from the first part) = 125.

Therefore, $88^2 = 125\ 44$

The above can be summarized as

First Part Second Part

1. $112^2 = (112 + 12)\ /\ 12^2 = 124\ /\ 44 = (124 + 1)\ /\ 44 = 125\ 44$
C.F

2. $113^2 = (113 + 13) \ / \ 13^2 = 126 \ / \ 169 = 127 \ 69$

C.F

3. $106^2 = (106 + 6) \ / \ 6^2 = 112/36 = 112 \ 36$

By observing and remembering a few properties regarding the behaviour of the last digits of numbers and of their squares and cubes, it is sometimes possible to solve certain kinds of questions. Hence, some of the important properties of the last digits of numbers are given below:

Last digit of any number 0 1 2 3 4 5 6 7 8 9
Last digit of its square 0 1 4 9 6 5 6 9 4 1
Last digit of its cube 0 1 8 7 4 5 6 3 2 9

Some important observations:

1. The square of a number can never end with 2, 3, 7 or 8.
2. Any power of any number ending in 0, 1, 5 or 6 ends with 0, 1, 5, 6, respectively.
3. If the last digits of two numbers are 10's complements, then the last digits of their squares will be equal. Hence, if the last digit of the square of a number is given, it is not possible to determine the last digit of that number uniquely. For example if n^2 ends with 9, n may end with 3 or 7.
4. We can uniquely determine the last digit of a number given the cube of that number, for example $(\ldots)^3 = \_\_3$, the number can end only in 7.
5. If the last digits of two numbers are 10's complements, then last digits of their cubes will be also 10's complements.
6. The square of a number with only n 1's, where n is a single digit number, will always be a palindrome. For example:
   112 = 121; 1112 = 12321; 11112 = 1234321. In general, $(1111\ldots \ n \text{ times})^2 = 123\ldots n^{n-1}\ldots 1$
7. The last two digits of any power of a number ending in 25 or 76 always end in 25 and 76, respectively.
8. (a) If the square of any number is ending in 1, then the ten's digit of that square should be an even number.
   (b) If the square of any number is ending in 4, then the ten's digit of that square should be an even number.
   (c) If the square of any number is ending in 5, then the ten's digit of that square should be 2.
   (d) If the square of any number is ending in 6, then the ten's digit of that square should be an odd number.
   (e) If the square of any number is ending in 9, then the ten's digit of that square should be an even number.

❐ Powers of 2 and 3

Remembering powers of 2 up to 12 and powers of 3 up to 8 will be of great help. It has been observed that various competitive examinations have direct questions on the application of these.

Property for the powers of 2:

$2^0 = 1$, $2^1 = 2$, $2^2 = 4$, $2^3 = 8$
$2^4 = 16$, 25 = 32, $2^6 = 64$, $2^7 = 128$
$2^8 = 256$, $2^9 = 512$, $2^{10} = 1024$, $2^{11} = 2048$

By observing the following, we can see that

$2^0 + 2^1 = 3 = 2^2 - 1$
$2^0 + 2^1 + 2^2 = 7 = 2^3 - 1$
$2^0 + 2^1 + 2^2 + 2^3 = 15 = 2^4 - 1$

Similarly,

$2^0 + 2^1 + 2^2 + 2^3 + 2^4 = 31 = 2^5 - 1$

That is, the sum of powers of 2 from 0 to any number k will be equal to $2^{k+1} - 1$.

The above concept can be used in the following example:

For example, the sum $2^0 + 2^1 + \ldots 2^n$ is equal to $(2^{n+1} - 1)$. This can help us arrive at the answer to a question like 'If ten brothers have some marbles each, such that every brother, except the youngest, has twice the number of marbles than that the brother immediately younger to him has, then find the least possible total number of marbles with the ten brothers'.

To have the least total, the youngest should have the least number of marbles, i.e., only one marble.

The second youngest will have 2 (i.e., 2^1), the next brother will have 4 (i.e., 2^2) and so on.

The eldest will have 2^9. The sum of all the marbles with them will be $2^0 + 2^1 + 2^2 + \ldots 2^9 = 2^{9+1} - 1 = 2^{10} - 1 = 1024 - 1 = 1023$.

Property for the powers of 3:

$3^0 = 1$, $3^1 = 3$, $3^2 = 9$, $3^3 = 27$
$3^4 = 81$, $3^5 = 243$, $3^6 = 729$, $3^7 = 2187$
$3^0 + 3^1 + 3^2 + 3^3 = 40$

Using a combination of these numbers, each occurring at the most once, we can obtain all the numbers from 1 to 40 by using the operation of only addition and/or subtraction.

The above concept can be used in the following example.

SOLVED EXAMPLES

2.01. $342 + 557 + 629 + 746 + 825 = ?$

Sol: When we are adding three-digit numbers, first add two-digits at a time (units and tens place).

$42 + 57 + 29 + 46 + 25 = 199.$

To add 42 and 57, mentally treat 57 as $50 + 7$ (50 would facilitate quick addition).

Thus, $42 + 57 = (42 + 50) + 7 = 92 + 7 = 99.$

Similarly, $99 + 29 = (99 + 20) + 9 = 128.$

$128 + 46 = (128 + 40) + 6 = 174.$

$174 + 25 = (174 + 20) + 5 = 199.$

The last two digits (the units place and the tens place) of the addition are 99, while the digit 1 is to be carried forward).

Now add

1(carried) $+ 3 + 5 + 6 + 7 + 8 = 30.$

$\therefore$ The result of the addition is 3099.

The same logic can be extended to four-digit additions.

2.02. $6965 + 3246 + 1234 + 9847 + 8238 = ?$

Sol: Part II

[2* + 69]	= 71	69
[(71 + 30) + 2]	= 103	32
[(103 + 10) + 2]	= 115	12
[(115 + 90) + 8] **	= 213	98
[(213 + 80) + 2]	= 295	82
		295

Part I

65		
46	[(65+40)+ 6]	= 111
34	[(111+30) + 4]	= 145
47	[(145+40) + 7]	= 192
38	[(192+30)+ 8]	= 230 from here, we carry forward 2
30		

[* The 2 shown here is the carry forward indicated at bottom-right].

[** Alternatively, this calculation can be performed as $115 + 100 - 2 = 215 - 2 = 213$].

2.03. $1598 + 5423 + 4627 + 7953 + 8675 = ?$

Sol: Part II

(2* + 15)	= 17	15
[(17 + 50) + 4]	= 71	54
[(71 + 40) + 6]	= 117	46
[(117 + 70) + 9]	= 196	79
[(196 + 80) + 6]	= 282	86
		282

Part I

98		
23	[(98 + 20) + 3]	= 121
27	[(121 + 20) +7]	= 148
53	[(148 + 50) + 3]	= 201
75	[(201 + 70) + 5]	= 276
76		

2.04. $987 - 256 = ?$

Sol: Instead of taking a single digit at a time, subtractions would be faster by taking two digits i.e.,

$87 - 56 = 31.$

$900 - 200 = 700$

$\therefore$ The result of $987 - 256 = 731$

2.05. $824 - 587 = ?$

Sol: Take 100s complement of 87 (i.e., $100 - 87$) which is 13 and add it to 24. The result is 37. This gives the units and tens digits of the result. Since $24 < 87$, we have actually subtracted 87 from 124, i.e., we have borrowed 1 from 8 (of 824). Therefore we now do $(7 - 5) = 2$. The result is 237.

2.06. $9217 - 858 = ?$

Sol: Adding 100s complement of 58 (which is 42) to 17, we get $(42 + 17) = 59$ which gives the units and 10s digits of the result.

Since 58 is greater than 17, we have to borrow 1 from 92 which leaves us with 91. So, the first part of the answer is $91 - 8 (= 83)$

Hence, the result is 8359.

2.07. $934 - 286 + 847 - 798 = ?$

Sol: When we have a combination of additions and subtractions, first add all the numbers with + sign before them and add all the numbers with – sign before them.

i.e., $(934 + 847) - (286 + 798) = 1781 - 1084.$

By applying the method explained in previous examples, $1781 - 1084 = 697.$

2.08. Find the product of 113 and 118.

Here, both the numbers are greater than 100 and the base here is 100. Taking the difference of the two numbers 113 and 118 from the base, we get +13 and +18 and write them as below.

113	→	+ 13
118	→	+ 18
131		234

The first part of the answer is the cross-total of 113 and +18 which is 131. The second part of the answer, i.e., the product of the deviations (+13 and +18) is equal to 234. But we said there should be as many digits in this product as the number of zeroes in the base (which is 100 here). Since the base has two zeroes, the second part of the answer should also have two digits. Since 234 has three digits, we should retain two digits 4 and 3 and carry forward the third digit 2 to the first part of the answer. Hence, the first part of the answer now becomes 133 and the second part is 34. The product of 113 and 118 is thus equal to 13334.

2.09. Find the product of 109 and 93.

Here, one number is greater than 100 and the other is less than 100. Write the differences from 100 (the closest power of 10) along with the sign of the deviation.

109	→	+9	
93	→	–7	
102		–63	Ans. 10137

The first part of the answer is the cross-total (of 109 and –7 or of 93 and +9) 102. The second part of the answer is the product of +9 and –7 which is –63. Since we cannot have a negative figure as a part of the answer, we need to convert this to a positive number. For this purpose, we borrow the necessary figure from the first part of the answer. Each unit borrowed from the first part of the answer, when it is brought to the second part, becomes equal in value to the base used. If we borrow 1 from the first part (102 here), we are left with 101 for the first part and the 1 that is borrowed becomes 100 for the second part. The second part now is 100 (borrowed) plus -63 (originally there) which is equal to 37. The final result is obtained by putting the first and the second part together. Hence, the product of 109 and 93 is 10137.

2.10. Find the product of 117 and 88.

117	→	+17	
88	→	–12	
105		–204	Ans. 10296

Please note that to take care of –204 of the second part, borrowing a 1 from the first part is not sufficient (because the 100 it becomes when it comes to the second part is not numerically greater than –204). So, we should borrow 3 from 105 (leaving 102 as the first part) which becomes 300 in the second part to which –204 should be added giving us 96. Hence, the product of 117 and 88 is 10296.

2.11. Find the product of 997 and 983.

Here, both the numbers are close to 1000 – they are both less than 1000.

997	→	–3	
983	→	–17	
980		+51	Ans. 980051

The second part 51 has only two digits whereas the base 1000 has three zeroes—so, 51 will be written as 051. Hence, the product is 980051.

2.12. Find the product of 1013 and 981.

1013	→	+13	
981	→	–19	
994		–247	Ans. 993753

The second part is –247 and if we borrow 1 from the first part (the first part itself will then become 993), it becomes 1000 in the second part. So, the second part will effectively be 1000 – 247 = 753. Since the base is 1000, the second part should have three digits and 753 has three digits. Hence, the product of 1013 and 981 is 993753.

We can also extend this method to find the product of two numbers which may not be close to a power of 10 but both of which are close to a multiple of a power of 10. This requires a little bit of modification to the method as discussed in the examples below.

2.13. Find the product of 297 and 292.

Here, the numbers are not close to any power of 10 but are close to 300 which is a multiple of 100 which itself is a power of 10. So, we adopt 300 as a 'temporary base'. This temporary base is a multiple (or a sub-multiple) of the main base 100. Here, the temporary base 300 = 3 × 100. Then, the procedure of finding out the deviation from the base, getting the cross-totals and the product of the deviations should be done in a manner similar to the previous cases except that the deviations will be taken from the temporary base.

297	→	–3	(289 × 3 = 867)
292	→	–8	
289		+24	Ans. 86724

We have got the first part of the answer as 289 and the second part of the answer as 24. But before we put these two parts together to get the final result, one more step is involved. The first part of the answer is not the final figure—this is an intermediate stage of the first part. This first part should be multiplied by the same figure with which the power of 10 is multiplied to get the temporary base. In this case, we multiplied 100 (which is the power of 10) by 3 to get the temporary base 300. So, the intermediate stage figure of the first part (289) will also have to be multiplied by 3 to get the final figure for the first part. Hence, the first part will be 867 (= 3 × 289). Now putting the first and the second parts together, the product of 297 and 292 is 86724 (Please note that the product of the deviations should still have as many digits as the number of zeroes in the base—in this case two because 100 has two zeroes).

2.14. Find the product of 287 and 281.

287	→	–13	(268 × 3 = 804)
281	→	–19	
——		——	
268		247	Ans. 80647

Here, the product of the deviations is 247—there are three digits in this whereas the base has only two zeroes. So, the digit 2 has to be carried forward to the first part of the answer but this carrying forward should be done only after the intermediate stage figure of the first part is multiplied suitably to get the final figure of the first part (in this case, 268 multiplied by 3 gives 804 as the first part of the answer). To this add 2 which is the carry forward digit from the second part and we get 806. Hence, the product of 287 and 281 is 80647.

2.15. Find the product of 317 and 291.

317	→	+17	(3 × 308 = 924)
291	→	–9	
——		——	
308		–153	Ans. 92247

Here, since one number is greater than 300 and the other is less than 300, the product of the deviations is negative. To make the second part positive, we need to borrow from the first part. But the borrowing should be done only after the intermediate stage figure of the first part is multiplied by the suitable digit to get the final figure of the first part. In this case, we get 308 × 3 = 924 as the final form of the first part. Now to take care of the negative second part of –153, we need to borrow 2 from the first part because the main base is 100, 2 borrowed becomes 200). The final form of the second part is 200 – 153 = 47. So, the product of 317 and 291 is 92247.

2.16. Find the product of 513 and 478.

513	→	+13	(491 × 5 = 2455)
478	→	–22	300 – 286 = 14
——		——	
491		–286	Ans. 245214

We can look at one more extension of this method where the numbers are not close to the same power of 10 but are close to two different powers of 10. We can multiply such numbers by making a simple modification to this method.

2.17. Find the product of 979 and 92.

(by adding 0 to the number 92, it becomes 920)

979	→	–21	
920	→	–80	
——		——	
899		+1680	Ans. 900680

Here, 979 is close to 1000 and 92 is close to 100. For finding the product, we force 92 also close to 1000 by taking it as 920.Then, apply our regular method and find the product of 979 and 920. From the resulting product drop the zero at the units place to give the correct result for the product of 979 and 92.

So, drop the 0 in units place. Hence, the product of 979 and 92 is 90068.

In some cases, the algebraic rule $a^2 - b^2 = (a - b)(a + b)$ will be very helpful to find the product of two numbers. For example, if we have to find the product of 132 and 118, rather than applying the method discussed in detail above, we can use the algebraic rule discussed just now.

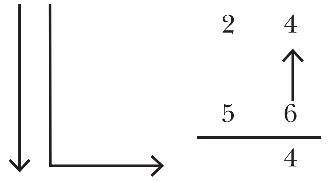
132 can be written as (125 + 7) and 118 can be written as (125 – 7). So, the product of 132 and 118 will be $125^2 - 7^2$. Since we have already discussed methods for calculating squares faster, this method can thus prove to be of immense help in a number of situations provided the student practices sufficiently.

There will be other short cut methods also for a variety of calculations, but the student has to note that none of these will be useful to him in an examination situation unless regular practice is there in using such methods. The student himself should take figures and keep applying various methods for practice on a regular basis.

2.18. Find the product of 24 and 56.

Sol: Step 1:

$6 \times 4 = 24$

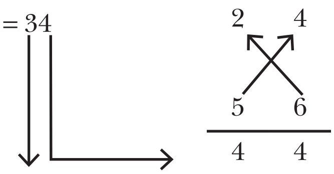


to be carried forward (C.F.) to the next step.

Step 2:

$(2 \times 6) + (4 \times 5) + 2$ (C.F.)

$= 34$

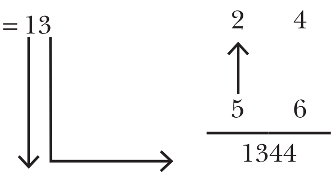


to be carried forward (C.F.) to the next step.

Step 3:

$(5 \times 2) + 3$ (C.F.)

$= 13$



$\therefore$ The product of 24 and 56 is 1344.

By observing the above calculation, we summarise the calculations as:

Step 1: Multiply the right most digits vertically (i.e., 6 4)

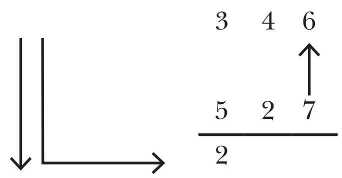
Step 2: Cross multiply and add the carry forward (C.F.) number $(6 \times 2 + 5 \times 4 + \text{C.F.})$

Step 3: Multiply the left most digits vertically and add the C.F. (i.e., $5 \times 2 + \text{C.F.}$)

2.19. Find the product of 346 and 527.

Sol: Step 1:

$7 \times 6 = 42$

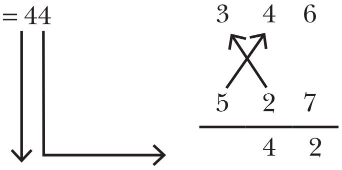


to be carried forward (C.F.) to the next step.

Step 2:

$(7 \times 4) + (2 \times 6) + $ (C.F.)

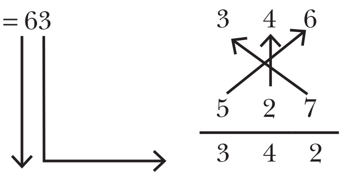
$= 44$



Step 3:

$7 \times 3 + 2 \times 4 + 5 \times 6 + 4$ (C.F.)

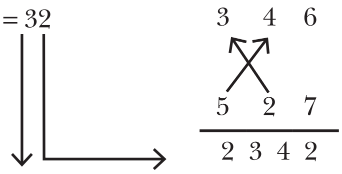
$= 63$



Step 4:

$2 \times 3 + 5 \times 4 + 6$(C.F.)

$= 32$

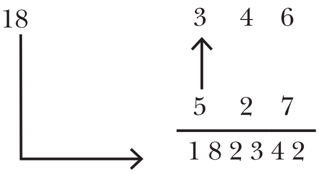


to be carried forward (C.F.) to the next step.

Step 5:

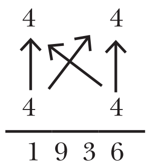
$5 \times 3 + 3$ (C.F.)

$= 18$



$\therefore$ The product of 346 and 527 is 182342.

With the help of the above methods, we can also find the square of any number. For example to find the square of 44,



2.20. $\frac{4}{9} + \frac{13}{18} + \frac{7}{54} = ?$

Sol: The L.C.M. of the denominators 9, 18 and 54 is 54. [The L.C.M. should be calculated mentally]

Let us find the numerators.

As 9 has to be multiplied by 6 to get 54, the numerator 4 is multiplied by 6, i.e., $4 \times 6 = 24$.

Similarly $13 \times 3 = 39$ and $7 \times 1 = 7$.

$$\therefore \frac{4}{9}+\frac{13}{18}+\frac{7}{54}=\frac{24}{54}+\frac{39}{54}+\frac{7}{54}$$

$$=\frac{24+39+7}{54}=\frac{70}{54}=\frac{35}{27}$$

2.21. $\frac{7}{18}-\frac{11}{24}+\frac{13}{36}=?$

Sol: The L.C.M. of the denominators 18, 24 and 36 is 72.
36 is divisible by 18, so, the L.C.M. of 18 and 36 is 36. To find the L.C.M. of 24 and 36, take the larger number, i.e., 36 and its multiples 72, 108, etc. 36 is not divisible by 24. So, L.C.M. is not 36. 72 is divisible by 24. So, the L.C.M. is 72. The denominator of the resultant fraction is 72.

$$\frac{7}{18}-\frac{11}{24}+\frac{13}{36}=\frac{28}{72}-\frac{33}{72}+\frac{26}{72}=\frac{21}{72}=\frac{7}{24}\ =$$

2.22. 37.5 % of 1248 =

Sol: 37.5% = 3/8

$\therefore$ 37.5% of 1248 = $3/8 \times 1248$
$= 3 \times 156 = 468$

2.23. 42.85% of 2114 =

Sol: 42.85% = 3/7

$\therefore$42.85% of 2114 = $3/7 \times 2114 = 3 \times 302 = 906$

2.24. 63.63% of 2233 =

Sol: 63.63% = 7/11

$\therefore$63.63% of 2233 = $7/11 \times 2233 = 7 \times 203 = 1421$

2.25. 58.33% of 2184 =

Sol: 58.33% = 7/12

$\therefore$ 58.33% of 2184 = $7/12 \times 2184 = 7 \times 182 = 1274$

2.26. 44.44% of 8127 =

Sol: 44.44% = 4/9

$\therefore$ 44.44% of 8127 = $4/9 \times 8127 = 4 \times 903 = 3612$

2.27. What is 20% of 1205?

Sol: Method 1
20% = 1/5
20% of 1205 = 1/5 of 1205 = 241
Method 2

$$10\% = \frac{10}{100} = 0.1$$

10% of 1205 = (0.1) (1205) = 120.5
$\therefore$ 20% of 1205 = $120.5 \times 2 = 241$

2.28. Find 22% of 4568

Sol: 20% ($10\% \times 2$) = $456.8 \times 2 = 913.6$
+ 2% = $1/10 \times 20\%$ = 91.36
22% = 1004.96

2.29. Find 36% of 183.5

Sol: Method 1
30% ($10\% \times 3$) = $183.5 \times 3 = 550.5$
+ 6% = $1/5 \times 30\%$ = 110.1
36% = 660.6
Method 2
40% ($10\% \times 4$) = 183.5×4 = 734
– 4% = 1/10 of 40% = – 73.4
36% = 660.6

2.30. Find the value of 26% of 496.

Sol: 26% = 25% + 1%
25% of 496 = 1/4 of 496 = 124

25% of 496	= 124
+	+
1% of 496	= 4.96
26% of 496	= 128.96

2.31. Find the value of 35.6% of 928.

Sol:

10% of 928	= 92.8
30% of 928	= 92.8×3 = 278.4
5% of 928	= 46.4
0.1% of 928	= 0.928
35.6%	= 30% = 278.4
+5%	= 46.4
+0.5%	= 4.6
+0.1%	= 0.9
	330.3

30% + 5% + 0.5% + 0.1% =
278.4 + 46.4 + 4.6 + 0.9 = 330.3

2.32. 39 is what percent of 186?

Sol: The number that follows 'of' should always come in the denominator.

So, $\frac{39}{186} \times 100$ is to be calculated.

10% of the denominator is 18.6
20% of the denominator is $18.6 \times 2 = 37.2$
1% of the denominator is 1.86
21% of the denominator is 37.2+ 1.86 ~ 39

$$\therefore \frac{39}{186} \simeq 21\%$$

2.33. 457 is what percent of 1382?

Sol: $1/3 \times 1382 \cong 461 = 33.33\%$
$461 - 457 = 4 \cong 3\ x\ 1.38 = 0.3\%$

$$\therefore \frac{457}{1382} = 33.33\% - 0.3\% = 33.03\%$$

2.34. Which of the following fractions is the smallest?

$$\frac{3}{5}, \frac{3}{7}, \frac{3}{13}, \frac{3}{8}$$

Sol: 13 is the largest denominator, hence, 3/13 is the smallest fraction. 5 is the smallest denominator, hence, 3/5 is the largest fraction.

2.35. Which of the following fractions is the smallest?

$$\frac{7}{5}, \frac{9}{5}, \frac{4}{5}, \frac{11}{5}$$

Sol: As 4 is the smallest numerator, the fraction 4/5 is the smallest.
As 11 is the largest numerator, the fraction 11/5 is the largest.

2.36. Which of the following fractions is the largest?

$$\frac{19}{16}, \frac{24}{11}, \frac{17}{13}, \frac{21}{14}, \frac{23}{15}$$

Sol: As 24 is the largest numerator and 11 is the smallest denominator, $\frac{24}{11}$ is the largest fraction.

2.37. Which of the following fractions is the largest?

$$\frac{64}{328}, \frac{28}{152}, \frac{36}{176}, \frac{49}{196}$$

Sol:
$$\frac{64}{328} = \frac{32}{164} = \frac{16}{82} = \frac{8}{41} \simeq \frac{1}{5}$$

$$\frac{28}{152} = \frac{14}{76} = \frac{7}{38} \simeq \frac{1}{5.5}$$

$$\frac{36}{176} = \frac{18}{88} = \frac{9}{44} \simeq \frac{1}{5}$$

$$\frac{49}{196} = \frac{7}{28} = \frac{1}{4}$$

As all the numerators are 1 and the least denominator is 4, the fraction $\frac{49}{196}$ is the largest.

2.38. Which of the following fractions is the largest?

$$\frac{71}{181}, \frac{214}{519}, \frac{429}{1141}$$

$$\frac{71}{181} = \frac{71 \times 6}{181 \times 6} = \frac{426}{1086}$$

$$\frac{214}{519} = \frac{214 \times 2}{519 \times 2} = \frac{428}{1038}$$

Sol: The numerators are now all almost equal (426, 428 and 429). The smallest denominator is 1038. Hence, the largest fraction must be

$$\frac{428}{1038}, \text{ i.e., } \frac{214}{519}$$

2.39. Which of the following fractions is the largest?

$$\frac{31}{37}, \frac{23}{29}, \frac{17}{23}, \frac{35}{41}, \frac{13}{19}$$

Sol: The difference between the numerator and the denominator of each fraction is 6. Therefore, the fraction with the largest numerals, i.e., 35/41 is the greatest and the fraction with the smallest numerals, i.e., 13/19 is the smallest.

2.40. Which of the following fractions is the largest?

$$\frac{31}{27}, \frac{43}{39}, \frac{57}{53}, \frac{27}{23}, \frac{29}{25}$$

Sol: As the difference between the numerator and the denominator is same, the fraction with the smallest values, i.e., $\frac{27}{23}$, is the largest.

We can also compare fractions as follows.
For example, to compare 5/13 and 9/20 make the numerator 1 for all the fractions by approximately dividing the denominator with the respective numerator (upto first decimal place).

$$\therefore \frac{5}{13} = \frac{1}{2.6} \text{ and } \frac{9}{20} \cong \frac{1}{2.2}$$

Now, clearly $\frac{1}{2.6} < \frac{1}{2.2}$ (from rule (ii) above)

$$\Rightarrow \frac{5}{13} < \frac{9}{20}$$

2.41. Which of the following fractions is the largest?

$$\frac{15}{17}, \frac{23}{29}, \frac{31}{34}, \frac{11}{15}$$

Sol: Comparing fractions

$$\frac{15}{17} \text{ and } \frac{23}{29}$$

The numerator of the fraction has increased from 15 to 23, i.e., $\frac{8}{15}$, i.e., a little more than 50%. The denominator of the fraction has increased from 17 to 29, i.e., 12/17, i.e., well over 50%. As the percentage increase in the numerator is less than the percentage increase in the denominator, the fraction $\frac{15}{17} > \frac{23}{29}$ Now compare. $\frac{15}{17}$ with $\frac{31}{34}$

As the change in the numerator is more than double (15 to 31), and the change in the denominator is exactly double, the fraction $\frac{15}{17} < \frac{31}{34}$.

$\left(\text{Alternately, } \frac{15}{17} = \frac{30}{34}\ \frac{15}{17} < \frac{31}{34}\right)$

Now compare $\frac{11}{15}$ and $\frac{31}{34}$.

The numerator has almost tripled from 11 to 31 whereas the denominator has just over doubled from 15 to 34. Since the increase in numerator is greater than the increase in the denominator,

$\frac{11}{15} < \frac{31}{34}$

So, $\frac{31}{34}$ is the largest fraction.

2.42. Find the value of x.

$$\frac{38}{154} = \frac{x}{190}$$

Sol: $x = \frac{38 \times 190}{154} = \frac{19}{77} \times 190 \simeq \frac{1}{4} \times 190 \simeq 47.5$

The second method is to find the approximate ratio of the numerators or denominators and arrive at the solution. This is illustrated in the following two examples.

2.43. Find the value of x.

$$\frac{54}{238} = \frac{11}{x}$$

Sol: $x = \frac{119}{27}(11) = \frac{108 + 11}{27}(11) = 44 + \frac{121}{27}$

$= 44 + 4\frac{13}{27} \cong 48.5$

2.44. Find the value of x

$$\frac{125}{220} = \frac{176}{x}$$

175 is 40% more than 125

$\therefore$ x is ~40% more than 220, i.e., 308.

2.45. $16 + \frac{3}{4}$ of

$\left[32 - 16 \div 4 \times 6 + \overline{23 - 11} + 3 - 2 \times 6\right] = ?$

(A) 89/4 (B) 77/4
(C) 97/4 (D) 81/4

Sol: $16 + \frac{3}{4}$ of $\left[32 - 24 + 12 + 3 - 12\right] = 16 + \frac{3}{4}$ of $[1]$

$= 16 + \frac{3}{4} \times 11 = 16 + \frac{33}{4} = \frac{97}{4}$

2.46. $\frac{1.7 \times 0.0028}{0.068 \times 0.014} = ?$

(A) 5 (B) 10 (C) 20 (D) 15

Sol: $\frac{1.7 \times 0.0028}{0.06 \times 0.012} = \frac{17 \times 28}{68 \times 14} \times 10 = 5$

2.47. $3\frac{2}{9} + 5\frac{1}{4}\left(16\frac{2}{3} \div 13\frac{4}{6}\right) \div 6\frac{3}{4} = ?$

(A) 3140/369 (B) 1342/369
(C) 1456/369 (D) 1539/369

Sol:

$3\frac{2}{9} + 5\frac{1}{4}\left(16\frac{2}{3} \div 3\frac{4}{6}\right) \div 6\frac{3}{4}$

$= \frac{29}{9} + \frac{21}{4}\left(\frac{50}{3} \times \frac{6}{82}\right) \div \frac{27}{4}$

$= \frac{29}{9} + \frac{21}{4} \times \frac{50}{3} \times \frac{6}{82} \times \frac{4}{27}$

$= \frac{29}{9} + \frac{350}{9 \times 41} = \frac{1539}{369}$

2.48. 40% of $\left[\left\{\left(\overline{16 - 8} + \overline{18 - 12}\right) \times 5 - 6\right\} \times 2 + 3\right] = ?$

(A) 262/5 (B) 271/5
(C) 267/5 (D) 313/5

Sol: 40% of $[\{(8 + 6) \times 5 - 6\} \times 2 + 3]$

$= \frac{2}{5}$ of $[64 \times 2 + 3] = \frac{2}{5} \times 131 = \frac{262}{5}$

2.49. $5\frac{7}{6} + 16\frac{2}{3} + 18\frac{4}{9} - 13\frac{5}{6} = ?$

(A) 187/9 (B) 247/9
(C) 319/9 (D) 419/9

Sol: $5\frac{7}{6} + 16\frac{2}{3} + 18\frac{4}{9} - 13\frac{5}{6} = \frac{37}{6} + \frac{50}{3} + \frac{166}{9} - \frac{83}{6}$

$= \frac{54}{6} + \frac{166}{9} = \frac{247}{9}$

2.50. Find the cube of 12.

Step 1: Cube the left most digit, i.e., 1 in this case, and write it down on the extreme left.

Step 2: Write three more numbers to its right such that the ratio of successive pairs of numbers is same as the ratio of the digits (1 : 2) in the original number. We get the following 1 2 4 8. (1 : 2 = 2 : 4 = 4 : 8)

Step 3: Double the second number (i.e., 2) and the third number (i.e., 4) of the above four numbers and write the result (i.e., 2 × 2 = 4 and 2 × 4 = 8) under the respective numbers.

Step 4: Add the two rows—one column at a time—such that each column contributes only one digit to the total. (If any column gives more than one digit, the additional digits are carried forward)

	1 ← carry forward			
	1	2	4	8
		4	8	
12^3 =	1	7	2	8

2.51. Find the cube of 23.

Step 1 : Cube the left most digit (i.e., $2^3 = 8$) and write it down on the extreme left.

Step 2 : Write three numbers next to the above, such that the ratio between any two successive numbers is the same as the ratio of the digits of the given number. (Therefore, in the number 23, the ratio of the digits is 2 : 3). We get 8 12 18 27

Note: It may sometimes be difficult to find the numbers, i.e., 12, 18 and 27. Note that these numbers are obtained, as 12 = 8 × 3/2; 18 = 12 × 3/2; 27 = 18 × 3/2. Therefore, to get any number, multiply the previous number by the units digit value (i.e., 3) and divide by the ten's digit value (i.e., 2).

Step 3 : Double the 2^{nd} number (i.e., 12) and the 3^{rd} number (i.e., 18) and write them down below the respective numbers.

8	12	18	27
	24	36	

Step 4: Add all the numbers, column wise, as shown below, each time carrying forward all digits except the units digit.

4	5	2 ← carry forward	
8	12	18	27
		24	36
12	1	6	7

∴ $23^3 = 12167$

2.52. Find the cube of 37.

23	47	34 ← carry forward	
27	63	147	343
	126	294	
50	6	5	3

∴ $37^3 = 50653$

2.53. A trader uses only five weights which together weigh 31 kg. With these five weights he can measure all integer weights from 1 kg to 31 kg, with the weight kept only in one pan of the weighing scale. Find the individual weights of the five pieces.

Sol: For measuring all integer weights up to 31 kg, the individual weights needed are the powers of 2, i.e., 2^0, 2^1, 2^2, 2^3 and 2^4

∴ if we have weights of 1 kg, 2 kg, 4 kg, 8 kg and 16 kg, we can measure all integer weights upto 31 kg.

For example, if we have to measure 23 kg, we have to use the weights 16 kg, 4 kg, 2 kg and 1 kg on one pan.

If you want to write any number from 1 to *M* as a sum of one or more of the integers of a given set of integers (each integer being used at the most once), it can be done by using the powers of 2. The set of integers we can use consists of all the powers of 2 starting from 1 (i.e., 2^0) to the largest power of 2 less than or equal to *M*. For example, if you want to build all the integers upto 255, the numbers 1, 2, 4, 8, 16, 32, 64, 128 are sufficient.

2.54. A trader uses only four weights, which together weigh 40 kg. With the four weights he could measure all integer weights from 1 kg to 40 kg, placing weights in both the pans. Find the weights of the four pieces.

Sol: For measuring all weights upto 40 kg, the weights needed are the powers of 3 whose sum adds upto 40, i.e., 3^0, 3^1, 3^2 and 3^3 … if we have weights of 1 kg, 3 kg, 9 kg and 27 kg, we can measure all weights from 1 to 40 kg.

For example, if we have to measure 33 kg, we have to keep the 27 kg and 9 kg weights on one pan and 3 kg weight on the other, i.e., 27 + 9 – 3 = 33 kg.

EXERCISE-1

***Directions for questions 1 to 55*:** For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the blank space provided.

1. $8563 + 3947 + 5760 + 5691 =$ ______ .

2. $99786 - 5584 - 934 - 88 - 9 =$ ______ .

3. $35408 + 81563 - 41341 - 51464 =$ ______ .

4. $1012 \times 98 =$ ______ .

5. $1372 \times 125 =$ ______ .

6. $2113 \times 2117 =$ ______ .

7. $239 \times 251 =$ ______ .

8. $7869 \times 982 =$ ______ .

9. 14.28% of 322 = ______ .

10. What percentage of 751 is 362 ______ .

11. 433 is what percentage of 1444 ______ .

12. 106×812 is what percent of 464×4 ______ .

13. 128.57% of 1694 = (to the nearest integer)______.

14. 84.71% of 742 = ______ .

15. $11 \times 4\%$ of 18×2 is what percentage more than $14 \times 1\%$ of 12×8 ______ .

16. $\frac{5}{24} + \frac{7}{36} + \frac{11}{48} =$ ______ .

17. $\frac{12}{17} - \frac{11}{15} + \frac{18}{19} =$ ______ .

18. $\frac{5}{7} + \frac{9}{10} + \frac{11}{14} + \frac{8}{35} =$ ______ .

19. $\frac{x}{540} = \frac{237}{681}$. Find x ______ .

20. $120.01 \times 4.99 + 5.99 \times 80.01 =$
(A) 1020 (B) 1040 (C) 1060 (D) 1080

21. $\sqrt[3]{91} \times 162\%$ of $48 =$ ______^{3}
(A) 4 (B) 5 (C) 7 (D) 9

22. $\frac{(8.99)(7.01)(2.05) + (17.05)(29.88)(15.01)}{(2.01)(16.01)(19.9)} =$
(A) 14 (B) 10 (C) 11 (D) 12

23. $534.95 - 15.23 + 35 \times 6.78 + 40\%$ of $478 =$
(A) 256 (B) 354 (C) 478 (D) 956

24. $\sqrt[3]{216400} + \sqrt{280} + \sqrt{322} =$
(A) 651 (B) 361 (C) 85 (D) 95

25. $\frac{\sqrt{532.69} + \sqrt{230.15}}{\sqrt{290.96} + \sqrt{364.56}} =$
(A) 1 (B) 3 (C) 5 (D) 6

26. $78927.95 \div 448.29 + 3425.6925 =$
(A) 1600 (B) 2600 (C) 4600 (D) 3600

27. $\sqrt{36.1} \times 34 + 15\sqrt{8.92} =$ ______ $\times (14.28\%$ of $217)$
(A) 10 (B) 6 (C) 8 (D) 11

28. ______$^3 + 23456 - 21246) \div 31421 = 25$
(A) 65 (B) 80 (C) 69 (D) 92

29. $\frac{1}{8}$ of $\frac{1}{7}$ of $\frac{7}{3}$ of $144 +$ ______ $= 13^2 - 5^2$
(A) 140 (B) 132 (C) 134 (D) 138

30. $[8 - (7 \text{ of } 16 \div 8 - 10 + 7)] \times \left[\frac{36 \times 14 \times 81}{7 \times 72 \times 3} + 30\right] =$
(A) –171 (B) –59 (C) –3 (D) –161

31. $\left\{\frac{1^2 + 2^2 + 3^2 - 11}{78 - \left(4^2 + 5^2 + 6^2 + 2\right)}\right\}\left\{\frac{-15 - \left(16 - \overline{12 + 12}\right)2}{18 \times 4 - 2 \times 6^2 - 1}\right\}$
(A) –1 (B) 3 (C) –1/3 (D) 1

32. 60% of 1300 = ______ % of $1700 + 30^2 \div 90 - 20\%$ of 400
(A) 30 (B) 50 (C) 60 (D) 80

33. $\frac{2}{5}$ of $\left\{\left(36 \div \overline{28 \div 7} \times \frac{1}{9}\right) + \frac{1}{9}\right\} =$
(A) $\frac{1}{9}$ (B) $\frac{2}{9}$ (C) $\frac{5}{9}$ (D) $\frac{4}{9}$

34. 45% of 60% of 80% of $\left[\frac{714}{8} \div \frac{17}{64}\right] \div 378 \times 1125 =$
(A) 116 (B) 126 (C) 184 (D) 216

35. $\overline{(24 - 20)^3 + (24 - 24)^3} \times 16\frac{2}{3}\%$ of ______ $= 84$
(A) 4 (B) 6 (C) 8 (D) 2

36. $[12 - (9 \text{ of } 24 \div 12 - 14 + 5)] \times \left[\frac{48 \times 22 \times 108}{11 \times 96 \times 9} + 36\right] =$

(A) 126 (B) 132 (C) 138 (D) 144

37. $\left\{\frac{30(7+4-12)}{-5+6+9}\right\} \div \left\{\frac{(8 \times 9 - 32)3}{(17+15-31)10}\right\}$

(A) –4 (B) –1 (C) –1 (D) –1/4

38. $\frac{9}{15} \text{ of } \frac{45}{81}\left\{\frac{49}{6} \times \left(\frac{16}{7} - 2\right)\right\} \text{ of } \frac{24}{5} \div \frac{16}{15} =$

(A) 7/2 (B) 2/7 (C) 3/7 (D) 5/9

39. $\sqrt[3]{\frac{\_\_\_\_}{32768}} = \frac{15}{32}$

(A) 3375 (B) 3500 (C) 4560 (D) 4975

40. $\sqrt{42025} \times \sqrt{3481} - (83)^2 = \sqrt{\_\_\_\_} + (72)^2$

(A) 488 (B) 484 (C) 464 (D) 488

41. $\sqrt{46656} + \sqrt{4096} \times 52 = (\_\_\_\_)^2 + 115$

(A) 48 (B) 52 (C) 57 (D) 68

42. $\sqrt{9218} \times \sqrt{2210} \div \sqrt{1028} =$

(A) 104 (B) 114 (C) 141 (D) 144

43. $(225)^2 \div \sqrt[3]{15625} = \_\_\_\_^2$

(A) 25 (B) 35 (C) 45 (D) 55

44. $19^3 - 18^3 = \_\_\_\_$.

45. $(84)^3 - (84)^2 =$

(A) 585468 (B) 558468 (C) 558648 (D) 585648

46. $324^2 + 576^2 + 324 \times 576 = \_\_\_\_$.

47. AB is a two digit number $(AB)^2$ = CDA, which is a three-digit number. How many values can AB take?

(A) 3 (B) 0 (C) 1 (D) 2

48. Among the options given below, which pair does not represent the combination of a number and its cube?

(A) 45 and 91125 (B) 72 and 373248
(C) 78 and 474552 (D) 87 and 658483

49. Which of the following is a perfect square?

(A) 4021025 (B) 1170875
(C) 1130375 (D) 9030025

50. A perfect square is added to twice of itself. The resulting number will

(A) never end with 8.
(B) never be a perfect square.
(C) never end with a 6.
(D) satisfy more than one of the above.

51. P and Q are natural numbers satisfying the equation $P^2 - Q^2 = 889$. How many integral values are possible for (P, Q)?

(A) 1 (B) 2 (C) 3 (D) infinite

52. Find the smallest natural number with which 9000 is to be multiplied to make it a perfect square.

(A) 10 (B) 2 (C) 5 (D) 45

53. What is the smallest natural number with which 1080 should be multiplied to make it a perfect cube?

(A) 50 (B) 75 (C) 100 (D) 25

54. $(132)^2 = \_\_\_\_$.

55. $(10.12)^2 = \_\_\_\_$.

ANSWER KEYS

Exercise-1

1. 23961	12. 23%	23. (D)	34. (D)	45. (D)
2. 93171	13. 2178	24. (D)	35. (C)	46. 623376
3. 24166	14. 628.55.	25. (A)	36. (D)	47. (D)
4. 99176	15. 15%	26. (D)	37. (D)	48. (D)
5. 171500	16. 91/144	27. (C)	38. (A)	49. (D)
6. 4473221	17. 4457/4845	28. (D)	39. (A)	50. (D)
7. 59989	18. 92/35	29. (D)	40. (B)	51. (B)
8. 7727358	19. 188	30. (A)	41. (C)	52. (A)
9. 45.98	20. (D)	31. (B)	42. (C)	53. (D)
10. 48.20%	21. (C)	32. (B)	43. (C)	54. 17424
11. 29.98%	22. (D)	33. (D)	44. 1027	55. 102.4144

SOLUTIONS

EXERCISE-1

1. The sum is 23961.

2. The answer is 93171.

3. The answer is 24166.

4. 1012×98

   $= (1000 + 12)\ (100 - 2)$

   $= 100000 + 1200 - 2000 - 24 = 99176$

5. $1372 \times 125 = 1372 \times \frac{1000}{8}$

   $= \frac{1372}{8} \times 1000 = 171.5 \times 1000 = 171500 = 171.5$

6. $2113 = 2110 + 3$ and $2117 = 2110 + 7$

   The base for 2113 and 2117 is 2110.

   And he sum of the units digits is $3 + 7 = 10$

   For such numbers, the ten's digit and units digit will be $3 \times 7 = 21$

   The other five digits starting from the ten lakh's place to the hundred's place will be $211 \times 212 = 44{,}732$

   (211 is common to both the numbers. Therefore the product of 211 and its successive positive integers should be considered)

   $\therefore$ The product of 2113 and 2117 will be 4473221.

7. $239\ (250 + 1)$

   $= 239 \times \frac{100}{4} + 239 \times 1$

   $= 59750 + 239 = 59989$

8. $7869 \times 982 = 7869 \times (1000 - (20 - 2))$

   $= 7869000 - 141642 = 7727358$

***Solutions for questions 9 to 11*:**

By using ten percent one percent concept, we can calculate the required percentage values.

9. 14.28% of 322 is 45.98

10. 362 is close to half of 751.

    $\therefore$ By using the ten percent one percent concept, we can find that 362 is 48.20% of 751.

11. $\frac{1}{4} \times 144 = 361$

    Now, $433 - 361$, i.e., 72 is very close to 5% of 1440.

    $\therefore$ The required answer is 29.98%

12. 25% of $464 \times 4 = 116 \times 10$

    106×812 is 9×288 less than 464×4. As 9×288 is 2% of 464×4, 106×812 is 23% of 464×4.

13. 128.57% of 1694 = 100% of 1694 + 28.57% of 1694

    $= 1694 + \frac{2}{7} \times 1694 = 1694 + 484 = 2178$

14. 84.71% of 742

    $= (85.71 - 1)\%$ of $742 = 628.55$.

15. $11{\times}4\%$ of $18{\times}2 = \frac{114}{1000} \times 18.2$

    $= \frac{114}{1000} \times (2 \times 9 + 0.2)$

    $= \frac{228 \times 9 + 228}{1000} = 2.0748$

    $14{\times}1\%$ of $12{\times}8$

    $= \frac{141}{1000} \times (10 + 2 + 0.8)$

    $= \frac{1410 + 282 + 11.28}{1000} = 1.804$

    $= \frac{2.07 - 1.80}{1.80} \times 100 = 15\ (\text{approximately})$

16. The L.C.M. of 24 and 48 is 48.

    By prime factorization $36 = 2^2 \times 3^2$

    And $48 = 2^4 \times 3^1$

    The L.C.M. of 36 and 48 $= 2^4 \times 3^2 = 144$

    $\frac{5}{24} = \frac{5 \times 6}{24 \times 6} = \frac{30}{144}$

    $\frac{7}{36} = \frac{7 \times 4}{38 \times 4} = \frac{28}{144}$

$$\frac{11}{48} = \frac{11\times 3}{48\times 3} = \frac{33}{144}$$

$$\therefore \frac{30}{144} + \frac{28}{144} + \frac{33}{144} = \frac{91}{144}$$

17. $\frac{12}{17} - 1 + \frac{4}{15} + 1 - \frac{1}{19}$

$$= \frac{4}{15} + \frac{12}{17} - \frac{1}{19} = \frac{4457}{4845}$$

18. $\frac{5}{7} = \frac{5\times 10}{7\times 10} = \frac{50}{70}$

$$\frac{9}{10} = \frac{9\times 7}{10\times 7} = \frac{63}{70}$$

$$\frac{11}{14} = \frac{11\times 5}{14\times 5} = \frac{55}{70}$$

$$\frac{8}{35} = \frac{8\times 2}{35\times 2} = \frac{16}{70}$$

$$\frac{50}{70} + \frac{63}{70} + \frac{55}{70} + \frac{16}{70} = \frac{184}{70} = 92/35.$$

19. $\frac{681}{540} \cong \frac{680}{540} = 1.26$

Let $x \times 1.25 = 237$

$$x = 237 \times \frac{4}{5} = 189.6$$

Since $x \times 1.25 = 237$

$x \times 1.26 < 237$

Hence, the answer will be approximately 188.

20. $? = 120 \times 5 + 6 \times 80 = 600 + 480 = 1080$

21. $?^3 \cong 4.5 \times 77.76$

$?^3 = 351$

$\Rightarrow ?^3 \cong 7^3 \Rightarrow \therefore ? = 7.$

22. $\frac{(8.99)(7.01)(2.05) + (17.05)(29.88)(15.01)}{(2.01)(16.01)(19.9)}$

$$\cong \frac{9\times 7\times 2 + 17\times 30\times 15}{2\times 16\times 20}$$

$$= \frac{7776}{640} \cong 12.$$

23. $? \cong 535 - 15 + 35 \times 7 + 191$

$\Rightarrow ? = 956.$

24. $\sqrt[3]{216400} + \sqrt{280} + \sqrt{322}$

$= 60 + 16.5 + 18 = 94.5 \cong 95$

25. $\frac{\sqrt{532.69} + \sqrt{230.15}}{\sqrt{290.96} + \sqrt{364.56}} = ?$

$$\Rightarrow ? \simeq \frac{23+15}{17+19} = \frac{38}{36}$$

$\Rightarrow ? \cong 1.$

26. $? = 176.06 + 3423.6925$

$\Rightarrow ? = 3599.75 \cong 3600.$

27. $\sqrt{36.1} \times 34 + 15\sqrt{8.92} = ? \times (14.28\% \text{ of } 217)$

$\Rightarrow 6 \times 34 + 15 \times 3 = ? \times 1/7 \times 217$

$$\Rightarrow 249 = ? \times 31 \Rightarrow ? = \frac{249}{31} = 8.$$

28. $(?^3 + 2210) = 25 \times 31421$

$\Rightarrow ?^3 = 785525 - 2210$

$\Rightarrow ?^3 = 783315 \Rightarrow ? = \sqrt[3]{783315}$

$\therefore ? \cong 43.$

29. $? = 13^2 - 5^2 - \frac{1}{8} \text{ of } \frac{1}{7} \text{ of } \frac{7}{3} \text{ of } 144$

$$169 - 25 - \frac{1}{8} \times \frac{1}{7} \times \frac{7}{3} \times 144$$

$= 144 - 6 = 138$

30. $8 - \{7 \text{ of } 16 \div 8 - 10 + 7\} \times \left[\frac{36\times 14\times 81}{7\times 72\times 3} + 30\right]$

$$\Rightarrow \left[8 - \left(\frac{7\times 16}{8} - 10 + 7\right)\right] \times 57 = -171$$

31. $\left(\frac{-15\left(16 - \overline{12+12}\right)2}{18\times 4 - 2\times 6^2 - 1}\right) \times \frac{1^2 + 2^2 + 3^2 - 11}{78 - \left(4^2 + 5^2 + 6^2 + 2\right)}$

$$= \left(\frac{15 - (16-24)2}{72 - 72 - 1}\right) \times \frac{3}{-1}$$

$= -1 \times -3 = 3$

32. $\frac{60}{100}\times 1300=\frac{?}{100}\times 1700+\frac{900}{90}-\frac{20}{100}\times 400$

$\Rightarrow ? \times 17 + 10 - 80 = 780$

$\Rightarrow ? \times 17 = 780 + 70 = 850$

$\Rightarrow ? = \frac{850}{17} = 50$

33. $? = \frac{2}{5}$ of $\left\{\left(36 \div 4 \times \frac{1}{9}\right)+\frac{1}{9}\right\}$.

$\Rightarrow ? \frac{2}{5}\times\left\{1+\frac{1}{9}\right\}$

$\Rightarrow ? \frac{2}{5}\times\frac{10}{9}=4/9$.

34. $? = \frac{45}{100}\times\frac{60}{100}\times\frac{80}{100}\times\left[\frac{714}{8}\times\frac{64}{17}\right]\times\frac{1}{378}\times 1125$

$\Rightarrow ? = \frac{9}{20}\times\frac{3}{5}\times\frac{4}{5}\times[42\times 8]\times\frac{1}{378}\times 1125$

$\Rightarrow ? = 24 \times 9 = 216.$

35. $\overline{(24-20)^3+(24-25)^3} + (24-25)^3 \times 16\frac{2}{3}\%$ of? = 84

$\Rightarrow (64-1)\times\frac{1}{6}$ of? $= 84 \Rightarrow ? = 8$

36. $? = [12 - (216 \div 12 - 14 - 5)] \times \left[\frac{48}{96}\times\frac{22}{11}\times\frac{108}{9}+36\right]$

$[12 - (18 - 14 + 5)] \times \left[\frac{1}{2}\times 2\times 12+36\right]$

$= [12 - 9] \times [12 + 36]$

$= 3 \times 48 = 144$

37. $\frac{30(7+4-12)}{-5+6+9} \div \frac{(8\times 9-32)3}{(17+15-31)10}$

$= -3 \div 12 = -1/4.$

38. $\frac{9}{15}$ of $\frac{5}{9}\left\{\frac{49}{6}\times\frac{2}{7}\text{ of }\frac{24}{5}\times\frac{15}{16}\right\}=?$

$\Rightarrow ? = \frac{9}{15}\times\frac{5}{9}\left\{\frac{7}{3}\times\frac{9}{2}\right\}$

$\Rightarrow ? = 7/2$.

39. $\frac{?}{32768}=\left(\frac{15}{32}\right)^3=\frac{3375}{32768}$

$\therefore ? = 3375.$

40. $205 \times 59 - 6889 = \sqrt{?} + 5184$

$\Rightarrow \sqrt{?} = 12095 - 6889 - 5184$

$\Rightarrow \sqrt{?} = 22 \Rightarrow ? = (22)^2$

$\therefore ? = 484.$

41. $\sqrt[3]{46656} + \sqrt{4096} \times 52 = (?)^2 + 115$

$\Rightarrow (?)^2 + 115 = 36 + 64 \times 52$

$\Rightarrow (?)^2 = 3364 - 115 = 3249$

$\Rightarrow ? = \sqrt{3249} = 57.$

42. $\frac{\sqrt{9218}\times\sqrt{2210}}{\sqrt{1028}} = ? \Rightarrow \frac{\sqrt{9216}\times\sqrt{2209}}{\sqrt{1024}} = ?$

$\Rightarrow ? = \frac{96\times 47}{32} = 141.$

43. $(?)^2 = 225 \times 225 \times \frac{1}{25}$

$\Rightarrow (?)^2 = 2025 \Rightarrow ? = \sqrt{2025}$

$\therefore ? = \pm 45.$

44. $19^3 - 18^3$ is of the form $a^3 - b^3$ where $a = 19$ and $b = 18$.

$a^3 - b^3$ is defined as $(a - b)\ (a^2 + b^2 + ab)$.

Hence, $19^3 - 18^3 = (19 - 18)$

$(19^2 + 18^2 + 19 \times 18) = 1027$

45. $(84)^3 - (84)^2 = ?$

$\Rightarrow ? = (84)^2\ [84 - 1] = 84 \times 83$

$\therefore ? = 585648.$

46. $324^2 + 576^2 + 324 \times 576$

$= 324^2 + 576^2 + 2 \times 324 \times 576 - 324 \times 576$

$= (324 + 576)^2 - (18^2 \times 24^2) = 900^2 - 432^2$

$= (900 + 432) \times (900 - 432) = 1332 \times (400 + 60 + 8)$

$= 623376$

47. $(AB)^2 = CDA$

CDA is a three digit perfect square.

$\therefore$ A cannot be 2, 3, 7, or 8 (1)

($\because$ No Perfect square ends in 2, 3, 7, or 8)

$(AB)^2 < 1000$. $\therefore$ $AB \leq 31$. (2)

From (1) and (2), $A = 1$,

$(1B)^2 = CD1$.

$B = 1$ or 9.

$AB = 11$ or 19.

$\therefore$ AB can take 2 values

48. Going by the options, as the numbers whose cubes are given are multiples of 3, the cubes of the numbers must be multiples of 3^3, i.e., 27.

Hence, the cubes must be multiples of 9.

The sum of the digits of 91125, 373248 and 474552 are multiples of 9 whereas the sum of the digits of 658483 is not a multiple of 9.

Hence, 658483 is not a cube of 87.

49. Any perfect square ending with a 5 must end with 25. Only choices (A) and (D) have the last two digits as 25.

Checking 4021025, using the rule of finding squares of numbers ending in 5, we need to find factors of the form $(n) \times (n + 1)$ [where n is a natural number] for 40210. Similarly for 90300. $200 \times 201 = 40200$. Hence, choice (A) is eliminated. But $90300 = 300 \times 301$. Therefore $9030025 = (3005)^2$.

50. Consider perfect square X^2

$X^2 + 2X^2 = 3X^2$

X^2 can end with 0, 1, 4, 5, 6, 9

$\therefore$ $3X^2$ can end with

0, 3, 2, 5, 8, 7

$\therefore$ Choice (C) is true, and choice (A) is false.

Besides the product of a non–perfect square and a perfect square can never be a perfect square.

$\therefore$ Choice (B) is also true.

51. $P^2 - Q^2 = 889$

$\therefore$ $(P + Q)\ (P - Q) = 127 \times 7 = 889 \times 1$

These are the only two ways of expressing 889 as a product of two natural numbers

Case 1: $P + Q = 127$, $P - Q = 7$

$\Rightarrow$ $P = 67$, $Q = 60$

Case 2: $P + Q = 889$, $P - Q = 1$

$P = 445$, $Q = 444$

52. $9000 = 3^2 \times 5^3 \times 2^3$

The smallest natural number to be multiplied with to make it a perfect square $= 5 \times 2 = 10$.

53. $1080 = 108 \times 10$

$= 18 \times 6 \times 10 = 2 \times 3^2 \times 2 \times 3 \times 2 \times 5 = 2^3 \times 3^3 \times 5$

The least natural number to be multiplied to make it a perfect cube $= 5 \times 5 = 25$.

54. $(132)^2 = (100 + 32)^2$

$= 10{,}000 + 6400 + 1024 = 17424$

55. $(10.\ 12)^2$

$= (10)^2 + (0.12)^2 + 2(10)\ (0.12)$

$= 100 + 0.0144 + 2.4 = 102.4144$

Unit 2

Data Interpretation

CHAPTER 1

Tables

Learning Objectives

In this chapter, you will:

- Gain understanding of various types of data presented in the form of tables.
- Develop calculation skills.
- Get exposed to different types of questions based on tables.
- Learn different shortcut methods and techniques in solving the questions.

❒ DATA TABLE

Here data is presented in the form of simple table. While any type of data can be presented in table form, that too in a very accurate manner, interpreting the data in table format is very difficult and time consuming than the other modes, all of which are basically pictorial or graphical in presentation.

Data tables can be of a number of types. They can be of a single-table variety or combination of tables. Some examples of tables are given below.

Table 1.1 Movement of Goods by Different Modes of Transport (in 000's of metric-ton-kms)

Year	Road	Rail	Air	Water	Total
1985	1000	1500	120	20	2640
1986	1600	2000	129	24	3753
1987	2907	3090	139	28	6164
1988	4625	5200	152	27	10004
1989	6346	7540	174	33	14093
1990	7920	10250	212	40	18422
1991	9540	13780	266	50	23636

Note: All figures are fictitious.

From the table we can deduce the following:

- Rate of growth by each mode of transport in successive years as well as cumulative annual growth.
- Rate of growth of total haulage by all modes of transport together in any year.
- Contribution by each mode of transport to the total haulage in any given year.
- Trends of growth over time for various modes of transport.
- Given the cost of transportation for each mode, we can calculate the total annual cost of transportation over the years for various modes of transport as well as make a cost comparison.
- Finding out the mode of transportation in any given year that forms the largest percentage of total haulage.
- For a given mode of transport, finding out the year in which the percentage increases in haulage over the previous year was the highest.

Table 1.2 Railway Time Table – Coromandel Express

Place	Cumulative mileage	Arrival Time (in hours)	Departure Time (in hours)
Madras	0	—	08.00
Nellore	200	11.20	11.30
Vijayawada	525	15.30	16.00
Rajahmundry	700	19.20	19.30
Visakhapatnam	1100	01.10	01.30
Bhubaneswar	1450	03.45	04.00
Kharagpur	1600	07.25	07.30
Calcutta	1925	09.30	—

From the above time table, we can obtain the following:

- Distance between various stations.
- Total idle time as a proportion of total travel time.
- Average speed between stations as well as over the entire journey.
- Minimum and maximum speeds of the train between two stations.

SOLVED EXAMPLES

Directions for questions 1.01 to 1.05: These questions are based on the following table, which gives the details of the sports liked by students in all the classes of a school.

The table gives the number of students in each class and the percentage of students in the class who like Cricket, Volleyball, Basketball and Football.

Class	Number of students	Cricket	Volleyball	Basketball	Football
6	120	60%	70%	50%	60%
7	140	50%	60%	60%	50%
8	160	40%	65%	55%	45%
9	180	65%	75%	65%	55%
10	240	70%	80%	75%	45%

1.01: How many students in the school like cricket?
(A) 436 (B) 432
(C) 491 (D) 511

Sol: Number of students who like cricket

$$= \frac{60}{100}(120) + \frac{50}{100}(140) + \frac{40}{100}(160) + \frac{65}{100}(180)$$

$$+ \frac{70}{100}(240) = 491$$

1.02: By what percentage is the number of students who like volleyball in class 6 more/less than those who like basketball in class 10?
(A) 40% less (B) 50% more
(C) 53.33% less (D) 56.67% more

Sol: Number of students who like volleyball in class 6

$$= \frac{70}{100}(120) = 84$$

Number of students who like basketball in class 10 $= \frac{75}{100}(240) = 180$

84 is less than 180 by $\frac{180 - 84}{180}(100)$

$= 53.33\%$

1.03: The number of students who like cricket in class 7 is what percentage of the number of students who like football in class 8?
(A) 88% (B) 93.5%
(C) 95.6% (D) 97.2%

Sol: Number of students who like cricket in class 7

$$= \frac{50}{100}(140) = 70$$

Number of students who like football in class 8

$$= \frac{45}{100}(160) = 72$$

Required percentage $= \frac{70}{72} \times 100 = 97.2\%$

1.04: In how many of the given classes can more than 90 students like all the four games?

(A) 4 (B) 3
(C) 1 (D) 2

Sol: In any class, the maximum value of the number of students who like all the four games would be the number of students who like the game liked by the least number of students.

In class 6, the percentage of students who like a game is the least for basketball. The number of students who like basketball

$$= \frac{50}{100}(120) = 60 < 90$$

In class 7, the percentage of students who like a game is the least for cricket and football. The number of students who like cricket

$$= \frac{50}{100}(140) = 70 < 90$$

In class 8, the number of students who like a game is the least for cricket. The number of students who like cricket

$$= \frac{40}{100}(160) = 64 < 90$$

In class 9, the percentage of students who like a game is the least for football. The number of students who like football

$$= \frac{55}{100}(180) = 99 > 90$$

In class 10, the percentage of students who like a game is the least for football. The number of students who like football

$$= \frac{45}{100}(240) = 108 > 90$$

$\therefore$ In two classes, more than 90 students can like all the games.

1.05: What can be the maximum percentage of students in class 6 who do not like any of the given games?

(A) 40% (B) 10%
(C) 50% (D) 30%

Sol: In class 6, the maximum percentage of students who like a game = the percentage of students who like volleyball, i.e., 70%. The percentage of students who like at least one game would be minimum when all students who like other games are also the same ones who like volleyball.

$\therefore$ Maximum percentage required

$= 100 - 70 = 30\%$

EXERCISE-1

Direction for questions 1 to 5: Answer these questions based on the information given below.

The below table gives the number of accidents and deaths in five districts P, Q, R, S and T during the period from 2008 to 2014.

	P		Q		R		S		T	
	Accidents	Deaths	Accidents	Deaths	Accidents	Deaths	Accidents	Deaths	Accidents	Deaths
2008	826	143	546	88	1465	189	1002	156	593	72
2009	802	139	589	94	1520	213	1089	161	526	78
2010	945	132	647	91	1680	221	1102	168	487	84
2011	765	141	672	105	1740	234	1162	174	462	81
2012	1012	173	745	112	1432	243	1175	183	493	76
2013	1126	168	685	132	1562	252	1286	189	514	87
2014	1432	195	720	127	1613	236	1315	191	554	98

Note: The maximum number of deaths in any accident was 2.

1. At least what percentage of the accidents in district P in the year 2009 resulted in deaths?
(A) 8.7 (B) 9.2
(C) 9.6 (D) 10.4

2. The total number of deaths in district P from 2008 to 2010 is approximately what percentage of the number of deaths in that district in 2014?
(A) 194.2 (B) 198.7
(C) 204.8 (D) 212.3

3. The total number of deaths in 2012 is approximately what percentage of all the accidents in that year?
(A) 14.1% (B) 15.0%
(C) 16.2% (D) 17.3%

4. In which year was the total number of deaths in all the five districts the highest?
(A) 2011 (B) 2012
(C) 2013 (D) 2014

5. What was the highest percentage increase in the number of deaths in district R in any year, when compared to the previous year?
(A) 11.8% (B) 12.7%
(C) 13.4% (D) 14.2%

Direction for questions 6 to 10: Answer these questions based on the information given below.

The table shows the number of fan regulators rejected and the percentage of those accepted by quality control department, out of the total number of fan regulators assembled by five machines P, Q, R, S and T.

Machine	Number of fan regulators rejected	Percentage of acceptance of fan regulators
P	600	90
Q	1000	80
R	550	90
S	840	86
T	1650	85

6. How many fan regulators assembled by machine R were accepted?

7. Which machine assembled the highest number of regulators? (Type 1 for P, 2 for Q and so on till S for T)

8. Approximately by what percentage is the number of rejections of regulators assembled by P more than that of R? (approximated to the closest integer)

9. By what percentage is the number of fan regulators assembled by T more than the total number of fan regulators assembled by P and Q?

10. What is the average number of regulators assembled by machines R and S?

Direction for questions 11 to 15: Answer these questions based on the information given below.

Number of employees in five companies

Name of the company	Classification based on the age of the employee: Above 20 years but below 30 years	Above 20 years but below 40 years	Above 20 years but below 50 years	50 years or above but below 60 years
Pallis	125	148	165	36
Dagritech	96	153	187	60
Starshine	108	127	141	72
Greenhorn	68	93	213	68
Farmtech	97	125	148	48

Note: The age of an employee lies between 20 years and 60 years.

11. In which of the following companies is the ratio of number of employees aged thirty years or above but below 50 years to the total number of employees of that company the highest?

(A) Pallis (B) Dagritech
(C) Greenhorn (D) Farmtech

12. In which of the following age groups is the number of employees of all the given companies, together the highest?

(A) More than 20 years but below 30 years.
(B) Thirty years or more but less than fifty years.
(C) Forty years or more but less than 50 years.
(D) Thirty years or more.

13. If 27 employees aged below 25 years but above 20 years join Greenhorn company, what percentage of the employees aged below 50 years but above 20 years, does the group of aged below 40 years but above 20 years constitute?

(A) 25% (B) 50%
(C) 45% (D) 200%

14. Which of the following could be the average age of the employees of Pallis?

(A) 26 years (B) 34 years
(C) 43 years (D) 51 years

15. Which of the following companies has the highest number of employees?

(A) Pallis (B) Agritech
(C) Greenhorn (D) Dagritech

Direction for questions 16 to 20: Answer these questions based on the information given below.

The below table gives the distribution of recognized educational institutions in a few states in India.

State / Union territory	Primary schools	Middle schools	High schools	Colleges for general education	Colleges for professional education	Deemed universities
Andhra Pradesh	58249	14472	14255	1080	279	22
Arunachal Pradesh	1303	333	184	7	1	1
Assam	33236	8019	4832	298	63	6
Bihar	53351	13571	5008	742	47	18
Goa	1267	443	448	20	10	1
Gujarat	15602	21143	6343	422	112	12
Haryana	11013	1892	4228	150	52	5
Himachal Pradesh	10877	1768	1954	65	12	3
Jammu & Kashmir	10926	3728	1504	33	12	4
Karnataka	22404	27712	8612	916	304	16
Kerala	6758	2973	4182	186	62	8
Madhya Pradesh	62530	25090	8471	413	78	16
Maharashtra	45971	24574	16059	1208	535	29
Manipur	1752	795	637	15	7	2
Uttar Pradesh	88927	20429	9063	758	189	27

16. If all the given states are arranged in the ascending order of the total number of primary and middle schools, then which state is the fourth from the last?
(A) Bihar (B) Andhra Pradesh
(C) Kerala (D) Maharashtra

17. A state in which the number of primary schools is more than the number of high schools, while the number of colleges for professional education is more than half that of the colleges for general education is said to be good in educational infrastructure. How many states are good in educational infrastructure?
(A) 0 (B) 1
(C) 2 (D) 3

18. In how many states is the number of high schools more than the number of colleges for general education by at least 300%?
(A) 15 (B) 14
(C) 11 (D) None of these

19. In all the states in which the number of primary schools is more than 50,000, the respective state governments own 50% of these primary schools. Find the least number of such states that should be clubbed together so that the total number of primary schools owned by the state governments of the states is more than 1,00,000.
(A) 2 (B) 3
(C) 4 (D) 5

20. Which state has the largest difference between the total number of primary and middle schools as compared to the number of Deemed Universities?
(A) Bihar (B) Madhya Pradesh
(C) Andhra Pradesh (D) Uttar Pradesh

Direction for questions 21 to 24: Answer these questions based on the information given below.

The details of the employees in five companies P, Q, R, S and T.

Company	No. of employees	Ratio of male to female employees	Avg. age of female employees	Avg. age of all employees
P	17865	2 : 1	28	32
Q	18183	8 : 11	31	30
R	21384	7 : 5	26	29
S	27185	3 : 2	29	28
T	16568	3 : 5	34	37

21. In which company is the number of female employees, the highest?

22. By how much does the number of male employees in companies R and S together exceed the number of male employees in the other three companies?

23. What is the average age (approximated to the closest integer) of the male employees in company R?

24. What is the difference between the number of female employees in the company with the highest number of male employees and the number of male employees in the company with the lowest number of female employees?

Direction for questions 25 to 28: These questions are based on the following table, in which the number of persons living in a residential area and their choices of certain brands of chocolates are given.

	Dairy Milk	Eclairs	Kit Kat	Perk	Amul	Bar One	Temptations	Munch
Dairy Milk	256	222	139	127	242	111	73	63
Eclairs	214	498	144	109	268	121	53	129
Kit Kat	118	152	178	102	149	892	47	112
Perk	128	134	76	144	82	74	48	108
Amul	214	232	151	112	286	124	34	123
Bar one	79	64	58	43	61	124	63	72
Temptations	43	29	31	68	48	53	88	59
Munch	74	83	91	69	107	121	118	132

(A) The number in each cell denotes the number of people who prefer different brands. The brand in the row represents first preference and the brand in the column represents second preference.

(B) Row 1, Column 1 = 256 (i.e., there are 256 people whose only choice of chocolates is Dairy Milk) Row 1, Column 2 = 222 (i.e., there are 222 people whose first preference is Dairy Milk and second preference is Eclairs)

25. The number of people who like only Bar One form approximately what per cent of those who like only Kit Kat?
(A) 150% (B) 143% (C) 70% (D) 67%

26. The price of a piece each of Perk and Munch is ₹12 and ₹8, respectively. What is the ratio of total sales of Perk and Munch, if the persons buying only Perk and only Munch are considered in the given area? Assume each person buys only one chocolate of his first preference.
(A) 3 : 2 (B) 96 : 37
(C) 5 : 6 (D) None of these

27. What is the total number of people who like only one brand of chocolates?
(A) 1706 (B) 1233
(C) 1126 (D) 2008

28. The number of people whose first preference is Eclairs and second preference is Munch is what percentage more than the number of people whose first preference is Munch and second preference is Eclairs?
(A) 35.65% (B) 55.42%
(C) 51.38% (D) 38.42%

Direction for questions 29 to 32: Answer these questions based on the information given below.

The percentage growth in sales turnover of five companies over the respective previous years.

Name of the Company	1998	1999	2000	Projected for 2001
A	10	20	30	45
B	15	15	30	38
C	8	20	30	30
D	5	30	20	40
E	12	20	28	42

29. In which of the following companies, is the percentage growth in sales turnover highest from 1997 to 2000?
(A) A (B) B
(C) D (D) E

30. Company D had earned a profit of ₹41 crores, which was 25% of its total sales in 2000. What was its sales in 1997 approximately?
(A) ₹64 crores
(B) ₹88 crores
(C) ₹164 crores
(D) ₹100 crores

31. In 1999, the sales of company B and company D are ₹100 crores and ₹130 cores, respectively. What was the ratio of their sales in 1997?
(A) 420 : 529
(B) 10 : 13
(C) 529 : 710
(D) 11 : 19

32. In 1999 the sale of each company was ₹130 crores. How many companies had sales below ₹100 crores in 1997?
(A) 2 (B) 4
(C) 5 (D) 1

Direction for questions 33 to 37: These questions are based on the following table, which shows the cumulative distribution of number of employees regarding the amount claimed as transportation expenses by employees.

The employees are from five departments HR, Marketing, Logistics, Accounts and Administration of Company XYZ. The number of employees in the departments in the same order are 20, 30, 15, 25 and 40, respectively. Every employee in the company has claimed transportation expenses in each of the given months.

	March				April				May			
Department	< ₹400	< ₹500	< ₹600	< ₹700	< ₹400	< ₹500	< ₹600	< ₹700	< ₹400	< ₹500	< ₹600	< ₹700
HR	8	10	14	16	12	15	16	17	11	14	15	18
Marketing	15	18	21	27	16	17	22	25	14	16	18	26
Logistics	9	10	12	13	8	9	13	14	8	11	12	15
Accounts	13	15	19	21	18	20	21	23	20	21	22	24
Administration	20	26	31	35	22	27	32	36	31	33	35	37

33. Considering any department in any month, what is the least number of employees who claimed at least ₹400 but less than ₹600 as transportation expenses?
(A) 2 (B) 3
(C) 4 (D) 6

34. What is the maximum number of employees in the company who claimed at least ₹450, but at most ₹750 as transportation expenses in the month of March?
(A) 52 (B) 58
(C) 65 (D) 67

35. Which of the following statements is true?

(A) The total number of employees who claimed more than ₹700 towards transportation expenses in the month of April is 15.

(B) Considering the Marketing department, the number of employees who claimed at least ₹600 towards transportation expenses in March is more than the number of employees who claimed at least ₹400 but less than ₹600 in the month of May.

(C) In the month of May, considering all the departments, the number of employees who claimed at least ₹600 but less than ₹700 as transportation expenses is 17.

(D) The total number of employees in the accounts department who claimed at least ₹400 but less than ₹500 as transportation expenses in all the given three months is 18.

36. For which of the following departments is the number of employees claiming at least ₹600 as transportation expenses the highest in any month?

(A) HR (B) Marketing
(C) Accounts (D) Administration

37. What is the maximum number of employees claiming exactly ₹600 as transportation expenses in the month of April?

(A) 8 (B) 9
(C) 10 (D) 11

Direction for questions 38 to 42: Answer these questions based on the information given below.

There was an inter-school competition of different events. The following table gives the details of students who participated and the prizes they won.

School	Sports		Essay writing		Painting		Music	
	A	B	A	B	A	B	A	B
P	100	20	16	4	100	30	60	10
Q	60	10	20	9	80	28	100	12
R	40	4	40	10	50	16	40	7
S	80	32	80	24	120	25	10	1
T	120	48	70	14	50	8	30	5

A – The number of students participating
B – The number of students who won prizes

In any event a school X is said to perform better than school Y, if the ratio of the number of prizes won to the number of students who participated in that event is greater for school X than school Y.

For all questions assume that no student takes part in more than one event.

38. What percentage of the participants won prizes in painting?

(A) 22.5% (B) 25%
(C) 26.25% (D) 26.75%

39. Which school performed the best in music?

(A) P (B) R
(C) S (D) T

40. What was the highest ratio of the number of prizes won per student participated for any single event for any of the five schools?

(A) 0.35 (B) 0.40
(C) 0.45 (D) None of these

41. What percentage of the students from school S did not win any prize?

(A) 71.7 (B) 74.2
(C) 76.8 (D) 78.5

42. The prizes won by students of schools Q and R in sports and essay writing is approximately what percentage of the prizes won by all the students?

(A) 9.7 (B) 10.4
(C) 10.8 (D) 11.5

Direction for questions 43 to 46: Answer these questions based on the information given below.

The following table gives the details of a project which is divided into ten tasks from I through X. The tasks can be done in any order as long as the other tasks which are required to be done before them are completed. The project is said to be completed when all the tasks are completed.

Task	Duration (in hours)	Other tasks to be completed before starting this task
I	2	—
II	3	—
III	5	I
IV	2	II, III
V	7	I, II
VI	1	IV
VII	5	II, V
VIII	2	I, II
IX	6	V, VII
X	4	II, IV, VIII

43. What is the shortest time (in hours), from the start, in which task IV can be completed?

44. What is the maximum number of tasks that can be completed in 10 hours from the start?

45. What is the minimum time (in hours), from the start, in which task X can be completed?

46. What is the minimum time taken (in hours) for the project to be completed?

Direction for questions 47 to 50: Answer these questions based on the information given below.

Prices of Cycles of Five Brands

Brand	Price per Cycle (in ₹)
Neon	1400
Ranger	2200
Hercules	2500
Atlas	1200
Avon	1100

Collections from the sales of cycles of the above five brands over a five-year period (in ₹ crore)

Brand	Year				
	2011	2012	2013	2014	2015
Neon	46	35	21	63	70
Ranger	77	33	27.5	49.5	115.5
Hercules	62.5	31.25	81.25	50	75
Atlas	30	48	54	66	30
Avon	27.5	38.5	44	55	71.5

There are two rating companies, namely TRICIL and CRICIL which rank the brands of cycles on the basis of two distinct parameters.

TRICIL ranks the brands from 1 to 5 on the basis of (decreasing order of) the collections in a year while CRICIL ranks brands from 1 to 5 on the basis of (decreasing order of) the number of cycles sold in that year.

If a brand is ranked first in TRICIL ratings, it gets 100 points for that year. If rated second, it gets 90 points. If rated third, 80 points and so on. The brand placed fifth, therefore, gets 60 points.

47. Which brand collected the maximum number of TRICIL points over the five-year period?
(A) Ranger (B) Hercules
(C) Atlas (D) Neon

48. Which brand is ranked 1st in the CRICIL charts the maximum number of times?
(A) Atlas (B) Ranger
(C) Neon (D) Avon

49. Which brand is ranked 2nd in the CRICIL ratings in the year 2015?
(A) Atlas (B) Avon
(C) Neon (D) Ranger

50. Which brand showed the highest percentage drop in the number of cycles sold in any year from 2012 to 2015 with respect to the year immediately preceding it?
(A) Neon (B) Avon
(C) Ranger (D) Atlas

EXERCISE-2

Direction for questions 1 to 4: The table below gives the population ratio of males to females and the percentage of literates in a region across six years.

Year	Population (lakhs)	Males : Females	Percentage of literate males	Percentage of literates
2010	7.3	1031 : 1000	61.2	52.6
2011	8.4	1073 : 1000	63.7	54.1
2012	8.7	1061 : 1000	64.5	55.2
2013	9.2	1089 : 1000	64.8	56.3
2014	8.1	1007 : 1000	65.2	56.5
2015	7.8	981 : 1000	64.5	55.7

1. What was the percentage increase in the number of males from 2011 to 2012?
(A) 2 (B) 3
(C) 4 (D) 5

2. In which year was the percentage increase in the number of females, when compared to the previous year, the highest?
(A) 2011 (B) 2012
(C) 2013 (D) 2014

3. What was the percentage of literate females in 2013?
(A) 44.8 (B) 47.0
(C) 48.2 (D) 49.6

4. In which of the given years was the ratio of literate males to literate females, the highest?
(A) 2010 (B) 2011
(C) 2012 (D) 2013

Direction for questions 5 to 9: Answer these questions based on the information given below.

External Debt and Debt Indicators of Various Countries for the Year 1994

S. No.	Country	EDT (US $ billion)	EDT as a percentage of GNP	PVD as a percentage of GNP	PVD as a percentage of XGS	TDS as a percentage of XGS	Concessions as a percentage of EDT
1.	Brazil	150	26.8	32	310	32.1	1.8
2.	China	120	19.3	16	75	8.9	15.9
3.	Indonesia	97	56.5	54	186	30	29.9
4.	Argentina	77	27.8	29	406	32	3.2
5.	India	99	34.2	26	214	26.3	48.2
6.	Turkey	66	57.4	38	188	31.2	12.1
7.	Thailand	61	43.1	39	96	15.6	11.3
8.	Malaysia	25	36.9	34	38	7.7	13.5
9.	Philippines	39	59.3	56	163	18.5	28.7
10.	Mexico	128	35.2	33	224	33.9	1.3
11.	Russia	94	25.0	13	158	6.2	7.0
12.	Korea	54	15.3	1.4	45	6.8	9.0

EDT → External Debt (Total)
PVD → Present Value of Debt
GNP → Gross National Product
XGS → Export of Goods and Services
TDS → Total Debt service

5. What is the approximate total debt service on the present value of the debt of Turkey (in US $ billion)?

6. For which of the given countries is the Gross National Product the greatest? (type the S. No. for the country)

7. What is the TDS of Malaysia as a percentage of its GNP (approximately)?

8. How many of the countries listed above have a GNP of at least US $200 billion?

9. For how many of the countries listed does the GNP exceed the XGS by 900% or more?

Direction for questions 10 to 13: Answer these questions based on the information given below.

The following table gives the percentage break up of expenses of the Sharma family on different items from 2012 to 2015.

Expense type	2012	2013	2014	2015
Rent	27.2	29.2	28.5	27.9
Food	14.3	15.1	16.2	15.6
Travel	6.2	5.8	6.0	5.7
Entertainment	5.3	4.8	5.1	5.8
Education	22.7	23.5	22.8	24.2
Others	24.3	21.6	21.4	20.8

10. If the expenses on rent increased by 10% every year from 2012, then what is the approximate percentage increase in the education expenses from 2012 to 2015?

11. If entertainment expenses showed a 50% increase from 2012 to 2015, then the expenses on food in 2015 is what percentage of that in 2012?

12. The expenses under which head showed the highest percentage increase from 2013 to 2015?

13. If the expenses on travel in the given period was the highest in 2014, then the total expenses in 2015 is at most what percentage of the total expenses in 2014?

Direction for questions 14 to 18: Answer these questions based on the information given below.

	PP_1	PP_2	PP_3	PP_4
GD_1	234	576	384	473
GD_2	346	278	272	968
GD_3	570	225	483	354
GD_4	425	840	372	527
GD_5	640	920	486	225
GD_6	725	386	680	324

	GD_1	GD_2	GD_3	GD_4	GD_5	GD_6
D_1	580	663	721	816	917	1127
D_2	640	1006	927	834	339	556
D_3	775	713	916	614	572	1157
D_4	480	576	812	537	528	911
D_5	574	847	1108	913	737	668
D_6	386	902	778	786	748	1248
D_7	447	853	883	488	497	625
D_8	533	912	525	845	1216	359

The two tables provide the information regarding the distances between four printing presses PP_1 to PP_4 and six godowns from GD_1 to GD_6 and the distances between the six godowns and the eight destinations from D_1 to D_8. All the distances are in kilometres and the cost of transporting one tonne of material for every kilometre is ₹250.

14. The maximum cost per tonne of transporting material from any printing press to any destination is
(A) ₹5.34 lakhs (B) ₹6.24 lakhs
(C) ₹4.70 lakhs (D) ₹7.80 lakhs

15. The approximate difference between the maximum and minimum cost per tonne of transporting the material from PP_4 to D_5 is
(A) ₹2.02 lakhs (B) ₹2.15 lakhs
(C) ₹3.56 lakhs (D) ₹3.05 lakhs

16. In how many distinct ways can the material be transported to D_7 from PP_1 or PP_3?
(A) 16 (B) 96
(C) 12 (D) None of these

17. The least total distance for transporting material between any printing press and any destination is
(A) 814 km (B) 497 km
(C) 722 km (D) 564 km

18. The number of distinct ways of transporting materials from any printing press to any destination is
(A) 18 (B) 24
(C) 48 (D) None of these

Direction for questions 19 to 23: Answer these questions based on the information given below.

The following tables give the details regarding 100 banks that were started after 1990. The details are the respective NPAs, profits and the number of branches.

Table-1		Table-2		Table-3	
NPAs	Number of Banks	Profit (₹Crore)	Number of Banks	Number of Branches	Number of Banks
10%	3	1000	4	700	6
9%	11	900	12	650	13
8%	16	800	18	600	19
7%	24	700	23	550	28
6%	39	600	34	500	36
5%	49	500	42	450	44
4%	71	400	58	400	53
3%	84	300	73	350	75
2%	93	200	86	300	82
1%	98	100	91	250	89

In each of the above three tables, the numbers in the second column give the number of banks for which the value of the relevant parameter is greater than the corresponding value mentioned in the first column. For example, from the second row of Table-2, there are 12 banks which have a profit of more than ₹900 crore. When any two banks are compared, the bank with the higher profit always has a lower level of NPAs but more number of branches than the bank with a lower profit.

19. How many of the 100 banks have a profit more than ₹400 crore but not more than ₹700 crore and also have more than 400 branches?

20. How many of the given 100 banks have more than 500 branches but NPAs of not more than 4%?

21. How many of the given 100 banks have a profit of not more than ₹100 crore and NPAs of more than 1% but do not have more than 250 branches?

22. How many of the given 100 banks have NPAs of more than 2% but not more than 8% and also have a profit of more than ₹200 crore but not more than ₹800 crore?

23. For how many of the given banks are the NPA's not more than 5% but has a profit more than ₹700 crore?

Direction for questions 24 to 27: Answer these questions based on the information given below.

The following table gives the marks scored by four students, namely Anand, Balu, Chetan and Deepak in the three areas Verbal, Quant and Reasoning of a mock CAT paper. The four students are disguised in the tables as A, B, C and D in no particular order.

	Student			
Section	**A**	**B**	**C**	**D**
Verbal	24	41	40	27
Quant	34	36	35	32
Reasoning	36	31	36	32

It is also known that, in reasoning, none of the other three students scored more than Chetan. Balu's total score in the three sections differs from that of Anand's by 3 marks.

24. What can be said regarding the following two statements?
Statement 1: Deepak scored the lowest marks in the reasoning section.
Statement 2: Anand's total score in the three sections is more than that of Deepak.
(A) If Statement 1 is true, then Statement 2 is necessarily true.
(B) If Statement 1 is true, then Statement 2 is necessarily false.
(C) Both Statement 1 and Statement 2 are true.
(D) Neither Statement 1 nor Statement 2 is true.

25. What can be said regarding the following two statements?
Statement 1: Balu's lowest score is in the reasoning section.
Statement 2: Anand's lowest score is in the quantitative section.
(A) If Statement 2 is true, then Statement 1 is necessarily false.
(B) If Statement 1 is false, then Statement 2 is necessarily true.
(C) If Statement 1 is true, then Statement 2 is necessarily true.
(D) None of the above

26. What can be said regarding the following two statements?
Statement 1: Anand had the highest score in the verbal section.
Statement 2: Balu had the highest score in the quant section.
(A) Both the Statements could be true.
(B) At least one of the Statements must be true.
(C) At most one of the Statements must be true.
(D) None of the above

27. If Deepak got his lowest score in the verbal section, then which of the following is true?
(A) Chetan's lowest score is in the reasoning section.
(B) Chetan's lowest score is in the quant section.
(C) Chetan's lowest score is in the verbal section.
(D) No definite conclusion is possible.

Direction for questions 28 to 31: Answer these questions based on the information given below.

The following table gives the number of students who passed in four subjects Maths, Physics, Chemistry and Biology in the three sections A, B and C in class X of a school. Each section had a student strength of 40.

Section	Maths	Physics	Chemistry	Biology
A	28	31	39	26
B	34	32	37	33
C	26	34	31	29

28. The number of students in section A who passed in all the four given subjects is at most __________.

29. The number of students in section C who passed in all the four subjects is at least __________.

30. At most how many students in section B passed in exactly one of the four subjects?

31. The number of students who passed in both Physics and Chemistry in the three sections combined is at most __________.

Direction for questions 32 to 35: The following table gives the information about the population and the literacy rate for different countries.

Country	Population (in lakhs)	% of male population	Female literacy rate (%)	% of illiterate population
A	500	54	35	52
B	280	56	42	46
C	740	50	32	40
D	360	58	28	56
E	410	55	36	50

$$\text{Per capita income} = \frac{\text{Total income}}{\text{Total population}}$$

32. The female population of country B is ________.
(A) 118.4 lakhs (B) 123.2 lakhs
(C) 126.4 lakhs (D) 128.8 lakhs

Difficulty Level-2: Moderate

33. The ratio of the number of female literates to that of total literates is the least for country

(A) A (B) B
(C) C (D) D

34. What is total the number of literates in country C (in million)?

(A) 40.6 (B) 41.2
(C) 47.2 (D) 44.4

35. In how many countries is the total number of illiterates more than two crore?

(A) 5 (B) 2
(C) 3 (D) 4

Direction for questions 36 to 39: Answer these questions based on the information given below.

Details of five families, namely A, B, C, D and E.

Detail	Family				
	A	B	C	D	E
Number of members	5	6	3	6	8
Annual income (₹lakhs)	3.6	4.8	6.0	4.0	7.2
Number of working members	2	1	2	1	2
Number of children	1	2	1	2	0
Own house / rented house	Rented	Own	Rented	Own	Own
Own a car / two-wheeler	Two-wheeler	Car	Car	Two-wheeler	Car

Activities taken-up by the families during weekends (Saturday and Sunday) in the month of January.

Family	Number of times an activity was taken up by the family				
	Hotel	Films	Courtesy calls	Temple	Resort
A	1	0	4	2	1
B	3	3	2	0	0
C	1	2	0	3	1
D	2	3	1	2	0
E	3	3	0	1	1

Note: Any family can take up at most two activities (not necessarily distinct) during a weekend. However, any family taking up the activity 'Resorts' on a weekend cannot take up any other activity on that weekend.

36. If January 3rd falls on a Sunday, at the most how many of the given families may not have taken up any of the given activities during at least one weekend of the month?

(A) 5 (B) 2
(C) 3 (D) 4

37. Find the number of families which satisfy each of the following three criteria.

(i) Average annual income per member is more than ₹70,000.
(ii) The number of visits made by the family to temple on a weekend is at least one in the month of January.
(iii) The family does not stay in a rented house.

(A) 1 (B) 2
(C) 3 (D) 4

38. If January 10th falls on a Monday, then how many of the given five families have an average annual income of at least ₹3.00 lakh per working member and also went to a resort on at least one weekend but definitely did not go to a film on more than two weekends in the month of January?

(A) 0 (B) 1
(C) 3 (D) 4

39. If January 30th falls on a Sunday, what is the maximum number of families going to a hotel, without clubbing any other kind of activity with it, during at least one weekend in the month of January?

(A) 5 (B) 2
(C) 3 (D) 4

Direction for questions 40 to 44: These questions are based on the following information. Four activities A_1, A_2, A_3 and A_4 are carried out daily in an organization. The following data represents the ten people, to whom the activities are assigned. The activities may be executed in any order.

Person	Activities			
	A_1	A_2	A_3	A_4
Rani	✓	×	×	✓
Ranjan	×	×	✓	×
Aakash	×	✓	×	✓
Biplab	✓	✓	✓	×
Menon	✓	✓	×	✓
Manoj	✓	×	×	×
Mukesh	×	×	×	✓
Amitabh	✓	×	✓	✓
Priya	×	✓	✓	✓
Priyanka	×	✓	✓	×

'✓' indicates that the activity is assigned to the person while '×' indicates that the activity is not assigned to the person. For example, Rani is assigned the activities A_1 and A_4 but is not assigned the activities A_2 and A_3.

The following data indicates the minimum and the maximum time period required to complete the activity by any person, to whom the work is assigned. The time (in hrs) taken to complete any activity is an integral number.

Activities	Minimum time (hrs)	Maximum time (hrs)
A_1	1	4
A_2	2	4
A_3	2	5
A_4	2	5

No person can execute two different activities at the same time. Any activity can be completed only by the person who started executing it.

40. If Amitabh is more efficient than Mukesh by X% in executing the activity A_4, then which of the following cannot be the value of X?

(A) 60 (B) 50
(C) 25 (D) 100

41. If on a day, every person started executing the activities assigned to him at 9 a.m., what could be the maximum possible number of persons who could have completed all the activities assigned to them in the next four hours?

(A) 9 (B) 1
(C) 6 (D) 5

42. What is the least possible time in which a team of exactly two persons working together can complete each of the four activities exactly once?

(A) 4 (B) 3
(C) 7 (D) None of these

43. If a group of three persons working together completed all the activities in less than 2 hours, which amongst the following can be the group?

(A) Rani, Ranjan and Aakash
(B) Aakash, Biplab and Menon
(C) Amitabh, Priya and Priyanka
(D) None of these

44. If Manoj and Mukesh started and completed the respective activities assigned to them at the same time, then which of the following cannot be the time (in hrs) taken by Manoj to complete the activities assigned to him?

(A) 5
(B) 1
(C) 2
(D) More than one of the above

Direction for questions 45 to 48: Answer these questions based on the information given below.

Six candidates who were interviewed for faculty positions in a reputed coaching institute are ranked according to their performances in six parameters, such as Educational qualification (E), Analytical Ability (A), Logical Ability (L), Communication Skills (V), Teaching Skills (T) and Creativity (C). No two candidates got the same rank in any single parameter and the ranks of a candidate in no two parameters are the same. The six candidates considered for selection were Vani, Pallavi, Raju, Martin, Asma and Scarlet and the ranks obtained by them in some of the parameters are given below.

	Person					
Parameters	**Vani**	**Pallavi**	**Rajiv**	**Martin**	**Asma**	**Scarlet**
Educational Qualification (E)	3			1		
Analytical Ability (A)	2					4
Logical Ability (L)	1	2	3	4	5	6
Communication Skills (V)			4			
Teaching Skills (T)	6				3	
Creativity (C)			1	3		2

A candidate gets 12 points for each parameter in which he/she gets the first rank, 8 points for rank 2, 5 points for rank 3, 3 points for rank 4, 2 points for rank 5 and 1 point for rank 6.

45. If only parameters A, C, V and T are considered, then which candidate got the fourth highest total score?

(A) Vani (B) Pallavi
(C) Rajiv (D) Martin

46. If only parameters E, A, L and T are considered, then which candidate got the second highest total score?

(A) Martin (B) Rajiv
(C) Vani (D) Asma

47. If the parameters other than L and T are considered, then the difference between the scores of the candidates getting the highest score and the least score is

(A) 6 (B) 8
(C) 10 (D) 14

48. If only parameters E, A and V are considered and only the candidates getting the top four scores are selected, then which of the candidates are not selected?

(A) Martin and Rajiv (B) Rajiv and Scarlet
(C) Scarlet and Martin (D) Rajiv and Vani

Direction for questions 49 and 50: Answer these questions based on the information given below.

Literacy rates (in percentages) among various classes in India.

Years	Scheduled Castes (C)	Scheduled Tribes (T)	Other Categories (O)
1961	10·27	8·54	28·30
1971	14·67	11·30	34·45
1981	21·38	16·35	43·57
1991	37·41	29·60	52·21
2001	54·69	47·10	64·80

Note: These three classes account for the whole of India's population.

49. If in the year 2001, the ratio of the population of the classes O, C and T was 7 : 3 : 1, then what was the approximate literacy rate in India in that year?
(A) 56%
(B) 58%
(C) 60%
(D) 62%

50. If none of the given classes constituted more than 50% or less than 25% of India's population in any of the years, in how many of the given years was the number of literates in India at least one-third of the total population?
(A) 4
(B) 3
(C) 2
(D) 1

EXERCISE-3

Direction for questions 1 to 5: Answer these questions based on the information given below.

A group of six colleges, 1 to 6, conducted a common written test with four different sections, each with a maximum of 50 marks. The following table gives the aggregate as well as section cut off marks fixed by the six colleges. A student will get admission only if he/she gets marks greater than or equal to the cut off marks in each of the sections and his/her aggregate marks are at least equal to the aggregate cut off marks as specified by the college.

	School cut off marks				Aggregate cut off marks
	Section A	Section B	Section C	Section D	
College 1	38			37	151
College 2	37		41		157
College 3			42		159
College 4		41		35	156
College 5		35		39	162
College 6	41	38	40		160

1. Shyam got calls from all the six colleges. What could be the minimum aggregate marks obtained by him?

2. Anand got calls from three colleges. What could be the maximum aggregate marks scored by him?

3. Priya got calls from two colleges. What could be the minimum marks obtained by her in a section?

4. Ramesh did not get calls from even a single college. What could be the maximum aggregate marks obtained by him?

5. What is the maximum difference between the aggregate marks obtained by two students who got calls from exactly one college?

Direction for questions 6 to 10: Answer these questions based on the information given below.

A company started manufacturing solar power generators in the year 2008. The panels used in the generators can be used for exactly one year and is discarded after that. The company provides panels, free of cost, for the first two years in which they have to be replaced. After that, the panels bought from company costs ₹2000 while third party vendors provide it for ₹1500. Every year, 30% of the customers buy the panels from the company while the remaining opt for the cheaper option available. The following table gives the number of filters sold or given free by the company and sold by third-party vendors in each year from 2012 to 2016. Three of the values have been left blank. Assume that all solar power generators manufactured from 2008 are working currently.

Replacement source	2012	2013	2014	2015	2016
Company				3900	4460
Third party	840	1610	2170	2800	3640

6. How many solar power generators were sold in 2013?

(A) 1000 (B) 1100
(C) 1200 (D) 1300

7. How many solar power generators were sold from 2008 to 2010?

(A) 1800 (B) 2000
(C) 2100 (D) 2300

8. In 2012, how many panels did the company sell or replace?

(A) 2260 (B) 2410
(C) 2640 (D) 2760

9. How many solar power generators were sold in 2010?

(A) 800 (B) 1000
(C) 1100 (D) 1200

10. How many solar power generators did the company sell in 2015?

(A) 1200 (B) 1400
(C) 1500 (D) Cannot be determined

Direction for questions 11 to 14: Answer these questions based on the information given below.

The below table represents the results of the annual performance appraisal in a company. Each individual's performance is rated on a scale of 1 – 20 and it is represented in the table under various parameters like designation, experience, etc. For example, 6 managers with an experience of 2 years, got an appraisal in the range from 12 – 14 with at least one of them getting a 12 and one of them getting a 14.

Experience (in years)	Performance appraisal score	
	Engineers	**Managers**
1	1(12, 12)	6(15 – 17)
2	4(10 – 13)	6(12 – 14)
3	3(12 – 16)	10(10 – 12)
4	5(9 – 15)	5(10 – 13)
Total	13	27

11. At the most, what percentage of the employees of the company have an appraisal score equal to or more than 15?

(A) $46\frac{2}{3}$ % (B) $33\frac{1}{3}$ %

(C) 30% (D) $43\frac{1}{3}$ %

12. The average performance appraisal score among the following is the highest for

(A) All the managers with 2 years' experience
(B) All the engineers with 3 years' experience
(C) All the managers
(D) Cannot be determined

13. Among those with three and four years of experience, the average performance appraisal score of managers differs from that of engineers by ______ points

(A) At least 3
(B) At least 4
(C) At most 2.12
(D) At most 3.79

14. What is the minimum average appraisal score for all the managers in the company?

(A) 12.25 (B) 12.06
(C) 11.89 (D) 11.72

Direction for questions 15 to 17: Answer these questions based on the information given below.

The table gives the details of the number of people (in the age group from 30 to 50 years) in five different companies, namely A, B, C, D and E in the years 2011 and 2016. No employee joined or left these companies or shifted to another company.

Company	2011	2016
A	27	38
B	46	39
C	50	60
D	74	82
E	110	110

15. What is the minimum possible number of employees who crossed the age of 50 years between 2011 and 2016?

(A) 7 (B) 29
(C) 22 (D) None of these

Difficulty Level-3: Advanced

16. If the number of employees who crossed the age of 50 years between 2011 and 2016 in company C is the maximum possible, then what is the number of employees who entered the age group of 30 to 50 years between 2011 and 2016?

(A) 50 (B) 10

(C) 60 (D) 40

17. What is the least possible number of employees who entered the 30 to 50 years of age group between 2011 and 2016?

(A) 20 (B) 29

(C) 22 (D) 26

Direction for questions 18 to 21: Answer these questions based on the information given below.

Details of the Indian widget industry

Ratio	**Year**					
	2010	**2011**	**2012**	**2013**	**2014**	**2015**
Profit Margin	0.27	0.3	0.24	0.3	0.33	0.36
DS Ratio	0.70	0.75	0.90	1.00	1.10	1.20
EXIM Ratio	0.60	0.64	0.72	0.50	0.60	0.68

$$\text{Profit Margin} = \frac{\text{Average selling price per widget}}{\text{Average cost price per widget}} - 1$$

$$\text{DS Ratio} = \frac{\text{Industry demand (by volume) for widgets}}{\text{Industry supply (by volume) for widgets}}$$

$$\text{EXIM Ratio} = \frac{\text{Volume of exports of widgets}}{\text{Volume of imports of widgets}}$$

Note:

(i) Industry demand includes domestic demand as well as export demand.

(ii) Industry supply includes domestic supply as well as imported supply.

(iii) The average export price per widget = The average selling price per widget.

(iv) The average import price per widget = The average cost price per widget.

18. In which of the given years was the average selling price per widget was the lowest, given that there was a uniform increase of 20% in the average cost price per widget every year?

(A) 2010 (B) 2012

(C) 2013 (D) 2014

19. Find the volume of widgets exported in the year 2012 as a percentage of the industry demand for widgets in that year.

(A) 64%

(B) 80%

(C) 83⅓%

(D) Cannot be determined

20. If the total value of widgets imported in the year 2013 was ₹200 crore, then what was the total value of the widgets exported in that year?

(A) ₹30 crore

(B) ₹60 crore

(C) ₹130 crore

(D) ₹220 crore

21. If the volume of widgets imported increased by a steady 16% every year, then during which of the following periods did the volume of widgets exported increase by the maximum percentage?

(A) From 2011 to 2012

(B) From 2013 to 2014

(C) From 2014 to 2005

(D) From 2010 to 2011

Direction for questions 22 to 25: These questions are based on the following information.

The following table provides the rankings of twenty companies based on eight parameters.

Ranks on different parameters

	Parameter							
Company	**Market capitalization as on 1st April, 2017**	**Percentage growth in market capitalization over 1st April, 2016**	**Sales in 2016-17**	**Percentage change in sales over 2015-16**	**Gross profit in 2016-17**	**Percentage change in gross profit over 2015-16**	**Net profit in 2016-17**	**Percentage change in net profit over 2015-16**
A	9	6	7	10	15	8	20	4
B	7	19	17	14	5	12	5	20
C	16	10	6	6	10	20	12	9
D	13	14	16	20	3	6	8	16
E	6	1	13	8	20	14	16	1
F	20	18	8	1	14	10	4	7
G	8	8	5	15	16	1	10	13
H	15	2	15	13	9	19	13	11
I	18	7	14	18	19	5	19	17
J	2	5	1	2	2	15	6	6
K	10	11	9	19	11	11	1	12
L	11	15	18	3	6	7	11	2
M	17	13	2	16	17	18	17	18
N	1	9	11	12	8	2	14	14
O	14	16	19	7	13	16	9	10
P	12	3	3	4	4	13	7	8
Q	3	17	20	17	18	9	2	15
R	19	12	10	9	7	3	18	3
S	4	4	4	5	1	17	15	19
T	5	20	12	11	12	4	3	5

For all questions, assume that the same twenty companies were ranked in all the given years.

22. How many of the given companies had a market capitalization which was definitely less than that of company K, as on 1st April 2016?

23. If profitability is the ratio of net profit to sales, how many of the given companies have a profitability which was definitely more than that of company M in 2016–17?

24. What could have been the best rank of company E in terms of sales in 2015–2016?

25. If Tax paid = Gross profit – Net profit, then how many of the given companies could have paid the highest tax in 2016–17?

Difficulty Level-3: Advanced

ANSWER KEYS

Exercise 1

1. (A)	9. 0	17. (A)	25. (C)	33. (A)	40. (C)	47. (C)
2. (D)	10. 5750	18. (A)	26. (D)	34. (C)	41. (A)	48. (A)
3. (C)	11. (C)	19. (B)	27. (A)	35. (B)	42. (B)	49. (D)
4. (D)	12. (D)	20. (D)	28. (B)	36. (B)	43. 9	50. (C)
5. (B)	13. (B)	21. S	29. (D)	37. (D)	44. 7	
6. 4950	14. (B)	22. 3006	30. (D)	38. (D)	45. 13	
7. 5	15. (C)	23. 31	31. (A)	39. (B)	46. 21	
8. 9	16. (D)	24. 1036	32. (B)			

Exercise 2

1. (B)	10. 38	18. (D)	27. (C)	36. (C)	45. (C)
2. (A)	11. 150	19. 30	28. 26	37. (A)	46. (A)
3. (B)	12. Entertain-ment	20. 29	29. 0	38. (B)	47. (A)
4. (D)		21. 9	30. 8	39. (A)	48. (B)
5. 8	13. 105.25	22. 66	31. 94	40. (A)	49. (C)
6. 2	14. (A)	23. 23	32. (B)	41. (C)	50. (C)
7. 7	15. (B)	24. B	33. (C)	42. (A)	
8. 7	16. (C)	25. (C)	34. (D)	43. (D)	
9. 3	17. (D)	26. (C)	35. (D)	44. (D)	

Exercise 3

1. 163	5. 24	9. (C)	13. (D)	17. (B)	20. (C)	23. 15
2. 189	6. (C)	10. (B)	14. (C)	18. (A)	21. (B)	24. 8
3. 6	7. (D)	11. (C)	15. (A)	19. (D)	22. 4	25. 3
4. 173	8. (A)	12. (D)	16. (C)			

SOLUTIONS

1. If we assume that every accident in district P resulted in 2 deaths, at least 70 accidents resulted in deaths.

The required value = $\frac{70}{802} \times 100 = 8.7\%$.

2. The total number of deaths from 2008 to 2010 = 143 + 139 + 132 = 414

The number of deaths in 2014 = 195

The required percentage = $\frac{414}{195} \times 100 = 212.3\%$

3. Total deaths in 2012 = 173 + 112 + 243 + 183 + 76 = 787

Accidents is 2012 = 1012 + 745 + 1432 + 1175 + 493 = 4857

The required percentage = $\frac{787}{4857} \times 100 = 16.2\%$

4. We can see that with every passing year, the number of deaths in each state is generally increasing. So, we need to check for only 2013 and 2014.

The total deaths in 2013 = 168 + 132 + 252 + 189 + 87 = 828.

Total deaths in 2014 = 195 + 127 + 236 + 191 + 98 = 847

The highest is in 2014.

5. The highest percentage increase was in 2009 = $\frac{24}{189} \times 100 = 12.7\%$

6. Number of fan regulators accepted = (550/10) (90) = 4950.

7. Percentage of assemblies rejected from machines P, Q, R, S, T is 10%, 20%, 10%, 14%, 15%, respectively.

So, 10% of production of assemblies of P = 600.

20% of production of assemblies of Q = 1000.

10% of production of assemblies of R = 550.

14% of production of assemblies of S = 840.

15% of production of assemblies of T = 1650.

So, 10% of production of T = 1650/15 × 10 = 1100.

By observation we can say that the number of assemblies by machine T is the highest.

8. Number of fan regulators from P, that were rejected = 600

Number of fan regulators from Q, that were rejected = 550

Required percentage = $\dfrac{600-550}{550} \times 100\% = 9\frac{1}{11}\%$

9. Number of fan regulators assembled by

T = (1650/15) (100) = 11,000.

Number of fan regulators assembled by

P = 600/10 × 100 = 6000.

Number of fan regulators assembled by

Q = 1000/20 × 100 = 5000.

Number of fan regulators assembled by T is equal to the total number of fan regulators assembled by P and Q. Therefore, required percentage is zero.

10. Number of fan regulators assembled by machine R and S are 550/10 × 100 = 5500 and 840/14 × 100 = 6000 respectively.

Their average number of assemblies

$$= \left(\frac{5500+6000}{2}\right) = 5750$$

11. In Pallis, the number of employees of age 30 years and above but below 50 years = 165 – 125 = 40.

Total number of employees in Pallis = Those below 50 years + Those aged 50 years and above = 165 + 36 = 201.

The required ratio = $\dfrac{40}{201} \approx \left(\dfrac{1}{5}\right)$

Similarly, the ratio for Dagritech

$$= \frac{187-96}{187+60} = \frac{91}{247}, \text{which lies between } \frac{1}{2} \text{ and } \frac{1}{3}.$$

For Greenhorn = $\dfrac{213-68}{231+68} = \dfrac{145}{299} \cong \dfrac{1}{2}$

For Farmtech = $\dfrac{148-97}{148+48} = \dfrac{51}{196} \approx \dfrac{1}{4}$

For Starshine = $\dfrac{141-108}{141+72} = \dfrac{33}{213} \cong \dfrac{1}{7}$

The highest ratio is $\dfrac{145}{299}$.

12. The total number of employees aged above 20 years but below 30 years for all the companies = 494.

Those aged above 20 years but below 50 years = 854. Those aged 30 years or above but below 50 years = 854 – 494 = 360.

Those aged forty years or above but less than 50 years = 854 –646 = 208.

Those aged 50 years or above but below 60 years = 284.

Those aged 30 years or more = 360 + 284 = 644.

In this age group, there are maximum number of employees.

13. In Greenhorn with the inclusion of 27 employees, those aged below 50 years but above 20 years = 213 + 27 = 240. Those aged below 40 years but above 20 years will be 93 + 27 = 120.

The required percentage = $\dfrac{120}{240} \times 100 = 50\%$.

14. In Pallis, the minimum age of the employees in the age group of above 20 years and below 30 years can be nearly 20 years and maximum age will be nearly 30 years. In the age group of 30 years to below 40 years, minimum age is 30 years. Maximum age will be 40 years. Considering the lower bounds in each case, we get the minimum average age as

$$= \frac{20\times125+30\times23+40\times17+50\times36}{125+23+17+36}$$

$$= \frac{5670}{201} = \frac{5670}{200}$$

= 28 years. (approximately)

Maximum average age

$$= \frac{30\times125+40\times23+50\times17+60\times36}{201}$$

$$= \frac{7680}{201} = \frac{7680}{200}$$

= 38 years.

(Actually 29 years should be considered, but by considering 30, the average will not be affected). Therefore, the average age of the employees of Pallis should lie between 28 years and 38 years and among the given choices only Choice (B) satisfies this condition.

15. By adding the number of employees aged below 50 years and those aged 50 years or above, the total number of employees of a company can be determined. By observation we can say that Greenhorn is the company with the highest number of employees, i.e., 213 + 68 = 281.

16. As the question asks for the state ranked 4th from the last after arranging the given states in ascending order, instead of arranging them in ascending order first and then finding the answer, the question can be easily answered by arranging the given states in descending order and finding the fourth state ranked from the top.

By observation, Uttar Pradesh has the highest number of primary schools as well as middle schools. Similarly, Madhya Pradesh is in second position with a total of 87,620. Andhra Pradesh is in third position at 72,721, whereas the state ranked fourth is Maharashtra with 70,545.

17. By observation, in all the given states, the number of primary schools is more than the number of high schools. Also observing the given data, in none of the states is the number of professional colleges more than half that of the colleges of general education.

18. As the number of high schools is to be more than the number of colleges of general education by more than 300%, they should be at least four times the number of colleges of general education. By observation, in all the states, the number of high schools is more than the number of colleges of general education by at least four times.

19. By observation Uttar Pradesh, Madhya Pradesh and Andhra Pradesh accounts for more than 2,00,000 schools. As the respective state governments own 50% of these schools, these three states must be clubbed together for the state governments to have more than 1,00,000 schools.

20. Observing the given data Uttar Pradesh has the highest number of **primary** and middle schools, whereas in all cases the number of deemed universities is negligible (or very small) when compared to the number of schools. Hence, the difference will be maximum for Uttar Pradesh.

Solution for questions 21 to 24: The numbers of male and female employees in the different companies are as follows.

Company	Male employees	Female employees
P	11910	5955
Q	7656	10527
R	12474	8910
S	16311	10874
T	6213	10355

21. The number of female employees is the highest in company S.

22. Male employees in companies R and S together = 28,785
The number of male employees in companies P and Q together = 25779
The difference = 3,006

23. Total age of employees in company
$$R = 21384 \times 29 = 620136$$
Total age of female employees in company
$$R = 8910 \times 26 = 231660$$
$$\therefore \text{Average age of male employees} = \frac{388476}{12474} = 31$$

24. The required difference = 11910 – 10874 = 1036.

25. Number of people who like only Bar one = 124.
Number of people who like only Kit Kat = 178.
$$\frac{124}{178} \times 100 = 69.6\% \text{ or } 70\%$$

26. Number of people who like perk = 144.
Number of people who like Munch = 132.
Ratio of sales = 144 × 12 : 132 × 8 = 18 : 11

27. In whichever square the 'brand' in the column and the brand in the row meet, that gives the exact number of chocolates of the brand. The number of each brand of chocolates is in the diagonal form starting with 256 and ending with 132.
Adding all these values, we get:
$$256 + 498 + 178 + 144 + 286 + 124 + 88 + 132 = 1706$$

28. The number of people whose first preference is Eclairs and second preference is munch are 129 and the number of people whose first preference is munch and second preference is Eclairs are 83.
$$\% = \frac{129-83}{83} \times 100 = \left(\frac{129}{83} - 1\right) \times 100$$
$$= (1.55 - 1)\ 100 \cong 55\%$$

29. Out of the given choices, the increase in B will be greater than that of A, since 15% of 115 will be greater than 20% of 110 which is greater than 20% of 108. The tie is between A and E only.
The overall percentage increase from 1997 to 2000 in case of A = 1.3 × 1.2 × 1.1 × 1 = 171.6
In case of E = 1.28 × 1.1 × 1.12 × 1 = 172.03
The increase in E is the greatest.

30. Profit in 2000 = 41 crore
$$\therefore \text{ sales in 2000} = \frac{100}{25} \times 41 = 164\,\text{cr}$$
Let the sales in 1997 be x.
$x \times 1.05 \times 1.30 \times 1.20 = 164$
$\therefore\ x = 100.12$ crore

31. Sales of B in 1999
100 crores = 1.15 × 1.15 × sales in 1997
$$\text{Sales of B in 1997} = \frac{100}{1.15 \times 1.15}$$
Sale of D in 1999 = ₹130 crores = 1.05 × 1.3 × sales in 1997
$$\text{Sales of D in 1997} = \frac{130}{1.05 \times 1.3}$$
Ratio of sales of B and D in 1997
$$= \frac{100}{1.15 \times 1.15} : \frac{130}{1.05 \times 1.3}$$
$$= 420 : 529$$

32. Had the sales of company A in 1997 been 100, the sales in 1999 would have been 100 × 1.1 × 1.2 = 132.
Given that the sales was 130.
$$\therefore \text{Actual sales of 1997} = \frac{130 \times 100}{132} \text{ which is less than 100.}$$
Similarly, sales turnover of B in 1997
$$= \frac{130 \times 100}{132.5} \text{ less than 100 crores}$$

Difficulty Level-1: Foundation

Sales turnover of C in 1997 = $\frac{130 \times 100}{129.6}$ more than 100

Sales turnover of D in 1997 = $\frac{130 \times 100}{136.5}$ less than 100

Sales turnover of E in 1997 = $\frac{130 \times 100}{134.4}$ less than 100

There is only one company, whose sales turnover was more than 100 crores.

33. Number of employees who claimed at least ₹400 but less than ₹600 in March for HR = 14 – 8 = 6
Marketing = 21 – 15 = 6
In this way we observe that in March the least number is 12 – 9 = 3.
Considered other months, we get the least number as 2 for the month of May for the employees of the accounts department.

34. Since we require the greatest number of employees, we can consider that all the employees other than those claiming less than ₹400 fall in this range.
The number of such employees for HR = 20 – 8 = 12
Marketing = 30 – 15 = 15
Logistics = 15 – 9 = 6
Accounts = 25 – 13 = 12
Administration = 40 – 20 = 20
The required number = 12 + 15 + 6 + 12 + 20 = 65

35. Statement (A) may or may not be true as we do not know the number of those have claimed ₹700 and the number of employees who claimed more than ₹700.
Number of employees in the marketing department who claimed at least ₹600 in March = 30 – 21 = 9.
Number of employees in the marketing department who claimed at least ₹400 but less than ₹600 in May = 18 – 14 = 4
Statement (B) is true.

36. The number of employees claiming less than ₹600 is the highest for HR in March. Those claiming at least ₹600 will be the highest, i.e., 20 – 14 = 6
For Marketing the corresponding value is 30 – 18 = 12 (in May).
The corresponding values for Logistics, Accounts and Administration departments are 3, 6 and 9.

37. In April, from HR department those who claimed less than ₹700 = 17 and those who claimed less than ₹600 = 16.
Those paying at least ₹600 but less than ₹700 = 17 – 16 = 1.
This one person could have paid ₹600.
Similarly, for Marketing = 25 – 22 = 3.
For Logistics = 14 – 13 = 1
For Accounts = 23 –21 = 2
For Administration = 36 – 32 = 4
The greatest number of employees paying exactly ₹600 is 1 + 3 + 1 + 2 + 4 = 11

38. The number of participants in painting = 400
Number of people who won prizes = 107

The required percentage = $\frac{107}{400} \times 100 = 26.75\%$.

39. As the ratio of prizes won to students participated is highest for school R, (the ratio of B/A for music) it has performed the best.

40. For school Q, 45% of the students who participated won prizes in essay writing.

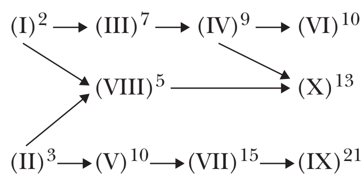
41. Total number of participants from school S = 290.
Number of students who won prizes = 82.
Number of students who did not win prizes = 208.

The required percentage = $\frac{208}{290} \times 100 = 71.7\%$.

42. Prizes won by students of Q and R in sports and essay writing = 33.
Total prize won by all the students = 114 + 61 + 107 + 35 = 317.

The required percentage = $\frac{33}{317} \times 100 = 10.4\%$.

Solution for questions 43 to 46: The shortest possible time to complete each task would be as follows:



(The number above each task denotes the minimum time (in hours) taken to complete that task.)

43. Task (IV) can be completed in 9 hours from the start.

44. The tasks that can be completed in 10 hours from the start are I, II, III, IV, V, VI and VIII, a maximum of seven tasks.

45. Task X can be completed in 13 hours from the start.

46. The project would be complete when all the tasks are over. The last task to be over would be task IX and it would be over in 21 hours from the start.

47.

Brand	Year and Points 2011	2012	2013	2014	2015	Total Points (TRICIL Rating)
Neon	80	80	60	90	70	380
Ranger	100	70	70	60	100	400
Hercules	90	60	100	70	90	410
Atlas	70	100	90	100	60	420
Avon	60	90	80	80	80	390

ATLAS collected the maximum TRICIL points.

48. Number of cycles sold (in lakhs).

Brand	Year 2011	2012	2013	2014	2015
Neon	3.3	2.5	1.5	4.5	5
Ranger	3.5	1.5	1.5	2.5	5.25
Hercules	2.5	1.25	3.25	2	3
Atlas	2.5	4	4.5	5.5	2.5
Avon	2.5	3.5	4	5	6.5

Atlas tops the CRICIL charts 3 times in the 5-year period.

49. RANGER was placed 2nd in the CRICIL ratings in the year 2015.

50. Ranger showed a maximum percentage decline in the number of cycles sold from the year 2011 to 2012.

EXERCISE-2

1. The number of males in 2011

$$= \frac{1073}{2073} x8.4 = 4.351 \text{ lakhs.} \times 8.4 = 4.351 \text{ lakhs}$$

The number of males in 2012

$$= \frac{1061}{2061} x8.7 = 4.48 \text{lakhs.} \times 8.7 = 4.48 \text{ lakhs}$$

The required percentage

$$= \frac{0.13}{4.35} x100 = 3\%. \times 100 = 3\%.$$

2. The number of females in the different years are

$$2010 = \frac{1000}{2031} x7.3 = 3.59 \times 7.3 = 3.59$$

$$2011 = \frac{1000}{2073} x8.4 = 4.05 \times 8.4 = 4.05$$

$$2012 = \frac{1000}{2061} x8.7 = 4.22 \times 8.7 = 4.22$$

$$2013 = 2013 \frac{1000}{2089} x4.2 = 4.40 \times 4.2 = 4.40$$

$$2014 = \frac{1000}{2007} x8.1 = 4.03 \times 8.1 = 4.03$$

$$2015 = \frac{1000}{1981} x7.8 = 3.93 \times 7.8 = 3.93$$

The percentage increase is the highest in 2011.

3. The number of males in 2013 = 9.20 – 4.40 = 4.80 lakhs.

The total literates

$$= \frac{56.3}{100} x9.2 = 5.18 \text{ lakhs.} \times 9.2 = 5.18 \text{ lakhs}$$

The number of literate males

$$= \frac{64.8}{100} x4.8 = 3.11. \times 4.8 = 3.11$$

The number of literates among females

$= 5.18 - 3.11 = 2.07.$

The percentage of literates among females

$$= \frac{2.07}{4.40} x100 = 47.0\%. \times 100 = 47.0\%.$$

4. As the ratio of males to females was the highest in 2013 and the total population was also the highest in that year, the ratio of literate males to literate females would be the highest in that year.

5. For Turkey $\frac{\text{PVD}}{\text{XGS}} = \frac{188}{100}$ and $\frac{TDS}{XGS} = \frac{31.2}{100}$

$$\frac{\text{PVD}}{\text{XGS}} \times \frac{\text{XGS}}{\text{TDS}} = \frac{188}{100} \times \frac{100}{31.2}$$

$$\Rightarrow \frac{\text{PVD}}{\text{TDS}} = \frac{188}{31.2} \Rightarrow \text{TDS} = \text{PVD} \times \frac{31.2}{188}$$

$$\text{But PVD} = \frac{\text{PVD}}{\text{GNP}} \times \text{EDT} \times \frac{\text{GNP}}{\text{EDT}}$$

$$\Rightarrow \text{TDS} = \frac{38}{100} \times \frac{100}{57.4} \times 72 \times \frac{31.2}{188}$$

$$\cong \frac{2}{3} \times 72 \times \frac{1}{6} \cong 8$$

6. From observation the answer must be either Brazil or China.

For Brazil,

$$\frac{\text{EDT}}{\text{GNP}} = \frac{26.8}{100}$$

EDT = \$150 billion

$$\therefore \quad \frac{150}{\text{GNP}} = \frac{26.8}{100}$$

$$\text{GNP} = \frac{150 \times 100}{26.8} = \text{US \$ 559 billion.}$$

Similar for China,

$$\text{GNP} = \frac{120 \times 100}{19.3} \cong 621.8$$

US\$ 621 billion.

7. For Malaysia,

$$\frac{\text{PVD}}{\text{GNP}} = \frac{34}{100}$$

$$\frac{\text{PVD}}{\text{XGS}} = \frac{38}{100}$$

$$\frac{TDS}{XGS} = \frac{7.7}{100}$$

$$\frac{\text{GNP}}{\text{PVD}} \times \frac{\text{PVD}}{\text{XGS}} \times \frac{\text{XGS}}{\text{TDS}} = \frac{100}{34} \times \frac{38}{100} \times \frac{100}{7.7}$$

$$\frac{\text{GNP}}{\text{TDS}} = \frac{1900}{130.9} = \frac{19000}{1309}$$

$$\frac{\text{TDS}}{\text{GNP}} = \frac{1309}{19000} \cong 7\%.$$

8. $\dfrac{\text{EDT}}{\text{EDS as \% of GNP}} \geq 200$ US \$ billion

This is true only for Brazil, China, Argentina, India, Mexico, Russia and Korea. A total of seven countries.

9. For GNP to exceed XGS by 900% or more:

$$\frac{\text{GNP}}{\text{XGS}} \geq 10 \Rightarrow \frac{\text{PVD as \% of XGS}}{\text{PVD as \% of GND}} \geq 10.$$

This is true for only Argentina, Russia and Korea.

10. Assume the expenses on rent in 2014 to be ₹100. The corresponding values in 2013, 2014 and 2015 would be 110, 121 and 133, respectively. As 27.2% of total = 100, the total expenses in 2012 would be $\dfrac{100}{27.2} \times 100 = 368$ cr

Similarly, total expenses in 2015 would be $\dfrac{133}{27.9} \times 100$ = 477 cr

Education expenses in 2012 = $\dfrac{22.7}{100} \times 368 = 83.5$

That in 2015 = $\dfrac{24.2}{100} \times 477 = 115.4$

The percentage increase = $\dfrac{32}{83.5} \times 100 = 38\%$

11. Assume that total expenses in 2012 to be 100. The expenses on entertainment = 5.3

∴ Expense on entertainment is 2015 = 5.3 × 1.5 = 7.95

$7.95 = \dfrac{5.8}{100} \times x$, when x is the total expenses in 2015.

∴ $x = 137$

Expenses on food in 2015 = $\dfrac{15.6}{100} \times 137 = 21.5$

Expenses on food in 2012 = 14.3

∴ Required value = $\dfrac{21.5}{14.3} \times 100 = 150\%$

Alternate solution:

As the percentage share of both food and entertainment increased by approximately the same percentage from 2012 to 2015, their corresponding increase also would be the same.

12. The percentage increase in the share of food and entertainment can be readily observed to be the highest.

Increase in share of food = $\dfrac{1.3}{14.3} \times 100 = 9.1\%$

Increase in share of entertainment = $\dfrac{0.50}{5.3} \times 100 = 9.45\%$

As the percentage increase in the share of entertainment is the highest from 2012 to 2015, the expenses under that head would have showed the highest percentage increase.

13. Assume that the total expenses in 2014 to be Rs. 100.

∴ Expenses on travel = Rs. 6

As expenses on travel in 2015 is at most Rs. 6, if the total expense that year is X.

$\dfrac{1.7}{100} \times X = 6$

∴ X = 105.25%

14. The maximum distance is from PP_2 to GD_5 and then from GD_5 to D_6.

∴ The maximum cost is (920 + 1216) 250 = ₹5.34 lakhs

15. The maximum cost is from PP_4 to GD_2 and then from GD_2 to D_5, i.e., (968 + 847) 250 = 1815 × 250

The minimum cost is from PP_4 to GD_5 and then from GD_5 to D_5, i.e., (225 + 737) 250 = 962 × 250.

∴ Required difference is 210 (1815 – 962) ≅ ₹2.15 lakhs.

16. Material can be transported from Printing Presses PP_1 or PP_3 to any godown and from any godown to D_7 in 2 × 6 = 12 ways.

17. The least distance between a printing press and godown is 225 km (PD_4 to GD_5). The least distance between a godown and the destination is 339 (GD_5 to D_2)

∴ The least distance between a printing press and destination is 225 + 339 = 564

18. The number of distinct ways is $4 \times 6 \times 8 = 192$.

19. Number of banks with a profit of more than ₹400 crore = 58
Number of banks with a profit of more than ₹700 crore = 23
Number of banks with more than 400 branches = 53
Of these 53 banks, every bank has a profit of more than ₹200 crore but only 30 have not more than ₹700 crore.

20. Number of banks with more than 500 branches = 36
Number of banks with NPAs of not more than 4%
$= 100 - 71 = 29$
Number of banks with more than 500 branches and NPAs of not more than 4% = Lesser of 29 and 36 = 29.

21. Number of banks with a profit of not more than ₹100 crore = 9
Number of banks with not more than 250 branches = 11
Banks with both the above attributes = 9
All these banks have more than 1% NPAs.
Hence, 9 banks satisfy the given conditions.

22. 82 banks have a profit of not more than ₹800 crore and also NPAs of more than 2%.
Of these 16 have NPAs not more than 8%.
Hence, 66 banks satisfy the given conditions.

23. Number of banks with NPA's not more than 5% = 100 – 49 = 51
Number of banks with a profit more than ₹700 crore = 23.
∴ 23 banks satisfy the given condition.

Solution for questions 24 to 27: As it is said that in reasoning none of the other three persons scored more than Chetan, Chetan is either A or C. From the second condition we can conclude that Balu and Anand is one among A or D in any order or one among B or C in any order.

24. If Deepak scored the lowest marks in the reasoning section, Deepak is student B which means Balu and Anand are one of A and D in any order and so statement 2 would be false.

25. If Balu's lowest score is in the reasoning section, Balu is student B and Anand is student C and the statement that Anand's lowest score is in the quantitative section is true.

26. If Anand gets the highest score in the verbal section, he is student B and Balu is student C.
∴ Both statements cannot be simultaneously true. Anand and Balu can also be A or D in any order in which case both statements would be false.
∴ At most one of the statements is true.

27. If Deepak gets his lowest score in the verbal section, he is student D in which case Chetan is student A.

28. The maximum number of students in section A who passed in all the four subjects is 26.

29. In section C, 14 students have failed in Maths, 6 students in Physics, 9 students in Chemistry and 11 students in Biology. If all these students are distinct, $14 + 6 + 9 + 11 = 40$ students would have failed in one subject each and so no student passed in all the four subjects.

30. For having the maximum number of students passing in exactly one subject, you should have the maximum number of students passing in all the four.
If x is the number of students in section B who passed in exactly one subject and y is the number of students who passed in exactly four subjects, then
$x + y = 40$ and $x + 3y = 136$
$x = 8$ and $y = 32$.

31. The maximum number of students who passed in both Physics and Chemistry in the different sections are A-31, B-32 and C-31, i.e., 31 + 32 + 31 = 94.

32. The female population of country B $= \frac{44}{100} \times 280$
$= 123.2$ lakhs

33. Let the population in each country be 100.
Literates in country A = 48

Female literates $= \frac{35}{100} \times 46 = 16.1$

Required ratio $= \frac{16.1}{48} = \frac{1}{3}$

Literates in country B = 54

Female literates $= \frac{42}{100} \times 44 = 18.48$

Required ratio $= \frac{18.48}{54} = 1/3$

Literates in country C = 60

Female literates $= \frac{32}{100} \times 50 = 16$

Required ratio $= \frac{16}{60} = 1/4$

Literates on country E = 50

Female literates $= \frac{36}{100} \times 45 = 162$

Required ratio $= \frac{16.2}{50} = 1/3$

It is lowest in country C.

34. Number of literates in country C $= \frac{60}{100} \times 740 = 444$ lakhs.

35. In countries A, C, D and E, the number of illiterates is more than two crores.

36. If January 3rd falls on a Sunday, there will be five Sundays or five weekends in the month. Family A has gone to a resort during a weekend. They cannot club any other

Difficulty Level-2: Moderate

activity with this. There are 7 more activities, which require four weekends. A has taken up at least one activity on every weekend. Analysing in this way we can find that B, C and D may not have taken up any activity during at least one weekend.

37. Families A, C, D and E have made at least one visit to a temple. Of these D and E are not staying in a rented house. D's average income per member per year is less than ₹70,000.

38. Families B, C, D and E have at least ₹3 lakhs income per working person per annum. Among these C and E go to a resort at least once in a month. But only C goes to films at the most twice a month. There is only one such family. E can go to a film on up to three weekends.

39. If January 30th falls on a Sunday, there will be five weekends during the month. For family A, one weekend is spent towards resorts. Barring 'Resorts' activity and hotel activity, the remaining 6 activities can be taken up during the remaining three weekends. In this way all the five families can go to a hotel without clubbing with any other activity.

40. Activity A_4 can be completed in either 2 or 3 or 4 or 5 hours. If Amitabh is more efficient than Mukesh, then he would take lesser time as compared to Mukesh.
Possibilities for the time taken

Amitabh	Mukesh	Value of x
2	3	50
2	4	100
2	5	150
3	4	33⅓
3	5	66⅔
4	5	25

So, Amitabh cannot be 60% more efficient than Mukesh.

41. The following persons cannot complete all the activities in exactly 4 hours. Biplab, Menon, Amitabh, Priya. (These persons are engaged in more than 2 activities) This implies exactly 6 persons can complete the activities assigned to them in 4 hours.

42. For the activities to be completed in the least time, by two persons where one person should work on two activities and the other person should work on the other two activities, i.e., (for example) Menon can work on activities A_1, A_2, while simultaneously Priya can work on A_3 and A_4. To complete A_1 and A_2, Menon will take 3 hours. While Priya completes A_3 and A_4 in 4 hours.

∴ A total of 4 hours is required to complete all the four activities.

43. No group of three persons can complete the work in 2 hours as to complete the activities A_2, A_3 and A_4 only, the group takes 2 hours and one of them takes one more hour to complete the work.

44. Manoj has to complete the activity A_1 while Mukesh has to complete A_4. Since Manoj should not take a time of 5 hours and Mukesh should not take a time of 1 hour, therefore, Manoj cannot do the work in 1 hour or 5 hours.

Solution for questions 45 to 48: Since no two candidates got the same rank in a single parameter and the rank of a candidate in no two parameters were the same, we can fill in the vacant ranks in the table given.

	Person					
Parameters	**Vani**	**Pallavi**	**Rajiv**	**Martin**	**Asma**	**Scarlet**
Educational Qualification (E)	3(5)	6(1)	2(8)	1(12)	4(3)	5(2)
Analytical Ability (A)	2(8)	3(5)	6(1)	5(2)	1(12)	4(3)
Logical Ability (L)	1(12)	2(8)	3(5)	4(3)	5(2)	6(1)
Communication Skills (V)	5(2)	1(12)	4(3)	6(1)	2(8)	3(5)
Teaching Skills (T)	6(1)	4(3)	5(2)	2(8)	3(5)	1(12)
Creativity (C)	4(3)	5(2)	1(12)	3(5)	6(1)	2(8)

The scores obtained by the candidates in different parameters are given within parenthesis.

(The ranks can be filled in the table with the fact that each of the 6 ranks will be present in each row and each column only once)

45. If only the parameters A, C, V and T are considered, the scores of the candidates will be as follows.

Candidate	Vani	Pallavi	Rajiv	Martin	Asma	Scarlet
Score	14	22	18	16	26	28

The candidate with fourth highest total score is Rajiv.

46. If only E, A, L and T are considered the candidates, then their scores will be as follows.

Candidate	Vani	Pallavi	Rajiv	Martin	Asma	Scarlet
Score	26	17	16	25	22	18

The candidate with the 2nd highest score is Martin.

47. If the parameters other than L and T are considered the candidates, then their respective scores will be as follows.

Candidate	Vani	Pallavi	Rajiv	Martin	Asma	Scarlet
Score	18	20	24	20	24	18

The difference between the highest score and the least score is 24 – 18 = 6.

48. With only parameters E, A and V being considered for selection the candidates and their total scores will be as follows.

Candidate	Vani	Pallavi	Rajiv	Martin	Asma	Scarlet
Score	1	18	12	15	23	10

Now candidates with the lowest scores will not be selected. The candidates with the lowest scores are Rajiv and Scarlet.

49. The literacy rate in India is

$$= \frac{7 \times 64.8 + 3 \times 54.7 + 1 \times 47.1}{7 + 3 + 1}$$

$$= \frac{453.6 + 164.1 + 47.1}{11} = 60.4\%.$$

50. For 1961 and 1971, the number of literates would definitely be less than one-third. While for 1991 and 2001 it would definitely be more than one-third.
$\therefore$ We must check only for 1981.
Considering the best case, the literacy rate in India would be only

$$= \frac{43.57 \times 50 + 21.38 \times 25 + 16.35 \times 25}{50 + 25 + 25}$$

$$= \frac{2178.5 + 534.5 + 408.75}{100} = \frac{3121.75}{100}$$

$$= \frac{3121.75}{100} = 31 \times 22\%, \text{ which is less than one-third.}$$

$\therefore$ It is at least one-third for only two years.

EXERCISE-3

1. To get calls from all colleges, he has to clear the cut off of all colleges and also clear the aggregate cut off (162).
$\therefore$ He has to score at least 41 marks in section A, 41 in section B, 42 in section C and 39 in section D.
The minimum aggregate score = 41 + 41 + 42 + 39 = 163.

2. If he scores 50, 50, 39 and 50 marks each in sections A, B, C and D, he would miss calls from colleges 2, 3 and 6.
$\therefore$ The maximum marks for three calls are 189.

3. To get calls from two colleges, she has to score at least 156 marks. If she scores 50, 50, 6 and 50 marks each in section A, B, C and D, respectively, then she would get calls from colleges 1 and 4.

4. If Ramesh scored 50, 50, 39 and 34 marks in sections A, B, C, and D, respectively, he would miss calls from all the six colleges.
The marks would be 50 + 50 + 39 + 34 = 173.

5. Minimum marks of a student who got one call is 151.
Maximum marks for a student with only one call is
50 + 50 + 39 + 36 = 175 marks.
The maximum difference is 175 – 151 = 24 marks.

Solution for questions 6 to 10: As the company started its operations in 2008, for solar power generators manufactured in 2008, the company will provide panels free of cost in 2009 and 2010. 70% of the people would take panels from third party vendors in 2011. The 2011 value is 70% of the sum of total production in 2008, 2009 and the total productions in 2010. In a similar way, we can determine the production in other years.

	2008	2009	2010	2011	2012	2013	2014	2015	2016
Sales	X	Y	1100	800	900	1200	1500	1400	
Company replacement		X	X + Y	0.3X + Y + 1100	360 + 1100 + 800 = 2260	360 + 330 + 800 + 900 = 2390	3030	3900	4460
Third party replacement			0	0.7X	840	1610	2170	2800	2640

As 0.7 $(x + y)$ = 840
X + Y = 1200
As the company replacement of 2014 is $\frac{30}{100}$ (1200 + 1100 + 800) + 900 + 1200 and the company replacement in 2015 is $\frac{30}{100}$ (1200 + 1100 + 800 + 900) + 1200 + 2014 production (X) = $\frac{30}{100}$ (4000) + 1200 + X = 3900
X = 3900 – 2400 = 1500.
Similarly, production in 2015 = 1400.

6. 1200 solar power generators were sold in 2013.

7. 2300 solar power generators were sold from 2008 to 2010.

8. In 2012, the company sold or replaced 2260 panels.

9. 1100 solar power generators were sold in 2010.

10. The company sold 1400 solar power generators in 2015.

11. Among those with 1-year experience, all 6 managers have 15 or more points.
Among those with 2 years' experience, no one has 15 or more points.
Among those with 3 years' experience, at most 2 engineers an have 15 or more (because the 3rd engineer has 12 points) points.
Among those with 4 years' experience, at most 4 engineers can have 15 or more points.
∴ At most 6 + 0 + 2 + 4 = 12 employees can have a performance appraisal of 15 or more points.

12. The average appraisal of each group varies within a range; therefore, nothing can be uniquely stated without knowing all the values.

13. The average performance appraisal of engineers with 3 and 4 years' experience lies in the range of 11.375 $\left(\dfrac{2(12)+16+4(9)+15}{8}\right)$ to 14.125

$\left(\dfrac{12+2(16)+9+4(15)}{8}\right)$.

Similarly, the average performance appraisal of engineers with 3 years and 4 years work experience lies in the range 10.33 to 12.
Maximum possible difference = 14.125 – 10.33 = 3.79.

14. The minimum average appraisal score of the managers would be

$$\frac{\begin{pmatrix}5\times15+1\times17+5\times12+1\times14+9\times10\\ +1\times12+4\times10+1\times13\end{pmatrix}}{27}$$

$$=\frac{75+17+60+14+90+12+40+13}{27}$$

$$=\frac{321}{27}=11.89$$

15.

Company	2011	2016	Change
A	27	38	+11
B	46	39	–7
C	50	60	+10
D	74	82	+8
E	110	110	0

16. A maximum of 50 people would have crossed the age of 50 years between 2011 and 2016.
This implies that 60 people must have entered the age group.

17. Least possible number of employees
= 11 + 10 + 8 = 29.

18. Average selling price (S) = (1 + profit margin) × Average cost price (C)
Given that the average cost price increased by 20% every year. Let C in 2000 = 100
⇒ S in 2000 = 1.27 × 100 = 127
S in 2001 = 1.3 × 120 = 156
S in 2002 = 1.24 × 144 ≅ 179
S in every other year will be more than that in 2002. Hence, S is lowest in the year 2000.

19. The ratio $\dfrac{\text{Exports}}{\text{Demand}}$ cannot be determined independent of the ratio $\dfrac{\text{Imports}}{\text{Supply}}$.

Therefore, in 2002, we can find that $\dfrac{\text{Supply}}{\text{Demand}}\times\dfrac{\text{Exports}}{\text{Imports}}$ as $\dfrac{0.72}{0.9}$. But we cannot find only $\dfrac{\text{Exports}}{\text{Demand}}$.
Hence, the question cannot be answered.

20. Given that the value of widgets imported in 2003 = ₹200 crore
= Average import price per widget × Volume of imports
= Average cost price per widget × Volume of imports
Now, value of widgets exported = Average selling price per widget × Volume of exports.

Hence, $\dfrac{\text{Average cost price}\times\text{Volume of imports}}{\text{Average selling price }\times\text{Volume of exports}}$

$$=\frac{1}{(1+0.3)}\times\frac{1}{0.5}$$

⇒ Value of exports = 0.65 × 200 = ₹130 crore.

21. Since the imports increased by a steady (i.e., equal) percentage every year, to find the year in which the exports increased by the highest percentage, we need to only consider the EXIM ratios. By observation, it is the highest from 2003 to 2004.

22. We can say that a company definitely had less market capitalization on 1st April 2016, when compared with company K, if both the following points are true.

(i) Market capitalization as on 1st April 2017 of the company is less than that of K, i.e., numerically higher rank in market capitalization happened for C, D, F, H, I, L, M, O, P and R.

(ii) Percentage growth in market capitalization of the company from 1st April 2016 to 1st April 2017 is more than that of company K, i.e., numerically lower rank in percentage growth in market capitalization happened for A, C, E, G, H, I, J, N, P and S. Both happened for C, H, I and P, i.e., a total of four companies.

23. Profitability of any company is definitely more than that of company M, if

(i) Net profit of the company is definitely more than that of company M, i.e., numerically lower rank compared to company M.

(ii) Sales of the company are definitely less than that of company M, i.e., numerically higher rank in sales compared to company M.

All companies (excluding M) except A, I, J and R satisfy. Hence, a total of 15 companies.

24. Looking at companies ranked lower than E in sales in 2016–17, it is possible that the difference in sales between E and the company just below it was very high in 2015–16 such that the companies which grew sales faster or slower than E was ranked below E. Now among companies which are ranked better than E in 2016–17, all companies which had a higher growth than E were ranked below E in 2015-16 and got ahead of E in 2016–17, i.e., companies C, F, J, P and S. So, E could have been ranked as high as 8th in 2015–16.

25. Company S had the highest gross profit while in terms of net profit, it was ranked 15th. So, it would have definitely paid more tax then all companies ranked from 1 to 14 in terms of net profit. Similarly, company R would have paid a higher tax than E and M. A also can be the company which paid the highest tax. Only one of these three companies could have paid the highest tax.

2 Bar Graphs

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:

- Understand about various types of data presented in the form of bar graphs.
- Get familiar with different types of bar graphs – horizontal bars, vertical bars, stacked bars.
- Learn how to convert bar graphs into tables and vice-versa.
- Get exposed to different types of questions based on bar graphs.
- Understand data which involves more than one data type – bar graph and line graph together, etc.

❒ BAR CHARTS

This is a type of graph which is widely used to depict data in a discrete way. They are accurate and the comparison of variables is very convenient.

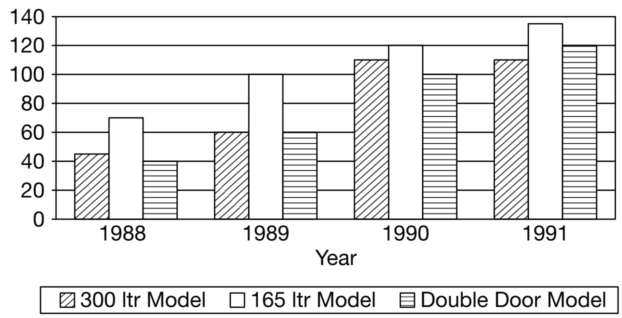


Fig. 2.1 Refrigerator sales of company abc (000's of units)

Figure 2.1 shows the model wise sales of refrigerators during four years. From this graph we can obtain the following:

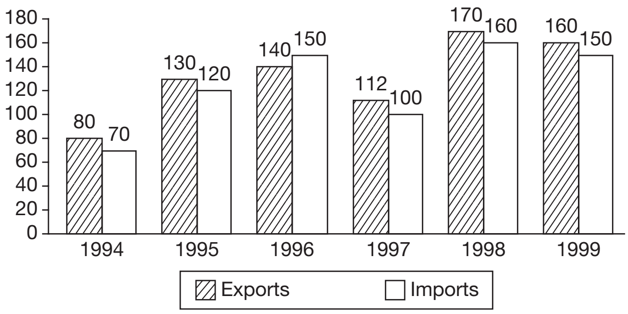
- Percentage contribution of each model to the company's total sales for four years.
- Relative increase or decrease in the share of each model.
- Sales trend of various models.

Using this bar chart one can carry out a detailed performance evaluation of the company with respect to the sales of the four year period from 1988 to 1991 for any given model. These bar charts can also be depicted horizontally. Another variation could be showing each product at one place (rather than each year at one place).

SOLVED EXAMPLES

Directions for questions 2.01 to 2.05: These questions are based on the information given below.

Exports and Imports of a country (in ₹ 000' crore)



2.01: In how many of the given years were the exports at least 10% more than the imports?
(A) 0 (B) 1
(C) 2 (D) 3

Sol: In 1994, exports = $80 > 70 + \frac{10}{100}(70) = 77$

In 1995, exports = $130 < 120 + \frac{10}{100}(120) = 132$

In 1996, exports < imports
∴ We need not consider this year.

In 1997, exports = $112 > 100 + \frac{10}{100}(100) = 110$

In 1998, exports = $170 < 160 + \frac{10}{100}(160) = 176$

In 1999, exports = $160 < 150 + \frac{10}{100}(150) = 165$

∴ The given condition was satisfied in two years.

2.02: What was the average exports for the given period (in '000 crore)?
(A) 145 (B) 132
(C) 126 (D) 119

Sol: Average exports

$$= \frac{80 + 130 + 140 + 112 + 170 + 160}{6} = 132$$

2.03: From 1995 to 1999, in which year was the percentage growth in exports, when compared to the previous year, the highest?
(A) 1995 (B) 1996
(C) 1997 (D) 1998

Sol: Exports in a year exceeded that in the previous year in 1995, 1996 and 1998. Percentages by which exports in 1995, 1996 and 1998 exceed the exports in the previous year were $\frac{50}{80}(100)\%$, $\frac{10}{130}(100)\%$ and $\frac{58}{112}(100)\%$, respectively.
The growth rate was the highest in 1995.

2.04: What is the simple average annual growth rate in the imports from 1994 to 1999?
(A) 15 (B) 18
(C) 19 (D) 23

Sol: Imports in 1994 (in '000 crore) = 70
Imports in 1999 (in '000 crore) = 150

Percentage growth = $\frac{150 - 70}{70} \times 100 = 115\%$

Average annual growth = $\frac{115}{5} = 23$

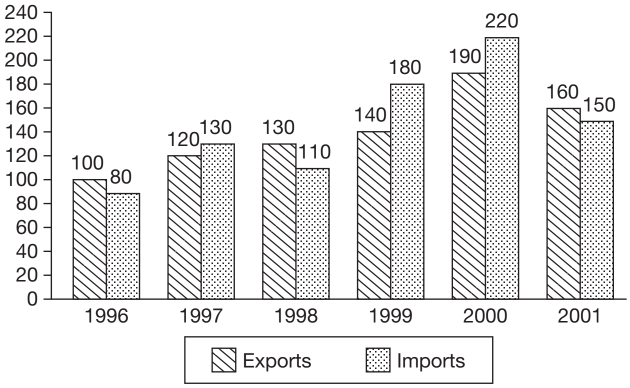
2.05: Among the years in which the imports as well as exports exceeded those in the previous years, in how many years was the percentage increase in imports less than the percentage increase in exports?
(A) 0 (B) 1
(C) 2 (D) 3

Sol: The imports as well as exports exceeded those in the previous years in 1995, 1996 and 1998.
∴ In none of the years was the given condition satisfied.

EXERCISE-1

Directions for questions 1 to 5: Answer these questions based on the information given below.

The value of exports/imports of a country (₹ in 000' crore)



Trade surplus = Exports – Imports.

Trade deficit = Imports – Exports.

1. For the period 1996 to 2001, what percentage of average exports is the cumulative trade deficit?

(A) 28.6% (B) 31.6%

(C) 27.3% (D) 32.3%

2. During the year 1999, the average cost of exports is ₹7000 per ton and that of imports is ₹6000 per ton. By what per cent is the total tonnage of exports less than the total tonnage of imports in that year?

(A) $66\frac{2}{3}\%$ (B) 50%

(C) 25% (D) $33\frac{1}{3}\%$

3. The percentage decrease of trade surplus from 2001 to 2002 is same as that from 1998 to 2001. Imports in 2002 increased by 20%. What is the value of exports in 2002 in thousands of crore?

(A) 180 (B) 185

(C) 190 (D) 195

4. It is decided to increase the exports by 10% every year over its previous year for the next three years from 2001 and also decrease the imports by 10% in the same way. What will be the value of total trade after three years, approximately in thousands of crore of rupees?

(A) 322 (B) 316

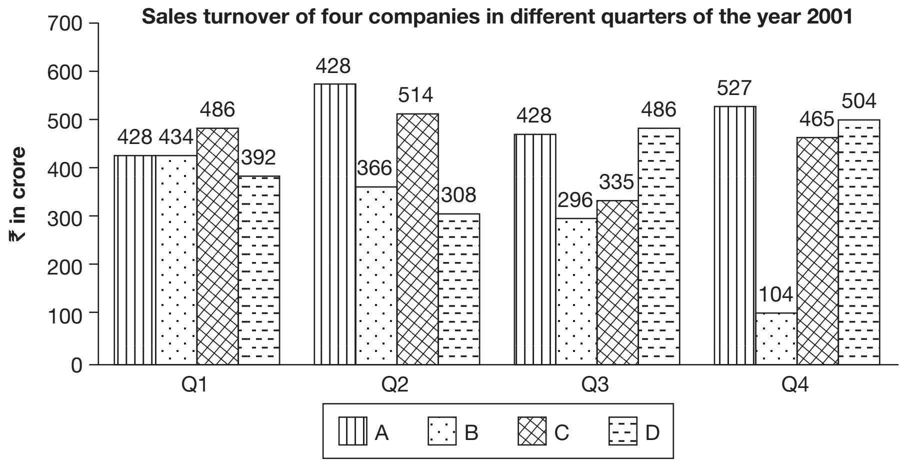
(C) 414 (D) 450

5. What was the approximate compounded annual growth rate of the exports from 1996 to 2001?

(A) 8% (B) 10%

(C) 12% (D) 14%

Directions for questions 6 to 10: Answer these questions based on the information given below.



6. Which company has the greatest sales turnover in the year 2001?

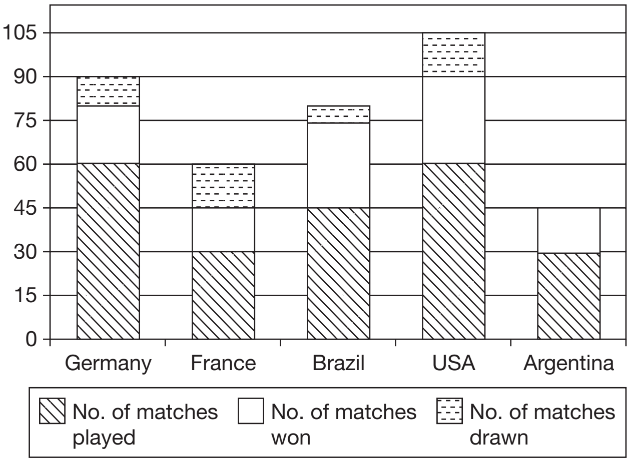
7. During the second quarter, the profits of A, B, C and D are in the ratio 8 : 7 : 6 : 5, respectively. For which of the four given companies, is profit as a percentage of sales turnover, the highest?

8. For how many quarters is the sales turnover of company A more than 25% of the total sales turnover of all the four given companies for that particular quarter?

9. What is the approximate percentage decrease in the total sales turnover of all the given companies from the second quarter to the third quarter?

10. For how many companies is the sales turnover consistently increasing or decreasing?

Directions for questions 11 to 15: The following questions are based on the stacked bar graph, which represents the overall record of matches played by football playing countries in the year 1998.



11. Among the given countries, which country lost maximum number of matches?

(A) Germany (B) USA
(C) Brazil (D) Argentina

12. If the number of matches played by Germany, Brazil and USA are doubled and the ratio of number of matches drawn and lost to total matches, respectively remains the same, then which of these teams have the highest success rate?

(A) Germany (B) Brazil
(C) USA (D) France

13. If the matches lost by Argentina to the other teams, namely to Germany, France, Brazil and USA are in the ratio 3 : 5 : 4 : 3, respectively then which of the following teams had the highest success rate against Argentina?

(A) France (B) Brazil
(C) Germany (D) Cannot be determined

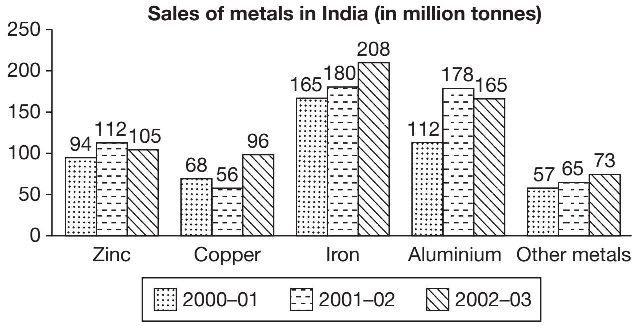
14. What is the ratio of matches lost by Germany to those lost by Brazil?

(A) 2 : 1 (B) 3 : 1
(C) 3 : 2 (D) 1 : 1

15. How many more matches were won by the three teams, which had the same success rate (matches won as a percentage of matches played) than the other two teams?

(A) 5 (B) 15
(C) 10 (D) 20

Directions for questions 16 to 20: Answer these questions based on the information given below.



16. The sales target for the sales of metals in the year 2002–2003 was 20% more than that of the actual sales in the year 2000–2001. What is the approximate percentage deficit or surplus achieved in the actual sales in the year 2002–2003?

(A) 8% deficit
(B) 9% surplus
(C) 30% surplus
(D) 120% deficit

17. The sales of Tin in 2000–2001 were 41.32% of the total sales of that of 'Other metals' and they increased by 10% every subsequent year. In 2002–2003, what percentage of the total sales were the sale of Tin?

(A) 3.8%
(B) 4.4%
(C) 7.7%
(D) Cannot be determined

18. If, for every year, the total sales of all metals are 80% of their quantity available and the total sales in the year 2003–2004 for every metal was 25% more than that in the year 2002–2003, then what was the quantity of copper available in the year 2003–2004 (in million tons)?

(A) 120 (B) 150
(C) 1480 (D) 164

19. What is the average annual percentage increase in the sales of all the metals from the year 2000–2001 to the year 2002–2003?

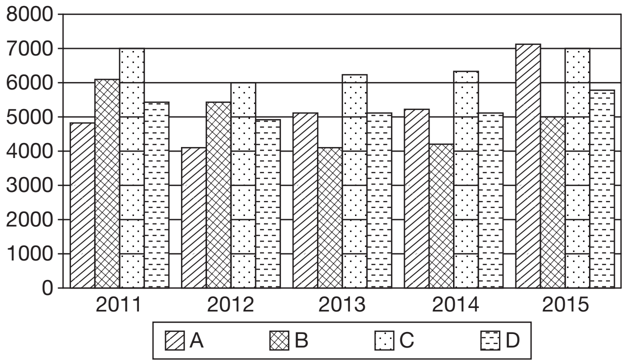
(A) 10.2% (B) 11.2%
(C) 30.6% (D) 15.2%

20. What is the approximate ratio between the sales of Gold, which is 25% of that of 'Other metals', in 2002–2003 to the sales of silver, which is 20% of that of 'Other metals' in 2000–2001?

(A) 0.625 (B) 0.976
(C) 1.6 (D) 1.976

Directions for questions 21 to 25: These questions are based on the following bar graph.

Production details of all the products manufactured by XYZ Ltd.



21. In how many of the given years was the production of D, as a percentage of that of B, more than 80% but less than 120%?

(A) 1 (B) 3
(C) 4 (D) 5

22. In which of the following years was the absolute change in the total production of XYZ Ltd. over that of the previous year, the highest?

(A) 2012 (B) 2013
(C) 2014 (D) 2015

23. In the year 2014, 37.5% of the production of D was exported. If the ratio of the total units exported of D to that of A was 5 : 6, then what percentage of the production of A was exported?

(A) 40% (B) 45%
(C) 50% (D) 60%

24. From 2015 to 2016, the production of B decreased by 40% and the production of every other product increased by 35%. In 2016, the production of B as a percentage of total production of all the four products is

(A) 10% (B) 15%
(C) 20% (D) 25%

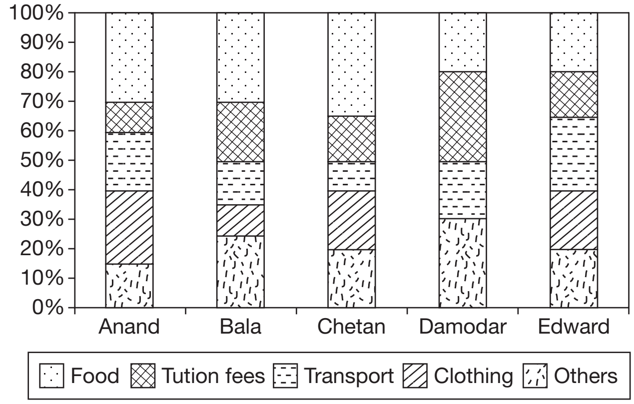
25. Which of the following statement/s is/are definitely true?

I. From 2011–2014, the production of only one of the products decreased and increased in alternate years.
II. For all the years put together, the production of the product D was the highest when compared to other products.
III. The percentage share of the production of D increased by about 1.7 percentage points from 2013–2015.

(A) Only II
(B) Only I
(C) Both I and II
(D) Both II and III

Directions for questions 26 to 30: These questions are based on the following stack bar.

Percentage breakup of expenses



26. If Bala's expenses on clothing are ₹3700, then how much did he spend on food and tuition fees together?

(A) ₹6250 (B) ₹6475
(C) ₹7200 (D) ₹7400

27. If Anand's expenses on food was ₹1620, then what was his expenses on the other four items together?

(A) ₹11400
(B) ₹10,800
(C) ₹9180
(D) None of these

28. If the total expenses of Bala and Chetan are in the ratio of 3 : 5, then what is the ratio of their 'other' expenses?

(A) 18 : 35 (B) 14 : 19
(C) 3 : 5 (D) 7 : 11

29. If Damodar spends 15% more on transport than Edward on clothing, then the total expenses of Damodar is what percentage more / less than that of Edward?

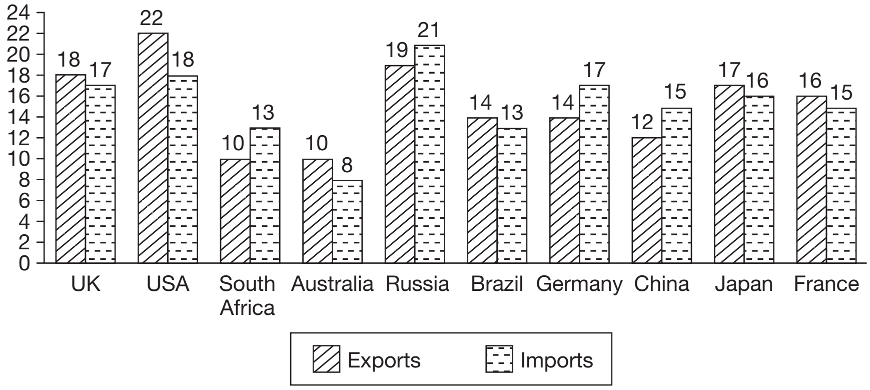
(A) 16.25% more
(B) 13.75% more
(C) 16.75% less
(D) 13.75% less

30. If Anand's expenses under each of the five heads is not less than that of Chetan's, then the total expenses of Anand is at least how many times that of Chetan's?

(A) 1.2 (B) 1.5
(C) 1.8 (D) 2.0

***Directions for questions 31 to 35*:** These questions are based on the following graph.

The country wise break up of exports/imports of country 'XYZ' in 1996 (in ₹ thousand crore)



Trade Surplus = Exports – Imports; Trade Deficit = Imports – Exports

31. The cumulative trade deficit of country XYZ is approximately what per cent of its average imports from each of the above-mentioned countries?

(A) 65% (B) 9%
(C) 6.5% (D) 0.6%

32. If the average cost of exports is ₹2000 per ton and that of imports is ₹3000 per ton, then by what per cent is the total tonnage of exports more/less than the total tonnage of imports?

(A) 33.3% more (B) 49% less
(C) 32.8% more/less (D) 49% more

33. By what percentage are the imports from the country to which the exports are the highest more than the exports to the country from which the imports are the least?

(A) 175% (B) 80%
(C) 55.55% (D) 125%

34. Which of the following statements is true?

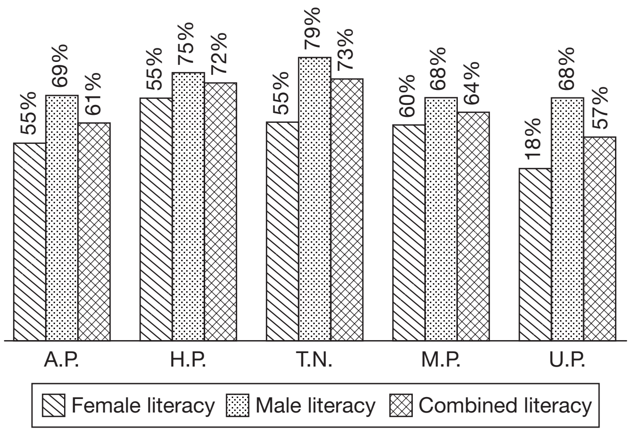
(A) Country XYZ has a cumulative trade surplus of ₹1 crore.
(B) The cumulative trade deficit of country XYZ is approximately one-fifteenth of its total imports.
(C) The trade deficit of country XYZ considering its trade with China alone is 300% more than its cumulative trade deficit/surplus.
(D) The difference between the highest exports to any country and the lowest imports from any country is equal to the average of the exports to Brazil and Germany.

35. What is the ratio of the total imports from Brazil, Japan, South Africa, Russia and China, to the total exports to the other five countries?

(A) 0.975 (B) 1.026
(C) 0.96 (D) None of these

Directions for questions 36 to 40: These questions are based on the bar graph and the table given below.

Literacy rate of several states in the year 2000



$$\text{Literacy rate} = \frac{\text{Number of literates}}{\text{Total population}}$$

Population (in crore) of the given states in the year 2000 is as follows:

A.P.	H.P.	T.N.	M.P.	U.P.
7	6.3	6.5	6	16

36. What is the ratio of number of females to the number of males in the state of T.N.?

37. By what percentage is the population of U.P. less than the total population of the other four states?

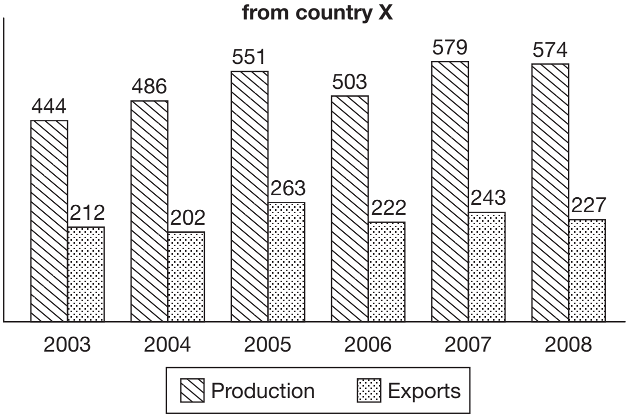
38. What is the total number of literate males and literate females in the state of U.P.?

39. In which of the following states is the number of literate males, the highest?

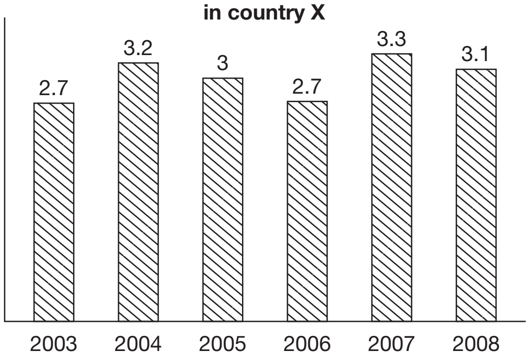
40. What is the number of males in the state of H.P.?

Directions for questions 41 to 44: Answer these questions based on the information given below.

Production and export of sugar (in million kg) from country X



Per capita consumption (in kg) of sugar in country X



Note: The country did not import sugar in any of the given years

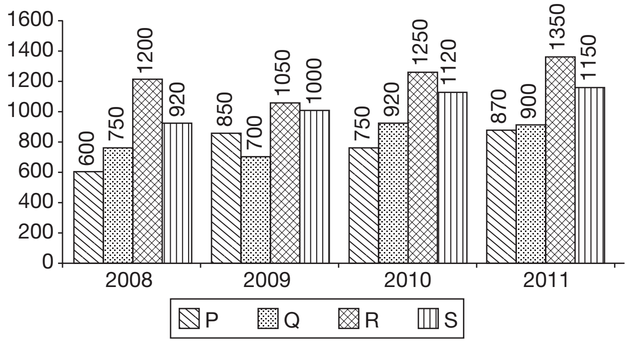
41. In which of the given years was the population of country X, the lowest?
(A) 2003 (B) 2004
(C) 2005 (D) 2006

42. In which year was the percentage increase in the population of country X, the highest?
(A) 2004 (B) 2008
(C) 2006 (D) 2007

43. If the government had planned to increase the exports by 10% each year from 2003 to 2008, what should have been the production (in million kg) in 2008 if the consumption was as given?
(A) 612 (B) 646
(C) 688 (D) 731

44. Which of the following has shown their greatest percentage increase from 2003 to 2008?
(A) Per capita consumption
(B) Production
(C) Population
(D) Consumption

Directions for questions 45 to 47: These questions are based on the information given below.



ABC Ltd produces four commodities, namely P, Q, R and S. The prices of P, Q, R and S, respectively in 2008 were ₹12, ₹15, ₹10 and ₹20 per kg. The price of each commodity goes up by 10% every year. The following bar graph gives the sales (in kg) of each of the four commodities from 2008 to 2011.

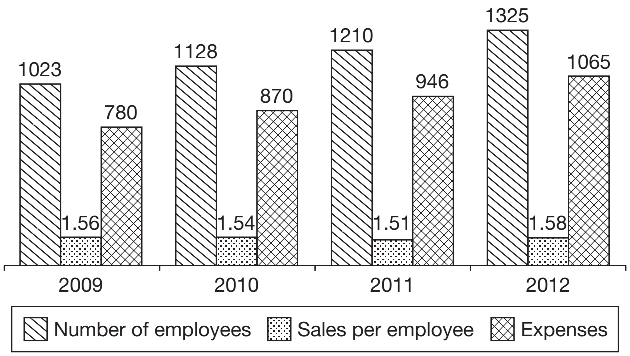
45. What was the sales (in ₹) of the company in 2009?
(A) ₹57,350 (B) ₹56,320
(C) ₹53,860 (D) ₹51,200

46. What was the percentage increase in the sales (in ₹) of the company from 2010 to 2011?
(A) 15.23 (B) 16.25
(C) 17.89 (D) 18.68

47. The percentage share of commodity P in the total sales (in ₹) of the company in 2011 is approximately
(A) 15.2 (B) 15.9
(C) 16.7 (D) 17.3

Directions for questions 48 to 50: Answer these questions based on the information given below.

The following bar graph gives the number of employees in a company, the sales per employee (in ₹ crore) and the expenses of the company (in ₹ crore) in four years from 2009 to 2012.



Sales of the company = (Number of employees) × (Sales per employee)

Profit = Sales – Expenses.

48. What was the highest percentage increase in sales of the company in any year, when compared to that of the previous year, in the given period?

(A) 8.5% (B) 10.2%
(C) 14.5% (D) 17.2%

49. Which of the following had the highest percentage increase in any year, when compared to that of the previous year, in the given period?

(A) Sales (B) Number of employees
(C) Expenses (D) Profits

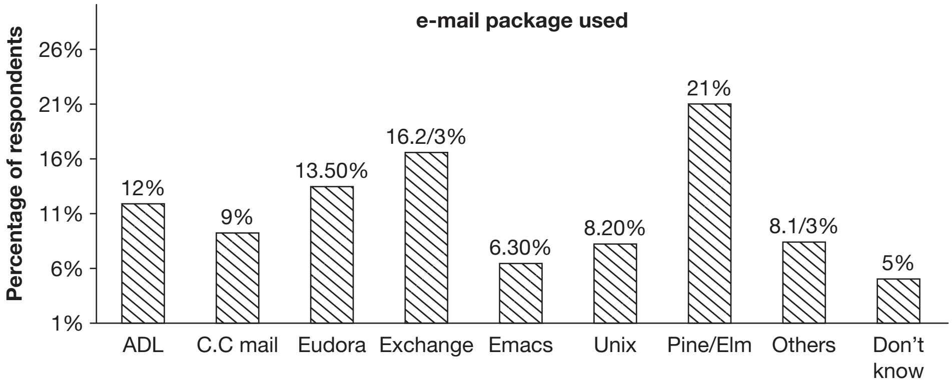
50. If profitability is defined as the ratio of profits to expenses, then in which year was the profitability, the least?

(A) 2009 (B) 2010
(C) 2011 (D) 2012

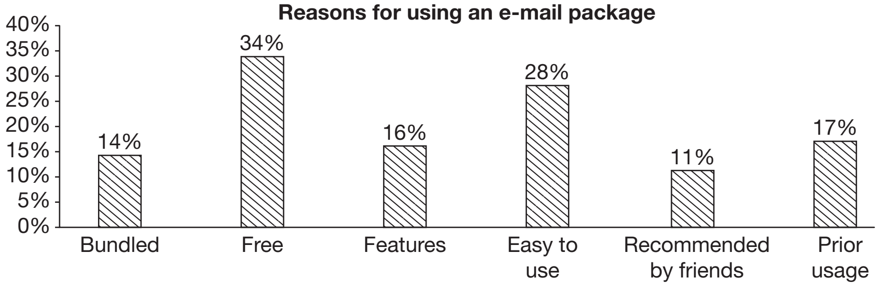
EXERCISE-2

Directions for questions 1 to 5: Answer these questions based on the information given below.

The graphs give the results of the market survey regarding various e-mail package used.



Note: No respondent uses more than one e-mail package.



Note: Each respondent claimed one or more than one of the above reasons.

Total number of respondents = 25500

1. If half of the number of users whose response was 'Don't know', use either AOL or Eudora, then what is the total number of respondents who use AOL or Eudora?

(A) 6120 (B) 7140
(C) 650 (D) 7850

2. Among the respondents, if the users who claim their reason for usage to be 'features' or due to 'prior usage', use only Pile/Elm, then how many Pine/Elm users could have claimed both the reasons?

(A) 3060 (B) 2550
(C) 5355 (D) 5510

3. If the users of C.C. mail shift to Eudora because Eudora is 'Easy to use', then what is the percentage increase in the number of users claiming the reason 'Easy to use'?

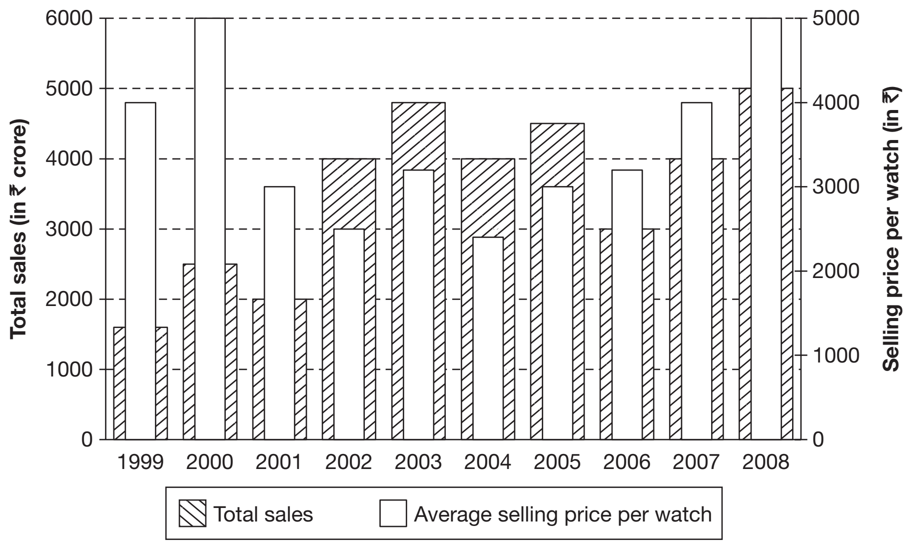
(A) 32%
(B) 66 2/3%
(C) 48.2%
(D) Cannot be determined

4. If all the respondents except the users of C.C. mail claim only one reason and the group of respondents who said 'don't know', were excluded from the survey of reasons for using an e-mail package, then what is the average number of reasons claimed by C.C. mail users?
(A) 43 (B) 3.8 (C) 1.2 (D) 10.6

5. Exactly 15% of the respondents claimed the combination of at least two reasons. A maximum of how many UNIX or AOL respondents claimed at most one reason?
(A) 5100 (B) 3825
(C) 5151 (D) None of these

Directions for questions 6 to 9: Answer these questions based on the information given below.

The following bar graph gives the total sales (in ₹crore) of watches sold by all the companies and the average selling price (in ₹) per watch sold by company ABC.



The following table indicates the percentage market share of company ABC and the profit as a percentage of its sales for the different years.

Year	Market share (by value)	Profit as a percentage of sales
1999	50%	12%
2000	40%	15%
2001	60%	10%
2002	37½%	20%
2003	33⅓%	15%
2004	45%	20%
2005	40%	16⅔%
2006	66⅔%	11%
2007	60%	16%
2008	50%	16%

6. If in the year 2005, the sales of watches of company ABC form 25% by volume of the total sales of watches, then what is the average selling price per watch of all the other companies in that year?
(A) ₹200 (B) ₹2000
(C) ₹1500 (D) ₹150

7. In how many years from 2000 to 2008, did the value of the sales of company ABC increase while its profit decreased over the previous year?
(A) 4 (B) 5 (C) 3 (D) 6

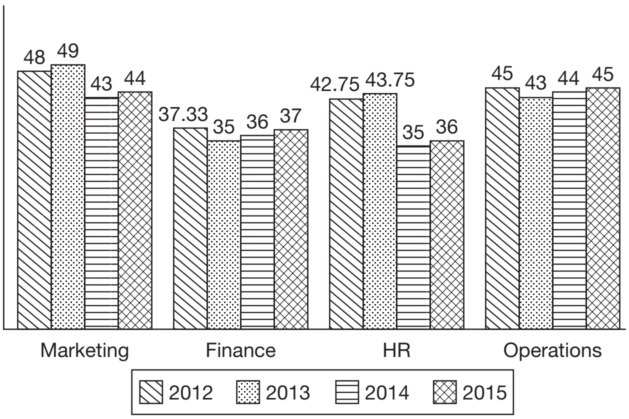
8. The ratio of the profits earned by company ABC in the year 2004 to that in 2008 is
(A) 5 : 4 (B) 7 : 6
(C) 8 : 9 (D) 9 : 10

9. If in each of the given years, the average selling price per watch of all the companies in the market was less than that of the average selling price per watch of company ABC, then in at least how many of the given years was the market share (by volume) of company ABC less than 50%?
(A) 4 (B) 5 (C) 6 (D) 7

Directions for questions 10 to 13: Answer these questions based on the information given below.

There are four departments, such as Marketing, Finance, HR and Operations of company XYZ had 5, 3, 4 and 6 employees as on 1 April 2012. In the next four years, the company recruited one employee in each of the four departments.

All the new employees who joined the company joined on 1 April and were 25 years of age at that point of time. During these four years, two employees who were aged 60 years and 64 years retired from two departments of the company. The following graph gives the average age of the employees in the departments as on 1 April 2012, 2013, 2014 and 2015.



10. From which department did the employee aged 64 years retire?

(A) Marketing (B) Finance
(C) HR (D) Operations

11. In which year did the new employee join the HR department?

(A) 2015 (B) 2014
(C) 2013 (D) 2012

12. From which department did the employee aged 60 years retire?

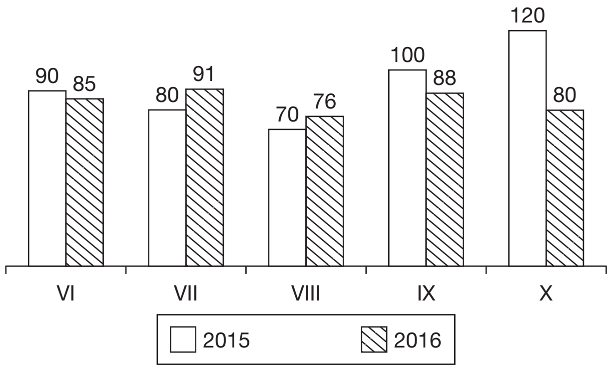
(A) Marketing (B) Finance
(C) HR (D) Operations

13. As on 1 April 2015 the age of the new employee who joined the operations department is _____ years.

(A) 28 (B) 26
(C) 25 (D) 27

Directions for questions 14 to 17: Answer these questions based on the information given below.

The following bar graph gives the number of students in all the classes at Champion school in the year 2015 and 2016, respectively. Government regulations ensure that not more than 25% students fail in a class and the school management has a policy of failing at least 10% students in each class. Students join the school only in Class VI and do not leave until they pass out of Class X.



14. Which class had the highest pass percentage in the year 2015?

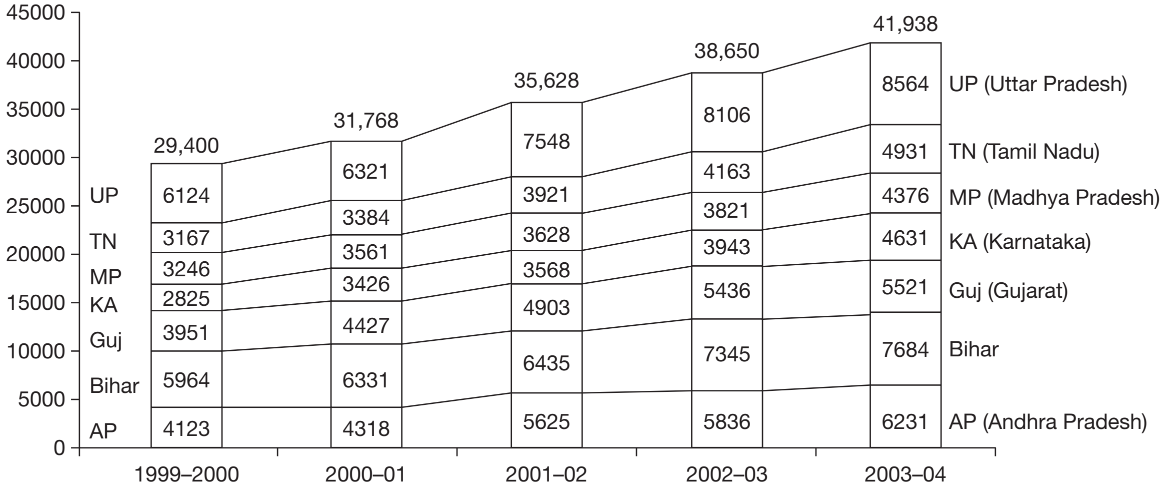
15. What is the number of students who joined the school in 2016?

16. What is the total number in students of the school who failed in the year 2015?

17. In the year 2015, in which class did the maximum number of students fail?

Directions for questions 18 to 21: Answer these questions based on the information given below.

The chart given below gives the land revenue collections (in crore of rupees) of the given states from 2000 to 2004. The values given at the top of each bar represents the total land revenue collections in the corresponding year.



18. If for each year, the states are ranked in terms of the descending order of land revenue collection, then how many states have got the same rank in four of the five given years?

(A) 2 (B) 3
(C) 4 (D) 5

19. The percentage share of land revenue collection of which state has increased by the maximum percentage points from 1999–2000 to 2000–01?

(A) Uttar Pradesh (B) Karnataka
(C) Bihar (D) Andhra Pradesh

20. Which pair of successive years shows the maximum growth rate of land revenue collection in Tamil Nadu?

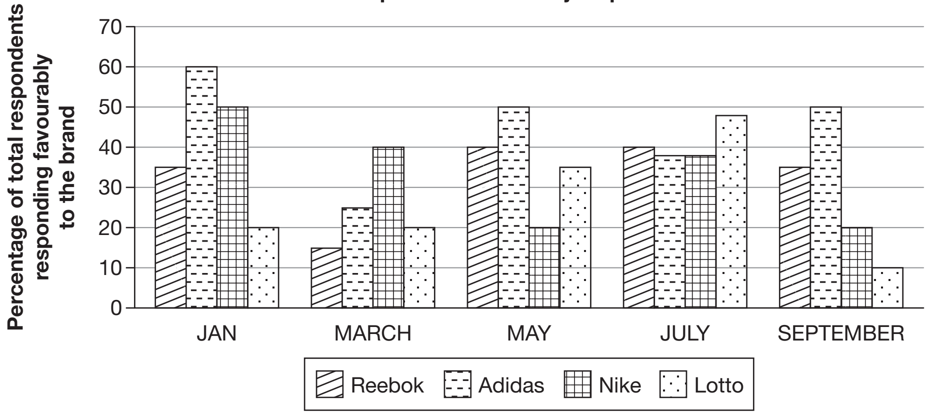
(A) 2000 to 2001 (B) 2001 to 2002
(C) 2000 to 2003 (D) 2003 to 2004

21. For which state has the land revenue increased by the same amount in two successive pairs of years?

(A) Andhra Pradesh (B) Karnataka
(C) Gujarat (D) Tamil Nadu

Directions for questions 22 to 25: Answer these questions based on the information given below.

Consumer preferences of major sportswear brands



Note: (i) A respondent could give a favourable response for more than one brand.

(ii) The percentage of total respondents responding favourably to a brand represents the consumer liking of that brand.

It is also known that, the sample size (i.e., the total number of respondents in the survey) in March was larger than that in January. The sample size in July was greater than that in March, while the sample size in September is the largest among all the surveys.

22. When the survey results of January and September are considered together, which brand was liked by the maximum number of consumers?

(A) Reebok (B) Adidas
(C) Nike (D) Cannot be determined

23. If the sample size in March was 1200, then the number of respondents who did not respond favourably to any of the four brands in that month could be at most

(A) 720 (B) 480
(C) 180 (D) None of these

24. If the number of respondents responding favourably to a brand is represented by N, then which of the following statements is/are false?

(I) The value of N for at least one brand in July was more than that in January.
(II) The value of N for at most one brand in September was less than that in July.
(III) The value of N for exactly one brand in March was the same as that in September.

(A) Only I (B) Only III
(C) Both II and III (D) None of the above

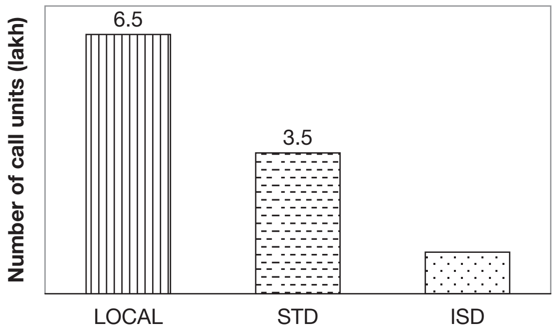
25. The sample size in January was 800 and in May it increased by 25%. The number of respondents, who responded favourably in May, to more than one brand (from among the given four brands), must be at least

(A) 100 (B) 150
(C) 250 (D) 400

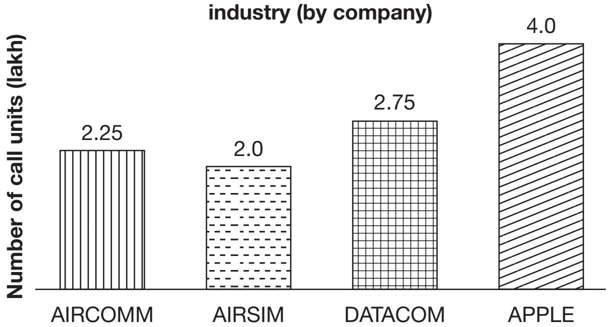
Directions for questions 26 to 29: Answer these questions based on the information given below.

The bar charts give the statistics regarding the telecommunications industry in a country. There are only three types of services, namely LOCAL, STD, and ISD – offered by the industry, and there are only four companies, such as AIRCOMM, AIRSIM, DATACOM and APPLE in the industry. Sales in this industry are measured in terms of number of call units.

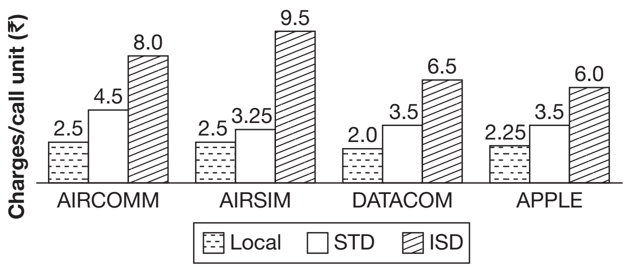
Distribution of the total sales volume of the industry (by type of service)



Distribution of the total sales volume of the industry (by company)



Charges customers pay for the different types of services offered by various companies



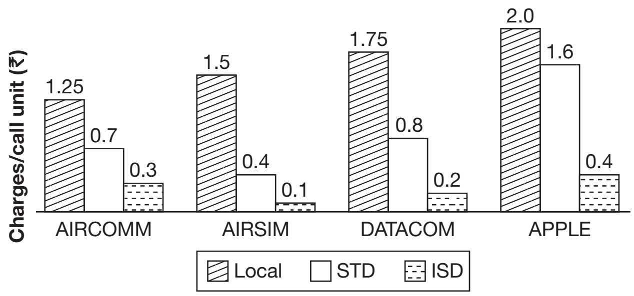
26. Given the above information, what could be the least possible amount spent by all the customers on all the LOCAL and STD calls put together? Assume that a customer can utilize the services of more than one company.

(A) ₹22 lakh
(B) ₹26.375 lakh
(C) ₹26 lakh
(D) ₹27.75 lakh

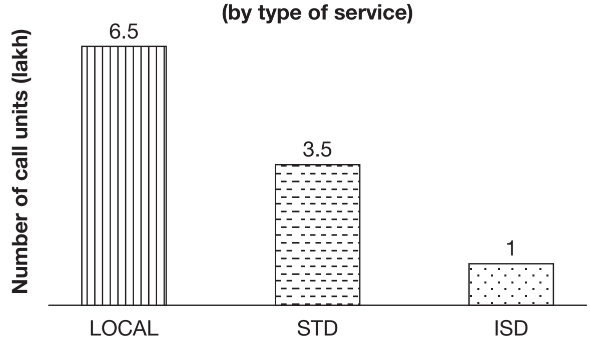
27. What would be the answer to the above question if the term STD is replaced with ISD?

(A) ₹18.875 lakh
(B) ₹20.125 lakh
(C) ₹22.175 lakh
(D) ₹22.875 lakh

Additional Data for questions 28 and 29:



Distribution of the total sales volume of the industry (by type of service)



28. If each company reduced its STD charges per call unit by ₹1, but to exactly compensate for the subsequent revenue loss, it simultaneously increased its ISD charges proportionately, then which company offers the second cheapest service for ISD calls? Assume that the total number as well as the distribution of call units remain the same.

(A) AIRCOMM (B) AIRSIM
(C) DATACOM (D) APPLE

29. The company whose total sales revenue is the highest is

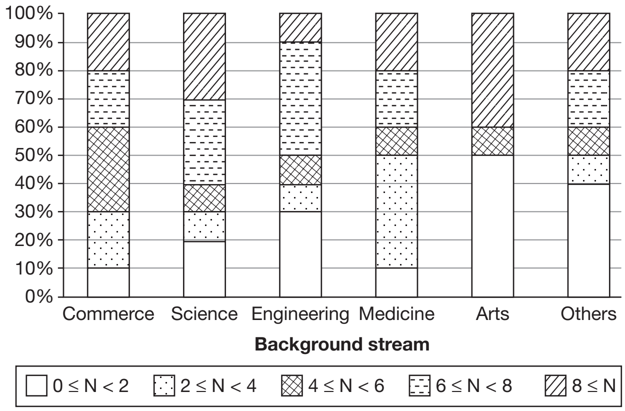
(A) AIRCOMM (B) AIRSIM
(C) DATACOM (D) APPLE

Directions for questions 30 to 33: These questions are based on the following table and bar graph.

Distribution of the total number of students enrolled in the top 10 Management Institutes based on their educational background.

Stream	Number of students
Commerce	1100
Science	2200
Engineering	3000
Medicine	1500
Arts	500
Others	400

Percentage distribution of the students of various backgrounds on the basis of their work experience (in years)



N = Number of years of work experience

30. The least number of students enrolled are from the group with work experience of
(A) At least four years but less than six years.
(B) At least six years but less than eight years.
(C) At least two years but less than four years.
(D) Less than two years.

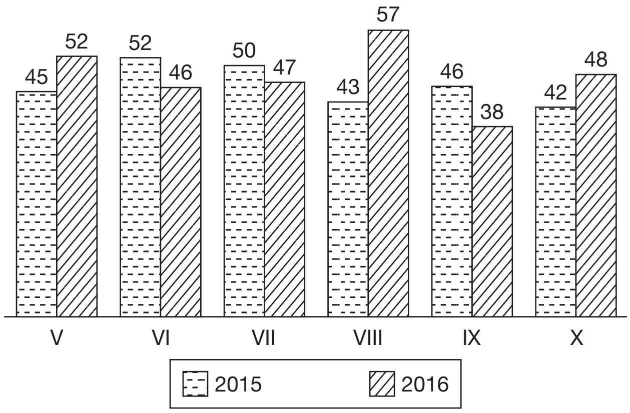
31. In which of the following categories is the number of students enrolled the highest?
(A) Number of Medical students with $2 \le N < 4$.
(B) Number of Arts students with $N \ge 8$.
(C) Number of Science students with $6 \le N < 8$.
(D) Number of Engineering students with $4 \le N < 6$.

32. If the stream wise break-up of the backgrounds of the total number of students was plotted on a pie chart, then the angle subtended by the sector representing the number of students from the Engineering stream would be approximately how many degrees more than that subtended by the sector representing the number of students from the Commerce stream?
(A) 78.6° (B) 75°
(C) 50° (D) 50.6°

33. Which of the following statements is true?
(A) The most number of enrolments in the work experience group $N \ge 8$ was from the Arts stream.
(B) The least number of enrolments in the work experience group $0 \le N < 2$ was from 'Other' streams.
(C) At least 30% of the total number of students enrolled in the top 10 management institutes are from the Engineering stream.
(D) None of the above

Directions for questions 34 to 37: Answer these questions based on the information given below.

The bar graph gives the details of students studying in classes V to X at Model Public School, for the years 2015 and 2016. Students join the school in Class V and leave the school only after they pass Class X. No student leaves or joins any other class. Students who pass the final examination in any class are promoted to the next higher class in the next year while students who fail have to continue in the same class in the next year also. It was also known that the pass percentage in class V in 2015 was at least 90.



34. How many students joined Class V in the year 2016?

35. In the year 2015, what is the number of students who failed in all the classes from V to X put together?

36. For which class was the pass percentage in the year 2015, the highest?

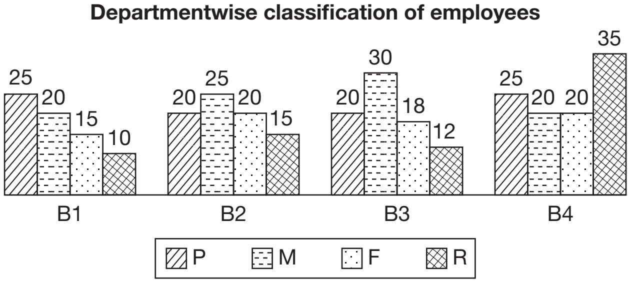
37. In the year 2015, for how many of the given classes was the pass percentage less than the overall pass percentage of the school?

Directions for questions 38 to 41: Answer these questions based on the information given below.

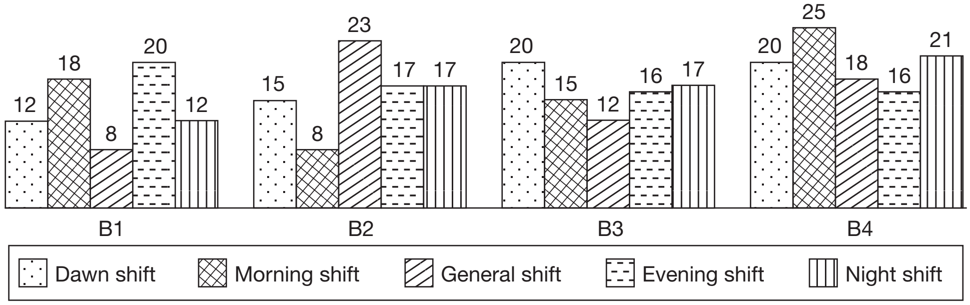
A company has four branches B_1, B_2, B_3 and B_4. There are four departments, such as Production (P), Marketing (M), Finance (F) and Research (R) at each of the four branches.

At each of the branches, every employee has opted for exactly one of the five shifts among Dawn shift, Morning shift, General shift, Evening shift and Night shift. Further, at each of the four branches, in each of the four departments, there was at least one employee working in each of the five shifts.

Departmentwise classification of employees



Shiftwise classification of employees



38. Considering all the four branches, the total number of marketing employees who work in the morning shift is at most

(A) 54 (B) 49
(C) 53 (D) 48

39. In Branch B_3, the difference between the number of Production employees who work in the morning shift and that of Finance employees who work in the general shift is at most

(A) 10 (B) 11
(C) 12 (D) None of these

40. Considering all the four branches, the number of employees from the Marketing department, who work in either the morning shift or the general shift is at most

(A) 75 (B) 77
(C) 78 (D) 79

41. The company shifted a total of x employees working in the evening shift into the night shift, so that the number of employees working in the night shift is more than those working in the evening shift in each branch. What is the least value of x?

(A) 3 (B) 4
(C) 5 (D) 6

Directions for questions 42 to 46: These questions are based on the following figures.

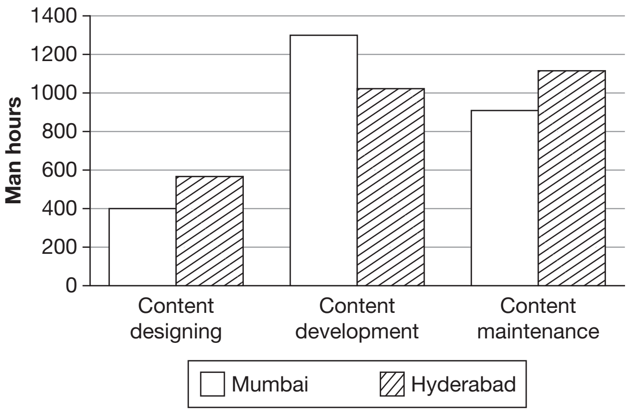
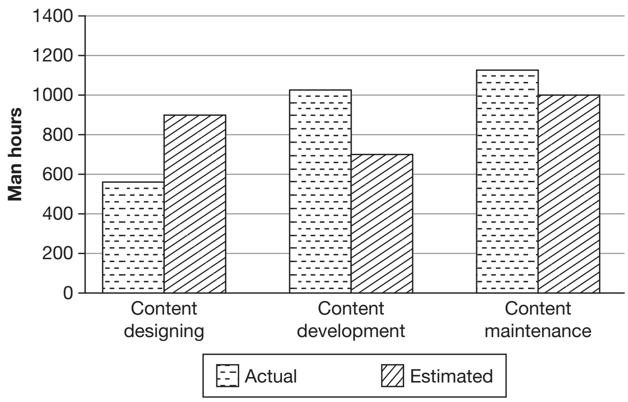


Figure shows the amount of work distribution, in man-hours for a book publishing organization between the activities conducted at two of its offices at Mumbai and Hyderabad.

The activities are content designing, content development and content maintenance.



Above figure shows the estimated and actual work involved in the same set of activities in the same organization (at the Hyderabad office) during the same period.

42. If the total working hours (for both offices) were 100, which of the following activities account for approximately 25 hours?

(A) Content development at Mumbai
(B) Content maintenance
(C) Content designing
(D) Content maintenance at Mumbai plus content designing at Hyderabad

43. Which of the work nearly requires as many man-hours as that spent in content development?

(A) Content maintenance and designing (at both the offices).
(B) Content development at Mumbai and content designing (at both the offices).
(C) Content maintenance (at both the offices).
(D) Content designing and maintenance at Hyderabad.

44. Roughly, what percentage of the total work is carried out at Hyderabad?

(A) 41 per cent (B) 21 per cent
(C) 51 per cent (D) 31 per cent

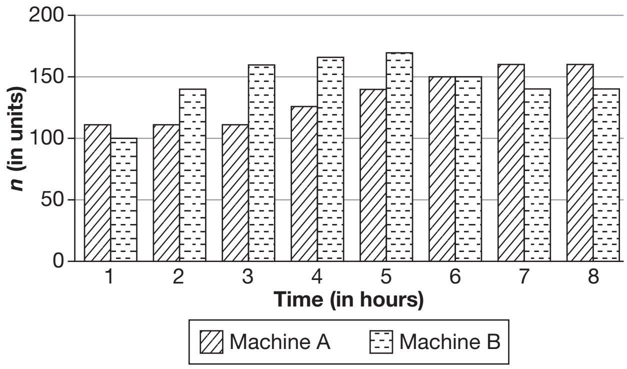
45. The total effort (in man-hours) spent at Mumbai is nearest to which of the following?

(A) The total estimated effort for Hyderabad.
(B) Twice the man-hours for the estimated content designing at Hyderabad.
(C) The actual man-hours at Hyderabad.
(D) The sum of the estimated and the actual effort for content development at Hyderabad.

46. If 40% of the work at Mumbai office were to be carried out at Hyderabad, with the distribution of effort between the tasks at Mumbai remaining the same, the percentage of total content maintenance carried out at Mumbai would be nearly

(A) 18 (B) 34
(C) 73 (D) 27

Directions for questions 47 to 50: The data given below is regarding the work pieces machined by two machines A and B. For each of the machines the average number of pieces machined per hour from the start to the end of different hours on a particular day is given.



For example, machine A averages 110 units per hour for the first 3 hours of its operation.

47. If machines A and B together take time t (in hours) from the start, to machine 1000 work pieces, which of the following is true of t?

(A) $t = 2$
(B) $2 < t < 3$
(C) $3 < t < 4$
(D) $4 < t < 5$

48. The difference between the total number of pieces machined by any machine by the end of any hour and that by the end of the previous hour is the number of pieces machined by that machine in that hour. During which of the following hours is the number of pieces machined by A the same as that of B?

(A) 1
(B) 2
(C) 3
(D) None of these

49. Considering each of the machines separately, in total, how many instances are there where the number of units machined by the machine is same as that by the machine in the previous hour?

(A) 2
(B) 3
(C) 4
(D) 5

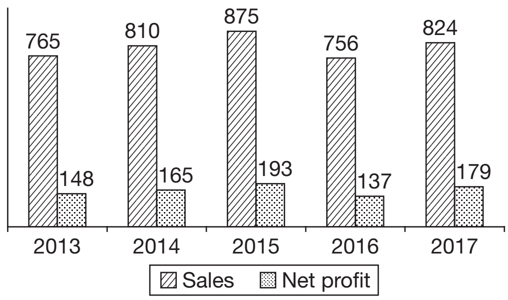
50. Approximately by what per cent is the number of work pieces machined by machine A in the last 3 hours, more/less than that manufactured by machine B in the same time period?

(A) 11% more
(B) 11% less
(C) 115% more
(D) 115% less

EXERCISE-3

Directions for questions 1 to 4: Answer these questions on the basis of the information given below.

The bar graph gives the sales and net profit (in ₹cr) of a company across five years.



Net profit = Gross profit – Taxes
Gross profit = Sales – Expenses

The tax was 25% of the gross profit in the first two years and 20% in the remaining three years.

1. What were the expenses in 2014?

(A) ₹680 cr
(B) ₹645 cr
(C) ₹590 cr
(D) None of these

2. What was the percentage increase in expenses from 2014 to 2015?

(A) 5.5 (B) 6.5
(C) 7.5 (D) 8.5

Difficulty Level-3: Advanced

3. In which of the given years was the ratio of sales to expenses, the lowest?
(A) 2013 (B) 2014
(C) 2016 (D) 2017

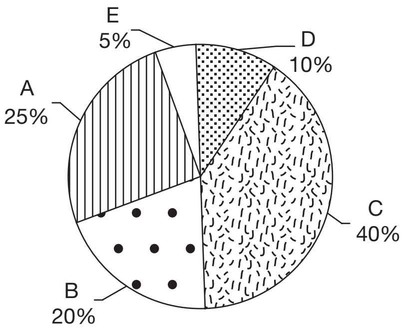
4. If probability is defined as $\frac{\text{gross profit}}{\text{sales}} \times 100$, what was the highest value of probability in any of the given years?
(A) 26.5 (B) 26.9
(C) 27.2 (D) 27.6

Directions for questions 5 to 8: Answer these questions on the basis of the information given below.

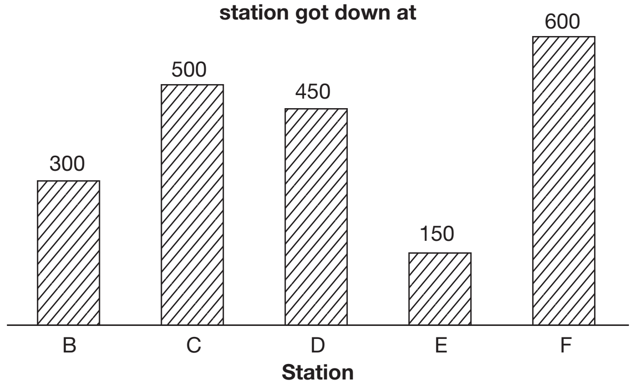
A local train started from station A at 10.00 a.m. in the morning and en route it stopped at exactly four intermediate stations, namely B, C, D and E and finally at station F, which was the destination station. It is known that the total number of passengers who boarded the train at stations A, B, C, D and E put together is 2000.

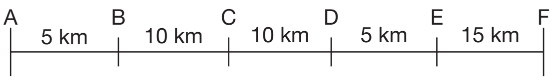
The pie chart given below shows the percentage distribution of the total number of passengers who boarded the train, according to the station at which they boarded the train. The bar chart shows the number of passengers who got down from the train at different stations. The distance between any two adjacent stations is given in the line diagram.

Percentage distribution of passengers according to station boarded at



Distribution of passengers according to station got down at





5. A group of villagers boarded the train at one of the stations and all of them got down together at another station. What is the maximum possible number of persons in that group, if the group travelled more than 15 km?
(A) 400 (B) 450
(C) 800 (D) 500

6. What is the maximum possible number of passengers who travelled for not more than 15 km and also had station C as either their boarding or destination station?
(A) 500 (B) 600
(C) 800 (D) 1100

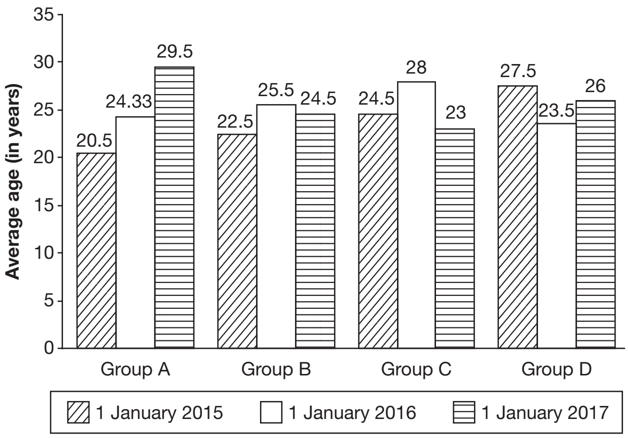
7. What is the maximum possible number of passengers who travelled for at least 30 km in the train?
(A) 100 (B) 300
(C) 150 (D) None of these

8. At 6 p.m. in the evening, the same train returns from station F to station A, with the condition that at each of the stations, the number of passengers who boarded the train in the evening is same as that of those who got down the train in the morning, while the number of passengers who got down the train in the evening is same as that of those who boarded the train in the morning. Which of the following statements is/are true?
(A) The number of passengers who travelled in the train between any two stations in the evening is same as that in the morning.
(B) In the evening, the train carried the maximum number of passengers while travelling between stations D and C.
(C) Maximum possible number of passengers who travelled for a distance of at least 30 km in the evening is the same as that in the morning.
(D) All the three

Directions for questions 9 to 12: These questions are based on the following information.

A company was established on 1st January 2015 with a total of 10 members. They were divided into four groups, namely A, B, C and D with each group containing at least two members. Each member had a distinct integral age as on 1 January 2015 with the minimum age being 20 years and the maximum age being 29 years. These groups were to continue for the whole year.

At the end of every year, reshuffling of these ten members takes place such that a different set of members (no common member) are there in any two years in a particular group. The total number of members in the company remained the same throughout. The following graph gives the average of the integral ages of the members of each group as on 1st January of the three years.



Note: It is known that group A had four members as on 1 January 2017.

9. What was the age of the eldest person in Group B as on 1 January 2017?
   (A) 23 years (B) 26 years
   (C) 25 years (D) 27 years

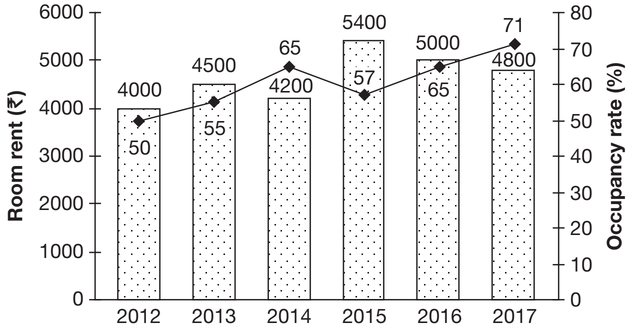
10. How many members are common between Group B for the year 2015 and Group D for the year 2016?
    (A) 0 (B) 1
    (C) 2 (D) 3

11. The total age of which group was the highest in the year 2017?
    (A) A (B) B
    (C) C (D) D

12. How many groups showed a consistent increase in the total of the age of all the members put together across the years?
    (A) 0 (B) 2
    (C) 3 (D) 1

Directions for questions 13 to 16: These questions are based on the following data.

The graph shows the details of the room rent (in ₹) and the occupancy rate (%) at a hotel across the years 2012 to 2017.



Occupancy rate = Percentage of available rooms occupied

$$\text{Profitability} = \frac{\text{Profit}}{\text{Revenue}} \times 100$$

Revenue in a year = Room rent × Number of rooms occupied × 365.

Number of rooms available increased by 10% every year from 2012.

The expenses of the hotel increased by 15% every year from 2012.

The hotel made a profit of ₹2.79 crore in 2012.

13. In which of the six years was the revenue of the hotel, the highest?

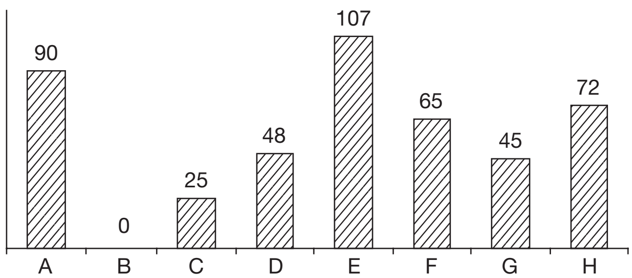
14. In which year was the percentage increase in revenue over the previous year, the highest?

15. In which year was the profit made by the hotel, the highest?

16. In which year was the profitability, the highest?

Directions for questions 17 to 20: Answer these questions on the basis of the information given below.

The following bar graph gives information about the runs scored by eight batsmen in a season. For each batsman, the runs scored relative (difference with respect to) to player B is given. The players are ranked from one to eight, with the player scoring the highest number of runs ranked first, the next one second and so on. It is known that player F was ranked eighth and player D was ranked immediately above player C.



17. What was the rank of player C?
(A) 4 (B) 5
(C) 6 (D) 7

18. How many players scored fewer runs than player H?
(A) 3 (B) 4
(C) 5 (D) 6

19. If the average number of runs scored by the eight players is 435, then how many runs did player C score?
(A) 431 (B) 410
(C) 460 (D) None of these

20. If player G scored 373 runs, how many runs did the eight players together score?
(A) 2966 (B) 3532
(C) 3476 (D) None of these

ANSWER KEYS

Exercise-1

1. (A)	**9.** 10	**17.** (B)	**25.** (B)	**33.** (B)	**40.** 3.6	**47.** (D)
2. (D)	**10.** 1	**18.** (B)	**26.** (B)	**34.** (D)	**41.** (A)	**48.** (C)
3. (B)	**11.** (A)	**19.** (D)	**27.** (C)	**35.** (A)	**42.** (B)	**49.** (D)
4. (A)	**12.** (B)	**20.** (C)	**28.** (A)	**36.** 1 : 2	**43.** (C)	**50.** (C)
5. (B)	**13.** (D)	**21.** (B)	**29.** (D)	**37.** 38	**44.** (D)	
6. (A)	**14.** (B)	**22.** (D)	**30.** (B)	**38.** 10.88	**45.** (B)	
7. (B)	**15.** (C)	**23.** (B)	**31.** (C)	**39.** U.P	**46.** (A)	
8. 3	**16.** (B)	**24.** ()	**32.** (D)			

Exercise-2

1. (B)	**9.** (D)	**17.** IX	**25.** (B)	**33.** (C)	**40.** (B)	**47.** (C)
2. (A)	**10.** (C)	**18.** (C)	**26.** (C)	**34.** 48	**41.** (D)	**48.** (D)
3. (D)	**11.** (B)	**19.** (B)	**27.** (B)	**35.** 22	**42.** (A)	**49.** (B)
4. (B)	**12.** (A)	**20.** (D)	**28.** (A)	**36.** VII	**43.** (B)	**50.** (C)
5. (C)	**13.** (D)	**21.** (C)	**29.** (D)	**37.** Four	**44.** (C)	
6. (C)	**14.** VIII	**22.** (B)	**30.** (A)	**38.** (D)	**45.** (A)	
7. (C)	**15.** 75	**23.** (A)	**31.** (C)	**39.** (B)	**46.** (D)	
8. (D)	**16.** 67	**24.** (B)	**32.** (A)			

Exercise-3

1. (C)	**4.** (D)	**7.** (D)	**10.** (A)	**13.** 2017	**16.** 2017	**19.** (A)
2. (C)	**5.** (B)	**8.** (D)	**11.** (A)	**14.** 2013	**17.** (B)	**20.** (D)
3. (C)	**6.** (D)	**9.** (B)	**12.** (D)	**15.** 2017	**18.** (C)	

SOLUTIONS

EXERCISE-1

1. Trade deficit is in 1997, 99 and 2000 = 10 + 40 + 30 = 80
Trade surplus is in 1996, 98 and 2001 = 10 + 20 + 10 = 40
Total trade deficit = 80 – 40 = ₹40 thousand crore
Average exports = 140 thousand crore.

30% of 140 = 42
– 2% of 140 = 2.8

28% of 140 = 39.2

+0.5% of 140 = 0.7

Hence, 28.5% = 39.9

2. In 1999, total tonnage of exports

$$= \frac{140\times10^3\times10^7}{7\times10^3} = 20\times10^7 \text{ tons}$$

Total tonnage of imports $= \dfrac{180\times10^3\times10^7}{6\times10^3} = 30\times10^7$

10^7 is common. Exports are less than imports by

$$\frac{30-20}{30}\times100 = 33\tfrac{1}{3}\%$$

3. Trade surplus in 1998 = 130 – 110 = 20000 crore.
Trade surplus in 2001 = 160 – 150 = 10000 crore.

Percentage decrease $= \dfrac{20-10}{20}\times100 = 50\%$

Trade surplus in 2002 = 50% of 10000 = 5000 crore.
Imports in 2002 = 150 + 20% of 150000 = 180000 crore.
Exports in 2002 = Imports + Trade surplus
= 180000 + 5000 = 185000 crore
= 185 thousand crore.

4. Exports after three years will become
$160(1.1)^3 = 212.96$ thousand crore.
Imports after three years will become
$150(0.9)^3 = 109.35$ thousand crore.
Total trade = 213 + 109 = 322 thousand crore.

5. The exports increased by 60%.
For a 10% compounded annual growth rate, 100 becomes $100\left(1+\dfrac{10}{100}\right)^5 = 161$ in 5 years.

∴ The growth is approximately 10%.

6. By keenly observing we can find that the turnover of A is the greatest.

7. Let the profits of the four companies for the second quarter be ₹8x, ₹7x, ₹6x and ₹5x, respectively.

Profit to sales ratio turnover of A, B, C and D are as follows.

$A = \dfrac{8x}{572} = \dfrac{x}{71}$; $B = \dfrac{7x}{366} = \dfrac{x}{52}$;

$C = \dfrac{6x}{514} = \dfrac{x}{86}$; $D = \dfrac{5x}{308} = \dfrac{x}{61}$

x is the common number for each of the fractions. For company B, the denominator is the lowest among all the denominators. Hence, B's profit percentage is the highest.

8. The ratio of A's turnover to that of the total turnover

in Q1 $= \dfrac{428}{1740} < \dfrac{1}{4}$ in Q2 $= \dfrac{572}{1760} > \dfrac{1}{4}$

in Q3 $= \dfrac{473}{1590} > \dfrac{1}{4}$ in Q4 $= \dfrac{527}{1600} > \dfrac{1}{4}$

Alternate method:
If A's turnover is greater than the average turnover of the other three companies, we can say that A's turnover is more than 25% of the total sales turnover of all the companies. In the first quarter by observation we find that A's turnover is less than the average turnover of the other three companies. And further we find that in II, III and IV quarters, A's turnover is greater than the average turnover of the other companies. Therefore, in three quarters A's turnover is greater than 25% of the total turnover of all the companies in that particular quarter.

9. Percentage decrease in the sales turnover of all the companies from Q_2 to Q_3

$$= \frac{1760-1590}{1760}\times100 \Rightarrow 9{\cdot}6\% \cong 10\%$$

10. By observation we find that the sales turnover of only company B is consistently decreasing.

11.

Country	Matches played	Matches won	Matches lost	Matches drawn
Germany	60	20	30	10
France	30	15	–	15
Brazil	45	30	10	5
USA	60	30	15	15
Argentina	30	15	?	?

By observation, Germany lost maximum number of matches.

12. Originally, Brazil had the highest success rate. Even after doubling, it will have the highest success rate. (Since the ratio of matches drawn and won to total matches remains the same)

13. As we do not know the number of matches played by each team against Argentina, the success rate cannot be determined.

14. Matches lost by Germany = 30
Matches lost by Brazil = 10
Required ratio = 30 : 10
= 3 : 1

15. By observing the table given in situation 1, we see that France, USA and Argentina had 50% success rate and won 60 matches in all. The other two teams, Germany and Brazil won 50 matches in all.

16. Total sales in 2002-03
$= 105 + 96 + 208 + 165 + 73 = 647 \rightarrow$(i)
Total sales in 2000-01 = 94 + 68 + 165 + 112 + 57 = 496
Let 647 ≅ 650 and 496 ≅ 500
Sales target in 2002–02 = 1.2 × 500 = 600 m tons
Hence, there is surplus in the sales target.

$\therefore$ Required percentage $= \dfrac{650 - 600}{600} \times 100 \cong 9\%$

17. Sales of Tin in 2000-2001 = 0.4132 × 57 = 23.55 million tons
In 2002-2003, Sales of Tin = 23.55 × 1.1 × 1.1
≅ 28.5 million tons
Total sales = 105 + 96 + 208 + 165 + 73 = 647 million tons

Required percentage $= \dfrac{28.5}{647} \times 100 = 4.4\%$

18. Sales of copper in 2002–2003 = 96
The sales increased by 25% in the next year, but still remained 80% of the quantity available.

$\therefore$ Quantity of copper available in 2003–04 $= \dfrac{96}{0.8} \times 1.25$
$= 150$

19. Sales of all metals in 2000–2001 = 496
Sales of all metals in 2002–2003 = 647
Annual percentage increase

$$= \frac{647 - 496}{496} \times \frac{1}{(x-1)} \times 100\% \; [x = 3] = 15.2\%$$

20. Sales of Gold in 2002-03 = 25% of 73 $= \dfrac{73}{4}$

Sales of Silver in 2000-01 = 20% of 57 $= \dfrac{57}{5}$

Required ratio $= \dfrac{73}{4} \times \dfrac{5}{57} \cong \dfrac{72}{4} \times \dfrac{5}{55} = \dfrac{18}{11} \cong 1.67$
The nearest value is given in Choice (C) as 1.6

21. We must check whether the production of D as a percentage of B is more than 80% but less than 120% or not, i.e., the difference between the production of D and that of B is less than 20% of B or not.

Year	Difference	20% of B
2011	700	1220
2012	500	1080
2013	1000	820
2014	900	840
2015	800	1000

$\therefore$ It happened for 2011, 2012 and 2015.

22. The total production in
2011 is 23300
2012 is 20400
2013 is 20500
2014 is 20800
2015 is 24900
By observing the above values, the change over the previous year is the highest in the year 2015.

23. In 2014:

Exports of D is $\dfrac{3}{8} \times 5100 = 1912$

Exports of A is $\dfrac{1912 \times 6}{5} = 2295$

$\therefore$ Percentage of exports $= \dfrac{2295}{5200} \approx 45\%$.

24. In 2015, B forms 20% of total production. (5000/24900)
Let the total production be 100; the production of B is 20.
Production of B in 2016 is 60% × 20 = 12
Total production in 2016 is 60% × 20 + 1.35 × 80 = 120.
Production of B as a percentage of total production is 10%. (12/120)

25. From the choices we can observe that II is present in all the four choices.
$\therefore$ By checking II (which is false), the choices (A), (C) and (D) can be eliminated. Now as the only option left is Choice (B) and though it appears that D in 2013 and 2014 is the same, there could be a marginal decrease (For example 5100 and 5090). So the answer is (B) as

Difficulty Level-1: Foundation

it is the only option left. Had the question had 'None of these' as an option, then it would have been the answer.

26. Bala's expenses on clothing = 20% his total expenses
∴ Bala's total expenses ₹18,500
Food and tution fees together constitute 35% of Bala's total expenses which is $\frac{35}{100} \times 18500 = ₹6475$

27. 15% of Anand's Expenses = ₹1620
∴ the remaining 85% = $\frac{85}{15} \times 1620 = ₹9180$

28. Let the total expenses of Bala and Chetan be ₹300 and ₹500, respectively.
The 'other' expenses are $\frac{30}{100} \times 300$ and $\frac{35}{100} \times 500$, i.e., 900 and 1750, respectively.
The required ratio = 18 : 35

29. Let Damodar's total expenses be D and that of Edward be E.
Given $\frac{20}{100} \times D = 115\%$ of $\frac{15}{100} \times E$
∴ 0.2 D = 0.1725 E
$\frac{D}{E} = \frac{0.1725}{0.2} \frac{86.25}{100}$
∴ D is 13.75% less than E.

30. 10% of Anand's expenses (on clothing) is at least equal to 15% of Chetan's expenses.
∴ 10% of A ≥ 15% of C
∴ $\frac{A}{C} \geq 1.5$
∴ Total expenses of Anand is at least 1.5 times that of Chetan.

31. Total imports = 17 + 18 + 13 + 8 + 21 + 13 + 17 + 15 + 16 + 15 = 153
Total exports = 18 + 22 + 10 + 10 + 19 + 14 + 14 + 12 + 17 + 16 = 152
Trade deficit = 153 – 152 = 1
Average imports = $\frac{153}{10} = 15.3$
∴ Required percentage = $\frac{1}{15.3} \times 100 = 6.5\%$

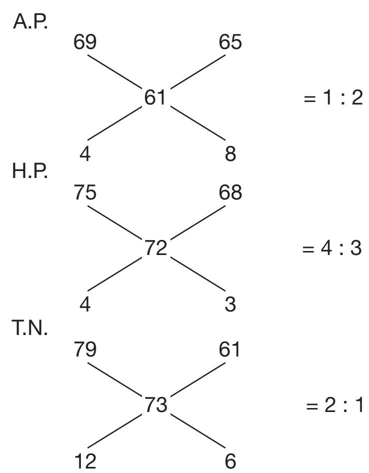
32. Total tonnage of exports = $\frac{152 \times 10^{10}}{2000} = 76 \times 10^7$
Total tonnage of imports = $\frac{153 \times 10^{10}}{3000} = 51 \times 10^7$
∴ Required percentage ≅ $\frac{76-51}{51} \times 100 \cong 49\%$ more

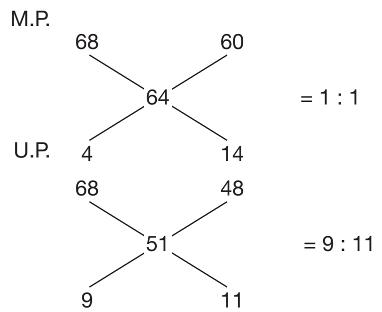
33. The highest exports are to USA and the least imports are from Australia.
Imports from USA = 18
Exports to Australia = 10
∴ Required percentage = $\frac{18-10}{10} \times 100 = 80\%$

34. Statement (A): As calculated before, the company's imports are ₹1 crore more than the company's exports. Therefore, it is a trade deficit (not trade surplus). Hence, this statement is false.
Statement (B): The cumulative trade deficit is ₹1 crore and the total imports of the company is ₹153 crore. 1/153 ≠ 1/15, hence this statement is false.
Statement (C): The trade deficit with China is (15 – 12) = 3, which is only 200% more than the cumulative deficit.
Statement (D): The difference between the highest exports and the lowest imports = 22 – 8 = ₹14 crore.
Average exports to Brazil and Germany $\frac{14+14}{2}$
= ₹14 crore
As both the figures are equal, hence, the only Statement (D) is true.

35. Total imports from Brazil, Japan, South Africa, Russia and China = 13 + 16 + 13 + 21 + 15 = ₹78 crore
Total exports to the other five countries = 18 + 22 +10 + 14 +16 = 80
Ratio = 78/80 = 0.975

Solutions for questions 36 to 40:





Number of males in A.P. = $\frac{1}{3} \times 7 = 2.3$ crore

Number of males in H.P. = $\frac{4}{7} \times 6.3 = 3.6$ crore

Number of males in T.N. = $\frac{2}{3} \times 6.5 = 4.3$ crore

Number of males in M.P. = $\frac{1}{2} \times 6 = 3$ crore

Number of males in U.P. = $\frac{9}{20} \times 16 = 7.2$ crore

Number of literate males in U.P. = $\frac{68}{100} \times 7.2 = 4.896$ crore

36. Let the number of females and males be x and y respectively.
61% of x + 79% of y = 73% of $(x + y)$

$\Rightarrow 61x + 79y = 73x + 73y\ 6y = 12x\ \frac{x}{y} = \frac{1}{2}$ or 1 : 2.

37. The population of other states is 25.8 crore and in U.P. it is 16 crore.

Required percentage = $\frac{25.8 - 16}{25 \cdot 8} \times 100$

$= \frac{9 \cdot 8}{25 \cdot 8} \times 100 \cong 38\%$

38. In U.P. the total number of literate males and literate females = $\frac{68}{100} \times 16$ crore = 1088,00,000 or 10·88 crore.

39. By alligation rule, the ratio of number of males to that of females in
Number of literate males in U.P. is more than number of males in remaining states so number of literate males in U.P. is the highest.

40. Let the number of females and males in H.P. be x and y.
68% of x + 75% of y = 72% of $(x + y) \Rightarrow x : y = 3 : 4$

Number of males = $\frac{4}{7} \times 6{\cdot}3$ crore = 3·6 crore

Solutions for questions 41 to 44: The values of consumption (in million kg) and the population in the different years is as follows.

In 2003, the consumption is 444 – 212 = 232 million kg.

Population = $\frac{232}{2.7}$ million = 8.6 million

Similarly, the values for other years can also be calculated as given in the table.

Year	2003	2004	2005	2006	2007	2008
Consumption	232	284	288	281	336	347
Population (in Cr.)	8.6	8.8	9.6	10.4	10.2	11.2

41. The population of country X was the lowest in 2003.

42. The percentage increase in the population of country X was the highest in 2008.

43. Had the exports in 2003 been 100 that in 2008 it should have been approximately 1.61.
Therefore, $212 \times 1.61 = 341.3$
∴ The required value of production = 688.3

44. It can be seen that consumption increased by nearly 50% while the percentage increase of all others is much less.

Solutions for questions 45 to 47:

45. The sales of the company in 2009
$= (850 \times 12 + 700 \times 15 + 1050 \times 10 + 1000 \times 20)\ 1.1$
$= (10{,}200 + 10{,}500 + 10{,}500 + 20{,}000)1.1$
$= (51{,}200)1.1 = 56{,}320$

46. As there is a 10% increase in the price of each commodity, we can take the prices of P, Q, R and S in 2010 to be ₹12, ₹15, ₹10 and ₹20, respectively.
Sales in 2010 = $750 \times 12 + 920 \times 15 + 1250 \times 10 + 1120 \times 20 = 57700$
Sales in 2011 = $870 \times 13.2 + 900 \times 16.5 + 1350 \times 11 + 1150 \times 22 = 66484$
The percentage increase = $\frac{8784}{57700} \times 100 = 15.23\%$

47. Sales of each commodity in 2011
$= 870 \times 12 + 900 \times 15 + 1350 \times 10 + 1150 \times 20$
$= 10440 + 13500 + 13500 + 23000 = 60400$

Share of commodity P = $\frac{10440}{60400} \times 100 = 17.3\%$

Solutions for questions 48 to 50:
The sales, expenses and the profits of the company in the different years are as follows:

	2009	2010	2011	2012
Sales	1596	1737	1827	2094
Expenses	780	870	946	1065
Profits	816	867	881	1029

48. The highest percentage increase was in 2012.
$$\frac{2094-1827}{1827}\times 100 \cong 14.5\%$$

49. The highest percentage increase for each of the items is as follows:
Sales = 14.5%
Expenses = 12.5%
Number of employees = 10.5% and Profit = 17%
The highest percentage increase in any year was in the profits.

50. The profitability was the least in 2011.

EXERCISE-2

1. Number of respondents who said, 'don't know' = 5%
Half of them = 2.5%
Number of AOL users = 12%
Number of Eudora users = 13.5%
Total number of users of either AOL or Eudora
= 12% + 13.5% + 2.5% = 28% [total respondents] = 7140

2. Respondents claiming 'features' = 16%
Respondents claiming 'prior usage' = 17%
But all of these are users of pine/elm = 21%
∴ There must be some respondents claiming both reasons.
Let the percentage of such people = x
∴ 21% = 16% + 17% – x
∴ x = 12% (25500) = 3060

3. We do not know if users of C.C. mail already claimed the reason of 'easy to use' for C.C. mail itself.
They can shift to Eudora claiming the same reason.
Thus, we cannot determine the change in number of respondents claiming 'Easy to use'.

4. Let the total number of users of e-mail packages = $100x$
Out of which user group saying 'don't know' is excluded
= $100 \times - 5x = 95x$
Now, among these $95x$ users, except for C.C.-mail users, others claim only one reason each.
Number of package users excluding C.C. mail
= $95 \times - 9x = 86x$
∴ These $86x$ users account for $86x$ reasons.
The total number of reasons by respondents
= $14x + 34x + 16x + 28x + 11x + 17x = 120x$
∴ Number of reasons claimed by C.C. mails users
= $120x - 86x = 34x$
The average number of reasons per C.C. mail user
= $34x/9x \cong 3.8$ approx.

5. UNIX + AOL = 12 + 8.20 = 20.20%
As maximum percentage of respondents who claimed at most one reason is greater than 20.2%, the required answer = 20.2% of 25,500 = 5151.

6. Sales of the company ABC in 2005
$$= \frac{40}{100}\times 4500 = 1800 \text{ crore}$$
$$\Rightarrow \text{Number of watches sold} = \frac{1800}{3000}\times 10^7 = 6\times 10^6$$
The total watches sold in the market = $4\times 6\times 10^6$
= 24×10^6
Let the average price of watches by all other companies be x.
$$\frac{(4500-1800)\times 10^7}{x} = 18\times 10^6 \Rightarrow x = ₹1500$$

7.

Year	Sales	Profit
1999	800	96
2000	1000	150
2001	1200	120
2002	1500	300
2003	1600	240
2004	1800	360
2005	1800	300
2006	2000	220
2007	2400	384
2008	2500	400

Over the given period, during 2001, 2003 and 2006 the sales increased while the profits decreased.

8. The ratio of profits in 2004 to 2008 is 360 : 400 = 9 : 10.

9. In the years 1999, 2000, 2002, 2003, 2004, 2005 and 2008 the market share (by volume) of the company is definitely less than 50% as in all these years the market share (by value) is 50% or less. As the average price of watches by all other companies is less than that of company ABC, they would have a higher share by volume than by value.

Solutions for questions 10 to 13: When the average age of any department increases by exactly one, that means no employee of 25 years joined the department and no employee of age 60 or 64 years left the department and the average age went up by one just because the same number of employees aged exactly by 1 year.

When any employee of age 25 years joins a department (where average age is more than 25-which is the case in all the departments in all the years), the average age will fall the next year and similarly when an employee of age 60 or 64 years retires from a department where the average age is much lower than 60 or 64 years the average age will fall the next year. Besides, when a new employee of age 25 years joins and an employee of age 60 or 64 years retires both in same department in the same year, then the average age of the next year will fall the maximum.

The number of employees in different departments in the beginning are 5, 3, 4 and 6, respectively.

The total ages of the employees in the marketing department = 5 × 48 = 240

The next year it becomes 240 + 5 = 245

Had a 25 years old person joined next year, the total age would become 245 + 5 + 25 = 275.

$\therefore$ Average age = $\frac{275}{6} = 45.8$

Had only a single person retired next year, the total age of people would become 245 + 5 – (60 or 64) = 190 or 186

and average age would become $\frac{190}{4}$ or $\frac{186}{4} = 47.5$ or 46.5.

As it is said that it became 43, the only possibility is that a 25-year-old person joined and a 60-year-old person retired.

Using the same logic, we can find when employees retired or joined other departments.

Solutions for questions 14 to 17: It is given that new students join the school only in Class VI and students leave the school only after they pass out of Class X. Also, the pass percentage in any class was from 75% to 90%.

In cases like this when the number of students who passed out from any class in the year 2015 is not given, we have to find the minimum and maximum number of students in each class in the year 2016.

	Students promoted from lower class		Students failed in 2015		Students in 2016	
Class	**Min**	**Max**	**Min**	**Max**	**Min**	**Max**
VII	68	81	8	20	76	101
VIII	60	72	7	17	67	89
IX	52	63	10	25	62	88

As the number of students in Class IX in the year 2016 is 88, which is the maximum possible, we say that 25 students failed in Class IX or the number of students who passed from Class VIII is 63 and 7 Students failed in Class VIII.

$\therefore$ 69 students passed from Class VII or 11 students failed in class VII. It means that 80 students passed from Class VI or 10 students failed in Class VI and 75 new students joined. In Class X, 14 students (89 – 75) failed and 106 students passed out of the school in 2015.

14. Class VIII had the highest (90%) pass percentage in the year 2015.

15. 75 students joined the school in 2016.

16. The total number of students who failed was 10 + 11 + 7 + 25 + 14 = 67

17. The maximum number of students failed in Class IX.

18. The states are ranked in the descending order of land revenue collections as tabulated below.

Rank	1999–00	2000–01	2001–02	2002–03	2003–04
1	U.P.	Bihar	U.P.	U.P.	U.P.
2	Bihar	UP	Bihar	Bihar	Bihar
3	A.P.	Gujarat	A.P.	A.P.	A.P.
4	Gujarat	AP	Gujarat	Gujarat	Gujarat
5	M.P.	M.P.	T.N.	T.N.	T.N.
6	T.N.	K.A.	M.P.	K.A.	K.A.
7	K.A.	T.N.	K.A.	M.P.	M.P.

Four states U.P., Bihar, A.P. and Gujarat have the same ranks in four of the five given years.

19. The percentage share of land revenue collection for the different states in the given periods are as follows:

State	1999–2000	2000–2001
U.P.	$\frac{161}{294} > 20\%$	$\frac{63}{317} < 20\%$
Karnataka	$\frac{28}{294} < 10\%$	$\frac{34}{317} > 10\%$
Bihar	$\frac{59}{294} > 20\%$	$\frac{63}{317} < 20\%$
A.P.	$\frac{41}{294} < 20\% = 13.9\%$	$\frac{43}{317} = 13.5\%$

The percentage has increased the maximum for Karnataka.

20. The percentage increased for Tamil Nadu from

$$2000 \text{ to } 2001 = \frac{3384-3167}{3167}\times 100\% = 7\%$$

$$2001 \text{ to } 2002 = \frac{3921-3384}{3384}\times 100\% = 16\%$$

$$2002 \text{ to } 2003 = \frac{4163-3921}{3921}\times 100\% = 6\%$$

$$2003 \text{ to } 2004 = \frac{4931-4163}{4163}\times 100\% = 18\%$$

21. From the year 2000 to 2001, and 2001 to 2002, the increase in the land revenue collection amount for the state of Gujarat was constant.
$4427 - 3951 = 476$
$4903 - 4427 = 476$

22. In both the months mentioned, Adidas is the most preferred. ∴ It has to be the most preferred when both months are considered together.

23. It is possible that the respondents who responded favourably to Nike (40%) in March were the same respondents who responded favourably to all other brands. Hence, (100 – 40) % = 60% is the maximum possible number of respondents who did not respond favourably to any of the brands.
∴ 60% of 1200 = 720

24. Let the total number of respondents in the months given be a, b, c, d, e, respectively.
Given

$b > a$ (1)
$d > b$ (2)
$e >$ each of a, b, c, d (3)

Considering Statement I:
Since from (1) and (2) above, $d > a$.
The N for two brands, i.e., Reebok and Lotto, definitely increased. Hence, Statement I is true.
Considering Statement II:
Since $e > d$, it is possible that the N for no brand or all the brands other than Adidas decreased. Hence, Statement II is not definitely false.
Considering Statement III:
Since $e > b$, and we need to compare the following four pairs of qualities:
15% of b vs 35% of e
50% of b vs 50% of e
25% of b vs 20% of e
20% of b vs 10% of e
We can observe that the first two pairs can never be equal while the second two pairs will always be either equal or unequal together, i.e., if 40% of b = 20% of e, then 20% of b = 10% of e and vice versa. Hence, the N for either zero brands or exactly two brands was the same in March and September.
Hence, Statement III is false.

25. Considering the sum of the percentage values of the given brands:
40% + 50% + 20% + 35% = 145%
Now, imagine that there are 145 'favourable responses' (henceforth referred to as 'votes'). These must be distributed among a total of 100 respondents or say voters. It is allowed for any voter to have more than one vote and up to a maximum of four votes. But these 145 votes must be distributed in such a way that the least possible number of respondents get more than one vote each. Hence, the extra 45 votes (left after first giving one vote to each voter) should be distributed among the least possible number of voters. This can be done by distributing away as many votes as possible to each voter. But, since the maximum possible votes allowed is only four, three more votes can be given to any voter. Hence, 45 votes can be distributed among at the least 15 voters. Hence, we can have only 15 voters who have more than one vote. Thus, our required percentage is 15%.
Therefore, 15% of 125% of 800 = 15% of 1000 = 150.
Hence, at least 150 respondents must have responded favourably to more than one of the given brands.

26. According to the question, all the LOCAL and STD calls have to be made such that the amount paid should be minimum. If customers have the choice from all the four companies, then they will go for the service which is cheapest.
For LOCAL calls, DATACOM is the cheapest but from DATACOM only 2.75 lakh calls can be made.
Similarly, the cheapest STD charges are when using AIRSIM and 2.0 lakh calls can be made from AIRSIM and the rest 1.5 lakh calls should be made using the second cheapest service (APPLE).

So, the total amount spent on STD calls is
$2 \times 3.25 + 1.5 \times 3.5 =$ ₹11.75 lakh.
Total amount spent on local calls is

$$\frac{2.0 \times 2.75}{\text{DATACOM}} + \frac{2.25 \times 2.5}{\text{APPLE}} + \frac{2.5 \times 1.25}{\text{AIRCOMM}} = 5.5 + 2.5(3.5)$$
$$= 5.5 + 8.75 = 14.25$$

Therefore, total amount spent $= 11.75 + 14.25 =$ ₹26 lakh.

27. When the charges of the companies are compared, the difference between the ISD charges is more when compared to the difference in the LOCAL charges. Therefore, ISD calls should be made first from the cheapest service.
1 lakh ISD calls should be from APPLE
Amount for ISD charges $= 1 \times 6 = 6$ lakh
For local calls, 2.75 lakh calls can be made from DATACOM at the rate of 2 rupees. 3 lakh calls can be made from APPLE at the rate of 2.25 rupees. The remaining 0.75 lakh calls can be made from either of AIRSIM or AIRCOMM at the rate of 2.5 rupees.
Total charges for LOCAL calls
$= 2 \times 2.75 + 3 \times 2.25 + 0.75 \times 2.5 \cong 14.13$ lakh
Total (LOCAL+ ISD) charges $= (14.13 + 6)$ lakh
$= 20.13$ lakh

28. Revenue loss due to reduction in STD changes
for AIRCOMM $= 0.7 \times (4.5 - 3.5) = 0.7$ lakh
for AIRSIM $= 0.4 \times (3.25 - 2.25) = 0.4$ lakh
for DATACOM $= 0.8 \times (3.5 - 2.5) = 0.8$ lakh
for APPLE $= 1.6 \times (3.5 - 2.5) = 1.6$ lakh
To compensate the revenue loss, increment in ISD charges per minute

$$= \frac{\text{Revenue loss for a company}}{\text{Number of ISD calls made from company}}$$

Increment in ISD charges

for AIRTEL $= \dfrac{0.7}{0.3} = 2.33$

for AIRSIM $= \dfrac{0.4}{0.1} = 4$

for DATACOM $= \dfrac{0.8}{0.2} = 4$

for APPLE $= \dfrac{1.6}{0.4} = 4$

New ISD charges per unit call for various companies:
AIRCOMM = 10.33, AIRSIM = 13.5, DATACOM = 10.5, APPLE = 10
The second cheapest ISD charges are for AIRCOMM

29. Revenue of AIRCOMM $= 2.5 \times 1.25 + 4.5 \times 0.7 + 8.0 \times 0.3$
$= 8.68$ lakh
Revenue of AIRSIM $= 2.5 \times 1.5 + 3.25 \times 0.4 + 9.5 \times 0.1$
$= 6$ lakh
Revenue of DATACOM $= 2.0 \times 1.75 + 3.5 \times 0.8 + 6.5 \times 0.2$
$= 7.6$ lakh
Revenue of APPLE $= 2.25 \times 2.0 + 3.5 \times 1.6 + 6.0 \times 0.4$
$= 12.5$ lakh
The company with the highest sales revenue is APPLE.

30. By observation, N must belong to one of the following two categories.
(i) $4 \le N < 6$
(ii) $2 \le N < 4$
Number of students enrolled for $4 \le N < 6$
$= (30 \times 11) + (10 \times 22) + (10 \times 30)$
$+ (10 \times 15) + 10(5) + 10(4) = 1090$
Number of students enrolled for $2 \le N < 4$
$= (20)\ (11) + (10)\ (22) + (10)\ (30)$
$+ (40)\ (15) + (10)\ (4)$
$= 220 + 220 + 300 + 600 + 40 = 1380$

31. By observation, the number of students enrolled is the highest for Science with $6 \le N < 8$.

32. Total number of students = 8700
Required number of degrees

$$= \frac{(360)\,(3000) - (360)\,(1100)}{8700} = \left(\frac{360}{8700}\right)(1900)$$

$$= \frac{(360)\,(19)}{(87)} \cong 78.6°$$

33. For $N \ge 8$,
Most number of enrolments is not from the Arts stream.
Least number of enrolments $(0 \le N < 2)$ is not form others but from the Commerce stream.
Percentage of enrolments from Engineering

$$= \frac{3000}{8700} \times 100 > 30\%$$

Solutions for questions 34 to 37: It is known that the pass percentage in Class V in the year 2015 was at least 90%.

∴ At least 41 of the 45 students in Class V would have passed in 2015. From the given graph, there are a total 52 students in Class VI in 2015 and 47 students in Class VII in 2016. Hence, at least 5 students of Class VI of 2015 must have failed and they must continue in Class VI in 2016 also. By this we can say that 41 students of Class V of 2015 are promoted to Class VI. This can be represented as follows.

Standard	Passed	Failed	Promoted/Joined
V	41	4	52 – 4 = 48
VI	52 – 5 = 47	46 – 41 = 5	41
VII	50 – 0 = 50	47 – 47 = 0	47
VIII	43 – 7 = 36	57 – 50 = 7	50
IX	46 – 2 = 44	38 – 36 = 2	36
X	42 – 2 = 38	48 – 44 = 4	44

34. 48 students joined Class V in the year 2016.

35. The total number of students who failed
$= 4 + 5 + 7 + 2 + 4 = 22$

36. The pass percentage was the highest in Class VII, i.e., 100%.

37. The overall pass percentage

$$= \frac{41 + 47 + 50 + 36 + 44 + 38}{45 + 52 + 50 + 43 + 46 + 42} \times 100 = \frac{256}{278} = 92\%$$

In classes V, VI, VIII and X, the pass percentage was less than the overall pass percentage.

38. The number of marketing employees working in the morning shift must be the maximum possible.
In B_1, 20 persons are working in the Marketing department and 18 persons are working in the morning shift.
Given, there must be at least one person from each department in each of the shifts.
From the Marketing department, at least one person has to work in each of Dawn, General, Evening and Night shifts. ∴ At most 20 – 4 = 16 persons of the Marketing department can be working in the morning shift.
Among the 18 persons working in the morning shift, at least one must be from each of Production, Finance and Research departments and hence, at most (18 – 3) = 15 persons can be working in the Marketing department.
Similarly, if we calculate for other branches.
∴ Only 15 employees of the Marketing department can be working in the morning shift.
B_1: Minimum of ((20 – 4), (18 – 3)) = 15
B_2: Minimum of ((25 – 4), (8 – 3)) = 5
B_3: Minimum of ((30 – 4) (15 – 3)) = 12
B_4: Minimum of ((20 – 4) (25 – 3) = 16
∴ The required answer is 15 + 5 + 12 + 16, i.e., 48.

39. In the branch B_3, the maximum possible number of Production employees working in the morning shift is 15 – 3, i.e., 12. The minimum number of Finance employees working in the general shift is only one.
∴ The required difference is 12 – 1 = 11.

40. Minimum of ((20 – 3), (26 – 6)) = 17
Minimum of ((25 – 3), (31 – 6) = 22
Minimum of ((30 – 3), (27 – 6) = 21
Minimum of ((20 – 3), (43 – 6)) = 17,
i.e., 17 + 22 + 21 + 17 = 77

41. From the evening shift to the night shift, minimum change can be done in the following way:
Shift 5 employees in B_1.
Shift 1 employee in B_2.
For B_3 and B_4, there is no need of a change.
∴ The required answer is 5 + 1 = 6.

42. Total man hours spent for both the offices
= (400 + 560) + (1300 + 1020) + (900 + 1120) = 5300
For every 100 hours, the activity should account for 25 hours.
For every 5300 hours then

$$\left(\frac{25}{100}\right)(5300) = 1325 \cong 1300 \text{ hours.}$$

Thus, the activity which accounts for 25 hours for a total of 100 hours spent is content development at Mumbai.

43. Total man hours spent in content development
= 1300 + 1020 = 2320.
From Choices
Choice (A) Man hours for content maintenance and designing = (900 + 1020) + (400 + 560) = 2980
Choice (A) is not true.
Choice (B) Content development (at Mumbai) and content designing (at both offices)
= 1300 + (400 + 560) = 2360

44. Required percentage

$$= \frac{560 + 1020 + 1120}{(1000) + (2300) + (2000)} = \frac{2700}{5300} \times 100 \cong 51$$

45. Total effort (in man hours) at Mumbai
= 400 + 1300 + 900 = 2600.
From Choices
Choice (A)
Total estimated efforts for Hyderabad
= 900 + 700 + 1000 = 2600.

46. If 40% of content maintenance (at Mumbai) have been transferred to Hyderabad, then the percentage of total content maintenance work carried out at Mumbai

$$= \frac{(100\% - 40\%)(900)}{2020} = \left(\frac{60}{100}\right)\frac{(900)}{2020}(100) \cong 27.$$

Solutions for questions 47 to 50:

	(Pieces/hour) at the end of t hours		Total pieces machined after t hour		Number of pieces machined in the t^{th} hour	
	A	B	A	B	A	B
$t = 1$	110	100	110	100	110	100
$t = 2$	110	140	220	280	110	180
$t = 3$	110	160	330	480	110	200
$t = 4$	125	165	500	660	170	180
$t = 5$	140	170	700	850	200	190
$t = 6$	150	150	900	900	200	50
$t = 7$	160	140	1120	980	220	80
$t = 8$	160	140	1280	1120	160	140

Difficulty Level-2: Moderate

47. The total number of pieces machined by A and B together at
$t = 1$ is $110 + 100 = 210$;
$t = 2$ is $220 + 280 = 500$;
$t = 3 = 330 + 480 = 810$
$t = 4$ is $500 + 660 = 1160$
$\therefore$ The time they will take lies between $t = 3$ and $t = 4$.

48. The number of pieces machined by A in any hour is not same as that of B.

49. It happened in three instances, i.e., at $t = 2$; $t = 3$; $t = 6$ for machine A but it never happened for machine B.

50. Number of pieces machined in the last three hours by machine A = 200 + 220 + 160 = 580;
machine B = 50 + 80 + 140 = 270.

$$\text{Required } \% = \frac{580-270}{270}\times 100 = \frac{3100}{270}\times 100$$
$$\cong 115\% \text{ more}$$

EXERCISE-3

Solutions for questions 1 to 4: The sales, net profit, gross profit and expenses in each of the given years is as follows:

Year	2013	2014	2015	2016	2017
Sales	765	810	875	756	824
Gross profit	197	220	241	171	224
Net profit	148	165	193	137	179
Expenses	568	590	634	619	645

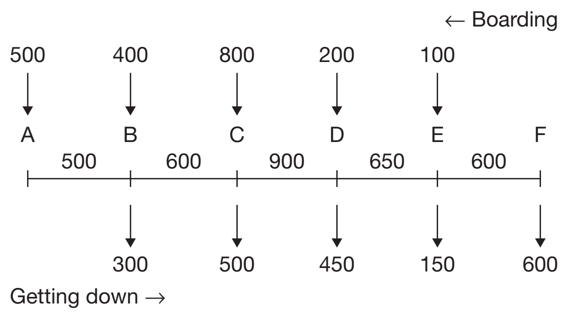
1. The expenses in the year 2014 was ₹590 cr.

2. The percentage increase in expenses from 2014 to 2015 is $\frac{44}{590}\times 100 = 7.46\%$

3. The ratios of sales to expenses in the given years are
$2013 - \frac{765}{568}$, $2014 - \frac{810}{590}$,
$2016 - \frac{756}{619}$ and $2017 - \frac{824}{645}$
It was lowest in 2016.

4. The probability in the different year are as follows.
$2013 - \frac{197}{765} = 25.8$, $2014 - \frac{220}{810} = 27.2$,
$2015 - \frac{241}{875} = 27.6$, $2017 - \frac{224}{824} = 27.2$.

Solutions for questions 5 to 8: From the pie chart, the number of passengers who boarded the train at different stations can be found. The total number of passengers travelling in the train between any two stations can also be found once we know the number of passengers who boarded the train and those who got down from the train at different stations.

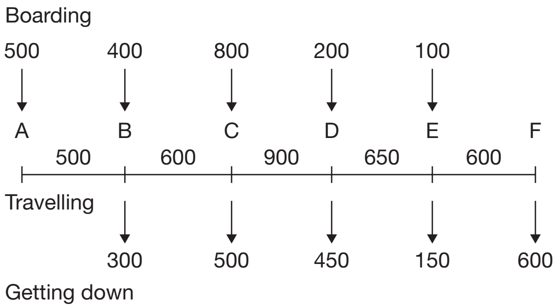


5. Maximum number of passengers boarded the train at station C. It can be observed that the difference in the number of passengers on board C to D and the number of passengers getting down at d is 900 – 450 = 450, which is the highest among the other such possible differences. Hence, it is possible that the maximum possible number of passengers satisfying the requirement may have boarded at C. Maximum number of passengers who boarded the train at C and got down at station F will travel a distance 30 km which is more than 15 km and these many passengers would satisfy the condition given in the questions. Therefore, maximum number of passengers in the group of villagers will be the number of passengers boarded at C and got down at F.
From C to D we have 900 passengers, out of which 800 were from C and 100 were from other stations. Number of passengers got down at D is 450 in which 100 passengers could be those who boarded at other stations (before C) and 350 will be those who boarded at C. So, the number of passengers travelling from D to E who boarded at C is (800 – 350), i.e., 450. At station E, 150 passengers got down, all of who could be from 200 passengers who boarded at D.
So, a maximum of 450 passengers who boarded at C can get down at F.

6. Maximum number of passengers who got down at C and travelled not more than 15 km are the passengers boarded at stations A and B.
So, maximum number of passengers boarded at A or B and got down at C = 500.
Maximum number of passengers who boarded at C and travelled not more than 15 km are passengers who got down at stations D and E which is equal to 600.
So, total such passengers = 1100.

7. If the passengers who boarded the train at A get down at E or F, then they travelled at least 30 km. Similarly, if the passengers boarded at B get down at F then they travelled at least 30 km.
Maximum number of passengers who boarded at either A or B who can travel till E or F is 100. If the passengers boarded at C gets down at F, then they travelled at least 30 km. The maximum number of such passengers is 350. So, the total passengers satisfying the condition is at most 450. Choice (D)

8. The line diagram for return journey is as given below.



Clearly, choice (A) is true, as number of passengers between any two stations in the evening is same as that in the morning.
Choice (B) is also true.
Choice (C) is true as the number of passengers who boarded at F and get down at C which is 30 km from F is 450. In the morning journey also, this number was 450, therefore this is also a true statement.

Solutions for questions 9 to 12: Let n = number of members in a group

$$10 = 2 + 2 + 2 + 4$$
$$10 = 2 + 2 + 3 + 3$$

Year 2015

Group	n	Total age
A	2	41
B	2	45
C	2	49
D	4	110
		245

Year 2016

Group	n	Total age
A	3	73
B	2	51
C	3	84
D	2	47
		255

Note: As the average age of group A is 24.33, the number of members cannot be 2 or 4. Hence, it must be 3.

Year 2017

Group	n	Total age
A	4	118
B	2	49
C	2	46
D	2	52
		265

Year	Sum of all the ages
2015	245
2016	255
2017	265

Year 2015:
Individual age of the members of the group can be evaluated (unique possibility)

Group	Individual ages
A	20, 21
B	22, 23
C	24, 25
D	26, 27, 28, 29

A different set of members is present in any group for any two years.

Year 2016

Group	Individual ages	Total
A	23, 24, 26	73
B	21, 30	51
C	27, 28, 29	84
D	22, 25	47

Year 2017

Group	Individual ages	Total
A	28, 29, 30, 31	118
B	23, 26	49
C	22, 24	46
D	25, 27	52

Note: Sum of the ages of members of group C (in 2017) must have been 42 as on 1st January, 2015. Which adds to a unique possibility of two members of age 20, 22 years. Similar reasoning is used to establish individual ages in the respective years.

9. Age of the eldest person in Group B as on 1st January 2017 = 26 years.

10. Group D:

Ages = 22, 25 (in 2016)
= 21, 24 (in 2015)

Group B, 2015
22, 23.
No person is common.

11. Group A had the maximum total age in the year 2017.

12. Only group A showed a consistent increase in the total age of all the members across the years.

Solutions for questions 13 to 16:

13. The revenues in the different years are (assuming 100 rooms in 2012) as follows.
2012 – 4000×50
2013 – $4500 \times 55 \times 1.1$
2014 – $4200 \times 65 \times 1.21$
2015 – $5400 \times 57 \times 1.33$
2016 – $5000 \times 65 \times 1.46$
2017 – $4800 \times 71 \times 1.61$
It can be seen that the revenue was the highest in 2017.

14. The percentage increase in revenue over the previous year was the highest in 2013, i.e., over 35%.

15. As it is given that the hotel made a profit in 2012, the revenue in that year is more than the expenses. As the expenses increased by 15% each year it can be seen that the growth rate of revenue was more than 15% in each year and so the profit would be the highest in 2017.

16. For any number of rooms, the profit increases at a faster rate than revenue. So, the profitability keeps an increasing and is the highest in 2017.

Solutions for questions 17 to 20: As it is known that player F was ranked eight and that player D was ranked immediately above player C, it can be concluded that both C and D scored more runs than player B, while player G scored fewer than player B. So also, as the difference between the runs scored by player A, E and H and player B is more than that of player F, all the three must have scored more runs than player B.

∴ If player B scored 0 'x' runs, the runs scored by the other players are as follows:

A	$x + 90$	Rank – 2
B	x	Rank – 6
C	$x + 25$	Rank – 5
D	$x + 48$	Rank – 4
E	$x + 107$	Rank – 1
F	$x - 65$	Rank – 8
G	$x - 45$	Rank – 7
H	$x + 72$	Rank – 3

17. The rank of player C was 5.

18. Five players scored fewer than player H.

19. The total runs scored by the eight players are

$$= \frac{8x + 232}{8} = x + 29 = 435$$

$\therefore\ x = 406$
Player C scored 406 + 25 = 431 runs.

20. $x - 45 = 373$
$\therefore x = 418$
Total runs = $8x + 232 = 8 \times 418 + 232 = 3576$

Difficulty Level-3: Advanced

CHAPTER 3

Pie Charts

Learning Objectives

In this chapter, you will:

- Understand various types of data presented in the form of pie charts.
- Get familiar with different types of pie charts.
- Get familiar with different types of questions based on pie charts.
- Learn shortcuts to solve questions involving two or more pie charts without finding all the values.
- Compare values (changes) across two or more pie charts using percentage techniques.
- Understand data which involves more than one data type – bar graph and pie chart together, pie chart and table, etc.

❐ PIE-CHARTS

This is probably the simplest of all pictorial forms of data presentation. Here, the total quantity to be shown is distributed over one complete circle or 360 degrees. In pie charts, data is essentially presented with respect to only one parameter (unlike in two and 3-dimensional graphs described later). This form essentially presents the shares of various elements as proportion or percentage of the total quantity. Each element or group in the pie chart is represented in terms of quantity (or value, as the case may be) or as the angle made by the sector representing the elements or as a proportion of the total or as a percentage of the total. Pie charts are also very frequently used in combination with other forms of data or along with other pie charts.

Figure 3.1 depicts the distribution of the population in different geographical zones.

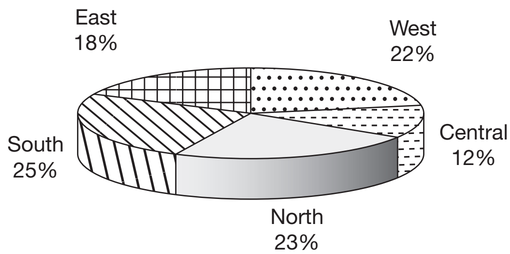


Fig. 3.1 Distribution of population in geographical zones

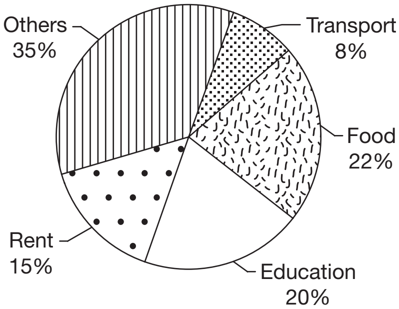
From the above pie chart, we can calculate the following:

- Population in any zone given the total population
- Population of any zone as a percentage of that of another zone.
- Percentage increase in the total population given the percentage increase in the population of one or more zones.

SOLVED EXAMPLES

Directions for questions 3.01 to 3.05: These questions are based on the information given below.

The following pie chart represents the break-up of Raju's monthly expenses.



3.01: If Raju spent ₹4500 more on food and transport together than he spent on rent, then find his monthly expenses (in ₹).

(A) 15,000 (B) 25,000
(C) 30,000 (D) 35,000

Sol: Percentage of his expenditure spent on rent = 15%

Percentage of his expenditure spent on transport and food = 30%

∴ 30% – 15% = 15% = ₹4500

Monthly expenses = 100%

$$= \frac{100}{15}(4500) = ₹30,000$$

3.02: If Raju increased his savings, which is currently 10% of his income by 20% and reduced his expenses by 20%, then his savings would be what percentage of his expenses?

(A) 10% (B) 12.5%
(C) 15% (D) $16^2/_3$%

Sol: Let his monthly income be ₹x.

$$\text{Original savings} = ₹\frac{10}{100}x$$

$$\text{New savings} = \frac{10}{100}x + \frac{20}{100}\left(\frac{10}{100}x\right) = ₹\frac{12}{100}x$$

$$\text{Original expenditure} = x - \frac{10}{100}x = ₹\frac{90}{100}x$$

$$\text{New expenditure} = \frac{90}{100}x - \frac{20}{100}\left(\frac{90x}{100}\right) = ₹\frac{72}{100}x$$

$$\text{Required percentage} = \frac{\frac{12}{100}x}{\frac{72}{100}x}(100) = 16\frac{2}{3}\%$$

3.03: Raju spent 20% of his expenditure on 'others' on entertainment. This amounted to ₹2100. Find his expenditure on education.

(A) 4500 (B) 5000
(C) 6500 (D) None of these

Sol: Expenditure on entertainment = 20% of 35%

= 7% = 2,100

1% = 300

Expenditure on education = 20% = ₹6000

3.04: Find the angle made by the expenditure on rent and 'others' put together.

(A) 120° (B) 160°
(C) 180° (D) 210°

Sol: Total expenditure on rent and 'others'

= 15% + 35% = 50%

$$\text{Required angle} = \frac{50}{100}(360°) = 180°$$

3.05: As prices dropped, Raju's expenditure on clothes dropped by 10%. As a result of this, his expenditure on 'others' decreased from ₹10,500 to ₹10,290. What percentage of his expenditure on 'others' was spent on clothes?

(A) 10% (B) 12%
(C) 15% (D) 20%

Sol: Decrease in expenditure on 'others'

= 10,500 – 10,290 = ₹210

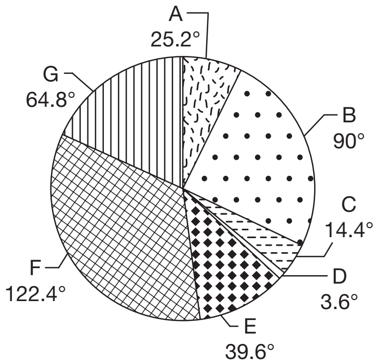
∴ Initial expenditure on clothes

$$= 210\left(\frac{100}{10}\right) = ₹2100$$

$$\text{Required percentage} = \frac{2100}{10500}(100) = 20\%$$

EXERCISE-1

Directions for questions 1 to 5: The pie chart given below shows the import of various commodities to India during 1999–2000.



Total = 36 billion dollars

A = Food and allied products
B = Fuel
C = Fertilizers
D = Paper and allied products
E = Capital goods
F = Bulk goods
G = Miscellaneous items

1. If 40% of the miscellaneous items imported are pharmaceutical products and this is one eighth of the total internal production, then the total value of internal pharmaceutical production in billion dollars is

(A) 2.592 (B) 25.92
(C) 20.736 (D) 3.24

2. Find the value of bulk goods, food and allied products imported.

(A) 14.76 billion dollars
(B) 16.74 billion dollars
(C) 11.46 billion dollars
(D) 17.46 billion dollars

3. In 1999–2000, the total exports of India is $24 billion. In this figure, 20% is contributed by the textile industry, whereas 50% of miscellaneous items are for imports of textiles. Find the value of net imports or exports of textile industry.

(A) $2.24 billion net exports
(B) $3.16 billion net imports
(C) $1.84 billion net imports
(D) $1.56 billion net exports

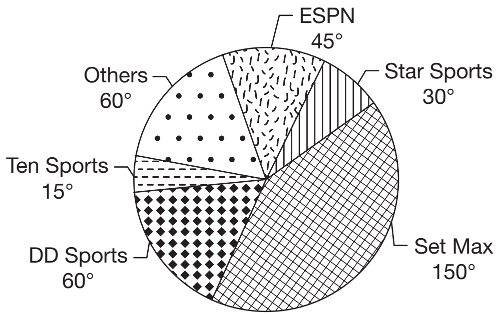
4. On account of the exploring of oil at two sites in Andhra Pradesh, India can save $2 billion on its fuel bill. Find the approximate ratio of imports of new fuel bill and capital goods bill.

(A) 7 : 5 (B) 7 : 4
(C) 3 : 1 (D) 5 : 2

5. The fuel bill in 2000–2001 increases by 40% and the bill for all other commodities remains the same as in 1999–2000. What is the angle made by capital goods sector imports if the pie chart is redrawn for the year 2000–2001?

(A) 36° (B) 39.6°
(C) 32.4° (D) 43.2°

Directions for questions 6 to 10: These questions are based on the following pie chart which shows the viewership of different sports channels in the month of February 2003 in India. There is no overlap in viewership of channels.



6. If 60,000 people watched Star Sports on an average per day in February 2003, then how many more people watched Set Max than Ten Sports on an average per day for the same period?

7. During the given period for how many sports channels is the viewership more than 20% of the total viewership?

8. If the viewership of DD Sports for the first half of February is half that of the second half of February, then what is the ratio of viewership of DD Sports for the second half of February to that of ESPN for the whole month?

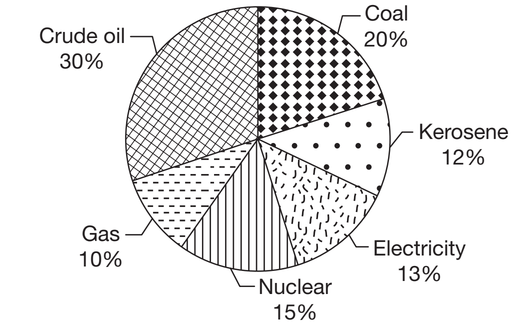
9. By mistake the viewership of DD sports has been underquoted by 20%. If this mistake is corrected then what is the correct share of viewership of Set Max?

10. If the viewership of ESPN on an average was 90,000 per day, then what was the viewership for all the sports channels on an average per day?

Difficulty Level-1: Foundation

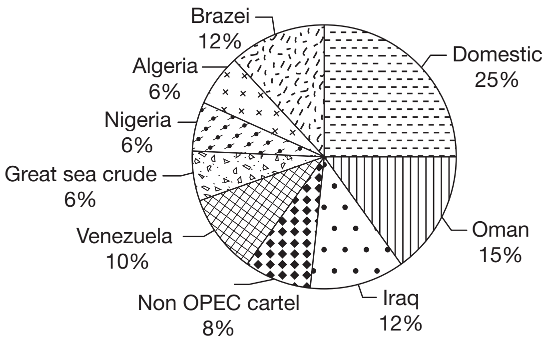
Directions for questions 11 to 15: These questions are based on the pie charts given below.

Break-up of various energy sources consumed by India (by value)



Total value = ₹ 60,000 crore

Break-up of crude oil supply in India (by volume)



Total = 250 million barrels

11. What is the value of 1 litre of kerosene given that the total kerosene consumed in India is 150 lakh kilolitres?

(A) ₹3.2 (B) ₹4.8
(C) ₹6.8 (D) Data inadequate

12. If 1000 kilowatts {= 1 gigawatt (1 GW)} of electricity costs ₹25 lakh, then what is the total amount of electricity generated in India?

(A) 31,200 GW (B) 25,000 GW
(C) 21,750 GW (D) Data inadequate

13. What is the domestic crude oil price per barrel in Oman given that Oman sells crude oil to India at a discount of 20% on its domestic price? (Assume that the price of crude oil to India from all the sources is the same)

(A) ₹900 (B) ₹750
(C) ₹720 (D) Data inadequate

14. If the Total Estimated Reserves (TER) of crude oil in India is 4000% more than the Total Recoverable Reserves (TRR) and the current domestic production of crude oil is 16% of the TRR, then what is the approximate TER of crude oil in India? (in million barrels)

(A) 11,500 (B) 13,700
(C) 15,000 (D) 16,000

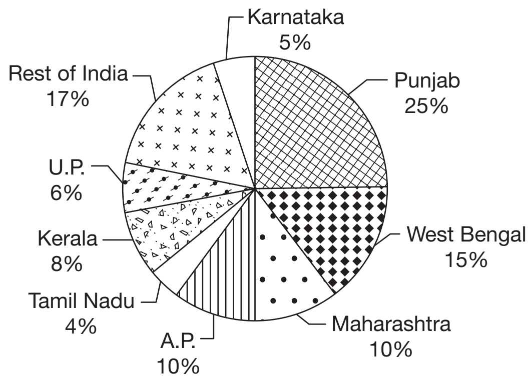
15. The current domestic price of crude oil is 25% less than the price of crude oil from all other sources (as shown in the pie chart), then what is the price of crude oil imported from Venezuela? (Assuming that prices of crude oil from all sources other than domestic are equal)

(A) ₹617 per barrel (B) ₹768 per barrel
(C) ₹917 per barrel (D) Data inadequate

Directions for questions 16 to 19: These questions are based on the following pie charts.

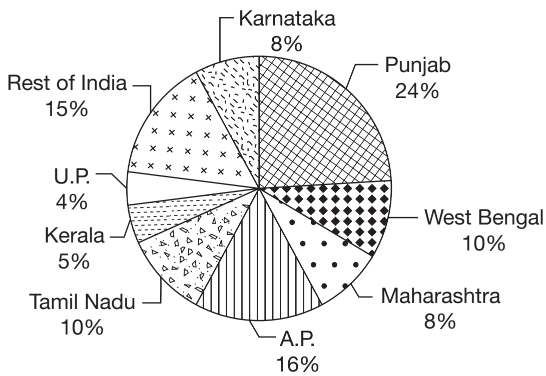
State wise Production of Food Grains in India

In 1999–2000



360 million tons

In 2000–2001



300 million tons

16. From 1999–2000 to 2000–2001, which of the following states showed the maximum percentage increase in the food grain production?

(A) A.P. (B) Tamil Nadu
(C) Kerala (D) Karnataka

17. In 1999–2000, rice and wheat form 60% of the total food grain production. In 2000–2001, they form 80% of the total food grain production. What is the percentage change in the production of food grains other than rice

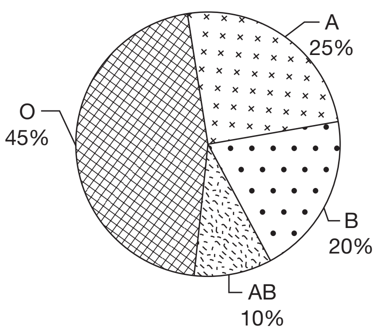
and wheat in Maharashtra, given that Maharashtra produced 4 million tons of rice and wheat in both the years?
(A) 20% increase
(B) 37.5% decrease
(C) 58.3% increase
(D) 33.3% decrease

18. If the ratio of the average price per kg of rice in 1999–2000 to that in 2000–2001 is 11 : 12, then the percentage of revenue gained/lost on account of rice by West Bengal from 1999–2000 to 2000–2001, given that 50% of its total production is from rice in both the years?
(A) 35% loss (B) 40% loss
(C) 40% gain (D) 35% gain

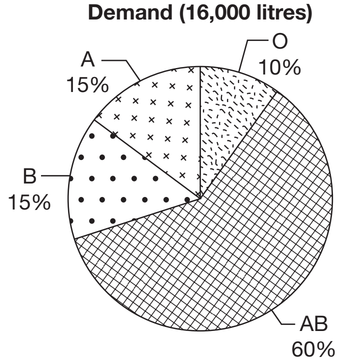
19. If the percentage increase in the production of food grains for A.P. and Karnataka in 2001–02 from the previous year is the same as the percentage increase from 1999–2000 to 2000–2001, then what is the difference in share of production between these two states in the year 2001–02?
(A) 11.11 percentage points
(B) 22.22 percentage points
(C) 33.33 percentage points
(D) Cannot be determined

Directions for questions 20 to 23: The following pie charts represents the demand and supply of blood in a city of different blood groups in a certain period.

Supply (4,000 litres)



Demand (16,000 litres)



20. If the ratio of supply of positive and negative combinations of the blood groups A, B, O and AB are 3 : 1, 4 : 1, 1 : 1 and 2 : 1, respectively, then the blood group having the least supply of blood is
(A) O^- (B) AB^-
(C) A^- (D) B^-

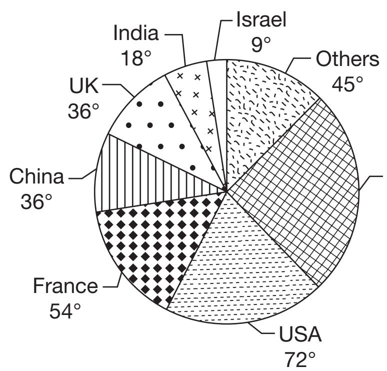
21. If total demand doubles and the demand percentage of blood group AB remains constant, then which of the following is true?
(A) Supply of blood groups A, B and O will decrease.
(B) Supply of blood groups A, B and O will increase.
(C) Demand of blood groups A, B and O will decrease.
(D) Demand of blood groups A, B and O will increase.

22. If the supply of blood group O^+ and O^- is in the ratio 3 : 2 and the demand of these blood groups is in the ratio 3 : 5, then how much excess supply of blood is there in blood group O^+?
(A) 620 litres
(B) 560 litres
(C) 480 litres
(D) 525 litres

23. If a person having blood group 'O' can donate maximum of up to 2 litres at a time and a person having blood group 'A' can donate maximum 1 litre at a time, then how many people are required to donate the blood for these two groups, so that the demand for these two groups is met. Assume that apart from the blood obtained from the above donors, there is no other supply of these blood groups.
(A) 3200 (B) 2900
(C) 2100 (D) 1600

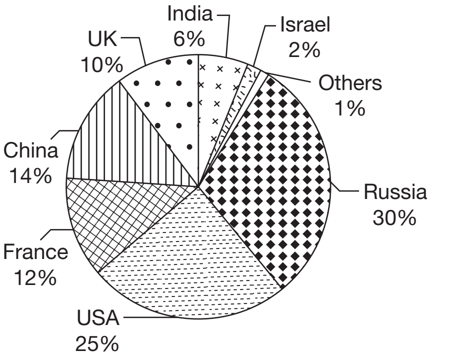
Directions for questions 24 to 28: These questions are based on the pie charts given below.

Distribution of Nuclear Warheads produced among different countries (in the year 2002)



Total = 80,000

Cost incurred in building and maintaining Nuclear Warheads by various countries (in the year 2002)



Total = $ 2500 million

Note: Assume that all the countries had no warheads at the end of the year 2001.

24. What is the ratio of the difference in the number of warheads possessed by Russia and USA and the difference in the number of warheads possessed by India and Israel?

(A) 1 (B) 1.5
(C) 2 (D) 2.5

25. What is the average cost of building and maintaining one nuclear warhead by China in 2002?

(A) $437.50 (B) $43,750
(C) $4375 (D) None of these

26. What is the difference in the average expenditure incurred in building and maintaining one warhead by Russia and that of USA?

(A) $1500
(B) $1562.50
(C) $1671.40
(D) None of these

27. If each warhead of the countries from the category 'Others' weighs 1000 kg, which is half the weight of the other countries given in the graph, then what is the weight of all the nuclear warheads of all countries put together?

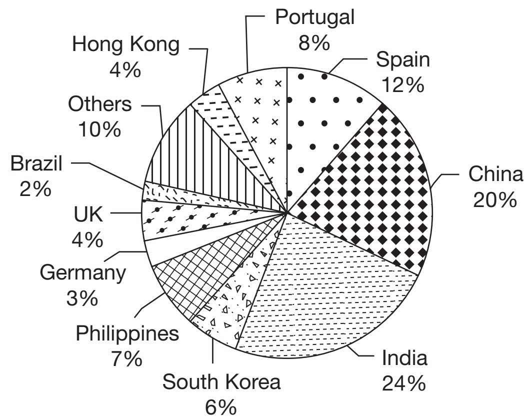
(A) 30,0000 tons
(B) 45,000 tons
(C) 1,40,000 tons
(D) 1,50,000 tons

28. If after signing the Nuclear Non-proliferation Treaty (NPT), Russia reduces the number of nuclear warheads with it by 40%, USA by 30%, UK and China by 20% each and all other countries by 10%, then what will be the total number of nuclear warheads in the world after this reduction?

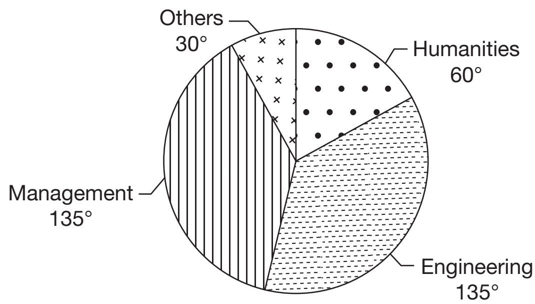
(A) 61,200 (B) 64,000
(C) 72,800 (D) None of these

Directions for questions 29 to 33: These questions are based on the pie charts which provides the statistics of international students pursuing post-graduation in the United States in the year 2000.

Distribution by nationality



Distribution by graduation



29. The number of students form India and China is what percent of the students form all the other countries (excluding the 'Others' category)?

(A) 96.78% (B) 95.12%
(C) 95.65% (D) 92.01%

30. How many students from the country which has the maximum representation in the year 2000 are pursuing a management course? (assuming that the nation wise distribution of the students of each graduation stream is as per the graph of distribution by nationality)

(A) 7200 (B) 9000
(C) 19200 (D) Cannot be determined

31. If the number of students pursuing post-graduation in USA in the year 2000 from Spain and Portugal is 25% of the number of domestic students pursuing MS in USA, then what is the total number of domestic students pursuing MS in USA?

(A) 16000 (B) 36000
(C) 60000 (D) None of these

32. If it is known that a total of 3,60,000 students are pursuing post-graduation in USA in 2000, then approximately what percentage of it is comprised of students from Asian countries (i.e., China, India, South Korea, Philippines and Hong Kong)?

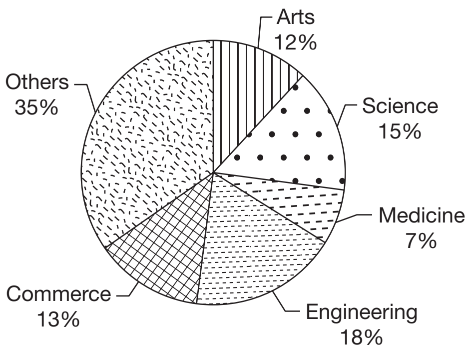
(A) 16.5% (B) 13.5%
(C) 19% (D) Cannot be determined

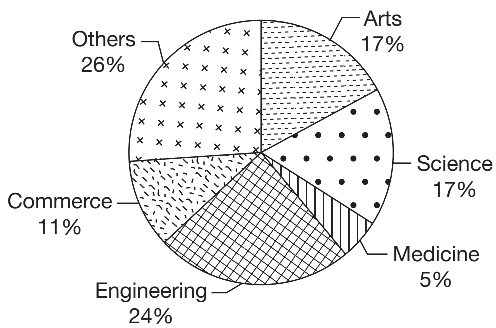
33. What is the number of Indian students pursuing either Engineering or Management as a percentage of the number of Chinese students pursuing any course other than Engineering and Management in USA in the year 2000? (use the data given in question 17)

(A) 120% (B) 220%
(C) 360% (D) 440%

Directions for questions 34 to 37: Answer these questions on the basis of the information given below.

The first pie chart gives the breakup of the total students doing graduation in a city according to their area of specialization. The second pie chat gives the breakup of the boys doing graduation according to their area of specialization. The ratio of the number of boys to girls doing graduation is 2 : 5.





34. If gender ratio = $\frac{\text{number of girls}}{\text{number of boys}}$, then what is the gender ratio of students doing Medicine?

(A) 3.9 (B) 4.3
(C) 3.5 (D) Cannot be determined

35. For how many areas of specialization is the number of boys at least half of the number of girls?

(A) 1 (B) 2
(C) 3 (D) 4

36. If the number of girls doing Medicine is 4056, then what is the total number of boys doing graduation?

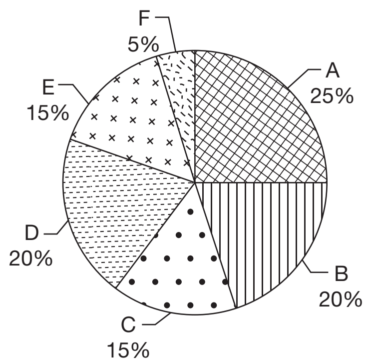
(A) 12,400 (B) 18,600
(C) 20,800 (D) None of these

37. What is the ratio of the number of girls in the area of specialization for which the number of boys is the second highest and the number of boys in the area of specialization for which the number of girls is the highest?

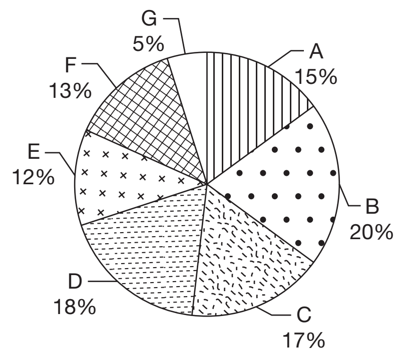
(A) 2 : 5 (B) 2 : 3
(C) 3 : 2 (D) Cannot be determined

Directions for questions 38 to 41: Answer the following questions based on the information given below.

The following figures represent the export performance of XYZ Ltd in the year 2015–16. The first pie chart represents the product-wise break-up of the exports to Germany, the main market of XYZ Ltd in Europe and the second pie chart represents the product-wise exports to Europe, which is the only region to which the company exports. 'Rest of Europe' refers to all regions in Europe excluding Germany.



Total = 40 million dollars



Total = 110 million dollars

38. For which of the given products is the percentage share of exports to Germany, out of its total exports to Europe, the highest?

(A) A (B) B
(C) D (D) E

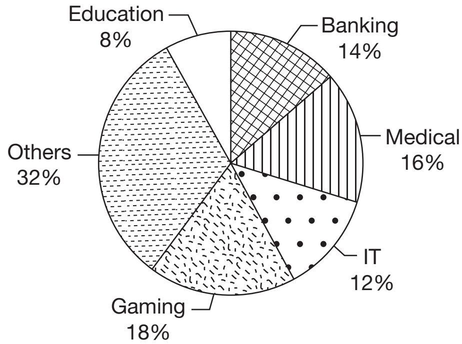
39. If for product A, the value of exports to France is accounted for 35% of that to 'rest of Europe', then what was its value (in million dollars)?
(A) 1.975 (B) 2.125
(C) 2.275 (D) None of these

40. For how many of the given products was the value of its exports to Germany more than two thirds of that to the 'rest of Europe'?
(A) 4 (B) 3
(C) 2 (D) 1

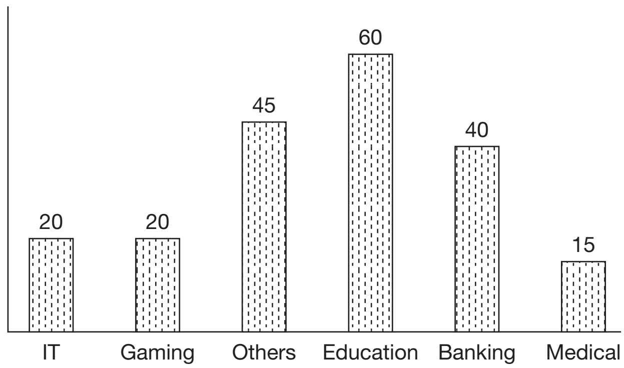
41. The exports of product B to the 'rest of Europe' is more than the exports of how many products to the whole of Europe?
(A) 0 (B) 1
(C) 2 (D) 3

Directions for questions 42 to 45: Answer these questions on the basis of the information given below.

The pie chart exhibits the break-up of all engineers in a city based on the industry they work for. The bar graph gives the percentage of females, among the engineers for each sector.



Total = 80,500



42. What is the average number of female engineers per sector?
(A) 2861 (B) 4273
(C) 4402 (D) 4454

43. The number of male engineers in the banking sector is what percentage more than the number of female engineers in the IT sector?
(A) 350% (B) 150%
(C) 250% (D) None of these

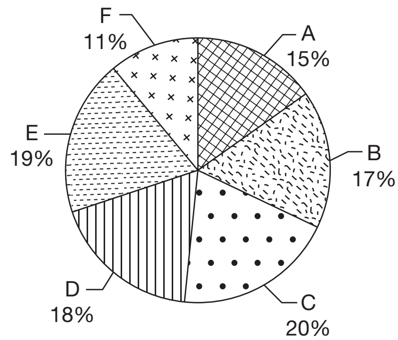
44. Had the number of female engineers in the gaming sector been 4628 and the number of male and female engineers in all other sectors remain the same, then what would have been the percentage of females among the total engineers?
(A) 35.5% (B) 34.6%
(C) 34.2% (D) 33.5%

45. What is the maximum difference between the number of male and female engineers in any single industry?
(A) 8126 (B) 8694
(C) 9016 (D) None of these

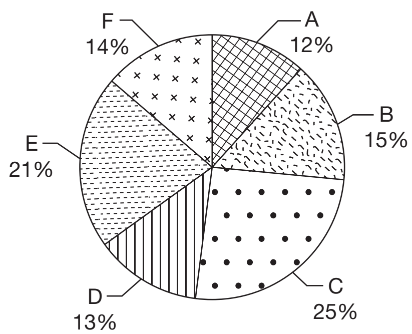
Directions for questions 46 to 50: Answer the questions based on the information given below.

The following pie chart gives the break-up of the costs and the sales value of all the six products, namely A, B, C, D, E and F manufactured and sold by company XYZ.

Cost



Sales value



46. If the company made an overall profit of 10%, then for which product was the profit percentage, the highest?
(A) B (B) C
(C) D (D) F

47. If the company made an overall loss, then for at most how many products was the sales value more than the costs?

(A) 0 (B) 1
(C) 2 (D) 3

48. If the company made an overall profit, then on at most how many products did it incur a loss?

(A) 4 (B) 3
(C) 2 (D) 1

49. On at least how many products should the company make a profit so as to make an overall profit?

(A) 1 (B) 2
(C) 3 (D) 4

50. If the company made a profit on each of the products, then the overall profit percentage is at least

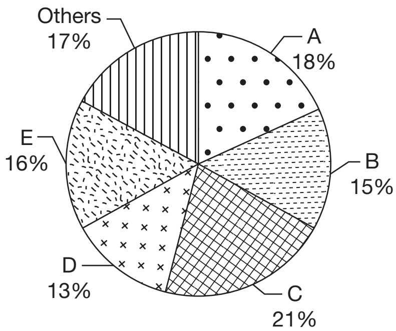
(A) 32% (B) 38.5%
(C) 43% (D) 48%

EXERCISE-2

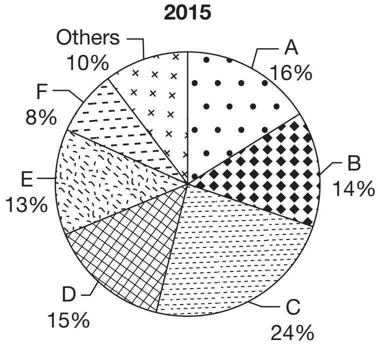
Directions for questions 1 to 4: These questions are based on the information given below.

The pie charts shows the market share of sales (by volume) of all major car manufactures in a country in two consecutive years. All manufacturers with less than 5% market share are classified under 'others'.

2014



2015



The total sales (by volume) increased by 5% from 2014 to 2015.

1. What was the percentage increase in the sales of company D?

(A) 15.4% (B) 18.4%
(C) 21.2% (D) 24.01%

2. If the average selling price of a car sold by company B increased by 8% from 2014 to 2015, then what is the percentage increase in sales (by value) from 2014 to 2015?

(A) 5.85 (B) 5.32
(C) 5.01 (D) 4.86

3. Which company had the highest percentage increase in the sales (by volume) from 2014 to 2015 (ignore companies with less than 5% share in both the years)?

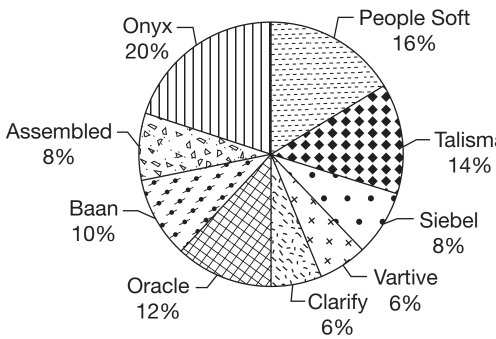
(A) C (B) D
(C) E (D) F

4. For how many companies is the growth in sales (by volume) greater than 20%? (Assume no company among 'others' had a volume growth more than 18%)

(A) 1 (B) 2
(C) 3 (D) 4

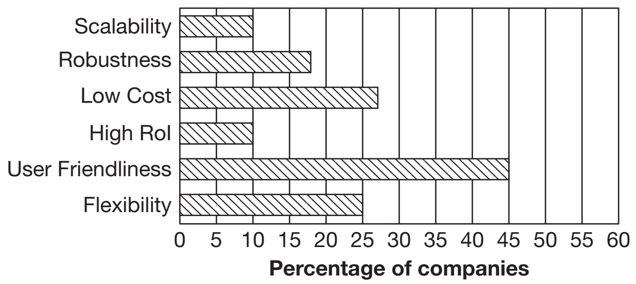
Directions for questions 5 to 9: The following charts exhibits the market survey data for various CRM software packages implemented by different companies. The total number of companies surveyed is 1000.

Percentage distribution of the number of companies (as a percentage of the total number of companies surveyed) implementing different CRM packages.



Note: Each company implements exactly one software package.

Percentage distribution of the number of companies (as a percentage of the total number of companies surveyed) quoting different reasons for implementing software CRM packages.



Note: Each company surveyed claimed at least one of the above reasons.

5. If $3\frac{1}{3}\%$ of the companies which implement Oracle, cited Robustness as a reason, then a maximum of how many companies which implemented either Clarify or Baan, could have cited Robustness as a reason?

(A) 140 (B) 40
(C) 160 (D) 100

6. Among the companies surveyed, if all the companies which implemented Talisma/People Soft/Onyx quoted either User Friendliness or Flexibility as a reason, then at least how many of these companies quoted both the reasons?

(A) 500 (B) 450
(C) 250 (D) 200

7. If all the companies which quoted either Low Cost or Scalability as a reason implement either Onyx or Oracle but not any other package, then at least how many of these companies implemented Oracle?

(A) 170 (B) 200
(C) 70 (D) 50

8. If all the companies that implemented either Siebel or Vartive quote the same reason, then which of the following cannot be that reason?

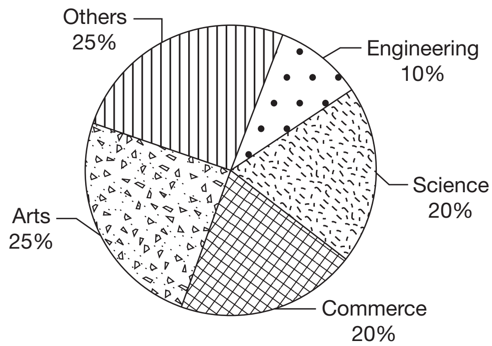
(A) Scalability (B) High RoI
(C) Low Cost (D) Both (A) and (B)

9. At the most how many companies which had implemented 'Assembled' could have quoted all the six reasons for implementation?

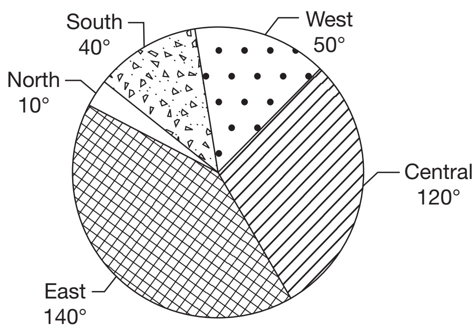
(A) 0 (B) 70
(C) 80 (D) 100

Directions for questions 10 to 14: These questions are based on the pie charts given below, which give the statistics regarding the total number of students (129600) graduating nationwide and their respective career details.

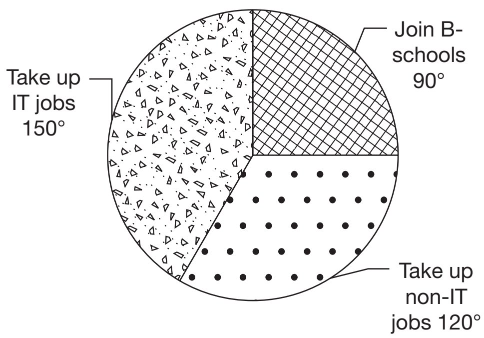
Discipline-wise distribution



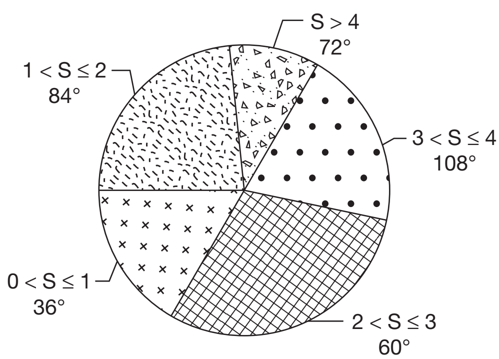
Region-wise distribution



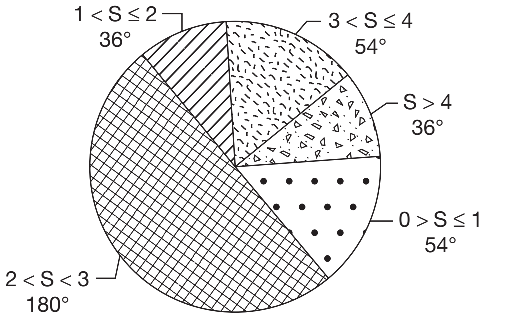
Career-wise distribution



Region-wise distribution of B-Schools Salary-wise distribution of students taking the students join up IT-jobs



Salary-wise distribution of students taking up non-IT jobs



10. The total number of students taking up either type of jobs (IT or non-IT) with a salary of more than ₹4 lakh per annum forms what percentage of the total number of students who joined a B-school in the Central region?
(A) 58% (B) 68%
(C) 140% (D) 168%

11. If a consolidated pie chart is drawn representing the distribution of the number of students taking up either type of jobs (IT or non-IT), then what is the approximate angle formed by the sector representing those with a salary of more than ₹1 lakh/annum but less than or equal to ₹2 lakh/annum?
(A) 100° (B) 120°
(C) 62.5° (D) 270°

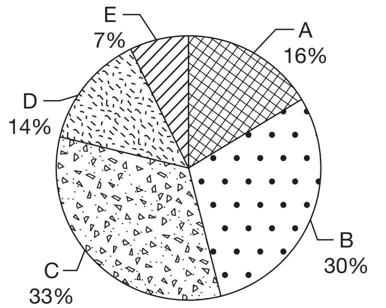
12. How many students from Commerce discipline joined a B-school located in the Western region?
(A) 360
(B) 3600
(C) 36000
(D) Cannot be determined

13. Find the number of students who took up a non-IT job with a salary of more than ₹2 lakh per annum.
(A) 32400 (B) 38880
(C) 21600 (D) 4320

14. If all students graduating in Commerce discipline join a B-school, then at the least how many students graduating in a discipline other than engineering joined B-schools located in the Central region?
(A) 10,800 (B) 7,200
(C) 4,320 (D) 1,240

Directions for questions 15 to 18: These questions are based on the pie chart and table given below.

There are five companies A, B, C, D and E which produce steel in country X. The following pie chart shows the percentage share of production costs of each company in the total production costs of all the five companies in the year 2007.



Total production cost = ₹ 90,000 million

The following table shows the ratio of production costs of each company in producing the three different varieties of steel and the corresponding profit % earned, for the year 2007. The five companies produce only those three varieties of steel.

Company	Ratio of production costs Variety I : Variety II : Variety III	Corresponding profit % Variety I	Variety II	Variety III
A	4 : 5 : 7	45%	45%	35%
B	3 : 5 : 2	33%	40%	48%
C	6 : 2 : 3	34%	38%	42%
D	7 : 5 : 2	35%	32%	46%
E	6 : 3 : 5	44%	48%	25%

Note: Profit is calculated on the production cost.

15. Which company has the maximum profit from the sales of variety II?
(A) B (B) D
(C) A (D) C

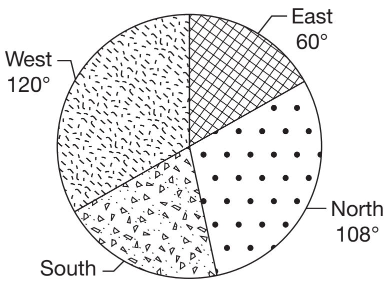
16. What is the overall profit of E (in ₹ millions)?
(A) 1162.50 (B) 2398.50
(C) 3164.50 (D) 4165.50

17. Production costs of company C are approximately how much percentage more than those of company D?
(A) 136% (B) 57.6%
(C) 52.3% (D) 48.6%

18. Which company has incurred the maximum production cost for variety III?
(A) B (B) D
(C) C (D) A

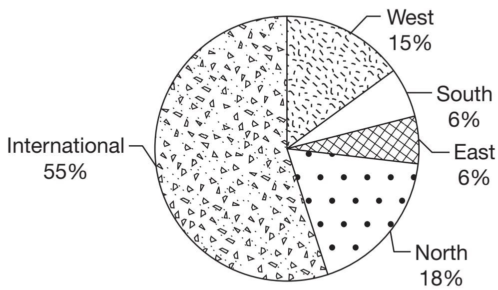
Directions for questions 19 to 23: Answer these questions based on the information given below.

Revenues earned by three Airways for the year ending 31.03.2002 Bharat Airways



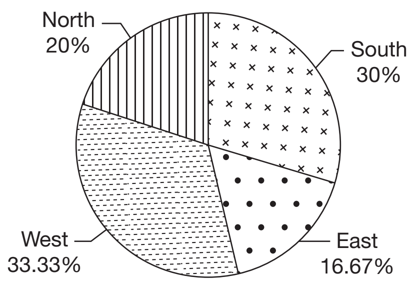
Total revenue = ₹ 6,000 crore

FUT Airways



Total revenue = ₹ 7,200 crore

Lahiri Airiways



Total revenue = ₹ 3,600 crore

Note: East, West, North and South are the four inland zones of the country.

19. Assuming that there are no other airways operating in the inland market, what is the inland market share of FUT Airways?

(A) 48.6% (B) 28.5%
(C) 25.2% (D) 21.8%

20. Considering the revenues generated by the three airways in each of the four inland zones, which of the following is the second highest?

(A) South zone for Bharat Airways
(B) North zone for FUT Airways
(C) West zone for Lahiri Airways
(D) North zone for Bharat Airways

21. If in the next year, the markets in the East, West, North and South zones increased by 10%, 20%, 10% and 30%, respectively and Bharat Airways captures half of the increase in the market in each of the respective zones, then by approximately what percentage does the total revenue of Bharat Airways increased when compared to that of the previous year?

(A) 15%
(B) 18%
(C) 21%
(D) Cannot be determined

22. Which of the following is not true?

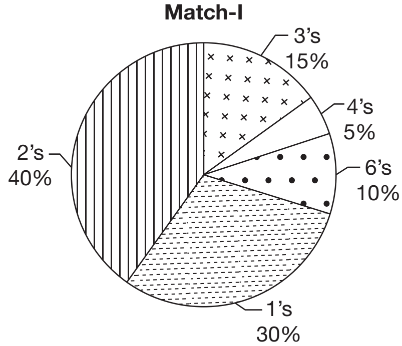
(A) The revenue generated from the North zone for Bharat Airways is equal to the revenue generated for Lahiri Airways from both North and South zones put together.
(B) For FUT Airways, the revenue generated from the North zone is 50% more than the revenue generated from the East and South zones together.
(C) The revenue generated from the South zone for Bharat Airways is 33.3% more than the revenue generated from the North zone for Bharat Airways.
(D) The revenue generated from the International zone for FUT Airways is more than the total revenue generated by Lahiri Airways.

23. If in the North zone, these three airways form 20% of the total market, then the total market (in ₹ crore) in the North zone is

(A) 19,080
(B) 76,920
(C) 1,908
(D) 7,69

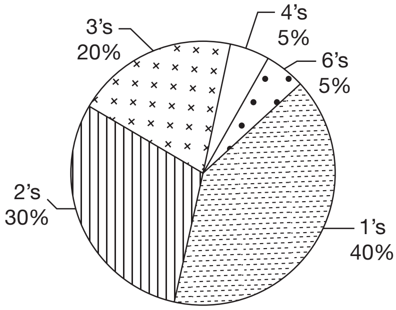
Directions for questions 24 to 29: These questions are based on the pie charts given below.

Match-I

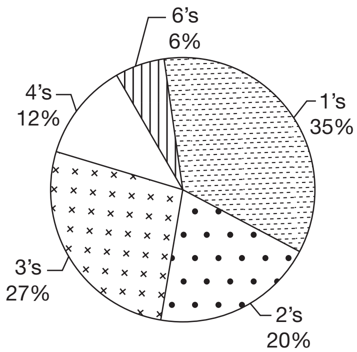


Difficulty Level-2: Moderate

Match-II



Match-III



A certain cricket team played three matches. In each match, the team faced a certain total number of balls, the runs scored off each ball being only in 1's, 2's, 3's, 4's or 6's. However, it is possible that the team scored no runs off some of the balls that it faced.

The pie charts given above shows the percentage distribution of the total number of runs scored by the team in the three matches. For example, 20% of the total runs scored by the team in Match-III were scored in 2s.

Further, the total number of runs scored by the team in each of the three matches are not known exactly, except for the ranges in which each of the totals lie. The following table gives this information.

Match	Total Number of Runs(T)
I	O < T < 300
II	100 < T < 300
III	100 < T < 300

24. Find the number of balls off each of which the team scored 4 runs in Match-I.

(A) 12 (B) 4

(C) 3 (D) 2

25. If every six balls that the team faces comprise an 'over', what is the total number of overs faced by the team in Match-II?

(A) $25\frac{1}{2}$ (B) $37\frac{2}{3}$

(C) $45\frac{1}{3}$ (D) Cannot be determined

26. What is the total number of runs scored by the team in Match-II?

(A) 248 (B) 256

(C) 280 (D) None of these

27. What is minimum number of runs scored by the team through 1s in any of the three matches?

(A) 72 (B) 70

(C) 144 (D) 140

28. Find the total number of runs scored by the team through 6s in Match-III.

(A) 12 (B) 18

(C) 24 (D) 6

29. If, in each of the three matches the team faced a total of exactly 300 balls, then the number of balls off which no runs were scored by the team was the least for

(A) Match-I (B) Match-II

(C) Match-III (D) Cannot be determined

Directions for questions 30 to 33: These questions are based on the pie charts and table given below. The two pie charts I and II give the distribution of three lakh graduates course-wise and state-wise.

Chart-I

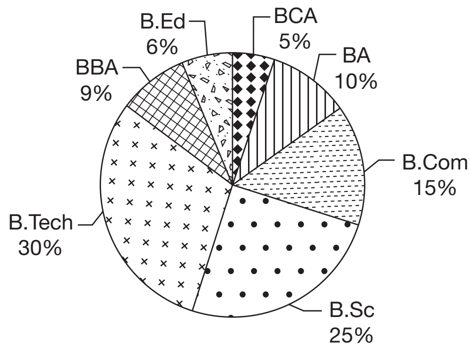
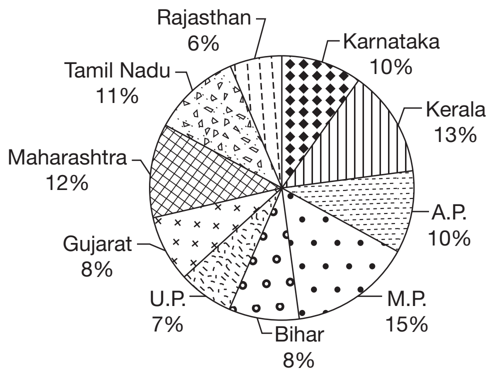


Chart-II



The table below gives the ratio of the number of males and females among the three lakh graduates in the different states.

State	Male : Females
A.P.	5 : 7
M.P.	8 : 7
Bihar	5 : 3
U.P.	11 : 10
Gujarat	2 : 3
Maharashtra	11 : 7
Tamil Nadu	13 : 9
Rajasthan	2 : 1
Karnataka	3 : 2
Kerala	6 : 7

30. In which state was the number of male graduates the highest?
(A) M.P. (B) Maharashtra
(C) Tamil Nadu (D) Bihar

31. Find the total number of female graduates in the four states with the highest number of total graduates.
(A) 6950 (B) 69500
(C) 22500 (D) 74500

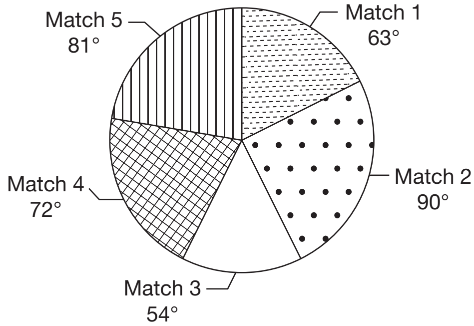
32. Which of the following statements is true?
(A) The difference in the number of B.Com. and B.Sc. graduates is at least 30,000.
(B) The number of male graduates in U.P. is more than the number of female graduates in M.P.
(C) The number of BBA graduates is more than the number of male graduates from Karnataka and Kerala put together.
(D) More than one of the above.

33. In how many states, is the number of male graduates not more than 5000?
(A) 10 (B) 9
(C) 0 (D) 1

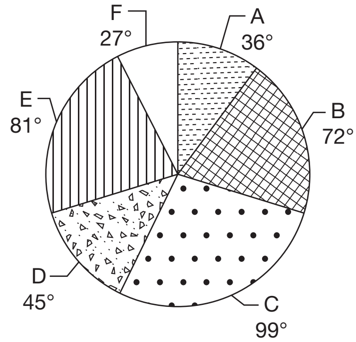
Directions for questions 34 to 37: Answer these questions based on the information given below.

The performance of 6 batsmen, namely A, B, C, D, E and F playing for a team in a recently conducted cricket series is given in the following pie charts. The series consisted of five one-day matches and all the six batsmen played in each one of them. The first pie chart gives the percentage distribution of total runs scored by the team in those five matches whereas the second pie chart gives the percentage distribution of the total runs scored by each batsman in the entire series. It is also known that the sum of the individual scores of the six batsmen in the entire series constitutes 90% of the total runs scored by the team.

Distribution of runs scored in the five matches



Distribution of individual scores in the five matches together



34. If C scored at least 20% of the total runs scored by the team in each of the 5 matches, then the runs scored by C in a match as a percentage of the total runs scored in that match is at most
(A) 43.75% (B) 36.66%
(C) 41.11% (D) 51.66%

35. If the difference between the total runs scored in Match 2 and the total runs scored by C in the entire series is 4, then what is the total runs scored by A and D together, in the series?
(A) 162 runs (B) 243 runs
(C) 324 runs (D) 360 runs

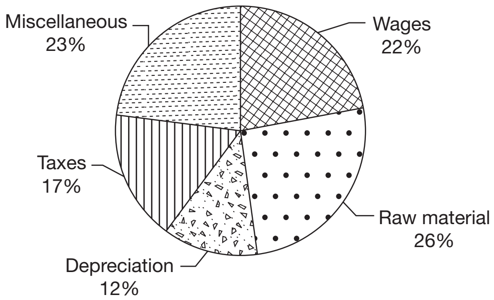
36. If each of the 6 batsmen contributed at least 16% of the total runs scored in Match 2, then the total runs scored by one of the 6 batsmen in Match 2 as a percentage of the total runs scored by that batsmen in the entire series is at most
(A) 44.44% (B) 74.07%
(C) 51.26% (D) 83.33%

37. If the minimum runs scored by the team in any single match in the given series was 180, then what is the total runs scored by B in the entire series?
(A) 216 runs (B) 144 runs
(C) 288 runs (D) 360 runs

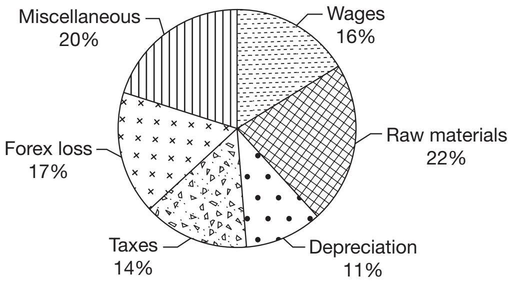
Directions for questions 38 to 41: Answer these questions based on the information given below.

The Agenta group was a diversified business house which had three companies X, Y and Z. As part of restructuring, the group decided to merge all the three companies. The following pie charts give the breakup of expenses of each company and that of the merged entity in the year 2007.

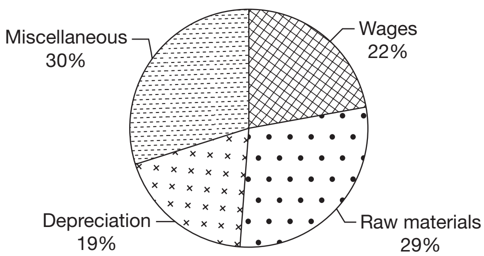
Company X



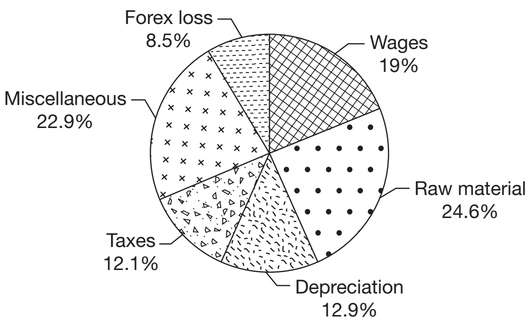
Company Y



Company Z



Merged entity



38. What is the ratio of the total expenses of companies X and Y?

(A) 3 : 2 (B) 2 : 5
(C) 3 : 5 (D) 2 : 3

39. The forex loses of company Y was what percentage of the depreciation expenses of company Z?

(A) 152% (B) 186%
(C) 205% (D) 223%

40. The wage bill of which of the three companies was the highest?

(A) X (B) Y
(C) Z (D) Both X and Y

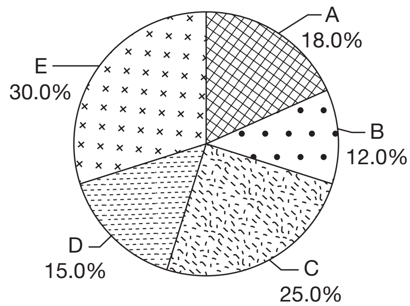
41. The expenses of company Z was under quoted by 20% because the taxes paid by it was not included in the given diagrams. If this figure is also included, then the taxes paid would account for what percentage of the expenses of the merged entity?

(A) 23.2% (B) 21.5%
(C) 18.6% (D) None of these

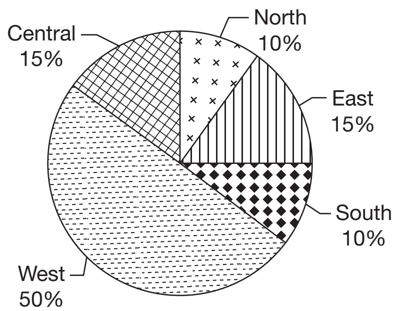
Directions for questions 42 to 46: Answer these questions based on the information given below.

A company named XYZ Ltd. which sells five different products, such as A, B, C, D and E gave misleading information to the press regarding some of its sales. The company exaggerated the sales of its product D in the Western region, while stating sales of all other products in all other regions accurately.

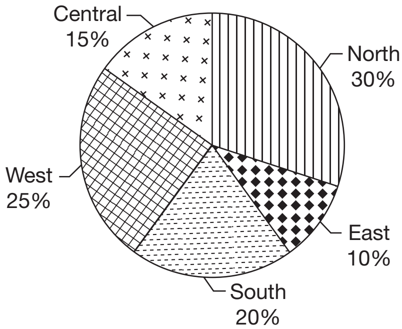
Product-wise distribution of total sales of XYZ Ltd.



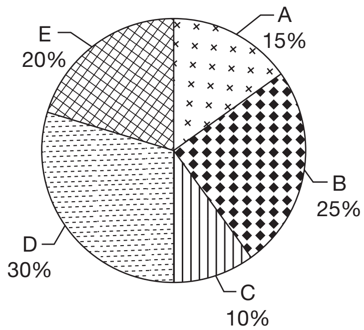
Region-wise distribution of the sales of product D by XYZ Ltd.



Region-wise distribution of the total sales of XYZ Ltd.



Product-wise distribution of the sales of XYZ Ltd., in the Western region



The above pie charts are drawn by Sunil, a business journalist, who is unaware of the above-mentioned discrepancy. Another set of four similar pie charts, (not shown here), representing the distribution of the same quantities mentioned above, are drawn by Anand, an employee of the company XYZ Ltd., who is aware of all the figures accurately and has drawn them accordingly.

42. If it is known that XYZ Ltd., exaggerated the value of the sales of product D in the Western region by 25%, then considering the pie charts drawn by Anand, what is the ratio of the total sales of product C and that of product E?

(A) 5 : 6 (B) 57 : 50
(C) 2 : 3 (D) 3 : 5

43. According to Sunil, what percentage of the total sales in the Northern region are the sales of product D in that region?

(A) 5% (B) 8⅓%
(C) 12.5% (D) 15%

44. According to the pie charts drawn by Anand, if the sales of product E in the Western region represent 25% of the total sales in the Western region, then what is the percentage share of the sales in the Northern region, in the total sales of the company?

(A) 28.8% (B) 30.2%
(C) 34.8% (D) None of these

45. If the actual total sales of the company is ₹14,000 crore and the actual sales of product B is ₹1800 crore, then by how many times are the sales of D in the Western region stated more than the actual?

(A) 7 times (B) 6 times
(C) 8 times (D) 9 times

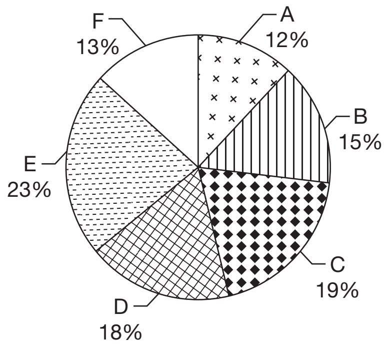
46. If the actual sales of product A in the Western region are known, which of the following additional data (considered independently along with the above information) is necessary and sufficient to find the actual total sales of product D?

I. The actual sales of product D in the Southern region.
II. The actual total sales in the Northern region.
III. The sum of the actual sales of product E and product D in the Western region.
IV. Difference between the actual total sales in the Northern and the Western regions.

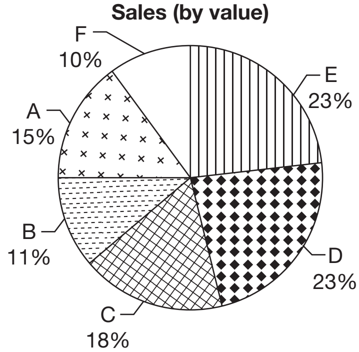
(A) Only I
(B) Only I or Only III
(C) Only I or Only IV
(D) Only III or Only IV

Directions for questions 47 to 50: Answer the questions based on the information given below.

Sales (by volume)



Sales (by value)



47. For which product is the selling price per unit the highest?

(A) A (B) B
(C) C (D) D

48. For how many of the given products is the selling price per unit more than the average selling price of all the six units?

(A) 1 (B) 2
(C) 3 (D) 4

Directions for questions 49 and 50: The break-up of costs is exactly the same as the breakup of sales (by volume).

49. If the company made an overall profit, then at most how many products did it incur a loss?

(A) 3 (B) 4
(C) 2 (D) 5

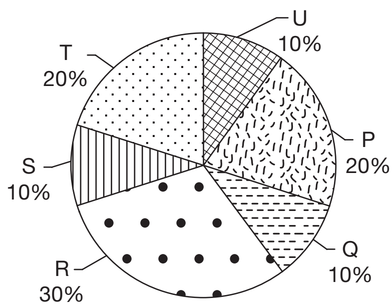
50. If the company did not make a loss on any of the six products, then the overall profit percentage is at least ________.

(A) 25% (B) 33.33%
(C) 40% (D) None of these

EXERCISE-3

Directions for questions 1 to 4: Answer these questions on the basis of the information given below.

Break-up production cost of six products-P through U



Total cost = ₹ 25 crore

Each of the six products is produced in two qualities, grade A and grade B. The ratio of the units produced for each product and the profit percentages is given in the table below.

	Ratio of production		Profit percentage	
Product	Grade A	Grade B	Grade A	Grade B
P	6	5	20	15
Q	1	2	30	15
R	3	4	20	25
S	4	5	15	20
T	2	3	20	25
U	1	1	10	15

Assume that for each item, the cost of production of grade A and grade B items are in the ratio 5 : 4.

1. What is the total profit made on product R (in ₹ lakh)?

(A) 135 (B) 152
(C) 170 (D) 190

2. For which product is the ratio of total profit to total cost, the highest?

(A) R (B) T
(C) P (D) S

3. What is the total approximate cost (in ₹ cr) incurred in producing items P, R and S of grade A?

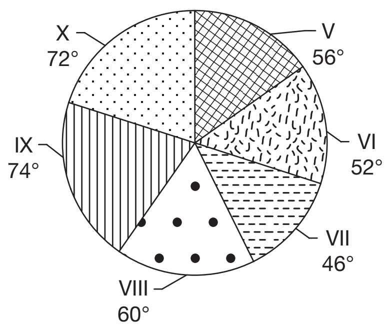
(A) 7.9 (B) 8.6
(C) 9.2 (D) 7.1

4. For how many of the given companies is the profit obtained on items of grade A more than that obtained on items of grade B?

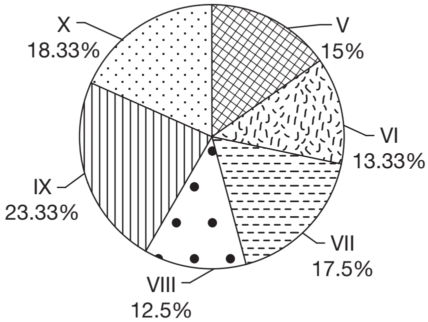
(A) 1 (B) 2
(C) 3 (D) 4

Directions for questions 5 to 8: These questions are based on the following data.

The first pie chart exhibits the breakup of students in various classes of a school and the second pie chart gives the breakup of boys who were selected for a scholarship. The table gives the breakup of students in the school, among those selected for the scholarship, according to the class they study.



Total = 1260



Total = 120

Class	Percentage of Students Selected
V	14.6%
VI	13.2%
VII	15.6%
VIII	19.5%
XI	18.5%
X	18.5%
Total = 205	

5. What is the number of girls in Class VII who were selected for the scholarship?
(A) 14 (B) 12
(C) 11 (D) 10

6. What percentage of students in Class VIII was selected for the scholarship?
(A) 22 (B) 19
(C) 17 (D) 15

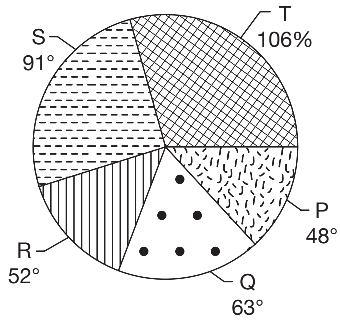
7. In which class was the percentage of students selected for the scholarship, the highest?
(A) Class VII
(B) Class VIII
(C) Class V
(D) None of these

8. In how many classes was the number of boys selected for the scholarship at least 50% more than the number of girls selected from the class?
(A) 0 (B) 1
(C) 2 (D) 3

Directions for questions 9 to 12: Answer these questions on the basis of the information given below.

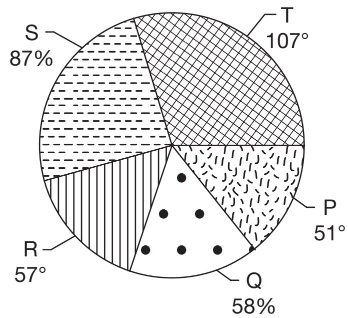
The following pie charts give the break-up of the number of products (in '000) sold by a company across two years 2015 and 2016. The first table show the price/unit of each product in 2015 and the second table show the percentage increase in the price/unit of each product from 2015 to 2016.

2015



Total = 79200 units

2016



Total = 86400 units

Price/unit in 2015	
Product	Price
P	132
Q	167
R	102
S	86
T	235

Percentage increase in 2016	
Product	Percentage increase
P	8
Q	14
R	2
S	0
T	7

9. What was the sales (by value) of P in 2016?

10. What was the sales (by value) of the five products together in 2015 (in ₹cr)?

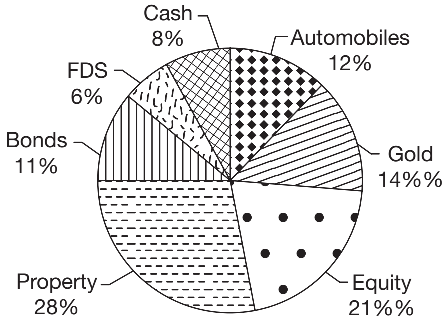
11. How many of the given five products had a more than 10% increase in the sales (by value) from 2015 to 2016?

12. What is the approximate increase in the sales (by value) of the five products together from 2015 to 2016?

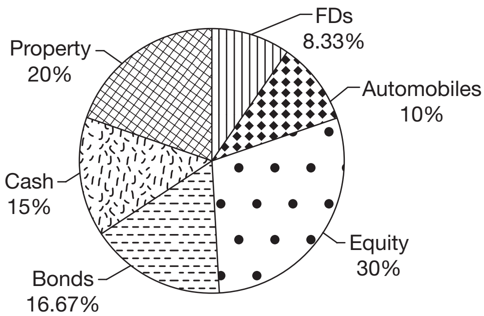
Directions for questions 13 to 16: Answer these questions on the basis of the information given below.

Ratan Lal, a businessman, decided to divide his assets among his three sons. The following pie charts give the breakup of the assets of Ratan Lal and the breakup of the assets received by the three sons. It is known that Ratan Lal divided his entire assets among the three sons.

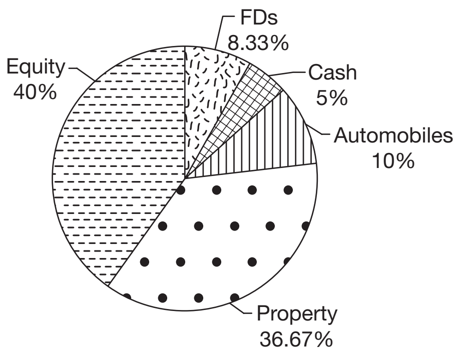
Assets of Ratan Lal



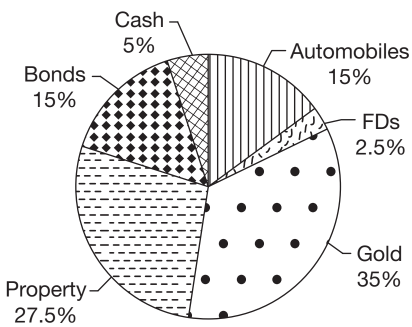
Share of first son



Share of second son



Share of third son



13. What percentage of his assets did Ratan Lal give his second son?

14. If the value of equity given to the first son was ₹5.25 crore, what was the value of assets (in ₹cr) given to the second son?

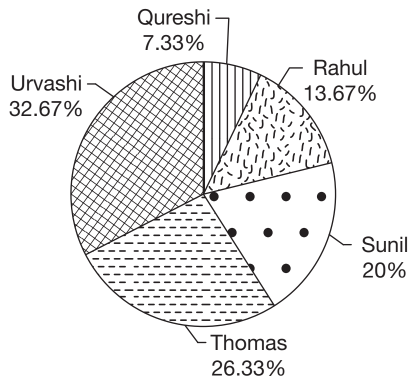
15. If the total assets of Ratan Lal were valued at ₹48 crore, what was the value of FDs (in ₹cr) received by the second son?

16. If the cash received by the first and the third sons together was ₹3.9 crore, what was the value of equity (in ₹cr) with Ratan Lal?

Directions for questions 17 to 20: Answer these questions on the basis of the information given below.

The ages of six children, namely Prakash, Qureshi, Rahul, Sunil, Thomas and Urvashi was 7, 8, 9, 10, 11 and 12, respectively. One day they went shopping, each carrying their pocket money with them. Before shopping it was found that the money with each of them was in the same ratio as their ages with Prakash having the lowest amount, Qureshi the next highest and so on till Urvashi who had the highest. Each of then spent a different amount while shopping. When they returned it was found that the amount left with Qureshi, Rahul, Sunil, Thomas and Urvashi were in the ratio of 5 : 4 : 3 : 2 : 1.

The following is the breakup of the total amount spent by these five children:



After Prakash came back, it was found that he had double the amount left with Sunil as he had spent only 15 rupees while shopping which was only 5% of what Sunil had spent.

17. What was the total amount spent by the children?

18. What was the maximum difference between the amount with any two children before they went shopping?

19. What percentage of the money with him did Sunil spent on shopping (approximated to the closest integer)?

20. How much more did Thomas spend on shopping than Sunil?

ANSWER KEYS

Exercise-1

1. (C)	10. 7,20,000	19. (D)	28. (A)	37. (C)	46. (D)
2. (A)	11. (B)	20. (B)	29. (C)	38. (A)	47. (D)
3. (D)	12. (D)	21. (D)	30. (A)	39. (C)	48. (B)
4. (B)	13. (A)	22. (C)	31. (D)	40. (B)	49. (C)
5. (A)	14. (D)	23. (A)	32. (B)	41. (C)	50. (B)
6. 2,70,000	15. (B)	24. (C)	33. (C)	42. (D)	
7. 1	16. (B)	25. (B)	34. (A)	43. (C)	
8. 8 : 9	17. (B)	26. (B)	35. (B)	44. (B)	
9. 40	18. (B)	27. (D)	36. (C)	45. (C)	

Exercise-2

1. (C)	10. (C)	19. (C)	28. (A)	37. (A)	46. (D)
2. (A)	11. (C)	20. (D)	29. (B)	38. (C)	47. (D)
3. (D)	12. (D)	21. (D)	30. (A)	39. (D)	48. (B)
4. (C)	13. (A)	22. (C)	31. (B)	40. (B)	49. (A)
5. (A)	14. (C)	23. (A)	32. (A)	41. (D)	50. (D)
6. (D)	15. (A)	24. (C)	33. (C)	42. (A)	
7. (C)	16. (B)	25. (D)	34. (D)	43. (A)	
8. (D)	17. (A)	26. (D)	35. (C)	44. (D)	
9. (B)	18. (C)	27. (B)	36. (B)	45. (C)	

Exercise-3

1. (C)	5. (C)	9. 175	13. 30	17. 1515
2. (B)	6. (B)	10. 1208	14. 17.5 crore	18. 225
3. (A)	7. (A)	11. 4	15. 1.2 crore	19. 67
4. (B)	8. (D)	12. 200	16. 12.6 crore	20. 95

SOLUTIONS

EXERCISE-1

1. Imports of pharmaceutical products = 40% of 64.8° = 25.92°.
Given 360° — 36 billion dollars
25.92° — ?

$? = \dfrac{25.92 \times 36}{360} = 2.592$ billion dollars

$\therefore$ Total internal production = 8 × 2.592
= 20.736 billion dollars.

2. Angle made by bulk goods, paper and allied products is 147.6°.
360° —— 36 billion dollars
147.6° —— ?
Clearly 1° — 0.1 billion dollars.
$\therefore$ 147.6° will be = 147.6 × 0.1 = 14.76 billon dollars.

3. Total value of exports of textile industry is 20% of 24 = 4.8 billion dollars.
Total value of imports of textile industry = ½ × 64.8 = 32.4° = 32.4 × 0.1 = 3.24 billion dollars.
$\therefore$ Net exports = 4.8 – 3.24 = 1.56 billion dollars.

4. India's fuel bill = ¼ × 36 = $9 billion.
New fuel bill = $(9 – 2) = $7 billion.

Capital good's bill = $\dfrac{39.6 \times 36}{360}$ = $3.96 billion.

$\therefore$ Ratio of imports of new fuel bill to the capital goods bill = 7 : 3.96 ≅ 7 : 4.

5. Fuel bill was 90° in 1999-2000 and has increased by 40%. Hence, the new angle is 90° + 36° = 126°. Since all the other commodities' bills have not changed, the total angle for these remains = 270°, i.e., the total angle is 396°. Angle for capital goods is 39.6°. Hence, for a new pie chart, the angle of capital goods is $\dfrac{39.6 \times 36}{360} = 36°$.

6. Given angle representing 30° is showing 60,000. Difference between the angles showing viewership of Set Max and Ten Sports is 150° – 15° = 135°

$\therefore$ The number equidistant to 135° = $\dfrac{135 \times 60{,}000}{30}$
= 2,70,000

7. Angle showing 20% viewership $= \frac{20}{100} \times 360 = 72°$

Only Set Max has shown angle more than 72°.

8. Viewership of DD Sports during second half of February

$= 60 \times \frac{2}{3} = 40°$

$\therefore$ Required ratio = 40° : 45° = 8 : 9

9. Given, 80% × (Actual value of DD sports) = 60°

$\therefore$ Actual value of DD sports = 75°

The total share is 360 + (75 – 60) = 375

% Share of Set Max $= \frac{150}{375} \times 100 = 40\%$

10. Given 45° = 90,000

360° = ?

$? = \frac{360 \times 90,000}{45} = 7,20,000$

11. Total value of kerosene consumed

$= 60,000 \times \frac{12}{100} = 7200$ crore

Total kerosene consumed by India (Quantity) = 150 lakh kilolitres.

$\therefore$ Value of 1 litre of kerosene $= \frac{7,200 \times 100 \text{ (in lakh)}}{150 \times 1000}$

Value of 1 litre of kerosene = ₹4.8

12. The graph here shows the consumption of electricity by India but does not state anything about the production of electricity. Hence, the amount of electricity generated in India cannot be estimated.

13. Price of 1 barrel of crude oil imported by India

$= \frac{60,000 \times 0.3}{25}$ (250 million = 2500 lacs = 25 crore)

$\therefore$ Price of 1 barrel of crude oil imported by India = ₹720.

Price of 1 barrel of oil that Oman sells to India = ₹720.

Price of crude oil in Oman (domestic) is 25% more than ₹720 = 900.

14. Current domestic production of crude oil

$= 250 \times \frac{25}{100} 62.5$ million barrels

16% of TRR = 62.5 million barrels

$\therefore$ TRR $= 62.5 \times \frac{100}{16} 390.625$ million barrels

We know that 100% more implies twice.

Similarly, 200% more implies thrice.

In the scene every 4000% more implies 41 times.

$\therefore$ TER are 41 times the TRR.

$\therefore$ TER = 41 × 390.655 ≃ 16000 million barrels.

15. Total value of crude oil consumed in India = 60,000 × 0.3 = 18,000 crore.

Domestic production = 250 × 0.25 = 62.5 million barrels

Crude oil from sources other than domestic = 187.5 million barrels

Assuming 1 barrel from other sources costs ₹ X, then the price of 1 barrel of domestic crude costs $= \frac{3X}{4}$

$\therefore 187.5\ X + 62.5 \times \frac{3X}{4} = 18000$

$234.375\ X = 18000$

$X = \frac{18,000 \times 100 \text{ (in lakh)}}{234.375 \times 10 \text{ (in lakh)}}$ $X = 768$

Solutions for questions 16 to 19: The food grains production in the two years are as follows:

State	1999–2000 (Million tons)	2000–2001 (Million tons)
Punjab	90	72
West Bengal	54	30
Maharashtra	36	24
A.P.	36	48
Tamil Nadu	14.4	30
Kerala	28.8	15
U.P.	21.6	12
Rest of India	61.2	45
Karnataka	18	24

16. The percentage increase in the food grain production for A.P. is

$$\frac{16\% \text{ of } 300 - 12\% \text{ of } 360}{12\% \text{ of } 360} \times 100$$

$$= \left(\frac{16\% \text{ of } 300}{12\% \text{ of } 3600} - 1\right) \times 100$$

Now, 300/360 will be common for all the states; we need not calculate the entire percentage. It would be sufficient if we observe the percentage increase in the percentages of the respective states.

Therefore, for A.P. 12% to 16% which is <100%,

for T.N. 4% to 10% which is >100%,

for Kerala 8% to 5% which is a decrease and

for Karnataka 5% to 8% which is <100%.

17. Share of Maharashtra in the total production of food grains in 1999–2000 and 2000–01 is 10% of 360, i.e., 36 and 8% of 300, i.e., 24 respectively.

$\therefore$ Required percentage $= \frac{32 - 20}{30} \times 100$

$= 37.5\%$

18. Rice production in West Bengal in
1999–00 = ½ × 15% of 360 = 27 million tons.
2000–01 = ½ × 10% of 300 = 15 million tons.
∴ Revenue from the sale of rice in 1999–2000 and that in 2000–01 is $27 \times 11X = 297X$ and
$15 \times 12X = 180X$.
∴ There is a decrease of 120 from 297.

Percentage loss = $\frac{120}{297} \times 100 \cong 40\%$

19. We can find the production of the food grains of A.P. and Karnataka for the year 2001–02 but the total food grain production in 2001–02 cannot be found.

Solutions for questions 20 to 23: The following table gives the supply and demand of blood for different blood groups.

Blood group	Supply (litres)	Demand (litres)
A	1000	2400
B	800	2400
AB	400	9600
O	1800	1600

20. Supply of various blood groups (in '00 litres) is as follows:

A	18.75%	6.25%
B	16%	4%
O	22.5%	22.5%
AB	6.66%	3.34%

∴ Blood group AB negative is having the least supply.

21. As demand percentage of AB blood group remained the same, it can be concluded that as the demand doubles, the demand for the other blood groups increases.

22. Supply of blood group O^+ is

$\frac{3}{5} \times \frac{45}{100} \times 4{,}000 = 1080$ litres and the demand of blood group O^+ is $\frac{3}{8} \times \frac{10}{100} \times 16000 = 600$ litres.

∴ Excess supply of O^+ blood group is 1080 – 600 = 480 litres.

23. The total demand of O and A blood groups is 10% of 16000 litres, i.e., 1600 litres and 15% of 16000 litres, i.e., 2400 litres.
As a person having blood group O can donate 2 litres, the number of persons required is $\frac{1{,}600}{2} = 800$ and that of A blood group is $\frac{2{,}400}{1} = 2400$.

∴ Total persons required are 800 + 2400 = 3200.

24. Difference in the number of warheads possessed by Russia and USA is 90 – 72 = 18° → A
Difference in the number of warheads possessed by India and Israel = 18° – 9° = 9° → B
A is 2 times B.

25. Number of nuclear warheads possessed by China

$= \frac{80{,}000 \times 36}{360} = 8000$

Cost incurred by China in building and maintaining these warheads = $\frac{2500 \times 14}{100}$ = \$350 million

∴ Average cost incurred on one warhead

$= \frac{350}{8000} = .04375$ Mn

.04375 × 1000,000 = 43,750 \$

26. Number of warheads possessed by Russia

$= \frac{80000 \times 90}{360} = 20000$

Expenditure incurred by Russia = $\frac{2500 \times 30}{100}$ = 750 million

Average cost per warhead for Russia = $\frac{750}{20000} \times 100000$

= 3750 \$

Number of warheads possessed by USA

$= \frac{80000 \times 72}{360} = 16000$

Expenditure incurred by USA = $\frac{2500 \times 25}{100}$ = 625 million

Average cost per warhead for USA = $\frac{625 \times 100000}{16000}$

= 3906.25\$

∴ Difference = 3906.25 – 3750 = \$1562.5

27. Warhead of 'Others' = 1000 kg
Warhead of other countries = 2000 kg

Number of warheads of 'Others' = $\frac{80000 \times 45}{360} = 10000$

∴ Total weight = 10000 × 1000 = 10000 tons.
Number of warheads possessed by other countries = 70000.
∴ Total weight = 70000 × 2000 = 1,40,000 tons.
Weight of all warheads put together = 1,50,000 tons.

28. Let the total number of warheads with all countries be 360.
The reduction by Russia = 40% of 90 = 36
That by USA = 30% of 72 = 21.6
That by UK and China put together = 20% of (36 + 36)
= 14.4
That by the other countries = 10% of (9 + 18 + 54 + 45)
= 12.6
Total reduction = 84.6

∴ The reduced number of warheads = 360 – 84.6 = 275.4

360 → 80,000

275.4 → ?

$= \frac{275.4}{360} \times 80,000 = 61,200$

29. Students from China and India = 20 + 24 = 44% → (A)
Students from all countries in the graph except 'others' category = 46% → (B)

(A) as a percentage of (B) = $\frac{44}{46} \times 100 = 95.65\%$

30. The country with the maximum representation in 2000 is India.

Number of students from India = $\frac{80,000 \times 24}{100} = 19200$

Students from India taking up Management

$= \frac{19200 \times 135}{360} = 7200$

31. Portugal + Spain = 20% = $\frac{80000 \times 20}{100} = 16000$

16000 is 25% of domestic students pursuing MS in USA.
∴ Domestic students pursuing MS in USA = 64000

32. Students of Asian Countries comprised
= 24 + 20 + 7 + 6 + 4 = 61%
Students of Asian Countries comprised

$= \frac{80000 \times 61}{100} = 48,800$

∴ 48800 as a percentage of 3,60,000 is

$\frac{48800}{360000} \times 100 = 13.5\%$

33. Indian students pursuing Engineering or Management

$= 24\% \text{ of } 80,000 \times \frac{270}{360}$ → (A)

Chinese students pursuing other courses

$= 20\% \text{ of } 80,000 \times \frac{90}{360}$ → (B)

(A) as percentage of (B) = 360%

Solutions for questions 34 to 37: As the rate of boys to girls is given as 2 : 5, assume that the number of boys is 200 and the number of girls is 500.

The values can be tabled as follows:

	Boys	Girls	Total
Arts	334	50	84
Science	34	71	105
Medicine	10	39	49
Engineering	48	78	126
Commerce	22	69	91
Others	52	193	245

34. Gender ratio of girls doing Medicine = $-\frac{39}{10} = 3.9$.

35. Only in Arts and Engineering is the number of boys at least half of the number of girls.

36. As the number of girls doing Medicine is $\frac{4056}{39} = 104$ times the value calculated in the table, the number of boys would be 104 × 200 = 20,800.

37. The number of boys is the second highest in Engineering. The number of girls in Engineering = 78
The number of girls is the highest in 'other'.
The number of boys in 'other' = 52
The required ratio = 78 : 52 = 3 : 2.

Solutions for questions 38 to 41: The value of exports of each product to Europe, Germany and the rest of Europe is given in the table below.

Europe		Germany		Rest of Europe	
Product	Export value	Product	Export value	Product	Export value
A	16.5	A	10.0	A	6.5
B	22	B	8.0	B	14.0
C	18.7	C	6.0	C	12.7
D	19.8	D	8.0	D	11.8
E	13.2	E	6.0	E	7.2
F	14.3	F	4.0	F	10.3
G	5.5			G	5.5

38. The percentage share is highest for the product A.

39. Exports of product A to rest of Europe (in million dollars) = $\frac{35}{100} \times 6.5 = 2.275$.

40. For products A, D and E, the value of exports to Germany was more than two thirds of that to the 'rest of Europe'.

41. Exports of product B to the rest of Europe = 14.0
It is more than the exports of products E and G to the whole of Europe.

42. The percentage of females in the total is

$12 \times \frac{20}{100} + 18 \times \frac{20}{100} + 32 \times \frac{45}{100} + 8 \times \frac{60}{100} + 14 \times \frac{40}{100} + 16 \times \frac{15}{100}$

= 2.4 + 3.6 + 14.4 + 4.8 + 56 + 2.4 = 33.2
Average number of female engineers per sector

$= \frac{33.2}{100} \times \frac{80,500}{6} = 4454$

43. Male engineers in the banking sector $= \frac{60}{100} \times 14 = 8.4\%$

Female engineers in the IT sector $= 12 \times \frac{20}{100} = 2.4\%$

The required percentage $= \frac{8. - 2.4}{2.4} \times 100 = 250\%$.

44. The number of female engineers in the gaming sector currently $= 18 \times \frac{20}{100} = 3.6\%$

$\frac{3.6}{100} \times 80500 = 2898$

As the number of females go up by 4628 – 2898 = 1730

The required percentage $= \frac{4454 \times 6 + 1730}{80500 + 1730} \times 100$

$= 34.6\%$.

45. Let the total number of employees be 100.
The difference between the number of male and female engineers would be more in gaming then in IT.
The difference in gaming $= 18 \times 6 = 10.8$
The difference in others $= 32 \times 1 = 3.2$
The difference in education $<$ that in banking.
The difference in banking $= 14 \times 2 = 2.8$
The difference in medical $= 16 \times 7 = 11.2$

The required value $= \frac{11.2}{100} \times 80500 = 9016$

46. Since the company has an overall profit of 10%, assume that the cost is ₹100 and the sales value is ₹110. The cost and the sales values of the different products are as follows:

Product	Cost	Sales value
A	15	13.2
B	17	16.5
C	20	27.5
D	18	14.3
E	19	23.1
F	11	15.4

We can see that product F had a profit percentage of 40, which is the highest.

47. To find the maximum number of companies, we have to assume that the companies made only a small loss, i.e., both cost and sales value are nearly equal. If we assume both of them to be 100, then

Product	Cost	Sales Value
A	15	12
B	17	15
C	20	25
D	18	13
E	19	21
F	11	14

∴ At most three products had a sales value more than the cost.

48. To find the maximum number of products on which there was a loss, we have to assume that the company makes only a small profit. Therefore, on products A, B and D it made a loss.

49. The company has to make a profit on at least products C, E and F to make an overall profit.

50. If the company made a profit on each of the products, then the profit percentage would be the least when the product on which the difference between the percentage share of costs and sales value is the highest, i.e., product D, makes a small profit.
∴ If the total cost is ₹100, then the sales value of product D would be 18 (its cost price) and so the total sales value would be $\frac{18}{13} \times 100 = 138.5$

EXERCISE-2

1. The total sales (by volume) has increased by 5%, so assume that sales in 2014 to be 100 and that in 2015 to be 105.
Sales of D in 2014 = 13% of 100 = 13
Sales of D in 2015 = 15% of 105 = 15.75

∴ The required percentage $= \frac{2.75}{13} \times 100 = 21.2\%. \times 100$

$= 21.2\%$.

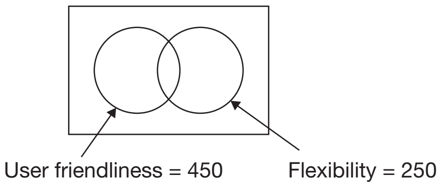
2. Let the total cars sold in 2014 be 100 and those in 2015 be 105, and the selling price of a car sold by B in 2014 be 100 that in 2015 be 108.
Sales (by value) of company B in 2014 $= 15 \times 100 = 1500$
Sales (by value) of company B in 2015 $= 14.7 \times 108 = 1587.6$
∴ The required percentage $= \frac{87.6}{1500} \times 100 = 5.85\% \times 100$
$= 5.85\%$.

3. Company F would have the maximum percentage increase as it has gone from 5% in 2014 to 8% in 2015 (minimum percentage growth of F would be 68%).

4. Companies C, D and F had a growth in sales by volume greater than 20%.

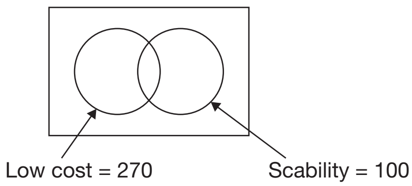
5. Number of companies which implemented Oracle = 12% of 1000 = 120.
$33^1/_3$% of 120 = 1/3 of 120 = 40
In total, 180 companies cited Robustness as one of the reasons.
∴ A maximum of (180 – 40) = 140 companies which implemented either Clarify or Baan could have cited Robustness as a reason.

6. Total number of companies which implemented Talisma/People Soft/Onyx = 140 + 160 + 200 = 500.



Minimum number of users who could have quoted both the reasons = (450 + 250) – (500) = 200.

7. Minimum number of companies which quoted either low cost or scalability as the reason = 270 = A
Minimum number of companies which implemented Oracle = A – B, where



B = Maximum number of companies which implemented Onyx.
Required answer = 270 – 200 = 70.

8. Total number of companies which cited only one reason for implementation = Number of Siebel users + Number of Vartive users = 80 + 60 = 140
Any reason for which the total number of companies < 140, is the required answer.

9. Considering each company quoting one reason as an instance, we have a total of (10% + 18% + 27% + 10% + 45% + 25%) = 135% instance. But there were only 100% of companies. Let, at most x% of the companies quote all the six reasons, that implies $(100 - x)$ of the companies could quote in reasons, where $(1 \le n < 6)$. Now, $6x + n(100 - x) = 135$ (By counting the total number of instances). Now, to maximize x, we must minimize 'n', hence, put $n = 1$, then $6x + 1(100 - x) = 135$
$\Rightarrow 5x = 135 - 100 = 35$
$\Rightarrow x = 7$
∴ 7% of 1000, i.e., 70 companies could have quoted six reasons.

10. Total number of students who took up any type of job (with a salary of more than ₹4 lakh/annum)

$$= \left[\left(\frac{15}{360}\right)\left(\frac{72}{360}\right) + \left(\frac{120}{360}\right)\left(\frac{36}{360}\right)\right](129600)$$

Total number of students who joined a B-school in Central region.

$$= \left(\frac{90}{360}\right)\left(\frac{120}{360}\right)(129600).$$

$$\text{Required percentage} = \frac{(150)(72) + (36)(120)}{(90)(120)}$$

$$= \frac{15120}{10800} \times 100 = 140\%.$$

11. Number of students taking up any type of job

$$= \left(\frac{270}{360}\right)(129600)$$

$$\text{Required angle} = (360)\left[\frac{\left(\frac{84}{360}\right)(150) + \left(\frac{36}{360}\right)(120)}{270}\right]$$

$$= \frac{(84)(150) + (36)(120)}{270}$$

$$= 16° + 46.5° \cong 62.5°.$$

12. Percentage distribution of the candidates (region-wise) for each of the disciplines is not given. Hence, it cannot be determined.

13. Number of students who took up non-IT jobs = $\left(\frac{120}{360}\right)(129600)$

$$\text{Required number} = \left[\frac{(180 + 54 + 36)}{360}\right]\left[\frac{120}{360}\right](129600)$$

$$= \left(\frac{270}{360}\right)(43200) = 32400.$$

14. 20% of total students graduating nationwide are from Commerce background and all of them join B-schools.
Number of students joining B-schools = $\frac{90}{360}$ = 25% of total students (T) graduating nationwide.
⇒ At the most 5% of T could be engineers.

⇒ Now students joining B-schools in the Central region

$= \frac{120}{360} \times \frac{90}{360}(\text{T}) = \frac{1}{12} \times (\text{T})$

Since $\frac{1}{12} \times (\text{T}) > 5\%$ of (T)

at least $\left(\frac{1}{12} - 0.05\right) \times (\text{T})$ students are from non-engineering disciplines, i.e., $\frac{1}{30} \times \text{T} = 4320$.

Solutions for questions 15 to 18: Let us tabulate the production cost and sales of different companies.

	Variety I	**Variety II**	**Variety III**	**Variety I**	**Variety II**	**Variety III**
Company	**Production cost**			**Sales**		
A	4×900	5×900	7×900	1.45×3600	1.45×4500	1.35×6300
B	9×900	15×900	6×900	1.33×8100	1.4×10500	1.48×5400
C	$6 \times 3 \times 900$	$2 \times 3 \times 900$	$3 \times 3 \times 900$	1.34×16200	1.38×5400	1.42×8100
D	7×900	5×900	2×900	1.35×6300	1.32×4500	1.46×1800
E	6×450	3×450	5×450	$1.44 \times 6 \times 450$	$1.48 \times 3 \times 450$	$1.25 \times 5 \times 450$

Company A's production cost is 16% of the total cost. Thus, cost of 3 varieties = production cost

$4K + 5K + 7K = 16\%$ of 90000. K = 900

$\text{Sales} = \left(\frac{100+45}{100}\right)\text{cost} = 1.45\ \text{cost(I)}$

Similarly, we can tabulate for A, B, C, D and E.

15. In variety II profit of companies (Sales – cost):
$4500\ (0.45) = 2025 \rightarrow$ A
$10500\ (0.4) = 4200 \rightarrow$ B
$6 \times 900\ (0.38) = 2052 \rightarrow$ C
$4500\ (0.32) = 1440 \rightarrow$ D
$1350\ (0.48) = 648 \rightarrow$ E
Maximum profit is for company B.

16. Profit = Profit (I) + Profit (II) + Profit (III)
Profit earned by E
$= 0.44 \times 6 \times 450 + 0.48 \times 3 \times 450 + 0.25 \times 5 \times 450$
$= 1188 + 648 + 562.50$
= ₹2398.50 million

17. Production cost of company C = ₹29,700
Production cost of company D = ₹12,600

$= \frac{297-126}{126} \times 100 = 136\%$

18. Production cost for variety III:
A → 6300
B → 5400
C → 8100
D → 1800
E → 2250

Solutions for questions 19 to 23: The table gives the revenue earned by the three airlines from the four zones (in ₹ cr)

Zone	**Bharat**	**FUT**	**Lahiri**
North	1800	1296	720
South	1200	432	1080
East	1000	432	600
West	2000	1080	1200
International	0	3960	0
Total	6000	7200	3600

19. Total inland market = 6000 + 45% of 7200 + 3600
= 12840

∴ Inland share of FUT Airways = $\frac{45\% \text{ of } 7200}{12840} \cong 25.2\%$.

20. By observing the pie charts, we can say that the revenue generated from the western region for Bharat Airways is the highest. Revenue generated from the South and the North zones for Bharat Airways is ₹1200 and ₹1800 crore, respectively, from the North zone for FUT Airways is 18% × 7200 = ₹1296 crore and from the West zone for Lahiri Airways is 33% × 3600 = ₹1188 crore.

21. Since it is not mentioned that there are no other airways operating in the inland market, the question cannot be answered.

22. Statement A: Revenue generated from the North zone for Bharat Airways is ₹1800 and the revenue generated for Lahiri Airways from the North and the South zones put together is 50% × 3600 = 1800 crore.

∴ Statement A is true.

Statement B: For FUT Airways the revenue generated from North zone is 18% and the South and the East zones together is 12%, i.e., revenue generated in the North zone is 50% more.

Statement C: Revenue generated in South zone is less than the revenue generated in the North zone.

∴ Statement C is not true.

23. Revenue generated in the North zone by Bharat, FUT and Lahiri Airways is 30% × 6000 + 18% × 7200 + 20% × 3600 = 3816

3816 is 20% of the total revenue generated in the North zone.

∴ Revenue generated in the North zone is 19080.

Solutions for questions 24 to 29: Let the number of balls off which 1s, 2s, 3s, 4s and 6s runs are made be a, b, c, d, e, respectively. Let the total runs in the 1st Match be T_1.

$$\frac{3T_1}{10} = a \qquad \frac{5T_1}{100} = 4d$$

$$\frac{4T_1}{10} = 2b \qquad \frac{10T_1}{100} = 6e$$

$$\frac{15T_1}{100} = 3c$$

$$\Rightarrow a = \frac{3T_1}{10} \qquad d = \frac{T_1}{80}$$

$$b = \frac{T_1}{5} \qquad e = \frac{T_1}{60}$$

$$c = \frac{T_1}{20}$$

T_1 = Common multiple of (80, 60, 20, 10 and 5).

Since $T_1 < 300$,

$T_1 = 240$

Similarly, let total runs in 2nd match be T_2.

$$\frac{4T_2}{10} = a \qquad \frac{5T_2}{100} = 4d$$

$$\frac{3T_2}{10} = 2b \qquad \frac{2T_2}{10} = 3c$$

$$\frac{5T_2}{100} = 6e$$

$$\Rightarrow a = \frac{4T_2}{10}, b = \frac{3T_2}{20}, c = \frac{2T_2}{30}, d = \frac{T_2}{80}, e = \frac{T_2}{120}$$

T_2 = Common multiple of (10, 20, 30, 80, 120) = 240

As $100 < T_2 < 300$

For the 3rd match, using similar logic,

T_3 = common multiple of (20, 10, 100, 25)

Number of runs in the 3rd match = Multiple of 100

Since $100 < T_3 < 300$, $T_3 = 200$

24. To find 'd' for 1st match $d = \frac{T_1}{80} = \frac{240}{80} = 3$.

25. Since it is not mentioned that runs were scored off each ball that the team faced, the total number of balls faced cannot be found out.

26. Total number of runs scored in the 2nd match = 240.

27.

Match	Runs (through1s)
I	$\frac{(3)(240)}{10} = 72$
II	$\frac{(4)(360)}{10} = 144$
III	$\frac{(35)(200)}{(100)} = 70$

28. Total number of runs scored in the IIIrd match = 200

Number of runs through 6s

$$= \left(\frac{6}{100}\right)(200) = 12.$$

29. In Match-I, the total number of balls off which runs were scored = $a + b + c + d + e$

$$= \frac{3 \times 240}{10} + \frac{240}{5} + \frac{240}{20} + \frac{240}{80} + \frac{240}{60}$$

= 72 + 48 + 12 + 3 + 4 = 139

Similarly, in Match-II,

$$\text{we get} = \frac{4 \times 240}{10} + \frac{3 \times 240}{20} + \frac{2 \times 240}{30} + \frac{240}{80} + \frac{240}{120}$$

= 96 + 36 + 16 + 3 + 2 = 153

In Match-III, we get

70 + 20 + 18 + 6 + 2 = 116

Comparing number of balls of which no runs were scored:

Match-I = (300 – 139)

Match-II = (300 – 153)

Match-III = (300 – 116)

Thus, it is least for Match-II.

30.

State	Number of male graduates as a fraction total number of graduates
M.P.	$\frac{8}{15}\times\frac{15}{100}=\frac{1}{12.5}$
Bihar	$\frac{5}{8}\times\frac{8}{100}=\frac{1}{20}$
Maharashtra	$\frac{11}{18}\times\frac{12}{100}=\frac{11}{150}$
Tamil Nadu	$\frac{13}{22}\times\frac{11}{100}=\frac{13}{200}$

By comparison, the fraction is maximum for M.P.

31. Top four states (number of graduates wise) are M.P., Kerala, Maharashtra and Tamil Nadu.
Required number of female graduates

$$=\left(\frac{7}{15}\times\frac{15}{100}+\frac{7}{13}\times\frac{13}{100}+\frac{7}{18}\times\frac{12}{100}+\frac{9}{22}\times\frac{11}{100}\right)\times\left(3\times10^5\right)$$

$$=\left(\frac{7+7+\frac{14}{3}+4.5}{100}\right)(3\times10^5)=\frac{69.5}{100}\times10^5=69500$$

32. Option (A): Difference = 10% × 3 lakh
= 30,000. This is a true statement.

Option (B): Male graduates in U.P. = $\frac{10}{21}\times\frac{7}{100}\times3\times10^5$

Female graduates in M.P. = $\frac{7}{15}\times\frac{15}{100}\times3\times10^5$

Number of male graduates is not more.

Option (C): BBA graduates = $\frac{9}{100}\times3\times10^5$

Male graduates from Karnataka and Kerala

$$=\left(\frac{3}{5}\times\frac{10}{100}+\frac{6}{13}\times\frac{13}{100}\right)\left(3\times10^5\right)=\frac{12}{100}\times3\times10^5$$

Number of BBA graduates is not more.

33. Minimum number of male graduates are in Gujarat

$$=\frac{4}{125}\times3\times10^5$$

$$=\frac{4}{125}\times3\times10^3\times10^2=9600$$

There is no state in which the male graduates are not more than 5000.

Solutions for questions 34 to 37: The runs scored in the matches are as follows.

Match I = $\frac{63}{360}\times100=17.5\%$

Similarly, in
Match 2 = 25%
Match 3 = 15%
Match 4 = 20%
Match 5 = 22.5%
The runs made in the different matches are in the ratio of 7 : 10 : 6 : 8 : 9

Let the total runs scored in Match 1, Match 2, Match 3, Match 4 and Match 5 be 7k, 10k, 6k, 8k and 9k, respectively.

Total = 40k

Similarly, the ratio of the total runs scored by the 6 batsmen are as follows:

A	B	C	D	E	F
10%	20%	27.5%	12.5%	22.5%	7.5%
4	8	11	5	9	3

Let the total runs scored by the 6 batsmen be $4k_2$, $8k_2$, $11k_2$, $5k_2$, $9k_2$ and $3k_2$, respectively.

Total = $\frac{90}{100}$ $(40k_1) = 36k_1$

So, depending on the value of k_1 we can have the following scores $k_1 = 10$, $k_1 = 30$, $k_1 = 40$, $k_1 = 50$.

Match 1	70	140	210	280	350
Match 2	100	200	300	400	500
Match 3	60	120	180	240	300
Match 4	80	160	240	320	400
Match 5	90	180	270	360	450

Similarly, we can find the total runs scored by the 6 batsmen.

34. Runs scored by C in the entire series = $11k_2$

$= 11\times\frac{9}{10}k_1 = 9.9k_1$

C scored 20% of the total runs in each of the 5 matches.

So, C scored $\frac{20}{100}$ $(40k_1) = 8k_1$

In addition, C scored $9.9k_1 - 8k_1 = 1.9k_1$
Now runs scored by C in a match as a percentage of the total runs scored in the match will be the highest when the total runs scored in that match is the least, i.e., in Match 3.

Out of a total of $6k_1$, C scored = $\frac{20}{100}$ $(6k_1) + 1.9k_1 = 3.1k_1$

Now $\frac{3.1\,k_1}{6\,k_1}\times100 = 51.66\%$.

35. Total runs scored in Match 2 = $10k_1$
Total runs scored by C in the entire series = $11k_2$

$= 11\times\frac{9}{10} = 9.9k_1$

Difference = $0.1\ k_1$
Now $0.1\ k_1 = 4$ or, $k_1 = 40$

Therefore, $k_2 = \frac{9}{10}\ (40) = 36$

Total runs scored by A = $4k_2 = 4(36) = 144$
Total runs scored by D = $5k_2 = 5(36) = 180$
Thus, total runs scored by A and D in the entire series = (144 + 180) runs = 324 runs.

36. Total runs scored in Match 2 = $10k_1$
Total runs scored by the 6 batsmen in the entire series.

A	B	C	D	E	F
$3.6k_1$	$7.2k_1$	$9.9k_1$	$4.5k_1$	$8.1k_1$	$2.7k_1$

Total runs scored by a batsman in Match 2 as a percentage of the total runs scored by the batsmen in the entire series will be maximum for the batsman who scored the least runs, i.e., F.
Since each batsman scored at least 16% of the total runs in Match 2, the remaining 5 batsmen scored at least (16% of $10k_1$) 5

$= 80\%$ of $10k_1 = \frac{4}{5}\ (10k_1) = 8k_1$

Thus, F scored the remaining $2k_1$ runs.

Now, $\frac{2k_1}{2.7k_1} \times 100 = 74\%$.

37. The minimum runs scored by the team in the entire series was 180.
Minimum runs was scored in Match 3, $6k_1 = 180$
So $k_1 = 30$
So, the total runs scored by B in the entire series was $7.2k_1 = 7.2(30) = 216$.

Solutions for questions 38 to 41: It can be seen that the expenses on account of forex loses is only for company Y. As it is 17% of the total expenses of company Y and 8.5% of the merged entity, the expenses of company Y is 50% of the total expenses of the merged entity. As the expense under taxes are only for companies X and Y, the taxes paid by company Y would account for $\frac{14}{2}$ = 7% of the taxes paid by the merged entity while the remaining 5.1% of the taxes paid was due to company X.

17% of X = 5.1% of $(x + y + z)$
$\therefore\ x = 30\%$ of $(x + y + z)$
$\therefore$ Ratio of expenses of X, Y and Z = 3 : 5 : 2.

38. Ratio of expenses of companies X and Y = 3 : 5.

39. Forex loses of company Y = 8.5% of total

Depreciation expenses of company Z = $19 \times \frac{2}{100}$
= 3.8% of total

Required ratio = $\frac{8.5}{3.8} \times 100 = 223\%$

40. As the total expenses of the companies are in the ratio 3 : 5 : 2, the wage bill of company Y would be the highest as 16% of 5 > 22% of 3 > 22% of 2.

41. Assume that the expenses of X, Y and Z are ₹300, ₹500 and ₹200, respectively.
$\therefore$ Taxes paid by company Z = ₹50

Share of taxes paid = $\frac{171}{1050} \times 100 = 16.3\%$.

42. The ratio between the value of the sales of C and that of E does not depend on the actual value of the sales of D in the western region.
$\therefore$ Required ratio = 25 : 30 = 5 : 6

43. Let the exaggerated total sales of product D = D
$\Rightarrow$ Sales of product D in the North = 10% of D

$\Rightarrow$ Total sales of all products = $\frac{100}{15}$ of D

$\Rightarrow$ Total sales in the Northern region = 30% of $\frac{100}{15}$ of D = 2D

$\therefore$ Required percentage = $\frac{10\%\text{ of D}}{2\text{D}} = 5\%$

44. Sales of E is 25% of the total sales in the western region.
$\therefore$ Total sales in the western region is four times the sales of E in that region.
New, total sales in western region is 80% of what is represented by Sunil.

Hence, West forms only $\frac{80}{100} \times 25 = 20\%$ of the total of 95%.

(The remaining 5% is the exaggerated value)

$\therefore$ Percentage share of the Northern region in the total sales = $\frac{30}{95} \times 100 = 31.58\% \cong 31.6\%$.

45. Sales of B = ₹1800 crore
$\therefore$ Total sales after the exaggeration of sales

$= \frac{1800}{12} \times 100 =$ ₹15000 crore

Sales of D is exaggerated by ₹1000 crore.

Exaggerated sales of D = $\frac{15}{100} \times 15000 =$ ₹2250 crore

Actual sales of D = ₹1250 crore
Exaggerated sales value in the Western region
= 50% of 2250 = ₹1125 crore
Actual sales value in the Western region
= 1125 – 1000 = ₹.125 crore
$\therefore$ It is exaggerated by 1000 crore.

Hence, $\frac{1000}{125} = 8$ times.

46. To find the actual total sales of D, we need any relation involving the actual sales of D in the Western region. Hence, either III or IV is sufficient.

47. The selling price per unit would be the highest for the product for which the ratio of $\frac{\text{sales (by value)}}{\text{sales (by volume)}}$ is the highest, i.e., for product D.

48. The products for which the share of sales (by value) is more than the share of sales (by volume), the selling price per unit is more than the average selling price of all the units, i.e., for products A and D.

49. If the company made an overall profit, the sales (by value) would be more than the cost and the company could have made a loss on products B, C and F.

50. If the company did not make a loss on any of the six products, then the overall profit has to be at least

$$\frac{15-11}{11}\times 100 = \frac{4}{11}\times 100 = 36.36\%.$$

EXERCISE-3

Solutions for questions 1 to 5: The cost of production and profits on both grades of the six items are as follows:

Item	Total cost	Grade A cost (₹ cr)	Grade B cost (₹ cr)	Grade A profit (₹ lakh)	Grade B profit (₹ lakh)
P	5 cr	3	2	60	30
Q	2.5 cr	0.96	1.54	29	23
R	7.5 cr	3.63	3.87	72.5	97
S	2.5 cr	1.25	1.25	19	25
T	5 cr	2.27	2.73	45.5	68
U	2.5 cr	1.39	1.11	14.0	17

1. The total profit made on item R is = 72.5 + 97 = 170 lakh.

2. The ratio of total profit to total cost is the highest for product T.

3. The total cost for producing grade A of products P, R and S = ₹7.88 crore.

4. Only for products P and Q is the profit obtained on grade A products more than that of grade B.

Solutions for questions 5 to 8: The number of students in each class, the number of students selected for the scholarship and the number of boys and girls selected is as follows:

Class	Number of students	Number of students selected	Number of boys selected	Number of girls selected
V	196	30	18	12
VI	182	27	16	11
VII	161	32	21	11
VIII	210	40	15	25
IX	259	38	28	10
X	252	38	22	16

5. The number of girls in class VII who were selected for the scholarship is 11.

6. The required percentage $\frac{40}{210}\times 100 = 19\%$

7. The percentage of students selected for the scholarship was the highest in class VII.

8. From the above table it can be seen that in class V, VII and IX the number of boys selected was at least 50% more than that of girls.

Solutions for questions 9 to 12: The number of units of the different products sold are as follows:

Product	2015 (in '000)
P	10560
Q	13860
R	11440
S	20020
T	23320

Product	2016
P	12240
Q	13920
R	13680
S	20880
T	25680

9. Sales of P in 2016 = 12,240 units

Price per unit = $132 \times \frac{108}{100} = 142.56$

Sales (by value) = $12240 \times 142.56 \times 10^3 \simeq 175$ crore

10. Sales (by value) in 2015
$= 10560 \times 132 + 13860 \times 167 + 11440 \times 102$
$+ 20020 \times 36 + 23320 \times 235$
$= 140 + 231 + 117 + 172 + 548$
$= 1208$ crore

11. The total sales increased by slightly more than 9%.
∴ If any product had a 1% or more increase in its share from 2015 to 2016, it would have more than 10% increase. We need to check for T as it is very close to 10%.
Sales of T in 2015 = 23320
10% = 2332
25652

As sales in 2016 are more than this, T also had a more than 10% increase.

12. Sales (by value) in 2016

$= 12240 \times 142.5 + 13920 \times 190.4 + 13680 \times 104 + 20880 \times 86 + 25680 \times 251.5$

$\simeq 175.0 + 265 + 142.0 + 180 + 646 = 1408$ crore

Difference = 1408 – 1208 = 200 crore

Solutions for questions 13 to 16: It can be seen that gold was only given to the third son. 35% of the total share of the third son equal to 14% of the total assets of Ratan Lal. Therefore, the third son got 40% of the assets of Ratan Lal. Now, bonds were given only to the first and third son and as the third son has got 40% of the share, we can conclude that the first son got 30% and so the second son also has to get 30%.

13. Ratan Lal gave 30% of his assets to his second son

14. As 30% of the share of first son is 5.25 crore, the total share of both the first and second sons would be 17.5 crore.

15. FDs received by the second son is 2.5% of the total assets of Ratan Lal, i.e., $\frac{2.5}{100} \times 48 = 1.2$ crore.

16. The value of cash received by the first and the third sons together is 6.5% of his assets. Equity with Ratan Lal is 21% of his assets.

The required value is $\frac{21}{6.5} \times 3.9 = 12.6$ crore.

Solutions for questions 17 to 20: As Sunil spent ₹300, the total amount spent by the five children is ₹1500. The amount spent by the different children are Prakash = 15, Qureshi = 100, Rahul = 205, Sunil = 300, Thomas = 395 and Urvashi = 490.

Now $7x - 15 = 6y$ and $8x - 100 = 5y$

$\therefore$ We get $x = 45$ and $y = 50$

Hence, the amount with them when they went for shopping was 315, 360, 400, 450, 495 and 540, respectively. The amounts with them when they returned is 300, 250, 200, 150, 100 and 50, respectively.

17. The total amount spent by the children was ₹1515.

18. The maximum difference in amounts was with Prakash and Urvashi, 540 – 315 = 225

19. Sunil spent $\frac{300}{450} \times 100 = 67\%$

20. Thomas spent ₹95 more than Sunil.

4

CHAPTER

Line Graphs

LEARNING OBJECTIVES

In this chapter, you will:

- Gain an understanding of various types of data presented and different types of line graphs.
- Learn to solve questions and in turn become familiar with area graphs.
- Get familiar with different types of questions based on line graphs.
- Learn how to compare values (changes) across two or more line graphs using percentage techniques.
- Understand how to decipher data which involves more than one data type—bar graph and line graph together, line graph and table, etc.

❒ TWO-DIMENSIONAL GRAPHS

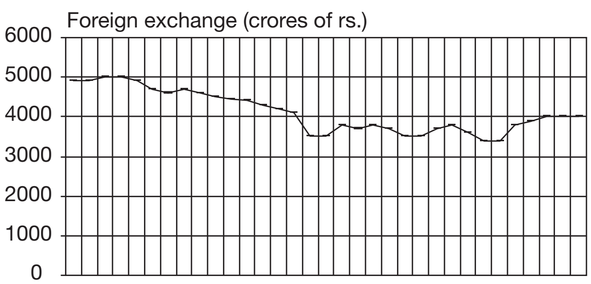


Fig. 4.1 Foreign exchange reserves of India

This is essentially used for continuous data but can also be used for depicting discrete data provided that we understand the limitation. The representation of this data is also known as Cartesian graphs, they represent the variation of one parameter with respect to another parameter each shown on a different axis. These types of graphs are useful in studying the rate of change or understanding the trends through extrapolations.

These graphs can be of various types and a few of them are shown below (Figures 4.1 to 4.3):

The graph in Figure 4.2 shows the changes in the foreign exchange reserves of our country during a period of time. One can find out the trends and the growth rates of foreign exchange reserves.

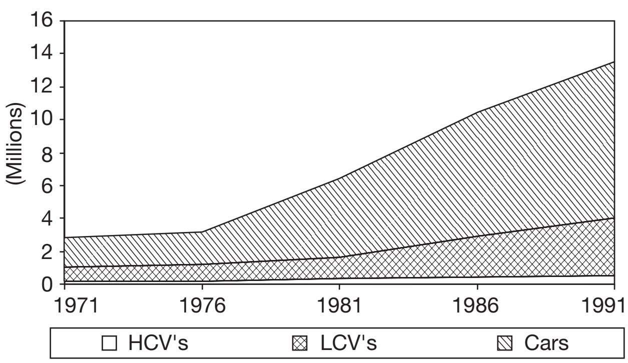


Fig. 4.2 Automobiles in India

Figure 4.2 shows a cumulative type of graph (stacked graph). This figure provides more information than the previous graph that you studied in Figure 4.1.

From the graph given in Figure 4.2, the relative proportion of different varieties of vehicles which constitute the total can be obtained along with the trends and growth rates, percentage variation, actual variations and trends for any period of time can be ascertained.

Figure 4.3 presents another type of 2-dimensional graph which is mostly used to depict scientific data like speed, velocity, vectors, etc. In the graph, the speed trends of three bodies Q1, Q2, Q3 is given.

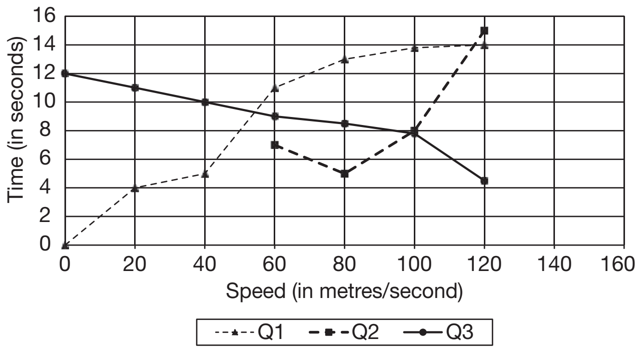
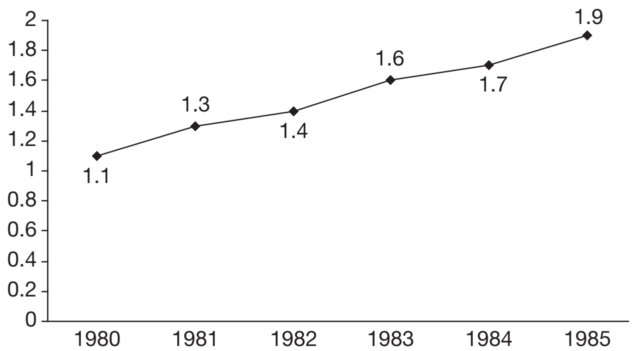


Fig. 4.3 Motion graph of Q1, Q2 and Q3

SOLVED EXAMPLES

Directions for questions 4.01 to 4.05: These questions are based on the information given below.

The following line graph represents the population (in millions) of country X in each year from 1980 to 1985.



4.01: Find the percentage increase in the population of X from 1980 to 1985.

(A) 60 (B) $66\frac{2}{3}$
(C) 70% (D) None of these

Sol: Required percentage

$$= \frac{1.9-1.1}{1.1}(100) = 72\frac{8}{11}\%$$

4.02: If in 1981, 60% of the population of X were men while in 1982 it was only 50%, find the percentage change in the male population from 1981 to 1982.

(A) 8% (B) 10%
(C) 12% (D) 13%

Sol: Number of men in 1981

$$= \frac{60}{100}(1.3\,\text{million}) = 0.78 \text{ million}$$

Number of men in 1982

$$= \frac{50}{100}(1.4 \text{ million}) = 0.7 \text{ million}$$

∴ The male population decreased.

$$\text{Required percentage} = \frac{0.78-0.7}{0.78}(100) \cong 10\%$$

4.03: If in the previous question, 60% of the female population in 1981 and 80% of those in 1982 were literate, then find the increase (in millions) in the number of literate women from 1981 to 1982.

(A) 0.16 (B) 0.192
(C) 0.213 (D) 0.248

Sol: Number of women in 1981

$$= \frac{40}{100}(1.3\,\text{million}) = 0.52 \text{ million}$$

Number of women who were literate

$$= \frac{60}{100}(0.52\,\text{million}) = 0.312 \text{ million}$$

Number of women in 1982

$$= \frac{50}{100}(1.4\,\text{million}) = 0.7 \text{ million}$$

Number of women who were literate

$$= \frac{80}{100}(0.7\,\text{million}) = 0.56 \text{ million}$$

The increase = 0.56 – 0.312 = 0.248 million

4.04: In which of the given years, from 1981 to 1985, did the population increase by the highest percentage over the previous year?

(A) 1981 (B) 1982
(C) 1983 (D) 1984

Sol: As the maximum increase in population between any two years is 0.2 million (1981, 1983 and 1985) and as the increase in population in 1981 is on a lower base, the percentage increase in 1981 would be the highest.

4.05: If the percentage of women in the population of country X in the years from 1980 to 1985 are 40, 45, 50, 55, 60 and 65, respectively, then in which of the given years when compared to its previous year, did the female population show the highest percentage increase?

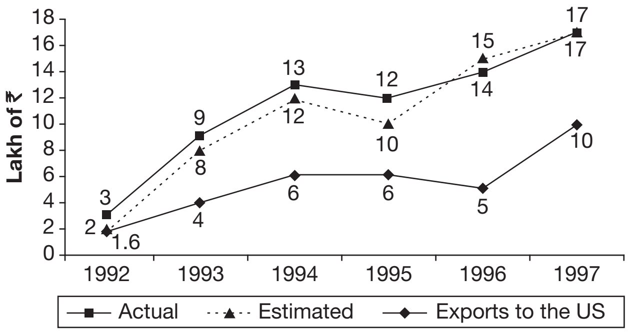
(A) 1985 (B) 1981
(C) 1983 (D) 1982

Sol: As the percentage increase in total population, and the percentage increase in the share of female population in 1981 (40 – 45, i.e., 12.5%) is the highest, the percentage increase in female population would be the highest in that year.

EXERCISE-1

Directions for questions 1 to 5: These questions are based on the following line graph.

Exports of tea of Niligiri tea Ltd., from 1992 to 1997



1. In which year is the ratio of actual exports of tea to the estimated exports of tea, the highest?

2. Total exports to the US is approximately what per cent of total actual exports for the given period?

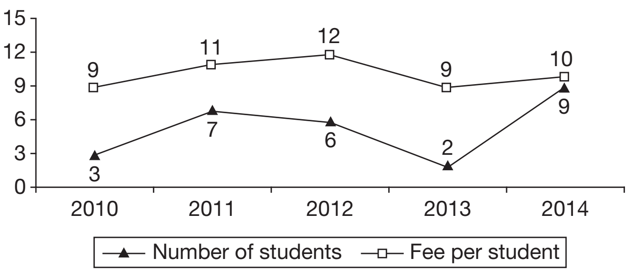
3. In the year 1996, the estimated exports to the US were ₹5.5. lakh when \$1 was estimated as ₹40. However, the actual exports to the US were only ₹5 lakh due to the appreciation of the value of rupee. What was the actual value of \$1 (other factors remained constant)?

4. In the year 1994, if \$1 was equal to ₹40, then find the quantity of tea exported (in kg) to the US where the cost of tea per kilo is \$8.

5. In which year is the ratio of exports of tea to the US when compared to the actual exports of Nilgiri tea Ltd, the highest?

Directions for questions 6 to 10: Answer these questions based on the information given below.

The line graph gives the percentage increase in the number of students and the fee per student at a coaching institute across five years. The increase in values for each year is with respect to that of the previous year.



6. What was the approximate percentage increase in the number of students from 2010 to 2013?

(A) 14 (B) 15
(C) 16 (D) 17

7. If the fee per student in 2011 was ₹26,000, then what was the fee in 2013?

(A) ₹30,600 (B) ₹31,740
(C) ₹32,600 (D) ₹33,250

8. If the number of students in the institute was 23,348 in 2014, how many students were there in 2012?

(A) 20,000 (B) 20,500
(C) 21,000 (D) 22,000

9. In 2012, the coaching institute had a 32% share of the total number of students who took coaching in the city. What was its share in 2014, given that the total number of students who took coaching in the city increased by 15% from 2012 to 2014?

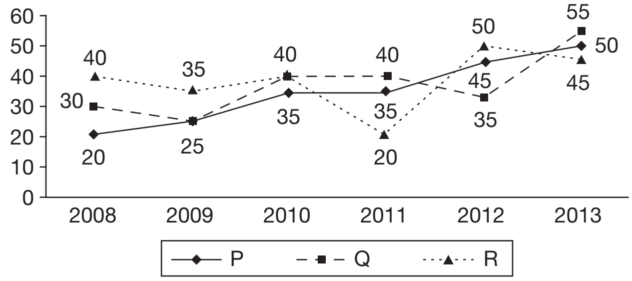
(A) 32.3% (B) 32.8%
(C) 33.1% (D) 31%

10. If the total fees collected by the institute in 2011 was ₹29.375 crore and the fee per student that year was ₹12,500, then what was the total fee collected in 2013?

(A) ₹38.8 crore (B) ₹36.2 crore
(C) ₹34.1 crore (D) ₹32.2 crore

Directions for questions 11 to 15: Answer these questions based on the information given below.

The graph gives the profit percentage of three companies P, Q and R across six years.



$$\text{Profit percentage} = \frac{\text{Profit}}{\text{Expenditure}} \times 100$$

Profit = Income – Expenditure

11. If, in the year 2011, the ratio of expenditures of companies P and Q is 6 : 8, then what is the ratio of their profits?

(A) 12 : 17 (B) 26 : 37
(C) 5 : 8 (D) 21 : 32

12. In the year 2009, the difference between the expenditure and the profit of company R is ₹2.99 crore. What is the income (in crore) of company R in that year?

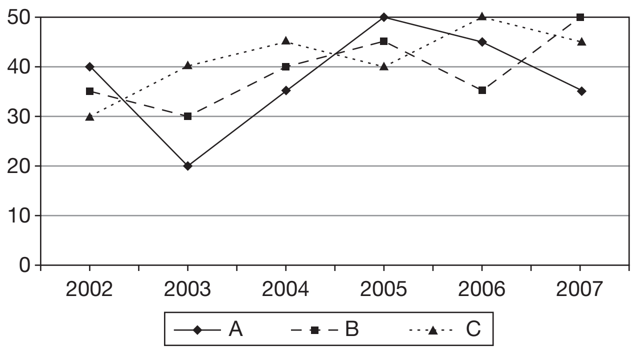
(A) ₹4.6 (B) ₹6.21
(C) ₹5.2 (D) ₹5.82

13. The income of company P in the year 2012 is ₹7.25 crore. What is the expenditure (in ₹ crore) of company P in that year?
(A) 5 (B) 5.25
(C) 6.25 (D) 6.75

14. Which company earns the highest profit in any of the given years?
(A) P (B) Q
(C) R (D) Cannot be determined

15. If in 2011, the ratio of the incomes of P, Q and R is 5 : 4 : 3, then the profit of P was approximately what percentage of the profit of Q and R together?
(A) 69 (B) 72
(C) 75 (D) 80

Directions for questions 16 to 20: These questions are based on the following graph which shows the profit percentage earned on three products A, B and C over the years 2002 through 2007.



Profit = Selling Price – Cost price

$$\text{Profit percentage} = \frac{\text{Profit}}{\text{Cost price}} \times 100$$

16. If the selling price of product B in 2003 was ₹3.9 lakh, then what was the profit earned on product B in rupees in that year?
(A) 60 thousand (B) 55 thousand
(C) 80 thousand (D) 90 thousand

17. The cost price of product A in 2003 is equal to the cost price of product C in 2007. What will be the ratio of the selling price of product A in 2003 to the selling price of product C in 2007?
(A) 24 : 29 (B) 29 : 25
(C) 25 : 29 (D) 29 : 24

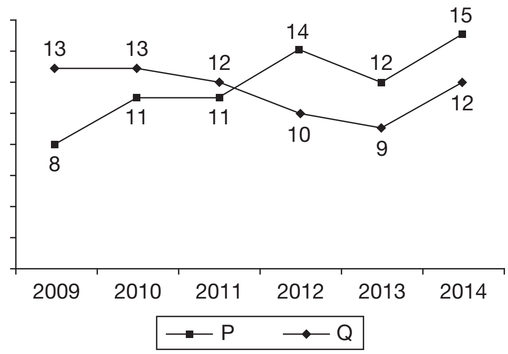
18. Selling price of product B in 2004 and 2005 were equal. What was the ratio of its cost prices in 2004 to that of 2005?
(A) 29 : 28 (B) 25 : 27
(C) 36 : 35 (D) 33 : 25

19. Selling price of product B in 2006 is equal to its cost price in 2007. What is the ratio of its selling price in 2006 to that in 2007?
(A) 1.5 (B) 0.66
(C) 0.8 (D) 1.2

20. In which year is the per cent rise/fall in the per cent profit earned by selling product A the least?
(A) 2003 (B) 2004
(C) 2005 (D) 2006

Directions for questions 21 to 25: Answer these questions based on the information given below.

The line graph gives the profit percentage of two companies P and Q from 2009 to 2014.



$$\text{Profit percentage} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$

21. If the expenditure of company P in 2012 was ₹85 lakh, then what was its income (in ₹ lakh) in that year?
(A) 95.4 (B) 96.9
(C) 98.2 (D) 99.3

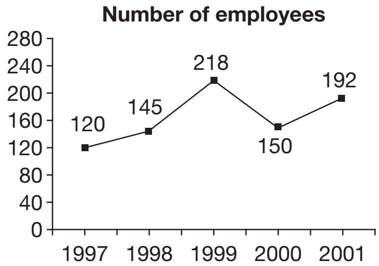
22. If in 2013, the ratio of the incomes of the companies P and Q was 224 : 98, then what was the ratio of their expenditures?
(A) 20 : 9 (B) 18 : 7
(C) 1 : 58 (D) None of these

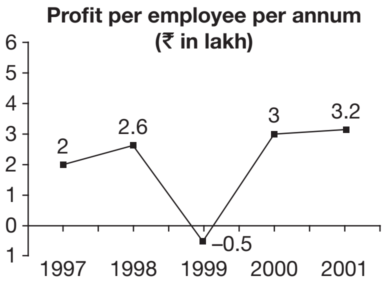
23. If in 2009, the expenditures of the companies P and Q were equal, then what was the ratio of their incomes?
(A) 98 : 99 (B) 73 : 79
(C) 108 : 111 (D) 108 : 113

24. If the profit of company Q in 2013 was ₹4.23 cr, then what was the income in that year?
(A) ₹48.76 cr (B) ₹50.21 cr
(C) ₹51.23 cr (D) ₹53.16 cr

25. If the ratio of the profits of P and Q in the year 2010 was 4 : 5, then what was the approximate ratio of their expenditures?
(A) 61 : 68 (B) 52 : 55
(C) 71 : 79 (D) 61 : 72

Directions for questions 26 to 30: The following two graphs show the number of employees and profit per employee per annum of company XYZ Ltd.





26. In which year was the profit of the company, the highest?

27. If the profit in the year 1998 was 30% of the total sales value, then what was the approximate total sales value (in cr) in that year?

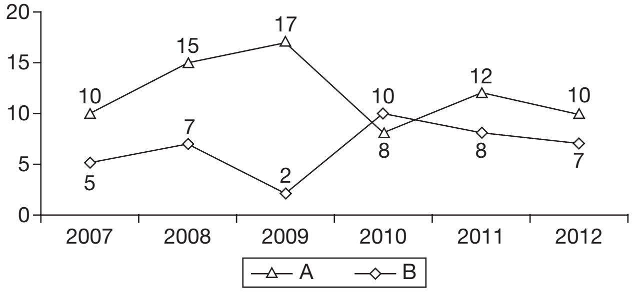
28. In how many years was the number of employees more than the average number of employees for the given period?

29. The company wants to achieve a profit in the year 2002 such that the profit in the year 2001 is 64% of the profit in the year 2002. What should be the profit (in cr) of the company in the year 2002?

30. What was the approximate percentage increase in the profit of the company from 1997 to 1998?

Directions for questions 31 to 33: Answer these questions based on the information given below.

The following line graph gives the percentage growth in the net profit of two companies A and B, over that of the previous year, for the period 2007 to 2012.



31. What is the approximate percentage growth in the net profit of company A from 2006 to 2012?

(A) 80% (B) 88%
(C) 97% (D) 105%

32. If the net profit of company B in 2006 was ₹180 crore, then what was its net profit in 2010?

(A) ₹210 crore (B) ₹227 crore
(C) ₹236 crore (D) ₹245 crore

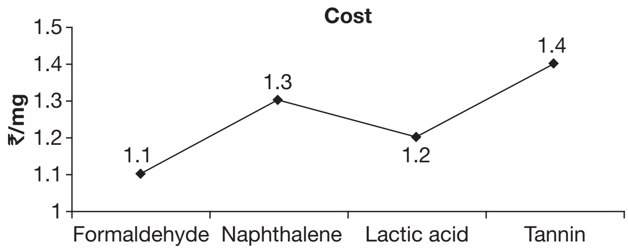
33. If in 2010, company A had a greater increase in net profit than company B, then which of the following cannot be the ratio of the net profit of A and B in 2006?

(A) 7 : 5 (B) 10 : 13
(C) 23 : 24 (D) None of these

Directions for questions 34 to 36: These questions are based on the information given below.

Five companies Johnson and Johnson, Pfizer, Roche, Sanofi and Ell Lilly make pills for a certain disease.

The composition and cost of the four chemicals Formaldehyde, Naphthalene, Lactic acid and Tannin used in making the pills is as follows:



Composition (in mg)

Company	Formaldehyde	Naphthalene	Lactic acid	Tannin
Johnson and Johnson	200	300	100	400
Pfizer	250	150	200	400
Roche	100	300	300	300
Sanofi	200	200	200	400
Ell Lilly	250	250	250	250

34. Which company incurs the least cost in producing the pill?

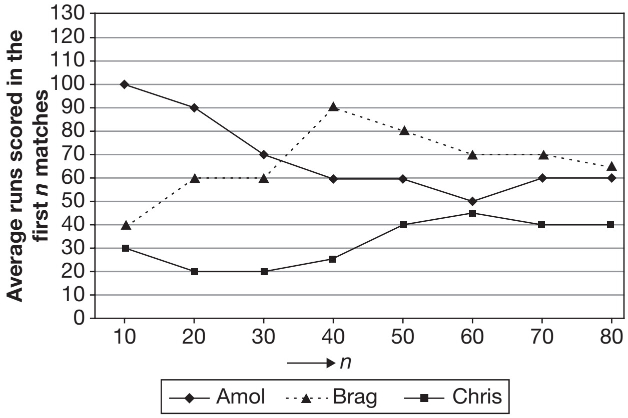
(A) Pfizer (B) Roche
(C) Sanofi (D) Ell Lilly

35. For how many of the given companies does Formaldehyde and Naphthalene form at least 50% and Naphthalene and Lactic acid form at least 40% of the pill by weight?

(A) 0 (B) 1
(C) 2 (D) 3

36. If the cost of formaldehyde increases by 10% and the cost of lactic acid increases by 20%, then for which company is the cost of making the pill the highest?
(A) Roche (B) Pfizer
(C) Sanofi (D) Johnson and Johnson

Directions for questions 37 to 40: Answer these questions based on the information given below.



The graph shown above represents the variation in the average runs scored by three leading cricket players, namely Amol, Brag and Chris. Assume that no player was 'not out' in any of the matches he played. All the three players are from the same team. Each of them played in all the 80 matches played by the team. Also, the runs scored by any of the players in any match was always a non-negative number.

For example, average runs scored by Amol in the first 40 matches played is 60, while those scored by Brag in the first 60 matches played is 70.

37. In which of the following groups of ten matches, is the number of runs scored by Amol the same as that scored by Brag?
(A) 41st to 50th (B) 51st to 60th
(C) 61st to 70th (D) None of these

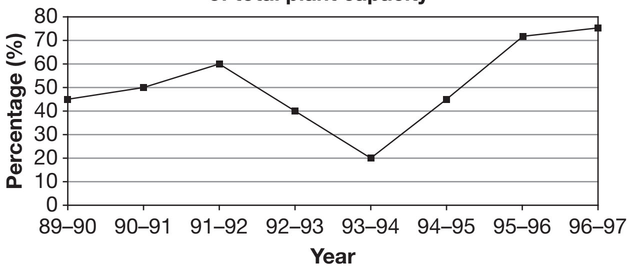
38. For which of the following groups of ten matches, is the average number of runs scored by Amol the highest?
(A) 21st to 30th (B) 31st to 40th
(C) 51st to 60th (D) 61st to 70th

39. If Brag did not score the same number of runs in any two matches, the highest score made by Brag in any of his first 60 matches is at least.
(A) 180 (B) 184
(C) 185 (D) 187

40. In which of the following groups of ten matches were the runs scored by Chris more than those scored by one of Amol and Brag, but less than the other?
(A) 11th to 20th (B) 31st to 40th
(C) 41st to 50th (D) 51st to 60th

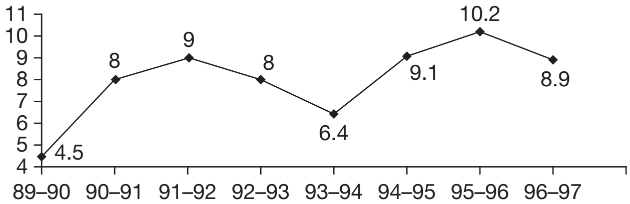
Directions for questions 41 to 45: These questions are based on the following line graph which show the statistics of Steel production and exports and imports of India.

Steel plant utilization as percentage of total plant capacity

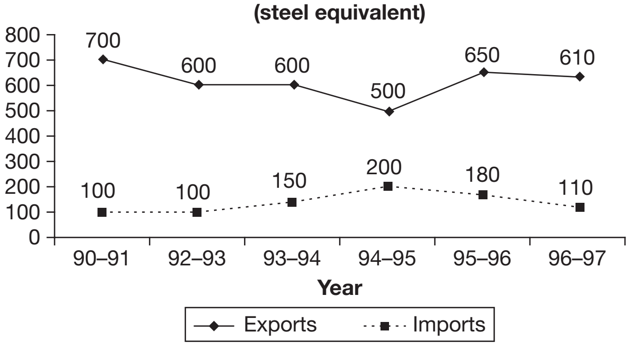


$$\text{Plant utilization} = \frac{\text{Output}}{\text{Capacity}} \times 100$$

Steel output in billion tons



Steel exports and imports in thousand tons (steel equivalent)



41. What was the total plant capacity of the steel plants in the year 1994–95?
(A) 20.2 billion tons (B) 15.0 billion tons
(C) 25.0 billion tons (D) Cannot be determined

42. In which year was the total plant capacity of the steel plants the lowest?
(A) 1996–97 (B) 1991–92
(C) 1994–95 (D) 1989–90

43. Which of the following statements is true?
I. A decrease in the plant utilization was accompanied by a proportionate decrease in the output of steel between 1991–1992 to 1993–1994.

II. The total plant capacity increased between 1991–1992 to 1993–1994.

III. Output of steel was the highest when the plant utilization was the highest.

(A) I, II, III
(B) I and II
(C) II Only
(D) II and III

44. In the steel industry, how many times is an increase in the imports accompanied by a decrease in the exports?

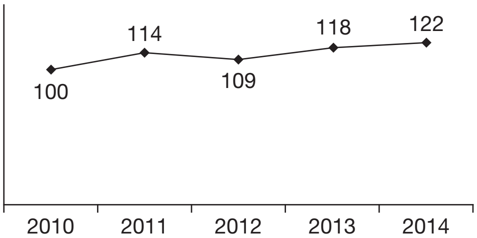
(A) 3 (B) 1
(C) 4 (D) Cannot be determined

45. The average annual increase in the total plant capacity for the given period is

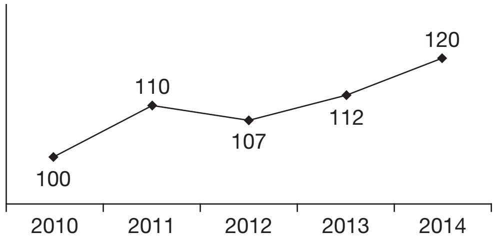
(A) 4% (B) 4.6%
(C) 3.1% (D) 2.7%

Directions for questions 46 to 50: Answer these questions based on the information given below.

The line graphs give the sales and expenditure of a company across five years, starting from 2010. The value of both sales and expenditure for the year 2010 is indexed to 100.



Sales



Expenditure

It is known that the profit in 2012 was ₹1 crore

46. In at least how many years did the company make a profit?

(A) 2 (B) 3
(C) 4 (D) 4

47. In which year was the profit the least?

(A) 2010 (B) 2012
(C) 2014 (D) Cannot be determined

48. In which year was the percentage income in sales the highest?

(A) 2011 (B) 2012
(C) 2013 (D) 2014

49. If the sales of the company in 2010 was ₹250 crore, then what was the profit in 2013?

(A) 8.6 crore (B) 10.8 crore
(C) 11.4 crore (D) 14.1 crore

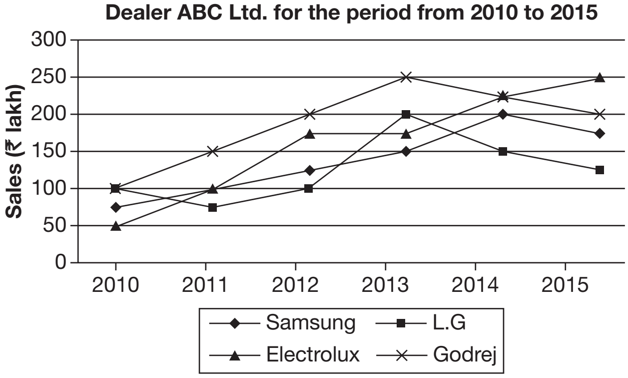
50. What was the approximate percentage decrease in profit in 2012, when compared to 2011?

(A) 6 (B) 8
(C) 9 (D) Cannot be determined

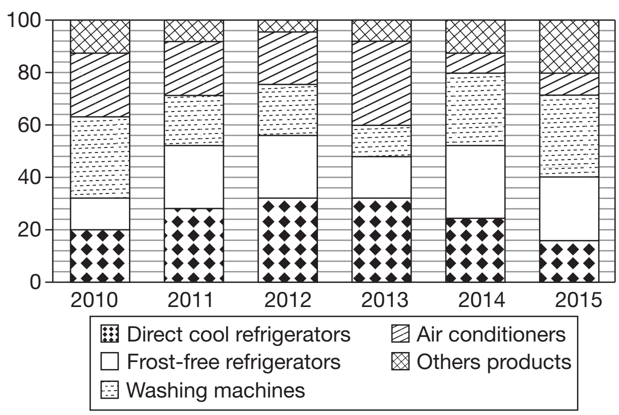
EXERCISE-2

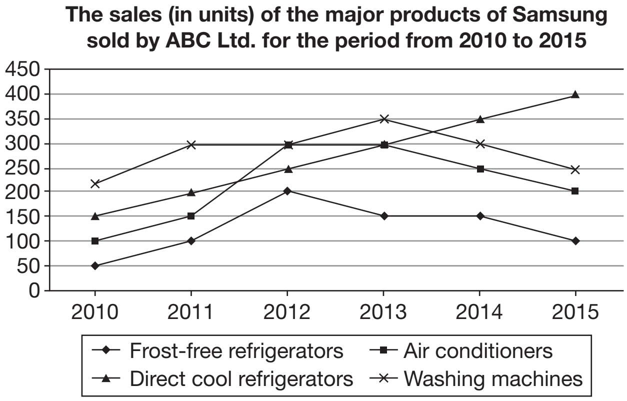
Directions for questions 1 to 4: Answer these questions based on the information given below.

The brand-wise sales of a Consumer Electronics Dealer ABC Ltd. for the period from 2010 to 2015



Product-wise split (by value) of the total annual sales of the products of Samsung sold by ABC Ltd. for the period from 2010 to 2015





1. What was the total sales of Samsung Frost-Free Refrigerators in the period from 2010 to 2012?
(A) ₹72 lakh (B) ₹81 lakh
(C) ₹69 lakh (D) ₹63 lakh

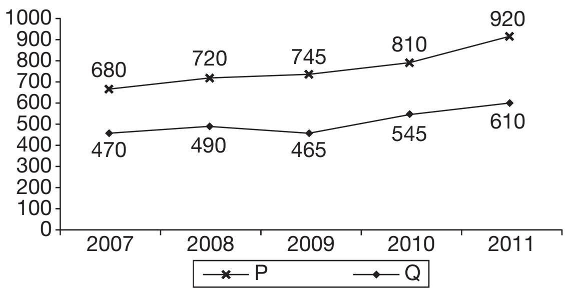
2. If in the year 2011, the ratio of the sales by value of air conditioners of the four brands is the same as the ratio of the total sales by value of the four brands, then find the sales (by value) of the air conditioners of LG in 2011.
(A) ₹15 lakh (B) ₹24 lakh
(C) ₹30 lakh (D) ₹18 lakh

3. In which year did the sales revenue from Samsung washing machines increased by the greatest percentage over that in the previous year?
(A) 2013 (B) 2014
(C) 2012 (D) 2015

4. Which of the following is the correct descending order of the four brands according to the percentage change in their respective sales (by value) from the year 2010 to the year 2015?
(A) Electrolux, Godrej, Samsung, LG
(B) Samsung, Electrolux, LG, Godrej
(C) Electrolux, Samsung, Godrej, LG
(D) Samsung, Electrolux, Godrej, LG

Directions for questions 5 to 8: Answer these questions based on the information given below.

The following line graph gives the sales of company P and expenses of company Q across five years.



The table gives the profit percentage of the companies in these years.

Year	P	Q
2007	17.2	20.3
2008	18.1	24.8
2009	17.5	19.5
2010	18.6	20.8
2011	19.2	21.6

$$\text{Profit\%} = \frac{\text{Profit}}{\text{Expenses}} \times 100$$

Profit = Sales – Expenses

5. In which year was the difference between the sales of P and Q, the least?
(A) 2007 (B) 2008
(C) 2010 (D) 2011

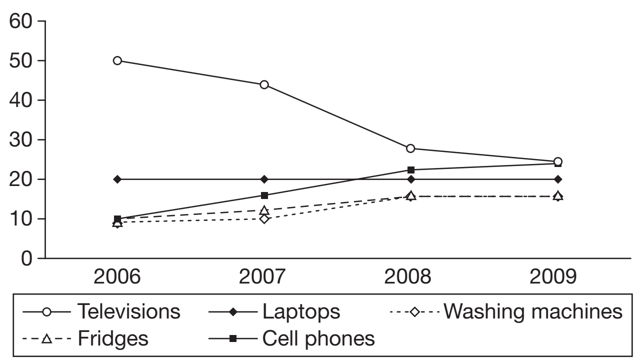
6. What is the percentage increase in the profit of company Q in the given period?
(A) 38.1 (B) 36.2
(C) 34.3 (D) 32.1

7. What is the average profit of company P from 2007 to 2011?
(A) 122.6 (B) 114.3
(C) 117.1 (D) 119.2

8. In how many years was the percentage increase in expenses of company P, more than that of company Q, for the same year?
(A) 1 (B) 2
(C) 3 (D) 4

Directions for questions 9 to 12: Answer these questions based on the information given below.

XYZ, an electronics company, sells five types of products. The following graph shows the percentage of revenue each product contributed to the total revenue of the company in four years from 2006 to 2009.



The total revenue in 2008 was more than that in 2006 while the total revenue in 2009 was less than that in 2007 and the total revenue in 2007 was more than that in 2008.

9. The sales of at most how many products could have increased continuously from 2006 to 2009?

(A) 1 (B) 2
(C) 3 (D) 4

10. At most how many of the following statements can be simultaneously true?

The revenue from sales of washing machines in the year 2008 is

(i) Less than that in 2009
(ii) Greater than that in 2007
(iii) Greater than that in 2006
(iv) Less than 80% of that in 2009

(A) 1 (B) 2
(C) 3 (D) 4

11. The revenue from sales of televisions in 2006 is at most how many times the revenue from sales of cell phones in 2007?

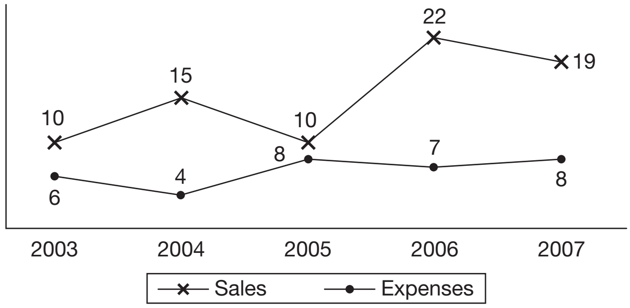
(A) 2.5 (B) 3.0
(C) 3.33 (D) 3.6

12. What is the least possible number of products which had a definite decrease in sales from 2007 to 2009?

(A) 0 (B) 1
(C) 2 (D) 3

Directions for questions 13 to 17: Answer these questions based on the information given below.

The following line graph gives the percentage increase in the sales and expenses, when compared to the previous year, of company XYZ Ltd, for a period of five years.



It is known that the company made a profit in 2002.

13. In how many of the given years did the company make a profit?

(A) 2 (B) 3
(C) 4 (D) 5

14. In which of the given years was the profit made by the company, the highest?

(A) 2003 (B) 2004
(C) 2007 (D) 2006

15. In which of the given years was the increase in the expenses of the company, over the previous year, the highest?

(A) 2005 (B) 2006
(C) 2007 (D) Both 2005 and 2007

16. In which of the given years was the increase in sales of the company, over the previous year, the highest?

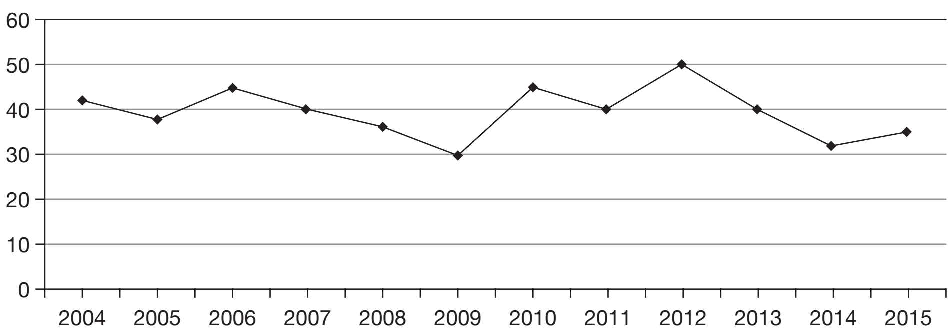
(A) 2007 (B) 2006
(C) 2004 (D) 2003

17. If the ratio of sales to expenses in 2002 was 2 : 1, then what is the ratio of the profits made by the company in 2007 and 2003?

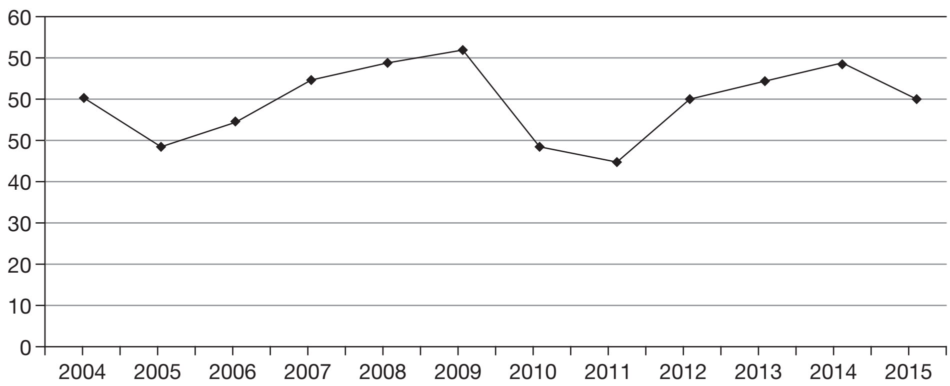
(A) 7 : 3 (B) 8 : 3
(C) 7 : 4 (D) 3 : 1

Directions for questions 18 to 22: Answer these questions based on the information given below.

Total petroleum production as a percentage of total crude production



Total petroleum consumption as a percentage of total crude consumption



Further, the total crude production (when compared to previous year) increased every alternate year starting from 2004 (i.e., 2006, 2008, …….) and in the remaining years it decreased.

The total crude consumption, when compared to the previous year decreased every year from 2004.

18. What is the maximum possible number of years from 2005 to 2015 in which the total petroleum production increased over the previous year?

19. In at most how many of the years from 2004 to 2011 could the petroleum consumption be the same as that in 2010?

20. What is the minimum possible number of years from 2005 to 2015 in which the total petroleum consumption decreased when compared to the previous year?

21. What is the minimum possible number of years in which the total petroleum production decreased when compared to the previous year?

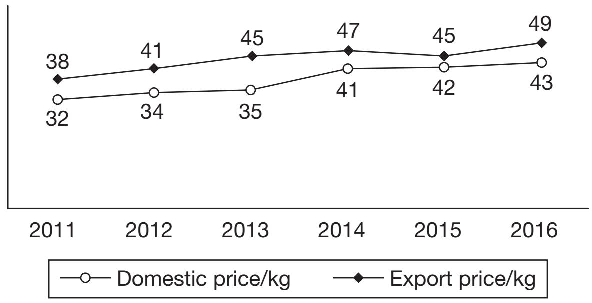
22. In at most how many of the given years can the petroleum consumption be equal to 5/4 times the petroleum consumption in 2005?

Directions for questions 23 to 26: These questions are based on the following information.

The products of company XYZ are sold both domestically and in the international market. Every year, the exports take place immediately after the production and the remaining products are sent to the godowns. Some quantity is damaged during this process of storing while the others are completely sold off in the domestic market.

The table gives the details of production, export and quantity lost during the storage process while the line graph gives the export and domestic prices for the years 2011 to 2016. Assume that both domestic and export prices remain constant in a year.

Year	Production ('000 kg)	Exports ('000 kg)	Damage during storing ('000 kg)
2011	137	42	5.0
2012	134	52	6.5
2013	161	73	7.2
2014	186	67	4.7
2015	172	75	8.1
2016	197	82	7.8



Note: The damaged quantity is discarded and the company incurs a cost of ₹3/kg in disposing it off.

Total revenue in a year = Amount obtained from sale of products – Cost incurred in disposing off the damaged products

Domestic revenue = Quantity sold in the domestic market × Domestic price/kg.

23. What was the percentage increase in the revenue of the company from 2012 to 2013?

(A) 28.3 (B) 29.1
(C) 30.2 (D) 31.4

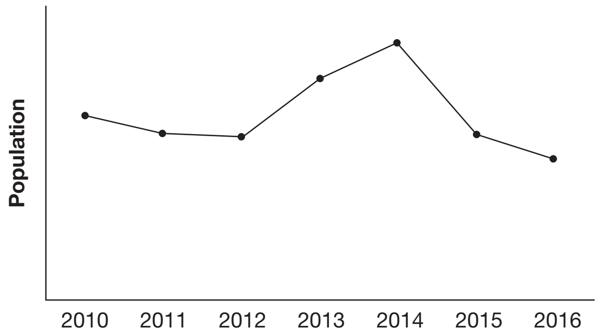
24. What was the highest value of the revenue per kg of the quantity produced, in any of the given years?

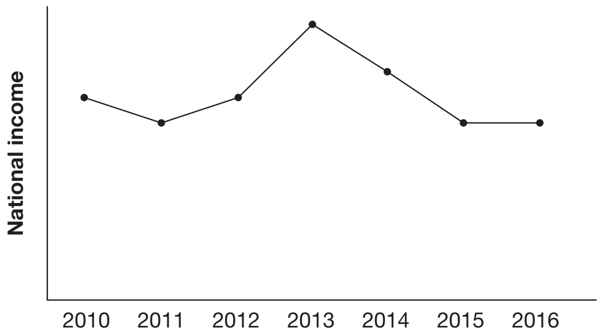
(A) 46.2 (B) 45.1
(C) 43.6 (D) 41.5

25. What was the highest percentage increase in domestic revenue when compared to the previous year?
(A) 65.7 (B) 58.4
(C) 59.4 (D) 60.2

26. The highest value of revenue from sales in the domestic market, as a percentage of the total revenue in that year, is
(A) 64% (B) 61%
(C) 59% (D) 56%

Directions for questions 27 to 29: Answer these questions based on the information given below.





The line graphs give the trend of population and the national income of a country.

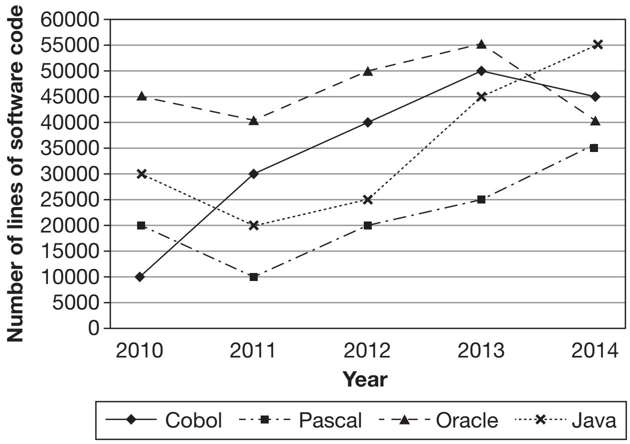
$$\text{Per capita income} = \frac{\text{National income}}{\text{Population}}$$

27. In at least how many years was there an increase in the per capita income when compared to the previous year?
(A) 0 (B) 1
(C) 2 (D) 3

28. In at least how many years was there a decrease in the per capita income when compared to the previous year?
(A) 0 (B) 1
(C) 2 (D) 3

29. In at most how many years did the per capita income remain the same as that in the previous year?
(A) 0 (B) 1
(C) 2 (D) 3

Directions for questions 30 to 33: These questions are based on the following line graph and table.



The line graph above represents the number of lines of software code written by four different programmers, namely Oracle, Pascal, Cobol and Java over five different years, from 2010 through 2014. The table given below indicates the rate (₹/line of code) charged by each of the four programmers over the given duration (2010–2014).

Rates (₹/line of code)

	Year				
Programmers	**2010**	**2011**	**2012**	**2013**	**2014**
Cobol	10	11	12	13	14
Pascal	11	12	12	13	14
Oracle	12	12	12	13	14
Java	13	14	15	16	17

Income of each of the programmer is the revenue realized from the lines of software code that the programmer writes.

30. The income of which programmer changed the maximum in percentage terms between any two consecutive years?
(A) Pascal (B) Java
(C) Cobol (D) Oracle

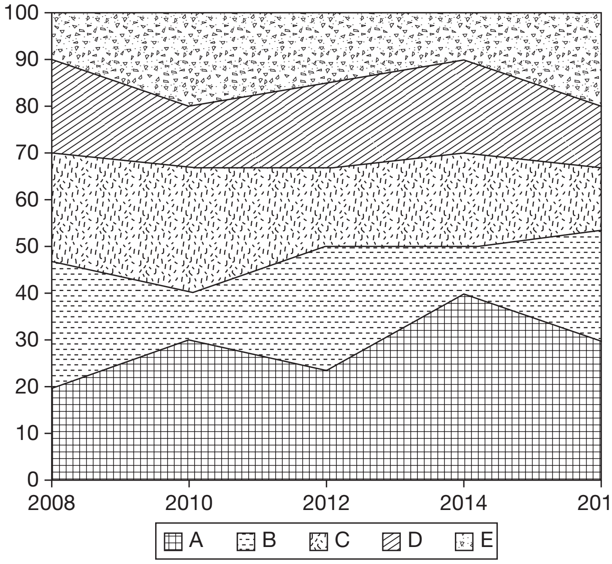
31. For how many years was the income of Java more than that of each of the other three programmers?
(A) 4 (B) 1
(C) 2 (D) 3

32. Income of Pascal in the year 2013 expressed as a percentage of that of Java in year 2010 is
(A) $83\frac{1}{3}\%$ (B) 120%
(C) 12% (D) None of these

33. In which year was the income of any of the programmers the maximum?
(A) 2010
(B) 2014
(C) 2013
(D) 2012

Directions for questions 34 to 37: These questions are based on the information given below.

The following graph gives the market share of five car companies in a country across five years. Assume that the total sales of these five companies increased by 20% every two years.



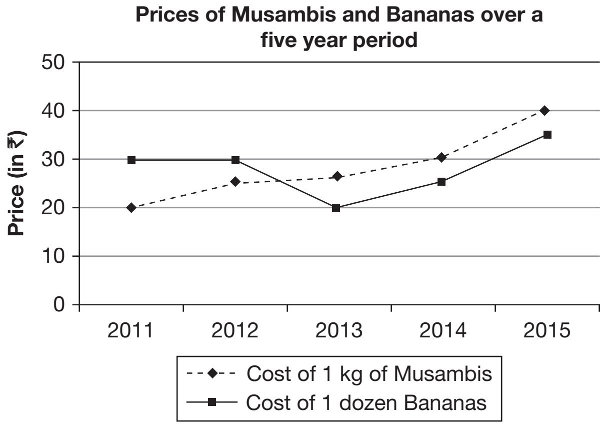
34. What was the percentage increase in the sales of company A from 2008 to 2012?
(A) 100 (B) 80
(C) 75 (D) 25

35. If the total sales of the five companies in 2008 were 2,85,000, then what were the sales of company D in 2014?
(A) 98,500
(B) 99,200
(C) 96,500
(D) 97,500

36. How many companies had an increase of more than 100% in their sales from 2008 to 2016?
(A) 2 (B) 3
(C) 1 (D) 4

37. Which company had the highest percentage increase in sales from 2010 to 2016?
(A) A (B) B
(C) D (D) E

Directions for questions 38 to 42: Answer these questions based on the information given below.



Due to the variation in the size of bananas produced in a particular year, the number of bananas per kg varies as per the table given below.

Average number of bananas per kg in various years	
2011	10
2012	10
2013	11
2014	12
2015	9

38. In 2011, what is the difference between the average price of 1 kg of musambis and 1 kg of bananas?
(A) ₹5 (B) ₹10
(C) ₹7.50 (D) None of these

39. If from 2015 to 2016, the cost of bananas increased by the same amount as it increased from 2014 to 2015, what would be the cost of 1 kg of bananas in 2016?
(A) ₹49 (B) ₹28
(C) ₹35 (D) Cannot be determined

40. Percentage increase in the cost of 1 kg of bananas from 2011 to 2015 is (negative signs show a decrease)
(A) 5% (B) 10%
(C) –5% (D) –10%

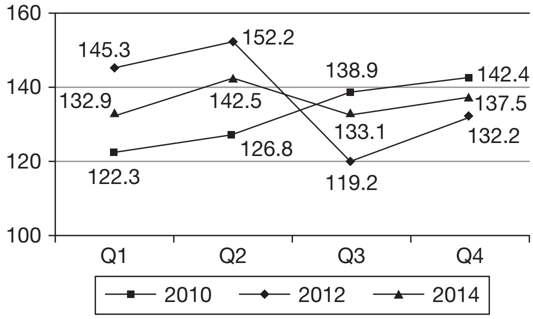
41. If the total production of musambis during 2011-15 was in the ratio of 1 : 2 : 1 : 3 : 2, then the average cost of 1 kg of musambis during the period would be
(A) ₹28 (B) ₹29.44
(C) ₹29 (D) ₹30

42. If the total production of bananas had remained constant throughout the given period, the average cost of 1 kg of bananas during the period would have been

(A) ₹28 (B) ₹29
(C) ₹24 (D) ₹31

Directions for questions 43 to 46: These questions are based on the information given below.

The following line graph shows the sales of mobile phones by company ABC (in ’000s), in the 4 quarters of the years 2010, 2012 and 2014.



The table below gives the percentage change in the sales of mobile phones of ABC in the different quarters of each year compared to the previous year from 2010 through 2015.

% Change	Q1	Q2	Q3	Q4
2010	–7.95	–2.0	–1.6	4.9
2011	8.25	4.8	–1.6	9.55
2012	9.77	14.52	–12.81	–15.23
2013	5.4	1.9	2.6	2.7
2014	–13.26	–8.08	8.87	1.22
2015	3.9	3.4	7.15	3.8

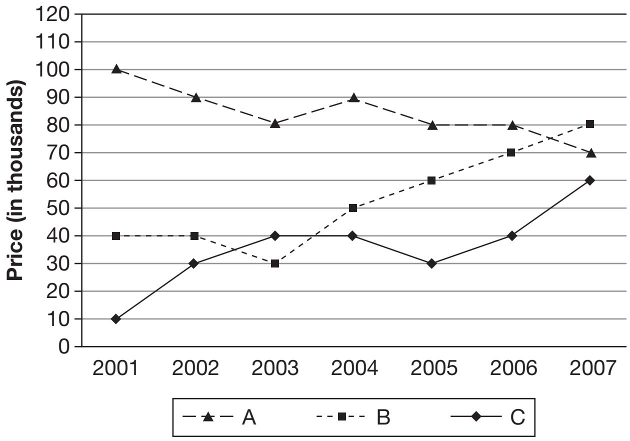
43. During the period 2009 to 2015, in how many years did the mobile phone sales of ABC exceeded 5,50,000?
(A) 1 (B) 2
(C) 3 (D) 4

44. In which of the following years during the given period is the change in the number of mobile phones sold compared to the previous year, the lowest?
(A) 2010 (B) 2012
(C) 2013 (D) 2015

45. What is the difference in the number of mobile phones sold by the company in the years 2009 and 2014?
(A) 6840 (B) 7120
(D) 6680 (D) 7280

46. What is the average number of mobile phones sold by the company in the quarter in which the most number of mobile phones were sold during the period 2010 to 2015?
(A) 136.74 (B) 142.8
(C) 140.9 (D) 141.1

Directions for questions 47 to 50: These questions are based on the graph given below.



The above graph gives the prices of products A, B and C for the period 2001–2007. The prices for products A and C are in rupees per cubic metre while that of B is expressed in rupees per ton. Assume that one ton is equal to 1000 kg and one cubic metre of B weighs 600 kg.

47. The maximum increase in price per cubic metre for any product over two successive years was

48. If one cubic metre of C weighs 800 kgs, the largest difference in price between one ton of C and that of one ton of B was in the year

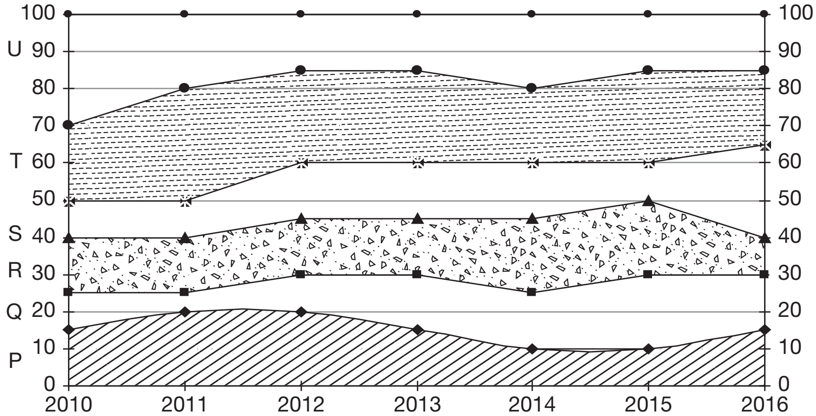
49. In 2005, the total sales of company (as measured in cubic metres) was made up of 50% A, 40% B and 10% C. The average realization per cubic metre in 2005 was

50. In 2008, the prices of A, B and C went up by 1%, 2% and 5%, respectively and the total sales measured in cubic metres were made up of 50% A, 30% B and 20% C. The average realization per cubic metre in 2008 was closest to

EXERCISE-3

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

A survey was done regarding the number of mobile phone subscribers of six companies in a country over the period 2010 to 2016. The graph gives the percentage break up of number of subscribers of these six companies P, Q, R, S, T and U. It is observed that, each year, the total number of subscribers of these six companies increased by 25% over the previous year.



1. What is the percentage increase in the number of subscribers of company P from 2011 to 2016?

(A) 110% (B) 121%
(C) 129% (D) 137%

2. The ratio of the number of subscribers of company R in 2013 and company T in 2015 was approximately

(A) 2 : 7 (B) 5 : 13
(C) 4 : 9 (D) 7 : 13

3. Which company had the maximum numerical increase in the number of subscribers from one year to the next?

(A) P (B) R (C) S (D) T

4. What was the percentage increase in the number of subscribers of all the six companies from 2010 to 2015?

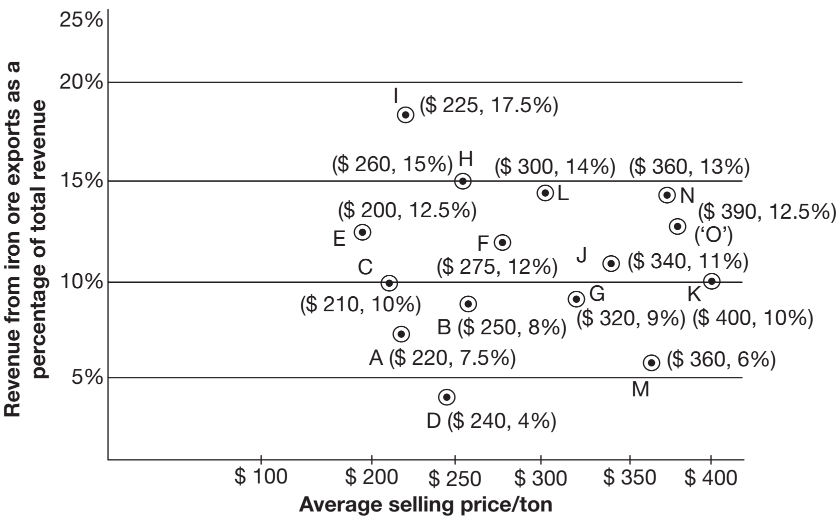
(A) 225% (B) 205%
(C) 180% (D) 175%

5. How many companies had a decrease in the number of subscribers from one year to the next, at least once during the given period?

(A) 2 (B) 4
(C) 5 (D) 6

Directions for questions 6 to 9: Answer the following questions based on the information given below.

The following table gives the average selling price per ton of iron ore and the revenue generated from iron ore exports as a percentage of the total revenue for 15 multinational companies among A, B, C, D, E, F G, H, I, J, K, L, M, N and O. The average selling price is given in US dollars.



6. It was found that the total volume of iron ore exported by companies B and E were equal. Which of the following is true?

(A) The total revenue in both the companies were same.
(B) Total revenue of B was 2 times that of E.
(C) Total revenue of B was 3 times that of E.
(D) Total revenue of E was 2 times that of B.

7. It is expected that in 2017, the revenue from iron ore exports as a percentage of the total revenue will double for C and quadruple for M. Assume that in 2017, the total revenue for C is twice that of M and the volume of iron ore exported by both the companies is same. What is the approximate percentage increase in the selling price per tonne of iron ore for company C if the percentage increase of the same for M is 25?

(A) 150% (B) 230%
(C) 260% (D) 360%

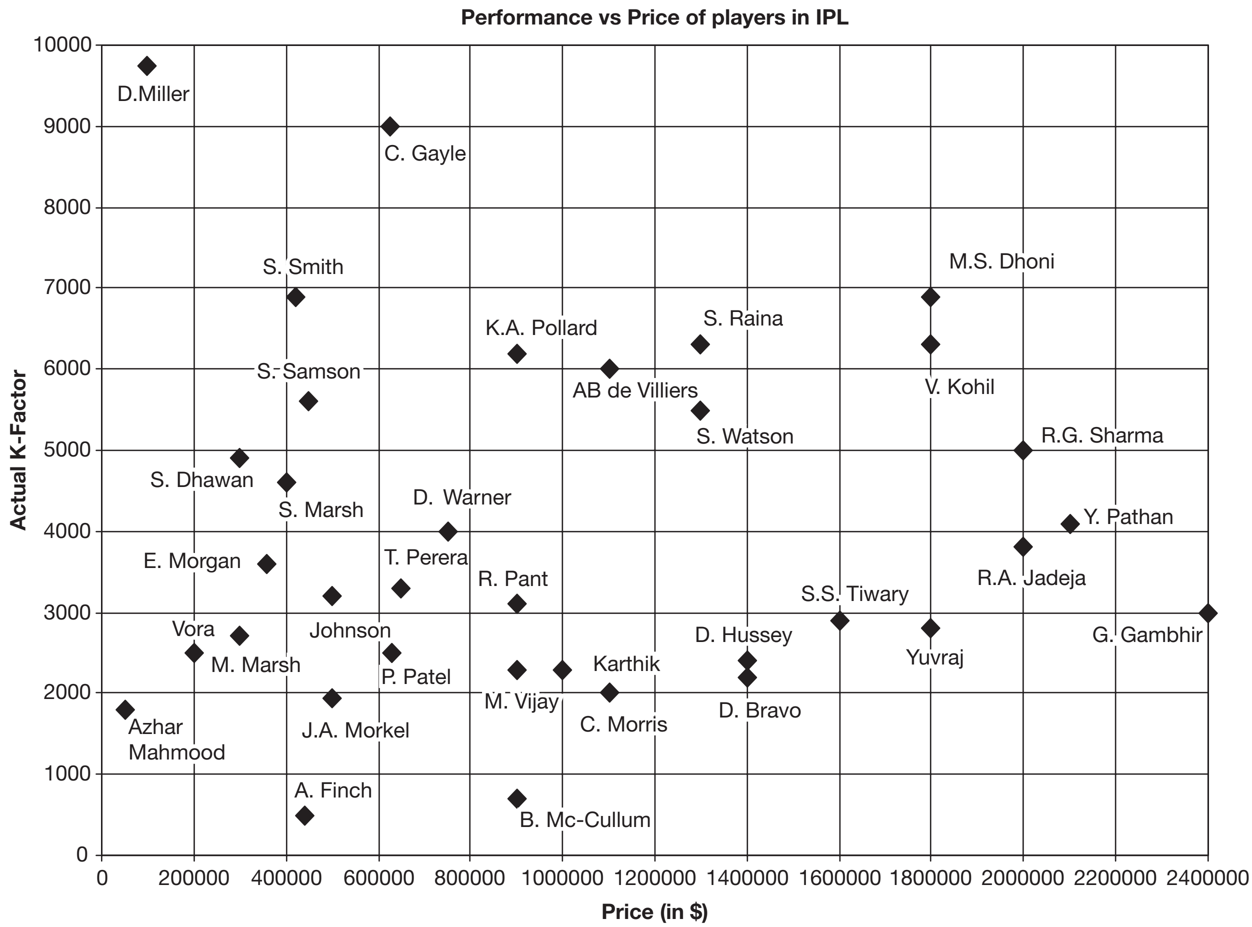
8. If the total revenue is the same for the pairs of companies listed in the choices below, choose the pair that has approximately the same volume of iron ore exports.

(A) A and C (B) F and N
(C) E and I (D) D and M

9. If the total volume of iron ore exported by all the companies were equal, which company had the highest total revenue?

(A) D
(B) I
(C) M
(D) More than one of the above

Directions for questions 10 to 13: These questions are based on the following information.



Note:

(1) For a player, the expected K-factor is directly proportional to the price.
(2) The return on investment (ROI) of a player is higher if his actual K-factor as a percentage of his price is higher.
(3) Expected (ROI) is the expected K-factor by the price.

10. Among players whose price is more than $1 million, whose ROI is the highest?
(A) AB de Villiers (B) S. Raina
(C) M. S. Dhoni (D) R. G. Sharma

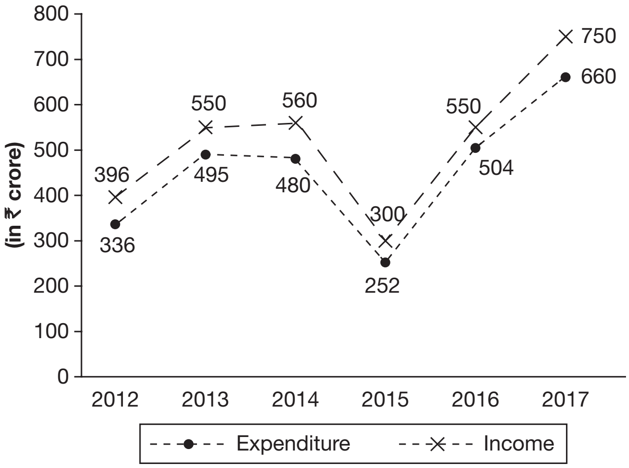
11. Which player was the biggest disappointment in terms of ROI?
(A) A. Finch (B) McCullum
(C) G. Gambhir (D) None of these

12. How many players had a higher ROI than Chris Gayle?
(A) 2 (B) 3
(C) 4 (D) 5

13. If the expected K-factor for a price of $600000 is 3000 and that for a price of $1800000 is 9000, then ROI for how many players was better than the expected ROI?
(A) 13 (B) 15
(C) 17 (D) 19

Directions for questions 14 to 16: Answer these questions on the basis of the information given below.

The income and expenditure of two companies A and B from 2012 to 2017



Ratio of expenditure and income of the two companies

Year	Expenditure	Income
2012	4 : 3	6 : 5
2013	6 : 5	6 : 5
2014	7 : 5	4 : 3
2015	3 : 4	2 : 3
2016	4 : 5	5 : 6
2017	5 : 6	7 : 8

Note:
(i) Profit = Income – Expenditure;
(ii) $\text{Profit \%} = \dfrac{\text{Profit}}{\text{Expenditure}} \times 100\%$

14. What is the profit earned (in ₹ crore) by company B in 2012, 2014 and 2017 together?

15. What is the difference (in ₹ crore) between the profit earned by company A and company B over the years 2013, 2014 and 2015 together?

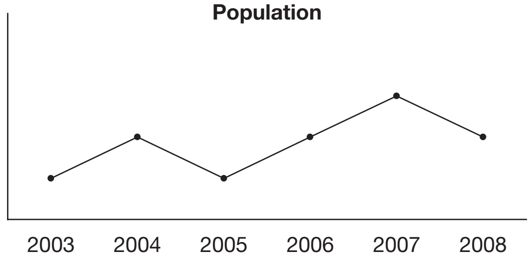
16. In which year, was the income of company A, the second highest?

Directions for questions 17 to 20: Answer the questions on the basis of the information given below.

The following graph gives the details of population of a country, its per capita consumption of rice and the production of rice in the country for a period of six years.

Production of rice

Population



If the total consumption of rice is more than the production of rice in the country, the extra rice is imported that year and if the total consumption is less than the production in that year, the excess rice is exported the very same year.

17. At least in how many of the given years did the country export rice?
(A) 0 (B) 1
(C) 2 (D) Cannot be determined

18. At most in how many years did the consumption of rice in the country when compared to previous year show an increase?
(A) 5 (B) 4
(C) 3 (D) 2

19. If it is known that the country exported rice in the year 2003, then at most in how many years did it import rice in the given period?

(A) 5 (B) 3
(C) 2 (D) 1

20. At most in how many years, even though the population of the country increased, did the total consumption of rice in the country decrease, both when compared to the previous year?

(A) 1 (B) 2
(C) 3 (D) 4

ANSWER KEYS

Exercise-1

1. 1992	**10.** (A)	**19.** (B)	**28.** 2	**37.** (D)	**46.** (B)
2. 47	**11.** (D)	**20.** (D)	**29.** 9.6	**38.** (A)	**47.** (D)
3. 36.36	**12.** (B)	**21.** (A)	**30.** 57	**39.** (C)	**48.** (A)
4. 1875	**13.** (A)	**22.** (A)	**31.** (C)	**40.** (B)	**49.** (B)
5. 1997	**14.** (D)	**23.** (D)	**32.** (B)	**41.** (A)	**50.** (D)
6. (C)	**15.** (D)	**24.** (C)	**33.** (B)	**42.** (D)	
7. (B)	**16.** (D)	**25.** (B)	**34.** (D)	**43.** (C)	
8. (C)	**17.** (A)	**26.** 2001	**35.** (C)	**44.** (B)	
9. (D)	**18.** (A)	**27.** 12.6	**36.** (A)	**45.** (D)	

Exercise-2

1. (D)	**10.** (D)	**19.** 0	**28.** (B)	**37.** (B)	**46.** (B)
2. (A)	**11.** (C)	**20.** 4	**29.** (D)	**38.** (A)	**47.** 20000
3. (B)	**12.** (C)	**21.** 5	**30.** (B)	**39.** (D)	**48.** 2001
4. (C)	**13.** (D)	**22.** 5	**31.** (A)	**40.** (A)	**49.** 57.4
5. (B)	**14.** (C)	**23.** (C)	**32.** (B)	**41.** (B)	**50.** 62.65
6. (A)	**15.** (C)	**24.** (C)	**33.** (D)	**42.** (C)	
7. (D)	**16.** (A)	**25.** (A)	**34.** (B)	**43.** (C)	
8. (C)	**17.** (A)	**26.** (A)	**35.** (A)	**44.** (A)	
9. (C)	**18.** 6	**27.** (C)	**36.** (B)	**45.** (A)	

Exercise-3

1. (C)	**4.** (B)	**7.** (C)	**10.** (A)	**13.** (B)	**16.** 2014	**19.** (A)
2. (B)	**5.** (D)	**8.** (D)	**11.** (B)	**14.** 116	**17.** (D)	**20.** (B)
3. (C)	**6.** (B)	**9.** (D)	**12.** (C)	**15.** 19	**18.** (C)	

SOLUTIONS

EXERCISE-1

1. The ratios of actual exports of tea to the estimated exports of tea from 1992 to 1997 are 3 : 2, (1.5), 9 : 8 (125), 13 : 12 (1.083), 12 : 10, (1.2), 14 : 15 (0.93), 17 : 17 (1)
So, in 1992 the ratio of actual exports is the highest when compared to estimated exports.
Hence, in 1992, the ratio of the actual exports to the estimated exports is the highest.

2. Total actual exports for the given period
= (3 + 9 + 13 + 12 + 15 + 17) lakh = 69 lakh
Total exports to the U.S. = 1.6 + 4 + 6 + 6 + 5 + 10
= 32.6 lakh
Required per cent = (32.6/69) (100) = 47%

3. Let the actual value of 1 dollar be ₹x. Number of dollars to be received when 1 dollar is ₹40 = $\frac{5.5\times10^5}{40}$

Number of dollars received when 1 dollar is ₹$x = \frac{5\times10^5}{x}$

$\Rightarrow \frac{5.5\times10^5}{40} = \frac{5\times10^5}{x} \Rightarrow x =$ ₹36.36

Alternate Solution
The decrease in exports is due to the appreciation of the rupee.

Percentage decrease in exports = $\frac{50000}{5,50,000}\times100 = 9.09\%$

The percentage appreciation in rupee would be the same, i.e., 9.09%.

$\therefore$ Value of \$1 = $40\times\frac{(100-9.09)}{100} = 40\times\frac{90.91}{100} = 36.36$

4. Quantity exported to the US = $\frac{6\times10^5}{8\times40} = 1875$ kg

5. The ratio of exports of tea to the US to actual exports from 1992 to 1997 are in the same order as 1.6 : 3 (i.e., 0.53) 4 : 9 (i.e., 0.44); 6 : 13 (i.e., 0.46) 6 : 12 (i.e., 0.50); 5 : 14 (i.e., 0.33); 10 : 17 (i.e., 0.58)
Hence, in 1997, the required ratio is the highest.

6. If the number of students in 2010 is 100, the number of students in 2013 would be

$100\times\frac{107}{100}\times\frac{106}{100}\times\frac{102}{100} = 115.68$

The percentage increase $\simeq$ 16.

7. The fee in 2013 = $24000\times\frac{112}{100}\times\frac{109}{100} = 31.740$

8. Number of students in 2012 = $\frac{23,348}{1.09\times1.02} = 21,000$

9. The number of students in the coaching institute went up by 20% from 2012 to 2014.
Assume that the total number of students who took coaching in 2012 were 100. The number in 2014 would be 115.
The share of the coaching institute in 2014

$= \frac{32\times1.11}{1.15}\times100 = \frac{35.58}{115}\times100 \cong 31\%$

10. The number of students in 2011 = $\frac{29.375\times10}{12,500} = 23,500.$

Number of students in 2013 = 23,500 × 1.06 × 1.02
= 25408
Fee collected per student in 2013
= 12,500 × 1.12 × 1.09 = 15260
Total fee collected = 38.77 crore

11. Let the expenditures of the companies P and Q in the year 2011 be $6x$ and $8x$, respectively.
The profit of company P in the year 2011

$= \frac{35}{100}\times6x = 2.1x$

The profit of company Q in the year 2011

$= \frac{40}{100}\times8x = 3.2x$

The required ratio = $2.1x : 3.2x = 21 : 32$.

12. Let the expenditure of company R in the year 2009 be ₹x crore.

Profit = $\frac{35}{100}x = 0.35x$

$x - 0.35x = 2.99$
$0.65x = 2.99$

$x = \frac{2.99}{0.65} = 4.6$

Income = Profit + Expenditure = 0.35 (4.6) + 4.6 = 6.21.

13. Profit percentage = $\frac{\text{Profit}}{\text{Expenditure}}\times100$

Profit = $\frac{\text{Expenditure}\times\text{Profit percentage}}{100}$

Income = Expenditure + $\frac{\text{Expenditure}\times45}{100}$

7.25 = 1.45 (expenditure)

Expenditure = $\frac{7.25}{1.45} = 5$ crore

14. We do not know the expenditures of the companies, so we can't find which company had the highest profit.

15. If the incomes of P, Q and R are in the ratio of 5 : 4 : 3, their expenditures approximately would be in the ratio 74 : 57 : 50.
The profits would be in the ratio 26 : 23 : 10.

The required percentage = $\frac{26}{33} \times 100 = 79\%$

16. Given S.P. = 3.9 lakh

$3.9 = \text{C.P.}\left(+1\frac{30}{100}\right)$

$\Rightarrow$ C.P. = 3 lakh
$\therefore$ Profit = S.P. - C.P. = 3.9 - 3 = 0.9 lakh

17. Let the C.P. of A and C in 2003 and 2007 respectively, be 100. Then the
S.P. of A in 2003 = 120
S.P. of C in 2007 = 145

$\therefore$ Required ratio = $\frac{120}{145} = \frac{24}{29}$

18. Let the S.P. of B in 2004 and 2005 be 100.
Therefore,

C.P in 2004 = $\frac{160}{1.4}$

C.P in 2005 = $\frac{100}{1.45}$

$\therefore$ Required ratio = $\frac{100}{1.4} : \frac{100}{1.45}$

= 1.45 : 1.4 = 29 : 28

19. Let the S.P. of B in 2006 and its C.P. in 2007 be ₹100. S.P. of B in 2007 = 150
$\therefore$ Required ratio = 100 : 150 = 0.66.

20. Percentage rise or fall in A is as follows:

2003: $\frac{20}{40} \times 100 = 50\%$

2004: $\frac{15}{20} \times 100 = 75\%$

2005: $\frac{15}{35} \times 100 = 42.85\%$

2006: $\frac{5}{50} \times 100 = 10\%$

2007: $\frac{10}{45} \times 100 = 22.22\%$

21. Expenditure of P = ₹85 lakh.

As profit percentage = 14, profit = $\frac{14}{100} \times 85 = 11.9$.

$\therefore$ Income = 85 + 11.9 = 96.9 lakh.

22. Income = Expenditure + Profit
The expenditure of P and Q in 2013 would be 200 and 90, respectively ($224 = 200 + 200 \times \frac{12}{100}$)

The required ratio = 20 : 9

23. Let the expenditure be 100 each in 2009.

Income of P in 2009 = $100 + 100 \times \frac{8}{100} = 108$.

Income of Q in 2009 = $100 + 100 \times \frac{13}{100} = 113$.

The required ratio = 108 :113.

24. As profit = Expenditure × Profit percentage

Expenditure = $\frac{4.23}{9}$ 100 = 47 crore.

Income = 47 + 4.23 = 51.23 crore.

25. Let the profit be ₹4 crore and ₹5 crore, respectively.
If the expenditure of P and Q are x and y, respectively.

$\frac{11}{100} \times x = 4$ crore $\therefore x = \frac{400}{11}$ crore

$\frac{13}{100} \times y = 5$ crore $\therefore y = \frac{500}{13}$ crore

The required ratio = 52 : 55.

26. In the year 2001, the number of employees and profit per employee per annum is also the maximum. So, in the year 2001, the profit of the company was the highest.

27. Profit in year 1998 = 145 × 2.6 lakh
30% of total sales value in the year 1998 = 145 × 2.6 lakh
Total sales value in the year 1998

$= \frac{145 \times 2.6 \times 100}{30}$ lakh = 1257 lakh = 12.57 crore

28. Average number of employees for the given period

$= \frac{120 + 145 + 218 + 150 + 192}{5} = 165$

In the years 1999 and 2001, the number of employees is more than the average number of employees.

29. Profit made in the year 2001 = 192 × 3.2 lakh. So, 192 × 3.2 lakh is 64% of the profit made in 2002.

Profit in 2002 = $\frac{192 \times 3.2}{64} \times 100$ lakh

= 960 lakh = ₹9.6 crore

30. Profit in the year 1997 = 120 × 2 lakh = ₹240 lakh
Profit in the year 1998 = ₹145 × 2.6 lakh = ₹377 lakh

Required percentage = $\frac{377 - 240}{240} \times 100 = 57\%$

31. The approximate percentage growth in net profit of company A from 2006 to 2012 was
$1.1 \times 1.15 \times 1.17 \times 1.08 \times 1.12 \times 1.1 = 1.969 \simeq 97\%$

32. If the net profit of company B in 2006 was ₹100, its net profit in 2010 would have been approximately
$100 \times 1.05 \times 1.07 \times 1.02 \times 1.1 = 126$ in 2010.
Hence, the value in 2010 would be 1.26 times the value in 2006. Since the actual profit in 2006 is ₹190 crore, the required value $= 180 \times 1.26 = 227$ crore

33. Assume that the net profits of company A in 2006 was $100x$ and that of company B was $100y$.
The net profit of company A in 2009 $= 148x$
That in 2010 $= 160x$
Increase in profit $= 12x$
The net profit of company B in 2009 $= 115y$
That in 2010 $= 126.5y$
Increase in profit $= 11.5y$
$12x > 11.5y$
$\therefore \frac{x}{y} > \frac{11.5}{12} = \frac{23}{24}$
Hence, the ratio of 10 : 13 is not possible.

34. The cost of pill for each company is as follows:
Johnson and Johnson $= 200 \times 1.1 + 300 \times 1.3 + 100 \times 1.2 + 400 \times 1.4 = 1290$
Pfizer $= 250 \times 1.1 + 150 \times 1.3 + 200 \times 1.2 + 900 \times 1.4 = 1270$
Roche $= 100 \times 1.1 + 300\ (1.3 + 1.2 + 1.4) = 1280$
Sanofi $= 200\ (1.1 + 1.3 + 1.2) + 400 \times (1.4) = 1280$
Ell Lilly $= 250\ (1.1 + 1.3 + 1.2 + 1.4) = 1250$
The cost is the least for Ell Lilly.

35. The weight of the pill is 1000 mg.
Johnson and Johnson $= 200 + 300$
$= 500$ and $300 + 100 = 400$
Pfizer $= 250 + 150 < 500$ and $150 + 200 < 400$
Roche $= 100 + 300 < 500$ and $300 + 300 > 400$
Sanofi $= 200 + 200 < 500$ and $200 + 200 = 400$
Ell Lilly $= 250 + 250 = 500$ and $250 + 250 > 400$
For two of the companies the given condition is satisfied.

36. Cost of the pill for each company is as follows:
Johnson and Johnson
$= 200 \times 1.21 + 300 \times 1.3 + 100 \times 1.44 + 400 \times 1.4 = 1336$
Pfizer $= 250 \times 1.21 + 150 \times 1.3 + 200 \times 1.44 + 400 \times 1.4 = 1345.5$
Roche $= 100 \times 1.21 + 300\ (1.3 + 1.44 + 1.4) = 1363$
Sanofi $= 200\ (1.21 + 1.3 + 1.44) + 400 \times 1.4 = 1350$
Ell Lilly $= 250\ (1.21 + 1.3 + 1.44 + 1.4) = 1337.5$
The cost will be the highest for Roche.

37. In none of the groups of ten matches, given in choices (1), (2), (3) and (4), did both Amol and Brag score the same number of runs. Number of runs scored by Amol from the 31st to the 40th match $= 40 \times 60 - 30 \times 70 = 300$
Similarly, values for others can be determined.

38. By similar calculations as above, Amol has the highest average from the 61st to the 70th match.

39. His least possible highest score from the 31st to the 40th match will be when he scored (the maximum) 1800 runs in 10 matches. The least possible highest score will be 185 and the other 9 scores will be 184, 183…175.

40. It happened for the 31st to the 40th match, when the number of runs scored by Chris is more than that of Amol but less than that of Brag.

41. Capacity $= \frac{\text{Output}}{\text{Utilization}} \times 100 = \frac{9.1}{45} \times 100 = 20.2$ bn tons

42. In 1996–1997 $\rightarrow \frac{8.9}{75} \times 100 \cong 11.86$

In 1991–1992 $\rightarrow \frac{9}{60} \times 100 = 15$

In 1994–1995 $\rightarrow \frac{9.1}{45} \times 100 \cong 20.2$

In 1989–1990 $\rightarrow \frac{4.5}{45} \times 100 = 10$

43. In Statement I, the decreases are not in the same proportion.
In Statement II:
1991–92, capacity = 15
1993–94, capacity = 32
Statement III is obviously wrong (by observation).

44. As shown in the graph during (93–94) to (94–95), imports increased from 150 to 200 and exports decreased from 600 to 500.

45. Capacity in 1989–1990 $= \frac{4.5}{45} \times 100$ billion tons
$= 10$ billion tons

Capacity in 1996–1997 $= \frac{8.9}{75} \times 100$ billion tons
$= 11.9$ billion tons

Average annual increase $= \left(\frac{11.9 - 10}{10}\right) \times \frac{1}{7} \times 100 \cong 2.7\%$

46. It is given that the profit in 2012 was ₹1 crore.
$\therefore 109$ (Sales) > 107 (Expenses)
As the corresponding ratio of sales and profit in 2011 and 2013 are more than $\frac{109}{107}$, we can definitely conclude that the company made a profit in these years.

47. The given information is not sufficient to find out in which year the profit was the least.

48. From 2010 to 2011, the percentage increase in sales was 14%, which was the highest.

49. Sales in 2012 = 272.5 crore
Profit in 2012 = 271.5 crore
$107x = 271.5 \qquad \therefore\ 112x = 284.2$

Sales is 2013 $= \frac{118}{100} \times 250 = 295$ crore.

Profit in 2013 $= 295 - 284.2 = 10.8$ crore.

50. As the value of sales and expenditure cannot be determined, we cannot find the percentage decrease in profits.

EXERCISE-2

1.

Year	Total sales	Frost-Free %
2010	75	12%
2011	100	24%
2012	125	24%

$\therefore$ Total sales of Samsung Frost-Free Refrigerators

$= (₹75 \times 0.12 + 100 \times 0.24 + ₹125 \times 0.24)$ lakh

$= (₹9 + ₹24 + ₹30) = ₹63$ lakh

2. **Year 2011:** Ratio of the total sales of the companies Godrej, Electrolux, LG and Samsung = 150 : 100 : 75 : 100 = 6 : 4 : 3 : 4
Given that it is same as the ratio of air conditioner sales.

A	E	L	S
6 :	4 :	3 :	4

Now we know that for Samsung in 2011, the value of the Air Conditioner sales = 24% of (Total sales) = (0 × 20) (100) = ₹20 lakh
From the above ratio, 4k = ₹20 lakh
$\therefore$ 3k = ₹15 lakh.

3. By observing the bar chart, the sales of Samsung Washing Machines as a percentage of the total sales of Samsung brand, experienced the steepest increase (from 12 percentage points to 28 percentage points) in 2014. Also, the total sales of Samsung experienced the steepest climb in the same year. 2014 must be the answer.

Alternative method:

Year	Value of sales of Washing M/Cs Samsung
2010	$75 \times 32\% = ₹24$ lakh
2011	$100 \times 20\% = ₹20$ lakh
2012	$125 \times 20\% = ₹25$ lakh
2013	$150 \times 12\% = ₹18$ lakh
2014	$200 \times 28\% = ₹56$ lakh
2015	$175 \times 32\% = ₹56$ lakh

By observation, the percentage increase is the highest in the year 2014.

4. By observation, Electrolux and LG must have the highest and the least percentage changes, respectively. By this we can eliminate choices (B) and (D). Among the other two brands we can observe that increase from 2010 to 2015 is the same but the base (in 2010) is lesser for Samsung. Hence, Samsung will experience higher growth rate. Thus choice (C) is correct.

5. The sales of Q in the different years are
$2007 - 470 \times 1.203 = 565$
$2008 - 490 \times 1.248 = 612$
$2009 - 465 \times 1.195 = 556$
$2010 - 545 \times 1.208 = 655$
$2011 - 610 \times 1.216 = 742$
The difference is the least in 2008.

6. The profit of company Q in 2007 = $20.3 \times 470 = 95.4$
The profit of company Q in 2011 = $21.6 \times 610 = 131.7$

$$\text{The percentage increase} = \frac{131.7 - 95.4}{95.4} \times 100$$

$$= \frac{36.3}{95.4} \times 100 = 38.1\%$$

7. The profit of company P in the different years are as follows:

Year	Profit
2007	$680 - \frac{680}{1.172} = 100$
2008	$720 - \frac{720}{1.181} = 110$
2009	$745 - \frac{745}{1.175} = 111$
2010	$810 - \frac{810}{1.186} = 127$
2011	$920 - \frac{920}{1.192} = 148$

$$\text{The average profit} = \frac{100 + 110 + 111 + 127 + 148}{5}$$

$$= \frac{596}{5} = 119.2$$

8. The expenses of companies P and Q and the percentage increase in the different years are as follows:

Year	P	%	Q	%
2007	580		470	
2008	610	5	490	4
2009	634	4	465	–5
2010	683	7.5	545	17
2011	772	13	610	12

It is true for three years, such as 2008, 2009 and 2011.

	Percentage			
Years	2006	2007	2008	2009
Cell phones	10	15	22	25
Laptops	20	20	20	20
Washing machine	10	10	15	15
Fridges	10	12	15	15
Televisions	50	43	28	25

9. It could have increased continuously for cell phones, for fridges (if the total revenue in 2009 was more than that in 2008) and for washing machines.

10. All the four statements can be simultaneously true.

11. Let the total revenue in 2006 be x and that in 2009 be y.
$y > x$
Revenue from sales of televisions in 2006 = $x/2 = 0.5x$
Revenue from the sales of cell phones in 2007 = 15% of the total sales = $0.15y$.
To find the maximum ratio, revenue from total sales in 2006 and 2007 must be nearly equal.
∴ It can be at most

$\frac{0.50}{0.15} = 3.33$ times the revenue from sales of cell phones in 2007.

12. The total revenue in 2007 is more than that in 2009.
As the share of laptops remain at 20% and the share of televisions decreased, these two items would definitely show a decrease.

13. As it is given that the company made a profit in 2002, sales is more than the expenses in that year. As in each year from 2003, the growth in sales is more than the growth in expenses, the company would have made a profit in each of the given years.

14. As it is given that the company made a profit in 2002 and the percentage increase in sales in each year is more than the expenses, the profit would increase each year and would be the highest in 2007.

15. Even though the percentage increase in sales is the same in 2005 and 2007, as the percentage increase in 2007 is on a higher base, the increase in expenses will be highest in 2007.

16. As the percentage increase in sales values are the highest in 2006 and 2007, and those being the years at the end, we need to only check for the years 2006 and 2007. Assume that the value of sales in 2005 is 100. The sales in 2006 will be 122 and that in 2007 will be $122 \times \frac{119}{100} =$ 145.18.
∴ The increase is the highest in 2007.

17. Assume that the sales in 2002 was 200 and the expenses is 100. The corresponding values in the following years are as follows:

Year	2003	2004	2005	2006	2007
Sales	220	253	278.3	339.5	404
Expenses	106	110.2	119.0	127.3	137.5

∴ The profit in 2007 = 404–137.5=266.5

The required ratio = $\frac{266.5}{114} = \frac{7}{3}$

Solutions for questions 18 to 22:

It is given that the crude consumption, compared to the previous year, decreased in every year and the share of petroleum consumption is also given for each of the years. From these two, regarding consumption of petrol we can say the following.

(i) Petroleum consumption will definitely decrease, if the share of petrol in crude consumption decreases.
(ii) Petroleum consumption may increase or decrease or remain the same, if the share of petrol in crude consumption increases.

The same applies for the production of petrol also, but in alternate years.

18. The maximum possible number of years in which the production of petroleum increased is 6, i.e., 2006, 2008, 2010, 2012, 2014 and 2015.

19. As the total crude consumption decreased every year from 2004, the crude consumption in 2011 was the lowest till that point, starting from 2004. As petroleum consumption is given as a percentage of crude consumption and as the percentage in 2010 is the lowest till that point, the petroleum consumption would be lowest in 2010. The consumption in 2011 would also be lower than in 2010.

20. The minimum possible number of years is 4, i.e., 2005, 2010, 2011 and 2015.

21. The minimum possible number of years is 5, i.e., 2005, 2007, 2009, 2011 and 2013.

22. The consumption in 2005 is 24%

$\frac{5}{4}$ times the consumption in 2005 = $\frac{5}{4}$ (24%) = 30%

But the consumption decreases every year.
∴ The per cent consumption should be more than 30 for the years following 2005 and it should be less than 30 for the years preceding 2005.
This happens only in years 2007, 2008, 2009, 2013 and 2014.

23. Revenue in 2012
= 52,000 × 41 + 75,500 × 34 – 6,500 × 3
= 21,32,000 + 25,67,000 – 19,500 = 46,79,500

Revenue in 2013
$= 73,000 \times 45 + 80,800 \times 35 - 7,200 \times 3$
$= 32,85,000 + 28,28,000 - 21,600 = 60,91,400$

The percentage increase $= \frac{14,11,900}{46,79,500} \times 100 = 30.2\%$

24. As the export as well as the domestic price is the highest in the year 2016, the average revenue realized per kg of the product will be the highest in 2016.
The revenue in 2016
$= 82,000 \times 49 + 107,200 \times 43 - 7,800 \times 3$
$=40,18,000 + 46,09,600 - 23,400$
$= 86,27,600 - 23,400 = 86,04,200$
The revenue per kg of production $= \frac{86,04,200}{1,97,000} = 43.6$

25. It can be seen that there is a significant increase in the domestic price in 2014 and the quantity in the domestic market also shows a significant increase.
∴ The percentage increase in domestic revenue would be the highest in 2014.
Domestic revenue in 2013 $= 80,800 \times 35 = 28,28,000$
Domestic revenue in 2014 $= 1,14,300 \times 41 = 46,86,300$

The required percentage $= \frac{18,58,300}{28,28,000} \times 100 = 65.7\%$

26. The approximate values for the different years would be as follows:

$$2011 = \frac{90 \times 32}{90 \times 32 + 42 \times 38}$$

$$2012 = \frac{75.5 \times 34}{75.5 \times 34 + 52 \times 41}$$

$$2013 = \frac{80.8 \times 35}{80.8 \times 35 + 73 \times 45}$$

$$2014 = \frac{114.3 \times 41}{114.3 \times 41 + 67 \times 47}$$

$$2015 = \frac{89 \times 42}{89 \times 42 + 75 \times 45}$$

$$2016 = \frac{107 \times 43}{107 \times 43 + 82 \times 49}$$

It can be easily seen that the ratio is the highest in 2011.
The required value would be

$$\frac{90 \times 32}{90 \times 32 + 42 \times 38 - 5 \times 3} \times 100$$

$$= \frac{2880}{2880 + 1596 - 15} = \frac{2880}{4461} \times 100 = 64\%$$

27. We can say that there is a definite increase in per capita income only during the below conditions:
(i) The national income increases and the population decreases or remains constant.
or
(ii) The national income remains constant and the population decreases.
This happens in 2012 and in 2016.

28. The per capita income definitely decreases only during the below conditions:
(i) The national income decreases and the population remains constant or increases.
or
(ii) The national income remains constant and the population increases.
This happens only in 2014.

29. The per capita income can remain constant only if both the national income and the population either simultaneously increase or simultaneously decrease or remain constant.
This happens in 2011, 2013 and 2015.

30. The number of males in 2011

$$= \frac{1073}{2073} \times 8.4 = 4.351 \text{ lakh} \times 8.4 = 4.351 \text{ lakh}$$

The number of males in 2012

$$= \frac{1061}{2061} \times 8.7 = 4.48 \text{lakh} \times 8.7 = 4.48 \text{ lakh}$$

The required percentage $= \frac{0.13}{4.35} \times 100 = 3\%$

31. The number of females in the different years are as follows:

$$2010 = \frac{1000}{2031} \times 7.3 = 3.59 \times 7.3 = 3.59$$

$$2011 = \frac{1000}{2073} \times 8.4 = 4.05 \times 8.4 = 4.05$$

$$2012 = \frac{1000}{2061} \times 8.7 = 4.22 \times 8.7 = 4.22$$

$$2013 = 2013\frac{1000}{2089} \times 4.2 = 4.40 \times 4.2 = 4.40$$

$$2014 = \frac{1000}{2007} \times 8.1 = 4.03 \times 8.1 = 4.03$$

$$2015 = \frac{1000}{1981} \times 7.8 = 3.93 \times 7.8 = 3.93$$

The percentage increase is the highest in 2011.

32. The number of males in 2013 $= 9.20 - 4.40 = 4.80$ lakh

The total literates $= \frac{56.3}{100} \times 9.2 = 5.18 \text{lakh} \times 9.2$
$= 5.18$ lakh
The number of literate males

$$= \frac{64.8}{100} \times 4.8 = 3.11 \times 4.8 = 3.11$$

The number of literates among females
$= 5.18 - 3.11 = 2.07$
The percentage of literates among females

$$= \frac{2.07}{4.40} \times 100 = 47.0\% \times 100 = 47.0\%$$

33. As the ratio of males to females was the highest in 2013 and the total population was also the highest in that year, the ratio of literate males to literate females would be the highest in that year.

34. Sales of A in 2008 = 20% of 100 (assuming total as 100)
Sales of A in 2012 = 25% of 144
∴ The required percentage

$$= \frac{36-20}{20} \times 100 = 80\%$$

35. Share of D = 20%
Total sales in 2014 = $2{,}85{,}000 \times (1.2)^3 = 4{,}92{,}480$
Sales of company D = $0.2 \times 4{,}92{,}480 \approx 98{,}500$.

36. The increase in total sales from 2008 to 2016 would be $100 \times (1.2)^4 = 207.36$
Any company which had an increase in share would definitely have a more than 100% increase companies A, B and E.

37. Just by observing the graph, company B has more than doubled its market share. Therefore, it would have the highest percentage increase in sales from 2010 to 2016.

38. In 2011, cost of 1 kg of musambis = ₹20
Cost of 1 dozen bananas = ₹30
1 kg = 10 bananas

∴ Cost of 1 kg of bananas = $\frac{10}{12} \times 30$ = ₹25

Difference = ₹5

39. As we do not know how many guavas we get per kg in 2016, the cost of 1 kg of guavas cannot be found.

40. In 2011, cost of 10 bananas = ₹30

∴ Cost of 1 kg = $\frac{10}{12} \times 30 = 25$

In 2015, cost of 12 bananas = 35
∴ Cost of 1 kg = 26.25

Percentage increase = $\frac{1.25}{25} \times 100 = 5\%$

41. Let the production be x, $2x$, x, $3x$, $2x$ kg, respectively.
∴ Average

$$= \frac{x \times 20 + 25 \times 2x + x \times 25 + 30 \times 30 + 40 \times 2x}{x + 2x + x + 3x + 2x} = 29.44$$

42. $\frac{25 + 25 + 18.3 + 25 + 26.25}{5}$

[∵ The production is constant; the average will now be the average of prices]
= 23.9

43. The mobile phone sales of ABC in each quarter can be calculated based on the sales values for the year 2010, 2012 and 2014 and the table given.

Sales	Q1	Q2	Q3	Q4	Total
2009	132.9	129.38	141.18	135.74	539.2
2010	122.33	126.79	138.92	142.39	530.43
2011	132.42	132.88	136.7	156	558
2012	145.36	152.18	119.18	132.24	548.96
2013	153.21	155.07	122.28	135.81	566.37
2014	132.9	142.54	133.13	137.47	546.04
2015	138.08	147.39	142.65	142.69	570.81

As can be seen from the table, the sales exceeded 5,50,000 in 2011, 2013 and 2015.

44. From 2009 to 2010, the difference in the annual sales is only 8770.

45. The difference is (546.04 – 539.2) × 1000 = 6840.

46. From the table above, we can see that the sales were highest in Q2 during 2010 to 2015. Average number of mobiles sold = (126.79 + 132.88 + 152.18 + 155.07 + 142.54 + 147.39)/6 = 142.81

47. For A: Maximum increase in price (₹/m^3) is 10000 (2003–2004).
For B: Maximum increase in price (₹/ ton) is ₹20000 in (2003–04).
Converting this amount in (₹/m^3) = (20000) (600)/1000 = 12000
For C: Maximum increase in price (₹/m^3) = 20000 (2001–02 and 2006–07)

48. Price of C is expressed in (₹/m^3). To convert it into (₹ / ton), it has to be multiplied with a constant factor k = (1000/800) = 5/4

Year	Difference in prices (in thousands of rupees)
2001	$40 - 10 \times \frac{5}{4} = 37.5$
2005	$60 - 30 \times \frac{5}{4} = 22.5$
2006	$70 - 40 \times \frac{5}{4} = 20$
2003	$40 \times \frac{5}{4} - 30 = 20$

The largest difference occurs in the year 2001.

49. Average realization
$= (0.5)\ (80) + (0.4)\ (60)\ (3/5) + (0.1)\ (30)$
$= 40 + 14.4 + 3 = ₹57.4/m^3$

50. Price in the year 2008 (in ₹/m^3).
(P)A = Price of A = $70 + (1/100)\ (70) = 70.7$

(P)B = Price of B = $[80 + (2/100)\ (80)]\ \frac{3}{5}$

$= 81.6 = (81.6)\ (3/5) \cong 49$
(PC) = Price of C = $60 + (5/100)\ (60) \cong 63$
Average realization
$= (0.5)\ (70.7) + (0.3)\ (49) + (0.2)\ (63)$
$= 35.35 + 14.7 + 12.6$
$= ₹62.65$

EXERCISE-3

1. Let the total subscribers in 2011 be 100.
The total subscribers in 2016 would be 305.
The subscribers of company P in 2011 = 20

Subscribers of company P in 2016 would be $\frac{15}{100} \times 305$

$= 45.75$

The percentage increase in subscribers = $\frac{25.75}{20} \times 100$

$= 128.75\%$.

2. Let the total subscribers in 2013 be 100.
Total subscribers in 2015 would be 156.25.

Number of subscribers of R in 2013 = $\frac{15}{100} \times 100 = 15$

Number of subscribers of T in 2015 = $\frac{25}{100} \times 156.25$

$= 39.06$.

The required ratio = 5 : 13.

3. We need to only check for the last two years as the number of subscribers is continuously increasing. The maximum increase would be for company S in 2016 as its share in the total doubled.

4. From the first question in the set we can see that over a period of five years, the total number of subscribers increased from 100 to 305. That is 205% increase.

5. Company P in 2013, Q in 2011, R in 2016, S in 2015, T in 2014 and U in 2011. All the six companies had a decrease in the number of subscribers at least once.

6. Selling price/ton of company B is \$225 and that of E is \$200.
If K tons are exported by both the companies, then the total revenue of the companies is

$\frac{255K}{0.08} = 3187.5$ K for company B and $\frac{200K}{0.125} = 1600K$ for company E.

Therefore, the total revenue of company B is approximately double that of company E.

7. In 2012, iron ore exports as a percentage of total revenue of the companies C and M will be 20% and 24%, respectively.
Let the volume of iron ore exported by both the companies be K ton.
Selling price /ton of M = 360 (1.25) = \$450

Total revenue for M = $\frac{450K}{0.24} = 1875K$

Therefore, total revenue of C = (1875K)2 = 3750K
Thus, selling price/tonne for company C = 3750(20%) = \$750
Therefore, the percentage increase in the selling price

for company C = $\frac{750 - 210}{210} \times 100\% \approx 257\%$

8. From the options we can check that only for companies D and M, the total revenue will be the same. Let us consider that the total volume of iron-ore exports for companies D and M be K ton.

Total revenue of company D = $\frac{240K}{0.04} = 6000K$

Total revenue of company M = $\frac{360K}{0.06} = 6000K$

9. We need to basically look for companies which are lower in the *y*-axis and have higher values along the *x*-axis. Assuming one unit of export for all the companies, we can see that the total revenue would be the highest for companies D and M.

10. The ROI of a player is higher if he has a higher value of $\frac{\text{actual k. factor}}{\text{price}}$ AB de Villiers would have the highest ROI among all players with a price more than \$1 million.

11. The rate of actual K-factor to price is the lowest for McCullum. So, he was the biggest disappointment in terms of ROI.

12. Four Players-David Miller, Azhar Mahmood, S. Dhawan and S. Smith gave a higher ROI than Chris Gayle.

13. A straight line drawn through (0, 0) and (2000000, 10,000) would give the expected ROI for each price. Any player above the line would have given a ROI which was better than expected, i.e., 15 players.

14. Required amount
= ₹(180 – 144) + (240 – 200) + (400 – 360) crore
= ₹116 crore

15. Profit gained in 2013, 2014 and 2015 together, by
company A:
₹(300 – 270) + (320 – 280) + (120 – 108) crore
= ₹82 crore
company B:
₹ ((250 – 225) + (240 - 200) + (180 – 144) crore
= ₹101 crore
∴ Required amount = ₹ (101 – 82)crore = ₹19 crore

16. Income of company A, over the years is as follows:
2012 : ₹216 crore; 2010: ₹300 crore
2014: ₹320 crore; 2012: ₹120 crore
2016 : ₹250 crore; 2014: ₹350 crore
∴ In 2014, it was the second highest.

17. As nothing is said with regards to whether the country exported or imported rice in 2003, we cannot find the answer for this.

18. The consumption of rice, when compared to the previous year, showed an increase when either the population or the per capita consumption or both show an increase, i.e., in 2004, 2006 and 2007.

19. Even though the country exported rice in 2003, it could have imported rice in each of the following years as the total consumption could have been more than the production.

20. The population of the country increased in 2004, 2006 and 2007. Of this, in 2004 and 2007, the total consumption of rice in the country could have decreased.

5 Caselets

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:

- Gain understanding of data presented in the form of paragraphs.
- Learn how to convert data given in caselets into equations.
- Learn how to jot down points and conditions simultaneously while reading the caselet and forming equations out of them.
- Get familiar with different kinds of cases – DI based and reasoning-based.
- Get familiar with venn-diagram based caselets, which frequently appear in MBA entrance exams.

❒ INTRODUCTION

At times the information in a DI set is not represented in the form of a bar graph, pie chart or any other graphical tool, it is described in a theoretical manner and the students need to interpret and organize the given data to solve the set.

Caselets appear frequently in the DI & LR sections in competitive exams like CAT, XAT, NMAT etc. While solving a caselet, the student should list down all the important points in the set and try and represent the given information in a table or some other systematic format.

SOLVED EXAMPLES

Directions for questions 5.01 to 5.05: These questions are based on the information given below.

Anurag, Anil and Anmol worked together to paint Arun's house. Arun gave an amount of ₹13,500 for the work. It was decided among themselves that Anurag should get one third more than Anil and Anmol should get 8/15th of the total amount.

Solution:

Let the amounts received by Anurag, Anil and Anmol for their respective works be x, y and z.

Given that $x + y + z = ₹13{,}500$ (1)

$$x = \left(1+\frac{1}{3}\right)y$$

$$\Rightarrow x = \frac{4}{3}y \qquad (2)$$

$$z = \frac{8}{15}(₹13{,}500) = ₹7{,}200 \qquad (3)$$

$(1) \Rightarrow x + y = ₹13{,}500 - ₹7200$

$\Rightarrow x + y = ₹6300$

From (2), we get: $\frac{4}{3}\,4 + y = ₹6300 \Rightarrow \frac{7y}{3} = ₹6300$

$\Rightarrow y = ₹2700$

$\Rightarrow x = 6300 - ₹2700 = ₹3600$

5.01: What is the amount received by Anmol and Anil together?

(A) ₹9300 (B) ₹9600
(C) ₹9900 (D) ₹10,200

Sol: Amount received by Anmol and Anil together $= y + z = ₹2700 + ₹7200 = ₹9900$

5.02: What is the difference between the amounts received by Anurag and Anil?

(A) ₹900 (B) ₹1000
(C) ₹1100 (D) ₹1200

Sol: The difference between the amounts received by Anurag and Anil $= x - y$
$= ₹3600 - ₹2700 = ₹900$

5.03: What is the ratio of the amounts received by Anurag and Anmol together to the amount received by Anil?

(A) 3 : 2 (B) 4 : 1
(C) 2 : 3 (D) 1 : 4

Sol: The ratio of the amounts received by Anurag and Anmol together to the amount obtained by Anil $= (x + z) : y = (₹3600 + ₹7200) : (₹2700)$
$= ₹10{,}800 : ₹2700 = 4 : 1$

5.04: The amount received by Anurag and Anil together is what percentage of the amount received by Anmol?

(A) 67.5% (B) 75%
(C) 80% (D) 87.5%

Sol: The total amount received by Anurag and Anil together $= x + y = ₹3600 + ₹2700 = ₹6300$
$\therefore$ The required percentage

$$= \frac{₹\,6{,}300}{₹\,7{,}200} \times 100\% = 87.5\%$$

5.05: The difference between the amounts received by Anurag and Anil is what part of the total amount received by the three together?

(A) $6\frac{2}{3}\%$ (B) 8%

(C) $9\frac{2}{3}\%$ (D) 10%

Sol: The difference between the amounts received by Anurag and Anil $= x - y$
$= ₹3600 - ₹2700 = ₹900$

$$\therefore \text{The required part} = \frac{₹900}{₹13{,}500} \times 100\% = 6\frac{2}{3}\%$$

EXERCISE-1

Directions for questions 1 to 5: Answer these questions based on the information given below.

Four crates of luggage A, B, C and D are measured for their weights and then loaded onto a cargo plane. The weight of crate A, which is 100 kg is less than that of crate C by the same amount by which the weight of crate C is less than that of crate D. The average weight of crates A, C and D is 300 kg. It is also known that the average weight of the 4 crates is 75 kg more than the weight of crate C.

1. The weight of how many crates is more than the average weight of the four crates?
(A) One (B) Two
(C) Three (D) Cannot be determined

2. Crate C weighs less than crate D by
(A) 40% (B) 150 kg
(C) 100 kg (D) 60%

3. The weight of crate B is
(A) 600 kg (B) 250 kg
(C) 350 kg (D) 100 kg

4. If the heaviest crate is not loaded on the plane, what is the average weight of the 3 loaded crates?
(A) 300 kg (B) $333\frac{1}{3}$ kg
(C) 400 kg (D) $466\frac{2}{3}$ kg

5. The ratio of the difference in weights of crates A and B and the sum of the weights of crates C and D is
(A) 5 : 2 (B) 1 : 2
(C) 5 : 8 (D) 3 : 8

Directions for questions 6 to 10: Answer these questions based on the information given below.

Hiralal wrote his will on his deathbed. The terms of his will are as follows:

(i) Hiralal's wife gets a third of his property originally worth ₹51,00,000.
(ii) After Hiralal's wife, his son Haralal gets 50% of the remaining property.
(iii) After Haralal, Hiralal's daughters Heera and Henna get the remaining property in the ratio 8 : 9, respectively.

6. The share of the property that Hiralal's wife gets, is what percentage of the property that Haralal and Heera together get?
(A) 60% (B) 58%
(C) 85% (D) 68%

7. What is the difference in the amounts received by Haralal and Henna (in ₹ lakh)?
(A) 8 (B) 9
(C) 11 (D) 12

8. The difference in the share of property that Heera and Henna gets is what percentage of the property that Hiralal's wife got?
(A) 5% (B) $5\frac{15}{17}\%$
(C) 6% (D) $6\frac{8}{17}\%$

9. The share of the total property that Hiralal's wife gets, is how many percentage points more than Heera's share?
(A) 33 (B) 17.64
(C) 15.69 (D) 13.5

10. If the shares of Henna and Haralal are interchanged, then Heera's share of the property would be what percentage (approx.) of the property share of Haralal and Hiralal's wife together?
(A) 29% (B) 31%
(C) 33% (D) 35%

Directions for questions 11 to 15: Answer these questions based on the information given below.

Amar, Akbar and Anthony sold their three cycles manufactured in different years to Kishanlal. Kishanlal gave a total of ₹1700 to the three and said that Amar should get ₹50 more than half of the total amount as his cycle was used less. Akbar's cycle being used more than Amar's, he should get about 6/17th the total amount and the last one gets the remaining amount. Each individual gets his amount only in denominations of ₹100.

11. What is the difference between the amounts received by Amar and Anthony?
(A) ₹900 (B) ₹700
(C) ₹800 (D) ₹600

12. The amount that Amar has is how much more than what Akbar and Anthony together have?
(A) ₹200 (B) ₹300
(C) ₹100 (D) ₹400

13. If the shares of Amar and Anthony are interchanged, then Akbar has how much more than what Amar has?
(A) ₹300 (B) ₹200
(C) ₹400 (D) ₹100

14. The ratio in which the amount shared among Akbar, Amar and Anthony is

(A) 2 : 9 : 6 (B) 9 : 6 : 2
(C) 2 : 6 : 9 (D) 6 : 9 : 2

15. The difference of the amount that Anthony and Akbar have is what percentage of the amount that Amar has?

(A) $25\frac{3}{5}$ (B) $44\frac{4}{9}$
(C) $33\frac{1}{3}$ (D) $66\frac{2}{3}$

Directions for questions 16 to 20: These questions are based on the following information.

In an MBA college $83^1/_3\%$ of the students like either Tennis or Formula 1 and their ratio is 7 : 4, respectively. Of those who like either Tennis or Formula 1, the ratio of boys and girls is 2 : 9, respectively. Of the boys who like Tennis or Formula 1, 60% like Tennis and 4 boys like Formula 1. Also the number of boys and girls who does not like either Tennis or Formula 1 is in the ratio 6 : 5. No student likes both the games.

16. The number of girls who like Formula 1 is what percentage of the boys who like Tennis?
(A) 220 (B) $266^2/_3\%$
(C) $233^1/_3\%$ (D) Cannot be determined

17. What is the total number of girls?
(A) 20 (B) 10
(C) 35 (D) None of these

18. What is the ratio of the total number of boys to the number of girls who like Tennis?
(A) 4 : 3 (B) 3 : 2
(C) 16 : 29 (D) 29 : 16

19. What is the total number of students in the college?
(A) 55 (B) 66
(C) 60 (D) Cannot be determined

20. The total number of girls who like Tennis is what percentage of the total strength of the college?
(A) 43.93% (B) 39.67%
(C) 41.72% (D) 47.33%

Directions for questions 21 to 25: Answer these questions based on the information given below.

Madhusudhan invested his savings of ₹10 lakh in 3 schemes, whose details are mentioned below:

Scheme 1: A fixed deposit scheme which gives a return of 8% per annum.

Scheme 2: A monthly recurring deposit scheme, which gives a return of 0.5% per month.

Scheme 3: Agri silver's land gold scheme guarantees a 10% appreciation in the value of the land over a years' time.

In the recurring deposit scheme, an investor invests a fixed amount (decided by him) every month and the amount earns simple interest on a monthly basis for the remainder of the period. For example, the first instalment invested in the first month of a year, earns interest for 12 months by the end of the year, the second instalment earns interest for 11 months and so on.

Madhusudhan invests equal sums of money in Scheme 2 and Scheme 3 and $33\frac{1}{3}\%$ more money in Scheme 1 than he invested in Scheme 2. The amount that he invested in Scheme 2 is spread over 12 instalments throughout the year.

21. What is the net worth of Madhusudhan's investments at the end of a year?
(A) ₹10.21 lakh (B) ₹11.4 lakh
(C) ₹10.2 lakh (D) ₹11.11 lakh

22. The return on Scheme 3 was what percentage higher/lower than the return on Scheme 1?
(A) 6.25% (B) 10%
(C) 20% (D) 12%

23. What would have been Madhusudhan's return from Scheme 2 if he had invested twice the money as he originally invested in the scheme?
(A) ₹9750 (B) ₹19,500
(C) ₹39,000 (D) ₹28,500

24. If the amounts that are invested in Scheme 1 and Scheme 3 are interchanged, then what will be the change in the net worth of Madhusudhan at the end of 1 year?
(A) ₹10,000 (B) ₹5000
(C) ₹2000 (D) ₹20,000

25. If Madhusudhan continues to invest the same amount in Scheme 2 for one more year, then how much would he earn more/less than the amount he earned through Scheme 1 and Scheme 3 together in the first year?
(A) ₹17,800 (B) ₹22,500
(C) ₹20,700 (D) ₹24,500

Directions for questions 26 to 30: Answer these questions based on the information given below.

After my retirement, I decided to invest my savings in four different categories.

I invested 20% of the amount in shares, 30% of the amount in National Savings Certificate (NSC), 40% of the remaining in land and the rest in FD's.

After the first year, the value of my shares increased by 20%. I get an 8% return, which is tax free, on my investment in NSCs. My FD's fetched me 6% returns on which I had to pay 5% TDS on the interest earned and the land prices appreciated by 10%.

At the end of the second year, due to the stock market boom, the value of my shares increased by a further 45% and

I sold off my shares and earned 100% tax free profits. NSCs again gave me an 8% tax free returns, the FD (my initial investment along with the interest earned after taxes in the first year) again gave me 6%, but again I had to pay 5% TDS on the interest earned in the second year and land prices appreciated by a further 10% when I sold it off and had to pay 10% tax on the gains I made.

At the end of two years I found that the difference between my gain from investment in shares and land was ₹55,100.

26. What was the amount invested (in lakh) by me in shares in the first year?

27. What was the total amount invested (in lakh) initially?

28. What is the value of all my investments (in lakh) at the end of two years (approximately)?

29. What is the approximate compounded annual return on my investment for the two-year period?

30. At the end of the two years, the difference in the amounts due to me for my investments in NSC and FD was approximately.

Directions for questions 31 to 35: Answer these questions based on the information given below.

Four friends Nikhil, Joy, Rohit and Binoy have their shops on the same lane and they sell plyboards, tyres, textbooks and jars of spices, respectively. On a given day, the four friends sell goods worth ₹2600 altogether. Rohit sold text books worth ₹600 and Rohit's revenue for that day was less than Joy's by the same amount by which Binoy's revenue was less than Rohit's. Nikhil's sales revenue for that day was ₹200 more than the average sales of Joy, Rohit and Binoy.

31. If each jar of spices sold by Binoy is worth ₹18, then which of the following cannot be Joy's sales on that day?
(A) ₹660 (B) ₹624
(C) ₹840 (D) ₹920

32. If the sales of Nikhil and Rohit together exceed those of Joy and Binoy together by ₹200 on that day, what were the sales of Binoy?
(A) ₹500 (B) ₹550
(C) ₹570 (D) Cannot be determined

33. If each plyboard that Nikhil sold is worth an integral number of rupees, then which of the following could be the number of plyboards Nikhil sold on that day?
(A) 30 (B) 25
(C) 35 (D) 15

34. If Nikhil, Joy, Rohit and Binoy pay 10%, 20%, 30% and 20% of their daily revenue respectively as trade tax to the trade association, how much did the four friends pay as tax on that day?
(A) ₹480 (B) ₹500
(C) ₹600 (D) ₹520

35. If Joy sold all tires at the same price of ₹20/tyre on that day and the sales of Nikhil and Binoy together equal the sales of Rohit and Joy together, then how many tyres did Joy sell on that day?
(A) 30 (B) 35
(C) 40 (D) 45

Directions for questions 36 to 40: Answer these questions based on the information given below.

In Indian Public School (IPS), 80% of the students who appeared for the Class X Board exams in 2014 passed the exam. Among these who passed the board exams, 60% joined Indian Intermediate college (IIC) for their Class XI. The students of IPS who joined IIC opted for Science, Commerce and Humanities streams in the ratio 3 : 4 : 5. Therefore, 60% of students in Science stream, 40% of students in Commerce stream and 50% of students in Humanities stream of Class XI in IIC happen to be students of IPS who passed in 2014. The total number of students in all the three streams of Class XI in IIC is 800. Each student opts for only one stream.

36. How many students failed in the board exams in IPS in 2014?
(A) 160 (B) 120
(C) 200 (D) 240

37. If 10% of the students who opted for Humanities in Class XI in IIC are awarded a scholarship, then how many students got the scholarship?
(A) 10 (B) 26
(C) 32 (D) 45

38. For every 3 students from IPS who opted for science stream in Class XI in IIC, there are 2 students from another school APS pursuing Science stream in Class XI in IIC. If the students of APS who joined Class XI in IIC opted for Science, Commerce and Humanities in the ratio of 2 : 3 : 4, then how many students pursuing Commerce stream in Class XI in IIC are from APS?
(A) 120 (B) 130
(C) 156 (D) 96

39. Using the data from the above question, what percentage of students pursuing Humanities in Class XI of IIC are from neither IPS nor APS?
(A) 15% (B) 18%
(C) 10% (D) 20%

40. If IIC collects a monthly fee of ₹1000, ₹1500 and ₹2000 from Class XI students of Science, Humanities and Commerce, respectively, then what would be the fee collected from the 3 streams in a month from Class XI of IIC?
(A) ₹10.2 lakh
(B) ₹10.8 lakh
(C) ₹12.8 lakh
(D) Cannot be determined

Directions for questions 41 to 45: Answer these questions based on the information given below.

Five students Ajay, Bharat, Kumar, Sanjay and Vishal, when asked about their scores in a quant test, replied as follows:

Ajay: Bharat, Kumar and I together scored 135 marks.
Bharat: Kumar, Sanjay and I together scored 137 marks.
Kumar: Sanjay, Vishal and I together scored 132 marks.
Sanjay: Vishal, Ajay and I together scored 138 marks.
Vishal: Ajay, Bharat and I together scored 133 marks.

41. Who scored the least marks among the given students?
(A) Vishal (B) Bharat
(C) Kumar (D) Sanjay

42. Who scored the highest marks among the given students?
(A) Ajay (B) Bharat
(C) Kumar (D) Sanjay

43. How many of the given students scored more marks than Ajay?
(A) 0 (B) 1
(C) 2 (D) 3

44. What is the maximum difference in the marks scored by any two of the five students?
(A) 6 (B) 8
(C) 9 (D) 10

45. Bharat's score is what per cent of Sanjay's score?
(A) 88.88% (B) 80%
(C) 90% (D) 125%

Directions for questions 46 to 50: Answer these questions based on the information given below.

A club had organized cricket, music and painting summer camps for children. The number of children who enrolled for both music and painting is 5. The number of children who enrolled for only painting and cricket and only cricket and music were 7 more and 4 more, respectively than the number of children who enrolled for all the three. The number of children who enrolled for cricket is the same as those who enrolled for more than one event. Eight more children enrolled for music than cricket and the number of children who enrolled for painting was twice that of cricket. It is also known that the highest number of children who enrolled for any one event alone was 29.

46. What is the number of children who enrolled in all the three camps?
(A) 2
(B) 3
(C) 4
(D) Cannot be determined

47. How many children enrolled in more than one camp?
(A) 14 (B) 18
(C) 22 (D) None of these

48. The number of children who enrolled in exactly one camp is
(A) 39 (B) 42
(C) 44 (D) 49

49. How many children enrolled for music?
(A) 32 (B) 30
(C) 28 (D) 26

50. How many children enrolled in exactly two camps?
(A) 19 (B) 21
(C) 23 (D) None of these

EXERCISE-2

Directions for questions 1 to 3: These questions are based on the information given below.

The following is the rate of income tax for different income groups:

Annual income (₹)	Tax rate
Up to 1 lakh	0%
1 – 1 × 5 lakh	10% of income in excess of ₹1 lakh
1 × 5 to 5 lakh	Up to ₹1×5 lakh as above + 20% of income in excess of ₹1×5 lakh
More than 5 lakh	Up to ₹5 lakh as above + 30% of income in excess of ₹5 lakh

Tax is always calculated on income, after deductions, if any.

The only allowed deductions are as follows.

(A) Maximum of ₹1 lakh per year on investments in specified securities.
(B) Maximum of ₹3 lakh per year as housing loan repayment.

The eligibility for housing loan is a minimum annual income of ₹1 lakh and annual repayment less than or equal to 20% of annual income. There is a service tax of 10% on the tax paid at the rate of 30%.

1. What can be the maximum income for which one need not pay any tax?
(A) ₹1 lakh (B) ₹2 lakh
(C) ₹2.5 lakh (D) ₹3 lakh

2. What is the minimum tax to be paid by a person with an annual income of ₹10 lakh?
(A) 1.2 lakh (B) 1.245 lakh
(C) 1.35 lakh (D) None of these

3. The difference between the minimum and the maximum tax that one has to pay on an annual income of ₹6 lakh is
(A) ₹80,000 (B) ₹54,000
(C) ₹57,000 (D) ₹83,000

Directions for questions 4 to 7: These questions are based on the information given below.

Murali sells samosas for a living. He sells each samosa for ₹4 and works from Monday to Saturday every week. He observes that he sells X samosas less than the previous day on 2 days of any week and X samosas more than the previous day on the other three days of the week. Every week, the highest number of samosas that he sells on any day is 150 and the least number of samosas he sells is 90.

4. If Murali sells 750 samosas in a particular week, then how many samosas does he sell on Friday?
(A) 90 (B) 120
(C) 150 (D) Cannot be determined

5. If Murali sold the minimum number of samosas on Friday and did not sell the maximum number of samosas on Thursday, then how much money did he earn that week?
(A) ₹2760 (B) ₹2880
(C) ₹3000 (D) ₹2840

6. How many samosas does Murali sell on Tuesday, if he sold 130 samosas on Friday?
(A) 90 (B) 110
(C) 120 (D) Cannot be determined

7. If Murali sells 130 samosas on Wednesday, then which of the following statement/s is/are necessary to find the number of samosas Murali sold on Saturday?
I. He sold 150 samosas on Thursday.
II. He sold 130 samosas on Friday.
III. $X = 20$
(A) Only III (B) Only II
(C) II and III (D) None of these

Directions for questions 8 to 12: Answer these questions based on the information given below.

The population of country XYZ is currently 100 million in Year 1. It is projected that by Year 25 the population will become 125 million. There are only 2 states A and B in XYZ. If the population of states A and B increases by 20% and 35% respectively, then the country will have the same population as it is predicted to have in Year 25.

Further, it is known that a third of the citizens of State A and a fourth of the citizens of State B are city-dwellers. By Year 25, the number of city dwellers in both the states is expected to double. Assume that all the predictions and expectations turn out to be true.

8. What is the total population of city dwellers in year 25?
(A) 61.1 million (B) 30.55 million
(C) 20 million (D) 15.28 million

9. Assuming that the population of both states grows at the same rate as that of the country, the percentage of citizens living in rural areas in Year 25 is approximately how many percentage points (approximately) more/less than the rural population in Year 1?
(A) 16 (B) 18
(C) 20 (D) 22

10. If State A grew by 35% and State B grew by 20%, then the population of the country would exceed the projections by
(A) 10 million (B) 7 million
(C) 5 million (D) None of these

11. In Year 1, 20% of the city dwellers are children and 30% of those living in rural areas are children. How many grown-ups are there in the country in Year 1?
(A) 70 million (B) 73 million
(C) 75 million (D) 78 million

12. If State A's population increased by 2% of the base value each year and State B's increased by 3% each year, in how many years will the projection be true?
(A) 7 (B) 9
(C) 10 (D) 11

Directions for questions 13 to 17: These questions are based on the information given below.

The information below gives partial details of the employees of ABC Pvt. Ltd. It tells us about the number of employees who had specialized in Finance, Marketing and HR during their MBA programme. It also tells us about the number of engineers and that of the non-engineers who had specialized in each of these streams. Each employee who was an MBA graduate specialized in exactly one stream.

	Specialization		
Educational background	**F**	**M**	**H**
Engineers			
Non-engineers	7		5
Total		35	

Note: F = Finance, M = Marketing and H = HR

The following details are known:

(i) The number of engineers is 20% more than the number of non-engineers.

(ii) The number of employees who specialized in Finance and the number of non-engineers are in the ratio 8 : 15.

(iii) The number of employees who specialized in HR and the number of engineers are in the ratio of 5 : 12.

13. How many non-engineers specialized in Marketing?

(A) 15 (B) 16
(C) 17 (D) 18

14. How many non-engineers did not specialize in Finance?

(A) 19 (B) 20
(C) 23 (D) 22

15. Find the number of employees who specialized in HR.

(A) 14 (B) 15
(C) 16 (D) 17

16. Which of the following is the least?

(A) The number of non-engineers who specialized in HR.
(B) The number of engineers who specialized in Finance.
(C) The number of non-engineers who specialized in Marketing.
(D) The number of engineers who specialized in Marketing.

17. What percentage of MBA graduates are engineers?

(A) $63^7/_{11}\%$ (B) $72^8/_{11}\%$
(C) $54^6/_{11}\%$ (D) $45^5/_{11}\%$

Directions for questions 18 to 22: These questions are based on the information given below.

The weights (in kg) of each of the five members of the Don Bosco school wrestling team, namely for Ajay, Bhushan, Chetan, Deepak and Emmanuel is a distinct integer. Before leaving for an inter-school competition, the coach decided to check the weights of all the five members. But in the only available weighing machine in the school, the weights from 0 to 100 were not properly visible. But while checking the weights was necessary to register them, he decided to weigh the students in groups of three making sure that no group of three students was repeated. The weights obtained while weighing them were as follows. 106 kg, 116 kg, 122 kg, 126 kg, 132 kg, 146 kg, 120 kg, 126 kg, 136 kg and 142 kg. It is also known that the weight of Ajay is the average of that of Bhushan and Emmanuel. Further, Emmanuel is heavier than Chetan but lighter than Deepak.

18. What is the weight (in kg) of the heaviest boy?

19. What is the weight (in kg) of Bhushan?

20. What is the weight (in kg) of Emmanuel?

21. What is the weight (in kg) of Chetan?

22. What is the average weight (in kg) of all the five boys?

Directions for questions 23 to 25: Answer these questions based on the information given below.

A survey was conducted among 180 employees in an organization to find out whether they were active on any of the three networking sites, such as Facebook, LinkedIn and Twitter. The number of employees who were active on any of the sites was 400% more than the number of employees who were not active on any of the three. The number of employees who were active on Facebook, LinkedIn or Twitter was 68, 61 and 59, respectively. The number of employees active on Twitter and exactly one other site was 21. 15 employees were active on Face book as well as LinkedIn. The number of employees active on Twitter alone was equal to the number of employees active on more than one of the sites.

23. If 15 employees are active on both Twitter and LinkedIn, then how many employees are active only on Facebook?

(A) 49 (B) 47
(C) 45 (D) 43

24. What percentage of the employees were active on exactly one of the sites?

(A) 72.5 (B) 76.0
(C) 67.5 (D) 63.33

25. What is the number of employees who are active only on Twitter?

(A) 30 (B) 36
(C) 39 (D) None of these

Directions for questions for 26 to 28: Answer these questions based on the information given below.

A survey was conducted among students to find out who was the person whom they admired the most among Mahatma Gandhi, Mother Teresa and Nelson Mandela. Each student had to vote for exactly one of the three persons. Off the 330 boys with an urban background, 40% voted for Mahatma Gandhi. The number of girls with a rural background was the average of the number of boys with a rural background and the number of girls with an urban background. The number of girls with an urban background who voted for Mother Teresa was 80, which was equal to the number of girls with an urban background who voted for Mahatma Gandhi or Nelson Mandela. The number of boys with an urban background who voted for Nelson Mandela was 78. The total number of boys with a rural background was equal to the number of boys with an urban background who voted for Mother Teresa. All the three persons got the same number of votes from boys with a rural background and the number of girls with a rural background who voted for Mahatma Gandhi and Mother Teresa were the same. A total of 290 students voted for Mother Teresa which was 23 more than those who voted for Mahatma Gandhi.

26. What is the total number of students surveyed?

(A) 750 (B) 600
(C) 900 (D) 800

27. The number of girls with a rural background who voted for Nelson Mandela was
(A) 30 (B) 35
(C) 40 (D) 45

28. Girls with a rural background who voted for Mahatma Gandhi was what percentage of the total girls?
(A) 15 (B) 16.67
(C) 17.5 (D) 20

Directions for questions 29 to 32: These questions are based on the following information.

Ramesh wrote his final exams which consists of four subjects, such as Maths, Physics, Chemistry and English. He expected 80, 40, 60 and 50 marks, respectively in those subjects (out of 300 in each subject). After the results, his friend asked him the marks he scored. Ramesh told that he got half the expected marks in one of the subjects, thrice the expected marks in the second one, twice the expected marks in the third and in the remaining subject he got exactly the expected marks.

29. If the average marks obtained by Ramesh is 95, then he definitely scored more than the expected marks in
(A) Maths (B) Physics
(C) English (D) Chemistry

30. Which of the following is true regarding the above statements?
(I) Ramesh's average marks is more than 100.
(II) Ramesh scored the expected marks in Maths.
(A) If (I) is true, (II) is false.
(B) If (I) is true, (II) is true.
(C) If (I) is false, (II) is false.
(D) None of these

31. If Ramesh got the same marks in two subjects, then which subject is definitely one of those?
(A) Maths (B) Physics
(C) English (D) Chemistry

32. If marks in one subject are thrice that in two other subjects, then the total marks of Ramesh is
(A) 330 (B) 350
(C) 375 (D) 380

Directions for questions 33 to 36: These questions are based on the information given below.

XYZ Ltd. had introduced a battery fitted car in the year 2006. The batteries would last a year and need to be recharged once a year after that. For the first two years, the company would recharge the batteries for free. From the third year onwards, it charges ₹20,000 for a recharge. However, there are third party vendors who do the same recharge for ₹17,000. Every year, 40% of the customers who opt for a battery recharge, do it from the company while the remaining 60% opt for the cheaper option. The following table gives the details of the number of paid recharges by the company and by third party vendors for all years from 2009 to 2013. Three values in the table have been intentionally left out. Assume that all cars from 2006 are in operation and are recharged after exactly one year.

Replacement Source	2009	2010	2011	2012	2013
Company	–	–	–	1880	2276
3rd party	1524	2214	2526	2820	3414

33. How many cars were sold by XYZ Ltd. in 2009?
(A) 470 (B) 490
(C) 510 (D) 540

34. How many cars were sold by XYZ Ltd. in the three years from 2006 to 2008?
(A) 3750 (B) 4050
(C) 4210 (D) None of these

35. How many batteries were recharged by the company in 2010?
(A) 1440 (B) 1476
(C) 1510 (D) Cannot be determined

36. How many cars were sold by the company in 2011?
(A) 630 (B) 720
(C) 690 (D) Cannot be determined

Directions for questions 37 to 39: Answer these questions based on the information given below.

In a class of 100 students, each student will play at least one of the games, such as Cricket, Hockey, Football and Chess. For each game or for any combination of the games there will be at least two students in the group who play only that game or that combination of games, i.e., there will be at least two students who play only Cricket, two students who play only Chess and at least two students who play both Cricket and Chess and so on. It is also known that at least 50 students play each game.

37. If each game is played by exactly 60 students, the number of students playing at most two games is at least
(A) 0 (B) 20
(C) 37 (D) 100

38. If the number of students playing any game is exactly 52, then the number of students who play exactly two games is at most
(A) 76 (B) 78
(C) 80 (D) 82

39. If each game is played by 50 students, then the number of students playing all the four games is at most
(A) 20 (B) 22
(C) 24 (D) 26

Difficulty Level-2: Moderate

Directions for questions 40 to 42: Answer these questions based on the information given below.

Ramesh was given a weighing balance and nine identical balls. One or more of the nine balls was faulty (weighed more or less than the others).

40. What is the minimum number of weighing's required to certainly identify the faulty ball if only one ball is faulty and it is known that it weighs more than the other balls?

41. If there are two balls which are faulty, but if it is known that both these balls weigh the same and are heavier than the others, then the number of weighing's required to certainly identify the faulty balls is at least

42. It is known that there is only one ball which is faulty and it is not known whether it weighs more or less than the other bells. What is the minimum number of weighing's required to certainly identify the faulty ball?

Directions for questions 43 to 46: These questions are based on the following information.

A company produces 4 different products, namely microwave ovens, refrigerators, ACs and washing machines. It produces two different types of each product, i.e., P and Q. The company produces a total of 1500 products. 20% of the total number of products are washing machines, of which 65% are of type Q. Three-twentieth of the total number of products are ACs. $33^1/_3$% of the ACs are of type P. One-fourth of the total number of products are refrigerators out of which 120 are of type Q. Nine-tenth of the number of microwave ovens are of type P.

43. Find the average number of products of type Q made by the company.

(A) 131.25 (B) 141.25
(C) 136.25 (D) 126.25

44. Find the ratio of the number of microwave ovens of type P to the number of washing machines of type Q.

(A) 48 : 17 (B) 17 : 48
(C) 36 : 13 (D) 13 : 36

45. Find the difference in the number of refrigerators of both the types.

(A) 215 (B) 225
(C) 135 (D) 245

46. Find the total number of ACs and microwave ovens of type P, washing machines and refrigerators of type Q produced by the company.

(A) 870 (B) 890
(C) 910 (D) 930

Directions for questions 47 to 50: These questions are based on the following information.

Anand invested in the shares of four companies, namely A, B, C and D. Each of these companies belonged to a different industry, such as Metals, IT, Automobiles and Infrastructure, in no particular order. At the time of investment, the price of each share was ₹400. Anand purchased ten shares of each of these companies. He was expecting returns of 25%, 10%, 20% and 45% from the shares of companies A, B, C and D, respectively. Returns are defined as the percentage change in the value of the share after one year. The returns for two companies were higher than the expected returns. One of these two companies belonged to the Metals or Automobile sector while the other one belonged to the IT or Infrastructure sector. For the company belonging to the Metals or Automobile sector, the returns were twice those of the expected returns and for the company belonging to the IT or Infrastructure sector, the returns were three times the expected returns. For the remaining two companies, the returns were the same as expected.

47. What is the minimum average return Anand could have earned during the year?

(A) 35% (B) 32%
(C) 30% (D) None of these

48. What is the maximum average returns Anand could have earned during the year?

(A) 47.5% (B) 50%
(C) 53.75% (D) None of these

49. If Anand earned 42.5% returns during the year, then which of these statements is definitely true?

I. Company A belonged to either IT or Infrastructure sector.
II. Company B belonged to either Metals or Automobiles sector.
III. For Company C, the returns were more than expected.
IV. For Company D, the returns were more than expected.

(A) I and III
(B) II and IV
(C) I and IV
(D) II and III

50. If Company C belonged to the IT or the Infrastructure sector and the returns from C were more than expected, then which of the following statements would necessarily be true?

I. Anand earned at most 45% returns.
II. Anand earned at least 37.5% returns.
III. If Anand earned 41.25% returns, then Company D belonged to the Metals or Automobile sector.
IV. If Anand earned 41.25% returns, then Company A gave more than the expected returns.

(A) I and II
(B) I and III
(C) III and IV
(D) II and IV

EXERCISE-3

Directions for questions 1 to 4: Answer these questions on the basis of the information given below.

CAT 2008 paper had three sections, namely Quantitative (Q), Verbal (V) and Data Interpretation (DI) with maximum marks in each section being 100. Each section had questions with 1, 2 and 3 marks. Each section had the same number of total questions and in each section had the number of one-mark questions was one more than the number of two-mark questions which in turn was one more than the number of three-mark questions. The penalty for each wrong answer was one fourth of the marks for that question.

1. The number of three-mark questions in the paper was
(A) 45 (B) 48
(C) 54 (D) 60

2. The total number of questions in the paper was
(A) 135 (B) 144
(C) 153 (D) 162

3. What is the maximum score possible if a person attempts less than half of the total questions in the paper with an accuracy of less than 80%?
(A) 108 (B) 120
(C) 146 (D) 164

4. What is the maximum possible score of a person with an accuracy of exactly 50%?
(A) 108 (B) 114.5
(C) 138.0 (D) 175.5

Directions for questions 5 to 8: Answer these questions on the basis of the information given below.

145 people who visited a bakery on a certain day ordered at least one and at most three items among burgers, pastries and bread. 103 customers ordered exactly two items. The number of customers who ordered only one item is six times the number of customers who ordered all the three. The number of customers who ordered only pastries and bread is two times the number of customers who ordered only burgers and pastries. The number of customers who ordered only bread is four more than those who ordered only burger and four less than those who ordered only pastries. The number of customers who ordered bread is 25 more than those who ordered burger.

The following table gives information about the average amount paid by customers who ordered different number of items.

Customers ordering	Average amount (in ₹)
Only one item	170
Only two items	290
Three items	?
All customers	266

5. How many customers ordered only pastries?

6. How many customers ordered both pastries and bread?

7. What is the average amount paid by customers who ordered all the three items?

8. Among customers who ordered only one item, the average amount paid by customers who ordered burgers, pastries and bread are in the ratio of 1 : 2 : 3. What is the average amount paid by customers who ordered only pastries (approximated to the closest integer)?

Directions for questions 9 to 13: Answer these questions on the basis of the information given below.

In a survey conducted among certain number of students, the ratio of the number of students who play Tennis, Football, Cricket and Hockey is 10 : 15 : 6 : 4. 25% of the students who play Tennis, play only Tennis. Among the students who play Football, boys are 12.5% less than girls. Among the students who play Cricket, boys are twice the number of girls. Among the students who play Hockey, boys are $33\frac{1}{3}\%$ of the number of girls.

No student plays only Hockey, only Hockey and Cricket, only Hockey and Tennis. The number of girls who play any combination of exactly three of the given four sports is equal to the number of girls who play all the four sports. No girls play only Tennis, only Tennis and Football, only Tennis and Cricket. The number of girls who play Tennis is 80 and the number of girls who play Hockey is 40 more than those who play Tennis. Among the boys, those who play only Tennis, those who play Tennis and Cricket, those who play Football and Cricket are equal. The number of boys who play only Tennis, Football and Hockey and the number of students who play all the four sports are 30 each. The number of boys who play only Tennis and Cricket are 20% less than the number of boys who play only Tennis, Football and Cricket. The number of students who play only Football is 260.

9. Find the number of boys who play at least two of the given games?
(A) 220 (B) 240
(C) 260 (D) 320

10. Find the number of girls who play at most two of the given games.
(A) 120 (B) 160
(C) 180 (D) Cannot be determined

11. Among the students who play Tennis and Hockey, what is the difference between the number of boys and the number of girls?
(A) 20 (B) 10
(C) 30 (D) 40

Difficulty Level-3: Advanced

12. If 10 boys and 10 girls do not play any of the given four sports, then among the students surveyed what is the ratio of the number of boys to the number of girls?

(A) 12 : 7 (B) 8 : 5
(C) 9 : 8 (D) None of these

13. Among the students who play Tennis, Football and Cricket, by what per cent was the number of girls less than the number of boys?

(A) 12.5% (B) $33\frac{1}{3}$ %
(C) 50% (D) 150%

Directions for questions 14 to 17: Answer these questions on the basis of the information given below.

XYZ Ltd. was in the business of providing coaching classes for CAT. The number of students joining the institute would depend on whether or not they provide videos for the chapters taught.

The fee charged by the institute is ₹40,000 when no videos are provided and ₹45,000 when videos are provided, and a student can enrol by paying 25% as the first instalment and the remaining after two months. At the time of joining, a student knows whether he/she is joining a course with or without videos and pay the fee accordingly. But it was observed that 20% of the students who enrolled drop out, and thus do not pay the second instalment.

The cost incurred by the Institute on each student who does not drop out (without considering the cost of videos) is ₹25,000 per student. It also incurs a cost of ₹6000 per student who drops out without paying the second instalment. The following table gives the number of students who are expected to enrol in the institute with and without videos, in the next four months. For each month, all students who join are provided videos or none are provided videos.

Month	With videos	Without videos
June	7400	4600
July	5300	3900
August	6500	4200
September	6800	4900

14. What is the profit expected from students who enrol in June if the institute decides to go ahead without videos?

(A) ₹5.21 crore (B) ₹5.62 crore
(C) ₹5.89 crore (D) ₹6.23 crore

15. What can be the maximum amount spent on producing videos for students who enrol in July, if the institute wants to make at least 10% more profit than it would have done without videos?

(A) ₹3.72 crore (B) ₹3.55 crore
(C) ₹3.38 crore (D) ₹3.21 crore

16. If the videos for students who enrolled in September is expected to cost ₹4.9 crore, what should the institute do to maximize its profit with the information available with it regarding the number of students expected to join?

(A) Do not provide videos
(B) Provide videos
(C) Both give equal profit
(D) Cannot be determined

17. If the institute provided videos for students who enrolled in all the four months and it earned a total profit of ₹21.6 crore in this time, then what was the total cost of providing videos?

(A) ₹19.21 crore (B) ₹20.34 crore
(C) ₹21.64 crore (D) ₹22.73 crore

Directions for questions 18 to 21: Answer these questions on the basis of the information given below.

The table given below shows 12 numerical values. Each numerical value represents exactly one of the following two data.

(a) Sales of any fruit on any particular day (in pieces).
OR
(b) Sales of any fruit on any particular day (as a % of the total sales of all the fruits on that particular day).

For example, sales of apples on Day 2 was either 24 pieces or 24% (of the total fruits sold on Day 2).

Sales (either in pieces or in percentage terms) of any fruit (on any day) was a two-digit natural number. Also, no two types of fruits had the same sales on any day. Further, no fruit had the same sales on any two days. On any day, at least one numerical figure was (in pieces) terms while at least one numerical figure was (in percentage) terms. Total sales of any fruit (across three days put together) was more than 100 and less than 200.

	Day 1	Day 2	Day 3
Apples	36	24	33
Oranges	38	41	27
Mangoes	31	68	22
Guavas	28	13	58

18. Find the total number of fruits sold across three days put together.

(A) 400 (B) 500
(C) 600 (D) None of these

19. How many of the following statements are false?

(i) There were exactly seven occasions when the total number of fruits (of any kind) sold on any day was more than 50 pieces.
(ii) The total number of oranges and mangoes sold across three days is more than the total number of apples and guavas sold across three days.

(iii) The total number of mangoes sold on Day 3 is not more than the total number of guavas sold on Day 2.

(A) 0 (B) 1

(C) 2 (D) 3

20. What was the maximum percentage increase in the sales (in pieces) of any fruit across two consecutive days?

(A) 115.79% (B) 123%

(C) 175% (D) 346.15%

21. Which fruit registered the maximum change in sales (in percentage points) from Day 1 to Day 3?

(A) Apples (B) Oranges

(C) Mangoes (D) Guavas

Directions for questions 22 to 25: Answer these questions on the basis of the information given below.

In a city, there are 13,600 employees. The ratio of number of males and females among them is 11 : 6. All the employees work in one establishment among Pizza point, Pasta centre, Frenzy ice-cream, Cookieyum, Caketake and Sandwichy. 20% of the total employees are employed at Pizza point. 24% of the females work in Sandwichy. $10\frac{2}{11}$% of the males work in Cookieyum. Females working in Caketake are 50% of the females working in Sandwichy. $20\frac{5}{11}$% of the males work in Frenzy ice-cream. Males working in Frenzy ice-cream are 20% more than females working there. 30% of the total employees work in Sandwichy. Number of females working in Sandwichy is equal to the number of males working in Pasta centre. The ratio of the number of males to females working in Pasta centre is 3 : 1 and that working in Caketake is 5 : 8. It is also know that all employees in the six establishments are from the city.

22. How many people work in Cookieyum?

23. The females working in Caketake form what percentage of the females working in Pasta centre?

24. How many people work in Frenzy ice-cream?

25. How many people work in Pasta centre?

ANSWER KEYS

Exercise-1

1. (B)	10. (B)	19. (B)	28. 6.35	37. (C)	46. (B)
2. (A)	11. (B)	20. (A)	29. 12	38. (D)	47. (C)
3. (A)	12. (C)	21. (C)	30. 7500	39. (C)	48. (D)
4. (A)	13. (C)	22. (A)	31. (D)	40. (C)	49. (B)
5. (C)	14. (D)	23. (B)	32. (D)	41. (A)	50. (A)
6. (D)	15. (B)	24. (C)	33. (B)	42. (D)	
7. (A)	16. (B)	25. (D)	34. (B)	43. (B)	
8. (B)	17. (D)	26. 1	35. (B)	44. (D)	
9. (B)	18. (C)	27. 5	36. (A)	45. (C)	

Exercise-2

1. (C)	10. (C)	19. 30	28. (B)	37. (C)	46. (D)
2. (D)	11. (B)	20. 50	29. (C)	38. (B)	47. (A)
3. (C)	12. (D)	21. 36	30. (A)	39. (C)	48. (C)
4. (D)	13. (D)	22. 42.4	31. (B)	40. 2	49. (A)
5. (A)	14. (C)	23. (C)	32. (B)	41. 4	50. (D)
6. (B)	15. (B)	24. (D)	33. (B)	42. 3	
7. (D)	16. (A)	25. (B)	34. (C)	43. (A)	
8. (A)	17. (C)	26. (A)	35. (B)	44. (C)	
9. (B)	18. 56	27. (C)	36. (D)	45. (C)	

Exercise-3

1. (B)	5. 16	9. (C)	13. (B)	17. (D)	21. (C)	24. 3300
2. (C)	6. 48	10. (D)	14. (C)	18. (C)	22. 1028	25. 1536
3. (D)	7. 430	11. (A)	15. (B)	19. (A)	23. 150	
4. (D)	8. 161	12. (D)	16. (B)	20. (C)		

SOLUTIONS

EXERCISE-1

1. Let A, B, C and D be the weights of the 4 crates.
A = 100 kg (Given)
C – A = D – C (given)
2C = D + 100 (1)
$\frac{A+C+D}{3} = 300$
C + D = 800 (2)
Solving (1) and (2), C = 300 kg, D = 500 kg
$\frac{A+B+C+D}{4} = 375 \Rightarrow B = 600$ kg
∴ A = 100 kg, B = 600 kg, C = 300 kg D = 500 kg.
Crates B and D weigh more than the average of the 4 crates.

2. $\frac{C}{D} = 60\%$
∴ Crate C weighs 40% less than crate D.

3. Crate B weighs 600 kg.

4. If crate B is not loaded, the average weight of crates A, C and D = $\frac{900}{3} = 300$ kg.

5. $\frac{B-A}{D+C} = \frac{500}{800} = \frac{5}{8}$

6. Hiralal's wife gets $\frac{1}{3}$ of 51 lakh = ₹17 lakh
Haralal gets $\frac{1}{2}$ of (51 – 17) lakh = ₹17 lakh
Heera's share : Henna's share = 8 : 9
Heera's share = $\frac{8}{17}(17)$ lakh = ₹8 lakh
Henna's share = 17 – 8 = ₹9 lakh
Hiralal's wife's share = ₹17 lakh
Heralal + Heera's share = ₹25 lakh
$\frac{17}{25} \times 100 = 68\%$

7. Haralal's share – Henna's share = ₹8 lakh

8. $\frac{(\text{Heera} - \text{Henna})}{\text{Hiralal's wife}} \times 100$
$\frac{1}{17} \times 100 = 5\frac{15}{17}\%$

9. Hiralal's wife got = $\frac{17}{51} = \frac{1}{3} = 33.33\%$ of the total property.
Heera got $\frac{8}{51} = 15.69\%$ of the total property.
∴ Hiralal's wife got 17.64 percentage points of the property more than Heera.

10. After interchanging their shares, Henna gets ₹17 lakh and Haralal gets ₹9 lakh.
$\frac{\text{Heera}}{\text{Haralal} + \text{Hiralal's wife}} \times 100$
$\frac{8}{9+17} \times 100 = \frac{8}{26} \times 100 = 30.77\%.$

11. Amar's share = ½ × 1700 + 50 = 900
Akbar's share = $\frac{6}{17} \times 1700 = 600$
Anthony's share = 1700 – 1500 = ₹200
Difference between the amount with Amar and Anthony = ₹900 – ₹200 = ₹700

12. The amount Amar has more than Akbar and Anthony = ₹900 – (₹600 + ₹200) = ₹100.

13. The amount Akbar has more than Amar = ₹(600 – 200) = ₹400.

14. Required ratio = 6 : 9 : 2.

15. Required percentage = $\frac{(600-200)}{900} \times 100 = 44\frac{4}{9}\%$

Solutions for questions 16 to 20: Given 60% of boys who either like Tennis or Formula 1, like Tennis, hence 40% of them like Formula 1. Let the number of boys be x.

Therefore, $\frac{40}{100}x = 4 \Rightarrow x = 10$

Hence, the number of girls who like either Tennis or Formula 1.

$\therefore \frac{10}{G} = \frac{2}{9} \Rightarrow G = 45$

$= \frac{9}{2} \times 10 = 45$

This simplification can be tabulated as below:

	Tennis	Formula 1	
Boys	6	4	10
Girls	29	16	45
	35	20	

Also, the number of students in the school = $\frac{6}{5} \times 55 = 66$.

Among the remaining students who did not like any of these games, the number of boys = $11 \times \frac{6}{11} = 6$ and the number of girls = 5

Number of boys who like Tennis = $\frac{60}{100} \times 10 = 6$

$\therefore$ Number of girls who like Tennis = 35 – 6 = 29

Number of boys who like Formula 1 = 4

Number of girls who like Formula 1 = 20 – 4 = 16

16. Required percentage = $\frac{16}{6} \times 100 = 266\frac{2}{3}\%$

17. Total number of girls = 45 + 5, i.e., 50

18. Total number of boys = 10 + 6 = 16
Number of girls who like Tennis = 29
$\therefore$ Required ratio = 16 : 29

19. Total number of students in the college = number of boys + girls = 66

20. Required percentage = $\frac{29}{66} \times 100 = 43.93$

21. Madhusudhan invested ₹3 lakh each in Schemes 2 and 3 and ₹4 lakh in Scheme 1.
Scheme 1: Amount at the end of 1 year
= 400000 (1.08) = ₹4,32,000.
Scheme 2: Amount invested every month

$= ₹25{,}000 \left(\because \frac{300000}{12}\right)$

Amount at the end of a year
= 25000 (1 + 12r) + 25000 (1 + 11r) + ... 25000 (1 + r)
= 25000 (12 + 78r)
= 25,000 (12 + 78 (0.5%)) = ₹3,09,750.
Scheme 3: Amount at the end of a year
= 300000 (1.1) = ₹3,30,000.
Net worth at the end of a year = ₹10,71,750.

22. Return on Scheme 3 = ₹30,000
Return on Scheme 1 = ₹32,000

$\frac{32000 - 30000}{32000} \times 100 = 6.25\%$

23. Madhusudhan's return on Scheme 2 was ₹9750. If he invests twice the amount he originally invested, then he would get a return of ₹19,500.

24. Instead of 4(1.08) lakh + 3.0975 lakh + 3(1.1)lakh, the total net worth would be 3(1.08) lakh + 3.0975 lakh + 4(1.1) lakh which comes to ₹10.7375 lakh.
$\therefore$ There is a change of ₹2000.

25. By investing a similar amount in Scheme 2 for another year, Madhusudhan's return can be determined by
25000 (1 + 24r) + 25000(1 + 23r) +..........25000 (1 + r)
= 25000 (24) + 25000 (300r)
= ₹6,37,500
$\therefore$ Return on Scheme 2 = ₹37,500
Return on Scheme 1 and Scheme 3 for the first year
= 32000 + 30000 = ₹62,000
$\therefore$ He earns ₹24,500 less.

Solutions for questions 26 to 30:

Let the total savings be 100%. Divisions of the savings will be as follows:
Stocks → 20%
NSC → 30%
Land → 40% of the remaining → 40% of 50% = 20%,
FD → (100 – (20 + 30 + 20)) = 30%

26. Since the same amount is invested in shares and land, let us assume this is 100x each

$\Rightarrow$ Total amount is $100x \times \frac{100}{20} = 500x$

Value of my shares after two years

$= 100x\left(1+\frac{20}{100}\right)\left(1+\frac{45}{100}\right) = 174x.$

The value of land is $100x\left(1+\frac{10}{100}\right)\left(1+\frac{10}{100}\right) = 121x.$

My gain from land is $121x - 100x = 21x$ on which I pay 10% tax.
$\therefore$ Net gain = $21x - 2.1x = 18.9x$
Differences in gain = $74x - 18.9x = 55.1x$
$55.1x = 55{,}100$
$\therefore$ $x = 1000$
Investment in shares and land = 1000 × 100 = 1 lakh

27. The total amount invested initially was ₹5 lakh.

28. Amount invested in NSC = 1,50,000

After 1 year it becomes $1{,}50{,}000\left(1+\frac{8}{100}\right) = 1{,}62{,}000$

After the second year it is $1{,}62{,}000 \times \left(1+\frac{8}{100}\right)$

= 1,74,960 ≈ 1.75 lakh
Amount from shares = 1,74,000
Amount from land = 1,18,900 ≃ 1,19,000
Amount from FD after 1 year

$= 1{,}50{,}000\left(1+\frac{6}{100}\right) = 1{,}59{,}000$

$= 1{,}59{,}000 - \left(\frac{5}{100} \times 9{,}000\right) = 1{,}58{,}550$

In the second year, I get another 8500 (which is same as the first year's net interest) and 5.7% on 8500.

$\frac{5.7}{100} \times 8{,}550 = ₹488 \approx ₹500$ (since 5% is paid as tax, 5% of 6% is 0.3%, therefore, 6 – 0.3 = 5.7)

∴ Total amount from FD's = ≃ 1,67,500

Total amount = 1.75 + 1.74 + 1.19 + 1.67 = 6.35 lakh.

29. 5 lakh invested becomes 6.35 lakh in two years.
Since the actual gain is $\frac{1.35}{5.00} \times 100 = 27\%$ for two years, the compounded annual rate of return should be a little less than half of it (i.e., $(\sqrt{1.27} - 1) = 12\%$

30. From solution to Q.18, the amount in NSC after two years = 1,74,960
Total amount from FD = 1,67,500
The difference = 1,74,960, – 1,67,500 = 7500

31. Let N, J, R, B represent sales of the 4 friends on that day.
N, R + x, R, R – x are the 4 friends sales
We also know, R = 600
N, 600 + x, 600, 600 – x are their sales.
Average of 600 + x, 600, 600 – x = 600
N = 600 + 200 (given) ⇒ N = ₹800
800, 600 + x, 600, 600 – x are the sales of Nikhil, Joy, Rohit and Binoy, respectively.
We do not know Joy's and Binoy's sales for that day but we know J + B = 1200.
And since each jar costs R18, B should be a multiple of 18.
In options A, B and C Binoy's sales namely 540, 576, 360) are multiplies of 18 but in option D, Binoy's sales are not a multiple of 18.

32. (800 + 600) exceeds (600 – x + 600 + x) by 200 irrespective of the value of x. Therefore, B cannot be determined.

33. Nikhil couldn't have sold 30, 35 or 15 plyboards as the cost of each plyboard will not be an integral number of rupees in that case.

34. 0.1(800) + 0.2(600 – x) + 0.3(600) + 0.2(600 + x) = ₹500.

35. 800 + 600 – x = 600 + x + 600
200 = 2x ⇒ x = 100
Sales of Joy = 600 + x = ₹700
∴ Joy sold 35 tyres at $\frac{700}{35}$ = ₹20/tyre

36. Let the total number of students who appeared for the board exams in IPS be N.
0.8N passed the board exams while 0.2N failed. Of the 0.8N who passed, 60% or 0.48N joined IIC.
The 0.48N students opted for Science, Commerce and Humanities in the ratio of 3 : 4 : 5.
∴ 0.12N from IPS opted for Science, 0.16N for IPS opted for Commerce and 0.2N from IPS opted for Humanities.
60% of Science stream = 0.12N
⇒ Total Science stream = 0.2N
Similarly, total Commerce stream = 0.4N
Total Humanities stream = 0.4N
Given, Science + Commerce + Humanities
= 0.2N + 0.4N + 0.4N = N = 800
Students who failed = 0.2N = 160

37. 10% of 0.4N = 10% of 0.4(800) = 32

38. Science stream has 0.2N = 160 students. Of the 160, 96 are from IPS and 64 are from APS.
$\frac{2}{3} = \frac{64}{x} \Rightarrow x = 96$
∴ 96 students in Commerce stream are from APS.

39. $\frac{2}{4} = \frac{64}{y} \Rightarrow y = 128$
In the Humanities stream, 50% or 160 students are from IPS and 128 students are from APS.
That leaves 32 students who are from neither school. Therefore, there are 10% students.

40. Total fees = 160 (1000) + 320 (1500) + 320 (2000)
= ₹12.8 lakh

Solutions for questions 41 to 45: Let the scores of Ajay, Bharat, Kumar, Sanjay and Vishal be denoted by A, B, K, S and V, respectively.

A + B + K = 135   B + K + S = 137
K + S + V = 132   S + V + A = 138
V + A + B = 133

Adding, we get
3(A + B + K + S + V) = 675
A + B + K + S + V = 225.
S + V = (A + B + K + S + V) – (A + B + K) = 225 – 135
A = (A + V + S) – (V + S) = 138 – 90 = 48
K = (K + S + V) – (S + V) = 132 – 90 = 42
B = (A + B + K) – (A + K)= 135 – (48 + 42) = 45
S = (B + K + S) – (B + K)= 137 – (45 + 42) = 50
V = (S + V) – S = 90 – 50 = 40
Thus, the scores are as follows:

A	B	K	S	V
48	45	42	50	40

41. Vishal scored the least marks among the given students.

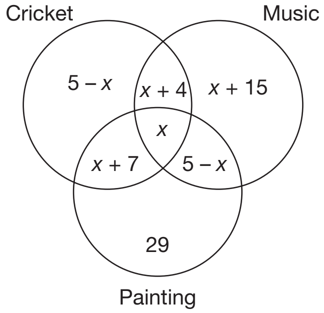
42. Sanjay scored the highest marks among the given students.

43. Only Sanjay scored more marks than Ajay.

44. The maximum difference in the marks scored by any two students = 10

45. Bharat scored 90% of Sanjay's score.

Solutions for questions 46 to 50: The Venn diagram for the given information is as follows:



As the maximum value of x is 5 and the maximum number of children who enrolled for any single event alone was 29, it has to be in painting alone.

Given, $5 - x + x + 4 + x + 7 + x = 2x + 16 = 1/2\ (x + 4)$
$4x + 32 = x + 41$
$\therefore\ x = 3$

46. The number of children who enrolled for all the three events is 3.

47. The number of children who enrolled for more than one event is the same as the number of children who enrolled for cricket $= 2x + 16 = 22$.

48. The number of children who enrolled for exactly one event is $29 + 5 - x + x + 15 = 49$.

49. The number of children who enrolled for music is $2x + 24 = 30$.

50. $22 - 3 = 19$. Therefore, 19 children enrolled for exactly two events.

EXERCISE-2

1. Maximum non-taxable income after deduction = 1 lakh
Maximum deduction on investments in securities = 1 lakh
Now he can also avail a housing loan such that the loan repayment is equal to 20% of his annual income.
If X is the annual income, $2 + 0.2X = X$
$X = 2.5$ lakh.

2. For an income of ₹10 lakh, maximum housing loan repayment of 20% of ₹10 lakh = 2 lakh
Maximum deductions = 2 + 1 = 3 lakh
Income on which tax is applicable = 7 lakh.
Tax on 7 lakh:
Up to 1×5 lakh = 5,000
1.5 to 5 lakh = 70, 000
5 lakh to 7 lakh = 60,000 + 0.1 (60,000)
Total tax (minimum) = 1,41,000

3. Maximum tax is applicable when the deductions are zero.
Up to 1.5 lakh = 5,000
1.5 to 5 lakh = 70,000
5 to 6 lakh = 30000 + 3000 ⇒ total = 1, 08,000
Minimum tax is when he has a deduction of 1 lakh on investments and 1.2 lakh (20% of 6 lakh) as housing loan repayment.
$\therefore$ Taxable income = 3×8 lakh
Up to 1.5 lakh = 5000
1.5 to 3×8 lakh = 46, 000
Total tax (minimum) = 51,000
Difference = 57,000

Solutions for questions 4 to 7: Let the number of samosas that Murali sells on the first day, i.e., on Monday be n.

It is given that the number of samosas that he sells on one day will be X less than the previous day twice during the week and X more than the previous day thrice during the week. The maximum and minimum number of samosas he sells on any day are 150 and 90, respectively.

So, we have to arrange 2 decrease by X and 3 increase by X from Tuesday to Saturday, which can be done in $\dfrac{5!}{2!3!} = 10$ ways

The 10 possible cases are below:

Case	Mon	Tue	Wed	Thur	Fri	Sat	Total
1	n	$n - x$	$n - 2x$	$n - x$	n	$n + x$	$6n - 3x$
2	n	$n - x$	n	$n - x$	n	$n + x$	$6n - x$
3	n	$n - x$	n	$n + x$	n	$n + x$	$6n + x$
4	n	$n - x$	n	$n + x$	$n + 2x$	$n + x$	$6n + 3x$
5	n	$n + x$	n	$n - x$	n	$n + x$	$6n + x$
6	n	$n + x$	n	$n - x$	n	$n + x$	$6n + 3x$
7	n	$n + x$	n	$n - x$	$n + 2x$	$n + x$	$6n + 5x$
8	n	$n + x$	$n + 2x$	$n + x$	n	$n + x$	$6n + x$
9	n	$n + x$	n	$n - x$	n	$n + x$	$6n + x$
10	n	$n + x$	n	$n - x$	n	$n + x$	$6n + x$

For Case I, $n - 2x = 90$ and $n + x = 150$
Difference = $3x$
$3x = 60$
$x = 20$

So, the number of samosas he sells are as follows.

Mon	Tue	Wed	Thu	Fri	Sat
130	110	90	110	130	150

Case	Mon	Tue	Wed	Thu	Fri	Sat	Total	Money earned in the week
1	130	110	90	110	130	150	720	2880
2	120	90	120	90	120	150	690	2760
3	120	90	120	150	120	150	750	3000
4	110	90	110	130	150	130	720	2880
5	120	150	120	90	120	150	750	3000
6	90	150	90	150	90	150	720	2880
7	90	120	90	120	150	120	690	2760
8	90	120	150	120	90	120	690	2760
9	90	120	150	120	150	120	750	3000
10	90	110	130	150	130	110	720	2880

4. If Murali sells 750 samosas in a particular week, it must be case 3, case 5 or case 9.
Number of samosas he sells on Friday:
Case 3: 120
Case 5: 120
Case 9: 150
So, he sells either 120 or 150 samosas on Friday.

5. Murali sells the minimum number of samosas on Friday. So, it is either Case 6 or Case 8.
Since he does not sell the maximum number of samosas on Thursday, it is not Case 6.
So, it must be Case 8 and he earns ₹2760 in that week.

6. Since Murali sells 130 samosas on Friday, it can be either Case 1 or Case 10.
In both the above cases, he sells 110 samosas on Tuesday.

7. Since he sells 130 samosas on Wednesday, the number of samosas he sells on Saturday can only be 110.
Thus, we do not need any other information in this case to answer the question.

8. $1.2a + 1.35b = 1.25a + 1.25b \Rightarrow a = 2b$
$\therefore\ a = 66.66$ million; $b = 33.33$ million
In State A, 22.22 million live in cities.
In State B, 8.33 million live in cities.
In Year 25, total city dwellers
$= 2(22.22 + 8.33) = 61.1$ million

9. Rural population in Year 25 = 125 – 61.1 = 63.9 million = 51.12%
Rural population in Year 1 = 69.45 million = 69.45%
Change = 69.45 – 51.12 = 18.33 percentage points

10. 1.35(66.66) + 1.2(33.33) = 130 million

11. 0.8(22.22) + 0.7(44.44) + 0.8(8.33) + 0.7(25) = 73 million

12. $66.66[1 + 0.02x] + 33.33[1 + 0.03x] \geq 125$
$\Rightarrow x \geq 11$ $\quad\therefore$ In 11 years it can happen.

Solutions for questions 13 to 17: Let the number of engineers and that of non-engineers be e and n, respectively.

(i) $e = n\left(1 + \frac{20}{100}\right)$. $e = \frac{6}{5}n$

(ii) Number of employees who specialized in Finance = $\frac{8}{15}n$

(iii) Number of employees who specialized in HR = 5/12

i.e., $= \frac{5}{12}\left(\frac{6}{5}n\right) = \frac{n}{2}$

Total number of employees = $\frac{8}{15}n + 35 + \frac{n}{2}$

$= e + n = \frac{6}{5}n + n$

$35 = \frac{11}{5}n - \left(\frac{8}{15}n + \frac{n}{2}\right) = \frac{35n}{60}$

$n = 30$

$\therefore\ e = 36$ and $e + n = 66$

$\frac{8}{15}n = 16$ and $\frac{n}{2} = 15$

The conclusions above are represented below:

EB	SP: F	M	HR	T
E	a	b	c	36
N.E	7	d	5	30
T	16	35	15	66

$a + 7 = 16.\ \therefore\ a = 9.$
$7 + d + 5 = 30.\ \therefore\ d = 18$
$b + d = 35.\ \therefore\ b = 17$
$c + 5 = 15.\ \therefore\ c = 10$

13. Required number = d = 18

14. Required number = Total number of non-engineers – Number of non-engineers who specialized in Finance
= 30 – 7 = 23

15. Required number = 15

16.

Choice (A) = 5,
Choice (B) = $a = 9$,
Choice (C) = $d = 18$,
Choice (D) = $b = 17$

17. Required percentage = Percentage that 36 forms of 66

$$= \frac{36}{66}(100)\ \% = \frac{6}{11}(100)\% = 54^{6}/_{11}\%.$$

Solutions for questions 18 to 22: Let the weights of the 1st lightest, 2nd lightest, 3rd lightest, 4th lightest and heaviest boys be ℓ_1, ℓ_2, ℓ_3, ℓ_4 and ℓ_5, respectively. When they are weighed in groups of three, the possible combinations are $\ell_1\ell_2\ell_3$, $\ell_1\ell_2\ell_4$, $\ell_1\ell_2\ell_5$, $\ell_1\ell_3\ell_4$, $\ell_1\ell_3\ell_5$, $\ell_1\ell_4\ell_5$, $\ell_2\ell_3\ell_4$, $\ell_2\ell_3\ell_5$, $\ell_2\ell_4\ell_5$ and $\ell_3\ell_4\ell_5$.

When we add, we get:

$$6(\ell_1 + \ell_2 + \ell_3 + \ell_4 + \ell_5) = 106 + 116 + 122 + 126 + 132 + 146 + 120 + 126 + 136 + 142$$

$$\therefore\ \ell_1 + \ell_2 + \ell_3 + \ell_4 + \ell_5 = \frac{1272}{6} = 212$$

Since the weights are in the order of ℓ_1, ℓ_2, ℓ_3, ℓ_4 and ℓ_5.

$\ell_1 + \ell_2 + \ell_3 = 106$ kg (lowest weight) and
$\ell_3 + \ell_4 + \ell_5 = 146$ kg (heaviest weight)
$\therefore\ \ell_1 + \ell_2 + 2\ell_3 + \ell_4 + \ell_5 = 252$ kg and
since $\ell_1 + \ell_2 + \ell_3 + \ell_4 + \ell_5 = 212$ kg
$\ell_3 = 40$ kg
$\therefore\ \ell_1 + \ell_2 = 106 - 40 = 66$ kg and
$\ell_4 + \ell_5 = 146 - 40 = 106$ kg

The group with the second heaviest weight must be $\ell_2\ell_4\ell_5$, which is 142 kg.

$\therefore\ \ell_2 = 142 - 106 = 36$ kg
and $\ell_1 = 66 - 36 = 30$ kg

The group with the second lightest weight is $\ell_1\ell_2\ell_4$ and $\ell_1 + \ell_2 + \ell_4 = 116$ kg.

$\therefore\ \ell_4 = 116 - 66 = 50$ kg and $\ell_5 = 106 - 50 = 56$ kg.
$\Rightarrow \ell_1 = 30$, $\ell_2 = 36$, $\ell_3 = 40$, $\ell_4 = 50$ and $\ell_5 = 56$

Given, Ajay = Average of Bhushan and Emmanuel and also Chetan < Emmanuel < Deepak

$\Rightarrow$ Bhushan = 30, Ajay = 40, Emmanuel = 50,
Chetan = 36, Deepak = 56

18. The weight of the heaviest boy is 56 kg.

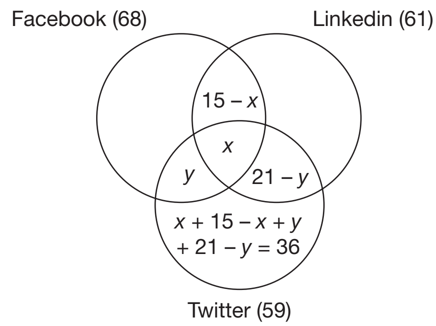
19. Bhushan's weight is 30 kg.

20. Emmanuel's weight is 50 kg.

21. Chetan's weight is 36 kg.

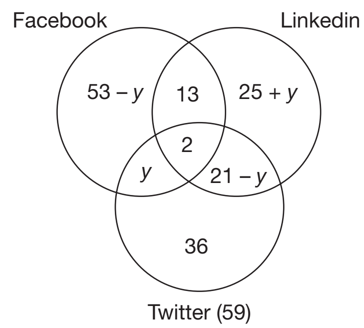
22. Average weight = $\frac{212}{5} = 42.4$ kg

Solutions for question 23 to 25: The number of employees who are active would be 150 and 30 employees are not active on any of the three sites.



As 36 + 21 + x = 59, x = 2 (employees active on all the sites).

$\therefore$ We shall draw the following Venn diagram.



23. Given, $21 - y = 13\ \therefore\ y = 8$
Only Facebook = 53 – 8 = 45

24. Exactly one site = 150 – (13 + 2 +21) = 114

The required percentage = $\frac{114}{180} \times 100 = 63.33\%$

25. The number of employees active only on Twitter is 36.

Solutions for question 26 to 28: The given information can be tabled as follows.

	Boys		Girls		
	Urban	Rural	Urban	Rural	Total
Mahatma Gandhi	132	40	45	50	267
Mother Teresa	120	40	80	50	290
Nelson Mandela	78	40	35	40	193
Total	330	120	160	140	750

26. The total number of students surveyed was 750.

27. 40 girls with a rural background voted for Nelson Mandela.

28. Girls with a rural background who voted for Mahatma Gandhi was one sixth of the total girls surveyed.

29. Total marks = 95 × 4 = 380
Expected total marks = 230

$\therefore$ Difference = 150

The possibilities are as follows:

(1) $50 \times 2 + 80 - \dfrac{60}{2} = 150$

(2) $60 \times 2 + 50 - \dfrac{40}{2} = 150$

In both the cases, he got more than the expected marks in English.

30. Let (I) be true.
Total marks > 400
Difference > 170
When Ramesh gets the expected marks in Maths, then the maximum possible difference

$$= 60 \times 2 + 50 - \frac{40}{2} = 150$$

$\therefore$ (II) is false.

31. Ramesh gets equal marks in two subjects if
(1) $40 \times 2 = 80 \times 1$
(2) $40 \times 3 = 60 \times 2$
(3) $40 \times 1 = 80 \times \dfrac{1}{2}$
$\therefore$ Physics is always one of the subjects.

32. It is possible only in the following case.
Here, $80 \times \dfrac{1}{2} = 40 \times 1$ and each of them multiplied by 3 gives 60×2.
In this case, he must have got thrice the expected marks in English.
$\because$ Total marks $= 40 \times 1 + 80 \times \dfrac{1}{2} + 50 \times 3 + 60 \times 2 = 350$

33. The cars which were sold in 2009 would come for a paid recharge in 2012. As 60% of the owners would go for a 3rd party replacement and the increase there is 294 (2820 – 2526), 294 is 60% of the cars sold, the number of cars sold would be 490.

34. All cars manufactured from 2006 to 2008 would come for a paid recharge in 2011. As 60% of them go to third-party vendors and 60% of the total = 2526
$\therefore$ Total = 4210.

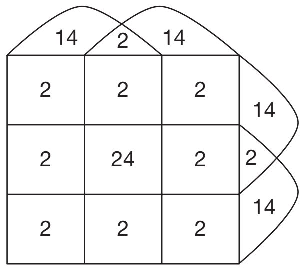
35. 60% were recharged by 3rd party vendors and that is equal to 2214. Then, 40% would be recharged by the company and that would be $\dfrac{40}{60} \times 2214 = 1476$.

36. As the cars sold in 2011 would come for a paid recharge in only 2014, we cannot determine the value.

37. As we have to allocate 2 to each of the 15 values, a total of 64 instances is covered.
Now the remaining 70 students must cover 176 ($60 \times 4 - 64$) instances.
Let us try to maximize the students who are playing three games and minimize the students who play one or two games. This happens only when all these students are playing exactly one game.
Let x students play one game and y students play 3 games.
$x + y = 70$
$x + 3y = 176 \Rightarrow y = 53$ and $x = 17$
$\Rightarrow$ Now, the total number of students playing at most two games is 20 (i.e., initial 10 values) + 17 = 37.

38. As each of the values is at least 2, the remaining number of students is 70.
Now, as each of the games is played by the same number of persons, each of the six letters which represent students who play exactly two games must be equal, i.e., each of b, d, f, l, n and k must be 11. The remaining four students must be distributed for e, h, m and j.
$\therefore$ The number of students playing exactly two games is $11 \times 6 + 2 \times 6 = 78$.

39. Let the number of students playing exactly one, two, three and four games be a, b, c and d, respectively.
Now, $a + b + c + d = 70$ (As all the 15 values take 2 each).
$a + 2b + 3c + 4d = 136$
d can at most be 22 and, in this case, $a = 48$, i.e., as follows:



$\therefore$ The total number of students playing all the four games is 24.

40. To identify the faulty ball in the minimum number of weighings, first divide the nine balls into three equal groups and keep three balls on each pan and if they balance, the faulty ball is in the other three balls kept aside and it can be identified with one more weighing. If the pans are not in equilibrium, the faulty ball is among the three balls in the heavier pan and can be identified in one more weighing.
$\therefore$ The faulty ball can be identified in two weighings.

41. Divide the nine balls, into groups of three each. The different situations that can arise are
(A) No faulty balls in both the group weighed
(B) One faulty ball in one of the groups weighed
(C) One faulty ball in each of the group weighed
(D) Both the faulty balls in one of the group weighed

When any two groups are weighed, we can have two cases (1) they are balanced (2) there is an imbalance.
Case (1) they are balanced:
If the two groups are balanced, it could either be case (A) or case (C).
Replace one group of balls with the group kept aside in the first weighing. If the new group of balls is heavier, then both the faulty balls are in the new group weighed and can be identified in one more weighing. If the new group is lighter, both the groups which were weighed first had a faulty ball each and both can be identified in one more weighing each.
$\therefore$ In four weighing's one can identity the faulty balls. In Case (2) there is an imbalance:
If there is an imbalance in the first weighing it could either be case (B) or (D).
Replace the heavier group with the third group of balls and if these is still an imbalance, the group removed after the first weighing and the group newly weighed would have a faulty ball each and can be identified in one more weighing each, i.e., a total of four weighings. If the pans are balanced in the second weighing, the replaced balls contain both the faulty balls and it can be identified in one more weighing.
$\therefore$ The faulty balls can be definitely identified in four weighings.

42. Divide the balls into groups of three each and weigh two groups. If they are balanced, then the faulty ball is in the third group and can be identified with two more weighing's, i.e., a total of these weighings. If the balls do not balance in the first weighing, replace one group of balls with the third group and depending on the outcome one can identify the group which contain the faulty ball. Once the group containing the faulty ball is identified, we can identify the faulty ball in one more weighing, i.e., a total of three weighings.

Solutions for questions 43 to 46: Number of washing machines produced $= \frac{20}{100}(1500) = 300$

Number of washing machines of type Q produced

$= \frac{65}{100}(300) = 195$ and that of type P produced

$= 300 - 195 = 105$

Number of AC's produced $= \frac{3}{20}(1500)$

$= 225$ number of ACs of type P produced $= \frac{33\frac{1}{3}}{100}(225)$

$= 75$ and that of type Q produced $= 150$.

Number of refrigerators produced $= \frac{1}{4}(1500) = 375$.
Number of refrigerators of quality Q produced = 120 and that of type P produced = 255.

Number of microwave ovens produced
$= 1500 - (300 + 225 + 375) = 600$.

Number of microwave ovens of type P produced

$= \frac{9}{10}(600) = 540$ and that of quality Q produced = 60.

The table below summarizes all the results obtained above:

	Type		
Product	**P**	**Q**	**Total**
Microwave ovens	540	60	600
Refrigerators	255	120	375
ACs	75	150	225
Washing machines	105	195	300

43. Average number of products of type Q
$= \frac{60+120+150+195}{4} = 131.25$

44. Required ratio = 540 : 195 = 36 : 13

45. Required difference = 255 – 120 = 135

46. Required total = (75 + 540) + (195 + 120) = 615 + 315 = 930.

47. For the minimum possible returns, Anand has to get three times the expected returns from the company and expected to give 10% returns and two times the expected return from the company expected to give 20% returns.

The returns would be $\frac{25+30+40+45}{4} = 35\%$

48. For the maximum possible returns, Anand has to get three times the expected returns from the company and expected to give 45% returns and two times the expected returns from the company expected to give 25% returns.

The returns would be $\frac{50+10+20+135}{4} = \frac{215}{4} = 53.75\%$

49. For 42.5% returns on average, Anand has to get three times the returns from the company expected to give 25% return (A) and double the returns from the company expected to give 20% return (C).
$\therefore$ Company A belonged to the IT or Infrastructure sector.

50. If Company C belonged to the IT or Infrastructure sector, the returns from it would be 60%.
The maximum returns would be
$$\frac{25+10+60+90}{4}=\frac{185}{4}=46.25\%$$
and the minimum returns would be
$$\frac{25+20+60+45}{4}=\frac{150}{4}=37.5\%$$

If Anand earned 41.25% returns, he would have got double the returns from the company that was expected to give 25% (Company A). Therefore, Company A belonged to the Metals or Automobile sector and gave more than the expected returns.

EXERCISE-3

Solutions for questions 1 to 4: Let the number of two-mark questions be x. Therefore, the number of one-mark questions is $x + 1$ and the number of three-mark questions is $x - 1$.

$(x + 1)\ 1 + 2x + (x - 1)\ 3 = 100$

$\therefore\ x = 17$

$\therefore$ The number of one-mark, two-mark and three-mark questions in each section are 18, 17 and 16, respectively.

$\therefore$ Total number of questions = 51 × 3 = 153 and total number of three-mark questions = 16 × 3 = 48.

3. Maximum number of attempts = 76
To score the maximum, he has to attempt more of two and three-mark questions with the mistakes being in the one-mark questions.
Minimum number of mistakes = 16
$$\therefore \text{Net score} = 48\times3+12\times2-16\times\frac{1}{4}=144+24-4=164$$

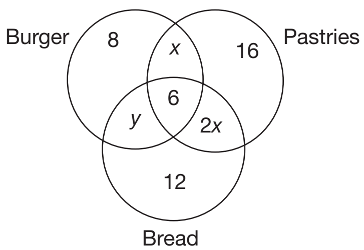
4. To have an accuracy of exactly 50%, he can attempt at most 152 questions. Therefore, he answered at most 76 questions correctly.
Maximum possible score
$$= 48\times3 + 28\times2 - 22\times\frac{1}{2} - 54\times\frac{1}{4}$$
$$= 144 + 56 - 11 - 13.5$$
$$= 200 - 11 - 13.5 = 175.5$$

Solutions for questions 5 to 8: As 103 customers ordered exactly two items, 42 customers opted for one for three items. If x is the number of customers who opted for all three items, then we have

$x + 6x = 42$

$\therefore\ x = 6$

Now, the given information can be represented as follows:



We have $18 + 2x = 14 + x + 25$

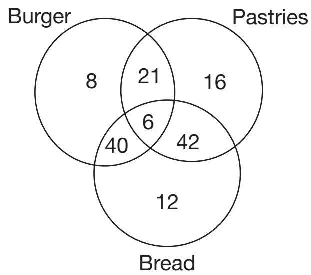
$4 + x = 25$

$\therefore\ x = 21$

and $2x + x + y = 103$

$y = 103 - 63 = 40$

The complete information is as given below.



5. 16 customers ordered only pastries.

6. 42 + 6 = 48 customers ordered both pastries and bread.

7. $170 \times 36 + 290 \times 103 + x \times 6 = 145 \times 266$

$6120 + 29870 + 6x = 38570$

$6x = 38570 - 35990$

$6x = 2580$

$x = 430$

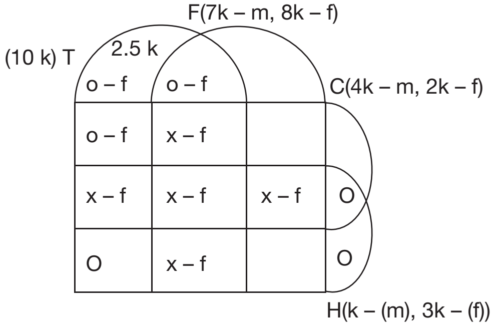
8. $$\frac{8x+16(2x)+12(3x)}{36}=170$$
$$\frac{76x}{36}=170$$
$x = 80.5$

The average amount paid by customers who ordered only pastries = $2x$ = 161.

Solutions for questions to 9 to 13: Let $10k$, $15k$, $6k$ and $4k$ be the number of students who play Tennis, Football, Cricket and Hockey, respectively.

Let x be the number of girls who play all the given four games.

The following Venn diagram depicts the data given.



Given, number of girls who play tennis = $4x = 80$

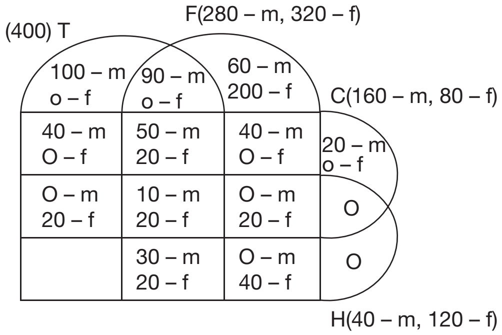
$\Rightarrow x = 20$: $3k = 80 + 40 \Rightarrow k = 40$.

Since, boys who play hockey is k, i.e., 40 and

given $n(T \cap F \cap C \cap H) = 10$: $n(T \cap F \cap H) = 30$

$\Rightarrow n(T \cap C \cap H) = n(F \cap C \cap H) = 0$

$\therefore$ The following Venn diagram can be drawn.



9. Required number of boys
= (90 + 40 + 50 + 40 + 10 + 30) = 260

10. As we do not know the number of girls who do not play any of the given sports, we cannot determine the required value.

11. Number of boys who play Tennis and Hockey
= 10 + 30 = 40
Number of boys who play Tennis and Hockey
= 3(20) = 60
Required difference = 60 – 40 = 20

12. Total number of boys = 440 + 10 = 450
Total number of girls = 340 + 10 = 350
Required ratio = 450 : 350 = 9 : 7

13. The number of girls who play Tennis, Football and Cricket = 40.
The number of boys who play Tennis, Football and Cricket = 60
Girls are less than boys by 60 – 40 = 20.

Required percentage = (20/60)*100% = $33\frac{1}{3}$ %

14. Number of students expected to join = 4600
Students expected to drop out = 920
Total fee collected = 920 × 10,000 + 3680 × 40,000
Total cost = 3680 × 25,000 + 920 × 6000
Profit = 3680 × 15,000 + 920 × 4000
= 5.52 crore + 36.8 lakh
= 5,88,80,000

15. Profit if no videos are provided
= 3120 × 15,000 + 780 × 4000
= 4.68 crore + 31.2 lakh
= 4.99 crore
$\therefore$ The institute wants to have a profit of at least 5.49 crore.
Profit if videos are provided (without considering video cost)
= 4240 × 20,000 + 1060 × 5250
= 8.48 crore + 55.65 lakh
= 9.036 crore
$\therefore$ The institute can afford to spend a maximum of 9.036 5.49 = 3.546 crore on videos.

16. Expected profit when no videos are provided
= 4120 × 15,000 + 780 × 4000
6.18 crore + 31.2 lakh = 6.492 crore
Expected profit when videos are provided
= 5440 × 20,000 + 1080 × 5,250 4.9 crore
= 10.88 crore + 56.7 lakh 4.2 crore
= 6.547 crore
$\therefore$ The institute must provide videos to maximize its profit.

17. Total profit 20,800 × 20,000 + 5200 × 5250
= 41.6 crore + 2.73 crore
= 44.33 crore
Expenses on videos = 44.33 21.6 = 22.73 crore

Solutions for questions 18 to 21: Total sales have to be less than 800 but more than 400.

Further, the sales on any day have to be more than or equal to 100. Further, the sales have to be a multiple of 100 (200 or 300). Further, a total of 300 fruits cannot be sold across each of the three days. For example, 100 fruits cannot be sold on Day 1 (as the sum total of the numbers itself is working out to be more than 100). Similarly, a case of 150 fruits on Day 1 can be rejected.

Total sales (in pieces):

	Day 1	Day 2	Day 3
Apples	72	24	66
Oranges	38	82	54
Mangoes	62	68	22
Guavas	28	26	58
Total	200	200	200

Percentages:

	Day 1	Day 2	Day 3
Apples	36	12	33
Oranges	19	41	27
Mangoes	31	34	11
Guavas	14	13	29
Total	100	100	100

18. Total number of fruits sold is 600.

19. Since it can be observed from the above table that all the statements are true. Hence, nothing is false.

20. As observed from the table, apples registered the maximum % increase in sales from Day 2 to Day 3. Increase is $((66 - 24)/24) \times 100 = 175$ %.

21. For mangoes, sales changed by 20 percentage points.

Solutions for questions 22 to 26: From the data given, the following table can be tabulated.

Establishment	Males	Females	Total
Pizza point	1664	1056	2720
Pasta centre	1152	384	1536
Frenzy ice-cream	1800	1500	3300
Cookieyum	896	132	1028
Cakestake	360	576	936
Sandwichy	2928	1152	4080

22. Number of persons working in Cookieyum is 1028.

23. Required percentage $\frac{576}{384} \times 100 = 150\%$

24. Number of people working in Frenzy ice-cream is 3300.

25. Number of people working in Pasta centre is 1536.

6 Games and Tournaments

CHAPTER

Learning Objectives

In this chapter, you will:

- Learn about some common tournament formats, like knockout tournaments, round robin tournaments etc.
- Learn to interpret the points table and complete the incomplete fields in a table.
- Learn about seeded players and scheduling matches based on seeding of players.
- Learn how to maximise and minimise scores to qualify for the finals/semi-finals etc.
- Learn how to solve sets based on rules defined in the question.

❒ INTRODUCTION

Games and tournaments is an area which has appeared quite a few times in the past few years, predominantly in the CAT and XAT exams as a part of the Logical Reasoning and the Data Interpretation section. The challenge that students face in these questions is that there is no standard method to approach every question and there can also be various formats in which the questions can be asked.

However, some standard models that one can expect in a games and tournament set are:

(i) **Knockout tournaments:** In this format, the winner of a match advances to the next round while the loser is eliminated from the tournament. Also, there can be a scenario where some of the players are given a bye and constraints like seed numbers, upsets are introduced to add complexity to the questions.

(ii) **Round robin tournaments:** In these questions certain number of teams participate in a tournament, where every team plays against every other team in the pool stage of the tournament. There is a certain rule for distributing points among the teams based on the result of a match i.e. win, loss or a tie. From the pool stage, based on the rankings certain number of teams are selected for the further rounds.

(iii) **Interpret the points table:** The third type of question is the one in which the final points table for a tournament is given along with some supporting data, the student is expected to identify facts like the teams and their opponents, on which day a match was played, the grouping of teams etc.

(iv) **Miscellaneous** Apart from the types mentioned above the examiner is free to mix and match rules of a particular tournament be it cricket, football, tennis or any other sport and ask questions.

SOLVED EXAMPLES

Directions for questions 6.01 to 6.05: These questions are based on the information given below.

A total of 64 players are scheduled to take part in a tennis tournament. The players are seeded from 1 to 64 with seed 1 being the top seed and seed 64 being the last seeded player. The tournament takes place in a knockout format with different rounds. In each round the winner of a match between two players advances to the next round while the loser is eliminated. This process is repeated till the finals. In the first-round player seeded 1 plays the player seeded 64, the player seeded 2 plays the player seeded 63 and so on. An upset is said to happen if a lower seeded player beats a higher seeded player. The matches are scheduled such that, in case of no upsets, in each round, the highest seeded player plays the lowest seeded player left in the tournament, the second highest seeded player plays the second lowest seeded player left and so on.

6.01: What is the total number of rounds in the tournament?

(A) 5 (B) 6
(C) 7 (D) 8

Sol: As there are a total of 64 players, in the first round, 32 players would be eliminated and 32 players would be left. In the next round half of 32, i.e., 16 players would be eliminated. In this way after 6 rounds there will be only one player left who is the winner of the tournament.

Note: Number of rounds is a factor of the power of 2. For 2 players, the number of rounds required is 1 as $2^1 = 2$, for 3 or 4 players the number of rounds required is 2 as $2^2 = 4$, for 5, 6, 7 or 8 players the number of rounds is 3 as $2^3 = 8$ and so on.

6.02: How many matches are played in the tournament?

(A) 52 (B) 60
(C) 63 (D) 64

Sol: As the tournament stated with 64 players and in the end only one player was undefeated, a total of 63 matches are played as one player is eliminated per match.

6.03: Which player faced the player seeded 3 in the quarter finals (round of 8) if the tournament had no upsets?

(A) 6 (B) 9
(C) 11 (D) 5

Sol: In any round, in case of no upsets, the sum of the seedings of the players is one more than the number of players left in the tournament.
As the match happens in the quarterfinals, the number of players left = 8.
∴ Sum of seedings of players = 8 + 1 = 9.
Since 9 – 3 = 6, player seeded 3 faced the player seeded 6 in the quarterfinals.

6.04: If the tournament had no upsets, in which round was the player seeded 15 eliminated?

(A) 6 (B) 5
(C) 4 (D) 3

Sol: In case of no upsets, the player seeded 15 would be in the pre-quarterfinals (round of 16). In the next round as 8 players are left he will be eliminated.
∴ The player seeded 15 was eliminated in the 3rd round.

6.05: How many matches did the player who lost in the semi-finals, win in the tournament?

(A) 2 (B) 3
(C) 4 (D) 5

Sol: As the player reached the semi-finals, he was among the last 4 players.
∴ He won four matches and reached the fifth round.

EXERCISE-1

***Directions for questions 1 to 5*:** Answer these questions based on the information given below.

Ten teams compete in the Master's cup. Each team plays exactly one match against every other team. The following scorecard, with some details missing was obtained at the end of the tournament.

Teams				**Result of match played against**						
Name	**Germany**	**Argentina**	**Colombia**	**Belgium**	**Netherlands**	**Brazil**	**Portugal**	**France**	**Spain**	**Uruguay**
Germany	X	D	L	W	D		W	D	L	
Argentina		X	D	D	D			W	W	
Colombia	W	D	X	L	L		D		D	L
Belgium		D		X				D	W	L
Netherlands				W	X		D	W	W	
Brazil	D	L	D	L	L	X		L	D	L
Portugal		D	D	W		D	X	D		
France	D	L	W	D		W		X	W	
Spain		L	D	L			D		X	D
Uruguay	D	D	W	W	W		L	L	D	X

W – Win (3 points) L – Loss (0 points) D – Draw (1 point)

The scorecard above gives the outcomes of some of the matches. For instance, Germany won the match against Belgium, winning 3 points while Colombia drew the match with Argentina and both got a point each.

At the end of the tournament, the teams are ranked from 1 to 10 based on the total points scored by them in the tournament. If two or more teams end up having the same total points, then the following tie-breaker rules are followed.

(a) Among any two teams having the same total score, the team which won the match played between the 2 teams is ranked higher.

(b) If (a) doesn't resolve the tie, the outcomes of the matches played by the two or more teams against the team ranked 1 are considered. A win is considered better than a draw, which is considered better than a loss.

(c) If (b) doesn't resolve a tie, the outcomes of the matches played by the two or more teams against the team ranked 2 are considered while everything else remaining the same.

1. Which team scored the highest points in the tournament?

(A) Netherlands (B) Germany
(C) Spain (D) Brazil

2. In which position did Colombia finish the tournament?

(A) 7 (B) 8
(C) 10 (D) 9

3. Which among the following teams drew the most number of matches?

(A) Argentina (B) Colombia
(C) Spain (D) Uruguay

4. How many points did Brazil score in the tournament?

(A) 5 (B) 4
(C) 6 (D) 8

5. To how many teams did Germany lose, which are ranked better than it at the end of the tournament?

(A) 1 (B) 0
(C) 2 (D) 3

Directions for questions 6 to 10: Answer these questions based on the information given below.

Teams A, B, C, D, E, F, G and H compete in a volleyball tournament. Teams A, B, C and D are in group 1 while teams E, F, G, and H are in group 2. In the group stage, every team plays a game of with every other team in its group. The winning team gets 2 points, the losing team gets no points and no game in the tournament ended in a tie. At the end of the group stage, the top two teams from each group reach the semi-final stage where the top team of group 1 plays against the top team of group 2 and the other two teams play against each other.

The winners of semi-finals clash in the finals and a winner is declared. When two or more teams have the same points in the group stage, the tie-breaker rules are applied.

6. How many games are played altogether in the tournament?
(A) 15 (B) 14
(C) 13 (D) 12

7. Team F can get at most how many points and still not reach the semi-finals?
(A) 2 (B) 4
(C) 6 (D) 8

8. If Team A won every game it played in the tournament and Team B won in the semi-finals, then what would be the highest possible total score of Team D at the end of the group stage?
(A) 6 (B) 4
(C) 0 (D) 2

9. The team which won the finals should have won at least how many games in the tournament?
(A) 3 (B) 4
(C) 5 (D) 2

10. What is the sum of the total points scored by teams A through H in the first stage of the tournament?
(A) 28 (B) 24
(C) 26 (D) Cannot be determined

Directions for questions 11 to 14: These questions are based on the following information.

The following is the table of points in the twenty club English premier league, which is played on a double round-robin basis. The points given below are after the clubs played around thirty games.

Club	Played	Won	Draw	Loss	Points
Chelsea	32	28	4	0	88
Arsenal	32	22	8	2	74
Manchester United	31	23	5	3	74
Liverpool	31	20	6	5	66
Newcastle United	32	18	8	6	62

Note: Only the top five clubs are shown in the table of points given.

Win → 3 points, Draw → 1 point, Loss → 0 points
In a double round-robin tournament, every team plays exactly two games with every other team.

11. What is the maximum points Arsenal can end up with?
(A) 88 (B) 92
(C) 93 (D) 95

12. If in the remaining matches Arsenal had at least three wins and at least two draws, Manchester United should win at least how many of the remaining games to guarantee itself at least the second place?
(A) 3 (B) 5
(C) 6 (D) 4

13. A minimum of how many more points should Chelsea score in the remaining matches, to guarantee itself the top most position?
(A) 6 (B) 8
(C) 10 (D) 12

14. Of the top five teams, if each team had to play the other four teams once more before the season is over and in the remaining matches between these teams, Newcastle wins four, Liverpool wins three, Manchester United wins two and Arsenal drew with Chelsea, which team finished fourth at the end of the season?
(Assume that none of the other teams can catch up with these five teams)
(A) Arsenal
(B) Manchester United
(C) Newcastle United
(D) Cannot be determined

Directions for questions 15 to 17: These questions are based on the following information.

43	44	45	46	47	48	HOME 49
42	41	40	39	38	37	36
29	30	31	32	33	34	35
28	27	26	25	24	23	22
15	16	17	18	19	20	21
14	13	12	11	10	9	8
START 1	2	3	4	5	6	7

You are playing a board game in which you start at 1 (START) and you have to reach 49 (HOME) in the least number of rounds.

In each round you throw a die exactly once and advance as many places as the number on the face of the die.

The following conditions apply in the game:

(a) If in three successive rounds, you throw three 6's consecutively, you should advance 5 more places at the end of that third round, i.e., say you are at 10, and in the next 3 rounds you get all 6's, so you reach $10 + (6 + 6 + 6) + 5 = 33$ at the end of the third round.
(b) If you reach 4 after any round, advance 13 places.
(c) If you reach 40 after any round, go back 12 places.
(d) If you reach 27 after any round, you can go to any number whose sum of digits is same as that of 27.
(e) If you reach 23, after any round, go to either the highest or the lowest number in that column.
(f) You can't reach home with a 1, 4 or 6 as the last try.

15. What is the least possible number of rounds required to reach 'HOME' from the start?

(A) 4 (B) 5
(C) 6 (D) 7

16. If your first try is 6, what is the least number of rounds in which you can hope to reach 'HOME'?

(A) 6 (B) 7
(C) 8 (D) 9

17. If condition (d) is removed, the minimum number of rounds required to reach 'HOME' from the start is

(A) 4 (B) 5
(C) 6 (D) 7

Directions for questions 18 to 22: Answer these questions based on the information given below.

A total of fifteen baseball teams compete in the Premier Baseball League (PBL), which is played on a double round-robin basis. In a double round-robin tournament each team plays exactly two games with every other team. The table of results below gives the updated results of the top 5 teams in the tournament at a point when some matches are still yet to happen.

Team	Played	Won	Draw
New York Yankees	23	21	1
Boston Red Sox	22	19	3
Chicago White Sox	24	19	1
Los Angeles Dodgers	22	17	2
San Francisco Giants	25	16	4

Win → 3 points, Draw → 1 point, Loss → 0 points

18. What is the highest number of points that New York Yankees can end the tournament with?

(A) 72 (B) 75
(C) 78 (D) 79

19. In the remaining matches, if Boston Red Sox won at least 2 matches and drew at least 2 matches, at least how many more matches should New York Yankees win in order to retain the top spot? Given that all the teams played at least 20 matches each currently.

(A) 2 (B) 5
(C) 3 (D) 4

20. If all the teams played at least 20 matches when the table of results was compiled, with a minimum of how many more matches should the Florida Cubs (currently outside the top five) win to reach the first place?

(A) 4
(B) 3
(C) 6
(D) Cannot be determined

21. With how many more wins can Los Angeles Dodgers be confident of second place in the final points tally?

(A) 4
(B) 5
(C) 6
(D) Cannot be determined

22. What is the best position in which San Francisco Giants can finish at the end of all the matches?

(A) First
(B) Second
(C) Third
(D) Cannot be determined

Directions for questions 23 to 26: These questions are based on the information given below.

The table given in below gives the listing of players, ranked from the highest (1) to the lowest (32), who are due to play in a Tennis Tournament. This tournament has four knockout rounds before the final, i.e., first round, second round, quarter-finals and semi-finals. In the first round, the highest ranked player plays with the lowest ranked player, which is designated as Match 1 of first round; the 2nd ranked player plays the 31st ranked player which is designated as Match 2 of first round and so on. Thus, for instance, Match 10 of first round is to be played between 10th ranked player and the 23rd ranked player.

Rank	Player	Rank	Player	Rank	Player	Rank	Player
1	Somdev	9	Anju	17	Peltsin	25	Nisha
2	Vijay	10	George	18	Peter	26	Murthy
3	Ramesh	11	Spassky	19	Sam	27	Swati
4	Prakash	12	Ramiz	20	Ramesh	28	Meenakshi
5	Anand	13	Sachin	21	Rajesh	29	Kapil
6	Mahesh	14	Amit	22	Rakesh	30	Arvind
7	Paes	15	Dibyendu	23	Roopesh	31	Niranjan
8	Sania	16	Stalin	24	Vidya	32	Sunil

In the second round, the winner of Match 1 of first round, plays the winner of Match 16 of the first round and is designated as Match 1 of the second round. Similarly, the winner of Match 2 of first round plays the winner of Match 15 of the first round and is designated Match 2 of the second round. Thus, for instance Match 5 of the second round is to be played between the winner of Match 5 of the first round and the winner of Match 12 of the first round. The same pattern is allowed for the next rounds as well. There is exactly one upset (a lower ranked player beating a higher ranked player) in each of the rounds, including the finals. Eventually, no match ended as a draw.

23. If Kapil reaches the final, then who will play with him in the finals if the number of upsets in the tournament is the minimum?

(A) Vijay (B) Ramesh
(C) Somdev (D) Anand

24. If Peter won the second round match, then who won the finals?

(A) Peter (B) Dibyendu
(C) Kapil (D) Cannot be determined

25. If Anand played the quarter-finals, then who amongst the following must not have played against him in quarter-finals?

(A) Ramesh (B) Prakash
(C) Sachin (D) Kapil

26. If Ramiz played the semi-finals, then who amongst the following could have played him in the semi-finals?

(A) Ramesh (B) Sania
(C) Meenakshi (D) Prakash

Directions for questions 27 to 31: Answer these questions based on the information given below.

A and B start playing a game in which a certain number of coins are placed on a table. Each player picks at least 'a' and at most 'b' coins in his/her turn unless there is less than 'a' coins on the table in which case the player has to pick all those coins left. A and B play alternately and intelligently so as to win the game. The game has 2 formats.

(A) Finders – Winners: In this format, the person who picks the last coin wins.
(B) Keepers – Losers: In this format, the person w h o picks the last coin loses.

27. A and B play a game of Finders – Winners with $a = 2$ and $b = 6$. If A starts the game and there are 74 coins on the table initially, then how many coins should A pick?

(A) 2 (B) 4
(C) 5 (D) 3

28. In a game of Keepers – Losers, B started the game when there were N coins on the table. If B is confident of winning the game and $a = 3$, $b = 5$, which of the following cannot be a value of N?

(A) 94 (B) 92
(C) 76 (D) 66

29. A and B play Finders–Winners with 56 coins, where A played first and $a = 1$, $b = 6$. What should B pick immediately after A?

(A) 2 (B) 4
(C) 3 (D) Cannot be determined

30. In a game of Keepers–Losers played with 126 coins where A plays first and $a = 3$, $b = 6$, who is the winner?

(A) A
(B) B
(C) Depends on A's first pick
(D) Cannot be determined

31. In an interesting variant of the game, B gets to choose the number of coins on the table and A gets to choose which format of the game it will be as well as pick coins first. If B chooses 144 and $a = 1$, $b = 5$, which format should A choose to win?

(A) Finders–Winners
(B) Keepers–Losers
(C) A will lose either way
(D) A will win either way

Directions for questions 32 to 34: Answer these questions based on the information given below.

India, Pakistan, Malaysia, South Korea, Japan and China are to take part in the Asian Hockey Championship. In the first round, each team plays each of the other teams exactly once. At these stage two points are awarded for a win, one point for a draw and zero points for a loss. After all the matches are played, the top two teams, in terms of the points scored, advance to the finals. In case two or more teams end up with the same number of points, the team with a better goal difference is placed higher.

32. The total number of matches in the tournament is

(A) 21 (B) 22
(C) 15 (D) 16

33. What is the minimum number of points with which a team can advance to the finals?

(A) 6 (B) 5
(C) 4 (D) 3

34. What is the maximum number of points that can be scored by a team, which failed to advance to the finals?

(A) 8 (B) 9
(C) 6 (D) 7

Directions for questions 35 to 38: Answer these questions based on the information given below.

A total of 128 players take part in a Grand Slam tennis tournament. The tournament is scheduled to be held in seven rounds and in each round, in a match between two players, the winner advances to the next round and the loser is eliminated. There are no draws or byes in the tournament. The players who take part in the tournament are seeded from 1 to 128, with seed 1 being the top seed, Seed 2 next and so on. The matches are scheduled in such a way that in any round, assuming there are no upsets, the highest seeded player plays against the lowest seeded player at that point, the next highest seeded player always plays against the next lowest seeded player and so on. An upset is said to happen when a lower seeded player beats a higher seeded player. The schedule of matches in the next round remains unchanged in case of an upset in a round, with the only difference that the player who caused the upset advances to the next round and takes the designated place of the player he upset.

35. In case of no upsets in the tournament, in which round would the player seeded 10 face a player seeded higher than him?

(A) 2nd round (B) 3rd round
(C) 4th round (D) 5th round

36. How many players in the tournament won exactly one match?

(A) 15 (B) 24
(C) 30 (D) None of these

37. Assuming no upsets, which player beat Seed 25?

(A) Seed 8 (B) Seed 6
(C) Seed 1 (D) Seed 14

38. If the player seeded 13 won the tournament, then what is the minimum number of upsets in the tournament?

(A) 2 (B) 3
(C) 4 (D) 5

Directions for questions 39 to 42: Answer these questions based on the information given below.

Geeta and Neeta are playing a game which involves picking up coins kept on a table. The players take turns alternately and each player in her turn has to pick at least two and at most five coins except when there is only one coin left on the table and the player has to pick that coin in her turn. Both players are equally intelligent and play to the best of their abilities to win the game.

Assume that the player who picks up the last coin loses the game.

39. During a game, when it was Geeta's turn to play, there were 32 coins left on the table. Which of the following can be the number of coins Geeta should pick up to win the game, no matter how Neeta plays?

(A) 1 (B) 2
(C) 4 (D) 5

40. During Neeta's turn if she removed four coins from the table which made sure that she won the game, then which of the following could have been the number of coins on the table before she removed the four coins?

(A) 45 (B) 52
(C) 76 (D) None of these

41. During a game when it was Neeta's turn to play, there were 28 coins left on the table. Which of the following is the number of coins she should pick up to ensure her win?

(A) 1 (B) 2
(C) 4 (D) Neeta cannot win

42. If during her turn Neeta had to remove two coins to ensure her win, then which of the following could have been the number of coins on the table before she removed the coins?

(A) 25
(B) 30
(C) 50
(D) More than one of the above

Directions for questions 43 to 46: Answer these questions based on the information given below.

Sixteen teams participating in a hockey tournament are divided into two pools, pool A and pool B, each having eight teams. In each pool, each team plays one match with every other team. Two points are awarded for a win, one point for a draw and zero points for a loss. At the end of the pool stage, the top two teams, in terms of the number of points scored advance to the semi-finals and the winners of the semi-finals play the finals. If two or more teams end up with the same number of points at the end of the pool stage, the team with the best goal difference is placed highest, the next one second and so on.

43. What is the total number of matches in the tournament?

(A) 51 (B) 56
(C) 58 (D) 59

44. What is the least number of points with which a team can advance to the semi-finals?

(A) 4 (B) 5
(C) 6 (D) 8

45. What is the maximum possible number of matches won by a team that was eliminated in the pool stage?

(A) 3 (B) 4
(C) 5 (D) 6

46. What is the minimum possible number of matches won by a team that reached the finals?

(A) 1 (B) 2
(C) 3 (D) 4

Directions for questions 47 to 50: Answer these questions on the basis of the information given below.

The World Chess Championship is conducted as an eight-player double round robin tournament, i.e., one in which each player plays every other player twice, they partake once with white and once with black pieces. One point is awarded for a win, half point for a draw and zero points for a loss. At the end of the tournament, the players are ranked according to the total points they won and the player with the highest number of points is crowned the World Champion.

In case two or more players end up with the same number of points, tiebreak rules are applied to determine their placings.

47. The total number of matches in the tournament is

(A) 28 (B) 36
(C) 56 (D) 72

48. Which of the following cannot be the points scored by the winner of the tournament?

(A) 7.0 (C) 8.0
(C) 9.0 (D) None of these

49. If there is only one person who finished with the least number of points, then the minimum difference between him and the winner of the tournament is

(A) 0.5 points
(B) 1 point
(C) 1.5 points
(D) 2 points

50. At the end of the tournament, if the points scored by all the players is different, then the minimum number of wins for the person who scored the highest number of points is

(A) 6 (B) 5
(C) 4 (D) 2

EXERCISE-2

Directions for questions 1 to 4: These questions are based on the following information.

For the first time in the history of the U.S. Open Men's Tennis Tournament, the organizers of the tournament decided to do away with the process of drawing lots for finalizing the fixtures. Instead, they seeded all the 128 players in the tournament, as per their present world ranking, given by the Association of Tennis Professionals (ATP), with Seed 1 being the highest seed and Seed 128 being the lowest seed. This tournament has six knockout rounds, i.e., first round, second round, third round, fourth round, quarter-finals and semi-finals, before the finals.

In the first round, the highest seeded player (i.e., Seed 1) plays the lowest seeded player (i.e., Seed 128) and this match is designated as Match 1 of the first round; the 2nd highest seeded player (i.e., Seed 2) plays the 2nd lowest seeded player (i.e., Seed 127) and this match is designated as Match 2 of the first round, and so on. Thus, for instance, Match 64 of the first round is played between the 64th seeded player and the 65th seeded player. In the second round, the winner of Match 1 of the first round plays the winner of Match 64 of the first round and this match is designated as Match 1 of the second round. Similarly, the winner of Match 2 of the first round plays with the winner of Match 63 of the first round and this match is designated as Match 2 of the second round. Thus, for instance, Match 32 of the second round is played between the winner of Match 32 of the first round and the winner of Match 33 of the first round. The same pattern is followed for all the subsequent rounds as well.

A match in which a lower seeded player beats a higher seeded player is termed as an upset.

1. If it was known that the player seeded third was *upset* in the third round, which of the following is not the seeding of the player who *upset* him?

(A) 35 (B) 94
(C) 99 (D) 29

2. If there were no *upsets* in the first two rounds, the lowest seeded player who could have won the tournament, by himself causing exactly one *upset* is

(A) Seed 27 (B) Seed 32
(C) Seed 24 (D) Seed 17

3. If the tournament was won by a player who was not among the first 50 seeds, the minimum number of *upsets* in the tournament was

(A) 5 (B) 6
(C) 11 (D) 17

4. If there were no *upsets* in the first two rounds and only matches 5, 8, 12 and 14 of the third round resulted in *upsets*, then who among the following reached the quarterfinals, given that there were no *upsets* in the fourth round?

(A) Seed 9 (B) Seed 28
(C) Seed 12 (D) Seed 20

Directions for questions 5 to 9: These questions are based on the following information.

Eight players from A through H qualified for the final round of the World Snooker Championship which is a round robin competition, i.e., each player plays with every other player exactly once. At the start of the tournament, these eight players were seeded from 1 to 8, with Seed 1 being considered as the top seed. No match ended as a tie and after each match, the winner was awarded one point and the loser was not awarded any points. At the end of the tournament, it was found that each player scored exactly one point less than his seeding. The following table gives the results of some of the matches played.

Player	A	B	C	D	E	F	G	H
A	×			W	W			
B		×				W		
C			×					W
D		W		×				
E			W		×			
F						×		
G	W				W		×	
H						W		×

W – Won

For example, in the match played between A and D, A won over D.

It was also known that C was seeded 3 and E was not among the top five seeds.

5. Which player was seeded 1 at the start of the tournament?

6. How many matches did B win?

7. How many points did A score?

8. At the end of the tournament, the eight players were ranked based on the number of points scored such that the person with the highest number of points is ranked 1, the one with the second highest number of points is ranked 2 and so on. For which player was his rank the same as the number of points he scored?

9. Who scored the maximum number of points in the tournament?

Directions for questions 10 to 14: Answer these questions based on the information given below.

Eight basketball teams P through W qualified for the final round of the state-level selections. The eight teams are seeded from 1 to 8, Seed 1 being the best team. Every team played match against every other team and no match ended in a draw. The winning team got 1 point and the losing team didn't get any points.

At the end of the 28 games, it was observed that each team got a total score which was one less than the initial seeding that the team was given. It was also known that:

(a) Team P beat team T
(b) Team Q beat team R
(c) Team R beat team W
(d) Team W beat team U
(e) Team T beat team S
(f) Team V beat teams P and T
(g) Team S beat team Q

10. Which team was seeded fourth at the start of the tournament?
(A) Q (B) S
(C) T (D) V

11. How many matches did team R win?
(A) 1 (B) 2
(C) 3 (D) 4

12. How many points did team T score?
(A) 4 (B) 5
(C) 6 (D) 7

13. Which team scored the highest points in the tournament?
(A) V (B) R
(C) T (D) S

14. If, at the end of the tournament, the teams were ranked 1 to 8 based on their points, with the team scoring the highest points getting rank 1, which team got the same rank as the number of points it scored in the tournament?
(A) R (B) T
(C) S (D) Cannot be determined

Directions for questions 15 to 19: Answer these questions based on the information given below.

Eight teams L, M, N, O, P, Q, R and S participated in a district-level volleyball competition.

The teams are divided into Pool-A and Pool-B, both having equal number of teams.

In the pool stage, all the teams play a match against every other team in the pool. The pool stage takes place over 6 days with two matches on each day and on any given day not more than one match happens between teams belonging to the same pool. The teams are awarded 2 points for a win, 1 point each for a draw and no points for a loss. The table gives the–day-score score sheet.

Days	D_1	D_2	D_3	D_4	D_5	D_6
L	2	2	2	3	3	3
M	1	3	3	3	3	3
N	0	1	3	3	5	5
O	0	0	1	1	3	3
P	0	1	1	2	2	2
Q	0	0	0	0	0	2
R	1	1	1	3	3	4
S	0	0	1	1	1	2

It is known that team Q lost a match on Day 3.

15. Which of the following teams are in the same pool as L?
(A) N, P (B) N, S
(C) M, R (D) M, Q

16. Who won the match on Day 3?
(A) M (B) N
(C) O (D) S

17. How many teams played matches on consecutive days?
(A) 3 (B) 4
(C) 5 (D) 6

18. Which of the following pair of teams played a match on the same day?
(A) L, Q and O, S (B) L, R and M, O
(C) P, Q and R, S (D) P, N and O, S

19. The complete details, such as match date, teams, outcome are not available for how many matches?
(A) 0 (B) 1
(C) 2 (D) 3

Directions for questions 20 to 23: Answer these questions based on the information given below.

Ten teams that are to take part in the Champions Challenge were divided into two pools as Pool A and Pool B. Each team in a pool was to play every other team in that pool. Two points are awarded for a win and zero points for a loss with there being no draws or ties. The top three teams in each pool would advance to the next stage, it is called the super six stage and they would carry forward all points they scored in matches against the other two teams which advanced to the super six stages from its pool. In the super six stage all teams that advance from Pool A are to be in Group 1 and all teams that advanced from Pool B are to be in Group 2. Each team in Group 1 plays against every team in Group 2, with points awarded as in the pool stage. At the end of the super six stage, the top two teams in each group, according to their total points (super six points + carry forward points) advance to the semi-finals with the top team in Group 1 playing the second team from Group 2 and vice versa. The winners of the semi-finals advance to the finals. If two or more teams end up with the same number of points at the end of the pool or the super six stage, the tie is resolved using tie-break rules.

20. The total number of matches in the tournament is

21. The number of points scored by a team which advanced to the super six stage is at least

22. The total points (super six points + carry forward points) of a team that advanced to the semi-finals is at least

23. The maximum number of wins by a team that failed to advance to the semi-finals is

Directions for questions 24 to 28: Answer these questions based on the information given below.

A total of five teams, namely Japan, USA, Cuba, Taiwan and Netherlands participated in the World Baseball Championship. The teams play a match against every other teams and the winning team gets three points; the losing team gets no points and both teams get a point each in case of a draw. In the final points tally, Japan, USA and Cuba got 10, 9 and 6 points, respectively. It is also known that in the final points tally Taiwan did better than Netherlands and Taiwan did not draw any match.

24. The final score of Netherlands at the end of the tournament is
(A) 1 (B) 2
(C) 3 (D) 4

25. How many matches in the tournament ended in a draw?
(A) 0 (B) 1
(C) 2 (D) 3

26. The result of how many matches in the tournament is not known?
(A) 0 (B) 1
(C) 2 (D) 3

27. Taiwan's overall points are less than that of USA by
(A) 3 (B) 5
(C) 4 (D) 6

28. How many matches did Cuba draw?
(A) 0 (B) 1
(C) 3 (D) Cannot be determined

Directions for questions 29 to 33: These questions are based on the following information.

The International Rugby Union (IRU) World Cup tournament has 16 teams seeded from 1 to 16, taking part in the first stage called the pool stage. They are divided into four pools such that seeds 1, 2, 3 and 4 are in pools 1, 2, 3, 4, respectively; seeds 5, 6, 7, 8 in pools 4, 3, 2, 1, respectively; seeds 9, 10, 11, 12 in pools 1, 2, 3, 4, respectively and seeds 13, 14, 15, 16 in pools 4, 3, 2 and 1, respectively. Each team in a pool plays all the other teams in the pool exactly once.

For the next stage called the Super Eight stage, the top two teams from each pool, based on their number of points, would qualify from each of the four pools. If at any stage, two or more teams end up with the same number of points, complex rules are applied to determine their placing. In the Super Eight stage, each team plays every other team except the team that qualified from its own pool. A team qualifying for the Super Eight stage carries forward only those points that it gained in its pool stage match against the other team that qualified from its pool. The top four teams in terms of points at the end of the Super Eight stage would qualify for the semi-finals, with the losers of the semi-finals playing for the third place and the winners of the semi-finals playing the final.

In the pool stage or in the Super Eight stage, a team is awarded two points for a win and zero points for a loss. An *upset* is caused when, in any match, a lower seeded team beats a higher seeded team. In any match, in case the scores are equal at the end of the normal duration of play, the teams play extra time till the winner is decided.

29. What is the number of matches in the Super Eight stage?

30. If the pool stage had only a single *upset*, then which is the lowest seeded team which can win the tournament?

31. What is the minimum number of points required for a team to reach the semi-finals?

32. For a team that has reached the Super Eight stage, what is the minimum total number of points required (including the points that it carried forward from the pool stage) such that it can guarantee itself a place in the semi-finals?

33. What is the total number of matches in the tournament?

Directions for questions 34 to 38: Answer these questions based on the information given below.

Sixteen teams qualified for the World Cup Cricket tournament. In the first stage called the pool stage, there are 16 teams, which are divided into two pools of eight teams each. In each pool, every team plays with every other team exactly once. Two points are awarded for a win and zero points for a loss with no match at any stage of the tournament ending in a tie.

At the end of the pool stage, the top three teams in each group, in terms of the points scored, advance to the next stage called the super six stage. The teams that advance to the super six stage also carry forward with them the points they scored against the other teams that advanced with them to the super six stage. In the super six stage, each of the three teams which advanced from a stage play a match with each of the other three teams which advanced from the other pool. The points for the matches are awarded as in the pool stage. At the end of all the matches in the super six stage, the four teams with the highest number of points advance to the semi-finals and the winners of the semi-finals play the finals. If at any stage of the tournament two or more teams end up with the same number of points, the team with better net run rate is placed higher.

34. What is the total number of matches in the tournament?
(A) 64 (B) 68
(C) 70 (D) 71

35. What is the minimum number of wins required for a team to reach the super six stage?
(A) 1 (B) 2
(C) 3 (D) 4

36. If Team X won the tournament, then the number of matches won by it is at least
(A) 5 (B) 6
(C) 7 (D) 8

37. If Team Y did not reach the semi-finals of the tournament, the number of matches won by it is at most
(A) 5 (B) 6
(C) 7 (D) 8

38. Which of the following statements is/are true?
(a) A team can advance to the semi-finals without winning a single match in the super six.
(b) The winner of the tournament won at least half of the matches it played.
(c) A team can fail to reach the semi-finals even after winning all the matches in the super six stage.
(A) Only (b) (B) Only (a)
(C) Only (c) (D) Both (a) and (c)

Directions for questions 39 to 41: Answer these questions based on the information given below.

The world series of poker had reached the final stage when the top eight players are left. The ranking of the players and the points they scored till the beginning of the final stage are as follows:

Player	Rank	Points
A	1	128
B	2	124
C	3	119
D	4	115
E	5	109
F	6	102
G	7	98
H	8	98

In the final stage, each of the players play exactly once with every other player. The points for the matches are awarded as follows. A maximum of 2 points will be awarded if one beats a player ranked higher than oneself and 1 point if one beats a player ranked lower than oneself. In the same way, points are deducted from the loser such that the net effect of a match is zero points. The points are computed after each match and the player with the higher number of points is ranked higher. If two players have the same number of points, the player who has played fewer number of matches in the final stage till that point would be ranked higher. If two players are still evenly matched, then the player who was ranked higher at the start of the final stage would be ranked higher. After all the matches are over, the player with the highest number of points is ranked first, the next one second and so on. No match in the tournament ends in a draw.

Note: The matches between the players can take place in any order.

39. What are the minimum points that player B could end up with at the end of all his matches?
(A) 112 (B) 111
(C) 110 (D) 113

40. What are the maximum points that player A could end up with at the end of all his matches?
(A) 142 (B) 135
(C) 136 (D) 137

41. How many of the given players has a chance of ending up as ranked first at the end of all the matches?
(A) 3 (B) 6
(C) 4 (D) 5

Directions for questions 42 to 46: Answer these questions based on the information given below.

75 tennis players who took part in an invitational tournament is seeded from 1 to 75. The tournament is played on a knockout basis such that a player once when he loses a match gets eliminated while the winner advances to the next round. In the first round, players seeded 1 to n were given byes such that there were no byes from the second round till the end of the tournament. A player is said to have received a bye when he advances to the next round, say the 2nd round, without playing a match in the previous round, i.e., the 1st round in this case. A match is said to be an upset if a lower seeded player, say seed 5, beats a higher seeded player, say seed 1.

The matches are scheduled such that, in any round, assuming there are no upsets, the highest seeded player plays the lowest seeded player left, the next highest seeded player plays the second lowest seeded player left and so on. In case of an upset, say Seed 25 beating Seed 5 in a round, the matches would proceed as usual with the only difference being Seed 25 advancing to the next round and meeting the opponent the player seeded 5 would have faced. No change happens in the seeding of a player because he causes an upset.

42. What is the total number of rounds in the tournament?
(A) 8 (B) 7
(C) 6 (D) 5

43. What is the total number of matches in the tournament?
(A) 68 (B) 71
(C) 72 (D) 74

44. How many players were given a bye in the first round?
(A) 6 (B) 13
(C) 11 (D) None of these

45. If the player seeded 1 was upset in the quarter-finals (round of eight), then what could be the seeding of the player who caused that upset?
(A) 6
(B) 8
(C) 9
(D) More than one of the above

46. If the player seeded 15 won the tournament, then what would be the minimum number of upsets in the tournament?
(A) 15 (B) 14
(C) 4 (D) 9

Directions for questions 47 to 50: Answer these questions based on the information given below.

The following table gives details about the performance of six tennis players in three Grand Slam tournaments, namely in The Australian Open, Wimbledon and The US Open.

Name	Total matches played	Matches played at the Australian Open	Matches played at Wimbledon	Matches played at the US Open	Total sets played
Federer	105	40	35	30	390
Nadal	93	35	28	30	362
Djokovic	80	35	25	20	337
Graf	115	33	40	42	260
Seles	60	21	20	19	198
Serena	140	25	70	45	325

(I) Nadal, Djokovic and Federer are male players while Graf, Seles, and Serena are female players.
(II) In the men's tournament, a player must win 3 sets to win a match.
(III) In the women's tournament, a player who wins 2 sets is the winner of the match.
(IV) Every year 128 participants take part in both the men's as well as women's section in each of the Grand Slams.
(V) All the matches are played between players of the same gender only, i.e., men play against men and women play against women.
(VI) Each tournament is played once in a year.
(VII) All the tournaments are played in the knockout format, i.e., the loser gets eliminated and the winner advances to the next round.

47. If Nadal and Djokovic never lost in the semi-finals, then what is the difference between the number of Grand Slams they participated, if they participated in the least possible number of tournaments?

48. What is the minimum number of matches that Federer lost in all the Grand Slams together?

49. If Serena never lost a match without winning a set, then what is the number of matches in which she won losing a set, if she lost the least possible number of matches in Grand Slams?

50. If Graf and Seles participated in all the three Grand Slams between the years 1990-95 and 1993-95, respectively, then what is the minimum number of matches they played against each other in these Grand Slams?

EXERCISE-3

Directions for questions 1 to 3: Answer these questions on the basis of the information given below.

A cricket tournament had three teams, such as India, Australia and Sri Lanka taking part in it. The format of the tournament was such that in the preliminary stage each of these teams would play the other teams four times. Four points are awarded for a win and in case a team beats another team by a huge margin, it is given a bonus point in addition to the four points. At the end of the preliminary stage, the top two teams, in terms of the points scored reaches the finals. No match in the tournament ends in a tie and if two teams end up with the same number of points at the end of the preliminary stage, the team with the better net run rate is placed higher.

1. If India reached the finals, then what is the minimum number of points it would have scored in the preliminary stage?

2. If Sri Lanka was eliminated in the preliminary stage, then what is the maximum number of points it could have scored?

3. If Australia had the highest number of points at the end of the preliminary stage, then at least how many points did it have?

Directions for questions 4 to 7: Answer these questions on the basis of the information given below.

A total of 128 players participate in each of the four Grand Slam tennis tournaments that take place in a year. Each tournament is played on a knockout basis such that in each round, the player who loses a match is eliminated from the tournament, while the player who wins, advances to the next round. No player receives a walkover/bye in any round in any of the Grand Slams.

For the questions below, assume that the same set of 128 players played in each of the four Grand Slams.

4. Player X had an overall win-loss record of 21-3 in the Grand Slams in a year. What is the earliest round in which he could have lost in a Grand Slam tournament in that year?

5. If the win-loss record in the Grand Slams in a year for the players ranked 1, 2, 3 and 4 in the world is 15-3, 17-3, 12-3 and 15-3, respectively, then what is the highest ratio of wins to losses for any player in the four Grand Slams in that year?

6. What could be the minimum number of matches won by player Y, if he had the best win and loss ratio among all the players in the four Grand Slams in a year?

7. At the end of four Grand Slams in a year, the top six players, in terms of the number of matches won in Grand Slams in that year qualify for the slam of champions. What could be the maximum number of matches won, in all the four Grand Slams put together, by a player who did not qualify for the slam of champions?

Directions for questions 8 to 12: Answer these questions on the basis of the information given below.

A total of 512 players participated in a knockout tournament with different rounds. Players are seeded from Seed 1 to Seed 512 such that Seed 1 is the top Seed while Seed 512 is the last seeded player. Matches are held in such a manner such that in the first round the 1st seeded player plays the last seeded player, similarly the 2nd seeded player plays the second last seeded player and so on. In each round, the winner of any match advances to the next round while the loser is knocked out of the tournament. This process is repeated till the finals when two players play amongst themselves to decide the winner. If at any point of time, a lower seeded player beats a higher seeded player, that match is defined as an upset. The matches are scheduled such that, in case of no upsets in each round, the highest seeded player plays the lowest seeded player left in the tournament, the second highest seeded player plays the second lowest seeded player left and so on. In case of an upset, the player who caused the upset (the lowest seeded player) would take the designated place of the player he upset (the highest seeded player) in the next round.

8. The player ranked 480 won the tournament. With which of the following seeded players, he must have not played in the semi-finals?

(A) Seed 100 (B) Seed 109
(C) Seed 93 (D) Seed 83

9. If player seeded 403 won the tournament, he must not have played with which of the following players in the finals?

(A) Seed 94 (B) Seed 84
(C) Seed 40 (D) Seed 93

10. Seed 100 must not have played with which of the following players in either the finals or the semi-finals?

(A) Seed 97 (B) Seed 83
(C) Seed 92 (D) Seed 88

11. Player ranked 200 won the tournament. With which of the following seeded players, he must have not played in the quarter-finals?

(A) Seed 18 (B) Seed 17
(C) Seed 113 (D) Seed 16

12. Seed 12 must not have played with which of the following players in the finals?

(A) Seed 91 (B) Seed 92
(C) Seed 98 (D) Seed 94

Directions for questions 13 to 16: Answer these questions on the basis of the information given below.

The following are the points scored by all the six teams. The points scored are P, Q, R, S, T and U at the end of a round robin tournament. Each team played five matches, one each with each of the other teams. Three points are awarded for a win, one point for a draw and zero points for a loss.

Team	Points scored
P	13
Q	7
R	9
S	–
T	1
U	6

The points scored by S is left blank.

13. How many matches did Q win?

(A) 1 (B) 2
(C) 3 (D) Cannot be determined

14. At most how may points did team 'S' score?

(A) 4 (B) 5
(C) 6 (D) 7

15. What is the maximum possible number of draws in the tournament?

(A) 5 (B) 6
(C) 7 (D) 8

16. At least how many points did team 'S' score?

(A) 0 (B) 1
(C) 2 (D) 3

Directions for questions 17 to 20: These questions are based on the following data.

Twelve teams took part in a football tournament, which is conducted in three stages. In the first stage, the teams are divided into two groups of six teams each. The teams within a group play with each other once and the top three teams of each group, in terms of the number of wins, go to the second stage. In the second stage, the three teams of each group play with each other once and the top two teams from each group then go to the third stage. In this stage, the two teams in each group play with each other and the winners from each group play with each other to decide the winner of the tournament. All games produce results. In case of a draw, a penalty shootout is used to decide the winner. In case of a tie, at the end of any of the first two stages the winner is decided by a set of complex tie breaking rules to ensure that only one team goes into the next round.

17. What is the minimum number of games a team should win to ensure that it goes into the second stage?

18. Of all the teams that reached the second stage, what is the minimum number of games a team could have won?

19. If a team gets ₹50,000 for each win in the first stage, ₹1,00,000 in the second stage and ₹1,50,000 in the third stage, find the maximum amount that any team can win.

20. What is the total number of matches in the tournament?

ANSWER KEYS

Exercise-1

1. (A)	10. (B)	19. (C)	28. (D)	37. (A)	46. (A)
2. (D)	11. (B)	20. (B)	29. (D)	38. (C)	47. (C)
3. (A)	12. (D)	21. (D)	30. (B)	39. (B)	48. (D)
4. (B)	13. (B)	22. (B)	31. (B)	40. (C)	49. (B)
5. (B)	14. (D)	23. (A)	32. (B)	41. (D)	50. (C)
6. (A)	15. (B)	24. (D)	33. (C)	42. (D)	
7. (B)	16. (B)	25. (A)	34. (A)	43. (D)	
8. (D)	17. (D)	26. (B)	35. (C)	44. (C)	
9. (A)	18. (D)	27. (A)	36. (D)	45. (D)	

Exercise-2

1. (D)	10. (A)	19. (A)	28. (A)	37. (D)	46. (C)
2. (C)	11. (B)	20. 32	29. 24	38. (C)	47. 1
3. (B)	12. (B)	21. 2	30. 16	39. (C)	48. 2
4. (A)	13. (A)	22. 2	31. 4	40. (C)	49. 43
5. F	14. (C)	23. 6s	32. 12	41. (B)	50. 1
6. 3	15. (A)	24. (A)	33. 52	42. (B)	
7. 6	16. (B)	25. (B)	34. (C)	43. (D)	
8. (D)	17. (C)	26. (A)	35. (B)	44. (D)	
9. G	18. (C)	27. (D)	36. (D)	45. (D)	

Exercise-3

1. 8	5. 6	9. (A)	13. (D)	17. 4
2. 20	6. 8	10. (C)	14. (D)	18. 2
3. 17	7. 20	11. (A)	15. (C)	19. 7,50,000
4. 3	8. (D)	12. (B)	16. (C)	20. 39

SOLUTIONS

EXERCISE-1

Solutions for questions 1 to 5: By filling the table, we get the following table:

Teams	Result of match played against										
	Germany	Argentina	Colombia	Belgium	Netherlands	Brazil	Portugal	France	Spain	Uruguay	Total
Germany	X	1	0	3	1	1	3	1	0	1	11
Argentina	1	X	1	1	1	3	1	3	3	1	15
Colombia	3	1	X	0	0	1	1	0	1	0	7
Belgium	0	1	3	X	0	3	0	1	3	0	11
Netherlands	1	1	3	3	X	3	1	3	3	0	18
Brazil	1	0	1	0	0	X	1	0	1	0	4
Portugal	0	1	1	3	1	1	X	1	1	3	12
France	1	0	3	1	0	3	1	X	3	3	15
Spain	3	0	1	0	0	1	1	0	X	1	7
Uruguay	1	1	3	3	3	3	0	0	1	X	15

1. Netherlands scored the highest points.

2. Colombia and Spain are tied for 7 points and they also drew the game between them. Both also lost to the team ranked 1 (Netherlands). Therefore, their outcomes against the team ranked 2 should be considered.
Argentina, France and Uruguay are tied for the 2nd rank with equal points and they also won against each other in a cyclic fashion.
Therefore, applying the breaker rules,
Uruguay is 2nd (Since it beat Netherlands).
Argentina is 3rd (Since it drew against Netherlands).
And France is 4th (Since it lost to Netherlands).
Now, among Colombia and Spain, Spain Uruguay (Rank 2) and is therefore ranked higher.
Colombia's rank = 9, Spain's rank = 8.

3. Argentina drew 6 matches which is the highest.

4. Brazil scored 4 points in the tournament.

5. Germany lost only to Colombia and Spain, both of whom are ranked lower than Germany.

6. Group stage = 6 + 6 = 12
Semi-finals = 2
Final = 1
There are 15 matches.

7. Teams E, F and G can win 2 matches each, getting 4 points each and still Team F might be left out because of tie breaker rules.

8. As teams A, B, C and D are in one group, the Team D could have won at most one game.
$\therefore$ It scored only two points.

9. The team which goes to the semi-final stage has to win at least one game in the group stage. After winning semi-final and final, the team wins at least three games.

10. Total matches = 6 + 6 = 12
Total points = $12 \times 2 = 24$

11. Since Arsenal has six more games left to play, it can score a maximum of $6 \times 3 = 18$ points more
$\therefore$Maximum points = 74 + 18 = 92

12. Since there are 20 teams and it is a double round-robin, each team plays 38 matches.
Arsenal has six games left. At most it could have had four wins and two draws getting 14 points.
Manchester united must get at least 15 points to guarantee itself the second place from seven matches. The minimum number of wins required is 4.
4 wins and 3 draws will give it 15 points.

13. Manchester united has seven games left and can score a maximum of 21 points. It can reach 95 points.
$\therefore$ Chelsea should get at least 96 points to be guaranteed of the title. It has to get a minimum of 8 more points.

14. Since the results of the remaining matches for these teams are not known, we can't determine who finishes third.

Solutions for questions 15 to 17: Since one can't reach home with 1, 4 or 6, one has to be at 44, 46 or 47 just before the last round.

Since, if one advance 13 places when you reach 4, always opt for that condition whenever possible.

15. The shortest route to 'HOME' would be as follows:
1st round $1 \to 4 \to 17$ (Advance 13)
2nd round $17 \to 22$ (can't go to 23 since we reach 48 and get stuck there)
3rd round $22 \to 7$ (Go to 45, because 2 + 7 and 4 + 5 are equal)
4th round $45 \to 46$ or 47 and
5th round 46 or $47 \to 49$

16. 1st round $\to 1 \to 6$
2nd round $\to 6 \to 12$ (maximum possible)
3rd round $\to 12 \to 17$ (can't go to 18 because 3 consecutive 6's takes you to 23 and then to 48).
4th round $\to 17 \to 22$ (can't go to 23 since we reach 48 and get stuck there).
5th round $\to 22 \to 27$ (go to 45) then 2 more rounds as in the previous question.

17. 1st round $1 \to 4 \to 17$
2nd round $17 \to 22$ (can't go to 23 since we reach 48 and get stuck there).
3rd round $22 \to 28$
4th round $28 \to 34$
5th round $34 \to 39$ (condition (a) or 31 to (40 + 5) and then two more rounds, as in the previous question)
6th round $39 \to 44$
7th round $44 \to 49$

18. If New York Yankees wins the remaining 5 matches, they will have 26 wins and 1 draw.
26(3) + 1(1) = 79 points

19. Boston Red Sox could at most win 4 and draw 2 matches, increasing their points to 74. New York Yankees have win at least 3 more matches and draw the remaining two.

20. As San Francisco Giants had a score of 52, Florida Cubs could've had a score of 51 at the most, with 17 wins and 3 losses. Assuming no one else in the top 5 wins any more points, Florida Cubs can win three more matches and draw five and reach the first place.

21. Cannot be determined as we have no data on the number of matches played by those teams which are not in top 5.

22. If San Francisco Giants wins the next three matches, they get 61 points, which is at best only 2nd position.

Solutions for questions 23 to 26:

Round	Number of Matches
I →	16
II →	8
Quarter-finals	4
Semi-finals	2
Finals	1

Round I:
Each match is played between the players, for whom the sum of the ranks = 33.
{(1, 32), (2, 31), (3, 30), (4, 29), (5, 28), (6, 27), (7, 26), (8, 25), (9, 24), (10, 23), (11, 22), (12, 21), (13, 20), (14, 19), (15, 18), (16, 17)}

Round II:
Exactly seven matches (since there was one upset in round I) are played between players (for whom sum of ranks is 17).

23. Kapil is ranked 29th, exactly one upset was there in every round. This was created by Kapil.
In the semi-finals, (1, 29) and (2, 3) matches were played.
The finals held will be between Kapil and Vijay.

24. Peter created upset in the first round. But, he may or may not have created an upset in the other rounds. We cannot determine the winner.

25. Ist round: (5, 28)
IInd round: (5, 12) or (5, 21)
QF: (5, 4)
or (5, 29)
or (5, 13)
Anand must not have played against Ramesh.

26. Ramiz reached the semi-finals.
1st round:
(12, 21)
2nd round:
(12, 5) or (12, 28)
Quarter-finals:
(12, 4) or (12, 29) or (12, 13) or (12, 20)
Semi-finals:
(12, 1) or (12, 32) or (12, 16) or (12, 8)

27. If A picks 2, B will be left with 72 and no matter how many B picks, A can always pick $(8 - x)$ and complete the 72.

28. In Keepers – Losers, the objective should be to give $8k + 1$ coins to the opponent to win for sure. With 66 wins, no matter what B does, he cannot give 65 coins to A. Therefore, B can't win.

29. Irrespective of what A picks, he will lose. However, what B picks first will depend on how many A picks. Hence, it cannot be determined.

30. To win, A should leave B, $9K + 1$ coins on the table, since it can't do that, he will definitely lose.

31. A should choose Keepers – Losers and pick 5 coins from the table, leaving 139 coins for B. As we get, $139 = 6K + 1$, A will win.

32. The total number of matches in the tournament $= 5 + 4 + 3 + 2 + 1 + 1$ (finals) $= 16$

33. The minimum number of points would be required when one of the teams scored the maximum points and the remaining teams equally shared all the remaining points.
Total points available $= 15 \times 2 = 30$
Maximum points for a team $= 5 \times 2 = 10$
The remaining 20 points can be shared by all the five teams and one of those teams with four points would advance to the finals based on the goal difference.

34. The worst case for a team happens when three teams score the highest points and one of those teams is eliminated.
Points scored by the last three teams (points only among themselves) = 6
Remaining points = 24
These points can be shared by the teams and one of the teams with 8 points could be eliminated based on goal difference.

35. A player seeded 10, assuming no upsets is expected to reach the last 16 stage, i.e., the fourth round. In this round he would face the player seeded 7 and would be eliminated.

36. The players who won exactly one match are those who won the first round but lost in the second round, i.e., 64 – 32 = 32 players.

37. The player seeded 25 would have lost to the player seeded 8.

38. The minimum number of upsets happen when the player seeded 13 caused all the upsets, i.e., in the fourth and fifth rounds, the semi-finals and the finals, a total of four upsets.

39. To ensure her win, Geeta must make sure that there is one or two coins left when it is Neeta's turn to play. In each round Geeta can make sure that seven coins $(2 + 5)$ leave the table.
$\therefore$ The number of coins left on the table before Neeta's turn has to be of the form $7n + 1$ or $7n + 2$, where n is an integer. As there are 32 coins left on the table, she must remove two or three coins, so that the coins left over is of the form $7n + 1$ or $7n + 2$.

40. The number of coins left after Neeta's turn must be of the form $7n + 1$ or $7n + 2$. As she removed four coins, the number of coins would have been of the form $7n + 5$ or $7n + 6$. Only Choice (C) satisfies this condition.

41. When the person who picks up the last coin wins the game, one must make sure that when it is the other person's turn to play there are six or seven coins left on the table so that whatever the other person plays, one can pick the last coin in his/her turn.
$\therefore$ The number of coins left on the table before Geeta's turn must be of the form $7n$ or $7n - 1$. As there are 28 coins left on the table, which is of the form 7n, whatever Neeta plays, the number of coins left will not be of the form '$7n$ or $7n - 1$' before Geeta's turn, i.e., Neeta cannot win no matter what she does.

42. As Neeta removed 2 coins, the number of coins was of the form $7n + 2$ or $7n + 1$. Choices (B) and (C) satisfy this condition.

43. The number of matches in the tournament is 28 + 28 (pool stage) + 2 (semi-finals) + 1 (final) = 59.

44. The least number of points required happens when one of the teams in a pool wins all the matches and the remaining seven teams evenly share the remaining points, i.e., $\frac{42}{7} = 6$.

45. A team can be eliminated with the maximum number of wins when three teams evenly share maximum points and the remaining teams only win points in matches among themselves, i.e., the top three teams end up with 12 points each and one of them would be eliminated.

46. A team can reach the finals with just six points in the pool stage, i.e., with only six draws and no wins. Therefore, just by a single win in the semi-finals a team can reach the finals.

47. As each player plays two matches, the total number of matches = $7 \times 8 = 56$.

48. As there is a total of 56 games, the total points available is 56. Even if all the players score the same number of points, each player would end up with seven points and one of them would end up as the winner.
$\therefore$ The winner can have any score from 7 to 14.

49. If only one person finished with the least number of points, then the minimum difference comes when he scores 6.5 points, the winner scores 7.5 points and all other players score 7 points each.
$\therefore$ The difference in points would be at least one.

50. If the number of wins of the winner of the tournament must be minimum, the points he scored must be minimum. If the scores of all players are distinct and the score of the winner is to be the minimum possible, the scores of the players are 9, 8.5, 8.0, 7.5, 6.5, 6.0, 5.5 and 5.0. If a player must score 9 points from 14 games, then he must have at least 4 wins, i.e., 4 wins and 10 draws.

EXERCISE-2

1. In the first round, the third seeded player would have beaten the player seeded 126. In the second round, he would have beaten the player seeded 62 or 67. In the third round, the third seeded player would have played against the player seeded 30 (in case of no *upset*) or player seeded 35 (potential opponent of player seeded 30 in the second round) or player seeded 99 (opponent of player seeded 30 in the first round) or player seeded 94 (opponent of player seeded 35 in the first round).
$\therefore$ Any of these players could have beaten seed 3 in the third round.

2. If there are no *upsets* in the first two rounds, the top 32 players will reach the third round. Now, we need to find the lowest seeded player who could have won the tournament by himself causing just a single *upset*. Now assume that the single *upset* happened in the third round, and along with the match, all other matches in the third round resulted in *upsets*. This would mean that players seeded from 17 to 32 would reach the fourth round. Now, Seed 32 would play Seed 17 (originally scheduled Seed 1 and Seed 16), Seed 31 would play Seed 18 and so on. As the only *upset* of the player who won the tournament already happened in the previous round, the lowest seed player who can reach the next round without an *upset* is Seed 24 (who plays Seed 25). Assume that all the matches in this round, except that involving Seed 24, also resulted in *upsets*. In that case, Seed 24 becomes the highest seeded player left and he can win without any further *upsets*.

3. For minimum number of *upsets*, we assume that only the matches of the winner of the tournament were *upsets*. If the winner of the tournament was any person with a seeding of 51 to 64, then he can win the tournament with six *upsets* (all rounds except the first round).

4. Had there been *upsets* in matches 5, 8, 12 and 14 of the third round, players seeded 28, 25, 21 and 19 would have reached the fourth round, instead of seeds 5, 8, 12 and 14. As there are no *upsets* in the fourth round, the quarterfinal line up would be Seed 1 Vs Seed 9, Seed 2 Vs Seed 7, Seed 3 Vs Seed 6 and Seed 4 Vs Seed 21.

5. It is given that each player scored exactly one point less than its seeding, i.e., seed 1 scored 0 points, seed 2 scored 1 point and so on such that the Seed 8 scored 7 points, which is possible only if he beats all the other teams in the tournament, the player seeded seventh won 6 points or he must have won against all the other players and so on and this can be tabulated as follows:

Seed	Won against	Lost to	Points
8	7, 6, 5, 4, 3, 2, 1	–	**7**
7	6, 5, 4, 3, 2, 1	8	**6**
6	5, 4, 3, 2, 1	8, 7	**5**
5	4, 3, 2, 1	8, 7,6	**4**
4	3, 2, 1	8, 7, 6, 5	**3**
3	2, 1	8, 7, 6, 5, 4	**2**
2	1	8, 7, 6, 5, 4, 3	**1**
1	–	8, 7, 6, 5, 4, 3, 2	**0**

From the tables it is clear that
A beats D and E, D beats B and B beats F. So also, G beats E and A, E beats C, C beats H and H beats F. All the other players except F has won at least one match.
∴ F must be Seed 1.

6. F lost against all other persons.
H lost against C.
C won against F and H.
Now, we have to find Seed 2, who has won only one match. But among the given person only H can win one match, as all others have at least two wins already. Similarly, we can find out the number of the wins of the other persons and we can tabulate the information as follows:

1) G	2) A
3) E	4) D
5) B	6) C
7) H	8) F

B won three points or he won three matches.

7. From the arrangement, it is clear that A scored 6 points.

8. D won 4 points and his rank was 4.

9. G who won all his matches scored the maximum number of matches in the tournament.

10. As each team scored one points less than it's seeding, therefore Seed 1 scored 0, Seed 2 scored 1 point and so on.

Seed	Won against	Lost to	Points
8	7,6,5,4,3,2,1	–	7
7	6,5,4,3,2,1	8	6
6	5,4,3,2,1	8,7	5
5	4,3,2,1	8,7,6	4
4	3,2,1	8,7,6,5	3
3	2,1	8,7,6,5,4	2
2	1	8,7,6,5,4,3	1
1	–	8,7,6,5,4,3,2	0

Since all players except U win a match, U is Seed 1. Among the other players, only W can win one match exactly, hence, W is Seed 2. Similarly, the others can be found out.

Seed	Team
1	U
2	W
3	R
4	Q
5	S
6	T
7	P
8	V

Seed 4 is team Q.

11. Team R is seeded 3, therefore, it won 2 matches.

12. Team T is seeded 6, so it scored 5 points.

13. Team V scored 7 points.

14. S won 4 points and its rank is 4.

15. We can see that the –day-wise matches are as follows:

	✓	
D_1	$L \times Q$	$\bar{M} \times \bar{R}$
		✓
D_2	$\bar{N} \times \bar{P}$	$M \times S$
	✓	
D_3	$N \times Q$	$\bar{O} \times \bar{S}$
		✓
D_4	$\bar{L} \times \bar{P}$	$O \times R$
	✓	✓
D_5	$L \times N$	$M \times O$
	✓	
D_6	$P \times Q$	$\bar{R} \times \bar{S}$

Here ✓ denote win and
— denote draw.
∴ Pool A → L, N, P, Q
Pool B → M, O, R, S

16. N won the match on day 3.

17. L, M, N, O, S played matches on consecutive days.

18. P, Q and R, S played on Day 6.

19. Complete details are available.

Solutions for questions 20 to 23: In the pool stage, as each team plays every other team once, the total matches in each pool is 10. In the super six stage, the total matches are 9.

$\therefore$ The total matches = 20 + 9 + 2 + 1 = 32

In the pool stage, the total number of points available in each pool is $10 \times 2 = 20$.

As there are five teams, each team can end with four points and two of the teams can be eliminated (or) one team can win all matches and get 8 points, another team 6 points and the three other teams 2 points each and one of them can advance to the super six.

20. The total number of matches in the tournament is 32.

21. The number of points scored by a team which advanced to the super six stage is at least 2.

22. Among the top three teams advancing to the super six stage from each group either all of them would carry forward 2 points each (or) one team would carry forward 4 points, second team 2 points and the third team zero points. Even if all the three teams lose all their super six matches, two teams would advance from this group and so a team with a total of 2 points can advance to the semi-finals.

23. Three teams can end up with 6 points each at the end of the pool stage, i.e., three wins each. Now these teams can win all their matches in the super six stage, but one of these teams would be eliminated.

$\therefore$ A team with 3 (in the pool stage) + 3 (in the super six stage) wins can be eliminated.

24. 10 points $\Rightarrow$ 3 wins, 1 draw (Japan)

9 points $\Rightarrow$ 3 wins, 1 loss (USA)

Cuba's 6 points can be 1 win and 3 draws or 2 wins and 2 losses. As 3 draws are not possible, because USA did not have any draw, Cuba won 2 matches and lost 2 matches.

✓	✓	✓
J – U	U – C	C – T
✓	✓	✓
J – C	U – T	C – N
✓	✓	✓
J – T	U – N	T – N
$\bar{J}$ – $\bar{N}$		

As Taiwan did better than Netherlands, Taiwan won against Netherlands.

J – 10, U – 9, C – 6, T – 3 , N – 1

Netherland's scored is 1 point.

25. Only one match was drawn.

27. 9 – 3 = 6 points.

28. Cuba did not draw any match.

Solutions for questions 29 to 33: The teams in the four pools are as follows:

Pool 1	Pool 2	Pool 3	Pool 4
Seed 1	Seed 2	Seed 3	Seed 4
Seed 8	Seed 7	Seed 6	Seed 5
Seed 9	Seed 10	Seed 11	Seed 12
Seed 16	Seed 15	Seed 14	Seed 13

29. There are 8 teams in the Super Eight stage. If each team plays all other teams, there must be 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28 matches. Since teams from the same pool don't play again in the Super Eight, the four matches between these teams must be deducted.

$\therefore$ Total matches = 28 – 4 = 24

30. In any pool, out of the four teams, the best seed; 2nd best seed, 3rd best seed and worst seed, even the worst seed can reach the next round by causing only one upset, i.e., by beating the 2nd best seed. In this case, the 2nd best, 3rd best and the worst seed will have one win, i.e., 2 points each. Hence, the worst seed can move to the next round.

If this happens in pool 1, all of the seeds 8, 9 and 16 would have one win each and seed 16 can enter the Super Eight stage and go on to win the tournament.

31. Since all teams that reach the Super Eight stage carry forward the points gained in the pool stage against the other qualifier in the group, for calculating the points it can be taken that all teams play every other team once.

$\therefore$ Total points in the Super Eight stage = $28 \times 2 = 56$.

As there are no ties, let us assume that the top three teams have won the maximum number of points, i.e., 14, 12 and 10, respectively and the remaining points are equally distributed, i.e., remaining five teams would have won 4 points each.

$\therefore$ A team that scores only four points in total can possibly advance to the semi-finals.

32. To find the maximum points a team can score and still be left out of the semi-finals, assume that five teams score evenly and the remaining three score points only in matches between themselves.

Total points = 56

Points scored by the bottom three teams = 6

All the five teams can score 10 points each and one of the teams is eliminated.

$\therefore$ To guarantee itself a place in the semi-finals, a team has to score 12 points (as there are no ties in the tournament, the team can't score 11)

33. Number of matches in the pool stage = $6 \times 4 = 24$.
Number of matches in the super six stage = 24
Then, semi-finals (2) + finals (1) + 3rd place match (1)
$\therefore$ Total = 24 + 24 + 2 + 1 + 1 = 52 matches

34. The number of matches in the tournament is 127 as 128 players take part in the tournament and in each match a player who is defeated is eliminated. As there is only a single player left undefeated in the tournament, 127 matches have to take place.

35. As there are 128 players the tournament has seven rounds. Therefore, the winner of the tournament won seven matches.

36. The player seeded fifth could have faced Seed 7 (in case of no upset till that round and seed 10 had he upset seed 7 in the round of 16.

37. All players who would have faced the player seeded 8 would have been eliminated, i.e., Seed 57, Seed 25, Seed 9, Seed 1 and Seed 4.

38. There are three players seeded above the player seeded 4. Of these, Seed 3 could have been eliminated by Seed 2 and so at least two players were upset, namely Seed 1 and Seed 2.

39. Assume that before B plays his first match, A had played and lost against F, G and H. Now B would have more points than any other player and if he loses all his matches, he could end up with 124 – 14 = 110 points.

40. As two points are awarded for a win against a player ranked higher, for A to get maximum points, player A should win all his matches and should face the maximum possible number of opponents when they are ranked higher than him. It can be seen that player C cannot be ranked higher than player A before they play each other.
$\therefore$ Only player B can be ranked above player A before they play.
The maximum points player A can have at the end of all his matches = 128 + 2 + 6 = 136

41. The minimum points for A at the end of all the matches would be 114.
Therefore, the only players who can score more than 114 points can end up being ranked as number 1.
Hence, all the players up to F can possibly end up as being ranked first at the end of all the matches.

42. As the number of players is 75 which lies between 2^6 and 2^7, there are seven rounds in the tournament.

43. As there are 75 players and only one player is undefeated, there are 74 matches in the tournament.

44. As there are no byes from the second round, there must be 64 players at the beginning of the second round. Therefore, only 11 players were eliminated in the first round, i.e., 11 matches. Thus, 75 – 22 = 53 players were given a bye.

45. Had there been no upsets till the quarter-finals, the player seeded 1 would have faced the player seeded 8th. The player seeded 8th would have faced the player seeded 9th in the round of 16, the player seeded 25 in the round of 32 and the player seeded 57 in the round of 64. Had the player seeded 8th been upset by any of these players in an earlier round, the player who caused the upset would have come in place of the player seeded 8th in the next round.
$\therefore$ Any of the given players could have faced the player seeded 1 in the quarter-finals.

46. For the minimum number of upsets, we have to assume that all the upsets were caused by the winner himself. The player seeded 15 can advance to the round of 16 without any upset. After that he need to cause four upsets to win the tournament.

47. If they participated in the least number of Grand Slams, they should have played the maximum number of matches in each tournament.
$\Rightarrow$ In a tournament containing 128 players, a player can play at most 7 matches.
Minimum number of tournaments in which Djokovic took part

$$= \frac{35}{7} + \frac{25}{7} + \frac{20}{7} = 5 + 4 + 4 = 13$$

For playing 25 matches he has to play at least four tournaments and for playing 20 matches again he has to play at least four (if he plays only three, it means that he lost in the semi-finals of one tournament).
Minimum number of Grand Slams in which Nadal took part

$$= \frac{35}{7} + \frac{28}{7} + \frac{30}{7} = 5 + 4 + 5 = 14$$

$\Rightarrow$ Difference = 14 – 13 = 1

48. To calculate the minimum number of matches lost by Federer, we must make his number of wins the maximum.
He won 5 Australian opens, 5 Wimbledon Opens, and 4 US Opens.
$\Rightarrow$ He lost at least 1 match each in the US Open and the Australian Open.
$\Rightarrow$ Minimum number of matches which Federer lost
= 1 + 1 = 2

49. Serena lost a minimum number of matches.
$\Rightarrow$ She won maximum number of matches.
She lost in the 4th round in one Australian Open and in the third round in one US Open.
$\Rightarrow$ 2 matches she lost, where she won 2 sets and lost 4 sets.
She won 138 matches $\Rightarrow 138 \times 2 = 276$ sets
But she played 325 sets.
She lost the 2 matches by playing 6 sets.
$\Rightarrow$ (325 – 6) – 276 = 43 sets were lost in the matches she won.

Difficulty Level-2: Moderate

50. To calculate the minimum number of matches they played against each other, we must assume that they met only in the finals.
Since we must calculate the minimum number of meetings, we have to make minimum possible appearances for both the players in the finals between 1993–95, for which one arrangement can be as follows.

Player	Year	Australian Open	Wimbledon	US Open
Graf	1990	7	7	7
	1991	7	7	7
	1992	7	7	7
	1993	5	7	7
	1994	4	6	7
	1995	3	6	7
Seles	1993	7	6	6
	1994	7	7	6
	1995	7	7	7

So, they will meet at least once in the final of the US Open.

EXERCISE-3

1. For India to reach the finals with the minimum number of points, one of the teams, say Australia should have won all the matches and India and Sri Lanka should have won two matches each in the four matches they played. So, India would have scored eight points, the same as Sri Lanka and can still advance to the finals on the basis of better net run rate.

2. This scenario happens when each team wins four matches that too all of them with bonus points, such that each team ends up with 20 points and one of the teams would be eliminated.

3. To top the preliminary stage with the least number of points, all teams should have the same number of wins, i.e., 4 and as Australia had the highest, it could have won one of the matches with a bonus point to have a total of 17 points.

4. As there are 128 players taking part in the game, each Grand Slam would have seven rounds. As it is said that player X has a 21-3 win – loss record, he lost three matches and so the maximum number of matches he could have won in the Grand Slam with the best performances are 7, 6 and 6.
∴ He won only two matches in the fourth Grand Slam and he reached at least the third round of each Grand Slam.

5. As all the top four players have only three losses each, they won a Grand Slam each. Therefore, the best record a player can have is if he has reached the finals of the four Grand Slams, for a win – loss record of 24-4, i.e., 6.

6. All the four Grand Slams would have exactly one winner, a player winning all the seven matches. As we need the minimum win – loss record for the player with the best record in all the four Grand Slam combined, it is to be assumed that the maximum number of matches won by these players is only seven each, except for one person who won one more match for a total of eight wins.

7. We have to find the highest number of wins possible for the player with the seventh highest number of wins. In any Grand slam, the winner wins seven matches, the losing finalist, six matches, the semi-finalists, five matches each and the quarter finalists, four matches each. If we consider that the top seven players reached the quarter finals or higher at each of the four Grand Slams and that they won the same number of matches in the four Grand Slams combined, they would have 20 wins each and one of the players would be eliminated.

8. Seed 480 is in the form of 8K. He would have played with any person whose seeding is of the form 8K + 4 or 8K + 5. Seed 100 is in the form of 8K + 4. Seed 109 is in the form of 8K + 5. Seed 93 can be represented in the form of 8K + 5. However, Seed 83 cannot be represented in any of the two forms → 8K + 4 or 8K + 5.

9. Any player with seeding $4x + 2$ or $4x + 3$ must have played with the player with seeding $4x$ or $4x + 1$ in the finals. So, the player seeded 403 can play with Seed 84, Seed 40 and Seed 93 (in the form of $4x + 1$) but Seed 403 must not have played with Seed 94 (in the form of $4x + 2$) in the finals.

Difficulty Level-3: Advanced

10. In the finals, Seed 100 could have played with any player whose seeding is in the form of $4x + 2$ or $4x + 3$. Further, in the semi-finals, Seed 100 could have played with any person whose seeding is in the form of $8x$ or $8x + 1$. Seed 92 cannot be represented in any of the four forms → $4x + 2$, $4x + 3$, $8x$ and $8x + 1$.

11. Seed 200 is in the form of 16K +8. He would have played with any person whose seeding is of the form 16K or 16K + 1. Seed 18 cannot be represented in any of the two forms → 16K or 16K + 1.

12. Seed 12 is of the form $4x$. He could have played with any player with Seed in the form of $4x + 2$ or $4x + 3$ in the finals. Seed 94 is of the form $4x + 2$. Seed 91 is of the form $4x + 3$. Seed 98 is of the form $4x + 2$. However, Seed 92 cannot be represented in any of the form $4x + 2$ or $4x + 3$.

Solutions for questions 13 to 16: As then are six terms in total, each team would play 5 matches. The total number of matches is 15. The points available in each match are either two or three. So, if all the matches end in draws, the total points would be 30 and if each match produces a decisive result, the total points of all the teams would be 45. For each draw, the total number of points would reduce by 1.

13. As Q had 7 points, it could have had two wins and one draw or one win and four draws. So, we cannot determine exactly how many matches it won.

14. The minimum number of draws in the tournament would be 1(P) + 1(Q) + 1(T) + 1(S) as the number of draws must be even. If two matches (4 draw counts) are drawn, the total points would be 43. Therefore, S can score at most 43 (13 + 7 + 9 + 1 + 6) = 7 points.

15. The minimum number of decisive matches in the tournament would be 4(P) + 1(Q) + 2(R) + 1(U) = 8
As there are 15 matches in total, 15 – 8 = 7 matches would end in draws as given in the table:

Team	Points scored	Wins	Loss	Draws
P	13	4	0	1
Q	7	1	0	4
R	9	2	0	3
S	2	0	3	2
T	1	0	4	1
U	6	1	1	3
Total	38	8	8	14

16. The number of points scored by S would be minimum when the number of draws is maximum so that the total points is minimum. As seen from the previous questions, these can be at most 7 draws and so at least 38 points were scored.
∴ Points scored by S = 38 – (13 + 7 + 9 + 1 + 6)
= 38 – 36 = 2

17. The wins of different teams can be as follows. The teams are arranged in descending order of the number of wins. If a team wins 4 matches, there can't be 3 other teams with a better performance.

5	4	2	2	2	0
5	4	3	2	1	0
4	4	4	2	1	0
4	4	3	2	2	0
4	4	3	2	1	1
4	4	2	2	2	1
4	3	3	3	2	0
4	3	3	3	1	1
4	3	3	2	2	1
4	3	2	2	2	2
3	3	3	3	3	0
3	3	3	3	2	1

From the last row we see that there can be five teams with three wins each. So, two teams with three wins will get eliminated. So, three wins are not enough.

18. If the top two teams win 5 and 4 matches, then out of the remaining 6 match results, a team which wins 2 matches can reach the second stage.

19. The top team can win at most 5 games in the first stage, 2 in the second and 2 in the third.
∴ The amount won by the top team in rupees
= (5 × 50,000) + (2 × 1,00,000) + (2 × 1,50,000)
= 7,50,000

20. The number of matches in the first stage is
$$15 \times 2 = 30$$
The number of matches in the second state is
$$3 \times 2 = 6$$
The number of matches is the third stage is
$$2 \times 1 + 1 \text{ (finals)} = 3$$
∴ Total number of matches = 30 + 6 + 3 = 39

Difficulty Level-3: Advanced

7

Networks and 3D Diagrams

CHAPTER

Learning Objectives

In this chapter, you will:

- Learn how to interpret and understand networks, routes, triangular graphs and spider charts
- Get exposed to different question types based on the same
- Learn to solve questions involving more than one way of representing data
- Learn how to optimize movement of goods in a network of pipelines, minimize slack, etc.
- Learn how to calculate shortest/most cost-effective path to take, balancing load in different branches of a network

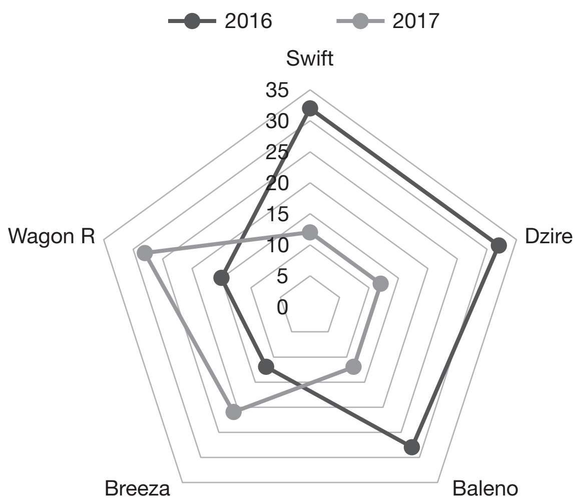
❒ INTRODUCTION

This chapter covers two topics – 3D Diagrams and Routes and networks. In the past, there have been questions in various entrance exams like CAT, XAT, NMAT, SNAP etc. from these chapters.

3D Diagram

A 3D diagram is a graph used to represent data with multiple parameters. In cases were a bar graph or a pie chart does not suffice or may complicate the data representation, a 3D graph comes handy.

An example of a 3D graph showing the sales (in 1000's units) of Maruti Suzuki Ltd. is given below:

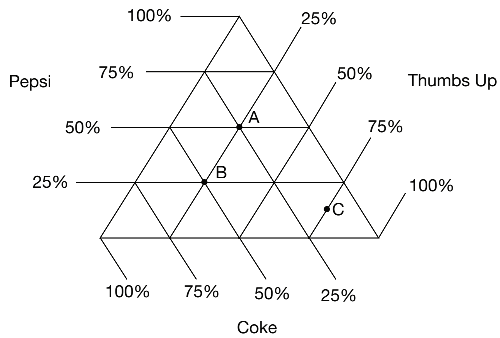


Routes and Network

A network diagram is given with nodes connected to each other by certain routes which are represented by either one-way or two-way arrows with constraints. The constraints can be capacity in case the arrows represent pipes or distance/toll rates in case the arrows represent roads. The diagram is such that the capacity of all the nodes needs to be fulfilled. In general, the network diagram represents a network of pipes supplying water/oil to cities, network of roads connecting various cities or a drainage system of a city. The questions in the solved examples section give a good idea of kind of question you can expect from this section.

❒ THREE-DIMENSIONAL GRAPH

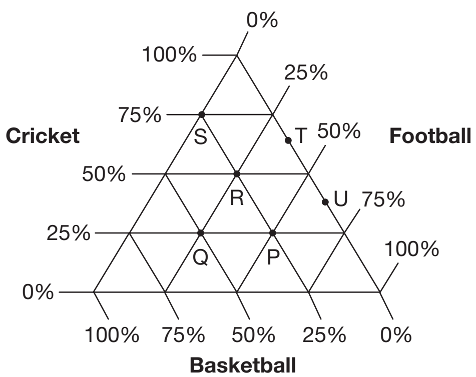
The data (parameters) in a triangular graph are given on each side of the triangle. Each point represents a particular parameter in terms of the percentage, the same represents.



This graph represents the percentage of students who like the three colas, such as Pepsi, Thumbs Up and Coke in three colleges A, B and C.

SOLVED EXAMPLES

Directions for questions 7.01 to 7.05: These questions are based on the information given below.



The above diagram shows the percentage of students in six classes, namely from P, Q, R, S, T and U and whose favourite game is cricket, football or basketball. The total number of students in these classes are P – 80, Q – 96, R – 120, S – 100, T – 80 and U – 120.

7.01: Find the number of students in classes R and S together whose favourite game is cricket?

(A) 95 (B) 110

(C) 125 (D) 135

Sol: Number of students in class R whose favourite game is cricket $= \frac{50}{100}(120) = 60$

Number of students in class S whose favourite game is cricket $= \frac{75}{100}(100) = 75$

$\therefore$ Required number = 135

7.02: How many more students in class Q had basketball as their favourite sport when compared to the number of students in class P whose favourite sport is football?

(A) 6 (B) 7

(C) 8 (D) 10

Sol: Number of students in class P whose favourite sport is football $= \frac{50}{100}(80) = 40$

Number of students in class Q whose favourite sport is basketball $= \frac{50}{100}(96) = 48$

$\therefore$ Required number = 8

7.03: The number of students in class Q whose favourite sport is cricket formed what percentage of the students in class R whose favourite sport is football?

(A) 60 (B) 75

(C) 84 (D) None of these

Sol: Number of students in Q whose favourite sport is cricket $= \frac{25}{100}(96)$

Number of students in R whose favourite sport is football $= \frac{25}{100}(120)$

$$\text{Required percentage} = \frac{\frac{25}{100}(96)}{\frac{25}{100}(120)}(100) = 80\%$$

7.04: Find the total number of students in the given six classes whose favourite sport is football.

(A) 208 (B) 199
(C) 196 (D) 191

Sol: Number of students in P whose favourite sport is football $= \frac{50}{100}(80) = 40$

Number of students in Q whose favourite sport is football $= \frac{25}{100}(96) = 24$

Number of students in R whose favourite sport is football $= \frac{25}{100}(120) = 30$

Number of students in S whose favourite sport is football $= \frac{0}{100}(100) = 0$

Number of students in T whose favourite sport is football $= \frac{37.5}{100}(80) = 30$

Number of students in U whose favourite sport is football $= \frac{62.5}{100}(120) = 75$

$\therefore$ Required number = 199

7.05: In the class if the percentage of students whose favourite sport is cricket was the highest, then what is the number of students whose favourite sport was not cricket?

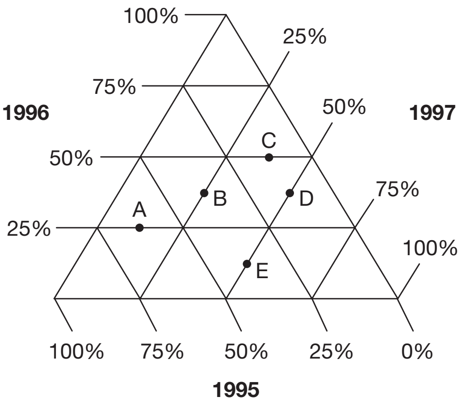
(A) 50 (B) 40
(C) 35 (D) 25

Sol: Class S had the maximum percentage of students whose favourite sport is cricket. In this class for 25% of the students, the favourite sport was not cricket.

$\therefore$ Required value $= \frac{25}{100}(100) = 25$

EXERCISE-1

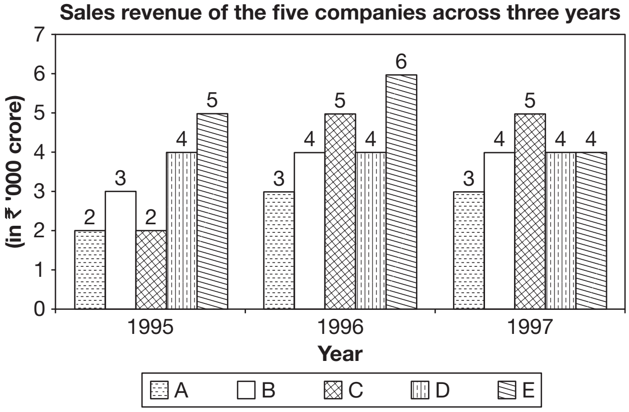
Directions for questions 1 to 5: These questions are based on the following data.



The above triangular chart represents the profit percentages earned by five companies A, B, C, D and E across three years.

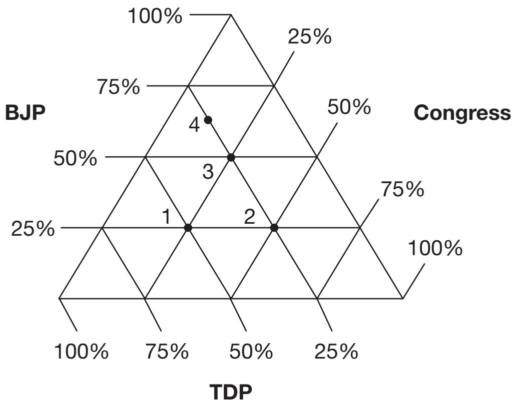
Note: (1) Profit percentage is calculated on sales revenue.
(2) Profit = Sales revenue – Expenditure

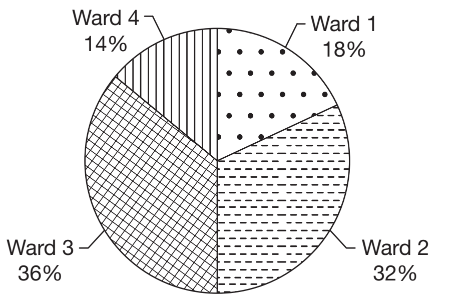
Sales revenue of the five companies across three years



1. Which of the following companies has earned maximum profit during the year 1997?
2. What is the total profit earned (in crore) by company A during the given years?
3. What is the highest profit earned (in crore) by any of the companies in any year?
4. What is the ratio of the expenditure of C to that of D during the year 1995?
5. During the year 1998, if the sales revenue of B increases by 25% and the expenditure decreases by 20%, then the profit percentage of B in 1998 is more (in percentage points) than that of 1997 by

Directions for questions 6 to 10: These questions are based on the following diagrams.





The numbers 1, 2, 3 and 4 in the triangle indicates the four wards, ward 1, ward 2, ward 3 and ward 4, respectively. The above triangle depicts the performance of three parties in the municipal elections of Hyderabad in four different wards. The pie diagram shows the distribution of total votes polled for these parties in the four wards. Total votes polled in all the four wards is 5 lakh. The number given in the pie chart indicates the ward number.

6. The number of votes polled for TDP in ward 1 is

(A) 4,50,000	(B) 45,000
(C) 67,500	(D) 72,500

7. In which ward did Congress secure the maximum number of votes?

(A) Ward 4	(B) Ward 3
(C) Ward 2	(D) Ward 1

8. How many votes did BJP secure in wards 3 and 4 together?

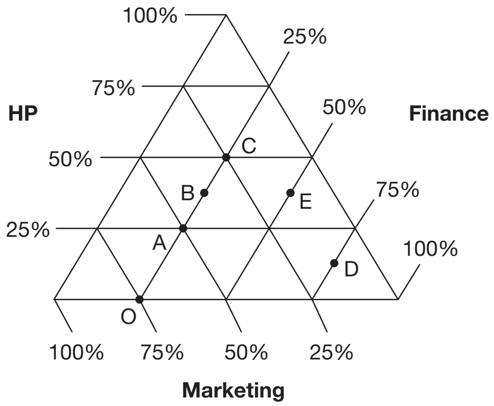
(A) 94375	(B) 137350
(C) 133750	(D) 157350

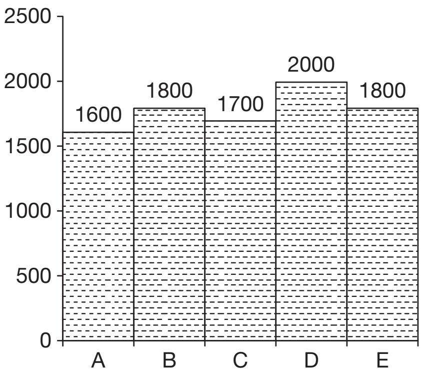
9. By what majority did Congress win over TDP in ward 2?

(A) 80000	(B) 20000
(C) 60000	(D) 40000

10. Which party secured a majority in ward 4?
(A) BJP
(B) TDP
(C) Congress
(D) BJP and Congress secured the same number of votes.

Directions for questions 11 to 15: These questions are based on the following diagrams.





The above diagram represents the composition of professionals in three departments, such as HR, Finance and Marketing of five different firms, A, B, C, D and E. Each of these firms consists of only these three departments. The bar chart shows the number of employees in each of these five firms.

11. The number of HR professionals in firm A is
(A) 1200 (B) 800
(C) 400 (D) 350

12. The difference in the number of Marketing professionals in firm B and that in firm C is
(A) 250 (B) 500
(C) 750 (D) 600

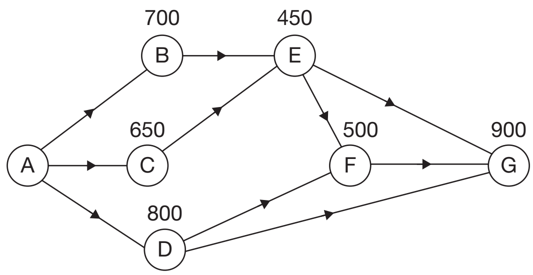
13. In the next year, in firm D the number of finance professionals increased by 30% while the increase in the number of HR and Marketing professionals is 10% and 20%, respectively. By what percentage does the total number of professionals in firm D increase?
(A) 26.25% (B) 52.5%
(C) 20.79% (D) 12.625%

14. If the Marketing professionals earn ₹7000/month on an average and the Finance and the HR professionals earn ₹6000/month and ₹5000/month, respectively in firm A, then the average monthly salary of all the professionals in firm A is
(A) ₹6250 (B) ₹6000
(C) ₹5250 (D) ₹4500

15. The ratio of HR professionals in firm B to that in firm E is
(A) 1 : 1 (B) 2 : 3
(C) 3 : 2 (D) 3 : 4

Directions for questions 16 to 19: Answer these questions on the basis of the information given below:

The diagram given below is the network for transporting oil from refinery A to depots B, C, D, E, F and G. The arrows indicate the direction in which oil flows and the value above each depot denotes its capacity (in thousand litres). The supply is arranged such that only after a depot is full will the excess oil be transferred to the depots next in the supply line. The maximum capacity (in thousands of litres) of the pipeline connecting the refinery with the depots is 1500 and those connecting the depots is 750. The slack in a pipeline is defined as the extra flow required to bring the pipeline to full capacity. The flow in the pipeline is such that the demand at each depot is completely met.



Note: All capacities mentioned are in thousand litres.

16. What is the minimum flow (in thousand litres) in the pipeline connecting A and D?

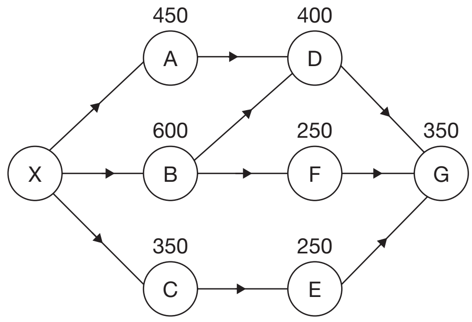
17. If the pipeline connecting A and B is under repair as a result of which its maximum capacity is reduced by 20%, what is the minimum flow (in thousand litres) in the pipeline connecting D and F?

18. What is the maximum value (in thousand litres) of the sum of the slacks in all the pipeline supplying oil to depot E?

19. If the slack in the pipeline connecting D and G is 50, then what is the minimum slack (in thousand litres) in the pipeline connecting E and F?

Directions for questions 20 to 24: Answer these questions on the basis of the information given below.

The following diagram gives the network used for supplying oil from a refinery X to depots A, B, C, D, E, F and G.



The flow of oil is in the direction shown and is such that only after the demand at an intermediate depot is completely met, oil is passed on to the next depot in the network. The values alongside the depots represent the demand (in units) at each depot. It is also known that the flow of oil is such that the demand at all the depots is exactly met and that the capacity of each pipeline is 1000 units.

Note: The slack in a pipeline is the excess flow that is required in the pipeline to bring it to full capacity.

20. What is the maximum slack in the pipeline connecting X and B?
(A) 100 units (B) 150 units
(C) 200 units (D) 250 units

21. What is the maximum flow in the pipeline connecting C and E?
(A) 750 units (B) 650 units
(C) 600 units (D) 550 units

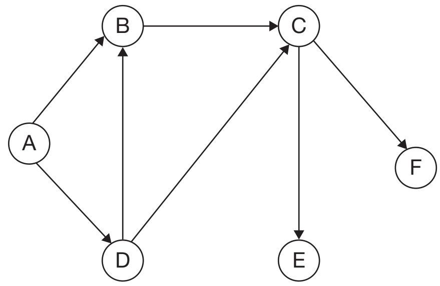
22. What is the sum of the slacks in all the pipelines put together?
(A) 5750 units (B) 5950 units
(C) 6100 units (D) 7350 units

23. What is the maximum flow in the pipeline connecting D and G?
(A) 350 units (B) 300 units
(C) 250 units (D) 200 units

24. If another depot H, located after G is connected to the network, then what is the maximum quantity of oil that can be supplied to H, all other values remaining the same?
(A) 500 units (B) 450 units
(C) 400 units (D) 350 units

Directions for questions 25 to 28: These questions are based on the following data.

The network shows the water pipelines connecting the 6 cities (A, B, C, D, E, F).



Through a pipeline, water can flow only in one direction as indicated by the arrow in the diagram.

- The maximum carrying capacity of any of the pipelines is 1000 m^3 per day.
- The daily requirement at C is 400 m^3.
- The slack in the pipeline is the difference between its maximum carrying capacity and the actual load carried by the pipeline.
- The slack in pipeline CE is 100 m^3 less than the slack in pipeline CF. The slack in pipeline AD is 300 m^3.
- The daily requirement at D = 100 m^3.
- The amount of water that flows through pipeline BC is twice the daily requirement at C.
- Slack in pipeline AB = 200 m^3.
- The ratio of the requirement at B to the slack in pipeline CE is 1 : 2.
- The ratio of the slacks in pipelines CE and CF is 6 : 7. The quantity of water flowing through the pipeline DB is the same as that flowing through pipeline DC.

25. Find the daily requirement (in m^3) at E, if it is known that its requirement is exactly met by the water flowing through the pipelines shown.
(A) 300 (B) 400
(C) 600 (D) 700

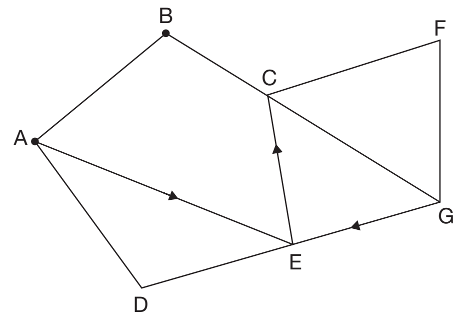
26. Find the daily requirement (in m^3) at F, if it is known that its requirement is exactly met by the water flowing through the pipelines shown.
(A) 300 (B) 400
(C) 800 (D) 900

27. If there exists a larger external pipeline of capacity 5000 m^3 that supplies water to city such that the requirements of all the 6 cities are met by the water supplied by it, then what is the slack in the external pipeline? It is given that the daily requirement at A = 500 m^3.
(A) 2500 m^3 (B) 3000 m^3
(C) 3500 m^3 (D) 2000 m^3

28. If on a particular day, the pipeline joining cities D and B is damaged and the amount of water that is intended to flow through pipeline DB gets wasted in the process, then find how much water is wasted on that day?

(A) 200 m^3 (B) 100 m^3
(C) 300 m^3 (D) 340 m^3

Directions for questions 29 to 31: Answer the questions on the basis of the information given below.



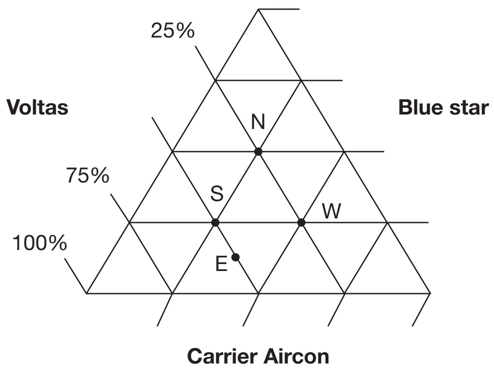
The above network shows 7 cities, such as A, B, C, D, E, F and G connected via two-way roads and one- way roads. For the one-way roads, the direction of travel is as indicated by the arrows in the diagram. Roads which do not contain any arrows are two-way roads.

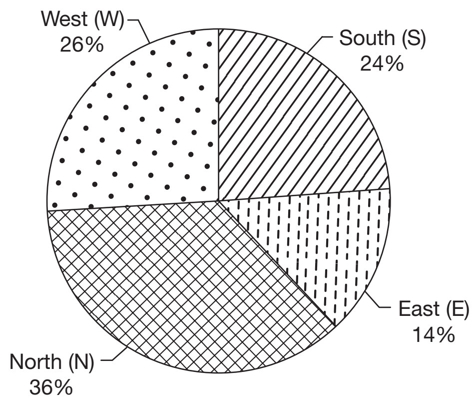
29. The number of distinct paths from city A to city G without going through any city more than once is
(A) 3 (B) 4
(C) 5 (D) 6

30. If the road connecting cities C and G is damaged and traffic cannot go along that road, then the number of distinct ways in which city G can be reached from city E without going through any city more than once is
(A) 3 (B) 1
(C) 4 (D) 2

31. In the previous question, if it is assumed that all the roads other than the damaged road allow two-way traffic, then the number of distinct ways of reaching city G from city E without going through any city more than once is
(A) 3 (B) 2
(C) 4 (D) 5

Directions for questions 32 to 36: These questions are based on the following data.





The above triangular chart depicts the split up of various data of companies and the sales of ACs, namely Voltas, Bluestar and Carrier Aircon, in the four regions of India in North (N), South (S), East (E) and West (W). The total number of ACs sold is 10 lakh and the pie chart indicates the split up of sale of ACs by different regions.

32. In which region did Voltas sell the maximum number of ACs?
(A) North (B) South
(C) East (D) West

33. What is the total number of ACs sold by Carrier Aircon in the given four regions?
(A) 3,22,500 (B) 3,32,500
(C) 3,45,000 (D) None of these

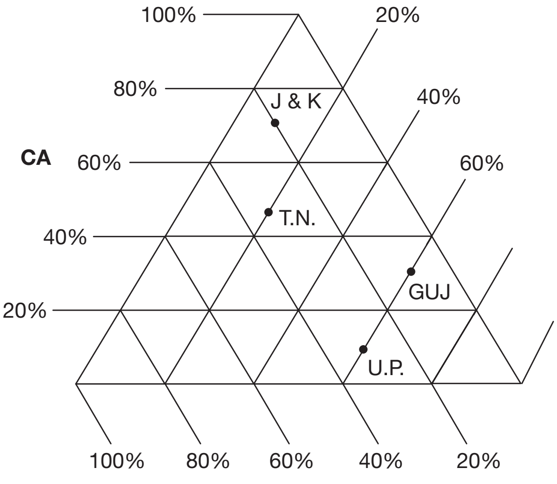
34. How many ACs did Blue Star sell in East and North regions, put together?
(A) 77,500 (B) 82,500
(C) 1,97,500 (D) 2,45,000

35. By what per cent is the number of ACs sold by Voltas in the South region more than Blue Star?
(A) 50% (B) 100%
(C) 200% (D) None of these

36. What is the ratio of the number of ACs sold by Voltas in East region to the total number of ACs sold in the North region by the three companies put together approximately?
(A) 1 : 2 (B) 1 : 6
(C) 2 : 7 (D) 1 : 5

Directions for questions 37 to 40: Answer these questions on the basis of the information given below.

The following figure gives the percentage of seats secured by Congress and its allies (CA), BJP and its allies (BA) and Others (O) in the assembly elections held in the four states, namely in Uttar Pradesh (U.P.), Jammu and Kashmir (J & K), Gujarat (GUJ) and Tamil Nadu (T.N.).



37. If the total number of seats won by CA in Gujarat and BA in Tamil Nadu are 54 and 48, respectively, then how many more seats were left in Tamil Nadu assembly than those in Gujarat? (Assume that elections were held in all the constituencies in both the states)

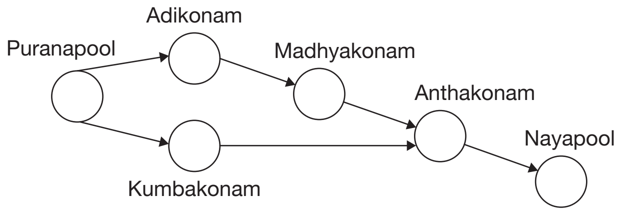
38. If the total number of seats in Uttar Pradesh is 330, then what is the number of seats won by CA in that state?

39. If the total number of seats won by Others in Uttar Pradesh assembly is 69, then how many seats are there in the Uttar Pradesh assembly, assuming that elections are held in all the constituencies?

40. Which of the following statements is/are definitely true?
(a) In Uttar Pradesh, the number of seats won by BA is more than that won by CA and Others together.
(b) BA has won an equal number of seats in both Gujarat and Uttar Pradesh.
(c) In Tamil Nadu, Others have won more number of seats than BA.
(A) Only a and c
(B) Only b and c
(C) Only a and b
(D) None of these

Directions for questions 41 to 45: Answer the questions on the basis of the information given below.

The following diagram shows the network of pipelines carrying the industrial waste from Puranapool to five recycling plants located at Adikonam, Madhyakonam, Kumbakonam, Anthakonam and Nayapool, respectively. Each recycling plant has a certain capacity for recycling of industrial waste, which is exactly met. The capacity at Adikonam is 600 units, at Madhyakonam 800 units, at Kumbakonam 500 units, at Anthakonam 900 units and at Nayapool 450 units. The arrow heads indicate the direction of flow of the industrial waste through the pipelines. The maximum capacity of each pipeline is 2000 units.



Note: The slack in a pipeline is defined as the extra flow that is required in the pipeline to bring it to its full capacity.

41. What is the maximum possible slack in the pipeline connecting Kumbakonam and Anthakonam?
(A) 1500 units (B) 600 units
(C) 1250 units (D) 1150 units

42. If the pipeline connecting Kumbakonam and Anthakonam has a slack of 800 units, then what is the supply of waste from Puranapool to Adikonam?
(A) 350 units (B) 1,200 units
(C) 450 units (D) None of these

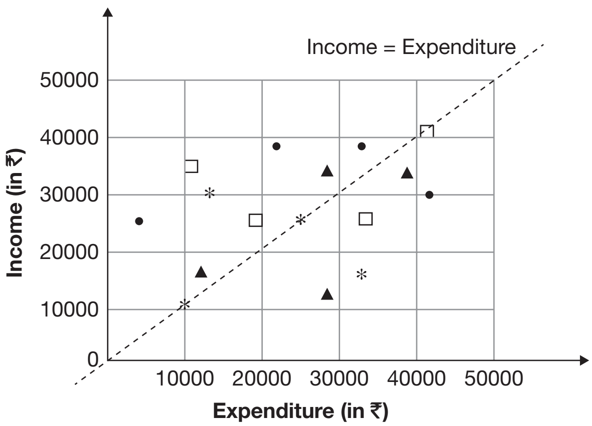
43. What is the minimum slack in the pipeline connecting Madhyakonam and Anthakonam?
(A) 850 units (B) 1100 units
(C) 1250 units (D) 1400 units

44. If the capacity at Nayapool alone is increased such that the total slack in all the pipelines is minimum, then what is the sum of the slacks in all the pipelines combined?
(A) 2700 units (B) 1100 units
(C) 3300 units (D) 2950 units

45. Due to a snag that developed in the pipeline from Puranapool to Adikonam, the maximum capacity of that pipeline fell by 20%. What is the slack in the pipeline connecting Kumbakonam and Anthakonam?
(A) 1200 units
(B) 800 units
(C) 850 units
(D) Cannot be determined

Directions for questions 46 to 50: Answer the questions on the basis of the information given below.

The data points in the given graph represent the monthly incomes and expenditures of the members belonging to four families, namely Arthur family (□), Menon family (•), Kaur family (∗) and Ambuja family (▲). Each symbol represents the income and the expenditure of a particular family member.



Note: Savings = Income – Expenditure.

46. Which family has the highest average income?
(A) Kaur (B) Arthur
(C) Menon (D) Ambuja

47. Which family has the lowest average expenditure?
(A) Arthur (B) Ambuja
(C) Kaur (D) Menon

48. For which family, is the income of any individual member more than the combined incomes of the other members of the family?
(A) Arthur (B) Menon
(C) Ambuja (D) None of the families

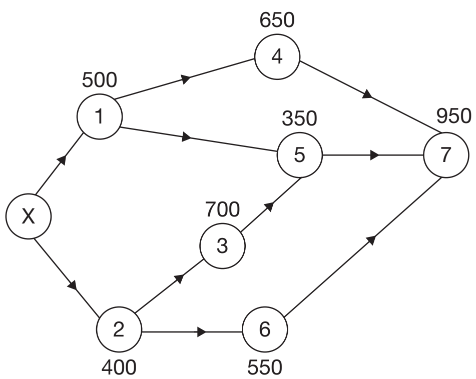
49. For how many families is the total income more than the total expenditure?
(A) 0 (B) 1
(C) 2 (D) 3

50. If the income of each member of the Menon family is increased by 20% and all other values remain the same, then which family has the highest savings?
(A) Menon (B) Kaur
(C) Arthur (D) Ambuja

EXERCISE-2

Directions for questions 1 to 5: Answer the questions based on the information given below.

The following diagram shows the flow of water from a reservoir X to seven water tanks. Water flows only in the direction of the arrows given and only after an intermediate tank is full, does the water flow to the next tank in the pipeline. The capacity of each tank (in kl) is given alongside it. The slack in a pipeline is the excess flow required to bring the pipeline to full capacity.



1. If the pipelines connecting the reservoir X to water tanks 1 and 2 have the same capacity, then what is the minimum capacity of each pipeline if the demand at all the tanks is met?
(A) 1850 kl (B) 1950 kl
(C) 2000 kl (D) 2050 kl

Additions information for questions 2 to 5: It is given that the capacity of the pipelines connecting X to 1 and 2 is 2500 kl and that of all other pipelines is 1500 kl.

2. What is the maximum flow in the pipeline connecting tanks 1 and 5?
(A) 900 kl (B) 1200 kl
(C) 1300 kl (D) 1350 kl

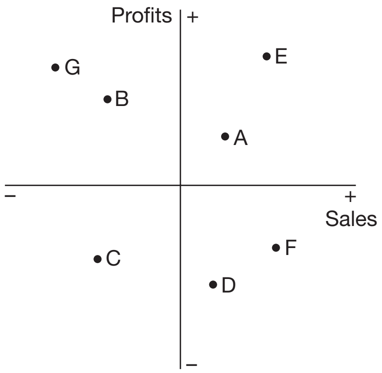
3. What is the minimum slack in the pipeline connecting tanks 2 and 6?
(A) 0 kl (B) 100 kl
(C) 200 kl (D) None of these

4. What is the maximum number of pipelines that can be shut down simultaneously without affecting the supply at any of the tanks?
(A) 1 B) 3
(C) 2 D) 4

5. What is the sum of the slacks in all the pipelines connected to tank 7?
(A) 3100 kl (B) 3300 kl
(C) 3450 kl (D) 3550 kl

Directions for questions 6 to 10: Answer the questions based on the information given below.

The graph gives the change in sales and profit of seven companies from 2013 to 2014.



Expenses = Sales – Profits

6. For how many companies did the expenses definitely increase from 2013 to 2014?

(A) 1 (B) 2
(C) 3 (D) 4

7. If the expenses of company E in 2013 were ₹165 crore, then the expenses incurred by it in the year 2014 could be

(A) ₹152 crore (B) ₹170 crore
(C) ₹175 crore (D) Any of the above

8. How many companies definitely had a decrease in expenses from 2013 to 2014?

(A) 2 (B) 1
(C) 3 (D) 4

9. If the expenses of companies B and F were the same in 2013, then what can be said about their expenses in 2014? (E_B – expenses of B, E_F – expenses of F)

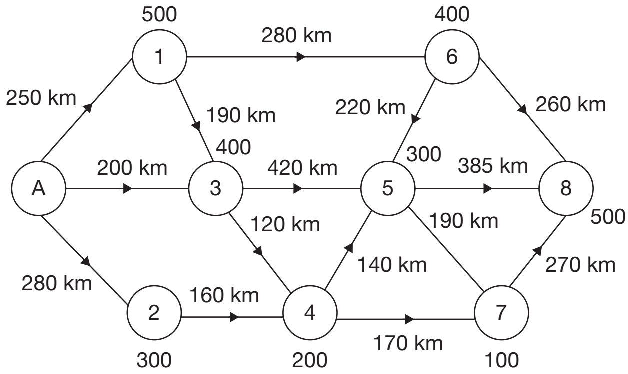
(A) $E_B > E_F$ (B) $E_B < E_F$
(C) $E_B = E_F$ (D) Either (B) or (C)

10. If profitability $= \dfrac{\text{Profit}}{\text{Sales}} \times 100$, then at most how many companies had an increase in profitability from 2013 to 2014?

(A) 3 (B) 5
(C) 6 (D) 7

Directions for questions 11 to 15: Answer the following questions based on the information given below.

The diagram shows the interconnections between a refinery A and eight depots. Oil from the refinery is to be transported to these depots using tankers. The capacity of the depots (in '000 litres) are given. The distance from the refinery to the depots and between two neighbouring depots are also given. Oil can be transported only in the direction in which the arrows point. So also, at each depot, only if it is filled to full capacity, the remaining quantity can be passed on to any of the next depots. Irrespective of the quantity transported, the cost of transportation is ₹.150/km. Oil can be transmitted only in quantities which are integral multiples of 50,000 litres. All depots currently hold 50% of their capacities.



11. What can be the minimum quantity (in litres) that is sent from A, for part of it to reach depot 8?

12. What is the maximum quantity (in litres) that can be sent from A, with a possibility that no part of it still reaches depot 8?

13. What is the minimum cost of transporting the required oil from A to depot 8?

14. If the pipelines between depots 1 and 3 and depots 4 and 7 are closed for repairs, what will be the minimum cost of transporting oil from the refinery to depot 8?

15. What should be the minimum quantity of oil at refinery A, such that all the depots can be filled from their existing level to their capacities?

Directions for questions 16 to 20: These questions are based on the following data.

	A	B	C	D
Roadways	250	400	350	200
Railways	400	200	250	300
Airways	350	400	100	400

The table and 3D chart alongside give the details of the total number of passengers travelling using different means in four different countries A, B, C and D and the distribution of passengers, among those using airways, using the three different airlines, namely P, Q and R airlines in those countries. There are only 3 airlines operating in these countries.

16. The difference between the number of passengers using Q airlines in country D and those using P airlines in country B is

(A) 300 (B) 80
(C) 50 (D) 20

17. The number of passengers using R airlines in country A is what percentage of those using the roadways in the same country?

(A) 70 (B) 20
(C) 17.5 (D) None of these

18. If all the passengers of country C travelling by rail start travelling in Q and P airlines equally, then in country C, the number of passengers using R airlines is approximately what percentage of the number of passengers using Q airlines? (Given that all other factors remain constant)

(A) 25% (B) $33\frac{1}{3}\%$
(C) 12% (D) 50%

19. The passengers using R airlines is what percentage of the total passengers in all the countries?

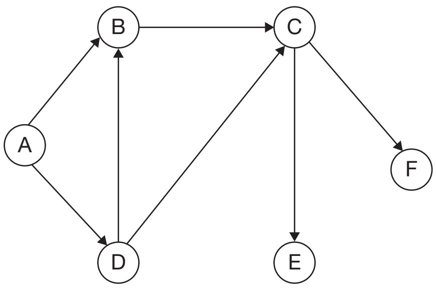
(A) 10% (B) 16.60%
(C) 8.75% (D) $33\frac{1}{3}\%$

20. If P and Q airlines of countries B and D are merged together, then what percentage of the total passengers travelling in those countries will they carry?

(A) 25% (B) 50%
(C) 75% (D) 35.7%

Directions for questions 21 to 24: Answer the questions on the basis of the information given below.

The network shows the water pipelines connecting the 6 major cities of India (A, B, C, D, E, F).



- Through a pipeline, water can flow only in one direction as indicated by the arrow in the diagram.
- The maximum carrying capacity of any of the pipelines is 1000 m^3 per day.
- The daily requirement at C is 400 m^3.
- The slack in the pipeline is the difference between its maximum carrying capacity and the actual load carried by the pipeline.
- The slack in pipeline CE is 100 m^3 less than the slack in pipeline CF. The slack in pipeline AD is 300 m^3.
- The daily requirement at D = 100 m^3.
- The amount of water that flows through pipeline BC is twice the daily requirement at C.
- Slack in pipeline AB = 200 m^3.
- The ratio of the requirement at B to the slack in pipeline CE is 1 : 2.
- The ratio of the slacks in pipelines CE and CF is 6 : 7. The quantity of water flowing through the pipeline DB is the same as that flowing through pipeline DC.

21. Find the daily requirement (in m^3) at E, if it is known that its requirement is exactly met by the water flowing through the pipelines shown.

22. Find the daily requirement (in m^3) at F, if it is known that its requirement is exactly met by the water flowing through the pipelines shown.

23. If there exists a larger external pipeline of capacity 5000 m^3 that supplies water to city A such that the requirements of all the 6 cities are met by the water supplied by it, then what is the slack (in m^3) in the external pipeline? It is given that the daily requirement at A = 500 m^3.

24. If on a particular day, the pipeline joining cities D and B is damaged and the amount of water that is intended to flow through pipeline DB gets wasted in the process, then find how much water (in m^3) is wasted on that day?

Directions for questions 25 to 29: Answer the questions as based on the information given below.

The following table gives the processes involved and the time taken for each process for the completion of a task. The task is said to be completed when all the processes are completed. Some of the processes can be executed simultaneously as long as the conditions are met.

Process	Time taken (in mins)	Processes to be completed before starting this
A	10	None
B	12	None
C	15	A
D	17	B, G, H
E	8	A, B
F	7	C
G	18	C
H	12	B
I	13	A, D
J	15	A, B
K	6	C
L	12	B, D

25. Which of the following processes must be completed before starting process I?

(A) C (B) E
(C) F (D) K

26. What is the earliest time by which process D can be started if the work stated at 10 a.m.?
(A) 10.24 a.m. (B) 10.43 a.m.
(C) 10.58 a.m. (D) None of these

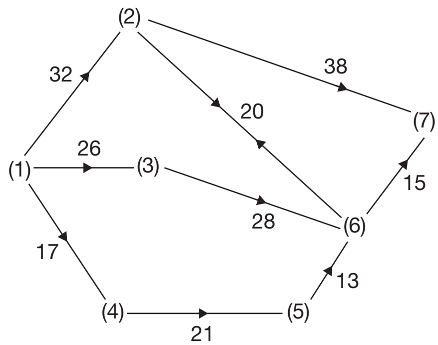
27. How many processes must be necessarily completed before starting process L?
(A) 4 (B) 5
(C) 6 (D) 7

28. At most how many processes can be completed within 35 minutes of starting the task?
(A) 5 (B) 7
(C) 8 (D) 9

29. What is the minimum time taken to finish the entire task?
(A) 56 minutes
(B) 68 minutes
(C) 72 minutes
(D) None of these

Directions for questions 30 to 32: These questions are based on the information given below.

A person has to go from station 1 to station 7. The arrows indicate the direction in which the person can travel. The number given along the arrow represents the distance (in kms) between the two connected stations.



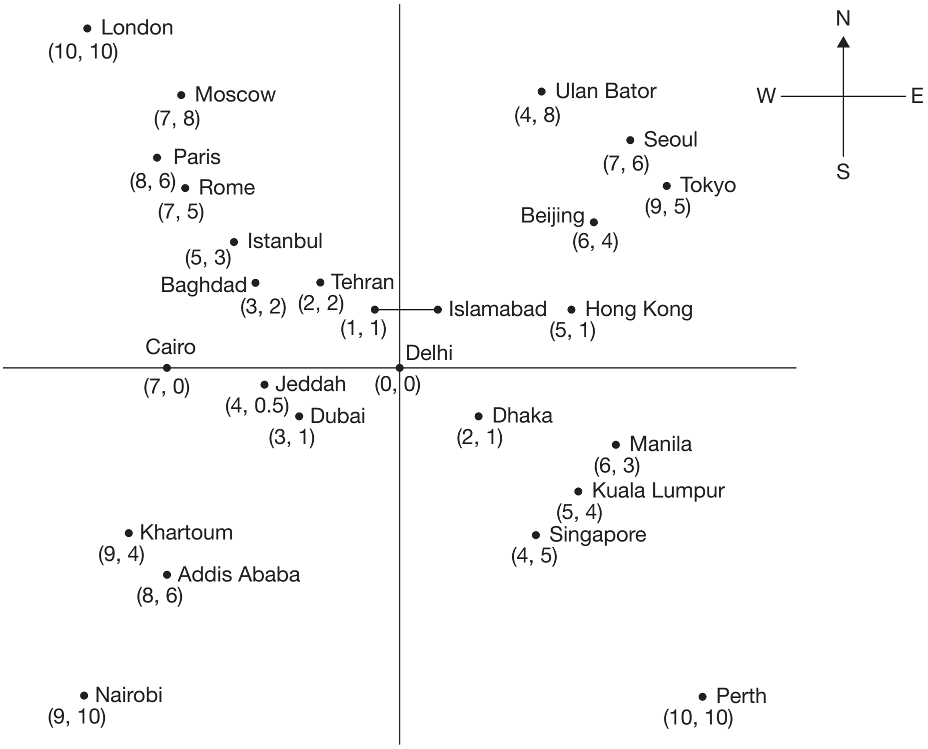
30. The length (in kms) of the longest path connecting stations 1 and 7 is

31. The number of different paths between station 1 and 7 is

32. What is the difference (in Kms) between the longest and the shortest path between stations 1 and 7?

Directions for questions 33 to 37: Answer these questions on the basis of the information given below.

The location of several important cities, their relative distances from and their bearing with respect to Delhi are represented in the figure given below. Each unit on the scale indicates a distance of 500 miles.



Difficulty Level-2: Moderate

Air India operates three types of aircrafts in order to reach the indicated destinations. The aircrafts are Airbus A–340, Boeing 737 and Airbus A–3XX having ranges of 1500 miles, 2000 miles and 4500 miles, respectively on a single refuelling. In all the questions that follow, every flight is assumed to start from Delhi, unless specified and flies in a straight-line path.

33. How many cities shown on the chart above cannot be reached by an Airbus A–340 on a single refuelling?
(A) 22 (B) 21
(C) 23 (D) None of these

34. What is the ratio of the number of cities that cannot be reached by Airbus A–3XX on a single refuelling to that of the number of cities that cannot be reached by an Airbus AAA–340, whose range is thrice that of an Airbus A–340?
(A) 2 : 9 (B) 10 : 13
(C) 3 : 1 (D) None of these

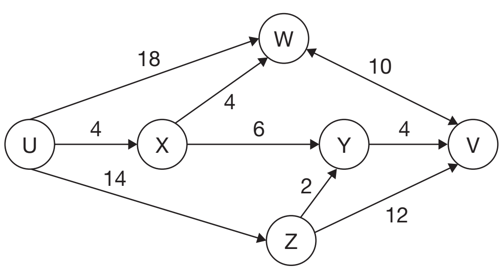
35. What is the ratio of the cities to the North of Delhi that cannot be reached by Boeing 737 to those to the South of Delhi that can be reached by Airbus A–3XX on a single refuelling?
(A) 5 : 3 (B) 2 : 5
(C) 3 : 5 (D) None of these

36. If after some technical improvements, Airbus A–3XX is upgraded to Airbus A–4XX, which has a range of 500 miles more than that of Airbus A–3XX, then of the cities shown, how many can be reached by an Airbus A–4XX in single refuelling?
(A) 18 (B) 20
(C) 21 (D) None of these

37. What is the number of refuellings, required for an Airbus A–3XX flight from London to Perth via Delhi?
(A) 1 (B) 2
(C) 3 (D) 4

Directions for questions 38 to 41: These questions are based on the information given below.

The network below represents a busy one-way street network starting at U and ending at V. Points W, X, Y and Z are junctions in the network and the arrows mark the direction of traffic flow. The time taken (in minutes) to travel between the points is indicated by the number adjacent to the arrow representing the street.



Motorists travelling from U to V would take the route for which the total time of travelling is the minimum. If two or more routes have the same least time of travel, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among all these routes.

The flow of traffic can be controlled only by having a checking point before a junction which would lead to a delay (of the traffic). For example, if a motorist takes the route U-W-V (using junction W alone), then the total time of travel would be 28 mins (i.e., 18 + 10) plus the delay at junction W.

38. If no traffic is to flow on the street from Z to V due to some repair work and equal amount of traffic is to flow through the junctions W and Y to avoid traffic jams, a feasible set of delay times (in mins) at junctions W, X, Y and Z, respectively would be
(A) 2, 10, 6, 6
(B) 2, 8, 8, 6
(C) 2, 10, 8, 4
(D) 0, 10, 4, 4

39. To ensure that all motorists travelling from U to V take the same time, (travelling and checking delays combined) regardless of the route they choose when the street from X to Y is under repair (and hence unusable), a feasible set of time delays (in mins) at junctions W, X, Y and Z, respectively would be
(A) 4, 10, 6, 4
(B) 0, 10, 6, 4
(C) 2, 10, 6, 4
(D) 4, 6, 10, 2

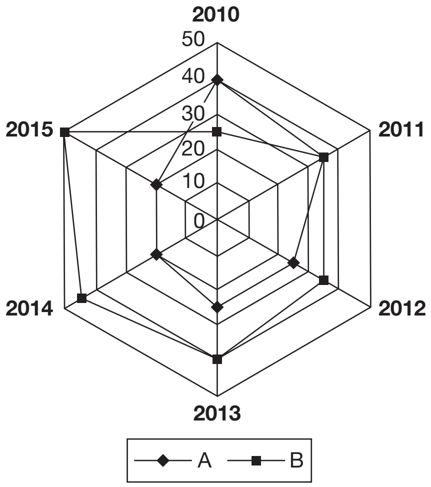
40. To ensure that the traffic at U gets evenly distributed along streets from U to W, from U to X and from U to Z, a feasible set of delays (in mins.) at junctions W, X, Y and Z, respectively would be
(A) 0, 10, 8, 2
(B) 0, 10, 4, 4
(C) 2, 10, 6, 6
(D) 2, 10, 6, 4

41. To ensure that all routes from U to V get the same amounts of traffic, then a feasible set of delay times (in mins) at junctions W, X, Y and Z, respectively would be
(A) 0, 10, 4, 4
(B) 0, 10, 8, 2
(C) 2, 10, 6, 6
(D) 2, 10, 6, 4

Directions for questions 42 to 45: Answer these questions on the basis of the information given below.

The following diagram gives the market share of the top two companies, for each of the years from 2010 to 2015, for a product for which there were four companies A, B, C and D in the market. No two companies had the same market share (in percentage) in a year other than companies A and B in

2011 and in none of the years did any company have a sales more than four times that of any other company.



Total sales of the product in different years

(in ₹ crore)

Year	2010	2011	2012	2013	2014	2015
Total product sales	1200	1400	1800	2000	2600	3000

Assume that market share (in percentage) of each of the companies in each of the given years was an integer.

42. What is the percentage increase in the sales for company B from 2010 to 2015?

(A) 100 (B) 250
(C) 400 (D) 500

43. The percentage increase in the market share of any company in a year, when compared to the previous year is at most

(A) 100 (B) 166.67
(C) 50 (D) None of these

44. If company C had the maximum percentage increase in the sales from 2013 to 2014, the increase in sales was

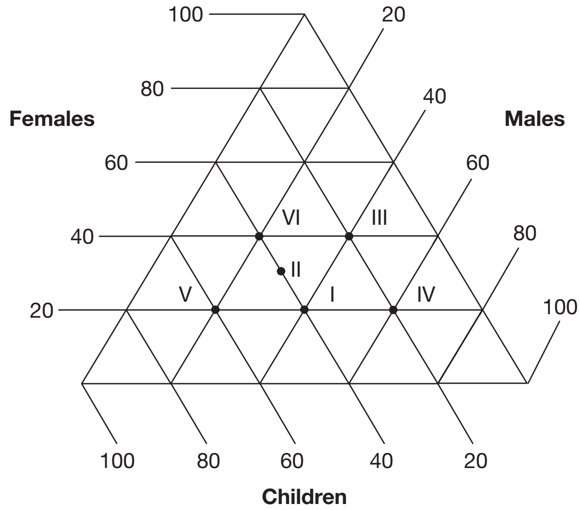
(A) ₹200 crore (B) ₹208 crore
(C) ₹274 crore (D) ₹294 crore

45. If company D had a decrease in sales from 2010 to 2011, the percentage decrease in its sales was at most

(A) 50 (B) 56.25
(C) 60 (D) 66.67

Directions for the questions 46 to 50: Answer these questions based on the information given below.

The following diagram gives the percentage of males, females and children who came for buffet at a restaurant on six consecutive days I to VI. The buffet charges for males, females and children are ₹500, ₹400 and ₹300, respectively and the restaurant only serves buffet.



The total number of people, who came for the buffet during the six days is as follows:

Day	Total people
I	460
II	520
III	540
IV	480
V	420
VI	380

46. What was the total number of males who came to the restaurant in these six days?

47. Children form what percentage of the people who visited the restaurant in these six days?

48. What was the revenue (in ₹) of the restaurant on day V?

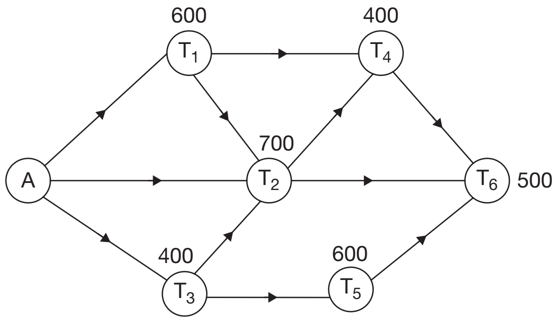
49. What was the highest revenue (in ₹) of the restaurant on any of the six days?

50. What is the average number of females (approximated to the closest integer) who visited the restaurant during these six days?

EXERCISE-3

Directions for questions 1 to 4: Answer these questions on the basis of the information given below.

The following figure gives the network connecting a main water supply tank to six smaller tanks via pipelines. The values given alongside the tanks give their capacity (in kl). The capacity of each pipeline is 1200 kl. Only after a tank is completely full, the water will be passed on to the next tank in the network. The arrows give the direction of water flow. The slack in a pipeline is the excess flow required to bring it to full capacity. The flow in the pipelines is such that each tank is completely filled.



1. What is the maximum slack in the pipeline connecting A and T_2?
(A) 600 (B) 700
(C) 500 (D) None of these

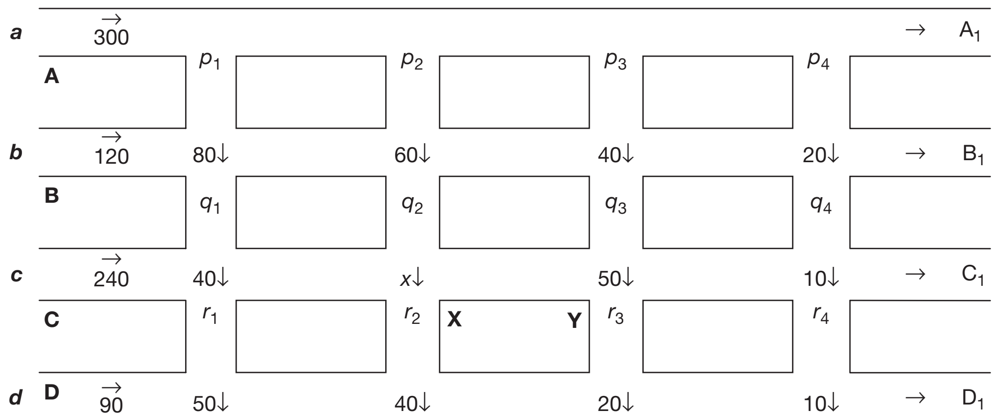
2. If there is no flow in the pipeline connecting T_1 and T_4, then what is the minimum flow in the pipeline connecting A and T_3?
(A) 700 kls (B) 800 kls
(C) 600 kls (D) None of these

3. What is the maximum sum of the slacks (in kls) in all the pipelines combined?
(A) 6200 (B) 6800
(C) 8500 (D) None of these

4. What is the minimum sum of the slacks (in kls) in all the pipelines combined?
(A) 6900
(B) 7200
(C) 7800
(D) None of these

Directions for questions 5 to 8: Answer these questions on the basis of the information given below.

Flow of water through a network of irrigation canals



There is a group of four canals, namely ***a, b, c*** and ***d*** through which water flows for the purpose of irrigation. The water flows into the canals at points A, B, C and D, and is discharged at the points A_1, B_1, C_1 and D_1, respectively. There are three sets of four pipes each, namely p_1, p_2, p_3, p_4; q_1, q_2, q_3, q_4 and r_1, r_2, r_3, r_4 through which water flows from ***a*** to ***b***, ***b*** to ***c*** and ***c*** to ***d***, respectively.

The rates of flow of water (in units of water flowing per unit time) flowing through the pipes and canals are given at their respective positions. For example, 300 units of water enters canal ***a*** per unit time, 80 units of which enters into canal ***b*** per unit time through pipe p_1 and 60 units of water enters into canal ***b*** per unit time through pipe p_2 and so on. For each of the three canals ***a***, ***b*** and ***c***, the quantity of water flowing can never exceed 1.5 times the total quantity of water entering into the respective canal. Assume that in any canal, water flows in only one direction, i.e., from left to right and at any junction, outflow precedes the inflow.

5. What is the minimum value of *x* (in units of water flowing per unit time)?
(A) 10 (B) 20
(C) 30 (D) 40

6. If pipes p_1 and q_2 are closed (blocked), what is the total quantity of water that flows per unit time out of canal ***c*** at point C_1?

(A) 140 (B) 180
(C) 220 (D) 240

7. What is the maximum possible amount of water that can flow per unit time between the points X and Y (shown in the diagram)?

(A) 330 units (B) 320 units
(C) 350 units (D) 360 units

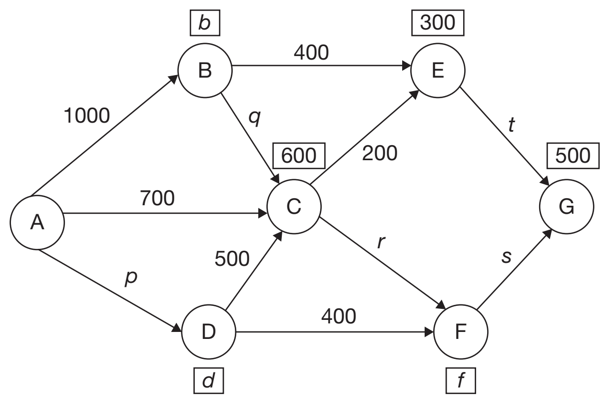
8. If $x = 40$ units, from which of the following points is the rate of outflow of water the least?

(A) A_1 (B) B_1
(C) C_1 (D) D_1

Directions for questions 9 to 12: Answer these questions on the basis of the information given below.

The following diagram represents the network supplying water in a locality. Water flows from the main tank A and is supplied to the secondary tanks B through G by pipelines as indicated in the figure. The direction of flow of water is indicated by arrows. The requirement at subsidiary tanks (in kls) is indicated by the values inside the box above the tank. The flow of water in the pipelines (in kls) is given above the pipeline. At any subsidiary tank, water is passed on to the next tank in the network only after the requirement at that place is completely met. The requirement at all subsidiary tanks are positive integral multiples of 100.

The capacity of each pipeline is 1000 kls. The slack in a pipeline is the extra flow required to bring the flow in it to full capacity.



The flow in the pipelines is such that the requirement at all the places is exactly met.

9. What is the difference (in kls) between *f* and *r*?

10. What is the least possible value of '*f*' in (kls)?

11. What is the value (in kls) of $p - s$?

12. What is the minimum slack (in kls) in all the pipelines together?

Directions for questions 13 to 16: Answer these questions on the basis of the information given below.

The following table gives the various tasks involved in completing a project. The project is complete when all the tasks are done. The tasks can be done in any order as long as the tasks to be done before it is completed.

Task	Time taken (hrs)	Other tasks to be completed before starting this
A	4	
B	7	D, G
C	2	
D	5	A, E
E	11	C
F	6	B
G	8	A, H
H	3	E
I	4	H
J	10	D

13. There are two persons to do the project such that each person can do any of the tasks but each one can do only one task at a time. What is the shortest time (in hrs) in which the project can be completed?

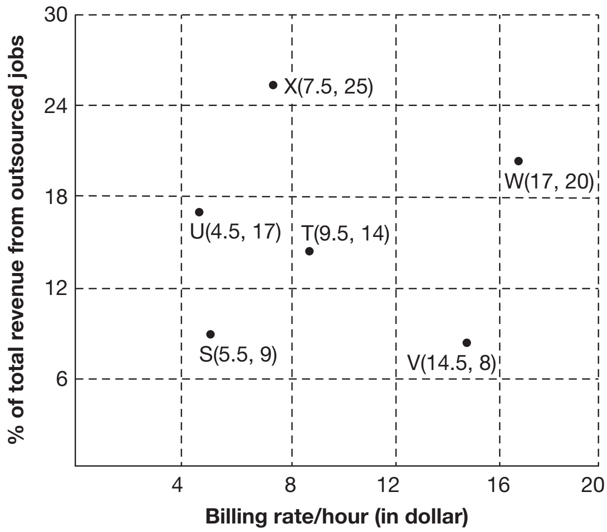
14. If there is no constraint on the number of people available to do the tasks, then what is the shortest time (in hrs) in which the project can be completed?

15. If the project is to be completed in at most 45 hours, then what can be the maximum time gap (in hrs) between the completion of task I and task J?

16. If a person can do only one task at a time, then what is the difference in the time taken (in hrs) to complete the project if there is only one person and when there are two persons, in both cases the project being done in the shortest possible time?

Directions for questions 17 to 20: Answer these questions on the basis of the information given below.

The following diagram gives the relation between the average billing rate per hour for six companies S, T, U, V, W and X which take up outsourcing jobs and the revenue from such jobs as a percentage of the total revenue for these companies.



17. If the total value of outsourcing jobs taken up by company T is \$125 million and the average billing rate for jobs other than outsourcing jobs done by the company is \$19 per hour, then what percentage of the volume of the work done by the company is outsourced work?

(A) 14 (B) 16
(C) 25 (D) 31

18. If the number of hours of outsourcing jobs taken up by all the six companies is the same, then which company had the highest total revenue?

(A) V (B) W
(C) T (D) X

19. Had the billing rate of company W for outsourcing jobs undertaken been 20% less and the quantity of outsourcing and non-outsourcing jobs and billing rate of non-outsourcing jobs done been the same as before, then what would be the percentage reduction in the total revenue of the company?

(A) 2 (B) 4
(C) 5 (D) 6

20. If the ratio of the number of hours spent in taking up outsourcing jobs for companies U and V is 2 : 3, then what is the approximate ratio of the total revenue of U and V?

(A) 1 : 5 (B) 2 : 7
(C) 1 : 10 (D) 3 : 13

ANSWER KEYS

Exercise-1

1. (D)	10. (A)	19. 50	28. (C)	37. 60	46. (C)
2. 2375	11. (C)	20. (B)	29. (D)	38. 3	47. (B)
3. 2500	12. (A)	21. (C)	30. (D)	39. 230	48. (D)
4. 1 : 2	13. (A)	22. (A)	31. (C)	40. (A)	49. (D)
5. 27	14. (A)	23. (B)	32. (B)	41. (C)	50. (A)
6. (B)	15. (A)	24. (D)	33. (B)	42. (D)	
7. (C)	16. 1150	25. (B)	34. (C)	43. (D)	
8. (C)	17. 0	26. (A)	35. (B)	44. (C)	
9. (D)	18. 3	27. (B)	36. (D)	45. (D)	

Exercise-2

1. (D)	10. (B)	19. (C)	28. (C)	37. (D)	46. 1004
2. (C)	11. 3,50,000	20. (D)	29. (D)	38. (D)	47. 35.7
3. (B)	12. 9,00,000	21. 400	30. 112	39. (C)	48. 1,51,200
4. (C)	13. 1,14,000	22. 300	31. 6	40. (A)	49. 2,26,800
5. (D)	14. 1,18,500	23. 3000	32. 46	41. (D)	50. 133
6. (B)	15. 13,50,00	24. 300	33. (B)	42. (C)	
7. (D)	16. (B)	25. (A)	34. (D)	43. (B)	
8. (A)	17. (A)	26. (B)	35. (A)	44. (C)	
9. (B)	18. (C)	27. (C)	36. (D)	45. (B)	

Exercise-3

1. (D)	4. (B)	7. (A)	10. 600	13. 37	16. 23	19. (B)
2. (B)	5. (D)	8. (A)	11. 800	14. 37	17. (C)	20. (C)
3. (C)	6. (C)	9. 200	12. 4900	15. 25	18. (A)	

SOLUTIONS

EXERCISE-1

1. During 1997, A's Profit = 12.5% of 3000 crore

$= \frac{1}{8} \times 3000 =$ ₹375 crore

B's Profit = 25% of 4000 crore = ₹1000 crore

C's Profit = 37.5% of 5000 crore

$= \frac{3}{8} \times 5000 =$ ₹1875 crore

D's Profit = 50% of 4000 crore = ₹2000 crore.

2. A's total profit = 62.5% of 2000 + 25% of 3000 + 12.5% of 3000= ₹2375 crore

3. From earlier solution, in 1997, D made a maximum profit of ₹2000 crore. Using this as reference, we see that in 1996 C exceeds this, C = ₹2500 crore and then in 1995 no company exceeds this.

4. Since the profit percentage of both C and D in 1995 are the same, the ratio of expenditure of C and D in 1995 will be same as that of their sales revenues.

$$\left(\text{Since expenditure} = \text{Revenue} \times \frac{(100 - \text{profit\%})}{100}\right)$$

5. During 1998, sales revenue of B = 125% of 4000 crore = 5000 crore

Expenditure in 1998 = 80% of 0.75 of 4000 = 2400 crore

$$\text{Profit} = \left(\frac{5 - 2.4}{5}\right) \times 100 = 52\%$$

Actual profit of B in 1997 = 25%

52 – 25 = 27 % points

Solutions for questions 6 to 10: Votes polled in various wards are as follows:

Ward 1 – 18% of 5,00,000 = 90,000

Ward 2 – 32% of 5,00,000 = 1,60,000

Ward 3 – 36% of 5,00,000 = 1,80,000

Ward 4 – 14% of 5,00,000 = 70,000

The following table can now be drawn to show the votes polled by various parties in different wards.

Party	Wards			
	1	2	3	4
TDP	$\frac{50}{100} \times 90{,}000 = 45{,}000$	$\frac{25}{100} \times 1{,}60{,}000 = 40{,}000$	$\frac{25}{100} \times 1{,}80{,}000 = 45{,}000$	$\frac{25}{100} \times 70{,}000 = 17{,}500$
Congress	$\frac{25}{100} \times 90{,}000 = 22{,}500$	$\frac{50}{100} \times 1{,}60{,}000 = 80{,}000$	$\frac{25}{100} \times 1{,}80{,}000 = 45{,}000$	$\frac{12.5}{100} \times 70{,}000 = 8{,}750$
BJP	$\frac{25}{100} \times 90{,}000 = 22{,}500$	$\frac{25}{100} \times 1{,}60{,}000 = 40{,}000$	$\frac{50}{100} \times 1{,}80{,}000 = 90{,}000$	$\frac{62.5}{100} \times 70{,}000 = 43{,}750$

6. 45,000

7. The choice will be between ward 2 and ward 3, because these words got the maximum number of votes polled.

8. 90,000 + 43,750 = 1,33,750

9. The total number of votes secured by Congress and TDP in ward 2 are 50% and 25%, respectively. Therefore, Congress got a margin of 25% over TDP in ward 2, i.e., 25% of 1,60,000 = 40,000

10. BJP

Solutions for questions 11 to 15:

	HR	Finance	Marketing
A	25% × 1,600 = 400	25% × 1,600 = 400	50% × 1,600 = 800
B	37.5% × 1,800 = 675	25% × 1,800 = 450	37.5% × 1,800 = 675
C	50% × 1,700 = 850	25% × 1,700 = 425	25% × 1,700 = 425
D	12.5% × 2,000 = 250	75% × 2,000 = 1,500	12.5% × 2,000 = 250
E	37.5% × 1800 = 675	50% × 1800 = 900	12.5% × 1800 = 225

11. By observing the above table, We get the number of HR professionals in firm A as 400.

12. (Number of Marketing professionals in B) – (Number of Marketing professionals in C)
= 675 – 425 = 250

13. Finance professionals in firm D are 1500 (1.3) = 1950
HR professionals in firm D are 250 (1.1) = 275
Marketing professionals in firm D are 250 (1.2) = 300
∴ Number of professionals are increased from 2000 to 2525.

$$\% \text{ Increase} = \left(\frac{2{,}525}{2{,}000} - 1\right)100 = 26.25\%$$

14. Average monthly salary is the weighted average salary of the three departments, i.e., 50% of 7000 (for Marketing) + 25% of 6000 (for Finance) + 25% of 5000 (for HR) = 3500 + 1500 +1250 = ₹6250

15. By observing the table, it can be concluded that the number of professionals in firms B and E are the same.
∴ Their ratio is 1 : 1

Solutions for questions 16 to 20: The capacities of the pipeline connecting A with B, C and D are 1500 each while the capacities of all other pipelines are 750 each.

16. To find the minimum flow in the pipeline connecting D and G, we have to maximize the flow in other pipelines.
The maximum flow in the pipeline connecting A and B is 1450 (as 700 is stored at depot B and the remaining 750 is the maximum that can flow through pipeline connecting B and E).
Similarly, the maximum flow in the pipeline connecting A and C is 1400.
∴ The minimum flow in the pipeline connecting A and D is 4000 – (1450 + 1400) = 1150.

17. The maximum flow in the pipeline connecting A and B is 1200, of which 500 litres will flow to E.
The flow through the pipeline connecting A and C can be at most 1400, of which 750 will flow to E.
Now E would receive 500 + 750 = 1250 of which 450 would be stored at E, 500 would be transferred to F and 300 to G, with G receiving the remaining 600 from D.
∴The flow in the pipeline connecting D and F can be zero.

18. The slack in the pipeline supplying oil to depot E would be maximum, when pipeline connecting A and D carries the maximum oil, i.e., 1500 litres.
Without any flow in the pipeline supplying oil to depot E, a maximum of 700 (A – B) + 650 (A – C) + 1500 (A – D - F or G) = 2850 can be transported. Therefore, a further 4000 – 2850 = 1150 is required which is to be supplied through depot E.
∴ Maximum slack = (750 + 750) – 1150 = 350

19. If the stack in the pipeline connecting D and G is 50, it means that oil flowing through the pipeline connecting D and G is 750 – 50 = 700. Of the 900 required at G, 700 is flowing through the pipeline connecting D and G. The stack in the pipeline connecting E and F is minimum, when the flow through it is the maximum.
The maximum flow through the pipeline connecting E and F is 500 (required at F) + 200 (further requirement at G), i.e., 700.
∴ The minimum slack is 750 – 700 = 50.

20. Even though 1000 units can flow through each of the pipelines XA and XC, it can be seen that the entire supply to depots B and F is through the pipeline connecting X and B. Therefore, at least 600 + 250 = 850 units should flow through X – B.
∴ The maximum slack is 150 units.

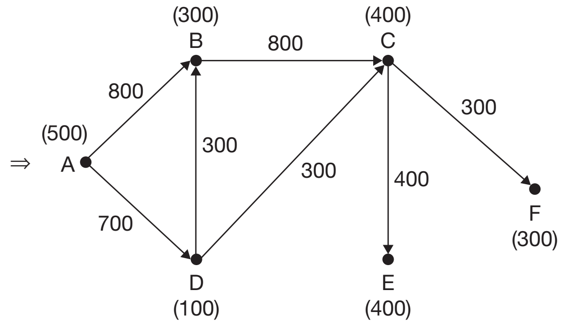
21. The pipeline connecting X and C supplies to depots C, E and G. Therefore, the maximum flow through it is 350 + 250 + 350 = 950 units. Thus, the maximum flow through the pipeline connecting C and E = 950 – 350 = 600 units.

22. The supply to each of A, B and C flows through only XA, XB and XC. The supply to each of D, F and E flows through two pipelines each, say XA and AD, XB and BF and XC and CE, respectively. The supply to G has to flow through three pipelines, i.e., EG or FG or DG.
∴ The total slack in all the pipelines put together.
= 10 (number of pipelines) × 1000 – [1 (450 + 600 + 350) + 2 (400 + 250 + 250) + 3 (350)]
= 10,000 – (1400 + 1800 + 1050] = 5750

23. As the total demand is 2650 units, at least 650 units must flow through the pipelines connecting X and C. Therefore, at least 50 units of demand at G must be supplied from E and so the maximum flow through the pipeline connecting D and G is 350 – 50 = 300 units.

24. The maximum quantity of oil that can be supplied from the refinery is 1000 + 1000 + 1000 = 3000 units. As the current demand is 2650 units, a maximum of 3000 – 2650 = 350 units can be supplied to the new depot.

Solutions for questions 25 to 28:



Difficulty Level-1: Foundation

⇒ Maximum capacity = 1000 m^3 per day – (1)
⇒ Requirement at C = 400 – (2)
⇒ Slack in CE – Slack in CF = 100 – (2)
⇒ AD = 1000 – 300 = 700 – (4)
⇒ D = 100, C = 400 – (5)
⇒ BC = 800 – (6)
⇒ Slack in CE : Slack in CF = 6 : 7 – (7)
⇒ DB : DC = 1 : 1 } (8)
⇒ DB = 300, DC = 300 }
⇒ Slack in AB = 200 – (9)

25. From (3) and (7), we get:
Slack in CE = 600
Slack in CF = 700
⇒ Water flowing through CE = Requirement at city E = 1000 – 600 = 400

26. Water flowing through CF = Requirement at city F = 1000 – 700 = 300

27. By using the above 9 conditions and drawing the diagram again, we get:
Water that should flow through the pipeline
= 500 + 800 + 700 = 2000 m^3
Slack = 5000 – 2000 = 3000 m^3

28. In the above figure, if the pipeline joining D and B is damaged, then the amount of water wasted = 300 m^3

29. Distinct paths from city A to city G are as follows:
A – B – C – F – G
A – B – C – G
A – E – C – G
A – D – E – C – G
A – E – C – F – G
A – D – E – C – F – G
∴ A total of 6 ways.

30. City G can be reached from city E in the following ways:
E – C – F – G
E – D – A – B – C – F – G
∴ A total of 2 ways.

31. If all the roads allow two-way traffic, then city G can be reached from city E in the following ways:
E – C – F – G
E – D – A – B – C – F – G
E – G
E – A – B – C – F – G
∴ A total of 4 ways.

Solutions for questions 32 to 36: We can get the following table but it is suggested that only the required data be calculated, in order to save time.

	Region				
Company	**North**	**South**	**East**	**West**	**Total**
Voltas	25% = 0.90	50% = 1.20	50% = 0.70	25% = 0.65	3.45L
Blue Star	50% = 1.80	25% = 0.60	12.5% = 0.175	25% = 0.65	3.225L
Carrier Aircon	25% = 0.90	25% = 0.60	37.5% = 0.525	50% = 1.30	3.325L
Total	**3.6L**	**2.4L**	**1.4L**	**2.6L**	**10L**

32. Voltas sold maximum number of ACs, i.e., 1.2 lakh in South region.

33. Total number of ACs sold by carrier Aircon in all four regions = 3,32,500

34. No. of ACs sold by Blue Star in East and North = 17,500 + 1,80,000 = 1,97,500

35. Required percentage = $\frac{1.2 - 0.6}{0.6} \times 100 = 100\%$

36. Required Ratio is 70,000 : 3,60,000
≃ 1 : 5 (approx)

37. In Tamil Nadu, BA won 20% of the total seats
⇒ 20% of total = 48
⇒ Total number of seats in Tamil Nadu = 240
In Gujarat, CA won 30% of the total seats.
⇒ 30% of total seats = 54
⇒ Total seats in Gujarat = 180
∴ Assembly seats in Tamil Nadu are more than those in Gujarat by 60.

38. Number of seats won by CA in Uttar Pradesh = 10% of 330 = 33

39. In Uttar Pradesh assembly, Others won 30% of the total seats ⇒ 30% of the total = 69
⇒ Total number of seats = 230

40. Statement (c) is true. Statement (a) is also true as BA has won 60% of the seats. As we do not know the number of assembly seats in each of the states, we cannot verify the data given in the statement (b).

41. When we require maximum slack in a pipeline, the minimum should be carried through that pipeline and maximum should be carried through the other pipelines.
The total requirement at all the five places is 600 + 800 + 900 + 500 + 450, i.e., 3250 units, of which 2000 units can be supplied through the Puranapool to Adikonam pipeline.
∴The remaining 1250 units should be sent through the Kumbakonam pipeline. The demand at Kumbakonam is 500 units.
∴ Slack in the Kumbakonam to Anthakonam pipeline is 20000 – (1250 – 500) = 1250 units.

42. The pipeline connecting Kumbakonam to Anthakonam has a slack of 800 units.
∴ The supply at Anthakonam 2000 – 800 = 1200 units.
The demand at Kumbakonam is 500 units.
∴ The supply from Puranapool to Kumbakonam
= 1200 + 500 units = 1700 units.
As the total demand is 3250 units and the supply from Puranapool to Kumbakonam is 1700 units, the supply from Puranapool to Adikonam is 3250 – 1700 = 1550 units.

43. For minimum slack, the supply from Madhyakonam has to be maximum, i.e., 2000 – 600 – 800 = 600.
∴ Slack would be 2000 – 600 = 1400 units.

44. For slack to be minimum, slack in both the lines from Puranapool should be zero.
Now, from Adikonam to Madyakonam, there will be a slack of 600 units and that from Kumbakonam to Anthakonam it is 500 units. The maximum supply from Madhyakonam to Anthakonam is 600 units.
∴The total supply at Anthakonam is 600 + 1500 = 2100 units. Since demand at Anthakonam is 900 units, maximum supply at Nayapool is 1200 units. If the demand at Nayapool is 1200 units, all these 1200 units can be supplied from Anthakonam.
∴ There will be a slack of 800 units from Anthakonam to Nayapool.
∴The total slack is 600 + 500 + 1400 + 800 = 3300 units.

45. The total demand at all the plants put together is 3250 units. The maximum possible flows in the pipelines connecting Puranapool to Adikonam and Kumbakonan are 1600 (i.e., 80% of 2000) units and 2000 units, respectively. Hence, the flow in the pipeline connecting Puranapool to Adikonam can vary from 3250 – 2000 = 1250 units to 1600 units, while the flow in the pipeline connecting Puranapool to Kumbakonan can vary from 1650 units to 2000 units. Since the exact pattern of flow cannot be determined, the slack in the pipeline from Kumbakonam to Anthakonam cannot be determined.

46. As the total income of the Menon family is the maximum, their average income is the highest.

47. As the expenditure of the members of the Ambuja family has lower values, they have the lowest average expenditure.

48. By observing the data points, we can say that in no family the income of any individual member is more than the combined income of the others.

49. The total income is more than the total expenditure for the Arthur, Menon and the Ambuja families.

50. The income of the members of the Menon family already is on the higher side and it has further increased by 20%. Also, the expenditures of the top three spending families are very close. Hence, Income – Expenditure will be the highest for the Menon family itself.

EXERCISE-2

1. The total demand at all the tanks = 500+ 400 + 700+ 650 + 350 +550 + 950 = 4100
∴ The capacities of the two pipelines connected should be at least 2050 kl.

2. The maximum flow would be when the entire demand at 5 and 7 is met by the flow through (1).
∴ Maximum flow = 350 + 950 = 1300 kl

3. For minimum slack, the flow must be maximum. If the maximum possible water flow is pumped through the pipeline connecting X and 2, then flow through the pipeline connecting 2 and 6 would be 2500 – (400 + 700) = 1400. Thus, the slack would be 100 kl.

4. We should only consider pipelines connecting 5 and 7 as only they have multiple input sources. The following cases are possible.
(1) The pipeline connecting 4 and 7 can be shut down and the entire demand at 7 can be met from either 5 or 6.
(2) The pipeline connecting 5 and 7 can be shut down along with or without shutting down the pipeline connecting 3 and 5.
(3) The pipeline connecting 6 and 7 can be shut down.
∴ The maximum number of pipelines that can be simultaneously shut down without affecting the demand at any place is 2, i.e., 3 – 5 and 5 – 7 or 4 – 7 and 6 – 7.

5. The maximum capacity of all the pipelines connected to tank 7 = 4500 kl.
The demand at 7 = 950 kl
∴ The slack = 4500 – 950 = 3550 kl.

6. For the expenses to definitely increase, the sales must have increased and the profit must have decreased. This happened only for companies D and F.

7. As the sales and profits of the company increased with respect to that in the previous year, we cannot say anything about its profit in 2014. Therefore, its profit can be any of the given values.

8. For a definite decrease in expenses, the sales should be less and the profits should be more when compared to the previous year. Only G and B satisfy the condition.

9. For B, the sales decreased and its profits increased.
∴ The expenses definitely decreased. For F, the sales increased and the profits decreased. The expenses definitely increased and we can say that the expenses of F would be more than that of B in 2014.

10. All companies except D and F could have had an increase in profitability.

11. All the refineries are currently filled to 50% of their capacity.
By taking the route A – 2 – 4 – 7 – 8 we can ensure minimum quantity to be sent to reach 8. We need to send (in ₹000 litres)
150 + 100+ 50 + 50 (for depot 8) = 350 (in ₹000 litres)

12. If we send through the route A – 1 – 3 – 4 – 5 – 6, we can send (in ₹000 litres)
250 + 200 + 100 + 150 + 200 = 900,
which will be consumed before reaching depot 8.

13. The shortest route is from A – 3 – 4 – 7 – 8, which is 760 km.
∴Cost = 760 × 150 = 1,14,000

14. Now the shortest route is A – 1 – 6 – 8, which is 790 km.
∴ Cost = 790 × 150 = 1,18,500

15. Total capacity of all depots (in ₹000) = 2700
Capacity to be filled (in litres)
= 50% of 2,700,000 = 13,50,000

16. In country D:
The total number of passengers using airways = 400
Passengers of country D using Q airlines = 30% = 120
In country B:
The total number of passengers using airways = 400
Passengers using P airlines = 10% = 40
⇒ Difference = 120 – 40 = 80

17. The number of passengers using R airlines in country A = 50% of 350 = 175
The number of passengers using roadways = 250

$$\Rightarrow \frac{175}{250} \times 100 = 70\%$$

18. The passengers of country C using railways = 250
Those using Q airlines currently = 40
Those using P airlines currently = 40
Those using R airlines currently = 20
Now after the addition of the passengers,
The number of passengers of Q airlines = 125 + 40 = 165
The number of passengers of P airlines = 125 + 40 = 165

$$\Rightarrow \text{Required } \% = \frac{20}{165} \times 100 = 12.1\%$$

19. The total number of passengers = 3600
Those using R airlines in each countries is as follows:

$$\text{Country A} \Rightarrow \frac{50}{100} \times 350 = 175$$

$$\text{Country B} \Rightarrow \frac{30}{100} \times 400 = 120$$

$$\text{Country C} \Rightarrow \frac{20}{100} \times 100 = 20$$

Country D ⇒ 0, Total = 315

$$\therefore \text{Required percentage} = \frac{315}{3600} \times 100 = 8.75\%$$

20.

	P	Q
B ⇒	40	240
D ⇒	280	120

⇒ Total = 680

$$\text{Required } \% = \frac{680}{1900} \times 100 = 35.7\%$$

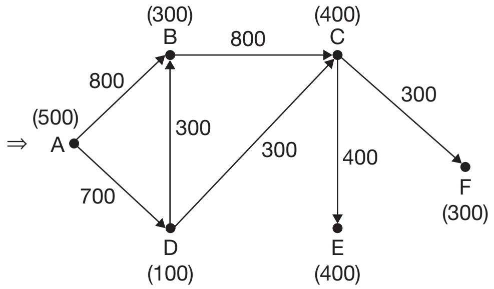
Solutions for questions 21 to 24:

⇒ Maximum capacity = 1000 m^3 per day – (1)
⇒ Requirement at C= 400 – (2)
⇒ Slack in CE – Slack in CF = 100 – (2)
⇒ AD = 1000 – 300 = 700 – (4)
⇒ D = 100, C = 400 – (5)
⇒ BC = 800 – (6)
⇒ Slack in CE : Slack in CF = 6 : 7 – (7)
⇒ DB : DC = 1 : 1
⇒ DB = 300, DC = 300
⇒ Slack in AB = 200 – (9)

21. From (3) and (7), we get:
Slack in CE = 600
Slack in CF = 700
⇒ Water flowing through CE =
Requirement at city E = 1000 – 600 = 400.

22. Water flowing through CF = Requirement at city F = 1000 – 700 = 300.

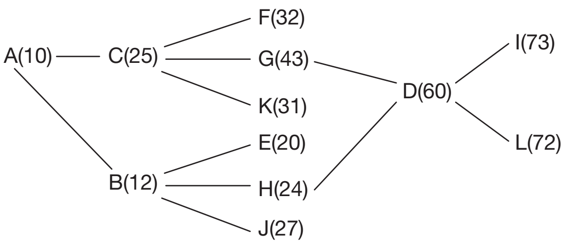
23. By using the above 9 conditions and drawing the diagram again, we get:



Water that should flow through the pipeline
$= 500 + 800 + 700 = 2000$ m^3
Slack $= 5000 - 2000 = 3000$ m^3

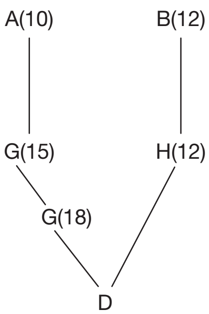
24. In the above figure if the pipeline joining D and B is damaged then the amount of water wasted $= 300$ m^3.

Solutions for questions 25 to 29: The order and total time taken for the tasks are as follows:



25. Before I, A and D must be completed. Before D is started B, G and H must be completed and before G, C must be completed.
∴ C must be definitely completed before I is started.

26. Process D can be started at $10 + 15 + 18 = 43$ minutes past 10 a.m.



27. Before starting process L, the ones that have to be necessarily completed are A, B, C, D, G and H.

28. Eight processes, i.e., A, B, C, E, F, H, J and K can be completed within 35 minutes of starting the task. (As given in the time figure above).

29. The time taken to finish the entire task is 73 minutes (As given in the figure above Q. 25)

30. The length of the path is as follows:
1 – 3 – 6 – 2 – 7
26 + 28 + 20 + 38 = 112 kms.

31. Path:
(1) 1– 4 – 5 – 6 – 7
(2) 1– 3 – 6 – 7
(3) 1 – 2 – 7
(4) 1 – 2 – 6 – 7
(5) 1 – 3 – 6 – 2 – 7
(6) 1 – 4 – 5 – 6 – 2 – 7.

32. The longest path is 1 – 3 – 6 – 2 – 7 which is 112 kms.
The shortest path is 1 – 4 – 5 – 6 – 7 which is 66 kms.
The difference = 112 – 66 = 46 kms.

33. Since 1 unit represents 500 miles, it means that Airbus A-340 can travel 3 units in any direction on a single refuelling. Thus, the only cities it can reach are Islamabad and Dhaka, all the other cities cannot be reached, which is a total of 21 cities.

34. The range of A-3XX is thrice that of A-340 and given that the range of AAA-340 is also thrice of A-340 the range of A-3XX and AAA-340 is the same. Therefore, the number of cities that cannot be reached by these two should be the same. Hence, the ratio must be 1 : 1.

35. The cities to the North of Delhi that can be reached by Boeing 737-400 on a single refuelling are Baghdad, Tehran and Islamabad. Therefore, the number of cities to the North of Delhi that cannot be reached are 13 – 3 = 10. The cities to the South of Delhi that can be reached by A-3XX on a single refuelling are Jeddah, Dubai, Dhaka, Maila, Kaula Lumpur and Singapore (i.e., a total of 6 cities).
∴ The ratio is 10 : 6 = 5 : 3.

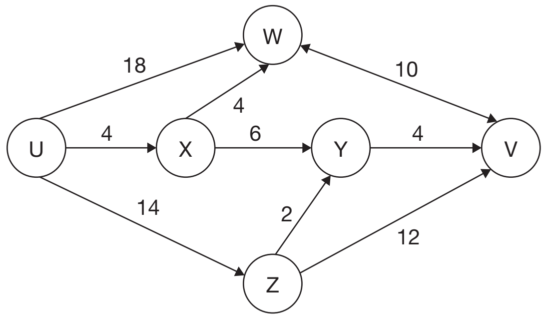
36. The range of Airbus A-4XX = 4500 + 500 = 5000 km
⇒ 10 units = radial distance of any reachable city
$= r \le 10 \Rightarrow r^2 \le 100$
$\therefore\ x^2 + y^2 \le 100$
Only London (10, 10), Perth (10,10), Nairobi (9, 10), Moscow (7, 8) and Tokyo (9, 5) is $x^2 + y^2 > 100$.
∴ 5 out of 24 cities cannot be reached.
⇒ 24 – 5 = 19 cities can be reached.

37. The distance between London to Perth via Delhi is

$$20\sqrt{2} \times 500 = 10{,}000 \times 1.41 \cong 14100$$

$$\text{Number of refuellings} = \frac{14100}{4500} = 4$$

Solutions for questions 38 to 41:



The possible routes are:

Route	Total time	Junction
U – W – V	28	W
U – X – W – V	18	X, W
U – X – Y – V	14	X, Y
U – Z – Y – V	20	Z, Y
U – Z – V	26	Z

It is given that, for each of the routes, the only way to increase the total time is to impose checking delays at junctions. Let the time delay due to checking at junctions W, X, Y and Z be w, x, y and z, respectively. Now the total time for each of the five routes will be as follows:

Route	Total time
U – W – V	$28 + w$
U – X – W – V	$18 + (x + w)$
U – X – Y – V	$14 + (x + y)$
U – Z – Y – V	$20 + (z + y)$
U – Z – V	$26 + z$

38. No traffic flows, from Z – V. Now applying each of the options, the total time would be as follows:

Route	Option A	Option B	Option C	Option D
U – W – V	30	30	30	28
U – X – W – V	30	28	30	28
U – Z – V	32	32	30	30

Route	Option A	Option B	Option C	Option D
U – X – Y – V	30	30	32	28
U – Z – Y – V	32	34	32	28
U – Z – V	32	32	30	30

As it is given that the traffic flow at junction W is the same as that at junction Y.

$\therefore$ Number of routes involving W that can be used must be the same as that involving Y.

Further only the routes with minimum time duration can be used.

This happens in only (D) as in the routes that can be used, the number of routes involving W is two (U – W – V) and (U – X – W – V) and that involving Y is also two (U – X – Y – V) and (U – Z – U – V).

39. As X – Y is unusable, U – X – Y – V is not possible. From the remaining, if we apply all the options:

Route	Option A	Option B	Option C	Option D
U – W – V	32	28	30	32
U – X – W – V	32	28	30	28
U – Z – Y – V	30	30	30	32
U – Z – V	30	30	30	28

Only in option (C), the total time taken is the same for each of the four routes.

40. From the given options:

Route	Option A	Option B	Option C	Option D
U – W – V	28	28	30	30
U – X – W – V	28	28	30	30
U – X – Y – V	32	28	30	30
U – Z – Y – V	30	28	32	30
U – Z – V	28	30	32	30

It is very likely that option (D) is selected. But if all the four routes take the same time, there will be an equal traffic in all the five routes, i.e., 20% in each route. But then the percentage of traffic in V – W = 20%, U – X = 40% as these are two routes involving U – X, U – Z = 40% (for the same reason as above).

But here the given condition is that time taken in U – W is equal to U – X, which in turn is equal to U – Z.
As V – W = U – Z
Of the routes, that can be used the number of routes involving U – W must be same as U – X, which in turn is same as U – Z. It happened in only option (A).

41. From the given options:

Route	Option A	Option B	Option C	Option D
U – W – V	28	28	30	30
U – X – W – V	28	28	30	30
U – X – Y – V	28	32	30	30
U – Z – Y – V	28	30	32	30
U – Z – V	30	28	32	30

As the time must be the same for all the routes, it must be option (D).

Solutions from questions 42 to 45: The percentage share in the market for the companies, (the range in case of companies C and D) in the different years are as follows:

Year	2010	2011	2012	2013	2014	2015
A	40	35	25	25	20	20
B	25	35	35	40	45	50
C	11-24	9-21	16-24	11-24	16-19	11-19
D	11-24	9-21	16-24	11-24	16-19	11-19

42. Sales of company B in 2010 = $\frac{25}{100} \times 1200 = 300$ crore

Sales of company B in 2015 = $\frac{50}{100} \times 3000 = 1500$ crore

Percentage increase = $\frac{1200}{300} \times 100 = 400\%$

43. The percentage increase in market share is maximum if company C or D had the minimum possible market share in 2011 (i.e., 9%) and the same company had the maximum possible market share in 2012 (i.e., 24%)

$\therefore$ The percentage increase = $\frac{15}{9} \times 100 = 166.67$

44. If company C had the maximum percentage increase in sales from 2013 to 2014.

Its sales in 2013 = $\frac{11}{100} \times 2000 = 220$ (minimum possible)

Its sales in 2014 = $\frac{19}{100} \times 2600 = 494$ (maximum possible)

$\therefore$ Increase in sales = ₹274 crore

45. Maximum value of company D's sales in 2010

$= \frac{24}{100} \times 1200 = 288$ crore

Minimum value of company D's sales in 2011

$= \frac{9}{100} \times 1400 = 126$ crore

Percentage decrease = $\frac{162}{288} \times 100 = 56.25\%$

Solutions for questions 46 to 50: The number of males, females and children who visited the restaurant and the total collection each day is as follows:

Day	Males	Females	Children	Total Revenue
I	184	92	184	92,000 + 36, 800 + 55200 = 1, 84,000
II	156	156	208	78,000 + 62, 400 + 62, 400 = 2, 02, 800
III	216	216	108	1, 08,000 + 86, 400 + 32, 400 = 2, 26, 800
IV	288	96	96	1, 44, 000 + 38, 400 + 28, 800 = 2, 11, 200
V	84	84	252	42, 000 + 33, 600 + 75, 600 = 1, 51, 200
VI	76	152	152	38, 000 + 60, 800 + 45, 600 = 1, 44, 400
Total	1004	796	1000	

46. The total number of males who came to the restaurant on these six days = 1004

47. The required percentage = $\frac{1000}{2800} \times 100 = 35.7\%$

48. The revenue was ₹1,51,200.

49. The highest revenue was on day III and it was ₹2,26,800.

50. The average number of females = $\frac{796}{6} = 132.87$.

EXERCISE-3

1. The total requirement at all the tanks – 3200 kls. As 2400 kls can flow through A – T_1, and A – T_3, at least 800 kl must flow through A –T_2. ∴slack = 1200 – 800 = 400.

2. If there is no flow in the pipeline connecting T_1 and T_4, the maximum flow in A – T_1, and A – T_2 can still be 1200 each.
∴ The minimum flow in the pipeline connecting A – T_3 would be 3200 – 2400 = 800 kl.

3. Let us assume that 1000 kls flow through A –T_1, and A – T_3, and 1200 kls through A – T_2.
From A → T_1, T_2, T_3 → Total capacity = 3600. Total flow = 3200 ∴ slack = 400 kls
From T_1, T_3 → T_2 → Total capacity = 2400.
Total flow = 0. Slack = 2400 kls
Similarly, slacks in $T_1 - T_4$, $T_3 - T_5$ and $T_2 - T_6$ are 800, 600 and 700, respectively. That in $T_4 - T_6$, $T_5 - T_6$ and $T_2 - T_4$ would be 1200 kls each.
∴ Slack = 400 + 2400 + 600 + 800 + 700 + 1200 + 1200 + 1200 = 8500 kls.
Alternate method:
Total capacity of all pipelines = 13,200 kls
The shortest path to reach T_4, T_5 and T_6, is through 2 pipelines and T_1, T_2, and T_3 is through one pipeline.
∴Total flow = (600 + 700 + 400) × 1+ (400 + 500 + 600) × 2 = 1700 + 3000 = 4700
Slack = 13,200 – 4700 = 8500 kls.

4. For minimum total slack, the water has to flow through the longest path. The longest route to reach the different tanks are as follows:
$T_1 \to A - T_1$, $T_2 \to A - T_2$, $T_3 \to A - T_3$
$T_4 \to A - T_1 - T_2 - T_4$
$T_5 \to A - T_3 - T_5$
$T_6 \to A - T_1 - T_2 - T_4 - T_6$ (200 kl)
$\to A - T_3 - T_2 - T_4 - T_6$ (200 kl)
$\to A - T_2 - T_4 - T_6$ (100 kl)
Total flow = (600 + 700 + 400) × 1 + (600 × 2) + (400 + 100) × 3 + 400 × 4 = 1700 + 1200 + 1500 + 1600 = 6000 kl
Slack = 13,200 – 6000 = 7200 kls.

5. Since the maximum quantity of water flowing through a pipe is 1.5 times the total quantity of water entering into each pipe, the maximum quantity of water that can flow through pipe B = 1.5 × 120 = 180.
Quantity of net flow of water flowing through pipe B = 120 + 80 – 40 + 60 + x = 180
⇒ x = 40

6. Since point P_1 is closed 80 units of water does not flow into pipe B from pipe A. Similarly, no water flows into pipe C from pipe B from Q_2. The quantity of water flowing through C at the point C_1 = 240 + (40 – 50) + (–40) + (50 – 20) + (10 – 10) = 220 units of water per unit of time.

7. The maximum flow at C_1 = 240 × 15
= 360 units/unit time
∴ The maximum flow between r_3 and r_4
= 360 – (10 – 10) = 360
The maximum flow between r_2 and r_3
= 360 – (50 – 20) = 330
∴ The maximum flow between X and Y can be 330 units per unit of time.

8. Quantity of water flown through point A_1
= 300 – (80 + 60 + 40 + 20) = 100 units.
Through B_1
= (120 + 80 – 40 + 60 – 40 + 40 – 50 + 20 – 10) = 180
Through C_1
= 240 + 40 – 50 + 40 – 40 + 50 – 20 + 10 – 10 = 260
Through D_1
= 90 + 50 + 40 + 20 + 10 = 210

Solutions for questions 9 to 12: As the minimum requirement at D is at least 100 kls, the value of p is 1000 kls. Similarly, as the value of 'b' lies between 100 and 600, 'q' can be from 500 to 0 kls.
Similarly, r = 400 + q and
F = r + 200 (as s = 500 – t = 500 – 300 = 200)

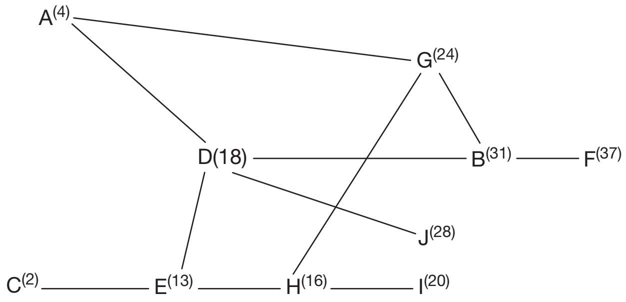
9. The difference between 'f' and 'r' is 200 kls.

10. The least possible value of 'f' would be when q = 0 and r = 400.
∴ 'f' = 400 + 400 – 200 = 600 kls

11. As the value of 'p' is 1000 kls and the value of 's' is 200 kls, $p - s$ would be 1000 – 200 = 800 kls

12. The slack would be minimum when the requirement at B is minimum so that requirement at F would be maximum. So also, water should reach the subsidiary tanks after passing through maximum number of pipelines.
The total capacity = 11 × 1000 = 11,000 kls
The maximum flow ((100 + 600 + 100) × 1 + 300 (E) × 2 + 400(F) × 2 + 700 (F) × 3 + 200 (G) × 3 + 300 (G) × 4)
= 800 + 600 + 800 + 2100 + 600 + 1200
= 6100 kls
The slack = 11000 – 6100 = 4900 kls

Solutions for questions 13 to 16: The order of doing the tasks can be diagrammatically represented as follows.



13. One person can do tasks A, D, J, and I while the other person can do C – E – H – G – B – F.
The first person will finish his task in 32 hours while the second person will finish his task in 37 hours.

14. The shortest time would be 37 hours as in the diagram given above.

15. Task I can be completed in 20 hours and task J can be completed in 45 hours. The maximum time gap is 25 hours.

16. If there are two persons, all the tasks can be completed in 37 hours. If it is only one person, it will take him 4 + 7 + 2 + 5 + 11 + 6 + 8 + 3 + 4 + 10 = 60 hours
The difference 60 – 37 = 23 hours.

17. It is given that outsourcing job's contribution is 14% of the total revenue of the company. As the billing rate for other jobs, $19 is double that for outsourcing jobs, the volume of outsourcing jobs should be nearly double the percentage revenue contribution, i.e., 25%.

18. Assume that the number of hours of outsourcing work done by each company is 1. The outsourcing revenue and total revenue of the six companies would be as follows:

Company	Outsourcing revenue	Total revenue
S	5.5	61
T	9.5	68
U	4.5	27
V	14.5	181
W	17	85
X	7.5	30

Company V would have the highest total revenue.

19. As outsourcing revenue is 20% of the total revenue of company W, a 20% reduction in this would lead to a 4% reduction in the total revenue of the company.

20. Let us assume that the number of hours spent on doing outsourcing jobs by companies U and V is 2 and 3, respectively.
$\therefore$Their outsourcing and total revenue would be as follows:

Company	Outsourcing revenue	Total revenue
U	9	53
V	43.5	544

$\therefore$The ratio of their total revenue is approximately 1 : 10.

8 Reasoning – Based DI

CHAPTER

Learning Objectives

In this chapter, you will:

- Learn how to solve questions involving both DI and LR
- Learn how to solve unconventional sets, venn-diagram based sets
- Learn maximisation and minimisation techniques
- Learn how to represent data in a systematic form, especially when it is fragmented and in bits and pieces

❐ INTRODUCTION

In the past few years especially CAT 15, 16 and 17, the difficulty level of the DI section had increased drastically. One of the noticeable characteristic of these papers was that most of the questions were a combination of different types. A major chunk of the papers were reasoning-based DI sets where concepts of logical reasoning and data interpretation were combined. This chapter gives a good exposure to a variety of reasoning-based DI sets.

SOLVED EXAMPLES

Directions for questions 8.01 to 8.05: These questions are based on the following information.

The following table shows the number of flights expected to fly between various cities in the month of Jan 2016.

From City	To City A	B	C	D	E	F	Total
A	–	680	450			240	1970
B	380	–		480		640	2430
C	420		–		720		1840
D		560	280	–			1960
E	680		440		–	320	
F	320			560		–	
Total	1860		1640		1520	2420	10,000

At least 10 flights are expected to travel from each city to any other city in Jan 2016.

City from	City to						
	A	B	C	D	E	F	Total
A	–	680	450	a	b	240	1970
B	380	–	c	480	d	640	2430
C	420	E	–	f	720	g	1840
D	h	560	280	–	i	j	1960
E	680	k	440	l	–	320	n
F	320	v	p	560	q	–	r
Total	1860	s	1640	t	1520	2420	10,000

$a + b = 1970 – (680 + 450 + 240) = 600$
$c + d = 2430 – (380 + 480 + 640) = 930$
$e + f + g = 1840 – (420 + 720) = 700$
$h + i + j = 1960 – (560 + 280) = 1,120$
$k + l = n – (680 + 440 + 320) = n – 1440$
$v + p + q = r – (320 + 560) = r – 880$
$h = 1860 – (380 + 420 + 680 + 320) = 60$
$e + k + v = s – (680 + 560) = s – 1240$
$c + p = 1640 – (450 + 280 + 440) = 470$
$a + f + l = t – (480 +560) = t – 1040$
$b + d + i + q = 1520 – 720 = 800$
$g + j + m = 2420 – (240 + 640) = 1,540$
$n + r = 10,000 – (1970 + 2430 + 1840 + 1960) = 1800$

8.01: What is the maximum number of flights expected to fly from city C to city D?
(A) 640 (B) 680
(C) 720 (D) 760

Sol: Maximum number of flights expected to fly from city C to city D
$= f_{max} = (700 – e_{min} – g_{min}) = 700 – 10 – 10 = 680$

8.02: What is the maximum number of flights expected to fly from city B to city E?
(A) 920 (B) 940
(C) 960 (D) 980

Sol: The maximum number of flights expected to fly from city B to city E = $d_{max} = 930 – C_{min}$
$= 930 – 10 = 920$

8.03: What is the minimum number of flights that are expected to land in city D?
(A) 920 (B) 970
(C) 1020 (D) 1070

Sol: The minimum number of flights that are expected to land in city D
$= t_{min} = (a_{min} + 480 + f_{min} + l_{min} + 560)$
$= 10 + 480 + 10 + 10 + 560 = 1070$

8.04: If it is planned that 320 flights will leave from city E to city C, then what could be the minimum number of flights that would leave from city E?
(A) 1380 (B) 1420
(C) 1460 (D) 1500

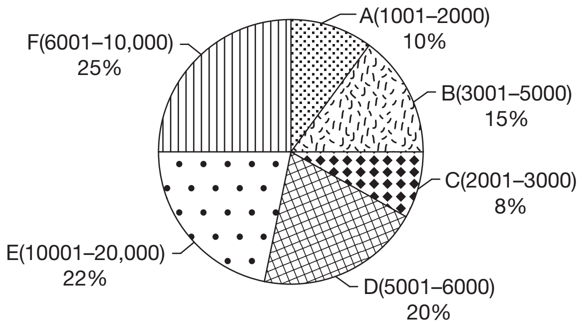
Sol: The minimum number of flights that could leave from city E = (680 + 10 + 440 + 320 + 10) = 1460

8.05: What could be the ratio of the minimum number of flights that will go to city B to the minimum number of flights that will leave from city F?
(A) 97 : 61 (B) 107 : 71
(C) 117 : 81 (D) 127 : 91

Sol: The minimum number of flights that will go to city B = S_{min} = 680 + 10 + 560 + 10 + 10 = 1270.
Minimum number of flights that will leave from city F = 320 + 10 + 10 + 560 + 10 = 910.
∴ The required ratio = 1270 : 910 = 127 : 91.

EXERCISE-1

Directions for questions 1 to 5: These questions are based on the following pie chart which shows the population-wise categorization of villages of a district into six groups A, B, C, D, E and F.



1. What is the ratio of the number of villages having a population in the range of 3001-5000 to the number of villages having a population in the range of 6001 – 10,000?

2. By what per cent is the number of villages having a population of at least 1001 but at most 3000 more than the number of villages having a population of more than 3000 but at most 5000?

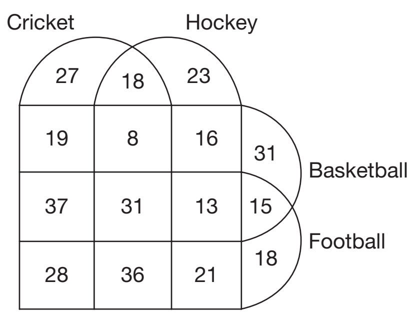
3. If the number of villages having a population in the range of 5001 to 6000 is 60, then what is the total number of villages having a population of more than 6000?

4. If the total number of villages in the district is 500, then how many villages have a population of at least 3001?

5. If the total population of all the villages in group-B is 45,000, then what is the minimum number of villages in group-B?

Directions for questions 6 to 10: Answer the questions based on the information given below.

The given figure provides the details of the number of students in a school who play any of the four games, such as cricket, basketball, football and hockey.



The total number of students in the class is 500.

6. How many of the students play at most one game?

(A) 99 (B) 60
(C) 159 (D) 258

7. The number of students who play either cricket or basketball but not football is

(A) 142 (B) 119
(C) 215 (D) None of these

8. How many students play at least three games?

(A) 125 (B) 57
(C) 98 (D) 112

9. The number of students playing at most one game exceeds those playing at least two games by

(A) 8 (B) 16
(C) 38 (D) None of these

10. How many students do not play Cricket, Football or Hockey?

(A) 178 (B) 190
(C) 196 (D) None of these

Directions for questions 11 to 13: Answer these questions based on the information given below.

These questions are based on the following table which gives the percentage distribution of the cars purchased in a city for the year 2006. The first table gives the company-wise distribution of the cars purchased for each category and the second table gives the category-wise distribution of cars for each company.

Company-wise distribution for each category

	Small	Economy	Mid size	Comfort	Luxury
Maruti	20%	10%	7.5%	25%	5%
Hyundai	16%	12%	22.5%	7.5%	35%
Ford	30%	40%	15%	7.5%	10%
Fiat	24%	18%	45%	22.5%	30%
Toyota	10%	20%	10%	37.5%	20%

Category-wise distribution for each company

	Maruti	Hyundai	Ford	Fiat	Toyota
Small	45%	24%	45%	24%	18%
Economy	7.5%	6%	20%	6%	12%
Mid-size	15%	30%	20%	40%	16%
Comfort	25%	5%	5%	10%	30%
Luxury	7.5%	35%	10%	20%	24%

11. What is the ratio of the number of luxury cars purchased to the number of Fiat cars purchased?
(A) 3 : 2 (B) 2 : 3
(C) 4 : 5 (D) 5 : 4

12. Of the total cars purchased in 2006, if the number of Luxury Fords were 150, then how many cars were either Hyundai or economy cars?
(A) 1480 (B) 1750
(C) 2250 (D) 2160

13. Of the Toyota cars purchased, the number of small cars is 10 more than the number of mid-size cars. How many cars were purchased in the city in 2006?
(A) 1000 (B) 12,000
(C) 3000 (D) Cannot be determined

Directions for questions 14 to 18: These questions are based on the following table, which gives the distribution of marks of 160 students in five subjects. The maximum marks in each subject is 100.

	Marks			
Subject	**70 and above**	**60 and above**	**50 and above**	**40 and above**
Maths	83	108	127	143
Marathi	91	113	139	151
Social	103	104	131	149
Statistics	108	139	149	156
English	93	105	117	139

14. How many students scored 50 marks or more but less than 60 marks in Social?
(A) 29 (B) 27
(C) 131 (D) 104

15. How many students scored less than 50% marks in English?
(A) 32 (B) 43
(C) 67 (D) 117

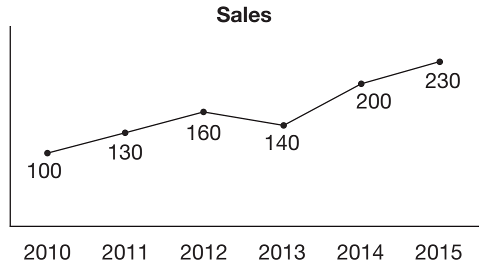
16. In which of the following subjects is the number of students who scored 50 or more but less than 70, the highest?
(A) English (B) Maths
(C) Social (D) Marathi

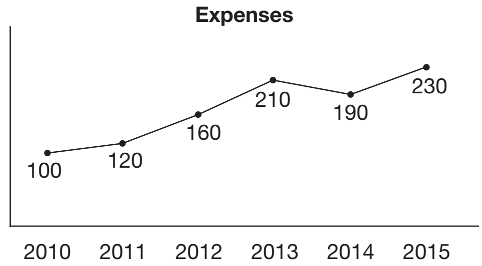
17. The number of students who scored more than 60 marks in all the given subjects is
(A) 104 (B) 105
(C) 113 (D) Cannot be determined

18. How many students scored less than 60 marks in Marathi?
(A) 56 (B) 68
(C) 108 (D) 47

Directions for questions 19 to 22: Answer the questions based on the information given.

The graphs give the trends of sales and expenses of ABC Corporation for the years 2010 to 2015. Both sales and expenses of the year 2010 are indexed to 100 and there was a profit in each of the given years.





Profit = Sales – Expenses

$$\text{Profitability } (\%) = \frac{\text{Profit}}{\text{Sales}} \times 100$$

19. At least in how many of the given years did all of sales, expenses and profit increase or decrease in unison?
(A) 2 (B) 3
(C) 1 (D) 4

20. At least in how many of the given years did the profits of ABC Corporation increase, when compared to the previous year?
(A) 4 (B) 3
(C) 2 (D) 1

21. If profitability in the year 2012 was 50%, then what was the profitability in the year 2014?
(A) 47% (B) 52.5%
(C) 60% (D) Cannot be determined

22. In which of the following years did ABC Corporation make the highest profit?
(A) 2011 (B) 2012
(C) 2015 (D) Cannot be determined

Directions for questions 23 to 27: These questions are based on the table below which gives the distribution of revenues generated by movies in different languages.

	Collection (₹)				
Language	**₹ 100 cr and above**	**₹ 80 cr and above**	**₹ 60 cr and above**	**₹ 40 cr and above**	**₹ 20 cr and above**
Hindi	63	88	107	123	145
Tamil	71	93	119	131	152
Telugu	83	84	111	129	143
Kannada	88	119	129	136	157
English	73	85	97	119	132

23. How many Tamil movies collected ₹ 40 crore or more but less than ₹ 80 cr?
(A) 28 (B) 36
(C) 38 (D) 46

24. How many English movies collected less than ₹ 40 crore?
(A) 21 (B) 13
(C) 11 (D) Cannot be determined

25. In which language is the number of movies which collected ₹ 60 crore or more but less than ₹ 100 crore, the highest?
(A) Hindi (B) Tamil
(C) Telugu (D) English

26. If each movie was released in only one language, then the number of movies which collected ₹ 60 crore or more but less than ₹ 100 crore in any of the given languages is
(A) 88 (B) 142
(C) 163 (D) None of these

27. If 170 movies were released in Hindi during the given period and no movie collected more than ₹ 140 crore, then the total collection (in ₹) of all the Hindi movies cannot be more than
(A) ₹ 11160 crore (B) ₹ 12730 crore
(C) ₹ 14180 crore (D) ₹ 15180 crore

Directions for questions 28 to 31: These questions are based on the following information.

The following table gives the number of students in Class I to IV in a school in two consecutive academic years, such as in year I and year II. New students join the school only in Class I and students leave the school only after passing out of Class IV. Each year, students who pass the annual exams are promoted to the next class while students who fail, have to stay in the same class the next year also and are joined by students who get promoted. It is known that 35 students passed out of Class IV at the end of year I.

Class	Year I	Year II
I	36	34
II	42	38
III	31	39
IV	38	32

28. How many students joined the school in year II?
(A) 23 (B) 26
(C) 29 (D) 31

29. How many students failed in Class I in year I?
(A) 1 (B) 2
(C) 3 (D) 4

30. How many students were promoted from Class III at the end of year I?
(A) 21 (B) 29
(C) 30 (D) 31

31. How many students in the school failed in the annual exams in year I?
(A) 5 (B) 8
(C) 10 (D) None of these

Directions for questions 32 to 35: These questions are based on the following information.

The table gives the partial data on the expected number of emails to be sent from one email account to another email account (in billion) in the year 2020.

Expected number of emails to be sent and received (in billions).

Server	Yahoo	Mailcity	Hotmail	Msn	Eudora	Rediff	Total sent
Yahoo	180	200				115	1182
Mailcity	100		137				1784
Hotmail		300		317			2074
Msn			386		198		
Eudora	85				372		
Rediff		215		273			1818
Total received	2183				1800	1000	10,000

It is expected that in the year 2020, any email account (out of the six given) will receive at least 1 billion emails from each of the six email accounts. Also, any email account will send at least 1 billion emails to each of the six email accounts.

32. Find the maximum number of emails (in billion) which is expected to be sent from Hotmail to Hotmail in the year 2020?

(A) 1454 (B) 1457
(C) 1400 (D) 1386

33. Find the minimum number of emails (in billion) expected to be sent from either msn or Hotmail to either Eudora or Msn.

(A) 515 (B) 520
(C) 517 (D) 508

34. Find the maximum number of emails (in billion) which is expected to be sent from Hotmail to yahoo in the year 2020.

(A) 1816 (B) 1454
(C) 1284 (D) 1586

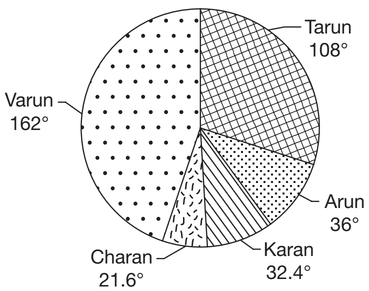
35. Find the maximum number of emails expected to be sent from any email account to any other email account.

(A) 2072 (B) 3805
(C) 2554 (D) None of these

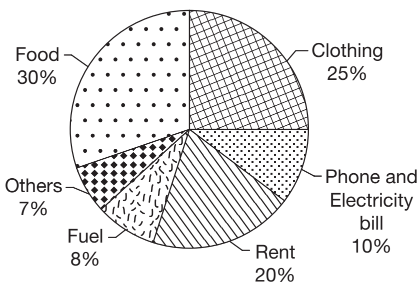
Directions for questions 36 to 40: Answer the questions based on the following information.

The following pie charts give the break-up of the income of all the five members, namely Varun, Tarun, Arvind, Karan and Charan of family XYZ and the break-up of the total family expenditure under different heads.

Split of total income of family XYZ



Split of total expenditure of family XYZ



Note: The total income of the family is equal to the total expenditure and the family has no other sources of income.

36. If Varun did not pay for 'Others', then his income can fully account for expenses under at most how many heads?

(A) 2 (B) 3
(C) 4 (D) 5

37. Whenever possible, if all the expenses under one head are paid for by a single person, then the number of heads under which more than one person shared the expenses is at least

(A) 1 (B) 2
(C) 3 (D) 4

38. If Varun does not spend any amount on food, then the expenditure of Varun on clothing and rent as a percentage of the total expenditure on rent and clothing cannot be less than

(A) 33.33% (B) 44.44%
(C) 25% (D) 66.66%

39. If at most 40% of the income of each person is paid for food, then the number of persons who did not pay for food is at most

(A) 1 (B) 2
(C) 3 (D) 4

40. If at least 5% of the total expenses under each head are paid from Karan's income, then the percentage share of Karan's payment under any head can be at most

(A) 22.5% (B) 90%
(C) 62.14% (D) $66^2/_3$%

Directions for questions 41 to 45: Answer the questions based on the information given below:

Prof. Bean has been tracking the number of visits to his homepage. His service provider has provided him with the following distribution of the number of visits as per the country and the university from which the visits were made. The data pertains to three days from Day 1, Day 2 and Day 3.

Number of Visits

Country	Day 1	Day 2	Day 3
China	1	0	0
Philippines	1	1	0
UK	1	0	2
USA	1	2	0
Germany	0	0	1

Number of Visits

University	Day 1	2	3
A	1	1	0
B	1	0	0
C	1	0	0
D	0	0	1
E	0	2	0
F	1	0	0
G	0	0	2

Note: Each university is in only one country.

41. In which country is University E located?
(A) USA
(B) UK or Philippines
(C) China or USA but not UK
(D) Germany or USA but not Philippines

42. In which country is University G located?
(A) UK (B) Germany
(C) China (D) USA

43. In which of the five countries mentioned, are three of the seven universities mentioned possibly located?
(A) USA (B) UK
(C) Philippines (D) None of the countries

44. Which of the following universities is not located in China?
(A) C (B) B
(C) A (D) F

45. Which of the following universities is in Philippines?
(A) A (B) B
(C) C (D) D

Directions for questions 46 to 50: Answer these questions based on the information given below.

The table gives the number of passengers getting in and out of a bus which travels from City A (starting point) to City B (destination) with four stops from Stop 1, Stop 2, Stop 3 and Stop 4 in between. The first set of passengers boarded the bus at City A and all the passengers who were in the bus got down at City B. Each passenger travelled at least from one stop to the next and no passenger who got down the bus at any stop got in again.

Stop	Number of passengers: Getting in	Getting out
City A	14	
Stop 1	12	5
Stop 2	7	10
Stop 3	10	12
Stop 4	8	11
City B		

46. How many passengers got down at City B?
(A) 8 (B) 12
(C) 13 (D) 15

47. At most how many passengers who got in at City A got down at Stop 2?
(A) 10 (B) 9
(C) 8 (D) 7

48. At least how many passengers who got in at Stop 3 got down at B?
(A) 0 (B) 1
(C) 2 (D) 3

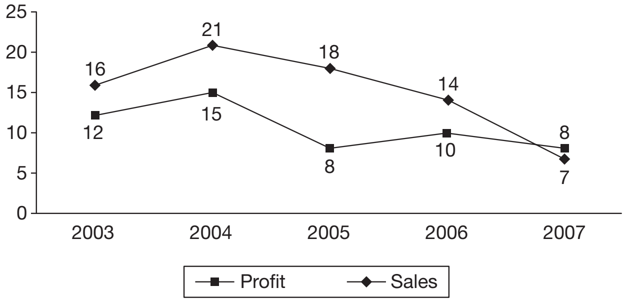
49. The number of passengers who got in at Stop 1 and got down at Stop 4 is at most
(A) 12 (B) 11
(C) 10 (D) 6

50. What is the maximum number of passengers who got down at exactly the third stop from where they got in?
(A) 10 (B) 11
(C) 12 (D) 15

EXERCISE-2

Directions for questions 1 to 3: These questions are based on the following information.

The following line graph shows the percentage increase in sales and profit, both when compared to that of the previous year, of company XYZ for five years starting from 2003.



Note: The company made a profit in each of the given years.

1. If the sales in the year 2003 were ₹120 crore, then what was the approximate value of the sales in 2006 (in crore)?
   (A) 162 (B) 176
   (C) 198 (D) 208

2. In which of the given years were the increase in sales, the highest?
   (A) 2004 (B) 2005
   (C) 2006 (D) Cannot be determined

3. If profitability is defined as $\frac{\text{profit}}{\text{sales}}$, then in which year was the profitability the least?
   (A) 2004 (B) 2005
   (C) 2006 (D) 2007

Directions for questions 4 to 7: Answer these questions based on the information given below.

260 students in a school had enrolled for the hobby classes. The students had the option of taking one, two or three hobbies among Karate, Skating, Abacus, Cricket, Chess, Painting, Music and Dance. The following table gives the number of students who enrolled for each of the eight hobbies.

Hobby	Number of students
Karate	72
Skating	46
Abacus	83
Cricket	41
Chess	65
Painting	32
Music	37
Dance	38

4. Which of the following can be the number of students who opted for exactly one hobby?
   (A) 197
   (B) 184
   (C) 120
   (D) 104

5. What is the minimum number of students who opted for at least two hobbies?

6. What is the maximum number of students who opted for three hobbies?

7. What is the minimum number of students who opted for three hobbies?

Directions for questions 8 to 12: These questions are based on the following information.

The following table gives the number of students who passed in four subjects, namely in Maths, Physics, Chemistry and Biology in the three sections A, B and C in Class X of a school. Each section had a student strength of 40.

Section	Maths	Physics	Chemistry	Biology
A	28	31	39	26
B	34	32	37	33
C	26	34	31	29

8. The number of students in section A who passed in all the four subjects is at most.

9. The number of students in section C who passed in all the four subjects is at least.

10. At most how many students in section B passed in exactly one of the four subjects.

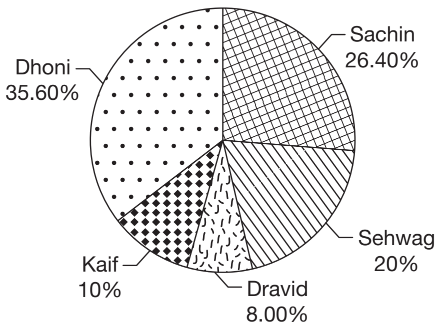
11. The number of students who passed in both Physics and Chemistry in the three sections combined is at most.

12. A student has to pass in all the four subjects to clear the Class X exam. The least number of students in the school who failed to clear the exam is

Directions for questions 13 to 17: Answer the questions based on the information given below.

The following pie chart gives the details of runs scored by five Indian batsmen in a match. Only five players batted during the innings and the table gives the details of the percentage of runs scored in 4s and 6s by these players. All other runs scored by each player were in an equal number of 1s and 2s. India's total score was the lowest number which satisfies all these conditions.

Difficulty Level-2: Moderate



Player	4s	6s
Sachin	36.36%	18.18%
Sehwag	16%	24%
Dravid	40%	0%
Kaif	64%	0%
Dhoni	22.47%	33.70%

13. How many runs did Dravid score in singles (1's)?
(A) 2 (B) 4
(C) 6 (D) 3

14. The player who has made the maximum number of boundaries (4s) is
(A) Sachin (B) Dravid
(C) Kaif (D) Cannot be determined

15. Which of the following is India's minimum possible total score?
(A) 100 (B) 125
(C) 250 (D) Cannot be determined

16. The total number of runs scored in boundaries (4s) and sixers (6s) by the Indian batsmen are
(A) 65 (B) 66
(C) 130 (D) 140

17. How many sixers (6s) did Dhoni score?
(A) 3 (B) 4
(C) 5 (D) Cannot be determined

Directions for questions 18 to 21: These questions are based on the following information.

The table below gives the percentage share of expenses of Mr. Dubey on different items in the years from 2005 to 2008.

Expense type	2005	2006	2007	2008
Rent	14	12	13	12
Food	15	17	15	12
Clothing	10	12	11	14
Entertainment	17	15	15	13
Medical	6	5	8	10
Education	27	29	25	26
Travel	11	10	13	13

18. If the total expenses in 2008 was more than that in 2005, then the expenses under which head showed the highest percentage increase from 2005 to 2008?
(A) Clothing (B) Entertainment
(C) Medical (D) Travel

19. If the expenses under each head in 2007 was more than the corresponding value in 2005, then the percentage increase in total expenses from 2005 to 2007 was at least
(A) 10% (B) 13.33%
(C) 15% (D) 17.5%

20. If the medical expenses in 2008 was 25% more than that in 2006, then the expenses on clothing in 2008 was what percentage of the entertainment expenses in 2006?
(A) 37.5% (B) 42.8%
(C) 52.5% (D) 58.3%

21. If the expenses on rent increased by 10% every year from 2005 to 2008, by what percentage did the expenses on food increase from 2005 to 2008?
(A) 18% (B) 24.3%
(C) 27.5% (D) 32%

Directions for questions 22 to 25: These questions are based on the following information.

The following table gives the marks scored by four students, namely Anand, Balu, Chetan and Deepak in the three areas, such as Verbal, Quant and Reasoning of a mock CAT paper. The four students are disguised in the tables as A, B, C and D in no particular order.

Section	Student A	B	C	D
Verbal	24	41	40	27
Quant	34	36	35	32
Reasoning	36	31	36	32

It is also known that, in reasoning, none of the other three students scored more than Chetan.

Balu's total score in the three sections differs from that of Anand's by 3 marks.

22. What can be said regarding the following two statements?
Statement I: Deepak scored the lowest marks in the reasoning section.

Statement II: Anand's total score in the three sections is more than that of Deepak.
(A) If Statement I is true, then Statement II is necessarily true.
(B) If Statement I is true, then Statement II is necessarily false.
(C) Both Statement I and Statement II are true.
(D) Neither Statement I nor Statement II is true.

23. What can be said regarding the following two statements?
Statement I: Balu's lowest score is in the reasoning section.
Statement II: Anand's lowest score is in the quantitative section.
(A) If Statement II is true, then Statement I is necessarily false.
(B) If Statement I is false, then Statement II is necessarily true.
(C) If Statement I is true, then Statement II is necessarily true.
(D) None of the above

24. What can be said regarding the following two statements?
Statement I: Anand had the highest score in the verbal section.
Statement II: Balu had the highest score in the quant section.
(A) Both the statements could be true.
(B) At least one of the statements must be true.
(C) At most one of the statements must be true.
(D) None of the above

25. If Deepak got his lowest score in the verbal section, then which of the following is true?
(A) Chetan's lowest score is in the reasoning section.
(B) Chetan's lowest score is in the quant section.
(C) Chetan's lowest score is in the verbal section.
(D) No definite conclusion is possible.

Directions for questions 26 to 29: Answer these questions based on the information given below.

The following table represents the number of cars sold (in thousands) by four companies in three countries. The companies Toyo Ltd., Mercedes Ltd., BWM Ltd., and Form Ltd., are disguised as Company A, Company B, Company C and Company D in the table, in no particular order.

Countries	Company A	Company B	Company C	Company D
USA	196	328	320	220
UK	276	288	280	260
Japan	288	252	288	260

Further it is known that, BWM Ltd. was one of the companies that had the highest sales in Japan.

Total cars sold by Mercedes Ltd. in the three countries differs from that of Toyo Ltd. by 20,000.

26. What can be said regarding the following two statements?
Statement 1: Mercedes Ltd. had its lowest sales in Japan.
Statement 2: Toyo Ltd. had its lowest sales in UK.
(A) If statement 2 is true, then statement 1 is necessarily false.
(B) If statement 1 is false, then statement 2 is necessarily true.
(C) If statement 1 is true, then statement 2 is necessarily true.
(D) If statement 1 is false, then statement 2 is necessarily true.

27. If Form Ltd. had its lowest sales in USA, then which of the following is necessarily true?
(A) BWM Ltd. had its lowest sales in Japan.
(B) BWM Ltd. had its lowest sales in UK.
(C) BWM Ltd. had its lowest sales in USA.
(D) BWM Ltd. is company B.

28. Which of the following additional information will help us to uniquely identify each of the four companies?
(a) BWM Ltd. is Company A.
(b) Toyo Ltd. is Company B.
(c) Form Ltd. is Company D.
(d) Mercedes Ltd. is Company C.
(A) Only (a)
(B) Only (b)
(C) Only (d)
(D) More than one of the above.

29. What can be said regarding the following two statements?
Statement 1: Toyo Ltd. had the highest sales in USA.
Statement 2: Mercedes Ltd. had the highest sales in UK.
(A) Both statements could be true.
(B) At least one of the statements must be true.
(C) At most one of the statements is true.
(D) Both statements are false.

Directions for questions 30 to 33: Answer these questions based on the information given below.

The table gives the ratio of the number of boys to the number of girls in different schools, in different cities, for the years 2015 and 2016.

	Jaipur		Mumbai		Pune		Hyderabad	
School	**2015**	**2016**	**2015**	**2016**	**2015**	**2016**	**2015**	**2016**
DPS	3 : 2	4 : 3	2 : 3	2 : 3	7 : 4	7 : 5	7 : 5	3 : 2
FPS	4 : 3	4 : 3	3 : 4	3 : 5	3 : 2	3 : 4	7 : 6	4 : 3
LSP	5 : 2	5 : 3	5 : 4	4 : 5	1 : 2	4 : 5	6 : 5	5 : 2
LPS	7 : 2	7 : 4	7 : 3	7 : 5	3 : 5	7 : 5	8 : 3	7 : 2
PDS	1 : 1	1 : 1	1 : 1	1 : 1	5 : 3	5 : 4	9 : 7	1 : 1
LFS	1 : 2	2 : 3	1 : 2	1 : 2	3 : 4	4 : 5	3 : 2	1 : 2

For any given school, in any year, assume that the number of students in Jaipur was more than that in Mumbai, which in turn was more than that in Pune, and in turn was more than that in Hyderabad.

Also, for any given city, in any year, the number of students followed this pattern n (DPS) > n (FPS) > n (LSP) > n (LPS) > n (PDS) > n (LFS) (Here, n (XYZ) denotes the number of students in the school XYZ in that year).

30. If in 2015, the difference between the number of boys and the number of girls in LSP was 120 and 90 in Jaipur and Pune, respectively, then what was the difference between the number of boys and the number of girls in LSP in Mumbai in that year?

(A) 29 (B) 30
(C) 31 (D) 33

31. If '*boyage*' is defined as the percentage of boys in the total students, then for how many of the given 24 campuses is the value of *boyage* in 2016, more than that in 2015? (Consider each school in each city as a campus.)

(A) 4 (B) 6
(C) 8 (D) 10

32. If the number of girls in LSP, Pune in 2015 is the same as the number of boys in DPS, Mumbai in 2016, then the total number of students in DPS, Mumbai in 2016 is more than the number of students in

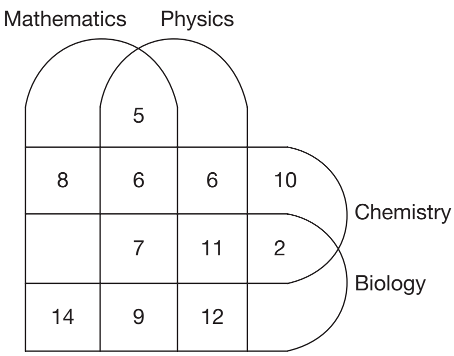
(A) FPS, Hyderabad in 2015.
(B) DPS, Pune in 2015.
(C) LFS, Jaipur in 2016.
(D) LPS, Pune in 2015.

33. In 2015, if the number of boys in DPS, Hyderabad is 315, then what is the maximum possible number of girls in LFS, Hyderabad?

(A) 210 (B) 212
(C) 204 (D) 196

Directions for questions 34 to 37: Answer the questions based on the information given below.

The Venn diagram below shows the number of students who study Mathematics, Physics, Chemistry and Biology. The total number of students studying the given subjects in the given order is 64, 76, 63 and 72, respectively.



34. How many students study only Physics?

(A) 11 (B) 20
(C) 12 (D) 14

35. How many students study only Mathematics?

(A) 1 (B) 2
(C) 3 (D) 4

36. How many students study all the four subjects?

(A) 7 (B) 6
(C) 12 (D) 22

37. How many students study either Physics and Mathematics or Chemistry and Biology?

(A) 67 (B) 99
(C) 53 (D) 45

Directions for questions 38 to 41: Answer these questions based on the information given below.

The table gives some information about the foreign exchange reserves of India for a period of ten years from 1990-91 to 1999-2000. Foreign exchange reserves comprise currency holdings and gold holdings. Currency holdings comprise reserves in three foreign currencies, such as in US Dollar, Pound Sterling and Euro.

The following table gives the prices (in ₹) of the three currencies and the price (in ₹) of gold by considering the year 1990–91 as the base year, in which the price of each of the currencies and the price of gold are taken as 100. The prices of each of these in the following years are given relative to that in the base year.

Holdings	Price with respect to the base year (1990–91)									
	1990–91	1991–92	1992–93	1993–94	1994–95	1995–96	1996–97	1997–98	1998–99	1999–2000
US Dollar	100	96	110	96	98	94	102	92	112	98
Pound Sterling	100	110	102	100	96	100	106	98	116	90
Euro	100	104	106	104	104	106	106	96	108	106
Gold	100	102	104	108	110	107	108	109	112	109

(i) Value (in ₹) of a currency (or gold) holding = Volume of the holding × Price (in ₹) of the holding.

(ii) Volume of a currency (or gold) holding = Number of units of that currency (or gold) held.

(iii) The quantity of each currency and that of gold with India remained constant throughout the given period and India had at least one unit of each of the three currencies and gold with it.

38. If the percentage increase in the total value of the foreign exchange reserves from 1990-91 to 1998-99 is $x\%$, then x cannot be equal to

(A) 10 (B) 12.5

(C) 8 (D) 11

39. If the total value of the currency holdings during 1995–96 was more than that in 1990–91, then what is the maximum possible number of years during which the total value of the currency holdings was less than that in 1990–91?

40. In how many of the given years was the total value of the currency holdings less than that in the year 1992-93?

41. During which of the given years was the total value of the foreign exchange reserves the highest?

Directions for questions 42 to 46: Answer the questions based on the information given below.

A total of six colleges, namely from A, B, C, D, E and F jointly conducted an entrance examination. The exam had four sections-I, II, III and IV. The following table gives the sectional cut off marks specified by the colleges and the overall cut off marks. A student will get a call from a college only if he scores at least the sectional and overall cut off marks specified by that college. The maximum marks in each section is 50.

College	Section				
	I	II	III	IV	Overall
A	40		43		160
B		40	42	45	168
C	41		43		165
D		45		42	170
E	43		45		175
F		43		40	167

42. Anuj did not get a call from any of the colleges. What is the maximum marks he could have scored?

(A) 155 (B) 159

(C) 180 (D) 183

43. Ram got a call from only one college. What is the maximum marks he could have scored?

(A) 160 (B) 164

(C) 177 (D) 185

44. Madhuri got calls from all the six colleges. What is the minimum marks she could have scored?

(A) 175 (B) 178

(C) 179 (D) 180

45. Ravi scored a total of 190 marks. What is the minimum number of colleges from where he could have got calls from?

(A) 1 (B) 2

(C) 3 (D) 4

46. Ravi and Raja scored 175 marks each. What could be the maximum difference in the number of calls they got?

(A) 6 (B) 5

(C) 4 (D) 3

Directions for questions 47 to 50: Answer these questions based on the information given below.

A study on population of eight cities was conducted by the human resources and social welfare department. These eight cities were ranked from 1 to 8 based on social welfare and it was found that for any City X, the number of cities with a population less than it was exactly one less than the rank of City X on the basis of social welfare.

The following table gives the comparison of populations of the eight cities.

	Indore	Pune	Bhopal	Shillong	Agra	Cochin	Patna	Mysore
Indore	X	M	L					L
Pune		X			M			
Bhopal			X			L	L	L
Shillong				X				M
Agra					X			
Cochin						X	L	M
Patna							X	M
Mysore								X

M in the table denotes that the population of that city was more and L in the table denotes that the population of that city was less than the corresponding city. For example, the table shows that the population of Indore was more than that of Pune and less than that of Bhopal. It was also known that Indore was 3rd and Shillong was not among the top 5 in the rakings based on social welfare.

47. Which city was ranked first?
(A) Pune (B) Agra
(C) Bhopal (D) Mysore

48. How many cities have less population than Mysore?
(A) 3 (B) 2
(C) 4 (D) 7

49. What was the rank of Shillong based on Social Welfare?
(A) 7
(B) 6
(C) 5
(D) Cannot be determined

50. The cities are ranked again based on the population such that the city with the highest population is ranked 1st that with the second highest population ranked 2nd and so on. Which city would have its rank and the number of cities with a population less than it, as equal?
(A) Shillong (B) Mysore
(C) Bhopal (D) Cochin

EXERCISE-3

Directions for questions 1 to 4: Answer these questions on the basis of the information given below.

Table 8.1 shows the number of ships that arrived at Mumbai port on different days of the week from October 9th to 15th (i.e., from Sunday to Saturday).

Table 8.1

Day	9 Oct Sunday	10 Oct Monday	11 Oct Tuesday	12 Oct Wednesday	13 Oct Thursday	14 Oct Friday	15 Oct Saturday
Number of ships arrived	28	47	40	45	40	35	25

Each of the ships mentioned in Table 8.1 departs from the port in the next week, starting from October 16 to October 22 (i.e., Sunday to Saturday). Table 8.2 shows the number of ships that departed from the port on different days.

Table 8.2

Day	16 Oct Sunday	17 Oct Monday	18 Oct Tuesday	19 Oct Wednesday	20 Oct Thursday	21 Oct Friday	22 Oct Saturday
Number of ships departed	37	43	50	45	35	30	20

Further, no ship arriving at the port can depart from the port on or before the 5th day after the day on which it arrived. Also, no ship can remain at the port after the 10th day after the day on which it arrived. For example, a ship which arrived on Wednesday cannot depart on or before the next Monday, but it must definitely depart on or before the next Saturday.

1. If, of the ships that arrived on Monday, October 10th, 22 ships departed on the next Sunday, then the number of ships that arrived on Sunday, October 9th and departed after the next Sunday is

(A) 13 (B) 15
(C) 22 (D) 24

2. If in the above table, all the ships that arrived on or before Tuesday, left on or before next Tuesday, then the number of ships that arrived on Wednesday and departed on the next Tuesday is

(A) 10 (B) 15
(C) 20 (D) 25

3. If 20 ships that arrived on Wednesday departed on Friday, the number of ships that arrived on Friday and departed on Thursday is at least

(A) 5 (B) 10
(C) 25 (D) 30

4. The number of ships that arrived on Tuesday and departed on Monday is at least

(A) 0 (B) 3
(C) 5 (D) 10

Directions for questions 5 to 8: Answer these questions on the basis of the information given below.

Six students of a class wrote Physics and Chemistry exams. Each exam had nine questions and, in each exam, marks are given based on the number of questions attempted correctly as follows:

Number of correct attempts (n)	Marks
0, 1	$n \times 1$
2, 3, 4	$n \times 2$
5, 6, 7	$n \times 3$
8, 9	$n \times 4$

Note: Assume that no marks are given for wrong answers or unanswered questions.

Further, the results of the exams are as follows:

Student	Number of questions answered correctly: Physics	Chemistry	Total	Total marks
A				
B			7	
C			13	
D				24
E				39
F				51
Total	36	30	66	

It is also known that,

(1) The number of questions answered correctly by D in Physics is the same as that of E in Chemistry.

(2) The number of questions answered correctly by F in Physics and Chemistry are equal to the number of questions answered correctly by B and C in Chemistry, not necessarily in the same order.

5. The total marks scored by A is

(A) 21 (B) 25
(C) 29 (D) 36

6. Who scored the second lowest total marks?

(A) A (B) B
(C) C (D) D

7. In Physics, how many students scored more marks than E?

(A) 0 (B) 1
(C) 2 (D) 3

8. How many students scored more marks in Physics than in Chemistry?

(A) 4 (B) 3
(C) 2 (D) 5

Directions for questions 9 to 13: Answer these questions on the basis of the information given below.

The following table gives the details of the number of mock CATs conducted by different institutes in 2004 and the number of these mock CATs written by different students.

Student	Institute P (30)	Q (40)	R (32)	S (24)	T (26)
Akshay	12	21	23	10	22
Bobby	16	20	16	20	11
Chahat	18	33	17	15	8
Daram	14	16	28	16	13
Emran	21	18	18	9	15
Feroz	16	21	15	12	10
Govinda	10	30	20	11	20
Hrithik	20	22	19	17	19

The number given in the brackets is the total number of mock CATs conducted by the respective institutes in 2004.

9. The number of students who wrote at least one mock CAT of each institute in common with Feroz is at least

(A) 0 (B) 1
(C) 2 (D) 3

10. Among the total mock CATs held, the number of mock CATs written by exactly one of Akshay and Hrithik is at least

(A) 18 (B) 20
(C) 23 (D) 26

11. Of the mock CATs conducted by institute R, the number of mock CATs which were written by more than one among Bobby, Emran and Govinda is at least
(A) 8 (B) 11
(C) 13 (D) 15

12. If Daram wrote all the mock CATs which were written by neither Chahat nor Feroz, then the number of mock CATs conducted by institute S and written by Daram, Chahat and Feroz is at most
(A) 6 (B) 7
(C) 8 (D) 9

13. Of the mock CATs conducted by institute Q, the number of common mock CATs written is the highest for
(A) Feroz and Akshay
(B) Hrithik and Chahat
(C) Chahat and Govinda
(D) Daram and Emran

Directions for questions 14 to 17: Answer these questions on the basis of the information given below.

Intra-State Migration Trends in India, 1991

Migration stream	X	Y
$R \rightarrow R$	49·67%	76.71%
$R \rightarrow U$	27·27%	11·95%
$U \rightarrow U$	15·38%	7.04%
$U \rightarrow R$	7·68%	4·3%

R = Rural; U = Urban

X = Male migrants in the stream as a percentage of total intra-state male migrants.
Y = Female migrants in the stream as a percentage of total intra-state female migrants.

	Male	Female
% of total intra-state migrants	24.89% (47.04)	75.11% (141.96)

Figures in brackets show absolute number of migrants in millions.

14. Which of the four migration streams mentioned alone has the highest gender ratio? Gender ratio of a stream is defined as the ratio of the number of male migrants to the number of female migrants in that stream.
(A) $R \rightarrow U$ (B) $R \rightarrow R$
(C) $U \rightarrow R$ (D) $U \rightarrow U$

15. Male migrants from urban to urban form what percentage of the female migrants from rural to urban?
(A) 37.8% (B) 42.6%
(C) 48.7% (D) 53.7%

16. What is the percentage of male migrants from urban to rural out of the total migrant population?
(A) 1.9% (B) 2.1%
(C) 2.3% (D) 2.5%

17. Total migrants from rural to rural areas form what percentage of total migrants in all the streams together?
(A) 55.62% (B) 54.89%
(C) 69.97% (D) 73.27%

Directions for questions 18 to 21: Answer these questions on the basis of the information given below.

Six friends, who are from six different cities, were asked about the cities to which each of them and their friends belong. Their replies were as follows.

	Bangalore	Chennai	Delhi	Hyderabad	Kolkata	Mumbai
Aman	Emma	Biswa	Dev	Aman	Charan	Fazal
Biswa	Aman	Fazal	Biswa	Emma	Charan	Dev
Charan	Emma	Fazal	Dev	Biswa	Aman	Charan
Dev	Charan	Biswa	Fazal	Dev	Aman	Emma
Emma	Emma	Biswa	Dev	Charan	Aman	Fazal
Fazal	Biswa	Dev	Fazal	Charan	Emma	Aman

It is known that no two persons gave an equal number of true replies, and that they all belong to a city from among, Bangalore, Chennai, Delhi, Hyderabd, Kolkata and Mumbai and no two persons belong to the same city.

18. Which of the following persons gave the highest number of true replies?
(A) Emma (B) Biswa
(C) Charan (D) Dev

19. The person who belong to Hyderabad is
(A) Aman (B) Biswa
(C) Charan (D) Dev

20. How many persons gave more true replies than Biswa?
(A) 1 (B) 2 (C) 3 (D) 5

21. How many persons gave his/her city name correctly?
(A) 0 (B) 1 (C) 2 (D) 3

Directions for questions 22 to 25: Answer these questions on the basis of the information given below.

A group of four experts, namely Anand, Babu, Charan and David were asked to rate three features—expressions, dialogue delivery and body language-of two artists, such as A_1 and A_2.

Table 8.3 gives the minimum, average and maximum rating given by the four experts on a scale of (0 to 10) where 0, 1, ... are integers.

	Expressions	Dialogue delivery	Body language
A_1	(5, 7.75, 10)	(6, 7.25, 8)	(6,7.5,9)
A_2	(5, 6.75, 8)	(2, 4, 8)	(4, 5.5, 7)

Table 8.4 gives the minimum and maximum rating across the three features for each expert artist combination.

	A_1	A_2
Anand	(5, 9)	(7, 8)
Babu	(6, 8)	(3, 6)
Charan	(6, 10)	(2, 7)
David	(8, 9)	(3, 8)

Table 8.5 given the average rating by experts features separately with average being compared across artists.

Expert	Expressions	Dialogue delivery	Body Language
Anand	6	7.5	8
Babu	6	5.5	6
Charan	8.5	4	6
David	8.5	5.5	6

22. The rating given by Babu for 'Expressions' for A_1 is
(A) 5 (B) 6
(C) 7 (D) 8

23. The rating given by Charan for 'Body language' for A_2 is
(A) 4 (B) 5
(C) 6 (D) 7

24. The rating given by Charan for 'Expressions' for A_2 is
(A) 7 (B) 6
(C) 8 (D) 9

25. The rating given by Anand for 'Dialogue delivery' for A_2 is
(A) 5 (B) 6
(C) 7 (D) 8

ANSWER KEYS

Exercise-1

1. 3:5	10. (B)	19. (B)	28. (D)	37. (A)	46. (C)
2. 20	11. (B)	20. (A)	29. (C)	38. (B)	47. (B)
3. 141	12. (D)	21. (B)	30. (B)	39. (C)	48. (A)
4. 410	13. (C)	22. (C)	31. (D)	40. (C)	49. (D)
5. 9	14. (B)	23. (C)	32. (A)	41. (A)	50. (D)
6. (D)	15. (B)	24. (D)	33. (C)	42. (A)	
7. (B)	16. (D)	25. (B)	34. (B)	43. (D)	
8. (A)	17. (D)	26. (D)	35. (D)	44. (C)	
9. (B)	18. (D)	27. (D)	36. (B)	45. (A)	

Exercise-2

1. (C)	10. 8	19. (B)	28. (D)	37. (C)	46. (B)
2. (B)	11. 94	20. (D)	29. (C)	38. (C)	47. (B)
3. (C)	12. 36	21. (B)	30. (C)	39. 3	48. (C)
4. (C)	13. (B)	22. (B)	31. (C)	40. 6	49. (D)
5. 77	14. (A)	23. (C)	32. (D)	41. 1998–99	50. (B)
6. 77	15. (C)	24. (C)	33. (C)	42. (D)	
7. 0	16. (C)	25. (C)	34. (B)	43. (D)	
8. 26	17. (C)	26. (C)	35. (B)	44. (B)	
9. 0	18. (C)	27. (C)	36. (A)	45. (B)	

Exercise-3

1. (A)	5. (D)	9. (B)	13. (C)	17. (C)	21. (B)	24. (A)
2. (B)	6. (C)	10. (C)	14. (A)	18. (A)	22. (C)	25. (D)
3. (D)	7. (A)	11. (B)	15. (B)	19. (C)	23. (B)	
4. (B)	8. (A)	12. (D)	16. (A)	20. (D)		

SOLUTIONS

EXERCISE-1

1. The ratio of the number of villages that have a population in the range of 3001 – 5000, to the total number of villages having a population in the range of 6001 to 10,000 = 15:25 = 3:5.

2. The percentage of villages having a population in the range of 1000 – 3000 = 10 + 8 = 18%.
The percentage of villages having a population in the range of 3001 – 5000 = 15%.
The required percentage increase = $\frac{3}{15} \times 100 = 20\%$.

3. The percentage of villages having a population in the range of 5001 – 6000 = 20%
The percentage of villages having a population in the range of (6001 – 20,000) = 25 + 22 = 47%.
Given that 20% = 60
$\therefore$ 47% = $\frac{47}{20} \times 60 = 141$

4. The percentage of villages having a population of at least 3000 = 15 + 20 + 25 + 22 = 82%.
The number of villages having a population of at least 3000 = $\frac{82}{100} \times 500 = 410$

5. Since the number of villages in group-B has to be the minimum, the population of each village has to be the greatest.
$\therefore$ Each village should have a population of 5000 people.
The number of villages = $\frac{45000}{5000} = 9$

6. The number of students playing at least one game = 341.
$\therefore$ The number of students playing none of the games = 159
The number of students playing exactly one game
= 27 + 23 + 31 + 18 = 99
$\therefore$The number of students playing at most one game = 159 + 99 = 258

7. The number of students who play either cricket or basketball but not football = 27 + 18 + 19 + 8 + 16 + 31 = 119.

8. Students playing at least three games = Students playing exactly three + Students playing exactly four.
= 8 + 13 + 36 + 31 + 37 = 125

9. The number of students playing at most one game = 258.
The number of students playing at least two games
= 500 – 258 = 242.
Difference = 16

10. The number of students who do not play Cricket, Football or Hockey is 31 + 159 = 190

11. We must find the ratio of luxury cars to Fiat cars.
Luxury Fiat cars from the 1st table is 30%.
From the second table, it is 20% of total Fiat cars.
Now,
30% of Luxury = 20% of Fiat

$$\frac{\text{Luxury}}{\text{Fiat}} = \frac{20\%}{30\%} = \frac{2}{3}$$

Similarly,
Small : Maruti = 45 : 20 = 9 : 4
Small : Hyundai = 24 : 16 = 3 : 2
Small : Ford = 45 : 30 = 3 : 2
Small : Fiat = 24 : 24 = 1 : 1
Small : Toyota = 18 : 10 = 9 : 5
Now,
Maruti : Hyundai : Ford : Fiat : Toyota = 4 : 6 : 6 : 9 : 5
Similarly,
Small : Maruti = 9 : 4
Economy : Maruti = 3 : 4
Mid Size : Maruti = 2 : 1
Comfort : Maruti = 25 : 25 = 1 : 1
Luxury : Maruti = 3 : 2
Now,
Small : Economy : Mid size : Comfort : Luxury
9 : 3 : 8 : 4 : 6

12. The number of Luxury Fords = 150
$\therefore$Total number of Ford cars = 1500
So also, the total number of Luxury cars = 1500
The number of Hyundai cars sold = 1500
(as Hyundai : Ford = 1 : 1)
The number of Economy cars sold = 750
(as Economy : Luxury = 1 : 2)
Of them, Economy Hyundai (90) are already counted.
$\therefore$Hyundai or Economy cars = (1500 + 660 = 2160)

13. Toyota as a percentage of total

$$= \frac{5}{4+6+6+9+5} = \frac{5}{30} = 16.67\%$$

Percentage difference between small and midsize cars among Toyota cars sold = 2%
$\therefore$2% of 16.67% of total = 10
0.33% of total = 10
$\therefore$Total = 3000

14. Number of students who scored 50 or more marks in Social = 131.
Number of students who scored 60 and above in Social = 104.

Those who scored 50 or more but less than 60 in Social = 131 – 104 = 27.

15. Total number of students = 160
Those who scored 50 and above = 117
Those who scored less than 50 = 160 – 117 = 43

16. Number of students who scored 50 or more but less than 70.
In Maths = 127 – 83 = 44
In Marathi = 139 – 91 = 48
In Social = 131 – 103 = 28
In Statistics = 149 – 108 = 41
In English = 117 – 93 = 24
The number of students is the highest for Marathi.

17. A student who scored more than 60 marks in Maths might not have not scored more than 60 marks in Marathi. Hence, we cannot determine the number of students who scored more than 60 marks in all the given subjects.

18. Total number of students = 160
Those who scored 60 or more in Marathi = 113
Those who scored less than 60 = 160 – 113 = 47

Solutions for questions 19 to 22: It is said that the company made a profit in each of the given years. That is in each year, Sales > Expenses. Let the value of the sales in the given years be $100x$, $130x$, $160x$, $140x$, $200x$ and $230x$, respectively and the values of expenses be $100y$, $120y$, $160y$, $210y$, $90y$ and $230y$, respectively. Since sales is greater than expenses in each year, $140x > 210y$ or $x > 1.5y$.

The minimum values of sales and the value of expenses and the minimum value of profit in terms of y are as follows.

Year	Sales	Expenses	Profit
2010	$150y$	$100y$	$50y$
2011	$195y$	$120y$	$75y$
2012	$240y$	$160y$	$80y$
2013	$210y$	$210y$	$0y$
2014	$300y$	$190y$	$110y$
2015	$345y$	$230y$	$115y$

19. In the years 2011, 2012 and 2015, the sales, expenses and profits of the company increased or decreased in unison.

20. At least in the years, 2011, 2012, 2014 and 2015, the profits of ABC Corporation increased when compared to the previous year.

21. If profitability in the year 2012 was 50%, $160x = 320y$ or $x = 2y$.
In 2014, Sales = $200 \times 2y = 400y$
Expenses = $190y$

$\therefore$ Profitability = $\frac{210}{400} \times 100 = 52.5\%$

22. Even for the minimum possible value, i.e., when $x = 1.5y$, we can see that the profit is the highest in the year 2015. For any higher value of sales also the profit is going to be the highest in 2015.

23. The number of Tamil movies which collected ₹ 40 crore or more but less than ₹80 crore= 131 – 93 = 38.

24. As we do not know the number of movies which collected less than ₹ 20 crore, we cannot find the answer.

25. The number of movies which collected ₹60 crore or more but less than ₹100 crore in the different language are as follows.
Hindi = 107–63 = 44
Tamil = 119 – 71 = 48
Telugu = 111 – 83 = 28
Kannada = 129 – 88 = 41
English = 97 – 73 = 24
The highest is for Tamil.

26. From the previous question, the value is
44 + 48 + 28 + 41 + 24 = 185.

27. For maximum collection, we have to assume that the lowest 25 (170 – 145) movies collected only slightly less than ₹ 20 crore and movie in each range collected the maximum possible revenues in that range.
The maximum collection = $25 \times 19.99 + 22 \times 39.99 + 16 \times 59.99 + 19 \times 79.99 + 25 \times 99.99 + 63 \times 140$
$\simeq 25 \times 20 + 22 \times 40 + 16 \times 60 + 19 \times 80 + 25 \times 100 + 63 \times 140$
$= 500 + 880 + 960 + 1520 + 2500 + 8820$
= ₹15180 crore

Solutions for questions 28 to 31: As it is mentioned that 35 students passed out of Class IV at the end of year I, three students who were in Class IV in year I failed in the class and as the number of students in Class IV in year II was 32, it means that 29 students got promoted from Class III at the end of year I.

$\therefore$ 2 students failed in Class III in year I, as there were 39 students in Class III in year II, 37 students were promoted from Class II.

$\therefore$ 5 students failed in Class II in year 1 and as there were 38 students in Class II in year II, 33 students were promoted from Class I. Therefore, 3 students failed in Class I in year I and 31 students newly joined in year II. The following can be represented in a table as follows.

Class	Students in Year I	Promoted	Failed	Students in Year II
I	36	33	3	34
II	42	37	5	38
III	31	29	2	39
IV	38	35	3	32

28. 31 students joined the school in year II.

29. 3 students failed in class I in year I.

30. 29 students were promoted from class III at the end of year I.

31. 13 students in the school failed in the annual exams in year I.

32. The maximum number of emails (in billion) expected to be sent from Hotmail to Hotmail account.
= (2074) – (300 + 317 + 3) = 2074 – 620 = 1454

33. The required number = (317 + 198 + 1 + 1) = 517.

34. The maximum number of emails expected (from yahoo to Hotmail)
= Min [(2074) – (300 + 317 + 3), (2183) – (180 + 100 + 85 + 2)]
= Min [(2074 – 620), (2183 – 367)]
= Min [1454, 1816] = 1454.
Choice (B)

35. Maximum number of emails can be sent from Msn or Eudora.
The maximum number of emails that could be sent from Msn = 10,000 – (1182 + 1784 + 2074 + 85 + 372 + 4 + 1818) = 10,000 – (7319) = 2681
The maximum number of emails that could be sent from Eudora = 10,000 (1182 + 1784 + 2074 + 386 + 198 + 4 + 1818) = 10,000 – (7446) = 2554
The maximum number of emails sent from Msn to Mailcity = 2681 – (386 + 198 + 3) = 2094
The maximum number of emails sent from Eudora to Hotmail or Msn = 2554 – (372 + 85 + 3) = 2094
∴The maximum number of emails sent from one email account to another = 2094

Solutions for questions 36 to 40: If we convert the distribution of income from degrees to percentages, we get the incomes of Varun, Tarun, Arun, Karan and Charan as a percentage of total income of the family as 45%, 30%, 10%, 9% and 6%, respectively.

36 If Varun did not pay for 'Others', he can fully pay for fuel (8%), phone and electricity bill (10%) and rent (20%) or clothing (25%).

37. To get the least number of heads of expenses paid by more than one person, Varun (45%) must pay for clothing (25%) and rent (20%), Tarun (30%) must pay for food (30%), Arun (10%) must pay for the phone and electricity bill (10%) and Karan (9%) must pay for fuel. Only 'Others' (7%) is paid by Charan (6%) and Karan (1%).

38. If Varun does not spend any amount on food, his expenditure will be only on the remaining items. As remaining items constitute 70%, out of which 30 percentage points are contributed by Varun. If Varun fully contributes to fuel, phone and electricity bill and others, then his contribution on rent and clothing will become the least.
∴ The required percentage

$$= \frac{45 - (10 + 8 + 7)}{45} \times 100 = 44.44\%$$

39. The bill for food is 30%, and at most 40% of each person's income can be paid for food. If we use 40% of each person's income, we get 40% of the total. As we need only 30%, i.e., 75% of 40%, 25% of the total income need not be used.
As the sum of the incomes of Arun, Karan and Charan is 25%, if we use 40% of incomes of only Varun and Tarun, then all expenses of food can be accounted for.

40. As 5% of each of the expenses is contributed by Karan, his contribution will become maximum for that item which has the least value and, in this case, it is 'Others'.

Required percentage = $5\% + \frac{4}{7} \times 100 = 62.14\%$

41. Let us fix the countries and the universities based on the information from the 3rd day to the 1st day.
On day 3, as there are 3 logins from different countries and from different universities, University D should be in Germany and University G should be in UK.
In the same way, based on the 2nd day's information, University E should be in USA and University A should be in Philippines.
In the same way, from the 1st day's information University B or F or C should be in China. University B or C or F should be in any other country other than Philippines or Germany.
∴ University E should be in USA.

42. University G is in UK.

43. Among the given countries, no country can host three universities.

44. As University A is in Philippines, it is not located in China.

45. University A is in Philippines.

46. As all passengers in the bus got down at city B, the required number is
(14 + 12 + 7 + 10 + 8) – (5 + 10 + 12 + 11) = 13

47. All the passengers who got in at stop 1 would have boarded the bus at city A.
∴ At most 14 – 5 = 9 passengers who got in at city A got down at Stop 2.

48. All the 10 people who got in at Stop 3 could have got down at the next stop and so none of them might have got down at the last stop.

49. Of the 12 passengers who got in at Stop 1, at least one got down at Stop 2 and at least 5 got down at Stop 3.
∴ At most 12 – (1 + 5) = 6

50. Of the 14 passengers who got in at city A, at most nine of them could have got down at Stop 3. Only two of the people who got in at Stop 1 could have got down at Stop 4 (as 10 of them would have got down at Stop 2). Among the seven people who got in at Stop 2, three of them would have got down at Stop 3 and four of them could have got down at city B, the third stop from where they got in
∴ At most 9 + 2 + 4 = 15 passengers could have got down at the third stop from where they got in.

Exercise-2

1. If the sales in the year 2003 were ₹120 crore, then its value in the year 2006 was
$120 \times 1.21 \times 1.18 \times 1.14 = 120 \times 1.65 = 198$ crore.

2. Let the value of sales in 2003 be 100.
The approximate values in the other years would be 2004–121, 2005–143, 2006–163, 2007–174.
The highest increase is in 2005.

3. As, till the year 2006 the growth in profit in each year is less than that of sales and the growth of profit in 2007 is more than that of sales, the profitability would be the least in 2006.

4. As the number of students is 260 and the total number of instances is 414, the minimum and maximum values of students taking only one hobby is 106 and 183.
∴Only 120 is possible.

5. The minimum number of students who opted for at least two hobbies would be when all the students opted for exactly one or three hobbies and is obtained as follows.

$$x + y = 260$$
$$x + 3y = 414$$
$$y = 77$$

6. As in the previous question, the maximum number of students who opted for three hobbies would be 77.

7. All the students could have opted for one or two hobbies and so the minimum number of students who opted for exactly three hobbies is 0.

8. The maximum number of students in section A who passed in all the four subjects is 26.

9. In section C, 14 students have failed in Maths, 6 students in Physics, 9 students in Chemistry and 11 students in Biology. If all these students are distinct, 14 + 6 + 9 + 11 = 40 students would have failed in one subject each and so no student passed in all the four subjects.

10. For having the maximum number of students passing in exactly one subject, you should have the maximum number of students passing in all the four.
If x is the number of students in section B who passed in exactly one subject and y is the number of students who passed in exactly four subjects, then
$x + y = 40$ and $x + 3y = 136$
$x = 8$ and $y = 32$

11. The maximum number of students who passed in both Physics and Chemistry in the different sections are as follows.

A-31, B-32 and C-31, i.e., 31 + 32 + 31 = 94

12. We need to find the maximum number of students who passed in all the four subjects in each of the three sections, the values are as follows.
Section A – 26
Section B – 32
Section C – 26

Total 84.
∴ At least 120 – 84 = 36 students in the school failed to clear the Class X exam.

Solutions for questions 13 to 17: Sachin scored 18.18% of his runs in sixers (6s)

If we assume he scored the minimum number of 6s, i.e., one six, he would have scored six runs in sixers. As the percentage of runs scored in sixers is 18.18, his total score would be $\frac{6}{18.18} \times 100 = 33$

∴ Sachin has scored at least 33 runs.

But if he had scored only 33 runs, the total score would be 125 and since Kaif scored 10% of the total, the total score cannot be 125. So, we have to consider the next possibility, i.e., 250. When the total is 250, all the conditions are satisfied.

13. Dravid scored 20 runs of which 8 are in boundaries. Of the remaining 12, he scored an equal number of 1s and 2s or he scored 4 runs in singles.

14. Of all the players, the one with the highest number of boundaries was Sachin.

16.

	4s	6s
Sachin	6	2
Sehwag	2	2
Dravid	2	0
Kaif	4	0
Dhoni	5	5
Total	19	9

Total runs scored = $19 \times 4 + 9 \times 6 = 76 + 54 = 130$

17. Dhoni scored five sixers (6s).

18. The medical expenses increased from 6% of total expenses to 10% of the total which is the highest increase along with expenses on clothing. But as the increase of medical expenses is on a lower base, the expenses under that head would have the highest percentage increase.

19. The percentage share of entertainment expenses has fallen the most from 2005 to 2007. As it is given that the expenses under each head in 2007 was more than the corresponding value in 2005, assuming that the total expenses in 2005 and 2007 to be x and y, respectively.
$0.15\, y > 0.17\, x$
$\therefore\ 15\, y > 17x$

$$\frac{y}{x} > \frac{17}{15}$$

$\therefore$ Total expenses in 2007 was at least 13.33% more than the corresponding value in 2005.

20. If the total expenses in 2006 and 2008 are x and y, respectively,

$$\frac{5}{4} \times 0.05x = .10y$$

$$\frac{0.25}{4} x = .10y$$

$$\frac{x}{y} = \frac{40}{25} = \frac{8}{5}$$

$\therefore$ Expenses on clothing in 2008 = $14 \times 5 = 70$
Expenses on entertainment in 2006 = $15 \times 8 = 120$

$\therefore$ The required percentage = $\frac{70}{120} \times 100 = 58.33\%$

21. Let expenses on rent in 2005 be ₹14.
$\therefore$ Total expenses in 2005 = ₹100
Expenses on rent in 2008
$= 14 \times 1.1 \times 1.1 \times 1.1 = 14 \times 1.331 = 18.65$
Expenses on food in 2005 = ₹15
Expenses on food in 2008 = ₹18.65

$\therefore$ The percentage increase = $\frac{18.65 - 15}{15} \times 100 = 24.3\%$

Solutions for questions 22 to 25: As it is said that in reasoning none of the other three persons scored more than Chetan, Chetan is either A or C. From the second condition we can conclude that Balu and Anand is one among A or D in any order or one among B or C in any order.

22. If Deepak scored the lowest marks in the reasoning section, Deepak is student B which means Balu and Anand are one of A and D in any order and so Statement II would be false.

23. If Balu's lowest score is in the reasoning section, Balu is student B and Anand is student C and the statement that Anand's lowest score is in the quantitative section is true.

24. If Anand gets the highest score in the verbal section, he is student B and Balu is student C.
$\therefore$ Both statements cannot be simultaneously true. Anand and Balu can also be A or D in any order in which case both statements would be false.
$\therefore$ At most one of the statements is true.

25. If Deepak gets his lowest score in the verbal section, he is student D in which case Chetan is student A.

26. If Mercedes Ltd. had its lowest sales in Japan, Mercedes Ltd. is Company B and Toyo Ltd. is Company C while BWM Ltd. is Company A.
$\therefore$ If Statement 1 is true, then Statement 2 is necessarily true.

27. If Form Ltd. had its lowest sales in USA, then it is either Company A or Company D.
Case 1: Form Ltd. is Company A, then BWM Ltd. is Company C and the second condition cannot be satisfied.
Case 2: Form Ltd. is Company D. Then BWM Ltd. is Company A, and Mercedes Ltd. and Toyo Ltd. are companies B and C in any order.
$\therefore$ BWM Ltd. has its lowest sales in USA.

28. Using either statement
(B) or statement (D) we can uniquely determine each of the four companies.

29. If Toyo Ltd. had the highest sales in USA, then it is Company B and Mercedes Ltd. is Company C and BWM Ltd. is Company A.
If Mercedes Ltd. had the highest sales in UK, then it is Company B and Toyo Ltd. is Company C in which case BWM Ltd. is Company A.
$\therefore$ If one of the statement is true, then the other must be false.
If Company B is Form Ltd., then Company C is BWM Ltd. and Toyo Ltd. and Mercedes Ltd. are one of companies A and D in any order.
$\therefore$ At most one of the two statements is true.

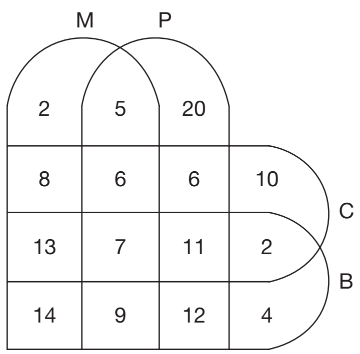
30. In 2015, the difference between the number of boys and the number of girls in LSP, Jaipur is 120.
$\therefore$ Number of students in the school is 280.
The difference between the number of boys and girls in LSP, Pune is 90.
The total number of students in LSP, Pune is 270.
The number of students in LSP, Mumbai must be between 270 and 280 and must be a multiple of 9.
It must be 279.
The difference between the number of boys and the number of girls is 31.

31. The value of *boyage* increased for DPS – Hyderabad, FPS – Hyderabad, LSP – Pune, LSP – Hyderabad, LPS – Pune, LPS – Hyderabad, LFS – Jaipur and LFS – Pune, i.e., a total of 8 schools.

32. As the number of girls (i.e., 2/3) in LSP, Pune in 2015 is same as that of boys in DPS, Mumbai in 2016, i.e.,
$\frac{2}{5}\left(\text{the number of students in DPS, in Mumbai, in 2016}\right)$.
$\Rightarrow$ Number of students in LSP, Pune in 2015 < Number of students in DPS, Mumbai in 2016.
Number of students in DPS, Mumbai in 2016 will be definitely more than LPS, Pune in 2015.

33. The number of students in Hyderabad in DPS = 540
Maximum possible number of students in FPS is as follows:
In Hyderabad = 533 (multiple of 7 + 6 = 13)
In LSP, in Hyderabad = 528 (multiple of 6 + 5 = 11)
In LPS, in Hyderabad = 517 (multiple of 8 + 3 = 11)
In PDS, in Hyderabad = 512 (multiple of 9 + 7 = 16)
In LFS, in Hyderabad = 510 (multiple of 3 + 2 = 5)
$\therefore$ Number of girls = 204.

Solutions for questions 34 to 37:



34. 76 – (5 + 6 + 6 + 7 + 11 + 9 + 12) = 20.

35. 64 – (5 + 8 + 6 + 13 + 7 + 14 + 9) = 2.

36. By observation, only seven students study all the four subjects.

37. Physics and Mathematics = 5 + 6 + 7 + 9 = 27
Biology and Chemistry = 13 + 7 + 11 + 2 = 33
Maths, Physics, Chemistry and Biology = 7
$\therefore$ 27 + 33 – 7 = 27 + 33 – 7 = 53.

38. The least growth from 1990–91 to 1998–99 is in the price of Euro, i.e., 8% and the highest is for Pound Sterling, i.e.,16%. The value must be greater than 8% and less than 16%.

39. Since the total currency holdings in 1995–96 were more than that of 1990–91, it can be concluded from the data that it can be possible only when the percentage share (by volume) of Euro in currency holding is more than that of US Dollar. Comparing the values of currency holding for different currencies in different years, we get only three possible years (1994–95, 1997–98 and 1999–2000).

40. Comparing the currency reserves with the currency reserves in 1992–93, we can observe that the total currency holdings were less than those in the year 1992–93 in the years (90–91, 93–94, 94–95, 95–96, 97–98, 99–00). Therefore, there are 6 such years.

41. By observation it can be seen that in the year 1998–99, the value of each of the currency and the price gold was the highest, therefore, irrespective of shares of individual currencies in total reserves, the year 1998–99 had the highest reserves.

42. If Anuj scores 50, 42, 41, and 50 marks in Sections I, II, III and IV, respectively, then he would miss the cut off of colleges D and F due to Section II and would miss the cut off of colleges A, B, C and E due to Section III score. So, he would not get a single call even with a score of 50 + 42 + 41 + 50 = 183 marks.

43. If Ram scores 50, 44, 41 and 50 in Section I, II, III and IV, respectively, then he would get a call from only college F with a score of 185 marks.

44. To get calls from all colleges, she needs to score at least 43 in Section I, 45 in Section II, 45 in Section III and 45 in Section IV. The total (43 + 45 + 45 + 45 = 178) is also more than the overall cut off of all the colleges.

45. If Ravi scores 50 marks each in Sections I, II and IV and scores only 40 marks in Section III, he would only get calls from colleges D and F.

46. With a minimum of 174 marks (41 in Section I, 45 in Section II, 43 in Section III and 45 in Section IV) one can get calls from all colleges other than E. As seen from the first question, one can score 175 marks and still not get a call from any college. Therefore, the maximum difference is 5.

47. Here, it is given that for any City X, the number of cities with a population less than it was exactly one less than its rank, i.e., the city with the lowest population is ranked as 1. So, the city with the highest population is ranked as 8, it means that its population is greater than all the other 7 cities.
From the given table, we can say that all cities except Agra have more population than at least one other city.
∴ Agra was ranked first.

48. Here, we can tabulate all the information as follows:

	Indore	Pune	Bhopal	Shillong	Agra	Cochin	Patna	Mysore
Indore	X	M	L	L	M	L	L	L
Pune		X	L	L	M	L	L	L
Bhopal			X	L	M	L	L	L
Shillong				X	M			M
Agra					X	L	L	L
Cochin						X	L	M
Patna							X	M
Mysore								X

We know that Agra has the lowest population. Accordingly, we can fill Agra's spaces.
After doing this we can see that except for Pune and Agra, all the cities have a population more than at least 2 cities. So, Pune was ranked second. Accordingly, we can fill the remaining.
It is given that Indore was ranked third.
∴ Bhopal was ranked fourth and Mysore was ranked fifth, i.e., four cities have a lesser population than Mysore.

50. The city ranked fourth in terms of population, i.e., Mysore would have its rank according to population and the number of cities having a population less than it as equal.

Exercise–3

Solutions for questions 1 to 4: From the given data the following table can be constructed.

		Departure Day							
		Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total
Arrival Day	Sun	a	b	c	d	-	-	-	28
	Mon	e	f	g	h	i	-	-	47
	Tue	-	j	k	l	m	n	-	40
	Wed	-	-	o	p	q	r	s	45
	Thu	-	-	-	T	u	v	w	40
	Fri	-	-	-	-	x	y	z	35
	Sat	-	-	-	-	-	a'	b'	25
	Total	37	43	50	45	35	30	20	

1. Given that: $e = 22$
$\Rightarrow a$ is $(37 - 22) = 15$
$\therefore\ b + c + d$ is $(28 - 15) = 13$
The number of ships that arrived on Sunday and departed after the next Sunday is 13.

2. From the table,
$d + h + i + l + m + n = 0$
$\Rightarrow a + b + c + e + f + g + j + k = 28 + 47 + 40 = 115$
And also, $a + b + c + e + f + g + j + k + o$
$= 37 + 43 + 50 = 130$
$\Rightarrow o = 15$
$\therefore\ o$ is 15.

3. From the table, $n + r + s + v + w + y + z + a' + b^1$
$= 30 + 20 = 50$
$a' + b' = 25$
$n + r + s + v + w + y + z = 25$
As $r = 20$, the maximum value of $y + z = 5$
As $x + y + z = 35$, the least possible value of x is 30.

4. From the table, $a + b + e + f + j = 80$
$(a + b + e + f)$ is at the most 75.
$\therefore\ j$ is at the least 5.

Solutions for questions 5 to 8: As D got 24 marks, the only possibility is
$3 \times 2 + 6 \times 3$ (=24)
$\Rightarrow$ 9 correct attempts.
Now as E got 39 marks, the only possibility is
$(7 + 6) \times 3$ (= 39)

⇒ 13 correct attempts.
But given that D and E got the same number of correct answers in Physics and Chemistry respectively (i.e., 6 correct answers)
⇒ D has 3 correct answers in Chemistry and E has 7 correct answers in Physics.
As F got 51 marks the only possibility is
$9 \times 4 + 5 \times 3$ (= 51)
⇒ F has 14 correct answers.
But given that B and C has the number of correct answers in Chemistry as the correct answers of F in different subjects.
⇒ B and C got 5 and 9 correct answers in Chemistry respectively.
(∴ Total correct answers of B is 7)
⇒ B and C got 7 – 5 = 2 and 13 – 9 = 4 correct answers in Physics.
As the total correct answers of B, C, D and E in Chemistry = 5 + 9 + 3 + 6 = 23.
F can have at most 7 correct answers in Chemistry.
⇒ F has 5 correct answers in Chemistry and 9 in Physics.
⇒ A has 8 and 2 correct answers in Physics and Chemistry, respectively.

	Physics	Chemistry	Total	Marks
A	8	2	10	36
B	2	5	7	19
C	4	9	13	44
D	6	3	9	24
E	7	6	13	39
F	9	5	14	51
Total	36	30	66	213

5. A scored 36 marks.

6. D scored the second lowest total marks.

7. In Physics, only A and F scored more marks than E

8. A, D, E and F got more marks in Physics than in Chemistry.

9. Feroz wrote 16 mock CATs conducted by institute P. Hence, the person who wrote more than 30 – 16 = 14 mock CATs conducted by P would have written at least one Mock CAT in common with Feroz.
Similarly, the values for Q, R, S and T are 19, 17, 12 and 16, respectively.
Only Hrithik satisfies the required condition.

10. The value will be the least when maximum number of mock CATs are written by both or none.
Among the tests conducted by P, at least 20 – 12 = 8 tests are written by exactly one of Akshay and Hrithik.
Similarly, the values for Q, R, S and T are 1, 4, 7 and 3, respectively.
∴ Required value = 8 + 1 + 4 + 7 + 3 = 23.

11. Among the 32 Mock CATs conducted by R, Bobby, Emran and Govinda wrote 16, 18 and 20, i.e., a total of 54 instances.
For the number of Mock CATs written by more than one of them to be minimum, maximum possible tests are to be written by one or three persons.
If each Mock CAT is assigned one person, 54 – 32 = 22 instances will be left.
Among these 22 instances, 2 instances per Mock CAT be assigned to 11 Mock CATs.

12. For the value to be maximum, each Mock CAT must be written by one or three persons.
Total instances = 16 + 15 + 12 = 43.

$$\therefore \text{ Required value} = \frac{43-24}{2} = 9.$$

13. A total of 40 Mock CATS were conducted by institute Q. Among these, Chahat and Govinda wrote at least 33 + 30 – 40 = 23 Mock CATs in common.
As no other person has written more than 23 Mock CATs in common with any person.
Hence, it is the highest for Chahat and Govinda.

14. By observation,
only R → U and U → U are the possible choices.

$$U \rightarrow U = \frac{15.38(\text{males})}{7.04(\text{females})} \cong 2.2$$

$$R \rightarrow U = \frac{27.27(\text{males})}{11.95(\text{females})} \cong 2.3$$

∴ Maximum gender ratio is R → U.

15. The ratio of total number of female immigrants to male immigrants is
$\frac{75.11}{24.89} \cong 3$. It is slightly more than 3.
Now the required value is:

$$\frac{15.38}{11.95 \times 3} \times 100 = \frac{11.38}{35.85} \times 100 \cong 43\%$$

The value will be slightly less than 43%.

16. Total male migrants from urban to rural areas is
= 7.68% of total male migrants

$$\cong 7.68\% \times \frac{1}{4} \times \text{total migrants}$$

≅ 1.9% of total migrants.

17. The ratio of total male migrants and total female migrants is 1 : 3.
Now total migrants from rural to rural areas as a percentage of total migrants from all streams can be found out by using allegations method, i.e.,

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$$= 76.71 - \frac{(76.71 - 49.67) \times 1}{(1+3)}$$

$= 76.71 - 6.76 = 69.95\%$

As the actual ratio is a bit more than 1 : 3, the value will be slightly more than 69.95%.

Solutions for questions 18 to 21: As given that no two persons gave equal number of true replies, the number of true replies is 0, 1, 2, 3, 4 and 6 (as if five replies are correct the sixth reply also has to be correct).

If all the replies of Aman are correct, then Biswa and Dev will have equal number of correct replies, i.e., 1 each.

Similarly, if all the replies of Biswa are correct, then Aman and Charan will have 1 each as correct replies.

If all the replies of Charan are correct, then Biswa and Dev will have 1 each as correct replies.

Similarly, Dev and Fazal also cannot have all the replies correct.

∴ If all the replies of Emma are correct, then

Aman – 4
Biswa – 0
Charan – 3
Dev – 2
Emma – 6
Fazal – 1

∴ The persons and the city pairs are as follows:

Emma – Bangalore; Charan – Hyderabad, Biswa – Chennai; Aman – Kolkata, Dev – Delhi; Fazal – Mumbai

21. Only Emma gave her city name correctly.

Solutions for questions 22 to 25: Taking data from the three tables, we can arrive at the following expert, actor and feature combination.

	A_1			A_2		
	Expressions	**Dialogue delivery**	**Body language**	**Expressions**	**Dialogue delivery**	**Body language**
Anand	5	7	9	7	8	7
Babu	7	8	6	5	3	6
Charan	10	6	7	7	2	5
David	9	8	8	8	3	4

22. The rating given by Babu for 'Expressions' for A_1 is 7.

23. The rating given by Charan for 'Body language' for A_2 is 5.

24. The rating given by Charan for 'Expressions' for A_2 is 7.

25. The rating given by Anand for 'Dialogue delivery' for A_2 is 8.

9 OMET Based DI

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:

- Learn about different types of questions which have appeared in Other Management Entrance Tests (OMETs)
- Learn how to convert data from one form of presentation to other forms
- Practise questions that involve a lot of calculations and data
- Learning shortcuts to approximate and find the required value, thereby avoiding cumbersome calculations

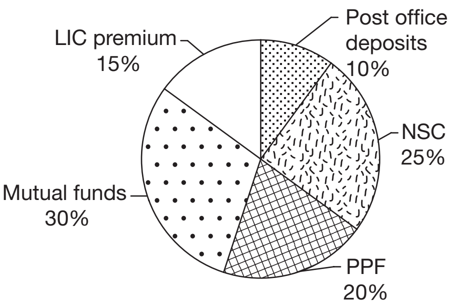
❒ INTRODUCTION

This chapter contains questions similar to the ones asked in other management entrance tests (OMET) like XAT, IIFT, SNAP, NMAT, etc. Exams like IIFT and NMAT are known to ask very calculation-intensive DI questions than the ones asked in CAT, also XAT is known to have asked unconventional and unorthodox sets which stand apart from the regular models asked in the CAT exam.

SOLVED EXAMPLES

These questions are based on the pie chart and the table given below.

The pie chart shows the breakup of the investment of a person in various schemes in 2000 and the table shows the investments in the same schemes in 2001.



Total investment = ₹50,000

Scheme	Percentage of total investment that is invested in the scheme
Post office deposits	8%
NSC	30%
LIC Premium	15%
PPF	15%
Mutual funds	32%

Total investment = ₹60,000

9.01: In how many of the schemes is his investment in 2000 less than that in 2001?

(A) 1 (B) 2

(C) 3 (D) 4

Sol: The share of LIC premium in both years were the same. His total investment increased from 2000 to 2001.

$\therefore$ His LIC premium in 2000 < His LIC premium in 2001.

His investments in post office deposits, mutual funds, PPF and NSC in 2000 were

$$\frac{10}{100}(50000), \frac{30}{100}(50000), \frac{20}{100}(50000) \text{ and}$$

$\frac{25}{100}(50000)$, i.e., 5000, 15,000, 10000 and 12500, respectively.

His investments in these schemes in 2001 were

$\frac{8}{100}(60000), \frac{32}{100}(60000), \frac{15}{100}(60000)$ and

$\frac{30}{100}(60000)$, i.e., 4800, 19,200, 9000 and 18000.

$\therefore$ A total of three schemes satisfied the given conditions.

9.02: In which scheme was his total investment in both years together the maximum?

(A) Post Office deposits
(B) NSC
(C) LIC premium
(D) Mutual funds

Sol: From the previous solution, his total investment in both years in Post Office deposits, mutual funds, PPF and NSC (in ₹) were 9800, 34,200, 19000 and 30500. His total investment in LIC Premium in both yeas $= \frac{15}{100}(50000 + 60000)$ = ₹16,500.

$\therefore$ Maximum total investment was in mutual funds.

Alternately: Percentage of his investments was maximum in mutual funds in each year. Therefore, his total investment must be maximum in mutual funds.

9.03: In 2002, his total investment was ₹70,000. His investment breakup was the same as that in 2000. In how many schemes in 2002 was his investment more than ₹12,600?

(A) 0 (B) 1
(C) 2 (D) 3

Sol: $12600 = \frac{12600}{70000}(100) = 18\%$ of his total investment. His investment was more than 18% of his total investment in three schemes, i.e., PPF, NSC and mutual funds.

9.04: In how many schemes in 2001 was his investment more than his average investment in the schemes?

(A) 2 (B) 3
(C) 4 (D) 0

Sol: There were 5 schemes.

$\therefore$ Average investment would be $\frac{100}{5} = 20\%$ of total investment. His investments in mutual funds and NSC exceeded this.

$\therefore$ Two schemes satisfied the given conditions.

9.05: If his combined investments in the various schemes in 2000 and 2001 were represented in a pie chart, for how many schemes would the angle be more than 90°?

(A) 0 (B) 1
(C) 4 (D) 2

Sol: $90° = \frac{90°}{360°}(100) = 25\%$.

As his investments in LIC premium, PPF and Post Office deposits in both years formed less than 25% of his total investment, the angle for these schemes in the pie chart formed will be less than 90°.

$\therefore$ For the other two schemes the angle would be more than 90°.

EXERCISE-1

Directions for questions 1 to 5: These questions are based on the table given below which shows the number of tons of fish caught through the traditional and modern methods across several years in Andhra Pradesh.

	Fishing in inland waters in different types of water bodies						Fishing in the seas	
	Artificial Tanks		Lakes		Rivers		Seas	
	Traditional	Modern	Traditional	Modern	Traditional	Modern	Traditional	Modern
1996	25	586	41	169	129	1348	569	5341
1997	47	631	72	201	181	1421	831	5583
1998	57	754	131	296	241	1639	947	6164
1999	61	836	129	354	297	1743	1152	6341
2000	63	929	129	421	324	1869	1181	6861
2001	60	1016	108	494	351	1931	1261	7146
2002	61	1089	121	528	407	2046	1734	7232
Total	374	5841	731	2463	1930	11997	7675	44668

1. During the year 1998, what is the total tonnage of fish caught in the inland waters?
(A) 10229 (B) 6164
(C) 4768 (D) 3118

2. Which of the following can be inferred from the given data for the period from 1999 to 2002?
I. The tonnage of fish caught in the inland waters by traditional methods is increasing every year over its value in the previous year.
II. The tonnage of fish caught by the traditional methods is continuously decreasing.
III. The tonnage of fish caught by the modern methods is continuously decreasing.
IV. There is a decrease in the quantity of fish caught by traditional methods in two successive years.
(A) I only (B) II only
(C) II and III only (D) I and IV only

3. During the given years, the quantity of fish caught by modern methods is approximately how many times that caught by traditional methods?
(A) 3 (B) 4
(C) 2 (D) 6

4. When compared to the previous year during which of the following years is the growth rate in quantity of fish caught, the least?
(A) 1998 (B) 1999
(C) 2000 (D) 2001

5. The average weight of the fish caught in the artificial tanks, lakes, rivers and the seas is in the ratio 2 : 3 : 4 : 5. The number of fish caught in which of the following water bodies is the least?
(A) Artificial tanks (B) Lakes
(C) Rivers (D) Seas

Directions for questions 6 to 10: These questions are based on the table and the bar graph given below.

Income plan of five persons for the year 2003–04

Rupees in Lakh

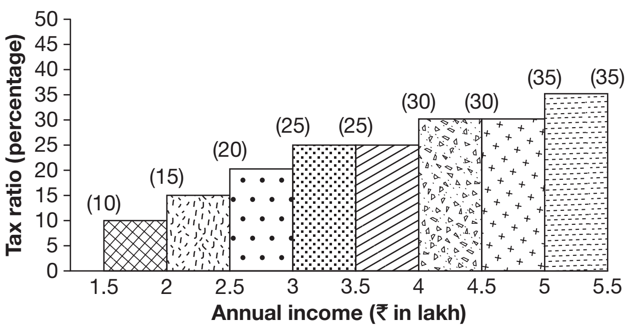
Name of the person	Sharma	Rao	Gupta	Solkar	Kuchroo
Profession	Doctor	Engineer	Business man	Doctor	Professor
Annual income	4.50	3.50	3.00	4.00	5.00
Annual expenditure	3.00	2.50	2.50	2.50	3.50
Annual savings	1.50	1.00	0.50	1.50	1.50

The annual savings are invested as shown below

Rupees in Thousand

Name of the person	Sharma	Rao	Gupta	Solkar	Kuchroo
PPF	50	50	25	50	80
Life insurance	20	8	7	20	15
Medical insurance	5	2	3	5	5
Pension plan	10	17	5	15	10
Debt funds	35	13	5	40	20
Monthly income plan (MIP)	30	10	5	20	20

The graph below shows various income slabs and the corresponding tax rates. For example, an annual income of ₹1.8 lakh falls in the range of ₹1.5 lakh – ₹2.0 lakh. Hence, the tax rate applicable is 10%. Similarly, for an income of ₹2 lakh, the tax rate is 15% as it falls in the slab of ₹2 lakh – ₹2.5 lakh.



Note: Income tax is calculated on the annual income.

6. How many of the given five persons paid an income tax exceeding rupees one lakh for the year 2003–04?

7. For which of the given persons, the ratio of the annual savings to that of the annual expenditure is least?

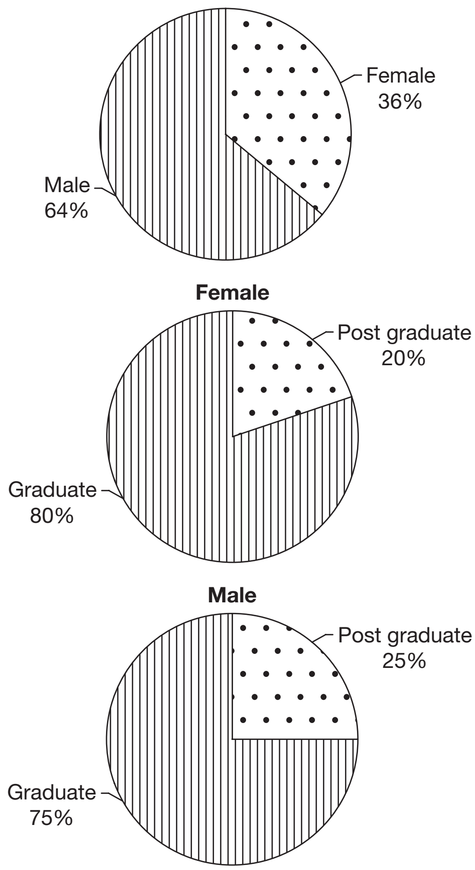
8. How much more should Gupta save so that his savings is 35% of his income?

9. An interest of 8% per annum is payable on the investment in PPF for the given year. Considering all the five persons, what is the average amount of interest payable per person?

10. If there is an exemption of tax on PPF, what is the total tax payable by the two doctors on their total taxable income?

Directions for questions 11 to 14: These questions are based on the pie charts below.

The pie charts give the breakup of graduates and postgraduates among males and females and the breakup of males and females in percentage.



Total number of persons = 5000

11. If 10% of the female postgraduates are married, the number of unmarried female postgraduates is
    (A) 36 (B) 54
    (C) 324 (D) 224

12. What approximate percentage of the students of the college are graduates?
    (A) 65% (B) 75%
    (C) 70% (D) 77%

13. If 496 male postgraduates are at least 30 years old, the number of male postgraduates aged below 30 years is
(A) 292 (B) 284
(C) 304 (D) 316

14. The total number of postgraduates is approximately what percentage of the total number of graduates?
(A) 32.4% (B) 28.6%
(C) 26.8% (D) 30.2%

Directions for questions 15 to 18: Study the following tables carefully and answer these questions.

Number of candidates who qualified in a competitive examination from five cities over the years

Year	City				
	Chennai	Delhi	Hyderabad	Kolkata	Mumbai
2004	4224	8670	8100	15600	4200
2005	6720	10500	6240	16240	1700
2006	4626	13000	14490	13230	3750
2007	3150	12960	13490	18900	3080
2008	7280	13340	11200	15870	4680
2009	6900	9120	13440	11800	4060
2010	9280	9800	13440	11590	5120

Percentage of candidates not qualified in the competitive examination from the five cities

Year	City				
	Chennai	Delhi	Hyderabad	Kolkata	Mumbai
2004	89	83	82	76	88
2005	86	79	88	72	90
2006	90	75	77	79	85
2007	94	76	81	73	89
2008	87	77	84	77	87
2009	88	81	79	80	86
2010	90	80	76	81	84

15. The difference in the number of candidates who qualified from Mumbai in any of the given years and that in the previous year was the highest in
(A) 2007 (B) 2009
(C) 2010 (D) 2006

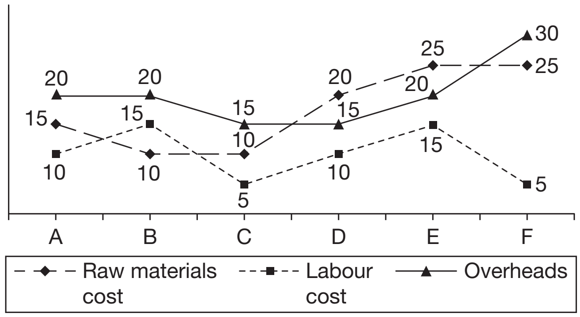
16. In which of the following years was the number of candidates who appeared from Chennai, the highest?
(A) 2010 (B) 2007
(C) 2008 (D) 2006

17. What is the number of candidates who did not qualify from Kolkata in 2009?
(A) 46,400 (B) 45,200
(C) 47,200 (D) 49,200

18. How many candidates appeared from Hyderabad in 2007?
(A) 73,000 (B) 76,000
(C) 71,000 (D) 77,000

Directions for questions 19 to 23: These questions are based on the following graph.

Six different companies A, B, C, D, E and F manufacture a similar product. The cost of raw materials, labour cost and overheads per unit are given below.



19. Which of the following products has the maximum cost per unit?
(A) A (B) B
(C) D (D) F

20. If company B produces 5000 units and sells them at ₹68, then the profit of the company is
(A) ₹1,65,000 (B) ₹1,40,000
(C) ₹1,15,000 (D) ₹1,55,000

21. Which of the following statements is true?
(A) The labour costs of E and F are same.
(B) The ratio of costs, overheads and labour is same for A, B and E.
(C) The ratio of total cost of A and D is same as the ratio of total cost of E and F.
(D) Both (A) and (B).

22. Company D can produce a maximum of 1000 units per day and company F can produce up to 800 units per day. If these companies sell their products at ₹60 and ₹80, respectively, then what percentage of D's profit is F's profit in the total maximum production of 10 days?
(A) 15 : 16 (B) 3 : 4
(C) 7 : 8 (D) 9 : 10

23. If the labour cost of B is the same as the labour cost of C, then what is the ratio of the total cost of the two companies?
(A) 1 : 1 (B) 1 : 2
(C) 1 : 3 (D) Cannot be determined

Directions for questions 24 to 28: These questions are based on the following table which represents the number of garments manufactured by four companies A, B, C and D for four segments of people, males (M), females (F), children (C) and sports persons (S) during quarters I, II, III and IV of the year 2003.

(in thousands)

	A				B				C				D				
	M	F	C	S	M	F	C	S	M	F	C	S	M	F	C	S	Total
I	33	60	72	20	18	25	36	12	26	35	41	21	41	71	88	30	629
II	44	76	80	25	23	31	43	15	40	59	71	28	55	78	98	35	801
III	45	75	85	30	30	34	48	18	52	68	81	26	60	80	95	41	868
IV	96	84	95	25	32	35	52	21	62	75	83	30	71	92	99	35	937
Total	168	295	332	100	103	125	179	66	180	237	276	105	227	321	380	141	3235

24. If the companies are arranged based on the number of garments manufactured by them in the year 2003, then which of the following is true?

(A) A > D > B > C (B) D > B > C > A
(C) D > A > C > B (D) A > D > C > B

25. For which of the following quarters is the percentage increase over the previous quarter in the number of garments manufactured by company C, for the female segment, the least?

(A) I (B) II
(C) III (D) IV

26. In the year 2003, the overall growth achieved by these companies in the number of garments manufactured is 25% more when compared to the number of garments manufactured in 2002. What was the total production of garments by these companies in the year 2002 in thousands?

(A) 2426 (B) 2588
(C) 2634 (D) 2432

27. For the year 2003, what is the ratio of the total number of garments manufactured for male segment to that of the female segment?

(A) 113 : 163 (B) 489 : 206
(C) 231 : 106 (D) 369 : 103

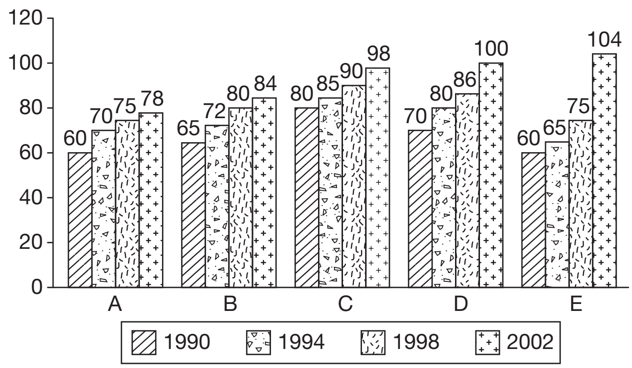
28. For how many companies is there a growth in the number of garments in every quarter and for every segment?

(A) 0 (B) 1
(C) 2 (D) 3

Directions for questions 29 to 32: These questions are based on the information given below.

The table represents the percentage of votes cast in an election in five small constituencies over four different election years and the bar graph shows the population (in thousands) of these five constituencies in the respective years.

Year	A	B	C	D	E
1990	45%	60%	75%	40%	55%
1994	50%	64%	80%	60%	70%
1998	65%	85%	60%	50%	80%
2002	70%	72%	65%	45%	60%



29. What is the total number of votes cast in the given five constituencies in the year 1998?

(A) 2,93,000 (B) 2,73,750
(C) 2,84,250 (D) 3,43,800

30. In the year 1994, what is the average number of votes cast in the given five constituencies?

(A) 48,516 (B) 44,295
(C) 42,248 (D) 48,446

31. In 2002, how many people in the five constituencies put together did not cast their vote?

(A) 1,67,246 (B) 1,77,820
(C) 1,79,840 (D) None of these

32. What is the ratio of the total number of votes cast in constituency D in the given four years to that in constituency E?

(A) 1231 : 2481 (B) 2481 : 1231
(C) 1640 : 2009 (D) 2009 : 1640

Directions for questions 33 to 36: Answer the following questions based on the table given below.

A, B, C and D are four different trains starting from the same station at different times with different average speeds and all are travelling on parallel tracks. Train A started at 05:00 hrs.

Time	Distance from the starting point (in kilometres)			
	Train A	**Train B**	**Train C**	**Train D**
6:10	100	–	–	–
6:30	100	–	30	–
7:00	130	40	60	–
8:00	200	100	130	–
12:00	400	260	330	187.5
16:00	600	460	530	437.5
18:00	680	560	610	492.5
19:00	720	600	650	562.5
19:15	720	600	650	562.5
19:30	740	650	690	562.5
20:00	750	700	730	656.25
21:00	800	760	800	718.75
22:00	840	840	820	778.25
23:30	850	910	900	875
00:00	900	940	900	906.25
01:30	1000	1000	1000	1000

Note: Some of the trains had maintained uniform speed.

33. The average speed of which train was the least among the four trains between 16:00 hrs to 22:00 hrs?
(A) Train A (B) Train B
(C) Train D (D) Train C

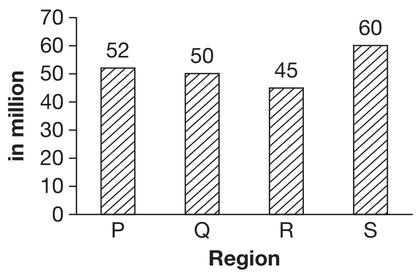
34. At 16:00 hrs, which two trains are farthest from each other?
(A) Train A and Train C
(B) Train A and Train B
(C) Train A and Train D
(D) Train C and Train D

35. At how many instances, is there a possibility of one train crossing another train after 12 noon?
(A) 1 time
(B) 0 times
(C) 2 times
(D) More than two times

36. If the stoppages of train A are not considered, then the approximate average speed of train A from 05:00 hrs to 01:30 hrs will be (assume it has only two stops).
(A) 48.8 kmph (B) 50.2 kmph
(C) 52 kmph (D) 54 kmph

Directions for questions 37 to 40: These questions are based on the graph and the table given below.

As per the National Readership Survey (NRS), the details regarding the readership (in million) of four major newspapers in regions P, Q, R and S are as follows:



The following table gives the percentage of readers for the newspapers in the four regions.

Newspapers	Region			
	P	**Q**	**R**	**S**
A	30%	25%	20%	30%
B	40%	25%	30%	30%
C	10%	30%	40%	20%
D	20%	20%	10%	20%

37. How many readers (in million) read newspaper B in region P?
(A) 20.8 (B) 30
(C) 16.5 (D) 22.5

38. What is the ratio of the number of readers who read newspaper A in region Q to the number of readers of newspaper B in region S?
(A) 11 : 12 (B) 25 : 36
(C) 18 : 17 (D) None of these

39. What is the total number (in millions) of readers who read newspaper C in all the four regions?
(A) 26 (B) 52.5
(C) 48.5 (D) 50.2

40. What is the ratio of the total readers of newspaper B to that of newspaper C?
(A) 341 : 171 (B) 324 : 251
(C) 3 : 5 (D) None of these

Directions for questions 41 to 44: These questions are based on the table given below.

Table 9.1 Details of Indian immigrants (people of Indian origin) in different countries as on 1980

Country	Persons of Indian origin (PIOs) (in 000s)	PIOs as a % of host country's population	No. of PIOs (in 000s) having foreign citizenship	% of total PIOs
(1) **AFRICA**				
(a) Kenya	70	0.31	1	0.55
(b) Mauritius	701	70.10	700	5.52
(c) Mozambique	21	0.14	11	0.16
(d) South Africa	850	2.57	850	6.69
(e) Tanzania	40	0.17	33	0.32
(2) **AMERICA**				
(a) Canada	229	0.89	129	1.80
(b) Guyana	300	30.30	300	2.36
(c) Jamaica	39	1.62	38	0.30
(d) Suriname	140	35.90	140	1.10
(e) Trinidad and Tobago	430	35.25	430	3.39
(f) USA	500	0.21	287	3.94
(3) **ASIA**				
(a) Afghanistan	46	0.30	45	0.36
(b) Bhutan	70	4.93	–	0.55
(c) Burma	330	0.84	50	2.60
(d) Malaysia	1170	7.07	1029	9.21
(e) Nepal	3900	27.12	2388	30.64
(f) Singapore	100	3.83	74	0.79
(g) Sri Lanka	1023	6.28	457	8.05
(h) Thailand	65	0.12	55	0.51
(4) **EUROPE**				
(a) France	42	0.8.	38	0.33
(b) Germany	32	0.50	8	0.25
(c) Netherlands	103	0.70	100	0.81
(d) UK	789	1.39	395	6.21
(5) **MIDDLE EAST**				
(a) Bahrain	48	11.16	–	0.38
(b) Iraq	35	0.21	–	0.28
(c) Kuwait	110	5.88	1	0.87
(d) Libya	36	0.88	–	0.28

(*Continued*)

Country	Persons of Indian origin (PIOs) (in 000s)	PIOs as a % of host country's population	No. of PIOs (in 000s) having foreign citizenship	% of total PIOs
(e) Oman	190	14.29	–	1.50
(f) Qatar	52	15.76	–	0.41
(g) Saudi Arabia	250	1.80	–	1.97
(h) UAE	240	16.55	1	1.89
(I) Yemen	103	1.41	100	0.81
(6) **OCEANIA INDONESIA**				
(a) Australia	99	0.61	87	0.78
(b) Fiji	339	47.75	339	2.67
(c) Indonesia	30	0.02	15	0.24
Total	12522	0.25	8101	100.00

41. In how many countries is the number of PIOs having citizenship of the country as a percentage of total PIOs in the country more than 90%?
(A) 5 (B) 11 (C) 12 (D) 14

42. The names of how many of the given countries which has the number of PIOs more than one lakh but less than five lakh, ends with a consonant?
(A) 2 (B) 3 (C) 4 (D) 5

43. For how many of the given countries is the country's total population less than the total number of PIOs in all the countries together?
(A) 16 (B) 18 (C) 19 (D) 17

44. The number of persons of Indian origin in ASIA is more/less than that in AMERICA by what percentage?
(A) 24.8 (B) 48 (C) 71.2 (D) 309.2%

Directions for questions 45 to 47: Answer the following questions based on the information given below.

The following table shows the growth in urban population since 1901 and the percentage of rural and urban population in the total population of India.

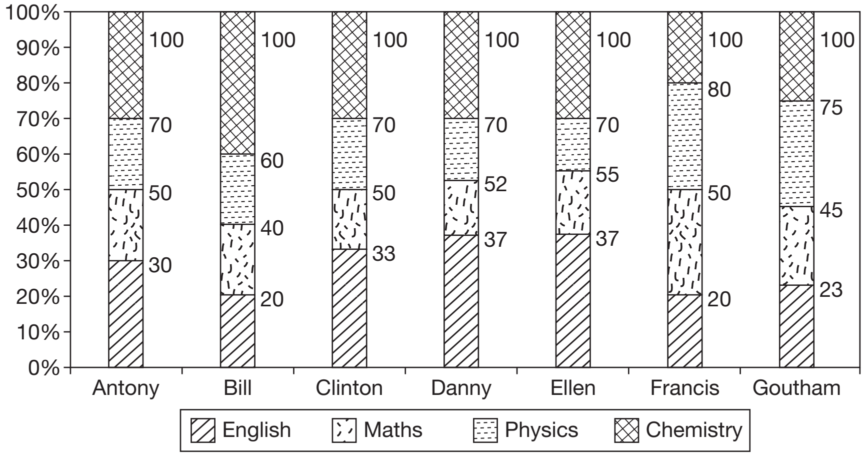
Year	Urban population (in million)	Percentage of total population	
		Rural	Urban
1901	25.8	89.0	11.0
1911	25.9	89.6	10.4
1921	28.0	88.7	11.3
1931	33.5	87.8	12.2
1941	44.1	85.9	14.1
1951	62.4	82.4	17.6
1961	78.9	81.7	18.3
1971	108.9	79.8	20.2
1981	162.2	76.3	23.7
1991	217.6	74.3	25.7

45. The approximate percentage increase in the rural population from 1901 to 1991 was approximately
(A) 130% (B) 160%
(C) 210% (D) 240%

46. For which of the following periods was the percentage increase in the total population, the highest?
(A) 1951–1961 (B) 1961–1971
(C) 1971–1981 (D) 1981–1991

47. In the time given, the only occasion when India's total population decreased was during
(A) 1901–1911 (B) 1911–1921
(C) 1921–1931 (D) 1931–1941

Directions for questions 48 to 50: The following graph shows the percentage of marks scored by Antony, Bill, Clinton, Danny, Ellen, Francis and Goutham in four subjects in their Class XII exam.



48. Who scored the highest marks in Maths?
(A) Goutham (B) Francis
(C) Danny (D) Anthony

49. In English, the marks scored by Ellen is approximately what percentage more than those by Goutham?
(A) 43% (B) 55%
(C) 65% (D) 63%

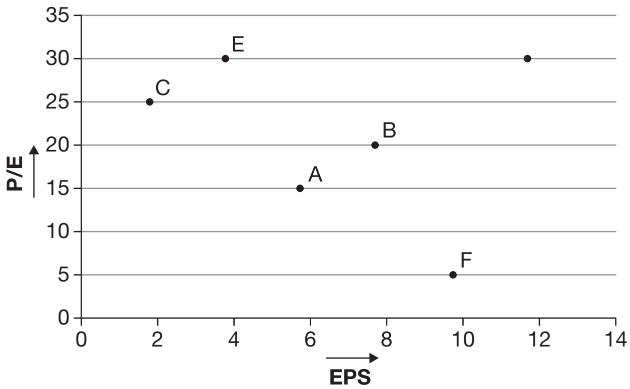
50. The total marks scored by all the seven students in Physics is approximately
(A) 672 (B) 650
(C) 480 (D) 553

EXERCISE-2

Directions for questions 1 to 5: Answer the following questions based on the information given below.

The following data gives the P/E ratio and EPS of six companies for two consecutive years.

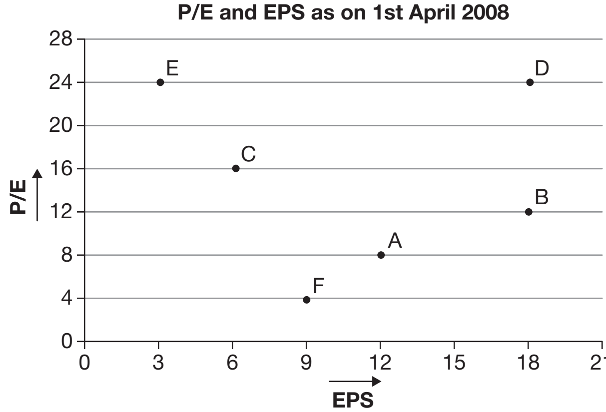
P/E and EPS as on 1st April 2007



Total number of shares on 1st April 2007

A	B	C	D	E	F
12.5	14.8	27.0	8.5	22	7.2

P/E and EPS as on 1st April 2008



Total number of shares as on 1st April 2008 (in lakh)

A	B	C	D	E	F
17.5	14.8	40.5	8.5	22.0	7.2

$$\text{P/E ratio} = \frac{\text{Price of the share in rupees}}{\text{Earnings per share in rupees (EPS)}}$$

$$\text{EPS} = \frac{\text{Total earnings of the company}}{\text{Total number of shares of the company}}$$

1. What is the percentage increase in total earnings of Company A from 1st April 2007 to 1st April 2008?

2. Which company had the highest increase in total earnings from 1st April 2007 to 1st April 2008?

3. What is the price of a share of Company B on 1st April 2008?

4. In percentage terms, the share price of which company appreciated the most from 1st April 2007 to 1st April 2008?

5. If the total earnings of Company E is same as that of Company D as on 1st April 2009, then what will be the EPS of Company E on 1st April 2009, given that the earnings of Company D decrease by 10% compared to that on 1st April 2008 and the number of shares of Company E as on 1st April 2009 is 27.0 lakh?

Directions for questions 6 to 10: Study the following table and answer the questions that follow. The table below compares five different countries on various aspects.

Country	Population (Lakh)	Literacy rate	Number of women per 1000 men	% of rural population
A	1321	70%	970	45%
B	2501	61%	951	62%
C	540	85%	1021	51%
D	91	90%	992	39%
E	832	80%	989	42%

6. Which country has the highest number of illiterate people?
(A) A (B) B
(C) C (D) D

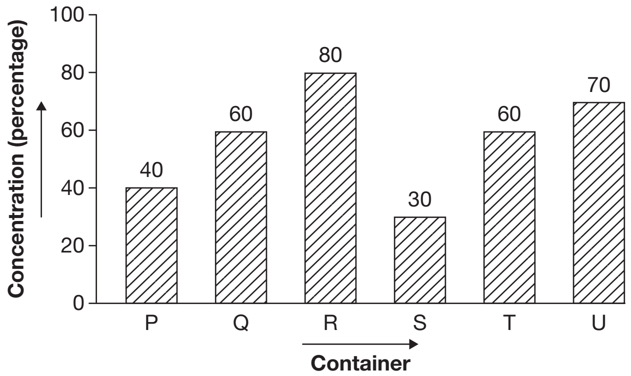
7. The urban population of Country A exceeds the urban population of Country E by
(A) 254.5 lakh (B) 377 lakh
(C) 244 lakh D) 223 lakh

8. What is the approximate number of women in the country which has the second highest number of men?
(A) 4140 lakh (B) 650 lakh
(C) 670 lakh (D) Cannot be determined

9. Which country has the least number of literate women?
(A) D (B) C
(C) E (D) Cannot be determined

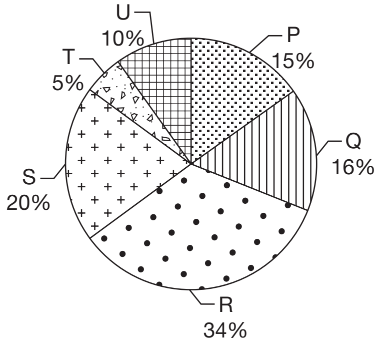
10. For which country is the number of men who are literate, as a percentage of the total number of men, the highest?
(A) E (B) C
(C) D (D) Cannot be determined

Directions for questions 11 to 15: Answer the following questions based on the information given below.



Each of six different containers, labelled from P to U contains some solution of milk and water with the concentration of milk (in percentage) as specified as above.

Percentage distribution of the total volume of six solutions



Total volume of all six solutions together = 100 litres

11. Which two solutions when mixed will yield a solution with maximum concentration?
(A) R and Q (B) Q and U
(C) R and T (D) R and U

12. Solutions P, Q, R are mixed in equal proportions and the resultant solution is labelled as X, while solutions S, T and U are mixed in equal proportions to give a solution labelled as Y. Which of the following is definitely true?
(A) The concentration of X is more than that of Y.
(B) The concentration of X is less than that of Y.
(C) The concentration of X is equal to that of Y.
(D) None of the above

13. What is the approximate concentration of the solution formed by mixing the entire volumes of R and S?
(A) 27% (B) 55%
(C) 61.5% (D) 64%

14. Which solution contains the maximum quantity of milk?
(A) Q (B) R
(C) S (D) T

15. How many pairs of solutions can be selected such that if their entire volumes are mixed, the concentration of the resulting solution will be more than 50%?
(A) 8 (B) 9
(C) 10 (D) 11

Directions for questions 16 to 20: Answer these questions based on the information given below.

In the recent past India has witnessed a mass transition to ATM enabled services. The following table shows the costs incurred by any bank, in rupees per transaction, when a customer of that bank uses an ATM of any of the banks.

For example, when an SBI customer uses an SBI ATM, then the cost incurred by SBI is ₹3 per transaction. When he uses an ICICI ATM, the cost incurred by SBI is ₹10 per transaction and when he uses a UTI ATM, the cost incurred by SBI is ₹12 per transaction.

(Transaction costs* in ₹)

	ATM of									
Customer	**SBI**	**ICICI**	**UTI**	**HDFC**	**PNB**	**IDBI**	**GTB**	**Citi Bank**	**HSBC**	**Corp Bank**
SBI	3	10	12	6	8	15	20	6	15	20
ICICI	12	2	21	7	9	19	11	14	17	22
UTI	16	10	5	14	10	14	9	17	19	18
HDFC	7	12	16	6	11	18	16	21	8	14
PNB	8	10	14	20	4	16	11	5	18	8
IDBI	22	11	12	8	17	8	12	7	11	14
GTB	13	8	12	13	22	17	11	25	13	20
Citi Bank	11	9	12	14	16	22	16	8	24	13
HSBC	28	16	17	22	22	23	17	18	12	15
Corporation Bank	14	14	17	10	17	27	14	13	11	9

* Transaction costs from January 2003 to June 2003.

From the month of July 2003 onwards, the banks in the above list formed two alliances, namely STAR and JUMBO, where HDFC and PNB are part of the STAR alliance while the remaining banks belong to the JUMBO alliance. Members of the STAR alliance get a 25% discount on transaction costs incurred for all transactions between themselves and similarly members of JUMBO alliance get a 20% discount on all transactions between themselves.

Transaction costs between members belonging to STAR alliance and those belonging to JUMBO alliance continue as per the table above (i.e., without any discounts).

16. If in a particular month in 2003, the customers of SBI bank made a total of 2 million ATM transactions, of which 75% were transacted through its SBI's own ATM's and the remaining through HDFC ATM's, then what is the total expenditure (in millions) incurred on ATM transactions by SBI in that month?

17. After the alliances are formed, for how many banks is it profitable to use the ATM's of another bank instead of its own?

18. If in June 2003, HSBC incurred a total expenditure of ₹252 lakh on account of its customers transacting through ATM's and of this, 50% was on account of transactions through HSBC ATM's, then what is the least possible number of ATM transactions (in lakh) by the customers of HSBC in June 2003?

19. If in the month of March 2003, all the banks had 10 million transactions each through their own ATM's, then which bank would incur the maximum additional expenditure in case its ATM's went out of operation for that month and its customers transacted through the ATM's of the next cheapest alternative bank?

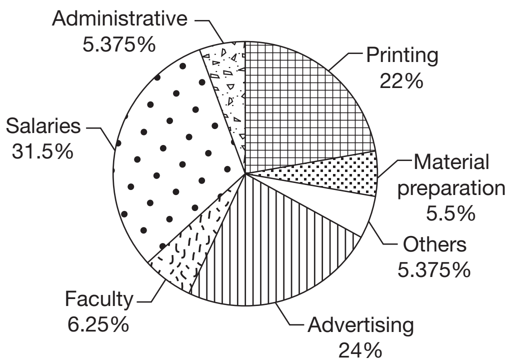
20. After the alliances were formed, for how many banks is the cost incurred per transaction, when it's customer uses the ATM of any other bank at least five and at most 18.

Directions for questions 21 to 25: These questions are based on the following information.

After every question there are two statements I and II. Mark your answer as Choice (A) if statement I alone is true. Choice (B) if statement II alone is true. Choice (C) if both statements are true. Choice (D) if neither of the statements is true.

Details of expenses of Excellent Management Training Institute (EMTI) are given in the pie chart and the table below.

Expenses of EMTI for 2015–16



Percentage increase in expenses from 2014–15 to 2015–16

Faculty	25%
Advertising	20%
Material preparation	10%
Printing	25%
Administrative	7.5%
Salaries	5%
Others	7.5%

Note: It is known that the expenses towards faculty in 2015–16 were ₹12.5 lakh.

21. I. The total expenses incurred towards advertising in 2014–15 were ₹10 lakh.
II. The total expenses incurred towards 'Others' in 2015–16 were ₹10.75 lakh.

22. I. In 2014–15, the total expenses incurred towards faculty and that incurred towards material preparation were equal.
II. In 2015–16, the total expenses incurred were ₹200 lakh.

23. I. The expenses incurred towards printing in 2015–16 are 4.4 times the expenses incurred towards material preparation in 2014–15.
II. The total expenses incurred in 2014–15 were ₹169.40 lakh.

24. I. The ratio of the expenses incurred towards administrative expenses in 2015–16 to that in 2014–15 is 47 : 40.
II. The increase in the total expenditure from 2014–15 to 2015-16 is more than 10%.

25. I. The total expenses incurred towards salaries in both the years put together are ₹126 lakh.
II. The total expenses incurred towards material preparation in both the years put together are ₹2.1 lakh.

Directions for questions 26 to 30: Answer these questions based on the information given below.

The following table gives the percentage share in the turnover of a business group by all the four companies, such as A, B, C and D of the group, for the period from 2003 to 2008. The table also gives the total turnover (in ₹ crore) of the group in these years.

	2003	2004	2005	2006	2007	2008
A	31	36	32	34	29	27
B	18	24	26	27	26	24
C	22	15	19	21	23	25
D	29	25	23	18	22	24
Total	850	1020	1165	1245	1380	1425

26. What was the percentage increase in the turnover of Company A from 2005 to 2006?
(A) 10.4% (B) 12.1% (C) 13.5% (D) 15.2%

27. How many of the companies of the group had an increase in turnover from 2003 to 2004?
(A) 0 (B) 1 (C) 2 (D) 3

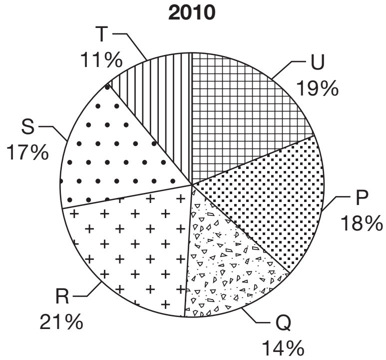
28. Which company had the highest percentage increase in turnover from 2003 to 2008?
(A) A (B) B (C) C (D) D

29. In which of the following years was the percentage increase in the turnover, over the previous year of all the four companies put together, the highest?
(A) 2004 (B) 2005 (C) 2006 (D) 2007

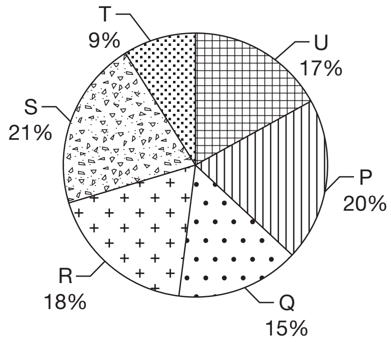
30. Which company of the group had an increase in turnover, when compared to the previous year in each year from 2004 to 2008?
(A) Only A (B) Only B
(C) Only D (D) None of the companies

Directions for questions 31 to 34: The following pie charts highlights the details of sales and expenses of six units, such as P, Q, R, S, T and U of a company across two years.

2010

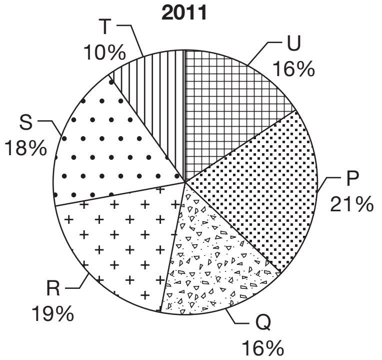


Total sales = ₹ 4800 crore

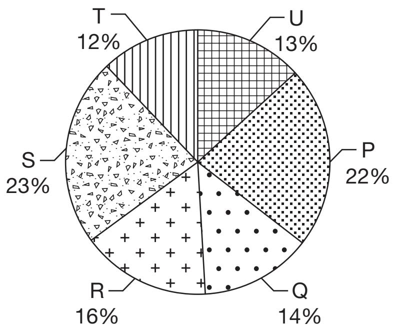


Total expenses = ₹ 3650 crore

2011



Total sales = ₹ 5600 crore



Total expenses = ₹ 4200 crore

$$\text{Profitability (\%)} = \frac{\text{Profit}}{\text{Expenses}} \times 100$$

31. Which unit of the company had the highest percentage increase in sales from 2010 to 2011?

(A) P (B) Q
(C) R (D) T

32. The highest percentage increase in profits for any unit from 2010 to 2011 is

(A) 131.6% (B) 138.3%
(C) 147.4% (D) 162.5%

33. Which unit of the company had the highest profitability in 2010?

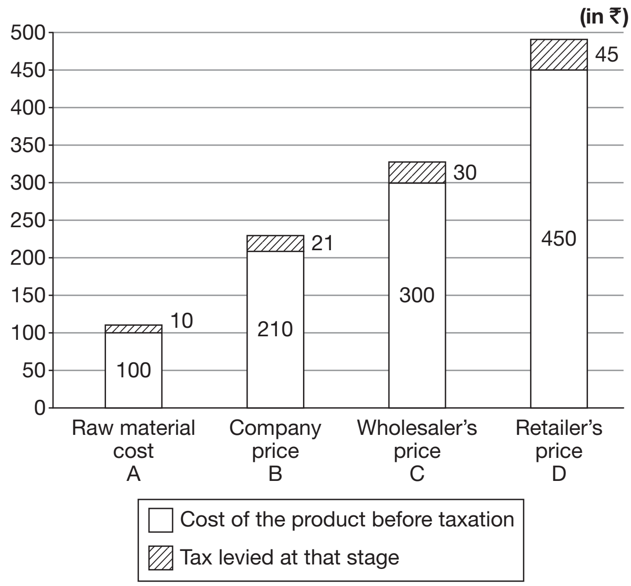
(A) U (B) T
(C) R (D) Q

34. How many units of the company had a profitability more than 50 in 2011?

(A) 5 (B) 4
(C) 3 (D) 2

Directions for questions 35 to 38: These questions are based on the following graph, which gives a particular system of taxation applicable upon a certain product.

Price of a product at various stages in the supply-chain



Note: (i) The company buys the product from the raw material supplier at the 'Raw material cost' and then sells the product to the wholesale dealer at the 'Company price'. The retailer buys the product from the wholesale dealer at the 'wholesaler's price' and then sells it to the customers at the 'Retailer's price'.

(ii) The rate at a stage =

$$\frac{\text{Tax levied at the stage}}{\text{Cost of product before taxation}} \times 100\%$$

The above taxation system is replaced by a new system of taxation in which only the 'value added' in any particular stage is taxed by the corresponding tax rate.

Value added in a stage = [cost of product before tax in that stage – cost of product after tax in the previous stage (according to new taxation)]. There will be no change in the tax at the first stage and in each of the other stages the price of the product will be reduced by the cumulative amount that is saved in all the earlier stages as well as that stage due to the new system of taxation. So that the value added in each of the stages remained same as that in the old taxation.

35. What is the least value added to the product in any stage (in ₹)?

36. What is the cost of the product after tax at the end of stage C as per the new system of taxation?

37. A retailer had bought a product when the old system of taxation was in place. After two days the new system of taxation came into act and he had to sell the same product at the new price. What is the approximate amount that the retailer would have saved, on each unit of the product if he waited for another two days before buying the product?

38. What is the final price that a customer needs to pay as per the new system of taxation?

Directions for questions 39 to 42: These questions are based on the information given below.

Indian Thermal Power Corporation (ITPC) has nine thermal power plants setup at various locations, such as in A, B, C, D, E, F, G, H, and I. The utilization factor is the ratio of the actual electricity generated (output) by a power plant to its installed capacity. The net capacity factor of ITPC is the ratio of net electricity generated (net output) by all the plants to the total installed capacity. The following table gives details about the installed capacity and the utilization factor of four of the nine plants in 2014.

Power Plant	Installed capacity (in MW)	Utilization factor
A	1200	0.6
B	600	0.8
C	1100	0.9
D	800	0.7

The following information is also given:

I. Plant B has the lowest installed capacity and plant A has the highest capacity.
II. Plants E and G have the same installed capacity whereas every other plant has a distinct installed capacity which is a multiple of 100 MW.
III. The average utilization factor of plants E and G is the same as that of plant B and the total output of plants E and G together is the same as that of plants B and D together.
IV. The installed capacity as well as the output of plant I is more than that of F, whose capacity as well as output is more than that of H.
V. The utilization factors of the five plants which are not mentioned in the table above are 0.4, 0.5, 0.5, 0.7 and 0.9.
VI. The average utilization factor of plants E, G and H is 0.66.

39. Find the net capacity factor of ITPC.
(A) 0.60 (B) 0.62
(C) 0.63 (D) 0.66

40. What is the percentage contribution of plants H and C in the net output?
(A) 16.7% (B) 23.33%
(C) 26.67% (D) 25.3%

41. In 2015, the average capacity utilization of plants A, C and E is 0.82 and there is no change in their installed capacity. What is the change in the net output of these plants over 2014?
(A) 254 MW (B) 384 MW
(C) 124 MW (D) Cannot be determined

42. What is the installed capacity of Plant F?
(A) 700 MW
(B) 900 MW
(C) 1000 MW
(D) Cannot be determined

Directions for questions 43 to 46: Answer these questions based on the information given below.

The following table gives the production and demand of a product from the year 2008 to 2015.

Year	Production (P) (in units)	Demand (D) (in units)
2008	500	500
2009	350	440
2010	400	400
2011	280	380
2012	480	510
2013	600	500
2014	600	600
2015	550	480

Every year a certain percentage of the production in that year is kept as storage(s) for the next year's use. The surplus, if any, is exported and additional quantity, if required, then it is imported.

43. If the storage is 20% every year and the export in 2008 was 10 units, then the production in 2007 was
(A) 750 units (B) 550 units
(C) 600 units (D) 650 units

44. From 2009 to 2015, in how many years was the product definitely exported or imported?
(A) 1 (B) 2
(C) 3 (D) 4

45. What should be the storage percentage at which there is neither import nor export in the year 2013?
(A) 83.33 (B) 66.66
(C) 50 (D) 25

46. If the storage is 10% in each of the years, then between the years 2009 and 2015, the exports in year A is equal to the imports in year B. A + B =
(A) 4010
(B) 4008
(C) 4009
(D) More than one of the above

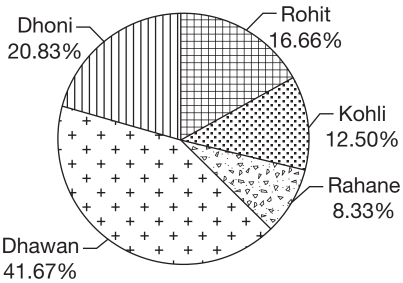
Directions for questions 47 to 50: Answer the questions based on the information given below.

India scored less than 1000 runs in a test match and most of the runs were scored by five batsmen, namely Rohit, Dhawan, Dhoni, Kohli and Rahane. Given below are the charts showing the distribution of the total runs scored by these batsmen and the runs scored by them in 4's and 6's.

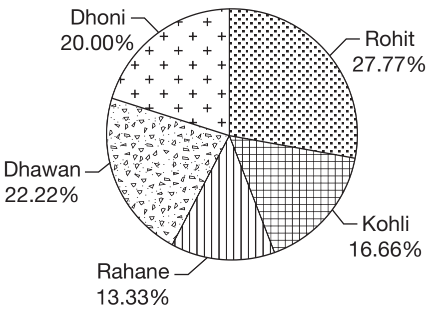
Further it is given that these players scored 77.77% of their combined total score in 4's and 6's and scored 90% of the total runs scored by India in that match.

Distribution of the total runs scored by the five batsmen

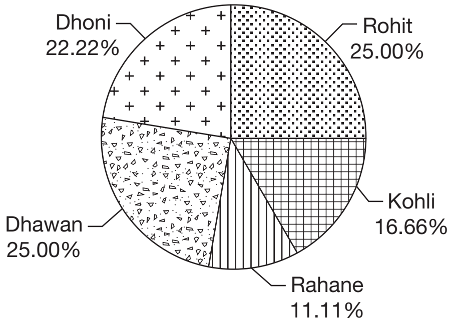
Runs scored in 60's



Runs scored in 4's



Total runs scored in the match



The remaining runs scored by these players were in 1's and 2's.

47. What per cent of the total runs scored by Rahane was in 4's and 6's?
(A) 37.5%
(B) 62.5%
(C) 75%
(D) 83.33%

48. The total number of 4's and 6's hit by Dhawan was
(A) 10 (B) 15
(C) 25 (D) 30

49. By what per cent is the total runs scored by Dhawan in 6's more/less than the total runs scored by Rohit in 4's?
(A) Less by 40%
(B) Less by 20%
(C) More by 25%
(D) Less by 20.83%

50. Kohli scored what percentage of his total score in 1's and 2's?
(A) 15%
(B) 20%
(C) 20.83%
(D) 27.77%

EXERCISE-3

Directions for questions 1 to 3: Answer these questions on the basis of the information given below.

Based on the diagram given below five terms, such as A, B, C, D and E are defined as follows:

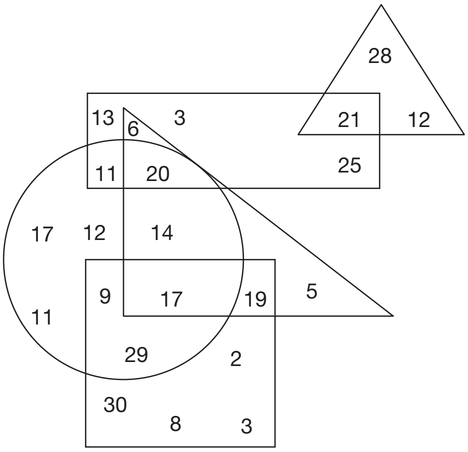
A → Sum of all the numbers inside the equilateral triangle.
B → Sum of all the numbers inside the rectangle.
C → Sum of all the numbers inside the right-angled triangle.
D → Sum of all the numbers inside the square.
E → Sum of all the numbers inside the circle.

The heights of five friends, namely Abhinav, Bindia, Chetan, Dravid and Enosh are as follows:

1. Abhinav's height is 3D – 3A.
2. Bindia's height is E divided by the only number inside the right-angled triangle that is not part of any other figure, then it is multiplied by the smallest number in the square.
3. Chetan's height is the sum of all the numbers that are part of exactly three figures.

4. Dravid's height is the sum of all the numbers that are part of at least two figures.

5. Enosh's height is the sum of all the numbers which are greater than eight and lie inside the square or the rectangle.



1. What is the difference in the heights of Bindia and Abhinav?
(A) 109 (B) 112
(C) 176 (D) 134

2. Who is the tallest among the five?
(A) Bindia (B) Chetan
(C) Enosh (D) Abhinav

3. If the height of Suman, Enosh's cousin, is the sum of all the numbers that are part of at least three figures, what is the difference in the heights of Dravid and Suman?
(A) 90 (B) 83
(C) 72 (D) None of these

Directions for questions 4 to 7: Answer these questions on the basis of the information given below.

The following table gives the percentage distribution of the runs scored by four cricketers in the first five matches of their career.

Match	Gaurav	Sheru	Tenchin	Drahul
1st	25	24	20	18
2d	18	22	26	15
3rd	20	24	21	22
4th	25	14	26	23
5th	12	16	7	22

Further, it is known that,

(1) The runs scored by Gaurav in his 2nd match was not more than that scored by Drahul in his 2nd Match.

(2) The runs scored by Tenchin in his 3rd match was not less than that scored by Sheru in his 3rd match.

(3) In the fifth match, the runs scored by Tenchin was not more than half of that scored by Gaurav.

4. If the number of runs scored by the given players in their fourth matches are compared, then who had the highest score?
(A) Gaurav (B) Sheru
(C) Tenchin (D) Drahul

5. If Drahul scored 126 runs in his first match, then the total runs scored by Tenchin in his first five matches is at most
(A) 350 (B) 400
(C) 450 (D) 500

6. The ratio of the runs scored by Gaurav in his second match to the runs scored by Sheru in his third match was at least
(A) 0.96 (B) 1
(C) 1.1 (D) 1.2

7. Among all the given players, the highest score made in a match in their first five matches was by
(A) Gaurav (B) Sheru
(C) Tenchin (D) Drahul

Directions for questions 8 to 12: Answer these questions on the basis of the information given below.

A total of eight teams, namely P, Q, R, S, T, U, V and W take part in a hockey tournament. In the first stage, the teams are divided into two groups of four teams each. Each team in a group played exactly one match with every other team in its group. The tournament is scheduled such that two group matches take place every day and all the group matches are over in six days. Three points are awarded for a win and one point for a draw. The following table gives the points of each team at the end of each of the six days.

Team	P	Q	R	S	T	U	V	W
Day 1	3	1	0	0	0	0	1	0
Day 2	3	1	0	3	0	1	1	1
Day 3	4	1	0	3	3	1	1	2
Day 4	4	1	0	4	4	4	1	2
Day 5	4	1	0	7	4	4	1	5
Day 6	4	2	0	7	5	7	1	5

It is known that no team played on three consecutive days and that team Q played on Day 2.

8. Which of the following teams are in one group?
(A) P, V, R, W (B) P, R, T, S
(C) P, R, U, W (D) None of these

9. Which of the following teams played on Day 3?
(A) T and R (B) T and V
(C) T and U (D) Cannot be determined

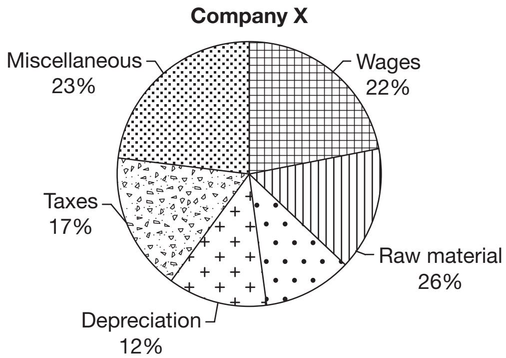
10. Which of the following teams lost a match on Day 5?
(A) V (B) P
(C) Q (D) T

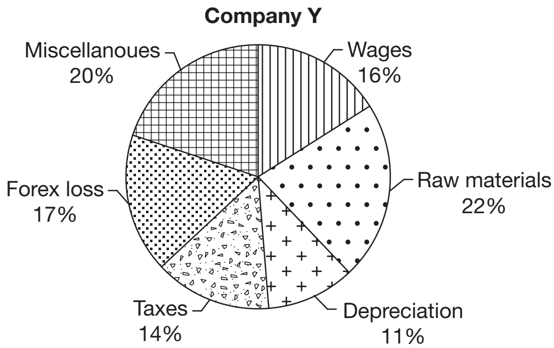
11. Which of the following teams played a match on Day 5?
(A) T, W (B) Q, V
(C) W, P (D) W, R

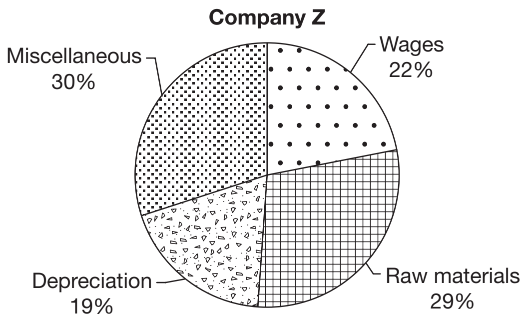
12. Which team did U beat on Day 4?
(A) P
(B) V
(C) R
(D) Cannot be determined

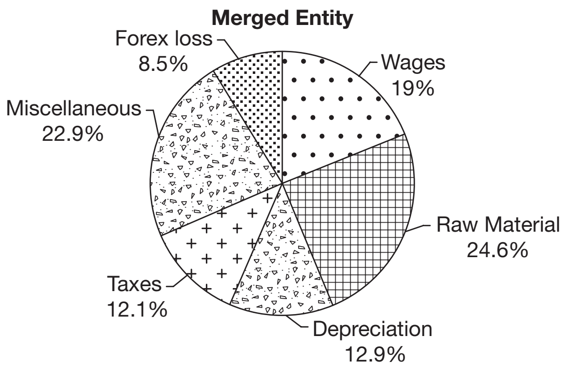
Directions for questions 13 to 16: Answer the questions on the basis of the information given below.

The Agenta group was a diversified business house which had three companies X, Y and Z. As part of restructuring, the group decided to merge all the three companies. The following pie charts give the break-up of expenses of each company and that of the merged entity in the year 2007.









13. What is the ratio of the total expenses of companies X and Y?
(A) 3 : 2 (B) 2 : 5
(C) 3 : 5 (D) 2 : 3

14. The forex loses of company Y was what percentage of the depreciation expenses of Company Z?
(A) 152% (B) 186%
(C) 205% (D) 223%

15. The wage bill of which of the three companies was the highest?
(A) X (B) Y
(C) Z (D) Both X and Y

16. The expenses of Company Z was under quoted by 20% because the taxes paid by it was not included in the given diagrams. If this figure is also included, then the taxes paid would account for what percentage of the expenses of the merged entity?
(A) 23.2% (B) 21.5%
(C) 18.6% (D) None of these

Difficulty Level-3: Advanced

Directions for questions 17 to 20: Answer these questions on the basis of the information given below.

The following are the runs scored by the players of the Indian cricket term in a match. The names are given in the order in which they came out to bat.

1.	Rahul	–	11
2.	Dhawan	–	27
3.	Pujara	–	35
4.	Kohli	–	53
5.	Rahane	–	6
6.	Pandya	–	27
7.	Saha	–	5
8.	Jadeja	–	39
9.	Ashwin	–	28
10.	Umesh	–	11
11.	Kuldip	–	5

The remaining runs in the team's total score are extras, which are not credited in the name of any batsman. It is also known that no batsman stopped his innings in between or retired hurt. The values below give the runs at which India lost its wickets, the fall of wickets occurred during 22, 54, 98, 121, 146, 157, 193, 212, 231, 252.

17. Who was the third batsmen to be out?
(A) Dhawan (B) Pujara
(C) Kohli (D) Cannot be determined

18. Who was the last batsmen to be out?
(A) Ashwin
(B) Umesh
(C) Kuldip
(D) Cannot be determined

19. By the time Pandya was out, how many wickets had India lost?
(A) 5 (B) 6
(C) 7 (D) 8

20. If during none of the partnerships India got more than two extra runs, who was the second last batsman to be out?
(A) Jadeja
(B) Ashwin
(C) Umesh
(D) Cannot be determined

ANSWER KEYS

Exercise-1

1. (D)	**10.** 207500	**19.** (D)	**28.** (B)	**37.** (A)	**46.** (C)
2. (A)	**11.** (C)	**20.** (C)	**29.** (B)	**38.** (B)	**47.** (B)
3. (D)	**12.** (D)	**21.** (C)	**30.** (A)	**39.** (D)	**48.** (B)
4. (D)	**13.** (C)	**22.** (A)	**31.** (B)	**40.** (B)	**49.** (B)
5. (B)	**14.** (D)	**23.** (D)	**32.** (C)	**41.** (B)	**50.** (D)
6. 3	**15.** (D)	**24.** (C)	**33.** (A)	**42.** (C)	
7. Gupta	**16.** (A)	**25.** (D)	**34.** (C)	**43.** (D)	
8. 55000	**17.** (C)	**26.** (B)	**35.** (D)	**44.** (D)	
9. 4080	**18.** (C)	**27.** (A)	**36.** (B)	**45.** (C)	

Exercise-2

1. 180	**10.** (D)	**19.** ICICI	**28.** (B)	**37.** 34	**46.** (A)
2. (C)	**11.** (D)	**20.** 3	**29.** (A)	**38.** 428	**47.** (D)
3. 216	**12.** (A)	**21.** (B)	**30.** (D)	**39.** (D)	**48.** (D)
4. (C)	**13.** (C)	**22.** (C)	**31.** (A)	**40.** (D)	**49.** (A)
5. 5.1	**14.** (B)	**23.** (A)	**32.** (C)	**41.** (D)	**50.** (D)
6. (B)	**15.** (C)	**24.** (B)	**33.** (B)	**42.** (D)	
7. (C)	**16.** 7.5	**25.** (D)	**34.** (C)	**43.** (B)	
8. (B)	**17.** 4	**26.** (C)	**35.** 69	**44.** (C)	
9. (A)	**18.** 15	**27.** (D)	**36.** 296	**45.** (A)	

Exercise-3

1. (B)	**4.** (D)	**7.** (D)	**10.** (A)	**13.** (C)	**16.** (D)	**19.** (C)
2. (C)	**5.** (D)	**8.** (C)	**11.** (D)	**14.** (D)	**17.** (B)	**20.** (C)
3. (D)	**6.** (B)	**9.** (B)	**12.** (D)	**15.** (B)	**18.** (D)	

SOLUTIONS

EXERCISE-1

1. The total tonnage of fish caught in the inland waters in 1998 = (Total quantity of fish caught in 1998) – (Total quantity of fish caught in the sea in 1998)
= (10,229) – (947 + 6164) = 3118 tons.

2. The tonnage of fish caught in the inland waters during 1999 = 61 + 129 + 297 = 487.
Similarly, the tonnage in 2000 = 516, in 2001 = 519 and in 2002= 589
There has been an increase in every year, statement (I) is true. Statement (I) is maintained in Choice (A) as well as Choice (D). By checking statement (IV), the quantity of fish caught, using the traditional method has decreased successively in two years.
$\therefore$ IV cannot be inferred.
Only statement (I) can be inferred.

3. During the given years, the total quantity of fish caught by the modern methods
= 5841 + 2463 + 11997 + 44668 = 64,969 ≅ 65,000 The quantity of fish caught by traditional methods = 374 + 731 + 1930 + 7675 = 10,710 ≅ 11,000; $\frac{65,000}{11,000} \cong 6$

4. The percentage increase in the quantity (tonnage) of fish caught over previous year in 1999 = $\frac{684}{10,229}$
In 1998 = $\frac{1262}{8,967}$, In 2000 = $\frac{864}{10,913}$,
in 2001 = $\frac{590}{11,777}$ and in 1997 = $\frac{759}{8208}$
Among the given fractions, in the fraction pertaining to 2001, the denominator is the greatest and the numerator is the least. Hence, the value of this fraction is the least.

5. Number of fish caught = $\frac{\text{Quantity of fish}}{\text{Average weight}}$
Number of fish caught in artificial tanks
$= \frac{374+5841}{2} = \frac{6215}{2} = 3017 \times 5$
In lakes = $\frac{731+2463}{3}$ =1064
In rivers = $\frac{1930+11997}{4}$ = 3481.
In seas = $\frac{7675+44668}{5} = \frac{52343}{5}$ = 10468.6
The number of fish caught in the lakes is the least.

6. Income tax paid by Sharma = (30/100) × 4.5
= 1.35 > 1 lakh
Income tax paid by Rao = (25/100) × 3.5
= 1/4 × 3.5 < 1 lakh
Income tax paid by Gupta = (25/100) × 3 < 1 lakh
Income tax paid by Solkar = (30/100) × 40 > 1 lakh
Income tax paid by Kuchroo = (35/100) × 5 > 1 lakh
There are three such persons.

7. The ratio of savings to expenditure for the persons are as follows:
Sharma $= \frac{1.5}{3} = \frac{1}{2}$
Rao $= \frac{1}{2.5} = \frac{2}{5}$
Solkar $= \frac{1.5}{2.5} = \frac{3}{5}$
Gupta $= \frac{0.5}{2.5} = \frac{1}{5}$
Kuchroo $= \frac{1.5}{3.5} = \frac{3}{5}$
We can observe that Gupta has the least ratio.

8. 35% of Gupta's Income = (35/100) × 3 lakh = ₹1.05 lakh
Savings of Gupta currently = ₹50,000
Gupta should save ₹55,000 more to reach the target of 35%.

9. Total investments in PPF by the five persons
= 50 + 50 + 25 + 50 + 80 = ₹2.55 lakh = ₹2,55,000
Interest payable per annum
= (8/100) × 2,55,000 = ₹20,400
Average interest payable per person = 20,400/5 = ₹4080

10. The two doctors are Sharma and Solkar.
Sharma's income = ₹4.50 lakh
PPF = ₹50000 or ₹0.50 lakh
Taxable income = 4.5 – 0.5 = ₹4.00 lakh
Tax payable = (30/100)₹ 4 = ₹1.20 lakh
Solkar's taxable income = 4.00 – 0.50 = ₹3.50 lakh
Tax payable = 25/100 × 3.50 = ₹87,500
Total tax payable by these two doctors
= 1,20,000 + 87,500 = ₹2,07,500

11. Number of female postgraduates
$= \frac{20}{100}.\frac{36}{100}.5000 = \frac{20}{100}.1800 = 360$
Number of unmarried female postgraduates
$= \frac{90}{100}.360 = 324$

12. Number of graduates = Number of female graduates + Number of male graduates = $\left(\frac{80}{100}\right)\left(\frac{36}{100}\right)(5000)$ + $\left(\frac{75}{100}\right)\left(\frac{64}{100}\right)(5000)$

$= \left(\frac{28.8}{100}\right)(5000) + \left(\frac{48}{100}\right)(5000) = \left(\frac{76.8}{100}\right)(5000)$

∴ 76.8% of the students of the college are graduates.

13. Number of male postgraduates $= \frac{25}{100}\frac{64}{100}.5000$

$= \frac{16}{100}5000 = 800$

Number of male postgraduates who are at least 30 years old = 496

Number of male postgraduates aged below 30 years = 800 – 496 = 304

14. Number of postgraduates = Number of male postgraduates + Number of female postgraduates

$= \frac{25}{100}\frac{64}{100}5000 + \frac{20}{100}\frac{36}{100}.5000$

$= \frac{16}{100}5000 + \frac{7.2}{100}5000 = \frac{23.2}{100}5000$

Required % $= \dfrac{\frac{23.2}{100}\times 5000}{\frac{76.8}{100}\times 5000}\times 100\% = 30.2\%$

15. The maximum difference occurred between the number of qualified candidates in 2005 and 2006.

16. The number of candidates who appeared from Chennai in 2004 $= \frac{4224\times 100}{100-89} = \frac{4224\times 100}{11}$

(∵ 4224 candidates qualified in 2004, 89% did not qualify)

∴ 11% qualified.

Also, the number of candidates who appeared from Chennai in 2006, 2007, 2008 and 2010 were $\frac{4626\times 100}{10}$, $\frac{3150\times 100}{6}$, $\frac{7280\times 100}{13}$ and $\frac{9280\times 100}{10}$, respectively.

The maximum of all these $= \frac{9280\times 100}{10}$

(∴ From all the above values, $\frac{9280\times 100}{10}$ has the highest numerator and proportionate denominator. Therefore, it is the highest of the values).

∴ Highest value occurred in 2010.

17. In 2009, 80% did not qualify.

∴ 20% qualified

20% = 11800

80% = 4(11800) = 47200

18. In 2007, 81% did not qualify.

∴ 19% qualified

19% = 13490

1% = 710

Number appeared = 100% = 71000

19. The total costs per unit of the different companies are as follows.

A = 10 + 15 + 20 = ₹45	B = 15 + 10 + 20 = ₹45
C = 5 + 10 + 15 = ₹30	D = 20 + 10 + 15 = ₹45
E = 25 + 15 + 20 = ₹60	F = 25 + 5 + 30 = ₹60

20. Sales income of the company:

B = 5000 × 68 = ₹3,40,000

Total cost of production of B for 5000 units

= 5000 × 45 = ₹2,25,000

∴ Profit = ₹1,15,000

21. If we observe the total cost of all the companies, then total costs of A and D are ₹45 and ₹45, respectively.

∴ Ratio of total costs = 1 : 1

Similarly, for E and F, the ratio of total costs = 1 : 1

∴ Statement C is correct.

22. Profit of company D for one-day production

= 1000 (60 – 45) = ₹1,50,000

Profit of company F for one-day production

= 800 (80 – 60) = ₹1,60,000

The ratio will be the same whether it is calculated for one day or for ten days.

∴ Ratio of profits = 15 : 16

23. The labour cost of B and C is same, say x.

∴ Total cost of B is $x + 20 + 10 = x + 30$

Total cost of C is $x + 15 + 10 = x + 25$

∴ The ratio of the total cost is $x + 30 = x + 25$

The solution cannot be determined.

24. During the year 2003, the total number of garments manufactured by A = 168 + 295 + 332 + 100 = 895.

Similarly, those manufactured by

B = 473, C = 798 and D = 1069.

Hence, D > A > C > B

Alternate method:

By observation, we can find that the total number of garments is the greatest for company D, and the next highest is for A.

25. The percentage increase in the number of garments manufactured by company C for female segment for the appropriate periods is as follows:

Second quarter $= \frac{59-35}{35} = \frac{24}{35}$

Third quarter $= \frac{68-59}{59} = \frac{9}{59}$

Fourth quarter $= \frac{75-68}{68} = \frac{7}{68}$

$\frac{7}{68}$ is the least as the numerator is the smallest and the denominator is the greatest.

Note: First quarter need not be considered since the number of garments for the previous quarter is not given.

26. If the total number of garments manufactured in 2002 was 100, then the total number of garments manufactured in 2003 was 125. Since the total number of garments manufactured in 2003 is 3235, those manufactured in 2002 = $\frac{100}{125} \times 3235$= 2588 thousands.

27. The ratio of the total number of garments manufactured for the male segment to that of female segment
= (168 + 103 + 180 + 227): (295 + 125 + 237 + 321)
= 678 : 978 = 113 : 163

28. Only for company B there is a growth in every segment for every quarter.

Solutions for questions 29 to 32: Let us calculate the total number of votes cast in different constituencies in different years.

Year	A	B	C	D	E
1990	$60{,}000 \times \frac{45}{100} = 27{,}000$	$65{,}000 \times \frac{60}{100} = 39{,}000$	$80{,}000 \times \frac{75}{100} = 60{,}000$	$70{,}000 \times \frac{40}{100} = 28{,}000$	$60{,}000 \times \frac{55}{100} = 33{,}000$
1994	$70{,}000 \times \frac{50}{100} = 35{,}000$	$72{,}000 \times \frac{64}{100} = 46{,}080$	$85{,}000 \times \frac{80}{100} = 68{,}000$	$80{,}000 \times \frac{60}{100} = 48{,}000$	$65{,}000 \times \frac{70}{100} = 45{,}500$
1998	$75000 \times \frac{65}{100} = 48{,}750$	$80{,}000 \times \frac{85}{100} = 68{,}000$	$90{,}000 \times \frac{60}{100} = 54{,}000$	$86{,}000 \times \frac{50}{100} = 43{,}000$	$75{,}000 \times \frac{80}{100} = 60{,}000$
2002	$78000 \times \frac{70}{100} = 54{,}600$	$84000 \times \frac{72}{100} = 60{,}480$	$98000 \times \frac{65}{100} = 63{,}700$	$100000 \times \frac{45}{100} = 45{,}000$	$104000 \times \frac{60}{100} = 62{,}400$

29. Required answer =
48750 + 68000 + 54000 + 43000 + 60000, i.e., 2,73,750.

30. In 1994, the required average number of votes cast
= (35000 + 46080 + 68000 + 48000 + 45500) ÷ 5 = 48,516.

31. In 2002, 464000 – 286180 = 1,77,820 people did not cast their vote.

32. Required ratio = $\frac{164000}{200900} = \frac{1640}{2009}$.

33. Trains A, B, C, D covered 240, 380, 290 and 341 km between 16:00 hrs. and 22:00 hrs. Hence, the average speed of Train A was the least during this period.

34. At 16:00 hrs
Train A is 600 km away from starting point.
Train B is 460 km away from starting point.
Train C is 530 km away from starting point.
Train D is 437.5 km away from starting point.
∴ Trains A and D are 162.5 km apart which is the farthest distance.

35. B crosses A at 840 kms at around 22:00 hrs.
C crosses A at 840 kms at around 22:00 hrs.
B crosses C at 840 kms at around 22:00 hrs.
Similarly, trains B, C and D could have crossed train A anytime between 12:00 and 16:00 before train A crossed them again and took the lead.
Hence, there are more than two cases possible.

36. The total stoppage times of Train A
= (6:10 to 6:30) + (19:00 to 19:15) = 35 minutes
Total distance = 1000 km
Total time = 05:00 hrs to 01:30 hrs – 35 minutes
= 19 hours 55 minutes ≃ 20 hours

Hence, the average speed = $\frac{1000}{20}$ = 50 kmph

(Slightly more than 50, actually)

37. In region P, the total number of readers = 52 million.
The number of readers reading newspaper B

$$= \frac{40}{100} \times 52 \text{ million}$$

$$= \frac{2}{5} \times 52 \text{ million}$$

$$= 20.8 \text{ million}$$

38. The number of readers reading newspaper A in region

$$Q = \frac{25}{100} \times 50 \quad \rightarrow \quad (1)$$

The number of readers reading newspaper B in region

$$S = \frac{30}{100} \times 60 \quad \rightarrow \quad (2)$$

The required ratio = 25 × 50 : 30 × 60 = 25 : 36

39. The required number of readers

$$= \frac{10}{100} \times 52 + \frac{30}{100} \times 50 + \frac{40}{100} \times 45 + \frac{20}{100} \times 60$$

$= 5.2 + 15 + 18 + 12 = 50.2$ million

40. The total readers of B

$$= \frac{40}{100} \times 52 + \frac{25}{100} \times 50 + \frac{30}{100} \times 45 + \frac{30}{100} \times 60$$

$= 20.8 + 12.5 + 13.5 + 18 = 64.8$

The total readers of C = 50.2

The required ratio = 64.8 : 50.2 = 324 : 251

41. The required percentage is more than 90%, in the countries Mauritius, South Africa, Guyana, Jamaica, Suriname, Trinidad and Tobago, Afghanistan, France, Netherlands, Yemen and Fiji, i.e., a total of 11 countries.

42. The countries are Netherlands, Kuwait, Oman and Yemen.

43. The given condition is satisfied for a country only when PIOs as a percentage of the country's population is more than the value of PIOs as a percentage of total PIOs. This happened for 17 countries, namely for Mauritius, Guyana, Jamaica, Suriname, Trinidad and Tobago, Bhutan, Singapore, France, Germany, Bahrain, Kuwait, Libya, Oman, Qatar, UAE, Yemen and Fiji.

44. PIO 's (ASIA)

$= 46 + 70 + 330 + 1170 + 3900 + 100 + 1023 + 65 = 6704$

PIO 's (America)

$= 229 + 300 + 39 + 140 + 430 + 500 = 1638$

$$\therefore \text{ Required } \% = \frac{6704}{1638} - 1 \ (100) = 309.2\%$$

45. Since the answer choices are not close, we can approximate the values.

Urban population in 1901 ≈ 26 million

Rural population in 1901 ≈ eight times the urban population

∴ Rural population = 26 × 8 = 208 million

In 1991, urban population ≈ 215

In 1991, rural population = three times urban population

∴ Rural population in 1991 = 215 × 3 = 645% increase =

$$\frac{645 - 208}{208} \times 100 = 210$$

46. The approximate population in the years are as follows:

$$1941: \frac{44.0}{19} \times 100 = 315$$

$$1951: \frac{62 \cdot 5}{17 \cdot 6} \times 100 = 355$$

$$1961: \frac{79}{18 \cdot 3} \times 100 = 430$$

$$1971: \frac{109}{20 \cdot 2} \times 100 = 540$$

$$1981: \frac{162}{23 \cdot 7} \times 100 = 685$$

$$1991: \frac{218}{25 \cdot 7} \times 100 = 847$$

Among the years mentioned only for 1961–71 and 1971–81 there is more than 25% increase

$$= \frac{540 - 430}{430} \times 100 = 25 \cdot 5\% \text{ and } \frac{685 - 540}{540} \times 100 \approx 27\%$$

47. Closely observing the values for urban population and the percentage of urban population in the total population we can see that only for the period 1911–1921, the actual increase in urban population

$= \frac{28 - 25 \cdot 9}{25 \cdot 9} \times 100 = 8$ which is less than the percentage increase in the percentage of urban population

$= \frac{11 \cdot 3 - 10 \cdot 4}{10 \cdot 4} \times 100 \approx 8 \cdot 5\%$ which means there is a decrease in the base, i.e., total population.

48. The marks scored by Antony in Maths $= \frac{20}{100} \times 370 = 74.$

The marks scored by Bill in Maths $= \frac{20}{100} \times 280 = 56$

The marks scored by Clinton in Maths $= \frac{17}{100} \times 380 = 64.6$

The marks scored by Danny in Maths $= \frac{15}{100} \times 365 = 54.75$

The marks scored by Ellen in Maths $= \frac{18}{100} \times 375 = 67.5$

The marks scored by Francis in Maths $= \frac{30}{100} \times 360 = 108$

The marks Goutham scored in Maths $= \frac{22}{100} \times 390 = 85.8$

Francis scored the highest in Maths.

49. The marks scored by Ellen in English $= \frac{37}{100} \times 375$ $= 138.75$

The marks scored by Goutham in English $= \frac{23}{100} \times 390$ $= 89.7$

The required percentage $= \frac{138.75 - 89.7}{89.7} \times 100 = 54.6\%$

50. The total marks scored in Physics by all the students

$$= \frac{20}{100} \times 370 + \frac{20}{100} \times 280 + \frac{20}{100} \times 380 + \frac{18}{100} \times 365 + \frac{15}{100} \times 375$$

$$+ \frac{30}{100} \times 360 + \frac{30}{100} \times 390$$

$= 74 + 56 + 76 + 65.7 + 56.25 + 108 + 117$

$= 553$

EXERCISE-2

1. Total earnings = EPS × Total number of shares
The number of shares of Company A has increased from 12 × 5L to 17 × 5L (L = lakhs)

$$\frac{17\cdot5-12\cdot5}{12\cdot5}\times100=40\%$$

Assume that Company A had only 10 shares in 2007.
∴ Total earnings = 10 × 6 = ₹60
Now, for the next year, the number of shares increases by 40% that is 10 becomes 14.
Total earnings = 14 × 12 = 168
∴ Required percentage increase

$$=\frac{168-60}{60}\times100=\frac{108}{60}\times100=180\%$$

2. For A it is 180% (from the previous question).
For B, as the number of shares hasn't changed, we can directly calculate the increase in total earnings from the increase in EPS.

$$\text{Increase}=\frac{18-8}{8}\times100=\frac{10}{8}\times100=125\%$$

For C, the number of shares increases from 27 × 0L to 40 × 5 lakh, that is a 50% increase.
Assume that C had only 10 shares in 2007.
Total earnings 10 × 2 = 20
The next year, the number of shares increases by 50%, that is 10 becomes 15.
Total earnings 15 × 6 = 90

$$\text{Increase}=\frac{90-20}{20}\times100=350\%$$

For D, as the number of shares does not change, we can use the same logic as used for Company B and find that the increase is 50%.

3. Share price = EPS × P/E = 18 × 12 = ₹ 216

4. Since the share price = P/E × EPS, the prices of companies on 1st April 2007 and 1st April 2008 can be found as follows.

Company	2007	2008
A	90	96
B	160	216
C	50	96
D	360	432
E	120	72
F	50	36

It is the highest for C.

5. Total earnings of D in 2008:
∴ EPS × The number of shares = 18 × 8× 5L = ₹153 lakh

$$\text{Earnings in 2009}=\frac{90}{100}\times153$$

Earnings of E in 2009:

$$\text{EPS}=\frac{90\times153}{100\times27}=5.1$$

6. For the number of illiterates to be maximum, the literacy rate should be minimum. By observing the data given, it is true for Country B.

7. 55% of 1321 – 58% of 832
Treating 55% as (50% + 5%) and 58% as (60% – 2%)
{660.5 + 66.05} – {499.2 – 16.64}
726.5 – 482.6 = 243.9

8. The second highest number of men is in Country A, as it has the second highest population.

$$\text{Number of women in A}=\frac{970}{1970}\times1321$$

$$\frac{970}{1970}\cong\frac{1}{2}\text{ but less than }\frac{1}{2}$$

$$\therefore\text{ Number of women}<\frac{1}{2}\times1321<660.5$$

Thus, it is slightly less than 660.

9. Even if all the women in Country D is literate, there will only be 91 × 1/2 = 45 lakh literate women in D.
Even if all the men in any other country (say C) are literate, out of a total of 540 × 0.85 = 460 lakh literates, only about 270 lakh (half the total population) can be men (since the male-female ratio is approximately 1) and the remaining 190 lakh will have to be literate women which is definitely higher than the number of literate women in D; thus, D has the least number of literate women.

10. Here, we need literacy rate among men and women separately which is not given.

11. We have to select amongst the solutions Q, R, T and U (as they have higher concentration as compared to others).

$$(R, Q): C=\frac{(80)(34)+(16)(60)}{16+34}$$

$$=\frac{2720+960}{50}=\frac{3680}{50}=73.6$$

$$(R, U): C=\frac{(80)(34)+(10)(70)}{44}$$

$$=\frac{2720+700}{44}=\frac{3420}{44}=77.7$$

Difficulty Level-2: Moderate

(R, T) : C = $\frac{(80)(34)+(60)(5)}{39}$

$= \frac{2720+300}{39} = \frac{3020}{39} = 77.4$

For Q and U, the concentration is less than 70%.
∴ Solutions R and U have to be mixed to get maximum concentration.

12. P + Q + R = X
S + T + U = Y
By observation, X has more concentration than Y.

13. (Concentration) R and S

$$= \frac{(80)(34)+(30)(20)}{34+20} = \frac{2720+600}{54}$$

$$\Rightarrow \frac{3320}{54} = 61.5\%$$

14. The solution which contains the maximum quantity of milk is R since its concentration as well as its total quantity is maximum.

15. Pairs of solutions are (P, Q), (P, R), (P, U)
(Q, R), (Q, T), (Q, U)
(R, S), (R, T), (R, U), (T, U).
A total of 10 pairs of solutions are there.

16. ATM transactions through SBI ATM's

$$= 2 \times \frac{75}{100} = 1.5 \text{ million.}$$

∴ Total expenditure incurred by SBI = 1.5 × 3
= 4.5 million rupees
ATM transactions of SBI through HDFC ATM's = 0.5 million
∴ Total expenditure incurred by SBI = 0.5 × 6
= 3 million rupees
Total cumulative expenditure = 4.5 + 3 = 7.5 million rupees.

17. Let us make a table to represent the information:

Bank	Cost per transaction	Next best alternative	Cost of this alternative	Profitable (Y/N)
SBI	3	HDFC	6	No
ICICI	2	HDFC	7	No
UTI	5	ICICI	$10 \times \frac{80}{100} = 8$	No
HDFC	6	SBI	7	No
PNB	4	Citi Bank	5	No
IDBI	8	Citi Bank	$7 \times \frac{80}{100} = 5.60$	Yes
GTB	11	ICICI	$8 \times \frac{80}{60} = 6.4$	Yes
Citi Bank	8	ICICI	$9 \times \frac{80}{100} = 7.2$	Yes
HSBC	12	Corporation	$15 \times \frac{80}{100} = 12$	No
Corporation	9	HSBC	$11 \times \frac{80}{100} = 8.80$	Yes

It can thus be seen that for four banks it is profitable to use another bank's ATM's after the alliances are formed.

18. Total expenditure of HSBC = 252 lakh
Total expenditure through its own ATM's = 126 lakh
∴ Total transaction through its own ATM's

$$= \frac{\text{Total expendidture}}{\text{Cost per transaction}} = \frac{126}{12} = 10.5 \text{ lakh}$$

For the transactions to be minimum the remaining 126 lakh expenditure must be through an ATM that has the highest cost per transaction for HSBC customers, which is the case of SBI at ₹28 per transaction.

∴ Total transactions through SBI = $\frac{126}{28}$ = 4.5 lakh

Total cumulative transactions = 10.5 + 4.5 = 15 lakh

19. In the month of March 2003, the alliance was not formed were no discounts are possible on member banks transactions, this has to be kept in mind while working out these questions. Let us now form a table and solve the question.

Bank	Cost	Least cost alternative	Additional expenditure per transaction
SBI	3	HDFC – 6	3
ICICI	2	HDFC – 7	5
UTI	5	GTB – 9	4
HDFC	6	SBI – 7	1
PNB	4	Citibank – 5	1
IDBI	8	Citibank – 7	*Saving of ₹1 *
GTB	11	ICICI – 8	* Saving of ₹3 *
Citi Bank	8	ICICI – 9	1
HSBC	12	Corporation – 15	3
Corporation Bank	9	HDFC – 10	1

Since transactions (ATM) are the same for all banks, ICICI will have the maximum additional expenditure.

20. Only for ICICI, PNB and IDBI was the cost of transaction at least five and at most 18.

Solutions for questions 21 to 25:

Given that the expenses towards faculty
= 6.25% of total expenses = ₹12.5 lakh
Total expenses = 12.5 × 100 / 6.25 = ₹200 lakh
Tabulating the expenses in lakh of rupees.

	2015–16	2014–15
Faculty	12.5	12.5/1.25 = 10.0
Advertising	48.00	48/1.2 = 40.0
Material preparation	11.00	11/1.1 = 10.0
Printing	44.00	44/1.25 = 35.20
Administrative	10.75	10.75/1.075 = 10.0
Salaries	63.00	63/1.05 = 60
Others	10.75	10.75/1.075 = 10
Total	200	175.20

21. Statement I is wrong because the expenses towards advertisement in 2014-15 were ₹40 lakh.
As per the table, Statement II is correct.

22. As per the table, Statements I and II are true.

23. As per the table, the ratio of printing expenses in 1995–96 to material preparation in 2014–15 = 44/10 = 4.4
Thus, statement I is true.
As per the table, total expenses in 2014–15 are not ₹169.40 lakh.
Thus, statement II is wrong.

24. As per the table, the ratio of amounts incurred towards the administrative expenses in 2015–16 and 2014–15
= 10.75: 10 = 1075 : 1000 = 43 : 40
Thus, statement I is wrong.
The percentage increase in total expenditure from 2014-15 to 2015–16 = (200/175.20 – 1) × 100 = 14.1%
Thus, statement II is correct.

25. As per the table, the total expenses towards salaries are 60 + 63 = ₹123 lakh
Thus, statement I is wrong.
The total expenditure towards material preparation in both the years = 11 + 10 = ₹21 lakh
Therefore, both the statements are wrong.

26. The turnover of Company A in 2005 = $\frac{32}{100} \times 1165$ = 372.8
The turnover of Company A in 2006 = $\frac{34}{100} \times 1245 = 423.3$
The percentage increase = $\frac{50.5}{372.8} \times 100 = 13.5\%$

27. The turnover of the group increased by 20% from 2003 to 2004. If the percentage share in 2003 is less than 120% of that in 2004, there will be an increase in turnover from 2003 to 2004, i.e., for companies A, B and D.

28. To find the company with the highest percentage increase, we need to find only the company which had the highest percentage increase in share in the group turnover, i.e., Company B.

29. The turnover of all the four companies put together is the group turnover and the highest percentage increase in the group turnover was from 2003 to 2004.

30. None of the companies had an increase in turnover over the previous year in each year from 2004 to 2008.

Solutions for questions 31 to 34: The sales, expenses and profits of the different units in 2010 and 2011 are as follows.

2010

Units	Sales (₹ crore)	Expenses (₹ crore)	Profit (₹ crore)	Profitability
P	864	730	134	14.3%
Q	672	547.5	124.5	22.7%
R	1008	657	351	53.4%
S	816	766.5	49.5	6.6%
T	528	328.5	199.5	60.9%
U	912	620.5	291.5	47.0%

2011

Units	Sales (₹crore)	Expenses (₹crore)	Profit (₹crore)	Profitability
P	1176	924	252	27.3%
Q	896	588	308	52.4%
R	1064	672	392	58.0%
S	1008	966	42	4.0%
T	560	504	56	11.1%
U	896	546	350	64.2%

31. We need to consider only P or Q as they are the only companies to have more than 25% increase in sales.

For P it is $= \frac{1176-864}{864} \times 100 = \frac{312}{864} \times 100 = 36\%$

For Q it is $\frac{896-672}{672} \times 100 = \frac{224}{672} \times 100 = 33.3\%$

Therefore, the highest is for P.

32. The highest percentage increase in profit would be for Q which is $\frac{308-124.5}{124.5} \times 100 = \frac{183.5}{124.5} \times 100 = 147.4\%$

33. In 2010, the profitability was the highest for unit T.

34. Units Q, R and U had a profitability more than 50% in 2011.

Solutions for questions 35 to 38: From the given data:

Stage	A	B	C	D
Total cost after tax	110	231	330	495
Total cost before tax	100	210	300	450
Value added	100	100	69	120
Tax rate	10%	10%	10%	10%

According to new system of taxation, only 'value added' is to be taxed.

Stage	A	B	C	D
Total cost after tax	110	220	295.9	428
Total cost before tax	100	210	289	416
Value added	100	100	69	120
Rate	10%	10%	10%	10%
Tax on value added	10/-	10/-	6.9/-	12/-

35. In stage 'C', the least value is added, i.e., ₹69.

36. From the above table, it is ₹295.9.

37. The operator in stage D had to buy from stage C operators.
In the old system, he paid ₹330.
In the new system he pays ₹295.9.
Thus, if he follows the old system, he will pay
330 – 295.9 = ₹34.1 more.

38. From the table, final cost = 427.9

39. The lowest installed capacity of a plant is 600 MW and the highest capacity is 1200 MW. As it is mentioned that every plant except E and G have distinct capacities which are multiples of 100 MW, of the remaining five plants three of them will have installed capacities of 700, 900 and 1000 MW.

Power plant	Capacity (in MW)	Utilization factor	Output
A	1,200	0.6	720
B	600	0.8	480
C	1,100	0.9	990
D	800	0.7	560
	700		
	900		
	1,000		
	X		
	X		

As the total output of plants E and G is same as that of plants B and D (1040 MW) and their average utilization factor is same as that of B (0.8), installed capacity of E and G together = 1040/0.8 = 1300 MW
∴ Installed capacity of plants E and G is 650 MW each.

Power plant	Capacity (in MW)	Utilization factor	Output
A	1200	0.6	720
B	600	0.8	480
C	1100	0.9	990
D	800	0.7	560
	700		
	900		
	1000		
E	650		
G	650		

As their installed capacities are the same, plants E and G must have utilization factors of 0.7 and 0.9 so that the average will be 0.8. But we don't know which one has a utilization factor of 0.7. Since the installed capacity of I, F and H are in a descending order, we know their respective installed capacities.

Power plant	Installed capacity (in MWs)	Utilization factor	Output
A	1200	0.6	720
B	600	0.8	480
C	1100	0.9	990
D	800	0.7	560
E	650	0.7/0.9	455/585
F	900		
G	650	0.9/0.7	585/455
H	700		
I	1000		

The installed capacity of E, G and H together is 2000. If their combined utilization factor is 0.66, their output must be (2000) (0.66), i.e., 1320 MW. Since we already know that the output of E and G together is 1040, the output of H is 280 MW. Since plant H's capacity and output are known, its utilization factor

= 280/700 = 0.4.

The utilization factor of the only two remaining plants, F and I will be 0.5 each.

Power plant	Installed capacity (in MWs)	Utilization factor	Output
A	1,200	0.6	720
B	600	0.8	480
C	1,100	0.9	990
D	800	0.7	560
E	650	0.7/0.9	455/585
F	900	0.5	450
G	650	0.9/0.7	585/455
H	700	0.4	280
I	1,000	0.5	500
Total	7,600		5,020

Net capacity factor = Total output / Total installed capacity = 5020/7600 = 0.66

40. Contribution of plants H and C = [(990 + 280)/ 5020] × 100 = 25.3%

41. The question cannot be answered as the utilization factor of E cannot be uniquely determined (it can be 0.7 or 0.9).

42. The installed capacity of Plant F is 900 MW.

43. Export in 2008 = P – D + $S_{2007} - S_{2008}$

$= 500 - 500 + S_{2007} - \frac{20}{100}(500)$

$= S_{2007} - 100 = S_{2007} - 100$

But it is given to be 10 units.

$\therefore S_{2007} = 110$

$\Rightarrow \quad \frac{20}{100}(P_{2007}) = 110$

$\Rightarrow \quad P_{2007} = 550$ units

44. Let x be the storage percentage.

Export in 2009 $= 350 - 440 + \frac{x}{100}(500 - 350)$

If there is neither export nor import, export = 0

$\therefore 0 = 350 - 440 + \frac{x}{100} 150$

$x = 60\%$

∴ When storage percentage is 60, there is neither export nor import.

But in 2010 the export $= 400 + \frac{x}{100}(350 - 400)$

The value can never be zero. Hence, there is an import. Similarly, there is either import or export in 2012 and 2015.

∴ There are three such instances.

45. Export $= 600 - 500 + \frac{x}{100}(480 - 600)$

$= 100 + \frac{x}{100}(-120)$

But there is neither an import nor an export.

$\therefore 0 = 100 + \frac{x}{100}(-120)$

$\Rightarrow x = \frac{10000}{120} = 83.33$

46. The export (E) or import (I) will be as follows:

Year	P	D	E	I
2009	350	440	–	75
2010	400	400	–	5
2011	280	380	–	88
2012	480	510	–	50
2013	600	500	88	–
2014	600	600	–	–
2015	550	480	75	0

The imports in 2009 are equal to the exports in 2015.
∴ 2009 + 2015 = 4010
But the imports in 2011 are equal to the exports in 2013.
∴ 2011 + 2013 = 4010

Solutions for questions 47 to 50:

6's		
Kohli	12.5%	$\frac{1}{8}$
Rohit	16.66%	$\frac{1}{6}$
Dhoni	20.83%	$\frac{5}{24}$
Dhawan	41.67%	$\frac{5}{12}$
Rahane	8.33%	$\frac{1}{12}$

4's		
Kohli	16.66%	$\frac{1}{6}$
Rohit	27.77%	$\frac{5}{18}$
Dhoni	20%	$\frac{1}{5}$
Dhawan	22.22%	$\frac{2}{9}$
Rahane	13.33%	$\frac{2}{15}$

Let the number of 4's and 6's be 90 K_1 and 24 K_2, respectively.
Choice total runs scored in 4's = 90 $K_1 \times 4$
Total runs scored in 6's = 24 $K_2 \times 6$

For K_1, K_2 = 1
Runs scored 4's 6's by the 5 bats men: 90×4 24×6
They scored 77.77% of their total runs in 4's and 6's.
360 + 144 = 504
Their total score = $\frac{9}{7}$ (504) = 648 runs.

Now this is 90% of India's total score.
$\frac{9}{10}$ (India's total Score) = 648
Therefore, India's total score = 720
We can check that no other combinations satisfy the given conditions.

Runs scored				
Batsmen	**4's**	**6's**	**In 1's and 2's**	**Total runs**
Kohli	60 (15)	18 (3)	30	108
Rohit	100 (25)	24 (4)	38	162
Dhoni	72 (18)	30 (5)	42	144
Dhawan	80 (20)	60 (10)	22	160
Rahane	48 (12)	12 (2)	12	72

47. Rahane scored 48 + 12 = 60 runs in 4's and 6's whereas his total score was 72.
$$\frac{62}{72} \times 100 = 83.33\%$$

48. The total number of 4's and 6's hit by Dhawan was 30.
20 boundaries and 10 over boundaries.
20 + 10 = 30

49. The total runs scored by Dhawan in 6's = 60
Total runs scored by Rohit in 4's = 100
Now, 60 is less than 100 by
$$\frac{40}{100} \times 100 = 40\%$$

50. Kohli scored 30 runs out of a total of 108 runs in 1's and 2's.
Now, $\frac{30}{108} \times 100 = 27.77\%$

EXERCISE-3

Solutions for questions 1 to 3:

Value of A is = 28 + 21 + 12 = 61
Value of B is = 13 + 11 + 6 + 20 + 3 + 25 + 21 = 99
Value of C is = 6 + 20 + 14 + 17 + 5 + 19 = 81
Value of D is = 9 + 29 + 17 + 2 + 17 + 30 + 8 + 3 = 117
Value of E is = 17 + 11 + 12 + 9 + 29 + 17 + 14 + 20 + 11 = 140
Abhinav's height = 3 × 117 – 3 × 61 = 168

Bindia's height = E ÷ Only no. inside the right-angled triangle X
The smallest number in the square
Bindia's height = 140 ÷ 5 × 2 = 56
Chetan's height = 20 + 17 = 37
Dravid's height = 21 + 6 + 11 + 20 + 14 + 17 + 9 + 29 + 19 + 30 = 194
Enosh's height = 13 + 11 + 20 + 21 + 25 + 9 + 17 + 29 + 19 + 30 = 194

1. Difference in the heights of Bindia and Abhinav is 168 – 56 = 112.

2. The tallest person is Enosh.

3. Suman's height = 20 + 17 = 37
   Dravids height = 146
   ∴ Difference = 146 – 37 = 109.

Solutions for questions 4 to 7: Let the total runs scored by Gaurav, Sheru, Tenchin and Drahul in their first five matches be G, S, T and D, respectively.

From (1), we get: $\frac{18}{100} G \le \frac{15}{100} D$

$\Rightarrow \frac{D}{G} \ge \frac{6}{5} \quad \rightarrow \quad (1)$

From (2), we get: $\frac{21}{100} T \ge \frac{24}{100} S \Rightarrow \frac{T}{S} \ge \frac{8}{7} \quad \rightarrow \quad (2)$

From (3), we get: $\frac{7}{100} T \le \frac{1}{1}\frac{12}{100} G$

$\Rightarrow \frac{G}{T} \ge \frac{7}{6} \quad \rightarrow \quad (3)$

4. $\frac{23}{100} D \ge \frac{23}{100}\frac{6}{5} G \ge \frac{27.6}{100} G$
   But Gaurav scored only 25% of the total runs in his fourth match. Hence, Drahul scored more than Gaurav in the fourth match. Similarly, Drahul scored more than Sheru and Tenchin, and hence, he is the highest scorer.

5. Given $\frac{18}{100} D = 126$
   $\Rightarrow D = 700$
   $\frac{D}{T} = \frac{D}{G}\frac{G}{T} \ge \frac{6}{5}\frac{7}{6}$
   $\Rightarrow T \le \frac{5}{7} D$
   $\Rightarrow T \le 500$

6. Required ratio = $\dfrac{\frac{18}{100} G}{\frac{24}{100} S}$
   $= \frac{3G}{4S} = \frac{3}{4}\frac{G}{T}\frac{T}{S} \ge 1$
   The ratio is at least 1.

7. For Drahul the highest score in any match is $\frac{23}{100} D$
   $\ge \frac{23}{100}\frac{6}{5} G$
   (i.e., 27.6%)
   $\ge \frac{23}{100}\frac{7}{5} T$
   (i.e., 32.2%)
   $\ge \frac{23}{100}\frac{8}{5} S$
   (i.e., 36.8%)
   But none of G, T and S had more than these percentages as scores in any match.
   ∴ Drahul scored the highest.

Solutions for questions 8 to 12: On day 1, Q and V both scored one point each. So, they would have drawn their match. On day 6, Q again drew with T and on day 4, T drew with S.

∴ Q, V, T and S are in one group and P, R, W and U are in the other group.
The matches on the different days which can be directly obtained from the table are:

Day 1	P	Q – V
Day 2	U – W	S
Day 3	P – W	T
Day 4	U	S – T
Day 5	W	S
Day 6	U	Q – T

Since U scored 7 points, it did not lose to P. As P won on Day 1 and played W on Day 3, P would have won against R on Day 1. U would have won against P or R on Day 4 and the other team (R or P) on Day 6.

In the second group as T had played on Day 4 and Day 6, it beat V on Day 3 and Q cannot play on Day 3 as it had already played on Day 1 and Day 2. The final table would be as follows:

Day 1	P – R	Q – V
Day 2	U – W	S – Q
Day 3	P – W	T – V
Day 4	U – P/R	S –T
Day 5	W – R	S –V
Day 6	U – R/P	Q – T

8. P, R, U and W are in one group.

9. T and V played on Day 3.

10. R and V lost on Day 5.

11. Among the given teams, W and R played on Day 5.

12. U could have won against P or R on Day 4.

Solutions for questions 13 to 16: It can be seen that the expenses on account of forex loses is only for Company Y. As it is 17% of the total expenses of Company Y and 8.5% of the merged entity, the expenses of Company Y is 50% of the total expenses of the merged entity. As the expense under taxes are only for companies X and Y, the taxes paid by Company Y would account for $\frac{14}{2}$ = 7% of the taxes paid by the merged entity while the remaining 5.1% of the taxes paid was due to Company X.

17% of X = 5.1% of $(x + y + z)$

$\therefore$ x = 30% of $(x + y + z)$

$\therefore$ Ratio of expenses of X, Y and Z = 3 : 5 : 2.

13. Ratio of expenses of companies X and Y = 3 : 5

14. Forex loses of Company Y = 8.5% of total

depreciation expenses of Company Z = $19 \times \frac{2}{100}$ = 3.8% of total

Required ratio = $\frac{8.5}{3.8} \times 100 = 223\%$

15. As the total expenses of the companies are in the ratio 3 : 5 : 2, the wage bill of Company Y would be the highest as 16% of 5 > 22% of 3 > 22% of 2.

16. Assume the expenses of X, Y and Z are ₹300, ₹500 and ₹200, respectively.

$\therefore$ Taxes paid by company Z = ₹50

Share of taxes paid = $\frac{171}{1050} \times 100 = 16.3\%$

Solutions for questions 17 to 20: As the name of the players are given in the order in which they came to bat, Rahul and Dhawan are the openers. As the first wicket fell at the score of 22, it has to be Rahul as Dhawan scored 27 runs. The next wicket has to be Dhawan as 32 runs were scored between the fall of the first and the second wickets and Pujara scored 35 runs. Using the same logic, the third wicket to fall was that of Pujara. When the fourth wicket fell, India's score was 121. As the first three batsmen scored 11 + 27 + 35 = 73 runs and Kohli scored 53 runs, had Kohli been out, India's score had to be at least 73 + 53 = 126 runs. Therefore, Rahane was the fourth batsman out. Similarly, Kohli was the fifth to be out. Now, the batsmen who were out have scored 11 + 27 + 35 + 53 + 6 = 132 runs. As the next wicket fell at 157, it was Saha (6th wicket). The seventh wicket was that of Pandya and the eighth Jadeja. The next wicket (9th wicket) can be Ashwin or Umesh. Had Ashwin been the 9th batsman out, the scores of all players till that point adds up to 231, which means all the 5 extra runs in India's innings happened during the last wicket partnership.

17. The third batsman to be out was Pujara.

18. The last batsmen to be out cannot be determined.

19. By the time Pandya was out, India had lost seven wickets.

20. In the case mentioned, Umesh was the 10th batsman to be out.

Challenge Your Understanding
Practice Set 1

Directions for questions 1 to 4: Answer these questions on the basis of the information given below.

The table below gives the process of manufacture of four different products A, B, C and D, each of which has to pass through all the four machines 1, 2, 3 and 4.

Machine \ Product	A	B	C	D
1	(6, 1)	(4, 1)	(3, 1)	(6, 2)
2	(8, 2)	(5, 1)	(6, 2)	(4, 1)
3	(3, 1)	(4, 2)	(6, 1)	(5, 1)
4	(5, 1)	(4, 1)	(5, 1)	(6, 2)

Each product should pass through machines 1, 2, 3 and 4 in that order, before it is ready. For each machine the first figure in bracket indicates the quantity (in units) of the corresponding product made, while the second indicates the time taken to produce that quantity.

For example, machine 1 can produce 6 units of product D in 2 hours or 1 unit every 20 minutes.

Each day only one type of product can be made and each day the working time is from 9.30 a.m. to 5.30 p.m. All machines are operated in such a way that any unit that is started on a day is finished by the end of that day. The selling prices and the percentage profit on each of the products is as given below.

Product	A	B	C	D
Selling price	275	504	234	345
Profit percentage *	10	12	30	15

* as a percentage of cost price.
All machines operate simultaneously.

1. What is the maximum number of units of any product that can be manufactured in a single day?

2. If the maximum revenue is to be realised on a day, then which product should be manufactured on that day? (All the quantity that is manufactured is sold on that same day)

3. To get the maximum profit on a single day which of the following should be produced?

4. Considering the day on which the maximum profit is earned, the total idle time of all the machines put together is (in mins)

Directions for questions 5 to 8: Answer these questions on the basis of the information given below.

Dandia Times, a national daily, decided to conduct a poll regarding the qualities which the people expect their Prime Minister to have. For this, Dandia Times surveyed exactly 100 respondents from each of the six major cities, namely Delhi, Mumbai, Kolkata, Chennai, Bangalore and Hyderabad.

In each city mentioned, each of the 100 respondents was given a response sheet listing out the five qualities that the Prime Minister was expected to have. Each respondent could mark one or more of the five given qualities that he expected in the Prime Minister, otherwise the respondent should mark a special option provided, which read 'None of the five qualities'. The following table gives, for each of the six cities, the number of people who expected each of the five given qualities in the Prime Minister.

Cities \ Qualities	Integrity	Political experience	Academic qualification	Public speaking skills	Leadership qualities
Delhi	35	63	35	48	26
Mumbai	42	32	42	25	27
Kolkata	55	26	38	37	48
Chennai	62	18	83	41	33
Bangalore	47	15	41	28	25
Hyderabad	51	29	47	39	36

For example, of the 100 persons surveyed in Delhi, the table shows that 35 persons expect the quality of 'integrity' in the Prime Minister.

5. Among the people surveyed in all the six cities put together, what is the maximum number of people who expected all the five listed qualities in the Prime Minister?

6. At least how many of the people surveyed in Bangalore expected a minimum of two of the five listed qualities in the Prime Minister?

7. If in Hyderabad, none of the persons surveyed expected more than three of the five listed qualities in the Prime Minister, then what is the minimum number of people surveyed in Hyderabad who expected exactly three of the five qualities in the Prime Minister?

8. At most how many of the people surveyed in Delhi and Chennai, put together, expected at most two of the five listed qualities in the Prime Minister?

Directions for questions 9 to 12: These questions are based on the following data.

Twelve teams took part in a football tournament, which is conducted in three stages. In the first stage the teams are divided into two groups of six teams each. The teams within a group play with each other once and the top three teams of each group go to the second stage. In the second stage, the three teams of each group play with each other once and the top two teams from each group then go to the third stage. In this stage, the two teams in each group play with each other and the winners from each group play with each other to decide the winner of the tournament. All games produce results. In case of a draw, a penalty shootout is used to decide the winner. In case of a tie, at the end of any of the first two stages the winner is decided by a set of complex tie breaking rules to ensure that only one team goes into the next round.

9. What is the total number of matches in the tournament?

10. What is the minimum number of games a team should win to ensure that it goes into the second stage?

11. Of all the teams that reached the second stage, what is the minimum number of games a team could have won?

12. If a team gets ₹50,000 for each win in the first stage, ₹1,00,000 in the second stage and ₹1,50,000 in the third stage, find the maximum amount that any team can win.

Directions for questions 13 to 16: Answer these questions on the basis of the information given below.

Mr. Anand is planning to set up a small-scale unit to manufacture water tanks. He can manufacture three types of tanks – 250 L, 500 L and 1000 L tanks. The costs involved are machinery rent and a variable cost which depends on the type and number of tanks made. The rent for machinery for making the 250 L and 500 L tanks is ₹1.2 lakh per month and the variable costs involved are ₹500 and ₹700 for a 250 L tank and a 500 L tank, respectively. For the manufacture of a 1000 L tank, the machinery would cost ₹1.7 lakh per month and the variable cost is ₹900 for a tank. He also has the option of renting machinery for ₹2.5 lakh per month, which can manufacture all the three types of tanks with the variable costs being ₹250, ₹400 and ₹700 for a 250 L, a 500 L and a 1000 L tank, respectively.

13. What is the least cost incurred per tank if Anand has to manufacture 900 tanks of 250 L capacity each month?

14. Anand decides to manufacture only 250 L tanks. What is the minimum number of tanks that he should manufacture so that the machinery which can manufacture all the three types of tanks is the economically viable option?

15. If Mr. Anand manufactures 2,500 units of 1000 L tanks per month, then what is the lowest price (in ₹) at which he should sell each tank to make a profit of 15%?

16. If Mr. Anand sells tanks of 500 L capacity at a price of ₹1300 per unit and makes a profit of ₹200 per unit, then how many 500 L tanks did he sell?

Directions for questions 17 to 19: These questions are based on the information given below.

A family consists of four persons, namely Rama, Jaya (Rama's wife), Hari (Rama's son) and Harini (Hari's wife). Rama's salary and Harini's salary put together is ₹24,000. Rama's salary is 25% of his son's and Harini's salary put together. Harini's salary is ₹15,000 less than Hari's salary. Jaya doesn't work and takes good care of her family and each member of the family rewards her by giving 10% of his/her salary and she donates 10% of what she receives to an old age organization and saves the rest.

17. What is Hari's salary (in ₹)?

18. What is the savings of Jaya (in ₹)?

19. The amount Jaya donates to the old age organization is what percentage of the total salary of all the members of the family?

Directions for questions 20 to 23: Answer these questions on the basis of the information given below.

The famous business school, IIM-P, has six branches, one in each of six different cities, such as in Delhi, Hyderabad, Mumbai, Bangalore, Chennai and Kolkata with an intake of 100 students in each branch. During the placement week, all the 100 students at each of the six branches attended job interviews. The distribution of the number of students with different specialisations and the number of students placed in various sectors is given in the following tables. Table P.1 gives the distribution of the students at each of the branches, based on their specialization and Table P.2 gives the distribution of the students at each of the branches, based on the sector in which they got placed.

Table P.1

Specialization	Branch					
	Delhi	Hyderabad	Mumbai	Bangalore	Chennai	Kolkata
Retail	24	15	10	16	20	20
HR	10	12	14	22	16	10
Information Systems	12	17	25	14	15	22
Finance	24	14	16	10	18	22
Operations	16	22	17	25	14	10
Marketing	14	20	18	13	17	16

Table P.2

Specialization	Branch					
	Delhi	Hyderabad	Mumbai	Bangalore	Chennai	Kolkata
Retail	34	20	22	20	22	27
IT	10	14	14	18	10	14
ITES	10	18	12	10	25	13
Banking	16	16	10	18	18	22
Telecom	18	19	25	14	9	14
Automobile	12	13	17	20	16	10

Further, it is also known that,

(i) All the students who had specialized in Retail, got placed in the Retail sector.

(ii) None of the students, who got placed in the Automobile sector, specialized in marketing.

(iii) All students who had specialized in Information Systems, got placed in either the IT or ITES sector.

(iv) None of the students who got placed in the Banking sector, specialized in either Finance or Operations.

Note: All students had exactly one job offer and each one of them had taken up exactly one specialization.

20. The number of students, at all the branches put together, who specialized in HR but got placed in the Retail sector, is at most

21. The number of students, at all the branches put together, who specialized in Marketing but got placed in the Banking sector, is at least

22. If none of the students who specialized in Finance got placed in the Automobile sector, the number of students, at all the branches put together, who specialized in Operations but got placed in the IT sector, is at most

23. What percentage of students who got placed in IT or ITES sector did not specialize in information systems?

Directions for questions 24 to 27: Answer these questions on the basis of the information given below.

The following table gives the distance (in kms) between 10 cities which Mr. Anand plans to visit. The cities A to J, not necessarily in that order are in a straight line and he plans, to start his journey from his home in city G. The distances given are the distances from the southernmost city C.

	A	B	C	D	E
F	160	280	65	190	305
G	105	225	120	135	250
H	180	60	405	150	35
I	30	150	195	60	175
J	65	55	290	35	80

24. What is the distance between cities G and I (in kms)?

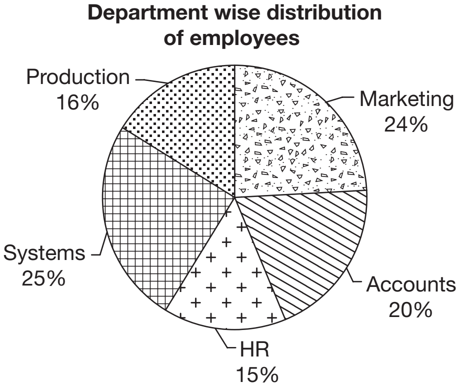
25. The northernmost city among the given ten cities is

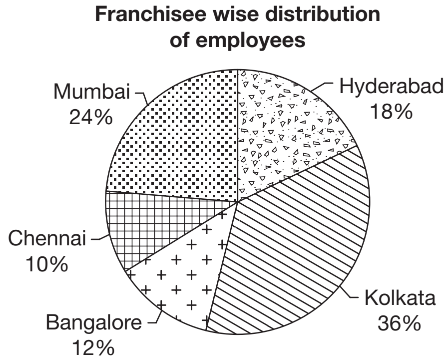
26. If Mr. Anand has to visit all the ten cities, then the total distance he has to travel is (in kms)

27. What is the least distance between any two of the given ten cities (in kms) is?

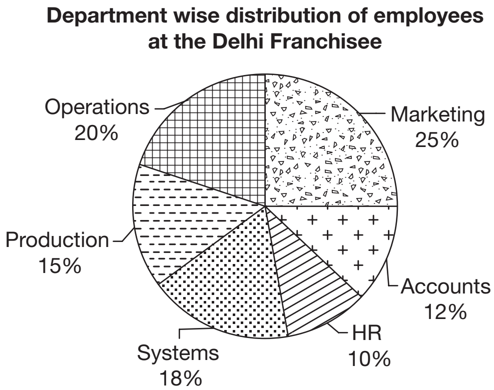
Directions for questions 28 to 30: Answer these questions on the basis of the information given below.

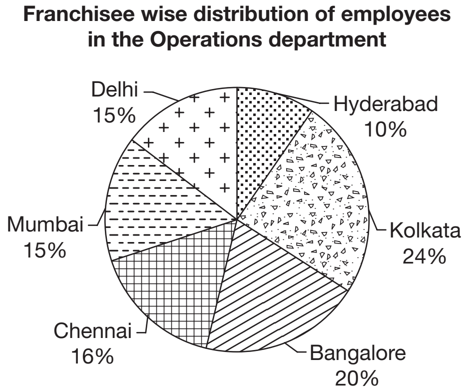
In company XYZ, which has a total of six franchisees, the head of the HR department wanted to find out the department wise as well as the franchisee wise distribution of all the employees in the company. The HR department started the task but as the number of employees working in the Operations department at the Delhi franchisee was not known, the employees of the Operations department at all the franchisees were left out while representing the department wise distribution. Further, all the employees at the Delhi franchisee were left out while representing the franchisee wise distribution of the total employees. The pie charts thus obtained were as follows.





After the exact number of employees in the Operations department of the Delhi franchisee was known, two more pie charts, one for the department wise distribution of the employees at the Delhi franchisee and the other one for the franchisee wise distribution of the employees in the Operations department, were drawn as shown below.





28. If the total number of employees at the Delhi franchisee is the same as that of the Bangalore franchisee, then what percentage of the employees working at the Mumbai franchisee work in the Operations department?

29. If the total number of employees at the Mumbai franchisee forms 15% of the total number of employees at all the franchisees put together, what is the ratio of the number of employees working in the Accounts department at all the franchisees put together to that of the total number of employees working at the Chennai franchisee?

30. If the total number of the employees in the company XYZ is known, then which of the following additional statements can enable one to find the total number of employees in each department?

(1) The number of employees in the Marketing department at the Hyderabad franchisee.
(2) The number of employees in the Systems department at the Chennai franchisee.
(3) The number of employees in the Operations department at the Mumbai franchisee.
(4) The number of employees in the Accounts department at the Chennai franchisee.

Mark your answer as the number of the required statement.

ANSWER KEYS

1. 22	6. 14	11. 2	16. 300	21. 24	26. 535
2. (D)	7. 2	12. 7,50,000	17. 28,500	22. 51	27. 25
3. (C)	8. 184	13. 528	18. 4725	23. 37.5	28. 10
4. 618	9. 39	14. 520	19. 1	24. 75	29. 1.6
5. 139	10. 4	15. 920	20. 35	25. H	30. 3

SOLUTIONS

Solutions for questions 1 to 4:

1. Let us first calculate the time taken to produce a unit of each product.
Product – A – Machine 1 takes 10 minutes.
From the 11th to the end of 25th minute, machine 2 is working on product A.
From the beginning of the 26th minute to the end of 45th minute machine 3 works on product A and for the next 12 minutes machine 4 works on A.
1 unit of product A is completed in 57 minutes.
When machine 2 is working on a unit of product A, machine 1 can operate on the next unit of product A and so on.
After the first unit of product A is completed, every succeeding unit is produced in 20 minutes and this time is determined by the slowest unit in operation. (machine 3 in this case).
So, one unit of product A is manufactured after 57, (57 + 20), (57 + 20 + 20) minutes and so on.
That is 57, 77, 97, …
∴ In 480 minutes, 22 units can be produced.
Similarly, for B, each unit is produced in 72, (72 + 30) minutes and so on.
∴ In 8 hours, 14 units can be completed.
For machine C first unit is produced after 62 mins and each succeeding one after a 20 minutes interval.
∴ In 8 hours, 21 units can be produced.
For machine D each unit is produced in 67, (67 + 20) minutes and so on.
∴ In 8 hours, 21 units can be produced.

2. Revenues per day:
For A – 22 × 275 = 6050
For B – 14 × 504 = 7056
For C – 21 × 234 = 4914 and
For D – 21 × 345 = 7245
'D' realises maximum revenue.

3. The profit for each unit is as follows:
A – ₹25
B – ₹54
C – ₹54 and
D – ₹45
Profit for A – 25 × 22
Profit for B – 54 × 14
For C – 54 × 21 and
For D – 45 × 21
By observation we can say that the profit is maximum for C.

4. For maximum profit, product C has to be manufactured.
For 21 units of C, machine I needs to operate for 21 × 20 = 420 minute.
Idle time = 480 – 420 = 60 minutes
For 2, idle time = 60 minutes.
For 3, idle time = 270 minutes.
For 4, idle time = 228 minutes.
Total = 60 + 60 + 270 + 228 = 618 minutes.

Solutions for questions 5 to 8:

5. The maximum number of people who expected all the five listed qualities in the Prime Minister is at most 26 (Delhi) + 25 (Mumbai) + 26 (Kolkatta) + 18 (Chennai) + 15 (Bangalore) + 29 (Hyderabad) = 139.

6. The number of people who expected a minimum of two of the five listed qualities occurs when the number of people expecting exactly one quality and the rest expect all the five qualities.
Total in Bangalore = 47 + 15 + 41 + 28 + 25 = 156
If x people expected exactly one quality and y people expected all the five qualities.
$x + y = 100$ and $x + 5y = 156$
∴ $x = 86$ and $y = 14$.

7. As no person expected more than three qualities, to find the minimum number of people who expected exactly three of the five qualities occurs when the number of people expecting exactly two qualities is the maximum.
Total instances in Hyderabad
= 51 + 29 + 47 + 39 + 36 = 202.
If 'a' people expected exactly two qualities and 'b' people expected exactly three qualities,

Practice Set 1

$a + b = 100$ and $2a + 3b = 202$
$\therefore a = 98$ and $b = 2$
At least two people in Hyderabad expected exactly three of the five listed qualities in the Prime Minister.

8. The people who expected at most two of the listed qualities would be maximum when a maximum number of people expect two qualities and the rest, all the five qualities.
Total instances in Delhi = 207.
If 'a' persons expects two qualities and 'b' persons expect five qualities,
$a + b = 100$ and $2a + 5b \geq 207$.
The maximum value of 'a' is 97.
Similarly, in Chennai, $a + b = 100$ and $2a + 5b \geq 237$
The maximum value of 'a' is 87.
$\therefore$ Maximum persons at both the places combined
= 97 + 87 = 184.

Solutions for questions 9 to 12:

9. There are 30 matches in the first stage, six matches in the second stage and three matches in the next stage. A total of 39 matches.

10. The wins of different teams can be as follows. The teams are arranged in descending order of the number of wins. If a team wins 4 matches, there can't be 3 other teams with a better performance.

5	4	3	2	1	0
4	4	4	2	1	0
4	4	3	2	2	0
4	4	3	2	1	1
4	4	2	2	2	1
4	3	3	3	2	0
4	3	3	3	1	1
4	3	3	2	2	1
4	3	2	2	2	2
3	3	3	3	3	0
3	3	3	3	2	1

From the last row we see that there are five teams with three wins each. So, two teams with three wins will get eliminated. So, three wins are not enough.

11. If the top two teams win 5 and 4 matches, then out of the remaining 6 match results, a team which wins 2 matches can reach the second stage.

12. The top team can win at most 5 games in the first stage, 2 in the second and 2 in the third.
$\therefore$ The amount won by the top team in rupees
= (5 × 50,000) + (2 × 1,00,000) + (2 × 1,50,000)
= 7,50,000

Solutions for questions 13 to 16:

13. The cost incurred would be
(a) 1.2 lakhs + 900 × 500 = 5.7 lakhs
(b) 2.5 lakhs + 900 × 250 = 4.75 lakhs
The least cost incurred per tank would be $\frac{4.75 \text{ lakhs}}{900}$
= ₹528

14. If he uses the machinery which costs 2.5 lakhs per month, he would incur an additional cost of ₹1.3 lakhs but his variable cost would be less by ₹250 per tank.
$\therefore$ Minimum tanks that has to be manufactured = $\frac{1.3 \text{ lakhs}}{250} = 520$.

15. For manufacturing 2500 tanks, the cost would be minimum if he rents the machinery which costs ₹2.5 lakhs. The cost incurred would be = 2.5 lakhs + 700 × 2500 = 20 lakhs.
$\therefore$ Amount per tank = $\frac{20 \text{ lakhs}}{2500} = 800$.
For a 15% profit he should sell it at ₹920.

16. The cost price of Mr. Anand = ₹1100 per tank.
In case he uses the first machinery, if x tanks are manufactured, 1.2 lakhs + $700x = 1100x$
$\therefore x = 300$
If he uses the second machinery, if x tanks are manufactured, 2.5 lakhs + $560x = 1100x$
$700x$ = 2.5 lakhs
x = (not an integer).

Solutions for questions 17 to 19:

17. Let the salary of Rama, Hari and Harini be R, A and H, respectively.
R + H = 24000 (1)
R = 1/4(A + H)
⇒ 4R = A + H (2)
A – H = 15000 (3)
Solving (1), (2) and (3), we get
A = ₹28,500.

18. A = ₹28,500, H = ₹13,500 and R = ₹10,500
Savings of Jaya (Mrs. Rama):
$$\left(\frac{90}{100}\right)\left(\frac{1}{10}\right)[(28{,}500 + 13{,}500 + 10{,}500)] = 4725$$

19. The amount Jaya donates is 10% of 10% of the salary of all the members of the family, i.e., 1%.

Solutions for questions 20 to 23:

20. Let us consider the HR students at each of the locations. As every Delhi student with a specialization in Retail got a job in Retail sector, 24 students who got jobs in Retail sector must have their specialization as Retail. Now of the 16 persons who got jobs in Banking sector, at most 14 can be from Marketing and the remaining 2 must be from

HR. Hence, a maximum of 8 persons of HR specialization got job in Retail sector.
Similarly,
Hyderabad = 5 students (i.e., 20 – 15 = 5)
Mumbai = 12 students (i.e., 22 – 10)
Bangalore = 4(20 – 16)
Chennai = 2 (i.e., 22 – 20)
Kolkata = As only 16 of the 22 students placed in Banking sector can be with Marketing specialization, remaining 6 must be with specialization in HR.
∴ Of the total 10 students placed in HR, only four can get the jobs in Retail sector.
∴ Total = 35

21. The minimum number of persons who specialized in Marketing and got a job in Banking sector at different locations is as follows.
(i) Delhi = 6
(ii) Hyderabad = 4
(iii) Mumbai = 0
(iv) Bangalore = 0
(v) Chennai = 2
(vi) Kolkata = 12
Therefore, there is a total of 24 students.

22. The required number of students at different locations is
Delhi = 8
Hyderabad = 14
Mumbai = 1
Bangalore = 14
Chennai = 10
Kolkata = 4
Therefore, there is a total of 51.

23. Total number of placements in IT or ITES sectors = 168
Number of students specializing not in information systems and placed in IT or ITES sector = $\frac{63}{168} \times 100 = 37.5\%$.

Solutions for questions 24 to 27: As C is the southernmost city and as the cities are in a straight line, we have to find the distance of each city from C.

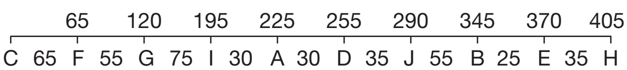
The corresponding distances are F – C = 65, G – C = 120, H – C = 405, I – C = 195, J – C = 290

To find the distance of cities A, B, D and E from C,
A – C
A – F = 160 and C – F = 65
A – G = 105 and C – G = 120
A – C = 160 – 65 = 95 or 160 + 65 = 225
and 120 – 105 = 15 or 120 + 105 = 225

As the common value is 225, the distance between A – C is 225.

Similarly, we can find the other values as B – C = 345, C – D = 255 and C – E = 370.

The ten cities according to their distance from C as



24. The distance between G and I is 75 kms.

25. As C is the southernmost city and the cities are in a straight line the city farthest from C would be the northernmost city, i.e., H

26. As he starts his journey from G, the shortest distance would be when he travels to C first and then travels northwards up to H, i.e., 405 + 120 = 525 kms

27. The least distance between any two cities is that between cities B and E, i.e., 25 kms.

Solutions for questions 28 to 35:

28. Given, 20% of the employees at the Delhi franchisee work in the Operations department and 15% of the employees in the Operations department work at the Delhi franchisee.
⇒ 20% of employees at the Delhi franchisee = 15% of the employees in the Operations department.

$$\frac{\text{Number of employees at the Delhi franchisee}}{\text{Number of employees in the Operations department}} = \frac{3}{4}$$

Let the number of employees in the company except at the Delhi franchisee be $100x$.
Total employees at the Bangalore franchisee = $12x$
Total employees at the Delhi franchisee = $12x$
Total employees at the Mumbai franchisee = $24x$
Total employees in the Operations department = $\frac{4}{3} \times 12x = 16x$

Of the employees in the Operations department, 15% of $16x = 2.4x$ work at the Mumbai franchisee.

Required percentage = $\frac{2.4x}{24x} \times 100 = 10\%$

29. Let the number of employees at all franchisee except at Delhi = $100x$
The employees working at the Mumbai franchisee = $24x$
The employees working at the Chennai franchisee = $10\ x$
Given that, employees at Mumbai form 15% of the total employees in the company.
Total employees in the company = $\frac{24x}{15} \times 100 = 160x$

Now the employees at the Delhi franchisee = $60x$
The employees in the Operations department = $80x$ (as 20% of the employees at the Delhi franchise form 15% of the employees in the Operations department)
Total employees in all the departments except Operations = $160x - 80x = 80x2$
Employees in the Accounts department = $\frac{20}{100}(80x) = 16x$
Required ratio = $16x : 10x = 8 : 5$

30. As the total number of employees in the company is known, we need either the number of employees in the Operations department or that at the Delhi franchisee. Hence, only (C) will be helpful in finding the required values.

Challenge Your Understanding
Practice Set 2

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

The table below gives the details about four different machines A, B, C and D. Each of the machines can produce four different types of products, X, Y, Z and W.

Machine \ Product	X	Y	Z	W
A	(20,2)	(30,3)	(40,1)	(30,2)
B	(30,3)	(40,2)	(60,1)	(30,2)
C	(30,1)	(40,2)	(80,4)	(30,1)
D	(15,1)	(20,1)	(40,2)	(80,4)

Each of the machines is allowed to operate for a total of 8 hours. The first figure in the bracket indicates the quantity (in units) of the product produced while the second numerical figure indicates the time taken (in hrs) to produce that quantity of that product. For example, machine A produces 20 units of X in 2 hrs.

Each machine operates independently and can produce only one type of product at any given point of time and produces the products X, Y, Z and W in that order and for the time period mentioned, i.e., machine A produces X for the first two hours, Y for the next three hours and so on. The same is the case with other machines. The machines produce any product at a uniform rate.

1. In the first 5 hours of operation, find the quantity of Z (in units) produced?

(A) 0 (B) 20
(C) 220 (D) 80

2. The quantity of W produced in the first x hours is at least 20. Which of the following is true?

(A) $x = 5$ (B) $x \geq 5$
(C) $x = 3$ (D) $x = 4$

3. In the first four hours, by what percentage is the quantity of Y produced by B more or less than that of C?

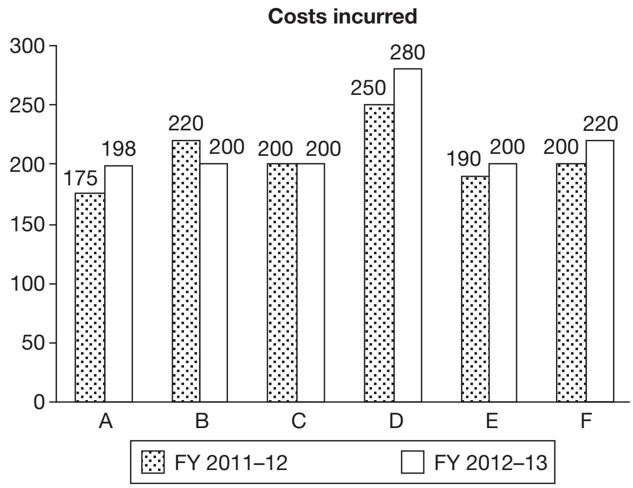
(A) 50% less
(B) 50% more
(C) 100% more
(D) 62.5% less

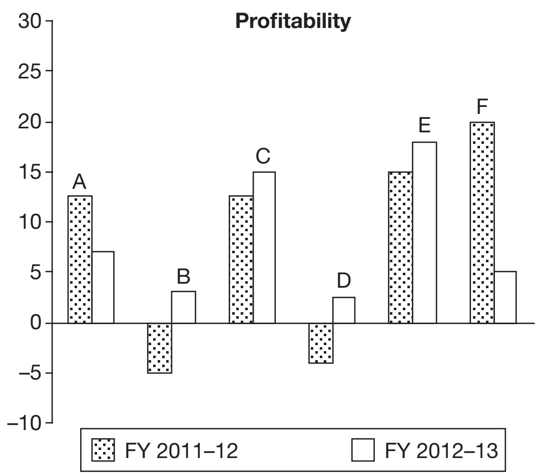
4. What is the maximum quantity of any product produced in the first 4 hours?

(A) 90 (B) 95
(C) 100 (D) 110

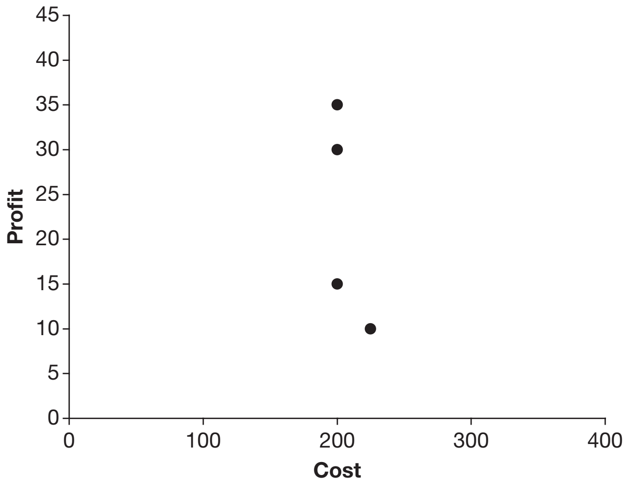
Directions for questions 5 to 8: Answer the questions on the basis of the information given below.

The following two charts show the costs (in ₹ cr) and the profitability of six companies in the financial years (FYs) 2011–12 and 2012–13. Profitability is defined as the ratio of profit obtained to the costs, incurred, typically expressed in percentage.





The profits of four of these companies are plotted against their respective costs for 2012–13.



5. Which of the following statements is not true?
(A) The company with the highest profitability in FY 2011–12 has the highest cost in FY 2012–13.
(B) The company with the highest cost in the two financial years combined has the highest cost in FY 2012–13.
(C) Companies with a higher cost in FY 2011–12 than in FY 2012–13 have higher profitability in FY 2012–13 than in FY 2011–12.
(D) Companies with profitability between 10% and 20% in FY 2011–12 also have costs between 150 crore and 250 crore in FY 2012–13.

6. Which company recorded the highest profit in FY 2012–13?
(A) A (B) E
(C) F (D) C

7. What is the approximate average profit in FY 2011–12 of the companies which are excluded from the third chart?
(A) –7.5 crore
(B) 3.5 crore
(C) –10.5 crore
(D) –12.0 crore

8. The average profit in FY 2012–13 of the companies with profitability exceeding 10% in FY 2012–13, is approximately
(A) 17.5 crore
(B) 25 crore.
(C) 27.5 crore
(D) 32.5 crore

Directions for questions 9 to 12: Answer the questions on the basis of the information given below.

The following table gives details about ten business groups in the country.

Business group	No. of companies in the group	Total group turnover (₹ crore)	% share in group turnover: of the largest company	% share in group turnover: of the smallest company
A	7	3000	25	8
B	5	2250	32	10
C	4	3100	40	15
D	8	2500	26	7
E	5	5200	32	6
F	6	3700	29	11
G	3	10200	45	25
H	5	6350	30	13
I	6	4200	27	11
J	4	4950	36	10

9. What could be the maximum turnover (in ₹400 crore) of the third largest company of business group A?
(A) 570 (B) 600
(C) 645 (D) 672

10. At most how many companies of group D had a turnover of more than ₹400 crore?
(A) 4 (B) 3
(C) 2 (D) 1

11. What is the maximum turnover (in ₹crore) of the second smallest company of business group I?
(A) 528 (B) 586
(C) 620 (D) 651

12. At most how many companies of business groups B, E or F had a turnover less than ₹500 crore?
(A) 6 (B) 9
(C) 8 (D) 10

Directions for questions 13 to 16: Answer the questions on the basis of the information given below.

Business process outsourcing (BPO) companies get their revenue from two sources, such as data processing and voice processing. The average revenue received from each hour of voice processing is called ABRH. In the table below, the revenue received from voice processing as a percentage of total revenue received and the ABRH in US Dollars (USD) are given for twenty companies from A through T.

Company	ABRH (in USD)	Revenue from voice processing as a percentage of total revenue
A	1	9
B	2	8
C	1	11
D	1	17
E	2	15
F	4	13
G	3	13
H	2	22
I	2	42
J	6	12
K	7	15
L	6	18
M	6	21
N	7	25
O	9	11
P	9	20
Q	8	23
R	9	28
S	10	23
T	11	20

13. If it is known that the number of hours of voice processing is the same for companies A and P, then which of the following statements is true?
(A) Total revenue is the same for both the companies.
(B) Total revenue of company P is about four times that of company A.
(C) Total revenue of company A is about four times that of company P.
(D) Total revenue of company A is about two times that of company P.

14. If the number of hours of voice processing is the same for all the companies given, then which company has the highest total revenue?
(A) A (B) J
(C) K (D) None of these

15. It is expected that in another two years, i.e., in 2011, revenue from voice processing as a percentage of total revenue will be tripled for company A and doubled for company L. Assume that in 2011, the total revenue of company A is twice that of company L and that the number of hours of voice processing is the same for both the companies. What is the percentage increase of ABRH of company A, if there is no change in ABRH of company L?
(A) 400 (B) 550
(C) 800 (D) 950

16. If the total revenue is the same for the pair of companies listed in the choices below, choose the pair that has approximately the same number of hours of voice processing.
(A) I and P (B) M and Q
(C) M and F (D) B and H

Directions for questions 17 to 19: Answer the questions on the basis of the information given below.

The following table provides the number of students writing MBA entrance exams in a year and the percentage of students, among those writing a particular exam, who write only that exam. The table gives data of all the MBA entrance exams in a year.

Exam	Students writing	
	That exam	Only that exam
CAT	1,95,000	56%
XAT	93,000	17%
IIFT	42,000	26%
SNAP	70,000	34%
CMAT	82,000	12%
MAT	36,000	24%

17. At most how many students wrote both CAT and XAT?
(A) 87,280 (B) 85,800
(C) 77,190 (D) None of these

18. The number of students who wrote all the six exams is at most _______.
(A) 31,080 (B) 37,180
(C) 24,820 (D) 27,360

19. What is the minimum number of students who wrote at least one of the six exams?
(A) 2,48,160 (B) 2,64,010
(C) 2,94,380 (D) None of these

Directions for questions 20 to 23: Answer these questions on the basis of the information given below.

There are 100 students in class XII of Model School. The following table gives the break-up of students with respect to the marks scored by them in the three core subjects, such as Maths, Physics and Chemistry and the two language subjects English and Hindi. The maximum marks in each subject is 100 and a student needs to score at least 50 marks to pass.

Number of students with marks	Maths	Physics	Chemistry	English	Hindi
90 or above	18	24	28	21	26
80 or above	35	42	39	42	38
70 or above	52	58	55	58	47
60 or above	61	75	72	65	60
50 or above	87	93	88	88	82

20. What is the maximum number of students who failed in all the five subjects?

(A) 5 (B) 7
(C) 9 (D) 12

21. The number of students who passed in all the five subjects is at least ________.

(A) 93 (B) 87
(C) 82 (D) None of these

22. Students who score 90 or more marks in at least two of the core subjects and at least one of the language subjects are eligible for a scholarship. At most how many students would be eligible for the scholarship?

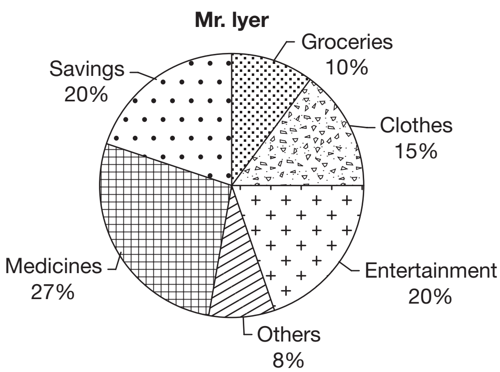
(A) 33 (B) 24 (C) 28 (D) 35

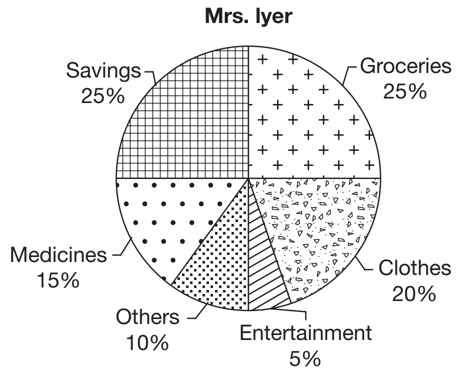
23. At most how many students scored 60 or more marks in at least four of the five subjects?

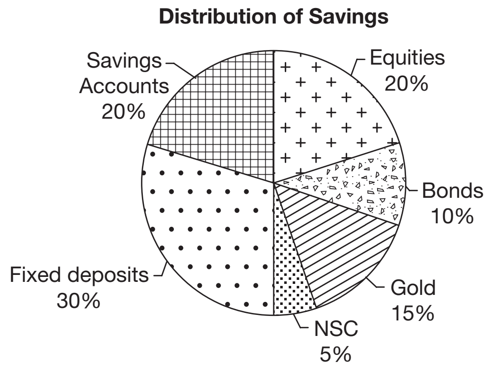
(A) 83 (B) 76 (C) 75 (D) 61

Directions for questions 24 to 27: These questions are based on the following information.

The pie charts show the distribution of the annual expenses and savings of Mr. and Mrs. Iyer and also the distribution of investment of their combined savings.







The total expenditure on any of the heads and also of the savings is the sum of the values of both the persons on that particular item.

24. If the total money invested by them in fixed deposits forms 15% of Mrs. Iyer's income, then what is the ratio of the incomes of Mr. and Mrs. Iyer?

(A) 4 : 5 (B) 3 : 2
(C) 7 : 5 (D) None of these

25. If the amount spent on clothes by Mr. Iyer is more than that by Mrs. Iyer, then the amount spent by Mr. Iyer on which of the following is definitely more than that by Mrs. Iyer?

(1) Groceries (2) Medicines
(3) Others (4) Entertainment
(A) Only 1 and 2 (B) Only 1, 3 and 4
(C) Only 2, 3 and 4 (D) Only 2 and 3

26. If the amount they invested in equities forms 4.55% of their combined total income, then Mrs. Iyer's salary is what percentage of Mr. Iyer's salary?

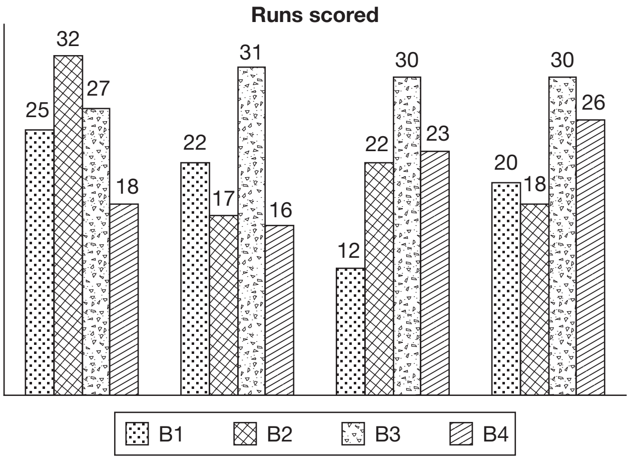
(A) 66.66 (B) 81.81
(C) 122.22 (D) 75

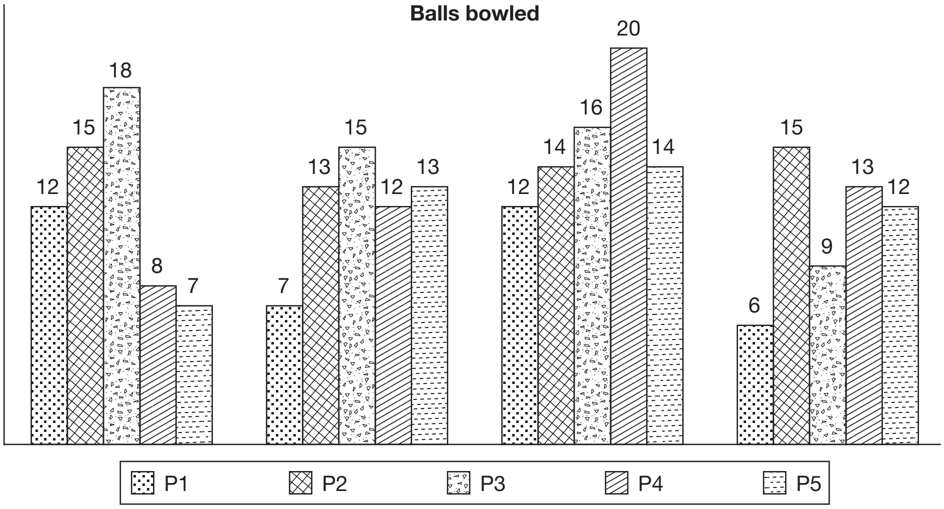
27. If the amount that is invested in savings accounts is the same as what Mr. Iyer spent on groceries, then the ratio of the amounts spent by Mr. and Mrs. Iyer towards entertainment is

(A) 10 : 3 (B) 3 : 1
(C) 5 : 3 (D) 2 : 1

Directions for questions 28 to 30: These questions are based on the following information.

While selecting the team for the Twenty20 World Cup, the selectors of the Indian team could not make up their mind regarding four batsmen namely B1, B2, B3 and B4 and five bowlers namely P1, P2, P3, P4 and P5 in the selection camp. Hence, they decided to conduct selection trials for these nine players. The trials were conducted in four sessions, such as S1, S2, S3 and S4 in a single day. The trials were conducted such that each batsman had to face the bowling of all the five bowlers in each of the four sessions. The following bar graphs exhibits the runs scored by the four batsmen in each session and the number of balls bowled by the five bowlers in each session.





Further, the following information is also known:

(1) Strike rate of a batsman = $\frac{\text{Runs Scored}}{\text{Balls faced}} \times 100$

(2) Only 0, 1, 2, 3, 4 or 6 runs can be scored off a single ball and if 0 runs are scored off a ball, it is called a dot ball.

(3) In none of the sessions did any of the bowlers bowl more than one dot ball to a batsman.

(4) The bowlers bowl only to these four batsmen and the batsmen face the bowling of only these five bowlers.

Directions for questions 28 to 30: Type your answer in the space provided below the question.

28. The number of sixers (6 runs off a ball) scored off the bowling of P1 was at most. ________

29. The number of balls faced by batsman B2 was at least ________ .

30. The number of balls off which a single run (i.e., 1 run) was scored was at least ________.

Directions for question 31: Select the correct alternative from the given choices.

31. If it is known that in each session, each of the bowlers bowled one dot ball to each batsman, then the minimum possible strike rate of B3 is______.
(A) 80.5 (B) 85.5
(C) 89.0 (D) 92.2

Directions for questions 32 to 35: Answer these questions on the basis of the information given below.

The cricket magazine Wisden, in its survey of the best one-day batsmen of the century, shortlisted six players, and then ranked them based on seven different parameters. In any parameter, these persons were ranked from 1 to 6 based on the decreasing order of the corresponding values for the players. In the following table, some of the values have been intentionally removed.

Batsman	Parameter						
	Total innings played	Total runs scored	Total balls faced	Number of times not out	Number of times out	Average	Strike rate
Ponting		6	2	1		6	
Richards		5		6		4	
Sachin	6		1		3		2
Inzamam	4		6		4	2	4
Lara	2				6		3
Dravid		2	4	5			

In the above table, in any parameter, the person with rank 1 has the highest value and the person with rank 6 has the least value in the corresponding parameter. For example, Sachin faced the highest number of balls and Inzamam faced the least.

For any player:

(i) Total Innings played = Number of times out + Number of times not out

(ii) Average = $\dfrac{\text{Total runs scored}}{\text{Number of times out}}$

(iii) Strike rate = $\dfrac{\text{Total runs scored}}{\text{Total balls faced}}$

Further, these six players are allotted points such that the player with the highest 'Average' will get 12 points, the second highest is 10, the third highest is 8 and so on. Similarly, the person with the highest 'Strike Rate' will get 6 points, the second highest 5 points and so on.

In the end, the points obtained by them in both the parameters put together, are considered as total points.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

32. Which player has the highest number of total points?
(A) Richards (B) Sachin
(C) Inzamam (D) Lara

33. Which player has the least number of total points?
(A) Ponting (B) Richards
(C) Lara (D) Inzamam

34. In how many of the given seven parameters did Sachin get rank 1? __________

35. In how many of the given seven parameters did Sachin get a better (numerically lower) rank than Dravid? _____

Directions for questions 36 to 39: Answer these questions on the basis of the information given below.

The following table gives the details of 100 boys and 100 girls who take a test. The test consisted of three areas, such as Quant, Reasoning and English. The tables give the breakup of the students according to their marks in each area. The maximum marks in each area is 100.

Quant

Boys	Girls
17 (7 – 25)	21 (6 – 20)
31 (26 – 49)	32 (24 – 60)
15 (54 – 75)	27 (63 – 80)
37 (76 – 100)	20 (81 – 98)

Reasoning

Boys	Girls
15 (11 – 27)	22 (7 – 32)
19 (29 – 48)	29 (35 – 61)
31 (51 – 72)	30 (62 – 78)
35 (74 – 95)	19 (80 – 97)

English

Boys	Girls
21 (7 – 27)	15 (11 – 32)
36 (29 – 53)	27 (34 – 58)
27 (57 – 74)	21 (62 – 77)
16 (77 – 88)	37 (78 – 92)

For example, the table under Quant shows that there are 17 boys who scored marks in the range of 7 to 25 with the lowest mark being 7 and the highest being 25 and there are 21 girls who scored marks in the range of 6 to 20, with the lowest marks being 6 and the highest being 20.

36. The average score of the given students in the quant section is at least

(A) 42.17 (B) 44.66
(C) 47.32 (D) 49.65

37. The number of students who scored more than 70 marks in each section is at most

(A) 88 (B) 92
(C) 99 (D) None of these

38. The average score of the girls in the English section is at most

(A) 67.26 (B) 69.93
(C) 61.84 (D) 72.35

39. The average score of the boys in the reasoning section is at least

(A) 48.13 (B) 48.76
(C) 49.64 (D) 50.27

Directions for questions 40 to 43: Answer these questions on the basis of the information given below.

A school has a total of 6720 students. The ratio of the number of boys to that of girls in the school is 9 : 7. All the students are enrolled in at least one of the three classes among painting, dancing and singing. One-twelfth of the boys are enrolled only for painting classes. Twenty-five per cent of the girls are enrolled in only singing classes whereas one-tenth of the boys are enrolled in only singing classes. Twenty per cent of the girls are enrolled in both painting and singing classes only. The number of girls enrolled only for painting classes is 100 per cent more than the number of boys enrolled in the same. One-ninth of the boys are enrolled in all the three classes together. The ratio of the number of boys enrolled in both singing and dancing classes only to the girls enrolled in the same is 7 : 5. One-tenth of the girls are enrolled in only dancing classes whereas $8\frac{1}{3}$ per cent of the girls are enrolled in both singing and dancing classes together but not in painting. None of the girls are enrolled in both painting and dancing classes only. The number of boys enrolled in both painting and singing classes only is equal to half the number of girls enrolled in the same. None of the boys are enrolled in both painting and dancing classes only.

40. Find the total number of boys enrolled in painting.

(A) 1008
(B) 996
(C) 1058
(D) None of these

41. The total number of boys enrolled in dancing forms approximately what percentage of the total strength of the school?

(A) 36% (B) 42%
(C) 48% (D) 54%

42. Find the total number of students enrolled in all the three classes.

(A) 820 (B) 836
(C) 848 (D) 868

43. The number of boys enrolled in only singing forms what percent of the girls enrolled in the same?

(A) 51.43% (B) 45.28%
(C) 56.57% (D) 62.63%

Directions for questions 44 to 46: These questions are based on the following information.

A car manufacturer has three race cars, such as car I, car II and car III. He fields them in car races. If car I wins he gets 80% more than the amount he had deposited on that car for the race. If car II wins, he gets 40% more and if car III wins he gets 40% more, then the respective amounts deposited.

If a car does not win a race, the manufacturer loses 20% of the money he had deposited on that car. In each race he fields only two cars and deposits an equal amount on each and in each race only one car wins. On a certain day there are three races. Car I wins race 1, car III wins race 2 and car II wins race 3. money deposited for races 2 and 3 is equal to the amount with the manufacturer at the end of races 1 and 2 respectively. He ends up with ₹31,460 in the end.

44. How much amount did the car manufacturer deposit for race 1?

(A) ₹10,000 (B) ₹30,000
(C) ₹15,000 (D) ₹20,000

45. In which race did the car manufacturer get the maximum amount as profit?

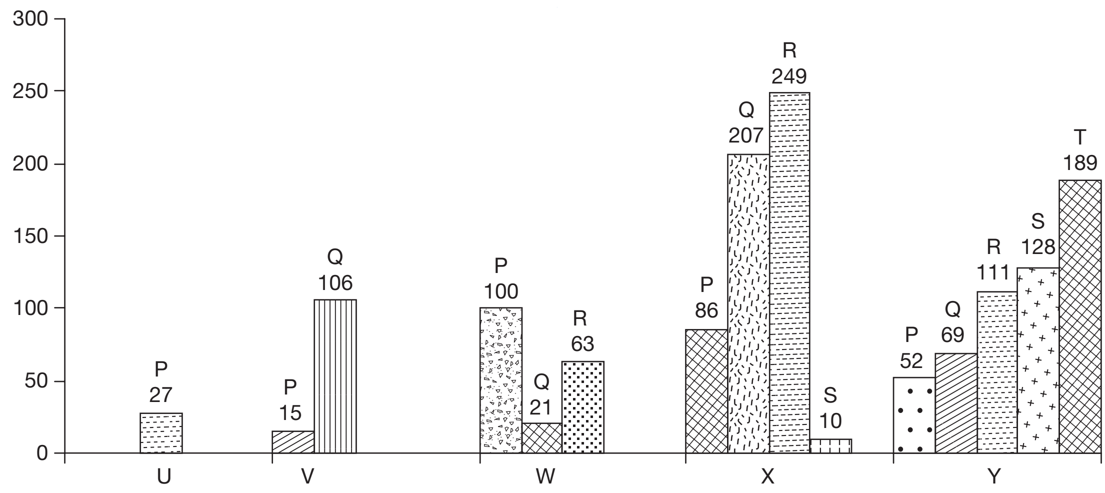
(A) Race 2 (B) Race 1
(C) Race 3 (D) Race 1 or Race 3

46. If the manufacturer had fielded only car I in race 1, only car II in race 2 and only car III in race 3 and the performance (winning or losing) of these cars that are fielded remains the same (as given earlier), what would be his net profit by the end of race 3? (Assume that in each race he deposits all the money he has)

(A) ₹1520 (B) ₹3040
(C) ₹4320 (D) ₹2160

Directions for questions 47 to 50: Answer these questions on the basis of the information given below.

The bar graph gives details of the marks scored by ten students – P, Q, R, S, T, U, V, W, and X in an exam. The graph gives details of the difference in marks of P, Q, R, S, and T with respect to U, V, W, X and Y.



It is known that student Q had the second highest marks among all the ten students.

47. Who scored the highest marks among the ten students?
(A) W (B) R
(C) P (D) Cannot be determined

48. If Q scored 450 marks, what is the sum of the marks scored by P and R?
(A) 821 (B) 786
(C) 852 (D) Cannot be determined

49. Who among U, V, W, X and Y scored the highest marks?
(A) U
(B) W
(C) Y
(D) Cannot be determined

50. If it is known that P scored 565 marks, which of the following can be the marks scored by U?
(A) 538
(B) 567
(C) 592
(D) Cannot be determined

Directions for questions 51 to 54: Answer these questions based on the following data.

In a hockey tournament, each one of the five teams named as A, B, C, D and E played exactly one match with every other team. For a team, if the number of goals made in a match is greater than that conceded, then the match is a win. If the number of goals made is equal to that conceded, then it is a draw. If the number of goals made is less than that conceded, it is a loss. For a win, a team gets 10 points, for a draw the team gets 4 points, while for a loss, the team loses 5 points. In addition, for every goal made, a team gets 2 points and for every goal conceded, it loses one point. The team getting the greatest number of total points wins the tournament.

		→ Goals conceded				
	Team	**A**	**B**	**C**	**D**	**E**
↓ Goals made	A	–	5	3	2	6
	B	3	–	4	2	3
	C	5	2	–	5	4
	D	3	4	3	–	2
	E	4	3	6	6	–

Note: For example, A made 3 goals against B while conceding 5 goals to it.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

51. How many points did team E score in the tournament?
(A) 13 (B) 15
(C) 18 (D) 20

52. Which team won the tournament?
(A) A (B) B
(C) C (D) D

53. What is the total number of draws in the tournament? ________

54. How many teams have two wins each? ________

Directions for questions 55 to 58: These questions are based on the following data.

A die, with the numbers 1 to 6 marked on its six faces was cast repeatedly. The table below gives the number of times that each individual number on the faces of the die turned up, in the first n casts, where $n = 20, 40, \ldots, 140$.

Casts	Number of times each number turned up					
	1	2	3	4	5	6
First 20	4	3	4	2	2	5
First 40	10	8	5	7	3	7
First 60	12	15	10	8	8	7
First 80	14	17	17	13	10	9
First 100	14	18	22	20	13	13
First 120	19	23	23	22	17	16
First 140	19	24	24	32	18	23

It was also observed that in no two consecutive casts did the same number turn up.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

55. If the number 4 turned up in the 140th cast, then which number/s could not have turned up in the 131st cast?
(A) 2 (B) 1
(C) 4 (D) 3

56. Which of the following numbers must have turned up for the maximum number of times in the first 65 casts?
(A) 2 (B) 1
(C) 5 (D) 3

57. What is the least number of times the number 4 could have turned up in the first 95 casts? _______

58. What is the maximum number of times any of the even numbers could have shown up from the 81st cast to the 130th cast? _______

Directions for questions 59 to 62: These questions are based on the following information.

In an examination, there are 100 questions. A student is awarded 12 marks for each correct answer. He loses 3 marks for each wrong answer and loses 2 marks for each unanswered question.

The net score of Hari in that test is 625.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

59. The number of ways in which Hari could have attempted the exam is _______ .

60. The maximum number of questions attempted by Hari could be _______ .

61. The number of questions that Hari got wrong if the number of correct answers that he got is a multiple of 5 is
(A) 12 (B) 15
(C) 20 (D) 25

62. Which of the following could be the number of questions Hari left unanswered?
(A) 5 (B) 15
(C) 25 (D) 35

Directions for questions 63 to 66: These questions are based on the following information.

Eight players from P to W take part in a double round robin chess tournament, in which each player plays every other player twice, once with white and once with black pieces. 3 points are awarded for a win, 1 point for a draw and 0 points for a loss. At the end of all the matches, the player with the highest points is awarded the first rank, the next one second and so on. If two or more players end up with the same number of points, they are given the same rank.

63. If it is known that player R was the sole winner of the tournament, then at least how many points did he score?
(A) 14 (B) 16
(C) 15 (D) 17

64. If all players end up with distinct points, what is the maximum points scored by player P, if it is known that he finished last?
(A) 16 (B) 15
(C) 14 (D) 17

65. Which of the following cannot be the points scored by all the players together at the end of the tournament?
(A) 108 (B) 121
(C) 132 (D) 164

66. If it is known that player R won the tournament and all players scored distinct points, then at least how many points did R score?
(A) 17 (B) 18
(C) 19 (D) 20

Directions for questions 67 to 70: The following table gives the national market share of an automobile company, ABC Ltd in different segments in the year 2011.

Segment	National share (%)
LCV	36
CV	42
Multi-axle	25
Hatchback	33
Sedan (compact)	56
Sedan (Luxury)	35
SUV	44

67. If the overall national sales of LCVs were 280% more than that of CVs, then for ABC Ltd, the sales of CVs was what percentage of LCVs?
(A) 40.5 (B) 38.1
(C) 35.2 (D) 30.7

68. If the sales of luxury sedans by ABC Ltd was 20% of its compact sedan sales, then the national sales of compact sedans forms what percentage of the national sales of compact and luxury sedans?
(A) 71.2% (B) 75.8%
(C) 68.2% (D) 67.2%

69. If all automobiles are classified in one of the seven segments and for ABC Ltd, multi-axle vehicles form 10% of its total sales, then the overall national sales of multi-axle vehicles form at least what percentage of the national sales of all vehicles?
(A) 15.1% (B) 12.8%
(C) 10% (D) 7%

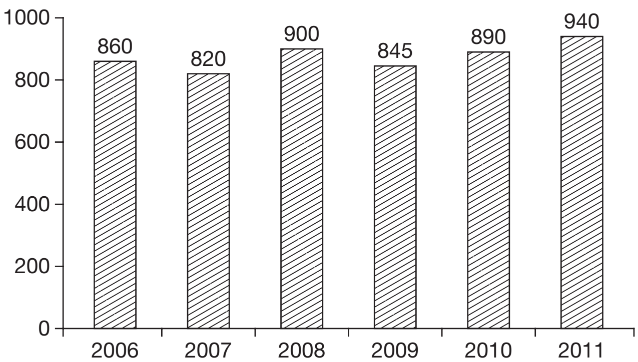
70. If only Hatchbacks, Sedans (both compact and Luxury) and SUVs come under the category of cars and the national share of ABC Ltd in cars is 47%, then what is the least percentage share of compact Sedans in total car sales in the country?
(A) 20 (B) 25
(C) $33\frac{1}{3}$ (D) 40

Directions for question 71 to 74: These questions are based on the following data.

The table shows the religion-wise break-up of the number of women per 1000 men in a locality in 2011.

Jains	802
Hindus	900
Muslims	875
Christians	795
Sikhs	1020

The line graph shows the number of women for every 1000 men in the locality, over the years.



71. If in 2011, the difference between the number of men and the number of women among Hindus is same as that among Muslims, then what is the ratio of the number of men who are Muslims to the number of women who are Hindus?
(A) 10 : 9 (B) 9 : 10
(C) 35 : 36 (D) 8 : 9

72. In 2011, at least what percentage of the people in the locality are Sikhs?
(A) 25 (B) 30
(C) $33^1/_3$ (D) 37.5

73. If the total number of Christians and Muslims in the locality in 2011 is the same, then by what percentage is the number of Christian men more then the number of Muslim men?
(A) 4.5 (B) 5.3
(C) 5.6 (D) 6.2

74. In 2011, the percentage of Sikhs in the locality was definitely less than _________.
(A) 52.8 (B) 59.6
(C) 64.4 (D) Cannot be determined

Directions for questions 75 to 78: These questions are based on the following data.

The following table gives the time taken (in hrs) by four machines, namely M_1, M_2, M_3 and M_4 to process four subtasks, such as T_1, T_2, T_3 and T_4 which constitute task T. Each machine can do any of the four subtasks and the only condition is that a machine cannot do more than one subtask at the same time and no subtask can be done simultaneously by two machines.

Machine \ Task	T_1	T_2	T_3	T_4
M_1	7	6	4	9
M_2	5	3	7	8
M_3	7	7	6	2
M_4	5	8	5	6

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

75. What is the minimum time (in hrs) in which the task can be completed if the subtasks can only be done one after the other in the order T_1, T_2, T_3 and T_4? _______

76. If machine M_3 is not working, then what is the minimum time required (in hrs) to finish the task, if the subtasks can be done simultaneously? _______

77. If only two machines can be used, then what is the shortest time (in hrs) in which all the subtasks can be completed? _______

(A) 10 (B) 9
(C) 8 (D) 7

78. If the subtasks can be done simultaneously, i.e., if two or more machines could work on a subtask at the same time, then what would be the approximate minimum time in which all the subtasks can be completed?
(A) 3.5 hrs (B) 3.7 hrs
(C) 4 hrs (D) 4.25 hrs

Directions for questions 79 to 82: Answer these questions on the basis of the following information.

50,000 units of brand X are being sold in the market at a price of ₹10 per unit. A competitive brand, Y enters the market. The courses of action available for the company marketing brand X are as follows:

(1) Cut the price of X by 50%, which would result in an increase in the number of units sold of X by 30% with a probability of 0.5 and by 20% with a probability of 0.5.

(2) Advertise, which would cost ₹2,00,000, but would result in an increase in the number of units of X sold by 50% with a probability of 0.1, by 20% with a probability of 0.5 and by 10% with a probability of 0.4.

(3) Remain silent, in which case, the probability of losing the market by 40% is 0.5 and the probability of retaining its market is 0.5.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

79. Which is the best course of action to follow?
(A) Cut down the price (B) Advertise
(C) Remain silent (D) Insufficient data

80. In which of the cases does the market of X (number of units sold) increase the most?
(A) Cut down the prices
(B) Advertising
(C) Remain silent
(D) None of these

81. What is the net loss (in ₹) if course 1 is followed?
(Net Loss = Money realized originally – Money realised now) _______

82. What is the revenue realised (net of advertising expenses) if course 2 is followed (in ₹)? _______

***Directions for questions 83 to 86*:** Answer these questions on the basis of the information given below.

The following table gives the average scores of the students of classes I to V.

Class	Average score of top 20% of the students	Average score of the lowest 20% of the students
I	82	36
II	76	31
III	68	24
IV	86	19
V	80	38

83. If there are 60 students in class IV and the average score of the class is 60, then the score of the student who got the 48th rank in the class is at most.
(A) 30 (B) 45
(C) 55 (D) 65

84. For how many of the given classes can the average score of the remaining 60% of the class be more than 45, if the average score of each class is 50?
(A) 1 (B) 2
(C) 3 (D) 4

85. If each class has the highest possible average, then the highest average is for which class?
(A) I (B) II
(C) III (D) IV

86. The least possible average marks of any class would be at least
(A) 28.6 (B) 30.4
(C) 32.4 (D) 34.6

Directions for questions 87 to 90: These questions are based on the following information.

In a school there are 135 students who play at most three sports, such as cricket, football and hockey. There is at least one student who plays all the three, at least one student who plays exactly two, and at least one student who plays exactly one of the above-mentioned sports.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

87. If the number of students who play all three sports is less than the number of students who play exactly two and if the number of students who play cricket is more than those who play hockey which in turn is more than the those who play football, while the number of students who play football is more than those who play exactly two sports, then what is the maximum number of students who do not play any of the given sports? _______

88. Using the data from the previous question, what is the maximum number of students who play all the three sports? _______

89. If the number of students who play the sports are as follows, only cricket < only hockey < only football < exactly two sports < exactly three sports, then the maximum number of people who play exactly two sports is

(A) 65 (B) 62

(C) 60 (D) 67

90. Using data from the previous question, the minimum number of people who play all the three sports is

(A) 1 (B) 2

(C) 3 (D) 4

ANSWER KEYS

1. (D)	**16.** (C)	**31.** (D)	**46.** (B)	**61.** (B)	**76.** 8
2. (B)	**17.** (C)	**32.** (D)	**47.** (B)	**62.** (C)	**77.** (C)
3. (A)	**18.** (D)	**33.** (A)	**48.** (A)	**63.** (B)	**78.** (B)
4. (C)	**19.** (B)	**34.** 2	**49.** (B)	**64.** (D)	**79.** (C)
5. (A)	**20.** (B)	**35.** 4	**50.** (D)	**65.** (A)	**80.** (A)
6. (B)	**21.** (D)	**36.** (B)	**51.** (B)	**66.** (C)	**81.** 187500
7. (C)	**22.** (D)	**37.** (A)	**52.** (B)	**67.** (D)	**82.** 395000
8. (D)	**23.** (A)	**38.** (B)	**53.** 1	**68.** (B)	**83.** (D)
9. (C)	**24.** (D)	**39.** (C)	**54.** 4	**69.** (B)	**84.** (C)
10. (B)	**25.** (C)	**40.** (D)	**55.** (B)	**70.** (B)	**85.** (A)
11. (D)	**26.** (C)	**41.** (B)	**56.** (A)	**71.** (D)	**86.** (C)
12. (D)	**27.** (A)	**42.** (D)	**57.** 17	**72.** (C)	**87.** 127
13. (B)	**28.** 30	**43.** (A)	**58.** 14	**73.** (A)	**88.** 68
14. (D)	**29.** 21	**44.** (D)	**59.** 3	**74.** (C)	**89.** (A)
15. (C)	**30.** 25	**45.** (B)	**60.** (C)	**75.** 14	**90.** (D)

SOLUTIONS

Solutions for questions 1 to 4:

1. The quantity of Z produced by different machines is as follows:

Machine	Quantity of Z
A:	0
B:	0
C:	$\frac{80}{4}(2) = 40$
D:	40
	Total : 80

2. In the first 5 hours, the quantity of W produced by A, B, C is 0. 20 units of W is produced by D. Thus, x ≥ 5.

3. In the first 4 hours,
the quantity of Y produced by B = 20
by C = 40.

$$\text{Percentage less} = \frac{20}{40}(100) = 50$$

4. In the first 4 hours, the quantity of X produced
= 20+30+30+15 = 95
Quantity of Y produced
=20+20+40+20 = 100.

Solutions for questions 5 to 8:

5. By observation we can say that option 1 is not true.

6. Profit for company

$$A = \frac{7.5}{100} \times 198 \cong 15 \text{ crore}$$

$$C = \frac{15}{100} \times 200 = 30 \text{ crore}$$

$$E = \frac{17.5}{100} \times 200 = 35 \text{ crore}$$

$$F = \frac{5}{100} \times 220 = 11.0 \text{ crore}$$

$$D = \frac{2.5}{100} \times 280 = 7.0 \text{ crore}$$

7. Required is the average profit of B and D profit in 2011–12 for

$$B = \frac{-5}{100} \times 220 = -11$$

$$D = \frac{-4}{100} \times 250 = -10$$

$\therefore$ Average profit $= -\dfrac{21}{2} = -10.5$ crore

8. Required is the average profit of C and E in 2012–13 which is $\dfrac{30+35}{2} = 32.5$ crore

Solutions for questions 9 to 12:

9. The maximum turnover for the third largest company occurs when the share of all companies from fourth to seventh has nearly the same values, i.e., 8% of group turnover and the remaining two companies, i.e., those ranked 2nd and 3rd have nearly the same share, i.e.,

$100 - (25 + 32) = \dfrac{43}{2} = 21.5$

$\therefore$ Maximum turnover $= \dfrac{21.5}{100} \times 3000$

$= 645$ crore

10. For maximum companies in group D to have a turnover of more than ₹400 crore, i.e., 16% of group turnover, we should assume that the smallest companies in the group have nearly the same turnover and those companies exceeding a turnover of 400 crores, had a turnover which is only slightly more than 400 crore. It can be seen that if we have four companies with 7% share, three companies with 16% share and one company with 26% share, the total adds up to 102% and so only two more companies (other than the largest company) can have turnover more than 400 crores.

11. For maximum turnover of the second smallest company, all companies of business group I ranked from second to fifth should have nearly the same turnover, i.e., $\dfrac{100-(27+11)}{4} = 15.5\%$.

$\therefore$ Maximum turnover $= \dfrac{15.5}{100} \times 4200 = 651$ crores

12. For business group B, 500 crore is 22.22% of the group turnover.
As the largest and smallest company accounts for 32 + 10 = 42% of the group turnover, the remaining 58% (100 – 42) can be distributed among the remaining three companies such that all the three are less than 22.22%.
$\therefore$ At most, four companies of business group B could have a turnover of less than ₹500 crore.
Similarly, we can find for others also.
For business group E, 500 crores is 9.65% of the group turnover.
$\therefore$ At most, two companies could have a turnover less than ₹500 crore.
For business group F, 500 crores is 13.5% of the group turnover.
$\therefore$ At most, four companies could have a turnover which is less than ₹500 crore.
$\therefore$ A total of 4 + 2 + 4 = 10 companies.

Solutions for questions 13 to 16:

13. Assume that the number of hours of voice processing for companies A and P is 1. The total revenue of company A would be 11.11. For company P it would be $\dfrac{9}{20}\times 100 = 45$.
Therefore, revenue for company P would be about four times that of company A.

14. Assume that the number of hours of voice processing in all the companies is one. The revenue earned by the companies would be equal to the ABRH. For company A, the revenue would be 9% of total revenue or the total revenue would be 11.11. Similarly, we can find the total revenue of all the companies and it can be found that the total revenue would be the highest for company 'O'.

15. In 2011, for company A, revenue from voice processing as a percentage of total revenue = 27.
For company L, it would be 36.
Let the total revenue in 2011 for companies A and L be 200 and 100, respectively. ABRH of company L = 6.
$\therefore$ Number of hours of voice processing for company L
$= \dfrac{36}{6} = 6\,hrs$
The revenue from voice processing for company A = $\dfrac{27}{100}\times 200 = 54$. $\therefore$ ABRH of company A $= \dfrac{54}{6} = 9$.
$\therefore$ Percentage increase $= \dfrac{9-1}{1}\times 100 = 800\%$

16. Given that the total revenue received is the same for the pair of companies given.
Choice (A) I and P: Let the total revenue be 100. The number of hours of voice processing for I and P are $\dfrac{42}{2}$ *and* $\dfrac{20}{9}$, respectively which is not equal.
Choice (B) M and Q: Let the total revenue be 100.
The number of hours of voice processing for M and Q are $\dfrac{21}{6} = 3.5$ and $\dfrac{23}{8} = 2.87$, respectively.
Choice (C) M and F: The values for M and F would be $\dfrac{21}{6} = 3.5$ *and* $\dfrac{13}{14} = 3.25$, respectively, i.e., nearly equal. It can be seen that the options in choice (D) do not give approximately equal value.

Solutions for questions 17 to 19:

17. The maximum number of students who wrote both CAT and XAT is the minimum of 44% of 1,95,000 and 83% of 93,000.
44% of 1,95,000 = 85,800 and
83% of 93,000 = 77190

18. The numbers of students who wrote the exams are as follows:

Exam	Total	Only	Others
CAT	1,95,000	1,09,200	85,800
XAT	93,000	15,810	77,190
IIFT	42,000	10,920	31,080
SNAP	70,000	23,800	46,200
CMAT	82,000	9,840	72,160
MAT	36,000	8,640	27,360

The maximum numbers of students who wrote all the six exams would be the minimum value in the 'others' column – 27360.

19. The minimum numbers of students who wrote at least one of the six exams would be the sum of the students who wrote only one exam and the maximum value in the 'others' column = 1,09,200 + 15,810 + 10,920 + 23,800 + 9,840 + 8,640 + 85,800 = 2,64,010.

Solutions for questions 20 to 23:

20. The number of students who failed in the different subjects are Maths – 13, Physics – 7, Chemistry – 12, English – 12 and Hindi – 18. The maximum number of students who failed in all the seven subjects is the minimum of these values, i.e., 7.

21. If we consider that the students who failed in the subjects are all unique, then we get the maximum number of students who failed, i.e., 13 + 7 + 12 + 12 + 18 = 62.
$\therefore$ At least 38 (100 – 62) students passed in all the five subjects.

22. The maximum number of students who scored 90 marks or more in at least two of the core subjects

$$= \frac{18+24+28}{2} = 35$$

The maximum number of students who scored 90 marks or more in at least one of the language subjects = 21 + 26 = 47
$\therefore$ Maximum number of students eligible for the scholarship = 35.

23. The number of students who scored 60 or more marks in at least four of the five subjects is the largest number less than $\frac{61+75+72+65+60}{4}$, i.e., 83.

Solutions for questions 24 to 27:

24. Let the total savings of the family be ₹100.
The money invested in fixed deposits = ₹30

Mrs. Iyer's income = $\frac{30}{15} \times 100 =$ ₹200

$\therefore$ Savings of Mrs. Iyer = $\frac{25}{100}\times 200 =$ ₹50

Savings of Mr. Iyer = ₹100 – ₹50 = ₹50

$\therefore$ Income of Mr. Iyer = $\frac{50}{20}\times 100 =$ ₹250

Ratio of their incomes = 5 : 4

25. Let the income of Mr. Iyer be $100x$ and Mrs. Iyer be $100y$.
Given that, $15x > 20y$
$\Rightarrow 3x > 4y$
Multiplying the above inequality with 9, we get
$27x > 36y \Rightarrow 27x > 15y$ (medicines)
Multiplying $3x > 4y$ with 2.66, we get
$8x > 10.64y$
$\Rightarrow 8x > 10y$, (others)
Multiplying $3x > 4y$ with 6.66, we get
$20x > 26.66y$
$\Rightarrow 20x > 5y$ (Entertainment)

26. Let the total incomes of Mr. Iyer and Mrs. Iyer be $100x$ and $100y$, respectively.
Now their total savings will be
$\therefore$ 20% of $100x$ + 25% of $100y$
Now the amount invested in equities from 20% of total savings, i.e., 20% of $(20x + 25y) = 4.55\ (x + y)$

$\therefore 0.45y = 0.55x \Rightarrow \frac{y}{x}\times\frac{11}{9}$

$\therefore$ Mrs. Iyer's salary forms $\frac{11}{9}\times 100 = 122.22\%$ of that of Mr. Iyer's.

27. Let the total savings be ₹100.
Money invested in savings accounts = 20
$\therefore$ Money spent by Mr. Iyer on Groceries = 20

Total income of Mr. Iyer = $\frac{20}{10}\times 100 =$ ₹200

Total savings of Mr. Iyer = ₹40
$\therefore$ Total savings of Mrs. Iyer = ₹60

Required ratio is $\frac{20}{20}\times 40 : \frac{5}{25}\times 60$

= 40 : 12 = 10 : 3

Solutions for questions 28 to 31:

28. The runs scored by the batsmen in the four sessions are 102, 86, 87 and 94, respectively. The number of balls bowled by the bowlers other than P_1 are 48, 53, 64 and 49, respectively. As a bowler bowled at most one dot ball to a batsman in a session, the number of balls off which runs were scored against bowlers other than P, was at least 48 – 16 = 32 (a bowler could have bowled at most four dot balls in a session. Therefore, four bowlers bowled at most $4 \times 4 = 16$ dot balls, 53 – 16 = 37, 64 – 16 = 48 and 49 – 16 = 33, respectively.

Practice Set 2

As at most 70 extra runs were scored in the first session it could have been due to a maximum of 11 sixers in the bowling of P_1, similarly it is 7, 6 and 6 in sessions 2, 3 and 4, respectively.

$\therefore$ At most 11 + 7 + 6 + 6 = 30 sixers were scored off the bowling of P_1.

29. The number of balls faced is the least when we assume that he has scored the maximum number of runs every ball, but as each batsman faced the bowling of each of the five bowlers in a session, he would have faced at least five balls in a session.

$\therefore$ The minimum number of balls faced by him was at least 6 (as he scored 32 runs) + 5 + 5 + 5 = 21 balls.

30. The number of runs scored, balls bowled and minimum number of balls in which runs were scored in different sessions are as follows.

Session	S_1	S_2	S_3	S_4
Runs	102	86	87	94
Balls	60	60	76	55
Min scoring balls	40	40	56	35

It can be seen that in all the sessions other than S_3, the minimum number of singles is zero, while in session 3, there were at least 25 singles.

$(31 \times 2 + 25 \times 1 = 87)$

31. For the minimum possible strike rate, we have to calculate the maximum number of balls he could have faced. As in each session each of the bowlers bowled a dot ball to each batsman, the minimum number of balls off which runs were scored in different sessions and the minimum number of balls faced by other batsmen in different sessions are as follows.

Session	S_1	S_2	S_3	S_4
Minimum scoring balls	40	40	56	35
Minimum balls faced by other batsmen	5 + 6 + 3 = 14	4 + 4 + 3 = 11	2 + 4 + 5 = 11	4 + 3 + 5 = 12
Maximum balls faced by B_3	40 – 14 = 26	40 – 11 = 29	30 (as he scored only 30 runs)	35 – 12 = 23

$\therefore$ Maximum number of balls faced by him = 26 + 5(dot balls) + 29 + 5 + 30 + 5 + 23 + 5 = 108 + 20 = 128

$\therefore$ Strike rate = $\dfrac{27+31+30+30}{128} \times 100 = \dfrac{118}{128} \times 100 =$ 92.2%.

Solutions for questions 32 to 35: Consider the 'strike rate' of Inzamam, i.e., rank 4. Now both Ponting and Richards scored less runs than Inzamam but faced more number of balls than Inzamam. Hence, both Ponting and Richards must be ranked worse (numerically higher) than Inzamam on 'strike rate'. Hence, Ponting and Richards must have ranks 6 and 5, respectively (since Ponting scored less runs, but definitely faced more number of balls than Richards). Now, Dravid must be ranked 1 in 'strike rate'.

Now, consider the total runs scored = Strike rate × Total balls faced. Though Inzamam faced the least number of balls, he was ranked 4th in strike rate. Therefore, at least three players must be ahead of Inzamam. Hence, Inzamam is ranked 4 in 'runs scored'. Similarly, it can be observed that Sachin and Lara must be ranked 1 and 3, respectively in 'runs scored'.

Now consider the 'Average'. Lara scored more runs than Inzamam and also was out for the least number of times. Hence, his average must be better than that of Inzamam. Hence, Lara is ranked 1 in 'average'. Therefore, Sachin and Dravid got ranks 3 and 5 (in any order) in 'average'.

Now the possible points for the six players are as follows:

Ponting = 2 + 1 = 3

Richards = 6 + 2 = 8

Sachin = (4 or 8) + 5 = 9 OR 13

Inzamam = 10 + 3 = 13

Lara = 12 + 4 = 16

Dravid = (8 or 4) + 6 = 14 OR 10

The final table obtained is as follows:

Batsman	Parameter						
	Innings played	Runs scored	Balls faced	Not outs	Outs	Average	Strike rate
Ponting	1	⑥	②	①	5	⑥	6
Richards	5	⑤	3/5	⑥	2	④	5

(*Continued*)

Batsman	Parameter						
	Innings played	Runs scored	Balls faced	Not outs	Outs	Average	Strike rate
Sachin	⑥	1	①	4	③	3	②
Inzamam	④	4	⑥	3	④	②	④
Lara	②	3	5/3	2	⑥	1	③
Dravid	3	②	④	⑤	1	5	1

Now points scored by them are as follows:

Player	Points
Ponting	3
Richards	8
Sachin	13
Inzamam	13
Lara	16
Dravid	10

Clearly, Lara and Ponting had the highest and the least number of points respectively. Using only this information, three of the questions (16, 17 and 18) can be answered. Now considering the relation 'Total innings played = Number of times out + Number of times not out' we get the ranks in 'number of times out', for Sachin, Inzamam and Lara as 4, 3 and 2, respectively.

Now considering the rank of Sachin in 'average', he got 3rd rank despite scoring the highest and getting out for the 3rd highest number of times. Hence, there are at least two players who scored less runs than Sachin and were out more number of times than Sachin. Hence, Sachin's rank in 'average' must be such that it has at least two ranks below it. Hence, Sachin is ranked 3 in 'average' and Dravid is ranked 5.

Now, similarly, it can be found that neither Richards nor Dravid can have fifth rank in 'number of times out'. Hence, Ponting is ranked fifth in 'number of times out'. Further, we get the ranks of Richards and Dravid in 'number of times out' as 2 and 1, respectively.

Hence, all the values, except the ranks of Richards and Lara in 'balls faced', are determined using which, the other two questions (i.e., 19 and 20) can be answered.

32. Lara got 16 points, which is the highest.

33. Ponting got 3 points, which is the lowest.

34. Sachin was ranked 1 in only 'runs scored' and 'balls faced'.

35. Sachin was ranked better than Dravid in 'runs scored', 'balls farced', 'not outs' and 'average'. Hence, four parameters.

Solutions for questions 36 to 39:

36. The minimum total score of the boys in the quant section is

$= 7 \times 16 + 25 \times 1 + 26 \times 30 + 49 \times 1 + 54 \times 14 + 75 \times 1 + 76 \times 36 + 100 \times 1$

$= 112 + 25 + 780 + 49 + 756 + 75 + 2736 + 100 = 4633$

The minimum total score of the girls in the quant section is

$= 6 \times 20 + 20 \times 1 + 24 \times 31 + 60 \times 1 + 63 \times 26 + 80 \times 1 + 81 \times 19 + 98 \times 1$

$= 120 + 20 + 744 + 60 + 1638 + 80 + 1539 + 98 = 4299$

The required average $= \dfrac{8932}{200} = 44.66$

37. The number of students who scored more than 70 marks in the different sections are

Quant

Boys	Girls
51	46

Reasoning

Boys	Girls
65	48

English

Boys	Girls
42	57

∴ At most 42 boys and 46 girls could have scored more than 70 marks in each section.

38. The maximum total marks scored by the girls in the English section is

$= 1 \times 11 + 14 \times 32 + 1 \times 34 + 26 \times 58 + 1 \times 62 + 20 \times 77 + 1 \times 78 + 36 \times 92$

$= 11 + 448 + 34 + 1508 + 62 + 1540 + 78 + 3312$

$= 69.93$

39. The minimum total marks of the boys in the reasoning section is

$14 \times 11 + 1 \times 27 + 18 \times 29 + 1 \times 48 + 30 \times 51 + 1 \times 72 + 34 \times 74 + 1 \times 95$

$= 154 + 27 + 522 + 48 + 1530 + 72 + 2516 + 95$

$= 4964$

Average = 49.64

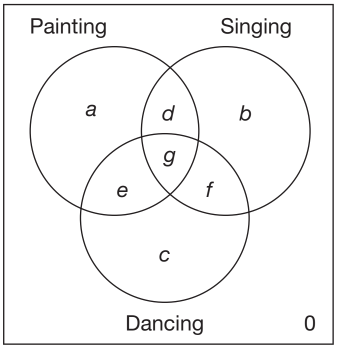
Solutions for questions 40 to 44:

Total number of boys = $\frac{9}{16}$ (6720) = 3780

Total number of girls = 6720 – 3780 = 2940

Boys

Total = 3780



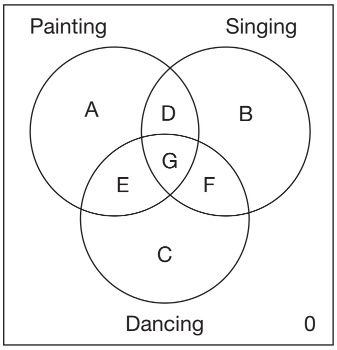
$a = \frac{3780}{12} = 315$

$b = \frac{3780}{10} = 378$

$g = \frac{1}{9}(3780) = 420$

Girls

Total = 2940



$\text{B} = \frac{25}{100}(2940) = 735$

$\text{D} = \frac{20}{100}(2940) = 588$

$\text{A} = \left(1+\frac{10}{100}\right) = 2a = 630$

$\frac{f}{F} = \frac{7}{5}$

$\text{C} = \frac{1}{10}(2940) = 294$

$\text{F} = \frac{8\frac{1}{3}}{100}(2940) = \frac{2940}{12} = 245$

$f = \frac{7}{5}(245) = 343$

G = 2940 – (A + B + C + D + E + F) where E = 0

∴ G = 448

$d = \frac{D}{2} = 294$

$c = 3780 - (a + b + d + e + f + g)$ where $e = 0$

∴ $c = 2030$

40. P = $a + d + e + g$ = 315 + 294 + 0 + 420 = 1029

41. Required % = $\frac{c+e+f+g}{6720} \times 100\%$

$= \frac{2030+0+343+420}{6720} \times 100\% = 41.56\% \approx 42\%$

42. Total number of students enrolled in all the three classes together = g + G = 420 + 448 = 868

43. Percentage that b forms of B = $\frac{b}{B} \times 100\%$

$= \frac{378}{735} \times 100\% \approx 51.43\%$

Solutions for questions 44 to 46: Let ₹100x be the amount deposited by the car manufacturer on each car in race 1. Since car I won race1, he gets ₹180x on that and only ₹80x on the other car that participated. The amount he wins at the end of the race 1 = ₹180x + ₹80x = ₹260x. Then he deposits ₹130x on each of the two cars that participated in race 2. Since car II won the race 2, he gets 130x (1.4), i.e., ₹182x on that and only 130 × (0.8), i.e., ₹104x on the other car that participated. The amount with him at the end of the race 2 = ₹182x + ₹104x = ₹286x. Then he deposits ₹143x on each of the two cars that participated in the race 3. Since car II won race 3, he gets 143 (1.4), i.e., ₹200.2x on that and only 143(0.8), i.e., ₹114.4x on the other car that participated.

∴ The amount with him at the end of race 3 = ₹200. 2x + ₹114.4x = ₹314.6x. Given that ₹314.6x = ₹31460

⇒ $x = 100$

44. The total amount he deposited in race 1 = ₹200x
= ₹200 × 100 = ₹20,000.

45. Amount gained in race1 = ₹260x – ₹200x = ₹60x
Amount gained in race2 = ₹286x – ₹260x = ₹26x
Amount gained in race3 = ₹314.6x – ₹286x = ₹28.6x
Hence, he gained the most in race 1.

46. At the end of race 3, he would have = 20000 (1.8) (0.8) (0.8) = ₹23040. Total money gained by the end of the three races would have been = ₹23040 – ₹20000 = ₹3040.

Solutions for questions 47 to 50: It is given that student Q had the second highest marks among all the students. As the difference between the marks of Q and R is always 42 with R being higher, it can be concluded that R scored the highest marks and it was 42 marks more than Q. Now comparing P and Q and as we know P scored less then Q, the difference between P and Q has to be 121, the common value among (106 + 15, 106 – 15), (100 + 21 or 100 – 21),(86 + 207 or 207 – 86) and (52 + 69 or 69 – 52).

In the same way we can get the value of S as Q – 197 (207 + 10 or 207 – 10), (69 + 128 or 128 – 69)

Similarly, V = Q – 106
W = Q – 21
X = Q – 207 and
Y = Q – 69.

Now taking the value of Q as A, we can get the other values as follows:

P – (A– 121)
Q – A
R – (A + 42)
S – (A – 197)
T – (A – 120) or (A – 258)
U – (A – 121) – 27 or (A – 121) + 27
V – (A – 106)
W – (A – 21)
X – (A – 207)
Y – (A – 69)

47. R scored the highest marks among the ten students.

48. If X = 450, P + R = X – 121 + X + 42 = 2X – 79 = 900 – 79 = 821.

49. Student 'W' scored the highest marks among U, V, W, X and Y.

50. U is P – 27, or P + 27, i.e., 538 or 592.

Solutions for questions 51 to 54:

51. Team E lost to teams C and D, drew against team B and won against team A.
$\therefore$ The points scored = 10 + 4 – (2 × 5) = 4 points.
For goals = 15 × 2 – 19 × 1 = 11.
$\therefore$ Total = 4 + 11 = 15 points.

52. Team A has conceded 16 goals.
16 × (–1) = –16 points
Team A made 15 goals.
15 (2) = 30 points
A has won against C and D.
Points won = 2 (10) = 20 points.
A loses two matches, points lost = (2) (5) = 10
Number of points won by A = 30 + 20 – 16 – 10 = 24
In this way.
The total points for B
= (14 × 2) – (12 × 1) + 2 (10) + (14) – 1 (5)
= 25 – 12 + 20 + 4 – 5 = 35
Total points for C
= (16 × 2) – 16 (1) + 2 (10) – 2 (5)
= 32 – 16 + 20 – 10 = 26
Total points for E
= (15 × 2) – (19 × 1) + (4 × 1) + (10 × 1) – (5 × 2)
= 30 – 19 + 4 + 40 – 10 = 15
Total points for D
= (15 × 2) – (12 × 1) + 2 (10) – 2 (5)
= 30 – 12 + 20 – 10 = 28
$\therefore$ B has won the tournament.

53. B has drawn with E. There is only one draw.

54. A, B, C and D have 2 win each.

Solutions for questions 55 to 58:

55. The number 4 appeared in cast 140.
The number 4 must appear 10 times from cast no. 121 to cast no. 140.
The cast numbers in which (4) must happen are
(1) 121, 123, 125, 127, 129, 131, 133, 135, 137,140.
(or)
(2) 122, 124, 126, 128, 130, 132, 134, 136, 138,140.
It can be observed that the number 4 may or may not appear in cast no. 131.
However, it is evident that the number 1 must not have appeared from cast no. 121 to 140.

56. The number 2 had already turned up for 15 times by the end of the first 60 casts. So, in the first 65 casts, the number 2 cannot appear for less than 15 times. However, other numbers must appear for less than 15 times.

57. To calculate the minimum possible number of times 4 turned up in 95 casts
= Number of times 4 turned up in 100 casts – Maximum possible number of times 4 could have turned up from cast 96 to cast 100 (both inclusive).
This maximum is 3 (i.e., in casts 96, 98 and 100).
$\Rightarrow$ Required answer = 20 – 3 = 17.

58. We only need to check for number 4 as it is the highest among 2, 4 and 6. 4 can turn up a maximum of 5 times between the 121st and 130th cast.
$\therefore$ The maximum possible value is (22 + 5) – 13 = 14 times.

Solutions for questions 59 to 62: Let x = Number of correct answers

z = Number of wrong answers
y = Number of unanswered questions
Given, $x + y + z = 100$ (1)
$12x - 2y - 3z = 625$ (2)
3(1) + (2) : $15x + y = 925$
$y = 925 - 15x = 5(185 - 3x)$
$\therefore$ y must be divisible by 5. Least $y = 0$.
When $y = 0$ or 5, x is not an integer.

When $y = 10$, $x = 61$.

$\therefore$ Least $y = 10$. This also means greatest $x = 61$.

$\therefore$ $x \leq 61$.

When $x = 61$, $y = 10$ and $z = 29$.

When $x = 60$, $y = 25$ and $z = 15$.

When $x = 59$, $y = 40$ and $z = 1$.

For every decrease of 1 in x, y increases by 15 and z decreases by 14.

$\therefore$ For $x < 59$, $z < 0$, but this is not possible.

$\therefore$ $x \geq 59$.

59. $(x, z, y) = (59, 1, 40)$ or $(60, 15, 25)$ or $(61, 29, 10)$.
There are three ways in which Hari could have attempted the exam.

60. Number of questions attempted = $x + z$. This is 60 or 75 or 90.
$\therefore$ Max$(x + z) = 90$.

61. x is a multiple of 5 only when $x = 60$.
$\therefore$ $z = 15$.

62. The three ways by which Hari could have scored 625 marks is as follows.
(1) 60 correct, 15 wrong and 25 unanswered.
(2) 59 correct, 1 wrong and 40 unanswered.
(3) 61 correct, 29 wrong, and 10 unanswered.
$\therefore$ The number of unanswered questions could be 10, 25 or 40.

Solutions for questions 63 to 66:

63. The total points would be the minimum, when all the matches are draws and each player scores 14 points. If one player scores one win, he will have 16 points and the player who lost would have 13 points. Therefore, if R is the sole winner, he should have at least 16 points.

64. The maximum points happen when the maximum number of matches produce a result. So, try to keep the number of draws to minimum starting with 7 wins and 7 loses for each player and assigning, we get the points as 24(8W), 23(7W, 2D), 22(7W, 1D), 21(7W), 20(6W, 2D), 19(6W, 1D), 18(6W) and 17(5W, 2D), with only 4 draws.
Note: We cannot take maximum points (168) and distribute as the points would be of the form 25,24, 23, 22, 20, 19, 18 and 17 as points like 25, 23 and so on can be obtained only with draws and for 168 points in total, there cannot be any draws.

65. The total points has to be between 56 (2) = 112 and 56 (3) = 168. It cannot be 108.

66. The minimum points for the winner would be 19 as follows:

3W	1L	10 D = 19
2W	0L	12 D = 18
2W	1L	11 D = 17
2W	2L	10 D = 16
1W	1L	12 D = 15
OW	OL	14 D = 14
OW	2L	12 D = 12
OW	3L	11 D = 11
10W	10L	

Solutions for questions 67 to 70:

67. Let the national sales of LCVs be 380 units and that of CVs be 100 units.
Sales of LCVs by ABC Ltd = 36% of 380 = 136.8
Sales of CVs by ABC Ltd = 42% of 100 = 42

$$\text{Required percentage} = \frac{42}{136.8} \times 100 = 30.7$$

68. Let the sales of luxury sedans by ABC Ltd be 35 units.
$\therefore$ Sales of compact sedans by ABC Ltd = 175 units
National sales of luxury sedans = 100
National sales of compact sedans

$$= \frac{175(100)}{56} = 312.5$$

$$\text{Required percentage} = \frac{312.5}{312.5+100}(100)$$

$$= \frac{312.5}{412.5}(100) = 75.8$$

69. Let the number of units of multi-axle vehicles sold by ABC Ltd be 10.
$\therefore$ National sales of multi-axle vehicles = 40.
The vehicles sold by ABC Ltd in all other segments (other than multi-axle) is 90. As the share of all other segments of vehicles sold by ABC Ltd is at least 33, the national sales of all vehicles (other than multi-axle) would be a maximum of $\frac{90}{33} \times 100 = 273$

$\therefore$ Multi-axle vehicles form at least $\frac{40}{313} \times 100$
$= 12.8\%$

70. If the number of SUVs sold is much more than hatchbacks and luxury sedans, then the national share of ABC Ltd in these three categories together would be very close to 44%. In this case the share of compact sedans can be as low as 25% and still the total share would be 47.

Solutions for questions 71 to 74:

71. Let the number of men and women among Hindus be $1000x$ and $900x$, respectively. And let the number of men and women among Muslims be $1000y$ and $875y$, respectively.

$\Rightarrow 1000x - 900x = 1000y - 875y$

$\Rightarrow 100x = 125y$

$$\Rightarrow \frac{x}{y} = \frac{5}{4}$$

$\therefore$ Required ratio 1000y : 900x

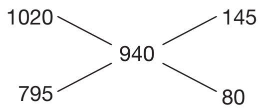
$$= 1000\left(\frac{4}{5}\right)x : 900x = 8 : 9.$$

72. Assume that in 2011, the number of Christians, Muslims and Jains in the locality is negligible compared to that of Hindus and Sikhs.
$\therefore$ The number of women per 1000 men (excluding Sikhs) would be 900 (same as that of Hindus). As the overall value is 940, the number of Sikhs and Hindus here should be in the ratio 1 : 2 or Sikhs form 1/3rd of the total number of people in the locality.

73. Let the number of Christian men be $100x$ and the number of Muslim men be $100y$.
Given $179.5x = 187.5y$

$$x = \frac{187.5}{179.5}y \text{ or } x = 104.46.$$

74. For the percentage of Sikhs to be maximum, we have to assume that Christians and Sikhs are significant in number while all others are close to zero.
The required percentage would be



$$\frac{145}{145+80} \times 100 = \frac{145}{225} \times 100 = 64.4\%$$

Solutions for questions 75 to 78:

75. To complete the task in the minimum time we should assign each subtask to the machine which will do it in the least possible time, i.e., $T_1 - M_4$, $T_2 - M_2$, $T_3 - M_1$ and $T_4 - M_3$. The time taken is 5 + 3 + 4 + 2 = 14 hrs.

76. If machine M_3 is not working, to do in the shortest possible time, M_2 must be assigned subtasks T_1 and T_2, with M_1 doing T_3 and M_4 doing T_4. The total time would be the time taken by M_2 to finish T_1 and T_2 by which time the other machines would have finished T_2 and T_4, i.e., 8 hrs.

77. We have to select two machines which could finish the four subtasks in the shortest possible time. Since M_2 does task T_2 and M_3 does task T_4 in much less time compared to other machines, these two machines must be selected. So M_2 would do tasks T_1 and T_2 and M_3 would do tasks T_3 and T_4. Time taken = 5 + 3 = 8 hrs.

78. First assign T_1 to M_4, T_2 to M_2 T_3 to M_1 and T_4 to M_3. After 2 hrs T_4 would be over and M_1 and M_3 can finish T_3 in 3.2 hrs. T_2 would be over in 3 hrs and M_2 can join in doing T_1 and it would be completed in slightly under 3.7 hrs.

Solutions for questions 79 to 82: Originally money realised by sales = 10 × 50000 = 5,00,000
Money realised from (3)
= 5,00,000 – 0.5 × 0.4 × 5,00,000 = 4,00,000
Money realised from (2)
= 5,00,000 [0.1 × 0.5 + 0.2 × 0.5 + 0.1 × 0.4] – 2,00,000 + 5,00,000 = 3,95,000
Money realised from (1)
= 0.5 × 50000 [1 + 0.5 × 0.3 + 0.5 × 0.2] = 3,12,500

79. Hence, the most profitable option would be to remain silent.

80. Decreases in (3); increases by 19% in (2); increases by 25% in (1).

81. Loss = 5,00,000 - 3,12,500 = 1,87,500

82. As calculated above, it is ₹3,95,000.

Solutions for questions 83 to 86:

83. The total score of the class = 60 × 60 = 3600
The total score of top 20% and the bottom 20% students

$$= (86 + 19) \times \frac{20\times 60}{100} = 105 \times 12$$

$\therefore$ The total score of other students = 3600 – 105 × 12
The 48th ranker will get the maximum possible rank when all these (except for 20% and bottom 20%) students get equal marks.

$$\therefore \text{ Required score} = \frac{3600 - 105\times 12}{36} = 65.$$

84. Average scores of 40% students (i.e., top 20% and bottom 20%) for class.

$$\text{I} - \frac{82+36}{2} = 59$$

II – 53.5
III – 46
IV – 52.5
V – 59
For I, let the remaining students get an average score of x.

$$\therefore \frac{59\times 2 + x\times 3}{5} = 50$$

$\Rightarrow x = 44$
Similarly, we can find the values for other classes among which classes II, III and IV have an average more than 45.

85. For I, the highest possible average of the remaining 60% students is 82.

$$\therefore \text{ Average of the class} = \frac{82(4)+36}{5} = 72.8$$

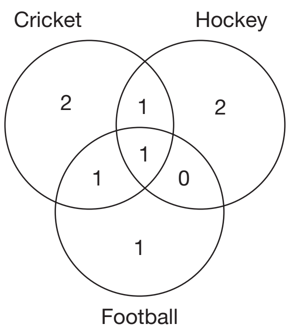
Similarly, the averages for II, III, IV and V are 67, 59.2, 72.6 and 71.6.

86. The least possible average of I $= \dfrac{82+36(4)}{5} = 45.2.$

The values for II, III, IV and V are 40, 32.8, 32.4 and 46.4.

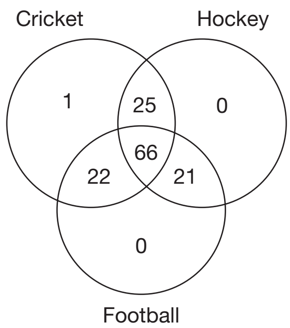
Solutions for questions 87 to 90:

87. There must be at least one student who played all the three. The Venn diagram for the minimum possible case satisfying all the given conditions can be as follows.



So, at least 8 students must play at least one of the three games. Maximum number of students who do not play any of the games is 135 – 8 = 127.

88. Since a maximum of 135 students play, considering the number of people playing all the sports is the least, it should be less than half. Maximum possible value is 66. 67, it is not possible as there should be 68 playing exactly two and at least one playing exactly one, making a total of 67 + 68 + 1 = 136. The following is one of the many possible cases that satisfy the above conditions.



Students playing all the three sports = 66
Students playing two sports = 68
Students playing football = 110
Students playing hockey = 112 and
Students playing cricket = 114, which satisfies all the conditions, including the condition that there are students who play exactly one sport.

89. The number of students who play only cricket can be zero, the number of students who play only hockey can be 1 and only football can be 2. Of the remaining 132, as people playing exactly two is less than people playing exactly three, the maximum number of people playing exactly two games is 65.

90. Here, consider the students who play only cricket to be 0, only hockey is 1, only football is 2 and exactly two is 3.
$\therefore$ The minimum value of students who play all the three is 4.

Unit 3

Logical Reasoning

CHAPTER 1

Linear Arrangement

Learning Objectives

In this chapter, we will:

- Understand how to interpret the information/statements given in the question and get a final arrangement out of it
- Learn to make arrangement of people/objects in horizontal rows
- Gain an understanding of arrangement of people/objects in vertical columns
- Grasp the concept of arrangement of people in rows and columns
- Learn to make linear arrangement with a distribution of multiple parameters
- Acquire knowledge of linear arrangement with people facing different directions

Linear Sequencing

LINEAR SEQUENCING is essentially arranging the items in a sequence (in a single line). Questions of this type are also referred to as **seating arrangement**. The word seating arrangement should not be misconstrued and it should not be treated as questions involving only people sitting as per specified conditions. Essentially, these questions involve arranging subjects (people or things) according to the given conditions. The arrangement is done only on one axis, and hence, the position of the subjects assumes importance here in terms of order, like first position, second position, etc.

Let us look at the examples.

SOLVED EXAMPLES

Directions for questions 1.01 to 1.05: Read the data given below carefully and answer the questions that follow.

Seven people Paul, Queen, Rax, Sam, Tom, Unif and Vali are sitting in a row. Rax and Sam sit next to each other. There are exactly four people between Queen and Vali. Sam sits to the immediate left of Queen.

1.01: If Paul and Tom are separated by exactly two person, then who sits to the immediate right of Vali?

(A) Paul (B) Tom
(C) Unif (D) Rax

1.02: If Queen is not sitting at either ends of the row, then who among the following has as many persons on the left as on the right?

(A) Sam (B) Unif
(C) Rax (D) Vali

1.03: If Queen sits at one end of the row, then who sits at the other end?

(A) Paul
(B) Tom
(C) Vali
(D) Cannot be determined

1.04: If Tom sits to the right of Queen, and Paul is separated from Tom by exactly three people, then who sits to the immediate right of Vali?
(A) Sam (B) Paul
(C) Tom (D) Unif

1.05: In how many different ways can this seven people sit in a row?
(A) 3 (B) 2
(C) 10 (D) 12

Solutions for questions 1.01 to 1.05: Let us write down the conditions given in short form and then represent them pictorially. Also, let us treat the people sitting at left as 'left' and their right as 'right' for interpreting the conditions.

Rax and Sam sit next to each other ➔ RS or SR.

There are exactly 4 people between Queen and Vali ➔ Q — — — — V or V — — — — Q.

Sam sits to the immediate left of Queen ➔ SQ.

Now let us analyse the data/conditions that we are given and then put the three conditions together. Let us number the seats from our left to right as Seat 1 to Seat 7.

Since S is to the left of Q and since R and S have to be next to each other, R can only be to the immediate left of S. Thus, R, S and Q, will be in the order RSQ. Since there are four people between Q and V, Q can be placed in seats 1, 2, 6 or 7. But if Q is in Seat 1 or 2, then there are no seats for R and S. Hence, there are only two seats available for Q. Let us fix the positions of R, S and V in each of these two positions of Q and write them down. The directions left and right is as shown below.

◀L ▶R

Arrangement I:

1	**2**	**3**	**4**	**5**	**6**	**7**
	V			R	S	Q

Arrangement II:

1	**2**	**3**	**4**	**5**	**6**	**7**
V			R	S	Q	

These are the only two arrangements possible for the four persons V, R, S and Q. The other three persons Paul, Tom and Unif can sit in the three vacant seats in any order, as no information is given about them. Now let us look at each of the questions.

1.01: Paul and Tom are separated by exactly two persons. Arrangement I is the only one possible as in Arrangement II, Paul and Tom cannot have exactly two persons between them. So, we have the arrangement as follows: T/P, V, U, P/T, R, S, Q So, Unif must be sitting to the immediate right of Vali.

1.02: If Queen is not at the end, then only Arrangement II mentioned above is possible. The person who has as many people seated on the left as on the right can only be the person who is sitting in the middle seat, i.e., Seat 4. In this arrangement, Rax is sitting in Seat 4.

1.03: 'Queen sits at one end' means that we should look at Arrangement I. In this arrangement, any one out of the three person Paul, Tom and Unif can be in Seat 1.

1.04: If Tom and Paul are separated by exactly three people, then only Arrangement II is possible. So, Tom and Paul have to be in seats 3 and 7. Since, it is also given that Tom is to the right of Queen. Tom has to be in Seat 7 and Paul in Seat 3. So, the arrangement must be as follows: V, U, P, R, S, Q, T
The person sitting to the immediate right of Vali is Unif.

1.05: We have two possible arrangements, such as 'Arrangement I and Arrangement II' that we have already looked at. In each arrangement, the remaining three people can sit in the remaining three seats in 6 ways. Thus, a total of 12 ways of seating the seven people is possible.

Directions for questions 1.06 to 1.10: These questions are based on the following information.

A group of eight people, namely G, H, I, J, K, L, M, and N are sitting in a row but not necessarily in the same order. Some of them are facing south and the remaining are facing north. No two person is sitting adjacent to each other facing south.

- J sits two places away from H and they both face different directions.
- M sits second to the right of G.
- The number of people to the right of M is one less than the number of people to the right of K.
- K sits to the immediate right of H, who faces the same direction as I face.

- I sits to the immediate left of N and one of them sits at an end.
- G sits to the left of N, who sits to the left of L.
- K and L face the same direction which is different from the direction which M faces.

1.06: Who sits second to the right of J?
(A) K (B) M
(C) L (D) N

1.07: Three of the following are alike in a certain way, and hence form a group. Which is the one that does not belong to that group?
(A) JH (B) IH
(C) MG (D) MK

1.08: Which of the following is/are definitely true?
(A) M and G face different directions.
(B) K sits three places away from J.
(C) Two people sit between L and H.
(D) Both (A) and (B).

1.09: Who sits to the immediate right of G?
(A) K (B) L
(C) H (D) J

1.10: How many people are facing north?
(A) Five
(B) Three
(C) Two
(D) Cannot be determined

Solutions for questions 1.06 to 1.10: It is given that M sits second to the right of G, who sits to the left of N. Thus, if N sits at the right end, N must face north, since I sits to the immediate left of N. As J sits two places away from H and they both face different directions and K sits to the immediate right of H, J must be sitting at the third place from the left end and H sits two places away from I. It is given that H and I face the same direction. For the above conditions to satisfy, I must face north. Thus, H also faces north and K sits adjacent to I. Since H and J face different directions, J faces south. Thus, the neighbours of J must face north as no two people sitting adjacent to each other face south. It is also given that M sits second to the right of G and the number of people to the right of M is one less than the number of people to the right of K. Hence, M must be sitting to the immediate left of J and G to the immediate right of J. As K and L face the same direction which is different from the direction in which M faces, K and L must face south as M faces north and L sits at the left end. Hence, the possible final seating arrangements are as below.

L↓ G↑ J↓ M↑ H↑ K↓ I↑ N↑

or

↓N ↓I ↑K ↓H ↓M ↑J ↓G ↑L

1.06: L sits second to the right of J.

1.07: Except MK, in all other options the second person sits second to the left of the first person.

1.08: K sits three places away from J.

1.09: J sits to the immediate right of G.

1.10: Since there are two arrangements, we cannot determine

EXERCISE-1

Directions for questions 1 to 9: Select the correct alternative from the given choices.

1. Five people L, M, N, O and P sit in a row, not necessarily in the same order. P sits exactly in between M and N. If L sits exactly in between M and O, then which of the following must be true?
(A) O sits to the immediate right of M.
(B) L and N always sit together.
(C) M sits exactly at the centre of the row.
(D) P sits between M and L.

2. Six people, namely Tanmay, Sanjay, Ganpat, Dhruv, Nagraj and Jivan are standing in a queue at a railway ticket counter. Further it is known that
(i) Ganpat is two positions ahead Jivan.
(ii) Only Nagraj is ahead Tanmay.
(iii) Neither Sanjay nor Jivan is standing at the end of the queue.
How many people are ahead of Dhruv but behind Tanmay?
(A) Zero (B) Two
(C) Three (D) Four

3. The Principal called five people, namely Srinivas, Murali, Raghu, Vijay and Krishna who are working as the Director, Secretary, Treasurer, Professor and Student Leader of a college, not necessarily in that order. They are seated in the five seats facing the Principal.
The Treasurer sat to the immediate left of Krishna who is one seat away from the Director.
Murali is two places away from the Secretary.
Vijay, who is the Student Leader, is one place to the right of Murali.
What is the position of Krishna with respect to the Professor?
(A) To the immediate right.
(B) Three places away to the left.
(C) Two places away to the left.
(D) None of the above

4. Seven men, A, B, C, D, E, F and G have parked their cars in a row. The cars of E and F should be next to each other. The cars of D and G should be parked next to each other. Whereas A and B cannot park their cars next to each other. But B and D must park their cars next to each other and C's car is parked to the immediate right of G's car. If E parks his car to the left of F, then which of the following statements is false?
(A) There are two cars in between B and G's cars.
(B) B and C's cars are not parked together.
(C) G's car is the only car in between D and C's cars.
(D) A's car is at the left extreme end.

5. Five people A, B, C, D and E are sitting in a row facing the same direction. A is two places away to the right of B. C is two places away to the left of D. E is not sitting at the extreme right. Who is sitting in the middle of the row?
(A) A (B) B
(C) C (D) Cannot be determined

6. Five people A through E are sitting in a row facing the same direction. A is three places away to the right of C. Two people are sitting between. B and D. who is sitting in the middle of the row?
(A) A (B) C
(C) E (D) D

7. A group of five people, namely Arnab, Ankur, Adi, Anush and Asraf are sitting in a row facing the same direction. There are at least two people sitting between Arnab and Asraf. There is at most one person sitting between Ankur and Anush. If Anush is sitting to the immediate right of Adi, who is adjacent to Arnab, then which of the following is true?
(A) Arnab is sitting at the extreme right.
(B) Asraf and Anush are adjacent to each other.
(C) Asraf is sitting between Anush and Arnab.
(D) Asraf and Ankur are adjacent to each other.

8. There are seven people, named M, N, O, P, Q, R and S sitting in a row, facing the same direction.
M is five places to the right of O.
P is four places to the right of Q.
R is three places to the right of S.
Who is sitting in the middle of the row?
(A) P (B) N
(C) S (D) R

9. A group of six people, namely Alpana, Brahma, Chetana, Drona, Ena and Fanna are sitting in a row. Alpana and Fanna are sitting adjacent to each other. Chetana is two places to the right of Ena and neither of them is sitting at the extreme ends. There is one person sitting between Alpana and Brahma. Who is sitting to the immediate right of Ena?
(A) Alpana (B) Fanna
(C) Drona (D) Brahma

Directions for questions 10 to 12: These questions are based on the following information.

A group of five people, namely Amit, Balram, Chetan, Deepak and Eswar are sitting in a row facing North. The following information is known about them.

(i) Only Deepak is sitting between Amit and Balram.
(ii) Neither Amit nor Balram is at the ends.
(iii) Chetan is sitting to the immediate left of Balram.

10. Who is sitting at the right end of the row?
(A) Amit (B) Balaram
(C) Chetan (D) Eswar

11. How many people are sitting between Amit and Chetan ?
(A) Zero (B) One
(C) Two (D) Three

12. What is the position of Eswar with respect to Balram?
(A) Immediate right (B) Second to the left
(C) Third to the right (D) Immediate left

Directions for questions 13 to 15: These questions are based on the following information.

Six people, namely P, Q, R, S, T and U are sitting in a row facing north. Further it is known that:
(i) Exactly two people are sitting between P and Q.
(ii) Exactly one person is sitting between T and U.
(iii) Q is sitting at the right end of the row.

13. If U is sitting adjacent to S, then how many people are sitting between U and R?
(A) One (B) Two
(C) Three (D) Cannot be determined

14. If S is sitting to the immediate right of T, then who is sitting second to the right of R?
(A) P (B) T
(C) U (D) S

15. Who among the following cannot be adjacent to T?
(A) P (B) R
(C) S (D) None of these

Directions for questions 16 to 18: These questions are based on the following information.

Eight books on different subjects, such as Biology, Chemistry, Physics, Maths, English, Hindi, Zoology, and Economics are stacked together. Further it is known that:
(i) Economics is above Biology which is just above Hindi, which is not at the bottom.
(ii) There are only two books between the Zoology and the English books.
(iii) Number of books above Chemistry is less than the number of books below it.
(iv) Only Maths book is above Zoology.

16. Which book is at the bottom of the stack?
(A) Physics (B) Hindi
(C) English (D) Economics

17. How many books are there between Economics and Hindi?
(A) Three (B) Two
(C) Four (D) Cannot be determined

18. Find the pair that does not exhibit a similar relationship as the other three pairs.
(A) Biology – Hindi
(B) Economics – Chemistry
(C) English – Hindi
(D) Maths – Zoology

Directions for questions 19 to 21: These questions are based on the following information.

Six buildings of different colours red, yellow, white, blue, green and orange are in a row. Each of these buildings belongs to a different person among Dubey, Sharma, Roy, Sanyal, Tiwari and Reddy. Following is the information known about them.
(i) Green building is three places to the right of Dubey's building.
(ii) Red building is three places to the right of Sharma's building.
(iii) White building is three places to the right of Reddy's building.
(iv) Roy's building is adjacent to the orange building.
(v) Sanyal's building is not green. Sharma's building is not blue.
(vi) Tiwari's building is not adjacent to Roy's building but three places away from Reddy's building.

19. What is the colour of Dubey's building?
(A) Blue (B) Yellow
(C) Green (D) Red

20. The red building belongs to
(A) Roy (B) Sanyal
(C) Tiwari (D) Dubey

21. Which of the following is true?
(A) There is at least one building between the orange and green coloured building.
(B) Reddy's building is to the left of Sharma's building.
(C) Reddy's and Sharma's buildings are not adjacent
(D) Sanyals' building is to the right of Sharma's building.

Directions for questions 22 and 23: These questions are based on the following information.

In a school there are five classes (Class I to Class V) and each class has two sections A and B. Each section is accommodated in a different classroom. The class rooms are in a row.
(i) The two sections, A and B of any class are not adjacent to each other.
(ii) Any four consecutive classrooms, accommodate two A sections and two B sections.
(iii) Class V A is three places away from Class I A and neither of these two is at any of the extreme ends.
(iv) Class III B is three places away from Class IV B and neither of these is at any of the extreme ends.

(v) Class II B is not at any of the extreme ends.
(vi) Class II A is at the extreme right.
(vii) Class V A is to the right of Class III B.

22. Which class is at the extreme left?
(A) I (B) V
(C) III (D) Cannot be determined

23. Which class is to the immediate right of Class III B?
(A) I A (B) II B
(C) V A (D) I B

Directions for questions 24 to 27: These questions are based on the following information.

Seven friends P, Q, R, S, T, U and V sit on a bench facing north. Each of them is of a different weight (in kg), their random bodily weights are 79, 83, 85, 87, 89, 92 and 96. The following information is known about them.

P sits third to the right of the heaviest person. The heaviest person sits exactly between R and the lightest person, who sits at an end. The third lightest person sits adjacent to R and that person is neither P nor adjacent to P. Q sits third to the left of the person whose weight is the next higher to R. R's weight is neither 83 kg nor 87 kg. P's weight is neither 92 kg nor 79 kg. T's weight is 83 kg. S is heavier than V but is not the heaviest.

24. Who is the third lightest?
(A) P (B) Q
(C) R (D) S

25. How many people sit between R and U?
(A) One (B) Two
(C) Three (D) Five

26. Who sits second to the right of the heaviest person?
(A) P
(B) S
(C) The person whose weight is 89 kg
(D) The person whose is weight is 87 kg

27. How many persons are lighter than S?
(A) Four (B) Five
(C) Three (D) Two

Directions for questions 28 to 30: These questions are based on the following information.

In a conference, five delegates A, B, C, D and E who are from different countries hailing from India, Pakistan, Sri Lanka, Bangladesh and Nepal are sitting in a row facing North.

(i) The delegate from Bangladesh is to the immediate left of the delegate from Sri Lanka.
(ii) A is the only person sitting between C and D. D is to the immediate right of E.
(iii) B, the delegate from Pakistan is sitting at one of the extreme ends.
(iv) The delegate from Nepal is sitting at the middle of the row.

28. In B is not adjacent to E, then who is the delegate from Nepal?
(A) D (B) C
(C) A (D) B

29. If D is not the delegate from Sri Lanka, then who is sitting at the extreme left end of the row?
(A) The delegate from Sri Lanka.
(B) The delegate from Pakistan.
(C) The delegate from Bangladesh.
(D) Cannot be determined

30. Which of the following is false?
(A) A is the delegate from Bangladesh.
(B) E is sitting at the extreme left end.
(C) C is the delegate from Nepal.
(D) None of these

Directions for questions 31 to 33: These questions are based on the following information.

Seven people, namely Akhil, Bhavya, Chaitra, Dinker, Eashan, Fallon and Geet are sitting in a row. The total number of people in the row is 28.

(i) Akhil is the eighth person from the left end.
(ii) Among the seven people Geet is in the right most position.
(iii) Eashan and Dinker are adjacent to each other. Except these two person no two among the given seven people are adjacent to each other.
(iv) There are thirteen people between Geet and Akhil.
(v) There are two people between Akhil and Chaitra.
(vi) Dinker and Eashan are sitting to the right of Chaitra and left of Fallon.
(vii) There are four people between Fallon and Geet.

31. If fourteenth position from the left is Bhavya, then at the most, how many people are there between Dinker and Chaitra?
(A) Five (B) Seven
(C) Six (D) Four

32. If Chaitra and Fallon interchange their positions, then how many people are there between Chaitra and Akhil?
(A) Eight (B) Ten
(C) Nine (D) Cannot be determined

33. If the people in the6th, 10th and 16th positions from the left leave the row, then how many people are there between Chaitra and Fallon?
(A) Five (B) Nine
(C) Six (D) Cannot be determined

Directions for questions 34 to 36: These questions are based on the following information.

Eight people, namely A, B, C, D, E, F, G, and H are sitting in a row facing north. There are exactly two people sitting between D and E. A is sitting third from the left end. B, A, C and G are sitting in that order from left to right but no two of them are in adjacent positions. D, F and H are sitting from left to right in that order, but not necessarily in adjacent positions.

34. If H and D interchange their positions and then D and E interchange their positions, who sits second to the right of A?

(A) H (B) D
(C) E (D) Cannot be determined

35. How many people sit between B and C?

(A) Three (B) Four
(C) Five (D) Cannot be determined

36. Which of the following pair of people sits at the ends?

(A) B, H (B) G, D
(C) B, G (D) H, D

Directions for questions 37 to 40: These questions are based on the following information.

There are six floors in an apartment (The ground floor is named as the first floor, the floor above the ground floor is named as the second floor and so on). There are 12 rooms from A through L. Each floor contains two rooms, which are adjacent to each other. These 12 rooms have two columns, with six rooms in each column.

(i) There are three floors between rooms H and G. G does not live on the top floor.
(ii) The rooms H and E are in the same floor.
(iii) Rooms B and E are in consecutive floors.
(iv) Room C is on the sixth floor and room D is on the first floor.
(v) The number of floors between room A and room C is equal to the number of floors between room G and room K.
(vi) Room F is above room G and in the same column.
(vii) Room A is in the odd numbered floor. Room I is above room L and below room F.
(viii) Room L is above room D in the same column and below room I in the same column.

37. Which of the following rooms are on the second floor?

(A) L, A (B) B, J
(C) L, J (D) Cannot be determined

38. How many floors are there between rooms H and J?

(A) One (B) Two
(C) Three (D) Cannot be determined

39. The room which is right above room J is

(A) I (B) A
(C) L (D) Cannot be determined

40. Which room is exactly between rooms I and D in the same column?

(A) L (B) J
(C) G (D) Cannot be determined

EXERCISE-2

Directions for questions 1 to 3: These questions are based on the following information.

Seven flags of different colours, such as Violet, Indigo, Blue, Green, Yellow, Orange and Red are placed in a row from left to right, not necessarily in that order. The Indigo flag and the Yellow flag have four flags between them. The Orange flag is not between the Indigo flag and the Yellow flag. The Violet flag and the Red flag cannot be next to the Blue flag.

1. What is the total number of possible arrangements?

(A) 12 (B) 8
(C) 4 (D) None of these

2. If the Black flag and the White flag are also to be placed in the row, such that they are adjacent to each other but neither of them is next to the Violet flag or the Red flag, and also neither of them is at any of the ends, then what is the total number of possible arrangements?

(A) 24 (B) 48
(C) 96 (D) None of these

3. Which of the following statements is true?

(A) Yellow flag is at one of the ends.
(B) The Indigo flag and the Orange flag are adjacent to each other.
(C) The Blue flag is adjacent to the Green flag.
(D) The Yellow flag or the Indigo flag is/are adjacent to both the Orange flag and the Blue flag.

Directions for questions 4 to 6: These questions are based on the following information.

A group of seven people, namely A, B, C, D, E, F, and G are standing in a queue in front of a ticket counter. The following information is known about them.

(i) The number of people standing in front of A is same as the number of people standing behind C.
(ii) The number of people standing in front of G is same as the number of people standing behind D.
(iii) Three people are standing between B and F.
(iv) B is standing behind A, but ahead of E.

4. Who is standing in the middle of the queue?
(A) A (B) E
(C) B (D) F

5. Who is standing at the front end of the queue?
(A) A (B) B
(C) C (D) D

6. Who is standing at the rear end of the queue?
(A) B (B) F
(C) D (D) C

Directions for questions 7 to 10: These questions are based on the following information.

A group of seven friends, namely Bipul, Lalita, Mihir, Naina, Deepa, Sushmita and Pradeep were sitting in a row in that order facing same direction. They rearrange themselves in another order, such that in the new arrangement,

(i) For any one of them neither of the neighbours is same as in the previous arrangement.
(ii) Only Mihir and Deepa remain at their previous positions.
(iii) Sushmita and Lalita do not sit adjacent to each other in the new arrangement.

7. How many people are sitting between Bipul and Pradeep?
(A) 0 (B) 1
(C) 2 (D) 3

8. The number of people sitting between Sushmita and Lalita is
(A) 1 (B) 2
(C) 3 (D) 4

9. Who among the following are adjacent to each other?
(A) Pradeep and Laltia
(B) Bipul and Deepa
(C) Deepa and Pradeep
(D) Sushmita and Mihir

10. Who is sitting at the right end of the row?
(A) Naina (B) Lalita
(C) Sushmita (D) Cannot be determined

Directions for questions 11 to 13: These questions are based on the following information.

A group of seven people, namely Amol, Bimal, Komal, Tamal, Kajol, Gopal and Mrinal were standing in a queue, not necessarily in the same order.

(i) Gopal is standing in front of only one person, i.e., Mrinal.
(ii) Kajol is the only person standing infront of Tamal.
(iii) Komal is standing immediately in front of Bimal.

11. If Tamal and Komal are adjacent to each other, then what is the position of Amol in the queue?
(A) Fourth (B) Fifth
(C) Third (D) Sixth

12. Who is/are definitely standing between Bimal and Tamal?
(A) Komal (B) Gopal and Amol
(C) Komal and Amol (D) Amol

13. Who is standing immediately behind Amol?
(A) Gopal (B) Bimal
(C) Komal (D) Cannot be determined

Directions for questions 14 to 16: These questions are based on the following information.

A group of nine people, namely A, B, C, D, E, F, G, H and I are seated in a row, not necessarily in the same order. Following is some information regarding the seating arrangement.

(i) A is seated as many places to the left of C as D is seated to the right of B.
(ii) The only person seated between F and H is seated two places to the left of I.
(iii) Neither C nor D is seated at any of the ends.
(iv) G, who is not seated at any of the ends is seated to the right of E.
(v) E and F are seated together.

14. In how many ways can these nine people be seated?
(A) Two (B) Four
(C) One (D) Three

15. Which of the following is definitely true?
(A) D and E are seated together.
(B) C is seated to the left of D.
(C) F and G are seated together.
(D) A and B are seated together.

16. Which one of the following may be a valid representation of the seating positions of the person?

(a)	B	A	D	C	E	F	G	H	I
(b)	B	D	A	C	E	F	G	H	I
(c)	A	B	C	D	E	F	G	H	I
(d)	A	C	B	D	E	F	G	H	I

(A) (a), (b), (c) and (d)
(B) (a) and (d) only
(C) (b) and (c) only
(D) (a), (b) and (d) only

Directions for questions 17 to 19: These questions are based on the following information.

A group of nine people from A to I are standing in a row. Each of E and G is next to exactly one person. There are two people between C and A. F is between I and B. B is to the immediate right of E and C is to the immediate left of G.

17. Which of the following additional statements is sufficient to determine the order of the person standing in the row from left to right?
(A) I is to the immediate left of A and C is to the immediate right of H.
(B) A is to the left of D and H.

(C) There are two persons between F and D.
(D) C is sitting to the left of G.

18. If I is to the immediate right of A, then who is to the immediate left of C?
(A) D (B) F
(C) H (D) Either D or H

19. How many arrangements are possible, given that, F is to the immediate right of B?
(A) Three (B) Four
(C) Six (D) Eight

Directions for questions 20 to 23: These questions are based on the following information.

A group of eight people from A through H sit on a bench in a multiplex and each of them has to go to a different screen among I to VIII, but not necessarily in the same order.

A sits second to the right of C, neither of them will be going to screen VIII and one of them is at an end. The person who will be going to screen VII is adjacent to C, but not to A. C will be not going to screen IV. B will be going to screen III and is to the immediate right of E. E is three places away from the person, who will be going to screen II and is second to the left of the person, who will be going to screen V. D will be going to screen II, but is not adjacent to F, who is two places away from H. H sits second to the left of the person who is going to screen I.

20. Who sits to the immediate right of D?
(A) H
(B) The person who will be going to screen V.
(C) The person who will be going to screen VI.
(D) G

21. Who is going to screen VI?
(A) A (B) C
(C) E (D) F

22. Who sits second from the right end?
(A) B
(B) F
(C) The person who sits to the immediate right of A
(D) The person who will be going to screen VIII

23. Three of the following four are alike in a certain way based on the given information and so form a group. Find the one which does not belong to that group.
(A) E, G
(B) H, the person who will be going to screen V
(C) The person who will be going to screen I, H
(D) F, D

Directions for questions 24 to 27: These questions are based on the following information.

Ten cars were parked in a parking lot which are of different colours among Red, Black, Green, White, Yellow, Pink, Blue, Gray, Violet and Orange, but not necessarily in the same order. These ten cars were parked in two rows in such a way that five cars in each row and each car from one row is exactly opposite a car from the other row. The cars were numbered from 1 to 10 and parked in such a way that odd numbered car is not opposite another odd numbered car and no two even numbered cars are adjacent to each other. Further information related to positions of the cars is given below.

The right and left are to be considered as if the driver is sitting in the car and these directions are as per the driver's left and right.

(i) Pink coloured car is numbered eight which is not in the same row as that of the cars numbered four or six.
(ii) White and yellow coloured cars are adjacent to the car numbered six, but white car is not at an end.
(iii) Blue coloured car is numbered 5 which is not in the same row as that of the cars numbered 3 or 6.
(iv) Red coloured car is neither adjacent nor opposite to Blue or Green coloured car.
(v) Pink coloured car is at the right end of the row.
(vi) The car numbered 6 is second from the right in its row.
(vii) Black coloured car is even numbered and is opposite to the car numbered 1.
(viii) Violet and Orange coloured cars are parked in the same row. The car numbered 6 is not to the left of car numbered 7.
(ix) Green coloured car is odd numbered but not 1 and is in the same row with Blue coloured car.
(x) Yellow coloured car is opposite to car numbered 2 and is numbered neither 3 nor 1.

24. Which of the following is the correct combination of cars parked in a row?
(A) 2, 3, 8, 9, 10 (B) 1, 3, 4, 6, 9
(C) 5, 7, 8, 2, 9 (D) 4, 1, 9, 7, 10

25. If Green coloured car is adjacent to Gray coloured car, then which numbered car is opposite to car number 4?
(A) 5
(B) 7
(C) 1
(D) Cannot be determined

26. Which pair of cars is adjacent to Black coloured car?
(A) Pink, Black (B) Black, Gray
(C) Blue, Gray (D) Blue, Green

27. Which pair cars is parked at the middle?
(A) Car numbered 10, Orange coloured car.
(B) Car numbered 1, Black coloured car.
(C) Car numbered 9, White coloured car.
(D) Cannot be determined

Directions for questions 28 to 30: Answer these questions based on the information given below.

Eight houses of eight different colours, such as red, yellow, green, blue, violet, pink, brown and white are on two sides of a road. There are four houses on each side of the road and each house is facing another house on the other side of the road. Further the following information is known about the houses.

(i) The pink coloured house is diagonally opposite to the brown coloured house.
(ii) The red coloured house is opposite to the yellow coloured house and is on the same side as the green coloured house.
(iii) The violet coloured house is opposite to the white coloured house but is not on the same side as the blue coloured house.
(iv) The blue and brown coloured houses are on the same side of the road.

28. What is the colour of the second pair of diagonally opposite houses?
(A) Yellow and green (B) Blue and red
(C) Blue and green (D) Red and white

29. Which of the following mentions the colours of two houses, which are on the same side of the road?
(A) Violet and yellow (B) Red and brown
(C) Brown and violet (D) Pink and green

30. How many houses are there in between the blue and brown coloured houses?
(A) None (B) One
(C) Two (D) Cannot be determined

Directions for questions 31 to 34: These questions are based on the given information.

A group of eight people, namley K, L, M, N, O, P, Q and R are sitting in a row (not necessarily in the same order).

Four of them are facing north and the remaining four are facing south. They belong to different professions, such as Professor, Scientist, Musician, Beautician, Teacher, Lawyer, Architect and Principal.

(i) P faces north and sits at the right end.
(ii) Either the Lawyer or Principal (but not both) is adjacent to Musician whose neighbours face south.
(iii) M is a Professor and sits adjacent to the Scientist. Beautician sits three places away to the right of the Lawyer.
(iv) N, the Principal, sits second to the right of O.
(v) Musician faces the same direction as P and sits second to the left of P.
(vi) R is the Scientist who is to the immediate left of the Architect and faces south.
(vii) K is neither a Principal nor a Lawyer. L and Scientist are neighbours of the Architect.
(viii) Q, the Musician, sits three places away from the Architect and is to the immediate right of K.

31. Who sits second to the left of N?
(A) The Architect (B) The Teacher
(C) O (D) K

32. Who are the neighbours of the Lawyer?
(A) N and Architect (B) O and Scientist
(C) O and Principal (D) Both (A) and (C)

33. Which among the following statements is true?
(A) Teacher sits at the left end.
(B) L is a not a Beautician.
(C) O is an Architect.
(D) Both (B) and (C)

34. Three of the following are alike in a certain way, and hence form a group. Which is the one that does not belong to that group?
(A) O – Architect (B) P – Beautician
(C) N – Principal (D) R – Scientist

Directions for questions 35 to 38: These questions are based on the following information.

Four boys K, L, M, and N sit in Row I facing north and they are from different professions, such as Engineer, Doctor, Professor and Actor. Four girls P, Q, R, and S sit in Row II facing south and they are of different professions, such as Lawyer, Teacher, Director and Collector. They sit in such a way that the distance between any two adjacent persons in a row is the same such that one person in one row faces the other person in the other row, but not necessarily in the same order

Collector sits opposite to the person who sits third to the right of N. Doctor and Teacher sit opposite to each other. Engineer sits opposite to the person who sits third to the left of R. Only one person sits between K and L. R sits second to the right of S, the Director. P does not sit opposite to Engineer and K is neither an Engineer nor a Doctor. Lawyer does not sit opposite to Actor.

35. Who is the Doctor?
(A) K (B) M
(C) L (D) N

36. Who sits opposite to the Director?
(A) Actor (B) Professor
(C) L (D) K

37. Which of the following 'person–profession' combination is correct?
(A) P – Director (B) S – Collector
(C) M – Actor (D) N – Engineer

38. What is the position of P with respect to R?
(A) Immediate right
(B) Immediate left
(C) Second to the right
(D) Second to the left

EXERCISE-3

Directions for questions 1 and 2: These questions are based on the following information.

Aksha, Bindu, Chandana, Deeksha, Harsha and Lasya have different number of years of work experience. They worked between the years 1988 to 2000. Each person has at least one-year experience. The following information is known about them.

(i) Harsha started working in 1988 and has 8 years of experience.
(ii) Aksha started working in 1991 and has 6 years of experience.
(iii) Bindu has 7 or 6 years of experience and her experience is overlapped with Harsha.
(iv) Chandana started working in 1988 and stopped in the same year when Deeksha started working.
(v) Lasya started working in 1989 and has 1 year more experience than Chandana.
(vi) No two persons stopped working in the same year.
(vii) Deeksha has one year less experience than Chandana.

1. How many pairs of persons stopped working in the consecutive years?
(A) 5 (B) 4
(C) 3 (D) 6

2. How many years are there in which more than three persons started working?
(A) 1 (B) 2
(C) 3 (D) None of these

Directions for questions 3 to 5: These questions are based on the following information.

A group of seven friends, namely Asha, Lata, Mahesh, Madhu, Mahima, Sandhya and Kavita are sitting in a row facing North. Lata is two places away to the left of Sandhya. Mahima has Kavita to her left and Madhu to her right. Asha is sitting to the immediate left of Madhu but not to the immediate right of Mahima. Mahesh is not sitting adjacent to Kavita.

3. If Madhu is not sitting at any of the ends, then how many people are sitting between Mahesh and Mahima?
(A) 0 (B) 1
(C) 2 (D) 3

4. Who among the following can be the one sitting to the left of Lata?
(A) Kavita (B) Madhu
(C) Asha (D) Mahesh

5. If Sandhya is sitting at the middle of the row, then who is sitting at the left end?
(A) Lata (B) Mahesh
(C) Kavita (D) Madhu

Directions for questions 6 to 8: These questions are based on the information given below.

A group of four friends, namely Aravind, Bharat, Chandrapaul and Daniel went for an excursion with their wives Preeti, Revati, Sravani and Vanita, not necessarily in the same order. Each couple hails from a different city, they are from Mumbai, Chennai, Kolkata and Hyderabad, not necessarily in that order. They went to Agra to visit the Taj Mahal, where they sat in a row. Each wife always sat to the immediate right of her husband.

(i) Bharat sat to the immediate right of Preeti.
(ii) Daniel is from Hyderabad and Preeti is not from Mumbai.
(iii) Revati and her husband were sitting to the immediate right of the couple that hailed from Chennai.
(iv) Chandrapaul and his wife were sitting to the immediate left of the couple from Kolkata and Chandrapaul was sitting to the immediate right of Sravani.
(v) Aravind sat to the immediate right of Revati, who is not from Mumbai.

6. Who is Daniel's wife?
(A) Sravani (B) Preeti
(C) Vanita (D) Revati

7. Which couple is from Chennai?
(A) Vanita and Bharat
(B) Daniel and Revati
(C) Chandrapaul and Preeti
(D) Aravind and Sravani

8. Which husband from the pair of couples was seated second in the row from left to right?
(A) Aravind (B) Bharat
(C) Chandrapaul (D) Daniel

Directions for questions 9 to 11: These questions are based on the following information.

A group of seven people, namely A, B, C, D, E, F and G wearing seven different coloured shirts, such as Red, Blue, Green, Yellow, Pink, White and Violet are sitting in a row not necessarily in the same order facing North. We know the following additional information.

(1) D is sitting as many places away to the right of the person wearing red coloured shirt as the person wearing white coloured shirt is sitting to the left of A.
(2) The person wearing pink coloured shirt is four places away to the right of G.
(3) The number of people to the left of C is same as the number of people to the right of the person wearing white coloured shirt.

(4) The person wearing blue coloured shirt is four places away to the left of the person wearing green coloured shirt.
(5) The people wearing violet coloured shirt and yellow coloured shirt are sitting at the second and seventh positions from the extreme left, respectively.
(6) F is sitting to the immediate left of E, who is adjacent to C.

9. Who is wearing green coloured shirt?
(A) A (B) C
(C) B (D) D

10. Who is wearing pink coloured shirt?
(A) B (B) F
(C) E (D) A

11. Who is sitting at the extreme right?
(A) D (B) A
(C) C (D) B

Directions for questions 12 to 14: Read the information given below and then answer the questions that follow.

Seven boxes of different colours, such as White, Indigo, Blue, Red, Yellow, Green and Violet have to be arranged in a row on a shelf in such a way that the Blue box and the Indigo box have only four boxes in between them, whereas the White box is not in between the Blue and the Indigo boxes and the Yellow box is to the immediate left of the Indigo box.

12. If the White and the Red boxes have two boxes between them, then which of these would be exactly in the middle of the row of boxes?
(A) Yellow box (B) Red box
(C) Violet box (D) Green box

13. If the green box is placed to the immediate left of the violet box and next to the blue box, wherein the white box is to extreme left, then which of the following boxes will be the fourth from the right end?
(A) Red box (B) Yellow box
(C) Green box (D) Violet box

14. Which of the following statements is definitely false?
(A) The violet box is exactly in the middle of the row.
(B) The white box is not at any of the extreme ends.
(C) The yellow box is in the third place from the right end.
(D) Each of the white and the indigo boxes are at the extreme ends.

Directions for questions 15 to 17: Read the data given below and then answer the questions that follow.

M, N, O, P, Q, R, S, T, U, V and W are eleven persons in a team. O is elected as their captain. O makes them sit in a row. P and R must sit together and V and W also sit together, whereas there are exactly four seats between the two pairs P, R and V, W. T and U sit together and T is to the immediate right of S, who is next to Q. M and N sit in that order only at one extreme end and no one sits to the left of P.

15. Who sits exactly at the fourth place to the right of U?
(A) N (B) M
(C) Q (D) P

16. How many people sit between S and N?
(A) 4 (B) 3
(C) 5 (D) 6

17. How many ways of arrangements are possible in the row with the given conditions?
(A) 2 (B) 3
(C) 4 (D) 5

Directions for questions 18 to 20: These questions are based on the following information.

Seven persons A through G sit in a row, not necessarily in the same order, some face north and the remaining face south. No two adjacent persons face the same direction. The following information is known about them.

Two persons sit between D and E, and E is at one of the ends. G is two places away to the right of E. B faces the same direction as C faces and is adjacent to both D and F. G faces north.

18. What is the position of D with respect to G?
(A) Immediate right (B) Immediate left
(C) Second to the left (D) Second to the right

19. Four of the following are alike in a certain way and so form a group. Find the one which does not belong to that group.
(A) E G (B) G C
(C) F D (D) A B

20. Which of the following is/are 'definitely true'?
(A) A sits adjacent to G
(B) E and C are not at the ends
(C) B is second to the left of C
(D) More than one of the above

Directions for questions 21 and 22: These questions are based on the following information.

Ten people are sitting in two parallel rows, which have five people each, in such a way that there is an equal distance between adjacent people. In Row I, A, B, C, D, E are seated (not necessarily in the same order) and all of them are facing north. In Row II, P, Q, R, S, and T are seated facing south. Therefore, each member seated in Row I faces exactly one member of Row II.

R and T are not sitting at any end. S is sitting opposite to the person who is not a neighbour of A or C. C is not sitting opposite to S. Neither B nor E is sitting opposite to T. Only one person is sitting between C and A. But neither of them is sitting at the left end. T is not a neighbour of S or R. E is not sitting opposite to S. A is not sitting opposite to the person who is sitting at the left end. At least one person sits to the left of Q.

21. Who is sitting opposite to the person who is not a neighbour of either S or Q?

(A) D (B) C
(C) E (D) A

22. If R and T interchange their positions, then who among the following will sit opposite to T?

(A) E (B) A
(C) B (D) D

Directions for questions 23 to 25: These questions are based on the following information:

A group of eight people K, L, M, N, O, P, Q and R are sitting in two rows facing each other, i.e., four people in each row not necessarily in the same order. They hail from various professions, such as Principal, Beautician, Doctor and Teacher, and they are facing south. Whereas, the other group of professionals, such as Architect, Politician, Singer and Scientist are facing north.

(i) The doctor is sitting opposite to the person, who is to the immediate right of Q, the politician.
(ii) Neither M nor O is a teacher.
(iii) Either R or K is the doctor.
(iv) The teacher and the scientist are sitting opposite to each other at an end.
(v) Neither Q nor O sits at an end.
(vi) The politician and the architect are adjacent to each other.
(vii) L is the singer and is sitting to the immediate left of P, who is a scientist.
(viii) K is the principal and he is sitting opposite to the architect.

23. Who among the following is the teacher?

(A) N (B) O
(C) R (D) K

24. Which of the following statements is definitely true?

(A) P is the scientist and sitting at left extreme.
(B) R is the doctor and sitting opposite the singer.
(C) N is the teacher and sitting to the immediate left of R.
(D) More than one of the above.

25. If Q interchanges his place with K, then who sits to the immediate left of Q?

(A) Architect (B) Singer
(C) Principal (D) Beautician

ANSWER KEYS

Exercise-1

1. (C)	8. (B)	15. (D)	22. (D)	29. (B)	36. (C)
2. (C)	9. (D)	16. (A)	23. (C)	30. (C)	37. (C)
3. (C)	10. (D)	17. (D)	24. (B)	31. (C)	38. (B)
4. (A)	11. (C)	18. (C)	25. (A)	32. (A)	39. (B)
5. (D)	12. (C)	19. (A)	26. (C)	33. (D)	40. (A)
6. (C)	13. (B)	20. (B)	27. (B)	34. (B)	
7. (D)	14. (A)	21. (D)	28. (C)	35. (B)	

Exercise-2

1. (D)	8. (D)	15. (C)	22. (B)	29. (D)	36. (A)
2. (D)	9. (C)	16. (A)	23. (D)	30. (C)	37. (C)
3. (C)	10. (D)	17. (A)	24. (B)	31. (D)	38. (B)
4. (B)	11. (B)	18. (D)	25. (A)	32. (D)	
5. (A)	12. (A)	19. (C)	26. (D)	33. (D)	
6. (D)	13. (D)	20. (A)	27. (B)	34. (B)	
7. (B)	14. (B)	21. (B)	28. (C)	35. (D)	

Exercise-3

1. (B)	5. (C)	9. (B)	13. (D)	17. (A)	21. (B)	24. (D)
2. (D)	6. (A)	10. (D)	14. (B)	18. (A)	22. (A)	25. (D)
3. (D)	7. (C)	11. (A)	15. (A)	19. (D)	23. (A)	
4. (A)	8. (C)	12. (B)	16. (C)	20. (D)		

SOLUTIONS

EXERCISE-1

1. Five people L, M, N, O, and P sit in a row, not necessarily in the same order.
P sits exactly between M and N which means that the following arrangements can be obtained:
M/N P M/N L/O L/O
(or)
L/O M/N P M/N L/O
(or)
L/O L/O M/N P M/N
It is also given that L sits exactly between M and O, which means that M sits at the center of the row.
N P M L O
(or)
O L M P N
'M sits exactly at the center of the row' is true.

2. From (ii), we can say that Nagraj is the first person in the queue while Tanmay is the second.
Now, from (i) and (iii) we can say that Ganpat is the 3rd person and Jivan is 5th.
Thus, Sanjay is the 4th person and Dhruv is the last person in the queue. Hence, the queue is as follows.

Nagraj
Tanmay
Ganpat
Sanjay
Jivan
Dhruv

There are 3 people (Ganpat, Sanjay and Jivan) ahead of Dhruv but behind Tanmay.

3. It is given that:

______	Krishna	______
Treasurer	Director	

As Murali is two places away from the Secretary, hence, Krishna must be the Secretary, as Vijay must sit to the immediate right of Murali.
Hence, we get the following arrangement:

______	Krishna	______	Murali	Vijay
Treasurer	Secretary	Director	Professor	Student Leader

4. A, B, C, D, E, F and G are seven people who parked their cars in a row. E's car and F's car should be next to each other, i.e., EF or FE. Similarly, the cars of D and G should be next to each other, i.e., DG or GD. A's car and B's car should not be next to each other, whereas the cars of B and D are parked next to each other, i.e., BD or DB. C's car is parked to the immediate right of G's, i.e., G → C. Hence, they can be parked in the following order.
(i) E F or F E
(ii) D G or G D
(iii) B D G
(iv) B D G C
As A and B are not parked together, the arrangement can be as follows.
(i) A __ __ B D G C
(ii) __ A __ B D G C
(iii) __ __ B D G C A
(iv) __ B D G C A __
(v) __ B D G C __ A
(vi) B D G C A __ __
(vii) B D G C __ A __
(viii) B D G C __ __ A
But as E and F must be together, the arrangement can be as follows. AFEBDGC or AEFBDGC,
FEBDGCA (or) EFBDGCA,
BDGCA (EF or FE) or BDGC (EF or FE) A.
If E and F park in that order only, i.e., E first and then F, then the order is EFBDGCA (or) BDGCAEF. In any case B and D have only one car between their cars. Choice (A) is FALSE. B and C do not park together. Choice (B) is TRUE. D and C have only G's car between their cars. Choice (C) is TRUE. A is at the extreme. Choice (D) is partly true.

5. As A is two places to the right of B and C is two places to the left of D, the possible arrangements are as follows.
(i) B C A D __
(ii) __ B C A D
(iii) C B D A __
(iv) __ C B D A
As E is not sitting at extreme right arrangements, the options (i) and (iii) are not possible. Hence, C or B is sitting at the middle.

6. Given that, A is three places away to the right of C. The possible arrangements are as follows.
(i) C __ __ A __
(ii) __ C __ __ A
Given that, two people are sitting between B and D. By combining with the above, we get the following possible arrangements.
(i) C B/D __ A D/B
(ii) B/D C __ D/B A
In either case E is the one who is sitting in the middle of the row.

7. As Anush is to the immediate right of Adi who is adjacent to Arnab, the possibilities are as follows:
(i) Arnab Adi Anush ____ ____
(ii) ____ Arnab Adi Anush ____
(iii) ____ ____ Arnab Adi Anush
As there is at most one person between Ankur and Anush, (iii) is not possible.
In (ii), Ankur must sit at the extreme right.
∴ There cannot be at least two people between Arnab and Asraf.
∴ Only case (i) is possible.
Arnab Adi Anush Asraf/Ankur Asraf/Ankur
The only choice (D) is true.

8. From the first two statements the possibilities are as follows:
(i) O __ Q __ __ M P
(ii) Q O __ __ P __ M
From the third statement the possible arrangements are as follows.
(i) O S Q N R M P
(ii) Q O S N P R M
∴ In either case N is sitting in the middle of the row.

9. As, Chetan is two places to the right of Ena neither of them is sitting at the extreme ends and Alpana and Fanna are adjacent to each other, the possibilities are as follows:
(i) __ E __ C A / F A / F
(ii) A / F A/ F E __ C __
As there is one person between Alapana and Brahma, the possibilities are as follows.
(i) D E B C A F
(ii) F A E B C D
∴In either case Brahma is sitting to the immediate right of Ena.

Solutions for questions 10 to 12: From (i) and (ii), we have
__ A/B D B/A ____
From (iii), we can say that C must be at the left end of the row. Thus, we get the following arrangement.
C B D A E

10. Eswar is sitting at the right end.

11. Two people are sitting between Amit and Chetan.

12. Eswar is third to the right of Balram.

Solutions for questions 13 to 15: From (i) and (iii), we get
__ __ P __ __ Q
Now, from (ii) and the above, we can say that the person sitting between T and U is P. Hence, we get 4 possible arrangements which are as follows.
R T P U S Q — Case (a)
R U P T S Q — Case (b)
S T P U R Q — Case (c)
S U P T R Q — Case (d)

13. U is adjacent to S in case (a) and (d), and in both the cases there are two people sitting between U and R.

14. S is sitting to the immediate right of T in case (b), Hence, P is sitting second to the right of R.

15. Any one of the three P, R and S, can be adjacent to T.

Solutions for questions 16 to 18: From (ii) and (iv), we can say that Maths is on the top, Zoology is 2nd from the top, and English is 5th from the top. Now from (i), we can say that Hindi is 2nd from the bottom and Biology is 3rd from the bottom. Now, from the above information and (iii), Economics and Chemistry are the 3rd and 4th from the top. Thus, Physics is the bottom one in the stack. Hence, we have the following arrangement.

Maths
Zoology
Economics/Chemistry
Chemistry/Economics
English
Biology
Hindi
Physics

16. Physics is at the bottom.

17. Position of Economics book is not certain, and hence, it cannot be determined.

18. Except English and Hindi all other pairs are in adjacent positions.

Solutions for questions 19 to 21:
From (i), Dubey's ______ ______ ______.
Green
From (ii), Sharma ______ ______ ______.
Red
From (iii), Reddy ______ ______ ______.
White

∴ The owners of the left three buildings are Dubey, Sharma and Reddy and the colours of the right three buildings are green, white and red.

The colours of the left three buildings are blue, yellow and orange.

The owners of the right three buildings are Roy, Sanyal and Tiwari.

From (iv), Roy's building is adjacent to the orange building. From the above explanation the arrangement will be as follows.

______ ______ ______ Roy ______ ______
Orange

From (vi), as Tiwari's building is three places away from Reddy's building, the following arrangement is possible.

______	______	Reddy	Roy	Sanyal	Tiwari
Orange	White				

From (v), since Sanyal's is not green, as well it is not white, it must be red.

Roy's building is green.

∴ The arrangement is as follows.

Dubey	Sharma	Reddy	Roy	Sanyal	Tiwari
Blue	Yellow	Orange	Green	Red	White

19. The colour of Dubey's building is blue.

20. The red building belongs to Sanyal.

21. Only choice (D) is true.

Solutions for questions 22 and 23: From (ii), as any four consecutive classrooms accommodate two A sections and two B sections, the following arrangements are possible.

	1	2	3	4	5	6	7	8	9	10
Arrangement A	A	B	B	A	A	B	B	A	A	B
Arrangement B	B	A	A	B	B	A	A	B	B	A

But it is given that Class II A is at the extreme right. Hence, arrangement (a) is not possible.

From (iv): III B/IV B ______ ______ IV B/III B

But neither of III B and IV B is at the extreme end.

∴ They have to be in positions 5 and 8 in any order.

From (vi), Class II A is in position 10.

From (v), Class II B has to be in position 4 or 9. But it is given that both sections A and B of the same are not adjacent to each other. Since, II A is in position 10, II B cannot be in position 9.

∴ II B is in position 4.

The arrangement obtained so far is as follows.

1	2	3	4	5	6	7	8	9	10
			II	III/IV			IV/III		II
B	A	A	B	B	A	A	B	B	A

From (vii), V A is to the right of III B.

∴ III B must be in position 5 and V A must be in position 6 or 7.

⇒ IV B is in position 8.

From (iii), V A is three places away from I A.

This is possible only when I A is in position 3 and V A in position 6.

The arrangement obtained so far is as follows.

1	2	3	4	5	6	7	8	9	10
		I	II	III	V		IV		II
B	A	A	B	B	A	A	B	B	A

Since, IV B is in position 8, IV A must be in position 2.

⇒ III A is in position 7.

∴ I B and V B are in positions 1 and 9 in any order.

∴ The final arrangement is as follows.

1	2	3	4	5	6	7	8	9	10
I/V	IV	I	II	III	V	III	IV	V/I	II
B	A	A	B	B	A	A	B	B	A

22. I B or V B can be at the extreme left.

23. V A is to the immediate right of class III B.

Solutions for questions 24 to 27: It is given that P sits third to the right of the person who is the heaviest. The heaviest person sits exactly between R and the lightest person, who sits at an end. P's weight is neither 92 kg nor 79 kg. Hence, the lightest person, i.e., the person who is of 79 kg should sit at the left end and R sits either third from the left end or third from the right end. Given the person whose weight is the third lowest (i.e., 85 kg) sits adjacent to R is neither P nor adjacent to P. Hence, we get the following arrangement.

__	__	__	__	R	P	__
79		96	85			

It is given that Q sits third to the left of the person whose weight is the next highest after R. R's weight is neither 83 nor 87 and P's weight is not 92. Hence, R's weight is 89, Q's weight is 85 and Q sits to the immediate left of R. The person whose weight is 92 kg sits at the right end. As T's weight is 83 and S is heavier than V, but is not the heaviest, the final arrangement is as follows.

V	T	U	Q	R	P	S
79	83	96	85	89	87	92

24. Q is the third lightest.

25. Only one person.

26. The person who is 89 kg.

27. Five persons

28. Form (ii), the arrangement of A, C, D and E must be in the order of E, D, A, C .... (a)

From (iv), we get:

___ ___ ___ ___ ___

Nepal

From (iii) and (a), we get the following cases.

Case (i):

B E D A C

Pakistan Nepal

Case (ii):

E D A C B

Nepal Pakistan

From (i), the final representations are as follows.

Case (i):

B	E	D	A	C
Pakistan	India	Nepal	Bangladesh	Sri Lanka

Case (ii):

E	D	A	Bangladesh	Sri Lanka	Nepal
C	B				
India	Pakistan				

If B is not adjacent to E, then A is the delegate from Nepal (Case (ii)).

29. If D is not the delegate from Sri Lanka, then the delegate from Pakistan is at the extreme left end of the row.

30. Choice (C) is false.

Solutions for questions 31 to 33: Let us represent the names of the seven people with their first letter.

From the given information the possible cases are,

(i)

— 7P — 8 — 2P — 11 12 13 14 15 16 17 — 4P — 22 — 6P —
A: 8, C: 11, F: 17, G: 22

(ii)

— 4P — 5 — 2P — 8 9 10 11 12 13 14 15 16 17 — 4P — 22 — 6P —
C: 5, A: 8, F: 17, G: 22

31. From the given information, case (i) is not possible. In case (ii), we have

— 4P — 5 — 2P — 8 9 10 11 12 13 14 15 16 17 — 4P — 22 — 6P —
C: 5, A: 8, E: 11, D: 12, B: 14, F: 17, G: 22

→ 6P ——

∴ Six people.

32. In both cases, if Chaitra and Fallon interchange their positions, 8 people sit between Chaitra and Akhil.

33. In case (i), there are four persons.
In case (ii), there are eight persons.

Solutions for questions 34 to 36: Given that, A is sitting third from the left end. B, A, C and G are sitting from left to right in that order but not in adjacent positions. Hence, B is sitting at the left end and the possible arrangements are as shown below.

(i) __ __ __ __ __ __ __ __
(ii) __ __ __ __ __ __ __ __
(iii) __ __ __ __ __ __ __ __

Given that, there are exactly two people sitting between D and E. Hence, case (i) and case (ii) are eliminated, as we cannot place D and E. In case (iii), D sits to the immediate right of B and E sits to the immediate left of C.

Given that D, F and H are sitting from left to right, in that order but not necessarily in adjacent positions.

Hence, the final row arrangement is as shown below.

B D A F E C H G

34. D sits second to the right of A.

35. Four people sit between B and C.

36. B and G.

Solutions for questions 37 to 40: From (iv), (ii) and (i), we can say that rooms H and E are on the fifth floor in any order. D and G are on the first floor in any order.

From (vii), we can say that room A is on the third floor.

From (iii) and (v), we can say that B and K are on the fourth floor.

From (vi), (viii) and (ix) and above, the possible arrangements are as shown below.

Floor Number	Rooms			
	Case (i)		Case (ii)	
6	C	F	F	C
5	H/E	E/H	H/E	E/H
4	B/K	K/B	B/K	K/B
3	I	A	A	I
2	L	J	J	L
1	D	G	G	D

37. L and J are on the second floor.

38. Two floors.

39. Room A.

40. Room L.

EXERCISE-2

1. In each arrangement above, we get 4 possibilities. Hence, the total number of possible arrangements is 16.

2. Let us consider Black and White flags as one single unit, they are represented as X. Now, X is not next to Violet or Red.

(i) I/Y B G V/R R/V Y/I O
↑
X

(ii) O I/Y B G V/R R/V Y/I
↑
X

(iii) I/Y V/R R/V G B Y/I O
↑
X

(iv) O I/Y V/R R/V G B Y/I
↑
X

Hence, the total number of arrangements
$= 8 \times 4 = 32$

3. (A) Yellow flag may or may not be at an extreme end as shown in various arrangements.
(B) Similarly, as shown in the figures above, Indigo and Orange flags may not be together.

(C) The Blue flag is always adjacent to the Green flag, as seen in every possible arrangement.

(D) This statement again may or may not be true, as it can be observed in the arrangements.

Solutions for questions 4 to 6: Let us indicate the positions in the queue from the front end to the rear end as follows.

1 –
2 –
3 –
4 –
5 –
6 –
7 –

From (i), the possible positions of A and C are positions (1 and 7) or (2 and 6) or (3 and 5).

From (ii), the possible positions of G and D are same as above.

From (iii), the possible positions of B and F are (1 and 5) or (2 and 6) or (3 and 7).

From the above discussion it is clear that any of A, B, C, D, F and G cannot be in position 4.

∴ E is in position 4.

If B and F are in positions 1 and 5, then A and C must be in positions 2 and 6. In such case we can find positions for G and D. Similar will be the situation if B and F are in positions 3 and 7.

Hence, B and F must be in positions 2 and 6.

Now, if A and C are in positions 1 and 7, then G and D must be in positions 3 and 5.

From (ii), B is standing ahead of E. Since, E is in position 4, B must be in position 2. It is given that B is standing behind A. Hence, A must be in position 1.

∴ The final arrangement is as follows.

1	A
2	B
3	G/D
4	E
5	D/G
6	F
7	C

4. E is standing in the middle of the queue.

5. A is standing at the front of the queue.

6. C is the last person in the queue.

Solutions for questions 7 to 10: Let the name of any person be represented with the first letter. The initial arrangement of the persons is as given below.

B L M N D S P

From (ii), the positions of M and D remain the same as given below.

__ __ M __ D __ __

Since, no person retained the previous positions and neither of the neighbours of any person remain same, N has to sit at one of the ends.

If N sits at the extreme left, then L can be at any of the two right positions at the extreme right and from (iii), S must be sitting to the immediate right of N as follows:

N S M P D L B

N S M B D P L

But the first case is not possible, as Bipul and Lalita are sitting adjacent to each other. Only the second case is possible.

Similarly, when N is sitting at the extreme right end, the possible arrangement is as follows:

S B M P D L N

∴ The possible arrangements are:

1) S B M P D L N
2) N S M B D P L

7. There is one person between Bipul and Pradeep in both the cases.

8. Four people are sitting between Sushmita and Lalitha in both the cases.

9. In both the cases Deepa and Pradeep are adjacent to each other.

10. Naina or Lalita is sitting at the right end of the row.

Solutions for questions 11 to 13: From (i): Mrinal is the last person in the queue and Gopal is the last but one person in the queue.

From (ii): Kajol is the first person in the queue and Tamal is the second person in the queue.

From (iii) and the above information the possible arrangements are as follows.

Case (A)	**Case (B)**
Kajol	Kajol
Tamal	Tamal
Komal	Amol
Bimal	Komal
Amol	Bimal
Gopal	Gopal
Mrinal	Mrinal

11. The given condition is satisfied in case (a), in which Amol is the fifth person in the queue.

12. In both the cases Komal is standing between Bimal and Tamal.

13. Gopal or Komal is standing immediately behind Amol.

Solutions for questions 14 to 16:

__	__	__	__	__	__	__	__	__
1	2	3	4	5	6	7	8	9

(a) Position 1 ≠ C, D, G, I

Position 9 ≠ A, B, C, D, E, F, H, G

(b) F – HI

OR

H – FI

(c) Since A, B, C, D, E, F, G, H are not seated at the extreme right ends, I must be seated there.

__	__	__	__	__	__	__	__	I

(d)

__	__	__	__	__	F	__	H	I

or

__	__	__	__	__	H	__	F	I

Hence, position 1 ≠ C, D, G, H, F, E, I, L so it is either A or B.

Since E and F are seated together, the following possibilities can exist.

(i)	A	C	B	D	E	F	G	H	I
(ii)	A	B	C	D	E	F	G	H	I
(iii)	B	D	A	C	E	F	G	H	I
(iv)	B	A	D	C	E	F	G	H	I

14. There are four ways.

15. Choice (A): D and E are not seated together in (iii) and (iv). Choice (B): C is to the right of D in (iii) and (iv). Choice (C): F and G are seated together in all the cases from (i) to (iv). Choice (D): A and B are not seated together in case (i) and (iii).

16. By observing the cases (i) through (iv), all the four are valid representations.

Solutions for questions 17 to 19: Both E and G are next to exactly one person.

⇒ E and G are at extreme ends of the row.

There are two persons between A and B and also between C and A.

⇒ A is at the middle place of the row.

B is to the immediate right of E.

⇒ G is at the extreme right end.

⇒ C is next to G.

1	2	3	4	5	6	7	8	9
E	B	__	__	A	__	__	C	G

F is between I and B.

⇒ F could be at 3, 4 or 6 and I could be at 4, 6 or 7.

17. From (A), we know that I is to the immediate left of A and C is to the immediate right of H, which gives us the following arrangement.

EB __ IA __ HCG

We already know that F is between I and B, which gives us the following arrangement.

EBFIADHCG

∴ Choice (A) is enough to determine the order of the persons from left to right, whereas, the other choices do not lead to a fixed and definite arrangement.

18. If I is to the immediate right of A, we get the following arrangement.

EB __ __ AI __ CG

1 2 3 4 5 6 7 8 9

This means that F is at either 3 or 4 which means that one of the remaining persons, i.e., D or H is at 7.

19. F is to the immediate right of B as given in the following arrangement.

E B F __ A __ __ C G

1 2 3 4 5 6 7 8 9

Positions 4, 6 and 7 are to be filled up by I, D and H. I can be placed at 4, 6 or 7, i.e., 3 ways. The remaining two D and H can then be placed in two ways.

∴ Total ways = 3 × 2 = 6.

Solutions for questions 20 to 23: Given, A sits second to the right of C, neither of them is going to screens VIII and one of them is at an end. The person who is going to screen VII is adjacent to C, but not to A. C is not going to IV. Hence, A sits at the right end, C sits third from the right end and the person who is going to screen VII sits fourth from the right end. Given, B is going to screen III and is to the immediate right of E. E is three places away from the person, who is going to screen II and is second to the left of the person, who is going to screen V. Hence, E sits at the left end, B sits second from the left end, the person who is going to screen V sits to the immediate right of B and the person who is going to screen II sits second to the right of B. Given, D is going to screen II, but is not adjacent to F, who is two places away from H. Hence, F is adjacent to C and A, H sits to the immediate left of C and G sits to the immediate left of D. Given, H sits second to the left of the person who is going to screen I. Hence, F is going to screen I, E is going to screen VIII, A is going to screen IV and C is going to screen VI.

∴ The final arrangement is as shown below.

E	B	G	D	H	C	F	A
VIII	III	V	II	VII	VI	I	IV

20. H sits to the immediate right of D.

21. C is going to screen VI.

22. F sits second from the right end.

23. Except in choice (D), in the remaining groups, there is exactly one person sitting between the first person and the second person.

Solutions for questions 24 to 27: It is given that, odd numbered cars are not opposite each other and no two even numbered cars are adjacent to each other. Hence, we can say that an odd numbered car is opposite to an even numbered car.

From (i), (ii), (v), (vi) and (x) and above, we get:

Pink	White	Yellow		
8	4	6	× 3, × 1	

From (vii), the only possibility is Black coloured car is numbered 10 and is opposite to White coloured car, which is numbered 1.

From (iii) and (ix), Blue and Green coloured cars are opposite to cars numbered 4 and 6 in any order.

From (iv), Red coloured car is opposite to Pink coloured car.

From (viii), a Violet and Orange coloured cars are numbered 4 and 6 in any order.

Hence, Gray coloured car is numbered 2.

Red coloured car is numbered 3.

Yellow coloured car is numbered 9 and Green coloured car is numbered 7.

The final arrangement is as shown below.

Pink	Blue/Green	Black	Green/Blue	Gray
8	5/7	10	7/5	2

Red	Voilet/Orange	White	Orange/Voilet	Yellow
3	4	1	6	9

24. 1, 3, 4, 6, 9.

25. Car numbered 5.

26. Blue and Green.

27. Car numbered 1 and the Black coloured car.

Solutions for questions 28 to 30: From (ii), the red and yellow coloured houses are opposite to each other. The red and green coloured houses are on the same side.

From (iii), the violet and white coloured houses are opposite to each other. Blue and white colour houses are on the same side of the road.

From (i), the pink and brown coloured houses are at diagonally opposite positions, the blue, white and yellow coloured houses must be on the same side of the road.

∴ The red, green and violet are on the other side.

The red and yellow coloured houses are opposite to each other and the violet and white coloured houses are opposite to each other, the green and blue coloured houses are diagonally opposite to each other.

From (iv) we get,

blue white yellow brown/blue yellow white brown

or

pink violet red green/pink red violet green

28. The blue and green coloured houses are diagonally opposite to each other.

29. The pink and green are on the same side.

30. There are two houses between the blue and brown coloured houses.

Solutions for questions 31 to 34: It is given that, P faces the north and sits at the right end. Musician faces the same direction as P and sits second to the left of P. From (4), we come to know that Q is the Musician and sits three places away from the Architect. Thus, Q also faces north. From (3), (4) and (5) we come to know that, K is to the immediate right of P and he is neither the Principal nor the Lawyer. Both the neighbours of Q face south.

___	___	___	___	↓	Q↑	K↓	P↑
		Architect			Musician		

From (5), L and the Scientist are neighbors of the Architect. Suppose, Scientist sits second to the left of the Q, from (8), Professor sits to the immediate left of Q. It is not possible because either Principal or Lawyer sits to the immediate left of Q (from (3) and (5)). Hence, R who is the Scientist who sits four places away from Q and faces south. M is the Professor who sits to the immediate right of R, and L sits second to the left of Q.

M	R↓	___	L
Professor	Scientist	Architect	

↓	Q↑	K↓	P↑
	Musician		

From (6), N who is the Principal, sits second to the right of O. So, N sits to the immediate left of Q and O is an Architect and he faces north. From (8), Beautician sits three places away to the right of the Lawyer. Thus, L is the lawyer and faces north and K is the Beautician. Hence, M has to face south and P must be the Teacher.

The final arrangement is as follows.

M↓	R↓	O↑	L↑
Professor	Scientist	Architect	Lawyer

H↓	Q↑	K↓	P↑
Principal	Musician	Beautician	Teacher

31. K, the Beautician, sits second to the left of N.

32. N, is the Principal and O, the Architect, are neighbours of the Lawyer.

33. L is the Lawyer and O is the Architect. Hence, both (B) and (C) are true.

34. Except in option (B), in all other the options person and their profession combination is true.

Solutions for questions 35 to 38: It is given that the Collector sits opposite to the person who sits third to the right of N, and the Engineer sits opposite to the person who sits third to the left of R and only one person sits between K and L.

R _ _ _ _ Collector

N K/L M L/K Engineer

R sits second to the right of S and P does not sit opposite to the Engineer and S is the Director.

R P S Q

N K/L M L/K

Doctor sits opposite to the Teacher and K is neither an Engineer nor a Doctor. Hence, the final arrangement is as follows.

Teacher Lawyer Director Collector

R P S Q

K M L

Doctor Professor Actor Engineer

35. N is the Doctor.

36. Actor sits opposite to the Director.

37. 'M – Actor' combination is true.

38. P sits to the immediate left of R.

EXERCISE-3

Solutions for questions 1 and 2: From (i), we can say that Harsha stopped working in the year 1996.

From (ii), we can say that Aksha stopped working in the year 1997.

From (ii) and (iii), we can say that Bindu has 7 years of experience. Since each person has different number of years of experience and Bindu started working in 1988 and stopped in 1995. From (iv) and (v), we can say that, Chandana and Lasya stopped working in the years 1989 and 1991 or 1990 and 1992 or 1991 and 1993 or 1992 and 1994 or 1993 and 1995 or 1998 and 2000, respectively.

Chandana did not stop working in 1994 or 1995, since no two persons stopped working in the same year. From (vii) and from (v), and above we can say that Deeksha started working in 1989 or 1990 or 1991 or 1992 or 1993 or 1998.

From (vii), we can say that Chandana did not have one-year experience. Hence, we can say that Chandana did not stop working in 1989. Since she started working in 1988 (from (iv).

If Chandana stopped working in 1991 (i.e., 3 years of experience) then Deeksha has 2 years of experience, i.e., Deeksha stopped working in 1993, which is not possible since Lasya stopped working in 1993. Similarly, Chandana did not stop working in 1992 or 1993 or 1997.

Hence, Chandana stopped working in 1990 and Lasya stopped working in 1992, and hence, Deeksha stopped working in 1991.

∴ The arrangement is as shown below.

Let us represent the names with their starting letters.

Year	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Started working	H, C, B	L	D	A									
Stopped working			C	D	L			B	H	A			

1. (Chandana, Deeksha), (Deeksha, Lasya), (Bindu, Harsha) and (Harsha, Aksha) stopped working in the consecutive years.

2. More than three persons started working in none of the years.

Solutions for questions 3 to 5: From the given information it is clear that Asha and Madhu are sitting adjacent to each other and Mahima is sitting to the left of Asha and at least there is one person sitting between them. Kavita is to the left of Mahima. As, it is given that Lata is two places away to the left of Sandhya, there is one person between Lata and Sandhya and the person can be Kavita or Mahima. As Mahesh is not sitting adjacent to Kavita.

If they are sitting adjacent to Kavita, then the possible case is:

I. Lata Kavita Sandhya Mahima Mahesh Asha Madhu

If they are sitting adjacent to Mahima, then the possible cases are:

II. Kavita Lata Mahima Sandhya Asha Madhu Mahesh

III. Kavita Lata Mahima Sandhya Mahesh Asha Madhu

If they are sitting adjacent to Mahesh, then the possible case is:

IV. Kavita Mahima Lata Mahesh Sandhya Asha Madhu

3. There are 3 people between Mahesh and Mahima, if Madhu is not sitting at any end (i.e., II).

4. Kavita can sit to the left of Lata (i.e., II, III and IV).

5. Kavita is sitting at the extreme left end, if Sandhya is at the middle of the row (II and III).

Solutions for questions 6 to 8: Males: Aravind, Bharat, Chandrapaul and Daniel

Females: Preeti, Revati, Sravani and Vanita

Cities: Mumbai, Chennai, Kolkata and Hyderabad

With reference to the couples, the wife always sits to the immediate right of her husband.

From (i), Bharat is not the husband of Preeti.

From (ii), Daniel is from Hyderabad.

From (iii), Revati is not from Chennai.

From (iv), Chandrapaul is not from Kolkata and he is not the husband of Sravani.

From (v), Aravind is not married to Revati and Revati is not from Mumbai.

Let us represent the persons with the first letter in their names.

From (iv),

$$[\underline{\text{Husband} - \text{S}}]\ [\underline{\text{C} - \text{Wife}}]\ \left[\dfrac{\text{Husband} - \text{Wife}}{\text{Kolkata}}\right]$$

From (iii) and (v),

$$\left[\dfrac{\text{Husband} - \text{Wife}}{\text{Chennai}}\right] \left[\dfrac{\text{Hubsane} - \text{R}}{\text{Not Mumbai}}\right]\ [\underline{\text{A} - \text{Wife}}]$$

By combining the above two arrangements, we obtain the following arrangement.

$$[\underline{\text{Husband} - \text{S}}]\ \left[\dfrac{\text{C} - \text{Wife}}{\text{Chennai}}\right] \left[\dfrac{\text{Hubsand} - \text{R}}{\text{Kolkata}}\right]\ [\underline{\text{A} - \text{Wife}}]$$

From (ii), we know that Daniel is from Hyderabad. Hence, he cannot be Revati's husband.

⇒ He is the husband of Sravani.

From (i), we know that Bharat is to the immediate right of Preeti.

⇒ Preeti is the wife of Chandrapaul and Bharat is the husband of Revati.

⇒ Vanita is the wife of Aravind. Thus, we obtain the following arrangement.

$$\left[\dfrac{\text{D} - \text{S}}{\text{Hyderabad}}\right]\left[\dfrac{\text{C} - \text{P}}{\text{Chennai}}\right]\left[\dfrac{\text{B} - \text{R}}{\text{Kolkata}}\right]\left[\dfrac{\text{A} - \text{V}}{\text{Mumbai}}\right]$$

6. Daniel is married to Sravani.

7. Chandrapaul and Preeti are from Chennai.

8. Chandrapaul and his wife are seated second in the row.

Solutions for questions 9 to 11: From (2), (5) and (6), we get:

____	G	____	__	_____	____	____
Blue	Violet			Green	Pink	Yellow

From (7), we have: F E C

From (3), we come to know that C must be wearing green coloured shirt.

∴ From (2), A is to the immediate right of C.

From (1), B is wearing blue coloured shirt and A and D are sitting adjacent to each other.

The final distribution is as follows:

B	G	F	E	C	A	D
Blue	Violet	White	Red	Green	Pink	Yellow

9. C is wearing green colour shirt.

10. A is wearing pink colour shirt.

11. D is sitting at extreme right.

Solutions for questions 12 to 14: The Blue box and the Indigo box have 4 boxes between them. The Yellow box is to the immediate left of the Indigo box. So, the possible arrangement may be as follows:

Yellow, Indigo, _ _ _ _ Blue.

(OR)

Blue, _, _, Yellow Indigo, White

(OR)

White, Blue, _ _ _ , Yellow, Indigo.

But the White box is not between the Blue and Indigo boxes. So, the arrangement can be as follows:

Blue, _, _, _, Yellow, Indigo, White

(OR)

White, Blue, _, _, _ , Yellow, Indigo.

12. If the White and Red boxes have two boxes between them, then the arrangement must be as follows:
White Blue – Red – Yellow Indigo.
The Red box must be in the middle of the row.

13. If the White box is to the left extreme end, then the Green box is placed to the immediate left of the Violet box and next to the Blue box, then the order of colours is White, Blue, Green, Violet, Red, Yellow, Indigo. The fourth box from the right end is the Violet box.

14. The arrangement can be as follows:
Blue, –, –, –, Yellow, Indigo, White
(or)
White, Blue, –, –, –, Yellow, Indigo.
Now, the Violet box may or may not be in the middle of the row. So, choice (A) is ruled out.
The Yellow box can be placed third from the right end. Now, choice (C) is ruled out.
White and Indigo boxes can be at the extreme ends. Choice (D) is ruled out. But the White box must be at one of the extreme ends.
So, choice (B) is definitely false.

Solutions for questions 15 to 17: Let us analyse all the given conditions. M, N, P, Q, R, S, T, U, V and W ten people sitting in a row. O is the eleventh person who makes them sit. It is also given that P and R sit together, whereas V and W sit together. There are exactly 4 seats between the two pairs P, R and V, W. Q, S and T, U sit in pairs together with T to the immediate right of S. Q, S, T, U must be the order of their sitting. M and N sit in that order only at one extreme end. No one sits to the left of P. So, P must be at the left extreme end.

P - - - - - - - M N

P R - - - - - - M N

P, R and V, W have exactly 4 seats between them.

P R - - - - V W M N

P R - - - - W V M N

The four spaces should be occupied by Q, S, T, U in that order only. So, the seating arrangement is, P, R, Q, S, T, U [V, W or W, V] M, N.

15. The order is as follows.
P R Q S T U V W M N or P R Q S T U W V M N
N is exactly at the fourth place to the right of U.

16. The order of seating is as follows.
P R Q S T U [V W or W V] M N
Between S and N there are 5 people and they are (T, U, V, W, M).

17. The possible arrangements of seating are as follows.
(i) P R Q S T U V W M N
or
(ii) P R Q S T U W V M N
Therefore, two possible arrangements can be made.

Solutions for questions 18 to 20: It is given that no two adjacent people face the same direction that means alternate people face the same direction. Two people sit in between D and E, and E sits at one of the ends. G is two places away to the right of E. We get the following cases.

Case (i):

E↑	↓	G↑	D↓	↑	↓	↑

Case (ii):

↓	↑	↓	D↑	↓	↑	E↓

Given that B faces the same direction as C faces and it is adjacent to both D and F.

Then the possibilities are as follows.

Case (i):

E↑	A↓	G↑	D↓	B↑	F↓	C↑

Case (ii):

C↓	F↑	B↓	D↑	G↓	A↑	E↓

Since G faces North, case (ii) is eliminated and the final row arrangement is shown below.

E↑	A↓	G↑	D↓	B↑	F↓	C↑

18. 'D' is to the immediate right of 'G'.

19. Except AB, in the remaining options both are facing the same direction.

20. A sits adjacent to G and B is second to the left of C are definitely true.
Therefore, more than one is true.

Solutions for questions 21 and 22: It is given that T and R are not sitting at any end and T is not adjacent to both S and R. Hence, we have the following possible seating arrangement.

____	T/R	____	R/T	____
____	____	____	____	____

S is sitting opposite to the person who is not a neighbour of both A and C. And only one person is sitting between A and C and neither of them is sitting at the left end. Hence, the arrangement is as follows.

S	R	____	T	____
____	____	A/C	____	C/A

A is not sitting opposite to the person who sits at the left end and at least one person is sitting to the left of Q. Neither B nor E is sitting opposite to T, hence, D has to sit opposite to T. E is not sitting opposite to S and C is not sitting opposite to S. Hence, the final seating arrangement is as follows.

S	R	Q	T	P	↓ Row II
B	E	A	D	C	↑ Row I

21. C is sitting opposite to P, who is not a neighbour of both S and Q.

22. If R and T interchange their positions, then E will sit opposite to T.

Solutions for questions 23 to 25: From (iv) and (vii), we come to know that P is a scientist and he is sitting opposite to the teacher at an end.

L is the singer and is sitting to the immediate left of 'P'.

From (i) and (vi), we come to know that Q is a politician and sits adjacent to the architect. From (v), we come to know that neither Q nor O sits at ends. Hence, the arrangement is as follows:

			Teacher
____	____	____	____
____	Q	L	P
Architect	Politician	Singer	Scientist

From (viii), we know that K is the principal and is sitting opposite to the architect. From (iii), either R or K is the doctor. Hence, R is the doctor.

From (ii), neither M nor O is the teacher and from (v) neither Q nor O sits at ends.

From (i), the doctor is sitting opposite to the person who is to the immediate right of Q.

Hence, R is the doctor and he must be sitting opposite to L.

Therefore, M is the Architect, O is the beautician and N is the teacher.

Hence, the final arrangement is as follows:

Principal	Beautician	Doctor	Teacher
K	O	R	N
M	Q	L	P
Architect	Politician	Singer	Scientist

23. N is the teacher.

24. Both the third and the fourth options are definitely true.

25. When Q interchanges his place with 'K', the beautician sits to the immediate left of Q.

CHAPTER 2

Circular Arrangement

Learning Objectives

In this chapter, we will:

- Extend the knowledge of solving linear arrangement puzzles to circular arrangement puzzles.
- Learn how to interpret the statements given and convert them into a circular arrangement.
- Understand the arrangement of even number of people/objects sitting diametrically opposite each other, around a circle.
- Apply knowledge to understand arrangement of odd number of people/objects around a circle.
- Understand and solve circular arrangement puzzles with multiple parameters.
- Understand and solve circular arrangement puzzles where people face different directions.

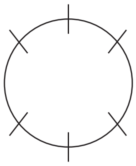
Questions on circular arrangement involve seating of people around a table or arrangement of things in a circular manner (for example, different coloured beads strung to form a necklace).

In case of people sitting around a table, the table could be of any shape, such as rectangular, square, circular or any other.

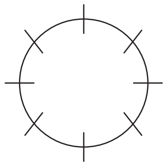
The data given in such sets of questions specify the positions of some or all of the individuals (or things) in the arrangement. The positions are specified through conditions involving specified people sitting (or not sitting) opposite to each other or a particular person sitting to the right or left of another person, etc.

Once you read the data, first draw the shape specified in the data and then draw the slots in the seating arrangement.

Six people around a circular table



Eight people around a circular table



Statements like 'A and B are sitting farthest from each other' or 'A and B sit across the table' imply that A and B sit opposite to each other.

On the other hand, you should remember that, unlike in straight-line arrangement, the words 'immediately' and 'directly' do not play any role in circular arrangement. In general, there is no left side or right side (unless we are talking of 'immediate right' or 'immediate left').

So, if it is given that C sits to right of B, then it is clear that C must be to the immediate right of B. A person's left is in clockwise direction and right is in anti-clockwise direction, when he or she sits facing the centre of a circular table.

To understand better, please go through the solved examples and try to solve them without going through the solution first.

SOLVED EXAMPLES

Directions for questions 1 to 5: These questions are based on the following information.

P, Q, R, S and T sit around a table.

P sits two seats to the left of R and Q sits two seats to the right of R.

2.01: If S sits in between Q and R, then who sits to the immediate right of P?

(A) T (B) S
(C) Q (D) R

2.02: Which of the following cannot be the correct seating arrangement of the five people in either the clockwise direction or the anti-clockwise direction?

(A) P, Q, R, S, T (B) P, S, R, T, Q
(C) P, Q, S, R, T (D) P, T, R, S, Q

2.03: If S is not sitting next to Q, then who is sitting between Q and S?

(A) R (B) P
(C) T (D) Both (R) and (P)

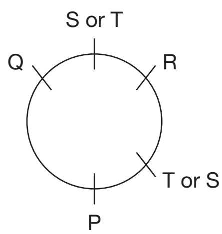
2.04: If a new person U joins the group such that the initial conditions for the seating arrangement should be observed and also a new condition that U does not sit next to R be satisfied, then which of the following statements is true?

(A) U sits to the immediate right of S.
(B) U sits to the immediate left of T.
(C) U sits to the immediate left of P.
(D) Either (A) or (B) above.

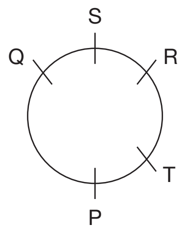
2.05: If a new person U joins the group such that the initial conditions for the seating arrangement should be observed and also a new condition that U does not sit next to P, S or T be satisfied, then who will be the neighbours of P (one on either side)?

(A) S and T (B) S and Q
(C) T and R (D) R and Q

Solutions for questions 2.01 to 2.05: P sits two seats to the left of R and Q sits two seats to the right of R. We can represent this information in the diagram below.

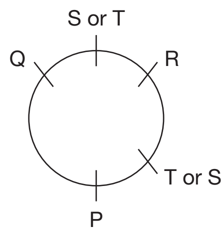


2.01: If S sits between Q and R, then the arrangement is as follows.



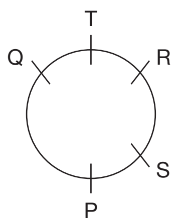
As can be seen from the diagram, T is to the immediate right of P.

2.02: We will take each choice and see whether it fits in the arrangement that we represented through a diagram in the analysis of the data (the same diagram is reproduced below).



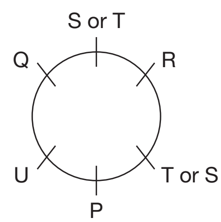
We can see that the arrangement given in choice (A) is not possible and hence, the answer choice is (A).

2.03: If S is not next to Q, then the seating arrangement is fixed as follows.



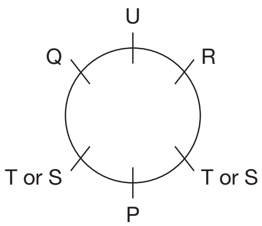
Now P is between Q and S.

2.04: On the basis of the diagram that we drew, we find that to accommodate U we have to create a new slot between P and Q.



Hence, choice (C) is the correct answer.

2.05: We create a new slot for the sixth person. But since U will not sit next to P, S or T, he will have to sit between R and Q. The arrangement will then look as follows:



As we can see from the diagram that the neighbours of P will be T and S.

Directions for questions 2.06 to 2.09: These questions are based on the following information.

There are 10 people at a round table conference, consisting of a Professor, a Lawyer, a Doctor, a Scientist, an Accountant, a Grocer, two Computer Specialists and two Marketing Executives. The Professor sits opposite to the Lawyer. The Scientist and the Doctor sits opposite to each other. The two Marketing Executives sit opposite to each other with one of them sitting to the immediate left of the scientist. The Professor sits to the immediate right of the Scientist.

2.06: If the two Computer Specialists sit opposite to each other but neither of them is immediately next to any Marketing Executive, who sits to the immediate right of the professor?
(A) Computer Specialist
(B) Marketing Executive
(C) Grocer
(D) Accountant

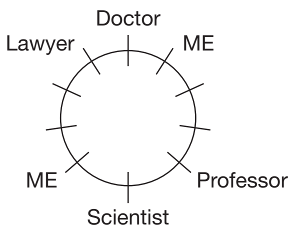
2.07: If the Grocer and Accountant do not sit opposite to each other, then which of the following must be TRUE?
(A) The Computer Specialist cannot sit beside the Lawyer.
(B) One of the Computer Specialists is next to a Marketing Executive.
(C) The Professor cannot have the Scientist and a Computer Specialist on his either side.
(D) The Computer Specialists must sit next to one another.

2.08: If a Computer Specialist is the immediate neighbour of a Marketing Executive and the Grocer is the immediate neighbour of the Lawyer, then how many different kinds of seating arrangements are possible? (Assume that the two Computer Specialists are indistinguishable from each other and the two Marketing Executives are indistinguishable from each other.)
(A) 3 (B) 6
(C) 16 (D) 8

2.09: The maximum number of persons you can count if you start counting with the Scientist and end with a Marketing Executive (excluding both) is
(A) 0 (B) 8
(C) 5 (D) 6

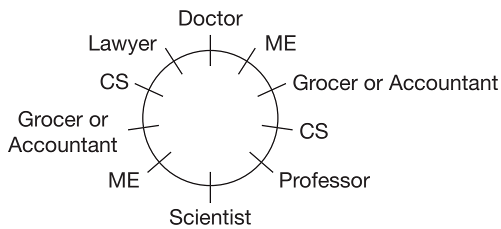
Solutions for questions 2.06 to 2.09: The Professor sits to the immediate right of the Scientist and opposite to the Lawyer. The Scientist sits opposite to the Doctor and one Marketing Executive is to the immediate left of the Scientist.

Choosing to place the Scientist in one of the 10 seats, we have the arrangement as follows.



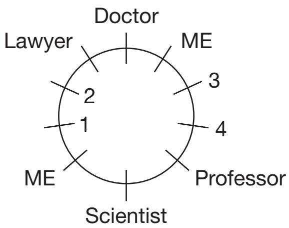
The vacant seats are one each for the two Computer Specialists, one for the Grocer and one for the Accountant.

2.06: The two Computer Specialists sit opposite to each other. Neither of them is next to any Marketing Executive. So, the arrangement must be as follows.



So, the Computer Specialist sits to the immediate right of the professor.

2.07: The Grocer and the Accountant do not sit opposite to each other. Then the arrangements can be as follows:

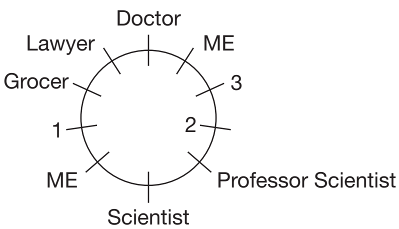


The Grocer and the Accountant can occupy the following pairs of seats: 3 and 4, 1 and 4, 1 and 2 or 2 and 3.

Then, the two computer specialists may occupy one of the pairs of seats 1 and 2, 2 and 3, 3 and 4 or 4 and 1.

We check for the choices given in the question, one by one, and find that whichever combination is taken, there is a Computer Specialist in Seat 1 or Seat 3, both of, which are next to the Marketing Executives seats. So, choice (B), which states that one of the Computer Specialists is next to a Marketing Executive is true.

2.08: Given that the Grocer is the immediate neighbour of the Lawyer, we have the three seats numbered 1, 2 and 3 (in the following diagram) free for the two Computer Specialists and the Accountant. Since a Computer Specialist has to be next to a Marketing Executive, he should be in Seat 1 or 3. By fixing the Accountant in any one of the three seats 1, 2 or 3, we can ensure that there is a Computer Specialist next to a Marketing Executive. Hence, there are three possible seating arrangements.



2.09: Based on the seating arrangement that we discussed, the number of persons between the Scientist and a Marketing Executive can be 3 or 8 (counted clockwise) or 0 or 5 (counted anti-clockwise). Maximum number that can be counted is 8.

Directions for question 2.10: Select the correct alternative from the given choices.

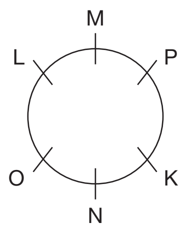
2.10: Six people K, L, M, N, O and P are sitting around a table. K and L do not sit next to each other. O and P are opposite to each other. M is sitting to the immediate right of P. If K is not between O and M, then N is not next to P. Which of the following is not an arrangement (in clockwise direction) satisfying the conditions given above?

(A) NKOLMP (B) PKNOLM

(C) LNOKMP (D) KMPNLO

Solution for question 2.10:

2.10: O and P are opposite to each other. M is to the right of P. Then we have two possible arrangements. In one case, when K is between O and M, the other two seats can be occupied by N and L. So, we cannot uniquely determine the seats of L and N. In the second case, if K is not between O and M (then, L has to be between O and M), then N is not next to P. This means that K has to be next to P and the only seat left is for N which is to the right of O.



From the choices, we can clearly see that choice (A) is the correct answer because that arrangement is not possible.

EXERCISE-1

Directions for questions 1 to 9: Select the correct alternative from the given choices.

1. Six people from A through F are seated around a circular table. A is opposite to D, who is two places to the right of F. If B is adjacent to A, then who is seated between E and C?
(A) B (B) A
(C) D (D) F

2. Six delegates are seated around a rectangular table such that two delegates are seated along each of the longer sides and one along each of the shorter sides. Vaibhav is seated at one of the shorter sides. Neither Vasu nor Varun is seated at any shorter side. Only one person is seated between Vasu and Vijay. Vallabh is opposite to Vagdev. If Varun is to the immediate right of Vagdev, then who is opposite to Vasu?
(A) Vijay (B) Vaibhav
(C) Varun (D) Vallabh

3. A group of six people from P through U are seated around a circular table. Among them three are boys and three are girls. No two girls are seated in adjacent places. S is three places away to the left of U. Q is neither adjacent nor opposite to P. Which of the following may represent the group of boys?
(A) S, T, U (B) Q, P, R
(C) Q, R, T (D) S, T, R

4. At buffet, six sweets are arranged in a circular order on a table. Burfi is opposite to Kova. There is one sweet between Laddu and Kaju Katli. Gulab Jamoon is to the immediate right of Kova. Rasagulla is one of the sweets. Which sweet is two places away to the left of Gulab Jamoon?
(A) Laddu (B) Rasagulla
(C) Kaju katli (D) (A) or (B)

5. Six people from A through F are sitting around a circular table such that C is sitting to the right of E and F is to the right of A. If B is sitting to the left of D, and A and C are sitting adjacent to each other, then who is sitting opposite to D?
(A) A (B) B
(C) C (D) E

6. A group of six friends from P through U are sitting around a hexagonal table. S is sitting adjacent to T and Q. Q is sitting to the left of U and P is sitting to the right of R. Who is sitting opposite to R?
(A) T (B) Q
(C) S (D) U

7. A group of six people from A through F are sitting in a circular table such that A is sitting two places away to the left of E, who is not adjacent to C and F. D is to the right of E and A is sitting between B and F. Now, who is sitting opposite to D?
(A) A (B) B
(C) C (D) F

8. A group of eight boys, namely A, B, C, D, E, F, G and H sit around a circular table, not necessarily in the same order. B and D sit neither adjacent to C nor opposite to C. A sits in between E and D, and F sits in between B and H. Which of the following is definitely true?
(A) H sits in between C and E.
(B) B sits in between A and G.
(C) C sits opposite to G.
(D) None of these

9. P, Q, R, S, T and U are the six corners of a table which has six sides, not necessarily in the same order. A group of six people, namely A, B, C, D, E and F are sitting along the sides joining the corners, not necessarily in that order. S is to the right of P and R is to the left of T. A is sitting opposite to E and to the immediate right of F. D is sitting between the corners of P and T and is opposite to C. If F is sitting between R and U, then who sits between Q and S?
(A) A (B) E
(C) C (D) B

Directions for questions 10 to 12: These questions are based on the following information.

Sameer, Sameep, Sandeep, Sangeet, Sanskar and Saarang are sitting around a hexagonal table in a library studying a book each. The cover of the book in their hands is of different colours, such as Red, Green, Blue, Orange, White and Yellow not necessarily in that order. Further, it is known that Sameer sits opposite to Sameep, who sits to the immediate left of the person holding the Blue cover book, who sits two places away from the person holding the White cover book. Sangeet sits two places away from the person holding the Yellow cover book and sits opposite to the person holding the white cover book. Sandeep and Saarang sit next to each other.

10. What is the colour of the book with Sanskar?
(A) Yellow (B) White
(C) Blue (D) Red

11. If the person holding the Orange cover book sits opposite to the person holding the Red cover book, then which statement among the following cannot be true?
(A) Sanskar does not hold the White cover book.
(B) Sangeet is not holding a Green cover book.
(C) Saarang holds the Yellow cover book.
(D) Sandeep holds the Yellow cover book.

12. If Sandeep sits opposite to the person holding the Green cover book, then who is sitting opposite to the person holding the Red cover book?
 (A) Sandeep (B) Sameer
 (C) Sanskar (D) Cannot be determined

Directions for questions 13 and 14: These questions are based on the following information.

Among the seven people, namely Maggi, Cherry, Prachi, Toto, Saxena, Gaddar and Basanti, sitting around a circular table.

(i) Toto is adjacent to neither Prachi nor Basanti.
(ii) Maggi is two places away to the right of Saxena.
(iii) Cherry is adjacent to both Basanti and Saxena.
(iv) Gaddar is not adjacent to Basanti.

13. Who is sitting four places away to the left of Cherry?
 (A) Gaddar (B) Maggi
 (C) Toto (D) Prachi

14. In which of the following combination is the second person sitting two places away to the left of the first person?
 (A) Gaddar, Cherry (B) Maggi, Prachi
 (C) Saxena, Basanti (D) Both (B) and (C)

Directions for questions 15 to 17: These questions are based on the following information.

Eight people A, B, C, D, E, F, G, and H are sitting around a rectangular table not necessarily in the same order. One person sits along the shorter side and three persons sit along the longer side.

A and G are sitting diagonally opposite to each other. D and E are sitting opposite to each other.

A is sitting to the immediate left of F, who is sitting at one of the shorter sides of the table.

15. If C is sitting to the immediate right of H, then who is sitting opposite to F?
 (A) B (B) C
 (C) H (D) D

16. If C is sitting opposite to A, then who among the following must be sitting on the same side as C?
 (A) G and E (B) G and D
 (C) G (D) H or B

17. If H is not sitting at one of the shorter sides, then how many arrangements are possible?
 (A) 24 (B) 8
 (C) 4 (D) 16

Directions for questions 18 to 20: These questions are based on the following information.

Each of the six people, namely John, Ted, Humpty, Dumpty, Jack and Jill is from one different country among India, Japan, China, Australia, America and England and are sitting around a circular table, they may not be in the same order. John, who is from China is sitting adjacent to an American, who is not Humpty. Ted is not an Indian, and the Chinese is not sitting adjacent to the Indian. The person from England is sitting one place away to the left of the Australian. Humpty is sitting opposite to the Indian, who is adjacent to the Japanese. The Australian and Dumpty are sitting opposite to each other. Jack is not from India and Ted is not from Japan and both are not adjacent to each other.

18. Who among them is from India?
 (A) Jill (B) Dumpty
 (C) Humpty (D) None of these

19. If Jack is the Japanese, then who is sitting opposite to the American?
 (A) Jill (B) Ted
 (C) Jack (D) Dumpty

20. To which country does, Humpty belong?
 (A) Japan (B) Australia
 (C) America (D) England

Directions for questions 21 to 23: These questions are based on the following information.

Eight friends, A through H are sitting around a circular table, playing a game of cards. They belong to two different teams X and Y. No two persons of the same team sit in adjacent seats.

(i) A sits neither opposite to D nor to H but is sitting in between C and G.
(ii) B sits neither opposite to A nor to G but is sitting in between F and D.
(iii) B and H belong to team X and D sits opposite to E.
(iv) A scored two points more than D, who scored three points more than F, who scored four points more than E. B scored twice as that of G, who scored twice that of C, who scored twice that of H.

21. Who are the members of team X?
 (A) A, D, F and E (B) B, H, C and E
 (C) B, D, H and G (D) B, H, C and G

22. If E and H scored one point each, then which team wins the game, given that the team with the minimum points wins the game?
 (A) Team X
 (B) Team Y
 (C) Both teams scored the same points
 (D) Cannot be determined

23. If team Y scores 92 points and team X scores 90 points, then which among the following is definitely true?
 (A) H and D scores equal points.
 (B) B scores twice as many points as F.
 (C) A scores three points less than G.
 (D) D scores one point more than G.

Directions for questions 24 to 26: These questions are based on the following information.

Eight people from A through H are sitting around a circular table, not necessarily in the same order. The following information is known about them.

(i) If C and F interchange their places, then each of them will have only one new neighbour.
(ii) H is sitting two places away to the left of A.
(iii) Two persons are sitting between C and G.
(iv) The one, who is adjacent to both C and F is not B. B is sitting opposite to G.

24. If A is not sitting adjacent to B, then who will be sitting three places away to the right of F?
(A) E (B) B
(C) D (D) Cannot be determined

25. If C is to the immediate left of E, then who will be sitting opposite to H?
(A) C (B) D
(C) F (D) Either (B) or (C)

26. If A is sitting between G and D, then who is sitting two places away to the right of B?
(A) A (B) F
(C) E (D) D

Directions for questions 27 to 29: These questions are based on the following information.

12 erasers named A through L are placed at a different hour division of a clock. B is at 7th hour division, E is opposite to K. L is at 60° from A. K is to the immediate left of H. H is at 90° from C which is 60° from D. F is at 11th hour division which is adjacent to K and J. G is at 30° from I.

27. What is the angle between E and H?
(A) 135° (B) 150°
(C) 120° (D) 170°

28. If A is at 5th hour division, then which is placed at 3rd hour division?
(A) L (B) G
(C) I (D) Cannot be determined

29. Which is opposite to G?
(A) C (B) D
(C) B (D) Cannot be determined

Directions for questions 30 to 33: These questions are based on the following information.

Among seven people, namely Tanuja, Divya and Vasudha are females and Srikanth, Ganesh, Sateesh and Appu are males and they are sitting around a circular table, but not necessarily in the same order. The following information is known about their seating.

No two females are adjacent to each other. Ganesh and Appu are not adjacent to each other. Srikanth is sitting to the immediate left of Vasudha, who is third to the left of Appu. Tanuja is not adjacent to Sateesh. Appu and Sateesh are adjacent to each other. All are facing the centre.

30. Who is sitting second to the right of Appu?
(A) Divya (B) Tanuja
(C) Srikanth (D) Ganesh

31. Who is sitting adjacent to Vasudha?
(A) Sateesh (B) Appu
(C) Divya (D) Ganesh

32. In a certain way, Vasudha is related to Divya. Sateesh is related to whom in the same way?
(A) Srikanth (B) Ganesh
(C) Tanuja (D) Appu

33. Three out of the following four follow a particular pattern and so form a group. Find the one which does not belong to the group.
(A) Appu, Divya (B) Vasudha, Ganesh
(C) Divya, Vasudha (D) Vasudha, Tanuja

Directions for questions 34 to 37: These questions are based on the following information.

A group of 12 people, from M through X are sitting around a circular table, but not necessarily in that order.

Starting from P in clockwise direction, O, V, W, N and Q are sitting in that order. Further, starting from P in anticlockwise direction, M, T, R, S, U and X are sitting in that order. Also, M is between P and Q, T is between Q and N, R is between N and W, S is between W and V and U is between V and O.

34. Who is sitting adjacent to both V and W?
(A) U (B) R
(C) S (D) Cannot be determined

35. Who is sitting opposite to R?
(A) O (B) P
(C) X (D) Cannot be determined

36. How many people sit between T and V when counted from the clockwise direction with respect to T?
(A) Five (B) Six
(C) Four (D) Cannot be determined

37. If all the persons are facing away from the centre, then who sits third to the left of R?
(A) V (B) Q
(C) U (D) Cannot be determined

Directions for questions 38 to 40: These questions are based on the following information.

There are two circular tables in a room. Six Russians, namely A, B, C, D, E and F are sitting at one table and six Frenchmen, such as M, N, O, P, Q and R are sitting at the other table. A and D are sitting opposite to each other. B and

E are sitting opposite to each other. C sits to the right of D. B is the only person who can translate Russian to French; C is the only person who can translate French to Russian, and none of them does the vice-versa, unless so stated. E and F sit adjacent to each other. Also, M sits opposite to P; Q sits to the right of R and R sits opposite to O. N and P sit adjacent to O. On the table, any person can talk to another person, only as stated below. The only conversations that took place are as given below:

A spoke to B; B to R; R to C; R to Q; Q to P; P to O; O to N; N to M; C to D; D to E; F to A; M to R; and E to F.

The conversations are one-sided, i.e., A spoke to B implies that A is the speaker and B is the listener and not vice-versa.

38. If C wants to send a message to M, then how many people must the message pass through? (excluding the first and the last)

(A) 1 (B) 6
(C) 10 (D) None of these

39. If the order of conveying messages is reversed at both the tables (i.e., 'A speaks to B' now becomes 'B speaks to A' and so on) and also B and C exchange their interpretory skills, then which of the following will be true?

(A) A can send a message to Q involving only 2 people.
(B) The person sitting to the right of F can send a message to the person sitting to the left of N, by involving only five people.
(C) The person sitting to the right of C, on the same table, can translate Russian into French.
(D) The maximum number of people involved in the longest message in this new arrangement is more than that in the previous arrangement.

40. If the person sitting to the right of B wants to send a message to the person sitting two places to the left of D, then what is the maximum possible number of people involved between them? (excluding the two people)

(A) 11 (B) 10
(C) 4 (D) None of these

EXERCISE-2

Directions for questions 1 to 5: These questions are based on the following information:

Six sofas of different colours are arranged in a circular order. On each sofa, a boy among O, P, Q, R, S and T and a girl among U, V, W, X, Y and Z are sitting.

(i) The red coloured sofa is in between the yellow coloured sofa and blue coloured sofa.
(ii) X is sitting on white sofa, which is opposite to the sofa where Z is sitting.
(iii) The orange coloured sofa is adjacent to the sofa where both O and V are sitting.
(iv) The sofa, where W is sitting is adjacent to the blue and pink coloured sofas.
(v) P is to the left of S, who is opposite to Y.

1. Who among the following is sitting on the sofa which is opposite to the pink coloured sofa?

(A) S (B) X
(C) Y (D) U

2. Who are sitting on the blue coloured sofa?

(A) Z, P (B) W, S
(C) O, V (D) T, Z

3. What is the colour of the sofa which is opposite to the sofa where Y is sitting?

(A) Yellow (B) Orange
(C) White (D) Blue

4. Which of the following represents the people sitting on the same sofa?

(A) Z, Q (B) W, P
(C) X, T (D) W, S

5. Which of the following represents the correct order of the colours of the sofas?

(A) Pink, Orange, Blue, Red, White, Yellow
(B) Pink, Blue, Red, Yellow, Orange, White
(C) Red, Blue, Orange, Pink, White, Yellow
(D) Pink, Orange, Yellow, Red, Blue, White

Directions for questions 6 to 9: These questions are based on the following information.

Seven children of a family are seated around a circular table to have their lunch. No two children finish their lunch at the same time and no two adjacent children finish their lunch immediately one after the other.

(i) Sujatha finishes her lunch immediately before Bhuvan finishes his lunch. Bhuvan is to the immediate left of Srilatha.
(ii) Pranav is three places away to the left of Anand but finishes his lunch after Anand finishes his lunch.
(iii) Krupa is to the immediate left of the child who finishes his/her lunch before two children.
(iv) Kruti finishes her lunch immediately after Bhuvan finished his lunch but not after Pranav finished his lunch.
(v) The number of children who finish their lunch before Srilatha finishes her lunch is same as the numbers of children who finish their lunch after.

6. How many children finish their lunch before Pranav finishes his lunch?
(A) 1 (B) 2
(C) 4 (D) 5

7. Who finishes his/her lunch immediately after Kruti finishes his lunch?
(A) Bhuvan (B) Krupa
(C) Sujatha (D) Srilatha

8. Who is to the immediate right of Pranav?
(A) Sujatha (B) Kruti
(C) Krupa (D) Srilatha

9. Which of the following represents the seating order?
(A) Anand, Sujatha, Pranav, Srilatha, Bhuvan, Kruti, Krupa
(B) Anand, Kruti, Sujatha, Pranav, Krupa, Srilatha, Bhuvan
(C) Anand, Krupa, Kruti, Pranav, Sujatha, Srilatha, Bhuvan
(D) Anand, Krupa, Bhuvan, Srilatha, Pranav, Sujatha, Kruti

Directions for questions 10 to 12: These questions are based on the following information.

A group of six boys, namely Prasad, Prakash, Prashant, Pranay, Praveen and Prabhat each wearing a T-shirt of a different colour, such as Indigo, Green, Blue, Orange, Violet and Yellow are sitting around a table in six equi-spaced chairs. Prakash is opposite to the boy wearing the Orange T-Shirt. Prabhat is opposite to the boy wearing the Green T-Shirt. Prashant is to the right of the boy wearing the Indigo T-Shirt and opposite to the boy wearing the Violet T-Shirt. Praveen is between the boys wearing Orange and Yellow T-Shirts and is not wearing the Violet T-Shirt. Pranay is opposite to the boy who is wearing the Yellow T-Shirt.

10. Which of the following statements is true?
I. The boys wearing Green and Orange T-Shirts are either next to each other or opposite to each other.
II. The boys wearing Indigo and Orange T-Shirts are either next to each other or opposite to each other.
III. The boys wearing Blue and Violet T-Shirts are always next to each other.
IV. The boys wearing Blue and Indigo T-Shirts are either opposite to each other or are next to each other.
(A) Only I and II (B) Only III
(C) Only III and IV (D) I, II, III and IV

11. If Prabhat is wearing the Orange T-Shirt, then who is wearing the Green T-Shirt?
(A) Prasad (B) Prakash
(C) Praveen (D) Pranay

12. If Prashant is wearing the Orange T-Shirt, then who is between Praveen and Pranay?
(A) Prashant (B) Prasad
(C) Prakash (D) Prabhat

Directions for questions 13 to 15: These questions are based on the following information.

Eight people, namely Ram, Ramesh, Mohan, Sohan, Seema, Saroj, Sakshi and Saloni are sitting around a circular table. Each of them is from different professions, such as Doctor, Engineer, Dancer, Singer, Teacher, Lawyer, Accountant and Pilot, not necessarily in the given order. Further it is known that

I. Pilot is sitting opposite to Ramesh, who is adjacent to the Accountant.
II. Dancer is sitting opposite to the Lawyer and is not adjacent to Sakshi who is not sitting adjacent to the Lawyer.
III. Saloni is sitting opposite to the Engineer, Ramesh is not a Lawyer or Doctor or Engineer.
IV. Sakshi, the Singer, is sitting one place away to the right of Saroj.
V. Seema is sitting opposite to the Lawyer and Ram is sitting opposite to the Dancer.
VI. Ramesh is sitting three places to the right of Singer. Mohan is neither the Accountant nor adjacent to the Dancer.

13. Who is the Doctor?
(A) Ramesh (B) Saloni
(C) Saroj (D) Cannot be determined

14. What is the profession of Mohan?
(A) Accountant (B) Pilot
(C) Engineer (D) Cannot be determined

15. Who is sitting opposite to Ramesh?
(A) Seema (B) Sakshi
(C) Saroj (D) None of these

Directions for questions 16 to 18: These questions are based on the following information.

Each of the eight boys from A through H has a different fruit with them and are seated around a square table such that two boys are seated along each side. The following information is known about the seating arrangement.

(i) G, who has Watermelon is four places away to the right of B. F is opposite to the boy who has Banana.
(ii) The boy who has Grapes and the boy who has Kiwi are on the same side of the table. C is to the immediate left of B but is not along the same side.
(iii) The boy who has Orange and the boy who has Guava are seated at opposite sides.
(iv) E has Litchi and H has Grapes. There are three boys between E and H.

(v) The boy who has Watermelon is to the immediate right of the boy who has Mango and is to the immediate left of F.

16. If A has Orange, then who will be sitting at the side which is opposite to the side where C is sitting?
(A) D (B) A
(C) H (D) F

17. Who is three places away to the left of F?
(A) A (B) E
(C) G (D) B

18. Which of the following is a correct combination of boy and the fruit that he has?
(A) C-Kiwi (B) G-Mango
(C) F-Orange (D) B-Watermelon

Directions for questions 19 to 21: These questions are based on the following information.

Eight people A, B, C, D, E, F, G and H are sitting around a square table, but not necessarily in that order. The people who are sitting at the corners, face the centre and the people who are sitting at the sides face away from the centre. Each of them likes a different colour among red, blue, green, yellow, pink, black, white and violet. The following information is known about their seating.

(i) B sits second to the left of the person who likes blue.
(ii) The person who likes violet sits at the corner, who is adjacent to both G and the person who likes blue.
(iii) A sit second to the right of the person who likes yellow and is not adjacent to G.
(iv) C sits adjacent to the person who likes yellow.
(v) C likes neither violet nor blue. The person who likes pink is adjacent to neither A nor B.
(vi) E sits to the immediate right of the person who likes pink.
(vii) F is not adjacent to the person who likes black.
(viii) The person who likes black sits at one of the sides.
(ix) D likes white.
(x) H does not like black.

19. Who likes red?
(A) C (B) E
(C) F (D) Cannot be determined

20. Who sits third to the right of F?
(A) H (B) C
(C) G (D) Cannot be determined

21. Three of the following four are alike in a certain way, based on the given information and so form a group. Find the one which does not belong to that group.
(A) The person who likes green.
(B) The person who likes violet.
(C) The person who likes yellow.
(D) The person who likes red.

Directions for questions 22 to 24: These questions are based on the following information.

Eight people, namely Anand, Brijesh, Chandak, Dweepesh, Sayan, Jagat Rupak and Palak are sitting around a square table such that two people are sitting along each side. The following information is known about them.

(i) Jagat, who is sitting to the immediate right of Rupak is sitting opposite to Chandak who is sitting to the immediate right of Brijesh.
(ii) Sayan is sitting opposite to Dweepesh, who sits along the same side as Brijesh.
(iii) Palak is not sitting along the same side as Sayan.

22. Who is sitting along the same side as Chandak?
(A) Anand (B) Palak
(C) Sayan (D) Rupak

23. Who is sitting opposite to Rupak?
(A) Palak (B) Anand
(C) Brijesh (D) Data inadequate

24. Who is sitting to the immediate right of Sayan?
(A) Anand (B) Rupak
(C) Chandak (D) Data inadequate

Directions for questions 25 to 27: These questions are based on the following information.

Eight people from P through W are sitting around a rectangular table, each of them facing the centre but not necessarily in that order. Three people sit along each of the longer sides of the table and one person sits along each of the shorter sides.

(i) If Q and S interchange their positions, then V sits to the immediate left of S.
(ii) If P and T interchange their positions, then R sits opposite to T.
(iii) If Q and U interchange their positions, then W sits third to the right of U.
(iv) P sits third to the left of Q, who sits at the longer side of the table.
(v) If W and S interchange their positions, then S sits third to the left of T.
(vi) U and S sit opposite to each other.

25. Who sits opposite to V?
(A) S (B) W
(C) N (D) Cannot be determined

26. Who among the following sit along the longer side of the table?
(A) Q, R, V (B) S, U, P
(C) Q, R, W (D) Cannot be determined

27. Who sits second to the left of S?
(A) Q (B) T
(C) U (D) Cannot be determined

Directions for questions 28 to 32: These questions are based on the following information.

Eight members of a family A through H are sitting around a circular table.

The following information is known about them:

(i) There are three married couples in the family.
(ii) One of A's sons is sitting opposite him while the other is adjacent to him.
(iii) H's sister-in-law is B, who is sitting to the immediate right of H's father-in-law.
(iv) The number of females in the family is less than the number of males in the family.
(v) Two of the married couples have two children each.
(vi) C, who is the eldest male in the family is sitting third to the left of his wife.
(vii) F, the youngest is not G's son and is sitting adjacent to H.
(viii) H is the aunt of E, who is sitting three places away from G.

28. Who is C's son?
(A) A (B) E
(C) D (D) None of the above

29. How is A's brother-in-law's nephew's grandmother related to B?
(A) Mother (B) Sister
(C) Niece (D) Aunt

30. What is the position of E's father with respect to C's daughter-in-law?
(A) Immediate left (B) Opposite to each other
(C) Second to the right (D) Second to the left

31. Based on information given three of the following four are similar in a certain way and, hence, form a group. Find the one that does not belong to the group.
(A) E (B) G
(C) F (D) B

32. How is F related to the person sitting third to the left of him?
(A) Son (B) Father
(C) Brother (D) Grandson

Directions for questions 33 to 36: These questions are based on the following information.

Eight people from A through H sit around a circular table, but not necessarily in that order. Some are facing the centre and the remaining are facing away from the centre. The following information is known about their seating.

(i) B sits third to the right of G.
(ii) There are two people sitting between G and A.
(iii) C sits second to the left of A.
(iv) C and G face the same direction.
(v) D sits third to the left of the person who is adjacent to C.
(vi) E sits third to the right of D, both of them face the same direction.
(vii) F faces E.
(viii) B and H face different directions.

33. How many people face the centre?
(A) 4 (B) 3
(C) 5 (D) Cannot be determined

34. If B faces the centre, then who sits third to the left of H?
(A) D (B) C
(C) G (D) E

35. If H and A face the same direction, then who sits second to the right of B?
(A) D (B) A
(C) G (D) H

36. Who sits to the immediate right of B?
(A) F (B) H
(C) D (D) Cannot be determined

Directions for questions 37 to 40: The following questions are based on the information given below:

Eight people, namely G, H, I, J, K, L, M and N are sitting around a square table. Some of them are facing the centre and others are facing away from the centre.

(i) I is sitting at one of the corners and is facing away from the centre.
(ii) Neither J nor M is a neighbour of I.
(iii) The neighbours of I face the same direction as I.
(iv) L and M face the same direction and sit opposite to each other.
(v) H is to the immediate right of M and G is to the immediate left of I.
(vi) J and K are neighbours of N and face different directions.
(vii) N sits opposite to I and faces the centre.
(viii) M and K are facing different directions and K is to the immediate right of N.

37. How many people are facing away from the centre?
(A) Five (B) Four
(C) Two (D) Three

38. Three of the four are alike in a certain way and so form a group. Which is the one that does not belong to that group?
(A) JH (B) NI
(C) MK (D) MN

39. Who is sitting to the immediate right of H?
(A) M (B) G
(C) I (D) K

40. Who is sitting in the opposite position of G?
(A) H (B) M
(C) K (D) J

EXERCISE-3

Directions for questions 1 to 3: These questions are based on the following information.

A group of eight people, namely A, B, C, D, E, F, G and H from eight cities, such as P, Q, R, S, T, U, V and W not necessarily in the same order are sitting around a circular table. We know the following additional information.

(1) Among A, E, G and F, no two people are adjacent to each other.
(2) Among the people from P, T, V and W, no two people are opposite to each other.
(3) A is to the immediate left of B, who is two places away to the right of the person from Q.
(4) The person from S is opposite to D, who is adjacent to E.
(5) The person from P is to the immediate left of G, who is from T.
(6) C, who is from W, is adjacent to the people from R and V.

1. Who is from R?
(A) A (B) B
(C) E (D) F

2. Who is to the immediate right of H?
(A) G (B) E
(C) A (D) Cannot be determined

3. Who is opposite to the person from V?
(A) A (B) G
(C) F (D) Cannot be determined

Directions for questions 4 to 6: These questions are based on the following data.

Eight chairs are arranged in a room. Four of them are exactly at the four corners while the remaining four are placed against the walls on the four sides in between each pair of chairs. In the corners P, Q, R and S are four boys and A, B, C and D are four girls who occupy the chairs all of which are facing the centre of the room. Q is in a corner chair and R is in a chair which is not along the same wall as either of the walls are adjoining Q. A and C are seated at corners, which is diagonally opposite to each other. B does not sit along any wall which is adjacent to the corner where A sits and is opposite to P. C sits to the immediate right of R, who is between C and D.

4. S must be seated between
(A) C and Q (B) A and D
(C) A and Q (D) C and A

5. If S and P interchange their seats, then who is to the immediate left of D?
(A) A (B) P
(C) R or C (D) S

6. Which of the following is not one of the correct arrangements of the corner seat occupants, either in clockwise direction or in anti-clockwise direction consecutively?
(A) Q, A, D and C (B) A, Q, C and D
(C) D, A, Q and C (D) D, Q, A and C

Directions for questions 7 to 9: These questions are based on the following data.

Four teachers Ranjan, Rajan, Raman and Raj, and four doctors Puneet, Piyush, Pratham and Pratima are sitting around a table. No two teachers sit adjacent to each other. Raj is two places to the right of Ranjan and adjacent to Pratima, who is two places to the left of Puneet, who is adjacent to Rajan.

7. If Raman is not opposite Ranjan, then who is seated two places to the left of Ranjan?
(A) Raman (B) Rajan
(C) Raj (D) Cannot be determined

8. If Pratima is adjacent to Raman, then who is seated opposite to Raj?
(A) Pratima (B) Raman
(C) Rajan (D) Ranjan

9. If Pratham is not opposite to Puneet, then who is seated opposite to Pratima?
(A) Puneet (B) Pratham
(C) Raman (D) Cannot be determined

Directions for questions 10 to 12: These questions are based on the following data.

A group of six people, namely Amit, Amitabh, Arnold, Aakash, Abhinav and Atul, each from a different profession, such as Doctor, Lawyer, Teacher, Manager, Business Analyst and Accountant, are seated around a table in six equi-spaced chairs. Atul is opposite to the Lawyer. Arnold is to the right of the Doctor and is opposite to the person who is the Business Analyst. Aakash is opposite to the Accountant. Abhinav is between the Manager and the Accountant and is not the Business Analyst. Amitabh is opposite to the Manager, who is to the left of Aakash.

10. Who is between the Lawyer and the Business Analyst?
(A) Amit (B) Amitabh
(C) Aakash (D) Cannot be determined

11. If Atul is not the Teacher, then who is the Doctor?
(A) Abhinav (B) Amitabh
(C) Akash (D) Amit

12. If Amitabh is the Business Analyst, then who is opposite to Amitabh?
(A) Amit (B) Atul
(C) Arnold (D) Cannot be determined

Directions for questions 13 to 15: These questions are based on the following data.

A librarian wishes to sit at the centre of his circular library hall with eight shelves arranged around him in a circle. There are books on eight subjects English, Physics, Sociology, Chemistry, Mathematics, French, German and History which are placed in the shelves, the books of one subject is only in one shelf. The books on French, German and English should be in three shelves placed side by side. The books on History should be in a shelf opposite to the shelf containing French books. The books on Physics and those on Chemistry should be in the shelves opposite to each other.

13. If the books on German are opposite to the shelf which has Mathematics books, and between the shelves containing books on Physics and French, then which of the following should be opposite to each other?
(A) English and Sociology shelves
(B) English and Physics books
(C) English and History books
(D) Sociology and Mathematics books

14. If the books on Sociology are between the shelves with Physics and History books, then the books on Mathematics would be between the shelves containing books on
(A) History and French
(B) History and Chemistry
(C) French and Chemistry
(D) French and Physics

15. If the English books are to the immediate left of the shelf with Physics books, then the shelf with German books is to the immediate right of shelf containing books of which subject?
(A) Only Physics (B) Chemistry or French
(C) Only French (D) French or German

Directions for questions 16 to 18: These questions are based on the following information.

A group of eight people, namely K, L, M, N, O, P, Q, and R sit around a circular table not necessarily in the same order. Some of them are facing the centre and the remaining are facing away from the centre.

(i) P sits to the immediate left of K.
(ii) M and R are sitting in the opposite places.
(iii) Either ‘Q’ or ‘O’ sits next to L but not both.
(iv) No two people are sitting next to each other facing the same direction.
(v) R is the neighbour of both N and Q.
(vi) L sits opposite to K and faces away from the centre.

16. Three of the following are alike in a certain way and so form a group. Which is the one that does not belong to that group?
(A) KM (B) ML
(C) NO (D) PO

17. Which among the following is ‘definitely true’?
(A) P is facing away from the centre.
(B) O and N are opposite to each other.
(C) K sits to the immediate right of P.
(D) L sits to the immediate right of O.

18. If M and N interchange their places, then who among the following sits to the immediate left of M?
(A) O (B) P
(C) L (D) None of these

Directions for questions 19 to 21: These questions are based on the following information.

A group of eight people from the same family sit around a rectangular table in such a way that four people, namely A, B, C and D sit along one of the longer sides of the table, facing north and the other four people, namely P, Q, R and S sit along the other longer side of the table facing south, not necessarily in that order. Each person faces exactly one person who sits on the opposite side. The following information is known about them.

P is the brother of Q, who is not adjacent to either P or S. A is the daughter of P and sits to the immediate right of B’s husband. D is opposite to neither P nor S. C is the niece of P but is not opposite to either P or S. B is not opposite to S but to opposite her brother. D has only one child who is a male. Q is the daughter of C’s grandfather, who is not R.

19. Who is to the immediate right of P?
(A) R’s father (B) P’s sister
(C) D’s father-in-law (D) R

20. Which of the following person sits at an end?
(A) A (B) R
(C) Q’s daughter (D) B’s father

21. Which of the following statements is true?
(A) Q is at the left end (B) B is at the right end
(C) R is the brother of A (D) All the above

Directions for questions 22 to 25: These questions are based on the following information.

A group of eight people, namely P, Q, R, S, T, U, V, and W belong to the same family and they sit around a circular table facing the centre (not necessarily in the same order). The following information is known about them.

(1) P sits second to the right of his nephew, whose neighbours are females.
(2) V is the wife of P and sits to the immediate right of her daughter T.
(3) U sits second to the right of his brother-in-law and opposite to his son Q.
(4) S sits third to the right of her sister-in-law and second to the left of her father R.
(5) W is the mother-in-law of U and is adjacent to her grandson.

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22. Who among the following is the wife of U?
(A) S
(B) The one who is opposite to T.
(C) The one who sits second to the right of R.
(D) Both (A) and (B)

23. Who among the following is the niece of S?
(A) The one who sits opposite to S.
(B) The one who sits to the immediate right of S's husband.
(C) T
(D) All the above

24. Which among the following is 'definitely true'?
(A) R and V sit adjacent to each other.
(B) T and her mother sit opposite to each other.
(C) W sits third to the right of her son.
(D) None of these

25. Four of the following are alike in a certain way, and hence form a group. Which is the one that does not belong to that group?
(A) TU (B) RS
(C) VW (D) QT

ANSWER KEYS

Exercise-1

1.	(C)	**8.**	(D)	**15.**	(C)	**22.**	(A)	**29.**	(D)	**36.**	(B)
2.	(C)	**9.**	(B)	**16.**	(C)	**23.**	(D)	**30.**	(A)	**37.**	(A)
3.	(D)	**10.**	(C)	**17.**	(B)	**24.**	(B)	**31.**	(D)	**38.**	(C)
4.	(B)	**11.**	(B)	**18.**	(B)	**25.**	(A)	**32.**	(C)	**39.**	(B)
5.	(A)	**12.**	(D)	**19.**	(A)	**26.**	(D)	**33.**	(B)	**40.**	(B)
6.	(C)	**13.**	(B)	**20.**	(B)	**27.**	(B)	**34.**	(C)		
7.	(A)	**14.**	(C)	**21.**	(D)	**28.**	(A)	**35.**	(C)		

Exercise-2

1.	(D)	**8.**	(B)	**15.**	(C)	**22.**	(B)	**29.**	(A)	**36.**	(D)
2.	(A)	**9.**	(C)	**16.**	(A)	**23.**	(A)	**30.**	(C)	**37.**	(B)
3.	(B)	**10.**	(D)	**17.**	(B)	**24.**	(B)	**31.**	(D)	**38.**	(D)
4.	(D)	**11.**	(B)	**18.**	(A)	**25.**	(B)	**32.**	(C)	**39.**	(A)
5.	(C)	**12.**	(A)	**19.**	(D)	**26.**	(B)	**33.**	(C)	**40.**	(C)
6.	(D)	**13.**	(B)	**20.**	(B)	**27.**	(D)	**34.**	(A)		
7.	(D)	**14.**	(C)	**21.**	(C)	**28.**	(D)	**35.**	(B)		

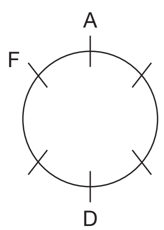
Exercise-3

1.	(D)	**5.**	(D)	**9.**	(B)	**13.**	(A)	**17.**	(D)	**21.**	(B)	**25.**	(C)
2.	(C)	**6.**	(D)	**10.**	(A)	**14.**	(B)	**18.**	(C)	**22.**	(D)		
3.	(A)	**7.**	(A)	**11.**	(A)	**15.**	(B)	**19.**	(C)	**23.**	(D)		
4.	(C)	**8.**	(C)	**12.**	(C)	**16.**	(D)	**20.**	(C)	**24.**	(D)		

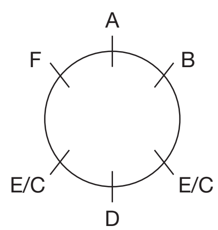
SOLUTIONS

EXERCISE-1

1. A is opposite to D, who is two places away to the right of F.

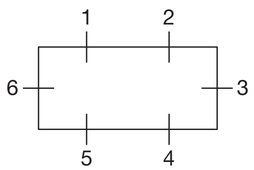


B is adjacent to A. The arrangement will be as follows.

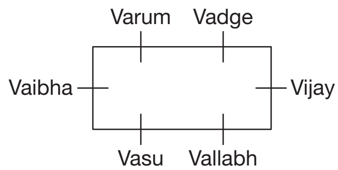


∴ D is in between E and C.

2. The group of six people are seated as follows:

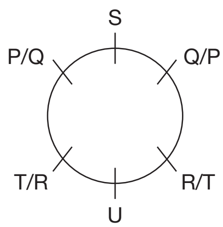


Let Vaibhav be seated at 6. Neither Vasu nor Varun is placed at 3.
Vallabh is opposite Vagdev.
∴ Neither Vagdev nor Vallabh is placed at 3. Vinay is at 3. Varun is to the immediate right of Vagdev. So, the arrangement will be as follows.

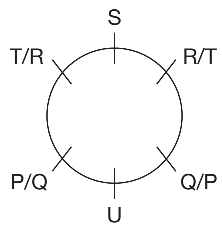


∴ Varun is opposite to Vasu.

3. As, S is 3 places away to the left of U,
S and U are opposite to each other. Given that Q is neither adjacent nor opposite to P.
∴ The following arrangements are possible.
Case – (i):

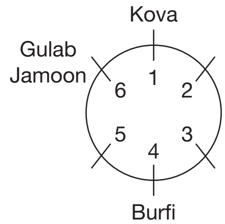


Case – (ii):



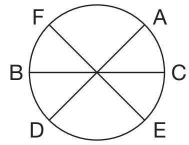
Case-(i):
∴ P, Q and U are of the same gender.
S, R and T are of the same gender.
Case-(ii):
S, P and Q are of the same gender.
U, R and T are of the same gender.

4. Burfi is opposite to Kova, which is to the immediate left of Gulab Jamoon. Hence, the following arrangement is possible.



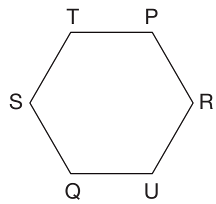
There is one sweet between Laddu and Kaju Katli.
∴ Laddu and KajuKatli are at 3 and 5 in any order and Rasagulla is at 2.
Rasagulla is two places away to the left of Gulab Jamoon.

5. According to the given information, the possible arrangement is as follows.



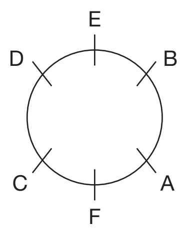
∴ A is sitting opposite to D.

6. The final arrangement is as follows.



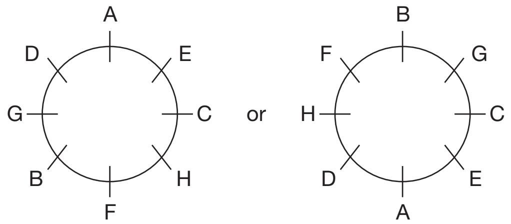
∴ S is sitting opposite to R.

7. It is given that A is two places away to the left of E, who is adjacent to B and D. D is to the right of E and A is setting between B and F. These conditions give us the following arrangement.



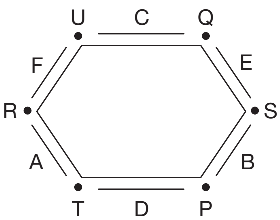
∴ A is sitting opposite to D.

8. It is given that A sits in between E and D, and F sits in between B and H.
It is also given that B and D sit neither adjacent to C nor opposite to C.

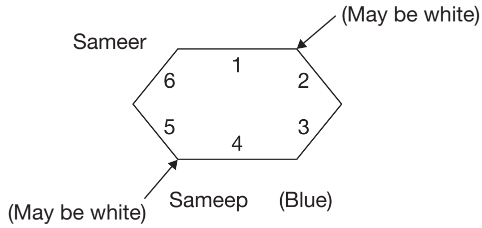


Hence, none of the choices is definitely true.

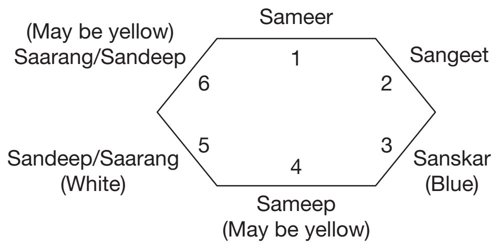
9. Given:
(i) Six corners (of a hexagonal table) P, Q, R, S, T and U.
(ii) Six people – A, B, C, D, E and F
(iii) $\xrightarrow[P]{\text{Right}}$ S; R $\xrightarrow[T]{\text{Left of}}$
(iv) E
↕ opp
F → A
(A is to the immediate right of F)
(v) D is sitting between the corners of P and T.
T/P _D_ T/P
(vi) D $\xleftrightarrow{\text{opp}}$ C
(vii) R/U _F_ U/R
Now, let us try to make an arrangement with the given information.
Clearly, E sits between the corners of Q and S.



Solutions for questions 10 to 12: Let us write the data as shown below:
(i) 6 people – Sameer (Smr), Sameep (Smp), Sandeep (Snd), Sangeet (Sgt), Sanskar (Skr) and Saarang (Srg) are sitting around a hexagonal table.
(ii) Colour of the books in their hands – Red, Green, Blue, Orange, White and Yellow.
(iii) Sameer $\xleftrightarrow{\text{opp}}$ Sameep
(iv) Sameep $\xleftrightarrow[\text{Left}]{\text{Immediate}}$ Blue
(v) Blue _____ _____ White (2 places away)
(vi) Sangeet _____ _____ Yellow (2 places away)
(vii) Sangeet is opposite to White.
(viii) Sandeep is adjacent to Saarang.
Based on (iii), (iv) and (v), we get the following arrangements:

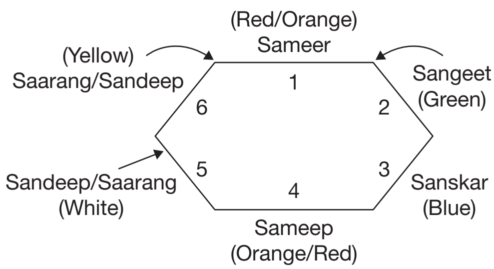


But from (vii), we get that Sangeet is opposite to White; in that case Sameer cannot have White coloured book in his hand, as he is opposite to Sandeep, not Sangeet. Hence, the person holding White coloured book must be to the immediate left of Sameep, i.e., at 5. Then Sangeet is opposite to White and to the left of Sameer, at 2. Now as Sandeep and Saarang are adjacent, hence, they must be accommodated at seat numbers 5 and 6, in any order. Then the only seat left for Sanskar is 3. Now, we get the following arrangement:



10. Hence, Sanskar holds the Blue cover book in his hand.

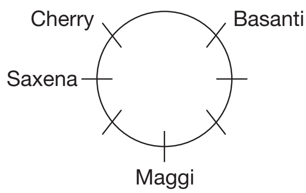
11. In the above figure, if the person sitting at 4 (i.e., Sameep) holds Yellow cover book, then the Red and Orange coloured books cannot be opposite to each other. Hence, the person sitting at 6 (i.e., either Saarang or Sandeep) must have Yellow cover book and Sameer and Sameep will have Red cover and Orange cover book in any order. Then, we get the following arrangement.



Hence, Sangeet must have Green book.
∴ Choice (B) is the only one that is definitely false.

12. Sandeep is sitting opposite to the person holding the Green cover book which implies that Sandeep is sitting opposite to Sangeet. Thus, either Sameer or Sandeep holds the Red cover book. Hence, it cannot be determined.

Solutions for questions 13 and 14: From (ii) and (iii), the arrangement of the persons can be represented as follows.

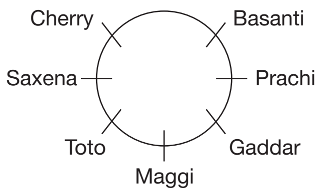


From (i) and (iv), neither Toto nor Gaddar is adjacent to Basanti.

∴ Prachi must be adjacent to Basanti.

As Toto is not adjacent to Prachi, Gaddar must be adjacent to Prachi.

The final arrangement of the person can be represented as follows.

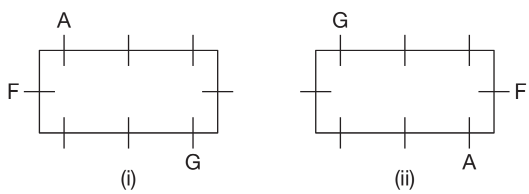


13. Maggi is sitting four places away to the left of Cherry.

14. Choice (C) is the correct representation.

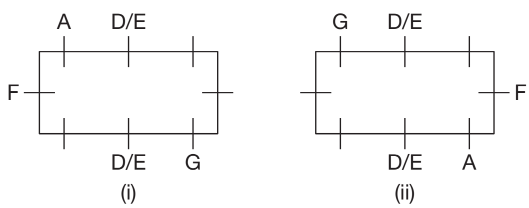
Solutions for questions 15 to 17: It is given that eight people, namely A, B, C, D, E, F, G and H are sitting around a rectangular table. One sits along the shorter side and three sits along the longer side.

A and G are sitting diagonally opposite to each other and A is sitting to the immediate left of F.

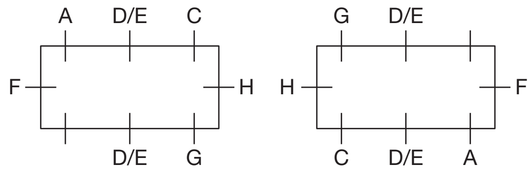


F is sitting along the shorter side.

D and E are sitting opposite each other, which is as follows.

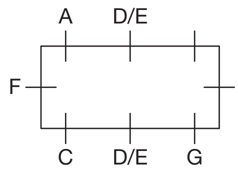


15. If C is sitting to the immediate right of H:



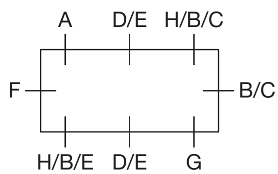
H sits opposite to F.

16. If C is sitting opposite to A, then the arrangement is as follows:



So, G sits on the same side as C.

17. If H is not sitting at one of the shorter sides, then the arrangement is as follows:



H is seated in two ways. B and C are seated in two ways.
D and E are seated in two ways.
Total ways = 2 × 2 × 2 = 8

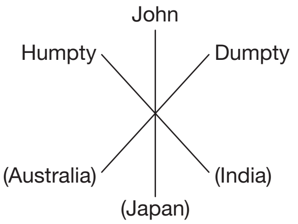
Solutions for questions 18 to 20: From the given information, John is from china and is adjacent to American who is not Humpty. But Humpty is opposite to the Indian who is not adjacent to Chinese but adjacent to Japanese, thus the following two arrangements are possible.

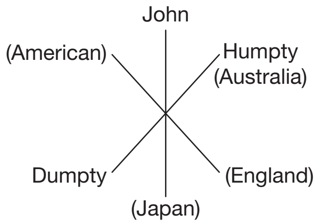
The English man is left to Australian thus in Case (i) Humpty can be from England and in Case (ii) Humpty can be from England and in Case (iii) Humpty can be from Australia. As Australian is opposite to the Dumpty.

Case (i), Dumpty is the Amercian and
Case (ii), Dumpty is the Indian.

Difficulty Level-1: Foundation

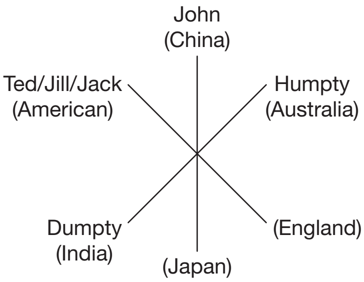
Thus, the arrangements are as follows.





Case (i): Now among the remaining people, i.e., Ted, Jill and Jack, as Ted and Jack are not from India Jill is the Indian. Ted and Jack are Australian and Japanese, respectively and are sitting adjacent to each other thus case (i) is not possible.
Case (ii): Either Jack or Jill is from the Japan. If Jack is from Japan, then Ted is the American and Jill is the English.

If Jill is the Japanese, then Ted is either from England or American and jack is either from England or America.



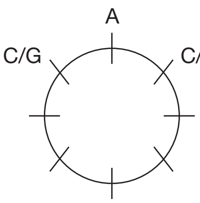
18. Dumpty is from India.

19. Jill is sitting opposite to the American.

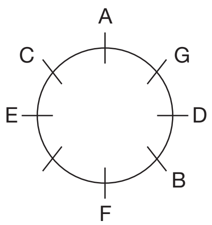
20. Humpty belongs to Australia.

Solutions for questions 21 to 23: It is given that eight friends, A through H are playing a game of cards and they belong to two different teams X and Y. No two persons of the same team sit adjacent to each other.

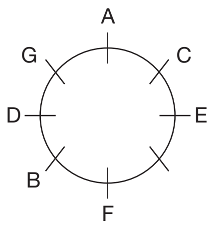
From (1), A sits between C and G but not opposite D or H.



From (1), (2) and (3), we get



or



Let the points scored by E = x, F = $x + 4$, D = $x + 7$ and A = $x + 9$.

Similarly, let the points scored by H = y, C = $2y$, G = $4y$ and B = $8y$.

21. B, H, C and G are the members of team X.

22. If E scores 1 point, then F scores 5 points, D scores 8 points and A scores 10 points.
A, D, E and F belong to team Y and their total points is $1 + 5 + 8 + 10 = 24$ points.
Similarly, H scores 1, C scores 2, G scores 4 and B scores 8. B, C, G and H belong to team X and the total sum of their points is $1 + 2 + 4 + 8 = 15$ points. Hence, team X wins the game.

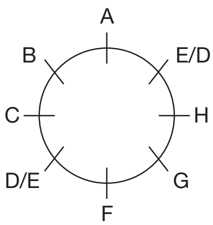
23. If team Y scores 92 points means E + F + D + A = 92.
$\Rightarrow x + (x + 4) + (x + 7) + (x + 9) = 92$
$\Rightarrow 4x + 20 = 92 \Rightarrow x = 18.$
E scores 18, F scores 22, D scores 25 and A scores 27. If team X scores 90 points, then
$y + 2y + 4y + 8y = 90$
$\Rightarrow 15y = 90 \Rightarrow y = 6$
H scores 6, C scores 12, G scores 24 and B scores 48. D scores one point more than G is definitely true.

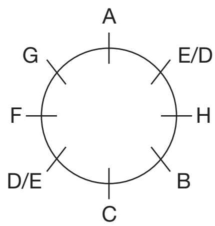
Solutions for questions 24 to 26: From (i), it can be said that there is exactly one person sitting between C and F.

From (ii), Two people are sitting between C and G, the remaining people E and D are sitting opposite to each other.

From (iii), the representation of people sitting is as follows.

From (iv), B and G are opposite to each other and B is not sitting between C and F.



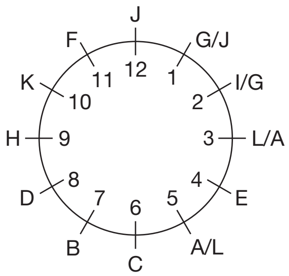


24. B will be sitting two places away to the right of F.

25. C will be sitting opposite to H.

26. D will be sitting two places away to the right of B.

Solutions for questions 27 to 29: It is given that, B is at 7th hour division and F is at 11th hour division. Given F is adjacent to K and J. K is to the immediate left of H. Hence, K is to the immediate right of F and J is to the immediate left of F. H is at 9th hour division. Given E is opposite to K. H is at 90° from C, which is 60° from D. Hence, C is at 6th hour division and D is at 8th hour division. Given L is at 60° from A. G is at 30° from I. Hence, the possible arrangements are as follows.

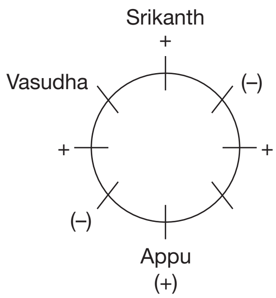


27. 150°

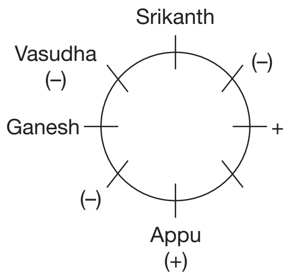
28. L is at 3rd hour division.

29. B or D is opposite to G.

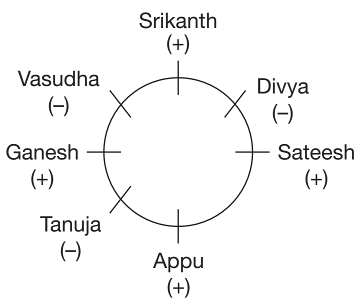
Solutions for questions 30 to 33: It is given that no two females are adjacent to each other. Vasudha (female) is third to the left of Appu (male) and Srikanth is to the immediate left of Vasudha.



Ganesh and Appu are not adjacent to each other. Hence, the arrangement will be as follows.



Appu and Sateesh are adjacent to each other, but Tanuja is not adjacent to Sateesh. Hence, the final arrangement will be as follows.



30. Divya is second to the right of Appu.

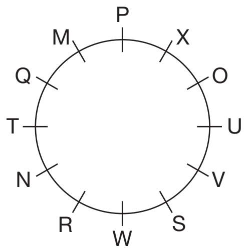
31. Ganesh is sitting adjacent to Vasudha.

32. Vasudha is sitting second to the right of Divya. Similarly, Sateesh is sitting second to the right of Tanuja.

33. Except Vasudha and Ganesh in all other options, the second person is sitting second to the right of first person.

Solutions for questions 34 to 37: From the given information, we can say that P, O, V, W, N and Q are sitting in alternate positions, starting from P in clockwise direction and M, T, R, S, U and X are sitting in alternate positions in anticlockwise direction and M is sitting adjacent to P.

The final circular arrangement is as shown below.



34. S is sitting adjacent to both V and W.

35. X is sitting opposite to R.

36. Six people.

37. V sits third to the left of R.

Solutions for questions 38 to 40: Let us take down the data as below;

(i) Six Russians, namely A, B, C, D, E and F and six French men M, N, O, P, Q and R.

(ii) A ← opp → D

(iii) B ← opp → E

(iv) D ← right → C

(v) B = Russian to French;
C = French to Russian.

(vi) E adj F

(vii) M ← opp → P

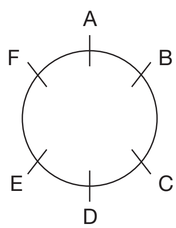
(viii) R ← right → Q

(ix) R ← opp → O

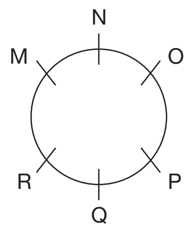
(x) (N and P) adj O

First of all, lets make the seating arrangement for the above data:

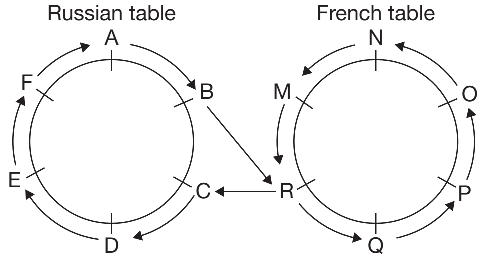
(A) Russian Table:



(B) French Table:



Also, it is given that:



38. C to M is:
(C) → D → E → F → A → B → R → Q → P → O → N → (M)
= Total of 10 people

39. (A) The order of sending message will be as shown below:
A → F → E → D → C → R → M → N → O → P → Q
As the number of people involved is more than 2, hence, this statement is false.

(B) To the right of F is E and to the left of N is O. The route will be as shown:
(E) → D → C → R → M → N → (O)
As there are 5 people between E and O involved in passing the message. Hence, this statement is true.

(C) B sits to the right of C, who now translates French to Russian. Hence, this statement is false.

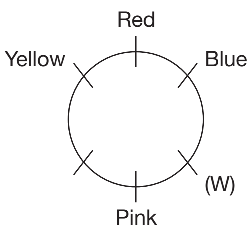
(D) The maximum number of people involved in sending message will remain the same in the new as well as the old arrangement. Hence, (D) is false.

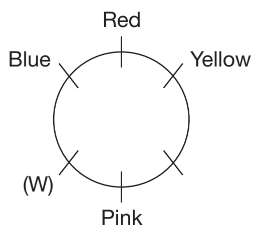
40. A is sitting to the right of B and F is sitting two places to the left of D. The message from A to F can be sent in the following ways:
[A], B, R, Q, P, O, N, M, R, C, D, E, [F]
So, all the 12 people are covered, subtract 2 from 12 (for A and F) and we get 10 people involved in between A and F. Also, in the route shown above, R is repeated twice, still the count is 10.

EXERCISE-2

Solutions for questions 1 to 5: From (A), we incur that red sofa is between blue and yellow sofas.

From (D), we incur that the sofa where W is sitting is adjacent to the blue and pink sofa. The pink sofa should be opposite to the red sofa. The partial arrangement will be as follows:





From (B), X is sitting on the white sofa and Z is opposite to X.

So, Z is either on yellow or blue sofa.

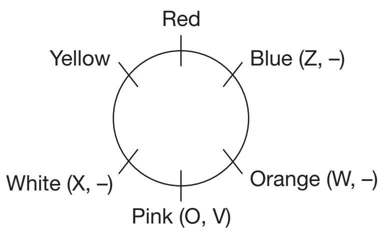
From (B) and (D), we incur that

as X and W cannot be seated on the same sofa and X is on white sofa, W is on orange sofa.

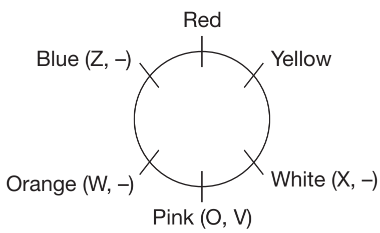
From (C), O and V are in pink sofa.

∴ The possible arrangements are as follows:

Case – (i):

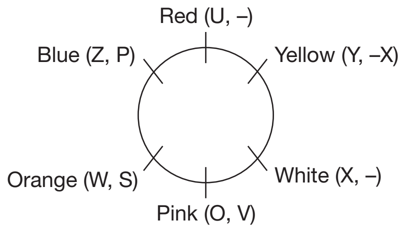


Case – (ii):



From (E), Y is opposite to S.

∴ Y is in yellow sofa and S is in the orange sofa. But in case (A) V is to the left of S, which contradicts (5). Hence, the following arrangement is possible.



1. U is sitting on red sofa.
2. Z and P are sitting on blue sofa.
3. Orange sofa is opposite the sofa where Y is sitting.
4. W and S are sitting on orange sofa.
5. Red, blue, orange, pink, white and yellow is the correct order.

Solutions for questions 6 to 9: The children are Sujatha, Bhuvan, Srilatha, Pranav, Krupa, Kruti and Anand.

From (v), Srilatha is the 4th person to finish her lunch.

From (i) and (iv), Sujatha, Bhuvan and Kruti finished their lunch immediately one after the other in that order. Hence, they can be either 1st, 2nd and 3rd children (or) 5th, 6th and 7th children to finish lunch.

From (iv), Pranav finished his lunch after Kruti. Hence, Sujatha, Bhuvan and Kruti are the first three people to finish their lunch.

From (iii), as Krupa is adjacent to the child who is the fifth to finish his/her lunch. Hence, Krupa cannot be the sixth person to finish her lunch. Hence, Krupa is the 7th child to finish her lunch.

From (ii), Pranav and Anand are the 6th and the 5th to finish their lunch.

The order of children from the child who finished his/her lunch first to that who finished his/her last is Sujatha, Bhuvan, Kruti, Srilatha, Anand, Pranav, Krupa.

The possible seating arrangement is as follows:

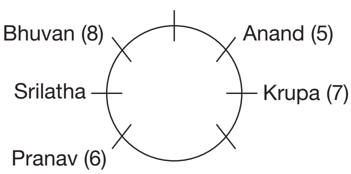
From (ii) and (iii), we incur that

Pranav is three places away to the left of Anand. Krupa is to the immediate left of the child who finished his/her lunch at the 5th place, which is Anand.

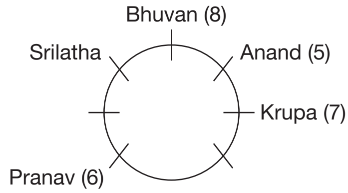
From (i), Bhuvan is to the left of Srilatha and both finished their lunch at the 3rd and the 2nd positions, respectively.

The possible arrangements are as follows:

Case (i):



Case (ii):

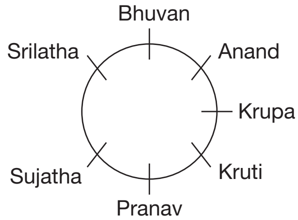


Let us consider case (i).

Here, neither Kruti nor Sujata can be adjacent to Bhuvan, because Kruti and Sujata are the 3rd and the 1st children to finish their lunch.

So, only case (ii) is possible.

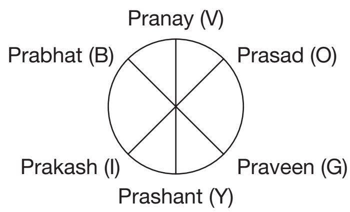
Kruti cannot be adjacent to Srilatha, so Sujata is adjacent to Srilatha.



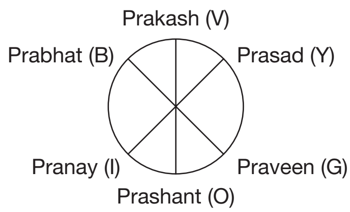
6. Five children finish their lunch before Pranav finishes his lunch.
7. Srilatha finished her lunch after Kruti finished his lunch.
8. Kruti is to the immediate right of Pranav.
9. Anand, Krupa, Kruti, Pranav, Sujata, Srilatha, Bhuvan.

Solutions for questions 10 to 12: By taking the data given in the problem, we get the following four different arrangements. The first letters of the names of the colours are used to denote the colours.

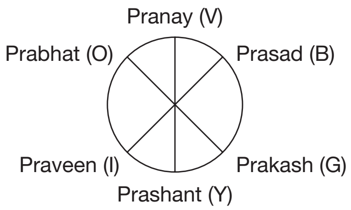
Case 1:



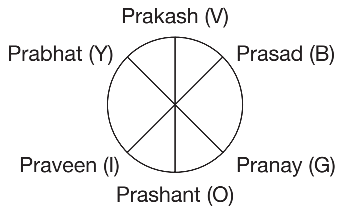
Case 2:



Case 3:



Case 4:



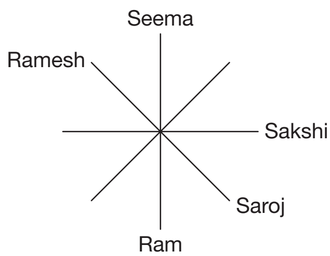
10. All the statements are true.

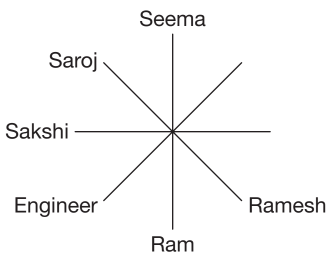
11. This question is referring to Case 3. In this case, it is Prakash who is wearing the Green T-shirt.

12. This question is referring to Cases 2 and 4. In both the cases it is Prashant who is between Praveen and Pranay.

Solutions for questions 13 to 15: From (II) and (V), we know that Seema, the Dancer is opposite to Ram, who is the Lawyer and Sakshi is not adjacent to anyone of these two.

From (IV) Sakshi, who is the Singer is at one place to the right of Saroj. And from (VI), Ramesh is sitting three places to the right of Singer.





From (I), Pilot is sitting opposite to Ramesh, thus Saroj is the Pilot and from (III), Saloni is opposite to the Engineer.

From (III) and above arrangement, Ramesh cannot be any one except Teacher. Saloni is the Doctor, Mohan is the Engineer and Sohan is the Accountant.

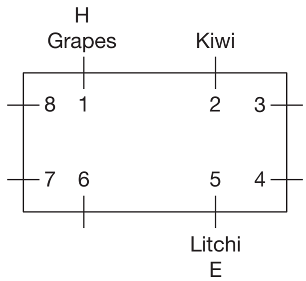
13. Saloni is the Doctor.

14. Mohan is the Engineer.

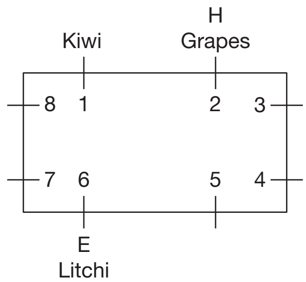
15. Saroj is sitting opposite to Ramesh.

Solutions for questions 16 to 18: From (ii) and (iv), we have two possibilities.

Case (i):



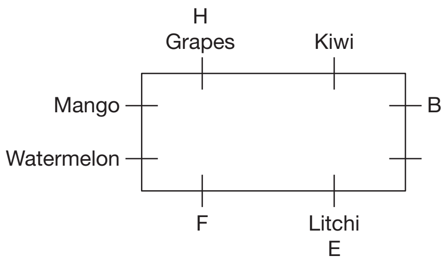
Case (ii):



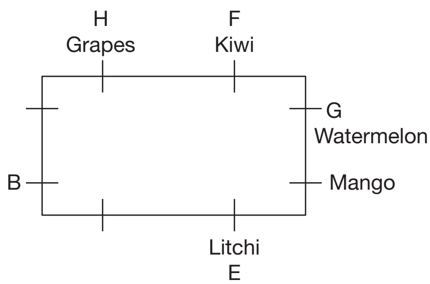
From (i), G has Watermelon and there are three boys between B and G.

From (v), the boys having Mango, Watermelon and F are seated one after the other in anticlockwise direction.

From case (i), the following arrangements are possible.



Or



But in either cases condition (iii) cannot be satisfied.

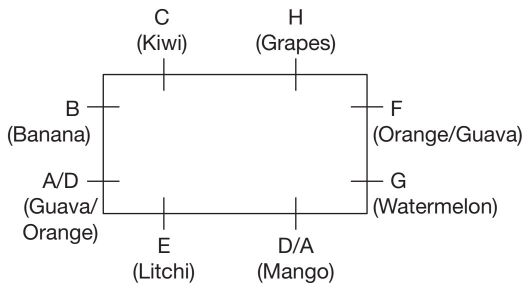
The possible arrangement from Case (ii), we incur the following:

From (iii), F has either Orange or Guava (As G has Watermelon and one boy on that side should have either Orange or Guava).

From (i), B has Banana (As, B is opposite to F).

Boy at position 7 has either Orange or Guava.

∴ The final arrangement is as follows.



16. If A has Orange, then C will be opposite to D.

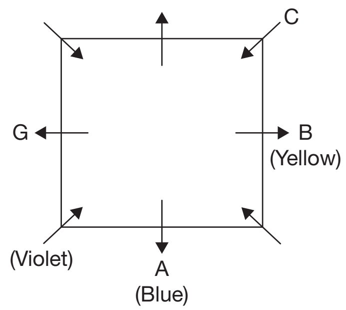
17. E is three places away to the left of F.

18. C – Kiwi is the correct combination.

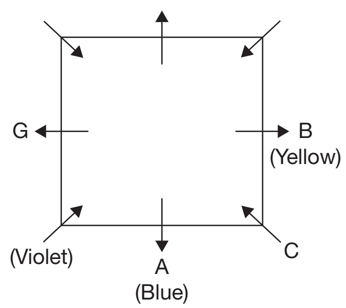
Solutions for questions 19 to 21: From (i) and (ii), G sits to the immediate left of the person who likes violet who sits to the immediate right of the person who likes blue.

From (iii), (iv) and (v), the possible cases are as follows.

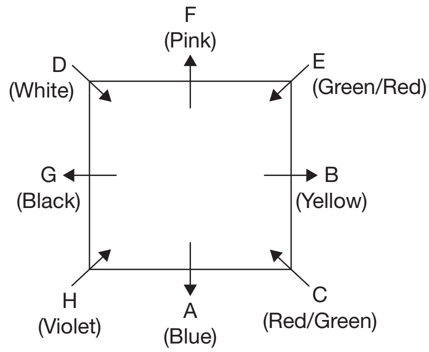
(i)



(ii)



From (vi), (vii), (viii), (ix) and (x), the possible arrangements are as shown below.



19. C or E

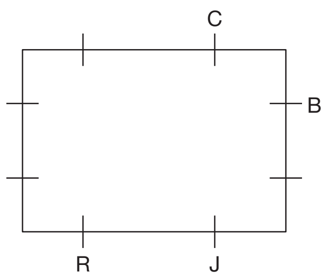
20. C

21. Except (C) remaining all sit at the corners.

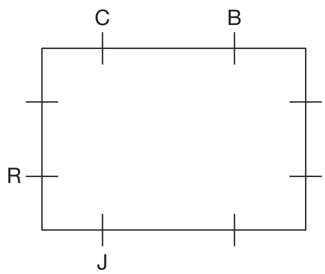
Solutions for questions 22 to 24: Let us represent the people by the first letters of each name.

From (i), we get the following possibilities.

Case (a):

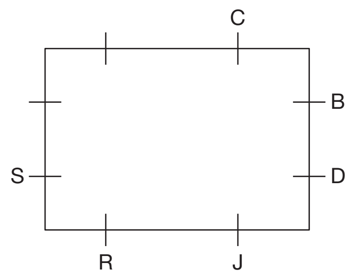


Case (b):

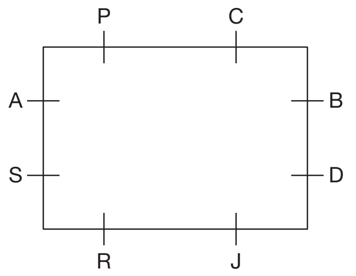


From (ii), as Brijesh and Dweepesh are sitting along the same side, Case (b) is not possible.

From (ii), we get



From (iii), the possibility is as follows.



22. Palak is sitting along the same side as Chandak.

23. Palak is sitting opposite to Rupak.

24. Rupak is sitting to the immediate right of Sayan.

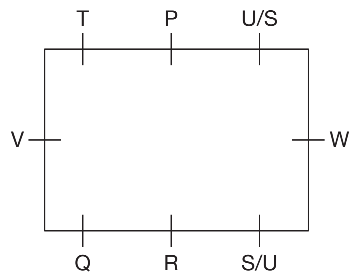
Solutions for questions 25 to 27: From (i), V sits to the immediate left of Q.

From (ii), R sits opposite to P.

From (iii), W sits third to the right of Q.

From (v), W sits third to the left of T.

From (iv), (vi) and above, the possible arrangements is as follows.

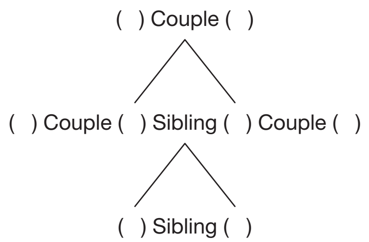


25. W sits opposite to V.

26. S, U and P.

27. Q or U.

Solutions for questions 28 to 32: As there are three married couples and two of them have two children each, the eight-member family is possible only with the following structure.



From (iv), we can say that there are 3 females and 5 males in the family.

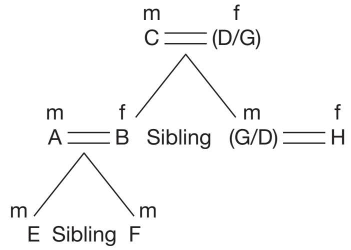
From (iii), we can say that H is from the second generation, as H has a sister-in-law as well as father-in-law. From (viii), we know that H is a female. Thus, one of the siblings in the second generation is male and the other is a female. Now from (ii), A is male and has two sons.

As H is the aunt of E,

A cannot be H's husband as only one of the two couples in the second generation has children.

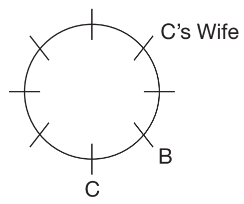
As F is the youngest and he is not G's son, he must be A's son.

Thus, from (ii), (iii), (vi) and (viii), we have the following.



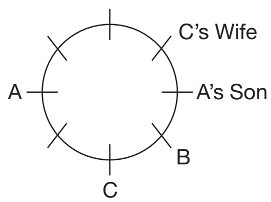
We do not have any information about D and G so let us leave it here.

Now, from (iii) and (vi), we get the following arrangement.

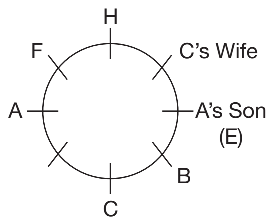


Now, only two seats that are opposite to each other are vacant. Thus, from (ii), A cannot be to the immediate right of B.

Thus, we get the following arrangement.

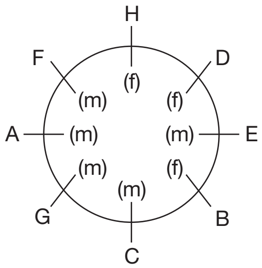


Since F is A's son and F is adjacent to both H. From (vii), (ii) and A, we get the following arrangement.



From (viii), E is three places away from G.

So, we get the following arrangement.



Thus, D is C's wife and G is H's husband.

28. G is C's son.

29. A's brother-in-law is G. G's nephew is E whose grand-mother is D.
D is B's mother.

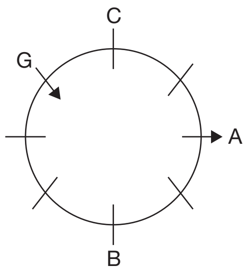
30. E's father is A. C's daughter-in-law is H. A is second to the right of H.

31. E, G and F are males while B is a female.

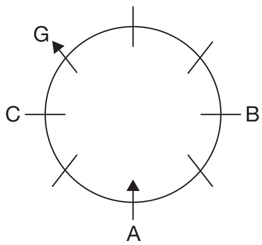
32. F is E's brother.

Solutions for questions 33 to 36: From (i), (ii) and (iii), the possible arrangements are as follows.

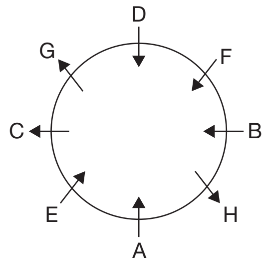
(i)

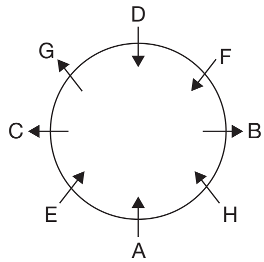


(ii)



From (iv), in case (i), C faces the centre.
In case (ii), C faces away from the centre.
From (v), in both the cases D sits to the immediate right of G.
From (vi) and (vii) and above, the possible arrangement is as follows.





33. 5 people

34. D (From case (i)).

35. A (From case (ii)).

36. H or F

Solutions for questions 37 to 40: From (i), I is sitting at one of the corners and facing away from the centre.

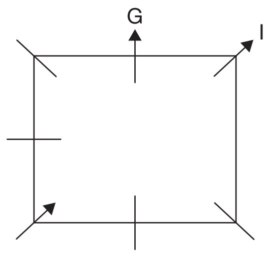
(ii) The neighbour of I faces the same direction as I faces, hence, they face away from the centre.

From (ix), the one who sits opposite to I faces a different direction from which I faces, hence, he/she faces the centre.

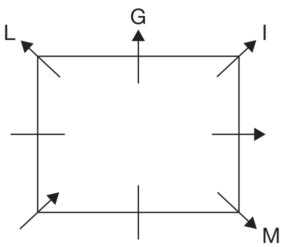
From (iv), L and M face the same direction and sits opposite to each other.

From (vii), 'G' is to the immediate left of I.

From (ii), neither J nor M is a neighbour of I.



Let us assume that L and M face away from the centre and L is to the immediate left of G.



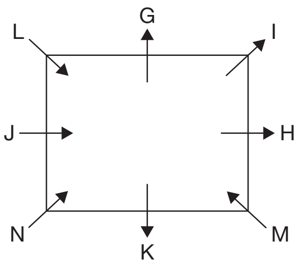
H is to the immediate right of M. If we place 'H' there we cannot place J and K, since from (viii), J and K are neighbours of N.

Hence, L and M must be facing the centre.

From (x), K must be facing away from the centre.

From (viii), J must be facing the centre.

Hence, the final arrangement is as follows:



37. Four people are facing away from the centre.

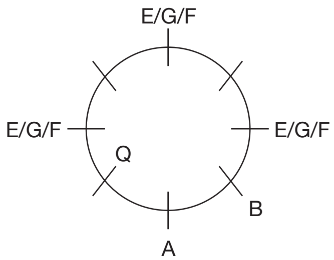
38. Except MN, in all other pairs JH, NI and MK and KN both are facing different directions.

39. M is sitting to the immediate right of H.

40. K is sitting opposite to G.

EXERCISE-3

Solutions for questions 1 to 3: From (1) and (3), we have:



From (2), no two people among P, T, V and W are opposite to each other and no two people from Q, R, S and U are opposite to each other.

From (4), the people from S cannot be opposite to the person from Q.

∴ The person from S must be B.

From (6), C is from W and H is from Q.

Also, C and H are opposite to each other.

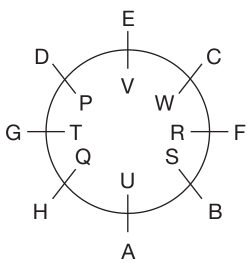
From (5), G must be to the immediate left of H.

∴ E is to the immediate left of D.

From (6) and (2), E must be from V.

F is from R and A is from U.

∴ The final arrangement will be as follows:



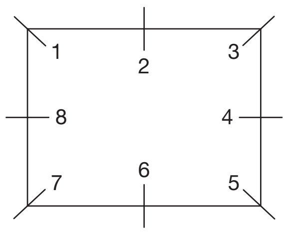
1. F is from R.

2. A is to the immediate right of H.

3. A is opposite to the person from V.

Solutions for questions 4 to 6: It is very clear from the given statements that the room (say table) has eight positions, of which four are at the corners and other four are at the four centres of the four sides of the table, which looks like the arrangement as follows.

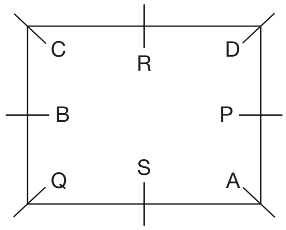
Let us number the chairs 1 to 8.



PQRS, four girls and ABCD, four boys are occupying these eight chairs. Let us analyse all the conditions.

Q is at the corner seat, so it can be anywhere either at 1 or 3 or 5 or 7. Let us say that Q is at seat 7, but as it is given that R is not along the same wall as Q, hence R must be at seat 2 or 3 or 4 whereas A and C are diagonally opposite. So, A and C must be at seat 1 and 5 not necessarily in that order. As B does not sit along any wall adjacent to the corner where A sits, but B is opposite to P. So, if A is at seat 1, then B is at 4 or 5 or 6. If so P must be at 8 or 1 or 2. Finally, as it is given that C is to the immediate right of R who is between C and D, the diagram must be as follows.

Table I



4. S is sitting between Q and A. Refer the table (I).

5. If S and P interchange their positions, then the arrangement will be as follows.

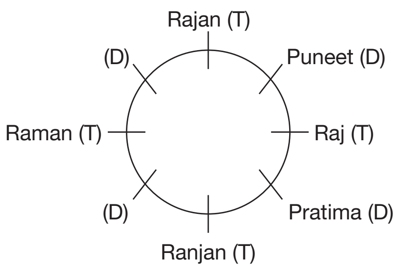
C	R	D
B		S
Q	P	A

S is to the immediate left of D.

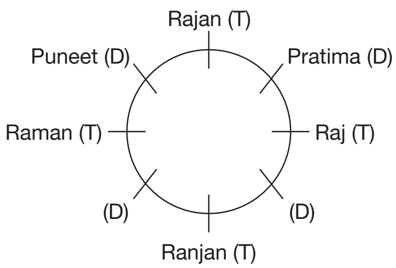
6. The corner seat arrangement clockwise is ⇒ Q, C, D, A (or) C, D, A, Q (OR) D, A, Q, C (OR) A, Q, C, D.
The anti-clockwise arrangement is:
A, D, C Q (OR) D, C, Q, A (OR)
C, Q, A, D (OR) Q, A, D, C.
Choice (A) in Q, A, D, C – Correct.
Choice (B) in A, Q, C, D – Correct.
Choice (C) in D, A, Q, C – Correct.
Choice (D) in D, Q, A, C – Incorrect.

Solutions for questions 7 to 9: The arrangements which can be made based on the data given.

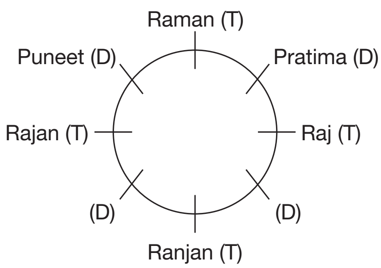
I



II



III



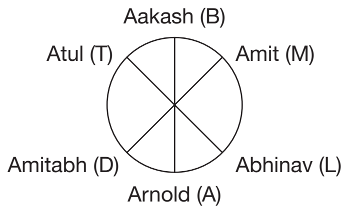
7. Here, cases I and II prevail and it is evident that Raman is two places to the left of Ranjan.

8. If Pratima is adjacent to Raman, then case III prevails and Rajan is opposite to Raj.

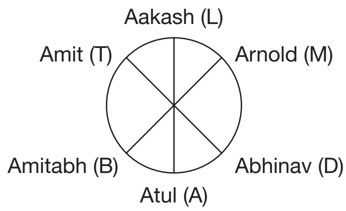
9. If Pratham is not opposite to Puneet, then in any of the two cases Pratham has to be opposite Pratima.

Solutions for questions 10 to 12: We arrive at the following two different arrangements. The first letters of the names of the professions are used to denote the profession.

Case 1:



Case 2:



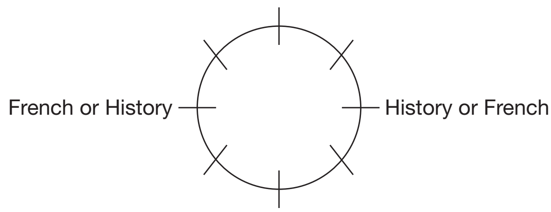
10. In all the cases, it can be observed that Amit is between the Lawyer and the Business Analyst.

11. This question refers to case (2), in which Abhinav is the Doctor.

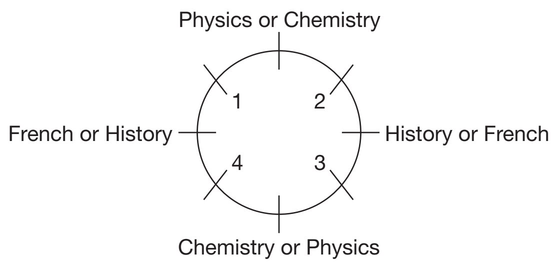
12. This question refers to case (2), where Arnold is sitting opposite Amitabh.

Solutions for questions 13 to 15: Let us analyse all the conditions. A person has eight shelves around him.

French books and History books are on opposite shelves. French, German and English books are in side by side shelves. Physics and Chemistry books are in opposite shelves.

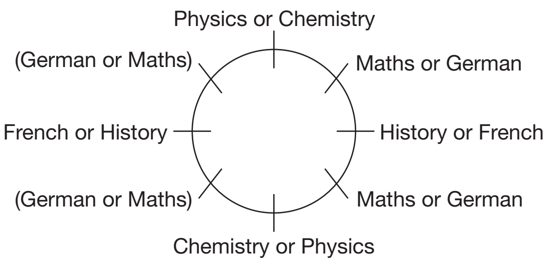


Physics and Chemistry should be on the perpendicular diagonal shelves. The final arrangement is



As French books must have English and German books on either side, these can be 1 and 4 or 2 and 3.

13. If the books on German are opposite to the shelf of Maths books, the arrangement will be as follows.

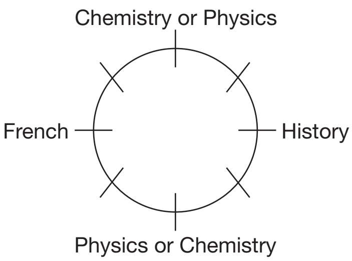


As German is between Physics and French, it can be at any of the two places as shown.

Then, Mathematics books may be between History and Chemistry shelves.

Then, English and Sociology shelves would be opposite to each other.

14. The shelf of Sociology is between the shelves with Physics and History books. Then the arrangement is

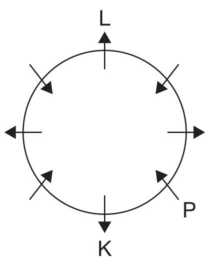


Hence, the books on Mathematics should be between books on History and Chemistry.

15. English books are to the immediate left of Physics books, so that the shelf must be between the shelves with French and Physics books. So, German books would be between the shelves of Chemistry and English. So, they are to the immediate right of Chemistry or French books.

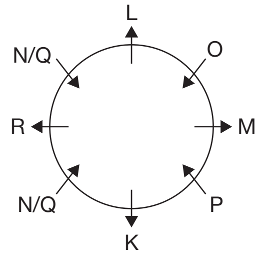
Solutions for questions 16 to 18: From (vi), L sits opposite to K and faces away from the centre. From (iv), no two people sitting next to each other face the same direction and from (i), P sits to the immediate left of K.

From the above points we get the following arrangement.



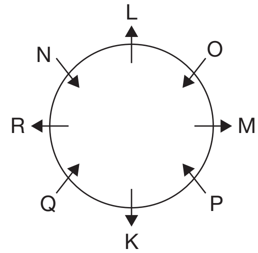
And from (ii), M and R are sitting opposite to each other. From (v), R is the neighbour of both N and Q.

Hence, the arrangement is



From (iii), either Q or O sits next to L.

Hence, the final arrangement is



16. Among KM, ML, and NO, the second person sits second to the left of the first person.

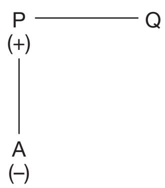
In PO, the second person is second to the right of the first person.

17. L sits to the immediate right of 'O' and is definitely true.

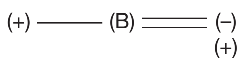
18. If M and N interchange their places, then L sits to the immediate left of M.

Solutions for questions 19 to 21: It is given that:

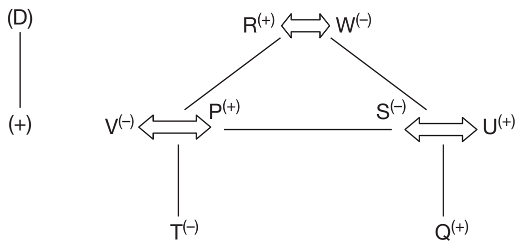
(i)



(ii)



(iii)

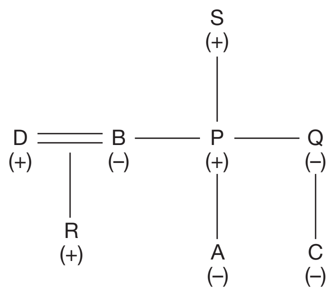


Given, P is the brother of Q, who is not adjacent to either P or S. Hence, Q sits either at the left end or right end and R sits adjacent to Q. Also given, A is the daughter of P and sits to the immediate right of B's husband. C is the niece of P. Hence, D is the husband of B and as D is not opposite to either P or S, we get the following cases.

(i)	P/S	S/P	R	Q
	—	—	D	A
(ii)	Q	R	S/R	P/S
	D	A	—	—
(iii)	Q	R	S/P	P/S
	—	D	A	—

Given, C does not sit opposite either P or S, hence, case (i) and (ii) can be eliminated. In case (iii), C sits at the left end, B sits at the right end. Given, B is not opposite to S, but is opposite to her brother. Hence, B is opposite to P, who is the brother of B. As, D has only one child who is a male, it is either R or S and C is the daughter of Q but, given Q is the daughter of C's grandfather who is not R. Hence, R should be the son of D and S is the grandfather of C. Therefore, the final arrangement and the family tree are as follows.

Q	R	S	P	↓
C	D	A	B	↑



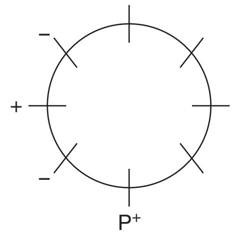
19. D's father-in-law (S) is to the immediate right of P.

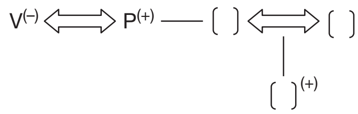
20. Q's daughter sits at an end.

21. Choice (B) is true.

Solutions for questions 22 to 25: It is given that, P sits second to the right of his nephew whose neighbours are females.

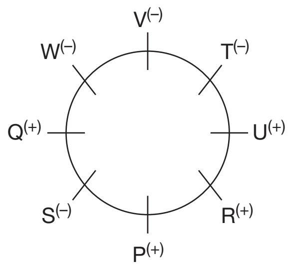
V is the wife of P and sits to the immediate right of her daughter T.





U sits second to the right of his brother-in-law and opposite his son Q. Hence, Q must be the nephew of P who is the brother-in-law of U. From (2), we come to know that V sits to the immediate right of her daughter T. We can place T only to the immediate right of U. From (4), we can say that S is the daughter of R and wife of U. From (5), we can say that W is the wife of R, and P and S are their children. Finally, the positions are, W sits to the immediate right of V, R sits to the immediate left of U, and S sits to the immediate right of P.

The final arrangement of the given family members is as follows.



22. Both the options (A) and (B) say about the wife of U.

23. All the options (A), (B) and (C) are saying about the niece of S.

24. None of the given options is true.

25. Except option (C), in all other options the opposite genders are given.

CHAPTER 3 Distributions

Learning Objectives

In this chapter, you will:

- Learn how to interpret the statements given and fill a table.
- Learn how to arrive at the final distribution of parameters among people/objects from the partially filled table.
- Apply knowledge of distributions to linear and circular arrangements.
- Understand and learn to solve puzzles that involve multiple parameters.

In this type of problems, you have to match two or more variables [Variable means a 'subject' as used in the discussion of linear arrangement]. In double line-up, the data given may talk of four people living in four houses each of a different colour. What we need to find out is the colour of the house of each of the four persons. There is no first position or second position of the houses.

Sometimes, double line-up is also known as distribution.

An example of data given for this variety of questions is:

Each of the four people A, B, C and D wears a different coloured shirt, such as red, pink, blue and white. A has a red shirt and D does not have a pink shirt. From the above statement, it becomes clear that no person among A, B, C and D can have shirts of two different colours among red, pink, blue and white.

As discussed in the questions on single line-up, questions can be solved easily by representing the given data pictorially. In case of double line-up, it will help us if we represent the data in the form of a matrix or a table.

Let us see how to draw a matrix for the data given above.

	Colours			
Names	**Red**	**Pink**	**Blue**	**White**
A	✓			
B				
C				
D		×		

As it is given that A has red colour shirt, it is clear that he does not have any other colour shirt. Similarly, B, C, D do not have red colour shirt. So, in all the other cells in the row belonging to A, we put a cross ('×'). Then, the table will look as follows:

	Colours			
Names	**Red**	**Pink**	**Blue**	**White**
A	✓	×	×	×
B	×			
C	×			
D	×	×		

In this manner, we can fill up the cells on the basis of the data given to us. Once, we use up all the data, we will draw any conclusions that can be drawn and then answer the questions given in the set. You will understand this better by going through the solved example section below, please try to solve them first without looking at the solution.

SOLVED EXAMPLES

Directions for questions 3.01 to 3.05: These questions are based on the following information.

P, Q, R, S, T, U, V and W are eight employees of a concern. Each of them is allotted a different locker, out of eight lockers numbered from 1 to 8 in a cupboard. The lockers are arranged in four rows with two lockers in each row.

Lockers 1 and 2 are in the top row from left to right, respectively while lockers 7 and 8 are in the bottom row arranged from left to right, respectively. Lockers 3 and 4 are in the second row from the top, arranged from right to left, respectively. So are lockers 5 and 6 arranged from right to left, respectively in the second row from the bottom. P has been allotted locker 1 while V has been allotted locker 8. T's locker is just above that of Q which is just above that of R, whereas W's locker is in the bottom row.

3.01: Which of the following cannot be the correct locker number–occupant pair?
(A) 3-Q (B) 7-W
(C) 4-U (D) 6-R

3.02: If U's locker is not beside Q's locker, whose locker is just above that of W?
(A) U (B) S
(C) R (D) Q

3.03: Which of these pairs cannot have lockers that are diagonally placed?
(A) P-Q (B) S-R
(C) U-R (D) Either (B) or (C)

3.04: Which of the following groups consists only occupants of odd numbered lockers?
(A) Q, R, W (B) R, V, W
(C) T, R, Q (D) P, T, Q

3.05: If U's locker is in the same row as that of R, and S exchanges his locker with V, then who is the new neighbour of V in the same row? (Assume that nothing else is disturbed from the original arrangement)
(A) P (B) Q
(C) R (D) U

Solutions for questions 3.01 to 3.05: Let us first try to locate the lockers in the cupboard as per the conditions given. Then, we will do the allotment to the people.

Lockers 1 and 2 are in the top row and lockers 7 and 8 are in the bottom-most row. In these two rows, the lockers are numbered from left to right. In the other two rows, the lockers are numbered from right to left.

L	R	
1	2	Top Row
4	3	
6	5	
7	8	Bottom Row

Now let us look at the conditions given for the allotment of the lockers.

P has locker 1. V has locker 8.

1-P	2
4	3
6	5
7	8-V

Locker of W is in the bottom row ➔ W's locker must be 7.

1-P	2
4	3
6	5
7-W	8-V

T's locker is just above that of Q, which is just above that of R ➔ The lockers of T, Q and R must be 2, 3 and 5, respectively (there are no other group of lockers which satisfy this condition).

1-P	2-T
4	3-Q
6	5-R
7-W	8-V

S and U have lockers 4 and 6 left for them.

Thus, on the basis of the data given to us, we can show the final arrangement of lockers as below:

1-P	2-T
4-S/U	3-Q
6-U/S	5-R
7-W	8-V

Now we can answer the questions easily on the basis of the above table.

3.01: By looking at the final arrangement of lockers above, we find that choice (D) does not represent the correct combination of locker number-occupant pair.

3.02: If U's locker is not beside Q's locker, then U's locker must be locker 6. So, it is U's locker that will be immediately above W's.

3.03: R's locker is in the same row as that of exactly one of S or U and diagonally placed to the other one. Hence, 'either S-R or U-R' is the answer.

3.04: The odd-numbered lockers 1, 3, 5 and 7 belong to P, Q, R and W, respectively. Of the choices, we find that Q, R, W appear in choice (A). Hence, this is the correct choice.

3.05: U's locker is in the same row as that of R which means that locker 6 belongs to U. So, locker 4 belongs to S. Now V and S exchange lockers. Then the new neighbour of V is Q.

Directions for questions 3.06 to 3.09: These questions are based on the following information.

There are four variety of trees, such as Lemon, Coconut, Mango and Neem each at a different corner of a rectangular plot. A Well is located at one corner and a Cabin at another corner. Lemon and Coconut trees are on either side of the Gate which is located at the centre of the side opposite to the side at whose extremes, the Well and the Cabin are located. The Mango tree is not at the corner where the Cabin is located.

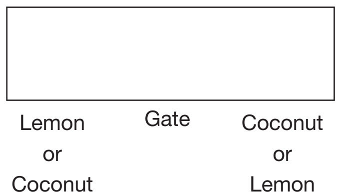
3.06: Which of the following pairs can be diagonally opposite to each other in the plot?
(A) Neem tree and Lemon tree
(B) Cabin and Neem tree
(C) Mango tree and Well
(D) Coconut tree and Lemon tree

3.07: If the Lemon Tree is diagonally opposite to the Well, then the Coconut tree is diagonally opposite to the
(A) Mango tree (B) Well
(C) Cabin (D) Gate

3.08: If the Coconut tree and the Neem tree cannot be at adjacent corners of the plot, then which of the following will necessarily have to be at diagonally opposite corners of the plot?
(A) Coconut tree and Well
(B) Lemon tree and Cabin
(C) Lemon tree and Coconut tree
(D) Lemon tree and Well

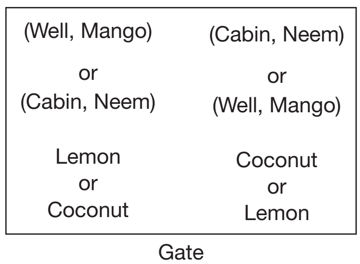
3.09: Which of the following is definitely false?
(A) Mango tree is adjacent to the Well at one corner.
(B) Neem tree is adjacent to the Cabin at one corner.
(C) Coconut tree is at the corner adjacent to the Well.
(D) Lemon tree is not on the same side of the plot as the Gate.

Solutions for questions 3.06 to 3.09: Lemon and Coconut are on either side of the gate.



The Well and the Cabin are at either end of the Well opposite to the Gate.

Mango tree and Cabin are not at the same corner. So, Neem tree and Well are not at the same corner. This means that Mango tree and the Well are at the same corner and Neem tree and the Cabin are at the same corner.



3.06: Let us take each choice and check with the above diagram to see if it is possible or not. Neem and Lemon can be diagonally opposite to each other. Hence, this is the correct answer choice. (In an exam, you do not need to check the other choices since the first choice is correct. But, for the sake of clarity and proper understanding, we will check all the choices).

From the diagram given above, we can see that Cabin and Neem cannot be located diagonally opposite to each other.
Mango and Well cannot be located diagonally opposite to each other.
Coconut and Lemon cannot be located diagonally opposite to each other.

3.07: If Lemon tree is diagonally opposite to the Well, then we can have the following two possible arrangements.

Neem	Well
Cabin	Mango

OR

Well	Cabin
Mango	Neem

The Coconut tree is diagonally opposite to the Cabin and Neem.

3.08: Since Coconut and Neem trees cannot be at adjacent corners, the following arrangements are possible.

(Well, Mango)	(Cabin, Neem)
Coconut	Lemon

OR

(Cabin, Neem)	(Well, Mango)

From the above diagrams, we find that choice (D) is the correct answer.

3.09: We check each statement with the diagram that we drew initially to find out which of the statements has to be false.
We find that choice (D) has to be false.

Directions for question 3.10: Select the correct answer from the given choices.

3.10: A, B, C and D play four different games among Baseball, Cricket, Kabaddi and Volleyball. A does not play Baseball or Cricket. B does not play Kabaddi or Volleyball. C plays Volleyball and D plays either Baseball or Volleyball. Who plays Cricket?

(A) A (B) B
(C) C (D) D

Solution for question 3.10:

3.10: C plays Volleyball. A does not play Cricket and D does not play Cricket as he plays either Baseball or Volleyball.
∴ B should play Cricket.

EXERCISE-1

Directions for questions 1 to 9: Select the correct alternative from the given choices.

1. Each of the five people, namely Bhanu, Lalit, Modi, Ravindra and Kamal is wearing a different coloured shirt among white, black, brown, indigo and yellow. Bhanu is wearing neither a yellow nor a black coloured shirt. Lalit is wearing neither an indigo nor a white coloured shirt. Modi is wearing neither indigo nor a yellow coloured shirt. If Ravindra and Kamal are wearing black and indigo coloured shirts, then which coloured shirt is Lalit wearing?

(A) Brown (B) Yellow
(C) White (D) Cannot be determined

2. Rama, Bhima and Bond have to take two articles each from the available six articles, such as Gun, Bow, Arrow, Mace, Sword and Dagger. Bhima will not take the Gun or the Bow, and Bond will not take any of the Sword, the Bow or the Dagger.
If one of the three people takes the Sword and the Mace, then what is the other item selected by the person who selects the Gun?

(A) Bow (B) Arrow
(C) Dagger (D) Cannot be determined

3. A group of four students, namely Sagar, Swaroop, Sachin and Suman went to four different cities, such as Mumbai, Vijayawada, Hyderabad and Nagpur to take 4 different tests XAT, JMET, CET, CAT. The following data is also known.

1. Suman did not take JMET.
2. Swaroop did not go to Hyderabad and he did not take CET and XAT.
3. CET was conducted in Nagpur.
4. Sagar did not go to Nagpur and he did not take CAT.
5. Sachin had gone to Mumbai.

If Sagar did not take XAT, then which of the following is the correct combination of the city visited and the test written by Swaroop?

(A) Vijayawada and JMET
(B) Nagpur and JMET
(C) Hyderabad and CAT
(D) Vijayawada and CAT

4. Five trains, such as GT Express, AP Express, Rajdhani Express, Goa Express and Bangalore Express travel to five different cities, like Delhi, Goa, Chennai, Bangalore and Hyderabad, not necessarily in the same order. No two trains travel to the same city and no two cities can be visited through the same train. GT Express travels to neither Delhi nor Chennai. Neither AP Express nor Goa Express pass through Hyderabad Bangalore can be visited by either Goa Express or AP Express. Rajdhani Express travels to Chennai. Either Goa Express goes to Hyderabad or the Bangalore Express goes to Delhi.
Which one of the following statements would help in completing the arrangement?

(A) Either Chennai is visited through Rajdhani Express or Goa is visited through AP Express.
(B) Only if Bangalore is visited through Bangalore Express, is Goa then visited through Goa Express.
(C) If Goa Express goes to Bangalore, then AP Express goes to Goa.
(D) If Bangalore is visited through AP Express, then Delhi is visited through the Bangalore Express.

5. Each of five men, such as A, B, C, D and E is married to a different female among P, Q, R, S and T, not necessarily in the given order. S is the wife of B and D is not the husband of P, who is not the wife of A. E is the husband of T. Who is the wife of C?

(A) D (B) P
(C) R (D) Cannot be determined

6. Each of the four people, namely Ramesh, Rajesh, Ramani and Ravan work for four different companies among TCS, CTS, Wipro and Accenture. Each of them belongs to a different city among Delhi, Kolkata, Mumbai and Chennai. The following information is known about them.

(1) Rajesh is from Kolkata but does not work in Wipro and the person who works in Wipro is not from Chennai.
(2) Ravan works neither in Wipro nor in Mumbai.
(3) Ramani works neither in Mumbai nor in Accenture.
(4) Ramesh works neither in TCS nor in Wipro and the person from Kolkata works neither in TCS nor in CTS.

Who is working in CTS?

(A) Ramesh (B) Rajesh
(C) Ramani (D) Ravan

7. Shiva brought 4 boxes, each of a different colour. Each of these boxes contains chocolates of a different brand. He distributed one box to each of his four friends. Manju received a red coloured box but not Eclairs. Either blue or orange coloured box is received by Sanju. Tanooj received Kit Kat. One of them received Bar One while Pooja received neither yellow coloured box nor Dairy Milk. If the orange coloured box contains Diary Milk, then which of the following is true?

(A) Sanju received Bar One.
(B) Bar One is in red coloured box.
(C) Eclairs is in yellow coloured box.
(D) Tanooj received blue coloured box.

8. The teacher in-charge of a class summarized the analysis of number of students who took tests in three different subjects, such asMaths, Physics and Chemistry. No student has appeared for more than one test. Half of the students who took the test in Chemistry passed it. The number of students who passed the Physics test is equal to the number of students who passed the Maths test. The number of students who failed the Maths test and the number of students who failed the Chemistry test are equal. 30% of the students who took the Physics test failed it. 100 students failed the Maths test. If the total number of students who took the tests is 470, then which of the following is true?

(A) Among the students who took tests in different subjects, the number of students who took the test in Physics is the minimum.
(B) The number of students who failed the Maths test is not the least.
(C) The number of students who passed the Maths test is more than those who passed the Chemistry test.
(D) More than one of the above.

9. In a campus recruitment each of the four friends studying in VIT College of Engineering were recruited for a different city and each of them is from any one of the two departments Electronics and Civil. Tinku is recruited for Kolkata. One of the Electronics students is recruited in Bhopal. Pinku belongs to Electronics but is not recruited for Hyderabad. Chinku is not recruited for Chennai and is not from Electronics department. Minku and Tinku belong to the same department. If Pinku is not recruited for Bhopal, then which of the following is true?

(A) Pinku is recruited for Chennai.
(B) Chinku is recruited for Hyderabad.
(C) Minku is recruited for Bhopal.
(D) All the above

Directions for questions 10 and 11: These questions are based on the following information.

A group of three people Karan, Manohar and Jamal own one of the cars from Zen, Alto and Indica. Each of these cars are parked in different parking spaces P1, P2 and P3. It is known that Alto is parked in P2 and it does not belong to Manohar. Jamal owns Indica and Zen is not parked in P1.

10. Which car does Karan owns?

(A) Zen (B) Alto
(C) Indica (D) Either Zen or Indica

11. In which parking space did Jamal park his car?

(A) P1 (B) P2
(C) P3 (D) Either P1 or P3

Directions for questions 12 and 13: These questions are based on the following information.

Each of the four people Ramesh, Rajesh, Ramani and Ravan work for four different companies among TCS, CTS, Wipro and Accenture. Each of them belongs to a different city among Delhi, Kolkata, Mumbai and Chennai. The following information is known about them.

(1) Rajesh is from Kolkata but does not work in Wipro and the person who works in Wipro is not from Chennai.
(2) Ravan works neither in Wipro nor in Mumbai.
(3) Ramani works neither in Mumbai nor in Accenture.
(4) Ramesh works neither in TCS nor in Wipro and the person from Kolkata works neither in TCS nor in CTS.

12. Who is from Chennai?

(A) Ramesh (B) Rajesh
(C) Ramani (D) Ravan

13. Who is working in CTS?

(A) Ramesh (B) Rajesh
(C) Ramani (D) Ravan

Directions for questions 14 to 16: These questions are based on the following information.

Ten monkeys from A through J visit a garden which has tree bearing fruits, such as Mango, Guava, Banana and Berry.

Further it is known that:

(i) Only one monkey visited all the trees.
(ii) Every tree is visited by six monkeys.
(iii) A, D, E, G, I, J visited the Guava tree.
(iv) D, F, G did not visit the Berry tree.
(v) B, C, D, G, I, J visited the Banana tree.
(vi) J, A, H did not visit the Mango tree.
(vii) Every monkey visited at least one tree and exactly two monkeys visited one tree only.

14. Which of the following monkeys visited all the trees?

(A) D (B) G
(C) I (D) J

15. How many monkeys visited exactly three trees?

(A) 3 (B) 4
(C) 5 (D) 2

16. Which of the following monkeys have visited the Berry tree?

(A) B (B) C
(C) A (D) F

Directions for questions 17 to 19: These questions are based on the following information.

Six people P, Q, R, S, T and U carry an umbrella and a bag of six different colours while going to school. The colours of each of the umbrellas and each of the bags are one among red, yellow, green, blue, pink and black. None among them carries an umbrella and a bag of the same colour. Further the following information is known.

(i) S carries a blue coloured umbrella but not a black coloured bag.

Difficulty Level-1: Foundation

(ii) The person who carries a pink coloured umbrella carries a green coloured bag.
(iii) P carries a red coloured bag but not a yellow coloured umbrella.
(iv) R carries a black coloured umbrella and T carries a yellow coloured bag.
(v) Q does not carry a black coloured bag.

17. Who carries a red coloured umbrella?
(A) T (B) U
(C) Q (D) Data inadequate

18. Which colour bag is carried by the person who carries a yellow coloured umbrella?
(A) Blue (B) Black
(C) Pink (D) Either (A) or (C)

19. Which of the following is the correct combination of the person, colour of umbrella and bag he/she carries, respectively?
(A) Q – Red – Blue (B) Q – Red – Black
(C) U – Red – Black (D) T – Red – Yellow

Directions for questions 20 to 22: These questions are based on the following information.

Eight boys from A through H gathered at a picnic. Each of them brought a different dish among P through W to the picnic. The following information is known about them.
(1) Neither A nor D brought S. Either B or E brought R.
(2) Either C or D brought P.
(3) Either G or F brought U.
(4) C brought neither S nor V. E brought W.
(5) Either H or D brought Q.
(6) Either A or F brought T. Neither G nor H brought S.

20. Which dish is brought by A?
(A) T (B) U
(C) W (D) V

21. Which of the following is a correct combination of the boy and the dish he brought?
(A) C – T (B) B – S
(C) C – Q (D) F – S

22. Who brought the dish V?
(A) A (B) B
(C) D (D) Cannot be determined

Directions for questions 23 to 26: These questions are based on the following information.

Each of the five people, namely Suman, Tarun, Uday, Yadav and Gopal is the owner of a field in a different city among A through E. Each of them planted different kinds of plants, such as guava, mango, apple, banana, and water-melon.

The following is known about them.
(1) Mango plants are not planted in the city A. Uday planted apple.
(2) One among banana and guava plants is planted by one among Tarun and Gopal in city E.
(3) Tarun planted watermelon and Suman does not own a field in the city C.
(4) Neither apple nor mango plants are planted either in city C or city D.
(5) Tarun does not have a field in D.

23. If Yadav planted guava, then who owns a field in city B?
(A) Uday (B) Yadav
(C) Suman (D) Tarun

24. Who owns a field in the city A?
(A) Uday (B) Yadav
(C) Tarun (D) Suman

25. Who owns a field in the city C?
(A) Yadav (B) Bhopal
(C) Tarun (D) Uday

26. If Suman planted mango plants, then which among the following is planted in the city D?
(A) Banana (B) Guava
(C) Water melon (D) Cannot be determined

Directions for questions 27 to 29: These questions are based on the following information.

A group of five friends Dweep, Manyata, Jagat, Poulami and Hemant, has at least one of the following items, such as pen, pencil, bag, ruler, calculator and eraser. Hemant has pen, pencil and calculator only. Jagat has calculator and pen only. Dweep has eraser and bag only. Poulami has eraser, bag and pencil only. Manyata has only one item. Each item is with at least one person.

27. Who has the ruler?
(A) Hemant (B) Poulami
(C) Manyata (D) Data inadequate

28. Which of the following gives the complete list of the people who have pencils?
(A) Hemant
(B) Poulami
(C) Hemant and Poulami
(D) Hemant, Dweep and Poulami

29. Which of the items is there with more than two friends?
(A) Pencil (B) Calculator
(C) Bag (D) None of these

Directions for questions 30 to 33: These questions are based on the following information.

A team of six professors, namely–Govind, Manoj, Prasad, Aravind, Bharath and Raman are scheduled to train newly appointed faculty members.

Each of the professors train the faculty members in a different subject, such as from Arithmetic, Logical Reasoning, Pure Maths, English, Currents Affairs and Communication

Skills on a different day among Monday, Tuesday, Wednesday, Thursday, Friday and Saturday of a week.

The following information is available about the schedule.

(1) Training in Pure Maths is scheduled on Tuesday but it is not by Aravind.
(2) Govind's session is scheduled on Wednesday but not in Logical Reasoning.
(3) The session on Current Affairs and Communication Skills are scheduled on two consecutive days.
(4) Aravind's session is scheduled on the day immediately after the day on which Manoj's session is scheduled.
(5) Prasad's session is on English but it is scheduled neither on Monday nor on Saturday.

30. Whose session is scheduled on Friday?
(A) Bharath (B) Raman
(C) Aravind (D) Manoj

31. Which subject is scheduled on Monday?
(A) Logical Reasoning
(B) Pure Maths
(C) Communication Skills
(D) English

32. If Aravind's session is on Current Affairs, then on which day of the week is the session on Communication Skills scheduled?
(A) Monday (B) Wednesday
(C) Thursday (D) Friday

33. On which day of the week is Prasad's session scheduled?
(A) Tuesday (B) Friday
(C) Thursday (D) Wednesday

Directions for questions 34 to 37: These questions are based on the following information.

In a garden there are seven different flower pots A, B, C, D, E, F and G, each of which is having a different flower among Buttercup, Carnation, Columbine, Crocus, Clover, Dahlia and Foxglove.

Each of the seven butterflies, such as P, Q, R, S, T, U and V feed on a different flower among the given but not necessarily in the same order.

(i) P feeds on the flower, which is in E.
(ii) Dahlia is not in pot F. S feeds on Crocus.
(iii) Columbine is in A, but neither U nor S feeds on that.
(iv) V and Q feed on Clover and Dahlia, which are in B and F.
(v) P and T feed on Carnation and Foxglove, but neither feeds on the flowers which is either in pot C or D.
(vi) If T feeds on Columbine, then Columbine is not in A.
(vii) E contains either Foxglove or Carnation.

34. Which butterfly feeds on Buttercup?
(A) P (B) R
(C) U (D) Cannot be determined

35. Foxglove is in which flower pot?
(A) B (B) E
(C) G (D) Cannot be determined

36. If Crocus is in C, then which flower is in D?
(A) Clover (B) Dahlia
(C) Foxglove (D) Buttercup

37. If V feeds on Dahlia, then Q feeds on
(A) Buttercup (B) Foxglove
(C) Clover (D) None of these

Directions for questions 38 to 40: These questions are based on the following information.

Twelve disciples Aman, Arjun, Arhan, Amith Akhil, Ajay, Bhuvan, Bharath, Balu, Bharani, Dharani and Danush were sent to different countries across the world to India, US, Bangladesh, Nepal, Bhutan and UK to spread the teachings of their Guru.

Exactly two disciples were sent to each country.

(i) Bharath and Dharani were sent to the same country, which is neither Bhutan nor the US.
(ii) Aman and Ajay were sent to a different country among UK and Bangladesh. Bhuvan was neither sent to Bhutan nor was he sent along with Ajay.
(iii) Aman and Balu were sent to the same country. Danush and Ajay were sent to different countries. Danush was not sent to the US.
(iv) One among Bharani, Balu and Akhil was sent to the US. The remaining two disciples were sent to different countries.
(v) The disciples having the same first letter in their names and also the same ending letters in their names were sent to Nepal.
(vi) The disciples who were sent to the same country, except Nepal, do not have the same starting letter of their names.

38. Which group of disciples were sent to India?
(A) Akhil, Bhuvan (B) Ajay, Bharani
(C) Bhuvan, Amith (D) Bharath, Dharani

39. If Aman was sent to Bangladesh, then Bharani was sent to
(A) India (B) UK
(C) US (D) None of these

40. Amith was sent to which country?
(A) India (B) Bangladesh
(C) Bhutan (D) UK

EXERCISE-2

Directions for questions 1 to 3: These questions are based on the following information.

A group of six friends A, B, C, D, E and F hailing from six different professions, such as engineer, doctor, professor, architect, lawyer and painter and they belongs to six different cities Kolkata, Bangalore, Hyderabad, Mumbai, Chennai and Delhi, may not be in the same order.

(i) The person from Bangalore is a doctor, who is not B.
(ii) A is an architect and C who is from Chennai, is an engineer.
(iii) The person from Delhi is a professor.
(iv) D is neither a professor nor the person from Bangalore is.
(v) E is from Mumbai and the person from Kolkata is neither an architect nor a lawyer.

1. What is the profession of F?
(A) Painter (B) Lawyer
(C) Professor (D) Doctor

2. What is the profession of the person from Kolkata?
(A) Architect (B) Doctor
(C) Painter (D) Lawyer

3. What is the profession of E?
(A) Professor (B) Lawyer
(C) Doctor (D) Painter

Directions for questions 4 to 7: These questions are based on the following information.

A group of five publishers, namely Prinston, Johnson, Holy Faith, Reprographics and Penguin published a book for competitive examinations. Each book contains three subjects, like Geography, Science, History, Polity and Mental ability. These books are arranged one over the other, three of these are second editions and two of these are first editions. The book published by Reprographic is the first edition and contains Geography. Only one book on science is the first edition. The three books containing science are stacked one over the other and both the first editions are stacked one over the other. All the second editions contain History. Neither of the first editions contains History. The book published by Holy Faith publishers contains Polity and Mental ability. The book published by Prinston publishers is at the top of the stack and the book published by Penguin publishers contains Mental ability and Geography. There is no book which contains both Science and Geography.

4. Which of the following is true?
(A) The 2nd book from the top is the 1st edition.
(B) The 3rd book from the top is the 2nd edition.
(C) The 2nd book from the bottom is the 1st edition.
(D) The bottommost book is the 1st edition.

5. Which of the following is definitely a correct combination of publishers of a book and its related subject?
(A) Penguin – History
(B) Holy Faith – History
(C) Reprographic – Science
(D) Johnson – Polity

6. Books of which publishers are on Polity?
(A) Prinston and Holy Faith
(B) Prinston and Reprographic
(C) Holy Faith and Reprographic
(D) Either (A) or (B)

7. Which among the following subjects can be contained in the least number of books?
(A) History (B) Science
(C) Polity (D) Geography

Directions for questions 8 to 10: These questions are based on the following information.

In a horse racing each of the six Jockeys, such as H, I, J, K, L and M is riding a different horse among Honey, Nunny, Pony, Rony, Tony and Sony but not necessarily in the same order. Each of the six gamblers Kamal, Krish, Kundan, Lohith, Rahul and Rohan bet on exactly one among the given horses. These horses were made to participate exactly in one race among the following races, such as Endurance race, Harness race, Flat race and Hunt race and finished the race in the top three positions only. No two of them got the same position in the same type of racing.

(i) Only Honey participates in Endurance race and its jockey is K. Only Tony finished the race in the 3rd position.
(ii) Kundan bets on the horse which participates in Harness race but did not finish the race in the 3rd position.
(iii) Lohith bets on the horse, whose jockey is M and finished the race either in the 1st or in the 2nd position.
(iv) Pony's jockey is H and participates in flat race. Nunny finished the race in the 1st position but its jockey neither I nor L.
(v) Among the given horses only two horses participate in Harness race. Neither of whose jockey is I.
(vi) Nunny, Honey and the horse whose jockey M finished the race in the same position.
(vii) I finished the race in the 2nd position. Rahul bets on a horse which finished the race in the 2nd position, but its jockey is not I.
(viii) Kamal bets on Sony, which participates with Rony in the same race. Rohan does not bet on Honey.

8. Which of the following statements is/are true?
(i) Sony participates in Hunt race.
(ii) Krish bets on Honey.
(iii) Sony finished the race in the 1st position.
(A) Only (i) (B) All the three
(C) Only (ii) (D) Only (i) and (ii)

9. Which jockey finished the race in the 1st position in Harness race?
(A) H (B) I
(C) J (D) M

10. Who bets on Rony?
(A) Kundan (B) Rahul
(C) Lohith (D) Cannot be determined

Directions for questions 11 to 14: These questions are based on the following information.

There are three students and three teachers, namely A, B, C, D, E and F. Each of these has taken three tests among the tests 1 through 6. Among which, two tests are only for students, two tests are only for teachers and two tests can be taken by both teachers and students. No two of them has taken the same set of tests.

(1) No test is taken by both A and D. Only test 6 is taken by both E and F.
(2) Test 5 is only for teachers while test 4 is only for students.
(3) E took the tests 1, 4 and 6 while A did not take test 5.
(4) Only test 3 is taken by both A and B.

11. What are the tests taken by D?
(A) 2, 5, 3 (B) 2, 5, 6
(C) 4, 1, 6 (D) 5, 1, 6

12. Which of the following tests are only for students?
(A) 1, 2 (B) 4, 1
(C) 5, 6 (D) 2, 5

13. How many people took the test 6?
(A) 3 (B) 2
(C) 4 (D) Cannot be determined

14. Which of the following tests is taken by C?
(A) 4 (B) 5
(C) 2 (D) 3

Directions for questions 15 to 17: These questions are based on the following information.

A group of three friends, namely Anand, Bhuvan and Chander have to select and buy some different music cassettes out of the six cassettes of six different singers, namely UB40, Vanessa, Williams, Xavier, Yellows and Zoloto.

Out of these three friends, one chooses five cassettes, another chooses four cassettes and the remaining person chooses three cassettes. There are exactly two different singers' cassettes, out of the six singers, which are bought by all the three people. No person has more than one cassette of the same singer. It is known that Bhuvan does not have UB40, Anand does not have Zoloto and Chander does not have Williams. The number of cassettes that Anand buys is more than the number of cassettes bought by Chander. Also, any person must have either UB40 or Zoloto, but not both. If a person has Vanessa, then he must have Yellows also. If a person has Yellows, then he must have Xavier also. Vanessa is bought by exactly one person. Also, each cassette is bought by at least one person.

15. Who has the least number of cassettes?
(A) Anand (B) Bhuvan
(C) Chander (D) Cannot be determined

16. Which among the following could be the group of cassettes belonging to only two owners?
(A) Xavier and Yellows (B) Zoloto and Williams
(C) UB40 and Zoloto (D) None of these

17. How many different arrangements are possible for the number of cassettes with the three friends?
(A) 6 (B) 3
(C) 2 (D) None of these

Directions for questions 18 to 20: These questions are based on the following information.

Eight students A, B, C, D, E, F, G, H went to four different places, such as Resort, Beach, Hotel and Cinema, such that each place was visited by two students each.

Each student visited exactly one place. After their return, their teacher asked them about the place visited by each of them. The following were their answers:

(i) A said 'I did not go with C or D and went to the Resort or the Cinema'.
(ii) B said 'I did not go with E or G and went to the Hotel or the Cinema'.
(iii) C said 'I did not go with D or F and went to the Beach or the Resort'.
(iv) D said 'I did not go with B or H and went to the Beach or the Hotel'.
(v) E said 'I went with B or C or D or F or H and went to the Cinema or the Beach'.
(vi) F said 'I did not go with A or G and went to the Resort or the Cinema'.
(vii) G said 'I went with B or D or E or F or H and went to the Beach or the Hotel'.
(viii) H said 'I did not go with C or A and went to the Resort or the Beach'.

18. Who went with A?
(A) E (B) B
(C) G (D) H

19. E went with _____ and visited the _____.
(A) C, Beach (B) F, Cinema
(C) D, Beach (D) G, Beach

Difficulty Level-2: Moderate

20. If only D and H lied about the places visited by them, then with whom did D visit the place of his choice?
(A) H (B) F
(C) G (D) Cannot be determined

Directions for questions 21 to 24: These questions are based on the following information.

A journalist organization which publishes secret information leaks information about the black money of seven people J, K, L, M, N, P and Q who have black money in various branches of Swiss Bank. Each of them has a different amount of black money among ₹19,898, ₹2436, ₹4537, ₹6734, ₹28,116, ₹3624 and ₹8697 (in crores), but not necessarily in the same order.

Each person kept his/her black money in one of the branches among Geneva, London and Switzerland.

Atleast two people kept their black money in each branch.

(i) L's black money is not kept in Geneva branch. M's black money is not ₹3624.
(ii) Either P's or J's black money is ₹19,898 and both of them kept their black money in London branch.
(iii) Neither L's nor P's black money is the highest.
(iv) K's black money is more than N's black money and they kept their black money in the same branch. No other person kept his/her money in this branch.
(v) K's black money is the third highest. Only those persons whose black money is the least and the second least kept their black money in Geneva.

21. Whose black money is 4537 crores (in ₹)?
(A) M (B) L
(C) N (D) Cannot be determined

22. Whose black money is the fourth highest?
(A) L (B) N
(C) Q (D) Cannot be determined

23. Which of the following statements is/are true?
(i) M's black money is the least.
(ii) Q's black money is the highest.
(iii) L's black money is in London.
(A) Only (i) (B) Only (ii)
(C) Only (i) and (iii) (D) All the three

24. Three of the following four pairs are alike in a certain way based on the given information and hence, form a group. Find the one that does not belong to the group.
(A) K, M (B) M, N
(C) P, Q (D) P, K

Directions for questions 25 to 27: These questions are based on the following information.

Six people A, B, C, D, E and F are wearing a different coloured dress among red, green, blue, yellow, violet and white. Following is the information known about them.

(i) Neither C nor F is wearing either a red or a yellow coloured dress.
(ii) Neither of D and E is wearing the dress coloured as white, red or blue.
(iii) B is wearing either a green or a blue coloured dress.
(iv) Neither D nor F is wearing a violet coloured dress.
(v) E is not wearing either a green or a violet coloured dress.

25. Who is wearing the green coloured dress?
(A) B (B) C
(C) D (D) F

26. What colour dress is A wearing?
(A) Green (B) Blue
(C) Red (D) White

27. Who is wearing the white coloured dress?
(A) A (B) F
(C) C (D) Data inadequate

Directions for questions 28 to 31: These questions are based on the following information.

Four channels W through Z telecast six films in three slots 8 to 10, 10 to 12 and 12 to 2. The films are categorized as comedy, horror and action. The number of films in any category is not the same and no channel telecast movies in two consecutive slots. No channel telecast the movies of same category. Kumphu and Tom are telecasted by the same channel while the movies Micky and Karate are telecasted in the same slot. Karate is the only film telecasted by the channel W in the slot 10 to 12, but it is not an action movie.

Micky is the only comedy film. The films Hanuman and Tom are not telecast in the same slot. Vali the horror film is not telecasted in the slot 8 to 10 and not in the channel Z or Y.

28. Which of the following is/are the correct combinations of film and the slot in which it was telecasted?
(A) Hanuman – (8 to 10)
(B) Hanuman – (10 to 12)
(C) Vali – (10 to 12)
(D) Micky – (12 to 2)

29. Which films are telecasted in the slot 12 to 2?
(A) Hanuman – Kumphu
(B) Vali – Tom
(C) Kumphu – Karate
(D) Vali – Micky

30. Which of the following is true?
(A) The film Micky is telecasted by X.
(B) The film Vali is telecasted by Y.
(C) The film Tom is telecasted by Z.
(D) The film Hanuman is telecasted by channel X.

31. Films of which category are maximum in number?
(A) Comedy (B) Horror
(C) Action (D) Either (B) and (C)

Directions for questions 32 to 34: These questions are based on the following information.

Six people Anju, Sanju, Raju, Manju, Billa and Sruthi went to a play station to play a different game among Gothic 2, Max Payne, Mirror's Edge, Mount & Blade, Star Craft and The Last Express. The owner of the play station allotted a different cabin among 1 to 6 to each of these six people. Each of the above-mentioned games were released in a different year among 1997, 1998, 2001, 2002, 2008 and 2009.

(i) The game played by Raju was released in 2001. He played the game in an odd numbered cabin.
(ii) Sanju played the game in Cabin 2. Gothic 2 was released immediately after Max Payne.
(iii) Billa played the game in Cabin 1. The game played by Sruthi was released after the game, The Last Express was released and was allotted Cabin 6.
(iv) The names of the games played by Raju and Manju start with the same alphabet. Mount and Blade was released in 2008.
(v) Mirror's Edge was released recently and was played by Anju in an even numbered cabin.
(vi) Neither 'The Last Express' nor 'Star Craft' was played in Cabin 2.

32. Who played Star Craft?
(A) Sanju (B) Billa
(C) Sruthi (D) None of these

33. In which year was 'The Last Express' released?
(A) 2002 (B) 1998
(C) 1997 (D) 2001

34. In which cabin did Manju play the game?
(A) 6 (B) 5
(C) 3 (D) Cannot be determined

Directions for questions 35 to 37: These questions are based on the following information.

Eight employees from P through W of a company went for a tour to different cities, such as Bangalore, Hyderabad, Mumbai and Delhi in different months during April, September, October and December in a year. Those employees who went to the same place did not go in the same month. Exactly two employees went in the same month and exactly two employees went to the same place.

T went to Delhi in either October or April. V went in September to neither Hyderabad nor Mumbai. One of the employees who went to Bangalore went in December. S went in December. U and R went to the same place, R and W went in the same month. P, R and Q went to different places, but not to Mumbai and they went in different months, but not in April. P did not go in September.

35. Who went to Delhi?
(A) P (B) Q
(C) R (D) V

36. Who went in October?
(A) P (B) Q
(C) R (D) T

37. Which among the following group of employees went to the same place?
(A) P, S (B) Q, T
(C) V, Q (D) S, T

Directions for questions 38 to 40: These questions are based on the following information.

In a horse racing, each of the six Jockeys named as H, I, J, K, L and M is riding a different horse Honey, Nunny, Pony, Rony, Tony and Sony but not necessarily in the same order. Each of the six gamblers Kamal, Krish, Kundan, Lohith, Rahul and Rohan bet on exactly one among the given horses. These horses were made to participate exactly in one race, they were Endurance race, Harness race, Flat race and Hunt race and finished the race in the top three positions only. No two of them got the same position in the same type of racing.

(i) Only Honey participates in Endurance race and its jockey is K. Only Tony finished the race in the 3rd position.
(ii) Kundan bets on the horse which participates in Harness race but did not finish the race in the 3rd position.
(iii) Lohith bets on the horse, whose jockey is M and finished the race either in the 1st or in the 2nd position.
(iv) Pony's jockey is H and participates in Flat race. Nunny finished the race in the 1st position but its jockey neither I nor L.
(v) Among the given horses only two horses participated in Harness race. Neither of whose jockey is I.
(vi) Nunny, Honey and the horse whose jockey M finished the race in the same position.
(vii) I finished the race in the 2nd position. Rahul bets on a horse which finished the race in the 2nd position, but its jockey is not I.
(viii) Kamal bets on Sony, which participates with Rony in the same race. Rohan does not bet on Honey.

38. Which of the following statements is/are true?
(i) Sony participates in Hunt race.
(ii) Krish bets on Honey.
(iii) Sony finished the race in the 1st position.
(A) Only (i) (B) All the three
(C) Only (ii) (D) Only (i) and (ii)

39. Which jockey finished the race in the 1st position in Harness race?
(A) H (B) I
(C) J (D) M

40. Who bets on Rony?
(A) Kundan (B) Rahul
(C) Lohith (D) Cannot be determined

EXERCISE-3

Directions for questions 1 to 4: These questions are based on the following data.

Each of five people A, B, C, D and E owns a different car among Maruti, Mercedes, Sierra, Fiat and Audi and the colours of these cars are black, green, blue, white and red, not necessarily in that order. No two cars are of the same colour. It is also known that:

(i) A's car is not black and it is not a Mercedes.
(ii) B's car is green and it is not a Sierra.
(iii) E's car is not white and it is not an Audi.
(iv) C's car is a Mercedes and it is not blue.
(v) D's car is not red and it is a Fiat.

1. If A owns a blue Sierra, then E's car can be a
(A) Red Maruti (B) White Maruti
(C) Black Audi (D) Red Audi

2. If A owns a white Audi, then E's car can be a
(A) Red Maruti (B) Blue Maruti
(C) Green Audi (D) Black Sierra

3. If A's car is a red Maruti and D's car is white, then E owns a
(A) Black Audi (B) Blue Sierra
(C) Black Sierra (D) Blue Audi

4. If E owns a red Maruti and A's car is white, then D owns a
(A) Green Fiat (B) Black Fiat
(C) Blue Fiat (D) Red Fiat

Directions for questions 5 and 6: These questions are based on the following data.

In a college, there are ten lecturers enrolled in a lecture program. These lecturers have been grouped in any one of the four subjects, such as Physics, Chemistry, Biology and Maths. One professor is assigned to each of these four subject groups. Kunal, Kapil and Kamal will give lectures on the same subject. Kapil and Karishma belong to the same subject group. Karan and Kamini belong to the same subject group. Kusum cannot be with Kamal and Kiran cannot be with Karan. Kapil will deliver a lecture on Maths and Kiran delivers a lecture on the same subject as Kate. Each of Kapil, Karan, Kusum and Kiran delivers lecture on a different subject. Kamal and Kiran are lecturers for Chemistry and Kusum is not a lecturer of Physics. Amar, Beena, Chander and Deepak are professors of subject groups with number of lecturers as 4, 3, 2 and 1, respectively.

5. Which of the following statements must be true?
(A) Amar is the subject group professor of Kamal for Chemistry.
(B) Deepak is the subject group professor of Kusum for Biology.
(C) Beena is the subject group professor of Karan for Maths.
(D) Chander is the subject group professor of Karishma for Physics.

6. Who among the following is a lecturer in Maths?
(A) Karan (B) Kiran
(C) Kamini (D) None of these

Directions for questions 7 to 10: These questions are based on the following information.

A group of eight people, namely P, Q, R, S, T, U, V and W are travelling by the following cars, such as Honda city, BMW and Honda Brio. Each of them belongs to a different city and they are from Hyderabad, Chennai, Kolkatta, Pune, Bengaluru, Cochin, Baroda and Noida but not necessarily in the same order. The number of people travelling by any car is minimum two and maximum three.

Only two people, P and the one from Pune are travelling by BMW. R and T are travelling by different cars but they are neither from Bengaluru nor from Baroda. V is from Kolkata but not travelling by Honda Brio. R and W are travelling by same car. R is not from Cochin, W is not from Bengaluru. T is not from Pune and S is from Hyderabad. U is from Chennai and travelling by Honda City.

7. Who is from Noida?
(A) P (B) W
(C) Q (D) R

8. Which of the following group of people are travelling by Honda Brio?
(A) RUW (B) RVW
(C) QRW (D) SRW

9. In which car is the person from Baroda travelling?
(A) BMW
(B) Honda Brio
(C) Honda City
(D) Either BMW or Honda Brio

10. Which of the following is true regarding the given information?
(A) P is from Noida.
(B) W is from Pune.
(C) T is from Cochin.
(D) R is travelling by Honda city.

Directions for questions 11 to 13: These questions are based on the following information.

Eight people A, C, E, G, H, K, M and P have eight different animals, such as camel, lion, monkey, horse, elephant, cat, dog and tiger. They went to three different Zoos, namely

Zoo – I, Zoo – II and Zoo – III. At least two people and at most three people went to each zoo. The following information is known about them.

Only G and M went to Zoo – III and one of them has a dog. E has a monkey and went to Zoo – I. The person, who went to Zoo – II has a tiger but is not C. C and the person, who has a cat went to Zoo – II. Neither A nor K has a tiger, but one of them went to Zoo – II. A and H went to the same Zoo. H has an elephant. The person, who has a camel did not go to either Zoo – II or Zoo – III. The person who went to Zoo – III does not have a lion. One among G and C has a horse.

11. Who went to Zoo – I?
(A) G, H, E (B) A, H, C
(C) A, K, M (D) A, H, E

12. Who has a lion?
(A) K (B) A
(C) C (D) G

13. Who has a tiger?
(A) G (B) C
(C) P (D) None of these

Directions for questions 14 to 16: These questions are based on the following information.

A group of eight people, namely A, B, C, D, E, F, G and H belong to different colonies named as P, Q and R and three different streets I, II and III, but not necessarily in the same order. No two people who belong to the same colony belong to the same street. At least two and at most three people belong to each colony and each street.

A belongs to street I and C belongs to colony R. A and E belong to neither the same colony nor the same street. D and F belong to the same colony. G and C belong to the same street. F belongs to colony P and G belongs to neither street I nor street II. E and C do not belong to the same street. B and C do not belong to the same colony. H belongs to neither colony P nor colony Q. C, D and E belong to different colonies and different streets. A and B belong to neither the same colony nor colony P. B and G belong to the same street. G does not belong to colony Q.

14. To which colony does G belong?
(A) Q (B) R
(C) P (D) P or R

15. Which of the following is the correct combination of person, colony and street respectively?
(A) F – P – II (B) G – P – II
(C) G – II – P (D) F – II – P

16. Which group of persons belong to the same colony?
(A) D, F, G (B) F, G, H
(C) D, E, F (D) None of these

Directions for questions 17 to 19: These questions are based on the following data.

A group of five men, namely Kambli, Kumble, Kamlesh, Kareem and Kishan are working in the same company but are earning different salaries. They are married to five women, namely Kunti, Kirti, Kamini, Kareena and Karishma not necessarily in that order.

(1) The person who is married to Kirti is neither earning the maximum nor the minimum salary.
(2) The husband of Kamini is earning ₹5 lakh/annum.
(3) Kishan earns ₹6 lakh / annum.
(4) Kumble, the husband of Karishma, is earning ₹1 lakh/annum more than Kamlesh, who earns less than Kishan.
(5) Kambli, who is not married to Kamini is earning ₹4 lakh/annum more than Kareem.
(6) Each of the five men earns at least 1 lakh/annum. Each man's earnings is a natural number.

17. Who earns ₹5 lakh per annum?
(A) Kareem (B) Kamlesh
(C) Kambli (D) Kishan

18. Who is married to Kirti?
(A) Kamlesh (B) Kambli
(C) Kishan (D) Kareem

19. If Kambli is not married to Kunti, then whose husband is earning the minimum salary?
(A) Kunti (B) Kareena
(C) Karishma (D) Kamini

Directions for questions 20 to 22: These questions are based on the following information.

Six people A, B, C, D, E, and F belong to six different professions, each of them being Accountant, Doctor, Engineer, Editor, Painter, Teacher and they are sitting around a circular table not necessarily in the same order. The following information is known about their professions and seating arrangement.

The Doctor and the Teacher are adjacent to each other. B is either the Engineer or the Editor. Neither A nor D is a Doctor but one of them is an Accountant. The Engineer is sitting second to the right of A. The Doctor is sitting opposite to F. Either F or E is the Painter. C is either the Editor or the Accountant. The Editor is not sitting opposite to the Engineer.

20. Who is sitting opposite to the Engineer?
(A) A (B) B
(C) D (D) C

21. If the doctor is to the immediate left of B, then who is sitting to the immediate left of the Accountant?
(A) Painter (B) Editor
(C) Engineer (D) Cannot be determined

22. Which of the following statements is definitely true?
(A) The Teacher is sitting second to the left of the doctor.
(B) D is an Accountant.
(C) A is the Teacher.
(D) A is sitting opposite to the Editor.

Directions for questions 23 to 25: These questions are based on the following information:

Seven people A, B, C, D, E, F and G live on seven floors (ground floor is considered the first floor and the floor just above the first floor is considered the second floor and so on.) of an apartment building. Each person takes an exam on each of the days Sunday, Monday, Tuesday, Wednesday, Thursday, Friday and Saturday, not necessarily in the same order. The following information is known about them:

(i) There are three floors between C's floor and D's floor from top to bottom in that order.
(ii) Either B or E lives on the top floor. The person who lives on the top floor takes an exam on Wednesday.
(iii) Neither F nor G takes an exam on Tuesday and there is one person between F's floor and G's floor who takes an exam on Saturday.
(iv) The person who takes an exam on Tuesday does not live on an even-numbered floor.
(v) There are only two floors below A's floor.
(vi) C takes an exam either on Sunday or on Wednesday. Only two people live between B and G and one of them takes an exam on Monday.
(vii) The person who takes an exam on Thursday is adjacent to the person who takes an exam on either Saturday or Monday.

23. Who lives on the sixth floor?
(A) A (B) B
(C) C (D) E

24. A takes an exam on which day?
(A) Friday (B) Saturday
(C) Monday (D) Wednesday

25. How many people live between E and F?
(A) None (B) One
(C) Two (D) Three

ANSWER KEYS

Exercise-1

1. (B)	8. (D)	15. (B)	22. (D)	29. (D)	36. (D)
2. (B)	9. (D)	16. (C)	23. (C)	30. (D)	37. (C)
3. (D)	10. (B)	17. (A)	24. (A)	31. (A)	38. (D)
4. (B)	11. (A)	18. (B)	25. (C)	32. (D)	39. (B)
5. (B)	12. (D)	19. (D)	26. (D)	33. (C)	40. (C)
6. (A)	13. (A)	20. (A)	27. (C)	34. (C)	
7. (B)	14. (C)	21. (D)	28. (C)	35. (D)	

Exercise-2

1. (D)	8. (D)	15. (C)	22. (D)	29. (B)	36. (C)
2. (C)	9. (C)	16. (D)	23. (C)	30. (D)	37. (B)
3. (B)	10. (C)	17. (C)	24. (B)	31. (B)	38. (D)
4. (C)	11. (B)	18. (B)	25. (C)	32. (C)	39. (C)
5. (A)	12. (B)	19. (A)	26. (C)	33. (C)	40. (C)
6. (C)	13. (D)	20. (B)	27. (B)	34. (D)	
7. (D)	14. (D)	21. (D)	28. (A)	35. (B)	

Exercise-3

1. (A)	5. (B)	9. (B)	13. (C)	17. (A)	21. (A)	24. (B)
2. (D)	6. (D)	10. (C)	14. (C)	18. (C)	22. (D)	25. (D)
3. (B)	7. (D)	11. (D)	15. (A)	19. (A)	23. (D)	
4. (C)	8. (D)	12. (C)	16. (A)	20. (C)		

SOLUTIONS

EXERCISE-1

Solutions for questions 1 to 9:

1. The given information can be represented as follows:

Name	Colours
Bhanu	x yellow x black
Modi	x yellow x indigo
Ravindra	Black/indigo
Kamal	Indigo/black
Lalit	x indigo x white

Since neither Bhanu nor Modi is wearing a yellow coloured shirt, Lalit must be wearing a yellow coloured shirt.

2. Bond will not take a Sword or a Bow or Dagger. Bhima will not take a Gun or a Bow.
∴ Bow must be taken by Rama.
Sword and Mace can be taken by Bhima only and Rama has to take the Dagger, as Bond will not take the Dagger.
∴ Gun and Arrow is taken by Bond.

3. From 1, we know that Suman did not write JMET. From 2 and 3, we know that Swaroop did not go to Hyderabad or Nagpur (as he did not write CET which was conducted in Nagpur). From 5, we know that Sachin went to Mumbai which means Swaroop did not go to Mumbai, which implies that she must have gone to Vijayawada and written either JMET or CAT.
From 4, we know that Sagar did not write CAT or CET (as he did not go to Nagpur) and from the last statement we also know that Sagar did not write XAT, which means that he must have written JMET. This implies that Swaroop wrote CAT.
∴ Swaroop goes to Vijayawada and writes CAT.

4. Forming a grid between the trains and the cities, we get the following arrangement:
(i) $GT_{XP} \neq$ Delhi, Chennai
(ii) Hyd $\neq AP_{XP}$, Goa_{XP}
(iii) Bangalore = Goa_{XP} or AP_{XP}
$\Rightarrow$ Bangalore $\neq Raj_{XP}$, GT_{XP}, $Bangalore_{XP}$
(iv) Raj_{XP} = Chennai
(v) Either Goa_{XP} = Hyd
or $Bangalore_{XP}$ = Delhi
As $Goa_{XP} \neq$ Hyd, $\Rightarrow Bangalore_{XP}$ = Delhi
$\Rightarrow GT_{XP}$ = Hyd
Now, let us analyse the choices:

	Delhi	Goa	Chennai	Bangalore	Hyd
GT_{XP}	× (i)	× (v)	× (i)	× (iii)	✓ (v)
AP_{XP}	× (v)	•	× (iv)	•	× (ii)
Raj_{XP}	× (iv)	× (iv)	✓ (iv)	× (iii)	× (iv)
Goa_{XP}	× (v)	•	× (iv)	•	× (v)
$B'lore_{XP}$	✓ (v)	× (v)	× (iv)	× (iv)	× (v)

(A) **Either p or q:**
Implications: ~p ⇒ q and ~q ⇒ p.
But p is true (i.e., Chennai is visited by Raj_{XP}), hence, q may or may not be true (i.e., Goa may or may not be visited by AP_{XP}). Hence, (A) will not help in completing the arrangement.
(B) **Only if p, then q:**
Implications: q ⇒ p; ~p ⇒ ~q
Here, Bangalore is not visited by the $Bangalore_{XP}$ (~p), which implies that Goa is not visited by the Goa_{XP} (~p ⇒ ~q). Hence, this statement helps in completing the arrangement.
(C) **If p, then q:**
Implications: p ⇒ q; ~q ⇒ ~p
Both p and q are not decided here, as it is not known whether Goa_{XP} visits Bangalore or not, and AP_{XP} visits Goa or not.
(D) **If p, then q:**
Here, Delhi is visited by $Bangalore_{XP}$, means that q is true, but can't say whether p is true or not (since p ⇒ q).
Hence, it is only (B) which helps in completing the arrangement.

5. Given S is the wife of B, D is not the husband of P, P is not the wife of A, E is the husband of T. Since P's husband is neither D nor A, P's husband should be C. So, P is the wife of C.

6. From clues (A) and (D), Rajesh is from Kolkata and is working in Accenture. From (B) and (C), Ravan and Ramani are not from Mumbai. So, Ramesh is from Mumbai. From (D) we can say Ramesh is from CTS. Hence, Ramani is from Wipro. Since the person from Wipro is not from Chennai, Ravan is from Chennai.
∴ The final arrangement is as follows.

Name	Company	Place
Ramesh	CTS	Mumbai
Rajesh	Accenture	Kolkata
Ramani	Wipro	Delhi
Ravan	TCS	Chennai

Ramesh is working in CTS.

7. The given information can be tabulated as follows.

Name	Chocolate	Colour
Manju		Red
Sanju		Blue/Orange
Tanooj	Kit Kat	
Pooja		

Orange coloured box contains Dairy milk, it is possible only when Sanju received Orange coloured box. The complete distribution is as follows.

Name	Chocolate	Colour
Manju	Bar one	Red
Sanju	Dairy milk	Orange
Tanooj	Kit Kat	Yellow
Pooja	Eclairs	Blue

Choice (B) is true statement.

8. Let the analysis be as follows.

	Number of student		
Subject	Passed	Failed	Total
Physics	a	b	g
Maths	c	d	h
Chemistry	e	f	i
Total			

From the given information, we derive:

$e = \frac{i}{2}$, $f = \frac{i}{2}$, $a = c$, $d = f$, $b = \frac{3g}{10}$, $g + h + i = 40$

$d = 100$, $f = 100$, $i = 200$, $e = 100$,

$b = \frac{3g}{10} \Rightarrow a = \frac{7g}{10} \Rightarrow c = \frac{7g}{10}$

$h = c + d = \frac{7g}{10} + 100$

$\therefore\ g + h + i = g + \frac{7g}{10} + 100 + 200 = 470$

$\frac{17g}{10} = 170$

$g = 100$

$\therefore\ a = 70$, $b = 30$, $c = 70$, $h = 170$

$\therefore$ (A), (B) and (C) are true.

9. The given information can be tabulated as follows.

Name	City	Department
Minku		
Tinku	Kolkata	
Pinku		Electronics
Chinku		

From the given information Chinku does not belong to Electronics, hence, she / he is from Civil, and Chinku is not from Bhopal. Minku is from Bhopal and is an Electronic student. Chinku is from Hyderabad while Pinku is from Chennai.

$\therefore$ All the given statements are true.

Solutions for questions 10 and 11: Given that the Alto is parked in P2 and it does not belong to Manohar and Jamal owns an Indica. Since Jamal owns an Indica, the Alto should be owned by Karan. So, Manohar owns a Zen and it is parked in P3.

The final arrangement is as follows.

Name	Car	Parking Space
Karan	Alto	P2
Manohar	Zen	P3
Jamal	Indica	P1

10. Karan owns an Alto.

11. Jamal parked his car in P1.

Solutions for questions 12 and 13: From clues (A) and (D) Rajesh is from Kolkata and is working in Accenture. From (B) and (C), Ravan and Ramani are not from Mumbai. So, Ramesh is from Mumbai. From (D), we can say that Ramesh is from CTS. Hence, Ramani is from Wipro. Since the person from Wipro is not from Chennai, Ravan is from Chennai.

$\therefore$ The final arrangement is as follows.

Name	Company	Place
Ramesh	CTS	Mumbai
Rajesh	Accenture	Kolkata
Ramani	Wipro	Delhi
Ravan	TCS	Chennai

12. Ravan is from Chennai.

13. Ramesh is working in CTS.

Solutions for questions 14 to 16: Given, every tree is visited by 6 monkeys, total sum of the number of trees visited by all the monkeys is 24, i.e., 4×6.

14. The monkey that visited all the trees cannot be D as it did not visit the Berry tree, it cannot be G as it did not visit the Berry tree, it cannot be J as it did not visit the Mango tree.
∴ The monkey that visited all the trees is I.

15. It is known that the number of monkeys that visited all the trees is 1 and those that visited only one tree is 2. Let the number of monkeys that visited 3 trees be x and those that visited 2 trees be y.
$1(4) + x(3) + y(2) + 2(1) = 24$
$\Rightarrow 3x + 2y = 18$
and we know that $x + y = 7$
$\therefore\ x = 4$.

16. From the given data, the monkeys that visited only one tree must be F and H as F did not visit Guava, Berry and Banana trees and H did not visit Guava, Banana and Mango trees.
∴ A must visit at least two trees as it did not visit the Banana and Mango tree, means it should visit the Berry and Guava trees. We cannot say anything about the remaining monkeys.

Solutions for questions 17 to 19: The given information is represented in the following table.

Name	Umbrella	Bag
P	× Yellow	Red
Q		× black
R	Black	× black
S	Blue	× black
T		Yellow
U		

From the above table it is clear that U carries a black coloured bag.
From (ii) and the above table, S does not carry a blue, black, red, green or yellow coloured bag.
∴ S carries a pink coloured bag.
From (ii) the pink coloured umbrella and green coloured bag are carried by Q.
∴ R carries a blue coloured umbrella.
As P is not carrying a yellow coloured umbrella, P is carrying the green coloured umbrella.
∴ T cannot carry a yellow coloured umbrella.
U is carrying a yellow coloured umbrella.
T is carrying the red coloured umbrella
The final table is as follows.

Name	Umbrella	Bag
P	Green	Red
Q	Pink	Green
R	Black	Blue
S	Blue	Pink
T	Red	Yellow
U	Yellow	Black

17. T is carrying a red coloured umbrella.

18. U is carrying a yellow coloured umbrella and the colour of his bag is black.

19. T – Red – Yellow is the correct combination.

Solutions for questions 20 to 22: From (1) and (4), E brought W, B brought R.
From (3), C did not bring U.
From (4), C brought neither S nor V.
From (5) and (6), C did not bring Q or T.
C did not bring S, Q, V, U, R, W or T.
∴ C brought dish P.
From (1) and (6), none among A, D, G and H brought S.
∴ F brought S and A brought T. G brought U and D and H brought Q and V in any order.

Boy	Dish
A	T
B	R
C	P
D	Q/V
E	W
F	S
G	U
H	Q/V

20. A brought the dish T.

21. F – S is the correct combination

22. Either D or H brought the dish V.

Solutions for questions 23 to 26: From (1) and (4), mango is not planted in the city A or C or D.
From (2), either banana or guava is planted in city E. Hence, mango is not planted in city E.
∴ Mango is planted in city B. From (2), apple is planted by Uday in city A.

From (2) and (3), Tarun planted watermelon and hence, Gopal owns a field in City E.

From (3), Suman owns a field either in the city B or in city D. Tarun owns a field in city C. From (2), Gopal planted banana or guava. We get the following distribution.

Field	Plant	Person
A	Apple	Uday
B	Mango	Yadav/Suman
C	Water melon	Tarun
D	Guava/Banana	Suman/Yadav
E	Banana/Guava	Gopal

23. Yadav planted Guava plants. Hence, he owns a field in city D.
∴ Suman owns a field in city B.

24. Uday owns a field in the city A.

25. Tarun owns a field in city C.

26. Mango is planted in the city B.
Either Guava or Banana is planted in city D.

Solutions for questions 27 to 29:

Name	Item
Hemant	Pen, Pencil, Calculator
Jagat	Calculator, Pen
Dweep	Eraser, Bag
Poluami	Eraser, Bag, Pencil

As none among Hemant, Jagat, Dweep and Poulami has a ruler, Manyata has ruler.

27. Manyata has a ruler.

28. Hemant and Poulami have pencils.

29. None of the items is present with more than two people.

Solutions for questions 30 to 33: Let us use the first letter of the names to represent respective professors and first two letters to represent the subjects.

From (1) and (2), we have

Mon		
Tues	Pu	A^x
Wed	Lo^x	G
Thu		
Fri		
Sat		

∴ From (5) Prasad's session is on English but it is scheduled neither on Monday nor on Saturday. It cannot even be on Friday because if it is Friday, (4) cannot be satisfied. Hence, Prasad's session is scheduled on Thursday. Hence, Manoj's session and Arvind's session are scheduled on Friday and Saturday respectively.

And from (3), the distribution is as follows.

Day	Subject	Faculty
Monday	Lo	B/R
Tuesday	Pu	R/B
Wednesday	Ar	G
Thursday	En	P
Friday	Cu/Co	M
Saturday	Co/Cu	A

30. Manoj's session is scheduled on Friday.

31. Logical Reasoning is scheduled on Monday.

32. In the given case Communication Skills is scheduled on Friday.

33. Prasad gives his orientation on Thursday.

Solutions for questions 34 to 37: The given information can be tabulated as shown below.

Flower pot	Flower	Butterfly
A	Columbine	✘ U ✘ S
B		
C		✘ T
D		✘ T
E		P
F	✘ Dahlia	
G		

From (iv) and the above, Dahlia is in B and Clover is in F. V and Q feed on Clover and Dahlia is in any order.

From (vi) and the above, T feeds on the flower which is in G. Hence, R feeds on Columbine and S and U are in C and D in any order.

From (v), (vii) and the above, E and G contains Foxglove and Carnation in any order. U feeds on Buttercup.

∴ The final distribution is as shown below.

Flower pot	Flower	Butterfly
A	Columbine	R
B	Dahlia	V/Q
C	Buttercup/Crocus	S/U
D	Crocus/Buttercup	U/S
E	Foxglove/Carnation	P
F	Clover	Q/V
G	Carnation/ Foxglove	T

34. U feeds on Buttercup.

35. E or G

36. Buttercup

37. Clover

Solutions for questions 38 to 40: The given information can be tabulated as shown below.

Country	Group of disciples
India	
US	✘ Bharath ✘ Dharani ✘ Danush
Bangladesh	(Aman, Balu)/(Ajay, ✘ Bhuvan, ✘ Danush)
UK	(Ajay/Aman, Balu) ✘ Bhuvan ✘ Danush

Country	Group of disciples
Nepal	
Bhutan	✘ Bharath, ✘ Dharani, ✘ Bhuvan
UK	(Ajay/Aman, Balu) ✘ Bhuvan ✘ Danush

From (v) and the above, Arjun and Arhan were sent to Nepal. Bharath and Dharani were sent to India. Bhuvan was sent to US and Danush was sent to Bhutan.

From (vi) and the above, Bharani and Ajay were sent to the same country.

From (iv) and the above, Akhil was sent to US and Amith was sent to Bhutan.

∴ The final distribution is as shown below.

Country	Group of disciples
India	Bharath, Dharani
US	Bhuvan, Akhil
Bangladesh	(Aman, Balu)/(Ajay, Bharani)
Nepal	Arjun, Arhan
Bhutan	Danush, Amith
UK	(Ajay, Bharani)/(Aman, Balu)

38. Bharath and Dharani were sent to India.

39. Bharani was sent to UK.

40. Amith was sent to Bhutan.

EXERCISE-2

Solutions for questions 1 to 3: The given information is as follows.

Name	Profession	Place
A	Architect	
B	Not Doctor	Not Bangalore
C	Engineer	Chennai
D	Not Professorr	Not Bangalore
E		Mumbai
	Not Architect Not Lawyer	Kolkata

The doctor from Bangalore cannot be any one among A, B, C, D and E. Hence, it is F.

Since D is not a professor, he is not from Delhi.

⇒ B is the professor from Delhi.

Since, the person from Kolkata is not an Architect, A is from Hyderabad and D is from Kolkata.

⇒ D is the Painter and E is the Lawyer.

The final distribution is as follows.

Name	Profession	Place
A	Architect	Hyderabad
B	Professor	Delhi
C	Engineer	Chennai
D	Painter	Kolkata
E	Lawyer	Mumbai
F	Doctor	Bangalore

1. F is the doctor.

2. The painter is from Kolkata.

3. E is the lawyer.

Solutions for questions 4 to 7: From the given information, the publishers are as follows:

(1) Reprographic publishers
(2) Holy Faith publishers
(3) Prinston publishers
(4) Penguin publishers and
(5) Johnson publishers

Three books are on History, three books are of 2nd edition, and two books are of 1st edition.

Three books are on Science out of which one is 1st edition and two are 2nd editions.

The books published by Penguin and Reprographic publishers contain Geography. No book contains both Geography and Science.

∴ Books published by Johnson, Prinston and Holy Faith publishers contain Science and are stacked together.

Since no book contains Geography and Science and all the Science books are stacked together, the books containing Geography, i.e., Reprographic and Penguin are stacked together. The Reprographic book is 1st edition. Since the other 1st edition books contain Science, Penguin must be the second edition book, with History, Geography and Mental ability.

∴ Reprographics book is on Geography, Polity and Mental ability.

Holy Faith's book is on Polity, Mental ability and Science; hence, it is 1st edition and is stacked together with Reprographic book. Hence, the order of the books from top to bottom are Prinston, Johnson, Holy Faith, Reprographic and Penguin.

Johnson and Prinston publisher's books are 2nd editions and both of them are on Science, History and either Polity or Mental ability. The final distribution is as follows.

Order from Top	Publishers	Edition	Subjects
1	Prinston	2	History, Science, Polity or Mental ability.
2	Johnson	2	Science, History, Polity or Mental ability.
3	Holy faith	1	Science, Polity and Mental ability.
4	Reprographic	1	Geography, Polity and Mental ability.
5	Penguin	2	Geography, History and Mental ability.

4. The second book from the bottom is 1st edition.

5. The book published by Penguin publishers is on History and it is the only correct combination.

6. Reprographic and Holy Faith books are on Polity.

7. Geography is published by the least number of publishers.

Solutions for questions 8 to 10: The given information can be tabulated as shown below.

Jockey	Horse	Race type	Gambler	Position
H	Pony	Flat		
I	✘ Nunny	✘ Harness	✘ Rahul	2nd
J				
K	Honey	Endurance	✘ Rohan	
L	✘ Nunny			
M			Lohith	1st/2nd

From (iv), (vi) and the above, J rides on Nunny. Honey and M finished the race in the 1st position.

From (i) and the above, Tony's jockey is L.

From (vii), (ii), (v) and the above, Rahul bets on Pony; Nunny participates in Harness race.

From (viii) and the above, I rides on Sony, M rides on Rony and Rohan bets on Tony. I and M participate in Hunt Racing. Krish bets on Honey.

∴ The final distribution is as shown below.

Jockey	Horse	Race type	Gambler	Position
H	Pony	Flat	Rahul	2nd
I	Sony	Hunt	Kamal	2nd
J	Nunny	Harness	Kundan	1st
K	Honey	Endurance	Krish	1st
L	Tony	Harness	Rohan	3rd
M	Rony	Hunt	Lohith	1st

8. Only (i) and (ii) are true.

9. J finished the race in the 1st position.

10. Lohith bets on Rony.

Solutions for questions 11 to 14: It is given that, there are three students and three teachers among six persons A through F.

Each one of them took three tests among six tests 1 through 6.

2 tests are only for teachers.
2 tests are only for students.
2 tests are common for all.

From (1), no test is taken by both A and D. So, among A and D one is a student and the other one is a teacher in any order. From (3), A did not take test 5.

From (2), Test 5 is for only teachers. Hence, A is a student and D is a teacher.

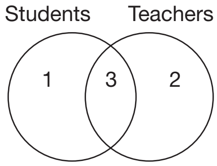
From (1) and (4), we can conclude that tests 3 and 6 are common for both teachers as well as students.

From (2), test 4 is only for students.

From (3), as E took tests 1, 4 and 6.

Test 1 is only for students and test 2 is only for teachers.

The above results can be represented as follows.



The different possible combinations are:

For students	For teachers
143	253
146	256
361	236
364	536

As, E took the tests 1, 4 and 6, E is a student.

A is a student and D is a teacher.

From (1), B is a Teacher.

As only test 6 is given by both E and F. E is a student and F is a teacher, C is a student.

A – Student
B – Teacher
C – Student
D – Teacher
E – Student
F – Teacher

From (4), A and B took only test 3 together.

A took tests 1, 4 and 3.

D took tests 2, 5 and 6 (from (1)).

Hence, C took either 1, 3, 6 or 3, 4, 6, i.e., 3, 6 and (1 or 4).

Similarly, F took 3, 6 and (2 or 5).

B took test 3.

The final combination is as follows.

			Tests	
A	Student	1	3	4
B	Teacher	3		
C	Student	3	6	1/4
D	Teacher	2	5	6
E	Student	1	4	6
F	Teacher	3	6	2/4

11. D took the tests 2, 5 and 6.

12. Test 1 and 4 are only for students.

13. Either four or five persons took test 6.

14. Among the given test C took test 3.

Solutions for questions 15 to 17: Let us take down the data as below:

(i) Three friends – Anand, Bhuvan and Chander (A, B, C).

(ii) Six singers/cassettes: UB40, Vanessa, Williams, Xavier, Yellows and Zoloto (U, V, W, X, Y, Z).

(iii) The three friends choose 5, 4 and 3 cassettes in any order.

(iv) Exactly 2 different singers' cassettes belong to all the three (2 singers common to all the three friends).

(v) $B \neq U$; $A \neq Z$; $C \neq W$.

(vi) $A > C$ (Number of cassettes)

(vii) Each person has either U or Z, but not both.

(viii) If a person has Vanessa, he must have Yellows. (If $V_{\checkmark} \Rightarrow Y_{\checkmark}$)

(ix) If a person has Yellows, he must have Xavier. (If $Y_{\checkmark} \Rightarrow X_{\checkmark}$)

(x) Vanessa is bought by exactly one person.
$\Rightarrow$ Vanessa is owned only by the person who has 5 cassettes.

(xi) Each cassette must be bought by at least one person.
Let us analyse the given data. We assume that V is bought by all the three friends (condition (iv), then Y must also be bought. But if Y is bought by all the three, then they must buy X also [From conditions (viii) and (ix)]. Now there will be three different cassettes (V, Y, X), which are bought by all the three, which violates condition (iv) (exactly two cassettes should be common among all the three friends). Hence, these two singers' cassettes, which are common among all the three friends must be Y and X. Also, A does not have Z, then he must have U [condition (vii)]. Similarly, B does not have U, then he must have Z. Now we get the following arrangement:

	U	V	W	X	Y	Z
$A \neq 3$	✓	..	..	✓	✓	✗
B	✗	..	..	✓	✓	✓
$C \neq 5$	..	✗	✗	✓	✓	..

Now, as A has more cassettes than C, hence, A cannot have least (i.e., 3) number of cassettes. Also, C cannot have maximum (i.e., 5) number of cassettes.

According to condition (vii), C must have exactly one cassette out of U and Z (hence, C's total now is 3 cassettes) and he will not have V. Thus

C = 3 cassettes
A = 4 or 5 cassettes
$\Rightarrow$ B = 4 or 5 cassettes.

15. C has the least number of cassettes, i.e., three.

16. X and Y are not only bought by two people, but by three people. UB40 and Zoloto cannot be bought together. A does not have Zoloto and C does not have W. Hence, only two people having same two cassettes is not possible.

17. There are only two arrangements which are possible. They are:

A	:	5	4
B	:	4	5
C	:	3	3

Solutions for questions 18 to 20: It is given that A did not go with C or D means A went with B or E or F or G or H (i)

Similarly,

B went with A or C or D or F or H (ii)
C went with A or B or E or G or H (iii)
D went with A or C or E or F or G (iv)
E went with B or C or D or F or H (v)
F went with B or C or D or E or H (vi)
G went with B or D or E or F or H (vii)
H went with B or D or E or F or G (viii)

From (i) and (v), we get:
A and E do not go with each other.
From (i) and (vi), we get:
A and F do not go with each other.
From (i) and (vii), we get:
A and G do not go with each other.
From (i) and (vii), we get:
A and H do not go with each other.
Hence, A and B go together.

Similarly, we can determine the names of the other people who go together. After this is done we get the following table.

	A	B	C	D	E	F	G	H
A	✗	✓	✗	✗	✗	✗	✗	✗
B	✓	✗	✗	✗	✗	✗	✗	✗
C	✗	✗	✗	✗	✓	✗	✗	✗
D	✗	✗	✗	✗	✗	✗	✓	✗
E	✗	✗	✓	✗	✗	✗	✗	✗
F	✗	✗	✗	✗	✗	✗	✗	✓
G	✗	✗	✗	✓	✗	✗	✗	✗
H	✗	✗	✗	✗	✗	✓	✗	✗

We can find out the places visited by them using the same logic as given above.

The places visited by them are as follows.

	A	B	C	D	E	F	G	H
Resort	✗	✗	✗	✗	✗	✓	✗	✓
Beach	✗	✗	✓	✗	✓	✗	✗	✗
Cinema	✓	✓	✗	✗	✗	✗	✗	✗
Hotel	✗	✗	✗	✓	✗	✗	✓	✗

A and B went to the Cinema.
C and E went to the Beach.
D and G went to the Hotel.
F and H went to the Resort.

18. B goes with A.

19. E went with C and they visited the Beach.

20. If D and H lied about the places they visited, then H goes along with G to the Hotel and D goes along with F to the Resort.

Solutions for questions 21 to 24: The given information can be tabulated as shown below.

Person	Bank Branch	Black Money (in cores)
J	London	
K	✗ Geneva	₹8697
L	✗ Geneva	✗ ₹28,116
M		✗ ₹3624
N		
P	London	✗ ₹28,116
Q		

From (iv) and (v), M and Q kept their black money in Geneva. N's black money is either ₹6734 or ₹4537. K and N kept their black money in Switzerland. L kept his black money in London. J's black money is ₹28,116.

From (ii) and the above, P's black money is 19,898.

From (i) and above, M's black money is ₹2436 and Q's black money is ₹3624. L's black money is either ₹4537 or ₹6734. Therefore, the final distribution is as shown below.

Person	Bank Branch	Black Money (in cores)
J	London	₹28,116
K	Switzerland	₹8697
L	London	₹4537/₹6734
M	Geneva	₹2436
N	Switzerland	₹6734/₹4537
P	London	₹19,898
Q	Geneva	₹3624

21. Either L's or N's black money is ₹4537 crores.

22. Either L's or N's black money.

23. Only (i) and (iii).

24. Except in (B), in the remaining pairs, the first person's black money is more than the second person's black money.

Solutions for questions 25 to 27: From (i), (ii) and (iii) none among C, F, D, E and B is wearing a red coloured dress. Hence, A is wearing the red coloured dress.

∴ From (iv), A, B, D or F is not wearing a violet coloured dress.

From (v) E is not wearing a violet coloured dress.

C is wearing the violet coloured dress.

Again, E is not wearing a white, blue or green coloured dress.

E is wearing the yellow coloured dress.

None among B, C and E is wearing a white coloured dress,

∴ F is wearing white coloured dress.

As D is not wearing a blue coloured dress, D is wearing the green coloured dress.

∴ B is wearing the blue coloured dress. The final distribution table is as follows.

A – Red
B – Blue
C – Violet
D – Green
E – Yellow
F – White

25. D is wearing a green coloured dress.

26. A is wearing a red coloured dress.

27. F is wearing a white coloured dress.

Solutions for questions 28 to 31: It is given that six different films are telecasted in four channels, in three slots. The films are Karate, Micky, Tom, Vali, Hanuman and Kumphu.

The channels – W, X, Y, Z.

Slots – 8 to 10, 10 to 12, 12 to 2.

Categories – Comedy, Horror, Action.

As the number of films in any category is not the same, the number of movies is 1, 2 and 3 in different categories.

As no channel telecast movies in two consecutive slots, a channel which telecasts in (10 to 12) slot does not telecast any other movie. Kumphu and Tom are telecast by the same channel, hence, neither of these movies were telecast in (10 to 12) slot. The movies Micky and Karate are telecast in the same slot. It is also given that Karate is telecast by the channel W in (10 to 12) slot. Hence, Micky is also telecasted in the same slot by another channel. Micky is the only comedy film. Hanuman and Vali are telecasted by the same channel in the slots (8 to 10) and (12 to 2). As Vali is not telecasted in (8 to 10) slot, it is telecasted by the channel X in (8 to 10) slot and Vali in (12 to 2). As Hanuman and Tom are not telecasted in the same slot, Tom is telecasted in (12 to 2) slot and Kumphu in (8 to 10) slot by either Y or Z channel. As Karate is not an action movie, it is a horror film as there is only one comedy film. As Vali is a horror movie, Hanuman is an action movie (as no channel telecast two films of the same category).

Between the films Tom and Kumphu, one is a horror film and the other one is an action movie, in any order. The above results can be represented as follows.

	8 to 10	**10 to 12**	**12 to 2**
W		Karate	
X	Hanuman		Vali
Y/Z	Kumphu		Tom
Z/Y		Micky	

Comedy	**Horror**	**Action**
Micky	Karate Vali Tom/Kumphu	Hanuman Tom/Kumphu

28. Hanuman – (8 to 10) is the correct combination.

29. Vali and Tom are telecasted in the slot 12 to 2.

30. Hanuman is telecasted by channel X is true.

31. Horror movies are maximum in number.

Solutions for questions 32 to 34: The given information can be tabulated as shown below.

Person	**Cabin**	**Game**	**Year of release**
Anju	4	Mirror's Edge	2009
Sanju	2	✗ The Last Express ✗ Star Craft	✗ 1997
Raju	3/5	Max Payne	2001
Manju		Mount & Blade	2008
Billa	1		
Sruthi			

From (i) and the above, Gothic 2 was released in 2002.

From (iii) and the above, Manju played the game in either Cabin 5 or Cabin 3. Sruthi played Star Craft in Cabin 6 and it was released in 1998. Hence, Billa played 'The Last Express' and it was released in 1997. Sanju played 'Gothic 2'.

∴ The final distribution is as shown below.

Person	**Cabin**	**Game**	**Year of release**
Anju	4	Mirror's Edge	2009
Sanju	2	Gothic 2	2002
Raju	3/5	Max Payne	2001
Manju	5/3	Mount & Blade	2008
Billa	1	The Last Express	1997
Sruthi	6	Star Craft	1998

32. Sruthi.

33. 1997.

34. Either 5 or 3.

Solutions for questions 35 to 30: The given information can be tabulated as shown below.

Employee	Place	Month
P	✗ Mumbai	✗ September ✗ April
Q	✗ Mumbai	✗ April
R	✗ Mumbai	✗ April
S		December
T	Delhi	October/April
U		
V	✗ Hyderabad ✗ Mumbai	September
W		

Given, U and R went to the same place. Hence, U should not go to Mumbai and S and W went to Mumbai, U and R went to Hyderabad, since only one of P, Q and R went to Hyderabad and exactly two employees went to the same place. Given, R and W went in the same month. Hence, it is not April and also neither December nor September, since, exactly two employees went in the same month. Hence, it is October. Therefore, T went in April, P went to Bangalore in December, Q went in September, U went in April and V went to Bangalore.

∴ The final distribution is as shown below.

Employee	Place	Month
P	Bangalore	December
Q	Delhi	September
R	Hyderabad	October
S	Mumbai	December
T	Delhi	April
U	Hyderabad	April
V	Bangalore	September
W	Mumbai	October

35. Q went to Delhi.

36. R went in October.

37. (Q, T) went to the same place.

Solutions for questions 38 to 40: The given information can be tabulated as shown below.

Jockey	Horse	Race type	Gambler	Position
H	Pony	Flat		
I	✗ Nunny	✗ Harness	✗ Rahul	2nd
J				
K	Honey	Endurance	✗ Rohan	
L	✗ Nunny			
M			Lohith	1st/2nd

From (iv), (vi) and above, J rides on Nunny. Honey and M finished the race in the 1st position.

From (i) and above, Tony's jockey is L.

From (vii), (ii), (v) and above, Rahul bets on Pony; Nunny participates in Harness race.

From (viii) and above, I rides on Sony, M rides on Rony and Rohan bets on Tony. I and M participate in Hunt Racing. Krish bets on Honey.

∴ The final distribution is as shown below.

Jockey	Horse	Race type	Gambler	Position
H	Pony	Flat	Rahul	2nd
I	Sony	Hunt	Kamal	2nd
J	Nunny	Harness	Kundan	1st
K	Honey	Endurance	Krish	1st
L	Tony	Harness	Rohan	3rd
M	Rony	Hunt	Lohith	1st

38. Only (i) and (ii) are true.

39. J finished the race in the 1st position.

40. Lohith bets on Rony.

EXERCISE-3

Solutions for questions 1 to 4: We have the following table which can be filled as we go through the statements.

	A	B	C	D	E
Car					
Colour					

From statements (i), (ii), (iii), (iv) and (v), we can fill up the following.

	A	B	C	D	E
Car			Mercedes	Fiat	
Colour		Green			

We have used all the information given and hence, we can now start answering the questions.

1. If A owns a blue Sierra, then B owns Audi (because E cannot own Audi) and hence, E owns a Maruti. Similarly, if A's car is blue, then E's car will be red or black. Hence, E will have red Maruti or black Maruti.

2. If A owns white Audi, then E will own Sierra (because B cannot own a Sierra). Only choice (D) has Sierra. (Also note that if A owns a white car, then the colour of E's car can be blue, red or black).

3. If A's car is Maruti, then E can only own a Sierra. If A's car is red and D's white, then E can only own a blue car (because C's car cannot be blue). Hence, E has a blue Sierra.

4. If E owns a red car and A owns a white car, then D can have only a blue car (because C cannot have a blue car).

Solutions for questions 5 and 6:

5. It is given that each of Kapil, Karan, Kusum and Kiran is a lecturer in different subject. Kapil teaches Maths, Kiran is a lecturer in Chemistry and Kusum is not a lecturer in Physics.

$\Rightarrow$ Kusum is a lecturer in Biology and Karan is a lecturer in Physics.

It is given that Kunal, Kamat and Karishma belong to the same subject group as Kapil, i.e., Maths.

It is given that Kamal is a lecturer of Chemistry and Kiran and Kate teach the same subject.

Hence, Kamal, Kiran and Kate teach Chemistry.

Kamini is in the same subject group as Karan, i.e., Physics.

It is given that Amar, Beena, Chander and Deepak are the professors of the subject groups with a lecturer strength of 4, 3, 2 and 1, respectively.

As per the given instructions, we get the following arrangements:

Maths	Physics	Biology	Chemistry
Kapil Kamat Kunal Karishma	Karan Kamini	Kusum	Kiran Kate Kamal
4	2	1	3
Amar	Chandar	Deepak	Beena

Only (B) is correct.

6. None of the given names belongs to the subject group Maths.

Solutions for questions 7 to 10: The given information can be represented in the tabular form as follows.

Name of the people	City	Car	
P		BMW	
Q			
R	Cochin (×) Bengaluru (×) Baroda (×)		
S	Hyderabad		Different cars
T	Pune (×) Bengaluru (×) Baroda (×)		Same cars
U	Chennai	Honda City	
V	Kolkata	Honda Brio	
W	Bengaluru (×)		

From the given information, V is not travelling by Honda Brio, we can say that V is travelling by Honda City, since the person who is travelling by BMW is from Pune.

Also, we can say that only two people are travelling by BMW.

From the given information, R and W are travelling by the same car, we can say that they are travelling by Honda Brio.

∴ T, U and V are travelling by Honda City.

As S is from Hyderabad, he cannot travel by BMW.

S, R, W are travelling by Honda Brio.

Q is from Pune and is travelling by BMW.

P is from Bengaluru, since R, T and W cannot be from Bengaluru.

∴ W is from Baroda.

As R is not from Cochin, T is from Cochin.

∴ R is from Noida.

The final arrangement is as follows.

Name of the person	City	Car
P	Bengaluru	BMW
Q	Pune	BMW
R	Noida	Honda Brio
S	Hyderabad	Honda Brio
T	Cochin	Honda City
U	Chennai	Honda City
V	Kolkata	Honda City
W	Baroda	Honda Brio

7. R is from Noida.

8. R, S and W are traveling by Honda Brio.

9. W is travelling by Honda Brio and is from Baroda.

10. Choice (C) is true.

Solutions for questions 11 to 13: Given that only G and M, went to Zoo – III and one of them has a dog. E has a monkey and went to Zoo – I. The person, who went to Zoo – II has a tiger, but is not C. C and the person, who has a cat, went to Zoo – II. Neither A nor K has a tiger, but one of them went to Zoo – II. A and H went to the same Zoo. H has an elephant. So, A cannot go to Zoo – II as already the person who has a tiger and C went to Zoo – II and A cannot go to Zoo – III as only G and M went to Zoo – III. Hence, A and H went to Zoo – I and hence, K went to Zoo – II. As K does not have a tiger, K has a cat. Three people A, E and H went to Zoo – I, so P went to Zoo – II and therefore, P has a tiger. The person, who has a camel did not go to either Zoo – II or Zoo – III. So A, who went to Zoo – I has a camel. The person, who went to Zoo – III, does not have a lion, hence, C has a lion. One among G and C has a horse. As C already has a lion, G has a horse and therefore, M has a dog.

∴ The tabular form of the people, their animals and the Zoo to which they went is as follows.

Person	Zoo	Animal
A	I	Camel
C	II	Lion
E	I	Monkey
G	III	Horse
H	I	Elephant
K	II	Cat
M	III	Dog
P	II	Tiger

11. A, H and E went to Zoo – I.

12. C has a lion.

13. P has a tiger.

Solutions for questions 14 to 16: The given information can be tabulated as shown below.

Name of the person	Colony	Street
A	xP	I
B	xP	
C	R	
D		
E		
F	P	
G	xQ	III
H	R	

From the given information, A and E belong to neither the same colony nor the same street. So, E belongs to either street II or street III. D and F belong to the same colony. Hence, D belongs to colony P. G and C belong to the same street, hence, C belongs to street III. E and C do not belong to the same street. Hence, E belongs to street II as E cannot belong to the same colony as A and C. B and C do not belong to the same colony, hence, B belongs to colony Q. C, D and E belong to different colonies and different streets. Hence, D belongs to street I as C and E belong to street III and street II, respectively and E belongs to colony Q as C and D belong to colony R and P, respectively. A and B neither belongs to the same colony nor colony P. Hence, A belongs to colony

R. B and G belong to the same street. Hence, H belongs to street III. The given condition is that the people who belong to the same colony do not belong to the same street. Hence, H belongs to street II as A and C belong to street I and III, respectively and at least two and at most three people belong to each colony and each street. Hence, G belongs to colony P and F belongs to street II.

∴ The final arrangement is as shown below.

Name of the person	Colony	Street
A	R	I
B	Q	III
C	R	III
D	P	I
E	Q	II
F	P	II
G	P	III
H	R	II

14. G belongs to colony P.

15. The correct combination of person, colony and street respectively is 'F – P – II'.

16. D, F and G belong to the same colony.

Solution for questions 17 to 19:

Males	Females	Salary (₹ lakh/annum)
Kambli		$x + 4$
Kumble	Karishma	$x + 1$
Kamlesh		X
Kareem		
Kishan		6

$x + y$ is a natural number and $x \geq 1$

$\Rightarrow x$ is a natural number.

Also, from (4), $x < 6$

If $x = 5$, then salary (in lakh/annum) of husband of Karishma

$= x + 1 = 6$.

This is not possible as each earns a different salary

$\Rightarrow x \neq 5$.

If $x = 4$

Salary (in lakh/annum) of Kumble $= x + 1 = 5$.

This contradicts rule (2).

If $x = 2$

Salary of Kambli (in lakh/annum) $= x + 4 = 6$ salary of Kishan.

This is not possible as each earns a different salary.

If $x = 1$,

Salary (in lakh/annum) of Kambli $= x + 4 = 5$,

This implies, from rule (2), that Kambli must be married to Kamini, which however violates the condition (5) $\Rightarrow x \neq 1$

The only possible value of $x = 3$.

Males (husband)	Females (wives)	Salary (₹ lakh/annum)
Kumble	Karishma	4/3/2
Kamlesh	(Kareeena/Kunti)	3/2/1
Kareem	Kamini	5
Kishan	Kirti	6
Kambli	Kunti/Kareena	9

17. Kareem earns ₹5 lakh/annum.

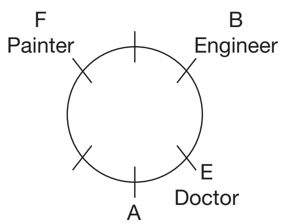
18. Kishan is married to Kirti.

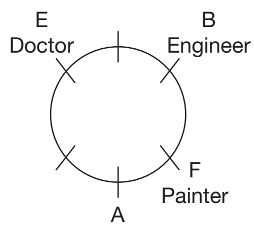
19. Kambli is not married to Kunti, implies Kamlesh is married to Kunti, then the husband of Kunti earns the minimum salary.

Solutions for questions 20 to 22: B is either Engineer or Editor. Neither A nor D is a Doctor. But exactly one of them is an Accountant. Either E or F is Painter and C is either an Editor or an Accountant. But either A or D is an Accountant. Hence, C is the Editor and B is the Engineer. The information can be represented in the following table. Since E is the Doctor, F is the Painter.

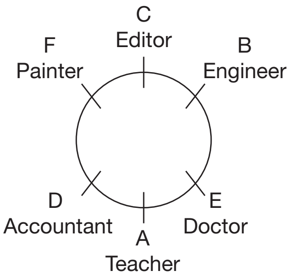
Person	Profession
A	Teacher/Accountant
B	Engineer
C	Editor
D	Accountant/Teacher
E	Doctor
F	Painter

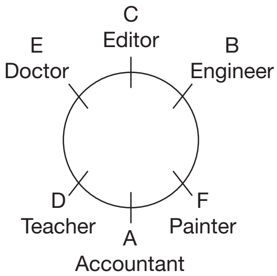
It is given that the Engineer is sitting second to the right of A and the Doctor and F are sitting opposite to each other. We have the following possible arrangements.





Since, the Editor is not sitting opposite to the engineer. Hence, the final arrangements will be as follows.





20. D is sitting opposite to the engineer.

21. The Painter is sitting to the left of the Accountant.

22. 'A is sitting opposite to the Editor' is definitely true.

Solutions for questions 23 to 25: It is given that there are three floors between C's floor and D's floor from top to bottom. Either B or E lives on the top floor and takes exam on Wednesday. Let us represent these points in the following floor arrangement.

Also given that neither F nor G takes the exam on Tuesday and there is one person between them who writes exam on Saturday.

Since there are only two floors below A's floor which means that A lives on the third floor. The possible arrangements are as follows:

Floor	Person	Day
7	B/E	Wednesday
6	F/G	Saturday
5	C	
4	G/F	
3	A	
2		
1	D	
1	D	

Floor	Person	Day
7	B/E	Wednesday
6		
5	C	
4	F/G	
3	A	Saturday
2	G/F	
1	D	

Since either B or E takes an exam on Wednesday, C takes an exam on Sunday and therefore, case (i) is eliminated. Also given that only two people live between B and G, B should live on the top floor and G should live on the fourth floor and as B takes an exam on Wednesday, G takes an exam on Monday.

Floor	Person	Day
7	B	Wednesday
6	E	
5	C	Sunday
4	G	Monday
3	A	Saturday
2	F	
1	D	

Since, the person who takes an exam on Tuesday does not live on an even – numbered floor. Hence, D takes an exam on Tuesday. Also given that the person who takes an exam on Thursday is adjacent to the person who takes an exam on either Saturday or Monday. The only possibility is F takes an exam on Thursday and E takes on Friday. The final arrangement is as follows.

23. E lives on the sixth floor.

24. A takes an exam on Saturday.

25. Three people live between E and F.

4 Selections

CHAPTER

Learning Objectives

In this chapter, we will:

- Understand how to interpret the conditions given in the question and write down the same in an unambiguous form.
- Find the number of teams that can be formed subject to the constraints/conditions.
- Learn to choose a team(s) without violating the given constraints.
- Learn how to choose a team(s) where multiple parameters are specified.

In this category of questions, a small group of items or people have to be selected from a larger group satisfying the given conditions. The conditions will specify as to when a particular item or person can be included or cannot be included in the subgroup. For example, the condition may specify that two particular people should always be together or that two particular people should not be together.

Sometimes, the conditions given for selection or non-selection of items or people may be based on logical connectives like if-then, either-or, unless, etc. You should be careful in interpreting the logical connectives used in the conditions.

Method of answering the questions:

- Step (i): Analyse the given conditions.
- Step (ii): Try to combine conditions.
- Step (iii): Apply the conditions to the questions.

SOLVED EXAMPLES

Directions for questions 4.01 to 4.05: These questions are based on the following information.

Amit, Bittu, Chintu, Dumpy, Falgun, Hitesh, Ronit, Purav and Saurav are nine players from among whom three teams consisting respectively of 4 members, 3 members and 2 members must be formed subjecting to the following conditions.

Chintu must have three more players with him while Dumpy must have only two more with him.

Chintu and Saurav cannot be in the same team.

Purav and Bittu cannot be in the same team.

Ronit and Hitesh must be in the same team.

4.01: If Dumpy, Falgun, Purav form the team of 3 members, then which of the following must be true?

(A) Hitesh must be in a team with Bittu.

(B) Saurav must form a two-member team with Amit or Chintu.

(C) Saurav must form a two-member team with Bittu or Amit.
(D) Chintu should form a team of 4 members with Hitesh, Ronit and Amit.

4.02: If Dumpy takes Amit as a part of his three-member team, which of the following must go into Chintu's team?
(A) Bittu and Hitesh
(B) Hitesh and Ronit
(C) Purav and Ronit
(D) Purav and Falgun

4.03: If Chintu and Falgun are together and Saurav is in the team of two members, then how many sets of different teams are possible?
(A) 4 (B) 3
(C) 2 (D) 1

4.04: If Chintu does not have Purav in his team and the two-member team consists of Saurav and Amit, then Chintu should take
(A) Hitesh, Bittu and Ronit
(B) Bittu but not Ronit
(C) Bittu and Falgun
(D) Hitesh and Ronit

4.05: If Purav is in the same team as Chintu and Falgun, then Saurav must be in the same team as
(A) Bittu (B) Bittu and Amit
(C) Amit (D) Bittu and Dumpy

Solutions for questions 4.01 to 4.05: It is given that:

Chintu must form a team of 4 members only.
Dumpy must form a team of 3 members only.

Since Chintu and Dumpy are in two different teams, let us for a convenience, denote the two teams as the respective teams of these two persons. Let us call the team with four members as the first team and the team with three members as the second team. The third team should have two people.

Number of members		
4	3	2
Chintu	Dumpy	Saurav
	Saurav	

Now let us take the other conditions and fill them up in the table above.

Chintu and Saurav cannot be in the same team.
Saurav will be in the second or the third team.
Purav and Bittu cannot be in the same team.
Hitesh and Ronit must be in the same team.

We cannot represent these two conditions right now in the table above but we will use them as we go along.

4.01: If Dumpy, Falgun, Purav form the team of 3 members, then Saurav should be in the third team.
Since Hitesh and Ronit must be in the same team, they have to be in the first team. That leaves only Amit or Bittu to be with Saurav in the third team.
(Also, note that we can eliminate choice (B) easily.)

4.02: Dumpy takes Amit as a member of his team.
If we take Hitesh and Ronit as the two members of the third team, then Saurav has to be in the second team, in which case we will have both Purav and Bittu coming into the same team, the first team, which is not possible.
Since Saurav cannot be in Chintu's team and Purav and Bittu cannot be in the same team, the three people required for Chintu's team will have to be Hitesh and Ronit, Falgun or Purav or Bittu.

4.03: Let us analyse the conditions. It is given that Chintu and Falgun are together, whereas Saurav is in the team of two members. Let us fill up these details in the box that we made above and then see in how many ways we can fill up the remaining cells in the box.

Chintu	Dumpy	Saurav
Falgun		

First let us look at Hitesh and Ronit who must be in the same team.
They can go into the first team or the second team. Let us consider these two cases.
Case 1: Hitesh and Ronit go into the first team. Then, one out of Bittu and Purav will go into the third team and the other into the second team. This gives rise to two ways of forming the teams, one with Bittu in the second team and the other with Bittu in the third team.
Case 2: Hitesh and Ronit go into the second team.

In this case too, one out of Bittu and Purav will go into the third team and the other into the second team. Hence, this will also give rise to two ways of forming the teams.
Hence, there are total four ways of forming the teams.

4.04: Let us use the table that we built in the initial analysis and fill up the details that we have in this problem.
Since the two-member team is already formed and Chintu does not take Purav, hence, Purav will have to go into the second team.

Chintu	Dumpy	Saurav
	Purav	Amit

Since Ronit and Hitesh have to be in the same team, they should go into the first team. Since Bittu cannot go with Purav, he should also be in the first team. This leaves Falgun for the second team. Thus, we can fill up the table as follows:

Chintu	Dumpy	Saurav
Ronit	Purav	Amit
Hitesh	Falgun	
Bittu		

4.05: If Purav is with Chintu and Falgun, then Bittu cannot be with them. Since Ronit and Hitesh should be together, the only other person left is Amit. These four members form the first team.
If Hitesh and Ronit together form the two-member team, then Bittu and Saurav will be part of the three-member team.
Instead, if Hitesh and Ronit are in the three-member team, then Saurav and Bittu will form the two-member team.
In either case, Saurav and Bittu are together in one team.

Directions for questions 4.06 to 4.09: These questions are based on the following information.

A, B, C, D, E, F and G are seven players. They form two teams of two players each and one team of three players. A and B cannot be in the same team. B and C cannot be in the same team whereas E and F must be in the same team. G and D cannot be in the same team.

4.06: If C, D and A form a team of three players, which of the following can be the members of one of the other teams?
(A) A and E (B) G and B
(C) E and F (D) Both (B) and (C)

4.07: If E, F and G form a team of three members, then in how many ways can the remaining two teams of two players each be formed?
(A) 2 (B) 4
(C) 3 (D) 1

4.08: If D and A are not in the same team, then altogether in how many ways can the teams of two members be formed?
(A) 4 (B) 7
(C) 8 (D) 5

4.09: If B, E and F form a team of three members, which of the following cannot be the two teams of two members each?
(A) AC, GD (B) AD, CG
(C) AG, CD (D) Both (A) and (B)

Solutions for questions 4.06 to 4.09: Let Team I be of 3 players, Team II be of 2 players and Team III be of 2 players.

It is given that A and B cannot be together. We will represent it as A × B.

Similarly, we have B × C and G × D.

E and F must be in the same team. So, E and F can form a team of 2 members on their own or can form a team of 3 members with another person.

Let us now take up the questions and work them out.

4.06: Given that C, D, A form a team of 3 members, one of the other teams has to have E and F together. Hence, B and G should form one team. Choice (D)

4.07: Given that E, F, G form a team of 3 players. Since A and B or B and C cannot be in the same team, we must necessarily have A and C together in one team and B and D in the other team. So, the teams can be formed only in one way.

4.08: Given that A and D are not in the same team.
Hence, A × B, B × C, G × D and A × D.
We already know that E and F must be in the same team. They may form a team of 3 members or they themselves be a team of 2 members. Let us consider the above two possibilities and then fill up the other teams. They can be formed as follows:

	Team I	Team II	Team III
1.	A E F	B D	C G
2.	A E F	C D	B G
3.	B E F	A G	C D
4.	C E F	A G	B D
5.	D E F	A C	B G
6.	G E F	A C	B D
7.	A C G	B D	E F

Thus, the teams can be formed in 7 ways.

4.09: If B, E, F form a team of 3 members, then the two members teams must be formed from A, C, D, G. The teams can be AD and CG or AG and CD. As D and G cannot form a team, AC and GD cannot be formed. (Please note that we can answer this question from the answer choices.)
From choice (A), we find that G and D are together in one team which is not possible. Thus, choice (A) is the answer.

Directions for question 4.10: Select the correct alternative from the given choices.

4.10: At least two boys out of A, B, C and D and at least two girls out of P, Q, R and S have to be chosen to form a group of 5 members.
Neither A nor C can go with Q.
Neither P nor S can go with B.
Q and R cannot be together.

Which of the following is an acceptable team?
(A) ARCQP (B) ASQPD
(C) ASQRP (D) PSRAD

Solution for question 4.10:

4.10: The required group of 5 members must be formed with at least two boys from A, B, C, D and at least 2 girls from P, Q, R, S.
Answers (A), (B) and (C) can be ruled out as A and Q cannot be together.
In choice (D), P, S, R, A, D can be together without violating any of the given conditions.

EXERCISE-1

Directions for question 1 to 6: Select the correct alternative from the given choices.

1. Adam, Andy, Anil, Ann, Jack, John, James and Jill want to go to a nearby city. Only two vehicles, a van and a car are available. Only Alen and Jack know how to drive the van, hence at least one of them must be in the van.

Each vehicle has a seating capacity of exactly four people. Adam and Anil cannot go in the same vehicle and John and James must go in the same vehicle.

Which of the following cannot be the list of people who are in the car?

(A) John, James, Andy, Alen
(B) Adam, James, John, Jack
(C) Anil, James, John, Jill
(D) Anil, Andy, Jill, Alen

2. A group of five is to be formed from a group of nine students A, B, C, D, E, F, G, H and I.

If A is selected, then F is selected. If F is selected, then D is not selected.

G is selected only if I is selected.

If H is selected, then C is not selected. If B is not selected, then C is selected.

If D is selected, then who among the following must be selected?

(A) C (B) G
(C) D (D) I

3. Each of P, Q and R has to select two items from the six items, such as A, B, C, D, E and F.

If P selects A, then Q does not select E.

Only if R selects E, Q does not select B.

If P selects D, then R will not select C.

If P does not select F, then R will select B.

If Q selects E, then P selects

(A) B and D (B) F and D
(C) C and D (D) F and C

4. A team of three students is to be selected for a quiz competition, a group of students, namely Ankita, Chanchal, Surbhi, Neha and Kanchan such that if Chanchal is selected, then Kanchan should not be selected. Unless Surbhi is selected Neha is selected. Which of the following students must be selected?

(A) Ankita (B) Chanchal
(C) Surbhi (D) None of these

5. A team of four is to be selected from three boys, namely Ajay, Sujay, Vijay and three girls Ena, Meena, Deepa such that exactly two boys are selected. Ajay and Ena should not be selected together. If and only if Meena is selected, then Vijay is selected. In how many ways can the team be selected?

(A) 4 (B) 3
(C) 2 (D) 1

6. A team of three is to be selected from six people, namely from Pavan, Sravan, Raghavan, Aman, Dawan and Bhavan such that if one of Pavan and Sravan is selected then the other must not be selected. If one of Raghavan and Dawan is not selected, then the other must not be selected. If Aman is not selected, then who among the following will not be selected?

(A) Pavan
(B) Bhavan
(C) Raghavan
(D) More than one of the above

Directions for questions 7 to 9: These questions are based on the following data.

A team of five players is to be selected from a group of ten players, such as A, B, C, D, E, F, G, H, I and J.

(i) Exactly one of G and H must be selected.
(ii) H and A must be selected together, if selected.
(iii) B and F must be selected together, if selected.
(iv) F and J cannot be selected together.
(v) C and D cannot be selected together.

7. Which of the following statements must be true?

(A) If G is selected, then B is selected.
(B) If G is selected, then at least one of E and I is selected.
(C) If H and B are selected, then E cannot be selected.
(D) If J is not selected, then B is selected.

8. If G is selected, then which of the following can be the group of players who are not selected?

(A) H, A, F, D, I
(B) H, A, D, E, I
(C) H, C, D, J, A
(D) H, D, J, E, I

9. If G is not selected and J is selected, then the total number of possible selections are

(A) Four (B) Five
(C) Two (D) Six

Directions for questions 10 to 13: These questions are based on the following information.

From a group of five batsmen P, R, S, U and X and five bowlers Q, T, V, W and Y, a group of five players is to be selected. The group must consist of exactly two batsmen.

It is also known that:
(i) At most one among S and Q must be selected.
(ii) Exactly two among R, U, X and V must be selected.
(iii) If R or X is selected, then none among Q, V and T are selected.
(iv) If P is selected, then neither T nor W is selected.

10. Among the batsmen, who must be selected?
(A) R (B) U
(C) X (D) None of these

11. Among the bowlers who must be selected?
(A) Q (B) Y
(C) T (D) None of these

12. If X is selected, then who among the following must be selected?
(A) D (B) U
(C) P (D) Such a case is not possible

13. How many different groups of players can be selected?
(A) 2 (B) 3
(C) 4 (D) 6

Directions for questions 14 to 16: These questions are based on the following information.

A team of three people is to be selected from a group of five people, namely from A, B, C, D and E under the following constraints.
(i) If A is selected, then B must be selected.
(ii) If C is not selected, then E must be selected.

14. In how many ways can the team be selected?
(A) Eight (B) Six
(C) Seven (D) None of these

15. If D is not selected, then who must always be selected?
(A) A (B) B
(C) C (D) E

16. Which of the following is not a possible team?
(A) C, E, D (B) E, A, B
(C) C, B, D (D) A, B, D

Directions for questions 17 to 21: These questions are based on the following information.

A team is to be selected from nine people, namely from R, S, T, U, V, W, X, Y and Z under the following constraints.
(i) If either R or S is selected, then Y must not be selected.
(ii) At least one of W and Z must be selected.
(iii) Unless both T and U are selected then V is selected.
(iv) If and only if W is selected, then Y is selected.
(v) Whenever X is selected, then S must also be selected.

17. What can be the maximum possible number of people selected in a team?
(A) 4 (B) 5
(C) 6 (D) 7

18. What is the minimum possible number of people selected in a team?
(A) 2 (B) 3
(C) 4 (D) 0

19. Which among the following groups can form a team?
(A) TWYZ (B) XYWVU
(C) TUSX (D) None of these

20. In how many ways can a team of four people be selected?
(A) 12 (B) 13
(C) 14 (D) 15

21. If X is selected, then in how many ways can a team of four people be selected?
(A) 4 (B) 2
(C) 3 (D) 1

Directions for questions 22 to 25: These questions are based on the following information.

A team of delegates is to be formed from a group of ten people, N through W by subjecting to the following conditions.
(i) If Q is selected, then none among U, V or W can be selected. Also, U, V and W cannot be selected together.
(ii) If R is selected, then either S or T must be selected. But S and T cannot be selected together.
(iii) At least one out of N, O and P must be selected.
(iv) If P is selected, then neither N nor O can be selected.
(v) N and R cannot be selected together.
(vi) N and Q cannot be selected together.
(vii) P and V cannot be selected together.
(viii) P and W cannot be selected together.

22. If a team of four is selected and Q being one of them, then which of the following must be selected?
(A) R (B) O
(C) T (D) S

23. What is the maximum possible size of a selected team?
(A) Four (B) Five
(C) Six (D) Seven

24. What is the maximum possible size of the team if P is selected?
(A) Four (B) Five
(C) Six (D) Seven

25. In how many different ways can the team be selected if Q is selected?
(A) Seven (B) Eight
(C) Nine (D) Ten

Directions for questions 26 to 28: These questions are based on the following information.

Vijay asked Ajay to select 6 pens of different colours from the available ten colours, such as Orange, Red, Blue, White, Pink, Yellow, Black, Grey, Violet and Brown.

Vijay has laid down some conditions for Ajay, as given below:

(i) If Ajay selects the Blue pen, then he must select the Orange pen also and vice versa.
(ii) If Ajay selects the Grey pen, then he must select the Black pen also and vice versa.
(iii) If Ajay selects the Yellow pen, then he cannot select the Grey pen.
(iv) Exactly one of Red and Violet pens must be selected.

26. If Ajay does not select the Grey or the Brown pen, then among the following choices he can reject the

(A) Pink pen (B) White pen
(C) Red pen (D) Orange pen

27. Which of the following can be the list of colours of pens selected by Ajay?

(A) Yellow, Pink, White, Blue, Brown, Orange.
(B) Red, Blue, Orange, White, Grey, Yellow.
(C) Black, Grey, Brown, Violet, Pink, White.
(D) Red, Pink, Blue, Orange, Violet, White.

28. Which of the following can confirm the selection of pens?

(A) Blue and Yellow pens are selected.
(B) Red and Grey pens are selected.
(C) Orange and Red pens are not selected.
(D) Grey pen is not selected, but the Orange pen is selected.

Directions for questions 29 to 33: These questions are based on the following information.

A team of five members is to be selected from four boys, namely Arjun, Sreekar, Bhavan and Dawan and four girls, namely from Sheela, Rama, Karuna and Nayana under the following constraints.

(i) At least two girls and at least two boys must be selected.
(ii) Sreekar and Karuna cannot be selected together.
(iii) Unless Bhavan is selected, Nayana cannot be selected.
(iv) At most two of Arjun, Dawan and Rama can be selected.
(v) If Sheela is selected, then at most one of the other three girls can be selected.

29. If three girls are selected, then in how many ways can the team be selected?

(A) 5 (B) 4
(C) 3 (D) 2

30. If three boys are selected, then in how many ways can the team be selected?

(A) 7 (B) 8
(C) 9 (D) 10

31. Who must be selected?

(A) Arjun (B) Bhavan
(C) Sreekar (D) Dawan

32. If Sheela is selected, then who is the other girl who must be selected?

(A) Rama (B) Karuna
(C) Nayana (D) None of these

33. If Karuna and Sheela are selected, then who must be selected?

(A) Arjun
(B) Dawan
(C) Nayana
(D) More than one of the above

Directions for questions 34 to 37: These questions are based on the following information.

A company gives an opportunity to its employees to go for a vacation among the months April, May, August, November and December in a year. The employees are A, B, C, D, E, F, G, H, I, J, K, L, M and N. The company gives choice to the employees for selecting the month in which they want to go on a vacation.

But it is stipulated that at least two and at most three persons go on vacation in a month.

(i) F, I, and K want to go on vacation in the same month, but after the month in which J wants to go.
(ii) In May, only two employees want to go on a vacation, but they are neither D nor N.
(iii) M, N, A and E wanted to go in different months.
(iv) If M chooses to go in April, then L wants to go in November and G wants to go in May.
(v) J and H do not want to go in the same month, but one of them wants to go in November.
(vi) A and B want to go in consecutive month in that order. C and E want to go in consecutive months in that order. No two among A, B, C and E wants to go in the same month.
(vii) Only if G wants to go in August, then H and C want to go in the same month.
(viii) If L and N want to go in the same month, then J does not want to go in May.

34. If the company wanted to select five employees for a project, then who wanted to go for the vacation in different months, then which of the following can be the team?

(A) AHMKG
(B) ECHLD
(C) MNHKE
(D) None of these

35. The company has identified three of these employees for promotion and it so happened that these three employees wanted to go on vacation in December, then who among the following will be one among the three?

(A) D (B) L
(C) B (D) M

36. The employees want to go on vacation in November are selected as the three best employees of the company, then who among the following will be in that group?

(A) L, H (B) D, G
(C) A, H (D) G, A

37. The company selects a team of four employees, two of whom want to go on vacation in August, one in April and the other one in May.

Then which of the following cannot be a possible team?

(A) CFKE (B) JGIF
(C) IKFJ (D) FILM

Directions for questions 38 to 40: These questions are based on the following information.

A film-maker wants to select five child artists from a group of children, namely Sakshi Meghana, Neerav, Nihit, Kunal, Kundan, Nabhya and Sagar.

(i) Nihit and Neelam are enemies. So, if one of them is selected, the other one should not be selected.
(ii) Sakshi and Neerav are the best friends. So, if any one of them is selected, then the other one should also be selected.
(iii) The film-maker wants to select exactly one of Sakshi and Sagar.
(iv) Nabhya cannot be selected without Meghana. If both Meghana and Sagar are selected, then Nabhya will not be selected.
(v) Kundan and Nabhya are identical twins. So only one of them must be selected.
(vi) Kunal is selected only if both Meghana and Neerav are selected.

38. In how many ways can the five children be selected?

(A) Five (B) Three
(C) Two (D) Six

39. Who among the following will be selected?

(i) Neerav (ii) Nabhya
(iii) Meghana

(A) (i) (B) (ii) and (iii)
(C) (iii) (D) (i) and (iii)

40. Which of the following is true?

(i) Kundan is not selected.
(ii) Sagar is selected.
(iii) Both Neelam and Kunal are selected.

(A) Only (i)
(B) Only (iii)
(C) Both (i) and (ii)
(D) None of the three

EXERCISE-2

Directions for questions 1 to 3: These questions are based on the following information.

A team of three people is to be selected from six people, namely from Praveen, Rahul, Qureshi, Swathi, Tarun and Umesh confirming the following conditions.

(i) If at least one of Qureshi and Tarun is selected, then Umesh cannot be selected.
(ii) If at least one of Umesh and Praveen is selected, then either Swathi or Rahul must be selected.

1. If Praveen is selected, then who must not be selected?

(A) Qureshi (B) Tarun
(C) Umesh (D) None of these

2. If Qureshi is selected, then in how many ways can the team be selected?

(A) 4 (B) 5
(C) 6 (D) 7

3. If Tarun is selected, then who must be selected?

(A) Praveen (B) Swathi
(C) Rahul (D) Either Rahul or Swathi

Directions for questions 4 to 6: These questions are based on the following information.

In a beauty pageant, the judges have to select five contestants for final round from three different groups of contestants I, II and III. Each group contains five contestants. Group I contains A, B, C, D, and E group II contains G, H, I, J and K, group III contains F, L, M, N and P.

(i) At least one contestant and at most two contestants should be selected from each group.
(ii) B and D should not be selected together. H and M should be selected together.
(iii) Either E or P must be selected. If H is selected, then J must not be selected.
(iv) Only if C is selected then A should be selected. Unless G is selected, F can not be selected.
(v) Neither I nor K should be selected with either B or M.
(vi) Exactly one among C, L and J should be selected.

4. If M and P are selected, then which of the following is a valid group?

(A) MPCBJ (B) MPHCA
(C) MPJAB (D) MPDAH

5. If only K is selected from group II, then in how many different ways can the five be selected?
(A) Six (B) Five
(C) Four (D) None of these

6. If H, M and E are selected, then in how many different ways can the five be selected?
(A) Four (B) Six
(C) Five (D) Seven

Directions for questions 7 to 9: These questions are based on the following information.

A film-maker wants to select five child artists from a group of children, namely Sakshi Meghana, Neerav, Nihit, Kunal, Kundan, Nabhya and Sagar.

(i) Nihit and Neelam are enemies. So, if one of them is selected, then the other one should not be selected.
(ii) Sakshi and Neerav are best friends. So, if any one of them is selected, then the other one should also be selected.
(iii) The film-maker wants to select exactly one of Sakshi and Sagar.
(iv) Nabhya cannot be selected without Meghana. If both Meghana and Sagar are selected, then Nabhya will not be selected.
(v) Kundan and Nabhya are identical Twins. So only one of them must be selected.
(vi) Kunal is selected only if both Meghana and Neerav are selected.

7. In how many ways can the five children be selected?
(A) Five (B) Three
(C) Two (D) Six

8. Who among the following will be selected?
(i) Neerav
(ii) Nabhya
(iii) Meghana
(A) (i) (B) (ii) and (iii)
(C) (iii) (D) (i) and (iii)

9. Which of the following is true?
(i) Kundan is not selected.
(ii) Sagar is selected.
(iii) Both Neelam and Kunal are selected.
(A) Only (i) (B) Only (iii)
(C) Both (i) and (ii) (D) None of the three

Directions for questions 10 to 12: These questions are based on the following information.

Eleven players out of a total of sixteen players have to be selected for a cricket match. Among these sixteen players, there are twelve batsmen, nine bowlers and two wicket-keepers. There are six all-rounders (players who can both bat and bowl are known as all-rounders) and one wicket-keeper who is also a batsman. An ideal eleven consists of at least 6 batsmen, at least six bowlers and exactly one wicket-keeper. No wicket-keeper bowls.

10. If the wicket-keeper, who is also a batsman is selected in the ideal eleven, then what is the minimum possible number of players in the team who can only bowl?
(A) Zero (B) One
(C) Two (D) More than two

11. If the number of all-rounders should be kept at a minimum while selecting the team, then what is the least number of players who can only bat?
(A) Six (B) Five
(C) Four (D) None of these

12. Which of the following statements is never true?
(A) All the eleven players can bat in a team.
(B) In a team, 8 players can bat and 8 players can bowl.
(C) In a team, 7 players can bat and 7 players can bowl.
(D) In a team, when the number of all-rounders is kept at a minimum, then the number of players who can only bat is less to those who can only bowl.

Directions for questions 13 to 16: These questions are based on the data given below.

Four teams are to be formed from fourteen people. A team must consist of at least two people and no two teams can have the same number of people. Each person can be a member of exactly one team. Each of Rama, Ramya, Radha and Raksha must be a member of a different team. Each of Rohini, Padma, Priya and Priyanka must be a member of a different team. Pratima, Pratibha and Sudha must be in the same team. Shreya and Shalini must be in the same team. Rama cannot be in the same team with any of Padma, Priya and Priyanka. Ramya can be in the same team with neither Priyanka nor Priya. Radha cannot be in the same team with Priyanka. Swetha is in one of the teams.

13. How many possible ways are there to form the four teams?
(A) 24 (B) 120
(C) 64 (D) 6

14. Which of the following statements is not definitely true?
(A) Swetha is in a three-member team.
(B) There is a two-member team.
(C) Sudha is in a five-member team.
(D) Shreya is in the two-member team.

15. Which of the following additional statements is sufficient to know the composition of teams?
(A) Rohini and Padma are in teams with five and four members, respectively and Radha is not in a three-member team.

(B) Sudha and Shalini are in teams with five and four members, respectively and Swetha is not in a two-member team.
(C) Pratima and Swetha are in teams with five and three members, respectively and Shreya is not in a two-member team.
(D) None of these

16. Who of the following must be a member of a five-member team?
(A) Priya (B) Priyanka
(C) Pratima (D) Shreya

Directions for questions 17 to 19: These questions are based on the following data.

In a class of ten students, namely A, B, C, D, E, F, G, H, I and J, ranks are given to the top five students, such that the student who gets the highest marks will get the 1st rank, the student getting the second highest marks will get the 2nd rank and so on. It is also known that no two students get equal marks. D gets less marks than G and H gets less marks than I.

If F gets a rank, then D will not get a rank. Exactly one of B and G gets a rank.

If I gets a rank, then C will get a rank and vice-versa.

17. If E did not get a rank and I got less marks than F, then which of the following is definitely false?
(A) J or A gets a rank. (B) D gets a rank.
(C) B gets a rank. (D) F gets a rank.

18. If D and H got two consecutive ranks, then C would not get the
(A) 1st rank (B) 2nd rank
(C) 3rd rank (D) 4th rank

19. If F is not ranked and H gets more marks than G, then who among the following must be ranked?
(A) C (B) G
(C) A or E or J (D) B

Directions for questions 20 to 23: These questions are based on the following data.

A cricket team consisting of 11 players has to be selected from amongst 16 players, A through P.

Among these 16 players
(A) A, C, E, G, I, K, M, J and O are batsmen.
(B) B, D, F, G, H, J, M and P are bowlers.
(C) L and N are wicket-keepers.
(D) Any player who is both a bowler and a batsman is called an all-rounder. The Captain and the Vice-Captain are the all-rounders. The team is selected as per the following restrictions:
(1) The team should contain 5 batsmen, 3 bowlers, 2 all-rounders and a wicket-keeper and the Captain and the Vice-Captain must be selected.
(2) Neither G nor M is the Captain and neither J nor G is the Vice-Captain.
(3) The players mentioned in the following pairs must not get selected together:

J and N; B and F; D and H; D and P; E and I; A and E; and B and C.

20. Which two players are the Captain and the Vice-Captain of the team respectively?
(A) J and G (B) G and M
(C) J and M (D) G and K

21. Who are the three bowlers selected in the team?
(A) F, H and P (B) D, F and H
(C) F, H and B (D) B, D and F

22. Which of the following is definitely false?
(A) J, O, P, A, L and I are selected.
(B) L, O, F, H and C are not selected.
(C) B, D, E and N are not selected.
(D) M, I, K, P and O are selected.

23. If after the first match, in every following match, the Captain and the Vice-Captain exchange their job responsibilities (i.e., Captain takes up the Vice-Captaincy and the Vice-Captain takes up the Captaincy), then who would be the Vice-Captain in the 86th match?
(A) J (B) M
(C) Either J or M (D) L

Directions for questions 24 to 26: These questions are based on the following information.

Two groups of four people each are to be selected from a group of eight people A, B, C, D, E, F, G and H.

The following conditions are to be followed while forming the teams.
(i) If B is selected into a team, then D should also be selected in the same team.
(ii) If E is selected into a team, then C must be selected into the other team.
(iii) G and H should not be selected in the same team.
(iv) Each person is selected into exactly one team.

24. If B is selected into one team, then who among the following pairs must be selected into the other team?
(A) AF (B) CH
(C) EH (D) EG

25. If E and D are selected in one team, then who are the other two people who are selected into the same team?
(A) BH (B) GF
(C) GB (D) Either (BG) or (BH)

26. Which of the following is a valid team?
(A) DEGA
(B) CHFD
(C) EHAF
(D) More than one of the above

Directions for questions 27 to 31: These questions are based on the following information.

Two teams are to be selected from twelve people A, B, C, D, E, F, G, H, I, J, K and L under the following constraints.

(i) Each team must contain at least four people.
(ii) If G is selected in a team, then H must be selected in the other team.
(iii) If I or J is selected in any team, then L must not be selected in any of the teams.
(iv) Unless D or E is selected in a team, K is selected in any team.
(v) F can be selected in a team only if A is selected in the other team.
(vi) If B is selected in a team, then A should not be selected in that team.
(vii) No two of A, C and E can be selected in the same team.

27. If H is selected in a team, then which of the following cannot be the other team?
(A) K, D, L, G
(B) G, D, I, A
(C) A, G, L, F
(D) More than one of the above

28. If B and E are selected in a team and each team has five members, then in how many ways can the other team be selected?
(A) 8 (B) 6
(C) 5 (D) 9

29. If A is selected in a team and J is selected in the other team, then who among the following must be selected in the team where A is selected?
(A) G (B) H
(C) I (D) None of these

30. If X and Y are the names of the two teams and K is selected into the Team X and L is selected, then in how many ways can the Team Y be selected?
(A) 1 (B) 2
(C) 3 (D) 4

31. If F is selected in a team, then how many of the remaining people can be selected in the other team?
(A) 11 (B) 10
(C) 9 (D) 8

Directions for questions 32 to 34: These questions are based on the following information

Each of the six delegates A through F has to give a presentation on different topics that ranges from Terrorism, Trafficking, Poverty, Unemployment, Environmental Protection and Illiteracy, not necessarily in that order, at the World Congress. Each of the six delegates is from a different nation and they are from Brazil, China, India, Russia, South Africa and the USA, not necessarily in that order. Each of the delegates is carrying a folder of different colour among red, green, black, blue, orange and white, not necessarily in that order.

It is known that three delegates have to give presentation in one slot and the remaining three in the other slot. It is also known that:

(i) D is from South Africa and has a presentation on Unemployment.
(ii) The delegate carrying the red folder and the delegate giving a presentation on Traffcking cannot be in the same slot.
(iii) F is not from China or Russia and has to be in the same slot along with the delegate carrying the white folder.
(iv) The delegate giving a presentation on Environmental Protection and the delegate carrying the orange folder must be in the same slot.
(v) B is from India, C is giving a presentation on Illiteracy but she is not carrying the white folder.
(vi) A is carrying neither the red folder nor the white folder but should be with the delegate from China in the same slot.
(vii) E is carrying the green folder.

32. If B is carrying the orange folder, then which of the following must be false?
(A) B is selected for the same slot as A.
(B) F and D are selected for the same slot.
(C) The delegate giving presentation on Environmental Protection is carrying the green folder.
(D) The delegate giving presentation on Unemployment and A are selected for the same slot.

33. If A is presenting on Trafficking and F is carrying the red folder, then which of the following must be true?
I. The delegate carrying the white folder and the delegate from China are selected for different slots.
II. The delegate from China is carrying the orange folder.
III. The delegates giving presentation on Unemployment and Environmental Protection are selected for the same slot.
(A) I only (B) I and II only
(C) III only (D) II and III only

34. If E is from the USA and F is giving presentation on Trafficking, then which of the following must be false?
(A) F is from Brazil and he is not in the same slot with the delegate from Russia.
(B) The delegate carrying the white folder is same slot as the delegate from China.
(C) B is carrying the white folder and he is selected with the delegate carrying the orange folder.
(D) The delegate from Russia is selected with the delegate carrying the blue folder.

Difficulty Level-2: Moderate

Directions for questions 35 to 37: These questions are based on the following information.

A group of three girls, namely Anjali, Bharathi and Chandrika and four boys, namely Kiran, Lala, Manoj and Naveen are to be divided into two teams under the following constraints.

(i) Each team must have at least one girl and at least one boy and at least three people in total.
(ii) If Anjali and Bharathi are selected in a team, then the team must have only one boy.
(iii) Kiran and Lala cannot be in the same team.
(iv) Chandrika and Naveen can be in the same team, only if Bharathi is selected in that team.

35. If Kiran and Chandrika are in the same team, then in how many ways can the other team be selected?
(A) Six (B) Three
(C) Four (D) Five

36. If Manoj is not in the same team as Bharathi, then in how many ways can the teams be selected?
(A) Three (B) Four
(C) Five (D) Six

37. If three boys are selected into one team, then in how many ways can the teams be selected?
(A) Four (B) Five
(C) Three (D) Six

Directions for questions 38 to 40: These questions are based on the following information.

A team of four people is to be selected from seven people, namely Anuj, Bindu, Chanti, Dheeraj, Eswar, Farhaan and Ganesh under the following constraints.

(i) At most two of Chanti, Eswar and Ganesh can be selected.
(ii) At least one of Anuj and Bindu must be selected.
(iii) If Farhaan is selected, then neither Anuj nor Chanti can be selected.

38. If Dheeraj is selected, then in how many ways can the team be selected?
(A) Eight (B) Ten
(C) Nine (D) Eleven

39. If at most one of Farhaan and Ganesh can be selected, then in how many ways can the team be selected?
(A) 11 (B) 12
(C) 13 (D) 14

40. If Eswar is not selected, then in how many ways can the team be selected?
(A) Six (B) Seven
(C) Eight (D) Nine

EXERCISE-3

Directions for questions 1 to 4: These questions are based on the following information.

Ten candidates appear for an interview and six of them are selected. There are two M.As, two M.B.As, two M.C.As and four B.Techs among the candidates. If at least one M.B.A. candidate is selected, then exactly two B.Tech candidates must be selected and vice versa. Of the six selected candidates, exactly one must be an M.A. candidate.

1. Which of the following statements is definitely true, if two B.Tech candidates are selected?
(A) Two M.C.As and two M.As are selected.
(B) Only two M.B.As and only one M.C.A are selected.
(C) One M.B.A and two M.As are selected.
(D) Two MBAs are selected.

2. If two M.C.A. candidates are selected, then which of the following statements can be true?
(A) One M.B.A. and one B.Tech candidate is selected.
(B) Three B.Tech candidates are selected.
(C) Only one M.B.A. and two B.Tech candidates are selected.
(D) One M.A. and three B.Tech candidates are selected.

3. Which of the following statements is definitely FALSE?
(A) If four B.Tech candidates are selected, then two M.B.A. candidates must be selected.
(B) One M.A. candidate, one M.B.A. candidate and two M.C.A. candidates can be selected.
(C) One M.A., one M.B.A., two M.C.As and two B.Techs is a possible combination of selection.
(D) More than one of the above

4. Which of the following statements, if true, will make the selection of six candidates impossible?
(A) Two M.B.As are selected.
(B) Two M.C.As are selected.
(C) Two B.Techs are selected.
(D) No M.C.A. is selected.

Directions for questions 5 to 9: Read the information given below and answer the questions that follow.

Ajay, Bony and Chetan are three people who go to buy six items, such as P, Q, R, S, T and U. Each one of them buys two different items in such a way that if Ajay buys R, then Bony buys neither P nor S. If Bony buys Q, then Chetan buys neither U nor T.

5. If Ajay buys R and T, then Bony buys
(A) P and S (B) Q and U
(C) P and Q (D) S and U

6. If Bony buys Q and S, then Ajay must buy
(A) P and R (B) T and U
(C) P and T (D) R and U

7. If Chetan has to buy P and S, then which of the following must be true?
(A) Ajay bought R (B) Bony bought Q
(C) Ajay bought T (D) None of these

8. If Ajay buys P and Bony buys Q, then which of the following is true?
(A) Chetan buys R and S.
(B) Chetan can buy any two of P, R and S.
(C) Chetan can buy any three of P, R, S and T.
(D) Chetan can buy any two of P, R, S, T and U.

9. Which of the following is definitely true?
(A) Ajay buys R and Bony buys Q.
(B) If Chetan buys T or U, then Bony buys Q and S.
(C) If Ajay buys R, then Bony buys T.
(D) If Ajay buys R and Bony buys Q, then Chetan has to buy P and S.

Directions for questions 10 and 11: These questions are based on the following information.

Out of seven people from A, B, C, D, E, F and G, four are to be selected.

1. At least one of E or B must be selected and at most one between A or D can be selected.
2. Either C or D must be selected.
3. A, F and G cannot be selected together.

10. If A is selected, then who must be selected?
(A) D (B) F (C) G (D) C

11. If neither A nor D is selected, then in how many different ways can the four people be selected?
(A) Three (B) Four (C) Five (D) Two

Directions for questions 12 to 15: These questions are based on the following information.

A group of six students, namely Jagan, Karan, Madan, Pavan, Rajan and Savan are to be divided into three teams of two students each for quiz competitions in Physics, Chemistry and History under the following constraints.

(1) Pavan does not want to be in the same team as Rajan.
(2) Savan does not want to be in Physics team.
(3) If Jagan is selected for Chemistry, then Madan must be selected for History.
(4) Karan and Madan must be selected in the same team.

12. If Rajan is selected for History team, then who must be the team mate of Pavan?
(A) Jagan (B) Savan
(C) Madan (D) Either (A) or (B)

13. If Rajan is selected for Chemistry, then who must be his team mate?
(A) Jagan (B) Savan
(C) Pavan (D) Madan

14. If Rajan wants to be in Physics, then in how many ways can the teams be selected?
(A) 5 (B) 4 (C) 3 (D) 2

15. If Savan wants to be in Chemistry, then in how many ways can the teams be selected?
(A) 5 (B) 4 (C) 3 (D) 2

Directions for questions 16 to 18: Answer these questions based on the data given below.

A team is to be selected from seven members A through G. In that team at least one among B, D and F must be selected. If B is selected, then neither C nor G can be selected. A and F cannot be selected together. If D is selected, then E must be selected and if C is selected, then A must be selected.

16. If a team of four members is to be selected, then in how many ways can the team be selected?
(A) 3 (B) 4 (C) 5 (D) 2

17. If a team of five members is to be selected, then who among the following cannot be selected?
(A) C (B) A (C) G (D) B

18. If a team of three members is to be selected, then in how many ways can the team be selected?
(A) 7 (B) 6 (C) 5 (D) 4

Directions for questions 19 to 21: These questions are based on the following information.

Rahul has to select five books from nine books. Among those nine books, five books are printed in the following different years, in 2004, 2005, 2006, 2007 and 2008 and the remaining books are written by different authors A, B, C and D. Rahul has to select at least two books which were printed in the above given years. Further, it is known that:

(i) If the book which was printed in either 2004 or 2008 is selected, then the book which was written by A must not be selected.
(ii) The book which was printed in 2006 cannot be selected with the books which were printed in the previous years of 2006.
(iii) The book which was written by D cannot be selected with the books which were printed in the previous years of 2007.
(iv) The books which were printed in 2005 and 2007 should not be selected together.
(v) If the book which was written by C is selected, then only two books are to be selected which were printed in the consecutive years.

19. Which among the following books cannot be selected?
(A) The book written by D
(B) The book printed in 2005
(C) The book printed in 2008
(D) The book written by A

20. Which among the following books must be selected, if the book written by A is selected?
(A) The book written by D
(B) The book printed in 2005
(C) The book printed in 2006
(D) None of the above

21. Which among the following books must not be selected, if the book written by D is selected?
(A) The book written by A
(B) The book written by B
(C) The book printed in 2007
(D) The book printed in 2008

Directions for questions 22 to 25: These questions are based on the following information.

A group of people, namely A, B, C, D, E, F, G, H, I, J and K are football players. Each of them can play in at least one of the positions, such as defence, mid-field, forward and goal keeping. Among them A, B, C and K are defenders. C, D, E and F are mid-fielders. F, G, H and I are forwards. J and K are goalkeepers.

A team of six players is to be selected for an exhibition match with the following restrictions. The team should consist of one goalkeeper, two defenders, one mid-fielder and two forwards. The following is known about selecting the players.

(i) Only one among J and K is selected.
(ii) If D is selected, then neither C nor K is selected.
(iii) If B is selected, then neither F nor I is selected.
(iv) If F or C is selected, then J is not selected.
(v) Among D, F and C, only one is selected.

22. If J is selected, then who will be selected as forwards?
(A) F along with G or I (B) Only G and H
(C) G along with H or I (D) Only F and H

23. If F is selected as the mid-fielder, in how many ways can the team be selected?
(A) Eight (B) Nine
(C) Seven (D) No such team is possible

24. If D is selected as the mid-fielder, then in how many ways can the team be selected?
(A) One (B) Three
(C) Six (D) Five

25. Who among the following cannot be selected as forward?
(A) F (B) G
(C) H (D) I

ANSWER KEYS

Exercise-1

1. (A)	8. (C)	15. (B)	22. (A)	29. (D)	36. (C)
2. (D)	9. (B)	16. (D)	23. (B)	30. (C)	37. (D)
3. (D)	10. (B)	17. (D)	24. (A)	31. (B)	38. (D)
4. (D)	11. (D)	18. (A)	25. (D)	32. (D)	39. (D)
5. (B)	12. (D)	19. (D)	26. (C)	33. (D)	40. (D)
6. (D)	13. (C)	20. (A)	27. (C)	34. (C)	
7. (B)	14. (B)	21. (B)	28. (C)	35. (C)	

Exercise-2

1. (D)	8. (D)	15. (A)	22. (B)	29. (D)	36. (D)
2. (B)	9. (D)	16. (C)	23. (A)	30. (B)	37. (A)
3. (D)	10. (A)	17. (B)	24. (A)	31. (D)	38. (D)
4. (B)	11. (C)	18. (D)	25. (D)	32. (D)	39. (C)
5. (B)	12. (D)	19. (A)	26. (A)	33. (A)	40. (A)
6. (D)	13. (A)	20. (C)	27. (D)	34. (B)	
7. (D)	14. (D)	21. (A)	28. (A)	35. (D)	

Exercise-3

1. (D)	6. (B)	11. (B)	16. (C)	21. (A)
2. (C)	7. (D)	12. (D)	17. (D)	22. (B)
3. (A)	8. (A)	13. (B)	18. (A)	23. (D)
4. (D)	9. (D)	14. (D)	19. (B)	24. (A)
5. (B)	10. (D)	15. (B)	20. (C)	25. (A)

SOLUTIONS

EXERCISE-1

1. Alen or Jack or both must be in the van.

Adam × Anil

John, James

(A) is the answer as none of Adam and Anil is in the car.
∴ They both are in van which violates the given condition.

2. ('✓' means selected, '✘' means not selected)

If $A_{✓} \Rightarrow F_{✓}$
If $F_{✓} \Rightarrow D_{✘}$
Only if $I_{✓} \Rightarrow G_{✓}$
If $H_{✓} \Rightarrow C_{✘}$
If $B_{✘} \Rightarrow C_{✓}$

If D is selected, then F is not selected. As F is not selected, A is also not selected.
Only one of H and C can be selected. So, we must select I.
If I is not selected, then G cannot be selected, there will be only four people left, which means that the group cannot be formed.

3. It is given that: (✓ is select and x is not selected)

(1) If P ✓ A, then Q ✘ E.
(2) Only if R ✓ E, then Q ✘ B.
(3) If P ✓ D, then R ✘ C.
(4) If P ✘ F, then R ✓ B.
(5) Q ✓ E.

As we know that Q selects E, it means P does not select A (from 1). From 2, we know that Q selects B as R does not select E.
As R does not select B, it means P selects F.
Let us now represent the above information on a table.

	P	Q	R
Selects	F	E, B	
Does not select	A		1

As P does not select A, it means R selects A. From 3, we know that if P selects D, then R does not select C which means that no one selects C, which is not possible. Hence, R selects D and P selects C.
The final table is as follows.

P	Q	R
F	E	A
C	B	D

4. From the given information at most one of Chanchal and Kanchan is selected, and at least one of Surbhi and Neha must be selected.

∴ Some of the possible teams are:
Chanchal, Surbhi, Ankita
Neha, Surbhi, Kanchan
Neha, Chanchal, Ankita.
∴ No one must be selected.

5. From the given information at least one of Ajay and Ena must not be selected, exactly one of Meena and Vijay must not be selected. As four have to be selected, both Meena and Vijay are to be selected.
If Ajay is the other boy, then Deepa is the other girl.
If Sujay is the other boy, then Deepa or Ena is the other girl.
Hence, in three ways we can select the team.

6. Aman is not selected.
At least one of Pavan and Sravan must not be selected.
⇒ At least one of Raghavan and Dawan must be selected.
⇒ Raghavan and Dawan must be selected.
One of Bhavan, Pavan and Sravan is the third person.

Solutions for questions 7 to 9: A team of five players is to be selected from a group of ten players A, B, C, D, E, F, G, H, I and J. It is given that:

(i) One of G and H must be selected.
(ii) H and A must be selected together and B and F must be selected together ⇒ HA and BF.
(iii) F and J cannot be selected together, and C and D cannot be selected together ⇒ $F \neq J$, $C \neq D$.

7. If G is selected, then H and A cannot be selected but B can be selected or cannot be selected.
∴ (A) is not correct.
If H and B are selected, then we cannot say anything about E.
∴ (C) is incorrect.
If J is selected, then we cannot say anything about B.
∴ (D) is not correct.
If G is selected, then H and A cannot be selected which means that the selection can be done as follows:

G	C	F	B E I
	D	J	

One of C and D can be selected and one of F and J can be selected. If F is selected, then B is also selected. We can select atmost one of C and D which means that one of E or I must be selected. If J is selected, then F and B are not selected. We can select one of C or D in the team which means at least one of E or I must be selected.

8. If G is selected, then H and A are not selected and F is definitely selected because either H and A or B and F must be selected.
So, (A) is not the answer.
(B) cannot be the answer because both F and J are selected.
(D) cannot be the answer because if G is selected then A definitely is not selected.
So, (C) is the correct answer.

9. If G is not selected and J is selected, then the total possible selections are
(1) H A J E I
(2) H A J C I
(3) H A J D I
(4) H A J C E
(5) H A J D E

Solutions for questions 10 to 13: From (i), if R or X's selected then none of Q, V and T are selected, then three bowlers cannot be selected. Hence, neither R nor X can be selected.

Hence, from (ii) U and V must be selected.

∴ One among S and P must be selected.

If P is selected, then the bowlers, Q, V and Y must be selected.

If S is selected, then the other two bowlers that are to be selected must be from Y, T and W.

∴ The possible teams are

(U, P, Y, Q, V), (U, S, V, Y, T), (U, S, V, Y, W) and (U, S, V, T, W).

10. U must be selected.

11. V must be selected always.

12. Such a case is not possible.

13. The different ways in which a team can be selected is four.

Solutions for questions 14 to 16: From (i), A and B can be selected as follows:
→ Both A and B are selected.
→ Only B is selected.
→ Neither A nor B is selected.
From (ii), C and E can be selected as follows:
→ Only E is selected.
→ Only C is selected.
→ Both E and C are selected.

14. From the above explanation, it is clear that at least one among C and E must be selected. If only C is selected, then the possibilities are
(1) C, A, B
(2) C, B, D
If only E is selected, then the possibilities are
(3) E, A, B
(4) E, B, D
If both C and E are selected, then the possibilities are
(5) C, E, D
(6) C, E, B
∴ There are 6 possible ways.

15. From the above solution, if D is not selected, then B must always be selected.

16. A, B, D is not a possible team.

17. From (i), it is clear that if Y is selected then only one of R and S should not be selected
∴ To maximize the strength of the team, 'Y' must not be selected. Hence, from (iv), W must not be selected.
∴ At most seven members can be selected into a team, i.e., R, S, T, U, V, X, Z.

18. From the rule (iii), either T and U or V must be selected. From the rule (ii), either W or Z must be selected but if W is selected, then Y must be selected. Hence, to minimize the team W and Y should not be selected. Hence, the team with minimum strength is Z, V and its strength is 2.

19. Choice (A) is violating the rule (iii). Choice (B) is violating the rule (v). Choice (C) is violating the rule (ii).

20. From (ii), among W and Z, we must select:
(a) only W or
(b) only Z or
(c) both W and Z.
(a) Only W is selected and Z is not selected.
From (iv), Y is selected.
From (i), R and S are not selected and from (v) X is not selected.
As already three are not selected, at least two of T, U and V must be selected.
∴ T and U are to be selected.
Hence, one team can be selected, i.e., W, Y, T, U.
(b) only Z is selected but not W.
From (iv) Y is not selected and
∴ R or S or both can be selected.
From (iii), among T, U and V either T or U cannot be selected alone.
Also, X and S together can also be selected, but not only X.
∴ Hence, along with Z the other three members can be as follows.
R, S, X
R, S, V
T, U, R
T, U, S
T, V, R
T, V, S
U, V, R
U, V, S
X, S, V
T, U, V
∴ Hence, in ten different ways a team with Z but not W can be selected.

(c) If W as well as Z is selected.
⇒ Y must be selected.
∴ Only one of the remaining six has to be selected.
From (iii), V must be selected.
∴ Only one team, i.e., W, Z, Y, V can be selected.
Total number of ways of selecting a team of size 4 is 1 + 10 + 1 = 12 ways.

21. From the above solution if X is selected then a team of strength four can be selected in two different ways, i.e., (A) ZRSX and (B) ZSXV

Solutions for questions 22 to 25: From (i), the pairs QV, QU, QW cannot be selected together. Also, UVW cannot be selected together.

From (ii), the possibilities are RS, RT, only S or only T.

From (iii) and (iv), among the possibilities only N, only O, only P or only N and O one must happen.

From (v) to (viii), the pairs NR, NQ, PV and PW should not be selected together.

22. If Q is selected, then none among N, V, U or W can be selected.
Among the remaining, one among O and P will be selected. The remaining two members have to be selected from R, S and T. S and T cannot be selected. Hence, R must be a part of the team.

23. The maximum number of selections in the team can be as shown below.
(A) N, O, two people out of (U, V, W) and one person out of (S, T).
(B) O, two people out of (U, V, W), R and one person out of (S, T).
Hence, the maximum team size is five.

24. If P is selected, then the maximum selections in the team can be as shown below.
(A) P, Q, R, S
(B) P, Q, R, T
Hence, the maximum team size is four.

25. If Q is selected, then the number of ways the team can be selected is shown below.
(A) O, Q, R, S (i) Q, O
(B) O, Q, R, T
(C) O, Q, S (j) Q, P
(D) O, Q, T
(e) P, Q, R, S
(f) P, Q, R, T
(g) P, Q, S
(h) P, Q, T
Hence, the team can be selected in ten ways.

Solutions for questions 26 to 28: It is given that:
(i) If Ajay selects the Blue pen, then he selects the Orange pen also and vice versa.
(ii) If he selects the Grey pen, then he selects the Black pen also and vice-versa.
(iii) If Ajay selects the Yellow pen, then he cannot select the Grey pen.
(iv) Exactly one of Red and Violet pens must be selected.

26. If he did not select Grey and Brown, then he must select White, Pink, Orange, Blue, Yellow, Red/Violet.
∴ He can reject Red.

27. (A) is not the answer, as there is neither the Red nor the Violet coloured pens.
(B) is not the answer, as both the Yellow and the grey coloured pens are selected.
(D) is not the correct answer, as both the Red and the Violet coloured pens are selected.

28. If the Orange coloured pen is not selected, then the Blue coloured pen also cannot be selected.
If the Red coloured pen is not selected, then the Violet coloured pen should be selected.
Hence, there are seven coloured pens out of which Ajay has to select six coloured pens.
He must select the Grey coloured pen and he cannot select the Yellow coloured pen.
The six coloured pens he selects are White, Pink, Black, Grey, Violet and Blue.

29. Since at least two girls are to be selected, from (v), Sheela must not be selected.
∴ Rama, Karuna and Nayana are selected.
From (iii), Bhavan must be selected.
From (ii), Sreekar must not be selected.
∴ Arjun or Dawan is the other boy.
In two ways we can select the team.

30. As three boys are selected, the teams are:
(A) If Arjun, Sreekha, Bhavan are the boys, then any two of Sheela, Rama and Nayana can be the girls, i.e., the team can be selected in three ways [From (ii)].
(B) If Arjun, Sreekhar, Dawan are the boys, then Karuna, Rama and Nayana cannot be selected [From (ii), (iii) and (iv)].
(C) If Arjun, Bhavan, Dawan are the boys, then any two of Sheela, Karuna and Nayana can be the girls [From (iv)], i.e., the team can be selected in three ways.
(D) If Sreekar, Bhavan, Dawan are the boys, then any two of Sheela, Rama and Nayana can be the girls [From (ii)],
i.e., the team can be selected in three ways.
Hence, in total, there are nine ways to select the team with three boys.

31. Let Bhavan not be selected then,
from (ii), Nayana is not selected.
From (v), at most two of the other three girls can be selected.
⇒ Arjun, Sreekar and Dawan must be selected.
⇒ Rama must not be selected [From (iv)]
⇒ Karuna and Sheela must be selected which is a contradicting statement (ii).

Hence, no team can be selected without Bhavan.
From solution (B), without Arjun or Sreekar or Dhawan we can select a team.

32. If Sheela is selected, then three boys must be selected.
From solution (B), any one of the remaining three girls can be the other girl.

33. Given that, Karuna and Sheela is not selected.
From (ii), Sreekar is not selected.
⇒ Arjun, Dawan and Bhavan must be selected.

Solutions for questions 34 to 37: From (i) and (vi), we can say that F, I and K want to go in August. J wants to go in either April or May.

From (v) and above, H wants to go in November.

From (vii) and above, C and E want to go in April and May, respectively. A and B wants to go in November and December, respectively.

From (ii), (iii), (iv), (viii) and above, the possible cases are as follows.

Case	April	May	August	November	December
(i)	C, J, M	E, G	F, I, K	A, H, L	B, D, N
(ii)	C, G, N	E, J	F, I, K	A, H, D,/L	B, M, L/D
(iii)	C, J, N	E, L	F, I, K	A, H, D/G	B, M G/D
(iv)	C, J, N	E, G	F, I, K	A, H, D/L	B, M, L/D

34. A and H are in the same month. Hence, (A) cannot be the answer. In option (B), one among the three pairs EL, HL and HD will be in the same month. MNHKE is a possible team.

35. None among D, L and M is definitely in December. But B is definitely in December.

36. A and H are definitely in November.

37. Each of (A), (B) and (C) is a possible team but not (D).

Solutions for questions 38 to 40: From (i), only Nihit, only Neelam or neither of them is selected.

From (ii), both Sakshi and Neerav or neither of them is selected.

From (iii), only Sakshi or only Sagar is selected.

From (iv), only Meghana, both Meghana and Nabhya or neither of them is selected. Only Sagar, only Meghana and Sagar or none of them is selected.

From (v), only Kundan or only Nabhya is be selected.

From (vi), only Meghana or only Neerav or Meghana, Neerav and Kunal or only Meghana and Neerav or none of them is selected.

∴ The final possible selections are as follows.
(i) Sakshi, Neerav, Meghana, Nabhya, Nihit.
(ii) Sakshi, Neerav, Meghana, Nabhya, Neelam.
(iii) Sakshi, Neerav, Meghana, Nabhya, Kunal.
(iv) Sakshi, Neerav, Kundan, Meghana, Kunal.
(v) Sakshi, Neerav, Kundan, Meghana, Nihit.
(vi) Sakshi, Neerav, Kundan, Meghana, Neelam.

38. The group can be selected in six different ways.

39. Among the given, Neerav and Meghana will be selected.

40. None of the statements is true.

EXERCISE-2

Solutions for questions 1 to 3: Let each person be denoted by the first letter of his name

1. If P is selected, then some of the possibilities are PSR, UPS and PQS, PTS.

2. If Q is selected, then U should not be selected.
Now from (ii), the possible selections are QTS, QTR, QPS, QPR, QSR.
∴ A total of 5 possibilities.

3. As T is selected, then U should not be selected.
If P is selected, then S or R must be selected.
If Q is selected, then P must not be selected and vice versa.
∴ To select a team of three, S or R must be selected.

Solutions for questions 4 to 6: From (ii), only B or only D or neither of them is selected. Both H and M or neither of them is selected.

From (iii), only E, only P or both E and P are selected. Only H, only J or neither of them is selected.

From (iv), C and A, only C or neither of them is selected. Only G, G and F or neither of them is selected.

From (v), only I, only K, both I and K, only B, only M, both B and M or none of them is selected.

From (vi), only C or only L or only J should be selected.

4. MPHCA is a valid group.

5. The group can be selected in five different ways.
(i) KCENP (ii) KDELN (iii) KDELP
(iv) KDENP (v) KPCDN

6. The group can be selected in six different ways.
(i) EHMCG (ii) EHMCN (iii) EHMCP
(iv) EHMLG (v) EHMLB (vi) EHMLD

Solutions for questions 7 to 9: From (i), only Nihit, only Neelam or neither of them is selected.

From (ii), both Sakshi and Neerav or neither of them is selected.

From (iii), only Sakshi or only Sagar is selected.

From (iv), only Meghana, both Meghana and Nabhya or neither of them is selected. Only Sagar, only Meghana and Sagar or none of them is selected.

From (v), only Kundan or only Nabhya is be selected.

From (vi), only Meghana or only Neerav or Meghana, Neerav and Kunal or only Meghana and Neerav or none of them is selected.

∴ The final possible selections are as follows.

(i) Sakshi, Neerav, Meghana, Nabhya, Nihit
(ii) Sakshi, Neerav, Meghana, Nabhya, Neelam
(iii) Sakshi, Neerav, Meghana, Nabhya, Kunal
(iv) Sakshi, Neerav, Kundan, Meghana, Kunal
(v) Sakshi, Neerav, Kundan, Meghana, Nihit
(vi) Sakshi, Neerav, Kundan, Meghana, Neelam

7. The group can be selected in six different ways.

8. Among the given, Neerav and Meghana will be selected.

9. None of the statements is true.

Solutions for questions 10 to 12: Following is the given information:

(i) From a total of 16 players, out of which 11 are to be selected.

(ii) Among these 16 players:

Number of batsmen = 12
Number of bowlers = 9
Number of wicket-keepers = 2
Number of all-rounders = 6 (out of 12 batsmen and 9 bowlers)
Number of wicket-keeper + Batsman = 1
Now, number of only wicket-keeper = 2 – 1 = 1
Number of only batsman = 11 – 6 = 5
(Eleven, because one batsman is wicket keeper)
Number of only bowlers = 9 – 6 = 3

(iii) Ideal Team

At least 6 batsmen, at least 6 bowlers and at least one wicket-keeper.

10. Ideal Eleven: (W.K. = Wicket-keeper)

Batsmen	Bowlers (6)	All-rounders
1 + 5 or more W.K.	6 or more	?

If all the 6 bowlers are also batsmen (i.e., 6 all-rounders), plus one wicket-keeper batsman gives 7 players. Thus, the least number of only bowlers will be zero.

11. As there are 3 players who can only bowl, but at least 6 bowlers are required, hence, minimum number of all-rounders = 3. The wicket-keeper selected is a batsman. Hence, there are three bowlers who can only bowl. Therefore, the team consists of 3 bowlers + 3 all-rounders + 1 wicket-keeper = 7.

But the total number of players who can only bat is (11 – 7) = 4

Total players (as shown below):

Only			Wicket-keeper	
Batsmen	Bowlers	All-rounders	Not Batsman	Batsman
4	3	3	–	1

12.

(A) If all eleven are batsmen, then the wicket-keeper, who can also bat must be selected, along with 6 all-rounders (so we can have six bowlers) and 4 only batsman.

(B) For 8 players to bat and 8 players to bowl, we can have the following arrangement:

Only batsman = 2
All-rounders = 6
Only bowler = 2
Only wicket-keeper $= \dfrac{1}{\underline{11}}$

(C) For 7 players to bat and 7 to bowl, we can have the following arrangement:

All-rounders = 4
Only batsman = 3
Only bowler = 3
Only wicket-keeper $= \dfrac{1}{\underline{11}}$

(D) The number of all-rounders = 3

(3 is the minimum number of all-rounders, as found earlier). Now, the number of only batsman can NEVER be less than the number of only bowlers, as out of these 8 remaining players, one is a wicket-keeper and the number of only bowlers cannot be more than 3.

Solutions for questions 13 to 16: Let us list out the conditions specified in the question.

(1) Each team must consist of a minimum of two members and each of the four teams must have a distinct number of members. This implies that the number of people in different teams must be 2, 3, 4 and 5.
(2) Each of Rama, Ramya, Radha and Raksha must be in a different team.
(3) Each of Rohini, Padma, Priya and Priyanka must be in a different team.
(4) Pratima, Pratibha and Sudha must be in the same team.
(5) Shreya and Shalini must be in the same team.
(6) Rama cannot be paired with any of Padma, Priya and Priyanka.
(7) Ramya cannot be paired with Priyanka or Priya.
(8) Radha cannot be paired with Priyanka and Swetha is in one of the teams.

From (1) we know that the teams should consist of 2, 3, 4 and 5 members respectively.

From (2) and (3), we know that each team should have two members out of Rama, Ramya, Radha, Raksha, Rohini, Padma, Priya and Priyanka, since as per (2) each of Rama, Ramya, Radha and Raksha should be in a different team and same is the case with (3).

From (2) and (3), we know that one each of Rama, Ramya, Radha, Raksha must be paired with one each of Rohini, Padma, Priya and Priyanka.

From (6), (7) and (8), we know that Priyanka cannot be paired with Rama, Ramya or Radha, which implies that Priyanka is paired with Raksha. Using a similar logic we find that Rama and Rohini, Ramya and Padma, Radha and Priya are paired together.

From (4), we know that Pratima, Pratibha and Sudha must be in a team together but since all teams already have two members each, these people can only be accommodated in the team that has five members in it.

From (5), we know that Shreya and Shalini must be in a team together which means that can only be accommodated in a team that has four members in it.

From (8), we know that Swetha is one of the members and she can only be placed in the team that has three members in it.

Teams	I (2 members)	II (3 members)	III (4 members)	IV (5 members)
Members		Swetha	Shreya Shalini	Pratima Pratibha Sudha

13. There are 4 pairs, i.e., Raksha–Priyanka, Rama–Rohini, Ramya–Padma, Radha–Priya who must be allocated to 4 teams which can be done in 4! ways.4! = 24 ways.

14. We can see that statements made in choices (A), (B) and (C) are true as per the above discussion. Shreya is in a four-member team, hence, choice (D) is definitely not true.

15. As per choice (A), if Rohini is in a team of five members, it means that Rama is also in that team. Padma is in a four-member team implies that Ramya is also in that team and Radha is not in a three-member team implies that she is in a two-member team along with Priya which leaves us with Priyanka and Raksha in a three-member team which completes the arrangement. Hence, choice (A) gives us the complete arrangement.

16. Pratima is a member of the five-member team.

Solutions for questions 17 to 19: The given data can be represented as follows.

(1) D < G and H < I

⇒ If D gets rank, then G also gets a rank and if H gets a rank, then I also gets a rank.

(2) F × D

(3) B / G (Only one of the two)

(4) IC (If I gets a rank, then C will also get a rank. If I gets a rank, both I and C get ranks.)

17. It is given that E did not get a rank and that F got more marks than I, which means that F > I > H. So, if F does not get a rank, then none of I or H gets a rank. If I does not get a rank, then C also does not get a rank, which means that A, B, D, G and J get ranks. However, this is not possible as only one out of B or G gets a rank, but not both, as per the given conditions. This means that F has to get a rank, which means that D will not get a rank.

18. If D and H get ranks, then G, I and C should also get ranks because G got more marks than D and I got more than H. If I gets a rank, then C also should get a rank. As D and H got two successive ranks, G and I must always be ranked higher than D and H. So, D and H gets either 3rd or 4th rank or 4th or 5th rank. Hence, C would not get the 4th rank.

19. If F is not ranked and H got more marks than G, then H and I both are selected. If I is selected C, must be ranked.

Solutions for questions 20 to 23: It is given that A, C, E, G, I, K, M, J and O are batsmen, B, D, F, G, H, J, M and P are bowlers, L and N are wicket-keepers. The players who are both bowlers and batsmen are known as all-rounders. Hence, G, M and J are all-rounders. The Captain and the Vice-Captain are all-rounders.

Neither G nor M is the Captain, that means J is the Captain. Similarly, M is the Vice-Captain. As only two all-rounders are to be selected, J and M are selected and G cannot be selected. Five batsmen are to be selected from A, C, E, I, K and O. It is also given that if E is selected, then A and I cannot be selected which means that 5 batsmen cannot be selected. Therefore, E should not be selected and the batsmen selected are A, C, I, K and O. It is given that three bowlers are to be selected from B, D, F, H and P.

It is also given that if D is selected, P and H cannot be selected and the bowlers selected would be B, D, F which is not a feasible combination (as B and F cannot be together). Therefore, D is not selected. We already know that C is selected, hence, B cannot be selected. The bowlers who are selected are F, H and P.

J and N cannot be selected. As J is already selected as the Captain. Hence, N cannot be selected into the team. Therefore, L is selected as the wicket-keeper.

The 11 players who are selected into the team are J (Captain), M (Vice-Captain), L (Wicket-keeper), O, F, H, P, A, C, I, K.

20. J and M are the Captain and the Vice-Captain of the team, respectively.

21. F, H and P are the three bowlers selected in the team.

22. L, O, F, H and C are not selected is definitely false.

23. In the 2nd match J is the Vice-Captain and M is the Captain. As, 86th match is an even numbered match, J will be the Vice-Captain in that match.

24. If B is selected into a team, then D must also be selected into that team also one person from E and C should be in this team and one person among G and H should be in this team. So, A and F should be in the other team.

25. (Let the teams be Team 1 and Team 2) Given that E and D are in one team.
So, C is in team 2. Now if B is in team 2, then D should also be in team 2 which is not possible. So, B is in team 1. Also, exactly one person among G and H is in team 1.
∴ B and G (or) B and H are in team 1.

26. B and D should be together (as described in previous question). So option (A) and (B) are not valid teams. Option (C) does not violate any rule.
∴ EHAF is a valid team.

27. Let H be the member of a team, and let us consider choice (A) i.e. KDLG as the other team. From (iii) we get that, since L belongs to one of the teams, neither I nor J can be selected into any of the teams.
Among the remaining A, B, C, E and F only one among A, C and E can be selected.
∴ B and F must be selected and from (iv), A cannot be selected. The possible teams are H, B, F, C / E.
Choice (B) does not violate any rule to form a team. Consider
choice (C), as A, G, L, F is a team, H must be there in the other team. But A and F both are in the same team, which violate (v).

28. We need to eliminate only two people.
From the rule (vii), one of A, C and E must not be selected. Hence, we can select from only 11 people.
From the rule (iii), If L is selected none or I and J should be selected, hence, L cannot be selected.
∴From (v), if A is not selected then F cannot be selected.
Let B and E belong to team α.
∴ A must be selected in team β, C should not be selected and F must be selected in team 'α'.
From (ii), G and H should be in two different teams.
∴ The possible teams are as follows:
α: B, E, F, G/H, I/J/K/D
β: A, H, G and any two among I, J and K.
Therefore, in 8 different ways, the other team can be selected.

29. The two teams can be ADIG and JFBH or A, D, K, H and JFBG or AIKG and JFBH.
Hence, none must be selected with A.

30. As K is selected, neither D or E can be selected.
As L is selected then I and J must not be selected.
∴ A, B, C, F, G, H, K and L are selected.
From (v), (vi) and (vii), B, F and C are into one team, which does not contain A.
From (ii), G and H are in different teams.
∴ Y has B, F and C and X has K, L and A.
Now, G and H can be distributed in two ways.

31. From (v), A must be selected into the other team.
∴ From (vi) and (vii), B, C and E cannot be selected into the other team.
Of the remaining, each person can be selected into the other team.

Solutions for questions 32 to 34: Based on the given information, the following inferences can be made:
(a) From (III), F is not carrying the white folder.
(b) From (VI), A is not from China.
We have the following distribution:
From the above, we can say that either B or D is carrying the white folder and either C or E is from China. There is no condition for (Brazil, USA), (Black, Blue) and (Terrorism, Poverty). All these can be taken in any combination.

32. If B is carrying the orange folder, then D will be carrying the white folder.
As per (iii), F and D must be in the same slot. Hence, (B) is true.
As per (vi), A and the delegate from China must be selected together.
$\Rightarrow \begin{pmatrix} \text{F} \\ \text{Brazil/USA} \end{pmatrix}$ and $\begin{pmatrix} \text{D.} \\ \text{S. Africa} \\ \text{Unemployment} \end{pmatrix}$ cannot be selected with A, since only 3 people can be allowed for any one slot. Hence, (D) is false.
We can get the following arrangement, without violating any of the given conditions.

Slot – I	Slot – II
A – Red	F – Trafficking
B – Orange – India	D – White – SA – Unemployment
E – Green – Ching – EP	C – Illiteracy

From the above, we infer that (A) and (C) are true.
∴ (D) is false.

33. Including the additional condition, we have:

Delegate	**A**	**B**	**C**	**D**	**E**	**F**
Country	x China	India		S. Africa		x China x Russia Brazil / USA
Area of presentation	Trafficking		Illiteracy	Unemployment		
Colour	x White x Red		x White		Green	Red

	Slot (a)					Slot (b)				
Delegate -------	F	+		+		A	+	C or E	+	
Country -------	Brazil/USA					?		China		
	?					Trafficking				
Presentation -------	Red		White			?				

F and A must be selected for different slots as per (ii).
So, statement I is definitely true.

Statement II and III are not true in the following arrangement.

Slot - I	**Slot - II**
A – Orange – Trafficking	F – Brazil/USA - Red – Terrorism/Poverty
E – China – Green – EP	D – SA – White – Unemployment
C – Blue/Black – Illiteracy	B – India – Black/Blue – Poverty/Terrorism

∴ Only I is true.

34. Since E is from the USA, F is from Brazil and A is from Russia.
Since, F and A have to be in different slots (A) is true.
Since A and the one from China are in one slot and F and the one carrying white folder are in the other slot, (B) is false.
(C) and (D) are true in the following:

Slot – I	**Slot – II**
A – Russia – Black – Trafficking	F – Brazil – Red – EP
C – China – Blue – Illiteracy	D – SA – White – VE
E – USA – Green –	B – India – Orange

In the above, (C) and (D) are true.
∴ Only (B) is false.

Solutions for questions 35 to 37: Let each person be denoted by the first letter of his / her name.

35. K and C are in the same team.
∴ L must be in the other team. [From (iii)]
If A and B are in the same team as L, then it violates (iv) [From (ii)].
Only one of A and B is in the same team as L.
∴ Using other conditions, the possibilities are:
(a) K, C, A, M and L, B, N
(b) K, C, B, M and L, A, N
(c) K, C, A and L, M, N, B
(d) K, C, B and L, M, N, A
(e) K, C, B, N and L, M, A

36. M and B are not in the same team.
From (iii), either K or L is with B.
Now, if A is with B, then C and N must be with M, which violates (iv).
∴ A is with M.
Now, at least one of C and N is with B.
The teams are:
B, K/L, C, N and M, L/K, A
B, K/L, C and M, L/K, A, N
B, K/L, N and M, L/K, A, C
∴ There are six possibilities.

37. From (iii), M and N are in the same team.
One of K and L is with them.
C cannot be with them as it violates (iv).
From (ii), only one of A and B is with them.
∴ The possibilities are:
M, N, K/L, A/B and L/K, C, B/A.
Therefore, there are four possibilities.

Solutions for questions 38 to 40: Let each person be denoted by the first letter of his/her name.
From (iii), if F is selected, then A and C are not selected.
From (ii), one among A and B must be selected.

∴ The possibilities are:

If F is selected, then

(a) F, B, E, G
(b) F, B, E, D
(c) F, B, D, G

If F is not selected, then A or B or A, B must be selected in the team as we have to select 4, we have to reject 3 and among C, E and G at least one must be rejected.

From (i), at most two of C, E and G can be selected.

∴ The possibilities are as follows.

(d) A, B, C, E
(e) A, B, C, G
(f) A, B, G, E
(g) A, B, D, C
(h) A, B, D, E
(i) A, B, D, G
(j) A, D, C, E
(k) A, D, C, G
(l) A, D, G, E
(m) B, D, C, E
(n) B, D, C, G
(o) B, D, G, E

38. Except possibilities (a), (d), (e) and (f), all other team have Dheeraj.

39. Except (a) and (c), all have at most one of F and G.

40. (c), (e), (g), (i), (k) and (n) are the possibilities.

EXERCISE-3

Solutions for questions 1 to 4: It is given that there are ten candidates, out of which 2 are M.A.s, 2 are M.B.As, 2 are M.C.As and 4 are B.Techs. The conditions are as follows.

1 M.B.A $\Rightarrow$ 2 B.Techs

2 B.Techs $\Rightarrow$ M.B.A ≥ 1

Exactly one M.A. candidate must be selected.

1. If two B.Tech candidates are selected, then one M.B.A. and one M.A. are selected. More than one M.As cannot be selected according to the data. One M.B.A. and only M.C.A. with one M.A. does not make a total of six. One M.A. is already there. Now 2 B.Techs, one M.B.A. and one M.A make a total of four candidates. Now 2 more candidates must be there. Those two can be 1 M.B.A. + 1 M.C.A. So, the correct option is choice (D).

2. Two M.C.A. candidates are selected. So, the choices must be 2 M.C.As + 1 M.A. + 1 M.B.A. + 2 B.Techs 1 M.B.A. and 1 B.Tech is not accepted. 3 B.Techs implies 2 M.C.As cannot be selected.

3. The given data says that 1 M.B.A. is followed by 2 B.Tech candidates. If four B.Techs are selected, 2 M.B.As cannot be selected because 1 M.A. must be selected and the total has to be six candidates only.

4. Let us validate each choice.
Choice (A): If 2 M.B.As are selected, then 2 B.Techs must also be selected and 1 M.A. is in every selection, which makes a total of five people and the 6th can be an M.C.A., which is an acceptable selection.
Choice (B): If 2 M.C.As are selected, then any way 1 M.A. is there and the remaining three will be M.B.As and B.Tech, i.e., 1 M.B.A. and 2 B.Tech.
Choice (C): If 2 B.Techs are selected, then it does not violate any rule as explained in choice (B).
Choice (D): If no M.C.A. is selected, then we need to select 6 out of M.B.As, M.As and B.Techs, i.e., 1 M.A and 5 out of M.B.A and B.Tech. The maximum of 2 M.B.A. can be selected and then only 2 B.Tech should be selected. If so, we have to take 1 M.C.A. also. So, this choice is false.

Solutions for questions 5 to 9: A group of three people, namely Ajay, Bony and Chetan buy two each out of 6 items P, Q, R, S, T and U. If Ajay buys R, Bony does not buy P or S or both. If Bony buys Q, Chetan does not buy U or T or both.

5. If Ajay buys R and T, then Bony cannot buy P or S or both. So Bony buys Q and U.

6. If Bony buys Q and S, then Chetan cannot buy T and U. So, Ajay must buy T and U, as each one has to buy two each.

7. If Chetan bought P and S, Ajay and Bony have to choose any two each of Q, R, T, U. Ajay may have any pair of QR, QT, QU, RT, RU or TU. So, we cannot say anything about their purchases as all the choices (A), (B) and (C) be true always.

8. If Ajay buys P and Bony buys Q, then Chetan buys neither T nor U. So, Chetan can buy the pair of R and S only.

9. Let us validate each choice.
Choice (A): If Ajay buys R, Bony cannot buy P and S but he can buy Q or T or U. So, it is not necessary to buy Q.
Choice (B): If Chetan buys T or U, that means Bony cannot buy Q. So, this is also false.
Choice (C): Ajay bought R, then it is not necessary for Bony to buy T as explained in choice (A).
Choice (D): If Ajay buys R and Bony buys Q, then Chetan has to buy only P and S, as he cannot buy T and U. Chetan has to buy P and S.

Solutions for questions 10 and 11: From (1), at least one of E or B must be selected.

E can be selected.

B can be selected.

B and E can be selected.

At most one of A or D can be selected.

One of A or D can be selected or none of A or D is selected.

From (3), A, F and G together cannot be selected.

From (2), either C or D must be selected.

10. If A is selected, then D cannot be selected. D is not selected implies that C must be selected.

11. If neither A nor D is selected, then the four people can be selected in four ways.
1.CEFG
2.CBFG
3.CBEF
4.CBEG

Solutions for questions 12 to 15: Let each student be denoted by the first letter of his name.

From (1) and (4), the teams can be (K, M), (P, J), (R, S) or (K, M), (P, S), (R, J).

From (3), in any case, if J is in Chemistry team, then M is in History team.

S must be in Physics team, which is violating (2).

J cannot be in the Chemistry team.

∴ We have the following possibilities:

	Physics	Chemistry	History
1	R, J	P, S	K, M
2	R, J	K, M	P, S
3	K, M	P, S	R, J
4	P, J	R, S	K, M
5	P, J	K, M	R, S
6	K, M	R, S	P, J

Solutions for questions 16 to 18: We will represent the instructions as below.

(1) B, D and F → at least 1 must be selected.
(2) If B → not C and not G
(3) A and F is not together
(4) If D → E
(5) If C → A

16. The possible combinations of a team with 4 members are as follows:
(1) B, A, D, E
(2) B, D, E, F
(3) D, E, A, C
(4) D, E, A, G
(5) D, F, E, G
∴ There are five ways to select a team of four members.

17. A team of five members is ACDEG.
Hence, B cannot be selected.

18. The possible combinations of the team with three members are as follows:
(1) A, B, E
(2) B, D, E
(3) D, E, F
(4) B, E, F
(5) D, E, A
(6) D, E, G
(7) E, F, G
∴ There are 7 ways possible.

Solutions for questions 19 to 21: From (i), the book printed in either 2004 or 2008 is selected or the book written by A is selected or the book printed in 2004 and 2008 are selected or none among them is selected.

From (ii), only the book printed in 2006 or the books printed in 2006 and 2007 or 2006 and 2008 or 2006, 2007 and 2008 can be selected or none among them is selected.

From (iii), (the book written by D and the book printed in 2007) or (the book written by D and the book printed in 2008) or (the book written by D and the books printed in 2007 and 2008) can be selected or none among them is selected.

From (iv), the book printed in only 2005 or only 2007 or none among them is selected.

From (v), the book written by C and the books printed in 2004 and 2005 or the book written by C and the books printed in 2005 and 2006 or the book written by C and the books printed in 2006 and 2007 or the book written by C and the books printed in 2007 and 2008 can be selected or none among them is selected.

∴ The possible selections are:
(i) A, B, C, 2006, 2007
(ii) B, C, D, 2007, 2008

19. The book printed in 2005 cannot be selected.

20. The book printed in 2006 must be selected, if the book written by A is selected.

21. The book written by A must not be selected, if the book written by D is selected.

Solutions for questions 22 to 25: ***From the given data, the*** possible combinations for each position are as follows.

Goalkeeper: J or K
Defenders: A + B, A + C, A + K, B + C, B + K, C + K
Mid-fielder: C or D or E or F
Forward: F + G, F + H, F + I, G + H, G + I, H + I

22. Given that J is selected. From (i), K will not be selected. According to (iv), neither F nor C will be selected. Hence, only A and B can be selected as defenders. According to (iii), when B is selected neither F nor I will be selected. Thus, only G and H can be selected as forwards.

23. Given that F is selected as the mid-fielder. From (iii), B will not be selected. From (iv), J will not be selected.

Hence, A and C both have to be in the team as defenders, but it violates (v), i.e., only one among C and F can be selected.

∴ No such team is possible.

24. When D is selected, from (ii) neither C nor K will be selected. Then J will be the goalkeeper and there is only possible combination of defenders to select, i.e., A + B. Since B is selected, from (iii), neither F nor I will be selected. Then there is only one possible combination of forwards to be selected, i.e., G + H. Thus, there is only one way in which the team can be selected.

25. When F is selected as forward, from (iv) K will be the goalkeeper and from (v) C will not be selected. In such case only A and B can be selected as defenders. But from (iii), even B cannot be selected. Thus, F cannot be selected as forward.

5 Comparisons

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:

- Understand how to interpret the data given in the question and get a final sequence or ranking out of it based on the given constraints
- Understand how to rank people/objects based on multiple parameters
- Learn how to deal with questions which are a combination of comparisons and other topics.

Questions based on Order Sequencing appear frequently in MBA entrance exams either as simple comparison questions or along with other topics like Linear/Circular Arrangements, Distributions etc. These puzzles involve comparison of persons or objects in various parameters like height, age, marks scored etc. The term 'Order Sequence' is self-explanatory. In questions for this category, you will be asked to deal with relative positions of subjects. The absolute values of the subjects is not what you should be interested in. It is the comparison between different subjects that you have to deal with. The data also specifies the relationships like 'A is greater than B' or 'C is not less than D' and so on. You have to decide the positions of the subjects in ascending or descending order on the parameters given. The subjects of comparison can be people or things.

In short, data will be given to compare the quality or quantity. The parameters on which the subjects are compared can be heights or weights of people, the money with them, complexion, sizes of things, etc.

In such questions, you will come across typical statements like 'A is taller than B', 'B is not shorter than C' and so on.

You may use the following symbols to symbolically represent the conditions given and then later, represent all the subjects pictorially.

Greater than	$>$
Less than	$<$
Greater than or equal	$\geq$
Less than or equal	$\leq$

'Not greater than' is the same as 'less than or equal to'. Similarly, 'not less than' is the same as 'greater than or equal to'.

Words like 'Who, And, Which, But' used in the data play a significant role in analysing the data. 'AND' and 'BUT' play the same role whereas 'Who' and 'Which' play the same role.

Let us illustrate with one statement.

'A is taller than B, who is shorter than C and taller than D but shorter than E, who is taller than F and G but shorter than H'.

By using appropriate symbols, the above statement can be represented as follows.

$A > B; B < C; B > D; B < E; E > F; E > G; E < H$

Questions on the above data can be as follows.

(i) Who is the tallest?
(ii) Who is the shortest?
(iii) Who is the second tallest in the group?

Let us take some examples.

SOLVED EXAMPLES

Directions for questions 5.01 to 5.05: These questions are based on the following information.

A, B, C, D and E are five cars while P, Q and R are three motorcycles. A is the fastest of the cars and R is the slowest of the motorcycles. C is costlier than D and Q but cheaper than B. Among cars, A is not the costliest. D is cheaper than E and there is no car whose cost lies between the cost of these two. E is faster than three of the cars and all the motorcycles. Q is costlier than R but cheaper than P, who is faster than Q.

5.01: Which of the following cars cannot stand exactly in the middle position among cars as far as their cost is concerned?
(A) A (B) C (C) E (D) D

5.02: Which of the following statements is true about the motorcycles?
(A) P is the costliest as well as the fastest motorcycle.
(B) The fastest motorcycle is not the costliest motorcycle.
(C) The slowest motorcycle is also the cheapest motorcycle.
(D) Both (A) and (C)

5.03: If P is costlier than E, how many cars are cheaper than P?
(A) 1
(B) 2
(C) 3
(D) Cannot be determined

5.04: If P is cheaper than A which is not costlier than E, which of these is the cheapest of all the cars and motorcycles put together?
(A) R
(B) Q
(C) E
(D) Cannot be determined

5.05: Which of these is the slowest of the cars, if B and C are faster than D?
(A) B (B) D (C) E (D) A

Solutions for questions 5.01 to 5.05: Let us first write down all the comparisons given for costs and speeds. Then we will tabulate them.

Speed:

A ➔ Fastest car
E ➔ Faster than three of the cars ➔ E is the second fastest car
R ➔ Slowest motorcycle
P > Q

Cost:

C > D
C > Q
B > C
A ➔ Not the costliest among cars
E > D ➔ No other car lies between these two
Q > R
P > Q
Now let us tabulate this data.

Speed:

Cars

Fastest	A	E				Slowest

Motorcycles

Fastest	P	Q	R	Slowest

Cost:

Cars

Costliest	B C E D	Cheapest

Here, we know that A is not the costliest car but we do not know where it will fit in. It can come anywhere after B except between E and D.

Motorcycles

Costliest	P	Q	R	Cheapest

In addition to the above, we also have to keep in mind that C > Q in cost. (From this we can conclude that B > Q, B > R, C > R in cost).

5.01: In terms of cost of the cars, A can come between B and C or between C and E or to the right of E. In each of the above cases, the middle car will be C, A and E, respectively. Hence, among the cars given, D cannot be in the middle.

5.02: By looking at the tables above, we can make out that choices (A) and (C) are both correct and hence, the correct answer is (D).

5.03: If P is costlier than E, we can also conclude that it is costlier than D but we cannot conclude anything about the relationship between the cost of P and that of B, C and A.

5.04: Since A is not costlier than E, it means that A is at the same level of E or cheaper than E. We cannot conclude which of these two positions A is in. Hence, we cannot conclude which is the cheapest of all the vehicles. [Please note that if A is the cheapest car, then R will be the cheapest of all the vehicles. However, if A is at the same level as E in cost, then there is a possibility of R or D being the cheapest of all the vehicles.]

5.05: If B and C are faster than D, then the order will be as follows:

1	2	3	4	5
A	E	B/C	C/B	D

Hence, D is the slowest of all the cars.

Directions for questions 5.06 to 5.09: These questions are based on the following information.

J, K, L, M and N are five boys in a class. They are ranked in the order of heights ranked from the tallest to the shortest and in the order of cleverness ranked from the cleverest to the dullest. K is taller than N, but not as clever as J and L, whereas M is the cleverest of all but shorter than J. While L is shorter than M but taller than K, L is not as clever as J. No two people got the same ranks in any of these parameters.

5.06: Who is the third in the order of heights?

(A) J (B) N
(C) K (D) L

5.07: If N is not the last in at least one of the two comparisons, which of the following is the dullest of all the five?

(A) K (B) L
(C) M (D) J

5.08: If L is the third in order of cleverness, then who is the dullest of all?

(A) M
(B) N
(C) L
(D) Cannot be determined

5.09: Who among the following is cleverer as well as taller than K?

(A) L and J only (B) N
(C) L and N (D) J, L and M

Solutions for questions 5.06 to 5.09: Let us first write down all the conditions given and then tabulate the data.

Cleverness:

J > K
L > K
M is the cleverest.
J > L

Height:

K > N
J > M
M > L
L > K

Now let us put together all the information we have.

Cleverness

Cleverest	M J L K	Dullest

We do not know where N will come in the order of cleverness but he will definitely be after M.

Height

Tallest	J M L K N	Shortest

5.06: From the table above, we can clearly see that L is ranked third in order of heights.

5.07: N is the last in terms of height. Since we are given that he is not the last in at least one of the lists, he cannot be the last in cleverness. So, K is the dullest of all. Choice (A)

5.08: If L is the third in the order of cleverness, as can be seen from the table above, either N or K can be the dullest. Choice (D)

5.09: By looking at the tables we made above and from the answer choices, we find that L, J and M are taller as well as cleverer than K.
Choice (D)

Directions for question 5.10: Select the correct alternative from the given choices.

5.10: P, Q, R, S, and T are five girls competing in a running race. R and P have at least two girls ahead of each of them. T and P do not have more than one girl behind each of them. Who arrives at the finishing line after two girls as well as before two other girls, if no two girls finish the race at the same time?

(A) Q (B) S
(C) T (D) R

Solution for question 5.10:

5.10: R and P have at least two girls before them. ➔ R and P have to be in two out of 3rd, 4th and 5th positions.
T and P have not more than one girl behind each of them ➔ T and P have to be in the 4th or 5th positions.
The above two statements together mean that R will have to be in the third position.

EXERCISE-1

Directions for questions 1 to 9: Select the correct alternative from the given choices.

1. Each of the five people, such as K, L, M, P and Q is of a different weight. It is known that the number of people heavier than P is the same as the number of people lighter than Q. L is the heaviest and K is not the lightest. Who is the lightest?

(A) P (B) K
(C) Q (D) M

2. Each of the five people from A, B and C is ranked 1 to 3 in the order of their heights as well as in the order of weights. No person got the same rank in both height and weight. C is heavier than B but shorter than A, who is not the tallest. Who is the heaviest?

(A) A (B) B
(C) C (D) Cannot be determined

3. A group of five boys, namely Lalit, Mohan, Naveen, Omi and Pavan are compared with each other in terms of their heights. Lalit is taller than Mohan but shorter than Pavan, who is shorter then Naveen, who is taller than Omi. Who among these five friends is the second tallest?

(A) Omi (B) Naveen
(C) Naveen or Pavan (D) Cannot be determined

4. A, B, C, D and E are five people working in an office. C comes to the office before B, but after A. E comes after D, but not immediately after him. The number of people who came between D and E is the same as those who come between C and B. Who was the first person to come to the office?

(A) A (B) D
(C) A or D (D) C

5. Each of the four students, namely Gopi, Hari, Murali and Anil did exactly one project out of four different projects, such as P.T., T.D., D.M.E. and H.T. No two students got the same marks. The project H.T. secured more marks than the project T.D. and the project P.T. did not get the first or the fourth rank. The projects are ranked from first to fourth from the maximum to minimum marks in that order.

Anil secured more marks than Murali and Hari got less marks than Gopi, who did the project H.T.

If the project P.T. got more marks than H.T., then who did the project P.T.?

(A) Murali (B) Anil
(C) Hari (D) Hari or Anil

6. The following are the comparisons made between five business tycoons Mukesh, Aditya, Prem, Murthy and Raju.

Mukesh is richer and younger than Murthy. Aditya is poorer than Prem but richer than Mukesh. The poorer of Mukesh and Prem is the younger of the two. Murthy is richer than Raju, who is older than Aditya but younger than Mukesh. How many people are richer but younger than Mukesh?

(A) 1 (B) 2
(C) 3 (D) 0

7. A group of seven people, namely Rama, Ramana, Rana, Ravi, Raju, Ramesh and Raman finished a race, not necessarily in the same order. No two people finished the race at the same time.

Rana finished the race before Raju but after Raman.

Rama finished the race after Ramana but before Rana.

Only Ramesh finished the race between Ramana and Raman. Ravi finished the race before Raman.

If no other person finished the race between Ravi and Raman, then who is the 6th person to finish the race?

(A) Rama (B) Ravi
(C) Ramesh (D) Rana

8. Anand, Mohan, Ravi and Kamal together have ten apples. Each person has at least one apple and no two people have the same number of apples. Kamal has more number of apples than Anand but does not have the highest number of apples. Mohan has more number of apples than Ravi. How many apples does Mohan have?

(A) Two (B) Three
(C) Four (D) Cannot be determined

9. Vinit, Karan, Santosh and Sid participated in a race. No two people ran with the same speed. Speed of Karan is more than that of Santosh and Sid finished the race before Vinit finished. It is known that Karan did not finish the race before Vinit. The speed of which person is the lowest?

(A) Vinit (B) Karan
(C) Santosh (D) Sid

Directions for questions 10 to 12: These questions are based on the following information.

A group of seven people, namely P, Q, R, S, T, U and V, who are of different ages, are comparing their ages. We know the following information.

(i) P is younger than R, who is not older than S.
(ii) S is younger than only two people.
(iii) Q is not the oldest but older than the fourth youngest person.
(iv) T is older than only U.

10. Who is the oldest?

(A) S (B) T
(C) U (D) V

11. Who is the third youngest?
(A) V (B) P
(C) R (D) S

12. Who is the fourth eldest?
(A) R (B) P
(C) S (D) V

Directions for questions 13 to 15: These questions are based on the following information.

A group of five people,namely P, Q, R, S and T are of different heights and different weights. Further, it is known that

(1) Either P or Q is the tallest.
(2) T is taller as well as heavier than both S and R.
(3) The heaviest person is the third tallest whereas the second tallest is the lightest.
(4) Only one person is lighter than Q, R is heavier and shorter than S.

13. Who is the lightest?
(A) P (B) Q
(C) R (D) S

14. Who is / are taller and heavier than S?
(A) P and T (B) Only T
(C) Only T and Q (D) Only R

15. How many people are heavier than P?
(A) One (B) Two
(C) Three (D) Four

Directions for questions 16 to 18: These questions are based on the following information.

A, B, C, D, and E are the top five rankers in each of the subjects Maths and Physics, respectively. No two people got the same rank in any subject and no person got the same rank in both subjects. The following information is known about them.

(1) B's rank in Maths is the same as that of D's rank in Physics.
(2) C got a better rank than at most one person in both subjects, respectively.
(3) E got the least rank in Maths.
(4) D got second rank in Maths and B got third rank in Physics.
(5) In Physics A's rank is better than E's rank.

16. What is the rank of D in Physics?
(A) 1 (B) 2
(C) 3 (D) 4

17. What is the rank of A in Physics?
(A) 1 (B) 2
(C) 3 (D) 4

18. The rank of A in Maths is the same as the rank of _____ in Physics.
(A) B (B) C
(C) D (D) E

Directions for questions 19 and 20: These questions are based on the following information.

Pavan, Sravan, Charan, Tarun and Kiran are the top five rankers in a class, not necessarily in the same order. Each of these five is of a different height. The tallest person is the fourth ranker while Kiran is the second ranker. Tarun is taller than at least two people and is the third ranker. The shortest person is the first ranker but he is not Charan. Sravan is taller than only one person and Tarun is taller than Kiran.

19. Who is the fourth ranker?
(A) Pavan (B) Charan
(C) Sravan (D) Tarun

20. How many people are taller than Pavan?
(A) One (B) Two
(C) Three (D) Four

Directions for questions 21 to 23: These questions are based on the following information.

Each of the six children, namely Amit, Sumit, Kamat, Namit, Ranjit and Charit has a different number of chocolates among 3, 4, 5, 6, 7 and 8, not necessarily in the same order. We know the following information.

(i) The difference between the number of chocolates with Charit and Ranjit is the same as that between the number of chocolates with Kamat and Ranjit.
(ii) The number of chocolates with Charit is less than that with Sumit which in turn is less than that with Ranjit.
(iii) The number of chocolates with Sumit is more than that with Namit.

21. Who has 6 chocolates?
(A) Sumit (B) Ranjit
(C) Amit (D) Charit

22. What is the number of chocolates with Sumit?
(A) 5 (B) 6
(C) 7 (D) 4

23. What is the difference between the number of chocolates with Namit and Kamat?
(A) 2 (B) 3
(C) 4 (D) 5

Directions for questions 24 to 26: These questions are based on the following information.

A group of six students, namely Anand, Brijesh, Charan, Deepti, Gopal and Hriday are the top six rankers of a class. No two people got the same rank. We know the following information regarding their ranks.

(i) Deepti got a better rank than at least two students.
(ii) Gopal got a better rank than Brijesh.
(iii) The number of persons who got better rank than

Anand is the same as the number of persons who got worst rank than Charan.
(iv) Anand got a better rank than Deepti.
(v) Only one person got a rank between the ranks of Hriday and Brijesh.

24. If Hriday got the third rank, then the only person whose rank is between the ranks of Deepti and Charan is
(A) Anand (B) Gopal
(C) Hriday (D) Brijesh

25. Who got the sixth rank?
(A) Charan (B) Brijesh
(C) Hriday (D) Cannot be determined

26. If Deepti got the second rank, then who got the fifth rank?
(A) Brijesh (B) Charan
(C) Hriday (D) Cannot be determined

Directions for questions 27 to 29: These questions are based on the following information.

A group of eight people, namely Anurag, Bhadri, Chakri, Dayanand, Eleena, Firoz, Goutam and Hemant who got different marks are comparing their marks. We know the following information regarding their marks.
(i) Anurag got more marks than Bhadri and the number of people who got less marks than Anurag is the same as the number of people who got more marks than Bhadri.
(ii) Chakri got more marks than Dayanand, but less marks than Eleena.
(iii) Firoz got the fifth highest marks.
(iv) Goutam got more marks than Hemant, who did not get the lowest marks.
(v) Dayanand got more marks than Goutam.

27. Who got the fourth highest score?
(A) Anurag (B) Eleena
(C) Chakri (D) Dayanand

28. Who got the third lowest score?
(A) Firoz (B) Dayanand
(C) Goutam (D) Hemant

29. Who got the highest score?
(A) Eleena (B) Anurag
(C) Girish (D) Cannot be determined

Directions for questions 30 to 34: These questions are based on the following information.

Each of the four athletes, namely Johnson, Bolt, Lewis and Powell competed in the World Athletic Meet in each of the four different events, such as 100 m, 200 m, 400 m and 800 m. In each event, these athletes finished in the top four positions.

No athlete finished any two events in the same position.
(1) The athlete who finished first in 100 m finished fourth in 800 m.
(2) The athlete who finished second in 200 m finished third in 400 m and first in 800 m.
(3) Bolt finished second in 100 m and Lewis is not the last one to finish in 200 m.
(4) Johnson finished after Bolt in 200 m and 800 m.

30. Who is the fourth to finish 200 m?
(A) Johnson (B) Bolt
(C) Lewis (D) Powell

31. Who is the first to finish 400 m?
(A) Johnson (B) Bolt
(C) Lewis (D) Powell

32. Who is the third to finish 800 m?
(A) Johnson (B) Bolt
(C) Lewis (D) Powell

33. Who is the second to finish 200 m?
(A) Johnson (B) Bolt
(C) Lewis (D) Powell

34. Who is the second to finish 800 m?
(A) Johnson (B) Bolt
(C) Lewis (D) Powell

Directions for questions 35 to 37: These questions are based on the following information.

An employee is recruited on the basis of eight parameters, such as Honesty, Communication, Sense of humour, Confidence, Commitment, Positive attitude, Creativity and Intuition.

These parameters are arranged in the order of importance.
(i) Confidence is ranked higher than Sense of Humour, which is ranked higher than Creativity.
(ii) Honesty is an important factor and only one parameter is ranked above it.
(iii) Commitment is ranked higher than Intuition, which is ranked higher than Positive Attitude.
(iv) Creativity is ranked higher than Communication but lower than Positive Attitude.
(v) Commitment is ranked lower than Confidence.

35. Consider the following statements:
(x) Sense of Humour is ranked as the fourth most important factor.
(y) Positive Attitude is ranked as the sixth most important factor.
(z) There are exactly three parameters between Creativity and Commitment.
(A) x is always true.
(B) z is always true.
(C) x and z can be true simultaneously.
(D) y and z can be true simultaneously.

Difficulty Level-1: Foundation

36. Which of the following statements is true?
   (A) There are at least 3 parameters which are ranked lower than Positive Attitude.
   (B) There are at most 3 parameters which are ranked higher than Intuition.
   (C) There are at least 3 parameters which are ranked higher than Intuition.
   (D) There are at least 3 parameters which are ranked lower than Sense of Humour.

37. Which of the following is true if Intuition is the fourth most important factor?
   (A) There are exactly four parameters which are ranked higher than Sense of Humour.
   (B) There are at least three parameters which are ranked lower than Positive Attitude.
   (C) There are at least three parameters which are ranked lower than Sense of Humour.
   (D) There are at least four parameters which are ranked higher than Sense of Humour.

Directions for questions 38 to 40: These questions are based on the following information.

A group of eleven students A, B, C, D, E, F, G, H, I, J and K are given ranks according to their total marks in a final exam. The student who got the highest marks is given Rank 1. The following information is known:

(i) The only student whose rank is between H and I is D.
(ii) C scored more than G.
(iii) No one scored more than B.
(iv) H is eight ranks above F.
(v) The only student whose rank is between C and G is E.
(vi) The only student whose rank is between A and F is J.
(vii) C scored more than K.

38. Who scored the fifth highest marks?
   (A) G (B) H
   (C) C (D) J

39. How many students scored less than J?
   (A) One (B) Two
   (C) Four (D) Six

40. How many students scored less than D but more than A?
   (A) Two (B) Four
   (C) Six (D) None of these

EXERCISE-2

Directions for questions 1 to 3: These questions are based on the following information.

A group of five people, namely Alfa, Beta, Gamma, Delta and Zeta have different efficiencies in completing a work. Each of them is of different height. All of them are given the same amount of work. It is known that Beta is more efficient but shorter than Alfa. Gamma takes the least time to complete the given work and is the shortest. Delta takes more time than Zeta to complete the work but is shorter than Zeta. Alfa is less efficient than Zeta but takes less time than Delta to complete the work. Zeta is taller than Beta who is taller than Delta.

1. Who is the least efficient?
   (A) Beta (B) Zeta
   (C) Alpha (D) Delta

2. Who is the second tallest?
   (A) Alpha (B) Zeta
   (C) Beta (D) Cannot be determined

3. If the people are ranked as per their height and efficiencies, such that the tallest will be given first rank, the second tallest will get second rank and so on and the most efficient will get first rank, the second most efficient will get second rank and so on, it can be observed that no person got same rank in two categories, then who is the third tallest and the third efficient, respectively?
   (A) Zeta, Delta (B) Beta, Zeta
   (C) Delta, Alpha (D) Zeta, Beta

Directions for questions 4 to 8: These questions are based on the following information.

Six people A, B, C, D, E and F are of different heights and they were made to stand in a queue in the decreasing order of their heights. They made the following statements regarding their position in the queue where all of which are true.

(1) A says, 'I am taller than both B and C. The number of people standing ahead of D is the same as the number of people standing behind me.'
(2) B says, 'I am taller than D but not as tall as E.'
(3) C says, 'I am not standing at any end of the queue.'
(4) D says, 'F is shorter than at least two people.'
(5) E says, 'I am not standing adjacent to A.'

4. Who is the tallest?
   (A) A (B) B
   (C) C (D) D

5. Who is standing between A and E?
   (A) Either B or C (B) Only C
   (C) Only F (D) Either C or F

6. Who is the shortest?
   (A) A (B) B
   (C) C (D) D

7. How many people are standing behind C?
(A) One (B) Two
(C) Three (D) Four

8. How many people are standing ahead of B?
(A) One (B) Two
(C) Three (D) Cannot be determined

Directions for questions 9 to 12: These questions are based on the following information.

Six employees, namely P, Q, R, S, T and U are comparing their income and expenditure. The following information is known about them.

(i) The income of P is more than that of U and his expenditure is more than that of T.
(ii) The income of Q is more than that of S, but his expenditure is less than that of R.
(iii) The income of U is less than that of R, but his expenditure is more than that of R.
(iv) No two people have the equal income and the equal expenditure.
(v) The person whose income is the second highest has the second lowest expenditure, but it is not Q.
(vi) The person whose expenditure is the highest has the second lowest income, but it is not S.
(vii) T's expenditure is the third highest and his income is less than that of U.
(viii) The person whose expenditure is the lowest has the third highest income.

9. Whose income is less than that of S but the expenditure is more than that of T?
(A) P (B) U
(C) Q (D) R

10. Who has the lowest income?
(A) P (B) Q
(C) S (D) T

11. How many people have the expenditure less than that of R?
(A) One (B) Two
(C) Three (D) Cannot be determined

12. How many people have more income than that of S but less expenditure than that of T?
(A) One (B) Two
(C) None (D) Three

Directions for questions 13 to 16: These questions are based on the following information.

A group of seven cousins, namely Abhay, Bhargav, Chandini, Deekshit, Ekta, Falgun and Giri was born in a different year from 1981 to 1989. The following information is known about them.

(i) Deekshit was not born before 1985.
(ii) The oldest among them was born in 1981 and the youngest in 1989.
(iii) Abhay and Chandini were born two years apart and neither of them is the oldest.
(iv) There are as many people older than Falgun as there are people younger than him. Falgun is also as many years younger than the oldest cousin as he is older than the youngest cousin.
(v) Both Ekta and Deekshit were born in leap years.
(vi) Giri was born five years before Chandini.

13. Who is the oldest among the seven?
(A) Giri (B) Bhargav
(C) Deekshit (D) Ekta

14. In which year was Chandini born?
(A) 1983 (B) 1985
(C) 1987 D) 1989

15. Who is four years older than Abhay?
(A) Bhargav (B) Giri
(C) Falgun (D) Ekta

16. Who among them is the fifth youngest?
(A) Deekshit (B) Chandini
(C) Ekta (D) Abhay

Directions for questions 17 to 20: These questions are based on the following information.

A group of four boys A, B, C, D and four girls W, X, Y, Z compare their heights. It is found that there are two boys each of whom is taller than exactly two girls. Similarly, there are two girls each of whom is taller than exactly two boys.

Also, it is known that B is taller than W, who is taller than A, who is taller than X. Y is taller than X, but shorter than D, who is shorter than Z, who is not as tall as C, who is taller than B.

17. Which of the following statements can be false?
(A) The shortest among girls is X.
(B) The shortest among boys is A.
(C) The tallest among boys is C.
(D) The tallest among girls is W or Z.

18. If W is taller than Z, then which of the following statements is not definitely true?
(A) X is the shortest among all.
(B) B is the second tallest and Y is the second shortest.
(C) W is the third tallest and D is the third shortest.
(D) Z is taller than four people.

19. If D is shorter than A, then which of the following statements can be false?
(A) W and Z are the two girls each of whom is taller than exactly two boys.
(B) A and D are the two boys each of whom is taller than exactly two girls.
(C) D and X are the shortest among the boys and the girls respectively.
(D) W is the tallest among the girls.

20. Which choice among the following has the names of the shortest boy and the tallest girl, respectively, given that D is taller than A?

(A) C and X (B) A and X
(C) C, and Z or W (D) A, and Z or W

Directions for questions 21 to 24: These questions are based on the following information.

In a cricket match, eleven players, from A to K, scored different number of runs against the opposite team. The first two top scorers are called openers and the four lowest scorers are tailenders.

(i) G is a tailender but did not score the lowest runs.
(ii) Only three people scored more runs than K, who scored more runs than A and C.
(iii) C did not score more runs than A, who scored less runs than E. C is not a tailender.
(iv) E did not score more runs than F, who did not score the highest runs.
(v) There are at least three people who scored more runs than E.
(vi) The scores of four people are between the scores of D and B. D scored more runs than B.
(vii) H scored more runs than I, who did not score less runs than G.

21. Who scored the lowest runs?

(A) B (B) J
(C) C (D) E

22. How many people scored more runs than G and less runs than K?

(A) Two (B) Three
(C) Four (D) Five

23. If C's score is 68 and F's score is 100, then what can be the score of E?

(A) 56 (B) 67
(C) 96 (D) 105

24. In a certain way, if A is related to B and D is related to E, then in the same way, who is related to K?

(A) B (B) A
(C) F (D) H

Directions for questions 25 to 27: These questions are based on the following information.

A group of six websites, such as gmail, rediff, eBay, LinkedIn, Ask and MSN have different average number of visitors per hour and the uploading speed is different for each website. The first three websites with the most number of visitors have the lowest uploading speeds not necessarily in the same order. No two websites have the equal uploading speed.

(i) 'Ask' has the lowest uploading speed but not the highest number of visitors.
(ii) 'eBay' has less number of visitors than 'Ask' and the uploading speed of 'eBay' is more than only two websites.
(iii) The number of visitors for 'Gmail' is greater than the number of visitors for 'LinkedIn' and the uploading speed is greater for 'LinkedIn' when compared to the uploading speed of 'Gmail'.
(iv) The uploading speed of 'gmail' is not less than that of 'eBay'.

25. Which of the following is false?

(A) The number of visitors for 'LinkedIn' is the lowest.
(B) The uploading speed of 'MSN' is the second lowest.
(C) The number of visitors for 'eBay' is less than that of 'LinkedIn'.
(D) The uploading speed of 'Gmail' is not the highest.

26. Consider the following statements and chose the appropriate answer choice.

(i) The number of visitors for 'eBay' is less than that of 'MSN'.
(ii) The uploading speed of 'MSN' is less than that of 'eBay'.

(A) If (i) is true, (ii) is false.
(B) If (ii) is true, (i) is false.
(C) If (i) is true, (ii) is also true.
(D) None of these

27. Which of the following is true, if the uploading speed of 'rediff' is less than 'eBay' and the number of visitors for 'gmail' is the second lowest?

(A) The number of visitors for 'MSN' is the lowest.
(B) The uploading speed of 'rediff' is the second highest.
(C) The uploading speed of 'MSN' is the highest.
(D) The number of visitors for 'LinkedIn' is the lowest.

Directions for questions 28 to 30: These questions are based on the following information.

A group of four brothers, namely A, B, C, and D bought shares of three different companies P, Q and R. Each of them has different number of shares in each of the companies.

The following information is known about them.

(i) A has the most number of shares in Company P and least number of shares in Company Q.
(ii) B's shares in each of Company P and Q are greater than C's shares, but lesser in Company R.
(iii) D's shares are more than A in two of the given companies but D's shares are not the highest in any of the companies.
(iv) B's shares are not the second lowest in any of the companies.
(v) No person has the same rank in any two of the companies.

28. Who has the highest number of shares in Company Q?

(A) B (B) C
(C) D (D) Either C or B

29. Who has less number of shares than B in Company Q and more number of shares than B, in Company R?

(A) A (B) C
(C) D (D) All of A, C, and D

30. How many people have more number of shares than C in Company Q?

(A) One (B) Two
(C) Three (D) None

Directions for questions 31 to 33: These questions are based on the following information.

A group of five people, namely Abanti, Bhabani, Chandan, Deeptam and Fahrook were comparing their expenditure and savings. It is known that:

(i) Expenditure of no two of them is the same but their income is the same.
(ii) For every person, income is the sum of his/her expenditure and savings.
(iii) Chandan's savings are more than Deeptam's savings.
(iv) Abanti's expenditure is more than Fahrook's expenditure.
(v) The savings of Bhabani are more than that of Fahrook and the expenditure of Bhabani is more than that of Deeptam.

31. Whose expenditure is the highest?

(A) Abanti (B) Bhabani
(C) Deeptam (D) Fahrook

32. Whose savings are the highest?

(A) Fahrook (B) Bhabani
(C) Deeptam (D) Chandan

33. Savings of how many people are more than that of Fahrook?

(A) 1 (B) 2 (C) 3 (D) 4

Directions for questions 34 to 37: These questions are based on the following information.

Each of the five women, namely Amala, Kamala, Nirmala, Parimala and Vimala are of different ages and each of them has exactly one child. The five children are of different ages from 1 year through 5 years. If all the women were given ranks according to the decreasing order of their ages (i.e., the eldest woman gets the first rank) and all the children were given ranks in the similar manner, no woman has the same rank as her child. The names of their children are Chinna, Kanna, Munna, Rinku and Tinku. We know the following information about them.

(i) Chinna is the eldest but his mother is not the youngest and Nirmala is not the eldest.
(ii) Nirmala, who is the mother of the three-year-old child is elder than Vimala and Amala is elder than Kamala.
(iii) The number of years by which Tinku is elder than Munna is same as the number of years by which Chinna is elder than Rinku.
(iv) Parimala's child is Tinku and the number of women elder than Parimala is same as the number of children younger than Tinku.

34. Who is the child of Nirmala?

(A) Chinna (B) Kanna
(C) Rinku (D) Cannot be determined

35. If Munna is the child of Vimala, then Parimala is elder than

(A) Vimala (B) Kamala
(C) Nirmala (D) Cannot be determined

36. If Nirmala is elder than only one woman, then who is the mother of Rinku?

(A) Kamala (B) Vimala
(C) Amala (D) Cannot be determined

37. Which of the following statement is true?

(A) Rinku is three years old.
(B) Amala's child is five years old.
(C) Nirmala got the same rank as Tinku.
(D) Kanna is the child of either Vimala or Kamala.

Directions for questions 38 to 40: These questions are based on the following information.

There is a group of five friends A, B, C, D and E. It is known that A is heavier and shorter than D, who is richer and younger than C, who is older and shorter than E. B is lighter, shorter and richer than E, but is neither the shortest nor the youngest. The person who is the richest is also the youngest and the person who is the heaviest is also the shortest. The person who is the second eldest is also second poorest. The person who is the second heaviest is also third shortest. The person who is the second shortest is also third poorest. The person who is second richest is third eldest. The ranks are from most to least, for instance, the heaviest is ranked first and the lightest is ranked fifth or last and so on for other parameters. In any of these four comparisons, D is never ranked least and E is never ranked first, and also no person gets the same rank in any of the two comparisons. Based on the above information, solve the following questions.

38. Which of the following statements would be required to complete the arrangement?

(I) A is shorter than E.
(II) E is lighter than A.
(III) The youngest person is heavier than the poorest person.

(A) Only I and II
(B) Only I and III
(C) Only II and III
(D) Any one of I, II and III

39. What is the total sum of the ranks obtained by all the five people?

(A) 50 (B) 60
(C) 90 (D) 120

40. Which choice consists of the correct order of the names of the people with the following characteristics, such as second richest, fourth richest, fourth heaviest, second shortest?

(A) D, E, D, (A or E)
(B) (A or E), C, B, D
(C) D, C, D, B
(D) B, D, (A or E), (A or E)

EXERCISE-3

Directions for questions 1 to 4: Read the given data carefully and answer the questions that follow.

A, B, C, D and E are five students in a class. A is cleverer than B but scores less marks than D. C is cleverer than B and also scores more marks than B. E is the least clever of all but scores more marks than C. The order of the five students is 1 to 5 from the cleverest to the least clever and from the highest scorer to the least scorer.

1. If D is the cleverest, then which of the following can be the order of the five students starting from the cleverest to the least clever?

(A) D, C, B, E and A (B) D, B, A, C and E
(C) D, C, A, B and E (D) D, A, B, C and E

2. If B is cleverer than D, then who can be the cleverest of all?

(A) B (B) C
(C) A (D) A or C

3. If C stands second in terms of marks scored, then who gets the third position?

(A) D (B) B
(C) A (D) Cannot be determined

4. Which of the following students is cleverer than and also scores more marks than two other people?

(A) A (B) B
(C) C (D) Cannot be determined

Directions for questions 5 to 8: These questions are based on the following information.

A group of four students, namely Praneeth, Rajesh, Sravan and Tarun got the top four ranks in Quant, Reasoning and Verbal. For each student, the ranks in no two subjects is the same. In each subject, no two students got the same rank. We know the following additional information.

(1) The sum of the ranks of no two students is the same.
(2) Rajesh got the first rank in Quant, Praneeth got the third rank in Reasoning and Tarun got the fourth rank in Verbal.
(3) The sum of the ranks of Sravan is the highest.
(4) The rank of Rajesh in Reasoning is not same as the rank of Sravan in Verbal.

5. Who got the third rank in Quant?

(A) Tarun (B) Praneeth
(C) Sravan (D) Either (A) or (B)

6. What is the sum of the ranks of Praneeth?

(A) 8 (B) 7
(C) 6 (D) Either (A) or (B)

7. Who got the second rank in Verbal?

(A) Praneeth (B) Rajesh
(C) Sravan (D) Either (A) or (B)

8. What is the sum of the ranks of Rajesh?

(A) 6 (B) 7
(C) 8 (D) Either (A) or (B)

Directions for questions 9 to 12: These questions are based on the following information.

A group of six people, namely Anil, Sunil, Bunty, Chanty, Tarun and Varun are of different heights and weights. They are given ranks according to the descending order of their heights and weights such that the heaviest person is the first ranker and lightest person is the sixth ranker in weight category and the tallest person is the first ranker and the shortest person is the sixth ranker in the height category.

(1) The rank of Bunty in each of the categories is the same as the rank of Tarun in the other category.
(2) Varun is heavier as well as taller than both Sunil and Chanty.
(3) No person got the same rank in both the categories.
(4) Anil is the fifth shortest and Chanty is thefourth heaviest.
(5) Sunil is taller than at least two people.
(6) Tarun is shorter than Sunil and Bunty is heavier than Anil.

9. Who got the third rank in weight?

(A) Sunil (B) Tarun
(C) Varun (D) Either (A) or (B)

10. What is the rank of Chanty in weight?

(A) 2 (B) 3
(C) 4 (D) 5

11. What is the rank of Bunty in height?
(A) 4 (B) 3
(C) 2 (D) 1

12. What is the sum of the ranks of Varun?
(A) 7 (B) 6
(C) 3 (D) 5

Directions for questions 13 to 15: These questions are based on the following information.

A green grocer sells five types of vegetables, such as Carrot, Tomato, Brinjal, Cabbage and Cauliflower. Tomato is more fresh and heavier than Cauliflower. Carrot is heavier than Brinjal and more fresh than Cabbage. Cabbage is heavier than Tomato, but less fresh than Cauliflower. Brinjal is heavier than Tomato, but less fresh than it.

13. Which of the following must be the least fresh of all the vegetables?
(A) Cabbage (B) Carrot
(C) Tomato (D) Cabbage or Brinjal

14. If Cabbage is the heaviest of all, then the second heaviest can be
(A) Brinjal (B) Cauliflower
(C) Tomato (D) Carrot

15. If Carrot is not the freshest of all the vegetables, then which of the following is the most fresh of all of them?
(A) Cabbage
(B) Tomato
(C) Cabbage or Brinjal
(D) Brinjal or Tomato

Directions for questions 16 to 18: These questions are based on the following information.

A group of seven boys, namely A, B, C, D, E, F and G are standing in a row in alphabetical order from left to right in the increasing order of their weights (in kgs). The weights of all the seven boys are distinct 2-digit numbers. The following is the additional information known about them.

(i) E's weight is the average of D's, F's and G's weights.
(ii) D's weight is the average of the weights of two boys, one whose weight is a perfect square and the other, whose weight is a perfect cube.
(iii) G's weight is the sum of B's weight and D's weight.
(iv) B's weight is 10 kg less than the weight of the person whose weight is a perfect square.
(v) A's weight is a multiple of 9.

16. F's weight is ____ (in kgs)
(A) 67 (B) 65
(C) 66 (D) Cannot be determined

17. What is the difference between G's weight and A's weight? (in kgs)
(A) 48 (B) 58
(C) 68 (D) 52

18. What is the weight of all the boys together? (in kgs)
(A) 336 (B) 436
(C) 326 (D) Cannot be determined

Directions for questions 19 to 21: These questions are based on the following information.

Six teams A, B, C, D, E and F play a game. In the first round of the game every team plays with every other team exactly once. If a team wins, it scores 40 points, if it loses, it loses 10 points and a draw results in 20 points for each team. After the first round, the top two teams advance to the finals.

The following are the results of the first round:

(i) Team C neither won nor lost a match.
(ii) Teams B and E lost exactly one match.
(iii) Team F lost exactly three matches.
(iv) Team D won as well as lost exactly two matches.
(v) Team A lost exactly two matches.
(vi) The match played between team E and team F was drawn.

19. Which of the following teams advanced to the finals?
(A) A, B (B) A, E
(C) B, E (D) B, C

20. Which of the following teams scored the same number of points at the end of the first round?
(A) B, D (B) A, D
(C) D, E (D) None

21. The total number of winners in the first round is
(A) 10 (B) 9
(C) 8 (D) Cannot be determined

Directions for questions 22 to 25: These questions are based on the following information.

Four people A, B, C and D participated in a bike racing competition on Road W, which is a North-South road. The race distance is 100 km. In every stretch of 20 km, there is one signal post, which controls the roads in four directions for a maximum time of 9 minutes. At exactly 8:30 a.m., all signal turn green towards North on Road W. At all signals, in any direction the signal will be red for a duration of 9 minutes. In each stretch, between any two signals, a person travels with uniform speed. The race begins at signal 0 and ends at signal 5. Signal 0, Signal 1, Signal 2 up to Signal 5 are consecutive signals on Road W from South to North. Race will be towards the north direction from Signal 0 and will begin at exactly 8.30 a.m. To travel any stretch between any two signals by any person, the time taken is 15 min, 16 min, 20 min and 30 min.

22. If D is travelling with 75 kmph initially, by what earliest time will he reach Signal 3?

Difficulty Level-3: Advanced

(A) 9:30 a.m. (B) 9:33 a.m.
(C) 9:42 a.m. (D) 9:36 a.m.

23. If B has reached Signal 5 at 10:45 a.m., then at what speed did he travel in all stretches respectively?

(A) (40, 60, 80, 40, 60)
(B) (40, 40, 80, 80, 80)
(C) (40, 40, 80, 40, 80)
(D) (40, 60, 40, 80, 75)

24. If C has only 4 min halting time at Signal 3 and travelled with the initial speed of 80 kmph, then at what time will he reach Signal 3?

(A) 9:50
(B) 9:38
(C) 10:02
(D) Either (A) or (B)

25. If A travelled with greater speed on Stretch 1 than on Stretch 2 and starts at Signal 2 at 9:30 a.m., then what is minimum and maximum halting timings?

(A) (10, 15)
(B) (10, 14)
(C) (14, 15)
(D) (12, 14)

ANSWER KEYS

Exercise-1

1. (D) **2.** (D) **3.** (D) **4.** (A) **5.** (A) **6.** (A) **7.** (D) **8.** (C) **9.** (C) **10.** (D) **11.** (B) **12.** (A) **13.** (A) **14.** (B) **15.** (D) **16.** (A) **17.** (B) **18.** (A) **19.** (B) **20.** (D) **21.** (B) **22.** (A) **23.** (D) **24.** (D) **25.** (D) **26.** (A) **27.** (D) **28.** (C) **29.** (B) **30.** (D) **31.** (D) **32.** (B) **33.** (C) **34.** (D) **35.** (C) **36.** (C) **37.** (D) **38.** (C) **39.** (B) **40.** (B)

Exercise-2

1. (D) **2.** (D) **3.** (B) **4.** (A) **5.** (D) **6.** (D) **7.** (D) **8.** (D) **9.** (B) **10.** (D) **11.** (A) **12.** (B) **13.** (B) **14.** (C) **15.** (C) **16.** (C) **17.** (B) **18.** (C) **19.** (D) **20.** (D) **21.** (B) **22.** (D) **23.** (C) **24.** (C) **25.** (C) **26.** (C) **27.** (D) **28.** (A) **29.** (D) **30.** (A) **31.** (A) **32.** (D) **33.** (C) **34.** (D) **35.** (A) **36.** (B) **37.** (C) **38.** (D) **39.** (B) **40.** (C)

Exercise-3

1. (C) **2.** (D) **3.** (D) **4.** (D) **5.** (A) **6.** (C) **7.** (B) **8.** (B) **9.** (A) **10.** (C) **11.** (D) **12.** (D) **13.** (D) **14.** (D) **15.** (B) **16.** (C) **17.** (B) **18.** (A) **19.** (C) **20.** (B) **21.** (B) **22.** (B) **23.** (B) **24.** (D) **25.** (A)

SOLUTIONS

EXERCISE-1

1. Given L is the heaviest and K is not the lightest.
Also, the number of people heavier than P is same as the number of people lighter than Q.
∴ The possible arrangements are (in decreasing order).
L P/Q K Q/P M
Therefore, M is the lightest.

2. Given C is heavier than B but shorter than A. A is not tallest. So, A has to be the second tallest and B is the third tallest. The different possibilities are:

	Height	Weight			Height	Weight
A	2	1		A	2	3
B	1	3	OR	B	1	2
C	3	2		C	3	1

∴ Either A or C is the heaviest. Hence, the answer cannot be determined.

3. The greater than sign '>' used here means 'taller than'. Using the first letter of the names of the boys, we get the following arrangements.
L > M; P > L; N > P; N > O
On collating the above data, we get:
$N >^{(o)} P^{(o)} > L^{(o)} > M^{(o)}$ and N > O
[Here, 'O' can be placed anywhere, as indicated].
As it is not known whether P or O is taller, hence, the person who is the second tallest cannot be determined.

4. C comes before B and after A.
A > C > B (A)
D comes before E.
D > E (B)
From (A) and (B), we get that either D or A comes first. We also know that there is at least one person between D and E.
Case I: D comes first.
The possible order could be
(A) DAECB
(B) DACEB
(C) DACBE
But all of the above orders violate the condition that there must be the same number of people between D and E as between C and B.
This means that A comes first and the order is ACDBE or ADCEB.
∴ A comes first.

5. H.T. > T.D. (H.T. got more marks than T.D.)
We also know that P.T. did not get I or IV rank.
∴ It would get either II or III rank.
Anil > Murali
Gopi > Hari (H.T.)
We also know that P.T. got more marks than H.T which in turn got more marks than T.D.
⇒ P.T. > H.T. > T.D
∴ P.T. should get the 2nd rank and DME the 1st rank. Since Gopi's project got more marks than Hari, it means that Hari did project T.D. Since we also know that Anil's project got more marks than Murali's project, it means that Anil did project DME and Murali did project P.T.

6. In terms of richness:
Muk > Mur
Pre > Ad > Muk > Mur > Ra
Age:
Mur > Muk
Pr > Muk > Raj > Adi
∴ The number of people richer and older than Mukesh is only one, i.e., Aditya.

7. Raman > Rana > Raju
Ramana > Rama > Rana
Ramana Ramesh Raman
Ravi > Raman
If no other person finished the race between Ravi and Raman, then we get the following arrangement:
Ravi > Raman > Ramesh > Ramana > Rama > Rana > Raju.
The 6th person to finish the race is Rana.

8. Since each person has at least one apple and no two people have the same number of apples, we can say number of apples as 1, 2, 3 and 4. Given that Kamal has more apples than Anand. Mohan has more apples than Ravi. But Kamal does not have the highest number of apples. So, Mohan has the highest number of apples that is four apples with him.

9. Given Karan has more speed than Santosh and Sid finished the race before Vinit.
So, Karan > Santosh
Sid > Vinit
Since Karan did not finish before Vinit, we can say that Santosh finished the race last or his speed is the lowest.

Solutions for questions 10 to 12: From (i), P < R < S.
From (ii), S is the third eldest.
From (iii), Q is elder than fourth youngest [i.e., fourth eldest]
∴ Q is the second eldest.
From (iv), T is the second youngest and U is the youngest.
∴ V must be the eldest.
Also, R is the fourth eldest and P is the fifth eldest.
∴ We have,
U < T < P < R < S < Q < V.

10. V is the oldest.

11. P is the third youngest.

12. R is the fourth oldest.

Solutions for questions 13 to 15: A group of five people, namely P, Q, R, S and T are of different heights and different weights.

From (1), we have either P or Q is the tallest.

From (2), T is taller as well as heavier than R and S.

∴ T is either the 2nd or the 3rd tallest and is not the lightest.

From (3), the lightest and the 2nd tallest are the same.

∴ T is 3rd tallest and from (3), T is the heaviest.

From (3), the second tallest, i.e., either P or Q is the lightest. But from (4), Q is the fourth heaviest, i.e., he is not the lightest.

∴ P is the lightest and the 2nd tallest, Q is the heaviest.

From (4), R is heavier and shorter than S.

∴ The final order is as follows.

	Height	Weight
1	Q	T
2	P	R
3	T	S
4	S	Q
5	R	P

13. P is the lightest.

14. Only T is taller as well as heavier than S.

15. Four people are heavier than P.

Solutions for questions 16 to 18: Given that E got last rank, i.e., 5th rank in maths. D got second rank in Maths and got third rank in Physics.

From (2), we can say that C got fourth and fifth rank in Maths and Physics, respectively. (∵ E got fifth rank in Maths)

The data can be represented as follows.

	Maths	Physics
A		
B	x	3
C	4	5
D	2	x
E	5	

So, x cannot be 2, 3, 4 or 5. Now, $x = 1$ also from (5), in Physics A's rank is better than E's rank. Hence, the ranks are as follows:

	Maths	Physics
A	3	2
B	1	3
C	4	5
D	2	1
E	5	4

16. D got 1st rank in Physics.

17. A got 2nd rank in Physics.

18. The rank of A in Maths is same as the rank of B in Physics.

Solutions for questions 19 and 20: A group of five students, namely Pavan, Sravan, Charan, Tarun and Kiran are the top five rankers in a class. Each one of these is of a different height. It is given that, the tallest person is the fourth ranker, Kiran is the 2nd ranker, Tarun is taller than at least two people. Hence, Tarun could be either the tallest, the second tallest or the third tallest. Tarun is the third ranker. As Tarun is the third ranker he cannot be the tallest because the fourth ranker is the tallest.

The shortest person is the first ranker but he is not Charan.

Charan is neither the first ranker nor the shortest.

Sravan is taller than only one person. Hence, Sravan is the fourth tallest.

∴ Sravan cannot be the first or the fourth ranker. As Tarun is taller than Kiran, Kiran is not the shortest. Pavan is the first ranker and the shortest.

Tarun is the second tallest and Charan is the tallest.

The final order is as follows.

Name	Rank	Height
Pavan	1	5
Sravan	5	4
Charan	4	1
Tarun	3	2
Kiran	2	3

19. Charan is the fourth ranker.

20. As Pavan is the shortest, the remaining four people are taller than Pavan.

Solutions for questions 21 to 23: Let the number of chocolates with each of them be denoted by the first letter of his name.

From (ii), $C < S < R$.

From (iii), $S > N$

From (iv), $R - C = K - R$

As $R - C$ is at least two, and no number has a difference of three with more than one of the given numbers.

$R - C = 2$

∴ $K - R = 2$

From (ii) and (iii), we get:
N < C < S < R
As K – R = 2,
A must be greater than R.
∴ The final arrangement will be as follows.

N	<	C	<	S	<	R	<	A	<	K
3	4	5	6	7	8					

21. Ranjit has 6 chocolates.

22. Sumit has 5 chocolates.

23. The difference is = 8 – 3 = 5.

Solutions for questions 24 to 26: From (i) and (iv), Anand got a better rank than at least three people, i.e., Anand's rank can be 1 or 2 or 3.

From (iii), Charan's rank can be 6 or 5 or 4.
If Anand's rank is 3, then Charan's rank must be 4, in this case, condition (v) is violated.
∴ Anand's rank is either 1 or 2.
Here we have three possibilities:

(A)	1	2	3	4	5	6
	Gopal	Anand	Deepti	Hriday/Charan	Brijesh/Brijesh	Harish

(B)	1	2	3	4	5	6
	Anand	Gopal	Hriday/Deepti	Brijesh/Charan	Brijesh	Hriday

(C)	1	2	3	4	5	6
	Anand	Deepti	Hriday	Gopal	Brijesh	Charan

24. It is possible in case (b) and (c).
But only in case (b), we have one person between Deepti and Charan, i.e., Brijesh.

25. We have more than one possibility.

26. It is possibly (c), in which Brijesh got the fifth rank.

Solutions for questions 27 to 29: Let the marks scored by each person be denoted by the first letter of his name.

From (ii), E > C > D
From (iv), G > H
From (v) D > G
Combining the above, we get.
∴ E > C > D > G > H
From (iii), we get

1	2	3	4	5	6	7	8

F
As H did not get the lowest score, either A or B got the lowest score.
From (i) and the above data, A got the first rank and B got the eighth rank.
∴ The final arrangement will be as follows.
A > E > C > D > F > G > H > B

27. Dayanand got the fourth highest score.

28. Goutam got the third lowest score.

29. Anurag got the highest score.

Solutions for questions 30 to 34: It is given that four athletes, namely Bolt, Johnson, Lewis and Powell competed in four different events 100 m, 200 m, 400 m and 800 m. In each event these athletes finished in four different timings. No athlete finished any two events in the same position.

From (1), we have the same athlete (say x) finished first in 100 m and fourth in 800 m.

From (2), we have the same athlete (say y) finished 2nd in 200 m, 3rd in 400 m, 1st in 800 m and hence, 4th in 100 m events.

We can conclude that the person x finished 2nd in 400 m and 3rd in 200 m event.

	100 m	200 m	400 m	800 m
1	x			y
2		y	x	
3		x	y	
4	y			x

From (3), we have Bolt finished second in 100 m. Hence, Bolt is neither x nor y and hence, Bolt finished 3rd in 800 m.

Let z be the remaining athlete. z is the 2nd in 800 m, 3rd in 100 m.

From (4), as Johnson finished after Bolt in 200 m.

Bolt is the first to finish 200 m and is the fourth to finish in 400 m.

z is the first to finish in 400 m.

As Johnson finished after Bolt in 800 m, from this we can determine that x is Johnson.

As Lewis is not the last one to finish 200 m, Powell finished fourth in 200 m and y is Lewis.

The final order is as follows.

	100 m	200 m	400 m	800 m
1	Johnson	Bolt	Powell	Lewis
2	Bolt	Lewis	Johnson	Powel
3	Powell	Johnson	Lewis	Bolt
4	Lewis	Powell	Bolt	Johnson

30. Powell is the fourth to finish 200 m.

31. Powell is the first to finish 400 m.

32. Bolt is the third to finish 800 m.

33. Lewis is the second to finish 200 m.

34. Powell is the second to finish 800 m.

Solutions for questions 35 to 37: From (i), Confidence > Sense of Humour > Creativity.

From (ii), Honesty is ranked 2 from the top.

From (iii), Commitment > Intuition > Positive Attitude.

From (iv), Positive Attitude > Creativity > Communication.

From (v), Confidence > Commitment.

Combining the above statements, we get the following possibilities:

(i) Confidence > Honesty > Sense of Humour > Commitment > Intuition > Positive Attitude > Creativity > Communication.

(ii) Confidence > Honesty > Commitment > Sense of Humour > Intuition > Positive Attitude > Creativity > Communication.

(iii) Confidence > Honesty > Commitment > Intuition > Sense of Humour > Positive Attitude > Creativity > Communication.

(iv) Confidence > Honesty > Commitment > Intuition > Positive Attitude > Sense of Humour > Creativity > Communication.

Solutions for questions 38 to 40: From (iv), (iii), (i) and (vi), we get the following cases:

(A) $\underline{B} > \underline{H} > \underline{D} > \underline{I} > - > - > - > \underline{A} > \underline{J} > \underline{F} > -$

(B) $\underline{B} > - > \underline{H} > \underline{D} > \underline{I} > - > - > - > \underline{A} > \underline{J} > \underline{F}$

But from the above data, (ii), (v) and (vii) we can eliminate case (b). The final arrangement is $\underline{B} > \underline{H} > \underline{D} > \underline{I} > \underline{C} > \underline{E} > \underline{G} > \underline{A} > \underline{J} > \underline{F} > \underline{K}$.

38. C scored the fifth highest marks.

39. Two students scored less than J.

40. Four students scored less than D but more than A.

EXERCISE-2

Solutions for questions 1 to 3: Let us represent Alfa, Beta, Gamma, Delta and Zeta with familiar symbols as α, β, γ, δ and z, respectively.

For efficiency of doing the work:

Given that β is more efficient than α. γ takes least time, so he has the highest efficiency. Also δ is more efficient than z. α is less efficient than z and more efficient than δ.

So, in terms of efficiencies $\gamma > \beta > z > \alpha > \delta$ or $\gamma > z > \beta > \alpha > \delta$

For heights:

Given that β is shorter than $\alpha \Rightarrow \beta < \alpha$

γ is shortest.

δ is shorter than $z \Rightarrow \delta < z$

Also, z is taller than β who is taller than δ.

$\Rightarrow \alpha > z > \beta > \delta > \gamma$ or $z > \alpha > \beta > \delta > \alpha$

1. Delta is the least efficient.

2. Either Alpha or Zeta is the second tallest.

3. Under the given condition, the order of efficiencies is as follows.

$\gamma > \beta > z > \alpha > \delta$

The order of heights is as follows:

$z / \alpha > \alpha / z > \beta > \delta > \gamma$

$\therefore$ Beta and Zeta are the third tallest and the third efficient, respectively.

Solutions for questions 4 to 8: It is given that six people, namely A, B, C, D, E and F are standing in a queue and each of them is of a different height. These six stands in the decreasing order of their heights.

From (1), the different possible arrangements are as follows:

(i)			D	A	B/C	C/B
(ii)			A	D	B/C	C/B
(iii)		A			D	
(iv)	A					D

From (2), B is taller than D. Hence, the arrangements (i) and (ii) are not possible.

Possibility (iii): From (1) and (2), B is taller than D and shorter than A. Hence, B is between A and D.

From (1) and (3), C is shorter than A but does not stand at any end. Hence, B and C are in between A and D. From (2), E is taller than B. Hence, E should be the tallest and next to A which contradicts the condition (5). Hence, (iii) is not possible.

The only possible way is (iv).

A ____ ____ ____ ____ D

The different possible ways are:

A C E B F D

A C E F B D

A C F E B D

4. A is the tallest.

5. Either C or F stands between A and E.

6. D is the shortest.

7. Four people are standing behind C.

8. Either three or four people stand ahead of B.

Solutions for questions 9 to 12: From (i), Income: P > U

Expenditure: P > T

From (ii), Income: Q > S

Expenditure: R > Q

From (iii), Income: R > U

Expenditure: U > R

From (vii), Income: U > T and from the other given conditions, we derive the following data.

	Income	Expenditure
Highest	P	U × S
	R × Q	P
	Q	T
	S	S
	U × S	R × Q
Lowest	T	Q

9. U's income is less than that of S and more than that of T.

10. T has the lowest income.

11. One person has expenditure less than that of R.

12. Two people have income more than that of S but less expenditure than that of T.

Solutions for questions 13 to 16 : From (ii), the oldest was born in 1981 and the youngest was born in 1989.

From (iv), Falgun is the fourth oldest, who was born in 1985.

From (i) and (v), Ekta and Deekshit are born in 1984 and 1988, respectively. Ekta is the third oldest and Deekshit is the second youngest.

From (iii), Abhay and Chandini are the fifth and seventh oldest, respectively but not necessarily in the same order. They are born in 1987 and 1989, respectively.

From (vi), Chandini was born in 1987 (and is the fifth oldest) so that Giri was born in 1982 (and is the second oldest). Abhay is the youngest, who was born in 1989. Hence, Bhargav is the oldest being born in 1981.

The final order, from the oldest to the youngest is as follows. A D

Bhargav (1981) > Giri (1982) > Ekta (1984) > Falgun (1985) > Chandini (1987) > Deekshit (1988) > Abhay (1989)

13. The oldest is Bhargav.

14. Chandini was born in 1987.

15. Falgun is four years older than Abhay.

16. The fifth youngest or third oldest is Ekta.

Solutions for questions 17 to 20: Given that:

(i) Four boys – A, B, C, D
Four girls – W, X, Y, Z

(ii) There are two boys, each of whom is taller than exactly two girls.

(iii) There are two girls, each of whom is taller than exactly two boys.

(iv) ('>' symbol represents taller than)
B > W > A > X

(v) C > Z > D > Y > X and C > B

From (ii) and (iii), we can judge that there are two possibilities.

(I) 2 boys > 2 girls > 2 boys > 2 girls

(II) 2 girls > 2 boys > 2 girls > 2 boys

But from (iv) and (v), we can judge that C is the tallest. Hence, (I) is the acceptable arrangement.

Analysing (iv):

B	>	W	>	A	>	X
(boy)		(girl)		(boy)		(girl)

Analysing (v):

C	>	Z	>	D	>	Y	>	X
(boy)		(girl)		(boy)		(girl)		(girl)

Hence, we conclude that:

2 boys > 2 girls > 2 boys > 2 girls

$\boxed{C > B}$ > $\boxed{W, Z}$ > $\boxed{A, D}$ > $\boxed{Y > X}$

Also, among boys, we have:

C > B > A and C > D

Hence, the tallest boy is C (C anyhow is the tallest among all the eight people).

Among girls, we have:

Z > Y > X and W > X

Hence, the shortest girl is X, who also happens to be the shortest among all eight. Hence, the shortest boy and the tallest girl is unknown. Also, the person who is taller among (W, Z) and (A, D) is not known.

17. As discussed above, the shortest among boys could be D. Hence, (B) can be false.

18. Given that W > Z, but the person who is taller out of A and D is not known. Hence, statement (C) is not definitely true.

19. Given that D < A or A > D, but the person who is taller between W and Z is not known.

20. Given that D > A, hence, the shortest boy is A. The tallest girl is either W or Z.

Solutions for questions 21 to 24: From (ii), (iii) and (v), K scored the fourth highest, E scored the fifth highest. A scored the sixth highest and C scored the seventh highest runs.

From (i), (vi) and (vii), D scored the third highest runs, B scored the fourth lowest runs. H scored the highest runs or the second highest runs and I scored the third lowest runs or the second highest runs.

From (iv) and the above data, J scored the lowest, F scored the second highest runs, I scored the third lowest runs and H scored the highest runs.

Therefore, the final arrangement is as follows:

$$\underline{H} > \underline{F} > \underline{D} > \underline{K} > \underline{E} > \underline{A} > \underline{C} > \underline{B} > \underline{I} > \underline{G} > \underline{J}$$

21. J scored the lowest runs.

22. Five players scored more than G and less than K.

23. E's score can be 96.

24. The number of people who scored between A and B is one and A scored more runs than B.
Similarly, F is related to K.

Solutions for questions 25 to 27: From the given information we can say that the top three websites which have the most number of visitors will have the lowest speed, i.e., the first lowest, the second lowest and the third lowest.

From (i) and (ii), 'Ask' has the second highest number of visitors and 'eBay' has the third highest number of visitors.

From (iii) and (iv), the uploading speed of 'gmail' is the highest or the second highest or the third highest and the number of visitors is the lowest or the second lowest or the third lowest. 'LinkedIn' has the lowest or the second lowest number of visitors and uploading speed is the highest or the second highest.

	Number of visitors	Uploading speed
1		
2	Ask	
3	eBay	
4		eBay
5		
6		Ask

25. Statement (C) is false.

26. If (i) is true, (ii) is also true.

27. Statement (D) is true.

Solutions for questions 28 to 30: From (i), (ii) and (iii), D could have bought the second highest or the third highest number of shares in company Q.

B could have bought the second highest or third highest number of shares in P and the highest or the second highest number of shares in Q.

From (iv) and (v), the number of shares bought by B is the second highest in P, the highest in Q and the lowest in R.

∴ The final distribution is as shown below.

P	Q	R
A	B	C
B	C	D
C	D	A
D	A	B

28. B has the highest number of shares in Q.

29. All of A, C and D.

30. One person has one more share than C in company Q.

Solutions for questions 31 to 33: Let each person be represented by the first letter of his respective name. As the expenditure of no two of them is the same and their income is the same, the person who has the maximum expenditure would have the least savings, the person who has the second highest expenditure would have the second least savings and so on.

From (iii) C's savings are more than D's savings.
∴ C's expenditure is less than D's expenditure.
From (iv) A's expenditure is more than F's expenditure.
∴ A's savings are less than F's savings.
From (v) B's savings are more than F's savings.
and B's expenditure is more than D's
∴ B's savings are less than D's.
In terms of savings the order is as follows:
C > D > B > F > A
In expenditure, the order is as follows:
A > F > B > D > C

31. Abanti's expenditure is the highest.

32. Chandan's savings is the highest.

33. Savings of three people are more than that of Fahrook.

Solutions for questions 34 to 37: Let us consider case (I).
From (i) and (ii), Nirmala does not have Rank 1, 3 or 5.
From (iv), Parimala's Rank is 2.
Hence, Nirmala's Rank is 4.
⇒ Nirmala's Rank is 5 (From (ii)).
From (ii), Amala's rank is 1 and Kamala's rank is 3.

Case (I)

Rank	Child	Woman
1	Chinna	Amala
2	Rinku	Parimala
3	Kanna	Kamala
4	Tinku	Nirmala
5	Munna	Vimala

From the above table, Kanna's mother is Nirmala and Chinna's mother is neither Amala nor Vimala.
Since Tinku's mother is Parimala, Chinna's mother is Kamala.
⇒ Munna's mother is Amala and Rinku's mother is Vimala.
Hence, the mother child pairs are as follows.

Amala – Munna, Parimala – Tinku, Kamala — Chinna, Nirmala – Kanna, Vimala – Rinku

Let us consider case (II):
Nirmala's rank is none of 1, 3 or 5.
From (iv), Parimala's rank is 4.
Hence, Nirmala's rank is 2.
From (ii), Rank 3 and 5 are for Vimala and Kamala in any order. Hence, Amala's rank is 1.

Case (II)

Rank	Child	Woman
1	Chinna	Amala
2	Tinku	Nirmala
3	Rinku	Vimala/Kamala
4	Munna	Parimala
5	Kanna	Kamala/Vimala

From the above table, Rinku's mother is Nirmala, Tinku's mother is Parimala, Chinna's mother is not Amala. Hence, Chinna's mother is either Vimala or Kamala, the woman whose rank is 3.
Hence, Kanna's mother is neither 3rd nor 5th ranked women.
Hence, Kanna's mother is Amala.
The following are the mother – child pairs.

Amala – Munna, Parimala – Tinku, Kamala — Chinna, Nirmala – Kanna, Vimala – Rinku

Let us consider case (III).
Parimala's rank is 4.
Nirmala's rank is none of 1, 3 and 5.
Hence, Nirmala's rank is 2.
From (ii), the rank's of Vimala and Kamala is 3 and 5 in any order.
Hence, Amala's rank is 1.

Case (III)

Rank	Child	Woman
1	Chinna	Amala
2	Tinku	Nirmala
3	Kanna	Vimala/Kamala
4	Rinku	Parimala
5	Munna	Vimala/Kamala

From the above table, Nirmala's child is Kanna and Parimala's child is Tinku. Chinna's mother is 3rd ranked woman.
Hence, Munna's child is Amala.
The following are the mother – child pairs.

Amala – Kanna, Nirmala – Rinku, Vimala — Munna/Chinna, Parimala – Tinku, Kamala – Munna/Chinna.

34. Nirmala's child is either Rinku or Kanna.

35. If Munna is the child of Vimala, cases (II) and (III) prevail. But in case (III) Vimala is the mother of Munna, then Vimala should have Rank 3. But Chinna's mother got Rank 3. Hence, case (III) is does not hold good. In case (II) Vimala has Rank 5 and Kamala has Rank 3.
⇒ Parimala is elder than Vimala.

36. If Nirmala is elder than only one person, then case (I) prevails. Then Vimala is Rinku's mother.

37. Statement (C) is true.

Solutions for questions 38 to 40: There is a total of four comparison parameters, such as Weight, Height, Richness and Age. Let us put down the data given for the comparison (greater than symbol '>', which represents a person being heavier, taller richer or older than the other).

	Weight	Height	Richness	Age
(i)	A > D	D > A	D > C	C > D
(ii)	E > B	E > C	B > E	C > E
(iii)	E > B	B ≠ Youngest		

(iv) B ≠ Shortest
(v) Richest = Youngest
(vi) Heaviest = Shortest
(vii) Second Oldest = Second poorest (or Fourth richest)
(viii) Second Heaviest = Third shortest (or Third tallest)
(ix) Second Shortest = Third poorest (or Third richest)
(x) Second Richest = Third eldest
(xi) E ≠ Heaviest, Tallest, Richest, Eldest
(xii) No person gets the same rank in any two comparisons
Now, neither B nor C can be the tallest, as E is taller than

these two. Also, A cannot be the tallest, as D is taller than A. E anyhow cannot be ranked first.

Hence, D must be the tallest.

It is given that E is never ranked first. Hence, the first rank in these 4 different comparisons must belong to A, B, C and D. Also, the last ranks must belong to A, B, C and E as D is never ranked last. Now B is neither the youngest nor the shortest and as B > E in richness. Here, B is not the poorest. Therefore, B must be the lightest.

Now the richest person cannot be D (already D is the tallest and no one can have the same rank in any two comparisons), nor E (B > E). Also, the youngest person is neither C (C > D, E) nor B (given B is not the youngest) nor D (D is never ranked last). Also, the richest and the youngest person is the same. As E is not the richest (E is never ranked first). Hence, the remaining person 'A must be the richest as well as the youngest'.

Now, the shortest person can neither be E (E > C, B) nor B nor A.

(They are already ranked fifth in some other comparison) nor D (D can never be ranked last). Hence, C must be the shortest as well as the heaviest person. This means that E must be the poorest person. Also, B must be the eldest person.

Now let us represent the ranks as given below

Person	Weight	Height	Richness	Age
A			1	5
B	5			1
C	1	5		
D		1		
E			5	

Also, Second poorest = Fourth richest, Third shortest = Third tallest, and Second shortest = Fourth tallest; Third poorest = Third richest.

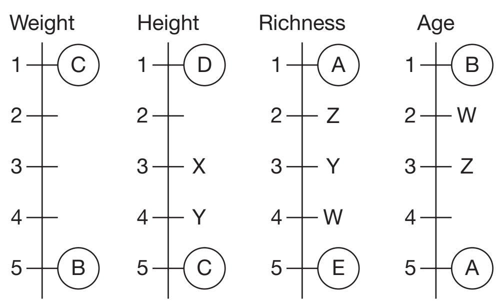
→ Second oldest, Fourth richest = W

→ Second heaviest, Third tallest = X

→ Fourth tallest, Third richest = Y

→ Second richest, Third eldest = Z

We now represent the ranks on a scale as shown below:



Now D is neither X nor Y. Hence, D ≠ Third poorest. Then D must be Z (as D ≠ 4, ∴ D > C) this means that D must be the second richest and third oldest.

Now D has first, second and third ranks in three of the parameters and as D ≠ 5. Hence, D must be the fourth heaviest.

As D is the third oldest, C must be the second oldest and the fourth richest (∴ C > D, E in age).

Then E must be the fourth eldest and B must be the third richest as well as the fourth tallest.

Now A or E is second or third heaviest. Similarly, A or E is second or third tallest. We complete the table as given below:

Person	Weight	Height	Richness	Age
A	Second or Third	Third or Second	1	5
B	5	4	3	1
C	1	5	4	2
D	4	1	2	3
E	Third or Second	Second or Third	5	4

38. Statement I (A is shorter than E) can complete the arrangement and fix the positions of A and E.
Statement II (E is lighter than A) makes A the second heaviest and the third tallest and E as the third heaviest and the second tallest thus completing the arrangement. Similarly, 'The youngest person (A) is heavier than the poorest person (E)', also helps in completing the arrangement. Hence, any one of these statements is sufficient to complete the arrangement.

39. Since in every comparison ranks 5, 4, 3, 2 and 1 are awarded, the sum of the ranks in any comparison is 5 + 4 + 3 + 2 + 1 = 15. Total sum of all ranks awarded is 15 × 4 (since there are 4 comparisons) = 60.

40. Second richest = D
Fourth richest = C
Fourth heaviest = D
Second shortest = B
Hence, the order is DCDB.

EXERCISE-3

Solutions for questions 1 to 4: In terms of cleverness, it is known that

A is cleverer than B, C is cleverer than B, E is the least clever of all, i.e., A > B and C > B, but there is no relation between A and C.

So, both A and C are cleverer than B and E is the last in order of cleverness. So, it is clear that neither A nor C can be 4th and B cannot be 1st or 2nd, as A and C are cleverer than him.

1	2	3	4	5
-	-	-	-	E

So, nothing can be decided about the exact order. In terms of scores, it is given that A scores less marks than D, i.e., D > A and C gets more marks than B, i.e., C > B, whereas E scores more than C, i.e., E > C so E > C > B. Now the order cannot be decided. It can be only said that E can be neither 4th nor 5th. B cannot be 1st and 2nd. D cannot be 5th and A cannot be 1st in the order of score.

1. If D is the cleverest of all, then the order is as follows:

1	2	3	4	5
D	A or C	C Or A	B	E

Then one of the orders can be, D, C, A, B, E.

2. If B is cleverer than D, then A or C stands first in terms of cleverness.

1	2	3	4	5
A/C	C/A	B	D	E

3. C stands second in order of marks. Then the third score may be of B or D. So, who gets the third position cannot be determined.
4. The order of cleverness or marks is not clearly given. So, it cannot be determined who is cleverer and also scores more marks than two people.

Solutions for questions 5 to 8: As, the total sum = 30 and least sum (1, 2, 3) = 6, and the highest sum is (2, 3, 4) = 9, the sum of the ranks must be 6, 7, 8 and 9.

From the given information, we have the following table.

	Rajesh	Tarun	Praneeth	Sravan
Quant	1			
Reasoning			3	
Verbal		4		
Total				9

As the sum of the ranks of Sravan is 9, the ranks must be 2, 3 and 4.

∴ Praneeth got the first rank in Verbal.

⇒ Tarun got the first rank in Reasoning.

In Reasoning, if Rajesh got the second rank and Sravan got the 4th rank, then Praneeth must have got the fourth rank in Quant. As only Rajesh can get the sum (1, 2, 3) as 6, Rajesh's rank in Verbal must be 3.

∴ Sravan's rank in Verbal must be 2nd, which is violating (4).

∴ In Reasoning, Rajesh got the fourth rank and Sravan got the second rank.

Now, only Praneeth can get the sum as 6.

∴ Praneeth's rank must be two in Quant.

Tarun's sum cannot be 7, Rajesh's rank in Verbal must be two.

∴ Sravan's ranks in Verbal and Quant are 3 and 4, respectively.

∴ Tarun got the third rank in Quant.

	Rajesh	Tarun	Praneeth	Sravan
Quant	1	3	2	4
Reasoning	4	1	3	2
Verbal	2	4	1	3
Total	7	8	6	9

5. Tarun got the third rank in Quant.
6. Sum of the ranks of Praneeth is 6.
7. Rajesh got the second rank in Verbal.
8. Sum of the ranks of Rajesh is 7.

Solutions for questions 9 to 12: Let each person be denoted by the first letter of his name.

From the given information we have:

	A	S	B	C	T	V
Height	2					
Weight				4		

From (2) and (6), we have

Height: V > S, V > C, S > T

⇒ V > S > T, V > C

Weight: V > S, V > C, B > A

From (1), neither B nor T got 2nd or 4th rank in any category.

∴ T's rank in height must be 5 or 6.

If T's rank in height is 6, then B's rank in weight will be 6, which violates (6).

T's rank in height is 5 and B's rank in weight is 5.

∴ A's rank in weight will be 6.

Also, as none of B, S, T, V can get the 6th rank in height, then C gets the 6th rank in height.

B and V got the first and the third ranks in height, respectively.

∴ S got the 4th rank in height.

Now, in weight, if V gets the first rank, then T must get the third rank.

In height, B must get the third rank.

⇒ V gets the 1st rank in weight as well, which violates (3).

∴ In weight, V gets the second rank.

∴ S gets the third rank.

⇒ T gets the first rank.

In height, B gets the first rank and V gets the third rank.

∴ The final results will be as follows:

	A	S	B	C	T	V
Height	2	4	1	6	5	3
Weight	6	3	5	4	1	2

9. Sunil got third rank in weight.

10. Chanty's rank in weight is 4.

11. Bunty got the 1st rank in height.

12. Five

Solutions for questions 13 to 15: The given data is that the five types of vegetables are Carrot, Tomato, Brinjal, Cabbage and Cauliflower.

In terms of freshness, it is given that:

Tomato is more fresh than Cauliflower, i.e., Tomato > Cauliflower.

Carrot is more fresh than Cabbage, i.e., Carrot > Cabbage.

Cauliflower is more fresh than Cabbage, i.e., Cauliflower > Cabbage.

Tomato is more fresh than Brinjal, i.e., Tomato > Brinjal.

So, Tomato is more fresh than Brinjal and Cauliflower.

Carrot and Cauliflower are more fresh than Cabbage.

Let us denote Tomato by To, Brinjal by Br, Cauliflower by Cl, Carrot by Cr and Cabbage by Cb.

The following things are known.

Freshness wise	To > Br To > Cl > Cab Cr > Cab
Weight wise	Cr > Br > To > Cl Cab > To

So, the order of freshness cannot be decided.

Tomato is heavier than Cauliflower, i.e., Tomato > Cauliflower.

Carrot is heavier than Brinjal, i.e., Carrot > Brinjal.

Cabbage is heavier than Tomato, i.e., Cabbage > Tomato.

Brinjal is heavier than Tomato, i.e., Brinjal > Tomato.

So, the order can be as follows:

Table I

1	2	3	4	5
Carrot	Brinjal	Cabbage	Tomato	Cauliflower
Cabbage	Carrot	Brinjal	Tomato	Cauliflower
Carrot	Cabbage	Brinjal	Tomato	Cauliflower

13. The order of freshness is:

Tomato, Brinjal, Cauliflower, Carrot, Cabbage

Tomato, Carrot, Cauliflower, Brinjal, Cabbage

Tomato, Carrot, Cauliflower, Cabbage, Brinjal

So, either Cabbage or Brinjal is the least fresh of all the vegetables.

14. If Cabbage is the heaviest, the order is in terms of the heaviest to the least heavy.

1	2	3	4	5
Cabbage	Carrot	Brinjal	Tomato	Cauliflower

So, Cabbage is the heaviest and Carrot is the 2nd heaviest.

15. If Carrot is not the freshest of all, then only Tomato is the freshest.

Solutions for questions 16 to 18: From (ii), the square and cube numbers can be (16, 64), (25, 27), (36, 64), (49, 27), and (81, 27). The weight of all the seven boys is a 2-digit number. (64, 64) is not possible, since no two boys have the same weight. Hence, D's weight can be 40, 26, 50, 38 and 54, respectively. From (iv) and the above data, the possible weights are as shown below.

Person	A	B	C	D	E	F	G
Case (i)		6		40			
Case (ii)		15		26			
Case (iii)		26		50			
Case (iv)		39		38			
Case (v)		71		54			

Case (i) is eliminated, since the weight of B is a single digit number. Case (iv) and case (v) are eliminated, since the people are standing in increasing order of their weights from left to right.

In case (ii), C's weight is 25 and E's weight is 27.

In case (iii), C's weight is 36 and E/F/G's weight is 64.

From (v), case (ii) is eliminated, since A's weight can be 9 or 18 or 27 and so on. A's weight cannot be 9, as 9 is a single digit and A's weight cannot be > 15.

In case (iii), A's weight is 18.

Persons	A	B	C	D	E	F	G
Case (i)	18	26	36	50	64		
Case (ii)	18	26	36	50		64	
Case (iii)	18	26	36	50			64

From (iii), in case (i) and case (ii) G's weight is = 26 + 50 = 76.
Case (iii) is eliminated as G's weight ≠ 76.
From (i):

In case (i), E's weight = $\frac{50 + \text{F's weight} + 76}{3}$

$$64 = \frac{126 + \text{F's weight}}{3}$$

∴ F's weight = 66.

In case (ii), E's weight = $\frac{50 + 64 + 76}{3}$ = 63.3 not a 2-digit number.

Hence, case (ii) is eliminated.
∴ The seven people and their weights are as shown below.

People	A	B	C	D	E	F	G
Weights	18	26	36	50	64	66	76

16. F's weight is 66 kgs.

17. The difference between G's weight and A's weight is 58 kgs.

18. The weight of all the boys together is 336 kgs.

Solutions for questions 19 to 21: From the given information, we derive the following table.

Team	Win	Loss	Draw
A		2	
B		1	
C	–	–	5
D	2	2	1
E		1	1
F		3	1

As C had drawn a match with every other team the five matches should reflect in the other's score sheets also. Hence, the number of matches drawn by A and B should be at least '1' each and E and F should be at least '2'. It can be further seen that the number of matches drawn is 6, as we already have 9 losses in the loss column, we need to have 9 wins in the wins column.
Hence, the final score sheet is as follows.

Team	Win	Loss	Draw	Score
A	2	2	1	80
B	3	1	1	130
C	–	–	5	100
D	2	2	1	80
E	2	1	2	110
F	–	3	2	10

19. B and E advance into the finals.

20. A and D have the same score.

21. The total number of winners in the first round is 9.

Solutions for questions 22 to 25: From the information on time taken to traverse a stretch, we can determine their speeds in km/h. Further, at all signals, the maximum wait time in any direction is 9 minutes, this implies that the signal remains green for three minutes each direction. It also implies that the signal turns green after every 12 minutes in each direction: Towards north, signal turns green at 8.30, 8.42, 8.54, 9.06, 9.18, 9.30 and so on. Thus, in any hour, at any signal, the signal turns green at 6th, 18th, 30th, 42nd and 54th minutes invariably.

No given speed or travel matches the cycle of green signal at the signal posts.

Thus, invariably each person will wait at each signal post irrespective of his speed.

Speed KMPH	Travel time to reach next signal	Waiting time	Time to start next stretch
80	15 minutes	9 minutes	24 minutes
75	16 minutes	8 minutes	24 minutes
60	20 minutes	4 minutes	24 minutes
40	30 minutes	6 minutes	36 minutes

22. D's initial speed = 75 kmph
⇒ Time on 1st stretch = 16 minutes
Earliest time
⇒ Speed has to be increased = 80 kmph
⇒ Time taken on second and third signal stretches = 15 minutes.
⇒ Time taken = (16 + 8) + (15 + 9) + (15 + 0) = 63 minutes.
Earliest time = 8.30 + 63
= 9.33

23. 10 : 45 – 8 : 30 = 2 : 15 = 135 minutes
Choice (A) = 36 + 24 + 24 + 36 + 20
= 140 minutes
Choice (B) = 36 + 36 + 24 + 24 + 15
= 135 minutes

Choice (C) = 36 + 36 + 24 + 36 + 15
= 157 minutes
Choice (D) = 36 + 24 + 36 + 24 + 16
= 136 minutes.
Speeds in choice (B) takes B to Signal 5 by 10.45 p.m.

24. Initial speed = 80 kmph ⇒ Time to reach Signal 3
= (15 + 9) + x + 20 x can be 24 or 36
∴ Time to reach Signal 3 = 44 + 24 or 44 + 36
= 68 or 80 minutes
⇒ Time = 9.38 or 9.50

25. Starts at Signal 2 = 9.30
⇒ Time = 60 minutes = 24 + 36 (24 > 36) minimum waiting time = 4 + 6 = 10 minutes. Maximum waiting time = 9 + 6 = 15 minutes

CHAPTER 6

Binary Logic

Learning Objectives

In this chapter, you will:

- Understand the concept of Binary Logic, truth tellers, liars and alternators.
- Figure out how to reject a person based on his/her statements.
- Understand how to arrive at facts by identifying whether statements made by a person are true or false and identify whether a person is a truth teller or a liar or an alternator.
- Learn how to solve questions which have a combination of Binary Logic and other topics.

In some of the competitive exams, we come across questions which are to be answered based on the truth or falsity of statements given in the question. In these questions we come across three kinds of terms. They are 'Truth-Teller', 'Liar' and 'Alternator'. These terms may or may not be explicitly defined in the question. The following are the definitions of these terms, which can be applied while answering these questions, unless otherwise defined.

Truth-teller: A person whose each and every statement is true.

Liar: A person whose each and every statement is false.

Alternator: If the first statement of the person is true, then the next statement is false or vice versa.

The question does not specify as to which statement is true or false. It has to be found out by trial and error and by checking for consistency in the given statement.

If the question states that there is a truth teller among the given persons, we assume one person as the truth teller and list down statements stated by that person as facts. We compare the statements of the other persons with the facts listed out. If we do not come across any contradiction, we can conclude that our assumption holds good, otherwise we continue by assuming another person to be the truth teller.

In case there is no indication whether or not there is a truth teller, we go by assuming a fact. The following illustrations help in understanding the concept.

SOLVED EXAMPLES

6.01: Among the three people A, B and C, one is a truth-teller, one is a liar and the other is an alternator. Each of them made the following statements in reply to the question asked about them.

A: I am not the liar. C is the liar.
B: I am the liar. A is the truth-teller.
C: I am the alternator. B is the liar.

Find out who the truth-teller is, who the liar is and who the alternator is.

Sol: The statement, 'I am not the liar', could be that of a truth-teller or of a liar or of an alternator. Hence, we cannot find out the nature of A. From the statement of B, i.e., 'I am a liar', it can be concluded that B is the alternator, because neither a truth-teller nor a liar would say that he is a liar. Hence, the second statement of B, i.e., 'A is the truth-teller' is true. From the above, it can be concluded that the statement of C, i.e., I am the alternator is false. Hence, C is the liar. Let us check the truth in each of the statements of A, B and C with the help of the conclusions made.

A: I am not the liar (True). C is the liar (True).
B: am the liar (False). A is the truth-teller (True).
C: I am the alternator (False). B is the liar (False).

From the above, it is clear that the conclusions made, i.e., A is the truth-teller (both the statements are true), B is the alternator (one statement is true and the other is false) and C is the liar (both the statements are false) are correct. The questions asked may have one, two, three or even four statements made by each person.

Directions for questions 6.02 to 6.04: These questions are based on the following information.

Each of the boys Raman, Raghu and Rajan likes a different colour among red, blue and green. One of them always speaks the truth, one of them always lies and the third one alternates between the truth and lie. They made the following statements.

Raman : I like green.
Raghu likes red.
I am an alternator.
Raghu : Rajan likes Blue.
Raman does not like green.
I am a liar.
Rajan : I do not like red.
Raghu does not like red.
Raman is a liar.

6.02: Who is the alternator?
(A) Raman
(B) Raghu
(C) Rajan
(D) Cannot be determined

6.03: Who likes red colour?
(A) Raman
(B) Raghu
(C) Rajan
(D) Cannot be determined

6.04. Who is the liar?
(A) Raman
(B) Raghu
(C) Rajan
(D) Cannot be determined

Solutions for questions 6.02 to 6.04: We cannot take Raman as a truth-teller, since his third statement is contradicting.

Similarly, we cannot take Rajan as a truth-teller since his third statement is contradicting.

Rajan is a truth-teller.

Person	Statement 1	Statement 2	Statement 3	Colour
Raman	F	F	F	Red
Raqhu	F	T	F	Blue
Raian	T	T	T	Green

Hence, Rajan is a truth-teller.

6.02: Raghu is the alternator.

6.03: Raman likes red colour.

6.04: Raman is the liar.

6.05: Each one of the three friends Divya, Bhanu and Ravi went to a different place among Goa, Ooty and Shimla for summer vacation. They replied to the question, "Who went to Goa?", in the following manner:

Divya : I went to Shimla;
Ravi did not go to Ooty.
Bhanu : Divya did not go to Shimla;
Ravi did not go to Shimla.
Ravi : Bhanu did not go to Ooty;
I went to Goa.

If exactly two of the friends always tells the truth then who went to Ooty?
(A) Divya
(B) Bhanu
(C) Ravi
(D) Cannot be determined

Solution for question 6.05:

6.05: (i) Let us assume that Divya is truth-teller.

Name	Statement 1	Statement 2	Place
Divya	T	T	
Bhanu	F	T	
Ravi	F	T	Goa

It we take Ravi as a truth-teller, then we won't have two truth tellers.

Therefore, Divya went to Ooty.

(ii) If Bhanu is a truth-teller.

Name	Statement 1	Statement 2	Place
Divya	F	F	Ooty
Bhanu	T	T	Shimla
Ravi	T	T	Goa

6.06: In the Dhola award ceremony, Salman, Abhishek and Shah Rukh were nominated for the awards– Hero No. 1 and Zero No. 1. The awards were given to two persons. When the reporters asked them about the awards, each of them made two statements as follows. It is known that one among them always speaks truth, one always lies and the other one alternates between truth and lie in any order.

Salman: I did not get any award.
Shah Rukh got the Hero No. 1 award

Abhishek: I got the Hero No. 1 award
Shah Rukh got the Zero No.1 award

Shah Rukh: I got the Hero No. 1 award
Salman got the Zero No. 1 award

If at least one of the statements made by Abhishek is a lie, then who got the Hero No. 1 award?

(1) Salman
(2) Abhishek
(3) Shah Rukh
(4) Cannot be determined

Solution for question 6.06: Assuming both the statements of Salman as true, we get

Name	I	II	Award
Salman	T	T	-
Abhishek	F	F	Zero No. 1
Shah Rukh	I	F	Hero No. 1

Here Shah Rukh got the Hero No 1. award.

Assuming both the statement of Shah Rukh as true, we get,

Name	I	II	Award
Salman	F	T	Zero No. 1
Abhishek	F	F	–
Shah Rukh	T	T	Hero No. 1

In this case also Shah Rukh got the Hero No.1 award.

Directions for questions 6.07 to 6.08: These questions are based on the following information.

I met four students A, B, C and D. They have four different nick names among, Kaka, Mama, Baba and Lala. Each of them is from a different class among 1 through 4, not necessarily in the same order. When I asked them about their nick, name and the respective classes, each of them made three statements and no two of them made the same number of true statements. The statements made by them are as follows.

A: Baba is in class 2.
Kaka is in class 1.
Mama is in class 3.

B: My nick name is Mama.
I am in class 2.
Baba is in class 1.

C: My nick name is Lala.
I am in class 4.
Mama is in class 3.

D: My nick name is Lala.
I am in class 3.
Kaka is in class 1.

All the statements made by C are true and the second statement of either B or D is true but not both. Answer the following questions.

6.07: Whose nick name is Baba?

(1) A (2) D
(3) B (4) C

6.08: Who is in class 2?

(1) A (2) B
(3) D| (4) C

Solution for question 6.06: As none of them made the same number of true statements, one of them made three true statements, one made two true statements, one made one true statement and other one made no true statement.

It is given that all the statements of C are true. We get, C's name as Lala and he is in class 4.

Also Mama is in class 3.

∴ The third statement of A is also true.

So Kaka and Baba are in class one and two (in any order).

Therefore Third statement made by B or D is lie. Let us assume that the third statements of B is a lie. So , the third statement of D will be true, i.e. Kaka is in class 1. Then the first two statements of A become true which violates the conditions.

∴ The third statement of B is true. So, the third statement of D is a lie. One among A, B, C and D has made all false statements. As A, B and C has made at least one true statement all the statements made by D are false.

∴ D's nick name is not Lala and D is not in class 3. Kaka is not in class 1.

As D's second statement is false, B's second statement must be true. Hence B is in class 2 and his nick name is Kaka. Also the nick name of A is Mama and he is in class 3.

D's name is Baba, and he is in class 1.

	I	II	II	Class	Name
A	F	F	T	3	Mama
B	F	T	T	2	Kaka
C	T	T	T	4	Lala
D	F	F	F	1	Baba

6.07: D's nick name is Baba

6.08: B is in class 2.

EXERCISE-1

Directions for questions 1 to 8: Select the correct alternative from the given choices.

1. Amit, Ashok and Azad made one statement each. The following are the statements made by them:

Amit: I am a liar.
Ashok: I am not a truth-teller.
Azad: I neither always make true statements nor always make false statements.

It is known that a truth-teller is one who always speaks the truth, while a liar is one who always lies (or makes false statements) and an alternator is one who alternates between a truth and a lie.

Whose statement is definitely true?

I. Amit's
II. Ashok's
III. Azad's

(A) Only I
(B) Only II
(C) Only III
(D) Only II and III

2. On 'Kya-Kya' island, there are two tribes, they are truth-tellers and liars. Truth-tellers always speak the truth and liars always lie. One day I met three Kya-Kya islanders A, B and C, and asked them 'Who among you is the truth-teller?' Following were their replies.

A : I am not a liar.
B : C is not a liar.
C : B is a truth-teller.

If it was known that exactly one person among the three was a truth-teller and the other two were liars, then who among them must be the truth-teller?

(A) A (B) B
(C) C (D) Cannot be determined

3. One day I met three people, named Anand, Bharat and Chandu each of whom belonged to a different tribe amongst truth-tellers, liars and alternators. When asked to introduce themselves, each of them gave two replies, as given below. Also, it is known that the truth-tellers always speak truth, the liars always lie and the alternators alternate between truth and lie, in any order.

Anand: I am the truth-teller. Bharat is the liar.
Bharat: I am not the liar. Chandu is the truth-teller.
Chandu: I am not the liar. Anand is not the truth- teller.

Who among these three people is the alternator?

(A) Anand (B) Bharat
(C) Chandu (D) Cannot be determined

4. A group of three players, namely Aalu, Kachaalu and Bhalu were playing poker and suddenly started to quarrel among themselves by blaming each other for cheating. It was found out that at least one person among the three cheated. When they were asked who cheated, their replies were as follows:

Aalu: I did not cheat.
Kachaalu cheated.
Kachaalu: I did not cheat.
Both Aalu and Bhalu cheated.
Bhalu: I did not cheat.
Only Kachaalu did not cheat.

If exactly one person among them always spoke truth, another always lied and the third alternated between the truth and lie, then which of the following statements can never be true in any case?

(A) Only Aalu and Bhalu cheated.
(B) Only Aalu and Bhalu did not cheat.
(C) Bhalu always spoke the truth.
(D) Bhalu alternated between truth and lie.

5. Sameer, Sameep and Sumer participated in a quiz contest and each one of them received exactly one title among the three titles, such as the winner, the 1st runner-up and the 2nd runner-up. When asked, 'Who among you three is the winner?', following were their replies:

Sameer: I am the winner.
Sameep is not the 1st Runner-up.
Sameep: I am the winner.
Sameer is the 2nd runner up.
Sumer: I am the winner.
Sameep is the 2nd runner-up.

It is also known that one among them always tells the truth, one always lies and one alternates between the truth and lie (not necessarily in that order).

Who can never be the 1st runner-up?

(A) Sameer (B) Sameep
(C) Sumer (D) Cannot be determined

6. Each person out of A, B and C had exactly one different title amongst the Good, the Bad and the Ugly. Also, each person always gave two answers to any question. Exactly one among them always spoke the truth, another always lied and the last person always alternates between truth and lie (in any order). When asked about their titles, following were their replies:

A: B is 'the Good'.
I am 'the Ugly'.
B: C is 'the Bad'
A is not 'the Good'.
C : B is 'the Ugly'.
A is not 'the Bad'.

Which among the following choices has the names of the persons who had the title the Good, the Bad and the Ugly, respectively?

(A) A, B, C (B) C, A, B
(C) B, C, A (D) A, C, B

7. John, Johny and Janardan participated in a race and each won a different medal among Gold, Silver and Bronze, not necessarily in that order. Each person among them gives two replies to any question, one of which is true and the other is false (in any order). When asked about the details of the medals obtained by them, the following were their replies:

John: I won the Gold medal.
Johny won the Bronze medal
Johny: John won the Silver medal.
I won the Gold medal.
Janardan: Johny won the Silver medal.
I won the Gold medal.

Which among the following is the correct order of the people who won the Gold medal, the Silver medal and the Bronze medal, respectively?

(A) John, Johny, Janardan
(B) Janardan, John, Johny
(C) Johny, Janardan, John
(D) Janardan, Johny, John

8. The inspector of police, Chulbul Pandey is questioning five suspects, namely Sheroo, Santhosh, Bhayankar, Chola, and Bhala about a bank robbery. They made the following statements.

Sheroo: Bhayankar robbed the bank.
Santhosh: Sheroo did not rob the bank.
Bhayankar: Santhosh is telling the truth.
Chola: Sheroo is telling the truth.
Bhala: Exactly one of us is telling the truth.

Chulbul Pandey just received evidence that Bhala is telling the truth. Then which of these statement is true?

(A) Sheroo robbed the bank.
(B) Bhayankar is telling the truth.
(C) Both Santhosh and Chola robbed the bank.
(D) Sheroo did not rob the bank.

Directions for questions 9 to 11: These questions are based on the following information.

Truth-teller: A person who always speaks the truth.
Liar: A person who always lies.
Alternator: A person who alternates between truth and lie, in any order.

9. Who among the above three kinds of person can make the following statement – 'I am not an alternator'?

(A) A truth-teller
(B) A liar
(C) An alternator
(D) Cannot be determined

10. Who among the above three kinds of person can make the following statement – 'I am not a truth-teller?'

(A) A truth-teller
(B) A truth-teller or a liar
(C) A liar or an alternator
(D) An alternator

11. Who among the above three kinds of persons can make the following statement – 'I am a truth-teller?'

I. A truth-teller
II. A liar
III. An alternator

(A) Only I
(B) Only I and III
(C) Only II and III
(D) Any one of I, II, III

Directions for questions 12 to 14: These questions are based on the following information.

A group of three people, namely Mohit, Nitin and Jayesh gave one statement each. There is one person who speaks the truth, one who tells lies and another whose statement cannot be classified as either true or false. Following are the statements made by them:

Mohit: I am not a liar.
Jayesh: I am a Liar.
Nitin: I neither speak the truth nor do I lie.

12. What is the name of the person whose statement can be classified as neither true nor as false?

(A) Mohit (B) Jayesh
(C) Nitin (D) Cannot be determined

13. Who tells lies?

(A) Jayesh (B) Mohit
(C) Nitin D) Cannot be determined

14. What is the nature of the statement given by Mohit?

(A) True
(B) False
(C) Cannot be classified as true or false
(D) Cannot be determined

Directions for questions 15 to 17: These questions are based on the following information.

There are four people, namely A, B, C and D each of whom plays exactly one game from a variety of sports, like Cricket, Football, Table-Tennis and Tennis. No game is played by two people. Each person gives two replies to any question asked to them. At least one person among them always speaks the truth and at least one person always tells lies. There is at least one person who always alternates between the truth and lie in any order.

When asked about the names of the people and the respective games played by them, following were their replies:

A: I play Cricket. C plays Cricket.
B: I play Tennis. D plays Tennis.

C: A plays Table-Tennis. B plays Cricket.
D: C plays Football. I play Table-Tennis.

It is also known that D plays Tennis and a definite arrangement can be obtained from the statements given by each person.

15. Who among the following plays Cricket?
(A) A (B) B
(C) C (D) D

16. Who among the following always speaks the truth?
(A) A (B) B
(C) C (D) D

17. Who are the two people who always alternate between the truth and lie?
(A) A and D (B) B and C
(C) D and B (D) A and C

Directions for questions 18 to 20: These questions are based on the following information.

In a country, there are three categories of people, such as truth-tellers, liars and alternators. Praveen met three people Honey, Bunny, and Cherry from that country. Each of the three people belongs to a different category. When asked about the details of the cites in the country, each of them made two statements.

Honey: City X is 30 km to the north of city Y. I am a liar.
Bunny: City W is 50 km away to the west of city Z. City M is 50 km away to the south of city W.
Cherry: City Y is not to the west of city Z. Honey is a truth-teller.

18. In which direction is city X with respect to city M?
(A) North-east (B) North-west
(C) North (D) Cannot be determined

19. If the distance between city Y and city Z is 10 km, then how far is city W from city X?
(A) 40 km (B) 50 km
(C) $30\sqrt{5}$ km (D) $40\sqrt{5}$ km

20. Who among them is a truth-teller?
(A) Honey (B) Bunny
(C) Cherry (D) Either (B) or (C)

Directions for questions 21 to 24: These questions are based on the following information.

Mahesh met four siblings of a family. When he asked them about their ages, their replies were as follows.

Tablo: I am the oldest.
I am older than Hablo.
Gablo: Pablo is not the oldest.
My age is less than 20 years.
Hablo: Tablo is the youngest.
The age of each one of us is a perfect square.
Pablo: Hablo is the second oldest.

The difference between the ages of any two consecutive siblings is not more than 10 years.

Mahesh also knew that no two among them has the same age and that Pablo is younger than Gablo. Each of them made one true statement and one false statement.

21. Who is the oldest?
(A) Tablo (B) Gablo
(C) Hablo (D) Cannot be determined

22. What is the age of the third oldest?
(A) 4 (B) 15
(C) 9 (D) 20

23. If Pablo is 16 years old, then what is the age of Tablo?
(A) 25 (B) 15
(C) 4 (D) 9

24. Who is the youngest?
(A) Tablo (C) Hablo
(B) Pablo (D) Cannot be determined

Directions for questions 25 to 27: These questions are based on the following information.

While travelling in a train, I met four people, each one of them belongs to a different state and they are from Uttar Pradesh (UP), Madhya Pradesh (MP), Tamil Nadu (TN) and Andhra Pradesh (AP). When I asked them about the state to which they belonged, each one of them made two statements. At least one person among them is a truth-teller (who always speaks the truth). At least one person among them is a liar (who always lies). At least one among them is an alternator (who alternates between the truth and lie in any order). Their replies were as follows.

Puneet: I am from AP.
Velu is from UP.
Velu: Navin is from MP.
Rajni is from AP.
Navin: I am from TN.
Puneet is from TN.
Rajni: I am from MP.
Velu is from MP.

It is also known that Puneet is from TN.

25. How many true statements are made by the four together?
(A) 3 (B) 4
(C) 5 (D) 3 or 4

26. The person from MP is a/an
(A) Truth-teller (B) Liar
(C) Alternator (D) Either (A) or (B)

27. Who are the two people who made the same number of true statements?
(A) Rajni and Navin (B) Puneet and Velu
(C) Velu and Rajni (D) Puneet and Navin

Directions for questions 28 to 30: These questions are based on the following information.

There is a group of three people, namely Ramu, Raman and Rajan-hailing from a different city, like Delhi, Mumbai and Chennai. Each person always gives two replies to any question asked. Out of these three, one person always speaks the truth, one always lies and the third one always alternates between truth and lie, in any order. When each was asked 'Which city do you belong to?', the following were their replies:

Ramu: I am from Delhi. Raman is from Mumbai.
Raman: I am from Delhi. Rajan is from Chennai.
Rajan: Ramu is from Mumbai. Raman is from Delhi.

Based on the above, answer the following questions.

28. Who among the three must be from Chennai?
(A) Ramu (B) Raman
(C) Rajan (D) Cannot be determined

29. To which city does Raman belong?
(A) Delhi (B) Mumbai
(C) Chennai (D) Chennai or Delhi

30. If there are exactly two people who always tell the truth, and the third person either always lies or alternates between truth and lie, then which of the following statements must be false?
(A) Rajan is not from Mumbai.
(B) Ramu is not from Delhi.
(C) Rajan is not from Chennai.
(D) Raman is from Delhi.

Directions for questions 31 to 33: These questions are based on the data given below.

To save our solar system from the attacks of the ETs from other galaxies, the representatives of all nine planets gathered at the 'Galaxy Hall' for a meeting. While entering the hall, each alien (representative of each planet) had to show his ID card to Mr. Gurkha, the gatekeeper. But three aliens, namely Eena, Meena and Deeka forgot their ID cards at the hotel. When asked, who represented which planet, the following were their replies.

Eena: Deeka is from Mars. Meena is from Saturn.
Meena: Eena is from Jupiter. Deeka is not from Saturn.
Deeka: Meena is from Saturn. I am from Jupiter.

It was known that exactly one amongst them belonged to the planet Jupiter, another belonged to the planet Saturn and the third belonged to the planet Mars. It was also known that each of them made at least one true statement.

31. Who is from Saturn?
(A) Eena (B) Meena
(C) Deeka (D) Cannot be determined

32. Which planet is Eena from?
(A) Mars (B) Jupiter
(C) Saturn (D) Cannot be determined

33. Which of following statements can never be true?
(A) Each of Eena and Meena always spoke the truth.
(B) Each of Eena and Meena spoke one truth and one lie.
(C) Deeka always spoke the truth.
(D) The first statement given by each person was always false.

Directions for questions 34 to 36: These questions are based on the following information.

I went to a bank, where I met three employees A, B and C of that bank. When I asked them 'Who is the manager?', their replies were as follows.

A : I am the manager.
B is a clerk.
C is a peon.
B : C is the manager.
I am not a peon.
A is a clerk.
C : Exactly one of my statements is false.
A is not a peon.
B makes exactly one false statement.

One among these three employees is a truth-teller. One of them is the manager, one is a clerk and the other is a peon. Each one of them is one among truth-teller, liar and an alternator.

Truth-teller is one who always speaks truth; alternator is a person who alternates between truth and lie and liar is a person who always lies.

34. Who is the truth-teller?
(A) B (B) C
(C) A (D) Cannot be determined

35. What is the designation of C?
(A) Manager (B) Clerk
(C) Peon (D) Cannot be determined

36. Who is a liar?
(A) C (B) A
(C) B (D) None of these

Directions for questions 37 to 40: These questions are based on the following information.

A group of three people, namely Achu, Babo and Chiki went to a stationery shop. Each of the three bought a different item from among Eraser, Pen and Pencil, each of which is of a different colour among Green, Red and Blue. Their friend Disha asked them 'What did you buy from the stationery shop?' Their replies were as follows.

Achu: I did not buy an eraser.
I bought a Red coloured item.
Babo bought a pencil.
Babo: I bought a pen.
Chiki bought a Green coloured item.
Achu bought a pencil.

Chiki: Babo bought an eraser.
I did not buy a pencil.
Achu bought a Green coloured item.

Each one of them is a truth-teller or a liar or an alternator. Truth-teller is one who always speaks truth; alternator is a person, who alternates between truth and lie and liar is a person who always lies. Disha knows that Babo bought a pencil.

37. If Achu is a truth-teller, then which item is in Red?
(A) Pen (B) Pencil
(C) Eraser (D) Cannot be determined

38. If Babo is an alternator, then who bought the Green coloured item?
(A) Achu (B) Babo
(C) Chiki (D) Cannot be determined

39. Which of the following statements is/are true?
(i) Achu is a truth-teller.
(ii) Babo is an alternator.
(iii) Chiki is a liar.
(A) Only (i) (B) (i) and (ii)
(C) (ii) and (iii) (D) None of these

40. If in all six false statements were made, then Chiki bought which coloured item?
(A) Red (B) Blue
(C) Green (D) Green or Blue

EXERCISE-2

Directions for questions 1 and 2: Select the correct alternative from the given choices.

1. There are three Pundits named Dwivedi, Trivedi and Chaturvedi sitting in a row from left to right in some order, consisting of three seats, such as extreme left, centre and extreme right. Each person gives two replies to any question asked to them, at least one of which is true. There is exactly one person who always speaks the truth. When asked about their respective positions in the row, the following were their replies:

Dwivedi: I sat at the extreme left end.
Trivedi sat at the centre.
Trivedi: Dwivedi sat between me and Chaturvedi.
I sat at the extreme right end.
Chaturvedi: I did not sit at the extreme left end.
Dwivedi did not sit at the extreme right end.

It is also known that a definite arrangement can be obtained by assessing their statements. What is the order in which they sat from the extreme left end to the extreme right end of the row?
(A) Dwivedi, Trivedi, Chaturvedi
(B) Chaturvedi, Trivedi, Dwivedi
(C) Dwivedi, Chaturvedi, Trivedi
(D) Trivedi, Dwivedi, Chaturvedi

2. Which of the following can be inferred from the given information?
I. Dwivedi's two statements are true.
II. Trivedi's second statement is false.
III. Chaturvedi's first statement is false.
(A) Only I
(B) Only I and III
(C) Only II and III
(D) None of the statements can be inferred.

Directions for questions 3 to 5: These questions are based on the following information.

A group of five people, namely Rahul, Ajay, Sandeep, Dhanush and Madhav are sitting in a row facing North. When their friend Charan called them, each of them made a statement about their seating positions.

Rahul: Sandeep is sitting two places away to the right of me.
Sandeep: I am sitting in the middle of the row.
Ajay: Madhav is sitting to the right of Sandeep.
Dhanush: Rahul is sitting two places away to the right of me.
Madhav: I am not at any of the ends.

Charan knows that on any day, exactly one of his five friends lie while others speak truth.

3. Who among them made a false statement?
(A) Rahul (B) Sandeep
(C) Dhanush (D) Cannot be determined

4. Who is sitting at the right end of the row?
(A) Rahul (B) Ajay
(C) Dhanush (D) Cannot be determined

5. Who is sitting in the middle of the row?
(A) Sandeep (B) Rahul
(C) Madhav (D) Cannot be determined

Directions for questions 6 to 9: These questions are based on the following information.

While going on a road, I met three people, namely Mona, Roma and Koma. Each of them belongs to a different profession among Engineer, Doctor and Professor and each of them owns a different car, such as Swift, Indigo and Micra. Each of them made three statements as given below.

Mona: Roma is a Doctor.
I am an Engineer.
Koma owns Micra.

Roma: I am not a Professor.
Mona owns Swift.
Koma does not own Indigo.

Koma: Mona is not a Professor.
I am an Engineer.
Roma does not own Indigo.

Among the three people, one of them is a truth-teller who always speak truth; one is a lair, who always lies and the other person is an alternator, who alternates between truth and lie, in any order.

6. Who owns a Swift?
(A) Mona (B) Roma
(C) Koma (D) Cannot be determined

7. Who is an Engineer?
(A) Mona (B) Roma
(C) Koma (D) Cannot be determined

8. Who is the alternator?
(A) Roma (B) Koma
(C) Mona (D) Either (A) or (B)

9. Who is the liar?
(A) Mona (B) Koma
(C) Roma (D) Either (B) or (C)

Directions for questions 10 to 12: These questions are based on the following information.

A group of four people, namely Chibi, Cakora, Ceasar and Chaila have accounts in a different social networking site. One of their friends asked them, 'Who is on WhatsApp?' Their replies were as follows.

Chibi: I am on WhatsApp.
Chaila is on Facebook.
Ceasar is on Twitter.
Cakora is on Instagram.

Cakora: Chibi is not on Facebook.
Chaila is on Twitter.
I am not on WhatsApp.
Ceasar is on neither Facebook nor Instagram.

Ceasar: I am on either WhatsApp or Facebook.
Chibi is on Twitter.
Chaila is not on WhatsApp.
Cakora is not on Facebook.

Chaila: Cakora is not on Twitter.
I am on WhatsApp.
If Cakora is on Facebook, then only Chibi is on Instagram.
Ceasar is on Facebook.

Among these four people, one of them is a truth-teller who always speaks truth; one is a liar, who always lies; one is an alternator, who alternates between truth and lie and the remaining one is any one of these three (i.e., truth teller, liar and alternator).

10. Who is a truth-teller?
(A) Chibi (B) Cakora
(C) Ceasar (D) Chaila

11. Who is on Twitter?
(A) Chibi (B) Cakora
(C) Ceasar (D) Chaila

12. Liar is on
(A) Instagram (B) WhatsApp
(C) Twitter (D) Facebook

Directions for questions 13 to 17: These questions are based on the following information.

Four people Pavan, Naveen, Madan and Sravan are the top four rankers in each of the subjects, such as Maths, Physics, Chemistry and Commerce, not necessarily in the same order. No person got the same rank in any two subjects. When asked about their ranks in each of these subjects, they made the following statements.

Pavan: Madan is the fourth ranker in Chemistry.
Sravan is the fourth ranker in Physics.
Naveen is the first ranker in Maths.

Naveen: I am the first ranker in Commerce.
Sravan is the first ranker in Chemistry.
Madan is the first ranker in Physics.

Madan : I am the first ranker in Commerce.
Sravan is the fourth ranker in Chemistry.
Pavan is the third ranker in Maths.

Sravan : Pavan is the third ranker in Commerce.
I am the fourth ranker in Maths.
Naveen is the second ranker in Physics.

It is known that, each of them made a true and a false statement alternately and in total, they made equal numbers of true and false statements.

13. Who is the third ranker in Chemistry?
(A) Pavan (B) Naveen
(C) Madan (D) Cannot be determined

14. What is the rank of Sravan in Physics?
(A) 4 (B) 3
(C) 2 (D) Cannot be determined

15. In which subject did Naveen get a better rank than Madan but a worse rank than Pavan?
(A) Maths (B) Commerce
(C) Physics (D) Chemistry

16. What is the rank of Naveen in Commerce?
(A) 2 (B) 3
(C) 4 (D) 1

17. In which subject did Pavan get a worse rank than both Naveen and Sravan?
(A) Maths (B) Physics
(C) Chemistry (D) Commerce

Directions for questions 18 to 22: These questions are based on the following information.

Four boys Abhay, Bharat, Chandu, and David are inhabitants of an island, each studying in a different class among VII, VIII, IX and X, not necessarily in the same order. On that island, each person belongs to one of the categories: Truth-tellers (who always speak the truth), liars (who always lie) and alternators (who alternate between true and false statements, in any order). When asked about their studies and categories, they made the following statements.

Abhay : David is not a truth-teller.
Chandu is not studying in IX.
I am studying in VII.
Bharat: Chandu is not a truth-teller.
Abhay is studying in VIII.
David is not studying in IX.
Chandu: Bharat is studying in VIII.
Bharat is a liar.
Abhay is not studying in X.
David: Abhay is not an alternator.
Bharat is studying in X.
Chandu is not studying in VIII.

18. Who is studying in Class X?
(A) Abhay (B) Bharat
(C) Chandu (D) Cannot be determined

19. Who among them is an alternator?
(A) Only Abhay (B) Only Bharat
(C) Only Chandu (D) Only David

20. How many of Abhay's statements is/are true?
(A) 3 (B) 2
(C) 1 (D) 0

21. In which class did David study?
(A) X (B) IX
(C) VIII (D) VII

22. What is the difference between the total number of true and false statements?
(A) 4 (B) 2
(C) 0 (D) Cannot be determined

Directions for questions 23 to 25: These questions are based on the data given below.

On True Lies island, there are four categories of people namely Trues, who always speak the truth; Liars, who always lie; Altrues, whose first statement is true and make alternate true and false statements and Allies, whose first statement is false and make alternate false and true statements.

A person from True Villa who visited the True Lies island has to make a telephone call to True Villa. He comes across a group of four people, where each member of the group belongs to a different category. He asks them about the availability of telephone with them. Each person has a badge attached to his shirt with the name of a category to which he does not belong. The following are the statements made by each of them. The category on their badges is given in brackets.

A (Trues):
I. Allies have telephones.
II. I have a telephone.
B (Liars):
I. I belong to Trues.
II. I do not have a telephone.
C (Altrues):
I. I do not have a telephone.
II. Only one of us has a telephone.
D (Allies):
I. The Liars have telephones.
II. I belong to Altrues.

In any category, either each member has a telephone or none has a telephone.

23. Who belong to the Trues category?
(A) A (B) B (C) C (D) D

24. Which of the following is true?
(A) B has a telephone.
(B) C has a telephone.
(C) Both B and C have telephones.
(D) Neither B nor C has a telephone.

25. How many categories have a telephone?
(A) 0 (B) 1 (C) 2 (D) 3

Directions for questions 26 to 28: These questions are based on the data given below.

John, James, Jack and Jeromy are compared with each other in terms of height, weight, age and wealth. Each of them gives three statements to any question, such that the three statements given by each person are alternately true and false in any order. The first statement made by exactly two people is false. When asked about the characteristic possessed by each of them, the following were their replies:

John: I am the tallest.
Jack is the heaviest.
James is the richest.
Jack: Jeromy is the shortest.
I am the youngest.
John is the second tallest.
Jeromy: James is the lightest.
Jack is the poorest.
John is the eldest.
James: I am the tallest.
Jack is the lightest.
Jeromy is the second youngest.

It is also known that no person gets the same rank (or position) in any two of the four comparisons and no comparison has two persons having the same rank.

26. Who among the following is the second heaviest?
(A) John (B) Jack
(C) Jeromy (D) James

27. Who is elder, wealthier, heavier, but shorter than James?
I. Jack
II. Jeromy
III. John
(A) Only I (B) Only II
(C) Only III (D) II and III

28. How many people weigh more than James?
(A) Zero (B) One
(C) Two (D) Three

Directions for questions 29 to 31: These questions are based on the following information.

Alex, Bhavna and Charan are standing, not necessarily in that order but in different positions in a queue of 10 people. The numbering in the queue starts with position 1 at the beginning to position 10 at the end of the queue.

When asked about their positions in the queue, they gave the following replies:

Alex: Charan is three places ahead of me.

The number of people behind Bhavna is one more than the number of people ahead of Charan.

If I interchange my position with Charan, I would be in position seven.

Bhavna: I am fourth from the end of the queue.

The sum of the numbers of our positions is a unit multiple of 5.

I am exactly between Alex and Charan.

Charan: All of Alex's statements are false.

If I interchange my position with Alex, I would be in position seven.

Bhavna's first and second statements are true.

It is known that exactly two of the three people alternate between truth and lie in any order and Charan is one of them.

29. What is the position of Bhavna with respect to Alex?
(A) Immediately behind Alex.
(B) Two positions behind Alex.
(C) Three positions ahead of Alex.
(D) Cannot be determined

30. Which of the following statements is true?
I. Alex is in position seven.
II. All the statements of Bhavna are false.
III. Alex alternates between true and false statements in that order.
(A) Only II (B) Only I and II
(C) Only I and III (D) I, II and III

31. Which of the following additional conditions, if true, gives the exact positions of the three people?
(A) Bhavna is not behind Alex.
(B) Bhavna is immediately behind Alex.
(C) Charan is not ahead of Bhavna.
(D) None of the above

Directions for questions 32 to 34: These questions are based on the following information.

P, Q and R when asked a question give three statements as reply in the following manner.

P – always replies in only one type of statements, such as in truth or lie.

Q – never replies in the same type of statements as P.

R – is neither consistent in all his statement types nor alternates between the two types of statements.

At the bus – stop, I asked 'Which bus goes to the airport?'

Their replies were as follows:

P: Take the north – bound bus.
The next bus arrives in 15 minutes.
It takes 45 minutes to reach the airport.

Q: The north – bound bus comes from the airport.
You have to wait for 20 minutes for the bus.
It takes 30 minutes to reach the airport.

R: Take the south – bound bus.
You have to wait for another quarter of an hour for the bus.

It takes anywhere between 1 hour to 1 hour 30 minutes to reach the airport.

32. Which of the following represents the reply that R gave?
(A) Truth – Truth – Lie
(B) Lie – Truth – Truth
(C) Lie – Lie –Truth
(D) Truth – Lie – Lie

33. Which of the following is definitely true?
I. P always lies.
II. Q always lies.
III. The north – bound bus goes to the airport.
(A) Only I (B) Only II
(C) Only I and II (D) Only II and III

34. How long does it take before I reach the airport from the bus-stop?
(A) 45 minutes
(B) 50 minutes
(C) Between 1 hour and 1 hour 30 minutes
(D) 30 minutes

Directions for questions 35 to 37: These questions are based on the data given below.

On the eve of Army Day Parade, I met five Army men, namely Ranjeet, Ranmeet, Ranpreet, Randheer and Ranveer each of whom had exactly one different rank from amongst Lieutenant, Captain, Colonel, Major and Brigadier, not necessarily in the same order. Each of these people always gave three replies to any question asked to them. Except one person, all the other four speak at least one true statement. Except one person, all the other four tell at least one lie. Exactly one person among them alternates between the truth and lie, in any order. When I asked them about their respective ranks, following were their replies:

Ranjeet:	Ranmeet is the Brigadier.
	Ranpreet is the Major.
	Ranveer is the Colonel.
Ranmeet:	Randheer is the Major.
	Ranjeet is the Captain.
	Ranpreet is the Lieutenant.
Ranpreet:	Ranjeet is the Colonel.
	Randheer is the Brigadier.
	Ranmeet is the Captain.
Randheer:	Ranveer is the Lieutenant.
	Ranjeet is the Major.
	Ranpreet is the Captain.
Ranveer:	Ranmeet is the Colonel.
	Ranpreet is the Major.
	I am the Lieutenant.

Only one definite arrangement exists based on their statements, which gives out their ranks.

35. Who among the following always makes one true statement and two false statement but does not alternate between truth and lie?

(A) Ranjeet (B) Ranpreet
(C) Randheer (D) Ranmeet

36. What is the rank of the person who alternates between the truth and lie, in any order?

(A) Brigadier (B) Captain
(C) Major (D) Colonel

37. Match the following:

Army men	**Ranks**
(i) Ranjeet	(A) Lieutenant
(ii) Ranmeet	(B) Captain
(iii) Ranpreet	(C) Colonel
(iv) Randheer	(D) Major
(v) Ranveer	(E) Brigadier

(A)	(i)	–	(a)	(B)	(i)	–	(b)
	(ii)	–	(b)		(ii)	–	(c)
	(iii)	–	(c)		(iii)	–	(e)
	(iv)	–	(d)		(iv)	–	(a)
	(v)	–	(e)		(v)	–	(d)
(C)	(i)	–	(c)	(D)	(i)	–	(c)
	(ii)	–	(e)		(ii)	–	(b)
	(iii)	–	(d)		(iii)	–	(d)
	(iv)	–	(b)		(iv)	–	(e)
	(v)	–	(a)		(v)	–	(a)

Directions for questions 38 to 40: There questions are based on the following information.

In a family, there are four members, namely Bingo, Tingo, Mingo and Pingo. Each member of that family belongs to one of the following category, such as truth-tellers (who always speaks truth), liars (who always lies) and the alternators (who alternates among truth and lie, in any order). They made the following statements when they are asked about their professions.

Bingo:	I am a truth-teller.
	I am the Manager.
	Tingo is the Chartered Accountant.
Mingo:	I am a truth-teller.
	I am the Engineer.
	Professor is an alternator.
Tingo:	I am a truth-teller.
	I am the Professor.
	Pingo is the Engineer.
Pingo:	I am an alternator.
	I am the Professor.
	Bingo is a liar.

It is also known that their professions are one among Manager, Engineer, Chartered Accountant and Professor (may not be in that order). And no two people are of the same profession.

38. If Tingo is the Manager, then who is the Chartered Accountant?

(A) Bingo
(B) Mingo
(C) Pingo
(D) Either (A) or (B)

39. If there is only one Liar and he is not Bingo, then who is the Engineer?

(A) Pingo
(B) Bingo
(C) Mingo
(D) Cannot be determined

40. From which of the given conditions, we will get a complete idea about them?

I. There are exactly two liars.
II. There are exactly two alternators.
III. There is no truth-teller.

(A) Only I
(B) Only II
(C) Any two of the above three
(D) Only III

EXERCISE-3

Directions for questions 1 to 3: These questions are based on the information given below.

A group of four friends, namely A, B, C and D sit around a square table not necessarily in the same order. Exactly one person sits on each side and everyone is opposite to exactly one person. Each person makes two statements for any question asked to him. It was known that there is exactly one person who always speaks the truth and exactly one person who alternates between truth and false. When asked about their respective positions around the table, the following were their replies:

A: D sits opposite to me. C sits to the left of D.
B: A sits to my left. D sits to the right of C.
C: B sits to my right. D sits to my left.
D: C sits opposite to me. B sits to my right.

It was also known that a definite arrangement can be obtained by assessing their statements.

1. Who always speaks the truth?
(A) A (B) B
(C) C (D) D

2. Who among them are liars?
(A) A and B (B) C and D
(C) A and D (D) Cannot be determined

3. Who sits opposite to D?
(A) A (B) B
(C) C (D) Cannot be determined

Directions for questions 4 to 6: These questions are based on the information given below.

Among the four members of a family K, L, M and N, there is one couple, their son and their daughter. When asked about their relationships, the following were their replies.

K: N is my husband.
M is my daughter.
L: K is my mother.
M is my son.
M: K and L are of the same gender.
L is my sister.
N: L is of the same gender as I.
M is my son.

It was also known that only one of them always speaks the truth.

4. Among the four, who cannot be the truth-teller?
(A) K (B) L
(C) M (D) Cannot be determined

5. Among them, if there are two people who always tell lies, then who always speaks the truth?
(A) K (B) M
(C) N (D) Cannot be determined

6. Among them, if there are two people who always alternate between the truth and lies, then who speaks the truth always?
(A) K (B) L
(C) M (D) Cannot be determined

Directions for questions 7 to 10: These questions are based on the information given below.

There is a group of four players, namely Abhinav, Bipash, Chandar and Danny, each of them hailing from different cities, like Hyderabad, Mumbai, Delhi and Kolkata. Each of them plays a different game among Chess, Badminton, Tennis and Bridge and they are of different heights. Each of them gives three statements to any question such that the three statements given by each person are alternately true and false in any order. The second statement made by exactly two people is true and Abhinav is one of them. When asked about them, the following were their replies.

Abhinav: Bipash is from Delhi.
Chandar plays Chess.
Danny is not the tallest.
Bipash: Abhinav is from Kolkata.
Chandar is not the shortest.
Danny plays Badminton.
Chandar: Abhinav is the 2nd tallest.
Bipash plays Tennis.
Danny is from Delhi.
Danny: Abhinav plays Chess.
Bipash is from Mumbai.
Chandar is not from Hyderabad.

7. Who is from Delhi?
(A) Abhinav (B) Bipash
(C) Chandar (D) Danny

8. Who is the Chess player?
(A) Abhinav (B) Bipash
(C) Chandar (D) Cannot be determined

9. Who is the shortest person?
(A) The Badminton player.
(B) The Chess player.
(C) The person from Delhi.
(D) The person from Kolkata.

10. Which two persons' second statements are true?
(A) Abhinav, Bipash
(B) Abhinav, Chander
(C) Abhinav, Danny
(D) Cannot be determined

Directions for questions 11 to 15: These questions are based on the following information.

A group of five people, namely Amar, Bharat, Chandu, Dinesh and Eswar are of different heights, weights and ages. It is also known that each of them speaks truth and lie in alternate fashion. Five people are ranked from 1 to 5 according to the decreasing order of their heights, weights and ages. No person is ranked the same in any two parameters. The following are the statements made by four people.

Amar: Bharat is the 3rd tallest.
Dinesh is the 3rd youngest.
Eswar is the lightest.

Bharath: Chandu is the tallest.
Amar is the 3rd oldest.
Eswar is the oldest.

Chandu: Bharath is 2nd youngest.
Bharath is the 2nd tallest.
Amar is the 3rd heaviest.

Dinesh: Eswar is the shortest.
Bharath is 4th lightest.
Amar is the 2nd tallest.

11. Who among them is the tallest?
(A) Chandu (B) Dinesh
(C) Bharath (D) Amar

12. Who among them is the oldest?
(A) Amar (B) Dinesh
(C) Eswar (D) Chandu

13. Who among them is the lightest?
(A) Eswar (B) Dinesh
(C) Bharath (D) Chandu

14. Who among them is the 2nd shortest?
(A) Dinesh (B) Amar
(C) Bharath (D) Eswar

15. Who among them is the 3rd youngest?
(A) Dinesh (B) Eswar
(C) Chandu (D) Bharath

Directions for questions 16 to 19: These questions are based on the following information.

Three people Gretta, Fischer and Schindler are talking about their heights. It is known that each of them is of a different height. Each of them belongs to one of the three categories, truth-tellers (who always speak truth), liars (who always tell lies) and the alternators (who alternately make a true and a false statement). They made the following statements.

Gretta: I am the tallest.
I always speak truth.
Fischer always lies.

Fischer: I am not the tallest.
Gretta is a liar.
Schindler is the shortest.

Schindler: Gretta is a liar.
I am not an alternator.
I am not the shortest.

16. Who is/are truth teller(s)?
(A) Gretta or Fischer (B) Schindler
(C) None of them (D) Cannot be determined

17. Who is the tallest?
(A) Fischer (B) Schindler
(C) Gretta (D) Cannot be determined

18. How many true statements have they together made?
(A) Four (B) Five
(C) Three (D) Three or four

19. Who is/are alternator(s)?
(A) Schindler and Gretta
(B) Fisher and Schindler
(C) Fisher
(D) Schindler

Directions for questions 20 to 23: These questions are based on the following information.

Four very naughty siblings Pringle, Qusac, Robert and Swank are going to meet their father's childhood friend who is meeting them after 30 years. Their father's friend doesn't have any information about these siblings. Each of them wrote the following two statements on a piece of paper. They mentioned only the first letter of their respective names on the paper so that the friend cannot find out the gender by name.

P: I have two sisters and a brother.
S is my brother.

Q: Both the statements of S are false.
R is my brother.

R: One statement of P is true and the other one is false.
I am S's sister.

S: I have two brothers and a sister.
Q is my sister.

It is known that exactly one person among them made two true statements and that person is not R, and exactly one person made both false statements.

20. Who definitely made one true and one false statement?
(A) Pringle (B) Qusac
(C) Robert (D) Swank

21. Who is/are definitely male?
(A) Pringle and Robert
(B) Swank
(C) Swank and Pringle
(D) Robert and Qusac

22. Who made two false statements?
(A) Pringle (B) Qusac
(C) Robert (D) Qusac or Swank

23. Which of the following is true?
(A) Swank's second statement is false.
(B) Robert's second statement is true.
(C) Qusac's first statement is true.
(D) Pringle's second statement is true.

Directions for questions 24 and 25: These questions are based on the following information.

Karuna and Sharmila are studying in the same class. Their teacher came to know that today is the birthday of either Karuna or Sharmila, but not of both.

The teacher asked them whose birthday is it today? They made the following statements.

Karuna: Today it is not my birthday.
Today it is Sharmila's birthday.
Sharmila: Both the statements of Karuna are false.
It is not the birthday of any one of us.

It is known that each of their statements is either true or false.

24. If Sharmila's first statement is false, whose birthday is on that day?
(A) Karuna's
(B) Sharmila's
(C) Both of them
(D) Cannot be determined

25. Which of the following is true?
(A) Karuna did not make even one true statement.
(B) Either both the statements of Karuna are true or both are false.
(C) Sharmila's second statement is definitely true.
(D) Neither Karuna nor Sharmila made two false statements.

ANSWER KEYS

Exercise-1

1. (B)	8. (A)	15. (B)	22. (C)	29. (B)	36. (D)
2. (A)	9. (D)	16. (C)	23. (D)	30. (C)	37. (A)
3. (C)	10. (D)	17. (C)	24. (D)	31. (B)	38. (C)
4. (C)	11. (D)	18. (D)	25. (B)	32. (D)	39. (D)
5. (B)	12. (B)	19. (B)	26. (C)	33. (D)	40. (A)
6. (D)	13. (C)	20. (B)	27. (D)	34. (C)	
7. (B)	14. (A)	21. (B)	28. (C)	35. (C)	

Exercise-2

1. (C)	8. (A)	15. (D)	22. (A)	29. (D)	36. (D)
2. (D)	9. (A)	16. (A)	23. (B)	30. (B)	37. (D)
3. (C)	10. (B)	17. (D)	24. (B)	31. (B)	38. (C)
4. (D)	11. (D)	18. (C)	25. (B)	32. (D)	39. (C)
5. (A)	12. (A)	19. (D)	26. (C)	33. (A)	40. (C)
6. (A)	13. (B)	20. (A)	27. (B)	34. (B)	
7. (C)	14. (C)	21. (B)	28. (C)	35. (C)	

Exercise-3

1. (B)	5. (C)	9. (B)	13. (D)	17. (A)	21. (B)	24. (B)
2. (B)	6. (A)	10. (C)	14. (B)	18. (D)	22. (D)	25. (B)
3. (B)	7. (D)	11. (A)	15. (A)	19. (C)	23. (D)	
4. (B)	8. (C)	12. (C)	16. (D)	20. (C)		

SOLUTIONS

EXERCISE-1

1. Amit makes the statement, 'I am a liar'. If Amit is a truth-teller, he would never call himself a liar. Also, if Amit is a liar, he would never admit that he is a liar. Hence, Amit is an alternator and his statement is False. Ashok's statement 'I am not a truth-teller'. If Ashok is a truth-teller, he would not tell otherwise. If Ashok is a liar, then his statement can imply that his statements is either a lie or cannot be classified as a truth or lie. But a liar would not admit that he is a liar. Hence, Ashok is an alternator and his statement is true. Azad's statement – I neither make a true statement nor makes a false statement. If Azad is a truth –teller, then he would not claim that he does not make true statements.
Hence, Azad could either be a liar or an alternator.
If Azad is a liar, his statement is false and if he is an Alternator, his statement is true. Thus, only Ashok's statement is definitely true.

2. Let us analyse the statements given by A, B and C as below.
A: 'I am not a Liar'.
If A is the truth-teller, then this statement given by him is true, hence, A must be the truth-teller.
B: 'C is not a liar'.
If B is the truth-teller, then C must be a truth-teller. But we cannot have two truth-tellers. Hence, B cannot be a truth-teller. ⇒ B is a liar.
C: 'B is a truth-teller'.
If C is the truth-teller, then B is also a truth-teller which means there are two truth-tellers. As it is given that exactly one person among them is a truth-teller, hence, C cannot be a truth-teller. ⇒ C is a liar.
∴ We find that A is a truth-teller and B and C are liars.

3. Turn by turn, we should assume each person to be the truth-teller and then analyse the arrangement.
(i) Assuming Anand to be the truth-teller: If Anand is the truth-teller, then Bharat must be the liar, which means that C must be the alternator. As Bharat is a liar, both his statements must be false (Bharat's first statement is 'I am not the Liar' which is false as he is a Liar and his 2nd statement that 'Chandu is the truth-teller' is false, because as per Anand, Chandu is the alternator). Hence, Chandu is the alternator, whose first statement is true and second is false.
(ii) Assuming Bharat to be the truth-teller: If both the statements given by Bharat are true, then both Bharat and Chandu must be the truth-tellers, which is not possible – as there is exactly one truth-teller among these three people.
(iii) Assuming that Chandu is the truth-teller: If both the statements given by Chandu are true, then even Bharat becomes the truth-teller, which is not desired.
Hence, Anand is the truth-teller, which means that Chandu must be the alternator.

4. ('✓' means cheated; '✘' means did not cheat)
(i) Assuming Aalu always spoke the truth:

	I	II	Cheated
Aalu	T	T	✘
Kach	F	F	✓
Bhalu	T	F	✘

In this case only Kachalu cheated. Hence, (2) is possible.
(ii) Assuming Kachalu always spoke the truth:

	I	II	Cheated
Aalu	F	F	✓
Kach	T	T	✘
Bhalu	F	T	✓

In this case, only Aalu and Bhalu cheated. Hence, (1) can be true.
(ii) Assuming Bhalu always spoke the truth:

	I	II	Cheated
Aalu			
Kach			?
Bhalu	T	T	✘

Both the statements made by Bhalu contradict each other. Hence, Bhalu can never be the person who always spoke the truth.

5. (T → True; F → False)
(i) Assuming that Sameer is the truth-teller: we get the following arrangement:

	Statement I	Statement II		
Sameer	T	T	Truth-teller	Winner
Sameep	F	F	Liar	2nd runner-up
Sumer	F	T	Alt.	1st runner-up

(ii) Assuming Sameep is the truth-teller, we get the following arrangement:

	Statement I	Statement II	
Sameer	F	T	2nd runner-up
Sameep	T	T	Winner
Sumer	F	F	1st runner-up

(iii) Assuming Sumer is the truth-teller, we get the following arrangement:

	Statement I	Statement II		
Sameer	F	T	Alternator	1st runner-up
Sameep	F	F	Liar	2nd runner-up
Sumer	T	T	Truth-teller	Winner

Hence, either Sameer or Sumer can be the 1st runner-up, but not Sameep.

6. If A always speaks the truth, then B is 'the Good', A is 'the Ugly'. Hence, C is 'the Bad'; which also means that B is also the truth-teller. Hence, neither A nor B is the truth-teller. Therefore, C must be the truth-teller, which gives us the following arrangement.

A → the Good

C → the Bad

B → the Ugly

7. Each among John, Johny and Janardan is an alternator. Let John's first statement be true and second be false.

	I	II	Medal
John	T	F	Gold
Johny	F	F	Silver
Janardan	T	F	Bronze

Hence, Johny is not an alternator, which means our initial assumption was false.
Let John's first statement be false and second be true:

	I	II	Medal
John	F	T	Silver
Johny	T	F	Bronze
Janardan	F	T	Gold

Hence, Janardan → Gold medal

John → Silver medal

Johny → Bronze medal

8. Since Bhala is telling the truth, there is exactly one among the five who is telling the truth and it has to be Bhala. The rest of them are lying.
We have the following:

Suspect	Statement	Implication
Sheroo	F	Bhayankar did not rob the bank.
Santhosh	F	Sheroo robbed the bank.
Bhayankar	F	Santhosh is lying ⇒ Sheroo robbed the bank.
Chola	F	Bhayankar did not rob the bank.
Bhala	T	

Hence, it can be inferred by Chulbul Pandey that Sheroo robbed the bank.

9. 'I am not an alternator' could be the statement of a truth-teller. Can it be a statement of a liar? If so, the statement for a liar this statement would be correct / true, which a liar would not make. Hence, it cannot be the statement of a liar. It can surely be that of an alternator when that particular statement is false and either his /her prior or later statement is true. Thus, both (A) and (C) are possible.

10. Refer to the explanation given for Ashok's statement in question 1.
Here, the person is an alternator.

11. 'I am a truth-teller' can be the statement of
– a truth-teller since he never lies.
(or)
– a liar since he always lies
(or)
– an alternator since he alternates in this fashion: Truth, Lie, Truth, Lie… (or) Lie, Truth, Lie, Truth etc.
Hence, any one of I, II, III is possible.

Solutions for questions 12 to 14: The statement given by Jayesh is 'I am a Liar'. If Jayesh always speaks the truth, he will not call himself a liar. Similarly, if Jayesh is a liar, then he will not speak the truth by admitting that he is a liar. Hence, Jayesh is the person whose statement is neither true nor false. Then, Nitin's statement must be false, as only one person whose statement cannot be classified as true and false and that is Jayesh. Hence, Mohit always speaks the truth. Therefore,

Mohit → Always speaks the truth.
Jayesh → Neither speaks the truth nor tells lies.
Nitin → Always tells lies.

12. Jayesh is the one whose statement cannot be classified as either true or false.

13. Nitin always tells lies.

14. Mohit always tells the truth.

Solutions for questions 15 to 17: It is known that D plays Tennis, which means that the 2nd statement made by B must be true, whereas the 2nd statement made by D must be false. As A says that he and C both play cricket, A cannot be the person who always speaks the truth, as each person plays exactly one game and exactly one game is played by each person. This means only C can always speak the truth, as there must be at least one person who always speaks the truth.

We take this as the basis and get the arrangement as given below:

	I	II	Game played
A	F	F	Table-tennis
B	F	T	Cricket
C	T	T	Football
D	T	F	Tennis

15. B plays Cricket.

16. C always speaks the truth.

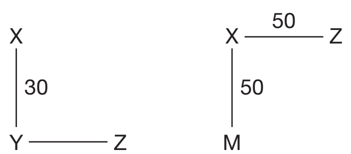
17. D and B alternate between truth and lie.

Solutions for questions 18 to 20: From, the second statement made by Honey, it can be said that Honey is an alternator and the second statement must be false and first statement must be true.

Now, from the second statement made by Cherry, it can be said that Cherry is a liar. So, Bunny is a truth-teller.

	I	II
Honey	T	F
Bunny	T	T
Cherry	F	F

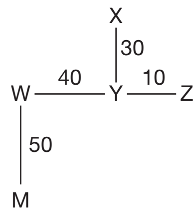
∴ The representation of cites is as follow.



(i) (ii)

18. City X can be in North-east, North-west or North with respect to city M.

19. Under the condition, the representation of cites is as follows.



∴ The distance between X and W is $\sqrt{30^2 + 40^2} = 50$ km.

20. Bunny is a truth-teller.

Solutions for questions 21 to 24: First let us assume that the first statement of Tablo is true, then the second statement must be false. But in this case the second statement also becomes true, as Tablo is the oldest.

So, the first statement of Tablo is false and the second statement is true.

Therefore, Tablo is not the oldest and Tablo will be either the second or the third oldest.

So, the first statement of Hablo, i.e., Tablo is the youngest is false.

∴ The second statement of Hablo, i.e., the age of each one among them is a perfect square is true.

As Tablo is older than Hablo, Hablo will be either the third or the fourth oldest.

The first statement of Pablo, i.e., Hablo is the second oldest is false.

∴ The second statement of Pablo, i.e., the difference between the ages of any two consecutive siblings is not more than 10 years is true.

Their ages can be among 1, 4, 9, 16 and 25.

It is given that Pablo is younger than Gablo.

The first statement of Gablo is true, i.e., Pablo is not the oldest.

That means, his second statement is false, i.e., Gablo's age is not less than 20 years.

∴ Gablo's age is 25 years.

The age of the second and the third oldest are 16 and 9 years, respectively.

The youngest child can be either one or four years old.

The above discussion is represented in the following table.

	I Statement	II Statement	Remarks	
Tablo	F	T	Not the oldest	Second or third oldest
Gablo	T	F	Oldest	Age 25 years
Hablo	F	T	Not the oldest or the second oldest	Third or the fourth oldest
Pablo	F	T	Not the oldest	Second, third or fourth oldest

21. Gablo is the oldest.

22. The third oldest is 9 years old.

23. Given, Pablo is 16 years old.
Hence, he is the second oldest.
In this case, Tablo has to be the third oldest.
∴ Tablo is 9 years old.

24. The youngest is either Pablo or Hablo.

Solutions for questions 25 to 27: Given that Puneet is from TN.

∴ The first statement of Puneet, i.e., Puneet is from AP is false.

The first statement of Navin is false and the second statement of Navin is true.

It is also given that there should be at least one person whose both statements are true.

That person cannot be Rajini as both the statements of Rajni cannot be true simultaneously.

∴ Velu's both the statements are true.
Navin is from MP.
Rajini is from AP.
Velu is from UP.
∴ Puneet's second statement is true.
Thus, both the statements of Rajni are false.
The final table is as follows:

Name	Place	1st Statement	2nd Statement
Puneet	TN	F	T
Velu	UP	T	T
Navin	MP	F	T
Rajni	AP	F	F

25. There are four true statements.

26. The person from MP is an alternator.

27. Puneet and Navin made same numbers of true statements.

Solutions for questions 28 to 30:

(i) Assuming Ramu always speaks the truth:

	I	II	City
Ramu	T	T	Delhi
Raman	F	T	Mumbai
Rajan	F	F	Chennai

This arrangement works out for answering the first two questions.

(ii) Assuming Raman always speaks the truth:

	I	II	City
Ramu	F	F	Mumbai
Raman	T	T	Delhi
Rajan	T	T	Chennai

(iii) Assuming Rajan always speaks the truth:

	I	II	City
Ramu	F	F	Mumbai
Raman	T	T	Delhi
Rajan	T	T	Chennai

Arrangement (ii) and (iii) help in answering the third question.

28. Rajan is from Chennai.

29. Raman belongs to Mumbai.

30. Rajan is not from Chennai is false.

Solutions for questions 31 to 33: Let us analyse the cases one by one for the person who always speaks the truth.

(T → Truth; F → False)

(i) Eena always speaks the truth:

	I	II	Planet
Eena	T	T	Jupiter
Meena	T	T	Saturn
Deeka	T	F	Mars

(ii) Deeka always speaks the truth:

	I	II	Planet
Eena	F	T	Mars
Meena	F	T	Saturn
Deeka	T	T	Jupiter

In the above two cases, Eena and Meena always speak the truth in case (i) and Deeka always speaks the truth in case (ii).

31. In either of the cases (i) or (ii), Meena is from Saturn.

32. In case (i), Eena is from Jupiter and in case (ii), Eena is from Mars. Hence, it cannot be determined.

33. (1) is true in case (i)
(2) is true in case (ii)
(3) is true in case (ii)
(4) is true in neither (i) nor (ii).

Solutions for questions 34 to 36: Let us assume, A is truth-teller, then

	Statements			
	I	II	III	Designation
A	T	T	T	Manager
B	F	T	F	Clerk
C	F	T	F	Peon

Let us assume, B is a truth-teller. Then, B's third statement is contradicting. Hence, B is not a truth-teller.
Let us assume, C is a truth-teller then C's first statement is contradicting. Hence, C is not a truth-teller.

34. A is the truth-teller.

35. C is a peon.

36. None of these

Solutions for questions 37 to 40: Achu's third statement is true since Babo bought a pencil. Hence, Babo's first and third statement and Chiki's first statement are false. Chiki's second statement is true. Hence, Chiki's third statement should be false. Now, Babo's second statement is either true or false and Achu's first statement should be true since Achu is either a truth-teller or an alternator. Hence, the possible cases are as follows.

(i)

	Statements				
Name	**I**	**II**	**III**	**Item**	**Colour**
Achu	T	T/F	T	Pen	Red/Blue
Chiki	F	T	F	Eraser	Green

	Statements				
Name	**I**	**II**	**III**	**Item**	**Colour**
Babo	F	T	F	Pencil	Blue/Red
Chiki	F	T	F	Eraser	Green

(ii)

	Statements				
Names	**I**	**II**	**III**	**Item**	**Colour**
Achu	T	T/F	T	Pen	Red/Blue
Babo	F	F	F	Pencil	Green
Chiki	F	T	F	Eraser	Blue/Red

37. If Achu is a truth-teller, then the Pen is in Red colour.

38. If Babo is an alternator, then Chiki bought Green coloured item.

39. None of the statements is true.

40. If a total of six false statements are made, then Achu should be an alternator, Babo should be a liar. Hence, Chiki bought a Red coloured item.

EXERCISE-2

Solutions for questions 1 and 2:

1. Let Dwivedi be the truth-teller:

	I	II	Position
Dwivedi	T	T	Extreme left
Trivedi	F	F	Centre
Chaturvedi	T	T	Extreme right

In this case, we get two truth-tellers (Dwivedi and Chaturvedi) and one liar (Trivedi) which violates the given conditions.
Let Trivedi be the truth-teller:

	I	II	Position
Dwivedi	F	F	Centre
Trivedi	T	T	Extreme right
Chaturvedi	F	T	Extreme left

Hence, Dwivedi's both statements are false, which is against the conditions given. Hence, Chaturvedi must be the truth-teller, in which case we get the following arrangements.

Case (i):

	I	II	Position
Dwivedi	T	F	Extreme left
Trivedi	F	T	Extreme right
Chaturvedi	T	T	Centre

Case (ii):

	I	II	Position
Dwivedi	T	T	Extreme left
Trivedi	F	F	Centre
Chaturvedi	T	T	Extreme right

Case (iii)

	I	II	Position
Dwivedi	F	F	Centre
Trivedi	T	F	Extreme left
Chaturvedi	T	T	Extreme right

In case (ii) and case (iii), both statements of Trivedi and Dwivedi, respectively are false. This is against the given condition.
Hence, the correct order from extreme left to extreme right is Dwivedi, Chaturvedi and Trivedi.

2. Based on the explanation to the previous questions, we have only one possible arrangement:

	I	II	Position
Dwivedi	T	F	Extreme left
Trivedi	F	T	Extreme right
Chaturvedi	T	T	Centre

Hence, none of the statements can be inferred.

Solutions for questions 3 to 5: The statement made by Rahul, Sandeep and Dhanush cannot be true at the same time. If the statement made by Rahul and Dhanush are true, then Rahul will be sitting in the middle of the row and so, the statement made by Sandeep will be false.

And the statements made by Ajay and Madhav are true. If the statements made by Rahul and Dhanush are true, then Sandeep will be at the right end and Sandeep's statement will be false.

Now, the statement made by Ajay must be false, but Ajay's statement is true.

∴ Sandeep's statement must be true and one among Rahul and Ajay must have made a false statement.

Now, from the statements made by Sandeep, Ajay and Madhav, the arrangement of the persons will be as follows.

________ ________ Sandeep Madhav _______

From the above arrangement the statement made by Dhanush must be false.
∴ The final arrangement will be as follows.

Rahul Dhanush/Ajay Sandeep Madhav Ajay/Dhanush

3. Dhanush made a false statement.

4. Either Ajay or Dhanush is sitting at the right end of the row.

5. Sandeep is sitting in the middle of the row.

Solutions for questions 6 to 9:
Case I: Let Mona be a truth-teller.
Now, Mona is an Engineer, Rama is a Doctor. So, Koma must be a Professor. Koma owns Micra.
Now, the first statement made by Rama must be true. So, Rama must be an alternator. Hence, her second statement must be false and third statement must be true.
∴ Mona owns Indigo and Rama must own Swift.
Koma must be a liar.
But, her third statement is true.
Hence, case I is invalid.

Case II: Let Rama be a truth-teller. Then Mona owns swift, Rama owns Indigo and Koma owns Micra.
Now, Mona's third statement is true and Koma's third statement is false.
So, Mona must be an alternator and Koma must be a liar, which is not possible because if Koma is a liar Mona is a truth teller.
Hence, case II is invalid.

Case III: Let Koma be the truth-teller, then the first two statements made by Mona must be false and so she must be a liar and Roma must be an alternator.

	Professor	Car	Category
Mona	Doctor	Swift	Liar
Roma	Professor	Micra	Alternator
Koma	Engineer	Indigo	Truth-teller

6. Mona owns Swift.

7. Koma is an engineer.

8. Roma is the alternator.

9. Mona is a liar.

Solutions for questions 10 to 12: Let us assume, Chibi is a truth-teller, then Cakora is none among the truth-teller, alternator and liar. Hence, Chibi is not a truth-teller.

Let us assume that Cakora is a truth-teller, then

	Statements				
Name	**I**	**II**	**III**	**IV**	
Chibi	F	F	F	F	Instagram
Cakora	T	T	T	T	Facebook
Ceasar	T	F	T	F	WhatsApp
Chaila	T	F	T	F	Twitter

Let us assume that Ceasar is a truth-teller, then Cakora is none among the truth-teller, alternator and liar. Hence, Ceasar is not a truth-teller. Let us assume that Chaila is a truth-teller, then Chibi is none among the truth-teller alternator and liar. Hence, Chaila is not a truth-teller.

10. Cakora is the truth-teller.

11. Chaila is on Twitter.

12. The liar is on Instagram.

Solutions for questions 13 to 17: As they made equal number of true and false statements, for two of them, the first statements are true and for the other two, the first statements are false.

Let us assume that Pavan's first statement is false. Therefore, his statements must be false, true, false in that order. Thus his 2nd statement being true, Sravan is the fourth ranker in Physics. Now, from Madan's 2nd statement, Sravan being the fourth ranker in Chemistry must be false. Therefore, the remaining 2 statements of Madan must be true. Sravan's 2nd statement, that he is the fourth ranker in Maths must be false, which means Sravan's remaining 2 statements are true. But Sravan's 1st statement and Madan's 3rd statement cannot be true simultaneously. So, our assumption that Pavan's first statement being false was wrong.

So, Pavan's first statement is true. Pavan's statements are true, false and true. Madan is the fourth ranker in Chemistry and Naveen is the first ranker in Maths. Madan's 2nd statement is false, which proves that Madan's other 2 statements are true. Madan's statements are true, false, true. So, Madan is the first ranker in Commerce and Pavan is the third ranker in Maths.

Naveen's first statement as well as Sravan's first statements are false. So, the statements made by Naveen and Sravan must be of the type: False, true and false.

	Maths	Physics	Chemistry	Commerce
Pavan	3	1		
Naveen	1			
Madan			4	1
Sravan	4		1	

From the above table it can be concluded that Madan is the second ranker in Maths and the third ranker in Physics.
⇒ In Chemistry, Naveen is the third ranker and Pavan is the second ranker.
⇒ In Commerce, Pavan is the fourth ranker, Sravan is the third ranker and Naveen is the second ranker.
∴ In Physics, Naveen is the fourth ranker and Sravan is the second ranker.

	Maths	Physics	Chemistry	Commerce
Pavan	3	1	2	4
Naveen	1	4	3	2
Madan	2	3	4	1
Sravan	4	2	1	3

13. Naveen is the third ranker in Chemistry.

14. Sravan is the second ranker in Physics.

15. Naveen got better rank than Madan only in Chemistry.

16. Naveen got second rank in Commerce.

17. In Commerce, Pavan got worse rank than both Naveen and Sravan.

Solutions for questions 18 to 22: Let the first statement of Abhay be false.
⇒ David is a truth-teller.
∴ Bharath is studying in Class X.
Also, Abhay must not be an alternator which implies that Abhay is a liar.
⇒ Chandu is studying in Class IX.
As Abhay is studying in Class VII is false, Abhay is studying in Class VIII and David is studying in Class VII.
But the above results are contradicting the first and the third statements of Chandu, the first being true and the third being a lie.
∴ Abhay's first and third statements are true.
⇒ Abhay is studying in Class VII.
⇒ Chandu's third and first statements are true.
⇒ Bharath is studying in Class VIII.
⇒ David's third and first statements are true and second statement is false.
∴ David is an alternator.
⇒ Abhay is not an alternator.
∴ Abhay is a truth-teller.
⇒ Chandu is not studying in Class IX.
Chandu is studying in Class X and David is studying in Class IX.
∴ Bharath's second and third statements are false.
⇒ Bharath is a liar.
⇒ Chandu's second statement is also true. The final arrangement is as follows.

	I	II	III	Studying
Abhay	T	T	T	VII
Bharath	F	F	F	VIII
Chandu	T	T	T	X
David	T	F	T	IX

18. Chandu is studying in Class X.

19. Only David is an alternator.

20. Abhay made three true statements.

21. David is studying in Class IX.

22. Required difference = 8 – 4 = 4.

Solutions for questions 23 to 25: It is given that no person belongs to the category that is mentioned on his badge.

Consider the second statement made by D. If it is true, then his second statement should be false (which is a contradiction). Therefore, his second statement should be false and he belongs neither to Altures nor to Allies.

⇒ He belongs to Liars. Therefore, his first statement should be false. Hence, Liars do not have a telephone.

Consider the first statement of A as false. This implies that Allies do not have a telephone and that A belongs to Allies. Now the second statement cannot be true as it leads to contradiction.

Therefore, A belongs to Altrues (As D belongs to Liars and A does not belong to Trues or Allies) and Allies have telephone.

Now B and C belong to Trues and Allies not necessarily in the same order. Therefore, the second statement of each of B and C should be true. As B does not have telephone, he cannot belong to Allies. Therefore, he belongs to Trues and C belongs to Allies.

	I	II
A	T	F
B	T	T
C	F	T
D	F	F

23. B belongs to Trues.

24. As C belongs to Allies, he has a telephone.

25. Only one category has telephone, i.e., Allies.

Solutions for questions 26 to 28: Here, the only way the arrangement works is by assuming that the three statements given by each of John and Jeromy must be False, True, False.

Then we get the following arrangement.

	I	II	III
John	F	T	F
Jack	T	F	T
Jeromy	F	T	F
James	T	F	T

As each pattern is followed by exactly two people, the statements given by Jack and James must be in the order of True, False and True.

Then, the final arrangement as per the individual rankings is as given below:

	Height	Weight	Age	Wealth
John	2	4	1	3
Jack	3	1	2	4
Jeromy	4	2	3	1
James	1	3	4	2

Hence, the questions can be answered based on the above table.

26. Jeromy is the second heaviest.

27. Jeromy is the desired person.

28. James is the third heaviest, which means there are two people who weigh more than James.

Solutions for questions 29 to 31: It can be clearly inferred that Bhavna's second statement cannot be true because sum of three different numbers starting with 1 cannot be a unit multiple of 5. It is possible only when two numbers are equal ⇒ – 1 + 1 + 3 or 1 + 2 + 2 – which is a violation that all the three people are in different positions.

We have the following about Bhavna's statements and the impact on Charan's statements:

	I	II	III
Alex			
Bhavna		F	
Charan	F	T	F

This is because Charan alternates between truth and lie statements in any order.

Now, Charan's IInd statement (True) ⇒ Alex is currently is position seven.

So, Alex's IIIrd statement would become false because Alex is already in position seven, implying he would not be in position seven after interchanging position with Charan.

This also implies that Bhavna's Ist statement is false since the fourth position from the end of the queue would be the seventh position from the beginning and Alex is already in position 7.

So, we have the following:

	I	II	III	Positions
Alex			F	7
Bhavna	F	F		
Charan	F	T	f	

Now, Bhavna cannot alternate between true and lie statements. So, Alex must be the second person to alternate between true and lie statements.

So, the final arrangement is as follows.

	I	II	III	Positions
Alex	F	T	F	7
Bhavna	F	F	F	?
Charan	F	T	F	?

The positions of Bhavna and Charan cannot be uniquely determined as illustrated below.

Position	(i)	(ii)	(iii)	(iv)	(v)
1				B	C
2	C		B		
3					
4		B			
5					

Position	(i)	(ii)	(iii)	(iv)	(v)
6		C			
7	A	A	A	A	
8	B		C		
9				C	B
10					

29. The position of Bhavana with respect to Alex cannot be uniquely determined.

30. I – Is true.
II – Is true.
III – Is false since Alex alternates in the order as false, true, false.

31. Choice (B) gives us a unique arrangement as shown earlier in case (i).

Solutions for questions 32 to 34: If P always speaks the truth, then Q always lies R.

If P always lies R, then Q always speaks the truth.
In either case, R cannot be speaking all truths or all lies.
R cannot alternate between truth and lies either.
So, we have the following:

Case (i)

	I	II	III
P	T	T	T
Q	F	F	F
R	F	T	f

Case (ii)

	I	II	III
P	F	F	F
Q	T	T	T
R	T	F	LF

In case (i), R becomes an alternator which violates the given condition.
Hence, case (ii) in the correct arrangement and the south – bound bus goes to the airport.

32. R replies in the manner: Truth – lie – lie

33. Only statement I is true.

34. (20 minutes to wait for the bus) + (30 minutes of travel time) = 50 minutes to reach the airport.

Solutions for questions 35 to 37: Let us rename the five army personnel as below:

Ranveer → Veer
Ranjeet → Jeet
Randheer → Dheer
Ranmeet → Meet
Ranpreet → Preet

It is known that except one person (who must be a Liar whose all 3 statements are all false) each of the other four spoke at least one true statement. Also, except one person (who must be the truth-teller whose all 3 statements must be true) each of the other four tells at least one lie, i.e., one false statement.

Also, exactly one person always alternated between the truth and lie (an alternator) in any order – (True False True) or (False True False).

This means that there must be two people, each of whom speaks at least one true and at least one false statement, but none of them is an alternator. Let us assume each person to be a truth-teller and find a definite arrangement.

Let Ranjeet be the truth-teller.

	I	II	III	Ranks
Jeet	T	T	T	Lieut./Capt.
Meet	F	?	F	Brigadier
Preet	F	F	F	Major
Dheer	F	F	F	Capt./Lieut.
Veer	F	T	F	Colonel

In this arrangement, both Preet and Dheer tell lies (all 3 statements are false), which violates the conditions as there is exactly one liar among them. By similar approach, when we assume that Ranpreet always speaks the truth, then we get the following arrangement.

	I	II	III	Ranks
Jeet	F	?	F	Colonel
Meet	F	F	?	Captain
Preet	T	T	T	Major /Lieut.
Dheer	?	F	F	Brigadier
Veer	F	?	?	Lieut./Major

Now, either Veer or Jeet is the alternator, Veer cannot be the alternator, because if his last statement is false (which means that Veer is not the Lieutenant but he is the Major), then his second statement must be true (i.e., Preet is Major), which is not possible, as exactly one person between Preet and Veer must be the Major.

Hence, Jeet must be the Alternator and his second statement must be true.

Now, we rearrange the above deductions as below.

	I	II	III	Ranks
Jeet	F	T	F	Colonel
Meet	F	F	F	Captain
Preet	T	T	T	Major
Dheer	T	F	F	Brigadier
Veer	F	T	T	Lieutenant

Based on this arrangement (which is the only definite arrangement), we answer the questions as below.

35. First statement given by Randheer is true and the rest are false.

36. Ranjeet, the Colonel, alternates between the truth and lie, in the order lie – truth – lie.

37. From the table, choice (D) gives the correct order.

Solutions for questions 38 to 40: From the given statement it is clear that Pingo cannot be the truth-teller. (As truth-teller cannot say that, I am an alternator).

He can be either an alternator or a liar.

If Pingo is the Professor, then his first statement is contradicting.

Hence, he cannot be the Professor.

If Bingo is a truth-teller, then Tingo's second statement must be false.

$\Rightarrow$ Tingo's first statement cannot be true. Therefore, Tingo must be a liar.

Now, Bingo is the Manager, Tingo is the Chartered Accountant (CA) and Pingo is not the Engineer. Therefore, Pingo must be a Professor, which is not possible.

$\therefore$ Bingo is not a truth-teller.

If Mingo is the Engineer, we have two possibilities, they are:

Case (A1):

Professor	Name	I	II	III
CA	B	F	F	F
Engineer	M	T	T	T
Professor	T	F	T	F
Manager	P	T	F	T

Case (A2):

Professor	Name	I	II	III
Manager	B	F	T	F
Engineer	M	T	T	T
Professor	T	F	T	F
CA	P	F	F	F

If Tingo is a truth-teller, then we get two possible cases, they are:

Case (B1):

Profession	Name	I	II	III
Manager	B	F	T	F
CA	M	F	F	F
Professor	T	T	T	T
Engineer	P	F	F	F

Case (B2):

Profession	Name	I	II	III
CA	B	F	F	F
Manager	M	F	F	F
Professor	T	T	T	T
Engineer	P	T	F	T

It is also possible that there is no truth-teller, but all of them cannot be liars, there must be at least an alternator (according to the statements of Pingo), i.e., if Pingo is a liar, then Bingo cannot be a liar and vice versa.

If Bingo is an alternator, we have one possibility:

Case (C):

Profession	Name	I	II	III
Manager	B	F	T	F
Professor	M	F	F	F
Engineer	T	F	F	F
CA	P	F	F	F

If only Pingo is an alternator, we have two possibilities:

Case (D1):

Profession	Name	I	II	III
Not Manager	B	F	F	F
Not Engineer	M	F	F	F
Manager/Engineer	T	F	F	F
Manager/CA	P	T	F	T

If Mingo is an alternator, then we get the following:

Case (D2):

Profession	Name	I	II	III
Professor	B	F	F	F
Engineer	M	F	T	F
Manager	T	F	F	F
CA	P	T	F	T

38. In case D2, Tingo is the Manager. Pingo is the Chartered Accountant. In case D1, if Tingo is the Manager, then Pingo will be the Chartered Accountant.

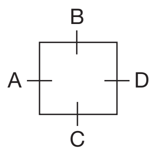
39. Here, Pingo's III statement is false and hence, his first statement is also false. So, Pingo is a liar. As no other is liar, the II statements of all others must be true.
∴ Mingo is the Engineer. This is case A2.

40. If we take any two of the statements, we will get case D2.

EXERCISE-3

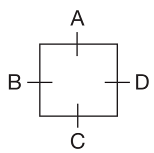
Solutions for questions 1 to 3:

Case (i):
Let us assume that A speaks the truth.
The arrangement is:



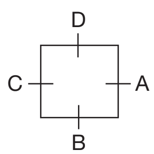
	I	II	
A	T	T	Truth-teller
B	F	T	Alternator
C	F	F	Liar
D	F	T	Alternator

Case (ii):
Let us assume that B is the truth-teller, then the arrangement is:



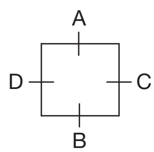
	I	II	
A	F	T	Alternator
B	T	T	Truth-teller
C	F	F	Liar
D	F	F	Liar

Case (iii):
Let us assume that C is the truth-teller, then the arrangement is:



	I	II	
A	F	F	Liar
B	F	F	Liar
C	T	T	Truth-teller
D	F	F	Liar

Case (iv):
Let us assume that D is the truth-teller, then the arrangement is:



	I	II	
A	F	F	Liar
B	F	F	Liar
C	F	F	Liar
D	F	F	Truth-teller

As it was given that there is exactly one person who alternates between the truth and lies, the arrangement in case (ii) is the valid one.

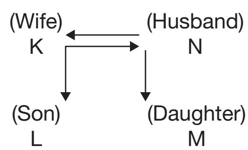
1. B is the truth-teller.

2. C and D are the liars.

3. B sits opposite to D.

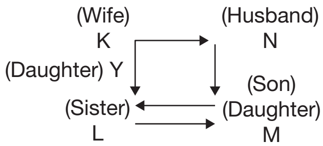
Solutions for questions 4 to 6: Among the four members, there is one couple, their son and their daughter. The statements given by L is definitely false because, according to his statements there are three generations. So, L cannot be the truth-teller.

Case (i):
Let us assume that K is the truth-teller, then



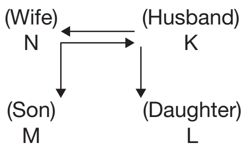
	I	II	
K	T	T	Truth-teller
L	T	F	Alternator
M	F	F	Liar
N	T	F	Alternator

Case (ii):
Let us assume that M is the truth-teller, then the arrangement is as follows.



	I	II	
K	T	F	Alternator
L	T	F	Alternator
M	T	T	Truth-teller
N	F	T	Alternator

Case (iii):
Let us assume that N is the truth-teller.



	I	II	
K	F	F	Liar
L	F	F	Liar
M	F	T	Alternator
N	T	T	Truth-teller

4. L can never be the truth-teller.

5. In case (iii), N is the truth-teller, and K and L are the liars.

6. In case (ii), N and L are the alternators and K is the truth-teller.

Solutions for questions 7 to 10: It is given that each of the four people alternates between truth and lies in any order and that the second statement of exactly two people, one of them being Abhinav is true. This implies that the first and the last statement of each of these two people is false. Hence, from Abhinav's statements, which are in the order of false, true and false, we know the following.

Bipash is not from Delhi.
Chandar plays Chess.
Danny is the tallest.
Now, with the above information, let us check the first statement of each of the other three people.
From the above information, the truth or falsity in the first statement of Bipash and Chandar cannot be found out. But we know that Danny's first statement is false. Hence, the second statement of Danny is true. Since, there are only two people, whose second statement is true, we get the following information.
Abhinav: Bipash is from Delhi - False
Chandar plays Chess – True ⇒ Chandar – Chess
Danny is not the tallest – False ⇒ Danny – Tallest
Bipash: Abhinav is from Kolkata – True
⇒ Abhinav – Kolkata
Chandar is not the shortest – False
⇒ Chandar – Shortest
Danny plays badminton – True
⇒ Danny – Badminton

Chandar: Abhinav is the 2nd tallest – True
⇒ Abhinav – 2nd tallest
Bipash plays Tennis – False
⇒ Bipash does not play Tennis
Danny is from Delhi – True ⇒ Danny – Delhi

Danny: Abhinav plays Chess – False
Bipash is from Mumbai – True ⇒ Bipash - Mumbai
Chandar is not from Hyderabad - False
⇒ Chandar – Hyderabad
From the above information, we get the following arrangement.

7. Danny is from Delhi.

8. Chandar is the Chess player.

9. The person who plays Chess is the shortest.

10. The second statement made by Abhinav and Danny are true.

Solutions for questions 11 to 15: From the given data:
As Amar and Dinesh belong to same group, their last statements cannot be simultaneously true.
Hence, their 2nd statements must be true.
Similarly, Bharath and Chandu's second statements are false.
The final table of heights and weights is as follows.

	Height	Weight	Age
1st	Chandu	Dinesh	Eswar
2nd	Dinesh	Bharath	Chandu
3rd	Eswar	Amar	Dinesh
4th	Amar	Eswar	Bharath
5th	Bharath	Chandu	Amar

11. Chandu is the tallest.

12. Eswar is the oldest.

13. Chandu is the lightest.

14. Amar is the 2nd shortest.

15. Dinesh is the 3rd youngest.

Solutions for questions 16 to 19: Schindler said 'I am not an alternator'. From this, we can conclude that he is either a truth-teller or an alternator.

Case (i): Schindler is a truth-teller.
Since all the statements of Schindler are true, we can conclude from his statements that (a) Gretta is a liar and (b) he is not the shortest. Considering that Gretta is a liar, we can conclude that Gretta is not the tallest and that Fischer is not a liar. From the facts derived so far, we can conclude that Fischer's second statement is true and the third one is false. Hence, he must be an alternator. Thus, his first statement is false. This implies that Fischer is the tallest. Since Schindler is not the shortest, he must be the second tallest and Gretta the shortest. Thus, the final arrangement is as follows.

Tallest	Fischer	Alternator
2nd tallest	Schindler	Truth-teller
Shortest	Gretta	Liar

Case (ii): Schindler is an alternator. In this case, his first and the third statements are true and the second one is false. From his statements, we get the same facts as we derived above. As a result, all other information that we have derived will also be the same except that Schindler is an alternator.

Tallest	Fischer	Alternator
2nd tallest	Schindler	Alternator
Shortest	Gretta	Liar

16. Either Schindler is the truth-teller or none of them is a truth-teller.

17. Fischer is the tallest.

18. They together made either three or four true statements.

19. Fischer is definitely an alternator.C)

Solutions for questions 20 to 23:

Case (i): Both the statements of P are true.
Facts from P's statements: S is male, Q and R are females. Hence, R's first statement is false and the second one is true. S's second statement is true and the first one is false. Now, whether P is male or female, the above deductions hold good.

Thus, in this case, both the statements of Q are false. Each of R and S made one true statement and one false statement.

Case (ii): Both the statements of Q are true.
Facts from Q's statements: R is male and both the statements of S are false.
Facts from S's statements: Q is male and S has three brothers or three sisters or two sisters and a brother.
Hence, for each of P and R, one statement should be true and the other one is false. Clearly, P's first statement is false. Hence, his second statement is true, which implies that S is also male. Now, R's first statement is true and second one is true. Since Q and R are males and the first statement S is false, P should also be male.
Thus, in this case, both the statements of S are false. Each of R and P made one true statement and one false statement. All of P, Q, R and S are male.

Case (iii): Both the statements of S are true.
Facts from S's statements: Q is female and both P and R are males.
In such case, Q's first statement is false and the second one is true. R's second statement is false. Now R's first statement cannot be true, because in such case there will be three people for each of whom one statement is true and the other one is false. Hence, R's second statement must also be false. This means that either both the statements of P are true or both are false. But in both the cases, the given data is violated. Hence, S cannot be the one with both true statements. Thus cases (i) and (ii) are valid.

20. R definitely made one true and one false statement.

21. Swank definitely is male.

22. Either Qusac or Swank made two false statements.

23. Statement (D) is true.

Solutions for questions 24 and 25: It is given that it is the birthday of one of the two. Hence, Sharmila's second statement is false.

Let Sharmila's first statement be true or false.

Case (i): Sharmila's first statement is true.
It implies that Karuna is lying. Hence, it is Karuna's birthday but not Sharmila's.

Case (ii): Sharmila's first statement is false.
It implies that either both the statements of Karuna are true or one of them is true and the other one is false.
If the first statement of Karuna is true and the second one is false, it implies it is not the birthday of any one of them. If the first statement is false and the second one is true, then it is the birthday of both of them. But both the cases violate the given information. Thus, it can be concluded that both the statements of Karuna must be true. Hence, it is the birthday of Sharmila.

24. If Sharmila's first statement is false, it would be Sharmila's birthday.

25. Statement (B) is true.

7 Venn Diagrams

CHAPTER

Learning Objectives

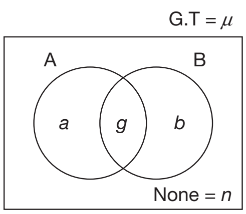
In this chapter, you will:

- Understand the concept of Venn diagrams – the need for representing sets in a graphical manner and properties of a set.
- Understand the use of Venn diagrams to explain the logical relationships between two or more sets.
- Learn how to interpret the statements (such as 'at least', 'at most', 'exactly', 'A or B but not C', 'A and B but not C' etc.) given in the questions and how to assign values to each category of variables within a Venn diagram.
- Gain knowledge of how to deal with Venn diagrams having two, three and four sets.

❒ INTRODUCTION

A **VENN DIAGRAM** is a diagrammatical representation of two or more sets/groups, which may/may not have some common elements (like women and athletes – some women may also be athletes), using geometric shapes to represent each set. Venn diagrams illustrate a logical relationship between the sets. In the theory given below we illustrate Venn diagrams with two, three and four sets.

Venn Diagrams Involving Two Variables

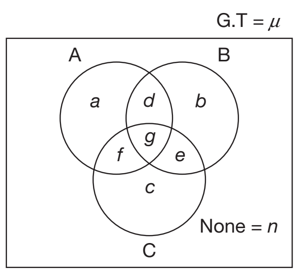


In the above diagram, A and B represent two different sets and the various regions can be referred to as given below.

$$A = a + g; B = b + g$$

Only A = a; Only B = b
Exactly one set = $a + b$
A and B = g; Only A and B = g
Exactly two sets = g
At least one set = Exactly one + Exactly two
$= a + b + g = T$
Grand total (G.T = μ) = $a + b + g + n = T + n$
A + B = $a + b + 2g = T + g$
A or B = $a + b + g = T$
Does not belongs to A = $b + n$
Does not belongs to B = $a + n$

Venn Diagram with Three Variables



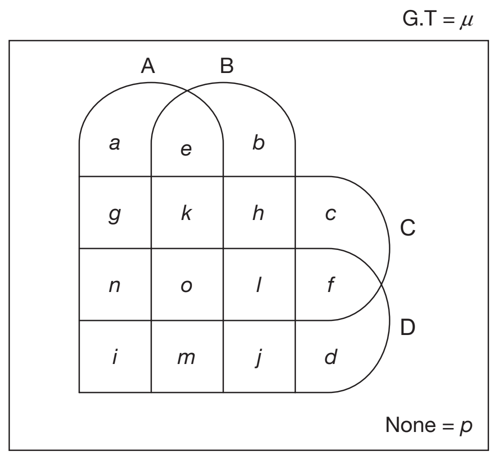
Here A, B and C are three different sets and the various regions can be referred to as given below.

A = a + $d + g + f$; Only A = a
B = $b + d + g + e$; Only B = b
C = $c + f + g + e$; Only C = c
Exactly one set = $a + b + c$
A and B = d + g; B and also C = e + g; C as well as A = $f + g$
Only A and B = d ; A and B but not C = d
Only B and C = e ; B and C but not A = e
Only C and A = f; C and A but not B = f
Exactly two sets = $d + e + f$
A, B and C = All the three = Only A, B and C = g;
Exactly three sets = g
None among A, B and C = n
At least one set = Exactly one + Exactly two + Exactly three = a + $b + c + d + e + f + g = \mu - n$
At least two sets = Exactly two + Exactly three = $d + e + f +$ g
At least three sets = Exactly three = g
At most one sets = Exactly one + None = $a + b + c + n$
At most two sets = Exactly two + Exactly one + None = $d + e + f +$ a + $b + c + n = \mu - g$
At most three sets = Exactly three + Exactly two + Exactly one + None = $g + d + e + f +$ a + $b + c + n = \mu$
A + B + C = $a + b + c + 2(d + e + f) + 3g$
= Exactly one + 2 (Exactly two) + 3 (Exactly three)
= (Exactly one + Exactly two + Exactly three) + Exactly two + 2 (Exactly three)
= At least one + Exactly two + 2 (Exactly three)
= At least one + (Exactly two + Exactly three) + Exactly three
= At least one + At least two + At least three
Does not belong to A = $b + e + c + n$
A or B or C = $a + b + c + d + e + f + g$ = At least one.
A or B = $a + b + d + e + f + g$
A or B but not C = a + $d + b$
Neither A nor B = $c + n$
(A and B) or C = $d + c + f + g + e$
A and (B or C) = $d + g + f$

Venn Diagram Involving Four Variables

Here, A, B, C and D are four different sets and the various regions can be referred to as given below.

A = $a + e + g + k + n + o + i + m$; Only A = a
B = $b + e + h + k + l + o + j + m$; Only B = b
C = $c + f + h + l + k + o + g + n$; Only C = c
D = $d + f + j + l + m + o + i + n$; Only D = d



Exactly one set = $a + b + c + d$
A and B = $e + k + o + m$; Only A and B = e
A and C = $g + k + o + n$; Only A and C = g
A and D = $n + o + i + m$; Only A and D = i
B and C = $k + h + o + l$; Only B and C = h
B and D = $m + j + o + l$; Only B and D = j
C and D = $n + o + l + f$; Only C and D = f
Exactly two sets = $e + f + g + h + i + j$
A, B and C = $k + o$; Only A, B and C = k
B, C and D = $l + o$; Only B, C and D = l
A, B and D = m + o; Only A, B and D = m
A, C and D = $n + o$; Only A, C and D = n
Exactly three sets = $k + l + m + n$
A, B, C and D = All the four = Exactly four set = o;
None among A, B, C and D = p

NOTE

Note the following for a n-set Venn diagram:

Name of the region (pocket of intersection of the sets)	Number of regions (pockets of intersections of the sets)	For a 5-set situation
Exactly 1, X	${}^nC_1$	5
Exactly 2, Y	${}^nC_2$	10
Exactly 3, Z	${}^nC_3$	10
Exactly 4, A	${}^nC_4$	5
Exactly 5, B	${}^nC_5$	1
None, N	${}^nC_0$	1
Total number of regions	2^n	32

SOLVED EXAMPLES

Directions for questions 7.01 to 7.04: These questions are based on the information given below.

The class teacher has posed two questions A and B to the 160 students of her class. 65 students could not answer question A, 80 students could not answer question B and 40 students answered both the questions.

7.01: How many students could not answer any of the two questions?
(A) 30 (B) 25
(C) 40 (D) 15

7.02: How many students answered only question A?
(A) 95 (B) 60
(C) 50 (D) 55

7.03: How many students could not answer exactly one question?
(A) 130 (B) 55
(C) 95 (D) 11

7.04: The number of students who answered only question B is what percentage of the number of students who answered question B?
(A) 50% (B) 60%
(C) 25% (D) $33\frac{1}{3}$ %

Solution for questions 7.01 to 7.04

Total number of students = 160

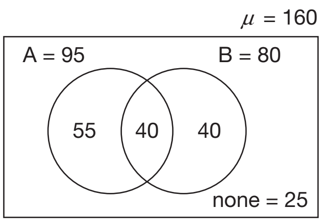
65 could not answer A

$\Rightarrow$ 160 – 65 = 5 answered question A.

80 could not answer B $\Rightarrow$ 160 – 80 = 80 answered question B.

40 students answered both the questions.

Venn diagram based on the above data is as follows.



$\Rightarrow$ 25 students could answer neither of the questions.

7.01: 25 students could not answer any of the two questions. Hence, the correct option is (B)

7.02: 55 students answered only A. Hence, the correct option is (D)

7.03: 40 + 55 = 95 students could not answer exactly one question. Hence, the correct option is (C)

7.04: 40 students answered only B.

Required % = $\frac{40}{80} \times 100 = 50\%$

Hence, the correct option is (A).

Directions for questions 7.05 to 7.08: These questions are based on the following data.

In a colony, the residents read different newspaper, among The Hindu. The Times of India and Dainik Bhaskar. It is known that 52% of the residents read at most one newspaper. 42% of the residents read The Times of India or Dainik Bhaskar but not The Hindu. 54% of the residents read The Times of India. 24% of the residents read both The Hindu and The Times of India. 36% of the residents read exactly two newspapers. 10% of the residents read only The Hindu and Dainik Bhaskar. The number of residents, who read all the three newspapers is twice the number of residents who read none of these newspapers. 2,800 residents read only the Times of India and Dainik Bhaskar.

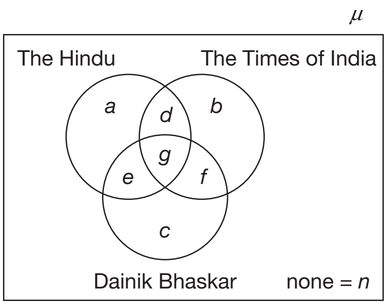
7.05: How many residents read only The Hindu?
(A) 4000 (B) 600
(C) 3200 (D) 3400

7.06: How many residents do not read any of the newspapers?
(A) 1800 (B) 1600
(C) 1450 (D) 1200

7.07: How many residents read at least two newspapers?
(A) 9600 (B) 8000
(C) 8400 (D) 10200

7.08: How many residents are there in the colony?
(A) 22000 (B) 16000
(C) 20000 (D) 18000

Solution for questions 7.05 to 7.08: Let us represent the given information in a Venn diagram, as follows.



Given. 52 % of the residents read at most one newspaper,

$$a + b + c - n = 52\% \quad (1)$$

42 % of the residents read Times of India or Dainik Bhaskar but do not read Hindu.

$$f + b + c = 42\% \quad (2\}$$

54% of the residents read The Times of India

$$d + b + g + f = 54\% \quad (3)$$

24% of the residents read both the Hindu and The Times of India.

$$d + g = 24\% \quad (4)$$

35% of the residents read exactly two newspapers,

$$d + e + f = 36\% \quad (5)$$

10% of the residents read only The Hindu and Dainik Bhaskar

$$e = 10\% \quad (6)$$

The number of residents who read all the three newspapers is twice the number of people who read none

$$g = 2n \quad (7)$$

2800 residents read only the Times of India and Dainik Bhaskar.

$$f = 2800 \quad (8)$$

By subtracting eqn (4) from (3), we get

$$b + f = 30\% \quad (9)$$

By subtracting eqn (9) from (2), we get

$$c = 12\% \quad (10)$$

From equation (1) and (5), we get

$$a + b + c + n + d + e + f = 88\%$$

$$\therefore \quad g = 100 - 88 = 12\%$$

Thus, $n = 6\%$

From equation (4), we get $d = 12\%$

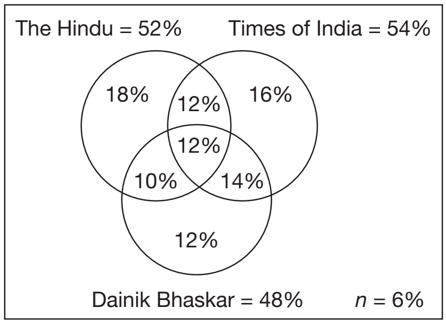
From eqn. (5), we get

$$f = 14\%$$

From eqn. (2), we get

$$b = 16\%$$

$\therefore$ The final diagram is as follows



Given

$$f = 14\% \text{ of total} = 2600$$

$\therefore$ Total no. of residents in the colony

$$= \frac{2800 \times 100}{14} = 20000$$

7.05: 18% of the residents read only Hindu.

$$\frac{2800 \times 6}{100} = 3600$$

Hence, the correct option is (B).

7.06: 6% of the residents do not read any of the news papers.

$$\frac{20000 \times 6}{100} = 1200$$

7.07: Residents who read at least two newspapers = (Residents who read exactly two + exactly three)

$$d + e + f + g = (12 + 12 + 10 + 14)\% = 48\%$$

$$\therefore \quad \frac{20000 \times 48}{100} = 9600$$

Hence, the correct option is (A).

7.08: Total number of residents = 100%

20,000 people are there in the colony.

Hence, the correct option is (C).

Directions for questions 7.09 to 7.12: These questions are based on the information given below.

A survey was conducted among some people It was found that 330 people watch Discovery channel, 330 people watch Star World, 315 people watch BBC and 285 people watch Star News. The number of people who watch each combination of exactly three channels is 40. The number of people, who watch only Discovery channel and BBC is 50. 80 people watch only Discovery channel, 100 people watch only Star World, 90 people watch only BBC and 70 people watch only Star News. 30 people watch only BBC and Star News, while

10 people watch ail the four channels. Each person watches at least one channel.

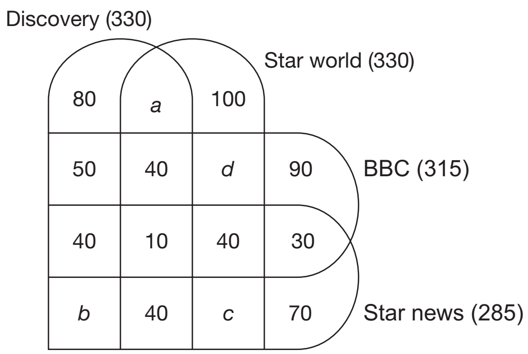
7.09: How many people watch only Discovery channel and Star World?
(A) 140 (B) 50
(C) 180 (D) 230

7.10. How many people watch BBC but not Discovery channel?
(A) 240 (B) 320
(C) 150 (D) 175

7.11. How many people watch exactly two channels?
(A) 200 (B) 220
(C) 310 (D) 180

7.12. How many people took part in the survey?
(A) 1000
(B) 1260
(C) 710
(D) Cannot be determined

Directions for questions 7.09 to 7.12: From the given information, we have the following venn diagram.



$$50 + 40 + d + 90 + 40 + 10 + 40 + 30 = 315$$

$\Rightarrow$ $d = 315 - 300$

$d = 15$

$a + b + b\,80 + 40 + 40 + 10 + 40 = 330$

$\Rightarrow$ $a + b = 70$ (1)

Similarly,

$$a + c + 100 + 40 + 10 + 40 + 40 + 15 = 330.$$

$\Rightarrow$ $a + c = 85$ (2)

and

$$b + c + 40 + 10 + 40 + 30 + 70 + 40 = 285$$

$\Rightarrow$ $b + c = 55$ (3)

By adding equations (1), (2) and (3), we get

$$2(a + b + c) = 210$$

$\Rightarrow$ $a + b + c = 105$ (4)

From equation (4) and (1), we get

$$c = 35$$

Similarly,

$$b = 20$$
$$a = 50.$$

7.09: People watch only Discovery and Star World $= a = 50$.

7.10: $90 + 30 + 40 + 15 = 175$ people watch BBC but do not watch Discovery.

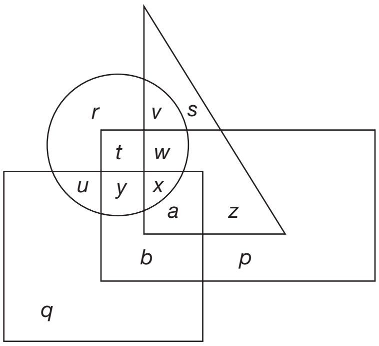
7.11: $50 + 50 + 15 + 30 + 20 = 200$ people watch exactly two channels.

7.12: 710 many people took part in the survey.

EXERCISE-1

Directions for questions 1 to 5: These questions are based on the following diagram.

In the following diagram, the circle represents all the people who like Maaza, the square represents all the people who like Thumbs Up, the Triangle represents all the people who like Mirinda and the Rectangle represents all the people who like Coca-Cola.



1. Which of the following represents the people who like Coca-Cola and Thumbs Up?
(A) *r* (B) *u*
(C) *b* (D) *q*

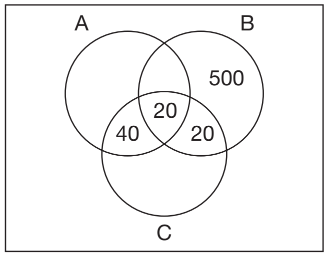
2. Which of the following represents the people who like Mirinda but not Thumbs Up?
(A) *v, s, w, x* (B) *v, s, z, a*
(C) *v, w, x, a* (D) None of these

3. Which of the following represents the people who like Maaza and Thumbs Up?
(A) *u, t, w* (B) *v, w, x*
(C) *b, a, x* (D) *u, y, x*

4. Which of the following represents the people who like both Maaza and Mirinda but not any of the other two?
(A) *y* (B) *v*
(C) *u* (D) None of these

5. Which of the following represents the people who like Mirinda, Maaza, Coca-Cola but not Thumbs Up?
(A) *b* (B) *r*
(C) *s* (D) *w*

Directions for questions 6 to 10: The following Venn diagram represents the 1200 employees of a company. Each of the employees is a member of at least one of three clubs, such as A, B and C. Using the given data, answer the questions that follow.



Total member of Club A = 420
Total member of Club B = 590
Total member of Club C = 340

6. How many employees are member of Club C only?
(A) 250 (B) 240
(C) 180 (D) 260

7. How many employees are member of both Club A and Club B?
(A) 50 (B) 80
(C) 70 (D) 60

8. How many employees are not member of Club B?
(A) 340 (B) 610
(C) 630 (D) 290

9. How many employees are member of Club A or Club C?
(A) 850 (B) 700
(C) 975 (D) 675

10. How many employees are member of exactly two clubs?
(A) 110 (B) 130
(C) 98 (D) 78

Directions for questions 11 to 15: These questions are based on the following information.

In a class of 150 students, 45 take History, 65 take Geography and 10 take both History and Geography.

11. How many students take only Geography?
(A) 45 (B) 10
(C) 55 (D) 65

12. How many students take only History?
(A) 65 (B) 35
(C) 10 (D) 45

13. How many do not take either History or Geography?
(A) 10 (B) 35
(C) 100 (D) 140

14. How many students take at least one subject?
(A) 10 (B) 90
(C) 50 (D) 100

15. How many students do not take any of the two subjects?
(A) 90 (B) 10
(C) 50 (D) 100

Directions for questions 16 to 18: These questions are based on the following information.

In a class of 150 students, 50 students passed in Mathematics, 40 students failed only in Chemistry and 20 students failed in both the subjects.

16. How many students passed in both the subjects?
(A) 20 (B) 15
(C) 10 (D) 16

17. How many students passed exactly in one subject?
(A) 25 (B) 120
(C) 140 (D) 145

18. How many students failed in at least one of the subjects?
(A) 130 (B) 125
(C) 145 (D) 140

Directions for questions 19 to 23: These questions are based on the following information.

In a survey conducted among 200 mobile phone using families, it was found that 140 use Panasonic, 120 use Nokia and 143 use Siemens. 95 use both Panasonic and Nokia, 85 use both Nokia and Siemens and 93 use both Panasonic and Siemens. 70 families use mobile phones of all the three companies.

19. How many families use mobilephones of only Siemens?
(A) 50 (B) 35
(C) 70 (D) 143

20. How many families use mobilephones of both Panasonic and Nokia but not Siemens?
(A) 25 (B) 57
(C) 165 (D) 95

21. How many families use mobilephones of exactly one company?
(A) 63 (B) 67
(C) 70 (D) 200

22. How many families use neither Panasonic nor Siemens?
(A) 40 (B) 120
(C) 110 (D) 10

23. How many families use none of the mobile phones?
(A) 10 (B) 70
(C) 0 (D) Cannot be determined

Directions for questions 24 to 28: Study the following data and the table to answer the questions that follow.

A survey was conducted among 100 students in a class who read detective novels written by Conan Doyle or Agatha Christie or both. Due to some recording error in the computer most of the figures were missing. The following table shows the remaining data.

	Doyle	Christie	Both	Total
Male				
Female	40			
Total		70		100

Further it is known that
(A) 37% of the students read both Doyle and Christie.
(B) The ratio of males to females is 1 : 1.
(C) 50% of the females read books of both the authors.

24. How many males read books by both the authors?
(A) 10 (B) 12
(C) 37 (D) 45

25. How many students read books by only Christie?
(A) 70 (B) 10
(C) 33 (D) 23

26. How many females read books by only Doyle?
(A) 25 (B) 40
(C) 35 (D) 15

27. How many students do not read books by both the authors?
(A) 12 (B) 27
(C) 37 (D) 63

28. How many males read books by Doyle?
(A) 27 (B) 67
(C) 12 (D) 15

Directions for questions 29 to 33: These questions are based on the following information.

Out of 200 people who attended a birthday party 120 had cool drink, 100 had ice cream, 80 had cake and 10 had none of these three. 100 people had exactly one of the three items.

29. How many people had exactly two of the three items?
(A) 70 (B) 80
(C) 110 (D) 85

30. If 20 people had only cool drink and ice cream, then how many people had only cake?
(A) 12 (B) 18
(C) 20 (D) 10

31. If 15 people who had only cool drink, had cake also and 5 people who had only ice cream and cake had cool drink also, then how many people had at least two of the three?
(A) 135 (B) 105
(C) 78 (D) 119

32. If 30 people had only ice cream, then how many people had at least one of cake and cool drink but not ice cream?
(A) 105 (B) 98
(C) 87 (D) 90

33. What is the maximum possible number of people who had only cool drink?
(A) 90 (B) 87
(C) 118 (D) 120

Directions for questions 34 to 37: These questions are based on the following information.

A total of 800 people participated in a consumer survey. The survey was conducted to get an idea about the popularity of the four soaps, such as Dove, Pears, Lux and Liril. Each participant liked at least one of the four products. The number of people who liked Liril is 450, those who liked Lux is 270 and those who liked Dove is 325. 100 liked only Pears, 50 liked only Lux and 90 liked only Liril. 50 liked only Dove and Pears, 60 liked only Dove and Lux, 90 liked only Dove and Liril, 30 liked only Pears and Lux, 120 liked only Pears and Liril, 60 liked only Lux and Liril. 20 liked only Dove, Pears and Lux, 15 liked only Dove, Lux and Liril, 25 liked only Pears, Lux and Liril.

34. How many people liked Dove and Pears but not Lux?
(A) 50 (B) 90
(C) 100 (D) 70

35. How many people liked at least two products?
(A) 480 (B) 560
(C) 600 (D) 520

36. How many people liked neither Dove nor Pears?
(A) 200 (B) 250
(C) 180 (D) 150

37. How many people liked either Lux or Liril?
(A) 590 (B) 630
(C) 570 (D) 610

Directions for questions 38 to 40: These questions are based on the following information.

Each of 780 bags contains at least one of raisins, almonds and peanuts. 375 bags contain raisins and 315 bags contain almonds. 75 bags contain both raisins and almonds, 90 bags contain both raisins and peanuts, 45 bags contain both almonds and peanuts.

38. Find the maximum possible number of bags which contain exactly two of raisins, almonds and peanuts.
(A) 240 (B) 180
(C) 210 (D) 150

39. Find the minimum possible number of bags which contain either peanuts or almonds.
(A) 525 (B) 620
(C) 640 (D) 600

40. If the number of bags which contain exactly two of raisins, almonds and peanuts is four times the number of bags which contain all the three, then how many bags contain only raisins?
(A) 210 (B) 240
(C) 270 (D) 180

EXERCISE-2

Directions for questions 1 to 4: These questions are based on the following information.

In a colony, 30 families read both The Hindu and Indian Express. 40 families read neither of these two news papers. 40% of the families read The Hindu and 55% of the families read Indian Express.

1. How many families are there in the colony?
(A) 150 (B) 200
(C) 160 (D) 250

2. How many families read at least one of the newspapers?
(A) 110 (B) 120
(C) 65 (D) 160

3. What percentage of the total number of families read at most one newspaper?
(A) 60% (B) 30%
(C) 70% (D) 85%

4. What percentage of the total number of families read The Hindu only?
(A) 25% (B) 30%
(C) 10% (D) 15%

Directions for questions 5 to 8: These questions are based on the following information.

In a class, 60% of the students passed in Finance. Among those passed in Finance, 33⅓ passed in Marketing. 150 students failed in both the subjects. 1/3rd of the students who passed in Marketing failed in Finance.

5. How many students are there in the class?
(A) 350 (B) 660
(C) 550 (D) 500

6. What percentage of the students failed in Finance only?
(A) 40% (B) 30%
(C) 50% (D) 10%

7. All the students who failed in at least one subject are given grace marks and it was found that the number of students who failed in both the subjects is decreased by 60% and the number of students who failed in exactly one subject went up by 20%. How many students passed in both the subjects?

(A) 120 (B) 140
(C) 80 (D) 60

8. After all the students who failed in exactly one subject have taken a re-exam in the subject in which they failed it was found that the number of students who passed in both the subjects increased by 60. What is the least value for the percentage of students in the class who failed only in Marketing?

(A) 20% (B) 28%
(C) 38% (D) 40%

Directions for questions 9 to 13: These questions are based on the following information.

Among all the students of a school half of the students learn exactly one martial art, such as Kung Fu, Karate and Judo. Half of the students who learn Karate do not learn any other martial art. The number of students who learn all the three martial arts is equal to 50/3% of those who learn exactly two among the three martial arts and is equal to 1/3rd of those who learn none of the three martial arts.

9. If 50 students learn all the three arts, then how many students learn exactly one of the three arts?

(A) 480 (B) 500
(C) 700 (D) 550

10. If 30 students learn none of the three arts, then what is the maximum possible number of students who learn Karate?

(A) 95 (B) 85
(C) 120 (D) 140

11. If there are 500 students in the school, then what is the maximum possible number of students who learn Kung Fu?

(A) 400 (B) 350
(C) 300 (D) 275

12. If 90 students learn exactly two of the three martial arts, then how many students are there in the school?

(A) 360 (B) 220
(C) 200 (D) 300

13. If out of 600 students of the school, 150 students learn Karate, then how many students learn only Kung Fu and Judo?

(A) 135 (B) 165
(C) 210 (D) 180

Directions for questions 14 to 16: These questions are based on the following table.

	Number of magazine readers						Total including non-readers	
	Business World		Business Times		Both			
Age group	Male	Female	Male	Female	Male	Female	Male	Female
< 15 years	145	65	155	65	50	30	260	115
15–34 years	175	125	105	85	40	50	265	190
≥ 35 years	115	135	120	100	35	45	215	195

14. How many males in 15–34 years age group do not read any of the 2 magazines?

(A) 15 (B) 45
(C) 75 (D) 25

15. Approximately, what percentage of the BT readers are above 15 years of age?

(A) 75% (B) 45%
(C) 65% (D) 80%

16. What percentage of females, who read neither BT nor BW are below 15 years of age?

(A) 30% (B) 60%
(C) 40% (D) 20%

Directions for questions 17 to 20: These questions are based on the following information.

A group of 1000 students at a summer camp are engaged in at least two of the activities, such as painting, swimming, dancing, singing or karate. It is further known that the number of students engaged in every combination of exactly two activities is three times the number of students who are engaged in every combination of exactly three activities. Also, the number of students engaged in all the five activities is a third of that engaged in exactly four activities. The number of students engaged in every combination of exactly four activities is the same.

17. If the number of students engaged in all the five activities is 100, then the number of students engaged in only painting and karate is

(A) 150 (B) 45
(C) 450 (D) Cannot be determined

18. If the number of students engaged in exactly three activities is 1.5 times the number of students engaged in all the five activities, then which of the following is true?
(A) The number of students engaged in only swimming, dancing and painting is 15.
(B) The number of students engaged in all the five activities is 100.
(C) Both (A) and (B)
(D) Neither (A) nor (B)

19. If the number of students enrolled in painting, swimming, dancing, singing and karate are 750, 800, 400, 900 and 600 respectively, then which of the following is definitely false?
(A) The number of students engaged in only painting, dancing, singing and karate is 90.
(B) The number of students engaged in exactly two activities is 300.
(C) The number of students engaged in exactly three activities is lesser than that engaged in only dancing, singing, swimming and karate.
(D) The number of students engaged in all the five activities is 150.

20. In the previous question, if the number of students engaged in painting is not known then which of the following can be that value?
(A) 270 (B) 90
(C) 360 (D) 120

Directions for questions 21 to 24: These questions are based on the following data.

A survey of 300 respondents showed that 135 of them read Business India, 125 read Business Today and 115 read Business World. Further, 42 of the respondents read Business India and Business Today, 48 read Business Today and Business World, 43 read Business India and Business World and 30 of the respondents read all the three magazines.

21. How many respondents read Business India or Business World?
(A) 199 (B) 272
(C) 207 (D) 175

22. If seven of the respondents who were previously reading only Business India now start reading a second magazine also and five of the respondents who were previously reading only Business India now stop even that, then how many respondents read Business India now?
(A) 75 (B) 132
(C) 142 (D) 130

23. If 15 respondents who were reading Business India stop reading Business India and instead start reading Business Today, then what is the maximum number of respondents who will now be reading only Business India?
(A) 120 (B) 65
(C) 78 (D) 93

24. If 16 of the respondents, who were reading Business Today, stop reading Business Today and instead start reading Business World, then what is the maximum number of respondents who will now be reading Business India and Business World?
(A) 59 (B) 55
(C) 75 (D) 63

Directions for questions 25 to 27: These questions are based on the following data.

In a colony, a survey was conducted regarding the ownership of three different types of vehicles, such as car, scooter and bicycle.

- The number of residents owning all three vehicles is the same as those owning none.
- The number of residents owning any two out of the three vehicles is the same as those owning any other two which in turn is the same as those owning none of the three.
- The number of residents owning scooters alone is the same as those owning cars alone and each in turn is twice those owning bicycles alone.
- Half the number of residents who own a bicycle also own at least one of the other two vehicles.

25. If the number of residents who own only bicycles is 150, then what is the total number of residents in the colony?
(A) 500 (B) 1000
(C) 750 (D) 1250

26. If 15 residents do not own any of the three vehicles, then how many residents are there in the colony?
(A) 100 (B) 200
(C) 500 (D) 300

27. What percentage of the residents own a scooter or a car but not a bicycle?
(A) 65% (B) 55%
(C) 75% (D) 45%

Directions for questions 28 to 30: These questions are based on the following information.

Among the 450 employees of a company 195 are members of Club A. 175 are members of Club B and 185 are members of Club C. 55 are members of clubs A and B. 40 are members of clubs B and C. 45 are members of clubs A and C. 25 employees are members of all the three clubs.

28. If 10 employees who are members of Club A take the membership of Club B also and 15 employees who are members of Club A withdraw from it and take the membership of Club C, then how many employees have membership of Club A?
(A) 120 (B) 140
(C) 180 (D) 145

Difficulty Level-2: Moderate

29. If 50 employees who are members of Club B withdraw from it and take the membership of Club C, then what is the maximum possible number of employees that are members of clubs A and C?

(A) 60 (B) 62
(C) 58 (D) 75

30. If 20 employees who have the membership of Club A withdraw from it and take the membership of Club C, then what is the least possible number of employees who are members of clubs B and A?

(A) 20 (B) 10
(C) 15 (D) 35

Directions for questions 31 to 33: These questions are based on the following data.

In a college library, four different business newspapers, such as Economic Times (ET), Business Standard (BS), Business Line (BL) and Financial Express (FE) are available. All students visit the library regularly but 20% of them do not read any business newspaper.

The four newspapers given in the above order are read by 230, 180, 180 and 220 students, respectively. The number of students reading exactly 2 newspapers for any two newspapers is 20. There are 30 students who read all the four newspapers but there is nobody who reads exactly three out of the four newspapers.

31. How many students do not read any newspaper at all?

(A) 75 (B) 100
(C) 225 (D) 150

32. What percentage of the people reading Business Standard also read at least one other newspaper?

(A) 35% (B) 55%
(C) 50% (D) 65%

33. If all the students in the college including those who do not read any newspaper read at least one newspaper, (out of the four newspapers above) which he is not reading at present, then what is the least number of students reading all the four newspapers?

(A) 60 (B) 25
(C) 15 (D) 30

Directions for questions 34 to 36: These questions are based on the following information.

Each of N students participated in at least one of the track events, such as in high jump, long jump and 100 m dash. 27 students participated in high jump, 27 students participated in long jump and 52 students participated in 100 m dash.

34. The number of students who participated in exactly one of the three events is 51, find the maximum possible number of students who participated in exactly two of the three events.

(A) 22 (B) 30
(C) 34 (D) 26

35. The number of students who participated in exactly one of the three events is atleast equal to that who participated in exactly two of three events which is atleast equal to that of those who participated in all the three events. Find the maximum possible number of students who participated in all the three events.

(A) 16 (B) 18
(C) 17 (D) 19

36. If the number of students who participated in at least two of the three events is 26, then find the minimum possible value of N.

(A) 50 (B) 54
(C) 52 (D) 46

Directions for questions 37 to 40: These questions are based on the following information.

A group of 100 students participated in at least one of the following events. The events are 100 m dash, 200 m dash, 400 m dash and 800 m dash. 40 students participated in each of the 100 m dash, 200 m dash, 400 m dash and 800 m dash. An equal number of students participated in only the 100 m dash, only the 200 m dash, only the 400 m dash and only the 800 m dash. An equal number of students participated in each pair of 100 m dash, 200 m dash, 400 m dash and 800 m dash. An equal number of students also participated in each combination of three of the events. 15 students participated in all the four 100 m dash, 200 m dash, 400 m dash and 800 m dash.

37. Find the maximum possible number of students who participated in only the 400 m dash.

(A) 20 (B) 25
(C) 15 (D) 30

38. Find the maximum possible number of students who participated in all the three 100 m dash, 200 m dash and 400 m dash.

(A) 27 (B) 19
(C) 23 (D) 29

39. If 16 students participated in only the 100 m dash, find the number of students who participated in the 100 m dash and at least one other.

(A) 24 (B) 20
(C) 16 (D) 28

40. Using the information in the previous question, find the maximum possible number of students who participated in only the 100 m dash and 200 m dash.

(A) 1 (B) 2
(C) 3 (D) 4

EXERCISE-3

Directions for questions 1 to 3: These questions are based on the following data.

A group of people went for a pilgrimage tour, out of 245 pilgrims, 105 visited Badrinath, 95 visited Kedarnath and 95 visited Somnath. Fifteen of them visited all three shrines, while 190 visited exactly one of the three shrines. The number of pilgrims who visited exactly two out of the three shrines is three times as many as those who have not visited any one of the three shrines.

1. If the number of pilgrims who have visited at least one of the two shrines Kedarnath and Somnath is 165, then how many pilgrims visited only Kedarnath and Somnath?

(A) 20 (B) 30
(C) 10 (D) 15

2. If 180 pilgrims visited at least one of the two shrines Kedarnath or Badrinath, then how many pilgrims visited only Somnath?

(A) 55 (B) 40
(C) 35 (D) 60

3. If there is nobody who visited only Badrinath and Somnath, then how many people visited only Kedarnath?

(A) 90 (B) 80
(C) 70 (D) 50

Directions for questions 4 to 7: These questions are based on the following information.

Each of the students, who are residents of Kalpana Chawla Bhawan, likes at least one among the four different brands of cool drinks, such as Coca-Cola, Thumbs Up, Limca and Sprite. 65 students like Thumbs Up and Coca-Cola. 77 students like Sprite and Thumbs Up. 73 students like Coca-Cola and Limca. 76 students like Limca and Thumbs Up. 74 students like Sprite and Coca-Cola. There are 67 students who like exactly one brand. The number of students who like only Limca, Thumbs Up and Coca-Cola is same as the number of students who like only Sprite, Thumbs Up and Coca-Cola. The number of students who like Sprite, Limca and Thumbs Up but not Coca-Cola is same as the number of students who like Sprite, Limca and Coca-Cola but not Thumbs Up. The number of students, who like only Coca-Cola and Sprite is 14. The number of students who like only Sprite and Limca and only Thumbs Up and Coca-Cola are 10 and 15, respectively. The sum of the number of students who like Thumbs Up, the number of students who like Coca-Cola, the number of students who like Sprite and the number of students who like Limca is 557.

4. How many students like both Sprite and Limca but not all the four?

(A) 60 (B) 50
(C) 55 (D) 44

5. What is the total number of students?

(A) 247 (B) 250
(C) 235 (D) 252

6. How many students like only Coca-Cola and Limca?

(A) 16 (B) 17
(C) 13 (D) 15

7. If 25 students like only Thumbs Up or only Sprite, then how many like Coca-Cola or Limca?

(A) 42 (B) 208
(C) 152 (D) 210

Directions for questions 8 to 12: These questions are based on the following information.

A survey was conducted among a group of football fans to know how many of them like the football teams Barcelona, Liverpool, Real Madrid, Manchester United and Bayern Munich.

(i) 57 out of the 125 fans who like Real Madrid also like Manchester United.
(ii) 10 fans like exactly three clubs. 10 fans like only Liverpool.
(iv) It is known that no one who likes Barcelona likes Real Madrid or Manchester United.
(v) The number of fans who like Bayern Munich and Barcelona is the same as those who like only Real Madrid and Manchester United and 4 more than those who like Barcelona and Liverpool.
(vi) The number of fans who like Bayern Munich and Manchester United is the same as that who like only Liverpool and Real Madrid, which in turn is one third of those who like only Real Madrid.
(vii) The number of fans who like Barcelona, Bayern Munich, Manchester United and Liverpool are 112, 75, 88 and 92, respectively.
(viii) No one who likes Bayern Munich likes Liverpool or Real Madrid.

8. How many like exactly two clubs?

(A) 183 (B) 152
(C) 137 (D) 154

9. How many like only Bayern Munich or only Barcelona?

(A) 45 (B) 51
(C) 33 (D) Cannot be determined

10. How many like the club Manchester United but not Liverpool?

(A) 66 (B) 72
(C) 81 (D) 71

11. How many like exactly one club?

(A) 102 (B) 94
(C) 123 (D) 96

12. Among the Barcelona fans, how many like at least two more clubs?
(A) 90 (B) 57
(C) 64 (D) None

Directions for questions 13 to 16: These questions are based on the following data.

In a class, 30% of the students gave their names to participate in the NSS and 75% to participate in the NCC. Three students participate in neither of these, where two and six students wanted to participate in both.

13. How many students are there in the class?
(A) 100 (B) 75
(C) 60 (D) 80

14. What percentage of students wants to participate only in the NSS?
(A) 30% (B) 25%
(C) 15% (D) 20%

15. What percentage of students wants to participate in only one programme either in NSS or NCC?
(A) 85% (B) 90%
(C) 75% (D) 20%

16. How many students want to participate in at least one programme?
(A) 97 (B) 87
(C) 147 (D) 57

Directions for questions 17 to 20: These questions are based on the following data.

In a school, 60% of the students passed in English and 25% of the students who passed in English passed in the foreign language also, whereas $66\frac{2}{3}\%$ of the students who passed in the foreign language failed in English. Twenty students failed in both English and the foreign language.

17. What is the total strength of the school?
(A) 250 (B) 150
(C) 200 (D) 100

18. What per cent of the students passed in exactly one of the two subjects, such as in English and the foreign language?
(A) 15% (B) 65%
(C) 45% (D) 75%

19. The students who failed in exactly one subject are allowed to take a re-exam and it was found that the number of students who passed in both the subjects increased by 20%. What is the least value for the percentage of students in the school who pass only in English?
(A) 42% (B) 46%
(C) 34% (D) 28%

20. All the students who failed in one or more subjects are given grace marks and it was found that the number of students passing in exactly one subject went up by 4 and the number of students who failed in both the subjects dropped by 40%. What per cent of the school now pass in both subjects?
(A) 40% (B) 15%
(C) 12% (D) 17%

Directions for questions 21 to 25: These questions are based on the following data.

There are three trade unions Viram, Vishram and Be-kam and 3600 workers in a company. Becoming a member of a trade union is optional. A worker can be a member of more than one of the three trade unions also.

There are 500 workers who are members of at least two trade unions while Vishram has 1400 members. There are 100 workers who are members of only Viram and Be-kam, whereas 200 Vishram members also are Be-kam members; 550 workers are members of only Be-kam where as 20% of Viram members are members of exactly one more union. An eighth of all the workers in the company are members of exactly two unions.

21. How many workers are members of all the three unions?
(A) 150 (B) 75
(C) 50 (D) 100

22. How many workers are not members of any union?
(A) 100 (B) 200
(C) 300 (D) 400

23. How many workers are members of only Viram or only Be-kam?
(A) 3200 (B) 2700
(C) 1400 (D) 1700

24. If 10 workers give up their Be-kam membership and take up Vishram membership, then how many workers will now have membership of all the three unions?
(A) 40 (B) 50
(C) 60 (D) 45

25. How many workers are members of Vishram but not members of Be-kam?
(A) 400 (B) 800
(C) 1200 (D) 1600

ANSWER KEYS

Exercise-1

1. (C)	8. (B)	15. (C)	22. (D)	29. (A)	36. (A)
2. (D)	9. (B)	16. (C)	23. (C)	30. (D)	37. (D)
3. (D)	10. (A)	17. (B)	24. (B)	31. (B)	38. (C)
4. (B)	11. (C)	18. (D)	25. (C)	32. (D)	39. (A)
5. (D)	12. (B)	19. (B)	26. (D)	33. (A)	40. (B)
6. (D)	13. (D)	20. (A)	27. (D)	34. (B)	
7. (C)	14. (D)	21. (B)	28. (A)	35. (A)	

Exercise-2

1. (B)	8. (B)	15. (C)	22. (D)	29. (C)	36. (B)
2. (D)	9. (B)	16. (A)	23. (C)	30. (D)	37. (B)
3. (C)	10. (D)	17. (B)	24. (B)	31. (D)	38. (C)
4. (A)	11. (A)	18. (C)	25. (B)	32. (C)	39. (A)
5. (D)	12. (D)	19. (C)	26. (D)	33. (D)	40. (C)
6. (D)	13. (A)	20. (A)	27. (A)	34. (D)	
7. (B)	14. (D)	21. (C)	28. (C)	35. (C)	

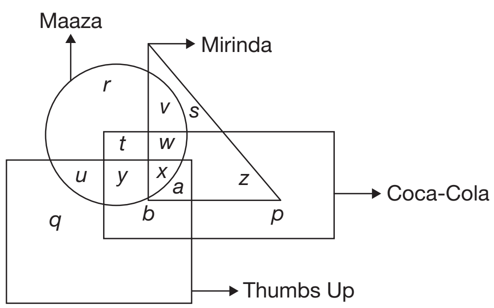
Exercise-3

1. (C)	5. (D)	9. (C)	13. (C)	17. (C)	21. (C)	24. (B)
2. (A)	6. (C)	10. (A)	14. (D)	18. (D)	22. (D)	25. (C)
3. (D)	7. (D)	11. (D)	15. (A)	19. (A)	23. (D)	
4. (A)	8. (A)	12. (D)	16. (D)	20. (D)		

SOLUTIONS

EXERCISE-1

Solutions for questions 1 to 5:



1. In the diagram, b represents the people who like only Coca – Cola and Thumbs Up.

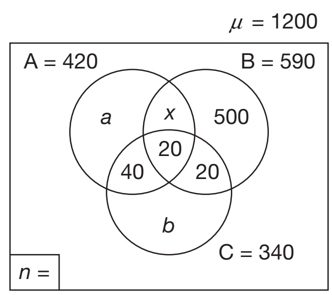
2. 'v', 's', 'w' and z are in the triangle but not in square. Hence, those represent the people who like Mirinda but not Thumbs Up.

3. u, y, x belongs to both circle and square. Hence, they like Maaza and Thumbs Up.

4. The letter 'v' belongs to circle and triangle but neither to square nor rectangle. Hence, v represents the people who like Maaza and Mirinda but not the other 2.

5. w belongs to circle and rectangle but not square. Hence, w represents the people who like Maaza, Mirinda, Coca-Cola but not Thumbs Up.

Solutions for questions 6 to 10:



$C = b + 40 + 20 + 20 = 340$

$\Rightarrow b = 260$

B = 500 + 20 + 20 + x = 590
$\Rightarrow x = 50$
A = $a + x$ (= 50) + 40 + 20 = 420
$\Rightarrow a = 310$
$n = 0$ [given]

6. Number of employee that are member of Club C only = $b = 260$.

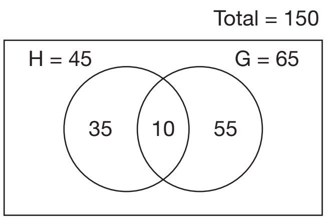
7. Number of members of Club A and B = $x + 20 = 50 + 20 = 70$.

8. $a + 40 + b + n = 310 + 40 + 260 + 0 = 610$ are not the members of Club B.

9. Members of Club A or Club C = $a + x + 40 + 20 + b + 20 = 310 + 50 + 40 + 20 + 260 + 20 = 700$.

10. $x + 40 + 20 = 50 + 40 + 20 = 110$ are the members of exactly two clubs.

Solutions for questions 11 to 15: As per the given data, we get the following diagram:



H → History G → Geography

11. 55 students take only Geography.

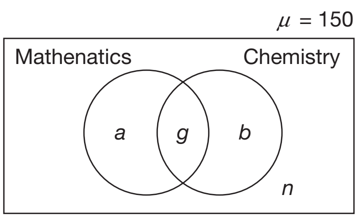
12. 35 students take only History.

13. 10 students take both History and Geography out of a total of 150 students. Hence, 150 – 10 = 140 students do not take either History or Geography.

14. 35 + 10 + 55 = 100 students take at least one subject.

15. (150 – 100) = 50 students take neither of the two subjects.

Solutions for questions 16 to 18: Let us represent the given information in the following Venn diagram.



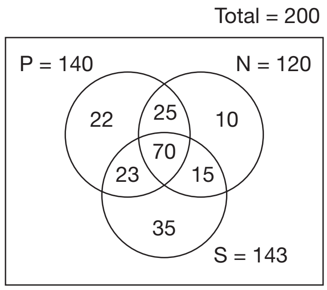
Given, $a + g = 50$.
The number of students who failed only in Chemistry = $a = 40$
$\Rightarrow g = 10$
$n = 20$
$\therefore\ b = 150 - (40 + 10 + 20) = 150 - 70 = 80$
$b = 80$

16. The number of students who passed in both the subjects, $g = 10$.

17. The number of students who passed in exactly one subject, $a + b = 40 + 80 = 120$

18. The number of students who failed in at least one subject $= a + b + n = 40 + 80 + 20 = 140$ or $\mu - g = 150 - 10 = 140$

Solutions for questions 19 to 23: From the given data, we get the following the diagram.



P → Panasonic
S → Siemens
N → Nokia

1. 22 families use only Panasonic phones.
2. 10 families use only Nokia phones.
3. 35 families use only Siemens phones.
4. 25 families use both Panasonic and Nokia but not Siemens.
5. 15 families use both Nokia and Siemens but not Panasonic.
6. 23 families use both Panasonic and Siemens but not Nokia.
7. All the 200 families use mobile-phones of at least one company.

19. 35 families use mobile phones of only Siemens.

20. 25 families use mobile phones of both Panasonic and Nokia but not Siemens (Region common to Panasonic and Nokia but not Siemens).

21. Exactly one company
= Only Panasonic + Only Nokia + Only Siemens
= 22 + 10 + 35 = 67 families

22. Neither Panasonic nor Siemens implies only Nokia.
So, 10 families use mobile phones of neither Panasonic nor Siemens.

23. All the families use mobile phones of at least one out of the three mentioned companies. So, there is no family which did not use any mobile phone.

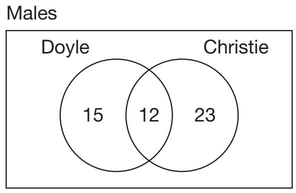
Solutions for questions 24 to 28: Given that 100 students were surveyed who read novels of Christie or Doyle or both.

From the given table we get the information that 40 females read Doyle and 70 students read Christie.

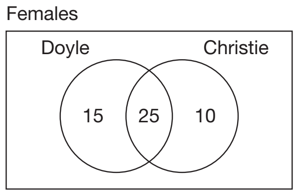
Further, we derive the following:

1. 37% of the total students read both, i.e., 37 students.
2. The ratio of males and females is 1 : 1, i.e., number of males = number of females = 50.
3. 50% of the females, i.e., 50% of 50 = 25 females read both.

With the above information, we get the following data:



So, Doyle = 27
Christie = 35
Both = 12



So, Doyle = 40
Christie = 35
Both = 25

Now after filling the gaps in the table, we get the following data:

	Doyle	Christie	Both	Total
Male	27	35	12	50
Female	40	35	25	50
Total	67	70	37	100

24. 12 males read the books by both the authors.

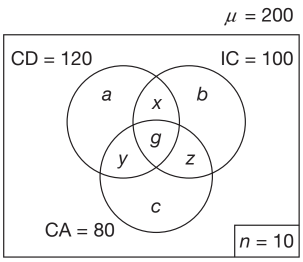
25. The number of students who read books by only Christie is 23 + 10 = 33.

26. 15 females read books by only Doyle.

27. 37 students read the books by both the authors. Hence, those who do not read both the books is 100 – 37 = 63.

28. 27 males read books by Doyle.

Solutions for questions 29 to 33: The given information can be represented in the following Venn diagram.



CD – Cool drink
IC – Ice Cream
CA – Cake
It is given that, $a + b + c = 100$
$a + b + c + x + y + z + g + n = \mu = 200$
$100 + (x + y + z) + g + 10 = 200$
$x + y + z + g = 90$ (A)
$CD + IC + CA = (a + b + c) + 2(x + y + z) + 3g = 300$
$\Rightarrow 2(x + y + z) + 3g = 200$ (B)
From (A) and (B), we get $g = 20$
$\therefore\ x + y + z = 70$

29. The number of people who had exactly two of the three = 70

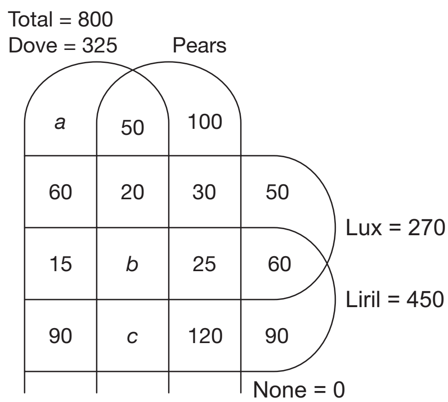
30. It is given that $x = 20$.
$\therefore\ y + z = 50$
The number of people who had only cake = 80 – 50 – 20 = 10

31. 15 people who had only cool drink had cake also, value of x reduces by 15 and value of y increases by 15.
5 people who had only ice cream and cake had cool drink also, i.e., value of z decreases by 5 and that of g increases by 5.
The number of people who had at least two of the three items $= x + y + z + g = x + (y + 15) + (z - 5) + (g + 5)$
$= x + y + z + g + 15 = 70 + 20 + 15 = 105.$

32. It is given that $b = 30$
$\Rightarrow x + z = IC - b - g = 100 - 30 - 20 = 50$
$y = 70 - (x + z) = 20$ as,
$a + b + c = 100,\ a + c = 100 - b = 70$
$\therefore\ a + c + y = 70 + 20 = 90$

33. What is the maximum possible value of a.
The value of a can be maximum when $x + y$ is minimum.
$x + y$ can be minimum when z is maximum.
As $g = 20$ and $y + g + z + c = 80$
$z_{max} = 60$ ($c_{min} = 0$)
As $x + y + z = 70$, $x_{min} + y_{min} = 10$
$a_{max} = CD - y_{min} - x_{min} - g = 120 - (10) - 20 = 90.$

Solutions for questions 34 to 37: The given data can be represented in the following diagram.



$b = 270 - (60 + 20 + 30 + 50 + 15 + 25 + 60) = 10$
$c = 450 - (15 + b + 25 + 60 + 90 + 120 + 90) = 40$
$a = 325 - (50 + 20 + b + c + 90 + 15 + 60) = 40$

34. 50 people liked only Dove and Pears.
40 people liked only Dove, Pears and Liril.
50 + 40 = 90 people liked Dove and Pears but not Lux.

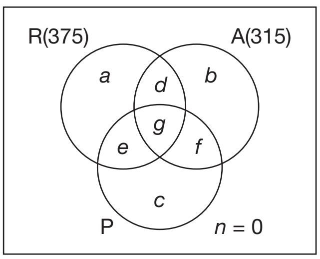
35. A total of 280, i.e., (40 + 100 + 50 + 90) people liked exactly one product. The remaining 520 people liked at least two products.

36. 50 people liked only Lux.
90 people liked only Liril.
60 people liked only Lux and Liril.
(50 + 90 + 60) = 200 people liked neither Dove nor Pears.

37. The number of people who liked Lux = 270
The number of people who liked Liril but not Lux
= 90 + 40 + 120 + 90 = 340
The number of people who liked either Lux or Liril
= 270 + 340 = 610

Solutions for questions 38 to 40:

Total = 780



$a + b + c + d + e + f + g = 780$
$d + g = 75,\ e + g = 90,\ f + g = 45$

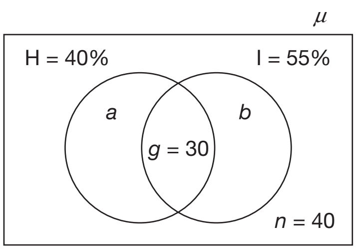
38. The number of bags which contain exactly two of raisins, almonds and peanuts = $d + e + f$.
Max $(d + e + f)$ = Max $(75 - g + 90 - g + 45 - g)$ = Max $(210 - g)$
$210 - g$ is maximum, when each of $75 - g$, $90 - g$, $45 - g$ is maximum, i.e., when g is 0.
The maximum possible number of bags which contain exactly two of raisins, almonds and peanuts is 210.

39. The number of bags which contain either peanuts or almonds = $315 + c + e$
$c = 78$ (R + A – 75) = 165
For Peanuts or Almonds to be minimum e must be minimum. Since $e + g = 90$, to get minimum value for e, should be maximum. g cannot be more than 45.
$\therefore\ e = 45$
The minimum possible number of bags which contain either peanuts or almonds = 315 + 165 + 45 = 525.

40. $d + e + f = 4g$
$75 - g + 90 - g + 45 - g = 4g$
$g = 30$
The number of bags which contain only raisins
$a = 375 - (d + e + g) = 375 - (75 - g + 90 - g + g) = 240$

EXERCISE-2

Solutions for questions 1 to 4: The given information can be represented in a Venn diagram as follows.



H – The Hindu
I – Indian Express
$a + g = 40\%$
$b + g = 55\%$
$a + b + 2g = 95\% \rightarrow$ (A)
$a + b + n + g = 100\% \rightarrow$ (B)
(B) – (A)
$\Rightarrow n - g = 5\%$
$\therefore$ 40 – 30 = 5% of μ
$\Rightarrow \mu = 200$
$a = 50,\ b = 80$

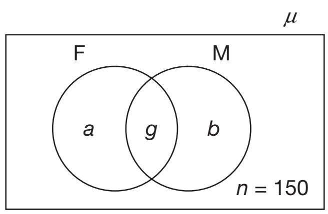
1. There are 200 families in the colony.

2. The number of families that read at least one of the news papers = $a + b + g = 50 + 80 + 30 = 160$.

3. The number of families that read at most one news paper = $a + b + n = 170 = 85\%$ of μ.

4. The number of families that read only Hindu, $a = 50 = 25\%$ of μ.

Solutions for questions 5 to 8:



F = Finance

M = Marketing

Let us represent the given information in the following Venn diagram.

Given, $n = 150$ and $F = \frac{6\mu}{10}$

$$g = \frac{1}{3} \times \frac{6\mu}{10} = \frac{2\mu}{10}$$

passed in marketing = M
out of them 1/3rd failed in Finance.
i.e., passed only in Marketing = b.

$\therefore\ b = ⅓\ \text{M} \Rightarrow g = ⅔\ \text{M}$

$\therefore\ g = 2b$

$b = \frac{\mu}{10}$

Among the students who passed in Finance, 33⅓% passed in Marketing also.

$\therefore\ g = \frac{a+g}{3} \Rightarrow a = 2g \Rightarrow a = \frac{4\mu}{10}$

$a + b + g = \frac{4\mu}{10} + \frac{\mu}{10} + \frac{2\mu}{10} = \frac{7\mu}{10}$

$a + b + g + n = \mu$

$\frac{7\mu}{10} + 150 = \mu$

$\frac{3\mu}{10} = 150 \ \Rightarrow \mu = 500$

$\therefore\ b = 50,\ a = 200,\ g = 100,\ n = 150$

5. There are 500 students in the class.

6. There are 200 students, who failed in Finance only.
$\therefore\ b = 50 = 10\%$ of μ

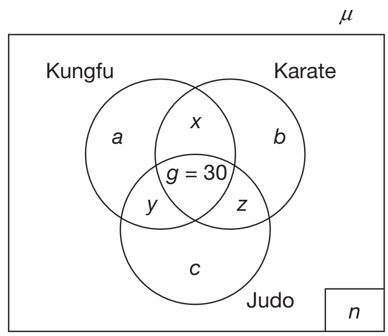
7. The students who failed in at least one subject are a, b and n.
n is decreased by 60%, i.e., these 60% of n passed in at least one of the subjects.

$\therefore\ n = \frac{60 \times 150}{100} = 90$

a + b increased by 20%, i.e., these 20% = (50) is transferred from n.
$\therefore$ The remaining 40 (transferred from n) is added to g.
$g = 100 + 40 = 140$.

8. The number of students who failed in exactly one subject = $a + b$.
g is increased by 60. This 60 is transferred from a and b.
Failed only in Marketing, $a = 200$.
By transferring 60 from a alone, the value of a will be the least.
$\therefore\ a = 140$, i.e., 28% of μ

Solutions for questions 9 to 13: Let us represent the given information in the following Venn diagram.



It is given that half of the students learn exactly one martial art.

$\therefore\ a+b+c = \frac{\mu}{2} \rightarrow$ (A)

Half of the students learning Karate are not learning any other martial art.
$\therefore\ b = x + g + z \rightarrow$ (B)
The number of students learning all the three arts = $\frac{50}{3}$ % of those who learn exactly two.

$g = \frac{50}{3} \times \frac{x+y+z}{100} = \frac{x+y+z}{6} \Rightarrow x+y+z = 6g \ \rightarrow$ (C)

and also, it is given that $g = \frac{1}{3} n \Rightarrow n = 3g$

As $a+b+c = \frac{\mu}{2}$, $x + y + z + g + n = \frac{\mu}{2}$

$6g + g + 3g = \frac{\mu}{2} \Rightarrow g = \frac{\mu}{20}$, $x + y + z = \frac{6\mu}{20}$, $n = \frac{3\mu}{20}$

9. It is given that $g = 50 \Rightarrow \mu = 20 \times 50 = 1000$ and $a + b + c = \frac{1000}{2} = 500$

10. It is given that $n = 30 \Rightarrow \mu = 200$ then $(b + x + g + z)_{max} = ?$

$x + y + z = \frac{6 \times 200}{20} = 60$

$(x + z)_{max} = 60$, $g = \frac{\mu}{20} = 10$ ($\because\ y_{min} = 0$)

From (B), $b_{max} = (x + z)_{max} + g = 70$
The maximum possible number of students who learn Karate = 70 + 70 = 140.

11. The value of $a + x + y + g$ can be maximum, when a is maximum and x, b, c and z are minimum, i.e., $c = 0$, $z = 0$ but, b cannot be zero.
($\because\ b = x + g + z$)

$g = \frac{\mu}{20} = 25$, $b_{min} = 25$ and $x = 0$

$a+b+c = \frac{\mu}{2} = 250$

$a_{max} + b_{min} + c_{min} = 250$

$a_{max} = 250 - 25 - 0 = 225$

$$x+y+z=\frac{6\mu}{20}=\frac{6\times 500}{20}=150$$

As $x = 0$ and $z = 0$, $y = 150$.

$\therefore$ The maximum possible number of students who learn Karate $= a + x + y + g = 225 + 0 + 150 + 25 = 400$.

12. It is given that $x + y + z = 90 \Rightarrow \frac{6\mu}{20}=90 \Rightarrow \mu = 300$.

13. It is given that $\mu = 600$

$b + x + z + g = 150$

$\Rightarrow x + z + g = 75$ and $b = 75 \Rightarrow x+z+\frac{\mu}{20}=75$

$\Rightarrow x + z = 75 - 30 = 45$ and we have $x+y+z=\frac{6\mu}{20}=180$

$\therefore y = 180 - 45 = 135$

14. From the 15 – 34 years age category in the table:
Number of males who do not read either BW or BT
= 265 – [175 + 105 – 40] = 25

15. Total BT readers = Total males + Females reading BT = 380 + 250 = 630.

Number of BT readers over 15 years = 225 + 185 = 410

$\therefore$ The percentage of BT readers over 15 years

$$=\frac{410}{630}\times 100=65\%$$

16. Females who do not read any of the 2 magazines

in < 15 years group = 115 – [65 + 65 – 130] = 15

15 – 34 years group = 190 – [125 + 85 – 50] = 30

> 35 years group = 195 – [135 + 100 – 45] = 5

$\therefore$ The percentage of females (below 15 years) who do not read any magazine

$$=\frac{15}{15+30+5}\times 100=30\%$$

Solutions for questions 17 to 20: This is a 5-set / variable Venn diagram puzzle, which can be solved without illustrating the sets – P, SW, D, S, K.

Let us assume that:

X denote – Number of students engaged in exactly 1 activity.

Y denote – Number of students engaged in exactly 2 activity.

Z denote – Number of students engaged in exactly 3 activity.

A denote – Number of students engaged in exactly 4 activity.

B denote – Number of students engaged in exactly 5 activity.

Note the following for a n-set Venn diagram:

Name of the region (pocket of intersection of the sets)	Number of regions (pockets of intersections of the sets)	For a 5-set situation
Exactly 1, X	${}^nC_1$	5
Exactly 2, Y	${}^nC_2$	10
Exactly 3, Z	${}^nC_3$	10
Exactly 4, A	${}^nC_4$	5
Exactly 5, B	${}^nC_5$	1
None, N	${}^nC_0$	1
Total number of regions	2^n	32

Note: ${}^nC_r=\frac{n!}{(n-r)!r!}$ and ${}^nC_r={}^nC_{n-r}$

From the given information, we have the following:

$$X + Y + Z + A + B + N = 1000$$

Also, X = N = 0 (a)

$$\text{So, } Y + Z + A + B = 1000 \quad (1)$$

Also, it is given that:

$$Y = 3Z \quad (2)$$

Since there are 10 pockets of Y and the above Equation (2) is based on the given information that each of the pockets in Y is three times that of each of the pockets in Z.

This is possible only if all the pockets in Y are equal and all the pockets in Z are equal.

$$B=\frac{A}{3} \quad (3)$$

Substituting (2) and (3) in (1), we get

$$Z + B = 250 \quad (4)$$

17. B = 100 $\Rightarrow$ From (4), Z = 150 and
From (2), Y = 450,
Hence, each pocket in Y is 45.

18. $Z=\frac{3}{2}B$ or $B=\frac{2}{3}Z$

$\Rightarrow$ From (4), Z = 150, B = 100 and
From (2) and (3), Y = 450, A = 300

Choice (A) — Any one pocket of Exactly 3 $=\frac{150}{10}=15$ is true

Choice (B) — is true.

Hence Both A and B are true

19. P = 750
SW = 800
D = 400
S = 900
K = 600
These numbers imply that
X + 2Y + 3Z + 4A + 5B = 3450

Difficulty Level-2: Moderate

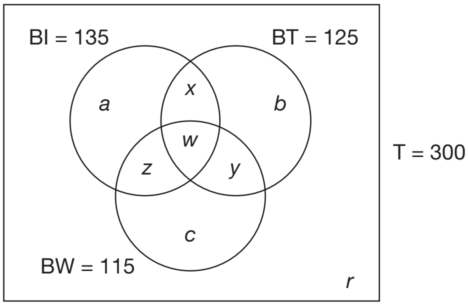
Or 9Z + 17B = 3450 $\rightarrow$ (5), From (a), (2) and (3)
Solving (4) and (5), we get
B = 150, Z = 100 and A = 450, Y = 300
Choice (A) — Any one pocket of Exactly 4 = $\frac{450}{5}$ = 90
Choice (B) — Y = 300 is true.
Choice (C) — Z = 100 > 90 is false.
Choice (D) — B = 150 is true.

20. From (5) in the previous question,
9Z + 17B = 2700 + x;
$x \rightarrow$ Number of students
From (4), Z + B = 250 engaged in painting
Solving the above two equations,
8B = 450 + x

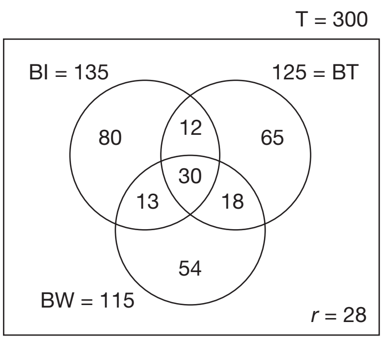
Or B = $\frac{1}{8}(450 + x)$

Hence, B is a multiple of 8.
So, from the choices only $x = 270$ satisfies.

Solutions for questions 21 to 24: Referring to the Venn diagrams given below, we derive the following:



Also, $w = 30$ (1)
$x + w = 42$ (2)
$y + w = 48$ (3)
$z + w = 43$ (4)
So, we get $x = 12$; $y = 18$; $z = 13$
Redrawing the diagram, we get:



Number of respondents reading at least one magazine = 135 + (125 – 42) + (115 – 43 – 18) = 272.
Number of respondents reading none of the magazines = T = 300 – 272 = 28.

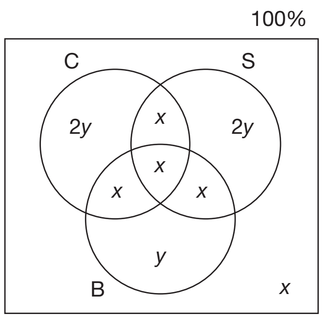
21. Business India or Business World is
80 + 12 + 13 + 30 + 54 + 18 = 207.

22. When 7 respondents reading Business India alone, start reading a second magazine also, it does not affect the number of respondents reading Business India because those 7 respondents read Business India plus one more magazine.
But, when 5 people who read Business India alone stop even reading that magazine, then the number of respondents reading Business India will come down by 5. Hence, the answer is 130.

23. The respondents, who shift from Business India to Business Today can be from those who were previously reading Business India alone or Business India and Business World alone. If we want the maximum number of respondents reading Business India alone (which was 80 initially) after the shift, the least number should shift from the category 'Business India alone', i.e., the maximum possible number should shift from the category 'Business India and Business World alone'. In this category, there are 13 respondents and maximum number of shifting means 13 respondents are shifting. Since the total number of respondents shifting is 15, at least 2 respondents reading Business India alone should shift. Hence, the maximum number of respondents reading Business India after the shift = 80 – 2 = 78.

24. By the similar logic which is explained in the above problem, the maximum number of respondents reading Business India and Business World will come if the maximum number of respondents reading Business India and Business Today alone shift and that is 12. So, the maximum number of respondents reading Business India and Business World = 43 + 12 = 55 (Because already, 30 + 13 = 43 respondents read Business India and Business World).

Solutions for questions 25 to 27:



The first three statements can be represented as shown in the diagram, and hence, $5x + 5y = 100$
(because we have taken x and y as percentages)
or $x + y = 20$ (1)
From the fourth condition, we get $(y + 3x) = B$
i.e., $y = 3x$ (2)

From equations (1) and (2), we get
$x = 5\%$ and $y = 15\%$

25. $y = 15\% = 150.$
Hence, total number % of residents

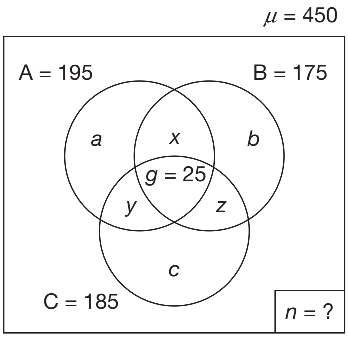
$$= x = \frac{150 \times 100}{15} = 1000$$

26. $x = 5\% = 15$

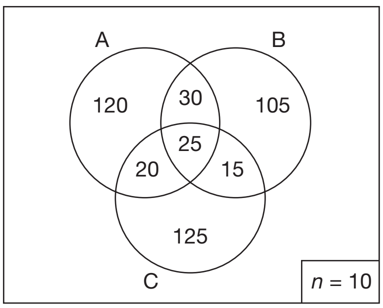
Hence, total $= \dfrac{15}{0.05} = 300$

27. $2y + 2y + x = 65\%$

Solutions for questions 28 to 30: The given data can be taken in the form of Venn diagram as follows.



Given, $x + g = 55 \Rightarrow x = 30,$
$z + g = 40 \Rightarrow z = 15$
$y + g = 45 \Rightarrow y = 20$
$\therefore$ A $= a + x + y + g = 195$
$\Rightarrow a + 30 + 20 + 25 = 195 \Rightarrow a = 120$
Similarly, $b = 105$, $c = 125$ and $n = 10$
$\therefore$ The complete Venn diagram is as follows.



28. 10 employees of Club A did not withdraw from it, but 15 members of A withdrew from their membership.
$\therefore$ Members of Club A = 195 – 15 = 180.

29. These 50 can be taken from b or x. The value taken from b gets added to c and the value taken from x gets added to y.
$\therefore$ To maximize $y + g$, take 30 from x.
$y + g = 20 + 25 + 30 = 75.$

30. These 20 can be taken from a or x. The value taken from a is shifted to c and the value taken from x is shifted to z. To minimize the number of members of clubs A and B, take as much value as possible from x. The entire 20 can be taken from x.
$\therefore$The minimum possible number of employees who are members of clubs A and B = (x + g) – 20 = 35

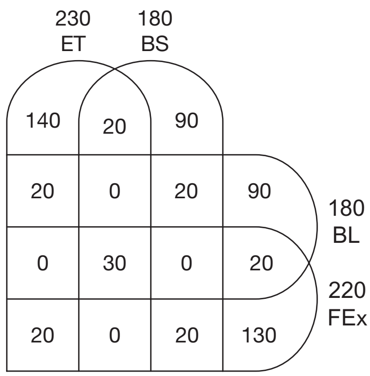
Solutions for questions 31 to 33: Out of the four newspapers, reading exactly two newspapers is possible in six different combinations.

($^4C_2 = 6$. They are ET – BS; ET – BL; ET – FE; BS – BL; BS – FE and BL – FE). Since each of these is 20 students, number of students reading exactly 2 newspapers = 6 × 20 = 120.

Also, there is nobody who reads exactly three out of the four newspapers. There are 30 students who read all the four newspapers.

To get the number of students reading one particular newspaper alone, we have to subtract 3 times 20 (because students reading two newspapers is 20 in number and for each newspaper, there will be three ways of pairing with one more newspaper) and 30 (which is the number of students reading all four newspapers).

Number of students reading only ET
= 230 – 3 × 20 – 30 = 140.
Number of students reading only BS
= 180 – 3 × 20 – 30 = 90.
Number of students reading only BL = 180 – 3 × 20 – 30
= 90.
Number of students reading only FE
= 220 – 3 × 20 – 30 = 130.
Total number of students reading exactly one newspaper
= 140 + 90 + 90 + 130 = 450.
Total number of students reading at least one newspaper
= 450 + 120 + 30 = 600.
This represents 80% of the total number of the students.
So, total number of students = 600/0.8 = 750
$\therefore$ Number of students who do not read any newspaper
= 0.2 × 750 = 150
This set can be solved with the help of the following diagram.



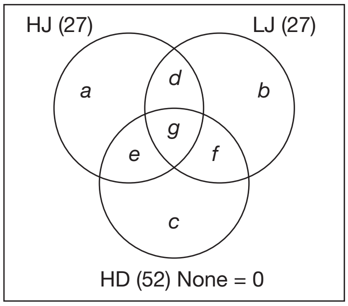
31. 150 students do not read any news paper.

32. Number of Business Standard students reading at least one more newspaper
= Exactly two newspapers + All four newspapers
$= 3 \times 20 + 30 = 90$
As a percentage of all the students reading Business Standard, this is $\frac{90}{180} \times 100 = 50\%$

33. Least increase in the number of students who read all newspapers will come only if each student reads exactly one additional newspaper.
But, since the number of students who read exactly three newspapers is zero, there will not be any addition to the figure of 30 students who read all four newspapers.
Hence, the answer is 30.

Solutions for questions 34 to 36:

Total = 780



34. $a + b + c = 51$

HJ + LJ + HD – $2(d + e + f) - 3g = 51$

$2(d + e + f) + 3g = 27 + 27 + 52 - 51 = 55$

$d + e + f$ is maximum when g is minimum.

When $g = 0$, $d + e + f$ is not an integer.

When g is 1, $d + e + f = 26$

∴ The maximum possible number of students who participated in exactly two of the three events is 26.

35. $a + b + c \geq d + e + f \geq g$

$27 + 27 + 52 - (2\ (d + e + f) + 3g) \geq d + e + f$ and $d + e + f \geq g$

$3(d + e + f + g) \leq 106$ and $d + e + f \geq g$

$d + e + f + g \leq 35\frac{1}{3}$ and $d + e + f \geq g$

$g + g \leq d + e + f + g \leq 35\frac{1}{3}$

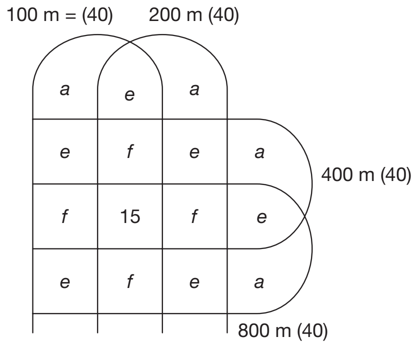
$g \leq \frac{106}{6}$

As g is an integer, $\max(g) = 17$

∴ The maximum possible number of students who participated in all the three events is 17.

36. $d + e + f + g = 26$
$n = 27 + 27 + 52 - (d + e + f + 2g) = 106 - (26 + g) = 80 - g$
N is minimum when g is maximum
Maximum value of g is 26.
∴ Minimum value of $n = 54$

Solutions for questions 37 to 40: 40 students participated in each of the 400 m dash and 800 m dash — (1)
No participant of the 3000 m dash participated in the 100 m dash —– (2)
No participant of the 1500 m dash participated in the 200 m dash —– (3)
60 students participated in each of 1500 m dash and 3000 m dash — (4)
From (2), (3) and (4), 40 students participated in each of 100 m dash and 200 m dash —– (5)
From (1), (5) and the given data, we obtain the diagram below.



37. $a + 3e + 3f + 15 = 40$
$a + 3(e + f) = 25$
Here, a is maximum when $e + f$ is minimum, i.e., 0
$\max(a) = 25$
The maximum possible number of students who participated in only the 400 m dash is 25.

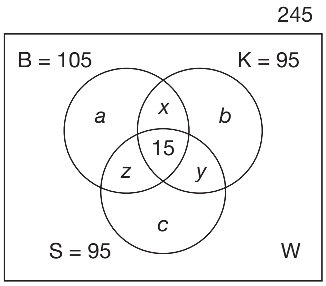
38. The number of students who participated in the 100 m dash, 200 m dash and 150 m dash = $f + 15$.
From the previous solution, $a + 3(e + f) = 25$
f is maximum when e and a are both minimum.
$\text{Min}(e) = 0$ and $\min(a) = 1$
∴ $\text{Max}(f) = 8$
The maximum possible number of students who participated in the 100 m dash, 200 m dash and 400 m dash = $8 + 15 = 23$.

39. 16 students participated in only the 100 m dash.
All the other students who participated in the 100 m dash participated in atleast one other.
24 students participated in the 100 m dash and atleast one other.

40. $16 + 3(e + f) + 15 = 40$
$e + f = 3$
Number of students who participated in only the 100 m dash and 200 m dash = e
$\text{Max}(e) = 3$

EXERCISE-3

Solutions for questions 1 to 3:



$$x + y + z = 3w,\ a + b + c = 190$$

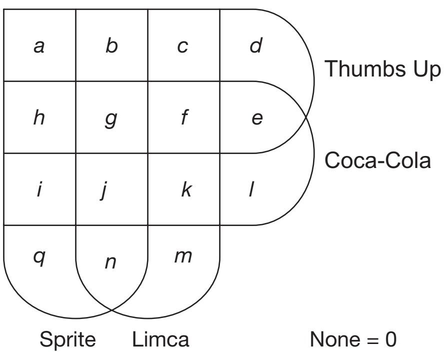
We have: $105 + \{95 - (x + 15)\} + \{95 - (y + z + 15\} + w = 245$
$\Rightarrow x + y + z - w = 20$
$2w = 20$
$\Rightarrow w = 10$
$x + y + z = 30$

1. $S + K - (S \cap K) = S \cup K$
$95 + 95 - p = 165$
Here, $p =$ (only S and K) + (S, K and B) $= y + 15$
$p = 25$
Since S, K and B = 15, $y = 25 - 15 = 10$.

2. $105 + 95 - q = 180$
Here, $q = x + 15 =$ (only B and K) + (S, K and B)
$q = 20$
Hence, $x = 20 - 15 = 5$
So, $y + z = 30 - 5 = 25$
Only Somnath $= 95 - 15 - (y + z)$
$= 95 - 15 - 25 = 55$

3. Given $z = 0$
Only Kedarnath $= 95 - 15 - x - y$
$= 95 - 15 - 30 = 50$

Solutions for questions 4 to 7:



The given information is as follows.
$h + g + f + e = 65$
$a + b + g + h = 77$
$g + f + j + k = 73$
$b + c + g + f = 76$
$h + g + i + j = 74$
It is also given that $h = f = x$ (assume)
$b = j = y$ (assume)
$\therefore$ The equations we can be written in the following way.

$$g + 2x + e = 65 \quad \text{(i)}$$
$$g + x + y + a = 77 \quad \text{(ii)}$$
$$g + x + y + k = 73 \quad \text{(iii)}$$
$$g + x + y + c = 76 \quad \text{(iv)}$$
$$g + x + y + i = 74 \quad \text{(v)}$$

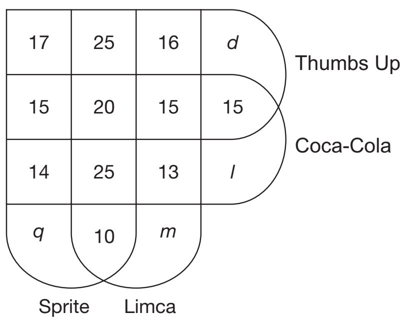
The other information given is:
Exactly 1 $= 67 = d + l + m + q$
$i = 14$
$e = 15$
$n = 10$
Exactly 1 + 2 Exactly 2 + 3 Exactly 3 + 4 Exactly 4 = 557 (vi)
Now substituting the value of i in equation (v), we get:

$$g + x + y = 60 \quad \text{(vii)}$$

Now from (vii), substituting the value of $g + x + y$ in equation (ii), (iii) and (iv), respectively, we get
$a = 17$
$k = 13$
$c = 16$
$\therefore$ Exactly 2 $= 17 + 14 + 10 + 13 + 15 + 16 = 85$
Exactly 3 $= 2(x + y) = 2(60 - g)$.
Substituting the above values in equation (vi), we get:
$67 + 2 \times 85 + 3 \times 2\ (60 - g) + 4g = 557$
$\Rightarrow 67 + 170 + 360 - 6g + 4g = 557$
$\Rightarrow 597 - 557 = 2g$
$\Rightarrow g = 20$
Now substituting the values of e and g in equation (i), we get:
$20 + 2x + 15 = 65$
$\Rightarrow 2x = 30$
$\Rightarrow x = 15$
Now, from equation (vii), we get

$$y = 60 - 15 - 20 = 25$$

$\therefore$ The final Venn diagram will be as follows:



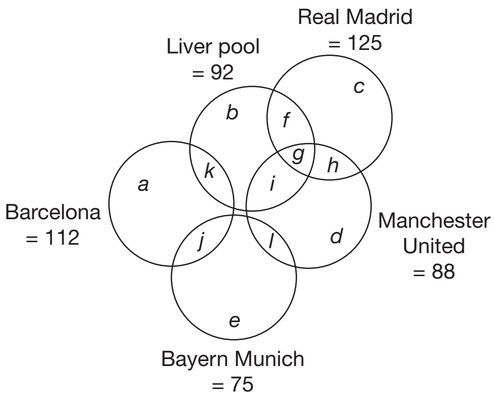
4. The number of students who like both Sprite and Limca but not all the four = $2y + n = 25 \times 2 + 10 = 60$.

5. The total number of students
= Exactly 1 + Exactly 2 + Exactly 3 + Exactly 4
= 67 + 85 +2(15 + 25) + 20
= 67 + 85 + 80 + 20 = 252

6. The number of students who like only Coca-Cola and Limca is 13.

7. Given, $d + q = 25$
$m + l = 67 - 25 = 42$
∴ The required number of students
$= 25 + 20 + 25 + 10 + m + 13 + 15 + 16 + 15 + 14 + 15 + l$
$= 168 + l + m$
$= 168 + 42 = 210$

Solutions for questions 8 to 12: From the given information it is clear that the set of people who like Barcelona has no intersection with Real Madrid or Manchester United. Similarly, the set of people who like Bayern Munich has no intersection with Liverpool or Real Madrid. Intersection between other sets of people is possible. Thus, we get the following Venn Diagram.



From (i), $h + g = 57$ and from (ii), $g = 10$ and $b = 10$
Hence, $h = 47$
From (ii), j = h= 47 and $j = k + 4$
Hence, $k = 43$
From (iv), $l = f = 1/3$ of c
It is given that Real Madrid = 125.
We know that $h = 47$ and $g = 10$.
Hence, $c + f = 68$.
Since, $f = 1/3$ of c, $f = 17$ and $c = 51$. Thus $l = 17$.
a = Barcelona - (j + k) = 22
e = Bayern Munich – (j + l) = 11
i = Liverpool – (b + f + g + k) = 12
d = Manchester United – (g + h + i + l) = 2

8. Exactly two clubs = $f + i + l + j + k = 183$.

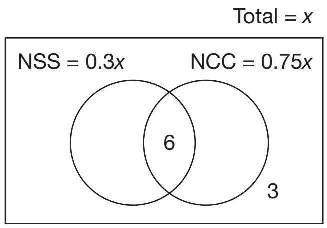
9. Only Bayern Munich or only Barcelona = a + e = 33

10. Manchester United but not Liverpool = $h + d + l = 66$

11. Exactly one club = a + $b + c + d + e = 96$

12. No one among the Barcelona fans likes at least two more clubs.

Solutions for questions 13 to 16:



If x is the total number of students in the class, then the number of students participating in NSS and NCC is $0.3x$ and $0.75x$, respectively.
Then, $\{0.3x + 0.75x - 6\} + 3 = x \Rightarrow 0.05x = 3 \Rightarrow x = 60$

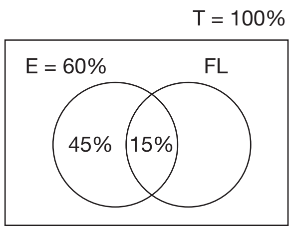
13. Total number of students = $x = 60$

14. Percentage of students who want to participate only in NSS $= 30\% - \dfrac{6}{60} \times 100 = 20\%$

15. Only in one programme
= 20% only in NSS + 75% – 6/60 × 100
Only in NCC = 85%.

16. At least in one programme = Total – Number of students participated in neither of these two
⇒ 60 – 3 = 57

Solutions for questions 17 to 20:



- 25% of 60%, i.e., 15% of the school passed in both English and foreign language.
- Since 66⅔% of the students who passed in foreign language failed in English, 33⅓% of students who passed in foreign language passed in English also, i.e., ⅓ Foreign Language = 15%

⇒ Foreign language = 45%
So, we have only passed in (English) = 60 – 15 = 45%,
Only foreign language passed = 45 – 15 = 30%
Passed both in English and foreign language = 15%
A total of 90% passed in at least one of the subjects. So, 10% failed in both.
Number of students in the school $= \dfrac{2}{0.10} = 200$

17. 200

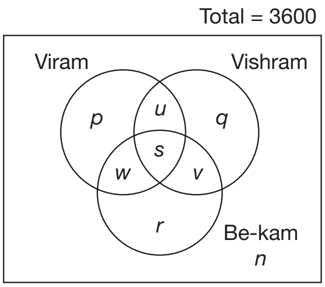
18. 45% + 30% = 75%

19. 20% of 15% = 3% increase in pass in both subjects. Hence, the least value of pass only in English will come when all the new people who pass in both subjects are from the group which passed only in English.
45 – 3 = 42%

20. 40% of 20 students = 8 students. Out of this 4 students passed in exactly one subject. Hence, the remaining 4 students (which is 2% of the school strength) pass in both subjects.
So, pass in both the subjects = 15 + 2 = 17%.

Solutions for questions 21 to 25:

Representation of various segments as in the following diagram:



Members of at least two Unions = $u + v + w + s = 500$ (1)
Vishram members = $q + s + u + v = 1400$ (2)
Only Viram and Be-kam = $w = 100$ (3)
Vishram and Be - kam = $s + v = 200$ (4)
Only Be - kam = $r = 550$ (5)

Members of Viram who are members of only one more union = $w + u = 20\%$ of $(p + u + s + w)$ (6)
$u + v + w = \frac{1}{8}$ (Total workers) = 450. (7)
From (1), (3) and (4), $u = 200$.
From equation (2), we get

$$q = 1400 - u - (s + v)$$
$$= 1400 - 200 - 200 = 1000.$$

From (7), $v = 450 - 200 - 100 = 150$
From (4), $s = 200 - 150 = 50$
From (6), $p = 1150$

$n = 3600 - (p + q + r + s + u + v + w)$
$= 3600 - (1150 + 1000 + 550 + 50 + 200 + 150 + 100)$
$= 3600 - (3200) = 400$

Now, we have all figures and the questions can be answered.

21. $s = 50$.

22. $n = 400$.

23. $p + r = 1150 + 550 = 1700$.

24. Since 10 workers have given up their Be-kam membership and taken Vishram membership, it means these 10 workers were initially Be-kam members but not Vishram members, i.e., they must be a part of r or w. When they give up Be-kam and take up Vishram, they will move to q or u, respectively. So, s does not undergo any change at all. Hence, 50 is the answer.

25. $q + u = 1000 + 200 = 1200$

8 Cubes

CHAPTER

Learning Objectives

In this chapter, you will:

- Learn to visualize a cube and how to cut a cube across different axes.
- Be able to relate the number of cuts to the number of smaller cubes/pieces that arise.
- Understand the different colored faces which form a part of a bigger cube.
- Understand how to deal with questions based on folding and unfolding of a cube.
- Learn how to get a given number of pieces using minimum number of cuts and how to get maximum number of pieces using a given number of cuts.

A cube is a three-dimensional solid having 6 faces, 12 edges and 8 corners. All the edges of a cube are equal, and hence, all the faces are square in shape.

In competitive exams a few questions may be asked based on cubes.

The questions on cubes may belong to any one of the following categories.

1. A cube is cut by making certain specified number of cuts. The directions in which the cuts are made may or may not be given. We are to find the number of identical pieces resulting out of the given cuts.
2. The number of identical pieces, into which a cube is cut is given and we need to find the number of cuts.
3. A cube could be painted on all or some of its faces with the same colour or different colours and then cut into a certain specified number of identical pieces. Then questions of the form 'How many small cubes have 2 faces painted?'. 'How many smaller cubes have only one face painted?' could then be framed.

SOLVED EXAMPLES

Directions for questions 8.01 to 8.04: A cube is painted and cut into 343 smaller but identical pieces by the minimum possible number of cuts.

8.01: How many smaller pieces have exactly three painted faces?

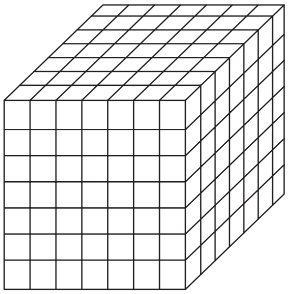
8.02: How many smaller pieces have exactly two painted faces?

8.03: How many smaller pieces have exactly one painted face?

8.04: How many smaller pieces have no painted face?

Solutions for questions 8.01 to 8.04: A cube's outer surface is painted and then it is cut into smaller pieces. Now the smaller pieces obtained have paint on some surfaces and the surfaces generated after the cut do not have any paint on them.

When a cube is cut into 343 smaller pieces by applying minimum number of cuts, it appears as follows



	The given cube $7 \times 7 \times 7 = 343$	No. of smaller pieces which have	Generalisation $n \times n \times n$
1.	8 corner pieces	(i) exactly three painted surfaces	8 corner pieces
2.	5 pieces at each edge i.e. $(7 - 2) \times 12$ edges = 60	(ii) exactly two painted surfaces	$(n - 2) \times 12 = 5 \times 12$
3.	25 pieces at the middle of each surface i.e. $(7 - 2)^2 \times 6$ surfaces = 150	(iii) exactly one painted surfaces	$(n - 2)^2 \times 6 = 5^2 \times 6$
4.	$5 \times 5 \times 5$ i.e., $(7 - 2)^3 = 125$	(iv) no painted surfaces	$(n - 2)^3 = 125$

Directions for questions 8.05 to 8.08: A cube is painted and cut into 210 smaller but identical pieces by making the minimum possible number of cuts.

8.05: How many smaller pieces have exactly three painted faces?

8.06: How many smaller pieces have exactly two painted faces?

8.07: How many smaller pieces have exactly one painted face?

8.08: How many smaller pieces have no painted face?

Solutions for questions 8.05 to 8.08: To cut a cube into 210 pieces i.e., 5, 6, 7 will be the number of pieces in each direction.

8.05: The number of pieces with exactly three painted surfaces is 8.

8.06: The number of pieces with exactly two painted surfaces:

For X-plane	$4(5 - 2)+$
For Y-plane	$4(6 - 2)+$
For Z-plane	$4(7 - 2)+$
Total =	48

8.07: The number of pieces with exactly one painted surface:

For X-Y plane	$2(5 - 2)(6 - 2)+$
For Y-Z plane	$2(6 - 2)(7 - 2)+$
For Z-X plane	$2(7 - 2)(5 - 2)$
Total =	94

8.08: The number of pieces having no painted face $= (5 - 2)(6 - 2)(7 - 2) = 60$

Directions for questions 8.09 to 8.14: A pair of opposite faces of a cube is painted is yellow, another pair of opposite faces, orange and the remaining two faces are painted white. The cube is then cut into 343 smaller but identical cubes.

8.09: How many of the smaller cubes have all the three colours on them?

8.10: How many of the smaller cubes have only white and orange on them?

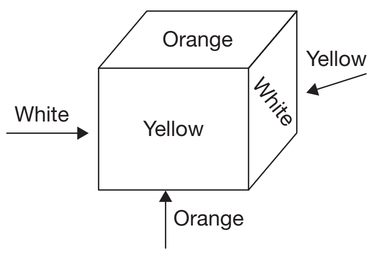
8.11: How many of the smaller cubes have exactly two colours on them?

8.12: How many of the smaller cubes have only white colour on them?

8.13: How many of the smaller cubes have exactly one colour on them?

8.14: How many of the smaller cubes have no colour on them?

Solutions for questions 8.09 to 8.14: The cube after cutting, has $7 \times 7 \times 7 = 343$ pieces. The patter of painting is as follows.



8.09: Since no two adjacent surfaces have the same colour, each corner piece has three painted faces and each face has a different colour. Hence the number of smaller cubes with all the three colours on them is 8.

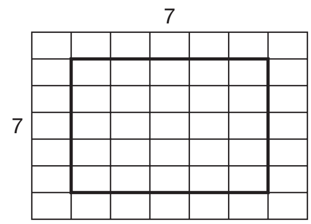
8.10: We find the smaller cubes with only white and orange colours on them at the edges which are common for white and orange surfaces. The number of smaller pieces with only white and orange on them is (white — orange edges) $\times$ 5 (at each edge) = 20.

Similarly, the number of small pieces with only white and yellow colours on them and those with only yellow and orange are 20 each.

8.11: The number of small pieces with exactly two colours is $(20 + 20 + 20) = 60$.

This can also obtained by applying the formula $(n - 2) \times 12$ i.e. $5 \times 12 = 60$.

8.12: We find that the smaller pieces with only white colour on them, are at the middle of the surfaces painted in white.



i.e. $5 \times 5 = 25$ pieces on one white surface $\times 2 = 50$ pieces. Similarly, the number of small pieces with only blue colour and those with only yellow colour are 50 each.

8.13: The number of small pieces with exactly one colour is $(50 + 50 + 50 = 150)$.

This car also be obtained by applying the formula $(n - 2)^2 \times 6$.

8.14: The number of small pieces with no colour on them is $(n - 2)^3$. i.e. $(7 - 2)^3 = 125$.

EXERCISE-1

Directions for questions 1 to 11: Select the correct alternative from the given choices.

1. What will be the maximum possible number of pieces when a cube is cut into 5 cuts?
(A) 18 (B) 6
(C) 25 (D) 5

2. What will be the maximum possible number of pieces when a cube is cut into 6 cuts?
(A) 37 (B) 36
(C) 42 (D) 27

3. What will be the maximum possible number of pieces when a cube is cut into 17 cuts?
(A) 250 (B) 160
(C) 270 (D) 294

4. What will be the maximum possible number of pieces when a cube is cut into 8 cuts?
(A) 36 (B) 48
(C) 45 (D) 40

5. What will be the maximum possible number of pieces when a cube is cut into 11 cuts?
(A) 100 (B) 90
(C) 84 (D) 54

6. What is the least possible number of cuts required to cut a cube into 80 identical pieces?
(A) 21 (B) 12
(C) 19 (D) 10

7. What is the number of ways in which two faces of a cuboid of dimensions 6 cm × 7 cm × 8 cm can be painted in green colour?
(A) 2 (B) 3
(C) 4 (D) 6

8. How many cubes of dimensions 1 cm × 1 cm × 1 cm are required to cover a cube of dimensions 7 cm × 7 cm × 7 cm completely?
(A) 169 (B) 294
(C) 386 (D) 488

9. How many cubes of dimensions 1 cm × 1 cm × 1 cm are required to cover a cuboid of dimensions 6 cm × 8 cm × 9 cm when it is placed at the corner of a room such that three faces of the cuboid are covered by two walls and the floor of the room?
(A) 288 (B) 261
(C) 198 (D) 448

10. 125 smaller cubes of dimensions 1 cm × 1 cm × 1 cm are stacked together to form a larger cube and then the cube is cut along two diagonals. How many of the smaller cubes are cut into smaller pieces?
(A) 10 (B) 25
(C) 45 (D) 50

11. 1000 smaller cubes of dimensions 1 cm × 1 cm × 1 cm are stacked together to form a larger cube and then the cube is cut along a diagonal. How many of the smaller cubes are cut into two pieces?
(A) 25 (B) 50
(C) 90 (D) 100

Directions for questions 12 to 14: These questions are based on the following information.

A large cube painted on all six faces is cut into 27 smaller but identical cubes.

12. How many of the smaller cubes have no faces painted at all?
(A) 0 (B) 1
(C) 3 (D) 4

13. How many of the smaller cubes have exactly one face painted?
(A) 3 (B) 6
(C) 12 (D) 15

14. How many of the smaller cubes have exactly two faces painted?
(A) 36 (B) 6
(C) 12 (D) 15

Directions for questions 15 to 17: These questions are based on the following information.

A large cube is painted on all six faces and then cut into a certain number of smaller but identical cubes. It was found that among the smaller cubes, there were eight cubes which had no face painted at all.

15. How many smaller cubes was the original large cube cut into?
(A) 27 (B) 48
(C) 64 (D) 125

16. How many small cubes have exactly one face painted?
(A) 12 (B) 24
(C) 16 (D) 32

17. How many small cubes have exactly two faces painted?
(A) 6 (B) 12
(C) 18 (D) 24

Directions for questions 18 to 20: These questions are based on the following information.

There is a cube in which one pair of adjacent faces is painted red, the second pair of adjacent faces is painted blue and a third pair of adjacent faces is painted green. This cube is now cut into 216 smaller but identical cubes.

18. How many small cubes are there with one face painted red?
(A) 64 (B) 81
(C) 60 (D) 120

19. How many small cubes are with both red and green on their faces?
(A) 8 (B) 12
(C) 16 (D) 32

20. How many small cubes are there showing only green or only blue on their faces?
(A) 64 (B) 72
(C) 81 (D) 96

Directions for questions 21 to 23: These questions are based on the following information.

A cube is painted in black and green, each on three faces such that any two faces with same colour are adjacent to each other. Now this cube is cut into 60 identical pieces using 2, 3 and 4 cuts parallel to different faces.

21. How many smaller pieces have exactly two faces painted in black?
(A) 5 (B) 9
(C) 18 (D) 27

22. How many smaller pieces have both the colours on them?
(A) 9 (B) 18
(C) 6 (D) 24

23. How many smaller pieces have no face painted?
(A) 6 (B) 9
(C) 11 (D) 1

Directions for questions 24 to 26: These questions are based on the following information.

Two opposite faces of a cube are painted in blue, another pair of opposite faces are painted green and the remaining faces are painted in red. The cube is now cut into 210 smaller but identical pieces using minimum possible number of cuts.

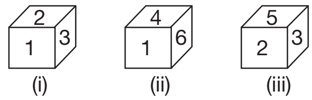
24. How many smaller pieces have exactly two colours on them?
(A) 48 (B) 36
(C) 24 (D) Cannot be determined

25. What is the maximum possible number of smaller piece, which have green and red colour on them?
(A) 20 (B) 24
(C) 26 (D) 28

26. What is the minimum possible number of pieces which have only blue colour on them?
(A) 24 (B) 12
(C) 30 (D) 60

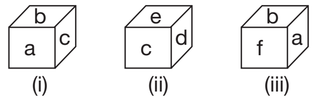
Directions for questions 27 to 33: In each of the following questions, three different views of a cube are given. Based on these diagrams answer the following questions.

27. Which of the following statements is true?



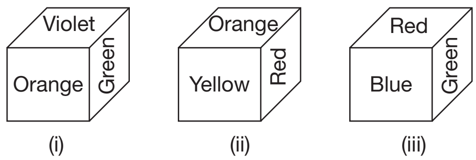
(A) 3 is opposite to 6 (B) 5 is opposite to 4
(C) 4 is opposite to 3 (D) D is opposite to 2

28. Which of the following indicates the correct pair of opposite faces?



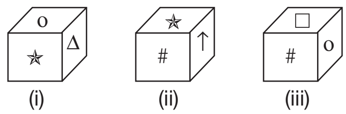
(A) a – d (B) a – f
(C) f – e (D) b – d

29. Which colour is at the bottom of the second figure?



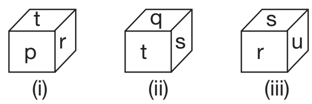
(A) Blue (B) Green
(C) Orange (D) Red

30. Which of the following are adjacent to Δ?



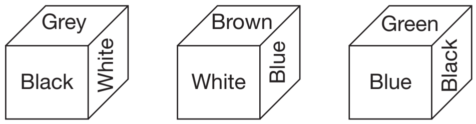
(A) ★, # (B) ↑, #
(C) o, ↑ (D) o, #

31. Which of the following are opposite to *r* and *t*, respectively?



(A) *u* and *q* (B) *p* and *s*
(C) *s* and *p* (D) *q* and *u*

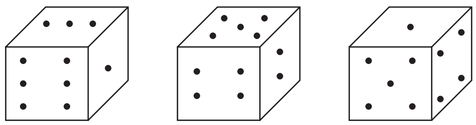
32. Which of the following is at the bottom of figure (i)?



(i) (ii) (iii)

(A) Blue (B) Green
(C) Black (D) Brown

33. What is the sum of the dots on the two faces which are adjacent to both the faces with two dots and five dots, if the number of dots on the six faces is 1, 2, 3, 4, 5 and 6, respectively?

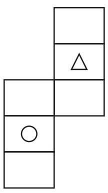


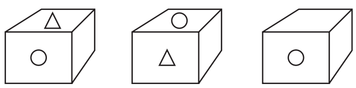
(i) (ii) (iii)

(A) 10 (B) 7
(C) 5 (D) 4

Directions for questions 34 and 35: In the following questions the figure is folded to form a box. Select from among the given alternatives, the box or boxes that can be formed by folding the figure.

34.

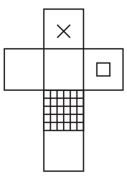


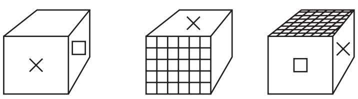


I II III

(A) Only I (B) Only II
(C) Only I and II (D) Only III

35.



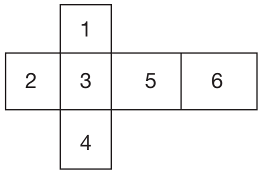


I II III

(A) Only I
(B) Both II and III
(C) Both I and III
(D) All of them

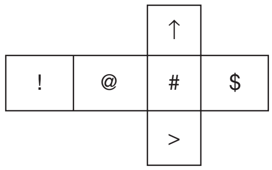
Directions for questions 36 to 40: Select the correct alternative from the given choices.

36. If the following figure is folded to form a cube, then what is the number on the face opposite to the face marked 3?



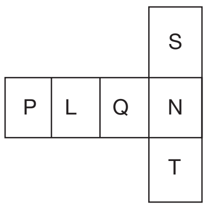
(A) 6 (B) 5
(C) 1 (D) 2

37. If the following figure is folded to form a cube, then what is the symbol on the face opposite to the face marked ‘@’?



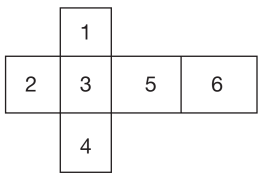
(A) ! (B) ↑
(C) > (D) $

38. If the following figure is folded to form a cube, what would be the letter on the face opposite to the face maked ‘L’?



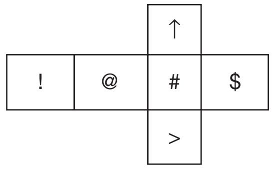
(A) Q (B) N
(C) T (D) S

39. If the following figure is folded to form a cube, then what is the number on the face opposite to the face marked 3?



(A) 6 (B) 5
(C) 1 (D) 2

40. If the following figure is folded to form a cube, then what is the symbol on the face opposite to the face marked '@'?



(A) ! (B) ↑
(C) > (D) $

EXERCISE-2

Directions for questions 1 and 2: Select the correct alternative from the given choices.

1. 216 small and identical cubes are placed together to form a big cube. What is the number of small cubes that get cut when the big cube is cut by one diagonal cut and two diagonal cuts, respectively?

(A) 64, 70 (B) 36, 70
(C) 36, 72 (D) 70, 64

2. 729 small and identical cubes are placed together to form a big cube. What is the number of small cubes that get cut by one diagonal cut and two diagonal cuts, respectively?

(A) 85, 150 (B) 81, 105
(C) 82, 162 (D) 81, 153

Directions for questions 3 to 7: These questions are based on the following information.

A cube is painted in such a way that a pair of adjacent faces is painted in green; a pair of opposite faces is painted in yellow and another pair of adjacent faces is painted in red. This cube is now cut into 125 small but identical cubes.

3. How many small cubes have exactly two faces painted in green?

(A) 10 (B) 7
(C) 5 (D) 8

4. How many small cubes have at least two different colours on their faces?

(A) 30 (B) 38
(C) 36 (D) 42

5. How many of the small cubes have exactly one colour on them?

(A) 60 (B) 45
(C) 54 (D) 15

6. How many of the small cubes do not have green colour but have yellow or red colours on them?

(A) 40 (B) 75
(C) 80 (D) 53

7. How many small cubes have exactly two painted faces and have exactly two colours on them?

(A) 36 (B) 30
(C) 24 (D) 34

Directions for questions 8 to 11: These questions are based on the following information.

One face of a cube is painted in green, two faces are painted in yellow and three faces are painted in white. This cube is now cut into 512 small but identical cubes.

8. What is the minimum possible number of small cubes that have two faces painted in yellow?

(A) 0 (B) 9
(C) 7 (D) 16

9. What is the maximum possible number of smaller cubes that have only green and yellow colours on them?

(A) 12 (B) 6
(C) 14 (D) 13

10. What is the maximum possible and the minimum possible number of small cubes, respectively that have exactly one colour on them?

(A) 241, 228 (B) 234, 228
(C) 241, 234 (D) 241, 226

11. What is the least possible number of small cubes that have at least two painted faces but have only white colour on them?

(A) 18 (B) 15
(C) 17 (D) 12

Directions for questions 12 to 16: These questions are based on the following information.

A cube is painted such that one of its faces is painted in black, one face is painted in white and one face is painted in red. The other three faces are left unpainted. The cube is now cut into 729 small and identical cubes.

12. What is the maximum possible number of small cubes that have all the three colours on them?

(A) 1 (B) 2
(C) 4 (D) 5

13. What is the minimum possible number of small cubes that have only black and white colours on them?

(A) 1 (B) 5
(C) 6 (D) 0

14. What is the minimum possible number of small cubes that have only red colour on them?

(A) 72 (B) 81
(C) 63 (D) 80

15. How many small cubes have white colour on them?

(A) 75 (B) 85
(C) 72 (D) 81

16. What is the minimum possible number of small cubes that have no colour on them?

(A) 343 (B) 512
(C) 504 (D) 269

Directions for questions 17 to 19: These questions are based on the following information.

A cube has all the six faces painted in six different colours, such as white, blue, red, yellow, green and pink in such a way that pink and green are on two opposite faces. The cube is placed on a table with the pink face touching the top of the table. Red is facing you, whereas white and blue faces are opposite to each other. The cube is cut into 120 identical pieces by making the least number of cuts possible where all the cuts are parallel to the faces of the cube. The least number of possible cuts are made in the horizontal direction and maximum number of possible cuts are made parallel to the red face.

17. How many small pieces have white colour on their faces?

(A) 36 (B) 42
(C) 30 (D) 24

18. How many small pieces have at least two different colours on their faces?

(A) 44 (B) 28
(C) 38 (D) 30

19. How many small pieces have no colour on their faces?

(A) 42 (B) 24
(C) 36 (D) 27

Directions for questions 20 to 22: These questions are based on the following information.

Two identical wooden cubes P and Q placed on a table facing you, have their faces painted as follows. One pair of opposite faces of cube P is painted with the same colour, i.e., red colour and another pair of opposite faces is painted blue. One of the remaining faces is painted yellow, whereas the other one is painted brown.

One pair of opposite faces of cube Q is painted blue. A second pair of opposite faces of Q are painted in such a way that the opposite face of brown is green. The other two opposite faces are painted black and yellow, respectively. In the following questions, 'two faces touch each other' implies that the complete area of one face touches the complete area of the second face.

20. The two cubes are placed next to each other on the table touching each other such that, whether the positions of P and Q are interchanged or left as they are, the two faces of P and Q touching each other are of the same colour. If the top faces of both P and Q have to be of the same colour, then which of the following must be true?

(A) The front faces of cube P and Q are red and yellow, respectively.
(B) The two faces of cube P and Q which are touching the table top are of brown and black colours, respectively.
(C) The front face of cube P is of red colour.
(D) The top faces of cubes P and Q are of red and yellow colours, respectively.

21. Q is placed on the top of P such that no blue face of either cube is horizontal. If brown and blue are the front faces of P and Q, respectively, then which of the following statements must be false?

(A) The faces of the two cubes touching each other are of red and green colour.
(B) The faces of two cubes which are touching each other are of red and brown colours.
(C) If blue and green are the colours on the right side faces of two cubes, respectively, then the left side faces of two cubes will be blue and brown, respectively.
(D) The faces of the two cubes which are touching each other are yellow and brown.

22. If cube Q is kept behind cube P in such a way, that the yellow face of P faces the brown face of cube Q and the faces touching the table are of red and black colours, then which faces of both the cubes have same colour?

(A) Top faces
(B) Top and bottom faces only.
(C) The faces to the left and the right only.
(D) Both top and front faces only.

Directions for questions 23 to 25: These questions are based on the following information.

Two colours, red and blue are used to paint a cube. Red is painted on three faces, each of which is adjacent to the other two and blue is painted on the remaining faces. Assume that one can see exactly three faces when the cube is kept on a plane.

23. What is the total number of ways in which the blue colour is not seen at all when the cube is kept on a table?

(A) 4 (B) 3
(C) 2 (D) 1

24. What is the total number of ways in which exactly one face painted blue is seen?

(A) 2 (B) 4
(C) 3 (D) 5

25. What is the total number of ways in which exactly two faces painted blue are seen?

(A) 3 (B) 2
(C) 5 (D) 1

Directions for questions 26 and 27: These questions are based on the following information.

Two faces of a cube are painted red, two faces are painted green and the remaining faces are painted blue. Now the cube is cut into 216 smaller but identical pieces with minimum possible number of cuts.

26. What is the minimum possible number of smaller pieces with exactly two different colours on them?

(A) 36 (B) 42
(C) 48 (D) 56

27. What is the maximum possible number of smaller pieces which have at most one colour on them?

(A) 160 (B) 172
(C) 198 (D) 208

Directions for questions 28 to 30: These questions are based on the following information.

Three different faces of a cube are painted in three different colours, such as red, green and blue. This cube is now cut into 216 smaller but identical cubes.

28. What are the least and the largest numbers of small cubes that have exactly one face painted?

(A) 75 and 86 (B) 64 and 81
(C) 64 and 72 (D) 75 and 84

29. What is the maximum number of small cubes that have one face painted green and one face blue and no other face painted?

(A) 2 (B) 4
(C) 6 (D) 8

30. What are the least and the maximum numbers of cubes that have no face painted at all?

(A) 125 and 130 (B) 120 and 125
(C) 115 and 120 (D) 100 and 125

Directions for questions 31 to 34: These questions are based on the following information.

Each face of a cube is painted in green, red or blue.

31. Totally in how many different ways can the cube be painted?

(A) 49 (B) 56
(C) 64 (D) 81

32. In how many different ways can the cube be painted with at least two faces blue?

(A) 24 (B) 30
(C) 34 (D) 42

33. In how many different was can the cube be painted such that all three colours are there on the cube?

(A) 32 (B) 29
(C) 25 (D) 30

34. In how many different ways can the cube be painted such that no two adjacent faces have the same colour?

(A) 3 (B) 1
(C) 2 (D) 4

Directions for questions 35 to 37: These questions are based on the following information.

A cube is painted red, blue and green in such a way that each face is painted with a single colour and each colour is painted on two adjacent faces. The cube is placed on a table and one can see exactly three faces of the cube.

35. What is the total number of distinct corners from where red and blue colours are visible?

(A) 5 (B) 4
(C) 6 (D) 8

36. What is the total number of ways in which all three colours can be seen?

(A) 2 (B) 3
(C) 1 (D) 5

37. What is the total number of distinct possible combinations of three colours that can be seen?

(A) 8 (B) 9
(C) 7 (D) 6

Directions for questions 38 to 40: These questions are based on the following information.

Each face of a die is marked with a different number from 1 to 6. The numbers on the faces of the die are marked in such a way that the sum of the numbers on any pair of opposite faces is seven. Two such dice are thrown. Assume that one can always see exactly three faces of each die.

38. What is the total number of distinguishably different ways in which the sum of the numbers on the visible faces of both the cubes together is 20?

(A) 2 (B) 6
(C) 3 (D) 5

39. What is the total number of distinguishably different ways in which the sum of numbers on visible faces is exactly 10 on at least one die?

(A) 12 (B) 17
(C) 15 (D) 19

40. What is the total number of ways in which a specified number is visible on both the dice?

(A) 32 (B) 16
(C) 14 (D) 18

Difficulty Level-2: Moderate

EXERCISE-3

Directions for questions 1 to 3: These questions are based on the following information.

A large cube is painted on only three of its faces with three different colours, such as red green and black. This cube is now cut into 125 smaller but identical faces.

1. What is the minimum number of cubes that have no face painted?

2. What is the maximum number of cubes that have exactly two faces painted?

3. What is the minimum number of cubes (respectively) that have exactly one face painted?

Directions for questions 4 to 6: These questions are based on the following data.

The faces of the cuboid are painted with three different colours, such as black, red, and yellow such that each colour is painted on at least one face. Now 4, 5 and 6 cuts are made in three different directions.

4. What is the maximum possible number of smaller pieces that have only black on their faces?

5. What is the maximum possible number of smaller pieces that have only black and yellow painted on their faces?

6. What is the maximum number of smaller pieces with three colours painted on them, respectively?

Directions for questions 7 to 10: These questions are based on the following information.

A large cube is formed by stacking 64 smaller and identical cubes. These smaller cubes are numbered 1 to 64 in the following manner. The four cubes in the front row of the bottom layer are numbered 1 to 4 from left to right. The cubes in the second row of the bottom layer are numbered 5 to 8. This pattern of numbering continued till all the 16 cubes in the bottom layer are numbered. The numbering of the second layer is done in a similar fashion, by numbering the cubes in the front row from 17 to 20 from left to right. This pattern of numbering continues for all the layers from the bottom layer to the top layer

7. What is the sum of numbers on all the cubes in the front row of the bottom layer?

8. What is the sum of numbers on all the cubes in the left column of the front face?

9. What is the sum of the numbers on the cubes along the left column of the back layer?

10. What is the sum of the numbers on the cubes along the second row of the top layer?

Directions for questions 11 to 14: These questions are based on the following information.

A large cube is formed by stacking 125 smaller and identical cubes. These smaller cubes are numbered 1 to 125 in the following manner. The five cubes in the front row of the bottom layer are numbered 1 to 5 from left to right. The cubes in the second row of the bottom layer are numbered 6 to 10. This pattern of numbering continues till all the 25 cubes in the bottom layer are numbered. The numbering of the second layer is done in a similar fashion, by numbering the cubes in the front row from 26 to 30 from left to right. This pattern of numbering continues for all the layers from the bottom layer to top layer.

11. What is the sum of numbers on the cubes along the column which has its base in the cube which is second from the left end of the bottom row of the layer behind the front layer?

12. What is the sum of the numbers on the cubes along the diagonal from the cube at the left end of the front row of the top layer to the cube at the right end of the last row of the top layer?

13. What is the sum of the numbers on the cubes along the diagonal from the cube at the left end of the bottom row of the front face to the cube at the right end of the last row of the top layer?

14. What is the sum of the numbers on the cubes along the diagonal from the cube at the right end of the bottom row of the front layer to the cube at the left end of the top row of the back layer?

Directions for questions 15 to 18: These questions are based on the following information.

A large wooden cube is painted with three different colours such that opposite faces are painted with the same colour. The cube is now entirely cut into 455 small and identical pieces by making the lowest possible number of cuts on the cube. All the completely unpainted smaller pieces are thrown away.

15. What is the total number of cuts made on the larger cube?

16. What is the highest possible number of pieces that have exactly one face painted with a particular colour?

17. What is the highest possible number of pieces that have exactly two faces painted with a particular combination of colours?

18. What is the lowest possible number of pieces which are painted with exactly one combination of two colours?

Directions for questions 19 to 22: These questions are based on the following information.

A large cube is built using 216 identical cubes which are engraved with numbers 1 to 216. The large cube is built in the following pattern.

(i) The cubes are stacked to form a large cube.

(ii) The cubes engraved 1 to 6 are placed in a column, one behind the other such that the smallest number is to the front. The six cubes numbered 7 to 12 are placed in the second column to the right of the cubes in the first column such that the smallest number is in the front. The same pattern is followed until the bottom layer is completely built.

(iii) Each other layer is built in the same pattern as the bottom layer. In these layers, smaller cubes are placed starting with the smallest available number being placed on the left most cube on the front row of the previous layer.

(iv) After the large cube is built with 216 cubes, its five visible faces are painted in blue.

19. What is the sum of the numbers on the cubes that have three sides painted?

20. What is the sum of the numbers on the small cubes that are resting on the floor but have no sides painted?

21. The two end cubes of one of the edges on the front face of the large cube have three sides painted. What is the sum of the numbers on the cubes that form this edge?

22. What is the sum of the numbers on the cubes that form the diagonal that connects the two corner cubes having three sides painted?

Directions for questions 23 to 25: These questions are based on the following information.

A cube is painted with blue on two adjacent faces. One of the remaining faces is painted with red colour such that its opposite face remain unpainted. The remaining two faces are painted with green colour. Now the cube is cut into 512 smaller and identical cubes.

23. How many smaller cubes have exactly one of their faces painted?

24. How many smaller cubes have at most two faces painted?

25. How many smaller cubes have only one colour painted on them?

ANSWER KEYS

Exercise-1

1. (A)	**8.** (C)	**15.** (C)	**22.** (B)	**29.** (A)	**36.** (A)
2. (D)	**9.** (C)	**16.** (B)	**23.** (A)	**30.** (C)	**37.** (D)
3. (D)	**10.** (C)	**17.** (D)	**24.** (A)	**31.** (D)	**38.** (B)
4. (B)	**11.** (D)	**18.** (C)	**25.** (D)	**32.** (A)	**39.** (A)
5. (A)	**12.** (B)	**19.** (C)	**26.** (A)	**33.** (A)	**40.** (D)
6. (D)	**13.** (B)	**20.** (B)	**27.** (A)	**34.** (D)	
7. (D)	**14.** (C)	**21.** (B)	**28.** (D)	**35.** (A)	

Exercise-2

1. (C)	**8.** (A)	**15.** (D)	**22.** (C)	**29.** (C)	**36.** (A)
2. (D)	**9.** (D)	**16.** (C)	**23.** (D)	**30.** (B)	**37.** (C)
3. (C)	**10.** (A)	**17.** (D)	**24.** (C)	**31.** (B)	**38.** (D)
4. (B)	**11.** (D)	**18.** (A)	**25.** (A)	**32.** (C)	**39.** (C)
5. (A)	**12.** (A)	**19.** (B)	**26.** (B)	**33.** (B)	**40.** (B)
6. (D)	**13.** (D)	**20.** (C)	**27.** (B)	**34.** (B)	
7. (B)	**14.** (C)	**21.** (D)	**28.** (D)	**35.** (B)	

Exercise-3

1. 60	**5.** 32	**9.** 148	**13.** 315	**17.** 44	**21.** 1176	**24.** 508
2. 12	**6.** 8	**10.** 218	**14.** 315	**18.** 12	**22.** 1191	**25.** 218
3. 48	**7.** 10	**11.** 285	**15.** 22	**19.** 794	**23.** 204	
4. 90	**8.** 100	**12.** 565	**16.** 110	**20.** 296		

SOLUTIONS

EXERCISE-1

1. The number of pieces obtained would be the maximum possible when the given number of cuts are made as equally as possible in the three directions.

Distribution	Number of pieces
2, 2, 1	$3 \times 3 \times 2 = 18$

2. The number of pieces obtained would be the maximum possible when the given number of cuts are made as equally as possible in the three directions.

Distribution	Number of pieces
2, 2, 2	$3 \times 3 \times 3 = 27$

3. The number of pieces obtained would be the maximum possible when the given number of cuts are made as equally as possible in the three directions.

Distribution	Number of pieces
6, 6, 5	$7 \times 7 \times 6 = 294$

4. The number of pieces obtained would be the maximum possible when the given number of cuts are made as equally as possible in the three directions.

Distribution	Number of pieces
3, 3, 2	$4 \times 4 \times 3 = 48$

5. The number of pieces obtained would be the maximum possible when the given number of cuts are made as equally as possible in the three directions.

Distribution	Number of pieces
4, 4, 3	$5 \times 5 \times 4 = 100$

6. We get the least possible number of cuts when the given number of pieces is factorized in such a way that the factors are as equal as possible.
$80 = 4 \times 4 \times 5 \Rightarrow 3 + 3 + 4 = 10$ cuts
$\therefore$ 10 is the minimum possible number of cuts required to cut the cube into 80 identical pieces.

7. The two faces of a cube to be painted can be either adjacent or opposite to each other.
The number of ways in which two adjacent faces of a cuboid can be chosen is 3 ways.
The number of ways in which two opposite faces of a cuboid can be chosen is 3.
$\therefore$ The total number of ways in which two faces of a cuboid to be painted in green colour = 3 + 3 = 6 ways.

8. After covering the cube of dimensions 7 cm × 7 cm × 7 cm, the dimension of the cube will be 9 cm × 9 cm × 9 cm.
$\therefore$ A total of $(9 \times 9 \times 9) - (7 \times 7 \times 7) = 729 - 343 = 386$.

9. The dimensions of $6 \times 8 \times 9$ cube which is kept at corner of a room, after covering with $1 \times 1 \times 1$ cubes will be $7 \times 9 \times 10$.
$\therefore$ A total of $(7 \times 9 \times 10) - (432) = 198$ are required.

10. $125 = 5 \times 5 \times 5$, here $n = 5$, which is odd.
When a cube is cut along two diagonals, the number of pieces that get cut $= 2n^2 - n = 2 \times 5^2 - 5 = 45$ (as n is odd).

11. $1000 = 10 \times 10 \times 10$. Here $n = 10$.
When a cube is cut along a diagonal, the number of pieces that get cut $= n^2 = 10^2 = 100$

12. Cutting the large cube into 27 smaller cubes will give us a $3 \times 3 \times 3$ configuration. Out of these, if we remove all the outer cubes to get the number of cubes not having any face painted at all, we have to remove one layer of cubes on each of the faces so that we are left with a $1 \times 1 \times 1$ cube which is not painted at all. Hence, the answer is one cube.

13. The cubes which are not along any edge are the ones that have only one face painted. On each face of the original cube, if we do not count the faces along the edges, then we have only one face at the middle which is painted only on one face. Hence, for six faces of the original cube, we get six cubes that have only one face painted.

14. The cubes along the edges but not at the corners will have two faces painted. Along each edge, if we remove the corner cubes, there is one cube that has two faces painted. Hence, for 12 edges of the cube, there will be 12 cubes which have only two faces painted.

15. When a cube is painted and cut into n number of smaller pieces along each edge, the total number of smaller cubes that we get will be $n \times n \times n$. From these, if we remove the complete outer layer of the cubes on all faces, we will have all smaller cubes with paint on them removed and we will be left with $(n - 2) \times (n - 2) \times (n - 2)$ cubes. In this case, if the number of cubes that do not have any face painted is 8, it is a $2 \times 2 \times 2$ cube. so, before painting, it must have been a $4 \times 4 \times 4$ cube so the original cube was cut into 64 smaller cubes.

Difficulty Level-1: Foundation

16. On each face of the original large cube, if we remove the outer row of cubes along all the four edges, the remaining 2×2 (= 4) cubes will have exactly one face painted. On all six faces together, there will be 24 cubes that will have exactly one face painted.

17. Along each edge, if we remove the corner cubes, the remaining cubes have two faces painted; since the original cube is cut into $4 \times 4 \times 4$ cubes, on each edge, we will have 2 cubes with exactly two faces painted on all twelve edges we have $2 \times 12 = 24$ cubes.

18. One face red $\Rightarrow$ out of 36 + 30 = 66 cubes (on both the red faces together), we need to remove 6 common cubes which have two faces painted red. Hence, 66 – 6 = 60.

19. There are 3 common edges giving 6 + 6 + 4 cubes which have green and red, i.e., 16.

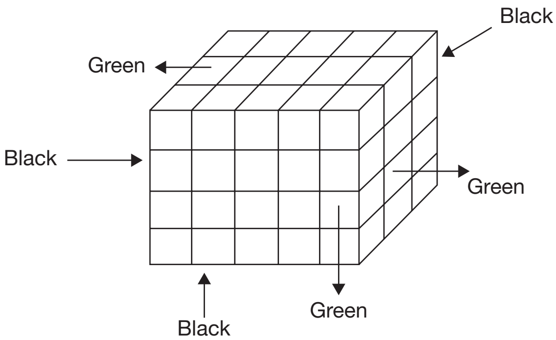
20. Only one colour (blue or green) $\Rightarrow$ We have to consider two possibilities:

(i) The central 4×4 square on a face (which gives 16 cubes).

(ii) The four middle-cubes along the common edge of two faces having the same colour (i.e., two green faces have a common edge of 6 cubes out of which four cubes have only green colour).

Thus, if we take the cubes which have only green colour on their faces, there are 16 cubes for each of the two green faces plus four common cubes – a total of (16 + 16 + 4) = 36 cubes with only green on their faces. Similarly, there will be 36 cubes which have only blue on their faces. Hence, a total of 72 cubes.

Solutions for questions 21 to 23:



Corners (8)	Edges (12)	Faces (6)
BBB – 1	BB – 3	B – 3
GGG – 1	GG – 3	G – 3
GGB – 3	BG – 6	
BBG – 3		

21. At the corners, three such pieces are there.
On the edges, the number of such pieces
= 1 (along the plane with 2 cuts) + 2 (along the plane with 3 cuts) + 3 (along the plane with 4 cuts)
$\therefore$ Required number of pieces = 3 + 6 = 9.

22. Except two corners, all the corners have both the colours $\Rightarrow$ 6 pieces.
On the edges, the number of pieces having both the colours according to the three different cuts in each plane.
$\therefore$ Required number of pieces = 6 + 12 = 18.

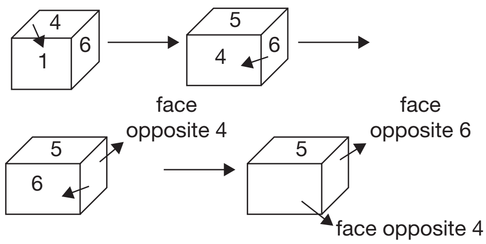
23. The number of pieces with no face painted = 1 (along the plane with 2 cuts) + 2 (along the plane with 3 cuts) + 3 (along the plane with 4 cuts) = 1 + 2 + 3 = 6.

24. $210 = 7 \times 6 \times 5$
The number of pieces that have exactly two colours on them = 4 [(7 – 2) + (6 – 2) + (5 – 2)] = 48.

25. The maximum possible number of pieces with green and red are obtained, when the pieces with green and red are along the side where the length is the highest.
Here, the maximum length is 7 cm.
$\therefore$ The number of pieces, in this case, with green and red $= 4 \times 7 = 28$ pieces.

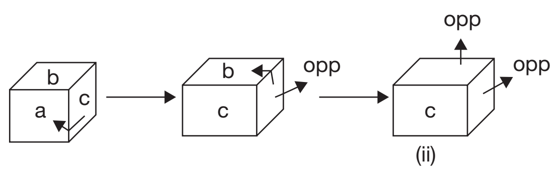
26. The minimum number of pieces with only blue colour on them is obtained when the smaller pieces are at the centre of the face with dimensions 5×6.
$\therefore$ The minimum possible number of required pieces
$= 2 \times (5 – 2)\ (6 – 2) = 24.$

27. From figures (i) and (ii), we understand that 1 is opposite to 5, because 2, 3, 4 and 6 are adjacent to 1. We obtain figure (iii) by rotating the die in figure (ii) as follows:



Hence, 2 is opposite to 4 and 3 is opposite to 6. The first option is correct.

28. From (i) and (ii), *c* is opposite to *f*. The second figure is obtained by rotating the first figure as follows:



Hence, *e* is opposite to *a* and *d* is opposite to *b*. Therefore, *b* – *d* is the correct pair of opposite faces.

29. From (ii) and (iii), red is opposite to violet, from (i) and (ii), blue is opposite to orange. Hence, yellow is opposite to green. The colour at the bottom in the second figure is blue.

30. From (i) and (ii), ★ is opposite to ☐. From (i) and (iii), ↑ is opposite to o . Hence, Δ is opposite to #. The faces adjacent to Δ are o and ↑.

31. From (i) and (ii), t is opposite to u and from (ii) and (iii), s is opposite to p. Hence, r is opposite to q. q and u are opposite to r and t, respectively.

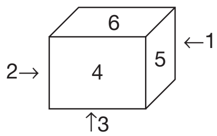
32. From (i) and (ii), 'White' is opposite to 'Green'. From (i) and (iii), 'Black' is opposite to 'Brown'. Hence, 'Grey' is opposite to 'Blue'. 'Blue' is at the bottom of figure (i).

33. From (i) and (ii), five dots are opposite to three dots. From figures (i) and (iii), two dots are opposite to one dot. Hence, six dots are opposite to four dots. The faces adjacent to both the faces with two dots and five dots are opposite to each other. Hence, they are faces with six dots and four dots, whose total is ten.

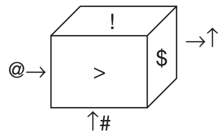
34. From the given figure, it is known that the circle and the triangle are opposite each other
Hence, Only III is correct.

35. From the given figure, it is known that the hatched face and the face with x on it, are opposite each other
Hence, Only I is correct.

36. It forms a cube as shown in the figure below. So, 6 is opposite to 3.

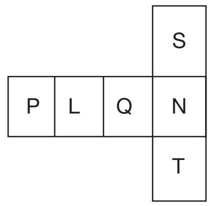


37. It forms a cube as shown in the figure below.



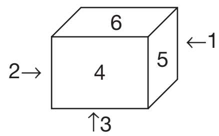
So, '$' is opposite to @.

38.

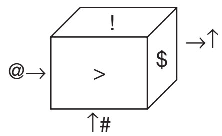


By folding the above figure, the alternate faces are opposite.
P is opposite to Q.
L is opposite to N.
S is opposite to T.

39. It forms a cube as shown in the figure below. So, 6 is opposite to 3.



40. It forms a cube as shown in the figure below.



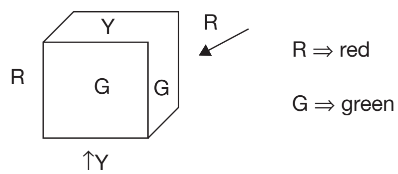
So, '$' is opposite to @.

EXERCISE-2

1. $216 = 6 \times 6 \times 6$
Here, $n = 6$, i.e., even.
(i) With one diagonal cut, the number of pieces that get cut $= n^2 = 6 \times 6 = 36$.
(ii) With two diagonal cuts, the number of pieces that get cut $= 2n^2 = 2 \times 6 \times 6 = 72$.

2. $729 = 9 \times 9 \times 9$
Here, $n = 9$, i.e., odd.
(i) With one diagonal cut, the number of pieces that get cut $= n^2 = 9 \times 9 = 81$.
(ii) With two diagonal cuts the number of pieces that get cut $= 2n^2 - n = 2 \times 9 \times 9 - 9 = 153$.

Solutions for questions 3 to 7: The cube can be painted in the following pattern:



The colour combination for the corners: GGY-2 pieces, RRY –2 pieces, RGY –4 pieces
Edges: GG-1 edge, RR-1edge, GY-4 edges, RY-4 edges, RG-2 edges

Middle of the face: G-2 faces, R-2 faces, Y-2 faces
Since the cube is cut into 125 small and identical cubes the number of small cubes along each edge is 5.

3. The number of small cubes with exactly two faces painted in green are found along the GG-edge, i.e., 5 small cubes.

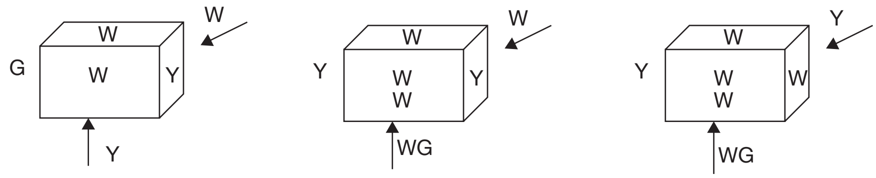
4. The small cubes with at least two different colours on their faces = Pieces at the corners with two colours or three colours + Pieces along the GY, RY and RG edges.
= 8 (corner pieces) + 10×3 (pieces along the edges) = 38.

5. The number of small cubes having exactly one colour on them = Pieces at the GG and RR edges + Pieces at the middle of the faces = $2 \times 3 + 6 \times 9 = 60$.

6. The number of small cubes with yellow or red but not green = Total number of small cubes – The number of small cubes with only green – The number of small cubes with no colour = $125 - (25 + 20) - 27 = 53$.

7. The number of small cubes having exactly two painted faces and have exactly two colours = The smaller cubes along the GY, RY and RG edges
= $10 \times 3 = 30$.

Solutions for questions 8 to 11: The cube can be painted in the following patterns:



The cube is cut into 512 small and identical cubes, i.e., each edge is cut into eight pieces. The colour combination for the pieces along the surface is as following:

	(i)	(ii)	(iii)
Corners	**WWY-2, WWG-2, YYW-2, YWG-2**	**WWY-4, YWG-4**	**WWW-1, WWY-2, YYW-1, WWG-1, YYG-1, YWG-2**
Edges	WW-2, WY-5 WG-3, GY –1 YY-1	WW-2, WY-6 GY-2, GW-2	WW-3, WY-4 YY-1, YG-2, GW-2
Middle of surface	W-3, Y-2, G-1	W-3, Y-2, G-1	W-3, Y-2, G-1

8. The number of small cubes with two faces painted in yellow is:
Case (i):
Corner: (WWY) 2 Cubes + Edge: (YY) 1 edge × 6 cubes = 8
Case (ii):
No such small cube is available.
Case (iii):
Corners: (YYW) 1 cube + (YYG) 1 cube + Edge: (YY) 1 edge × 6 Cube = 1 + 1 + 6 = 8
∴ The minimum possible number of small pieces with two faces painted in yellow is zero.

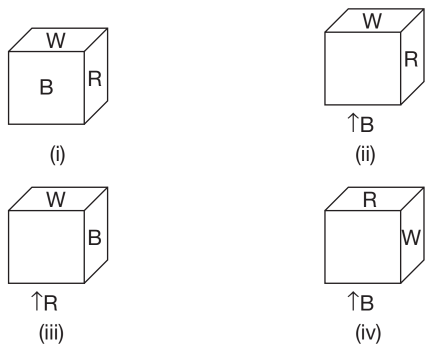
9. The number of small cubes with only green and yellow colours.
Case (i):
(GY) edge 1×6 Cubes = 6
Case (ii):
(GY) edge 2×6 Cubes = 12
Case (iii):
(YYG) corner: 1 cube + (GY) edge $2 \times 6 = 13$
∴ The maximum possible number of small cubes with only yellow and green colour is 13.

10. The number of small cubes with exactly one colour on them:
Case (i):
[(WW) edges 2 + (YY) edge 1] × 6 + middle part of surfaces $6 \times 36 = 234$
Case (ii):
(WW) edges 2×6 + middle parts of surface: $6 \times 36 = 228$
Case (ii):
(WW) corner 1 cube + [(WW) edge 3 + (YY) edge1] × 6 + middle part of surface $6 \times 36 = 241$
∴ The maximum and the minimum possible number of small cubes that have exactly one colour is 241 and 228, respectively.

11. The number of small cubes that have at least two painted surfaces but have only white colour on them:
Case (i):
(WW) edges $2 \times 6 = 12$
Case (ii):
(WW) edges $2 \times 6 = 12$
Case (ii):
(WWW) corner 1cube + (WW) edges $3 \times 6 = 19$

$\therefore$ The minimum possible number of small cubes that have only white colour on them is 12.

Solutions for questions 12 to 16: The painting can be done in the following patterns:



The cube is cut into 729 pieces by $(8 + 8 + 8) = 24$ cuts.

12. The maximum possible number of small cubes with all the three colours can be obtained in figure (i). The corner cube is having all the three colours on it.

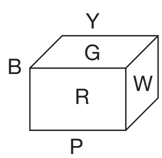
13. In figure (ii), white and black colours are on opposite faces. Hence, the minimum possible number of small cubes that have only white and black colours is zero.

14. The minimum number of the small cubes with only red face can be obtained from the figure (ii), i.e., $(9 - 2) \times 9 = 63$ cubes.

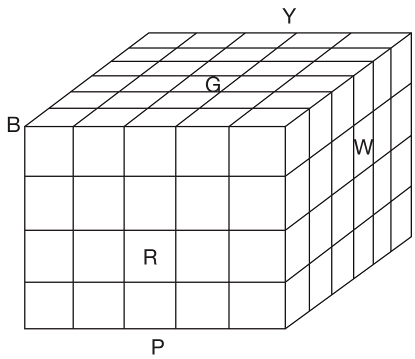
15. From any one of the figures (i), (ii) (iii) and (iv), cubes with white face are $= 9 \times 9 = 81 \Rightarrow 81$.

16. The number of small pieces with no colour = The pieces lying inside + The pieces from the three surfaces which have no colour.
From figure (i): $(9 - 2)^3 + (8 \times 8) + (7 \times 8) + (7 \times 7) = 512$.
From any one of the figures (ii), (iii) and (iv):
$(9 - 2)^3 + (7 \times 8) + (7 \times 8) + (7\times 7) = 504$
$\therefore$The minimum possible number of small pieces that have no colour on them is 504.

Solutions for questions 17 to 19: As per the given directions, the coloured cube will look as follows.



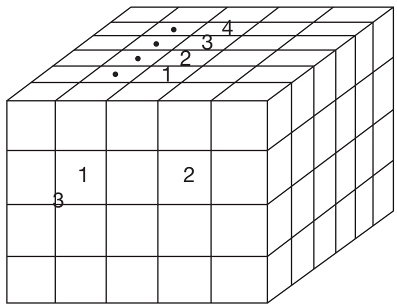
The first letters of all the colours are used in the above diagram.
Now this cube has to be cut into 120 identical pieces by using minimum possible cuts, so the combination can be 4, 5 and 6, i.e., $4 \times 5 \times 6$ and the cuts will be 3, 4 and 5.
Now the cut figure will be as follows.



17. Now the white colour pieces will be 6×4 as per the above diagram, i.e., 24.

18. At least two different colours, means two different colours or more than two colours. We know that more than two colours will be at 8 corners and two colours will be at the 12 edges, so 44 pieces at the edges including 8 pieces at the corners.

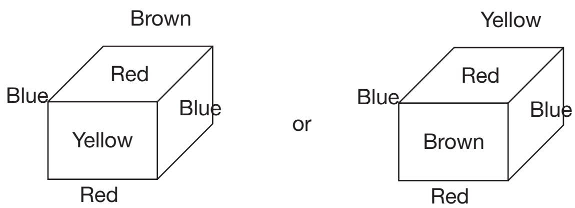
19. It is better to study the cube properly. Let's see the cube given below.



If we see the cube drawn above, then it is clear that the outer faces of the pieces will definitely be painted but the inner faces of the pieces will not be painted at all. So, the simplest approach is to count the pieces from outside to know how many pieces are to be considered, i.e., they are numbered from 1 to 6, leaving the pieces at the edges as they will be painted. Now the number of layers parallel to the front face, on which 1 to 6 are numbered inside will be 4 as shown above. So, 6×4 pieces will have no paint on their faces at all, i.e., 24 pieces.

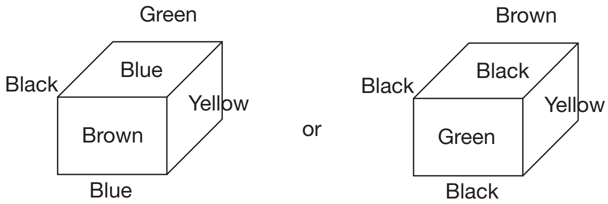
Solutions for questions 20 to 22: Let us draw the diagrams of cube P and Q.

Cube P:

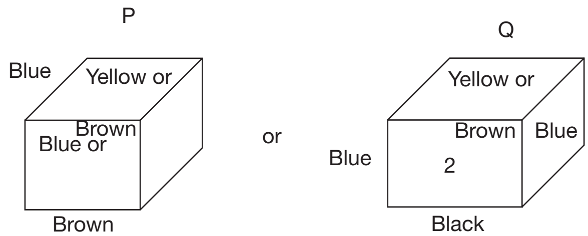


Difficulty Level-2: Moderate

Two opposite faces of blue can be replaced by two opposite reds and vice-versa.
Cube Q:

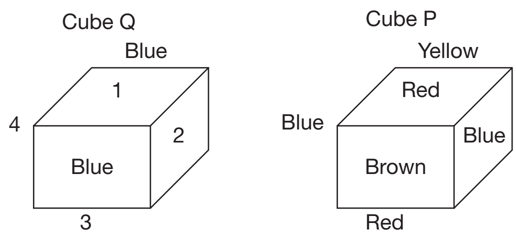


20.



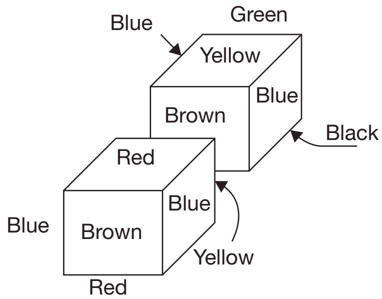
The faces marked with number 2 of cube Q can be of any colour among yellow, brown, black and green. But as far as the front face of the cube is concerned, it has to be of red colour, as the two faces of cubes P and Q which are touching each other are of the same colour even if they interchange their positions. So, those two faces must be of blue colour.

21. Cube Q is placed above P.
Cube Q



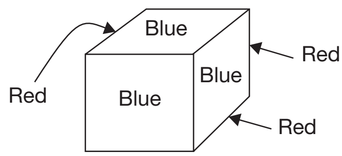
Now in cube Q the faces 1 and 3 can be of brown and green and vice-versa or they can be of black and yellow and vice-versa. On the other hand, faces 2 and 4 can also be of black and yellow and vice-versa or they can be of brown and green or vice-versa. Choice (A), (B) and (C) may or may not be true. Choice (D is definitely false. When Q is above P, then Q's bottom face and P's top face will touch each other and these two faces will have definitely red and any one of the colours, such as black, brown, green and yellow.

22. In all the possible arrangements, one thing is common, i.e., the front face will have brown colour.



If observed carefully, we find that the top faces do not have the same colours and only the two side faces have the same colour on them.

Solutions for questions 23 to 25:



23. It is possible in exactly one way. (Example: When we can see RRR, we cannot see blue.)

24. It is possible in 3 ways.

25. This is possible in 3 ways.

26. The minimum possible number of pieces with two colours on them will be obtained when the faces of the cube to be painted with same colour are on adjacent faces.
In this case, the number of pieces with exactly two faces painted in two different colours = $(12 - 3)\ (6 - 2) = 36$
[As there will be three edges with two faces painted but with only one colour]
The number of pieces with two colours but on three faces $= (8 - 2) = 6$
[As there will be two pieces at corner which have all the three colours]
$\therefore$ Total number of required pieces $= 36 + 6 = 42$

27. The number of pieces with no colours on them $= 4 \times 4 \times 4 = 64$.
For maximum possible number of pieces with one colour on them, the faces which are to be painted with same colour must be adjacent.
The number of pieces with only one colour on exactly one face $= 6 \times (6 - 2)^2 = 96$.
The number of pieces with only one colour but on two faces $= 3 \times (6 - 2) = 12$.
$\therefore$ The total number of pieces $= 64 + 96 + 12 = 172$

Solutions for questions 28 to 30: When three faces of a cube are painted, there are two possibilities, they are: (a) the three faces are continuous (in a row) or (b) the three faces meet at a corner.

Also, when the cube is cut into 216 smaller but identical cubes, we get a $6 \times 6 \times 6$ configuration.

28. Here, also we have to consider both the cases.

(A) When the three painted faces are continuous: On the two outer painted faces, there will be 6×5 (= 30) cubes with exactly one face painted (on two faces, there will be $2 \times 30 = 60$ such cubes) whereas, on the middle face, there will be 4×6 (= 24) cubes with exactly one face painted. Hence, a total of 84 cubes with exactly one face painted.

(B) When the three painted faces meet at one corner: On each of the painted faces, there will be 5×5 cubes (after we remove the cubes along the edges common with the other painted faces); since there will be 3 such edges, a total of 75 cubes.

Hence, the answer is 84 or 75 cubes.

29. Maximum number of two faces painted will be when the three painted faces are continuous in a row. Here, all the six cubes along the common edge between green face and blue face will satisfy the given condition. Hence, the answer is 6 cubes.

30. Here again, we have to consider both the cases of painting.

(A) When the three painted faces are continuous. Here, the number of cubes that have no face painted at all is as follows:

Middle cube – 6×6 cubes = 36

Two end cubes – $2 \times (5 \times 6) = 60$

A total of 96 cubes have painted face out of 216 cubes. Hence, the number of cubes that do not have any face painted at all in this case is 120.

(B) When the three painted faces meet at a corner: Here, the number of cubes having any paint at all is as follows:

Along the three common edges together, there will be $(3 \times 5 + 1 =)$ 16 cubes.

On each of the faces (after the above has been deducted), there will be $(5 \times 5 =)$ 25 cubes. On all three faces together, there will be 75 cubes.

Hence, a total of $(75 + 16 =)$ 91 cubes have any paint at all out of a total of 216 cubes. So, there will be 125 cubes that do not have any face painted at all. Hence, the answer is 120 or 125.

31. Different ways of distributing three colours on the six faces of a cube are as follows:

0-0-6; 0-1-5; 0-2-4; 0-3-3;

1-1-4; 1-2-3; 2-2-2

<u>0-0-6 combination:</u>

<u>Here, all face single colour. Hence, totally, 3 ways.</u>

<u>0-1-5 combination:</u>

1 way $\times$ 3 for selection of colours out of the three available colours $\Rightarrow$ 2 for distribution of these colours between 1 and 5 = <u>6 ways.</u>

<u>0-2-4 combination:</u>

Two faces having one colour can be adjacent or opposite 2 ways. For each of these ways, two colours can be arranged in 3! ways. Hence, 2×6 = <u>12 ways.</u>

<u>0-3-3 combination:</u>

Three faces of same colour on one corner or continuous $\Rightarrow$ 2 ways. For each of these, two colours can be arranged in 3 ways. Hence, 2×3 = <u>6 ways.</u>

<u>1-1-4 combination:</u>

2 single faces adjacent or 2 single face opposite $\Rightarrow$ 2 ways. For each of these, three colours can be arranged in 3 ways. Hence, 2×3 = <u>6 ways.</u>

<u>1-2-3 combination:</u>

3 faces of same colour can be in 2 ways, (A) continuous (B) at a corner.

1. <u>Continuous</u> - Under this, the 2 faces with same colour can be adjacent or opposite $\Rightarrow$ 2. For each of these, the three colours can be arranged in 3! or 6 ways.
Hence, 2×6 = <u>12 ways.</u>

2. <u>At a corner</u> - This gives one arrangement. The three colours can be arranged in 3! ways. Hence, $3! \times 1$ = <u>6 ways.</u>

<u>2-2-2 combination:</u>

Different possibilities are

(I) 2 opp - 2 opp - 2opp - <u>1 way</u>

(II) 2 opp - 2 adj - 2 adj - 1 way. The three colours can be arranged in 3! ways. Hence, <u>3 ways.</u>

(III) 2 adj - 2 adj - 2 adj - <u>1 way</u>

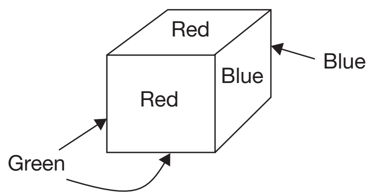
We get the total number of ways in which the cube can be painted by adding all the above values, which is 56.

32. For each of the combinations worked out, blue can come as $1 + 2 + 8 + 4 + 2 + 8 + 4 + 5 = 34$.

33. For the relevant combinations discussed above, the number of ways is $6 + 12 + 6 + 5 = 29$.

34. The only way that the cube can be painted such that no two adjacent faces have the same colour is where opposite faces have the same colour so that the three pairs of opposite faces can have three different colours. Hence, the answer is one way.

Solutions for questions 35 to 37: The following diagram shows the colours of the faces.



Difficulty Level-2: Moderate

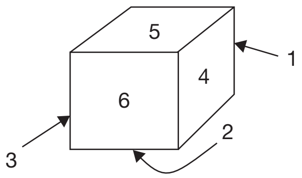
In total, there are eight possible ways of seeing three faces. They are: GGR, GGB, RRG, RRB, BBR, BBG, BGR and BGR.
Note: BGR can be seen in two ways.
B = Blue
G = Green
R = Red

35. Red and blue both are not visible in the following cases: GGR, GGB, RRG and BBG. Both of them are visible in the other four ways.
Th total number of ways is 4.

36. It is possible in two ways.

37. It is already explained above and the required number is 7 [BGR is possible in two ways].

Solutions for questions 38 to 40:



The dice is shown in the following diagram.
The possible ways of views are:
6, 4 and 5 [sum = 15]
6, 5 and 3 [sum = 14]
6, 4 and 2 [sum = 12]
6, 3 and 2 [sum = 11]
5, 4 and 1 [sum = 10]
5, 3 and 1 [sum = 9]
4, 2 and 1 [sum = 7]
3, 2 and 1 [sum = 6]

38. This sum can be obtained in the following ways.
6 + 14 = 20
14 + 6 = 20
9 + 11 = 20
11 + 9 = 20
10 + 10 = 20
Total number of possible ways = 5.

39. Let us take D_1 and D_2 – two dice.
Assume that it is 10 on D_1 and not 10 on D_2.
There are seven possible ways to have this.
Assume that it is 10 on D_2 and not 10 on D_1.
There are another 7 ways.
Assume that it is 10 on both the die.
There is exactly one way for this.
Total number of ways is 7 + 7 + 1 = 15.

40. Any specified number appears in exactly sixteen ways.

EXERCISE-3

Solutions for questions 1 to 3: When three faces of a cube are painted, there are two possibilities.

(i) One pair of opposite faces is painted. The third painted face will be adjacent to these two faces.
(ii) The three painted faces are mutually adjacent to each other. Thus, every painted face is opposite to an unpainted face.

1. By considering the painted cubes in case (i), the number of painted faces = $(5 \times 5) + 2\ (5 \times 4) = 40 + 25 = 65$.
Hence, the total number of cubes which are unpainted = $125 - 65 = 60$.

2. By considering the case (i), there are two edges common and on each of the two edges, there are 5 cubes that have exactly two faces painted.
Hence, total required cubes = 10
By considering the case (ii):
There are three common edges and along each edge, four cubes have exactly two faces painted = $3 \times 4 = 12$.

3. By considering the case (i), on the two opposite faces painted we have $2 \times 5 \times 4 = 40$ cubes which have exactly one face painted and in the other faces, we have $5 \times 3 = 15$.
Hence, total required cubes = 55.
By considering the case (ii), except the cubes at the edges and the corner remaining 4×4 cubes on each painted face have only = $3 \times 16 = 48$.

Solutions for questions 4 to 6: Given that 4, 5 and 6 cuts are made in three different directions.

We can visualize the 42 pieces on one pair of opposite faces, 35 on second pair of opposite faces and 30 on the third pair of opposite faces.

4. To get the maximum number of smaller pieces which have only back on them, the faces which have 42, 42, 35 and 35 pieces have to be painted in black.
Hence, the pieces which are painted only in black = 30 + 30 + 15 + 15 = 90.

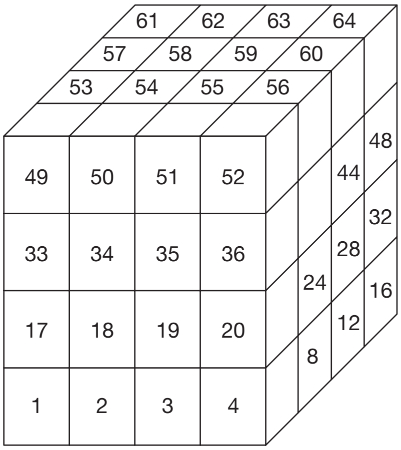
5. To get the maximum number of smaller pieces with black and yellow on them, the opposite faces which consists of 42 cuboids each are painted with black, the other pair of opposite faces which consists of 35 cuboids each are painted with yellow and the out of the remaining pair of opposite faces, one face is to be painted with black or yellow and the other has to be painted red.
Colours on faces: Opposite faces 7×6 = black/yellow.
Opposite face 7×5 = yellow/black.

One face of 5×6 = yellow/black (not black/yellow).
Other face of 5×6 = red.
There are six cuboids including corners (excluding those corners which have red paint) on 4 edges and 4 cuboids on 2 edges each which have only black and yellow on them.
Hence, the required number of cuboids

$$= 4 \times 6 + 2 \times 4 = 32.$$

6. By painting the three pairs of opposite faces of the larger cuboid with the three different colours, we have 8 corner pieces with three different colours on them.

Solutions for questions from 7 to 10:



7. The numbers on the referred cubes are 1, 2, 3 and 4, whose sum is 10.

8. The required numbers are 1, 17, 33 and 49. These are in AP with a common difference of 16, whose sum is 100.

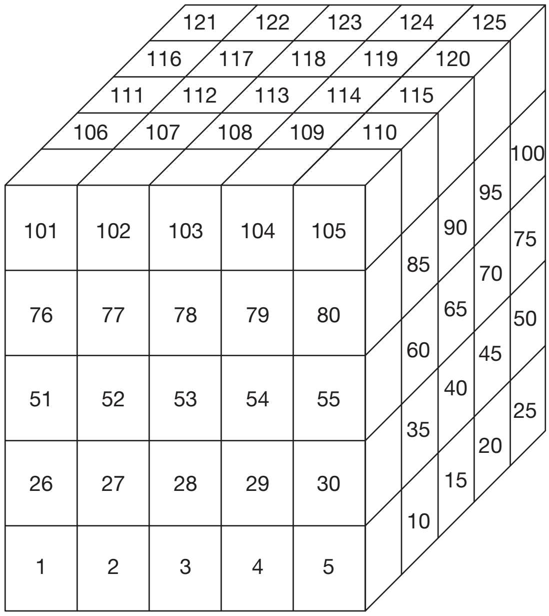
9. The required numbers are 13, 29, 45 and 61. These numbers are in AP with a common difference of 16, whose sum is 148.

Alternate Solution:

This scenario can be viewed in relation to that in previous question. Each required cube, now, is 12 cubes away from each required cube of previous questions (from bottom to top).
Therefore, sum = $100 + 12 \times 4 = 148$.

10. The required numbers are 53, 54, 55 and 56, whose sum is 218.

Solutions for questions from 11 to 14:



11. The required numbers are 7, 32, 57, 82 and 107. These numbers are in AP with a common difference of 25, whose sum is 285.

12. The required numbers are 101, 107, 113, 119 and 125. These numbers are in AP with a common difference of 6, whose sum is 565.

13. The required numbers are 1, 32, 63, 94 and 125. These numbers are in AP with a common difference of 31, whose sum is 315.

14. The required numbers are 5, 34, 63, 92 and 121. These numbers are in AP with a common difference of 29, whose sum is 315.

Solutions for questions 15 to 18:

Total number of identical wooden pieces
$= 455 = 5 \times 7 \times 13$ (All prime factors of 455)
Let the dimensions of the large wooden cube be 455 cm $\times$ 455 cm $\times$ 455 cm.
Then the dimensions of each identical piece would be 5 cm $\times$ 7 cm $\times$ 13 cm.
Cubes thrown away = $11 \times 5 \times 3 = 165$.
Cubes left with = $455 - 165 = 290$.

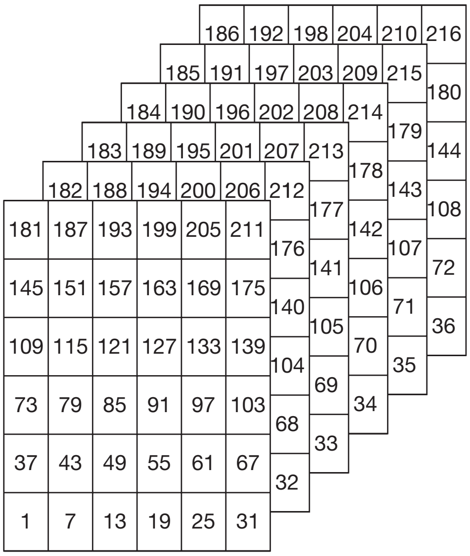
15. Total number of cuts made = $12 + 6 + 4 = 22$

16. Pieces that are on opposite faces are identical. Among them, the pieces that are painted on only one face are highest in number. Among them, the two opposite faces on which 12 × 6 cuts are made will be identical. Total number of such pieces = (11 × 5) × 2 = 110.

17. The edges which are cut into 13 pieces have 11 pieces in a particular pair of colours painted on exactly two faces. There are four such edges. Hence, in total there are 44 such pieces.

18. The edges which are cut into 5 pieces have three pieces, along each of these edges, which have the same pair of colours on them. There are four such edges. Hence, there are 12 such pieces.

Solutions for questions 19 to 22: The arrangement of the large cube is as follows:



19. Four corners of the top layer have three sides painted.
181 + 211 + 186 + 216 = 794

20. The following diagram shows the bottom layer (with shaded region representing cubes with at least one side painted).

6	12	18	24	30	36
5	11	17	23	29	35
4	10	16	22	28	34
3	9	15	21	27	33
2	8	14	20	26	32
1	7	13	19	25	31

Sum of all cubes (bottom layer) = $\frac{36}{2} \times (36 + 1) = 37 \times 18 = 666$

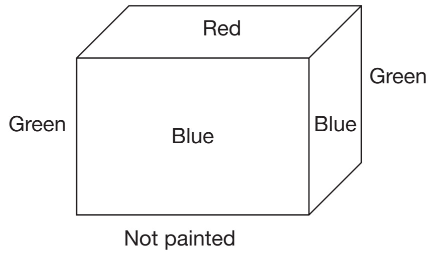
Sum of cubes of shaded region = $\frac{5}{2} \times ((1 + 5) + (6 + 30) + (36 + 32) + (31 + 7)) = \frac{5}{2} \times (6 + 36 + 38 + 66) = \frac{5}{2} \times (146) = 370$

Answer = 666 – 370 = 296

21. The cubes are on front row of the top layer; the numbers are in arithmetic progression.
Sum =181 + 187 + 193 + 199 + 205 + 211 = 3 × (181 + 211) = 1176

22. There are two such diagonals. There will be six cubes in a diagonal and the numbers will be in arithmetic progression. Sum of the numbers on any diagonal is the same though the common difference varies. The six numbers are in arithmetic progression from 211 to 186 or 181 to 216. Sum = 3 × (211 + 186) = 3 × (181 + 216) = 1191

Solutions for questions 23 to 25: The painted faces of the cube are shown below:



As the cube is cut into 512 small and identical cubes, there are 8 cubes along each edge.

23. Smaller cubes having exactly one of their faces painted
= The cubes along the edge [except corner cubes] of the not painted face + The cubes on the painted faces [except corners and edges] = 24 + 36 × 5 = 24 + 180 = 204.

24. At most two painted faces = All cubes – Three painted faces = 512 – 4 = 508

25. Cubes having only one colour on them
= Painted only in blue + Painted only in green + Painted only in red
= (36 × 2 + 6 × 3 + 1) + (36 × 2 + 6 × 3 + 1) + 36
= (72 + 18 + 1) + (72 + 18 + 1) + 36
= 182 + 36 = 218

9 Deductions

CHAPTER

Learning Objectives

In this chapter, you will:

- Understand the correct logical interpretation of terms like 'All', 'Some', 'No', 'Some not' etc.
- Learn how to derive logical conclusions from abstract statements using various methods.

Questions based on deductions are frequently asked in competitive exams. These types of questions are generally solved by using two methods:

1. Venn diagrams
2. Rules governing syllogisms

Syllogism rules are preferred for answering those questions with two statements and do not have 'either – or' answer choice. Whereas Venn diagram method is applicable to answering questions with any number of statements and any kind of answer choices.

We are going to discuss Venn diagram method first and the syllogism concept later.

The statements given in the questions and the conclusions that are derived may not confine to generally accepted facts. None of the three statements below is a fact, but they still may be a part of a question.

Example:

1. All cats are dogs.
2. Some birds are elephants.
3. Some flowers are not mountains.

To understand and analyse these statements and to draw conclusions, we can use symbolic logic for clear expression of our thoughts. The examinee has to understand the logical implications of the given statements and verify the truthfulness of each of the given conclusions, strictly within the preview of the given statements. Each of the given statements has to be taken as true, though they deviate from generally accepted facts and check whether the given conclusions logically follow the statements.

To achieve the above task, the given statements have to be represented in a combined format. Representation through Venn diagrams is an effective way to combine and to draw conclusions based on these hypothetical statements.

❒ VENN DIAGRAMS METHOD

Venn Diagrams: These are diagrammatic/pictorial representation of sets by using geometrical figures. The Venn diagram drawn to represent all the given statements should be a combined diagram. A set of given statements can be represented in several ways using Venn diagrams. We can conclude that a conclusion definitely follows the given statements only if that conclusion is true for all possible diagrammatic representations.

Quantifiers: A quantifier describes the extent (quantity) to which one kind (or term) is similar (or dissimilar) to another kind (or term). The main quantifiers are 'All', 'No', 'Some' and 'Some-not'. The following are few examples of statements/ conclusions consisting of each of the above four quantifiers.

All: All A's are B's, All animals are living things, All shoes are socks, etc.

No: No A is B, No boy is girl, No bat is rat, No weak is coward, etc.

Some: Some A's are B's, Some doctors are men, Most girls are brave, etc.

Some – not: Some A's are not B's, Some Cricketers are not Indians, etc.

Words like 'a few, most, many, more', etc., are treated as synonyms to 'Some', 'Not all' is equivalent to 'Some-not'.

The statements, which contain the qualifiers 'All' and 'Some' are called affirmative statements and those containing the qualifiers 'No' and 'Some-not' are called negative statements.

❐ COMPLEMENTARY PAIR

Certain combinations of conclusions, consisting of one negative and the other affirmative, negate each other. For example,

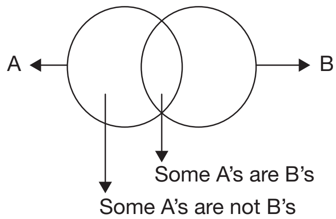
'**SOME** A's are B's' negates '**NO** A is B'.

'**ALL** A's are B's' negates **SOME** A's are **NOT** B's, i.e., when 'Some A's are B's' is true, 'No A is B' cannot be true and vice versa.

Similarly, when 'All A's are B's' is true, then 'Some A's are not B's' cannot be true and vice versa.

In each of the above pairs, only one statement can be true or false at a time but both cannot be true or false at the same time.

Thus, 'Some A's are B's' and 'Some A's are not B's' does not form a complementary pair, as both can be true at the same time, as in the following figure.

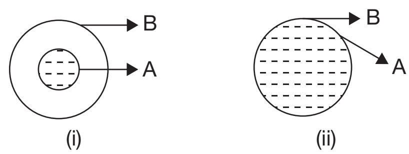
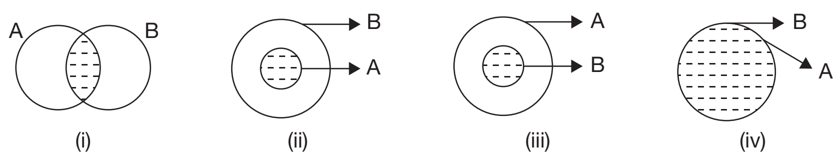
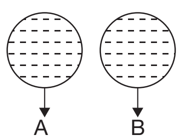
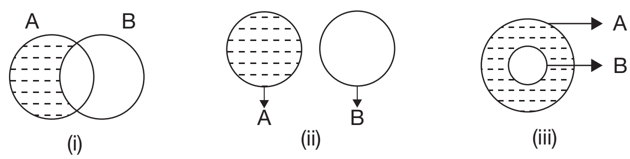


Similarly, 'All A's are B's' and 'No A is B' does not form a complementary pair because they both are false at the same time as for the above diagrams.

Thus, the pairs of qualifiers ('Some' and 'No') and ('Some-not' and 'All'), for the same terms, form complementary pairs. The existence of conclusions can be observed while reading the question itself.

The following table shows different ways of representing a statement consisting of a qualifier by using Venn diagrams.

Table – 1

Qualifier	Representations using venn diagram
1) ALL: Example: All A's are B's	
2) SOME: Example: Some A's are B's	
3) NO: Example: No A is B	
4) SOME, NOT: Example: Some A's are not B's	

From the above table, it is clear that a statement can be represented diagrammatically in several ways. Similarly, a diagram may represent more then one statement.

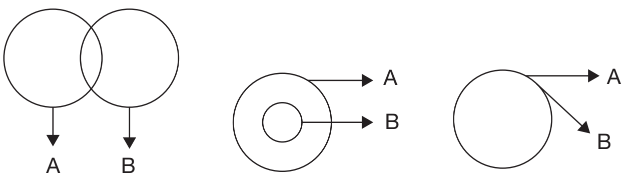
From the above table, we can draw the following possible conclusions.

Statement	Definitely true	Definitely false	May or may not be true
(1) All A's are B's	Some A's are B's Some B's are A's	No A is B No B is A	All B's are A's Some B's are not A's
(2) Some A's are B's	Some B's are A's	No A is B No B is A	All A's are B's All B's are A's Some A's are not B's Some B's are not A's
(3) No A is B	No B is A Some A's are not B's Some B's are not A's	Some A's are B's Some B's are A's All A's are B's All B's are A's	________
(4) Some A's are not B's	________	All A's are B's Some A's are B's Some B's are A's	Some B's are not A's No A is B No B is A All B's are A's

The diagrams are also classified as 'Basic Diagrams' (BD) and 'Alternate Diagrams' (AD), based on intersections and extent of overlap.

Basic Diagram (BD):

This is a diagram which represents the least possible situation for a given statement. To get the least possible representation, the diagram should contain minimum overlapping. The extent of overlap is of three kinds as shown below.



Minimum (a) Moderate (b) Maximum (c)

In Figure (a), circles (A) and (B) are overlapping with each other only to some extent, i.e., minimum for both.

In Figure (b), one circle, i.e., B is completely overlapped by A, but circle A is overlapped by B only to some extent. Here, the extent of coverage of one circle is full and the other is partial, i.e., the overlap is moderate on the whole.

In Figure (c), each of the circles is overlapped completely by the other, i.e., overlapping is the maximum.

In Table I for statement (1), Figure (i) has lesser overlapping than Figure (ii). Hence, Figure (i) forms the BD for statement (1). Similarly, Figure (i) for statement (2) has least overlapping among all possible diagrams for that statement. Hence, Figure (i) forms BD for statement (2). Similarly, Figure (ii) for statement (4) is the BD for it. For statement (3), only one diagram is possible.

Alternate Diagram (AD):

Any diagram, other than BD for the given statements is an alternate diagram. For each set of statements, several alternate diagrams are possible.

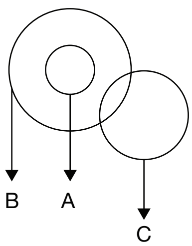
Method to draw Venn diagrams for the given statements:

Each question contains two or more statements. The Venn diagrams that we draw to represent these statements should be a combined diagram, i.e., the diagram should link all the given statements. The following examples show how a combined diagram is drawn.

Example 1:

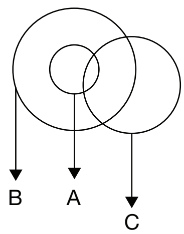
Statements: All A's are B's.
Some B's are C's.

(a)



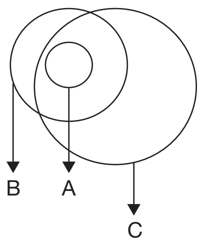
(B D)

(b)



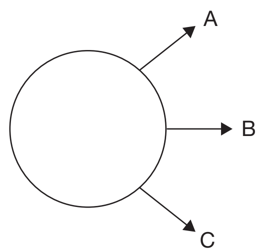
(A D)

(c)



(A D)

(d)



(A D)

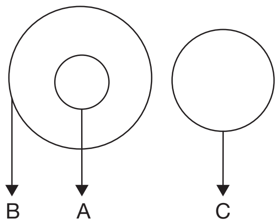
Figure (a) is the BD, as the overlapping for each of A, B and C is least in view of the given statements.

Figures (b) and (c) are the ADs because the overlap is moderate. Several other ADs are possible for the given statements. Figure (d) is the AD with the maximum overlap.

Example 2:

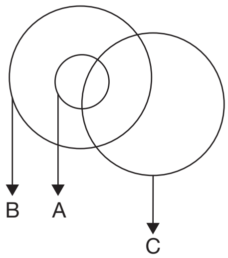
Statements: All A's are B's.
Some A's are not C's.

(a)



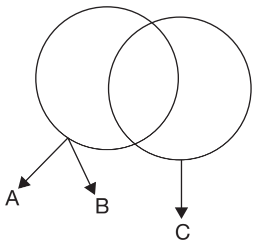
(B D)

(b)



(A D)

(c)



(A D)

Figure (a) is the BD, as it has minimum overlapping. Figure (b) is the AD with moderate overlapping and Figure (c) is the AD with maximum overlapping.

While drawing these diagrams it has to be ensured that no diagram contradicts the given statements.

Answering the questions:

In each question, the statements are followed by three or more conclusions. The student has to verify whether the given conclusions follow the statements or not. Conclusion is said to follow the given statement, if it is true for all possible Venn diagrams for a given set of statements. We can see that in Examples (1) and (2), several diagrams are possible. Instead of drawing all possible diagrams and verifying the truthfulness of each conclusion in each of these diagrams, it would be convenient if we can verify the truthfulness of these conclusions by using minimum number of diagrams. A student has to think logically in order to minimize the number of diagrams required to verify the truthfulness of a conclusion.

Guidelines to minimize the number of diagrams:

(i) We know that BD represents the least possible situation for the given set of statements. If an affirmative conclusion is true for BD, then it will be true for all ADs.

(ii) If an affirmative conclusion or a negative conclusion is false for BD, then it can be said that the conclusion does not follow the given statements.

(iii) If a negative conclusion is true for BD, it may or may not be true for ADs.

Let us understand the above with the help of the following example.

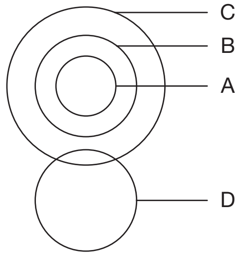
Example 1:

Statements: All A's are B's.
All B's are C's.
Some C's are D's.

Conclusions: (I) All A's are C's.
(II) Some B's are D's.
(III) No A is D.
(IV) No C is A.

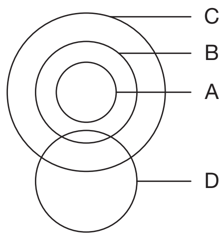
The following are some of the possible diagrams for the given set of statements.

(a)



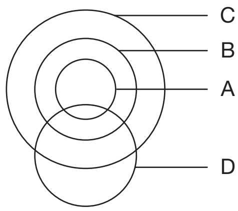
(B D)

(b)



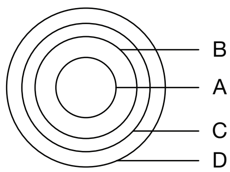
(A D)

(c)



(A D)

(d)



(A D)

Conclusion (I), 'All A's are C's', is affirmative and is true for the basic diagram (a). This means that it will be true for all the ADs. Hence, conclusion (I) follows the given statements. Conclusion (II), 'Some B's are D's', is affirmative and is false for the BD. Therefore, it does not follow the given statements. Conclusion (III), 'No A is D' is negative and is true for BD. This means that we will have to draw some ADs to verify its truthfulness. It observed that the statement is true for the AD (b), but false for the ADs (c) and (d). Hence, it can be concluded that conclusion (III) does not follow the given statements. Conclusion (IV), 'No C is A' is negative and is false in the BD. Hence, it does not follow the given statements.

Analysis:

Since BD represents the least possible situation, for the given statements, an affirmative conclusion, which is true for the least possible situation will always be true. On the other hand, a negative conclusion, though true for the possible situation, may or may not be true for the other situations.

From the above example, it is clear that we need to draw AD only when a negative conclusion becomes true for BD.

As per the above information, the truthfulness of such negative conclusion should be checked by drawing all possible ADs. But instead of checking in so many ADs, we need to draw only one AD, in which the statement, which is complementary to this particular conclusion is true. Hence, if the 'complementary conclusion' turns out to be true, then the conclusion under consideration is false. While trying to draw such AD, it has to be ensured that no given statement is negated in the AD. If such AD can be drawn, then the negative conclusion does not follow the given statements; otherwise such a conclusion is always true.

If a complementary pair exists in the given conclusions, then either negative conclusion of that pair is true and the affirmative conclusion is false for the BD, or the negative conclusion becomes false for the BD and the affirmative conclusion becomes true. In such circumstances, we choose the answer choice in terms of 'either-or'. There may be occasions where the negative conclusions always remain true for all ADs and the affirmative statements are always false. In such a case, the 'either-or' situation does not arise.

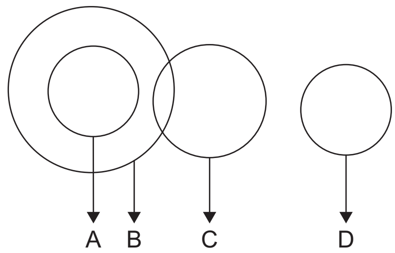
The following example will be helpful in understanding the above concept.

Example 2:

Statements: All A's are B's.
Some B's are C's.
No B is D.

Conclusion: (I) No C is D.
(II) No A is D.
(III) Some C's are not D's.

Solution:



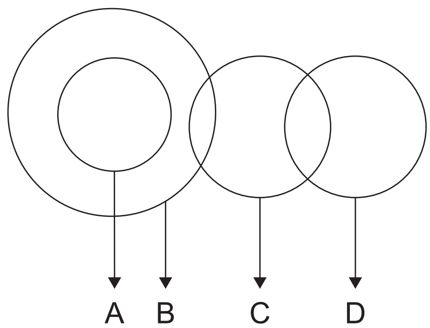
Above diagram is the BD for the given statements.

Conclusion (I) is negative and is true.

Conclusion (II) is negative and is true.

Conclusion (III) is negative and is true.

To prove conclusion (I), 'No C is D' to be false, we have to prove that its complementary conclusion, 'Some C's are D's' is true. Hence, the AD for this will be as shown below:



The above diagram does not defy any of the given statements. Conclusion (I), 'No C is D' is false for this diagram. Hence, it is not valid.

To negate conclusion (II), 'No A is D' and (III), 'Some C's are not D's', we have to prove that their respective complementary conclusions 'Some A's are D's' and 'All C's are D's' are true. This is possible, only if D encroaches into B. But this will violate statement (3). Hence, no diagram can be drawn to negate these two conclusions. Hence, only conclusion (II) and (III) follows.

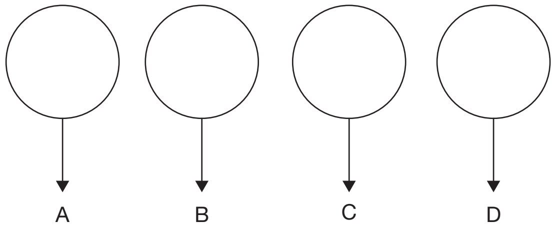
Thus, we are able to answer the question with only two diagrams. From the BD, we can verify the truthfulness of each statement and accordingly decide whether a conclusion follows the given statements or not. In case of a negative conclusion becoming true, we go for AD to prove the negative conclusion is false. If such AD is possible, the negative statement does not follow, otherwise it follows the given statements. In certain rare cases we may have to go for a second AD, as shown in the following example.

Example 3:

Statements: Some A's are not B's.
Some B's are not C's.
Some C's are not D's.

Conclusions: (I) Some A's are not D's.
(II) Some B's are not D's.
(III) Some A's are not C's.
(IV) Some C's are not A's.

Basic diagram:

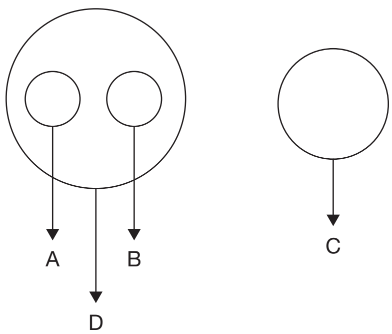


Conclusion (I) is negative and is true.
Conclusion (II) is negative and is true.
Conclusion (III) is negative and is true.
Conclusion (IV) is negative and is true.

We observe that:

'All A's are D's' is complementary to conclusion (I).
'All B's are D's' is complementary to conclusion (II).
'All A's are C's' is complementary to conclusion (III).
'All C's are A's' is complementary to conclusion (IV).

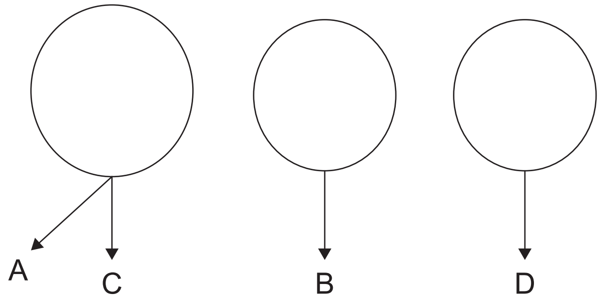
Alternate diagram 1:



Conclusion (I) is false.
Conclusion (II) is false.

To negate conclusions (III) and (IV), we have to draw another alternate diagram.

Alternate diagram 2:



Conclusion (III) is false.
Conclusion (IV) is false.
Hence, none of the conclusions follows.

From the above example, it is also clear that no conclusion can be drawn when all the statements are negative.

Let us take some more examples.

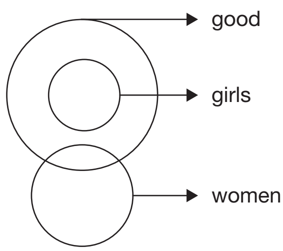
Example 4:

Statements: All girls are good.
Some good are women.

Conclusions: (I) Some women are girls.
(II) No woman is a girl.

Clearly, the above two conclusions form a complementary pair.

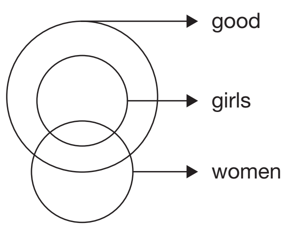
Basic diagram:



Conclusion (I) is affirmative and is false.
Conclusion (II) is negative and is true.

Now, we should draw an AD, which would make conclusion (II) false, i.e., which proves 'Some women are girls'.

Alternate diagram:



From the above diagram, conclusion (II) is false, but conclusion (I) is true at the same time. But they both cannot be true or false at the same time. Hence, either (I) or (II) follows.

Example 5:

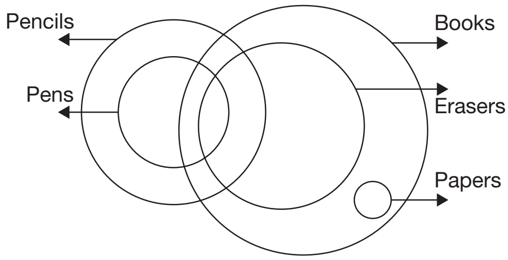
Statements: All pens are pencils.
Some pens are erasers.
All papers are books.
All erasers are books.

Conclusions: (I) Some pencils are books.
(II) Some books are pens.
(III) No paper is pen.
(IV) Some erasers are papers

(A) Only I, II and IV (B) Only I and II
(C) Either I or II (D) Only III and IV

Solution:

The given statements can be represented in the following basic diagram.

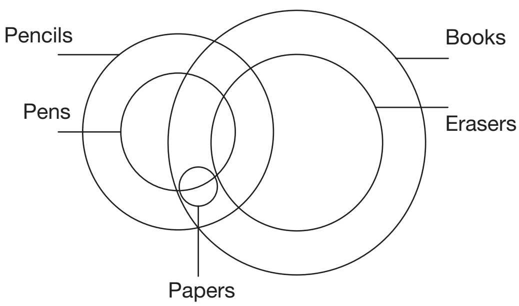


From the above diagram, we incur the following:

Conclusion I, affirmative, follows.
Conclusion II, affirmative, follows.
Conclusion III, negative follows.
Conclusion IV, affirmative, does not follow.

As the affirmative conclusions (I and II) are true in the basic diagram, they will always be true. The affirmative conclusion (IV) is false in the basic diagram. Even if it is true in other diagrams, it cannot be said to be true as there is a situation, where it is false.

The negative conclusion III, which is true in the basic diagram has to be checked whether it can be false in any alternate diagram. The following is such diagram.



There is a situation, where conclusion III is false. Hence, only I and II are true.

Example 6:

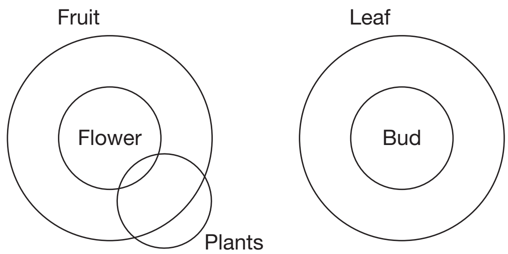
Statements: No leaves are fruits.
Flower is fruit.
Every bud is leaf.
Some plants are flowers.

Conclusions: (I) No flower is bud.
(II) Some plants are not fruits.
(III) Some plants are not leaves.
(IV) Some leaves are not buds.

(A) Only I (B) Only II
(C) Both I and III (D) Both II and III

Solution:

The given statements can be represented in the following basic diagram.



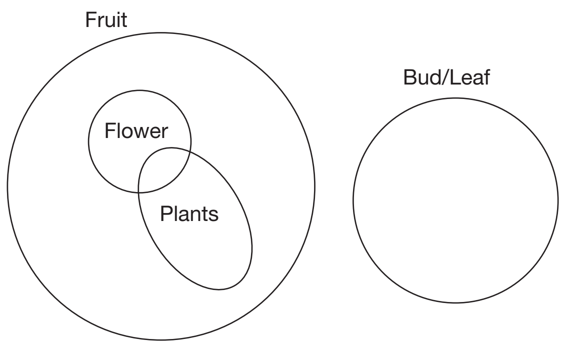
From the above diagram, we incur the following:

Conclusion I, negative, follows.
Conclusion II, negative, does not follow.
Conclusion III, negative, follows.
Conclusion IV, negative, follows.

As the negative conclusions are true for the basic diagram, let us try to negate them. As no leaf is fruit (given statement), no flower which is inside the circle fruit can ever be a bud which is inside the circle leaf. Hence, conclusion I cannot be negated. Similarly, some plants which are inside the circle flowers can never be leaves. Hence, conclusion III cannot be negated.

Conclusion IV can be negated in the following alternate diagram.

The alternate diagram is given as:



Hence, only I and III follow.

Summary:

1. The words 'Some', 'Some-not', 'All' and 'No' are called qualifiers.
2. Words like 'Many', 'More', 'a Few', 'Most', etc., are treated as synonyms to 'Some'. Words like 'Not-all' is treated equivalent to 'Some-not'.
3. Statements/conclusions with qualifiers 'Some' and 'All' are classified as affirmative statements/conclusions and those with qualifiers 'Some-not' and 'No' are classified as negative statements/conclusions.
4. ('Some', 'No') and ('Some-not', 'All') form complementary pairs.
5. Diagram with the least encroachment is called Basic Diagram (BD) and all other diagrams are called Alternate Diagrams (AD).
6. Truthfulness of conclusions are first tested for BD. Only when a negative conclusion is true for BD, AD is required to be drawn in such a way that this negative conclusion which was true for BD becomes false in the AD. If such AD is not possible, then the conclusion is always true.
7. Whenever there is a complementary pair of conclusions, check for 'either – or' as answer.
8. When all the statements are negative, no conclusion can be drawn. In such a case, the answer is always 'None of these'.
9. As a corollary, when all the conclusions are affirmative, BD alone is sufficient to answer the question

❒ SYLLOGISM METHOD

Here we will look at arriving at the deduction by using some simple rules.

First, let us look at some basic terms used in the rules and understand what they mean.

The two statements given in the question are called 'premises' and the answer, the conclusion.

Example: All dogs are cats (i)
All cats are pigs (ii)

These two statements are called 'premises.'

Conclusion: All dogs are pigs.

The premises normally start with the words All, No, Some and Some - Not.

The word 'All' has its synonyms, such as Every, Any, Each, whereas the word 'Some' can also be replaced by Many, Few, A little, Most of, Much of, More, etc.

These words are referred to as qualifiers (also termed as quantifiers).

A premise consists of a subject and a predicate wherein the first term [For example: 'dogs' in statement (i)] is the subject and the second term [For example: 'cats' in statement (i)] is the predicate. Similarly, in statement (ii), 'cats' is called the subject and 'pigs' is the predicate.

The word that occurs in both the premises is known as the 'middle term' ('cat' in the example, given above). The answer or 'conclusion' should consist of the other two words ('dogs' and 'pigs' in the above example) and the middle term should not appear in the answer.

The premises can be divided into:

1. Universal statements
2. Particular statements

This classification of the premises into the above categories is dependent on the qualifier used in the premise. For example, the statements where 'All' is used are called universal statements and the statements where 'Some' is used are called particular statements.

Premises can also be divided into:

1. Positive (affirmative) statements
2. Negative statements

If there is a negative term like 'not' or 'no' in the statement, it is called a negative premise. Otherwise it is called a positive premise or an affirmative statement.

The combination of the two different categories of classifications leads to four different premises as given in Table 2 below.

Table – 1

	Affirmative	Negative
Universal	All A	No E
Particular	some; many I	some not; many not O

The subject or the predicate can be either distributed or not distributed in the given premise.

The subject and the predicate are either distributed (✓) or not distributed (✗) depending upon the type of statement given (particular, affirmative, etc.). The following table shows the distribution pattern of the subject and the predicate.

Table – 2

	Subject	Predicate
Universal affirmative	✓	×
Universal negative	✓	✓
Particular affirmative	×	×
Particular negative	×	✓

Note: ✓ indicates distributed
× indicates undistributed

❒ RULES FOR DEDUCTIONS

1. Every deduction should contain three and only three distinct terms.
2. If both the premises are particular, no conclusion can be drawn.
3. If both the premises are negative, no conclusion can be drawn.
4. The middle term must be distributed at least once in the premises.
5. If one premise is particular, then the conclusion must be particular.
6. If one premise is negative, then the conclusion must be negative.
7. No term can be distributed in the conclusion, if it is not distributed in the premises.

We take examples of each type and look at them in detail.

Example 1:

All dogs are cats. (i)
All cats are pigs. (ii)

As the first statement is a universal affirmative statement, the subject (dogs) has to be distributed (✓) and the predicate (cats) is not distributed (×). As the second statement is also universal affirmative, the subject cat is distributed (✓) and the predicate pigs is not distributed (×). The above answer/logic is arrived at on the basis of Table II.

The middle term ('cats' is the middle term as it occurs in both the premises) is distributed once in the premises. Hence, it satisfies Rule 2. As 'dogs' is distributed in the premise and 'pigs' is undistributed in the deduction also, they should appear accordingly. The type of statement that satisfies both of them is universal affirmative statement, i.e., a statement with 'All'. Hence, the answer will be 'All dogs are pigs'.

The answer cannot be 'All pigs are dogs', because Rule 7 states that no term can be distributed in the conclusion if it is not distributed in the premises. As 'pigs' is not distributed in the premise, it cannot be distributed in the conclusion (because if we take 'All pigs are dogs', then the subject 'pigs' will be distributed). Hence, the conclusion 'All pigs are dogs' is wrong.

Example 2:

All cats are dogs. (i)
All cats are pigs. (ii)

Statement (i) is universal affirmative and hence, the subject 'cats' is distributed and the predicate 'dogs' is not distributed as per Table II.

Statement (ii) is also universal affirmative and hence, the subject 'cats' is distributed and the predicate 'pigs' is not distributed as per Table II.

Here, the middle term 'cats' ('cats' is the middle term as it is occurring in both the premises) is distributed. Hence, we can draw a conclusion.

The answer should contain the terms 'dogs' and 'pigs' and both the terms are not distributed. Referring to Table II, we find that this is possible only in particular affirmative [the conclusion cannot start with the qualifier 'All' as the subject in 'All' should be distributed]. According to Rule 7, a term cannot be distributed in the conclusion if it is not distributed in the premises. So, the answer will be as follows.

Some dogs are pigs.
or
Some pigs are dogs.

Example 3:

All dogs are cats. (i)
All pigs are cats. (ii)

Statement (i) is universal affirmative and hence, the subject 'dogs' is distributed and the predicate 'cats' is not distributed. In statement (ii), which is also a universal affirmative, the subject 'pigs' is distributed and the predicate 'cats' is not distributed. This is arrived at on the basis of Table II.

The middle term 'cats' ['cats' is the middle term as it occurs in both the statements] is not distributed in either of the two statements. From Rule 2, which states that the middle term should be distributed at least once in the premises for drawing a conclusion, we cannot draw any conclusion in this case.

Example 4:

All cats are dogs. (i)
Some cats are pigs. (ii)

The first statement is a universal affirmative premise and hence, the subject 'cats' is distributed and the predicate 'dogs' is not distributed (×). The second statement is particular affirmative and hence, both the subject 'cats' and the predicate 'pigs' are not distributed (×) as per Table II. As we have a particular premise, the conclusion should also be a particular one as per Rule 4. The middle term is distributed and we can draw a conclusion. So, the answer will be as follows.

Some dogs are pigs.
or
Some pigs are dogs.

Example 5:

All dogs are cats. (i)
No cats are pigs. (ii)

As the first premise is a universal affirmative, the subject (dogs) is distributed and the predicate (cats) is not distributed. In the second premise, which is a universal negative, the first term (cats) and the second term (pigs) are both distributed (as per Table II). As the middle term is distributed at least once in the premises, Rule 2 is satisfied and hence, we can draw a conclusion.

From Rule 3, which states that if one of the premises is negative the conclusion should be negative, the conclusion should be negative and as both the terms 'dogs' and 'pigs' are distributed, the conclusion should be a universal negative statement. Hence, the answer will be as follows.

No dogs are pigs.
or
No pigs are dogs.

Example 6:

All dogs are cats. (i)
Some cats are pigs. (ii)

Since the first statement is a universal affirmative, 'dogs' is distributed and 'cats' is not distributed. Since the second statement is a particular affirmative, 'cats' is not distributed and 'pigs' is also not distributed (as per Table II).

In the above given example, no conclusion can be drawn, as Rule 2 states that the middle term ('cats' in the above example as it occurs in both the premises) should be distributed at least once in the premises, which is not satisfied.

Example 7:

All dogs are cats. (i)
Some cats are not pigs. (ii)

The first statement is a universal affirmative and hence, the subject (dogs) is distributed and the predicate (cats) is not distributed.

The second statement is a particular negative and hence, the subject (cats) is not distributed and the predicate (pigs) is distributed (Table II).

But as the middle term (cats) is not distributed at least once in the premises, Rule 2 is not satisfied and hence we cannot draw any conclusion.

Example – 8

All cats are dogs. (i)
Some cats are not pigs. (ii)

The first statement is a universal affirmative and hence, 'cats' is distributed and 'dogs' is not distributed. The second statement is a particular negative and hence, 'cats' is not distributed and 'pigs' is distributed (as per Table II).

Here, the middle term (cats) is distributed and hence, we can draw a conclusion.

The conclusion should be particular negative as Rule 3 states that if a premise is negative, the conclusion should also be negative. Also Rule 4 states that if a premise is particular, the conclusion should also be particular. Hence, the conclusion should be a particular negative.

In particular negative, we know that the subject is not distributed and the predicate is distributed.

The terms 'dogs' and 'pigs' should come in the conclusion. Also, since 'dogs' is not distributed in the premise, it cannot be distributed in the conclusion, as per Rule 7.

As per the above reasoning, only 'pigs' can be the predicate in the conclusion and hence, 'dogs' will be the subject.

Thus, the answer will be 'Some dogs are not pigs'.

EXERCISE-1

Directions for questions 1 to 15: Select the alternative that logically follows from the two given statements, but not from one statement alone.

1. All cooks are drivers.
All drivers are dancers.
(A) All dancers are drivers.
(B) All cooks are dancers.
(C) No cook is dancer.
(D) Both (A) and (B)

2. All sane are men.
Some sane are insane.
(A) Some men are insane.
(B) No men are insane.
(C) All men are insane.
(D) None of the above

3. No plane is ship.
No ship is bus.
(A) No bus is plane.
(B) No plane is bus.
(C) Both (A) and (B)
(D) None follows

4. All cats are dogs.
No dogs are rats.
(A) All cats are rats.
(B) Some cats are rats.
(C) No cat is rat.
(D) None of the above

5. Some books are brooks.
Some brooks are not cooks.
(A) Some books are cooks.
(B) Some books are not cooks.
(C) Both (A) and (B)
(D) None follows

6. Some waters are rivers.
All rivers are oceans.
(A) Some waters are not oceans.
(B) Some oceans are not waters.
(C) Some waters are oceans.
(D) Both (A) and (C)

7. All papers are books.
Some books are not diaries.
(A) No paper is a diary.
(B) Some papers are not diaries.
(C) Some papers are diaries.
(D) None follows

8. No white is black.
No black is red.
(A) Some white is red.
(B) Some white is not red.
(C) No white is red.
(D) None follows

9. Some shirts are trousers.
Some trousers are pants.
(A) Some shirts are not pants.
(B) No shirt is pant.
(C) Some pants are not shirts.
(D) None follows

10. All Ws are Ps.
All Ws are Ks.
(A) All Ps are Ks. (B) All Ks are Ps.
(C) Some Ps are Ks. (D) Both (B) and (C)

11. All Asians are Indians.
No Asian is an African.
(A) Some Africans are not Indians.
(B) No Indian is an African.
(C) Some Indians are not African.
(D) Both (B) and (C)

12. Some animals are humans.
Some animals are not birds.
(A) Some humans are not birds.
(B) Some birds are not humans.
(C) No bird is a human.
(D) None follows

13. All mobiles are electronics.
Some mobiles are smart phones.
(A) Some electronics are smart phones.
(B) Some smart phones are electronics.
(C) Some electronics are not smart phones.
(D) Both (A) and (B) follow.

14. All roads are wide.
Some wide are not highways.
(A) Some highways are not roads.
(B) Some roads are highways.
(C) No highway is a road.
(D) None follows

15. Some vessels are bottles.
No vessel is a container.
(A) Some containers are not bottles.
(B) Some bottles are not containers.
(C) No bottle is a container.
(D) None follows

Directions for questions 16 to 20: Each of these questions consist of two statements followed by two conclusions marked I and II. Consider the statements to be true, even though they seem to be at variance with the commonly known facts and find out which of the given conclusion(s) logically follow(s) the statements, disregarding the commonly known facts. Mark your answer as follows:

(A) If only I follows.
(B) If only II follows.
(C) If neither I nor II follows.
(D) If both I and II follow.

16. Statements: All seas are bees.
Some teas are bees.
Conclusions: I. All teas are seas.
II. Some seas are teas.

17. Statements: All grapes are apples.
All apples are mangoes.
Conclusions: I. All grapes are mangoes.
II. All mangoes are grapes.

18. Statements: Some doctors are lawyers.
Some lawyers are architects.
Conclusions: I. Some doctors are architects.
II. All architects are doctors.

19. Statements: All weddings are writings.
All weddings are wirings.
Conclusions: I. Some writings are wirings.
II. All writings are wirings.

20. Statements: Some queues are rows.
No row is a circular.
Conclusions: I. All circular are queues.
II. Some circulars are queues.

Directions for questions 21 to 25: The questions given below have four groups of three statements each. Read the statements in each group carefully and identify the group/groups where the third statement logically follows the first two statements in the group.

21. (a) All books are copies. All copies are papers. All books are papers.
(b) All cubes are squares. All cubes are triangles. All triangles are squares.
(c) All singers are dancers. All dancers are musicians. All musicians are singers.
(d) No cock is hen. All hens are chickens. No hen is chicken.
(A) Only d (B) Only a
(C) Only b and c (D) Only a and d

22. (a) Some journals are magazines. Some magazines are periodic. Some journals are periodic.
(b) Some horror is ghost. All ghosts are faints. Some horror are faints.
(c) All scientists are researchers. All researchers are professors. Some professors are scientists.
(d) Many baggages are luggages. All luggages are packages. Some packages are not baggages.
(A) Only b, c and d (B) Only b and c
(C) Only a and d (D) Only b

23. (a) Many fountains are cascades. No waterfall is fountain. Some cascades are not waterfalls.
(b) No bag is pack. No pack is jack. No jack is bag.
(c) No good is bad. All bad is not good. Some good is not good.
(d) Scale is ruler. No ruler is pointer. No pointer is scale.
(A) Only a (B) Only d
(C) Only b and c (D) Only a and d

24. (a) No esthetic is an atheist. Some esthetics are monotheists. Some monotheists are not polytheists.
(b) All sentences are words. No word does not have meaning. Some which do not have meanings are not sentences.
(c) No river is sea. Some seas are oceans. Some oceans are not rivers.
(d) No MMTS is MRTS. All public transports are MRTS. No public transport is MMTS.
(A) Only b, c and d (B) Only c and d
(C) Only a and b (D) Only b and c

25. (a) No sitar is a guitar. No guitar is violin. No violin is a sitar.
(b) Ragas are songs. Some pops are not songs. Some pops are not ragas.
(c) Some costume designers are not hair designers. All designers are not hair designers. Some designers are not costume designers.
(d) AC's are not DCs. Some DCs are not BC's. Some AC's are not BC's.
(A) Only b and d (B) Only a and d
(C) Only b (D) Only a and b

Directions for questions 26 to 30: Each question below has four groups of three statements each. Read the statements in each group carefully and identify the group/groups where the third statement logically follows the first two statements in the group.

26. (a) All bohemians are transcendent. Some transcedents are vermishells. Some bohemians are vermishells.

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(b) Some milks are cheese. No cheese is butter. Some butters are not milks.
(c) All hookahs are smokes. All smokes are pungents. Some hookahs are pungents.
(d) No white is black. All blacks are silks. Some silks are not whites.
(A) Only c and d (B) Only a
(C) Only a and b (D) a, b, c and d

27. (a) Few peanuts are groundnuts. Many coconuts are peanuts. Some groundnuts are not coconuts.
(b) Every one is loyal. All honest are one. Some loyal are honest.
(c) No chings are changs. Some twangs are not changes. Some twangs are not chings.
(d) People are hardworking. Some hardworking are successful. All successful are people.
(A) Only c (B) Only b
(C) Only b and d (D) Only a and c

28. (a) Tigers are kings. Kings are deers. Tigers are deers.
(b) Palika is a very good neighbour. Palika is humble. Some humble are not very good neighbour.
(c) Some friends are best. No best is enemy. Some friends are enemy.
(d) All mothers are goodness. All mothers are females. All females are goodness.
(A) Only a and b (B) Only a
(C) Only b and c (D) Only a, c and d

29. (a) All pinks are purple. All purple are violet. All violet are pinks.
(b) No chord is scale. Some scales are minors. No chord is minor.
(c) Some pollutions are dusts. Some dusts are harmfuls. Some pollutions are harmfuls.
(d) No calcium is protein. All vitamins are calcium. No vitamin is protein.
(A) Only b (B) Only a and c
(C) Only d (D) Only a and d

30. (a) All formidable are fearless. All warriors are fearless. All formidable are warriors.
(b) Some pubs are casino. No casino is public. Some pubs are not public.
(c) Some sages are ages. All fages are sages. Some ages are fages.
(d) No wine is old. No gold is old. No wine is gold.
(A) Only a and c (B) Only a and d
(C) Only b (D) Only c

Directions for questions 31 to 35: In each of the following questions, three statements followed by four conclusions marked I, II and III are given. Consider the statements to be true, even though they seem to be at variance with the commonly known facts and find out which of the given conclusion(s) logically follow(s) the statements, disregarding the commonly known facts.

31. Statements: All digits are symbols.
All symbols are letters.
Some letters are elements.
Conclusions: I. All digits are letters.
II. Some symbols are elements.
III. All letters are digits.
(A) Only I and II follow.
(B) Only II and III follow.
(C) Only I and III follow.
(D) Only I follows.

32. Statements: All fats are mats.
Some mats are rats.
All rats are cats.
Conclusions: I. Some fats are cats.
II. Some mats are cats.
III. Some rats are fats.
(A) Only I follows. (B) Only II follows.
(C) Only III follows. (D) Both I and II follow.

33. Statements: All inputs are outputs.
Some outputs are results.
No result is good.
Conclusions: I. Some inputs are results.
II. Some goods are not outputs.
III. Some inputs are good.
(A) Only I follows.
(B) Only II follows.
(C) Both II and III follow.
(D) None follows

34. Statements: No pen is pencil.
No pencil is paper.
No paper is board.
Conclusions: I. No pen is paper.
II. Some pencils are not boards.
III. No board is pen.
(A) Only I follows. (B) Only II follows.
(C) Only III follows. (D) None follows

35. Statements: No one is two.
Some two are threes.
All four are two.
Conclusions: I. Some four are threes.
II. No one is a four.
III. Some four are not one.
(A) Only I and II follow.
(B) Only I and III follow.
(C) Only II and III follow.
(D) Only I follows.

Directions for questions 36 to 40: Each of these questions consists of six statements followed by sets of three statements each. Find the set in which the third statement can be logically concluded from the first two statements.

36. (a) No wolf is a tiger.
(b) No deer is tiger.
(c) Some bears are not tigers.
(d) Some deers are not bears.
(e) Some bears are wolves.
(f) No deer is wolf.
(A) aec (B) bfd
(C) efd (D) fbd

37. (a) Anything which is kind is gentle.
(b) Everything which is gentle is not hard.
(c) Nothing which is firm is gentle.
(d) Something which is firm is hard.
(e) Many things which are hard are not kind.
(f) Nothing which is kind is firm.
(A) acf
(B) dfe
(C) abe
(D) More than one of the above

38. (a) Screening is sedimentation.
(b) Sedimentation is purification.
(c) Purification is filtration.
(d) Filtration is screening.
(e) Screening is purification.
(f) Sedimentation is filtration.
(A) cde (B) bfc
(C) bcf (D) aeb

39. (a) No fish is bird.
(b) Flyers are gliders.
(c) No flyer is fish.
(d) Some gliders are birds.
(e) No flyer is bird.
(f) Some flyers are birds.
(A) dbf (B) ace
(C) efd (D) None of these

40. (a) Junk food contains more fat.
(b) Fast food is not healthy.
(c) Fast food does not contain more fat.
(d) Junk foods are fast foods.
(e) Some fast food are junk foods.
(f) Junk food is not healthy.
(A) ace (B) ade
(C) bfe (D) bdf

EXERCISE-2

Directions for questions 1 to 18: Each of these questions consists of six statements followed by sets of three statements each. Find the set in which the statements are logically related.

1. (a) Shed is not shelter.
(b) Roof is protection.
(c) Roof is shed.
(d) Roof is shelter.
(e) Some shelter is not protection.
(f) Shed is protection.
(A) cda
(B) aef
(C) bcf
(D) More than one of the above

2. (a) Engineers are not doctors.
(b) Some doctors are psychologists.
(c) Some doctors are not professors.
(d) Some engineers are professors.
(e) No professor is a psychologist.
(f) Some psychologists are not engineers.
(A) acd (B) def
(C) bfa (D) All the above

3. (a) All cricketers are footballers.
(b) All footballers are magicians.
(c) All magicians are cricketers.
(d) Some cricketers are footballers.
(e) Some footballers are magicians.
(f) Some magicians are cricketers.
(A) abc (B) efb
(C) bcd (D) def

4. (a) Some RCs are not DCs.
(b) All PCs are ACs.
(c) Some ACs are not RCs.
(d) Some ACs are not DCs.
(e) Many RCs are PCs.
(f) Some PCs are not DCs.
(A) ceb (B) fdb
(C) afe (D) dbe

5. (a) Truss is not roof.
(b) Truss is not timber.
(c) Post is roof.
(d) Timber is roof.
(e) Post is not truss.
(f) Timber is post.

(A) bfe
(B) cae
(C) cdf
(D) More than one of the above

6. (a) All plays are puzzles.
(b) All riddles are plays.
(c) Some games are puzzles.
(d) No puzzle is game.
(e) Some games are riddles.
(f) All riddles are puzzles.
(A) afd (B) cef
(C) fab (D) Both (B) and (C)

7. (a) Some figures are curves.
(b) Some curves are squares.
(c) Some squares are figures.
(d) Some squares are not triangles.
(e) All curves are figures.
(f) No figure is triangle.
(A) cdf (B) def
(C) acb (D) abe

8. (a) No shoe is black.
(b) Some black is leather.
(c) Some leather are shoes.
(d) All black is dark.
(e) Some shoes are black.
(f) No shoe is dark.
(A) afd (B) abd
(C) ced (D) def

9. (a) All roses are lillies.
(b) All lillies are jasmines.
(c) All jasmines are roses.
(d) Some orchids are lillies.
(e) Some orchids are roses.
(f) Some orchids are jasmines.
(A) abc (B) dea
(C) efd (D) bef

10. (a) Some directories are dictionaries.
(b) Some yellow pages are not dictionaries
(c) All directories are information.
(d) No dictionary is directory.
(e) All directories are yellow pages.
(f) All information are yellow pages.
(A) ade (B) def
(C) bde (D) abc

11. (a) All sad are anxious.
(b) Some mood are depressed.
(c) No sad is anxious.
(d) All moods are sad.
(e) All sad are depressed.
(f) All anxious are moods.
(A) bde (B) cdf
(C) adf (D) ade

12. (a) All prayers are tantras.
(b) Some mantras are prayers.
(c) Chantings are mantras.
(d) No mantra is tantra.
(e) All tantras are chantings.
(f) Some prayers are chantings.
(A) cde (B) efa
(C) bcf (D) Both (B) and (C)

13. (a) All gliders are smoother.
(b) All gliders are shiner.
(c) Some shiner are mirror.
(d) Some smoother are shiner.
(e) All smoother are shiner.
(f) No mirror is smoother.
(A) cdf (B) abe
(C) cef (D) acd

14. (a) All seconds are minutes.
(b) All minutes are hours.
(c) All hours are days.
(d) Some hours are minutes.
(e) Some minutes are seconds.
(f) Some seconds are hours.
(A) efb (B) fda
(C) ade (D) dfe

15. (a) Some rooms are dormitories.
(b) All suites are hotels.
(c) No building is a dormitory.
(d) Some suites are not dormitories.
(e) All hotels are buildings.
(f) Some dormitories are not hotels.
(A) cbf (B) eba
(C) cfe (D) adc

16. (a) No fibre is wood.
(b) Some fibre is rubber.
(c) Some wood is not fibre.
(d) Some rubber is not wood.
(e) No stick is fibre.
(f) All sticks are wood.
(A) eaf (B) cfe
(C) adb (D) All the above

17. (a) Some doors are not locks.
(b) Some locks are not glasses.
(c) Some glasses are not keys.
(d) Some doors are not glasses.
(e) All keys are locks.
(f) No key is a door.
(A) aef (B) cbe
(C) fcd (D) None of the above

18. (a) All lions are elephants.
(b) No pig is a tiger.
(c) Some elephants are not pigs.
(d) Some lions are not pigs.
(e) Some pigs are elephants.
(f) Some lions are tigers.
(A) fbd (B) dca
(C) fde (D) Both (A) and (B)

Directions for questions 19 to 23: In each of the following questions, four statements followed by four conclusions are given. Consider the statements to be true even though they appear to be at variance with the commonly known facts. Find which of the conclusion(s) logically follow(s) the given statements, disregarding the commonly known facts and choose appropriate answer choice.

19. Statements: All pedals are frames.
All frames are roses.
All hubs are roses.
All keys are hubs.

Conclusions: I. All roses are pedals.
II. All keys are roses.
III. Some hubs are frames.
IV. Some frames are keys.

(A) Only II follows.
(B) Only II and III follow.
(C) Only III and IV follow.
(D) Only I, II and III follow.

20. Statements: Some baskets are caskets.
Some caskets are trunks.
All trunks are fans.
All sweets are fans.

Conclusions: I. Atleast some baskets are trunks is a possibility.
II. Atleast some fans are caskets is a possibility.
III. All fans are baskets is a possibility.
IV. Atleast some sweets are not caskets is a possibility.

(A) Only I and III follow.
(B) Only II and IV follow.
(C) Only I, II and III follow.
(D) All follow

21. Statements: Some forks are spades.
Some spades are not shovels.
All chisels are shovels.
No potato is a chisel.

Conclusions: I. Some shovels are not potatoes is a possibility.
II. Atleast one chisel is a spade is a possibility.
III. All potatoes are shovels is a possibility.
IV. Some forks are chisels is a possibility.

(A) Only III and IV follows.
(B) Only II and IV follow.
(C) Only I, II and IV follow.
(D) All follow.

22. Statements: All bolts are nuts.
All chips are washers.
Some screws are nuts.
All nuts are washers.

Conclusions: I. All bolts are washers.
II. Some washers are screws.
III. Some washers are bolts.
IV. All chips are screws.

(A) Only I follows.
(B) Only I and II follow.
(C) Only I, II and III follow.
(D) All follow

23. Statements: Some doctors are actors.
Some actors are teachers.
All dancers are teachers.
All doctors are engineers.

Conclusions: I. Some actors are engineers.
II. Some teachers are engineers.
III. No engineer is a teacher.
IV. All teachers are doctors.

(A) Either II or III follows
(B) Only I follows
(C) Only I and III follow
(D) Only I and exactly one of II or III follows

Directions for questions 24 to 30: Select the alternative that logically follows from the two given statements, but not from one statement alone.

24. Some shirts are pants.
All pants are shorts.
(A) No shirt is shorts.
(B) Some shirts are shorts.
(C) All shirts are shorts.
(D) None of the above

25. Some gauges are cages.
Some cages are not catches.
(A) Some guages are not catches.
(B) No guage is a catch.
(C) Some guages are catches.
(D) None of the above

26. No red is black.
All blue are black.
(A) All red are blue.
(B) Some red are blue.
(C) No red is blue.
(D) None of the above

27. Some hammers are tools.
All tools are made of iron.
(A) Some hammers are made of iron.
(B) Some hammers are not made of iron.
(C) No hammer is made of iron.
(D) None of the above

28. Some tools are not hammers.
All tools are made of iron.
(A) Some hammers are made of iron.
(B) Some hammers are not made of iron.
(C) Some tools made of iron are not hammers.
(D) None of the above

29. All cigarettes are cigars.
Some cigarettes are not good for health.
(A) Some cigars are not good for health.
(B) Some cigars are good for health.
(C) No cigar is good for health.
(D) Both (A) and (B)

30. Some MBAs are CEOs.
All CEOs are Directors.
(A) Some MBAs are not Directors.
(B) Some MBAs are Directors.
(C) Both (A) and (B)
(D) None of the above

Directions for questions 31 to 35: Each of the following questions consists of four statements followed by four conclusions. Consider the statements to be true even if they vary from the normally known facts and find out which of the conclusion(s) logically follow(s) the given statements and choose the proper alternative from the given choices.

31. Statement: Some watches are clocks.
Some clocks are times.
Some times are fast.
Life is fast.
Conclusions:
I. Some watches are life.
II. Some lifes are time.
III. Some clocks are fast is a possibility.
IV. Some watches are fast is not a possibility.
(A) Only I and II
(B) Only III follows.
(C) Only II, III and IV
(D) All follow

32. Statements: Earth is tree.
All trees are branches.
All branches are leaves.
All branches are flowers.
Conclusions: I. Earth is a flower.
II. All flowers are leaves is a possibility.
III. Some trees are flowers.
IV. No flower is leaf is not a possibility.
(A) Only II and IV
(B) Only I, II and III
(C) Only I and III
(D) All follow.

33. Statements: Some boys are engineers.
All engineers are graduates.
All graduates are literate.
Some girls are literate.
Conclusions: I. No boy is a girl.
II. Some boys are literates.
III. Some girls are engineers.
IV. All engineers are literate.
(A) Only II and IV
(B) Only I, II and IV
(C) Only II, IV and I or III
(D) Only III and II

34. Statements: No rice is curd.
All rice is grain.
Oats are grains.
No flour is grain.
Conclusions: I. No curd is grain.
II. No rice is oat.
III. No flour is oat.
IV. Some oats are curds.
(A) Only I, II and III
(B) Only III
(C) Only II and III
(D) Only III and II or IV

35. Statements: Flowers are beautiful.
No beautiful is ugly.
Coal is ugly.
Beautiful is attractive.
Conclusions: I. Flowers are attractive.
II. No flower is coal.
III. Some flowers are not ugly.
IV. Some attractive are not coal.
(A) Only I
(B) Only I, II and III
(C) Only II, III and IV
(D) All follow

Directions for questions 36 to 40: Each of these questions consists of three/four statements and four choices. Consider the statements to be true, even though they seem to be at variance with the commonly known facts and find out which choice logically does not follow the given statements, disregarding the commonly known facts:

36. Statements: Some bazaar are beach.
No beach is a beauty.
All beauty is bean.

(A) Some beaches being beans is not a possibility.
(B) Some beauty is bean.
(C) Some bazaar being bean is a possibility.
(D) All bean being beauty is a possibility.

37. Statements: All chat are compete.
Some chat are cherry.
All clean are compete.

(A) Some compete are cherry.
(B) Some clean are compete.
(C) Some clean are chat.
(D) All cherry being compete is a possibility.

38. Statements: All giant are glow.
Some glow are not locks.
Some locks are music.

(A) Some giant are glow.
(B) Some music being glow is a possibility.
(C) Some locks are giant.
(D) All glows being giant is a possibility.

39. Statements: Some mute are sound.
Some sound are pink.
Some sweet are sound.

(A) Some pink being mute is a possibility.
(B) All mute being sound is not a possibility.
(C) Some pink are sound.
(D) Some mute being pink is a possibility.

40. Statements: No battle is paper.
Some paper are word.
All battle are amount.

(A) All paper being word is a possibility.
(B) Some amount are battle.
(C) Some word being amount is a possibility.
(D) Some battles are papers.

EXERCISE-3

Directions for questions 1 to 7: Each question below has four groups of three statements each. Read the statements in each group carefully and identify the group/groups where the third statement logically follows the first two statements in the group.

1. (a) A few gasoline are flammable. All flammable are ignition. Many gasoline are ignition.
(b) All suppers are breakfast. Some supper are not dinner. Some breakfast are not dinner.
(c) No electron is nucleus. All protons are electrons. Some protons are not nucleus.
(d) Some cars are not taxies. All cars are buses. Some buses are not taxies.

(A) Only b (B) Only c and d
(C) Only a and b (D) All of them

2. (a) All colours are walls. No wall is plain. No colour is plain.
(b) All sonic are cosmic. Some cosmic are superfast. All sonic are superfast.
(c) Every amount is refundable. Some amounts are retainable. Some refundable is retainable.
(d) No one is novice. Some novice is shrewd. Some shrewd are not one.

(A) Only a and b (B) Only a, c and d
(C) Only b and c (D) Only c and d

3. (a) All even are numbers. Some numbers are odd. Some odd are even.
(b) Some coins are rupees. No dollar is a rupee. Some coins are not dollars.
(c) All shirts are cloths. Some good are not cloths. Some shirts are not good.
(d) Some glasses are spectacles. Some fibre are not glasses. Some fibre are spectacles.

(A) Only a (B) Only b
(C) Only c (D) Only b and d

4. (a) All relations are friends. No relation is an enemy. Some friends are not enemies.
(b) No relation is an enemy. All relations are friends. Some enemies are not friends.
(c) All bikes are vehicles. All vehicles are useful. All useful are bikes.
(d) All bikes are vehicles. All useful are bikes. Some vehicles are useful.

(A) Only b and c (B) Only a and c
(C) Only a and d (D) Only b and d

5. (a) All protocols are rules. All protocols are mandatory. All rules are mandatory.
(b) All fruits are eatable. No eatable is vegetable. No vegetable is a fruit.
(c) No protocol is a rule. Some rules are mandatory. Some mandatory are not protocols.
(d) Some eatable are fruits. Some vegetables are eatable. Some fruits are vegetables.
(A) Only a and b (B) Only b and c
(C) Only c and d (D) Only a and d

6. (a) All imports are taxable. Some exports are not taxable. Some exports are not imports.
(b) Banks are financial institutions. Some financial institutions are not NBFC. Some NBFC are not banks.
(c) No gold is silver. Some metals are silver. Some gold are not metals.
(d) Some books are files. Some manuals are not books. Some manuals are not files.
(A) Only b (B) Only d and b
(C) Only a and c (D) Only a

7. (a) Bulbs are bright. No bright is flammable. Some bulbs are not flammable.
(b) Some black are not pens. All blue are pens. Some black are not blue.
(c) Some debts are NPA. All NPA are bad debts. Some debts are bad debts.
(d) Some laptops are desktops. No tab is a desktop. Some tabs are not laptops.
(A) Only b, c and d (B) Only b and c
(C) Only a, b and c (D) All follow

Directions for questions 8 to 17: Each of these questions consists of six statements followed by several sets of three statements each. Select your answer from the given sets in which the statements are logically related.

8. (L) Every ball is round.
(M) Some balls are rings.
(N) All which are round are spheres.
(P) All rings are round.
(Q) Some rings are spheres.
(R) Some rings are not spheres.
(A) PML (B) NPL
(C) PQL (D) QNP

9. (L) Some truths are lies.
(M) No false is true.
(N) Some false are truths.
(P) All lies are false.
(Q) Some lies are not false.
(R) All false are wrong.
(A) PQR (B) NQR
(C) LPN (D) MQR

10. (L) Some teams are great.
(M) No good is great.
(N) Some teams are not good.
(P) Some players are great.
(Q) All players are good.
(R) All players are teams.
(A) PQM (B) RNQ
(C) PRL (D) MPN

11. (L) Some women are old.
(M) Some men are not old.
(N) Some engineers are women.
(P) All men are young.
(Q) All engineers are old.
(R) Some men are engineers.
(A) NQL (B) PMR
(C) LNR (D) None of these

12. (L) No cup is a saucer.
(M) Some cups are not fly.
(N) All cups are big.
(P) Some saucers are flying.
(Q) No saucer is big.
(R) Some which fly are not cups.
(A) LMP (B) RPL
(C) NQL (D) Both (B) and (C)

13. (L) Some hexagons are not pentagons.
(M) No square is a rectangle.
(N) All rectangles are pentagons.
(P) No pentagon is an octagon.
(Q) Some hexagons are not rectangles.
(R) Some quadrilaterals are not squares.
(A) NPM (B) NQL
(C) MNR (D) PQL

14. (L) Some cubes are prisms.
(M) No prism is a pyramid.
(N) No cube is a pyramid.
(P) All prisms are pyramids.
(Q) Some prisms are not pyramids.
(R) All cubes are symmetrical.
(A) NQR (B) PRN
(C) LNP (D) LNQ

15. (L) Some chocolates are good.
(M) Some fats are not good.
(N) No chocolate is protein.
(P) All proteins are good.
(Q) No protein is a fat.
(R) Some chocolates are proteins.
(A) PNL (B) LPR
(C) MPQ (D) None of these

16. (L) All grass is brass.
(M) No brass is copper.
(N) Some copper are metals.
(P) All metals are gold.
(Q) Some copper are not grass.
(R) Some copper are gold.
(A) LMQ (B) NPR
(C) MNP (D) Both (A) and (B)

17. (L) No kite is a rhombus.
(M) All rhombuses are quadrilaterals.
(N) Some rectangles are squares.
(P) All squares are quadrilaterals.
(Q) No square is a rhombus.
(R) Some quadrilaterals are not kites.
(A) LMR (B) LNP
(C) PQN (D) None of these

Directions for questions 18 to 22: Each of these questions consists of six statements followed by four sets of three statements each. Select your answer in the given sets in which the statements are logically related.

18. (a) A few tufts are combs.
(b) All crests are combs.
(c) No crest is tuft.
(d) All crests are tufts.
(e) A few crests are not combs.
(f) A few tufts are not combs.
(A) abd (B) abc
(C) eda (D) bcf

19. (a) A few straps are not curbs.
(b) Some curbs are not chains.
(c) All curbs are chains.
(d) Many straps are chains.
(e) Many chains are curbs.
(f) Some straps are not chains.
(A) acf (B) abf
(C) dae (D) afe

20. (a) Some desks are not decks.
(b) No slope is desk.
(c) Some slopes are desks.
(d) No desk is a deck.
(e) No slope is decks.
(f) All desks are slopes.
(A) cde (B) def
(C) abe (D) bde

21. (a) No dogma is a belief.
(b) Some beliefs are dogmatic.
(c) Some dogmatics are not dogmas.
(d) Some dogmatics are dogmas.
(e) Many beliefs are not dogmatic.
(f) Some beliefs are dogmas.
(A) abd (B) ace
(C) cba (D) bdf

22. (a) No frontier is limit.
(b) Some margins are not frontiers.
(c) All margins are frontiers.
(d) Some margins are frontiers.
(e) No margin is limit.
(f) Some limits are margins.
(A) eba (B) eac
(C) adf (D) acf

Directions for questions 23 to 27: In each of the following questions, three statements followed by four conclusions marked I, II, III and IV are given. Consider the statements to be true, even though they seem to be at variance with the commonly known facts and find out which of the given conclusion(s) logically follow(s) the statements, disregarding the commonly known facts.

23. Statements: Some arguments are arrangements.
All arrangements are agreements.
Some agreements are achievements.
Conclusions: I. All arguments are agreements.
II. Some agreements are arguments.
III. Some arguments are achievements.
IV. Some arrangements are achievements.
(A) Only I and III follow.
(B) Only I, II and III follow.
(C) Only II and IV follow.
(D) Only II follows.

24. Statements: All even are odd.
Some even are prime.
All prime are digits.
Conclusions: I. Some odd are prime.
II. All odd are prime.
III. All odd are even.
IV. Some digits are even.
(A) Only I and II follow.
(B) Only II and III follow.
(C) Only I and IV follow.
(D) Only I and III follow.

25. Statements: Some shirts are trousers.
Some trousers are not shorts.
All shorts are costly.
Conclusions: I. Some shirts are shorts.
II. No shirt is costly.
III. Some trousers are shorts.
IV. Some costly are trousers.
(A) Only I follows.
(B) Only II follows.
(C) Only I and II follow.
(D) None follows

26. Statements: Some north are east.
No east is west.
All west are south.

Conclusions: I. No north is west.
II. Some east are west.
III. Some south are not east.
IV. All south are east.

(A) Only I follows.
(B) Only II follows.
(C) Only III follows.
(D) Either III or IV follows.

27. Statements: No cause is effect.
All weak are effect.
Some effect are strong.

Conclusions: I. Some strong are cause.
II. No cause is a weak.
III. Some strong are not cause.
IV. Some weak are strong.

(A) Only I follows.
(B) Only II and III follow.
(C) Only III follows.
(D) Only II, III and IV follow.

ANSWER KEYS

Exercise-1

1. (B)	8. (D)	15. (B)	22. (B)	29. (C)	36. (A)
2. (A)	9. (D)	16. (C)	23. (D)	30. (C)	37. (D)
3. (D)	10. (C)	17. (A)	24. (A)	31. (D)	38. (C)
4. (C)	11. (C)	18. (C)	25. (C)	32. (B)	39. (D)
5. (D)	12. (D)	19. (A)	26. (A)	33. (D)	40. (D)
6. (C)	13. (D)	20. (C)	27. (B)	34. (D)	
7. (D)	14. (D)	21. (B)	28. (B)	35. (C)	

Exercise-2

1. (C)	8. (A)	15. (C)	22. (C)	29. (A)	36. (A)
2. (C)	9. (B)	16. (D)	23. (D)	30. (B)	37. (C)
3. (C)	10. (C)	17. (D)	24. (B)	31. (B)	38. (C)
4. (B)	11. (A)	18. (D)	25. (D)	32. (D)	39. (B)
5. (D)	12. (D)	19. (A)	26. (C)	33. (A)	40. (D)
6. (D)	13. (B)	20. (D)	27. (A)	34. (B)	
7. (A)	14. (A)	21. (D)	28. (C)	35. (D)	

Exercise-3

1. (D)	5. (B)	9. (C)	13. (B)	17. (A)	21. (C)	25. (D)
2. (B)	6. (D)	10. (C)	14. (D)	18. (A)	22. (B)	26. (C)
3. (B)	7. (C)	11. (A)	15. (B)	19. (A)	23. (D)	27. (B)
4. (C)	8. (D)	12. (D)	16. (D)	20. (B)	24. (C)	

SOLUTIONS

EXERCISE-1

1. A. The term ‘dancers’ is not distributed in the premise. Hence, it should not be distributed in the conclusion. Also, the middle term ‘drivers’ cannot be included in the conclusion.
   B. No rule is violated, hence, it is correct.
   C. From two affirmative statements, we cannot get a negative conclusion, hence, it is incorrect.

2. The middle term ‘sane’ is distributed in the first premise and one premise is particular. Hence, the conclusion must be particular. Choice (A) is the correct option.

3. As both the premises are negative, no conclusion can be derived.

4. As one premise is negative, the conclusion should be negative. Hence, (A) and (B) are incorrect but (C) is correct.

5. As both the premises are particular, no conclusion can be derived.

6. Conclusions (A) and (B) are negative. Hence, they do not follow. Only conclusion (C) satisfies all the conditions.

7. The middle term ‘books’ is not distributed. Hence, no conclusion can be drawn.

8. Both the given statements are negative. Hence, no conclusion can be drawn.

9. Both statements are particular. Hence, no conclusion can be drawn.

10. Only option (C) does not negate any rule. In option (A) and (B), the term ‘P’ and ‘K’ are distributed respectively, which are not distributed in the statements.

11. In choice (A) and choice (B) ‘Indian’s’ is distributed which was not distributed in the premises. Choice (C) satisfies all the conditions.

12. The middle term is not distributed in the premises. Hence, none follows.

13. Choice (C) is negative whereas both given premises are positive. Choice (A) and choice (B) satisfy all the conditions.

14. Middle term is not distributed in the premises. Hence, none follows.

15. In choice (A) and choice (C) ‘bottle’ is distributed which was not distributed in the premises. Choice (B) satisfies all the conditions.

Solutions for questions 16 to 20: Refer the rules for deductions given in the introduction.

16. The middle term ‘bees’ is not distributed. Hence, no conclusion can be drawn.

17. The middle term ‘apples’ is distributed in the second premise. Hence, the conclusion that can be drawn is ‘All grapes are mangoes’.

18. Both statements are particular. No conclusion can be drawn.

19. The middle term ‘weddings’ is distributed in both the premises. Hence, the conclusion that can be drawn is ‘Some writings are wirings’.

20. The middle term ‘row’ is distributed in the second premise. Since one premise is particular and the other is negative, the conclusion that can be drawn must be particular negative, i.e., ‘Some queues are not circular’.

21. (a) → Does not violate any rule.
    (b) → Here, the term ‘triangle’ which is not distributed in the premise is distributed in the conclusion.
    (c) → Here, the term ‘musician’ which is not distributed in the premise is distributed in the conclusion.
    (d) → Here, the term ‘chicken’ which is not distributed in the premise is distributed in the conclusion.

22. (a) → Here the premises are particular, so no conclusions can be drawn.
    (b) → Does not violate any rule.
    (c) → Does not violate any rule.
    (d) → The term ‘baggages’ which is distributed in the conclusion is not distributed in the premise.

23. (a) → Does not violate any rule.
    (b) → All the statements are negative.
    (c) → There are only two terms.
    (d) → Does not violate any rule.

24. (a) → There are four terms.
    (b) → Does not violate any rule.
    (c) → Does not violate any rule.
    (d) → Does not violate any rule.

25. (a) → Both the premises are negative, so no conclusion can be drawn.
    (b) → Does not violate any rule.
    (c) → Both the premises are negative.
    (d) → Both the premises are negative.

26. (a) The middle term ‘transcendent’ is not distributed in any of the premises. Hence, no conclusion can be drawn.

(b) The term 'milk' is distributed in the conclusion. While it is not distributed in the premise.
(c) No rule is violated.
(d) No rule is violated.
Hence, both c and d.

27. (a) Both the premises are particular. Hence, no conclusion can be drawn.
(b) No rule is violated.
(c) Both the premises are negative. Hence, no conclusion can be drawn.
(d) When one of the premises is particular, the conclusion cannot be universal.

28. (a) No rule is violated.
(b) When both the premises are affirmative, the conclusion cannot be negative.
(c) When one of the premises is negative, the conclusion cannot be affirmative.
(d) The term 'females' is distributed in the conclusion while it is not distributed in the premise.

29. (a) The term 'Violet' is distributed in the conclusion, while it is not distributed in any of the premises.
(b) When one of the premises is particular, the conclusion cannot be universal.
(c) Both the premises are particular. Hence, no conclusion can be drawn.
(d) No rule is violated.

30. (a) The middle term 'fearless' is not distributed. Hence, no conclusion can be drawn.
(b) No rule is violated.
(c) The middle term 'sages' is not distributed in any of the premises. Hence, no conclusion can be drawn.
(d) Both the premises are negative. Hence, no conclusion can be drawn.

31. Using statements 1 and 2 as premises, the middle term 'Symbols' is distributed in the second statement. Hence, the conclusion is 'All digits are letters'.
Using statements 2 and 3 as premises, the middle term 'Little' is not distributed.
Hence, no conclusion can be drawn.
∴ Only conclusion I follows.

32. Using statements 1 and 2 as premises, the middle term 'mats' is not distributed. Hence, no conclusion can be drawn.
Using statements 2 and 3 as premises, the middle term 'rats' is distributed in the third statement.
Since the second statements is particular, the conclusion must be particular, i.e., 'Some mats are cats'.
∴ Only conclusion II follows.

33. Using statements 1 and 2 as premises, the middle term 'Outputs' is not distributed.
Hence, no conclusion can be drawn.
Using statements 2 and 3 as premises, the middle term 'results' is distributed in the third statements. Since the second statement is particular and the third statement is negative, the conclusion must be particular negative, i.e., 'Some outputs are not good'.
∴ None follows.

34. Using statements 1 and 2 as premises or statements 2 and 3 as premises, no conclusions can be drawn, since all the statements are negative.
∴ None follows.

35. Using statements 1 and 2 as premises, the middle term 'two' is distributed in the second statement since the second statement is particular and the first statements is negative the conclusion that must be drawn is particular negative, i.e., 'Some threes are not one'.
Using statements 2 and 3 as premise, the middle term 'two' is not distributed. Hence, no conclusion can be drawn.
Using statements 1 and 3 as premises, the middle term 'two' is distributed in the first statements. Since the first statements is negative, the conclusions drawn must be negative, i.e., 'No one is four', 'Some one are not fours' and 'Some fours are not ones'.
∴ Only II and III follows.

36. Choice (A) → Does not violate any rule.
Choice (B) → Both the premises are negative.
Choice (C) → The term 'bear', which is distributed in the conclusion is not distributed in the premise.
Choice (D) → All the statements are negative.

37. Choice (A) → Does not violate any rule.
Choice (B) → Does not violate any rule.
Choice (C) → The middle term 'gentle' is not distributed.

38. Choice (A) → The term 'screening' which is distributed in the conclusion is not distributed in the premise.
Choice (B) → The term 'purification' which is distributed in the conclusion is not distributed in the premise.
Choice (C) → Does not violate any rule.
Choice (D) → The term 'sedimentation' which is distributed in the conclusion is not distributed in the premise.

39. Choice (A) → The middle term 'gliders' is not distributed.
Choice (B) → A and C are negative.
Choice (C) → There are only two terms.

40. Choice (A) → Affirmative conclusion cannot be drawn from negative statement.
Choice (B) → Middle term appears in the conclusion.
Choice (C) → No conclusion can be drawn from two negative statements.
Choice (D) → Does not violate any rule.

EXERCISE-2

1. Choice (A) → There is only one negative statement.
Choice (B) → Here, only (e) can be the conclusion but the term 'protection' is distributed, which is not distributed in (f).
Choice (C) → Does not violate any rule in the order cfb.

2. Choice (A) → Only (c) can be the conclusion but the term 'professor' which is distributed in the conclusion is not distributed in the premise.
Choice (B) → Only (f) can be the conclusion, but here the term 'engineer' which is distributed in the conclusion is not distributed in the premise.
Choice (C) → Does not violate any rule in the order abf.

3. Choice (A) → In any order if we take the statements, the term which is not distributed in the premise is distributed in the conclusion.
Choice (B) → There are only two terms in the statements (b) and (e).
Choice (C) → Does not violate any rule in the order bcd.
Choice (D) → All the statements are particular.

4. Choice (A) → There is only one negative statement.
Choice (B) → Does not violate any rule in the order bfd.
Choice (C) → All are particular statements.
Choice (D) → There are four terms.

5. Choice (A) → Does not violate any rule in the order efb.
Choice (B) → Does not violate any rule in the same order.
Choice (C) → Does not violate any rule in the order cfd.

6. (A) afd → There is only one negative statement.
(B) cef → Here, efc is the correct order.
(C) fab → Here, abf is a right order.

7. (A) cdf → cfd is the correct order.
(B) def → Here, 'square' and 'curve' appear only once.
(C) acb → All are particular statements.
(D) abe → The term 'curve' appears in all the three statements.

8. (A) afd → Here, fda is the correct order.
(B) abd → There is only one negative statement.
(C) ced → The term 'dark' appears only once.
(D) def → There is only one negative statement.

9. (A) abc → For every possible order the seventh rule of syllogism is violated.
(B) dea → Here, aed is the correct order.
(C) efd → All the statements are particular.
(D) bef → The term 'lilly' and 'rose' appear only once.

10. (A) ade → There is only one negative statement.
(B) def → There is only one negative statement.
(C) bde → Here, deb is the correct order.
(D) abc → There is only one negative statement.

11. (A) bde → Here, deb is the correct order.
(B) cdf → There is only one negative statement.
(C) adf → For every possible order the seventh rule of syllogism is violated. Hence, it is incorrect.
(D) ade → The term depressed appears only once.

12. (A) cde → There is only one negative statement.
(B) efa → Here, aef is the correct order.
(C) bcf → Here, cfb is the correct order.

13. (A) cdf → There is only one negative statement.
(B) abe → Here, aeb is the correct order.
(C) cef → There is only one negative statement.
(D) acd → The term 'mirror' appears only once.

14. efb – Here, bef is the correct order.
fda – The term minutes has appeared three times.
ade – The term minutes has appeared three times.
dfe – Here, all are particular statements.

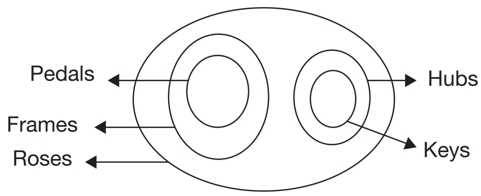
15. cdf – Here, all are negative.
eba – There are more than three terms.
cfe – Here, ecf is the correct order.
adc – There are more than three terms.

16. eaf – Here, fae is the correct order.
cfe – Here, efc is the correct order.
adb – Here, abd is the correct order.

17. aef – Here, the possible order is efa, but it violates distribution rule.
cbe – It violates distribution rule.
fcd – Here, all are negative statements.

18. fbd – Here, fbd is the correct order.
dca – Here, adc is the correct order.
fde – Here, all are particular statements.

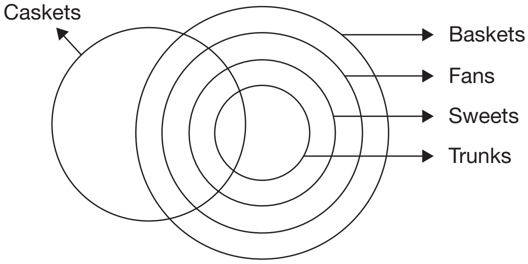
19. The basic diagram for the given statements is as follows.



From the above basic diagram, we incur the following.
Conclusion I, affirmative, does not follow.
Conclusion II, affirmative, follows.
Conclusion III, affirmative, does not follow.
Conclusion IV, affirmative, does not follow.
∴ Only II follows.

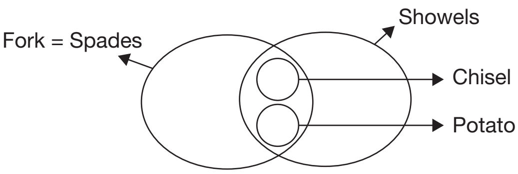
Difficulty Level-2: Moderate

20. The possible diagram that can be drawn from the above statements is as follows.



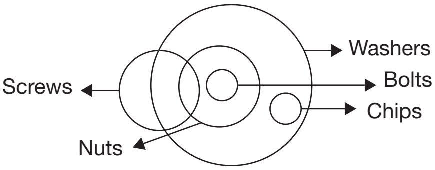
From the above possible diagram, we incur that:
Conclusion I, follows.
Conclusion II, follows.
Conclusion III, follows.
Conclusion IV, follows.
∴ All follow.

21. The possible diagram that can be drawn from the above statements is as follows.



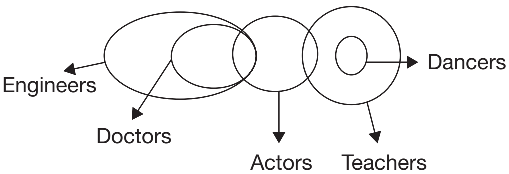
From the above possible diagram, we incur that:
Conclusion I, follows.
Conclusion II, follows.
Conclusion III, follows.
Conclusion IV, follows.
∴ All follow.

22. The basic diagram for the given statements is as follows.

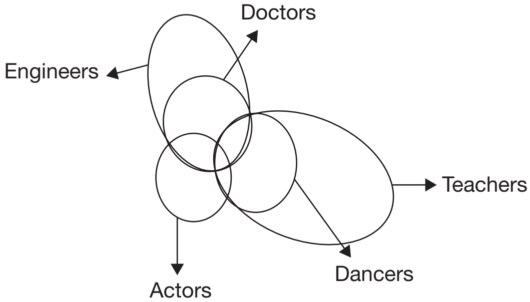


From the above basic diagram, we incur that:
Conclusion I, affirmative, follows.
Conclusion II, affirmative, follows.
Conclusion III, affirmative, follows.
Conclusion IV, affirmative, does not follow.
∴ I, II and III follow.

23. The basic diagram for the given statements is as follows.



From the above basic diagram, we incur that:
Conclusion I, affirmative, follows.
Conclusion II, affirmative, does not follow.
Conclusion III, negative, follows.
Conclusion IV, affirmative, does not follow.
Since conclusion III, negative, follows.
Let us try to draw an alternate diagram to negate it.
The required alternate diagram is shown below.



In the above alternate diagram, conclusion III does not follow, but II follows.
Here, conclusion II and III are contradictory to each other.
∴ Only I and either II or III follow.

Solutions for questions 24 to 30: Refer the rules for deductions given in the introduction.

24. As one premise is particular, the conclusion should be particular. (B) satisfies all the rules.

25. As both the premises are particular, no conclusion can be derived.

26. As one premise is negative, the conclusion must be negative. Therefore, (C) satisfies all the rules.

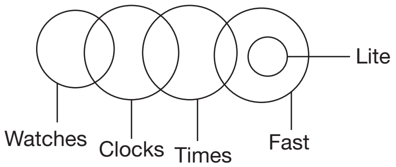
27. As both the premises are affirmative and also one premise is particular, the conclusion should be particular affirmative. Therefore, (A) satisfies all the rules.

28. All + Some not = Some not.
The middle term is distributed. Therefore, option (C) satisfies all the rules.

29. All + Some not = Some not.
The middle term is distributed. Therefore, option (A) satisfies all the rules.

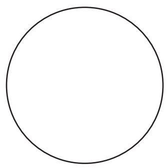
30. All + Some = Some.
The middle term is distributed. As both the premises are affirmative and one of them is particular, the conclusion should be particular and affirmative. Therefore, option (B) satisfies all the rules.

31. The basic diagram for the given statements is as follows.



From the basic diagram, we derive the conclusion as:
I. Affirmative, does not follow.
II. Affirmative, does not follow.

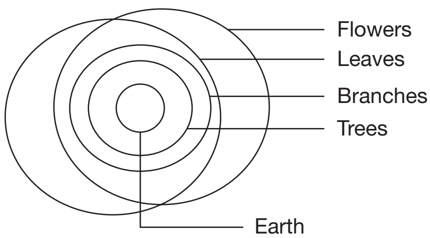
Fast = life = Time = dock = watch



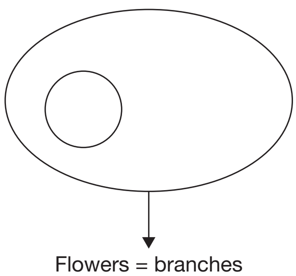
Possible diagram

From the above possible diagram, we derive that:
Conclusion III, possibility, follows.
Conclusion IV, possibility, does not follow.
Hence, only III follows.

32. The basic diagram for the given statements is as follows.

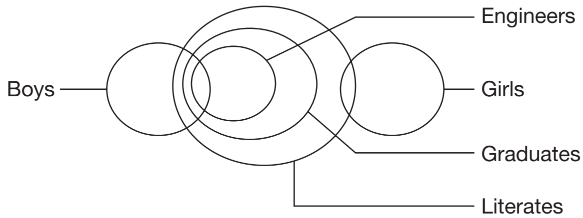


In the above basic diagram, the conclusion is as follows:
I. Affirmative, follow.
III. Affirmative, follows.
Conclusion II follows.
The possible diagram is as follows.



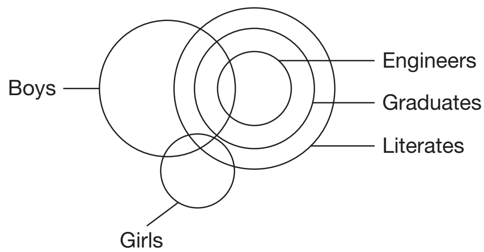
From the possible diagram:
Conclusion IV follows.
Hence, all follow.

33. The basic diagram for the given statements is as follows.



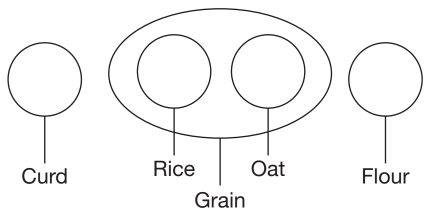
In the basic diagram, the conclusion is:
I. Negative, follows.
II. Affirmative, follows.
III. Affirmative, does not follow.
IV. affirmative, follows.

Conclusions II and IV are the definite conclusions as they are affirmative, whereas conclusion I may be false, as it is a negative conclusion. To prove that conclusion I is false, we have to prove that 'Some boys are girls'. The alternate diagram is as follows.



Hence, only II and IV follow.

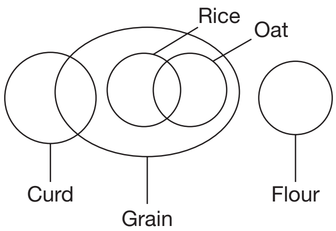
34. The basic diagram for the given statements is as follows:



From the basic diagram, the conclusion is as follows:
I. Negative, follows.
II. Negative, follow.
III. Negative, follows.
IV. Affirmative, does not follow.

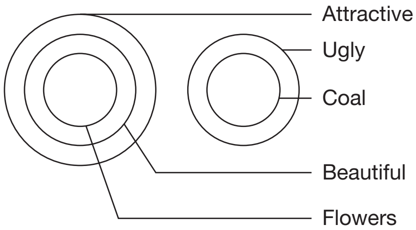
Conclusion I, II and III are true but negative. Hence, the conclusions may be false. Let us try to prove them false. So, we have to prove:

(I) Some curds are grains.
(II) Some rice are oats.
(III) Some flours are oats.
The alternate diagram for the statements is as follows:



Hence, it is not possible to draw a diagram to negate III. Only III follows.

35. The basic diagram for the given statements is as follows.

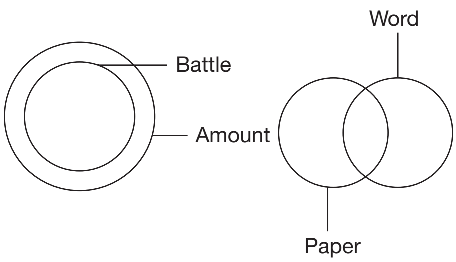


In the above basic diagram, we derive the conclusion as:
I. Affirmative, follows.
II. Negative, follows.
III. Negative, follows.
IV. Negative, follows.

To prove that conclusion II, III and IV is false, we have to prove that:
(I) Some flowers are coal.
(II) All flowers are ugly.
(III) All attractives are coal.

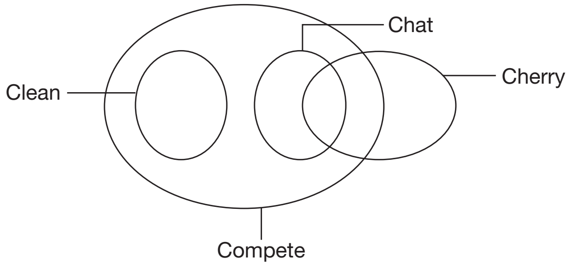
The alternate diagram for any of these three is not possible. Hence, all follow.

36. The given statements are represented in the following basic diagram.



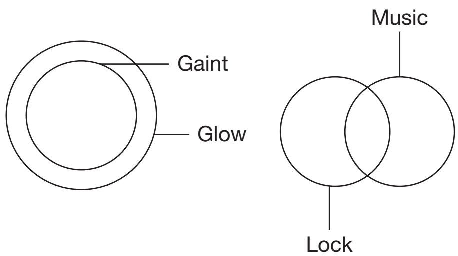
Choice (A), possibility, does not follow.
Choice (B), affirmative, follows.
Choice (C), possibility, follows.
Choice (D), possibility, follows.

37. The given statements are represented in the following basic diagram.



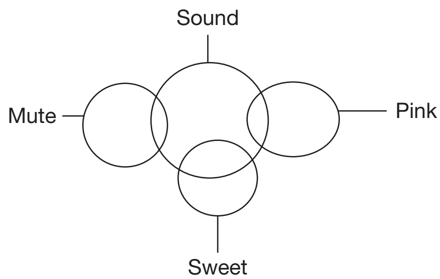
Choice (A), affirmative, follows.
Choice (B), affirmative, follows.
Choice (C), affirmative, does not follow.
Choice (D), IV, possibility, follows.

38. The given statements are represented in the following basic diagram.



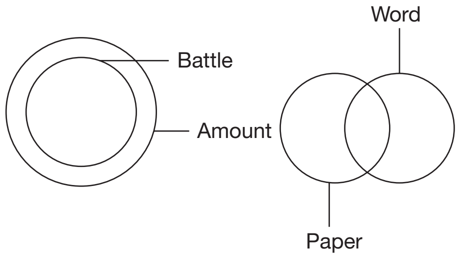
Choice (A), affirmative, follows.
Choice (B), possibility, follows.
Choice (C), affirmative, does not follow, since, some glow are not lock.
Choice (D), possibility, follows.

39. The given statements are represented in the following basic diagram.



Choice (A), possibility, follows.
Choice (B), possibility, does not follow.
Choice (C), affirmative, follows.
Choice (4), possibility, follows.

40. The given statements are represented in the following basic diagram:



Choice (A), possibility, follows.
Choice (B), affirmative, follows.
Choice (C), possibility, follows.
Choice (D), affirmative, does not follow.

EXERCISE-3

1. (a) No rule is violated.
(b) No rule is violated.
(c) No rule is violated.
(d) No rule is violated.

2. (a) No rule is violated.
(b) When one premise is particular, the conclusion cannot be universal.
(c) No rule is violated.
(d) No rule is violated.

3. (a) The middle term is not distributed. Hence, doesn't follow.
(b) It satisfies all the conditions. Hence, follows.
(c) The term 'good' is distributed in the conclusion which is not distributed in the premises. Hence, does not follow.
(d) Here, both premises are particular. Hence, does not follow.
Hence, only (b) follows.

4. (a) It satisfies all the conditions.
(b) The term 'friend's' is distributed in the conclusion which was not distributed in the premises. Hence, does not follow.
(c) The term 'useful' is distributed in the conclusion which was not distributed in the premises. Hence, it does not follow.
(d) It satisfies all the conditions.
∴ Only (a) and (d) follow.

5. (a) The term 'rules' is distributed in the conclusion which was not distributed in the premises. Hence, it does not follow.
(b) It satisfies all the conditions.
(c) It satisfies all the conditions.
(d) Here, both the premises are particular. Hence, it does not follow.
∴ Only (b) and (c) follow.

6. (a) It satisfies all the conditions.
(b) The middle term is not distributed. Hence, it does not follow.
(c) The term 'metals' is distributed which was not distributed in the premises. Hence, it does not follow.
(d) Here, both premises are particular. Hence, does not follow.
∴ Only (a) follows.

7. (a) It satisfies all the conditions.
(b) It satisfies all the conditions.
(c) It satisfies all the conditions.
(d) The term 'laptops' is distributed in the conclusion which was not distributed in the premises.
∴ Only (a), (b) and (c) follow.

8. (A) PML: By combining P and M or M and L we get a conclusion in particular.
Now, when we combine P and L then we do not get a conclusion as the common term is not distributed.
(B) NPL: No combination would give a valid conclusion.
(C) PQL: No combination would give a valid conclusion.
(D) QNP: By combining N and P, we get 'all rings are spheres'. Hence, 'some rings are spheres' is correct.
Hence, the correct combination is NPQ.

9. (A) PQR: Here no combination would give a conclusion.
(B) NQR: No combination gives a valid conclusion.
(C) LPN: By combining, L and P, the conclusion obtained is 'some false are truths'.
(D) MQR: There are more than 3 terms, hence, it is wrong.

10. (A) PQM: No combination would give a valid conclusion.
(B) RNQ: No combination would give a valid conclusion.
(C) PRL: When P and R are combined, we get the conclusion 'some teams are great'.
(D) MPN: No combination would give a valid conclusion.

11. (A) NQL: By combining N and Q we get the conclusion 'some women are old'.
Hence, choice (A) is logically related.
(B) PMR: No combination would give a valid conclusion.
(C) LNR: No combination would give a valid conclusion.

12. (A) LMP: No combination is logically related.
(B) RPL: No rule is violated; hence, it is logically related.
(C) NQL: No rule is violated; hence, it is logically related.

13. (A) NPM: No combination is logically related.
(B) NQL: No rule is violated hence it is logically related.
(C) MNR: No combination is logically related.
(D) PQL: No combination is logically related.

14. (A) NQR: No combination is logically related.
(B) PRN: No combination is logically related.
(C) LNP: No combination is logically related.
(D) LNQ: No rule is violated; hence, it is logically related.

15. (A) PNL: No combination is logically related.
(B) LPR: No rule is violated; hence, it is logically related.
(C) MPQ: No combination is logically related.

16. (A) LMQ: No rule is violated; hence, it is logically related.
(B) NPR: No rule is violated; hence, it is logically related
(C) MNP: No combination is logically related.

17. (A) LMR: No rule is violated; hence, it is logically related.
(B) LNP: No combination is logically related.
(C) PQN: No combination is logically related.

18. From the choices:
A. From b and d, we can conclude that, 'few tufts are combs', which is A.
B. It is not a valid group as exactly one statement is negative.
C. It is not a valid group as exactly one statement is negative.
D. Among b, c, f, b cannot be the conclusion of the other two as it is affirmative. As the term 'comb' is not distributed, f cannot be the conclusion. As 'tuft' is not distributed, c cannot be the conclusion.

19. From the choices:
A. From c and f, it can be concluded that, 'few straps are not curbs', which is a.
B. It is not a valid group as all the three statements are particular. For the same reason as above, (C) and (D) are also not possible.

20. From the choices:
A. Among c, d, e, as c is particular neither d nor e can be the conclusion. As both d and e are negative, C cannot be the conclusion.
B. From f and e, it can be concluded that, 'no desk is a deck', which is d.
C. It is not a valid group as all the three statements are negative.
D. It is not a valid group as all the three statements are negative.

21. From the choices:
A. It is not a valid group as exactly one statement is negative.
B. It is not a valid group as all the three statements are negative.
C. From a and b, we can conclude that, 'some dogmatics are not dogmas', which is C.
D. It is not a valid group as all the three statements are particular.

22. From the choices:
A. It is not a valid group as all the three statements are negative.
B. From c and a, we can conclude that, 'no margin is limit', which is e.
C. It is not a valid group as exactly one statement is negative.
D. It is not a valid group as exactly one statement is negative.

23. Using statements 1 and 2 as premises, the middle term 'arrangements' is distributed in the second statements. Since the first statements is particular the conclusions drawn must be particular, i.e., 'Some arguments are agreements'.
Using statements 2 and 3 as premises, the middle term 'agreements' is not distributed, hence, no conclusion can be drawn.
Using statements 1 and 3 as premises, both statements are particular, hence, no conclusion can be drawn.
∴ Only II follows.

24. Using statements 1 and 2 premises, the middle term 'even' is distributed in the first statement. Since the second statement is particular, the conclusion drawn, must be particular, i.e., 'Some odd are prime'.
Using statements 2 and 3 as premises, the middle term 'Prime' is distributed in the third statement. Since the second statement is particular, the conclusion drawn must be particular, i.e., 'Some even are digits'.
∴ Only I and IV follows.

25. Using statements 1 and 2 as premises, since both statements are particular, no conclusion can be drawn.
Using statements 2 and 3 as premises, the middle term 'Shorts' is distributed in both the premises. Since the second statement is particular and negative, the conclusion that can be drawn must be particular negative. But such a conclusion is not possible since the two terms 'Trousers' and 'Costly' are not distributed in the statements and hence, cannot be distributed in conclusion.
∴ None follows.

26. Using statements 1 and 2 as premises, the middle term 'Cast' is distributed in the second statement. Since the first statement is particular and the second statement is negative, the conclusion drawn should be particular negative, i.e., 'Some north are not west'.

Using statements 2 and 3 as premises, the middle term 'west' is distributed in the third statement. Since the second statement is negative, the conclusions drawn must be negative, i.e., 'No east is south', 'Some east is not South' and 'Some south is not east'.

∴ Only III follows.

27. Using statements 1 and 2 as premises, the middle term 'Effect' is distributed in the first statements, which is negative. Hence, the conclusions drawn must be negative, i.e., 'No cause is weak', 'Some cause are not weak' and 'Some weak are not cause'.

Using statements 2 and 3 as premises, the middle term 'effect' is not distributed. Hence, no conclusion can be drawn.

Using statements 1 and 3 as premises, the middle term 'effect' is distributed in the first statements. The first statement is negative. Hence the conclusion drawn, must be negative, i.e., 'No cause' and 'Some cause are not strong'.

∴ Only II and III follows.

CHAPTER 10 Connectives

Learning Objectives

In this chapter, you will:

- Understand the correct logical interpretation of various conditional statements.
- Understand the conclusions that can be drawn from conditional statements.

There are few questions that are frequently asked in entrance exams, which are based on logical statements and logical connectives. A proper understanding of some basics in 'Logic' will eliminate the difficulty in answering such questions. These questions can be answered easily and very quickly with a clear understanding of the basics that we shall look at in the following sections. The concepts discussed in this chapter, not only help in answering these questions, but also have lot of application value. The logical interpretations discussed can be applied, whenever we (irrespective of the test area) come across these kinds of statements.

In Logic, we deal with statements that are essentially sentences in the English language. However, in Logic we are not interested in or worried about the factual correctness of the sentence. We are interested only in the Logical 'truthfulness' of the statements.

For example, consider the following statement:

'If the sun rises in the west, then the moon rises in the north'.

Here, we are not concerned with whether the sun rises in the east or west or with the direction in which the moon rises. We will only look at whether the moon will rise in the north or not depending on whether the part of the statement 'The sun rises in the west' is true or not. If we are given that the sun rises in the west (which, incidentally, is factually incorrect), we can then conclude that the moon rises in the north (which again does not concern with the direction in which the moon actually rises).

We can represent statements in Logic using symbols, like p, q, etc., the way we represent variables/unknowns in Algebra using symbols like x, y, z, etc.

Statements like 'I will go for a movie', 'It is a sunny day', etc., are called simple statements. When two or more such simple statements are connected together to form a single statement, then such statement is called a compound statement.

The simple statements are combined using logical connectives to form compound statements. We should know some of the important logical operators/connectives to be able to effectively tackle questions that involve compound statements and logical operations on compound statements.

❑ NEGATION 'NOT'

Any statement can be negated by using the words 'not' or 'no'. In layman's language, negation is like the opposite of a statement.

For example, the negation of the statement 'It is raining' is 'It is 'NOT' raining'. The negation of the statement 'He will pass the exam' is 'He will not pass the exam'. This is equivalent to saying 'He will fail in the exam'. So, when you are looking at negating the given statement, you should keep in mind the English equivalents of the statements also.

Having defined simple statements, we shall now study about a few common operators (also called connectives) that can be used to combine (or operate upon) two or more simple statements and arrive at more complicated or compound statements.

❑ LOGICAL CONNECTIVE 'OR'

Two or more statements can be connected using the connective 'OR'. The following is an example using 'OR'.

It is raining or I will go to my friend's house.

The same statement can also be written as:

Either it is raining or I will go to my friend's house.

Both the statements above mean the same. The additional word 'either' does not change the meaning of the statement.

When two (or more) statements are connected using 'OR', at least one of them is true.

Suppose we have a statement 'Either p or q', since at least one of the two statements p, q must be true, we have p alone is true or q alone is true or both are true.

This is the interpretation to be given to an 'OR' statement (irrespective of the meaning of the sentence as per English language).

For example, the statement 'Either I will go for a movie or I will go to my friend's house' means:

I will go for a movie

or

I will go to a friend's house

or

I will go both for a movie and to a friend's house.

Let us take the statement 'Either he is dead or he is alive'. This statement means:

He is alive

or

He is dead

or

He is both alive and dead.

In this case, the possibility 'He is both alive and dead' does not make sense if we look at the meaning in English language because a person cannot be dead and alive at the same time. However, as discussed earlier, we will not be concerned about the meaning of the statements.

Hence, we will always interpret the statement '(Either) p or q' as 'p alone is true or q alone is true or both are true' (unless otherwise explicitly stated that both are not true at the same time). In other words, in a statement 'p or q', we can say that at least one of the two statements is true.

Given 'p or q', we get four different possibilities that follow:

1. **Given 'p or q', we are then told 'p is true':**
   Since we need at least one of the two statements p or q to be true and here we already know that p is true, we cannot conclude anything about q, i.e., we cannot conclude whether q is true or false but both possibilities exist.
2. **Given 'p or q', we are then told 'q is true':**
   Since we need at least one of the two statements p or q to be true and here we already know that q is true, we cannot conclude anything about p, i.e., we cannot conclude whether p is true or false but both possibilities exist.
3. **Given 'p or q', we are then told 'p is 'NOT' true':**
   Since we need at least one of the two statements p or q to be true and here we already know that p is not true, q **has** to be true so that at least one of the two statements will then be true. So, here we **can** conclude that q **is true.**
4. **Given 'p or q', we are then told 'q is 'NOT' true':**
   Since we need at least one of the two statements p or q to be true and here we already know that q is not true, p **has** to be true so that at least one of the two statements will then be true. So, here we **can** conclude that p **is true.**

There is one particular category of questions that has appeared in CAT for three years in a row. These questions are based on the concepts that we looked at in the previous section. We will take two or three examples to understand these questions.

The directions of the questions asked were as follows:

'Each question has a main statement followed by four statements labelled A, B, C and D. Choose the ordered pair of statements where the first statement implies the second, and the two statements are logically consistent with the main statement'.

10.1: Either the elephant is big or the lion is cruel.

(a) The elephant is big.
(b) The elephant is not big.
(c) The lion is cruel.
(d) The lion is not cruel.

(A) ac (B) db
(C) bc (D) ad

Sol: The main statement has two simple statements. 'The elephant is big' and 'The lion is cruel' connected by 'OR'. Let us call these two statements p and q, respectively for the purpose of our discussion. Then the main statement can be represented as 'p 'OR' q'.

First, let us look at each choice and understand the logic discussed above. Once we do that, we will also see how to answer such questions in a much shorter time.

At least one of these two statements have to be true in any ordered pair we look at. As per the discussion we had above, from among the choices, if we have an ordered pair where the first part of the ordered pair from the two statements in the choice is true, then we cannot conclude anything about the second part of the ordered pair. However, if the first part of the ordered pair in the choice is not true, then the second part should contain the second statement as given in the main statement (i.e., the second statement has to be 'true').

Take choice (A) for the above question. The first statement is A which says 'The elephant is big'. This is p (as we denoted above) which is given in the main statement. Since p is true, we cannot conclude whether q is true or not, i.e., q may be true or it may be false. So, we cannot have any statement following A which can be concluded from A and is consistent with the main statement. Hence, this cannot be the answer choice.

Take choice (B). The first statement is D which says 'The lion is not cruel'. This is the negation of statement q, that is to say, 'Not q' is the first of the two statements in the choice. Since q is negated, p must be true (for at least one of the two statements to be true). But the second statement in this choice is 'The elephant is not big' which is 'Negation p'. Hence, this is not the correct choice.

Take choice (C). The first statement is B which is 'The elephant is not big', i.e., Negation p. Since p is negated, q must be true (for at least one of the two statements to be true). The second statement in the choice is C which is 'The lion is cruel', i.e., q. Thus, in this choice, we have Negation p followed by q. So, this is the correct answer choice.

Let us also take a look at choice (D). The first statement in this choice is A, which is 'The lion is big', i.e., p is true. Since the first statement is true, we cannot conclude anything about statement q.

❐ APPROACH IN THE EXAM

In an exam, for these types of questions, we do not need to go from the answer choices and check each and every one of them. We can directly identify the correct combinations of statements that will satisfy the directions given.

We know that if the first statement out of the two statements in the choice is either p or q (that is one of the two statements given in the question), then we cannot draw any conclusion.

We also know that if p or q is negated, then the other statement should definitely be true. So, 'Negation p is followed by q' and 'Negation q is followed by p' will be the correct combination of statements. Hence, we directly check out for 'NOT' $p \to q$ or 'NOT' $q \to p$ in the answer choices.

In the above example not $p \to q$, is represented by *bc* and not $q \to p$, is represented by *da*. We should check which among *bc* and *da* is/are given in the answer choices.

❐ LOGICAL CONNECTIVE 'AND'

Two or more statements can be connected using the connective 'AND'. The following is an example using 'AND'.

It is raining **and** I will go to my friend's house.

The two statements connected by **and** have to be true for the compound statement to be true. In general, if we have a statement 'p and q', then we can conclude that p should be true as well as q, i.e., both the statements should be true. Even if one of the two statements is false, the compound statement is false.

Negation of compound statements formed with 'OR', 'AND'

A compound statement formed with 'OR' or 'AND' can be negated in the following manner:

'Negation (p 'OR' q)' is the same as 'Negation p 'AND' Negation q'.

'Negation (p 'AND' q)' is the same as 'Negation p 'OR' Negation q'.

As can be seen in the above example, when a compound statement consisting of two simple statements (connected with 'OR' or 'AND') is negated, the result will consist of each of the individual statements negated. In addition to that, the following will also have to be observed:

'OR' will become 'AND'

'AND' will become 'OR'

❐ LOGICAL CONNECTIVE 'IF-THEN'

This is a very important connective. This is represented by $p \rightarrow q$ (and is read as 'p implies q'). This means that if we know that p has occurred, q has to occur or must have occurred. For example, the statement 'If it is raining, then I wear a raincoat' means that if we know that it is raining, we can conclude that I must be wearing a raincoat.

The statement 'p implies q' is called an implication statement. The term on the left hand side in $p \rightarrow q$ is called the 'antecedent' and the term q is called the 'consequent'.

Let us look at the following cases when we are given that $p \rightarrow q$.

1. Given that $p \rightarrow q$, we are then told that q has occurred. Can we conclude that p must have occurred?

   We cannot conclude that p must have occurred. This is because while whenever p occurs, q will definitely occur, q may occur even otherwise, i.e., even without the occurrence of p. So, both p and Negation p are possible and hence, we cannot conclude anything when we know that q has occurred.

2. Given that $p \rightarrow q$, we are then told that p has not occurred. Can we conclude that q will also not occur?

   We cannot conclude that q will not occur. This is because while whenever p occurs, q will definitely occur, q may occur even when p does not occur (as discussed above). So, both q and Negation q are possible, and hence, we cannot conclude anything when we know that p has not occurred.

3. Given that $p \rightarrow q$, we are then told that q has not occurred. Can we conclude that p must not have occurred?

   We can conclude that p must not have occurred. This is because had p occurred, q would have occurred. But we know that q has not occurred, so p must not have occurred. So, we can conclude that 'Negation p' follows 'Negation q'.

So, if we are given that $p \rightarrow q$, then 'Negation $q \rightarrow$ Negation p'. This is a very important relationship. We can express it in words as:

'In an implication statement, negation of the right hand side will always imply the negation of the left hand side'.

We can summarize the above three points as follows:

$p \rightarrow q$	Given
$q \rightarrow p$	Cannot be concluded
$q \rightarrow$ Negation p	Cannot be concluded
Negation $p \rightarrow$ Negation q	Cannot be concluded
Negation $p \rightarrow q$	Cannot be concluded
Negation $q \rightarrow$ Negation p	Is always true

In certain CAT papers, there were questions on 'if—then' concepts discussed above and the questions similar to those on 'either—or' that we looked at above. Let us take an example and understand these questions. The directions are the same as that we looked at above:

'Each question has a main statement followed by four statements labelled as A, B, C and D. Choose the ordered pair of statements where the first statement implies the second and the two statements are logically consistent with the main statement'.

10.2: If the elephant is big, then the lion is cruel.

(a) The elephant is big.

(b) The elephant is not big.

(c) The lion is cruel.

(d) The lion is not cruel.

(A) ca (B) bd
(C) bc (D) db

Sol: The main statement has two simple statements 'The elephant is big' and 'The lion is cruel' connected by 'IF—THEN'. Let us call these two statements p and q, respectively for the purpose of our discussion. Then the main statement can be represented as 'p implies q' or '$p \to q$'.

First, let us look at each choice and understand the logic discussed above. Once we do that, we will also see how to answer such questions in a much shorter time.

Take choice (A). In terms of p and q, this can be represented as $q \to p$. As per the table above, we know that this cannot be concluded, given $p \to q$. Hence, this is not the correct answer.

Take choice (B). In terms of p and q, this can be represented as 'Negation $p \to$ Negation q'. Again, as per the table above, we know that this cannot be concluded, given $p \to q$. Hence, this is not the correct answer.

Take choice (C). In terms of p and q, this can be represented as 'Negation $p \to q$', as per the table above, we know that this cannot be concluded, given $p \to q$. Hence, this is not the correct answer.

Since we eliminated three answer choices, the fourth has to be the correct answer. Let us take choice (D) and look at it. In terms of p and q, it can be represented as 'Negation $q \to$ Negation p'. As per the table above, we know that this can definitely be concluded. Hence, this is the correct answer choice.

❒ APPROACH IN THE EXAM

In an exam, for these types of questions, we do not need to go from the answer choices and check each and every one of them. We can directly identify the combinations of statements that will satisfy the directions given.

Given that $p \to q$, we know that 'Negation $q \to$ Negation p'. Hence, the two correct combinations are $p \to q$ (because this is the given statement itself) and 'Negation $q \to$Negation p'.

So, in the above example, we should look for ac or db. Hence, the correct answer is choice (D).

❒ OTHER FORMS OF 'IF-THEN'

There are different types of statements which can be reduced to or represented as $p \to q$. Let us look at these statements in descriptive form and the representation by using '$\to$' sign.

S. No.	Statement	Representation using $\to$	Also equivalent to	Remarks
1.	If p, then q	$p \to q$	Neg. $q \to$ Neg. p	Already discussed above
2.	q, if p	$p \to q$	Neg. $q \to$ Neg. p	Identical to 1 above
3.	When p, then q Whenever p, then q	$p \to q$	Neg. $q \to$ Neg. p	Identical to "if p, then q"
4.	q, when p q, whenever p	$p \to q$	Neg. $q \to$ Neg. p	Same as 3 above
5.	Everytime p, q	$p \to q$	Neg. $q \to$ Neg. p	Same as "If p, then q"
6	q, everytime p	$p \to q$	Neg. $q \to$ Neg. p	Same as 5 above
7.	q, only if p	$q \to p$	Neg. $p \to$ Neg. q	
8.	Unless p, q	Negation $p \to q$	Neg. $q \to p$	
9.	q, unless p	Negation $p \to q$	Neg. $q \to p$	Same as 8 above
10.	p, otherwise q	Negation $p \to q$	Neg. $q \to p$	Same as "Unless p, q"

❒ ANOTHER MODEL OF QUESTIONS

There is one particular model of question that appeared in the XAT exam in the recent past. These questions are based on the logic that has been discussed above. We will take an example and see how to solve such questions.

Directions: Each question below consists of a main statement followed by four statements. From the choices, select the one that is logically consistent with the main statement.

(Please note that the directions, instead of asking you to find out the choice that is 'logically consistent with the main statement', may ask you to find out the choice that is 'logically equivalent to the main statement' or 'Which of the following statements is true?')

10.3: If it is raining, then I will go for a movie or I will visit my friend's house.

(A) It is not raining, means that I will not go for a movie or I will not visit my friend's house.
(B) It is not raining, means that I will not go for a movie and I will not visit my friend's house.
(C) I will not go for a movie or I will not visit my friend's house, means that it is not raining.
(D) I will not go for a movie and I will not visit my friend's house, means that it is not raining.

Sol: Solving this question involves the application of simple concepts/rules about IF—THEN, 'OR', 'AND' and NEGATION which we have already looked at.

EXERCISE-1

Directions for questions 1 to 20: Each question below consists of a main statement followed by four answer choices. From the answer choices, select the one that logically follows the main statement.

1. If movie is a superhit, then I watch it.
(A) I watched a movie means it is a superhit.
(B) The movie is a superhit. Hence, I do not watch it.
(C) I did not watch a movie, though it was a superhit.
(D) I did not watch a movie implies that the movie was not a superhit.

2. Whenever it is cold, I wear a jacket.
(A) It is cold implies I am wearing a jacket.
(B) I did not wear a jacket implies it was not cold.
(C) It is cold but I did not wear the jacket.
(D) Both (A) and (B)

3. Either Shaheen cooks or Salim brings the food parcel.
(A) Shaheen cooks means Salim will not bring the food parcel.
(B) Salim did not bring the food parcel. Hence, Shaheen cooked.
(C) Shaheen is cooking. Hence, Salim has brought the food parcel.
(D) Salim did not bring food parcel implies that Shaheen did not cook.

4. Only if Raj does not come, then Geetika will come to the party.
(A) Raj came, hence, Geetika will also come to the party.
(B) Geetika will not come to the party. Hence, Raj will come.
(C) Geetika has come to the party means Raj is not coming.
(D) Raj has not come. Hence, Geetika has come to the party.

5. Unless I have money, I cannot enjoy my weekend.
(A) I have money, so I can enjoy my weekend.
(B) I can enjoy my weekend means I have money.
(C) I do not have money implies I cannot enjoy my weekend.
(D) Both (B) and (C)

6. If I can swim, then I can clear the exam.
(A) I can swim. Hence, I can clear the exam.
(B) I cannot clear the exam implies I cannot swim.
(C) I can swim. Hence, I cannot clear the exam.
(D) Both (A) and (B)

7. India will talk, only if the terrorists are handed over.
(A) The terrorists are handed over; hence, India will talk.
(B) India will not talk implies that the terrorists have not been handed over.
(C) India will talk though the terrorists are not handed over.
(D) India will talk implies the terrorists are handed over.

8. Whenever there is demand, there will be supply.
(A) There is supply, hence, there is demand.
(B) There is demand, hence, there will be supply.
(C) There is no supply implies that there is no demand.
(D) Both (B) and (C)

9. Suhasita purchases either a cooler or a refrigerator.
(A) Suhasita is not purchasing a cooler implies that she is purchasing a refrigerator.
(B) Suhasita is not purchasing a refrigerator implies that she is purchasing a cooler.
(C) Suhasita is purchasing neither a cooler nor a refrigerator.
(D) Both (A) and (B)

10. I will not have enemies, unless I fight.
(A) I fight implies that I will have enemies.
(B) I do not fight implies that I will not have enemies.
(C) I have enemies implies that I fight.
(D) Both (B) and (C)

11. If you are good, then everyone loves you.
(A) You are good, hence, everyone does not love you.
(B) Everyone loves you means you are good.
(C) You are not good; hence, everyone does not love you.
(D) Everyone does not love you means you are not good.

12. Unless Pratik comes home, his son does not eat.
(A) Pratik did not come home so his son does not eat.
(B) Pratik came home, hence, his son eats.
(C) Pratik's son eats means Pratik came home.
(D) Both (A) and (C)

13. Only if you are qualified in written test, you get a call for interview.
(A) You did not qualify in written test means you do not get a call for an interview.
(B) You do not get a call for an interview means you did not qualify in written test.
(C) You are qualified in written test, hence, you get a call for an interview.
(D) Both (A) and (C)

14. If Raju plays well, then the team can win the match.
(A) Raju did not play well, implies that the team did not win the match.
(B) The team won the match implies that Raju played well.
(C) The team did not win the match implies that Raju did not play well.
(D) More than one of the above.

15. Either he plays cricket or eats biscuit.
(A) He is playing cricket implies that he is not eating biscuit.
(B) He is not eating biscuit implies that he is playing cricket.
(C) He is eating biscuit implies that he is not playing cricket.
(D) All the above

16. Unless the inflation is low, economic growth will not be high.
(A) The economic growth is low means that inflation is high.
(B) The inflation is not low implies that the economic growth will not be high.
(C) The economic growth is high; hence, the inflation is low.
(D) Both (B) and (C)

17. Only if the train does not leave late, he can attend the interview.
(A) He attended the interview implies that the train left on time.
(B) He could not attend the interview implies that the train left late.
(C) The train left late implies that he could not attend the interview.
(D) More than one of the above

18. Whenever I go to church, I pray to god.
(A) I prayed to god means I did not go to church.
(B) I did not pray to god implies that I did not go to church.
(C) I did not go to church implies that I did not pray to god.
(D) Both (B) and (C)

19. If you deposit money, then you will get interest.
(A) You did not get interest means you did not deposit money.
(B) You deposited money. Hence, you will get interest.
(C) You deposited money but did not get interest.
(D) More than one of the above.

20. Unless Kiran finds a soulmate, he will not marry.
(A) Kiran did not find a soulmate, hence, he will not marry.
(B) Kiran found a soulmate, hence, he will marry.
(C) Kiran found a soulmate, but he will not marry.
(D) Both (A) and (B)

Directions for questions 21 to 35: In each question, there is a main statement followed by four statements a, b, c and d. From the choices, choose the pair in which the first statement implies the second statement and the two are logically consistent with the main statement.

21. Teachers can teach, only if students are well mannered.
(a) Teachers can teach.
(b) Students are not well mannered.
(c) Teachers cannot teach.
(d) Students are well mannered.
(A) ab (B) bc
(C) da (D) cd

22. Prajakta is healthy, whenever she is happy.
(a) Prajakta is not happy.
(b) Prajakta is happy.
(c) Prajakta is healthy.
(d) Prajakta is not healthy.
(A) cb (B) bc
(C) da (D) bc and da

23. Rohit is suffering either from malaria or from typhoid.
(a) Rohit is not suffering from typhoid.
(b) Rohit is suffering from malaria.
(c) Rohit is not suffering from malaria.
(d) Rohit is suffering from typhoid.
(A) ab (B) ac
(C) bc (D) dc

24. Manjula works, unless she is married.
(a) Manjula is not married.
(b) Manjula is married.
(c) Manjula works.
(d) Manjula does not work.
(A) ca (B) bd
(C) bd and ac (D) ac and db

25. If tea is sweet, then Samarth cannot drink it.
(a) Samarth can drink tea.
(b) Tea is not sweet.
(c) Samarth cannot drink tea.
(d) Tea is sweet.
(A) cb (B) ab
(C) ab and bc (D) ab and dc

26. Nisha will do an MBA only if she gets admission into a good college.

(a) Nisha will not do MBA.
(b) Nisha got admission into a good college.
(c) Nisha did not get admission into a good college.
(d) Nisha will do MBA.

(A) bd (B) db
(C) ac (D) cd

27. Rajesh is wealthy, only if he is healthy.
(a) Rajesh is not wealthy.
(b) Rajesh is healthy.
(c) Rajesh is not healthy.
(d) Rajesh is wealthy.

(A) ab (B) db
(C) bc (D) ac

28. Madhuri is not a philosopher, unless she completes her Ph.D.
(a) Madhuri is a philosopher.
(b) Madhuri is not a philosopher.
(c) Madhuri completed her Ph.D.
(d) Madhuri did not complete her Ph.D.

(A) ca (B) db
(C) ad (D) Both (A) and (B)

29. Happiness is real, whenever it is shared.
(a) Happiness is not shared.
(b) Happiness is not real.
(c) Happiness is shared.
(d) Happiness is real.

(A) cd (B) ba
(C) bd (D) Both (A) and (B)

30. Ramu wants to be either a Manager or a Director.
(a) Ramu did not become a Director.
(b) Ramu becomes a Manager.
(c) Ramu becomes a Director.
(d) Ramu did not become a Manager.

(A) bd (B) da
(C) bc (D) ab

31. If I have money, I will buy a book.
(a) I do not have money.
(b) I will not buy a book.
(c) I will buy a book.
(d) I have money.

(A) ca (B) ab
(C) dc (D) bd

32. Swati would be selected in the first company, if she has an excellent academic record.
(a) Swati is selected in the first company.
(b) Swati is not selected in the first company.
(c) Swati has an excellent academic career.
(d) Swati does not have an excellent academic record.

(A) ac (B) bd
(C) db (D) ad

33. Only if Abhijeet has good knowledge in classical music, he would be elected as Musical Idol.
(a) Abhijeet does not have good knowledge in classical music.
(b) Abhijeet is elected as Musical Idol.
(c) Abhijeet is not elected as Musical Idol.
(d) Abhijeet has good knowledge in classical music.

(A) ca (B) db
(C) bd (D) ba

34. Unless the Indian government seals the borders illegal migration in India will not stop.
(a) Indian government sealed the border.
(b) Illegal migration in India stopped.
(c) Indian government had not sealed the borders.
(d) Illegal migration in India will not stop.

(A) cd (B) ab
(C) cb (D) ad

35. Whenever Sandeep receives a message from Sangeeta, he seems to be on cloud nine.
(a) Sandeep did not receive a message from Sangeeta.
(b) Sandeep is on cloud nine.
(c) Sandeep is not on cloud nine.
(d) Sandeep received a message from Sangeeta.

(A) ba (B) bd
(C) cd (D) ca

Directions for questions 36 to 40: Select the correct alternative from the given choices.

36. If Ali has good knowledge of JAVA, he will be selected in Satyam Computers. Unless Ali is not selected in Satyam Computers, he will not be selected in CTS.
Ali is selected in CTS implies that
(A) Ali has good knowledge in JAVA.
(B) Ali is selected in Satyam Computers.
(C) Ali does not have good knowledge of JAVA.
(D) None of these

37. The HR manager of TCS will come, if the strike does not affect the flight timings. Only if the HR manager of TCS comes, TCS will recruit people.
TCS is recruiting people implies that
(A) The strike affects the flight timings.
(B) The strike does not affect the flight timings.
(C) The HR manager of TCS does not come.
(D) None of these

38. Unless the coding is not tested, the company can implement it. If the company can implement the coding, the network system will work properly.
The network is not working properly, it implies that
(A) The coding is tested.
(B) The coding is not tested.

(C) The company implements the coding
(D) None of these

39. When the Infosys team's performance is excellent, then Infosys will become the top IT company. Either Infosys does not become the top IT company or TCS remains in the top rank.
The Infosys team's performance is excellent, means that
(A) TCS remains in the top rank.
(B) TCS will not remain in the top rank.
(C) Infosys will be in the top rank.
(D) None of these

40. If a person follows the conventional methods, he cannot be successful. Unless a person is successful, he cannot be a part of successful company.
Mr Prasad has become a part of P & G, a successful company. Hence, it can be concluded that
(A) Mr. Prasad is not successful.
(B) Mr Prasad follows conventional methods.
(C) Mr Prasad does not follow conventional methods.
(D) None of these

EXERCISE-2

Directions for questions 1 to 18: Each question given below is a statement followed by four different statements. Choose the one which is the correct negation of the given statement.

1. Either Anand marries Vandana or Madhavi marries Kollol.
(A) Anand does not marry Vandana, so Madhavi marries Kollol.
(B) Neither Anand marries Vandana nor Madhavi marries Kollol.
(C) Madhavi does not marry Kollol but Anand marries Vandana.
(D) None of these

2. Whenever Bhiru and Basanti go for a long drive, Joy follows them.
(A) Joy follows Bhiru and Basanti but they are not going for a long drive.
(B) Bhiru and Basanti are going for a long drive and Joy follows them.
(C) Joy does not follow Bhiru and Basanti even when they go for a long drive.
(D) None of these

3. Pratap Rana will attend the class, only if his father allows him to go by bike.
(A) Pratap Rana is not attending the classes even his father allows him to come by bike.
(B) Pratap Rana's father did not allow him to go by bike but he was attending the class.
(C) Pratap Rana is not attending the classes because his father did not allow him to go on bike.
(D) None of these

4. Unless Aiswariya plays the role of Paro, Madhuri will not play the role of Chandramukhi.
(A) Madhuri is not playing the role of Chandramukhi, but Aiswariya is playing the role of Paro.
(B) Aiswariya is playing the role of Paro, Madhuri is playing the role of Chandramukhi.
(C) Madhuri is playing the role of Chandramukhi but Aiswariya is not playing the role of 'Paro'.
(D) None of these

5. Unless the change happens, the problem will not be solved.
(A) The problem is solved and the change did not happen.
(B) The change happened but the problem is not solved.
(C) The change happened and the problem is solved.
(D) The problem is solved implies that the change happened.

6. The presentation was lengthy but simple.
(A) The presentation was not lengthy and not simple.
(B) The presentation was lengthy but not simple.
(C) The presentation was not lengthy or not simple.
(D) The presentation was simple but not lengthy.

7. Unless Tarun learns the basics, he cannot solve connectives.
(A) Tarun learned the basics but he could not solve connectives.
(B) Tarun did not learn the basics, but he could solve connectives.
(C) Tarun learned basics and solved connectives.
(D) Tarun did not learn basics and he did not solve connectives.

8. He either goes to US or he will join in a job.
(A) He went to US and did not join in a job.
(B) He went to US but joined in a job.
(C) He did not go to US and joined in a job.
(D) He did not go to US and did not join in a job.

9. If you share your sorrow with your friends, you will be happy.
(A) You did not share your sorrow with your friends and you are happy.
(B) You shared your sorrow with your friends but you are not happy.
(C) You did not share your sorrow with your friends and you are not happy.
(D) You shared your sorrow with your friends so you are happy.

10. Every mind works at its best, only if it is open.
(A) Mind worked at its best when it is open.
(B) Mind did not work at its best because it is not open
(C) Mind worked at its best even though it is not open.
(D) Mind did not work at its best even when it is open.

11. If it is the post of a manager, then Shastri will join the firm.
(A) It is the post of a manager, but Shastri did not join the firm.
(B) The post is not of a manager, but Shastri joined the firm.
(C) Shastri did not join the firm as the post is not of a manager.
(D) Shastri joined the firm as the post is that of a manager.

12. I cannot make tomato soup, unless I have some onions.
(A) I have onions but I cannot make tomato soup.
(B) I do not have onions; hence, I cannot make tomato soup.
(C) I made tomato soup though I do not have onions.
(D) I have onions; hence, I can make tomato soup.

13. Only if Tara is happy, then she does not go to work.
(A) Tara is not happy and she does not go to work.
(B) Tara is happy and she goes to work.
(C) Tara is not happy and she goes to work.
(D) Tara is happy and she does not go to work.

14. Paul is popular either as a lead guitarist or as a base guitarist.
(A) Paul is popular as a lead guitarist but not as a base guitarist.
(B) Paul is famous neither as a lead guitarist nor as a base guitarist.
(C) Paul is not popular as base guitarist but popular as a lead guitarist.
(D) Paul is popular as both a lead guitarist and as a base guitarist.

15. Kohli cannot score a hundred, unless Jhonson bowls.
(A) Kohli scored a hundred though Jhonson did not bowl.
(B) Jhonson did not bowl hence Kohli did not score a hundred.
(C) Jhonson bowled but Kohli did not score a hundred.
(D) Jhonson bowled but Kohli scored a hundred.

16. Either he wears shoes or a tie.
(A) Neither he wore shoes nor he wore a tie.
(B) He wore a tie but not shoes.
(C) He wore shoes but did not wear a tie.
(D) All of the above

17. If it is a holiday, I will sleep throughout the day.
(A) I slept throughout the day even though it is not a holiday.
(B) I did not sleep throughout the day even though it is a holiday.
(C) I slept throughout the day even though it is a holiday.
(D) Both (A) and (C)

18. Rajesh goes to college and attends classes.
(A) Rajesh either goes to college or attends classes.
(B) Rajesh will not go to college but attends college.
(C) Rajesh neither goes to college nor attends classes.
(D) None of these

Directions for questions 19 to 33: Each question below consists of a main statement followed by four answer choice. From the answer choices, select the one that logically follows the main statement.

19. If Ankita eats pastry, then it is a black forest or a pineapple.
(A) Ankita eats pastry but it is not a pineapple, means it is a black forest.
(B) Ankita eats pastry but it is not a black forest, means it is a pineapple.
(C) The pastry is neither a pineapple nor a black forest, means Ankita does not eat the pastry.
(D) All the above.

20. If you want to stay fit, then you must eat nutritious food and exercise.
(A) You did not eat nutritious food; hence, you cannot stay fit.
(B) You did not exercise which implies you cannot stay fit.
(C) You ate nutritious food and exercised which implies you have stayed fit.
(D) Both (A) and (B)

21. Yaseem plays cricket, only if he wears blue or white.
(A) Yaseem plays cricket which implies he wears blue and white.
(B) Yaseem plays cricket but he does not wear blue, hence, he wears white.

(C) Yaseem wears neither blue nor white implies that he may play cricket.
(D) None of these

22. Sunil cannot meet his friends or his family, unless he has a job.
(A) Sunil met his friend and his family implies he has a job.
(B) Sunil does not have a job and cannot meet his family; hence, he cannot meet his friends.
(C) Sunil does not have a job but he met his family implies that he cannot meet his friends.
(D) Both (A) and (C)

23. Only if there is a sale, I will buy clothes or cosmetics.
(A) I have bought clothes and cosmetics means there is a sale.
(B) I have bought cosmetics, hence, there is a sale.
(C) I have bought clothes, hence, there is a sale.
(D) All the above.

24. Unless you take medicines, you will not recover and will not be able to walk.
(A) You recovered means you have taken medicines.
(B) You are not able to walk means you have not taken medicines.
(C) You have not taken medicines; hence, you will not recover but you will be able to walk.
(D) You did not recover means you have not taken medicines.

25. If Ganshyam goes to the U.S.A., his mother or his brother will accompany him.
(A) Ganshyam is going to the U.S.A. but his brother is not accompanying him implies his mother will accompany him.
(B) Neither Ganshyam's mother nor his brother is accompanying him means Ganshyam is not going to the U.S.A.
(C) Ganshyam is not going to the U.S.A, hence neither his mother nor his brother is accompanying him.
(D) Both (A) and (B)

26. Whenever Preeti watches TV, then she watches movies and sports.
(A) Preeti is not watching sports implies that she is watching TV.
(B) Preeti is not watching movies implies she is not watching TV.
(C) Preeti is neither watching movies nor sports, hence, she must be watching something else on TV.
(D) Preeti is watching TV, but not movies, implies that she is watching sports.

27. Sagar will marry Sheela, only if she is a graduate and a good cook.
(A) Sheela is a good cook but not a graduate, hence, Sagar will not marry Sheela.
(B) Sagar will marry Sheela since she is a good cook though she is not a graduate.
(C) Sheela is a graduate and a good cook implies that Sagar will marry Sheela.
(D) Sagar did not marry Sheela implies that she is neither a graduate not a good cook.

28. The electricity supply will not be restored and we will not be able to watch TV unless you pay the bill.
(A) The electricity supply is restored and we are able to watch TV implies that you have paid the bill.
(B) We are able to watch TV but electricity supply is not restored implies that you have paid the bill.
(C) We are unable to watch TV but electricity supply is restored implies that you have paid the bill.
(D) All the above

29. If you try hard, then you can win or gain.
(A) You tried hard but did not win means you gained.
(B) You tried hard but did not gain, means you won.
(C) You neither won nor gained means you did not try hard.
(D) All the above

30. Rajesh cooks, only if it is Sunday or Saturday.
(A) Rajesh cooked implies that it is Sunday and Saturday.
(B) Rajesh cooked but it is not Sunday, hence, it is Saturday.
(C) Rajesh cooked but it is neither Sunday nor Saturday.
(D) None of these

31. Ravi does not meet Pranith or Mani, unless he goes to New York.
(A) Ravi met Pranith and Mani implies he went to New York.
(B) Ravi did not go to New York and did not meet Pranith, hence, he did not meet Mani.
(C) Ravi did not go to New York but he met Mani implies that he did not meet Pranith.
(D) Both (A) and (C)

32. Whenever David goes to church, he donates money and clothes.
(A) David did not donate clothes implies that he went to church.
(B) David did not donate money implies that he did not go to church.
(C) David donated neither money nor clothes, hence, he must have gone to church.
(D) David went to church but did not donate money implies that he donated clothes.

33. If Prashanth buys a book, he gives it to his brother or his friend.

(A) Prashanth bought a book, but he did not give it to his friend implies that he gives it to his brother.

(B) Prashanth gave the book to neither his brother nor his friend means he did not buy a book.

(C) Prashanth did not buy a book, hence, he gave the book neither to his brother nor to his friend.

(D) Both (A) and (B)

34. If I am not paid, I will not work and I will not take leave.

(A) If I have worked and I took leave, then I am paid.

(B) If I have worked and I have not taken leave, then I was paid.

(C) If I have worked or I have taken leave, then I was paid.

(D) More than one of the above.

35. If Rama leaves Ayodhya, then he will go to forest or to Sri Lanka.

(A) Rama did not go to forest and did not go to Sri Lanka, implies that he did not leave Ayodhya.

(B) Rama did not leave Ayodhya, implies that he will not go to forest or will not go to Sri Lanka.

(C) Rama went to forest or to Sri Lanka, implies that he did not leave Ayodhya.

(D) Rama did not leave Ayodhya, implies that he will not go to forest and will not go to Sri Lanka.

36. Unless the party gets a majority, the house will be dissolved and the President's rule will be imposed.

(A) The party got a majority, it means that either the house will not be dissolved or the President's rule will not be imposed.

(B) The house is not dissolved or the President's rule is not imposed, means that the party got a majority.

(C) The house is not dissolved and the President's rule is not imposed means that the party got a majority.

(D) Both (B) and (C).

37. If you plant trees, then there will be no pollution and you get fruits.

(A) If there is no pollution and you did not get fruits, then you planted trees.

(B) If there is pollution and you did not get fruits, then you did not plant trees.

(C) If there is pollution or you did not get fruits, then you did not plant trees.

(D) Both (B) and (C)

38. If there is no traffic, then I will not drive slow but I will go on a long drive.

(A) If there is traffic, then I will drive slow but I will not go on a long drive.

(B) If there is traffic, then I will not drive slow but I will not go on a long drive.

(C) If I drive slow or I do not go on a long drive, it means that there is traffic.

(D) If I did not drive slow and I went on a long drive, it means that there is traffic.

Directions for questions 29 and 30: Each question consists of a set of statements in alphabetical order. Assume that each one of these statements is individually true. Each of the four choices consists of a subset of these statements. Choose the subset as your answer where the statements therein are logically consistent among themselves.

39. (a) Only if the water level in the coastal areas rises, then the people change their life style.

(b) People change their life style only if they are rewarded.

(c) If people are rewarded, then they will not change their life style.

(d) If the temperatures rise, then the water level in the coastal areas rises.

(e) Whenever the water level in the coastal areas rises, then the temperature rises.

(f) Unless the people change their lifestyle, temperature rises.

(g) People are rewarded.

(h) Water level in the coastal area does not rise.

(A) c, d, f, g and h (B) g, f, d, *b* and h
(C) a, c, d, g and h (D) e, f, g, h and b

40. (a) If Gulam sings, then audience sleep.

(b) If Gulam sings, then audience dance.

(c) Unless audience do not dance, the concert will be successful.

(d) Only if audience dance, the concert will be successful.

(e) If Vani dances, then Gulam sings.

(f) Gulam sings, only if Vani dances.

(g) Vani dances.

(h) The concert is successful.

(A) c, f, g, *b* and h (B) a, c, f, g and h
(C) e, c, g, *b* and h (D) d, f, g, h and b

Difficulty Level-2: Moderate

EXERCISE-3

Directions for questions 1 to 10: Each question below consists of a main statement followed by some numbered statements. From the numbered statements, select the one that logically follows the main statement.

1. If it is a holiday, then I will go for a picnic or I will visit my uncle's house.
   (A) I will not go for a picnic or I will not visit my uncle's house implies that it is not a holiday.
   (B) If it is not a holiday, then I will not go for a picnic and I will not visit my uncle's house.
   (C) I will not go for a picnic and I will not visit my uncle's house implies that it is not a holiday.
   (D) If it is not a holiday, then I will not go for a picnic or I will not visit my uncle's house.

2. Whenever my mom scolds me, I either hide behind my dad or complain to my grandma.
   (A) If I complain to my grandma or I hide behind to my dad, then my mom must have scolded me.
   (B) If I did not complain to my grandma and I did not hide behind my dad, then my mom must not have scolded me.
   (C) If my mom does not scold me, I will neither hide behind my dad nor complain to my grandma.
   (D) Both (A) and (B)

3. Whenever it rains, I will either carry an umbrella or wear a raincoat.
   (A) It is not raining means that I will neither carry an umbrella nor wear a raincoat.
   (B) I am carrying an umbrella or I am wearing a raincoat, implies that it is raining.
   (C) I am not carrying an umbrella or I am not wearing a raincoat means that it is not raining.
   (D) If it is raining but I am not wearing a raincoat means that I must be carrying an umbrella.

4. If it is very hot outside, then I will carry an onion with me and I will return home by lunch time.
   (A) I will not carry an onion with me or I will not return home by lunch time means that it is not very hot outside.
   (B) It is not very hot outside means that I will not carry an onion with me and I will not return home by lunch time.
   (C) I will not carry an onion with me and I will return home by lunch time means that it is very hot outside.
   (D) I will carry an onion with me and I will return home by lunch time means that it is very hot outside.

5. Whenever Arpita's father is in town, she abstains from college and goes to her uncle's house.
   (A) If Arpita has not abstained from college or she has not gone to her uncle's house means that her father is not in town.
   (B) If Arpita has not abstained from college but her father is in town, then she will definitely go to her uncle's house.
   (C) If Arpita has abstained from college but she has not gone to her uncle's house, it means that her father is not in town.
   (D) Both (A) and (C) above.

6. If the tea is not hot, then I will not go to school and will not have dinner.
   (A) If I have gone to school or I have not had dinner, then the tea is not hot.
   (B) If I have gone to school and I had dinner, then the tea is hot.
   (C) If I have gone to school and I have not had dinner, then the tea is hot.
   (D) If I have gone to school or I have had dinner, then the tea is hot.

7. If Ramesh leaves his job, then he will join for an MBA course or for an MCA course.
   (A) Ramesh has neither joined an MBA course nor an MCA course implies that he has not left his job.
   (B) Ramesh has not left his job implies that he will not join an MBA course or he will not join an MCA course.
   (C) Ramesh has joined an MBA course or an MCA course implies that he has not left his job.
   (D) Ramesh has not left his job implies that he will not join an MBA course and he will not join an MCA course.

8. Unless we win the Assembly elections, we will lose the Rajya Sabha elections and the Presidential elections.
   (A) We have won the Assembly elections, it means that we will not lose either the Rajya Sabha elections or the Presidential elections.
   (B) We have not lost the Rajya Sabha elections or we have not lost the Presidential elections means that we have won the Assembly elections.
   (C) We have not lost the Rajya Sabha elections and not lost the Presidential elections means that we have won the Assembly elections.
   (D) Both (B) and (C).

9. If it is a Sunday, then on that day there is no college and I go to Church.

(A) If there is no college and I do not go to Church, then that day is a Sunday.
(B) If there is college and I do not go to Church, then that day is not a Sunday.
(C) If there is college or I do not go to Church, then that day is not a Sunday.
(D) Both (B) and (C).

10. If it is not raining, then I will not go for a movie but I will visit my friend's house.

(A) If it is raining, then I will go for a movie but I will not visit my friend's house.
(B) If it is raining, then I will not go for a movie but I will not visit my friend's house.
(C) If I go for a movie or I do not visit my friend's house, it means that it is raining.
(D) If I will not go for a movie and I will visit my friend's house, it means that it is raining.

Directions for questions 11 to 13: Each question below consists of a main statement followed by four numbered statements. From the numbered statements, select the one that logically follows the main statement.

11. Harish will get through the interview, if he is thorough with the basics.

(A) Harish got through the interview, hence, he was thorough with the basics.
(B) Harish is not thorough with the basics; hence, he will not get through the interview.
(C) Harish did not get through the interview means he was not thorough with the basics.
(D) Although he was not thorough with the basics, still Harish managed to get through the interview.

12. Either Pakistan or China will attack India, only if India supports Russia and the USA.

(A) Pakistan and China attacked India means India supported Russia and the USA.
(B) India neither supported Russia nor supported the USA means that only Pakistan attacked India.
(C) India supported the USA but not Russia, means that only China attacked India.
(D) All of the above

13. I will neither talk to you nor play with you, unless you apologize to me.

(A) I talked with you or played with you means that you apologized to me.
(B) I did not apologize to you means that you neither talked with me nor played with me.
(C) You apologized to me means that you neither talked nor played with me.
(D) Both (A) and (B)

Directions for questions 14 and 15: Each question has a main statement followed by four statements labelled as a, b, *c* and d. Choose the ordered pair of statements where the first statement implies the second and the two statements are logically consistent with the main statement.

14. Either Rajeev is a genius or he cheated in the exam.

(a) Rajeev cheated in the exam.
(b) Rajeev is a genius.
(c) Rajeev is not a genius.
(d) Rajeev did not cheat in the exam.

(A) Only ca	(B) Only bd
(C) cd and ba	(D) db and ca

15. Unless the politician took money, he is not good enough.

(a) The politician is not good enough.
(b) The politician took money.
(c) The politician is good enough.
(d. The politician did not take money.

(A) Only bc	(B) Only da
(C) Only ab	(D) cb and da

Directions for questions 16 to 20: Each question below consists of a statement followed by some numbered statements. From the numbered statements, select the one that logically negates the main statement.

16. Sravan will go to the movie, if his parents are not with him.

(A) Sravan did not go to the movie and his parents are with him.
(B) Sravan's parents are with him and he went to the movie.
(C) Sravan did not go to the movie and his parents are not with him.
(D) Sravan went to the movie and his parents are not with him.

17. Ramesh works very hard whenever there is an exam.

(A) Ramesh worked very hard and there is no exam.
(B) Ramesh did not work hard and there is no exam.
(C) Ramesh did not work hard and there is an exam.
(D) Both (B) and (C)

18. Either it is a flying saucer or the person is not telling the truth.

(A) It is not a flying saucer and the person is not telling the truth.
(B) The person is telling the truth and it is not a flying saucer.
(C) It is a flying saucer and the person is telling the truth.
(D) The person is not telling the truth and it is a flying saucer.

19. Sachin scores a century, unless he is paired with the Captain.
(A) Sachin is paired with the Captain and he did not score a century.
(B) Sachin scored a century and he is not paired with the Captain.
(C) Sachin is paired with the Captain and he scored a century.
(D) Sachin did not score a century and he is not paired with the Captain.

20. Bond will buy the car only if it is the costliest and fastest.
(A) Bond did not buy the car and it is neither the fastest nor the costliest.
(B) Bond bought the car and it is not the costliest or it is not the fastest.
(C) The car is the fastest and costliest, and Bond did not buy it.
(D) Bond bought the car and it is the fastest and costliest.

Directions for questions 21 to 23: Each question has a main statement followed by four statements labelled as a, b, *c* and d. Choose the ordered pair of statements where the first statement implies the second and the two statements are logically consistent with the main statement.

21. If the price of a good increases, then its consumption decreases.
(a) The price of a good increased.
(b) The price of a good did not increase.
(c) The consumption of the good decreased.
(d) The consumption of the good did not decrease.

(A) cb (B) ad
(C) ca (D) db

22. The exam's difficulty level increases only if the number of applicants increases.
(a) The exam's difficulty level increased.
(b) The exam's difficulty level did not increase.
(c) The number of applicants have increased.
(d) The number of applicants have not increased.

(A) ca (B) db
(C) ac (D) both (B) and (C)

23. Whenever the hero needs money, he acts in a new movie.
(a) The hero needed money.
(b) The hero did not need money.
(c) The hero acted in a new movie.
(d) The hero did not act in a new movie.

(A) ac (B) ad
(C) ca (D) bd

Directions for questions 24 and 25: Each question below consists of a main statement followed by four numbered statements, From the numbered statements, select the one that logically follows the main statement.

24. The chief guest will come on time, if the fog does not affect the flight timings. Only if the chief guest comes, then the meeting will be started.
The meeting started implies that
(A) The fog did not affect flight timings.
(B) The fog affected the flight timings.
(C) The chief guest did not come.
(D) None of these

25. Unless coding is done, the software project cannot be completed. If the company does not meet the project completion deadline, then the team working on it will be fired.
The team working in the project, is not fired implies
(A) The coding is done.
(B) The software project was not completed.
(C) The company did not meet the project completion deadline.
(D) None of these

ANSWER KEYS

Exercise-1

1.	(D)	8.	(D)	15.	(B)	22.	(D)	29.	(D)	36.	(C)
2.	(D)	9.	(D)	16.	(D)	23.	(A)	30.	(D)	37.	(D)
3.	(B)	10.	(D)	17.	(C)	24.	(D)	31.	(C)	38.	(B)
4.	(C)	11.	(D)	18.	(B)	25.	(D)	32.	(B)	39.	(A)
5.	(D)	12.	(D)	19.	(D)	26.	(B)	33.	(C)	40.	(C)
6.	(D)	13.	(A)	20.	(A)	27.	(B)	34.	(A)		
7.	(D)	14.	(C)	21.	(B)	28.	(B)	35.	(D)		

Exercise-2

1.	(B)	8.	(D)	15.	(A)	22.	(D)	29.	(D)	36.	(D)
2.	(C)	9.	(B)	16.	(A)	23.	(D)	30.	(B)	37.	(D)
3.	(B)	10.	(C)	17.	(B)	24.	(A)	31.	(D)	38.	(C)
4.	(C)	11.	(A)	18.	(D)	25.	(D)	32.	(B)	39.	(B)
5.	(A)	12.	(C)	19.	(D)	26.	(B)	33.	(D)	40.	(C)
6.	(C)	13.	(A)	20.	(D)	27.	(A)	34.	(D)		
7.	(B)	14.	(B)	21.	(B)	28.	(D)	35.	(A)		

Exercise-3

1.	(C)	5.	(D)	9.	(D)	13.	(A)	17.	(C)	21.	(D)	24.	(D)
2.	(B)	6.	(D)	10.	(C)	14.	(D)	18.	(B)	22.	(D)	25.	(A)
3.	(D)	7.	(A)	11.	(C)	15.	(D)	19.	(D)	23.	(A)		
4.	(A)	8.	(D)	12.	(A)	16.	(C)	20.	(B)				

SOLUTIONS

EXERCISE-1

1. If $\overset{p}{\boxed{\text{movie is a superhit,}}}$ then $\overset{q}{\boxed{\text{i watch it}}}$ The statement is of the form, 'If p, then q'.
The implications are (i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$
Choice (D) is according to (ii).

2. Whenever $\overset{p}{\boxed{\text{it is cold,}}}$ $\overset{q}{\boxed{\text{I wear a jacket.}}}$The statement is of the form 'whenever p then q'.
The implications are: (i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$.
Choice (A) is according to (i) and Choice (B) is according to (ii).

3. Either or
The statement is of the form 'either p or q'.
The implications are:
(A) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$
Choice (B) is according to (ii).

4. Only if $\overset{p}{\boxed{\text{Raj is not coming}}}$, then
$\overset{q}{\boxed{\text{Geetika will come to the party}}}$
The statement is of the form 'only if p, then q'.
The implications are:
(i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$
Choice (C) is according to (i).

5. Unless $\overset{p}{\boxed{\text{I have money,}}}$ $\overset{q}{\boxed{\text{I cannot enjoy my weekend}}}$
The statement is of the form 'unless p then q'.
The implications are:
(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$
Choice (B) is according to (ii) and Choice (C) is according to (i).

6. If $\overset{p}{\boxed{\text{I can swim,}}}$ then $\overset{q}{\boxed{\text{I can clear the exam}}}$

The statement is of the form, 'If p then q'.
The implications are:
(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$
Choice (A) is according to (i) and Choice (B) is according to (ii).

7. $\overset{q}{\boxed{\text{India will talk}}}$ Only if $\overset{p}{\boxed{\text{the terrorists are handed over.}}}$

The statement is of the form 'only if p then q'.
The implications are:
(i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$
Choice (D) is according to (i).

8. Whenever $\overset{p}{\boxed{\text{there is demand}}}$ $\overset{q}{\boxed{\text{there will be supply}}}$

The statement is of the form 'whenever p then q'.
The implications are:
(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$
Choice (C) is according to (ii) and Choice (B) is according to (i).

9. Suhasita purchases either $\overset{p}{\boxed{\text{a cooler}}}$ or $\overset{q}{\boxed{\text{a refrigerator}}}$

The statement is of the form 'either p or q'.
The implications are:
(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$
Choice (A) is according to (i).
Choice (B) is according to (ii).

10. $\overset{q}{\boxed{\text{I will not have enemies}}}$ Unless $\overset{p}{\boxed{\text{I fight}}}$

The statement is of the form 'unless p then q'.
The implications are:
(A) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$
Choice (B) is according to (i).
Choice (C) is according to (ii).

11. If $\overset{p}{\boxed{\text{you are good,}}}$ then $\overset{q}{\boxed{\text{everyone loves you.}}}$

The statement is of the form, 'if p, then q'.
The implications are: (i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$
Choice (D) is according to (ii).

12. Unless $\overset{p}{\boxed{\text{Pratik comes,}}}$ $\overset{q}{\boxed{\text{his son does not eat}}}$

The statement is of the form, 'unless p, q'.
The implications are: (i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$
Choice (A) is according to (i) and Choice (C) is according to (ii).

13. Only if $\overset{p}{\boxed{\text{you are qualified in written test}}}$

$\overset{q}{\boxed{\text{you get a call for an interview}}}$

The statement is of the form 'only if p, then q'.
The implications are:
(i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$
Choice (A) is according to (ii).

14. The given statement is of the form
If $\underset{p}{\boxed{\text{Raju plays well}}}$ then $\underset{q}{\boxed{\text{the team can win the match.}}}$

The implications are $p \Rightarrow q$ and $\sim q \Rightarrow \sim p$.
(C) logically follows the main statement.

15. The given statement is of the form
Either $\underset{p}{\boxed{\text{He plays cricket}}}$ or $\underset{q}{\boxed{\text{eats biscuit.}}}$

The implications are $\sim p \Rightarrow q$ and $\sim q \Rightarrow p$.
(B) logically follows the main statement.

16. The given statement is of the form
Unless $\underset{p}{\boxed{\text{the inflation is low}}}$

$\underset{q}{\boxed{\text{economic growth will not high.}}}$

The implications are $\sim p \Rightarrow q$ and $\sim q \Rightarrow p$
$\therefore$ Both (B) and (C) logically follow the main statement.

17. The given statement is of the form.
Only if:
$\underset{p}{\boxed{\text{the train does not leave late}}}$ then

$\underset{q}{\boxed{\text{he can attend the interview}}}$

The implications are $q \Rightarrow p$ and $\sim p \Rightarrow \sim q$.
(C) logically follows from the main statement.

18. The given statement is of the form:
Whenever $\underset{p}{\boxed{\text{I go to church}}}$ then $\underset{q}{\boxed{\text{I pray to god}}}$

The implications are: $p \Rightarrow q$ and $\sim q \Rightarrow \sim p$
Choice (B) logically follows the main statement.

19. If $\overset{p}{\boxed{\text{you deposit money,}}}$ $\overset{q}{\boxed{\text{you will get interest.}}}$

The statement is of the form 'if p, then q'.
The implications are: (i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$
Choice (A) follows (ii) and Choice (B) follows (i).

20. Unless $\overset{p}{\boxed{\text{Kiran finds a soul mate,}}}$ $\overset{q}{\boxed{\text{he will not marry}}}$

The statements of the form 'unless p, q'.
The implications are: (i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$
Choice (A) follows (i).

21. $\overset{q}{\boxed{\text{Teachers can teach,}}}$ only if $\overset{p}{\boxed{\text{students are well mannered}}}$

The statement is of the form 'only if p then q'. The implications are: (i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$

(a) $\Rightarrow q$ (b) $\Rightarrow \sim p$
(c) $\Rightarrow \sim q$ (d) $\Rightarrow p$

Here, 'ad' and 'bc' are proper pairs.

22. $\overset{q}{\boxed{\text{Prajakta is healthy}}}$ whenever $\overset{p}{\boxed{\text{she is happy}}}$

The statement is of the form 'whenever p then q'.
The implications are:

(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$

(a) $\Rightarrow \sim p$ (b) $\Rightarrow p$
(c) $\Rightarrow q$ (d) $\Rightarrow \sim q$

Here, 'bc' and 'da' are proper pairs.

23. Rohit is suffering either

$\overset{p}{\boxed{\text{from malaria}}}$ or $\overset{q}{\boxed{\text{from typhoid.}}}$

The statement is of the form 'either p or q'.
The implications are:

(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$

(a) $\Rightarrow \sim q$ (b) $\Rightarrow p$
(c) $\Rightarrow \sim p$ (d) $\Rightarrow q$

Here, 'ab' and 'cd' are the proper pairs.

24. $\overset{q}{\boxed{\text{Manjula works}}}$ unless $\overset{p}{\boxed{\text{she is married.}}}$

The statement is of the form 'unless p then q'.
The implications are:

(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$

(a) $\Rightarrow \sim p$ (b) $\Rightarrow p$
(c) $\Rightarrow q$ (d) $\Rightarrow \sim q$

Here, 'ac' and 'db' are the proper pairs.

25. If $\overset{p}{\boxed{\text{tea is sweet}}}$ then $\overset{q}{\boxed{\text{Samarth cannot drink it.}}}$

The statement is of the form 'if p then q'.
The implications are:

(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$

(a) $\Rightarrow \sim q$ (b) $\Rightarrow \sim \mathrm{p}$
(c) $\Rightarrow q$ (d) $\Rightarrow p$

Here, 'dc' and 'ab' are the proper pairs.

26. $\overset{q}{\boxed{\text{Nisha will do an MBA}}}$ only if

$\overset{p}{\boxed{\text{she gets admission into a good college.}}}$

The statement is of the form 'only if p then q'.
The implications are:

(i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$

(a) $\Rightarrow \sim q$ (b) $\Rightarrow p$
(c) $\Rightarrow \sim p$ (d) $\Rightarrow q$

Here, 'db' and 'ca' are the proper pairs.

27. $\overset{q}{\boxed{\text{Rajesh is wealthy}}}$, only if $\overset{p}{\boxed{\text{he is healthy}}}$

The statements of the form 'only if p, then q'.
The implications are:

(i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$

(a) $\sim q$ (b) p
(c) $\sim p$ (d) q

Here, 'db' and 'ca' are proper pairs.

28. $\overset{q}{\boxed{\text{Madhuri is not a philosopher,}}}$ unless

$\overset{p}{\boxed{\text{she completes her Ph.D.}}}$

The statement is of the form 'unless p, q'.
The implications are:
(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$

(a) $\sim q$ (b) q
(c) p (d) $\sim p$

Here, 'db' and 'ac' are proper pairs.

29. $\overset{q}{\boxed{\text{Happiness is real,}}}$ whenever $\overset{p}{\boxed{\text{it is shared}}}$

The statement is of the form 'whenever p, then q'.
The implications are:
(i) $p \Rightarrow \mathrm{q}$ (ii) $\sim q \Rightarrow \sim \mathrm{p}$

(a) $\sim \mathrm{p}$ (b) $\sim q$
(c) p (d) q

Here, 'cd' and 'ba' are proper pairs.

30. Ramu wants to be either

$\overset{p}{\boxed{\text{a Manager}}}$ or $\overset{q}{\boxed{\text{a Director}}}$

The statement is of the form 'either p or q'.
The implications are:
(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$

(a) $\sim q$ (b) p
(c) q (d) $\sim p$

Here, 'dc' and 'ab' are proper pairs.

31. If $\overset{p}{\boxed{\text{I have money,}}}$ $\overset{q}{\boxed{\text{I will buy a book}}}$

The statement is of the form 'if p, then q'.
The implications are:
(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$

(a) $\sim p$ (b) $\sim q$
(c) q (d) p
Here, 'dc' and 'ba' are proper pairs.

32. $\overset{q}{\boxed{\text{Swati would be selected in the first company,}}}$ if $\overset{p}{\boxed{\text{she has an excellent academic record.}}}$

The implications of the statement are:
(i) $p \Rightarrow$ q and (ii) $\sim q \Rightarrow \sim p$
$\therefore$ It can be ca or bd.

33. Only if $\overset{p}{\boxed{\text{Abhijeet has good knowledge is classical music,}}}$ $\overset{q}{\boxed{\text{he would be elected as Musical Idol.}}}$

The implications of the above statement are:
(i) $q \Rightarrow p$ and (ii) $\sim p \Rightarrow \sim q$
It can be bd or ac.

34. Unless $\overset{p}{\boxed{\text{The Indian government seals the borders,}}}$ $\overset{q}{\boxed{\text{illegal migration wil not stop.}}}$

The implications of the above statement are:
(i) $\sim p \Rightarrow q$ and (2) $\sim q \Rightarrow$ p
It can be cd or ba.

35. Whenever $\overset{p}{\boxed{\text{Sandeep receives a message from Sangeeta}}}$ $\overset{q}{\boxed{\text{he seems to be on cloud nine.}}}$

The implications of the above statement are:
(1) $p \Rightarrow q$ and (2) $\sim q \Rightarrow \sim p$
It can be db or ca.

36. If $\overset{p}{\boxed{\text{Ali has good knowledge of JAVA,}}}$ $\overset{q}{\boxed{\text{he will selected in Satyam Computers,}}}$

The implications for the above statement are:
(i) $p \Rightarrow q$ and (ii) $\sim q \Rightarrow \sim p$
Unless $\overset{\sim q}{\boxed{\text{Ali is not selected in Satyam Computers}}}$ $\overset{r}{\boxed{\text{he will not be selected in CTS.}}}$

The implications for the above statement are:
(iii) $q \Rightarrow r$ and (iv) $\sim r \Rightarrow \sim q$
The given statement:
Ali is selected in CTS [$\sim r$]
From (iv) $\sim r \Rightarrow \sim q$ and from (ii) $\sim q \Rightarrow \sim p$
$\therefore$ $\sim r \Rightarrow \sim p$, i.e., Ali does not have good knowledge of JAVA.

37. $\overset{q}{\boxed{\text{The HR manager of TCS will come,}}}$ if $\overset{p}{\boxed{\text{the strike does not effect the flight timings.}}}$

The implications of the above statement are:
(i) $p \Rightarrow$ q (ii) $\sim q \Rightarrow \sim p$
Only if $\overset{q}{\boxed{\text{The HR manager of TCS comes,}}}$ $\overset{r}{\boxed{\text{TCS will recruit people.}}}$

The implications of the above statement are:
(iii) $r \Rightarrow q$ and (iv) $\sim q \Rightarrow \sim r$
The given statement is TCS is recruiting people [r].
From (iii), $r \Rightarrow q$.
But there is no implication for q.
So, the answer will be none of these.

38. Unless $\overset{p}{\boxed{\text{the coding is not tested,}}}$ $\overset{q}{\boxed{\text{the company can implement it.}}}$

The implications of the above statement are:
(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow$ p
If $\overset{q}{\boxed{\text{the company can implement the coding}}}$ $\overset{r}{\boxed{\text{the network system will work properly.}}}$

The implications of the above statement are:
(iii) $q \Rightarrow r$ and (iv) $\sim r \Rightarrow \sim q$
The given statement, 'The network is not working properly' ($\sim r$).
From (iv) $\sim r \Rightarrow \sim q$ and from (ii) $\sim q \Rightarrow$ p
$\therefore$ $\sim r \Rightarrow p$ (The coding is not tested).

39. Whenever $\overset{p}{\boxed{\text{the Infosys team's performance is excellent,}}}$ $\overset{q}{\boxed{\text{they become the top IT company}}}$

The implications of the above statement are:
(i) $p \Rightarrow$ q (ii) $\sim q \Rightarrow \sim p$

$\sim q$

Either Infosys does not become the top IT company

r

or TCS remains in the top rank.

The implications for the above statement are:

(iii) $q \Rightarrow r$ (iv) $\sim r \Rightarrow \sim q$

The given statement, 'The Infosys team's performance is excellent' (p).

From (i) $p \Rightarrow q$ and from (iii) $q \Rightarrow r$

$\therefore$ $p \Rightarrow r$ [TCS remains in the top rank].

p

40. If a person follows conventional methods,

q

He cannot be successful.

The statement is of the form, if p then q.

The implications are:

(i) $p \Rightarrow q$; (ii) $\sim q \Rightarrow \sim p$

$\sim q$

Unless a person is successful,

r

he cannot be a part of successful company.

The statement is of the form unless ($\sim q$), r.

The implications are:

(i) $q \Rightarrow r$ (ii) $\sim r \Rightarrow \sim q$

The given statement, Mr Prasad has become a part of P & G, a successful company ($\sim r$).

From (ii) $\sim r \Rightarrow \sim q$ and from (ii) $\sim q \Rightarrow \sim p$

Now,

$\sim r \Rightarrow \sim q \Rightarrow \sim p$ (Mr Prasad does not follow conventional methods).

EXERCISE-2

p

1. Either Anand will marry Vandana or

q

Madhavi will marry Kollol.

The negation of the above statement is $\sim p$ and $\sim q$.

p

2. Whenever Bhiru and Basanti go for long drive,

q

Joy followed them.

The negation of the above statement is p and $\sim q$.

q

3. Pratap Rana will attend the class,

p

only if his father allows has to go by bike

The negation of the above statement is $\sim p$ and q.

p

4. Unless Aiswariya plays the role of 'Paro',

q

Madhuri will not play the role of 'Chandramukhi'.

The negation of the above statement is $\sim p$ and $\sim q$.

p

5. Unless The change happens,

q

The problems will not be solved.

The statement is of the form:

Unless p, q.

Negation for the above statement is $\sim p$ and $\sim q$.

6. The statement is of the form:

p and q

The negation is $\sim p$ or $\sim q$.

7. The statements is of the form:

Unless p, q.

The negation is $\sim p$ and $\sim q$.

8. The statement is of the form:

Either p or q.

The negation is $\sim p$ and $\sim q$.

9. The statement is of the form:

If p, then q.

The negation is p and $\sim q$.

10. The statement is of the form, p, only if q.

The negation is p and $\sim q$.

p

11. If it is the post of a manager, then

q

Shastri wil join the firm.

The statement is of the form 'if p then q'.

The negation is 'p and $\sim q$'.

q p

12. I cannot make tomato soup, unless I have some onions.

The statement is of the form 'unless p then q'.
The negation is $\sim p$ and $\sim q$.

p q

13. Only if Tara is happy, then she does not go to work
The statement is of the form 'only if p then q'.
The negation is $\sim p$ and q. p

14. Paul is popular either as a lead guitarist or

q

as a base guitarist.

The statement is of the form 'either p or q'.
The negation is $\sim p$ and $\sim q$.

q p

15. Kohli cannot score a hundred, unless Jhonson bowls.
The statement is of the form 'unless p, q'.
The negation is $\sim p$ and $\sim q$.

16. The given statement is of the form

He either wears shoes or a tie

p q

The negations are: $\sim p$ & $\sim q$ and $\sim q$ & $\sim p$.
Choice (A) is the correct negation of the given statement.

17. The given statement is of the form

If It is a holiday then I sleep throughout the day

p q

The negations are p & $\sim q$ and $\sim q$ & $\sim p$.
Choice (B) is the correct negation of the given statement.

18. The given statement is of the form

Rajesh goes to college and attends classes.

p q

The negations are $\sim p$ or $\sim q$ and $\sim q$ or $\sim p$.

19. The statement is of the form, 'If p, then q or r'.

p q

If Ankita eats pastry then It is a black forest

r

or a pinapple.

The implications are:

(i) $p \Rightarrow q$ or r (ii) $\sim q$ and $\sim r \Rightarrow \sim p$
(iii) p and $\sim q \Rightarrow r$ (iv) p and $\sim r \Rightarrow q$

Choice (A) is according to (iv).
Choice (B) is according to (iii).
Choice (C) is according to (ii).

20. The statement is of the form 'if p then q and r'.

p q

If you want to stay fit, then you must eat healthy food

r

and exercise.

The implications are:

(i) $p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow \sim p$
(iii) $\sim q \Rightarrow \sim p$ (iv) $\sim r \Rightarrow \sim p$
(v) $\sim q$ and $\sim r \Rightarrow \sim p$

Choice (A) is according to (iii) and Choice (B) is according to (iv).

21. The statement is of the form 'only if q or r then p'.

p q

Yaseem plays cricket only if he wears blue

r

or white

The implications are:

(i) $p \Rightarrow q$ or r (ii) $\sim q$ and $\sim r \Rightarrow \sim p$
(iii) p and $\sim q \Rightarrow r$ (iv) p and $\sim r = q$

Choice (B) is according to (iii).

22. The statement is of the form 'unless p then q or r'.

q r

Sunil cannot meet his friends or his family unless

p

he has a job.

The implications are:

(i) $\sim p \Rightarrow q$ or r (ii) $\sim q$ and $\sim r \Rightarrow p$
(iii) $\sim p$ and $\sim q \Rightarrow r$ (iv) $\sim p$ and $\sim r \Rightarrow q$

Choice (A) is according to (ii).
Choice (C) is according to (iv).

23. The statement is of the form 'only if p then q or r'.

p q

Only if there is a sale, I will buy clothes

r

or cosmetics.

The implications are:

(i) q or $r \Rightarrow p$ (ii) $q \Rightarrow p$
(iii) $r \Rightarrow p$ (iv) q and $r \Rightarrow p$
(v) $\sim p \Rightarrow \sim q$ and $\sim r$

Choice (A) is according to (iv).
Choice (B) is according to (iii).
Choice (C) is according to (ii).

24. The statement is of the form 'unless p then q and r'.

Unless $\overset{p}{\boxed{\text{you take medicines,}}}$ $\overset{q}{\boxed{\text{you will not recover}}}$ and $\overset{r}{\boxed{\text{will not be able to walk.}}}$

The implications are:

(i) $\sim p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow p$
(iii) $\sim q \Rightarrow p$ (iv) $\sim r \Rightarrow p$
(v) $\sim q$ and $\sim r \Rightarrow p$

Choice (A) is according to (iii).

25. The statement is of the form 'if p, then q or r'.

If $\overset{p}{\boxed{\text{Ganshyam goes to the U.S.A,}}}$ $\overset{q}{\boxed{\text{his mother}}}$ or $\overset{r}{\boxed{\text{his brother}}}$ will accompany him.

The implications are:

(i) $p \Rightarrow q$ or r (ii) p and $\sim q \Rightarrow r$
(iii) p and $\sim r \Rightarrow q$ (iv) $\sim q$ and $\sim r \Rightarrow \sim p$

Choice (A) is according to (iii).
Choice (B) is according to (iv).

26. The statement is of the form 'whenever p then q and r'.

Whenever $\overset{p}{\boxed{\text{Preeti watches T.V.,}}}$ then $\overset{q}{\boxed{\text{she watches movies}}}$ and $\overset{r}{\boxed{\text{sports}}}$

The implications are:

(i) $p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow \sim p$
(iii) $\sim q \Rightarrow \sim p$ (iv) $\sim r \Rightarrow \sim p$
(v) $\sim q$ and $\sim r \Rightarrow \sim p$

Choice (B) is according to (iii).

27. The statement is of the form 'only if q and r, then p'.

$\overset{p}{\boxed{\text{Sagar will marry Sheela,}}}$ only if $\overset{q}{\boxed{\text{she is a graduate}}}$ and $\overset{q}{\boxed{\text{a good cook.}}}$

The implications are:

(i) $p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow \sim p$
(iii) $\sim q \Rightarrow \sim p$ (iv) $\sim r \Rightarrow \sim p$
(v) $\sim q$ and $\sim r \Rightarrow \sim p$

Choice (A) is according to (iii).

28. The statement is of the form 'unless p, then q and r'.

$\overset{q}{\boxed{\text{The electricity supply will not be restored}}}$ and $\overset{r}{\boxed{\text{we will not be able to watch T.V.,}}}$ unless $\overset{p}{\boxed{\text{you pay the bill.}}}$

The implications are:

(i) $\sim p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow p$
(iii) $\sim q \Rightarrow p$ (iv) $\sim r \Rightarrow p$
(v) $\sim q$ and $\sim r \Rightarrow p$

Choice (A) is according to (v).
Choice (B) is according to (iv).
Choice (C) is according to (iii).

29. The statement is of the form, 'if p, then q or r'.

If $\overset{p}{\boxed{\text{you try hard}}}$ then $\overset{q}{\boxed{\text{you can win}}}$ or $\overset{r}{\boxed{\text{gain.}}}$

The implications are:

(i) $p \Rightarrow q$ or r (ii) $\sim q$ and $\sim r \Rightarrow \sim p$
(iii) p and $\sim q \Rightarrow r$ (iv) p and $\sim r \Rightarrow q$

Choice (A) is according to (iii).
Choice (B) is according to (iv).
Choice (C) is according to (ii).

30. The statement is of the form 'only if q or r then p'.

$\overset{p}{\boxed{\text{Rajesh cooks,}}}$ only if $\overset{q}{\boxed{\text{it is Sunday}}}$ or $\overset{r}{\boxed{\text{Saturday}}}$

The implications are:

(i) $p \Rightarrow q$ or r (ii) $\sim q$ and $\sim r \Rightarrow \sim p$
(iii) p and $\sim q \Rightarrow r$ (iv) p and $\sim r \Rightarrow q$

Choice (B) is according to (iii).

31. The statement is of the form 'unless p, then q or r'.

Whenever $\overset{p}{\boxed{\text{David goes to church,}}}$ $\overset{q}{\boxed{\text{he donates money}}}$ and $\overset{r}{\boxed{\text{clothes}}}$

The implications are:

(i) $\sim p \Rightarrow q$ or r (ii) $\sim q$ and $\sim r \Rightarrow p$
(iii) $\sim p$ and $\sim q \Rightarrow r$ (iv) $\sim p$ and $\sim r \Rightarrow q$

Choice (A) is according to (ii).
Choice (C) is according to (iv).

32. The statement is of the form 'whenever p, then q and r'.

Whenever $\overset{p}{\boxed{\text{David goes to church,}}}$ $\overset{q}{\boxed{\text{he donates money}}}$ and $\overset{r}{\boxed{\text{clothes}}}$

The implications are:
(i) $p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow \sim p$
(iii) $\sim q \Rightarrow \sim p$ (iv) $\sim r \Rightarrow \sim p$
(v) $\sim q$ and $\sim r \Rightarrow \sim p$
Choice (B) is according to (iii).

33. The statement is of the form 'if p, then q or r'.

If $\overset{p}{\boxed{\text{Prashanth buys a book,}}}$

then he gives it to his $\overset{q}{\boxed{\text{brother}}}$

or $\overset{r}{\boxed{\text{his friend.}}}$

The implications are:
(i) $p \Rightarrow q$ or r ii) p and $\sim q \Rightarrow r$
(iii) p and $\sim r \Rightarrow q$ (iv) $\sim q$ and $\sim r \Rightarrow \sim p$
Choice (A) is according to (iii).
Choice (B) is according to (iv).

34. If $\overset{p}{\boxed{\text{Iam not paid,}}}$ then $\overset{q}{\boxed{\text{I will not work}}}$ and

$\overset{r}{\boxed{\text{I will not take leave.}}}$

Statement: If p, then q and r.
Implications:
(i) $p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow \sim p$
(iii) q and $\sim r \Rightarrow \sim p$ (iv) $\sim q$ and $r \Rightarrow \sim p$
(v) $\sim q$ and $\sim r \Rightarrow \sim p$
Implication (v) is represented in Choice (A), (iv) is represented in Choice (B) and (ii) is represented in the Choice (C).

35. If $\overset{p}{\boxed{\text{Rama leaves Ayodhya,}}}$ then $\overset{q}{\boxed{\text{he will go to forest}}}$

or $\overset{r}{\boxed{\text{to Sri Lanka.}}}$

Statement: If p, then q or r.
Implications:
(i) $p \Rightarrow q$ or r (ii) $\sim q$ and $\sim r \Rightarrow \sim p$
(iii) p and $\sim q \Rightarrow r$ (iv) p and $\sim r \Rightarrow q$

$\sim p$ is 'Rama did not leave Ayodhya', $\sim q$ is 'Rama did not go to forest' and $\sim r$ is 'Rama did not go to Sri Lanka'.
Implication (i) is represented in the Choice (A).

36. Unless $\overset{p}{\boxed{\text{the party gets majority,}}}$

$\overset{q}{\boxed{\text{the house will be dissolved}}}$

and $\overset{r}{\boxed{\text{the President's rule will be impossed.}}}$

Statement: Unless p, then q and r.
Conclusions:
(i) $\sim p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow p$
(iii) $\sim q \;\&\; r \Rightarrow p$ (iv) $q \;\&\; \sim r \Rightarrow p$
(v) $\sim q \;\&\; \sim r \Rightarrow p$

Conclusions:
(i) The party did not get a majority, implies that the house will be dissolved and the President's rule will be imposed.
(ii) The house is not dissolved or the President's rule is not imposed means that the party got a majority, Choice (B).
(iii) The house is not dissolved but the President's rule is imposed means that the party got the majority.
(iv) The house is dissolved and the President's rule is not imposed means that the party got a majority.
(v) The house is not dissolved and the President's rule is not imposed implies that the party got a majority, Choice (C).

37. If $\overset{p}{\boxed{\text{you plant trees,}}}$ then $\overset{q}{\boxed{\text{there will be no pollution}}}$

and $\overset{r}{\boxed{\text{you get fruits.}}}$

Statement: If p, then q and r.
Implications:
(i) $p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow \sim p$
(iii) $\sim q$ and $r \Rightarrow \sim p$ (iv) q and $\sim r \Rightarrow \sim p$
(v) $\sim q$ and $\sim r \Rightarrow \sim p$
$\sim p$ is 'You did not plant trees', $\sim q$ is 'There will be pollution' and $\sim r$ is 'You do not get fruits'.
Implication (v) is represented in the Choice (B) and (ii) is represented in Choice (C).

38. If $\overset{p}{\boxed{\text{there is no traffic,}}}$ then $\overset{q}{\boxed{\text{I will not drive slow}}}$

but $\overset{r}{\boxed{\text{I will go on a long drive.}}}$

Statement: If p, then q and r.
Implications:
(i) $p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow \sim p$
(iii) $\sim q$ and $r \Rightarrow \sim p$ (iv) q and $\sim r \Rightarrow \sim p$
(v) $\sim q$ and $\sim r \Rightarrow \sim p$
Implication (ii) is represented in Choice (C).

39. (a) Only if $\overset{p}{\boxed{\text{the water level in the coastal areas rises,}}}$

then the $\overset{q}{\boxed{\text{people change their life style}}}$

The implications are:
(i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$

(b) , only if

q

People change their life style, only if

r

they are rewarded.

The implication are:

(iii) $q \Rightarrow r$ (iv) $\sim r \Rightarrow \sim q$

r

(c) If people are rewarded

$\sim q$

then they will not change their life style.

The implications are:

(v) $r \Rightarrow \sim q$ (vi) $q \Rightarrow \sim r$

s

(d) If the temperature rises,

p

then the water level in the coastal areas rises.

The implications are:

(vii) $s \Rightarrow p$ (viii) $\sim p \Rightarrow \sim s$

(e)

p

Whenever the water level in the coastal areas rises,

s

then the temperature rises.

The implications are:

(ix) $p \Rightarrow s$ (x) $\sim s \Rightarrow \sim p$

q

(f) Unless the people change their life style,

s

temperature rises.

The implications are:

(xi) $\sim q \Rightarrow s$ (xii) $\sim s \Rightarrow q$

(g) People are rewarded $\Rightarrow r$

(h) Water level in the coastal area does not rise.

$\Rightarrow \sim p$

Choice (A) $\Rightarrow h \rightarrow$ (viii) $\rightarrow$ (xii) $\rightarrow$ (vi) $\rightarrow \sim r$

$\therefore$ Hence, the statements are inconsistent.

or G $\rightarrow$ (v) $\rightarrow$ (xi) $\rightarrow$ (vii) $\rightarrow p$

$\therefore$ Hence, the statements are inconsistent.

Choice (B) $\Rightarrow h \rightarrow$ (viii) $\rightarrow$ (xii) $\rightarrow$ (iii) $\rightarrow r$, i.e., G.

$\therefore$ Hence, the statements are consistent.

Choice (C) $\Rightarrow$ Here s is mentioned only in statement (d).

$\therefore$ No consistency or inconsistency can be established.

Choice (D) $\Rightarrow$ There is no relation with either g or h for any of the other statements. So, no consistency can be established.

40. (a) then

p q

If Gulam sings, then audience will sleep.

The implications are:

(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$

(b) then

p r

If Gulam sings, then audience dance.

The implications are:

(iii) $p \Rightarrow r$ (iv) $\sim r \Rightarrow \sim p$

(c)

$\sim r$

Unless audience do not dance,

s

the concert will be successful.

The implications are:

(v) $r \Rightarrow s$ (vi) $\sim s \Rightarrow \sim r$

(d)

r

only if the audience dance,

s

the concert will be successful.

The implications are:

(vii) $s \Rightarrow r$ (viii) $\sim r \Rightarrow \sim s$

(e)

p t

Gulam sings, only if Vani dances.

The implications are:

(ix) $t \Rightarrow p$ (x) $\sim p \Rightarrow \sim t$

(f) only if

p t

Gulam sings, only if Vani dances.

The implications are:

(xi) $p \Rightarrow t$ (xii) $\sim t \Rightarrow \sim p$

(g) Vani dances $\Rightarrow t$

(h) The concert is successful $\Rightarrow s$

Choice (A) $\Rightarrow$ There is no implication for either s or t.

$\therefore$ Consistency cannot be established.

Choice (B) $\Rightarrow$ The term q is mentioned only in statement a.

$\therefore$ Consistency cannot be established.

Choice (C) $\Rightarrow g \rightarrow$ (ix) $\rightarrow$ (iii) $\rightarrow$ (v) $\rightarrow s \rightarrow h$.

$\therefore$ The statements are consistent.

Choice (D) $\Rightarrow$ We can only relate g to (vii).

$\therefore$ Consistency cannot be established.

EXERCISE-3

1. If $\overset{p}{\boxed{\text{it is a holiday,}}}$ then $\overset{q}{\boxed{\text{I will go for a picnic}}}$

or $\overset{r}{\boxed{\text{I will visit my Uncle's house.}}}$

Statement:
If p, then q or r.

Conclusions:

(i) $p \Rightarrow q$ or r (ii) $\sim q$ and $\sim r \Rightarrow \sim p$
(iii) p and $\sim q \Rightarrow r$ (iv) p and $\sim r \Rightarrow q$

Conclusions:
(i) It is a holiday means that I will go for a picnic or I will visit my uncle's house.
(ii) I did not go for a picnic and I did not visit my uncle's house means that it was not a holiday.
(iii) It was a holiday but I did not go for a picnic means that I visited my uncle's house.
(iv) It is a holiday but I did not visit my uncle's house means that I went for picnic.

But only statement (ii) is represented in Choice (C).

2. Whenever $\overset{p}{\boxed{\text{my mom scolds me,}}}$ I either

$\overset{q}{\boxed{\text{hide behind my dad}}}$ or

$\overset{r}{\boxed{\text{complain to my grandma.}}}$

Statement:
Whenever p, then q or r is same as if p, then q or r.

Conclusions:

(i) $p \Rightarrow q$ or r (ii) $\sim q \,\&\, \sim r \Rightarrow \sim p$
(iii) $p \,\&\, \sim q \Rightarrow r$ (iv) $p \,\&\, \sim r \Rightarrow q$

Conclusions:
(i) My mom scolded me, so I hid behind my dad or I complained to my grandma.
(ii) I did not hide behind my dad and I did not complain to my grandma means that my mom did not scold me.
(iii) My mom scolded me but I still did not hide behind my dad means that I complained to my grandma.
(iv) My mom scolded me but I did not complain to my grandma means that I hid behind my dad.

Questions numbered 3 and 7: These questions can be solved in the same way as questions numbered 1 and 2 are solved.

4. If $\overset{p}{\boxed{\text{it is very hot outside,}}}$ then

$\overset{q}{\boxed{\text{I will carry on onion with me}}}$

and $\overset{r}{\boxed{\text{I will return home by lunch time.}}}$

Statement:
If p, then q and r.

Conclusions:

(i) $p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow \sim p$
(iii) $\sim q \,\&\, r \Rightarrow \sim p$ (iv) $q \,\&\, \sim r \Rightarrow \sim p$
(v) $\sim q \,\&\, \sim r \Rightarrow \sim p$

Conclusions:
(i) It is very hot outside means that I will carry an onion with me and I will return home by lunch time.
(ii) I will not carry an onion with me or I will not return home by lunch time means that it is not very hot outside.
(iii) I will not carry an onion with me but I will return home by lunch time means it is very hot outside.
(iv) I will carry an onion with me but I will not return home by lunch time means that it is not very hot outside.
(v) I will not carry an onion with me and I will not return home by lunch time means that it is not very hot outside.

Question 5: This can be solved in the same way as question 4 is solved.

6. If $\overset{p}{\boxed{\text{tea is not hot,}}}$ then $\overset{q}{\boxed{\text{I will not go to school}}}$

nor $\overset{r}{\boxed{\text{will I have dinner.}}}$

Statement:
If p, then q and r.
Note: Here, 'nor' is same as 'and'
so, Questions 7 is similar to question 1.

8. Unless $\overset{p}{\boxed{\text{we win the Assembly elections,}}}$

$\overset{q}{\boxed{\text{we will lose the Rajya Sabha elections}}}$ and

$\overset{r}{\boxed{\text{the presidential elections.}}}$

Statements:
Unless p, q and r.

Conclusions:

(i) $\sim p \Rightarrow q$ and r (ii) $\sim q$ or $\sim r \Rightarrow p$
(iii) $\sim q \,\&\, r \Rightarrow p$ (iv) $q \,\&\, \sim r \Rightarrow p$
(v) $\sim q \,\&\, \sim r \Rightarrow p$

Conclusions:

(i) We did not win the Assembly elections means that we lost the Rajya Sabha elections and the Presidential elections.

(ii) We did not lose the Rajya Sabha elections or we did not lose the Presidential elections means that we won the Assembly elections.

(iii) We did not lose the Rajya Sabha elections but lost the Presidential elections means that we won the Assembly elections.

(iv) We lost the Rajya Sabha elections but did not lose the Presidential elections means that we won the Assembly elections.

(v) We did not lose the Rajya Sabha elections and did not lose the Presidential elections means that we won the Assembly elections.

Questions 9 and 10: These questions can be solved in the same way as question 8 is solved.

11. $\underbrace{\boxed{\text{Harish will get through the only interview}}}^{q}$ if $\overbrace{\boxed{\text{he is through with the basics.}}}^{p}$

Implications:

(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$

Choice (C) follows (ii).

12. $\overbrace{\boxed{\text{Either Pak or China attacks India,}}}^{q}$ only if $\overbrace{\boxed{\text{India supports Russia and USA.}}}^{p}$

Implications:

(i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$

The first choice follows the first implication.

13. $\overbrace{\boxed{\text{I will neither talk to you nor play with you,}}}^{q}$ $\overbrace{\boxed{\text{you apologize to me.}}}^{p}$

Implications:

(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$

Choice (A) follows (ii).

14. The given statement is of the form, p or q, i.e.,

either $\boxed{\text{Rajeev is a genius}}$ or

p

$\boxed{\text{he cheated in the exam.}}$

q

The possible implications are:

(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$

Therefore, the answer is ca or db.

15. The given statement is of the form, unless p, then q, i.e.,

unless $\underbrace{\boxed{\text{the politician took money}}}_{p}$

$\underbrace{\boxed{\text{he is not good enough.}}}_{q}$

The possible implications are:

(i) $\sim p \Rightarrow q$ (ii) $\sim q = p$

Therefore, the answer is cb and da.

16. $\overbrace{\boxed{\text{Sravan will go to the movie,}}}^{q}$ if $\overbrace{\boxed{\text{his parents are not with him}}}^{p}$

Statement: q, if p

Negation: p and $\sim q$

Sravan's parents are not with him and he did not go to the movie.

17. $\overbrace{\boxed{\text{Ramesh works very hard}}}^{q}$ whenever $\overbrace{\boxed{\text{there is an exam}}}^{p}$

Statement: q, whenever p

Negation: p and $\sim q$

There is an exam and Ramesh did not work hard.

18. Either $\overbrace{\boxed{\text{it is a Flying Saucer}}}^{q}$ or $\overbrace{\boxed{\text{the person is not telling the truth}}}^{p}$

Statement: Either p or q

Negation: $\sim p$ and $\sim q$

It is not a flying saucer and the person is telling the truth.

19. $\overbrace{\boxed{\text{Sachin scores a century,}}}^{q}$ unless $\overbrace{\boxed{\text{he is paired with the Captain}}}^{p}$

Statement: p unless q

Negation: $\sim p$ and $\sim q$

Sachin did not score a century and he is not paired with the Captain.

20. $\overbrace{\boxed{\text{Bond will buy the car}}}^{q}$ only if $\overbrace{\boxed{\text{it is the costliest}}}^{p}$ $\overbrace{\boxed{\text{fastest}}}^{r}$

Statement: p only if (q and r), i.e., $p \Rightarrow q$ and r

Negation: p and $\sim(q$ and $r) \Rightarrow p$ and $(\sim q$ or $\sim r)$

Therefore, Bond bought the car and it is not the costliest or it is not the fastest.

21. The statement is in the form of if p, then q.
The implications are:
(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$
Therefore, it can be ac or db.

22. The statement is in the form of q, only if p.
Implications:
(i) $q \Rightarrow p$ (ii) $\sim p \Rightarrow \sim q$
Therefore, it can be ac or db.

23. Statement: whenever p, then q.
Implications:
(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$
Therefore, the answer can be ac or db.

24. The chief guest will come on time = q
if
the fog does not affect the flight timings = p.
The implications of above statements are:
(i) $p \Rightarrow q$ (ii) $\sim q \Rightarrow \sim p$
Only if
the chief guest comes = q
then the meeting be started = r.
The implications of above statements are:
(i) $r \Rightarrow q$ (ii) $\sim q \Rightarrow \sim r$
Therefore, implications with respect to q is not possible here.

25. Unless
the coding is done = p
the software project cannot be completed = q.
The implications of above statements are:
(i) $\sim p \Rightarrow q$ (ii) $\sim q \Rightarrow p$
If
the company does not meet the project completion dead line = q,
the team working on it employees will be fired = r.
The implications of above statements are:
(iii) $q \Rightarrow r$ (iv) $\sim r \Rightarrow \sim q$
The given statement is 'the team working on the project are not fired' $\Rightarrow \sim r$.
From (iv) $\sim r \Rightarrow \sim q$ and from (ii) $\sim q \Rightarrow p$
Therefore, $\sim r \Rightarrow p$ 'the coding is done'.

Difficulty Level-3: Advanced

11 Quant Based Reasoning

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:

- Learn how to deal with questions which involve concepts of both logical reasoning and quantitative aptitude
- Learn how to interpret the given data and get the final outcome out of it.

❒ INTRODUCTION

In this section we deal with questions which are a mixture of both quantitative as well as reasoning section. Such, questions are important from the exam point of view as the CAT generally gives questions based on combination of multiple topics**.**

EXERCISE-1

Directions for questions 1 to 10: Select the correct alternative from the given choices.

1. The maximum possible number of squares that can be formed using 12 straight lines is ______.

(A) 45 (B) 55

(C) 60 (D) 65

2. Two candles of different lengths burn at different rates. Each candle burns exactly for one hour. Which of the following time intervals can we measure by burning these candles?

(A) 20 minutes (B) 15 minutes

(C) 10 minutes (D) 24 minutes

3. A group of three friends Rose, Jasmine and Tulip were watching butterflies. All three of them saw one butterfly in common. Any two of them saw two butterflies in common and each one of them saw one butterfly which is not common with others. Among the butterflies seen by Tulip, four have polka dots on them. Among the butterflies seen by Jasmine, three have polka dots on them and among the butterflies seen by Rose, two have polka dots on them. What is the minimum possible number of butterflies that do not have polka dots on them?

(A) 0 (B) 1

(C) 2 (D) 3

4. Mr Bhatnagar is an exporter of Champagne. While crossing the borders he bribes the guards by giving them a bottle of Champagne per box of Champagne to cross every border. Each box can contain a maximum of 15 bottles of Champagne. If he has to cross 10 borders to supply 50 bottles of Champagne, then what is the minimum possible number of bottles of Champagne he should start with?

(A) 110 (B) 99
(C) 105 (D) 107

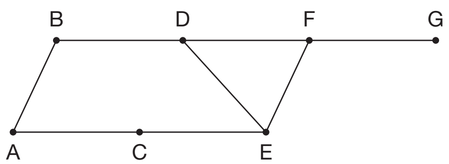
5. When I asked my friend about her age, she said that she has two siblings and the product of the ages of the three is 15 and the sum of their ages is an odd number, then what is the age of my friend if she is the eldest?
(A) 1 (B) 3
(C) 5 (D) Cannot be determined

6. A pile has 34 coins. Two friends, Neeta and Leena are playing a game in which each of them draws 2 or 3 coins from a pile of 34 coins. The person to draw the last coin is the loser. How many coins should Neeta draw to ensure her win if she is the first one to draw?
(A) 2
(B) 3
(C) Either (A) or (B)
(D) Neeta cannot win

7. Using a 4 litre vessel and 6 litre vessel which of the following quantities of water cannot be measured? (Assume there is infinite supply of water)
(A) 2 litre
(B) 3 litre
(C) 8 litre
(D) More than one of the above

8. Each of A, B and C is a different digit among 1 to 9. How many different values of the sum of A, B and C are possible, if ABA × AA = ACCA?
(A) 1 (B) 3
(C) 7 (D) 8

9. In the following diagram, A is the reserve station from which a lorry can take 100 units of liquid at a time to be supplied to B, C, D, E, F and G, whose requirements per day are 78, 61, 26, 42, 34 and 59 units, respectively. Every day a lorry starts at A and supplies the requirements. In this process it moves back and forth from A to the other places and the location of all places is as shown in the diagram given below:



In a day, how many times does the lorry start from A and what is the unused capacity of the lorry per day?
(A) 4 times and 100 units.
(B) 5 times and 200 units.
(C) 3 times and zero units.
(D) 5 times and zero units.

10. The following sets of weighing stones are available to measure weights.

Number of stones in the range	Weights in the range (in grams)	Interval between two successive weights
9	1.001, 1.002, …1.009	0.001 g
49	1.01, 1.02, 1.49	0.01 g
19	0.5, 1.0, 1.5, …9.5	0.5 g
9	10, 20, 30, …90	10 g

What is the minimum number of stones required to weigh an item which weighs 68.892 g?
(A) 6 (B) 5
(C) 3 (D) 4

Directions for questions 11 to 13: These questions are based on the following letter multiplication in which each letter represents a unique non-zero digit.

A B C
B A C
____
C A D E F C
____

Also, it is known that E = 3B, F = A + B and 3A = 2D

11. What is the value of C?
(A) 1 (B) 3
(C) 5 (D) 2

12. What is the value of A?
(A) 2 (B) 4
(C) 5 (D) 6

13. What is the value of F?
(A) 3 (B) 6
(C) 9 (D) 7

Directions for questions 14 and 15: These questions are based on the following letter multiplication in which each letter represents a unique non-zero digit.

A B
C D
____
F E B D
____

Also, it is known that D = E + A, B = 2E and 2B = 3F.

14. What is the value of D?
(A) 4 (B) 1
(C) 8 (D) 5

15. What is the value of C?

(A) 3 (B) 7
(C) 9 (D) 6

Directions for questions 16 and 17: These questions are based on the following letter multiplication in which each letter represents a unique non-zero digit.

A B
C D
———
CCEB
———

Also, it is known that B = 2C, 5D = 6E and A = B + E.

16. What is the value of A?

(A) 7 (B) 9
(C) 5 (D) 3

17. What is the value of C?

(A) 4 (B) 2
(C) 1 (D) 6

Directions for questions 18 to 21: These questions are based on the following information.

A computer helps in finding out a 4-digit code number in the following way.

If we select a number, the computer gives the number of digits present in the selected number. These numbers are also present at the same positions in the code number as shown in the column 'R' in the table given below. The computer also gives the number of digits present in the selected number. These numbers are not present in the code number, which is shown in the column W in the table.

To construct a code number only digits from 1 to 6 are used.

The following 3 numbers are selected to find out the code.

S. No.	Selected number	R	W
1	3425	2	2
2	3625	1	2
3	3426	2	1

18. If 6 is the first digit from the left in the code number then which of the following is the second digit from right?

(A) 1 (B) 2
(C) 3 (D) 4

19. If 6 is the second digit from the right, then which of the following is the first from the left in the code number?

(A) 1 (B) 2
(C) 3 (D) 4

20. Which of the following digits is not in the code number?

(A) 2 (B) 3
(C) 5 (D) 4

21. Which of the following cannot be the code number?

I. 6421 II. 3461
III. 3416

(A) Only I (B) Only III
(C) Only II (D) Both I and III

Directions for questions 22 to 25: These questions are based on the following information.

A group of five people, namely Govind, Mathew, Naveen, Jagdish and Preet travelled for a different time duration among 1 hour, 2 hour, 3 hour, 4 hour and 5 hour and at a different speed among 15 kmph, 20 kmph, 30 kmph 40 kmph and 60 kmph, not necessarily in the same order. The total distance travelled by each of them is also different. The following is the information known about them.

(i) Naveen travelled the maximum distance but he neither travelled at the maximum speed nor for the maximum duration.

(ii) Govind travelled more distance than Preet but less distance than Mathew.

(iii) Jagdeesh travelled a distance of 120 km.

(iv) None of them travelled a distance of 60 km and one of them travelled a distance of 30 km.

22. Who travelled at a speed of 15 kmph?

(A) Jagdish (B) Mathew
(C) Govind (D) Cannot be determined

23. For what time duration did Govind travel?

(A) 1 hour (B) 2 hour
(C) 3 hour (D) 5 hour

24. Who travelled for the least time duration?

(A) Jagdish (B) Govind
(C) Mathew (D) Preet

25. If everyone reached the picnic spot at the same time then who among them started earlier than anyone else?

(A) Jadish (B) Govind
(C) Mathew (D) Preet

Directions for questions 26 to 29: These questions are based on the following information.

A group of six teams from A to F participated in a football tournament. Each team played exactly one match against each of the other teams in the tournament. The tournament was planned in three weeks such that equal number of matches are played in each week and each team played at least one match but not more than two matches in a week. The following table gives week-wise performance of the teams.

Teams	Result at the end of the 1st week		Result at the end of the 2nd week		Result at the end of the 3rd week	
	Goals for	Goals against	Goals for	Goals against	Goals for	Goals against
A	6	3	9	5	14	8
B	1	3	1	9	7	13
C	4	6	10	8	11	8
D	1	3	6	7	8	11
E	5	4	7	7	10	12
F	6	4	10	7	12	10

(i) No match ended in a draw.
(ii) No team scored more than 3 goals in a match.
(iii) No match was played between A and E in the second week.

26. Who won the maximum number of matches?
(A) C (B) F
(C) A (D) B

27. Against whom did B play in the third week?
(A) Both A and F (B) Both D and F
(C) Only D (D) Both D and E

28. In the first week, D play (s) the match against
(A) A only (B) E only
(C) Both A and E (D) Both A and C

29. In how many matches was there a goal difference of more than 1?
(A) 3 (B) 4
(C) 5 (D) 6

Directions for questions 30 to 33: These questions are based on the following information.

Narayan, Michael and Rusell participated in a car race. All three of them could drive the car for distinct time periods (because their fuel tanks got empty) and at different speeds (in km/hr). The person who drove the car for the maximum time period had driven at a minimum speed compared to the other two but covered the maximum distance. The person who drove at the maximum speed covered the minimum distance of 360 km, but he was not Michael, who drove at 150 km/hr. Russell drove the car for a time period, which was the same as the sum total of the time periods taken by the other two contestants. The total distance covered by all the three contestants is 1620 km. The speeds and the time taken by the contestants in their respective units (km/hr and hr) are integral numbers.

30. What is the distance covered by Russell?
(A) 360 km (B) 600 km
(C) 810 km (D) 660 km

31. What is the time taken by Michael?
(A) 2 hours (B) 3 hours
(C) 4 hours (D) 5 hours

32. Which of the following statements may be true?
(A) Russell drove at a speed of 110 km/hr.
(B) Narayan drove the car for 2 hours.
(C) Russell drove at a speed of 132 km/hr.
(D) More than one of the above

33. If no contestant drove the car at a speed beyond 200 km/hr, then what was the speed of Narayan?
(A) 110 km/hr (B) 132 km/hr
(C) 180 km/hr (D) 150 km/hr

Directions for questions 34 to 37: These questions are based on the following information.

In a tournament, each of the six teams played every other team. In a match between two teams, the winner got two points, the loser got zero points and if it was a draw, then each team got one point. Scores of A, B, C, D, E and F were 9, 8, 7, 3, 2 and 1 point, respectively. There were exactly two draws.

34. D had a tie with which of the following teams?
(A) A (B) C
(C) F (D) Cannot be determined

35. A had a tie with which of the following teams?
(A) D (B) C
(C) F (D) B

36. Which of the following lost the maximum number of matches?
(A) D (B) E
(C) F (D) Both E and F

37. Out of the following teams, A did not win against
(A) B (B) C
(C) D (D) F

Directions for questions 38 to 40: These questions are based on the following information.

Ravi, a retail dealer of Air-Tel prepaid cards, asked his brother Gopi to buy cards of denominations ₹200, ₹700, ₹1000, ₹1500 and ₹2000. He asked Gopi to buy five cards each of exactly three of the above denominations and six cards each of the remaining denominations. However, Gopi forgot which denominations he was supposed to buy five and which he had to buy six of each. However, the wholesale dealer could figure out how many cards of each denomination were required as Ravi had sent an amount of ₹30,000, which was the exact amount required to buy the cards of Ravi's choice.

38. What is the ratio of the total number of cards of denominations ₹200 and ₹2000 to those of all other cards purchased by Gopi?

(A) 10 : 17 (B) 16 : 11
(C) 17 : 11 (D) 11 : 16

39. What is the total value of all those cards of which five each were bought?

(A) 14,400 (B) 12,000
(C) 18,000 (D) 16,000

40. If Gopi had told the shopkeeper that he required 6 cards each of the 3 denominations that his brother asked him to get 5 each and 5 cards each of the other denominations that his brother asked him to get 6 each, then what is the amount that Gopi would have left with him or fall short of from the total amount of ₹30,000 his brother had given him?

(A) He was left with ₹600.
(B) He fall short of ₹400.
(C) He fall short of ₹600.
(D) He was left with ₹400.

EXERCISE-2

Directions for questions 1 to 3: These questions are based on the following information.

There are 8 containers and each container has K number of balls. Each of the balls in seven containers weighs 2 kg, whereas each of the balls in the remaining container weighs 1 kg. A spring balance is used to weigh these balls and n is the number of minimum weighing required to find the container that contains 1 kg balls.

1. If $K = 12$, then $n = ?$

(A) 1 (B) 2
(C) 3 (D) 4

2. If $K = 4$, then $n = ?$

(A) 1 (B) 2
(C) 3 (D) 4

3. If $K = 3$, then $n = ?$

(A) 1 (B) 2
(C) 3 (D) 4

Directions for questions 4 to 6: These questions are based on the following information.

Mr Helpinghand has ₹x with him in the denomination of ₹1 only. Each time a beggar approaches him, he divides the money with him into four equal parts and a remainder (if any). He gives one part and the remainder (if any) to the beggar. This continues till he is left with less than ₹4, which he gives to the last beggar.

4. If Mr Helpinghand has ₹45 with him, then to how many beggars can he give money?

(A) 9 (B) 7
(C) 8 (D) 6

5. If Mr Helpinghand wants to give money to five beggars, then what is the minimum possible initial amount he has to carry with him?

(A) ₹12 (B) ₹24
(C) ₹18 (D) ₹16

6. If Mr Helpinghand wants to serve six people, then what is the maximum possible amount he might be carrying?

(A) ₹31 (B) ₹27
(C) ₹28 (D) ₹30

Directions for questions 7 to 10: These questions are based on the following information.

A group of five people, namely Sarvajeet, Manjeet, Paramjeet, Karamjeet and Biswajeet invested a different amount, such as ₹2000, ₹3000, ₹4000, ₹5000 and ₹6000 at a different simple rate of interest among 4, 5, 6, 7.5 and 8 per cent per year for 5 years. The following is the information known about them.

(i) The interest earned by each one of them was different.
(ii) The interest earned by Sarvajeet was more than that earned by Karamjeet, which was more than that earned by each of Paramjeet and Biswajeet. Manjeet earned the least interest among the five.
(iii) The interest earned by Biswajeet was ₹600.
(iv) The person who invested the maximum amount did not earn the highest interest.

7. What is the investment made by Paramjeet?
(A) ₹4000 (B) ₹5000
(C) ₹3000 (D) Cannot be determined

8. What is the interest earned by Sarvajeet?
(A) ₹1800
(B) ₹1875
(C) ₹2000
(D) Cannot be determined

9. Who invested at 6% per annum rate of interest?
(A) Paramjeet (B) Karamjeet
(C) Biswajeet (D) Sarvajeet

10. What is the difference between the interest earned by Biswajeet and Karamjeet?
(A) ₹1000 (B) ₹1500
(C) ₹1275 (D) ₹1200

Directions for questions 11 to 15: These questions are based on the following information.

In a game show called 'Graded Answer' there were five contestants, namely Kamal, Ranjit, Ajay, Varun and Sashank.

In each round a question was given to all the contestants. For each question the computer has ten predetermined answers. In each round, every contestant gave one answer from these predetermined answers and no contestant is aware of the answers given by the other contestants. Each set of answers were given district grades by the computer. In each round, the computer awards points to the contestants based on the grade of the answer given by them. The contestant whose answer has the lowest grade among the five answers, gets one point and the points for other contestants are increased if the grade goes on increasing. In the first round the five contestants gave five different answers. The table given below shows some of the cumulative scores of the contestants at the end of different rounds.

No.	Kamal	Ranjit	Ajay	Varun	Sashank
1	5	2			4
2	6		3		
3					5
4	7	7	7		7
5		9			11
6			9		
7	13	14		15	

In any round if a group of contestants (i.e., two or more) give the same answer, then the cumulative scores at the end of that round of each contestant in the group is reduced to the least of the cumulative scores of the contestants of this group at the end of the previous round the rest of the contestants get points starting from 1 depending on the grades as explained in earlier.

We know the following additional information about the proceedings of the game.

(i) The cumulative scores of at least two contestants are equal at the end of the second round onwards until at the end of the sixth round (both the rounds included).
(ii) If two or more people give the same answer in a round, in the next round the answers given by the contestants are district.
(iii) The game ended in the seventh round, at the end of which the cumulative scores are district.
(iv) Each contestant's answer matches with that of another contestant in at least one round in the game.
(v) The averages of cumulative scores for the five contestants at the end of each of the round 1 and 2 are the same.
(vi) In the sixth question, only Kamal and Varun gave the same answers.
(vii) Total of cumulative scores of the contestants at the end of the fifth round was 51 and that at the end of the seventh round was 71.
(viii) In the fourth round, Varun gave the least graded answer.
(ix) The person with the highest cumulative score at the end of the game was the winner.

11. Who was the winner?
(A) Ajay (B) Varun
(C) Sashank (D) Either (A) or (B)

12. Who gave the same answer in the second round?
(A) Varun and Sashank
(B) Ajay and Sashank
(C) Ranjit and Sashank
(D) Ranjit, Varun and Sashank

13. What was the score of Ranjit at the end of the third round?
(A) 3 (B) 5
(C) 4 (D) Either (3) or (5)

14. What was the score of Kamal and Varun at the end of the sixth round?
(A) 10, 10 (B) 11, 11
(C) 12, 12 (D) Either (A) or (C)

15. Who gave the highest graded answer in the fifth round?
(A) Kamal (B) Sasankh
(C) Varun (D) Ajay

Directions for questions 16 to 19: These questions are based on the following information:

In sports gambling's, fractional odds are often used. If a bookmaker is offering an odd of 10/1 on a particular team, it means that for every ₹1 that a gambler puts at stake, he earns ₹10, in addition to the original stake being returned to him. If the team loses, the gambler, of course, does not win anything and loses his stake on a particular day, when three football matches are taking place. Match I is between teams A and B, Match II is between teams C and D, Match III is between teams E and F. A bookmaker has offered the following odds on different teams:

Match	Odds	Odds
I	A – 1/4	B – 10/1
II	C – 2/5	D – 7/2
III	E – 20/1	F – 1/5

An 'upset' happens when a team beats an opposing team, which had better chance of winning. The bookmaker offers worse odds on teams that are expected to win.

For example: A team with odds 2/7 has a better chance of winning than a team with 7/2 odds.

16. Amit put ₹100 at stake in each of the three matches (one team per match). What is the maximum possible amount that he can receive, if there is only one match that results in an upset?

(A) 2365 (B) 2745
(C) 2435 (D) 2565

17. If there was no upset in any of the three matches and Bhaskar bet ₹20, ₹40 and ₹60 in each match (in any order) and he ended up earning the maximum possible amount, then what is his total earnings?

(A) 158 (B) 162
(C) 170 (D) 188

18. If Ravi bet equal amounts on all the six teams, then which results are most favourable to him if there are 2 upsets? Pick the option with the winning teams.

(A) B, C, E (B) A, D, E
(C) B, D, E (D) B, C, F

19. Suresh has a strong feeling that team F will win and he bought stake ₹50 on it. He bought a stake ₹50 in one of the teams playing Match II. What is the difference between the maximum and minimum earnings?

(A) 285 (B) 250
(C) 265 (D) 295

Directions for questions 20 to 23: These questions are based on the following information.

In a college, each of the 900 students participated in at least one of the six events, 50 m dash, 100 m dash, 150 m dash, 200 m dash, 250 m dash and 300 m dash. No student who participated in the 50 m dash participated in the 200 m dash or in the 250 m dash. No student who participated in the 300 m dash participated in the 100 m or 150 m dash. The same number of students participated in only 50 m dash, only 100 m dash, only 150 m dash, only 200 m dash, only 250 m dash and only 300 dash. The same number of students participated in each combination of exactly two events. An equal number of students participated in each combination of exactly three events. 20 students participated in exactly four events. The number of students who participated in only 100 m dash, in only 100 m and 50 m dash and in only 100 m, 50 m and 150 m dash are in 1 : 2 : 3 ratio.

20. How many students participated in the 50 m dash?

(A) 80 (B) 160
(C) 100 (D) 120

21. How many students participated in both the 100 m dash and the 150 m dash?

(A) 240 (B) 180
(C) 200 (D) 320

22. How many students did not participate in the 300 m dash?

(A) 160 (B) 640
(C) 580 (D) 740

23. How many students participated in at most two events?

(A) 680 (B) 520
(C) 600 (D) 440

Directions for questions 24 to 27: These questions are based on the following information.

P, Q and R played a game and each scored some points. The number of points is an integer. When I asked four individuals A, B, C and D about the scores of P, Q, and R, they made the following statements.

A: Exactly two of P, Q and R together scored 10 points.
B: Exactly two of P, Q and R together scored 11 points.
C: Exactly two of P, Q and R together scored 12 points.
D: Exactly two of P, Q and R together scored 13 points.

I understood that at least one of A, B, C and D was lying and later I found out the names of the people who could have lied.

24. Who was not lying?

I. A II. B
III. C IV. D

(A) Only I and III
(B) Only II and IV
(C) Only I and IV
(D) Only II and III

25. If the average score of P, Q and R is an integer, then who lied?

(A) A (B) B
(C) C (D) D

26. Which of the following is the highest score?

(A) 6 (B) 7
(C) 8 (D) 9

27. Which of the following is the least score?

(A) 3 (B) 4
(C) 5 (D) Cannot be determined

Directions for questions 28 to 31: These questions are based on the following information.

A group of four wealthy people, namely Ozwald Henry, Princess Stephanie, Gennady Yuganov and Henry Ford III each bought one of four different classic watches, such as a Louis Ulysse Chopard, a Breguet Dupuis, a Piaget Sunmaster and a Rolex Mercator at the annual Sotheby's auction. The following information is available about the person, the watch purchased and their prices.

(i) The total amount paid for these four watches was \$8,40,000 and the costliest watch was priced \$1,20,000 more than the cheapest.
(ii) Ozwald did not purchase the costliest watch and neither did he purchase a Piaget.
(iii) Gennady did not buy the costliest or the cheapest watch but had paid \$1,80,000 for his watch.
(iv) The Rolex Mercator is the costliest and the Chopard is the cheapest among the watches.
(v) Princess Stephanie purchased the Breguet Dupuis and had paid \$40,000 more than what Gennady Yuganov had paid.

28. Which watch did Ozwald Henry purchase and at what price?

(A) Louis Ulysse Chopard at \$2,20,000.
(B) Rolex Mercator at \$1,80,000.
(C) Louis Ulysse Chopard at \$1,60,000.
(D) Rolex Mercator at \$1,60,000.

29. What is the difference in the cost of the watches purchased by Henry Ford III and Gennady Yuganov?

(A) \$80,000 (B) \$1,00,000
(C) \$1,20,000 (D) \$1,10,000

30. Which of the following statements is true?

(A) Princess Stephanie bought the cheapest watch.
(B) Ozwald Henry did not purchase the Louis Ulysse Chopard.
(C) Henry Ford III bought a watch that was priced \$60,000 more than the Piaget Sunmaster.
(D) The watches bought by Ozwald Henry and Henry Ford III cost more than the watches bought by Gennady Yuganov and Princess Stephanie.

31. Which of the following watches was purchased by Gennady Yuganov?

(A) The watch that was priced \$40,000 less than the one bought by Henry Ford III.
(B) The watch that was the cheapest of all.
(C) The watch that was called Piaget Sunmaster.
(D) The watch that was called Breguet Dupuis.

Directions for questions 32 to 35: These questions are based on the following information:

These questions are based on the following information.
Six friends are comparing their expenses on a recent trip to Goa. Each of them spent a different amount. The following information is known about their expenses.

(i) Piyush spent ₹3783.
(ii) Saket spent ₹4640, which is ₹600 more than how much Uday spent.
(iii) The difference between the expenses of Uday and Tomar is ₹535.
(iv) The maximum difference between the expenses of any two of the six is ₹1135, whereas the minimum difference between the expenses of any two people is ₹167.
(v) The difference between the expenses of Qureishi and Raina is ₹246.
(vii) Raina spent ₹4227.

32. How much did Qureishi spend?

(A) ₹4473 (B) ₹4493
(C) ₹3981 (D) ₹4040

33. Whose expense was the highest?

(A) Qureishi (B) Saket
(C) Raina (D) Tomar

34. What is the difference between the expenses of Raina and Uday?

(A) ₹167 (B) ₹246
(C) ₹187 (D) ₹257

Difficulty Level-2: Moderate

35. Which of them spent the third least amount?

(A) Tomar (B) Uday

(C) Raina (D) Qureishi

Directions for questions 36 to 40: These questions are based on the following information.

A company named XYZ Ltd. manufactures a product 'Q' and sends it to five of its outlets A, B, C, D and E. The cost of production is ₹10,000 per unit. To transport one unit of Q to A, B, C, D and E, XYZ Ltd. spends ₹1000, ₹2000 ₹3000, ₹4000 and ₹5000, not necessarily in that order. The selling price of Q is ₹20,000 at three of the outlets, ₹21,000 at one of the outlets and ₹22,000 at another. Two of the outlets sell 40 units each per month and the remaining outlets sell 30 units, 45 units and 50 units per month. Sum of the cost of production and cost of transportation is subtracted from the selling price to arrive at profit per unit. The product of profit per unit and sales in units per month is profit per month. The following additional information is available.

I. One of the outlets earns ₹5,000 as profit per unit and it gets the least profit per month.

II. Exactly two outlets earn the same amount of profit per unit.

III. None of the outlets earn ₹11,000 profit per unit.

IV. Profit of outlet E per month is ₹3,20,000.

V. Profit per unit at outlet D is more than that at outlet E but its profit per month is less than that at outlet E.

VI. Profit of outlet B per month is ₹10,000, which is more than that of another outlet.

VII. Profit of outlet A per month is more than that of outlet C.

VIII. There is exactly one outlet which earns more profit per month than that of outlet E.

IX. Selling prices at outlets A, D and E are distinct.

36. What is the profit per month earned by the outlet B?

(A) ₹2,70,000 (B) ₹4,50,000

(C) ₹4,05,000 (D) ₹2,80,000

37. What is the selling price of the outlet E?

(A) ₹20,000 (B) ₹21,000

(C) ₹22,000 (D) Cannot be determined

38. If the selling price at outlet A is more than that at outlet D, then what is the transportation cost per unit at outlet A?

(A) ₹1000 (B) ₹2000

(C) ₹3000 (D) ₹4000

39. If the number of units sold by the outlet A is more than those sold by the outlet C, then what is the profit per month of C?

(A) ₹2,50,000 (B) ₹2,70,000

(C) ₹2,25,000 (D) ₹2,80,000

40. What is the profit per unit at the outlet E?

(A) 8000 (B) 9000

(C) 10,000 (D) None of these

EXERCISE-3

Directions for questions 1 to 4: These questions are based on the following information:

A chemical crusher unit has five different mills, such as P, Q, R, S and T of different capacities. The crusher unit operates 24 hours per day in three shifts 01st – 08th hour, 09th – 16th hour and 17th – 24th hours.

The time during which the mill is running is called uptime. For any mill each uptime is of exactly one-hour duration. In a period of 24 hours each mill has at least four hours of total uptime. The time period between two successive uptimes is called downtime. It is measured in hours and is always a whole number.

The downtime of a mill is directly proportional to its capacity and a constant. No two mills have the same downtime duration between successive uptimes.

During his visit in the third shift on a particular day, the new maintenance engineer observed that the mills P, Q , R , S and T were in uptime in the first five hours of the third shift, in that order. He was given a slip of paper indicating the history of functioning of the mills, observed during their uptime.

Mill	Day	Time
P	Yesterday	5th hour of the third shift
Q	Two days ago	6th hour of the third shift
R	Two days ago	3rd hour of the third shift
S	Yesterday	2nd hour of the day
T	Yesterday	Last hour of the day

The engineer kept thinking if he could determine the uptimes and downtimes of each of the mill.

1. Which mill has the highest capacity?
(A) P (B) S
(C) R (D) Cannot be determined

2. If different ranks from 1 to 5 were given to the mills in the descending order of their downtimes, then which mill will be ranked the second?
(A) P (B) T
(C) R (D) Cannot be determined

3. If mill R has lesser downtime than P, but not the lowest amongst all, then what is the downtime of mill R?
(A) 4 hours (B) 6 hours
(C) 3 hours (D) 8 hours

4. If mill R has greater downtime than mill S, which of the following is true for the mills P, Q, R, S and T to be in their uptime in the first five hours of a day respectively?
(A) This is possible at least once in a week.
(B) This is possible at most once in a week.
(C) Cannot be determined.
(D) Such a case is not possible.

Directions for questions 5 to 8: These questions are based on the following information.

A tour operator plans a one tour package each in four different circuits. Each tour starts at 7 a.m. from the office in a bus on the first day of the tour package and ends by dropping the tourists back at the office at 7 p.m. on the last day of the tour package. The four tour packages are (i) Circuit A – Seven days duration which starts every Wednesday and Thursday (ii) Circuit B – Three days duration which starts every Thursday and Friday (iii) Circuit C – Four days duration which starts every Wednesday and Saturday and (iv) Circuit D – A daily tour of 12 hour duration.

5. If a person has started his tour with Circuit A, then what is the minimum number of days required for him to completely tour all the circuits?
(A) 18 days (B) 17 days
(C) 16 days (D) 15 days

6. To completely tour all the four circuits in the shortest possible time, with which tour does a person shall start his touring?
(A) Circuit C on Saturday.
(B) Circuit B on Friday.
(C) Circuit C on Wednesday.
(D) Circuit B on Thursday.

7. On which day of the week, will there be the least activity at the tour operator's office?
(A) Sunday (B) Friday
(C) Monday (D) Saturday

8. If a person wants to complete all the circuits in the shortest possible time but with one day rest between any two tour packages, what is the best day for a person to start touring?
(A) Friday (B) Monday
(C) Sunday (D) Wednesday

Directions for questions 9 to 12: These questions are based on the following information.

A kid is promised by his father that starting the following Monday, a pocket money of five rupees per day will be given every day in the morning. The kid has school for five days in a week from Monday to Friday and wants to spend that amount for purchasing snacks during break in the school. He equally likes the chocolate (₹5), Samosa (₹10) and the pastry (₹15). He purchases not more than one item on any day. He makes a purchase if he has sufficient amount to purchase an item and he will not purchase the same item in the next two purchases.

(Assume that the kid did not have any other money and there are no holidays other than Saturdays and Sundays)

9. Which of the following is true with regard to the pattern in which the kid makes his purchases?
I. More data is required to identify a pattern.
II. The pattern of the purchases is repetitive.
III. If the first purchase of the kid is known, then the pattern will be repetitive.
(A) Only I and III (B) Only II
(C) Only I (D) Either I or III

10. What is the maximum possible amount available with the kid on any Monday?
(A) ₹15 (B) ₹20
(C) ₹25 (D) Cannot be determined

Additional information for questions 11 and 12: During the second week, the kid tasted a complimentary fruit worth ₹5, he decided to add to the fruit his purchase list from the following Monday, along with the other three such that the price of every next purchase increases and decreases alternately.

11. Which of the following is definitely true?
(A) The kid does not purchase on a Wednesday.
(B) The kid does purchase on a Thursday.
(C) The kid does not purchase on a Friday.
(D) The kid does purchase on a Tuesday.

12. What is the maximum possible amount available with the kid on any Monday (after he decided to eat the fruit)?
(A) ₹15 (B) ₹20
(C) ₹25 (D) Cannot be determined

Directions for questions 13 to 16: These questions are based on the following information.

Annie, Ben, Cain, Dan and Engel are five friends who purchased a book where each of them related to one of the following fields, such as Architecture, Biotechnology, Criminology, Demography and Economics. Further, the following information is known.

(a) No friends first letter of the name matches with the field to which the book purchased is related.
(b) Annie and Dan love to read books related to Criminology apart from the books they purchased.
(c) Engel hates Criminology and Biotechnology and hence, did not purchase them.
(d) The first letter of the field to which the book that Ben purchased pertains to, matches with the first letter of the name of the friend who purchased a book pertaining to Biotechnology.

13. After a month of reading, Annie exchanges her book with Dan and then Dan exchanges this book with Engel. The exchanges resulted in the first letter of the friends matching with the field to which the book belonged without violating conditions (b) to (d). Which of the following is true?
(A) Engel bought the book which is related to Demographics.
(B) Annie bought the book which is related to Demographics.
(C) Dan bought the book which is related to Economics.
(D) Engel bought the book which is related to Architecture.

14. Which of the following is not necessarily true?
(A) Either Annie or Ben bought a book which is related to Criminology.
(B) Dan bought the book which is related to either Architecture or Economics.
(C) Either Dan or Engel bought the book which is related to Architecture.
(D) Either Engel or Cain bought the book which is related to Demographics.

15. While delivering the books, the sales man interchanged the books of two friends in such a manner that the field to which the book belonged and the starting letter of only one of the friends matched. However, conditions (b) to (d) were not violated.
Which of the following conditions lets you to completely determine the fields of the books possessed by the five friends?
(A) Annie's and Engel's books were interchanged.
(B) Neither Annie nor Engel possess the book related to Economics after the sales man interchanged their books.
(C) Neither Dan nor Engel had books related to Biotechnology after the interchange.
(D) Interchange happened between the books of Annie and Dan.

16. While delivering the books, the sales man interchanged the books of two friends in such a manner that the field to which the book belonged and the starting letter of only one of the friends matched. However, conditions (b) to (d) were not violated. Then which pair of statements among (a), (b), (c) and (d) cannot be true simultaneously?
(a) Dan did not possess books related to Economics after the interchange.
(b) Engel did not possess books related to Architecture after the interchange.
(c) Neither Dan nor Annie possess the books related to Economics after the interchange.
(d) Interchange took place between Annie and Dan.
(A) (a) and (b)
(B) (b) and (d)
(C) (c) and (d)
(D) (a) and (c)

Directions for questions 17 to 20: These questions are based on the following information.

A test consists of two parts. Part I consists of five questions and for any question the student will score two marks for a correct answer and zero for a wrong answer. Part II consists of four questions, in which for any question, the student will score ten marks for a correct answer, five marks for a partially correct answer and zero for a wrong answer.

17. Which of the following scores is not possible for the test?
(A) 29 (B) 31
(C) 47 (D) 41

18. If Virat scored 33 marks then which of the following is not necessarily true?
(A) Virat got at least one question wrong in Part I.
(B) Virat answered less than six questions in all.
(C) Virat did not answer any question of Part II wrong.
(D) Virat gave wrong answer to at least one question of Part II.

19. Each of Rahul, Beena, Johan and Bijaya attempted seven questions and did not get zero in any question. No two of them scored the same marks. What is the maximum possible difference of the total marks scored by Rahul and Beena, and that of the marks scored by Bijaya and Johan?
(A) 51 (B) 43
(C) 52 (D) 44

20. If both U and V attempted six questions each, the marks scored by each of them are unique and the marks received for any question is other than zero, what is the minimum possible difference between their respective scores?
(A) 1 (B) 2
(C) 3 (D) 4

Directions for questions 21 to 25: These questions are based on the following information.

A library assistant has marked four racks for shelving the books each from different specializations, such as Marketing, Operations, Systems and Human Resource. Totally, there are 24 books. The librarian arranged the textbooks in such a way that each rack contained even number of textbooks and they are unique and are non-empty. The students often misplace the textbooks among any of the racks. The following information about the books in various racks on a particular day are as follows.

(i) Half the textbooks are placed incorrectly. But each rack has the same number of textbooks as there was originally.

(ii) The number of textbooks in Marketing rack is equal to the sum of the number of textbooks in the other three racks.

(iii) One third of the textbooks in the Marketing rack originally belong to the Operations rack.

(iv) All but two textbooks that are there in the Human Resource rack originally belonged to a different rack.

(v) The number of textbooks in the Operations rack is twice that of the number of textbooks in the Systems rack.

(vi) The textbooks from a rack are misplaced into at most one rack.

21. How many textbooks belonging to Operations rack are placed in Human Resource rack?

22. How many textbooks belonging to Systems rack is placed correctly?

23. How many textbooks belonging to Human Resource rack are in the Operations rack?

24. Which rack has all the books correctly placed?

(A) Marketing (B) Operations
(C) Systems (D) None of these

25. How many text books related to Operations are in the systems rack?

ANSWER KEYS

Exercise-1

1. (B)	**8.** (C)	**15.** (B)	**22.** (C)	**29.** (C)	**36.** (D)
2. (B)	**9.** (C)	**16.** (A)	**23.** (C)	**30.** (D)	**37.** (B)
3. (C)	**10.** (D)	**17.** (C)	**24.** (D)	**31.** (C)	**38.** (D)
4. (D)	**11.** (A)	**18.** (B)	**25.** (C)	**32.** (D)	**39.** (B)
5. (D)	**12.** (B)	**19.** (C)	**26.** (C)	**33.** (C)	**40.** (A)
6. (B)	**13.** (D)	**20.** (C)	**27.** (D)	**34.** (C)	
7. (B)	**14.** (C)	**21.** (B)	**28.** (B)	**35.** (B)	

Exercise-2

1. (A)	**8.** (D)	**15.** (A)	**22.** (D)	**29.** (B)	**36.** (D)
2. (B)	**9.** (B)	**16.** (A)	**23.** (B)	**30.** (D)	**37.** (C)
3. (B)	**10.** (D)	**17.** (A)	**24.** (B)	**31.** (C)	**38.** (B)
4. (C)	**11.** (C)	**18.** (A)	**25.** (A)	**32.** (A)	**39.** (C)
5. (D)	**12.** (D)	**19.** (A)	**26.** (B)	**33.** (B)	**40.** (A)
6. (A)	**13.** (C)	**20.** (B)	**27.** (D)	**34.** (C)	
7. (A)	**14.** (B)	**21.** (A)	**28.** (C)	**35.** (B)	

Exercise-3

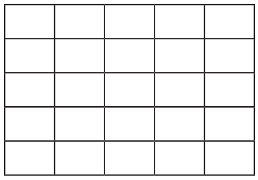
1. (D)	**6.** (A)	**11.** (D)	**16.** (C)	**21.** 0
2. (A)	**7.** (C)	**12.** (A)	**17.** (C)	**22.** 2
3. (C)	**8.** (D)	**13.** (A)	**18.** (B)	**23.** 4
4. (D)	**9.** (B)	**14.** (D)	**19.** (D)	**24.** (C)
5. (B)	**10.** (C)	**15.** (B)	**20.** (A)	**25.** 0

SOLUTIONS

EXERCISE-1

Solutions for questions 1 to 10:

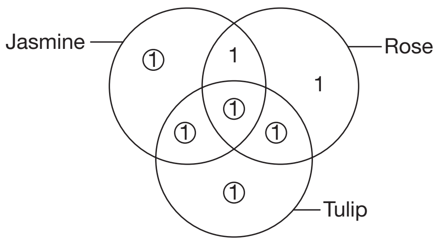
1. For the formation of maximum number of squares, six lines are to be horizontal and parallel to each other and the other six lines vertical are as shown below.



$\therefore$ The maximum possible number of squares $= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$

2. At first one candle is lighted from two sides and the second candle is lighted from one side.
The first candle is burnt in 30 minutes. Immediately, the second stick is burnt from the other side.
Now the remaining part of the second stick will be burnt in 15 minutes.

3. This question can be answered by using a Venn diagram as follows.



Here, the circled number are having polka dots.
$\therefore$ There are only two butterflies which do not have polka dots.

4. The details of the number of bottles while crossing the border is given below.

After crossing	Remaining number of bottles	Number of boxes	Number of bottles given as bribe
10th	$50 = 15 \times 3 + 5$	–	–
9th	$54 = 15 \times 3 + 9$	4	4
8th	$58 = 15 \times 3 + 13$	4	4
7th	$63 = 15 \times 4 + 3$	5	5
6th	$68 = 15 \times 4 + 8$	5	5
5th	$73 = 15 \times 4 + 13$	5	5
4th	$79 = 15 \times 5 + 4$	6	6
3rd	$85 = 15 \times 5 + 10$	6	6
2nd	$92 = 15 \times 6 + 2$	7	7
1st	$99 = 15 \times 6 + 9$	7	7
Initial	$107 = 15 \times 7 + 3$	8	8

(*Continued*)

$\therefore$ He should start with a minimum of 107 bottles of champagne.

5. There are two possible sets of values for their ages.
The product can be $5 \times 3 \times 1$ or $15 \times 1 \times 1$.
$\therefore$ The age cannot be uniquely determined.

6. Neeta should draw in such a way that always $n(2 + 3) + 1$ coins remain, so that if Leena draws 3 coins Neeta should draw 2 coins and vice versa.
So, there should be 30 + 1 coins remaining.
$\therefore$ She should draw 3 coins.
So, at the end there will be 6 coins from which Leena has to draw.
So, in any case Neeta will win if she draws 3 coins initially.

7. The process is tabulated below:

	4 litres	6 litres
Initial	0	0
1	0	6
2	4	2
3	4	0
4	0	4
5	4	4

Hence, 3 litres cannot be measured.

8. From the given expression, A must be 1, the expression must be:

1 B 1
$\times$ 1 1
1 C C 1

Here, B can be any value from 2 to 8.
$\therefore$ There will be a total of 7 different values.

9. The total requirement is 78 + 61 + 26 + 42 + 34 + 59 = 300 units.
$\Rightarrow$ The lorry starts 3 times and the unused capacity is zero units.

10. 68.892 g can be measured by using the following stones $1.002 + 1.39 + 6.5 + 60 = 4$ stones

Solutions for questions 11 to 13: The possible value for C is 1/5 (1 or 5).

As E = 3B, B = 1/2/3

As 3A = 2D, A = 2/4/6 and D = 3/6/9.

If A = 2, then D = 3, B = 1, C = 5, F = 3

Hence, A = 2 is not possible as values of D and F will be same. If A = 4, then D = 6 and B cannot be 1 or 2. Therefore, B must be 3. Then F = 7 and E = 9.

if C = 5, E = 9

$$\begin{array}{r} 435 \\ \times\ 345 \\ \hline 150075 \\ \hline \end{array}$$

Hence, C cannot be 5.

If C = 1

$$\begin{array}{r} 431 \\ \times\ 341 \\ \hline 146971 \\ \hline \end{array}$$

Therefore, A = 4, B = 3, C = 1, D = 6, E = 9 and F = 7.

11. The value of C is 1.

12. The value of A is 4.

13. The value of F is 7.

Solutions for questions 14 and 15: The possible values for B = 3/6/9 and F = 2/4/6

As B = 2E, B must be 6.

If B = 6, then E = 3, F = 4, D = 8, and A = 5.

$\Rightarrow 4368 \div 56 = 78 \Rightarrow C = 7$

$\therefore$ The values of A = 5, B = 6, C = 7, D = 8, E = 3 and F = 4.

14. The value of D is 8.

15. The value of C is 7.

Solutions for questions 16 and 17: As 5D = 6E, D must be 6 and E must be 5.

If D = 6, B must be either 2 or 4.

Case (i): B = 2.

$A = 7, C = 1 \Rightarrow 72 \times 16 = 1152$.

Hence, B = 2 is possible.

Case (ii): B = 4.

$A = 9, C = 2 \Rightarrow 94 \times 26 = 2444$.

Hence, B = 4 is not possible.

$\therefore$ A = 7, B = 2, C = 1 D = 6 and E = 5.

16. The value of A is 7.

17. The value of C is 1.

Solutions for questions 18 to 21: Assume that 2 and 5 are in the correct position for the 1st selection. Then it must indicate 2 in column R in the 2nd selection.

$\Rightarrow$ Both 2 and 5 are not at the correct position.

Assume that 5 is at the correct position.

$\Rightarrow$ 3 and 2 are wrong.

$\Rightarrow$ 4 is the 2nd digit in the code.

$\Rightarrow$ There cannot be ‘2’ in column ‘R’ in the 3rd selection.

$\Rightarrow$ 5 is not present in the code.

Assume ‘2’ is in the correct position.

$\Rightarrow$ 4 is also in the correct position.

$\Rightarrow$ 6 is present in the code but is not the 4th digit.

$\Rightarrow$ The number is ‘6421’.

Assume ‘2’ is not present in the code.

$\Rightarrow$ 3 and 4 are in the correct position.

$\Rightarrow$ 1and 6 are present in the code and 6 is not the 4th digit in the code.

$\Rightarrow$ The number is ‘3461’.

18. 2 is the second digit from right.

19. 3 is the first digit.

20. 5 is not present in the code number.

21. ‘3416’ cannot be the code number.

Solutions for questions 22 to 25: Let us represent the people with the first letter of their name.

From (i) and (ii), we get the comparison of the distance travelled by them N > M > G > P.

From (i), it is given that the speed and the number of hour travelled by Naveen is not the maximum.

$\therefore$ Speed cannot be 60 kmph and the number of hours travelled is not 5.

Hence, the distance travelled by N has to be less than 200 km.

As Jagdish travelled 120 km, Naveen travelled more than 120 km.

That can only be 160 km.

$\therefore$ Naveen’s Speed = 40 kmph and time = 4 hours.

Distance = 160 km

$\therefore$ Jagdish cannot travel for 4 hours.

To get 120 km distance, there is only one possibility. Jagdish has travelled at 60 kmph speed for 2 hrs.

Now, as one of them has travelled 30 km, that must be at 30 kmph speed for 1 hour.

The person who travelled at 20 kmph speed travelled for 5 hours, i.e., 100 km.

$\therefore$ The person who travelled at 15 kmph speed travelled for 3 hours, i.e., 45 km.

The table of the final result is as follows.

Name	Speed	Time	Distance
Naveen	40	4	160
Jagdish	60	2	120
Mathew	20	5	100
Govind	15	3	45
Preet	30	1	30

22. Govind travelled at 15 kmph speed.

23. Govind travelled for 3 hours.

24. Preet travelled for the least time.

25. Mathew travelled for maximum time.

Solutions for questions 26 to 29: There are 6 teams, so 15 matches are played in the tournament and 5 matches in each week.

As no team scored more than 3 goals in a match. From the result of the first week it is clear that, each of A, C, E and F plays two matches. There was no match between A and F. The goals in the matches for each team in the first week can be represented as follows.

	1st week	
	For	Against
A	3 + 3	2 + 1
B	1	3
C	2 + 2	3 + 3
D	1	3
E	3 + 2	1 + 3
F	3 + 3	2 + 2

From the above table, E played one match with one of A and F and the other match with one of B and D.
C must play a match with A and the other match with F.
∴ The result of these two matches are A – 3, C – 2, and F – 3, C – 2.
As E scores 2 goals in the other match, then that match must be between F and E.
The result of that match is F – 3, E – 2.
∴ B will play the match with either A or E and D will also play the match with A or E.
Similarly, for the second week after subtracting the goals of first week we will get the goals scored and conceded by a team.
The scoring pattern of the matches played by the 6 teams in the 3 weeks are tabulated below.

	1st Week		2nd Week		3rd Week	
Teams	Goals for	Goals against	Goals for	Goals against	Goals for	Goals against
A	3	1	3	2	3	2
	3	2			2	1
B	1	3	0	3	3	2
			0	3	3	2
C	2	3	3	2	1	0
	2	3	3	0		

(*Continued*)

	1st Week		2nd Week		3rd Week	
Teams	Goals for	Goals against	Goals for	Goals against	Goals for	Goals against
D	1	3	3	1	0	1
			2	3	2	3
E	3	1	2	3	2	3
	2	3			1	2
F	3	2	1	3	2	3
	3	2	3	0		

F played the 2nd match with D. Since A did not play with E in the 2nd week, so A's match was with D. The remaining match in 2nd week was between C and E.
Matches played in 2nd week were BC, BF, FD, AD and CE.
In the 1st week C played one match with A and the other with F. Since A played with D in the 2nd week, so A's other match in the 1st week was with B. E played on match with F and the other one with D. Matches played in 1st week were CA, CF, AB, EF and ED.
In the 3rd week C's only match was with D. A played one match with E and the other with F. B played one match with D and other with E.
Matches played in 3rd week were CD, AE, AF, BD and BE.
The results are as follows.

BC	B – 0	C – 3
BF	B – 0	F – 3
AD	A – 3	D – 2
EC	E – 2	C – 3
DF	D – 3	F – 1

∴ In the first week a match cannot be between A and D. Hence, the match must be between A vs B and E vs D.
Similarly, for 3rd week, F and C play exactly one match each. In the match which C plays, it scores 1 goal and concede 0 goals.
∴ Now the remaining matches are played in 3rd week.
The remaining matches are (A, F), (A, E), (B, D), (B, E), (C, D). By comparing the goals scored and conceded in the round, the goals scored by each team can be obtained.
The goals scored by each team will be as follows.

AE	A – 2	E – 1
AF	A – 3	F – 2
BD	B – 3	D – 2
BE	B – 3	E – 2
CD	C – 1	D – 0

26. A won the maximum number of matches.

27. B plays against D and E in the 3rd week.

28. In the 1st week the match was between D and E.

29. There are 5 such matches.

Solutions for questions 30 to 33: It is clear that Russell drove the car for a maximum time period. Hence, his speed was minimum but the distance covered was maximum.

Person	Time	Speed	Distance
Narayan	t_n	S_n	$S_n t_n$
Michael	t_m	150	$150t_m$
Russell	$(t_n + t_m)$	S_r	$(t_n + t_m)S_r$
1620			

Let t_n, t_m be the time taken by Narayan, Michael while S_n and S_r are the respective speeds of Narayan and Russell.

$S_r < S_n$

$S_r < 150$

Since, $S_n > 150 \Rightarrow S_r < 150\ s_n$

Since all the values in their respective units are integral numbers.

$\Rightarrow 150\ t_m \neq 360$

Also, $(t_n + t_m)s_r \neq 360$

(Since, 360 km is the minimum distance)

Hence, $s_n\ t_n = 360$

Here, $s_n > 150$

If $t_n = 1$ $\quad s_n = 360$

$t_n = 2$ $\quad s_n = 180$

$t_n = 3$ $\quad s_n = 60$ (not possible)

Let $t_n = 1 \Rightarrow s_n = 360$

Exploring possibilities, we get:

S. No.	t_n	t_n	$S_n t_n$	$(150\ t_m)$	$(t_n + t_m)s_r$
1	2	1	360	300	960
2	3	1	360	450	810
3	4	1	360	600	660
4	5	1	360	750	410
5	6	1	360	900	260

Only in S. No. (3) the conditions are satisfied.

$660 = (4 + 1)s_r$

$s_r = 132$

Case I:

Person	Time (hours)	Speed (km/hr)	Distance (km)
Naryan	1	360	360
Michael	4	150	600
Russell	5	132	660

Case II:

Let $t_n = 2$, $S_n = 180$

Over here also, $t_m + t_n = (4 + 2) = 6$ hours is the only possibility.

$\Rightarrow s_r = 660/6 = 110$ km/hr

Case (II):

Person	Time	Speed	Distance
Narayan	2	180	360
Michael	4	150	600
Russell	6	110	660

30. Distance covered by Russell is 660 km.

31. Time taken by Michael is 4 hours.

32. By observing the two possible cases, all the statements may be true.

33. This belongs to case (II). Narayan drove at a speed of 180 km/hr.

Solutions for questions 34 to 37: As the scores of A, C, D and F are odd, they must have a draw each.

$\Rightarrow$ B and E did not have any draw as only two draws are there in the tournament.

Now their scores are shown below:

Name	Won	Dram	Lost	Score
A	4	1	0	9
B	4	0	1	8
C	3	1	1	7
D	1	1	3	3
E	1	0	4	2
F	0	1	4	1

If D had a tie with either A or C, F must also have a tie with A or C.

$\Rightarrow$ Both D and E won against F.

$\Rightarrow$ D and E have a tie, this is not possible.

$\Rightarrow$ D had a tie with F.

$\Rightarrow$ A had a tie with C.

$\Rightarrow$ A won against B, D, E and F.

B won against C, D, E and F.

C won against D, E and F.

D won against E.

E won against F.

34. D had a tie with F.

35. A had a tie with C.

36. Both E and F lost four matches each.

37. A had a tie with C Hence, it did not win against C.

Solutions for questions 38 to 40: The total amount given is ₹30,000. 5 cards each of 3 of the given denominations and 6 cards each of the other denominations is required. The denominations are ₹200, ₹700, ₹1000, ₹1500, ₹2000. The key lies in identifying the cards of which 5 are bought.

Take for example, the ₹700 denomination card. If 6 of them are bought, it amounts to ₹4200 and whatever combination of other cards are bought, one cannot round off the hundreds to thousands (since exactly 30,000 is spent). It implies that 5 cards of ₹700 are bought which amounts to ₹3500. To round off the hundreds to the thousands Gopi needs to buy 5 cards of ₹1500 denomination (₹7500). 6 cards of ₹200 denomination will cause the same problem as discussed above. Hence, only 5 cards of ₹200 denomination is purchased. Therefore, 5 cards each of denominations 200, 700 and 1500 are purchased and 6 cards each of ₹1000 and ₹2000 denominations are purchased.

Denomination	Number purchased	Total amount
₹200	5	₹1000
₹700	5	₹3500
₹1000	6	₹6000
₹1500	5	₹7500
₹2000	6	₹12,000

38. The number of cards of denomination ₹200 are 11 and those of denomination ₹2000 are 16.
∴ Ratio is 11 : 16.

39. 5 cards each of ₹200, ₹700 and ₹1500 were bought.
∴ The combined value of these cards is
5 (200 + 700 + 1500) = 12000.

40.

Denomination	Number purchased	Total amount
200	6	1200
700	6	4200
1000	5	5000
1500	6	9000
2000	5	10000

Total amount = 29,400
∴ Gopi would have left with ₹600.

EXERCISE-2

Solutions for questions 1 to 3:

1. When K = 12 we have to take one ball from the first bag, two from the second and so on. If the total weight is 1 kg less then the required box is the first one. If the weight is 2 kg less, then it is the second box and so on.
∴ Only one weighing is required.

2. If K = 4 we have to take one ball each from the first two containers, two balls each from the third and fourth containers and so on.
Similar to the logic in the previous question we can find which group of containers consists 1 kg ball. Now by weighing one ball from one of these two containers we can find that the container having balls of 1 kg.
Thus, two weighings are required.

3. If K = 3 then we have to take one ball each from the first three containers and two balls each from the next three containers and three balls each from the last two containers.
We can find which group of containers has 1 kg ball.
Now one more weighing is required to find the exact container in which 1 kg balls are present.
∴ A total of two weightings are required.

Solutions for questions 4 to 6:

4. The money with Mr Helpinghand after giving to each beggar will be as follows:

	Initial	1st beggar	2nd beggar	3rd beggar	4th beggar	5th beggar	6th beggar	7th beggar	8th beggar
Amount given to the beggar	–	12	9	6	6	3	3	3	3
Amount with Mr Helpinghand	45	33	24	18	12	9	6	3	0

∴ Mr Helpinghand gives money to eight beggars.

5. Mr Helpinghand will always be left with money which is a multiple of 3 but the initial amount need not be a multiple of 3.

	Last beggar	Previous one	Previous one	Previous one	First beggar	Initial
Amount to the beggar	3	3	3	3	4	–
Amount with Mr Helpinghand	–	3	6	9	12	16

∴ The last beggar will always get ₹3. The total amount will be minimum, when there is a minimum possible remainder each time. The calculation is shown in the following table from the last to the first beggar.
∴ ₹16 is the minimum possible initial amount.

6. As already explained, the initial steps should be the same. Here, we will try to maximize the remainder.

	Last beggar	Previous one	Previous one	Previous one	Previous one	Previous one	Initial
To the beggar	3	3	3	6	6	10	–
With Mr Helpinghand	–	3	6	9	15	21	31

∴ ₹31 is the maximum possible amount value.

Solutions for questions 7 to 10: Let us represent the people with the first letter of their names.

As per the given information in (ii) the interest earned by them can be written in the following order.

S > K > P /B > M

From (iii), B earned ₹600 as interest which is possible for an investment of ₹2000 at 6% per annum or an investment of ₹3000 at 4% per annum.

B cannot be the least interest earner.

The least interest would be earned for an investment of ₹2000 at 5% per annum.

It has to be ₹3,000 at 4% per annum for B.

∴ B is the second lowest interest earner.

From (iv), ₹6000 is not invested at 7.5% per annum or at 8% per annum.

Hence, it is invested at 6% per annum.

∴ Interest earned would be ₹1800, which is not the highest interest.

The highest interest would be earned for an investment of ₹5000 either at 7.5% per annum or at 8% per annum and the third highest interest would be earned for the investment of ₹4000 either at 7.5% per annum or at 8% per annum.

∴ The following table shows the distribution of the investment, interest rate and interest earned.

	Investment	Rate of interest (per annum)	Interest earned
Manjeet	2000	5%	500
Biswajeet	3000	4%	600
Paramjeet	4000	7.5/8 %	1500/1600
Karamjeet	6000	6%	1800
Sarvajeet	5000	7.5/8%	1875/2000

7. Investment of Paramjeet is ₹4000.

8. Sarvajeet earned ₹1875 or ₹2000.

9. Karamjeet invested at 6% per annum.

10. The difference between the interest earned by Biswajeet and Karamjeet = ₹1200.

Solutions for questions 11 to 15: Let each contestant be denoted by the first letter of his name.

Given, in the first round all the contestants gave distinct answers.

Let the cumulative scores of K, R, A, V, S at the end of the nth round be denoted by $C(K)_n$, $C(A)_n$, $C(V)_n$ and $C(S)_n$, respectively.

Let the total of cumulative scores at the end of nth round be denoted by $C(\text{Total})_n$.

$\therefore$ The sum of the cumulative scores at the end of the first round is $1 + 2 + 3 + 4 + 5 = 15$.

$C(K)_1 + C(R)_1 + C(A)_1 + C(V)_1 + C(S)_1 = 15$

Given the averages of the cumulative scores at the end of the rounds 1 and 2 are the same.

$\therefore$ The total of the cumulative scores of the contestants at the end of the second round is 15.

We can say that at least two contestants given the same answer in the second round.

$\therefore$ The sum of the cumulative scores of R, V, S at the end of the second round is $15 - (9 + 3) = 6$.

$\Rightarrow C(R)_2 + C(V)_2 + C(S)_2 = 6$

As K's cumulative score is increased by 1 from round 1 to round 2, K have given the least graded answer.

$\therefore$ If S is not one of the contestants who gave the same answer, then his cumulative score at the end of the second round must be at least 6, if so the sum of the cumulative scores of R and V must be 0, which is not actually possible.

$\therefore$ S must have given the same answer as other in the second round.

If A given the same answer as S, then $C(S)_2 = 3$.

But $C(R)_2 + C(V)_2$ cannot be 3

$\therefore C(\text{Total})_2 > 15$, which is not possible.

Here, A given a district answer in the second round.

$\therefore$ C(A) is increased by at least two.

As $C(A)_2 = 3$, $C(A)_1$ must be equal to 1.

$\therefore C(V)_1 = 3$

Hence, V must have given the same answer as S in the second round.

If R given a distinct answer, then

$C(V)_2 + C(S)_2 = 3 + 3$, which results in $C(\text{Total})_2 > 15$.

$\therefore$ R given the same answer as V and S in the second round.

$C(R)_2 = C(V)_2 = C(S)_2 = 2$

From (ii), we can say that C(Total) is increased by 15 at the end of the third round comparing to that at the end of the second round.

$\therefore C(\text{Total})3 = 15 + 15 = 30$

From (vii) and (vi), we can say that,

$C(\text{Total})6 = C(\text{Total})7 - 15$

$= 71 - 15 = 56$

From (iv) and the above data, we can say that A has given the same answer as at least one of the other in the fourth round.

$\therefore C(\text{Total})4 = C(\text{Total})5 - 15$

$= 51 - 15 = 36$

$\therefore C(V)_4 = 36 - 28 = 8$

Also, in rounds 1, 3, 5 and 7 all contestants given district answers.

$\therefore C(K)_3$ is at least 7.

$\Rightarrow$ K have given the same answer as A in the fourth round.

$\therefore$ Each of $C(K)_3$ and $C(A)_3$ must be at least 7.

From (viii), we can say that V have got 1 point in the fourth round.

$C(V)_3 = C(V)_{4-1} = 8 - 1 = 7$

$\therefore$ V got five points in the third round.

A can get at most five points in the third round.

As $C(A)_2 = 3$, $C(A)_3 \geq 7$, $C(A)_3 = 7$

$\therefore$ A got four points in the third round.

As $C(R)_3 < 7$, R has given a distinct answer in the fourth round.

Also, we have S, which has got 2 points in the fourth round.

$\therefore$ R must have got 3 points in the fourth round.

$C(R)_3 = C(R)_4 - 3$

$= 7 - 3 = 4$

$\therefore$ R got 2 points in the third round.

$\Rightarrow$ K got 1 point in the third round.

From (vi), as A given a district answer in the sixth round,

$C(A)_6 \geq C(A)_4 + 2$

As $C(A)_6 = 9$ and $C(A)_4 = 7$, $C(A)_5$ must be equal to 8.

$\therefore$ A got 1 point each in the fifth round and sixth round.

R and S got 2 and 3 points in the sixth round.

$\therefore C(R)_6 + C(S)_6 = 9 + 11 + 2 + 3 = 25$.

As $C(\text{Total})_6 = 56$,

$C(K)_6 + C(V)_6 = 56 - C(R)_6 - C(S)_6 - C(A)_6$

$= 56 - 25 - 9 = 22$

As K and V given the same answer in the sixth round,

$C(K)_6 = C(V)_6 = \frac{22}{2} = 11$.

$\therefore C(K)_5$ or $C(V)_5$ must be equal to 11.

As S got four points in the fifth round, K cannot get four points in the fifth round.

$C(K)_5 \neq 11 \Rightarrow C(K)_5 > 11$

$\therefore C(V)_5 = 11$

V got 3 points in the fifth round.

$\therefore$ K got 5 points in fifth round and $C(K)_5 = 12$.

As $C(R)_5 = 9$ and $C(R)_7 = 14$, the sum of the points scored in the sixth and the seventh round by R is 5.

Also, R got 2 or 3 points in the sixth round

As K got 2 points in the seventh round, R can not get 2 points in that round.

$\therefore$ R cannot get 3 points in the sixth round.

R got 2 points in the sixth round and 3 points in the seventh round.

$\therefore$ S got 3 points in the sixth round and $C(S)_6 = 14$

From (iii), we can say that $C(A)_7 \neq 14$.

$\therefore$ A did not got five points in the seventh round

A must have scored 1 point and S must have got 5 points in the seventh round.

$\therefore C(S)_7 = 14 + 5 = 19$

The final scores will be as follows:

Round no.	Total of cumulative scores	The contestants who gave same answer	K	R	A	V	S
1	15	X	5	2	1	3	4
2	15	RVS	6	2	3	2	2
3	30	X	7	4	7	7	5
4	36	KA	7	7	7	8	7
5	51	X	12	9	8	11	11
6	56	KV	11	11	9	11	14
7	71	X	13	14	10	15	19
Cumulative scores are:			61	49	45	57	62

11. Sashank was the winner.

12. Ranjit, Varun and Sashank gave the same answer in the second round.

13. The score of Ranjit after at the end of the third round = 4.

14. The score of Kamal and Varun at the end of the sixth round are 11 and 11, respectively.

15. Kamal gave the highest graded answer in the fifth round.

Solutions for questions 16 to 19: The teams with better chances of winning are teams A, C and F. The teams that have worst chances of winning are B, D and E.

16. If there is only one upset, in order to maximize Amit's earnings, then let us assume that he put stakes on three winning teams, two that were expected to win and the one that caused an upset.
Among B, D, E, team E has the best odds of 20%.
Hence, if Amit bets ₹100 each on teams A, C and E, then he will maximize his eanings.

$\therefore$ His earnings = $\left(100+\frac{1}{4}\times 100\right)+\left(100+\frac{2}{5}\times 100\right)+\left(100+\frac{20}{1}\times 100\right)$

= 125 + 140 + 2100
= 2365

17. In order to maximize his earnings, let us assume that Bhaskar put more money at stake on teams with better odds (among the expected winners only).
Among A, C and F, Bhaskar has the best odds on C followed by A and then F.

$\therefore$ Bhaskar's earnings = $\left(60+\frac{2}{5}\times 60\right)+\left(40+\frac{1}{4}\times 40\right)+\left(20+\frac{1}{5}\times 20\right)$

= 84 + 50 + 24
= 158

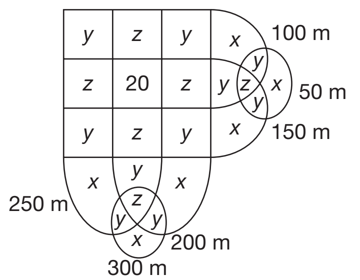
18. Among the team expected to lose, i.e., B, D and E, B and E have the best odds. Assuming that these two teams caused upsets and the other match went the expected way, teams B, C, E winning would be the most favourable to Ravi.

19. Suresh's minimum possible earnings are zero, if both his teams lose.
In Match III, the best possible outcome for Suresh is team F winning, which gives him an earning of $\left(50+\frac{7}{5}\times 50\right)$ = ₹ 60.

In Match II, an upset would maximize his earning. If team D wins and Suresh put his money on that team, his earnings would be $\left(50+\frac{1}{2}\times 50\right)$ ₹225.

$\therefore$ His maximum earnings = Difference between maximum and minimum possible earnings = 225 + 60 = 285.

Solutions for questions 20 to 23: The given information can be represented in the following Venn diagram.



x represents exactly one.
y represents exactly two.
z represents exactly three.
Since exactly four is 20, $6x + 10y + 6z = 880$.
Given $x : y : z = 1 : 2 : 3$
$6x + 20x + 18x = 880$
$\therefore\ x = 20$

20. The number of students who participated in 50 m dash $= x + 2y + z = 8x = 160$.

21. The number of students who participated in both 100 m and 150 m dashes $= 3z + y + 20 = 9x + 2x + 20 = 11x + 20 = 240$.

22. The number of students who did not participate in 300 m dash $= 900 - 160 = 740$.

23. The number of students who participated in at most two events $= 6x + 10y = 26x = 520$.

Solutions for questions 24 to 27: Assume that A is lying.

$\Rightarrow$ Total score is $(11 + 12 + 13) \div 2 = 36 \div 2 = 18$ (1)

Assume that B is lying.

$\Rightarrow$ Total score is $(10 + 12 + 13) \div 2 = 35 \div 2 = 17.5$

This is not possible.

$\Rightarrow$ B is not lying.

If C is lying:

Total score $= (10 + 11 + 13) \div 2 = 34 \div 2 = 17$

D cannot be wrong, because otherwise the total score will not be an integer.

$\Rightarrow$ Either A or C is lying.

(i) Assume that A lied.
$\Rightarrow$ Total score is 18.
Individual scores are:
$18 - 11 = 7$
$18 - 12 = 6$
$18 - 13 = 5$

(ii) Assume that C lied.
$\Rightarrow$ Total score = 17
Individual scores are:
$17 - 10 = 7$
$17 - 11 = 6$
$17 - 13 = 4$

In both the cases, the first two individual scores are the same, but the least score varies.

If A lied, P, Q, R = 5, 6, 7 (in any order)

If C lied, P, Q, R = 4, 6, 7 (in any order)

24. We can definitely say that B and D are not lying.

25. In this case, A lied as 18 is divisible by 3 but 17 is not.

26. 7 is the highest score.

27. Least score cannot be determined.

Solutions for questions 28 to 31: From (v), we know that Stephanie purchased the Breguet Dupuis and from (iv) and (vi) we know that Henry Ford purchased the Rolex Mercator. From (ii), we know that Ozwald did not purchase a Rolex or a Piaget (also, not a Breguet according to (v)). Hence, he must have purchased a Louis Ulysse Chopard which means that Gennady purchased the Piaget.

From (iii) and (v), we know that Gennady paid \$1,80,000 and Stephanie paid \$2,20,000 for their watches, respectively.

From (i) and (iv), we know that all four watches together cost \$8,40,000 of which Gennady had purchased a watch worth \$1,80,000 and Stephanie, a watch that was worth \$2,20,000.

$\therefore$ Cost of the other two watches
= 8,40,000 – 4,00,000 = \$4,40,000

From (i), we know that the Rolex costs \$1,20,000 more than the Chopard, which means that the Rolex costs \$2,80,000 and the Chopard costs \$1,60,000.

28. Ozwald Henry purchased the Chopard at \$1,60,000.

29. Henry Ford's watch costs \$2,80,000 where as Yuganov's watch costs \$1,80,000.
$\therefore$ The difference is \$1,00,000.

30. Choice (D) is the only one that's true from the data given.

31. Choice (C) is the only one that is correct.

Solutions for questions 32 to 35: From (i) and (ii), Piyush spent ₹3783, Saket spent ₹4640 and Uday spent ₹4040.

From (iii) and (iv), the maximum difference between the expenses must be between the highest and lowest spenders. Hence, Tomar's expenses must be the lowest at ₹3505.

From (i) and (v), the minimum difference between the expenses of any two persons is ₹167. The difference between the expenses of Piyush and Tomar is ₹278, whereas the difference between the expenses of Uday and Tomar is ₹535. Hence, Piyush must have spent the second least amount, whereas Uday must have spent the third least amount.

The difference between the expenses of Saket and Uday is ₹600, whereas that between the expenses of Qureishi and Raina is ₹246. Since Raina spent ₹4227, Qureishi must be the second highest spender at ₹4473 and Raina must be the third highest spender.

The descending order of their expenses is as follows:

Saket (₹4640) > Qureishi (₹4473) > Raina (₹4227) > Uday (₹4040) > Piyush (₹3783) >Tomar (₹3505)

32. Qureishi spent ₹4473.

33. Saket spent the highest amount.

34. The difference between the expenses of Raina (₹4227) and Uday (₹4040) is ₹187.

35. Uday spent the fourth highest or the third least amount.

Solutions for questions 36 to 40: The profit per month of E is ₹3,20,000. The only possibility is 8000×40, i.e., profit per unit is ₹8000 and the number of units sold is 40.

The profit per unit of D is greater than that of E and as profit per month is less than that E, the number of units sold is less than that of E, i.e., less than 40, i.e., 30.

Profit per month of B is greater than the profit per month of one of the other outlets by ₹10,000 and profit per unit of A is greater than that of C.

C is the only outlet that can have least profit per unit, i.e., 5000 and the least profit per month.

∴ C got ₹5,000 per unit.

Given that the selling prices of A, D and E are different.

∴ Selling prices of B and C are same, i.e., ₹20,000.

B's profit per month cannot be 10,000 more than that of E because we cannot get 3,30,000 by multiplying any of the profits per unit given (6000, 7000, 8000, 9000, 10,000) by the number of units sold (40, 45, 50).

If the profit per month of B is more than ₹3,30,000, then there will be a minimum of two outlets with profit per month more than that of outlet E. This is not possible as it is a contradiction of the given data.

Profit per month of B must be less than 3,20,000.

∴ The outlet with profit per month greater than that of E cannot be B, D or C. Therefore, it must be A.

The profit per month of C can be:

(1) $5000 \times 40 \to$ ₹2,00,000

(2) $5000 \times 45 \to$ ₹2,25,000

(3) $5000 \times 50 \to$ ₹2,50,000

Let us take the profit per month of B as 10,000 more than that of C.

B's profit per month can be 2,10,000 or 2,35,000 or 2,60,000. Number of units sold by B = 40 or 45 or 50.

For any of the values of profit per unit, this is not possible.

Hence, the profit per month of B is not 10,000 more than that of C.

B's profit per month is not 10,000 more than that of A or C or E. It should be 10,000 more than that of D.

The possible profits per month of D are:

(1) $10,000 \times 30 = 3,00,000$

(2) $9,000 \times 30 = 2,70,000$

B's profit per month can be 3,10,000 or 2,80,000. Only 2,80,000 is possible, i.e., 40×7000.

The profits per unit of D and E are 9000 and 8000, respectively. The profit per unit of A must be 8000 or 9000. The unit that is selling at 22,000 cannot have the cost price of 11,000 or 12,000, i.e., it has 14,000 as the cost price. The outlet that is selling the unit at ₹21,000 cannot have 11,000 as the cost price.

∴ It should have 12,000 as cost price, i.e., a profit of 9000.

The outlet that is selling at 20,000 has a cost price of 11,000, i.e., a profit per unit of 9000.

The profit of different outlets is

Outlets	Selling price(₹) and cost price (₹)respectively	Profit(₹)/ Unit	Unit/ Month
A	20,000 and 11,000 or 21,000 and 12,000	9,000	45/50
B	20,000 and 13,000	7,000	40
C	20,000 and 15,000	5,000	45/50
D	21,000 and 12,000 or 20,000 and 11,000	9,000	30
E	22,000 and 14,000	8,000	40

36. The profit per month of B is ₹7000×40 = ₹2,80,000.

37. The selling price at the outlet E is ₹22,000.

38. The selling price of A is more than that of D.

∴ The selling price of D is ₹20,000 and the selling price of A is ₹21,000 and the total cost price is ₹12,000, i.e., the transportation cost is ₹2000.

39. The number of units sold by A is more than that by C, i.e., the number of units sold by A is 50 and that by C is 45.

∴ The profit per month of C is 45×5000 = ₹2,25,000.

40. Profit per unit of E is ₹8000.

EXERCISE-3

Solutions for questions 1 to 4: Each mill has a different downtime and each mill has at least four hours of uptime in a day. Thus, the maximum downtime for any mill is 5 hours so that it has an uptime of one hour for six hours.

From the given information, the difference (in hours) between the two different uptimes of each mill are tabulated as below:

Mill	Difference in hours between the known uptimes	Possible frequency of uptimes (Every nth hour)
P	20 hours	2nd, 4th, 5th
T	21 hours	3rd
Q	44 hours	2nd,4th
R	48 hours	2nd, 3rd, 4th, 6th
S	42 hours	2nd, 3rd, 6th
T	21 hours	3rd

Every mill has a different downtime, it implies that the possible downtimes are 1, 2, 3, 4 and 5 hours.

Thus, mill T runs every 3rd hour ⇒ downtime = 2 hours.

Only P can run every 5th hour ⇒ downtime = 4 hours.

Thus, the possible downtime for the mills are as follows:

Mill	Possible frequency of uptimes (Every nth hour)
P	5th
Q	2nd,4th
R	2nd, 4th, 6th
S	2nd, 6th
T	3rd

1. Either R or S can have the highest capacity based on the possibility that both can have the highest downtime.
2. The second highest downtime is four hours. Mill P has a downtime of 4 hours.
3. If R is not of the lowest downtime but lesser than that implies that R runs every 4th hour.
Thus, the downtime of mill R is 3 hours.
4. If R has greater downtime than mill S, then mill S has to run every 2nd hour. The only possibility is Q has to run every 4th hour and R has to run every 6th hour.
The frequencies of the mills P, Q, R, S and T will be once every 5th, 4th, 6th, 2nd and 3rd hours, respectively.
The same pattern repeats every 60 hours (LCM of 2,3,4,5 and 6). For this pattern to repeat in the first five hours of a shift, it repeats every 120 hours (LCM of 2, 3, 4, 5, 6 and 8) = Once every 5 days but only in the third shift. Hence, the pattern repeating in the first five hours of the day is not possible.

Solutions for questions 5 to 8: The given information can be tabulated as below:
[**Note:** The subscript under the circuit name (A, B and C) is to differentiate the two tours in the same circuit that start on different days of the week]

Day	Tour package	
	First day	Last day
Sunday	D	D, B
Monday	D	D
Tuesday	D	D, A_m, C_q
Wednesday	D, A_m, C_p	D, A_n
Thursday	D, A_n, B_x	D
Friday	D, B_y	D
Saturday	D, C_q	D, B_x, C_p

The subscript under the circuit name differentiates the two tours in the same circuit that start on different days.

5. Circuit A starts on two days; hence, two cases arise:

Case (a)		Case (b)
A (Wednesday to Tuesday) 7 days	A (Wednesday to Tuesday) 7 days	A (Thursday to Wednesday) 7 days
D (Wednesday) 1 day	C (Wednesday to Saturday) 4 days	B (Thursday to Saturday) 3 days
B (Thursday to Saturday) 3 days	D (Sunday) 1 day	D (Sunday) 1 day
Wait 3 days	Wait 3 days	Wait 2 days
C (Wednesday to Saturday) 4 days	B (Thursday to Saturday) 3 days	C (Wednesday to Saturday) 3 days
18 days	18 days	17 days

Alternate solution: Starting with circuit A, it is clear that after taking two tours, one has to definitely wait (Wait period is from Sunday to Tuesday). This wait can be minimized if the daily trip is planned in one of these three days. Further, the tour after the wait has to be A or C which starts on a Wednesday. Since, we are starting with A, the last tour has to be C (for minimum wait). Because A ends on Tuesday or Wednesday, D shall not be immediately after A, thus in the order A, B, D and C, with A starting on Thursday, the tour can be completed in 17 days.

6. If we observe the given information, out of all the tours that end and start on two consecutive days, all the tours are available between Tuesday and Thursday. C ends on Tuesday, A starts on Wednesday, which ends on Tuesday, D ends on Wednesday and B starts on Thursday. Hence, the tour shall start with C_q which starts on a Saturday.
7. Only on a Monday, tourists of only one tour come to the office (either start or end the tour).
8. A (Wednesday – Tuesday) + B (Thursday – Saturday) + D (Monday) + C (Wednesday – Saturday).
The tour shall start on a Wednesday.

Solutions for questions 9 to 12: The kid receives ₹5 per day ⇒ earns ₹35 per week but earns a maximum of ₹25 in the first five days. The statement implies that the kid spends a maximum of ₹25 in the first five days ⇒ purchases only two items in the first week.

The amount available by next Monday is the sum of savings during first five days and amount earned during weekend.

The kid earns ₹10 on weekend which is available for next Monday. Thus, from the second week onwards, he can purchase all the items at least once.

Amount available on second Monday will be among ₹15 or ₹20 or ₹10.

Whatever items the kid purchases in the first week, the total amount the kid can spend up to the second week is income of the first week and first five days of the second week, i.e., ₹35 + ₹25 = ₹60.

By repeating the purchase of three items twice, the kid purchases all the items twice.

Starting from the second week, in n weeks the kid purchases all the items at least n times and at most $(n + k)$ times.

In order for the purchases to make a pattern:

n (35) = $(n + \mathrm{k})$ (30)

$\Rightarrow n = 6k$.

$\Rightarrow$ For every 6 weeks, $(k = 1)$, all the items would be purchased seven times.

Pastry = P = ₹15

Samosa = S = ₹10

Chocolate = C = ₹5

Week	Balance + Income weekday	Expenses	Weekend income	Balance	Purchases
	A	B	C	A – B + C	
1	0 + 25	15 + 5 5 + 10	10	15 10 + 10 = 20	C, S
2	20 + 25	15 + 5 + 10 + 15	10	0 + 10 = 10	P, C, S, P
3	10 + 25	5 + 10 + 15 + 5	10	0 + 10 = 10	C, S, P, C
4	10 + 25	10 + 15 + 5	10	5 + 10 = 15	S, P, C
5	15 + 25	10 + 15 + 5 + 10	10	0 + 10 = 10	S, P, C, S
6	10 + 25	15 + 5 + 10	10	5 + 10 = 15	P, C, S
7	15 + 25	15 + 5 + 10	10	10 + 10 = 20	P, C, S
8	20 + 25	15 + 5 + 10 + 15	10	0 + 10	P, C, S, P

9. From the above, the given data is sufficient to determine that the pattern of the purchases is repetitive. Hence, only II is true.

10. From the table, the maximum savings available on any Monday is ₹20, plus he would receive a pocket money of ₹5, thus the maximum amount available on any Monday is ₹25.

11. Given that the kid makes a purchase as soon as he accumulates sufficient money to make his next purchase as per the defined conditions.
With the new condition, the possible combinations in the five days of the week, for the kid to purchase the four items Chocolate (C), Fruit (F), Samosa (S) and Pastry (P) is:
F/C, S/P, C/F, __, P/S (No purchase on Thursday).
Only (D) is definitely true.

12. In the third week, opening balance = ₹10. Earns ₹25 during the five weekdays. Spends ₹35 during five days and earns ₹10 during weekend.

The kid starts with a balance of ₹10 every week. The maximum amount the kid has on any Monday is ₹15.

Solutions for questions 13 to 16: Let us represent the people name with A, B, C, D and E and also the fields to which each book belongs to A, B, C, D and E.

Person	Book
A	D/E
B	Criminology
C	Biotechnology
D	A/E
E	D/A

13. Annie exchanges with Dan and Dan exchanges with Engel so the book possessed by Annie goes to Engel. Hence, Annie purchased book related to Economics.

Don Purchased book related to Architecture and Engel purchased book related to Demographic.

14. Statement (A) is true.
Statement (B) is true.
Statement (C) is true.
Statement (D) is not necessarily true.

15. Condition (A) Annie and Engel books were interchanged. There can be two possibilities with this condition with Annie taking books related to Demography or Architecture.
Condition (B) states that neither Annie nor Engel possess books related to Economics.
Since Annie and Engel did not get the book related to Economics that means Dan possessed the book related to Economics, Annie passed the book related to Architecture and Engel possessed the book related to Demographics.

16. As condition (d) is not violated, the interchange was between any two among Annie, Dan and Engel. From the above table, and the possibility of interchange between any two among the three, six cases arise as shown below. The statements which are consistent in each of the case are listed accordingly.

Statements consistent			
	Interchange between		
Person = Book	A and D	D and E	A and E
A = D E = A D = E	a, d	a, b, c	b
A = E E = D D = A	b, d	a	a, b, c

From the above table, in various cases, the pairs of statements ab, ac, ad, bc and bd are consistent simultaneously while only statements c and d are not consistent simultaneously in any case.

Solutions for questions 17 to 20:

17. A. $29 = 4 + 25 = (2 \times 2) + (2 \times 10) + (1 \times 5)$
Therefore, it is possible.
B. $31 = 6 + 25 = (3 \times 2) + (2 \times 10) + (1 \times 5)$
Therefore, it is possible.
C. $47 = 40 + 7$ or $35 + 12$
Neither of which are possible.

18. X score 33 $\Rightarrow 25 + 8 = (1\times 10) + (3 \times 5) + (4 \times 2)$ or $= (2 \times 10) + (1 \times 5) + (4 \times 2)$

All choices except (B) is not necessarily true.

19. Maximum marks can be obtained as follows:

	Part I	Part II	Total
Rahul/Beena	3×2	4×10	46
Beena/Rahul	3×2	$3 \times 10 + 1 \times 5$	41

Minimum possible marks:

	Part I	Part II	Total
Johan/Bijaya	5×2	2×5	20
Bijaya/Johan	4×2	3×5	23

Maximum difference = 87 – 43 = 44.

20. By scoring positive marks in six questions, each of U and V must score two marks each in two questions of Part I. In the remaining four questions, a score of 10 or 5 or 2 is possible. For different marks possible for any of the questions, the difference in marks would be in multiples of $\pm (10 - 5)$ or $\pm (5 - 2)$, i.e., difference of ± 5 or ± 3 will be seen in their totals. To obtain minimum difference of '1', we have to check if the difference of $1 = + 5 - 3 - 3$ or $- 5 + 3 + 3$ is possible.
The possible marks obtained by U and V in the remaining four questions are as follows.

	Case I	Case II	
U	$(10 + 5 + 5 + 5)$	$10 + 5 + 2 + 2$	15
V	$(10 + 10 + 2 + 2)$	$(5 + 5 + 5 + 5)$	18
Difference	1	1	

The difference is 1.

Solutions for questions 21 to 25: It is given that there are 24 textbooks distributed in four racks. Each rack containing a distinct even number of textbooks. From (2) and (3), it can be concluded that the number of books of Marketing = 12, HR = 6, Operations = 4 and Systems = 2. It is also given that the number of textbooks in each rack remained the same even after misplacement.

From (3), Marketing rack has four textbooks from Operations rack and the rest are from Marketing rack only.

From (4) in HR rack, two books are of HR and the rest four are from a different rack. These textbooks have to be from Marketing rack as Systems rack has only two textbooks.

This implies that the four books from HR rack are placed in Operations rack. Thus, the present position is as follows.

Marketing:	Marketing-8 and Operations-4
Operations:	HR-4
HR:	HR-2 and Marketing-4
Systems:	Systems-2

21. None of the books related to Operations was placed in Human Resource.

22. Two textbooks related to Systems were placed correctly.

23. Four of the books related to Human Resource were in Operations rack.

24. Textbooks from Systems rack are in system rack only.

25. No textbook belonging to Operations rack is placed in System rack.

Challenge Your Understanding
Practice Set 1

Directions for questions 1 to 3: These questions are based on the following information.

Raju starts going to temple 60 days before the commencement of his exams. Raju offers flowers to God on each day. God accepts these flowers on the next day and leaves two of them, but before Raju offers flowers on that day. Raju takes the two flowers left and then offers flowers for that day. The number of flowers taken by God on the third day is 12 and it is known that on any day, the difference between the number of flowers offered by Raju and that taken by God is constant (No flower is taken by God on the first day).

1. On which of the following days does Raju offer 320 flowers?

2. How many flowers does God take in the first 30 days?

3. How many flowers does Raju offer in the first 45 days?

Directions for questions 4 to 7: Answer the questions based on the information given below. Type your answer in the space provided below.

The following table gives the details about the number of people entering and leaving an exhibition ground at different times on a particular day in a 7-hour period.

	3 p.m.–4 p.m.	4 p.m.–5 p.m.	5 p.m.–6 p.m.	6 p.m.–7 p.m.	7 p.m.–8 p.m.	8 p.m.–9 p.m.	9 p.m.–10 p.m.
Number of people entering	300	346	562	648	713	–	–
Number of people leaving	200	380	459	520	629	862	900

Also note that:

(1) People were allowed to enter the ground at 1 minute past 'n' p.m., where $n = 3, 4, \ldots, 9$, for only 30 seconds, after which they were not allowed to enter.

(2) People were allowed to leave the ground at 1 minute to 'm' p.m., where $m = 4, 5, \ldots, 10$. for only 30 seconds, after which they were not allowed to leave.

(3) The entry was allowed from exactly 3 p.m. and nobody was present in the ground before 3 p.m.

(4) The venue was closed at exactly 10 p.m. and everybody left the ground before 10 p.m.

(5) The number of people entering the ground in each of the time durations from 8 p.m.–9 p.m. and 9 p.m.–10 p.m. is unknown.

4. At any moment from 3 p.m. to 8 p.m., what was the highest number of people present in the ground?______

5. What is the minimum number of people who entered the ground from 8 p.m. to 9 p.m.? _______

6. What is the least number of people who were in the ground at any moment from 4 p.m. to 8 p.m.? ________

7. What is the minimum number of people entering the ground at 8 p.m.? _______

Directions for questions 8 to 10: These questions are based on the following information given below. Type your answer in the space provided below for each question.

A cube is dipped in a tank of red paint up to half its height. Then, it is reversed and then dipped in green paint up to half its height. Then the top and the bottom faces of the cube are coloured with blue and yellow, respectively. This cube is now cut into 27 smaller and identical cubes.

8. How many small cubes does not have red paint on any of its faces?

9. How many small cubes have two colours on them?

10. How many small cubes have three colours on them?

Directions for questions 11 to 13: These questions are based on the following information.

A survey was conducted among 100 candidates, each of whom passed in at least one paper among Physics, Chemistry and Maths. Among them, 45 failed in Physics, 40 failed in Chemistry, 40 failed in Maths and 40 failed in at least two subjects.

11. How many candidates passed in at most two subjects?

12. How many candidates passed in exactly two subjects?

13. How many candidates passed in all the three subjects?

Directions for questions 14 to 16: These questions are based on the information given below.

The following table gives the details of the number of new books and the number of 2nd editions of the books which is already in circulation both published by ABC publishers.

Year	2000	2001	2002	2003	2004	2005
New books published	140	180	210	220	260	300
2nd edition books published	90	90	120	145	190	160

Total books published = Fiction books published + Non-fiction books published.

For non-fiction books that are published in a year, 2nd editions will be published for exactly 50% of the books in the next year.

For fiction books published in a year, 2nd edition will be published for all the publications exactly two years later. Further in 2005, the 2nd edition of fiction books published is the same as the 2nd edition of non-fiction books published.

Directions: Type in your answer in the space provided in the question.

14. How many new fiction books were published in 1999? ________

15. How many new non-fiction books were published in 2004? ________

16. For how many non-fiction books published between 2002 and 2004 (both the years included), 2nd edition was not published? ________

Directions for questions 17 to 20: These questions are based on the information given below.

A group of four friends, namely P, Q, R, and S together spent a total amount of ₹9600. Each of them spent some money on shoes, some on clothes and the remaining on books. In total, P spent ₹200 more than Q but ₹200 less than R, who in turn spent ₹200 less than S. The total money spent on clothes is ₹200 less than that on books and ₹200 more than that on shoes. Further, it is also known that:

(i) Q spent ₹400 on clothes and twice of it on shoes.

(ii) The amount spent by P on clothes is the same as that spent by R on shoes.

(iii) The amount spent by S on books is the sum of the amount spent by P on books and the amount spent by S on shoes.

(iv) The amount spent by Q on shoes and that on books is same as that spent by S on clothes and R on clothes, respectively.

Directions: Type in your answer in the space provided in the question.

17. Among the four, what is the least amount spent in total? ________

18. How much did S spend on books? ________

19. How much did P spend on books? ________

20. How much did Q spend on shoes? ________

Directions for questions 21 to 24: Type in your answer in the space provided below each of the question. These questions are based on the following information.

A print media advertising agency has undertaken a readership survey in a habitat of less than 700 houses for formulating a tariff package for print advertisements in vernacular newspapers in that region. The following information is shared:

It is found that each house in the village is a subscriber to at least one of the dailies among JanSamachar, LokVichaar and Veekshan. No other newspaper is under circulation in that village.

The number of subscribers to LokVichaar alone is twice the subscriptions to only LokVichaar and JanSamachar, as well as only LokVichaar and Veekshan.

Subscriptions to only two dailies, one of which is JanSamachar is same as the subscriptions to all the three dailies which are found to be more than 110.

For every subscriber of only JanSamachar and LokVichaar, there are three subscribers of only Veekshan. Only 20% of the houses subscribed to all the three dailies, which is not a multiple of 20.

Subscription to only JanSamachar and LokVichaar is $41^2/_3$% of number of subscriptions to all the three.

21. What is number of subscriptions to the highest circulated newspaper?

22. What is the number of subscriptions to the least circulated newspaper?

23. What is the total number of houses in the village?

24. What is the total number of newspaper subscriptions in the village?

Directions for questions 25 to 27: These questions are based on the following information.

In a group of 50, every member can speak at least one of the four languages, such as English, Hindi, German and Spanish. The number of people who can speak all the languages except Spanish is same as those who speak only Spanish, which in turn is equal to 4. 25 members do not speak German and 7 members speak only German. Number of people who speak exactly two languages is twice that of those who speak exactly three, which in turn is twice that of those who speak all the languages. 15 people speak only one language. 18 people speak English and Hindi, of which 6 people speak no other language. German and Hindi are spoken by 11 people, of which only 1 person does not speak any other language. The number of people who speak Spanish and English is 13, of which the number of people who do not speak any other language is same as those who speak only German and Spanish. The number of people who speak English is 26.

Directions: Type in your answer in the space provided below the question.

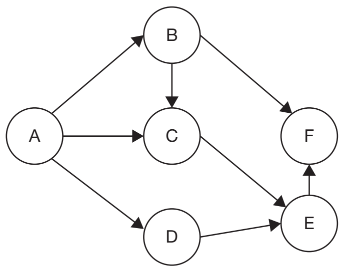
25. How many people speak only Spanish and Hindi? ______

26. How many persons speak only Hindi? ________

27. How many people speak at least three languages? _______

***Directions for questions 28 to 31*:** These questions are based on the following information.

A, B, C, D, E and F are close friends. A has a chocolate factory and he distributes some chocolates to his friends and his friends distribute the chocolates among themselves as shown below.



The following information is also known.

(i) No person receives the same number of chocolates from two different people.
(ii) After the distribution, each person has a different number of chocolates, which are consecutive natural numbers.
(iii) No person gives the same number of chocolates to two different people.
(iv) The number of chocolates with C after the distribution is four and that with A is the least.
(v) A gives six chocolates to D and C gives one chocolate to E.
(vi) The number of chocolates with F is twice that with E.
(vii) No person gives more than 10 chocolates to any other person.
(viii) At least one chocolate is passed between two people who are connected in the network.

Directions: Type in your answer in the space provided below the question.

28. What is the total number of chocolates with A before distribution?________

29. The number of chocolates with D after distribution is ________

30. The number of chocolates given by A to C is ________

31. The total number of chocolates with A, B and D after distribution is ________

Directions for questions 32 to 35: These questions are based on the following information.

A, B, C, D, E and F are six friends appearing for a Management entrance test. They applied together and were surprised to see that though they were all seated in the same row, their hall ticket numbers were not in serial order. Further, the following information was available.

(i) Neither A nor F is seated at the ends and the people sitting at any ends do not have hall ticket numbers whose last digit is 2.
(ii) A and F are adjacent to each other, while B and E are adjacent to each other and are to the left of A. The remaining two friends are also adjacent to each other. Also, A and F have hall ticket numbers whose first digit is 4.
(iii) Each hall ticket number is of two digits which is a multiple of 6, such that there are three pairs of consecutive multiples of 6. The sum of each pair is a multiple of 10.
(iv) They are seated in such a way that the hall ticket numbers that end with the same digit are in ascending order from left to right.
(v) E is not at any end and the same is true of D, who is not adjacent to A.

32. In how many ways can the six friends be seated?

33. If A's hall ticket number ends with the digit '2', then what is the hall ticket number of F?

34. The lowest of the hall ticket numbers of the six friends is

35. What is the hall ticket number of the person sitting to the immediate right of F?

Directions for questions 36 to 39: These questions are based on the following information.

Six MBA aspirants P, Q, R, S, T and U arrived in Hyderabad to participate in a group discussion (GD) for the 20 IIMs spread across India. The number of calls received by each of them is a multiple of either 3, 4 or 5. They are to be seated around a circular table during the GD. Further, the following information is known about them.

(i) No two of them received the same number of calls. The lowest number of calls that a candidate among them received is 5, while none of them received a call from all the 20 IIMs.
(ii) The candidates are seated in such a manner that the number of calls received by each person increases as we move in anticlockwise direction starting from the candidate who received the lowest number of calls.
(iii) P and S are opposite to each other and together they got a call from each of the 20 IIMs. But no IIM sent calls to both of them.
(iv) The person seated to the immediate left of P is Q, who received the lowest number of calls.
(v) R received 9 calls and is opposite to T.
(vi) The highest number of calls received by a person is an odd number.

36. What is the number of calls received by U?

37. What is the difference between the highest and the lowest number of calls received in the group?

38. What is the number of calls received by the person who sits two places away to the right of T?

39. The ratio of number of aspirants with even number of calls to that of odd number of calls is (Give ratio as decimal number)

Directions for questions 40 to 43: These questions are based on the following information.

A team of agricultural scientists is trying to figure out the banana production planning for the forthcoming year. A report on the same has provided the following information.

Andhra Pradesh, Tamil Nadu, Karnataka, Maharashtra and Gujarat are the five major producers of bananas in India. Each state has a production target (in tons) from among 1000, 2000, 3000, 4000, 5000, 7000 and 9000. Each state has a target area for cultivation, which is one among 500, 1000, 1500, 2000, 3000 and 3500 (in hectares). Further, the team gathered the following information.

(i) Every state has a different targeted yield (in tons per hectares) with Gujarat having the highest target (yield) of 2.5, while Karnataka has the least yield target of 0.86 (Approximately).

(ii) Andhra Pradesh has the lowest production and area targets among the five states with 2000 tons and 1500 hectares, respectively. Although its targeted yield is greater than that of Maharashtra but less than that of Tamil Nadu.

(iii) Tamil Nadu has a yield target of 2 and the current area needs to increase production target by 1000 tons to equate Gujarat's target yield.

(iv) Karnataka has the same production target as Maharashtra but plans to utilize 500 hectares of more land than that of Maharashtra.

(v) The states are ranked 1 to 5 in each of the parameters production target, area target and yield target, with the best rank being 1 and the last rank being 5. If two or more states have same value in production target or area target, the state with higher yield target is given better rank.

Yield target = Production target in tons/ Area target in hectares

Directions: Write the answer in the below each question.

40. What is Gujarat's rank in terms of area target? ________

41. What is the rank of Karnataka in terms of production target? ________

42. If Karnataka targets to move up three ranks and tie with the state currently holding that position in terms of yield, how much more should it produce (in tons) with the current targeted area? ________

43. If Tamil Nadu's actual yield is likely to fall to 50% of the targeted value, then how much area (in hectares) has to be increased or decreased to keep the production at the targeted level? ________

Directions for questions 44 to 47: These questions are based on the following information.

Santiago, Benjamin and Mateo are three friends from Brazil visiting India. Each of them purchased a different item, such as a shawl, Darjeeling tea and a saree from the Rajasthan State Tourism Development Handicraft outlet with each item bearing a different cost. Additionally, they had to pay a tax of 10% on the cost of the item purchased. While exiting the outlet they were required to show the bills of the purchased items, which could not be found.

When asked by the outlet security about the purchases each of them made three statements.

It is known that each of them belongs to a different group among truth teller, alternator and liar.

Truth tellers always speak the truth, liars always lie and alternators alternate between truth and lies in any order.

Santiago:

(i) Benjamin bought a saree.
(ii) Mateo paid a tax of ₹300.
(iii) I bought an item which is worth ₹2000.

Benjamin:

(i) Santiago purchased a shawl.
(ii) Mateo paid a tax of ₹300.
(iii) I bought an item for a price of ₹2000.

Mateo:

(i) I purchased Darjeeling tea.
(ii) Benjamin paid a tax of ₹100.
(iii) Santiago purchased an item which is priced for ₹2000.

Directions: Write your answer in the space provided below each question.

44. What is the total price paid for Darjeeling tea? ________

45. The cost of the saree is how many times the cost of the product purchased by Benjamin? ________

46. If the sum of the cost of product Benjamin purchased and the shawl is equal to the cost of the of product purchased by Mateo, then what is the total money spent by Santiago for the purchase of his product (tax included)? ________

47. What is the total amount spent by all the three at the outlet, if the conditions in the previous question remains the same? ________

Directions for questions 48 to 51: These questions are based on the following data.

A survey was conducted in a community of 350 people regarding three games, such as Chess, Carroms and Chinese Checkers. The following information is obtained in the survey.

(i) Thrice the number of people who play all the three games is equal to the number of people who play Chinese Checkers.

(ii) The number of people who play Chinese Checkers and Carroms is equal to the number of people who play Chess only.

(iii) For every three people who play Chess and Chinese Checkers only, there are five people who play none of the three games.

(iv) In every seven people who play Chinese Checkers, four people play Carroms also.

(v) For every six people who play only Carroms, there is one who plays Chinese Checkers only.

(vi) For every four people who play exactly two games, there is one who plays Carroms and Chinese Checkers only and two people who play none of the three games.

Directions: Fill the space below the questions with appropriate value.

48. How many people play exactly two games?__________

49. How many people play Chess but not Carroms? __________

50. How many people do not play Chinese Checkers?_______

51. How many people play Chess or Carroms? __________

Directions for questions 52 to 55: These questions are based on the following information.

In a colony of 280 families, which use mobile phones of different companies, like Panasonic, Sony Ericsson, Motorola and Nokia, 175 families use Sony Ericsson, 155 families use Panasonic, 165 families use Motorola and 150 families use Nokia. Each of the families use mobile phones of at least one company. The number of families using Sony Ericsson and Motorola is same as those using Nokia and Panasonic which in turn is same as those using mobiles of exactly three different companies, which is 75. Also, it is known that the sum of the number of families using Sony Ericsson and Motorola only, and the number of families using Nokia and Panasonic only is 25. The number of families using mobiles of exactly two companies is 100 more than that using mobiles of exactly one company.

Directions: Fill the space below each question with appropriate value.

52. How many families use mobiles of all the four companies?________

53. How many families use mobiles of at least two of the four companies but at most three of the four companies? ________

54. How many families use mobiles of exactly one of the four companies? ________

55. If the sum of the number of families using Sony Ericsson, Motorola and Nokia but not Panasonic and the number of families using Sony Ericsson, Motorola and Panasonic but not Nokia is 35, then what is the sum of the number of families who use Nokia only and the number of families using Panasonic only? ________

Directions for questions 56 to 60: These questions are based on the following information.

In a college of 500 students, each student belongs to either the first year or the second year only but not to both. Each student belongs to exactly one of the two streams, such as Commerce and Science, each student is either an NSF member or an SFI member but not both. 90 first year Commerce students are SFI members. There are 270 NSF members. 50 Commerce students are neither first year students nor NSF members. 140 first year students are NSF members. 150 NSF members are either first year students or Commerce students but not both. 120 second year students are not Science students. 100 Science students are either first year students or NSF members but not both.

Directions: Fill the space below each question with appropriate value.

56. Find the number of second year Science students who are SFI members._________

57. Find the number of first year Science students who are not NSF members. _________

58. Find the number of first year Commerce students who are NSF members. _________

59. Find the number of Commerce students who are either NSF members or second year students but not both. _________

60. Find the total number of students who are NSF members or second year students. _________

ANSWER KEYS

1. 53	**10.** 0	**19.** 700	**28.** 21	**37.** 10	**46.** 1100	**55.** 5
2. 2610	**11.** 85	**20.** 800	**29.** 2	**38.** 8	**47.** 6600	**56.** 50
3. 6300	**12.** 45	**21.** 429	**30.** 4	**39.** 1	**48.** 100	**57.** 40
4. 1010	**13.** 15	**22.** 330	**31.** 8	**40.** 3	**49.** 90	**58.** 60
5. 481	**14.** 60	**23.** 660	**32.** 1	**41.** 4	**50.** 245	**59.** 110
6. 65	**15.** 160	**24.** 1111	**33.** 48	**42.** 4000	**51.** 285	**60.** 370
7. 481	**16.** 195	**25.** 5	**34.** 12	**43.** 2000	**52.** 25	
8. 11	**17.** 2100	**26.** 3	**35.** 72	**44.** 2200	**53.** 215	
9. 24	**18.** 1300	**27.** 15	**36.** 10	**45.** 1.5	**54.** 40	

SOLUTIONS

Solutions for questions 1 to 3: Let x be the flowers offered by Raju on the first day. And it is given that God takes none of them on the first day.

$\therefore$ The difference between the number of flowers offered and that taken $= x - 0 = x$.

The flowers offered is x more than the flowers taken.

On the second day, God takes all the flowers that were offered on the first day except two, i.e., $x - 2$.

$\therefore$ The number of flowers offered by Raju on the second day $\Rightarrow x - 2 + x = 2x - 2$

On the third day God takes $(2x - 2) - 2 = 2x - 4$ flowers.

But given is 12.

$\therefore 2x - 4 = 12 \Rightarrow x = 8$

The flowers offered and taken is as follows:

Day →	1	2	3	4
Offered	8	14	20	26
Taken	0	6	12	18

The flowers offered is in arithmetic progression with a common difference of 6 and also the number of flowers taken is in arithmetic progression with a common difference of 6.

Flowers offered $= 2 + 6n$

Flowers taken $= 6\ (n - 1)$

When $n =$ the number of the day.

1. $2 + 6n = 320$

$\Rightarrow 6n = 318 \Rightarrow n = 53$

2. The sum of first n terms in an arithmetic progression $= \frac{n}{2}[2a + (n-1)d]$

Here, $n = 30$, $a = 0$ and $d = 6$

$\Rightarrow \text{Sum} = \frac{30}{2}[(30 - 1)6]$

$= 15\ [29 \times 6] = 2610$

3. The sum of the first n terms in an arithmetic progression $= \frac{n}{2}[2a + (n-1)d]$

Here, $n = 45$, $a = 8$, $d = 6 \Rightarrow \text{Sum} = \frac{45}{2}[16 + 44 \times 6]$

$= 45\ [8 + 132] = 6300.$

Solutions for questions 4 to 7: People enter the ground from n : 01 : 00 p.m. to n : 01 : 30 p.m., where $n = 3, 4 \ldots, 9$, and leave the ground from n : 59 : 00 p.m. to n : 59 : 30 p.m., where $n = 3, 4 \ldots, 9$. From n : 01 : 31 p.m. to n : 59 : 00 p.m., where $n = 3, 4, \ldots, 9$, the number of people in the mall remain constant.

	The number of people entering from n : 01:00 p.m. to n : 01:30 p.m.	The number of people leaving from n : 59:00 p.m. to n : 59:30 p.m.	The number of people present from n : 01:30 p.m. to n : 59:00 p.m.
$n = 3$	300	200	300
$n = 4$	346	380	(300 – 200) + 346 = 446
$n = 5$	562	459	(446 – 380) + 562 = 628
$n = 6$	648	520	(628 – 459) + 648 = 817
$n = 7$	713	629	(817 – 520) + 713 = 1010

4. From the above table, the highest number of people in the ground at any time would be 1010.

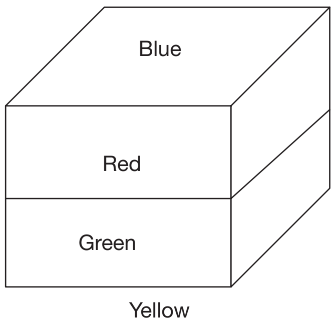
5. The number of people who were present in the ground after 7 : 59 : 30 = 1010 – 629 = 381

As 862 people left between 8 p.m. and 9 p.m., at least 862 – 381 = 481 people must enter the ground from 8 p.m. to 9 p.m.

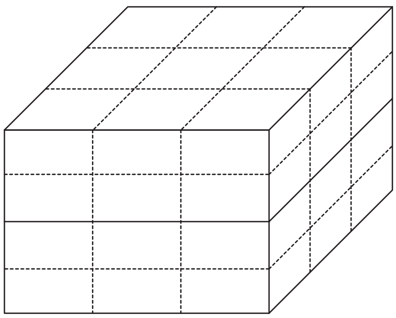
6. The number of people present in the ground from 3 : 59 : 30 to 4 : 01 : 00 = 300 – 200 = 100
The number of people present in the ground from 4 : 59 : 30 to 5 : 01 : 00 = 446 – 380 = 66.
Proceeding similarly, we find that the minimum number of people present at any moment in the ground from 4 p.m. to 8 p.m. was 66.

7. At 8 p.m., 381 people remain in the ground before the entry and we know that at 9 p.m. 862 people left the ground
∴ Minimum people entering the ground would be 862 – 381 = 481.

Solutions for questions 8 to 10: After painting, the cube would appear as given below.



After cutting the cube into 27 smaller and identical cubes, we get:

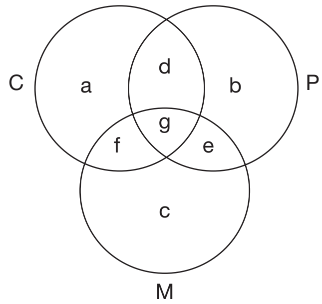


8. The cubes on the bottom plane, the one at the middle of the top face and the middle one does not have red colour on any of its faces. So, the number of cubes which do not have red paint on them is $3 \times 3 + 1 + 1 = 11$.

9. All the cubes on the outer surface except the centre cubes on the top and the bottom faces, i.e., 24 cubes.

10. None of the cubes have three different colours on them.

Solutions for questions 11 to 13:



Here P, C and M represent those who passed in Physics, Chemistry and Mathematics, respectively.

Given information is as follows:

$a + f + c = 45$ (1)

$b + e + c = 40$ (2)

$a + d + b = 40$ (3)

$a + b + c = 40$ (4)

$(a + b + c) + (d + e + f) + g = 100$ (5)

Adding (1), (2) and (3), we get

$2(a + b + c) + (d + e + f) = 125$

$d + e + f = 125 - 2 \times 40 = 45$ [From (4)]

From (5), $g = 100 - 45 - 40 = 15$

11. Passed in at most two subjects
= Exactly two subjects + Exactly one subject
= 45 + 40 = 85

12. Passed in exactly two subjects = 45

13. Passed in exactly three subjects = 15

Solutions for questions 14 to 16: It is given that the number of 2nd edition of fiction books is same as that of non-fiction books in 2005.

⇒ 80 books are fiction and 80 books are non-fiction.

⇒ 160 of the newly published books in 2004 are non-fiction and 100 are fiction.

⇒ In 2003, 140 are non-fiction and 80 are fiction.

⇒ In 2004, of the 2nd edition, 70 are non-fiction and 120 are fiction. Similarly, we can work out other values.

New publications

	2000	2001	2002	2003	2004	2005
Non-fiction	60	80	90	140	160	Y
Fiction	80	100	120	80	100	$300 - y$

2nd edition

	2000	2001	2002	2003	2004	2005
Non-fiction	x	30	40	45	70	80
Fiction	$90 - x$	60	80	100	120	80

14. As 60 fiction books had a 2nd edition in 2001, all these 60 were published in 1999.

15. The new non-fiction books published in 2004 is 160.

16. Of the total non-fiction books published, i.e., (90 + 140 + 160) = 390
195 have 2nd edition, but the remaining 195 do not have 2nd edition.

Solutions for questions 17 to 20: Assume that Q spent ₹x.

P spent ₹200 more than Q.
⇒ P spent $(x + 200)$
R spent ₹200 more than P.
⇒ R spent $(x + 200 + 200) = (x + 400)$
S spent ₹200 more than R.
⇒ S spent $(x + 400 + 200) = (x + 600)$
We have:
P + Q + R + S = 9600
⇒ $x + 200 + x + x + 400 + x + 600 = 9600$
or $x = 2100$
∴ P spent ₹2300; Q spent ₹2100;
R spent ₹2500 and S spent ₹2700.
Now, assume that amount spent on shoes = x.
Amount spent on clothes = $(x + 200)$
Amount spent on books = $(x + 400)$
⇒ $x + x + 200 + x + 400 = 9600$
or $x = 3000$
From (i) and the above data, we get

	Shoes	Clothes	Books	Total
P				2300
Q	800	400		2100
R				2500
S				2700
Total	3000	3200	3400	9600

Q spent 2100 – (800 + 400) = 900 on books.
From (iv), we get:
S spent ₹800 on clothes and
R spent ₹900 on clothes.
∴ P's expenditure on clothes = $\underset{\text{Total}}{3200} - (\underset{Q}{400} + \underset{S}{800} + \underset{R}{900})$
= ₹1100
Also, from (ii), the table will be

	Shoes	Clothes	Books	Total
P		1100		2300
Q	800	400	900	2100
R	1100	900		2500
S		800		2700
Total	3000	3200	3400	9600

Amount spent by R on books = 2500 – (1100 + 900) = 500.
Let the amount spent by S on books = x
Amount spent by P on books = a
Amount spent by S on shoes = b
From (iii), we get
$n = a + b$
The total of books purchased by P, Q, R and S is 3400.
Hence, $a + 900 + 500 + x = 3400$
$a + 900 + 500 + x = 3400$
or $a + x = 2000$ (1)
From row S,
$b + 800 + x = 2700$
or $b + x = 1900$ (2)
(1) + (2)
⇒ $a + b + 2x = 3900$
We have: $a + b = x$
⇒ $3x = 390$
(or) $x = 1300$
$a + x = 2000 \Rightarrow a = 700$
$b + x = 1900 \Rightarrow b = 600$
Hence the final table will be

	Shoes	Clothes	Books	Total
P	500	1100	700	2300
Q	800	400	900	2100
R	1100	900	500	2500
S	600	800	1300	2700
Total	3000	3200	3400	9600

17. Q spent ₹2100 only.

18. S spent ₹1300 on books.

19. P spent ₹700 on books.

20. Q spent ₹800 on shoes.

Solutions for question 21 to 24: Given information:

Total number of houses is less than <700.

From (ii), (Only LokVichaar) = 2 × (Only LokVichaar and JanSamachar)
= 2 × (Only LokVichaar and Veekshan).

From (iii), only JanSamachar and (LokVichaar or Veekshan) = (JanSamachar and LokVichaar and Veekshan)

i.e., (Only JanSamachar and Veekshan) + (Only JanSamachar and LokVichaar) = (JanSamachar and LokVichaar and Veekshan)

From (iv), (Only Veekshan) = 3 × (Only JanSamachar and LokVichaar).

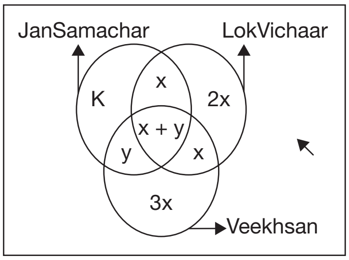
20% (Total number of houses) = Subscriptions to all the three dailies ≠ Any multiple of 20.

⇒ (Total number of houses) = 5 × (Subscriptions to all the three dailies)

(Only JanSamachar and LokVichaar) = $41^2/_3$% of number of subscriptions to all the three.

If we calculate by letting (Only JanSamachar and LokVichaar) = x.

The given information can be represented in the following Venn diagram:



From the above information, we can deduce the following:

$41^2/_3\%(x+y) = x \Rightarrow \frac{125}{300}(x+y) = x \Rightarrow 5y = 7x \Rightarrow y = \frac{7x}{5}$

$\Rightarrow$ Subscriptions to all the three dailies $= \left(x + \frac{7x}{5}\right) = \frac{12x}{5}$

Total number of houses $= 5 \times (x+y) = 12x$

$\Rightarrow k + x + 2x + \frac{7x}{5} + \frac{12x}{5} + x + 3x = 12x \Rightarrow k = 5x - \frac{19x}{5}$

$\Rightarrow k = \frac{6x}{5}$

Also, we know that $12x < 700$ and $\frac{12x}{5} > 110 \Rightarrow 550 < 12x < 700 \Rightarrow 48.5 < x < 58.3$

But each of x, y and k are integers and $k = \frac{6x}{5} \Rightarrow x$ is a multiple of 5.

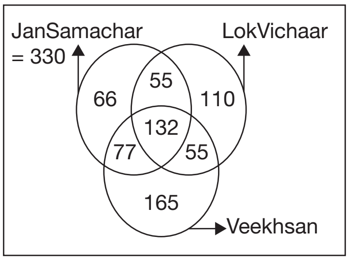
Thus, the possible values for x are $x = 50$ or $x = 55$.

If x =50, then $\frac{12x}{5} = 120$.

This is not possible because subscriptions to all the three dailies is not a multiple of 20.

Therefore, $x = 55; \Rightarrow k = 66$ and $y = 77$

The resultant Venn diagram would be as follows:



21. Veekshan is having the highest number of subscriptions = 429.

22. JanSamachar is having the lowest number of subscriptions = 330.

23. Total houses in village = $132 \times 5 = 660$.

24. Total subscriptions = 330 + 352 + 429 = 1111.

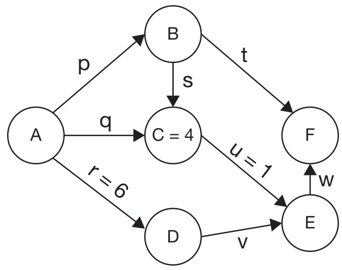
Solutions for questions 25 to 27:

25. 5 people speak only Spanish and Hindi.

26. 3 people speak only Hindi.

27. The number of people who speak at least three languages are $1 + 2 + 3 + 4 + 5 = 15$.

Solutions for questions 28 to 31: The network can be represented as follows.



Given that, $r = 6$, $u = 1$, C = 4 and F = 2E.

As C = 4 and the number of chocolates with them are consecutive natural numbers, the maximum number of chocolates with any person can be 8 (As A is the least).

As F = 2E, (E, F) can be (1, 2), (2, 4), (3, 6) or (4, 8).

E cannot be 1 as A has the least number of chocolates or E and F cannot have 4 chocolates (as C = 4).

$\therefore$ E = 3 and F = 6

As the number of chocolates are consecutive natural numbers, there should be five chocolates with A, B or D (E = 3, C = 4 and F = 6).

As A has the least number, he cannot have 5 chocolates.

If D = 5, r = D + v

$\Rightarrow v = 1$

Now $u = v = 1$, which does not satisfy the condition that a person should not receive the same number of chocolates from two different people.

$\therefore$ B = 5

Now D should be either 2 or 7 to form consecutive numbers. As $r = 6$, D cannot be 7.

$\therefore$ D = 2 and A = 1 (the least)

$v = r - \text{D} = 6 - 2 = 4$

$w = v + \text{E} - w = 4 + 1 - 3 = 2$

$t = \text{F} - w = 6 - 2 = 4$

Now $p = \text{B} + s + t = 5 + s + 4$

$\Rightarrow p = 9 + s$ and s should be at least 1 and p must be at most 10.

$\therefore s = 1$ and $p = 10$

$q + s = \text{C} = 4 \Rightarrow q = 4 + 1 - 1 = 4$

Solutions for questions 28 to 31: Type in your answer in the space provided below the question.

28. The number of chocolates with A before distribution is equal to the total number of chocolates with A, B, C, D, E and F after distribution = $1 + 2 + 3 + 4 + 5 + 6 = 21$.______

29. D = 2 ________

30. $q = 4$ ________

31. A + B + D = 1 + 5 + 2 = 8 ________

Solutions for questions 32 to 35: From condition (1), A and F are not seated at the ends, also the people sitting at the ends do not have hall tickets whose number ends with 2.

From condition (2), since neither A nor F is at the ends they form the middle pair.

B/E	E/B	A/F	F/A	C/D	D/C

Since the last two digits of the hall ticket numbers are multiples of 6: 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, 78, 84, 90, 96.

Using condition (4), we get

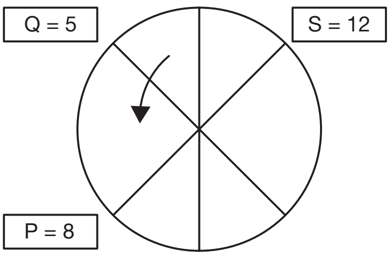
B/E	E/B	A/F	F/A	C/D	D/C
18		42/48	48/42		78

Using condition (5), we get

B	E	A	F	D	C
18	12	42/48	48/42	72	78

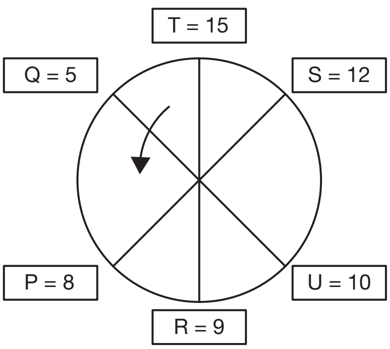
Solutions for questions 36 to 39: From (4), Q is the individual who has received 5 calls (minimum).

Therefore, using 1, 2, and 3 we have:



Calls are any number from of: 3, 6, 9, 12, 15, 18, 4, 8, 12, 16, 5, 10, 15.

With condition (3), P = 8 and S = 12 is the only possibility. Using condition (5), we get



Between 9 and 12 we have only 10, so U = 10. T = 15, 16 or 18 but since it has to be odd, only 15 is possible.

36. U received ten calls.

37. The difference is 15 – 5= 10.

38. P is second to the left of 8, who received eight calls.

39. The ratio is 3 : 3 = 1

Solutions for questions 40 to 43: From (2), we have

AP $\rightarrow$ Production target is 2000 tons (least) and area target is 1500 hectares (least).

Hence, from (1), yield target 2.5 for Gujarat's possible with yield target of 5000 tons and area target of 2000 hectares. Similarly, for Karnataka 2.86 = 3000/3500.

From (2), for Andhra Pradesh yield target
= 2000/1500 = 1.33

Tamil Nadu > 1.33 > Maharashtra

From (3), for Tamil Nadu yield target 2 is possible, under the given condition, only for a production target of 4000 tons and area target of 2000 hectares.

From (4), it can be said that for Maharashtra, the production target is 3000 tons, area target is 3000 hectares. Hence, yield target is 1.

The final table is as follows:

State	Yield	Production target	Area target
Gujarat	2.5 (1)	5000 (1)	2000 (3)
Karnataka	0.86 (5)	3000 (4)	3500 (1)
Andhra Pradesh	1.33 (3)	2000 (5)	1500 (5)
Tamil Nadu	2 (2)	4000 (2)	2000 (4)
Maharashtra	1 (4)	3000 (3)	3000 (2)

40. The rank of Gujarat in terms of area target is 3.

41. The rank of Karnataka in terms of production target is 4.

42. Kanataka's 2 × 3500 = 7000
New target = 7000
So, excess target = 7000 – 3000
= 4000

43. New yield = 1, so production is 2000.
$\Rightarrow$ 2000 hectares more required to maintain production target.

Solutions for questions 44 to 47: Let's assume Santiago is TT then all his statements would be true.

	S_1	S_2	S_3
Santiago	T	T	T

	Santiago	Benjamin	Mateo
Product	DT	Saree	Shawl
Cost	2000		3000
Tax @10%	200		300

So, now Benjamin's statement should be analysed (DT stands for Darjeeling tea) using the above table.

	S_1	S_2	S_3
Santiago	T	T	T
Benjamin	?	T	F

Since, there is one true and false statement Benjamin is an alternator.

$\therefore$ S_1 is false $\Rightarrow$ Mateo purchased Shawl.

Now, the above information is compared with Mateo's statement.

		S_1	S_2	S_3
Truth teller	Santiago	T	T	T
Alternator	Benjamin	F	T	F
Liar	Mateo	F		T

Here, S_3 is true. But Mateo must be a liar. We have come across a contradiction. Hence, Santiago cannot be the truth teller.

Let Benjamin be truth teller (TT).

	S_1	S_2	S_3
Benjamin	T	T	T

	Santiago	Benjamin	Mateo
Product	Shawl		
Cost		2000	3000
Tax @10%		200	300

With the above table let us analyse Santiago's statement.

	S_1	S_2	S_3
Benjamin	T	T	T
Santiago	?	T	F

So, Santiago S_1 = F

Therefore, Benjamin bought Darjeeling tea.

	Santiago	Benjamin	Mateo
Product	Shawl	DT	Saree
Cost		2000	3000
Tax @10%		200	300

Now, Mateo's statement can be analysed using the above table.

	S_1	S_2	S_3
Benjamin	T	T	T
Santiago	F	T	F
Mateo	F	F	F

So, Benjamin = Truth teller

Santiago = Alternator

Mateo = Liar

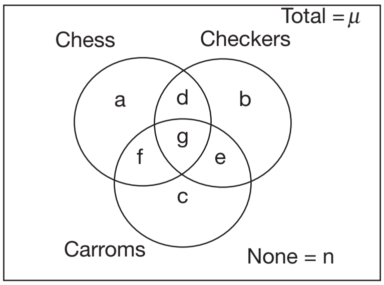
44. ₹2200 is paid for Darjeeling tea.

45. 1.5 times.

46. ₹1100 is paid by Santiago.

47. In all, ₹6600 is spent.

Solutions for questions 48 to 51:



Let us represent the given information in the form of a Venn diagram.

Let the number of people who play Checkers = $21x$

From (iv), $b + d + g + e = 21x$ and $g + e = 12x \Rightarrow b + d = 9x$

From (ii), $g + e = a = 12x$

From (i), $3g = 21x$ (or) $g = 7x \Rightarrow e = 5x$

From (vi), $e : n = 1 : 2$

$\therefore\ n = 10x$

From (iii), $d : n = 3 : 5$

$\therefore\ d = 6x \Rightarrow b = 3x$

From (vi), $d + e + f : e = 4 : 1$

$\therefore\ f = 9x$

From (v), $c : b = 6 : 1$

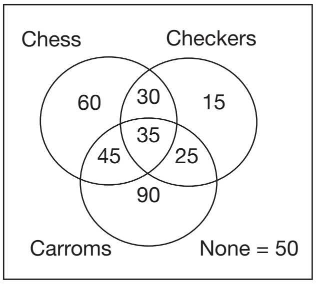
$\therefore\ c = 18x$

$a + b + c + d + e + f + g + n$

$= 12x + 3x + 18x + 6x + 5x + 9x + 7x + 10x = 70x$

Given that $70x = 350$

$\Rightarrow x = 5$



$\therefore$ The final Venn diagram is as follows:

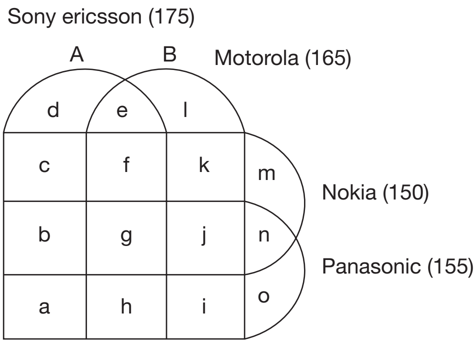
48. $30 + 45 + 25 = 100$ people play exactly two games.

49. Chess but not Carroms $= 60 + 30 = 90$.

50. Do not play Chinese Checkers = Total – Chinese Checkers $= 350 - (30 + 15 + 35 + 25) = 300 - 105 = 245$.

51. Chess or Carroms = Total – (None + Chinese Checkers only) $= 350 - (50 + 15) = 285$.

Solutions for questions 52 to 55:



It is given that Sony Ericsson and Motorola = 75

$\Rightarrow e + f + g + h = 75$ (1)

Also,

Nokia and Panasonic $= 75 \Rightarrow b + g + j + n = 75$ (2)

$(1) + (2) = e + b + f + j + h + n + 2g = 150$

Given that exactly three = 75

$\therefore f + b + j + h = 75 \Rightarrow e + n + 2g = 75$

Also, given that sum of the number of families using Sony Ericsson and Motorola only and the number of families using Nokia and Panasonic only is 25.

$\Rightarrow e + n = 25$

$\therefore g = 25$

52. All the four $= g = 25$

53. Exactly one + Exactly two + Exactly three + Exactly four = 280.

Given that: Exactly two = Exactly one + 100

As Exactly four = 25 and Exactly three = 75. Exactly two = 140.

$\therefore$ Exactly two + Exactly three = 215.

54. Exactly two + Exactly three = 215 and Exactly four = 25

$\therefore$ Exactly one = (Exactly one + 100) $= 280 - (75 + 25)$

$\Rightarrow$ Exactly one = 40

55. Given, $f + h = 35$.

We have, $e + f + g + h = 75 \Rightarrow e + g = 75 - 35 = 40$.

Also, $g = 25 \Rightarrow e = 15$

Also, we know that $e + n = 25 \Rightarrow n = 10$

Sony Ericsson + Motorola $= 175 + 165 = 340$

$\therefore a + b + c + d + i + j + k + l + 2(e + f + g + h) = 340$

$a + b + c + d + e + f + g + h + i + j + k + l = 340 - 75 = 265$ (1)

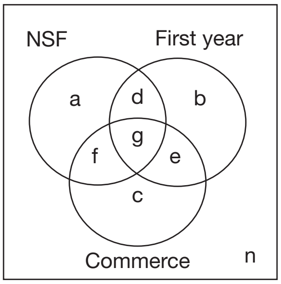
Total – (1) $= m + n + o = 280 - 265 = 15$

We have: $n = 10$

$\therefore m + o = 5$

The required value is 5.

Solutions for questions 56 to 60: Consider the following Venn diagram. Group total (GT) = 500



$a + b + c + d + e + f + g + n = 500$ (1)

In the above Venn diagram, the number inside each circle is the number of students who belong to that group and the remaining belong to the other group in the category.

Example: Number of NSF members is $a + d + f + g$.

and the number of SFI members is $b + c + e + n$.

Here, 'n' represents the second year Science students who are SFI members. Also, the number of the first year Science students who are NSF members is d.

Given $e = 90$ (2)

$a + d + f + g = 270$ (3)

$c = 50$ (4)

$d + g = 140$ (5)

$d + f = 150$ (6)

$f + c = 120$ (7)

$a + b = 100$ (8)

From (4) and (7), $f = 120 - 50 = 70$ (9)

From (9) and (6), $d = 150 - 70 = 80$ (10)

From (10) and (5), $g = 140 - 80 = 60$

$\therefore$ From (3), $a = 270 - (d + f + g)$

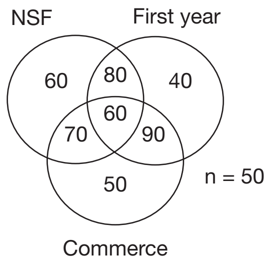
$= 270 - (150 + 60) = 60$

From (8), $b = 100 - 60 = 40$

From the above results, we calculate the final result:

$n = 500 - (270 + 90 + 40 + 50) = 50$

∴ The final Venn diagram is as follows.
GT = 500



56. The required number is n, i.e., 50.

57. The required number is b, i.e., 40.

58. The required number is g, i.e., 60.

59. The required number is $c + g$, i.e., $50 + 60 = 110$.

60. The required number is GT – $(b + e)$,
i.e., $500 - 40 - 90 = 370$.

Challenge Your Understanding
Practice Set 2

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

A group of six children, namely Sanjay, Srikanth, Mandira, Jeffery, Charu and Rameez each of a different height are asked to bring as many books as possible from a room in which exactly 21 books from A through U are kept distributed among six shelves, each of which contains a distinct number of books and is at a height equal to the height of exactly one of the six children.

The children bring the books from the room, one after the other, such that no two children are in the room at the same time and each child brings all the books contained in all the shelves which are accessible to him/her, i.e., all the shelves that are at his/her height and below. The following information is further available:

(i) There is at least one book in each shelf and the total number of books in all the shelves that are accessible to at most two children is the same as that in all the shelves that are accessible to at least five children, which in turn is the same as that in all the shelves that are accessible to less than five but more than two children.

(ii) Though Rameez is taller than Jeffery, he returns without bringing any book.

(iii) The number of books that each of Srikanth, Mandira and Sanjay bring is the same.

(iv) Rameez enters the room immediately before Srikanth, who is taller than Mandira.

(v) Though, Charu, who enters the room after Rameez and Srikanth is not the shortest, he does not bring any book.

(vi) Jeffery brings six books, while Sanjay is the last one to enter the room.

1. Who is the third to enter the room?

(A) Mandira (B) Rameez
(C) Srikanth (D) None of these

2. How many of the children are shorter than Charu?

(A) Two (B) One
(C) Three (D) Four

3. What is the number of books kept in the shelf which is at the height of Srikanth?

(A) Two (B) Three
(C) Four (D) Five

4. If the children are ranked according to their heights, such that the tallest child is ranked 1st, then who brings as many books as his/her respective rank in terms of height?

(A) Mandira (B) Sanjay
(C) Srikanth (D) Jeffery

Directions for questions 5 to 7: These questions are based on the following information.

Top 64 players participated in a knock out tennis tournament. This tournament has five knock out rounds before the final, i.e., first round, second round, third round, quarter finals and semi-finals. In the first round, the highest seeded player (Seed 1) plays the lowest seeded player (Seed 64) and this match is designated as Match 1 of the first round; the 2nd seeded player plays the 63rd seeded player and this match is designated as Match 2 of the first round and so on. Thus, for instance Match 32 of the first round is to be played between the 32nd seeded and the 33rd seeded players. In the second round, the winner of Match 1 of the first round plays the winner of Match 32 of the first round and this match is designated as Match 1 of the second round. Similarly, the winner of Match 2 of the first round plays the winner of Match 31 of the first round and this match is designated as Match 2 of the second round. Thus, for instance, Match16 of the second round is to be played between the winner of Match 16 of the first round and the winner of Match 17 of the first round. The same pattern is followed for later rounds as well. An upset is said to be taken place if a lower seeded player beats a higher seeded player.

5. What is the maximum possible number of upsets in the tournament?

(A) 64 (B) 63
(C) 127 (D) 32

6. If there is no upset in the tournament, with whom does Seed 3 play in the quarterfinals?

(A) Seed 6 (B) Seed 2
(C) Seed 4 (D) Seed 5

7. If Seed 43 reaches the third round, who among the following players could he have played in that round?

(A) Seed 52 (B) Seed 36
(C) Seed 54 (D) Seed 38

Directions for questions 8 to 10: These questions are based on the following information.

Each of the three people, namely Mamta, Madhuri and Mithu, belong to different city among Hyderabad, Delhi and Mumbai and also each work in a different city among the cities mentioned above. A person, who belongs to a city, does not work in the same city. When they were asked about themselves, they gave the following replies.

Mamta: I belong to Delhi. Mithu does not work in Delhi.
Madhuri: I work in Delhi. Mithu belongs to Mumbai.
Mithu: I do not work in Delhi. Mamta belongs to Mumbai.
It is also known that, each of them alternates between truth and lie in any order.

8. If Mamta works in Delhi, where does Mithu belong to?
(A) Hyderabad
(B) Delhi
(C) Mumbai
(D) Hyderabad or Delhi

9. If the first statement of the person, who belongs to Delhi is true, where does Mamta work?
(A) Mumbai (B) Delhi
(C) Hyderabad (D) Either (A) or (B)

10. Who belongs to Hyderabad?
(A) Madhuri (B) Mamta
(C) Mithu (D) Either (A) or (B)

Directions for questions 11 to 15: These questions are based on the following data.

In a medical college, the courses being offered in a semester are Anatomy, Physiology, Pathology, Ophthalmology, Microbiology, Pharmacology, Biochemistry, Forensic Medicine and Cardiology. Every student is expected to enrol for 6 subjects, subject to the following conditions:
(i) The student must always choose only one subject between Anatomy and Pathology.
(ii) The student must enrol for two and only two subjects from among Physiology, Ophthalmology and Cardiology.
(iii) The student must also opt for Anatomy, Physiology and Biochemistry, if he or she enrols for Pharmacology.
(iv) The student cannot opt for Microbiology, if he or she has not chosen Pathology or Biochemistry.
(v) The student who does not enrol for Physiology or Pathology cannot choose Forensic Medicine.

11. If a student enrols for Cardiology and Forensic Medicine, then he or she must enrol for
(A) Biochemistry and Microbiology
(B) Anatomy and Pharmacology
(C) Ophthalmology and Microbiology
(D) Biochemistry and Pharmacology

12. If a student enrols for Microbiology, then the courses he or she cannot opt for are
(A) Pathology and Physiology
(B) Microbiology and Biochemistry
(C) Forensic Medicine and Microbiology
(D) Anatomy and Pharmacology

13. The two courses which can never be taken up together are
(A) Pharmacology and Microbiology
(B) Physiology and Pathology
(C) Biochemistry and Ophthalmology
(D) Cardiology and Pathology

14. If a student enrols for Microbiology and Forensic Medicine, then he or she must enrol for
(A) Ophthalmology
(B) Cardiology
(C) Pharmacology
(D) Either (A) alone or (B) alone

15. If a student chooses Ophthalmology and Cardiology, then the maximum number of courses that he or she can enrol for are (given that the condition regarding number of courses to be selected could be violated)
(A) Six (B) Five
(C) Four (D) Three

Directions for questions 16 to 19: These questions are based on the following information.

A logician is telling his wife about four pieces of jewellery he has just seen in an exhibition. 'I have seen a watch, a ring, a necklace and a brooch. The heavier of the necklace and the brooch is the costliest, while the costlier of the watch and ring is the lightest. The cheaper of the ring and the brooch is the heaviest, while the heavier of the necklace and the ring is the cheapest. Also, if the brooch is costlier than the necklace, then it is lighter than the watch'. His intelligent wife immediately ranks them in terms of their cost and weight and says, 'We cannot determine the ranks of two of the pieces in one of the comparisons'.

16. Which of the following pairs is the wife of the logician talking about?
(A) Necklace and Brooch
(B) Brooch and Watch
(C) Watch and Ring
(D) Ring and Necklace

17. The logician then gave his wife an additional statement which was sufficient to determine the complete rank-list in both the comparisons. Which one of the following statements did the logician give his wife?
(A) If the ring is lighter than the necklace, then the necklace is costlier than the brooch.
(B) If the necklace is lighter than the brooch, then the brooch is costlier than the watch.
(C) If the watch is cheaper than the ring, then the ring is heavier than the watch.
(D) If the ring is not lighter than the watch, then the watch is costlier than the brooch.

18. Which of the following is the costliest?
(A) Necklace (B) Brooch
(C) Watch (D) Either (A) or (B)

Practice Set 2

19. Which of the following is the lightest?
(A) Ring (B) Watch
(C) Necklace (D) Either (B) or (C)

Directions for questions 20 to 24: These questions are based on the following information.

A group of six football teams from Maharashtra, Tamil Nadu, West Bengal, Delhi, Karnataka and Andhra Pradesh participated in a tournament. After the first two rounds, it is known that every team played with two different teams, one in each round and won one of the matches and lost the other.

Each team scored a different number of goals, such as 2, 3, 4, 5, 6 and 7 in the first round of tournament and each team scored a different number of goals, such as 1, 3, 4, 5, 6 and 8 in their second round of tournament. We know the following information about their scores.

(i) West Bengal scored nine goals in total, but it scored less number of goals in the match they won with respect to that in the match they lost.
(ii) The number of goals scored by Tamil Nadu in the first round is the same as that conceded by it in the second round.
(iii) The number of goals conceded by Karnataka is the same in both the rounds and in both the matches the difference of goals between the winner and the loser is two.
(iv) The total number of goals scored in each of the two matches of Maharashtra is 10 but the total number of goals scored by Maharashtra is not 10.
(v) Delhi is the only team that scored the same number of goals in both the matches. But in total, it scored less number of goals than any of the other teams.

20. What is the maximum number of goals scored in any match?
(A) 14 (B) 11
(C) 12 (D) 13

21. Which team scored the least number of goals in a match?
(A) Delhi (B) Karnataka
(C) Andhra Pradesh (D) West Bengal

22. Which team scored the maximum number of goals in both the matches together?
(A) West Bengal (B) Tamil Nadu
(C) Karnataka (D) Maharashtra

23. Against which team did Delhi win the match in the first two rounds?
(A) West Bengal (B) Maharashtra
(C) Tamil Nadu (D) Andhra Pradesh

24. What is the number of goals scored by Maharashtra in the match it won?
(A) Six (B) Five
(C) Seven (D) Four

Directions for questions 25 to 28: These questions are based on the following information.

Harish, Mahesh, Divya, Rohini, Dilip and Seema are a group of people, who are performing in six cities across India. The cities are Delhi, Mumbai, Kolkata, Chennai, Bangalore and Hyderabad. All six of them can sing, dance and play music. In each of the six cities they perform all the three activities one after the other. They stage three shows in each city and in each show, they perform one of the activity. Each of them has a fixed partner and perform only with their partners in any city or any show or in any activity. Each pair performs only one activity in a city. Further the following information is known.

(1) Divya sings in Delhi but does not dance in Bangalore.
(2) Rohini sings in Hyderabad and Harish who dances in Delhi sings in Chennai.
(3) Seema plays music in Delhi and Rohini plays music in Mumbai.
(4) Harish does not sing in Kolkata and Mahesh do not sing in Hyderabad but sings in Bangalore.
(5) Divya plays music in two cities only and Harish sings in three cities.
(6) In Chennai, Dilip is one of the dancers.

25. Which pair dances in Mumbai?
(A) Harish – Mahesh (B) Dilip – Divya
(C) Rohini – Seema (D) Cannot be determined

26. Which of the following is true?
(A) Harish and Mahesh will dance in Kolkata.
(B) Rohini and Seema dance only in two cities.
(C) Dilip and Divya play music in Bangalore.
(D) None of the above

27. If Harish and Mahesh play music in Kolkata, then who plays music in Hyderabad?
(A) Harish and Mahesh
(B) Dilip and Divya
(C) Rohini and Seema
(D) Cannot be determined

28. Which of the following statement is sufficient to know the complete schedule?
(A) Harish and Mahesh dance in Kolkata.
(B) Dilip and Divya play music in Hyderabad.
(C) Rohni and Seema dance in Kolkata.
(D) Harish and Mahesh dance in Hyderabad.

Directions for questions 29 to 33: These questions are based on the following information.

The Dean of a college, Prof. Himanshu, asked Prof. Deodhar to provide him with the analysis of the results of recently completed semester exams in the college, which was written by 108 students. Prof. Deodhar analysed the performance of students in five different subjects, such as Business Statistics (BS), Micro Economics (ME), Supply

Chain Management (SCM), Marketing Management (MM) and Consumer Behaviour (CB). The following are some of his observations.

(i) The students who passed in MM failed in all other subjects except CB.
(ii) The students who did not fail in BS, passed in ME.
(iii) The number of students who failed in four subjects is seven less than those who did not pass in ME. The number of students who passed in three subjects is 18.
(iv) The students who failed in CB passed in ME and none passed in both the subjects.
(v) The number of students who passed only in ME is 17 and those passed in MM is 13.
(vi) The number of students who passed in SCM is 46 and the number of students who passed only in CB is 10.

29. How many students passed in exactly two subjects?
(A) 46 (B) 52 (C) 41 (D) 63

30. How many students who passed in BS also passed in at least one of the other subjects?
(A) 22 (B) 40
(C) 18 (D) 57

31. How many students passed in all subjects except ME and MM?
(A) 21 (B) 11 (C) 10 (D) 0

32. How many students passed in SCM but not in BS?
(A) 18 (B) 28
(C) 38 (D) None of these

33. Which of the following statements is/are true?
I. The number of students who passed in at least two subjects is 81.
II. The number of students who passed in only SCM and ME is 17.
(A) Only I (B) Only II
(C) Both I and II (D) Neither I nor II

Directions for questions 34 to 38: These questions are based on the following information.

In the year 2012, KBC Inc., started a GK training institute for general public. It offers certifications in five grades. (In each grade there may be students who newly enrolled or students promoted from a lower grade or students retained from the same grade of previous year.)

A consultant who wanted to perform an audit of the institute's performance collected the following data:

New Enrolments:

(i) In the first year, 50 students each have enrolled in each of the grades. The number of enrolments in Grade I is the same every year.
(ii) In grades II, III, IV and V, there were no new enrolments in a grade whenever the number of students retained in that grade is more than or equal to the number of promotees to that grade. In all other cases, their number is equal to the difference of numbers of the students retained and the students promoted.
(iii) No student has left the institute.

Retention of Students:

(iv) The number of students retained in any grade is at most 50 and is a multiple of 10.
(v) Year 2015 has the lowest number of students retained, i.e., 140.
(vi) In any year, no two grades have the same number of retained students.
(vii) For any grade, no two years have the same number of retained students.

Promotions:

(viii) Every student who is not retained is awarded the grade certificate and is promoted to the next higher grade for the next year.
(ix) In no grade among I, II, III and IV, in any year, all the students have been promoted. In Grade V, only in 2014 all the students have been issued certificates.

Further, the following incomplete table is shared by one of the employees:

	2012	2013				2014				2015				2016			
	T	N	R	P	T	N	R	P	T	N	R	P	T	N	R	P	T
Grade I	50				100		40				30						70
Grade II	50				40						20		120		50		
Grade III	50						30										
Grade IV	50			30	60												
Grade V	50		30					10					20		10		40

N = New enrolments
R = Number of students retained from the same grade previous year
P = Students of one lower grade promoted from previous year
T = Total number of students

Directions: Type in your answer in the space provided below the question.

34. How many Grade V certifications were issued in all, during the given period?_________

35. What is the highest number of new enrolments in any grade across all years? _________

36. How many new enrolments were made in the year 2014? _________

37. What was the total strength of the students in the year 2015? _________

38. In which year KBC Inc. had seen the highest 'New total enrolments'? _________

Directions for questions 39 to 42: These questions are based on the following information.

A, B, C, D and E are five friends each of whom answered five questions in an examination. Each question has five answer choices a, b, c, d and e. The answers marked by the five friends are given in the following table:

	Q.no.1	Q.no 2	Q.no. 3	Q.no. 4	Q.no. 5
A	b	c	a	e	d
B	b	e	b	b	d
C	e	c	b	e	a
D	b	c	a	b	d
E	b	b	a	e	d

The following information is known regarding them:

(i) Each person has answered at least one question correct/right in the examination.

(ii) Only one person has got all the answers correct in the examination.

(iii) But for B and E, no two among the others have got the same number of questions correct in the examination.

(iv) Each question has only one correct answer.

39. Which of the following person has got all the questions correct?

(A) A (B) B

(C) D (D) C

40. What is the right choice/answer for Question 2?

(A) e (B) c

(C) b (D) a

41. How many questions were answered correctly by B?

(A) Two (B) One

(C) Four (D) Three

42. Which of the following combination is correct?

(A) A – 1 (B) D – 5

(C) B – 5 (D) C – 5

Directions for questions 43 to 45: These questions are based on the following information.

There are 10 boxes, each containing n balls. Each of the balls in nine boxes weigh 1 kg whereas each of the balls in the remaining box weighs 2 kg. A spring balance is used to weigh these balls and m is the minimum number of weighing required to find the box that contains the 2 kg balls.

43. If $n = 10$, then $m =$

(A) One (B) Two

(C) Three (D) Four

44. If $n = 2$, then $m =$

(A) Two (B) Three

(C) Four (D) Five

45. If $n = 3$, then $m =$

(A) Two (B) Three

(C) Four (D) Five

Directions for questions 46 to 48: These questions are based on the following information.

A group of five friends, namely A, B, C, D and E purchased exactly one fruit basket from among the five fruit baskets, such as P, Q, R, S and T. Each of the five baskets contained four apples or five bananas or six oranges or seven mangoes or eight guavas. No two baskets contained the same variety of fruits. Q contained six fruits in it. Each of the five friends, then transferred at least one and at most two fruits from her respective basket to each of the other baskets. None of the friends transferred any fruit that they received from another. After all the transfers, it is observed that

(i) R has no guavas in it.

(ii) P has a total of ten fruits, at least three of which are mangoes.

(iii) T has six fruits in all, none of which is an apple.

46. How many fruits were there in S initially?

(A) Six (B) Seven

(C) Four (D) Five

47. Which fruit is present in all the baskets after all the transfers?

(A) Banana (B) Orange

(C) Mango (D) Both Orange and Mango

48. Which of the following statements is true?

(A) The baskets having the same total number of fruits after all the transfers have an equal number of each kind of fruit.

(B) The two baskets which have a difference of four in the total number of fruits after all the transfers have an equal number of fruits of exactly three kinds.

(C) After all the transfers, no two baskets have an equal number of fruits of more than two kinds.

(D) The basket which was initially having the highest total number of fruits, now has the lowest total number of fruits.

Directions for questions 49 to 52: Answer these questions on the basis of the information given below.

The company XYZ Ltd. recruited 80 trainees, of which it was observed that every trainee has

(i) Exactly one qualification among
  (a) B.E. (b) M.Sc.
(ii) At least one feature among the following two
  (a) He is from a reputed college.
  (b) He has a good academic record.
(iii) Experience in exactly one of the following two domains
  (a) Programming (b) Testing

Further it is also known that:

(a) 20 trainees who are B.Es have good academic record as well as experience in Testing. These trainees form 50% of those with experience in Testing.
(b) No trainee with experience in Programming has both the features mentioned in (ii), i.e., being from a good college and having a good academic record.
(c) 30 trainees were M.Scs, of which 15 have experience in Programming.
(d) Of the trainees with experience in Testing, 15 were from a reputed college and they have good academic record.
(e) In total, 45 trainees have good academic record, of which two-thirds were B.Es and one-third have experience in Programming.

49. How many of the trainees with experience in Programming are B.Es and also have good academic record?
(A) 8 (B) 10
(C) 12 (D) Cannot be determined

50. How many trainees with experience in Testing have good academic record and also are B.Es and are from a reputed college?
(A) 5 (B) 7
(C) 12 (D) Cannot be determined

51. How many M.Scs are from a reputed college but have neither a good academic record nor any experience in Programming?
(A) 5 (B) 6
(C) 7 (D) Cannot be determined

52. If 18 B.Es have good academic record but are not from a reputed college, how many M.Scs are from a reputed college and with good academic record?
(A) 3 (B) 4
(C) 5 (D) Cannot be determined

Directions for questions 53 to 56: These questions are based on the following information.

A group of six people, namely Anand, Bindu, Charu, Dilip, Indu and Kiran stay in rooms numbered from 1 through 6. There are two types of products, such as books and mobiles. Among them, three people bought books and the others bought mobiles. The books are Maths, Physics and Chemistry and the mobiles are Nokia, LG and Samsung. The following is also known.

(1) No two people with consecutive room numbers bought the same type of product.
(2) Charu stays in room number 6 and Anand in room number 3.
(3) Bindu and Charu bought the same type of product; Anand and Dilip bought different types of product.
(4) Indu bought LG mobile and Bindu bought Maths book.
(5) The person who stays in room number 4 bought Chemistry book.
(6) The person in room number 5 bought Nokia mobile.

53. Who bought the Chemistry book?
(A) Anand (B) Charu
(C) Dilip (D) Kiran

54. Who stays in room number 2?
(A) Bindu (B) Dilip
(C) Indu (D) Kiran

55. The person who is in room number 1 bought ______.
(A) LG mobile (B) Maths book
(C) Samsung mobile (D) Physics book

56. The person who bought Samsung mobile is in the room number adjacent to that of ______.
(A) Indu
(B) Charu
(C) The person who bought Physics book.
(D) The person who bought Maths book.

Directions for questions 57 to 59: These questions are based on the following information.

A group of people, namely Ram, Shyam, Tarun and Uday belong to Zig-zag island. In that island, there are four different types of people. Yes-Yes type, who always speak truth, No-No type who always lie, Yes-No type who speak truth first followed by a lie and No-Yes type people who speak a lie first followed by a true statement.

No two people belong to the same category. The following are the statements made by them.

Ram: Shyam is No-No type.
Uday is Yes-No type.
Shyam: Tarun is No-No type.
I am Yes-Yes type.
Tarun: I am Yes-No type.
Shyam is Yes-No type.
Uday: Tarun is No-No type.
Shyam is No-No type.

57. Who belongs to Yes-No type?
(A) Ram (B) Shyam
(C) Uday (D) Tarun

Practice Set 2

58. Who belongs to No-No type?
(A) Ram (B) Shyam
(C) Uday (D) Tarun

59. Who belongs to No-Yes type?
(A) Ram (B) Shyam
(C) Tarun (D) Uday

Directions for questions 60 to 63: These questions are based on the following information.

In a cricket trophy, a group of six teams, such as Australia, India, New Zealand, South Africa, Sri Lanka, England are competing against each other.

Matches are held in 2 stages. In Stage-1, each team plays 3 matches and in the Stage-2, each team plays 2 matches. Each team plays against the other only once. Tie breakers are used to avoid draw matches.

The observations after Stage-1 and Stage-2 are as given below:

STAGE-1:
(i) One team won all the matches in this stage.
(ii) Two teams lost all the matches played in this stage.
(iii) England lost to Australia but won against Sri Lanka and New Zealand.
(iv) South Africa lost to India but won against Sri Lanka and New Zealand.
(v) India won at most 2 matches.
(vi) New Zealand did not play against the top-team in Stage-1.

STAGE-2:
(i) The top-team of Stage-1 lost all the remaining matches.
(ii) Of the two teams at the bottom after Stage-1, one team won both matches, while other lost both matches.
(iii) In all, 3 teams lost both matches in Stage-2.

60. The team which lost the highest number of matches is
(A) Sri Lanka
(B) New Zealand
(C) South Africa
(D) Sri Lanka and England

61. The team (s) that lost exactly two matches in the event is/are
(A) India (B) Australia
(C) New Zealand (D) Australia and India

62. The total number of teams that lost two matches in any stage is
(A) Two (B) Three
(C) Four (D) One

63. The team which won against Australia can be
(A) India (B) South Africa
(C) New Zealand (D) Both (B) and (C)

Directions for questions 64 to 67: These questions are based on the following information.

The Sports Authority of India (SAI), in its 'catch them young' drive selected 215 sprinters, from all over the country to be trained for the 2020 Olympics in three events, such as in 100 m, 200 m and 400 m. These sprinters were divided into three categories A, B and C.

'A' – The sprinters who are suitable for all the three events.
'B' – The sprinters who are suitable for only two events.
'C' – The sprinters who are suitable for only one event.

Further it is known that:
(1) The number of sprinters in categories A, B and C is 35, 60 and 120, respectively.
(2) Every sprinter is trained in at least one of the three events.
(3) From the sprinters in each category, an equal number of sprinters are trained in every possible combination of events, for which the sprinters of that category are suitable.

64. How many sprinters are trained in exactly two events?
(A) 60 (B) 45
(C) 30 (D) 50

65. If during training, 60% of the sprinters of Category A who are trained in only the 400 m were selected for training in the 100 m as well, then how many sprinters are trained only in the 100 m and the 400 m events?
(A) 21 (B) 15
(C) 18 (D) 24

66. It is known that VISAS can be issued to only 35 athletes to participate in the Olympics. If SAI wants to put its athletes to optimum utilization, then how many times the names of the Indian athletes appear in the list of participants in these three events together?
(A) 60 (B) 85
(C) 75 (D) 50

67. In view of the conditions imposed in the previous question, what is the least possible number of athletes who participate in the 200 m sprint event?
(A) 35 (B) 20
(C) 15 (D) 10

***Directions for questions 68 to 70*:** These questions are based on the information given below.

A group of five friends, namely Tushar, Bhupen, Jemmy, Arpit and Manas went to a shop with ₹75, ₹150, ₹40, ₹300 and ₹200, respectively. Five materials, such as Pen, Bracelet, Perfume, Wristwatch and Teddy Bear are available in that shop and the costs per unit of the above items are ₹20, ₹30, ₹50, ₹70 and ₹100, respectively. Each of the five people bought at least one of the items and on each item they together spent at least ₹100. At the end of their shopping, the shopkeeper received ₹610 in total. None of them bought more than one quantity of any item.

68. Who among them was left with the maximum amount of money at the end of the shopping?
(A) Bhupen (B) Arpit
(C) Manas (D) Tushar

69. Who among them did not buy the Bracelet?
(A) Tushar
(B) Bhupen
(C) Jemmy
(D) More than one of the above

70. Which of the following is false regarding the exact amount which they were left with?
(A) Exactly two among them was left with ₹30 each.
(B) Exactly one among them was left with ₹20.
(C) None of them was left with ₹40.
(D) Exactly two among them was left with ₹20 each.

Directions for questions 71 to 74: Answer the following questions based on the information given below.

Six teams (P, Q, R, S, T and U) are taking part in a cricket tournament. Matches are scheduled in two stages. Each team plays three matches in Stage-I and two matches in Stage-II. No team plays against the same team more than once in the event. No ties are permitted is any of the matches. The observations after the completion of Stage-I and Stage-II are as given below.

Stage-I:

- One team won all the three matches.
- Two teams lost all the matches.
- S lost to P but won against R and U.
- T lost to Q but won against R and U.
- Q lost at least one match.
- U did not play against the top team of Stage-I.

Stage-II:

- The leader of Stage-I lost the next two matches.
- Of the two teams at the bottom after Stage-I, one team won both matches, while the other lost both the matches.
- One more team lost both matches in Stage-III.

71. The two teams that defeated the leader of Stage-I are
(A) U and S (B) T and U
(C) Q and S (D) T and S

72. The only team(s) that won both matches in Stage-II is (are)
(A) Q (B) T and U
(C) P, T and U (D) Q, T and U

73. The teams that won exactly two matches in the event are
(A) S and U (B) S and T
(C) T and U (D) S, T and U

74. The team(s) with the most wins in the event is (are)
(A) Q and T (B) P and R
(C) U (D) T

Directions for questions 75 to 78: Answer the questions on the basis of the information given below.

Two person, Arjun and Madhu were involved in buying and selling of gold over five trading days. At the beginning of the first day, the price of a gram of gold was ₹1000, while at the end of the fifth day it was priced at ₹1100. At the end of each day, the gold price (per gram) went up by ₹100 or it came down by ₹100. Both Arjun and Madhu took buying and selling decisions at the end of each trading day. The beginning price of gold on a given day was the same as the ending price of the previous day. Arjun and Madhu started with the same quantity of gold and amount of cash and had enough of both. Below are some additional facts about how Arjun and Madhu traded over the five days.

- Each day, if the price went up, Arjun sold 10 grams of gold at the closing price. On the other hand, each day if the price went down, he bought 10 grams at the closing price.
- If on any day, the closing price was above ₹1100, then Madhu sold 10 grams of gold, while if it was below ₹900, he bought 10 grams, all at the closing price.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

75. If Arjun sold 10 grams of gold on three consecutive days, while Madhu sold 10 grams only once during the five days, the price of gold (per gram) at the end of day 3 was ₹.__________

76. If Arjun ended up with ₹13000 more cash than Madhu at the end of day 5, the price of gold (per gram) at the end of day 4 was ₹. __________

77. If Madhu ended up with 20 grams of gold more than Arjun at the end of day 5, the price of a gram of gold at the end of day 3 was ₹. __________

78. If Madhu ended up with ₹1000 less cash than Arjun at the end of day 5, what was the difference in the quantity of gold with Madhu and Arjun (at the end of day 5)?
(A) Madhu had 10 grams more than Arjun.
(B) Madhu had 20 grams more than Arjun.
(C) Arjun had 10 grams more than Madhu.
(D) Both had the same quantity of gold.

Directions for questions 79 to 82: These questions are based on the information given below:

In factory XYZ, the factory raw material passes through five stages during processing. The five stages of processing require machines A, B, C, D and E, respectively. The processing can be done only in the above-mentioned order. The capacity and time taken for processing by each machine is as given below.

Machine	Max capacity	Time taken
A	50 kgs	4 hours
B	25 kgs	3 hours
C	10 kgs	1 hours
D	20 kgs	1 hours
E	50 kgs	4 hours

Time taken by each machine to process the material is called a cycle.

Machine A needs a break of two hours after every cycle.

Machine B and E can run continuously. Machine C needs a break of 1 hour after a maximum of three cycles. After every break it can run for three continuous cycles.

Machine D needs a break of 1 hour after a maximum of two cycles.

A machine takes the same time to process irrespective of the capacity used.

There is 100 kg raw material in the factory which needs to be processed.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

79. The minimum time taken to process the 100 kg raw material is hours.________

80. By doubling the capacity of which of the following machines can the maximum time be reduced?
(A) Machine B (B) Machine D
(C) Machine E (D) Machine C

81. By doubling the capacity of any one machine, the processing time can be reduced by a maximum of ________ hours.

82. If the raw material available was only 50 kgs, then the minimum time required to process it is________hours.

Directions for questions 83 to 86: These questions are based on the following information.

A group of eight lecturers, from A to H are scheduled to teach five subjects, such as Maths, Physics, Chemistry, Biology, and English during a week starting on Monday and ending on Friday such that each lecturer teaches only one subject and is scheduled to teach only once during the week. At least 1 and at the most 2 lecturers teach each subject. Each day has two slots, whereas it is only the morning slot and the afternoon slot such that no subject is taught more than once in the same slot (morning or afternoon) during the week. In addition to that, the same subject is not taught twice during a single day. No two lectures are scheduled simultaneously.

Further:

(i) A and C are scheduled to teach on the same day. G and H are scheduled to teach on the same day.
(ii) C and F teach the same subject. Also, E and H teach the same subject.
(iii) B teaches Chemistry and D teaches Biology. No one else teaches the subjects B and D teach.
(iv) D and E alone are scheduled to teach on their respective days.
(v) F teaches Maths during the morning slot on Friday.
(vi) Biology is taught during the morning slot on Tuesday.
(vii) Maths is taught immediately after the day D teaches. Physics is taught both immediately before and immediately after the day D teaches.
(viii) G is scheduled to teach before A, not necessarily immediately.

83. Which subject does E teach?
(A) Maths (B) Physics
(C) English (D) Cannot be determined

84. Which subject is not taught during the morning slot?
(A) Chemistry (B) English
(C) Physics (D) Maths

85. Which of the following slots is free (i.e., no lecture is scheduled)?
(A) Monday – Morning slot
(B) Wednesday – Afternoon slot
(C) Thursday – Morning slot
(D) Thursday – Afternoon slot

86. Which of the following pairs of subjects is not taught on the same day?
(A) English and Physics
(B) English and Maths
(C) Chemistry and Maths
(D) Maths and Physics

Directions for questions 87 to 91: Answer the questions on the basis of the information given below.

Kalyan, Laxman, Mohan, Naveen, Pranav, Qureshi, Rahul, Sanjay, Uday and Watson are the only ten people working in the HR department of a company. There is a proposal to form a team from within the members of the department, subject to the following conditions.

- A team must include exactly one among Pranav, Rahul and Sanjay.
- A team must include either Mohan or Quereshi but not both.

- If a team includes Kalyan, then it must also include Laxman and vice versa.
- If a team includes one among Sanjay, Uday and Watson, then it must also include the other two.
- Laxman and Naveen cannot be the members of the same team.
- Laxman and Uday cannot be the members of the same team.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

87. Who cannot be a member of a team of Size 3?
(A) Laxman (B) Mohan
(C) Naveen (D) Pranav

88. Who can be a member of a team of Size 5?
(A) Kalyan (B) Laxman
(C) Mohan (D) Pranav

89. What is the maximum possible size of the team?________

90. What could be the size of a team that includes Kalyan?
(A) 2 or 3 (B) 2 or 4
(C) Only 4 (D) Only 2

91. What is the number of ways in which a team can be constituted so that the team includes Naveen? ________

Directions for questions 92 to 96: Answer the questions independently of each other.

A group of 21 sports people from four southern states (Kerala, Karnataka, Tamil Nadu and Andhra Pradesh) are selected to receive the Arjuna Award. Each of the 21 people had represented India in one of the four disciplines, such as in Athletics, Basketball, Football and Cricket. The following facts about the sports people are known.

(i) The number of Basketball players was exactly half the number of sports people in each of the three other disciplines.
(ii) There was no Basketball player from Kerala. Otherwise, every state, including Kerala sent at least one sports person in each discipline.
(iii) None of the states had more than three sports person in any discipline.
(iv) Had there been one less sports person from Tamil Nadu, then there would have been twice as many sports people from Karnataka than each of the other states.
(v) Selvan and Muthuraj are two Football players among the 21 selected. They are from Tamil Nadu

92. Which of the following cannot be determined from the information given?
(A) Number of Basketball players from Karnataka.
(B) Number of athletes from Andhra Pradesh.
(C) Number of athletes from Tamil Nadu.
(D) Number of cricket players from Kerala.

93. Which of the following combinations is not possible?
(A) 2 Football players from Karnataka and 2 athletes from Kerala were among those selected.
(B) 2 Football players from Karnataka and 1 athlete from Kerala were among those selected.
(C) 3 cricket players from Karnataka and 1 athlete from Kerala were among those selected.
(D) Exactly 1 Football player from each of Kerala and Karnataka were among those selected.

94. If Arjun is the only Football player from Karnataka, then which of the following is not true about the number of sports people from the four states?
(A) There is one athlete from Kerala.
(B) There is one cricket player from Kerala.
(C) There are two athletes from Karnataka.
(D) There are three cricket players from Karnataka.

95. If the number of Football players from Karnataka is equal to the number of cricketers from that state, then what is the number of Cricket players from Karnataka?
(A) Two (B) Three
(C) One (D) Cannot be determined

96. Which of the following is true about the number of Football players from Tamil Nadu?
(A) It is twice the number of athletes from Kerala.
(B) It is twice the number of Football players from Kerala.
(C) It is twice the number of Cricket players from Kerala.
(D) None of these

Directions for questions 97 to 101: These questions are based on the following information:

A group of five students Anil, Sunil, Pavan, Naveen and Sravan appeared for an exam consisting of five questions. Each question is having five answer choices A, B, C, D and E and only one of them is the correct answer choice. The answers given by them is as follows.

Q. No.	Anil	Sunil	Pavan	Naveen	Sravan
1	B	C	D	B	C
2	A	D	B	A	C
3	C	A	D	A	D
4	A	B	C		A
5	E	B	B	E	C

It is also given that four marks are awarded to each correct answer while one mark is deducted for every wrong answer. No marks are awarded or deducted for the questions that are not attempted.

Further, it is known that:

(i) Naveen did not attempt the fourth question and the total score of Anil is zero.

(ii) Only one student answered the first question correctly, only one student answered the second question correctly, two students answered the third question correctly, only one student answered the fourth question correctly while two students answered the fifth question correctly.

(iii) Naveen's score is the highest among the five students and choice (A) is the correct answer for exactly one question.

(iv) The number of questions for which the correct answer choice is (B) is the maximum.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

97. Who scored the lowest number of marks?
(A) Anil
(B) Sunil
(C) Pavan
(D) Sravan

98. What is the total marks scored by Naveen?__________

99. Which pair of students scored the same number of marks?
(A) Pavan and Sravan
(B) Anil and Sunil
(C) Anil and Sravan
(D) Pavan and Sunil

100. The difference between the highest score and the lowest score is__________.

101. If Naveen had marked the fourth question with an answer choice that he has not marked in any other question who would be scoring the second highest?
(A) Naveen
(B) Pavan
(C) Anil
(D) Both (A) and (B)

Directions for questions 102 to 104: These questions are based on the data given below.

A survey was conducted among 200 mobile subscribers of three different companies Airtel, IDEA and Cellone. It was found that 50 members do not subscribe to Cellone, 75 members do not subscribe to Airtel and 100 members do not subscribe to Idea. 125 subscribed to at least two of the three companies.

Directions: Type in your answer in the space provided in the question.

102. The number of customers who subscribe to exactly one service is__________.

103. The number of customers who subscribe to exactly two services is__________.

104. The number of customers who subscribe to exactly three services is__________.

Directions for questions 105 to 109: Answer the questions on the basis of the information given below.

Five friends met at a restaurant one evening. Each one brought some money. They had only ₹10 notes and ₹5 coins with them. Each had a different number of notes and coins. The number of notes they had were 2, 6, 8, 9 and 12. While the number of coins they had were 0, 3, 4, 8 and 12. Below are some more facts.

(i) The number of coins with Harish was three times the number of coins with the person who had six notes.

(ii) Three people, including the one who had eight coins, did not keep a wallet.

(iii) Dinesh did not keep a wallet

(iv) The one who had only two notes with him, did not have any coins or a wallet.

(v) Arpit had notes and coins and also had a wallet.

(vi) Manjeet, who did not have a wallet, had half as many coins as the person who had twice as many notes as he had.

(vii) Sohan had four more notes than Harish, but Harish had four more coins than Sohan.

105. Which of the following statements is true?
(A) Manjeet had 3 coins.
(B) Arpit had no coins.
(C) Harish had 4 coins.
(D) Sohan had 8 coins.

106. Which of the following statements is true?
(A) Manjeet had 8 notes.
(B) Arpit had 9 notes.
(C) Sohan had 6 notes.
(D) Dinesh had 9 notes.

107. Which of the following statements is true?
(A) Dinesh had 6 notes, no coins and no wallet.
(B) Sohan had 12 coins and 8 notes but no wallet.
(C) Harish had 8 notes and 12 coins but no wallet.
(D) Manjeet had 6 notes and 4 coins but no wallet.

108. Which of the following is the ratio of number of coins to that of notes with Arpit?
(A) 1 : 1 (B) 3 : 8
(C) 1 : 3 (D) 2 : 3

109. Which of the following statements is true?
(A) Sohan has twice as many coins as the number of notes that Mahesh has.
(B) Mahesh has twice as many coins as the number of notes that Arpit has.
(C) Harish has twice as many coins Mahesh has.
(D) Mahesh has twice as many notes as the number of coins that Arpit has.

Directions for questions 110 to 113: These questions are based on the following information.

A school allows its students to choose their courses according to certain restrictions. Each course has a certain number of credits. The respective credits for each course are given below.

For example: Hindi has 2 credits; Physics has 4 credits and so on.

Each student must pick exactly one course from Group 1, exactly two courses from Group 2 and exactly two courses from Group 3. The courses and their respective credits are shown in the tables below.

Group 1		
Hindi	English	French
2	3	1

Group 2			
Physics	Biology	Chemistry	Maths
4	4	3	5

Group 3			
Sports	Arts and Crafts	Music	Dance
5	3	3	3

110. Manju chose courses with a total of 17 credits. If she did not pick Hindi, then which of the following courses cannot be picked together?
(A) French and Physics
(B) English and Chemistry
(C) French and Music
(D) English and Sports

111. If Ajay picked French and Chemistry and his aim is to choose maximum possible number of credits, then which course should not be picked by him?
(A) Music (B) Dance
(C) Biology (D) Maths

112. Surabhi has chosen a total of 18 credits. Which of the following is a possible combination of courses she picked?
(A) French and Chemistry
(B) Hindi and Biology
(C) English and Maths
(D) Both (B) and (C)

113. Madhuri has picked Maths and Music. She wants to choose a total of 19 credits. Which of the following pairs of courses can she pick?
(A) Hindi and Chemistry
(B) English and Biology
(C) English and Chemistry
(D) French and Sports

Directions for questions 114 to 117: These questions are based on the following data.

Five colleagues from different divisions of a company met in the club discussing their Sunday's winnings at cards games.

(i) Mathur and the person from Engineering division together had won ₹1500.
(ii) Sastry and Saxena together won ₹1400.
(iii) Saxena and the senior Vice President together won ₹1200.
(iv) Verma and the Production Manager had together won ₹1000.
(v) The General Manager and the International Trading Division person together won ₹900.
(vi) The Foods Division man and the Soaps Division person together won ₹700.
(vii) The Vice President and the Oil Seeds Division person together won ₹600.
(viii) Rao and the Soaps Division person together won ₹400.
(ix) The Deputy General Manager together with the only person in Churidar-kurta won ₹800.
(x) The two people in three-piece suit together won ₹1100.
(xi) The person in Safari-suit has won more than the person in two-piece suit.

114. Which of the following statements are false?
I. The Production Manager is in Soaps Division.
II. The Deputy General Manager is in Oil Seeds Division.
III. Rao is the Vice President.
(A) Only I and II (B) Only II and III
(C) Only III and I (D) All three statements.

115. How much have Rao and Saxena together earned (in ₹)?
(A) 900 (B) 1100
(C) 1300 (D) None of these

116. Which of the following statements are not true?

I. The Vice President and the Senior Vice President together have earned more than the Production Manager and the General Manager put together.

II. The person from Foods Division earned less than the person from the Oil Seeds Division.

III. The person from International Trading earned the same amount as the person in two-piece suit.

(A) I and II
(B) II and III
(C) III and I
(D) All three statements

117. Who has won the maximum amount?

(A) Mathur (B) Saxena
(C) Verma (D) Rao

Directions for questions 118 to 121: Answer the questions on the basis of the following information.

In the manufacturing unit of a company, seven machines are used to manufacture a particular product. These machines are named as M1, M2, M3, M4, M5, M6 and M7. On observing the working of these machines, the following facts were known.

M1: It started first and the next two machines to start were M4 and M3. When it was stopped, M2 and M7 started running. M5 was stopped with it.

M2: It started with M7 at the time when M1 was already running. One more machine was running at that time.

M3: It was operated for a short duration and M1, M4 and M5 were running at that time.

M4: It was stopped immediately after M3 was stopped.

M5: It was used with M1, M2, M3, M4 and M7 for the first time, when it was started again, M2 and M7 were running.

M6: It was operated for a small duration. M2 and M5 were running during that time.

M7: Nothing is known about it.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

118. Which of the two, M2 or M5 was started first?

(A) M2
(B) M5
(C) Both started together
(D) Cannot be determined

119. Which machine was running with M1 when M2 was started?

(A) M3 (B) M4
(C) M5 (D) M6

120. The number of machines that were operated with M7 is_________.

121. Which were the two machines that were the last to be stopped?

(A) M2 and M6 (B) M6 and M7
(C) M5 and M6 (D) M2 and M5

Directions for questions 122 to 125: Answer the questions on the basis of the information given below.

Physicists have assigned a number called Einstein number (named after the famous physicist Albert Einstein). Only Einstein himself has an Einstein number of zero. Any physicist who has written a research paper with Einstein has an Einstein number of 1. For other physicists, the calculation of his/her Einstein number is illustrated below:

Suppose that a physicist M has co-authored papers with several other physicists. From among them, physicist N has the smallest Einstein number. Let the Einstein number of N be n. Then M has an Einstein number of $n + 1$. Hence, any physicist with no co-authorship chain connected to Einstein has an Einstein number of infinity.

In a seven-day long mini-conference organized in memory of Einstein, a close group of eight physicists, namely P, Q, R, S, T, U, V and W, discussed some research papers. At the beginning of the conference, P was the only participant who had an infinite Einstein number. Nobody had an Einstein number less than that of U.

(i) On the third day of the conference U co-authored a paper jointly with P and R. This reduced the average Einstein number of the group of eight physicists to 3. The Einstein numbers of Q, S, T, V and W remained unchanged with the writing of this paper. Further, no other co-authorship among any three members would have reduced the average Einstein number of the group of eight to as low as 3.

(ii) At the end of the third day, five members of this group had identical Einstein numbers while the other three had Einstein numbers distinct from each other.

(iii) On the fifth day, T co-authored a paper with U which reduced the group's average Einstein number by 0.5. The Einstein numbers of the remaining six were unchanged with the writing of this paper.

(iv) No other paper was written during the conference.

122. How many participants in the conference did not change their Einstein number during the conference?

(A) two (B) three
(C) four (D) five

123. The person having the largest Einstein number at the end of the conference must have had how many Einstein number (at that time)?

(A) Five (B) Seven
(C) Nine (D) Fourteen

124. How many participants had the same Einstein number at the beginning of the conference?

(A) Two (B) Three
(C) Four (D) Five

125. What could be the maximum possible number of people with distinct Einstein numbers at the beginning of the third day of the conference?

(A) Five (B) Six
(C) Two (D) Three

ANSWER KEYS

1. (B)	**19.** (B)	**37.** 470	**55.** (A)	**73.** (A)	**91.** 6	**109.** (D)
2. (C)	**20.** (D)	**38.** 2012	**56.** (D)	**74.** (A)	**92.** (D)	**110.** (D)
3. (A)	**21.** (C)	**39.** (C)	**57.** (C)	**75.** 1100	**93.** (D)	**111.** (C)
4. (D)	**22.** (B)	**40.** (B)	**58.** (D)	**76.** 1000	**94.** (C)	**112.** (D)
5. (B)	**23.** (D)	**41.** (D)	**59.** (A)	**77.** 900	**95.** (C)	**113.** (C)
6. (A)	**24.** (C)	**42.** (B)	**60.** (A)	**78.** (D)	**96.** (D)	**114.** (C)
7. (D)	**25.** (B)	**43.** (A)	**61.** (B)	**79.** 25	**97.** (D)	**115.** (D)
8. (C)	**26.** (C)	**44.** (B)	**62.** (C)	**80.** (D)	**98.** 6	**116.** (D)
9. (A)	**27.** (B)	**45.** (A)	**63.** (D)	**81.** 2	**99.** (D)	**117.** (B)
10. (B)	**28.** (C)	**46.** (D)	**64.** (B)	**82.** 18	**100.** 11	**118.** (B)
11. (A)	**29.** (D)	**47.** (C)	**65.** (C)	**83.** (C)	**101.** (C)	**119.** (C)
12. (D)	**30.** (B)	**48.** (D)	**66.** (C)	**84.** (A)	**102.** 75	**120.** 3
13. (A)	**31.** (D)	**49.** (B)	**67.** (B)	**85.** (C)	**103.** 75	**121.** (D)
14. (D)	**32.** (B)	**50.** (D)	**68.** (A)	**86.** (B)	**104.** 50	**122.** (D)
15. (B)	**33.** (C)	**51.** (A)	**69.** (C)	**87.** (A)	**105.** (D)	**123.** (B)
16. (B)	**34.** 120	**52.** (A)	**70.** (D)	**88.** (C)	**106.** (B)	**124.** (B)
17. (D)	**35.** 130	**53.** (C)	**71.** (B)	**89.** 5	**107.** (D)	**125.** (A)
18. (A)	**36.** 100	**54.** (A)	**72.** (D)	**90.** (C)	**108.** (C)	

SOLUTIONS

Solutions for questions 1 to 4: The number of books are 21 (i.e., A through U). It is given that at least one book is kept in each shelf and the number of books in different shelves are distinct. Hence, the number of books in different shelves are 1, 2, 3, 4, 5 and 6.

Let us indicate the six different shelves by I, II, III, IV, V and VI, respectively in decreasing order.

It is given that the sum of the number of books in shelves I and II (i.e., accessible to five or more children) is equal to the sum of the number of books in shelves V and VI, and also III and IV.

⇒ The sum of the number of books at levels III and IV = I and II = V and IV = 7.

From (vi), Jeffery brought six books, from (ii), Rameez did not bring any book, from (v), Charu did not bring even one book.

⇒ Srikanth, Mandira and Sanjay together brought fifteen books. From (iii), each of Srikanth, Mandira and Sanjay brought five books.

From (vi), Sanjay is the last one to enter the room and he brought five books.

⇒ Sanjay is the tallest among the six and he brought all the books kept at the highest level only.

> Hence, Sanjay is the tallest and the number of books at level I is five.

⇒ Number of items at level II is two.

⇒ Each of Mandira and Srikanth brought books kept at two different levels of height, i.e., 3 + 2 or 4 + 1.

⇒ Mandira and Srikanth together brought books kept at four different levels of height.

> ⇒ Jeffery brought all the books kept at only one level of height.

Since, Mandira and Srikanth brought books kept at different levels, neither of them is the shortest.

From (ii), Rameez is not the shortest.

From (v), Charu is not the shortest.

Hence, Jeffery is the shortest but still he brought 6 items.

> ⇒ Jeffery is the first to enter the room.

From (iv), Srikanth is taller than Mandira.

⇒ Mandira entered the room before Srikanth.

From (v), Charu entered the room after Rameez and Srikanth.

⇒ Charu entered after Mandira, Rameez and Srikanth.

Hence, Charu is the fifth to enter the room.

Since, Charu could not bring any book, he is not the second tallest.

⇒ Mandira is the second or the third to enter the room.

Since, six books are in shelf VI, the number of books at level V is one since five books are at level I, the number of books at level II is true.

Hence, we obtain the number of books at different levels as follows.

I	–	5
II	–	2
III	–	3/4
IV	–	4/3
V	–	1
VI	–	6

Since, Mandira and Srikanth brought five books each and Srikanth is taller than Mandira, Mandira is the fourth tallest and the number of books in shelf IV is four. Similarly, Srikanth is the second tallest and the number of books in shelf III is three.

⇒ Rameez is the third or the fifth tallest.

Since Rameez entered the room before Srikanth he must be the second or the third to enter the room.

⇒ Mandira and Rameez are the second and the third to enter the room in any order.

Since, Mandira is the fourth tallest and she brought books kept in two different shelves. Rameez who could not bring any book is shorter than Mandira. Hence, Rameez is the fifth tallest.

⇒ Charu is the third tallest. From (v), Rameez is the third to enter the room.

Hence, the final arrangement is as follows.

Order of entering the room	Name	Rank in height	Number of books brought
1	Jeffery	6	6
2	Mandira	4	5
3	Rameez	5	0
4	Srikanth	2	5
5	Charu	3	0
6	Sanjay	1	5

1. Rameez is the third to enter the room.

2. As Charu is the third tallest, there are three children shorter than him.

3. Two books are kept in the shelf which is at the height of Srikanth.

4. Jeffery's rank in height is six and he brought six books.

Solutions for questions 5 to 7: As it is a knockout tournament for eliminating any single player, one match is needed. Now only one of the 64 players has to be the winner. Hence, the remaining 63 players are to be eliminated. Therefore, 63 matches are required.

5. If each match is an upset, we will get a maximum of 63 upsets.

6. As there are no upsets, all the top 8 players will reach quarter finals. In the quarter-finals, Seed 1 plays Seed 8, Seed 2 plays Seed 7, Seed 3 plays Seed 6, Seed 4 plays Seed 5.

7. In the first round, Seed 43 plays Seed 22. In the second round, Seed 43 plays with the winner of the match between Seed 11 and Seed 54. In the third round, Seed 43 and the winner of the matches between 59th and 6th seeded players or the winner of 27th and 38th seeded players are played.

   ∴ 43rd seeded player would have played with 6th or 27th or 38th or 59th seeded player in the third round.

Solutions for questions 8 to 10: First, let us observe the second statements made by Madhuri and Mithu. Clearly, both of them cannot be true simultaneously. First, let us assume the second statement made by Madhuri is true, so the second statement of Mithu is false, which implies the first statement of Mithu is true.

∴ The second statement of Mamta is true.

Therefore, we get:

Case (1):

Name	I	II	Belongs to	Working in
Mamta	F	T	Hyderabad	Delhi
Madhuri	F	T	Delhi	Mumbai
Mithu	T	F	Mumbai	Hyderabad

Now, let us assume that the second statement of Madhuri is false. So, her first statement is true.

∴ The second statement of Mamta is true and also, the first statement of Mithu is true. So, we get:

Case (2):

Name	I	II	Belongs to	Working in
Mamta	F	T	Hyderabad	Mumbai
Madhuri	T	F	Mumbai	Delhi
Mithu	T	F	Delhi	Hyderabad

8. In Case (1), Mithu belongs to Mumbai.

9. In Case (2), the first statement of the person from Delhi is true. So, Mamta works in Mumbai.

10. Mamta belongs to Hyderabad.

Solutions for questions 11 to 15:

11. If a student enrols for Forensic Medicine, he must enrol for Physiology and Pathology.
Pathology enrolled ⇒ No Anatomy and hence, no Pharmacology.
Since Physiology and Cardiology are taken ⇒ Cannot enrol for Ophthalmology.
Hence, choices (B), (C) and (D) are eliminated and choice (A) does not violate any condition.

12. If a student takes up Microbiology, then he must also take Pathology and Biochemistry. This implies that Anatomy cannot be chosen. Since Anatomy cannot be chosen, even Pharmacology cannot be selected.

13. Since Pathology and Anatomy cannot be selected together, those courses, requiring these two as prerequisites, cannot be selected together. Hence, Microbiology or Forensic Medicine cannot be taken with Pharmacology.

14. If a student enrols for Microbiology and Forensic Medicine, then he must also enrol for Biochemistry, Physiology and Pathology.
Since Pathology is taken, Anatomy and Pharmacology cannot be taken.
Since he has to take Physiology, he cannot take both Ophthalmology and Cardiology but only one of these two.

15. If D chooses Ophthalmology and Cardiology, then he cannot choose Physiology.
⇒ He cannot choose Pharmacology or Forensic Medicine. He cannot enrol for 3 courses. Also, out of Anatomy and Pathology, he can enrol only for one course. Hence, he cannot enrol totally for 4 courses.
⇒ He can take up a maximum of 5 courses.

Solutions for questions 16 to 19:

Statement I:
Costlier of watch and ring is the lightest.
Let us assume that ring is the lightest.
Statement II:
Heavier of necklace and ring is the cheapest.
Statement III:
Heavier of necklace and brooch is the costliest.
Statement IV:
Cheaper of ring and brooch is the heaviest.
If ring is the lightest, it cannot be the heaviest.
⇒ Brooch is the heaviest [statement IV].
⇒ Brooch is the costliest [statement III].
⇒ Necklace is the cheapest [statement II].
But brooch is the heaviest.
⇒ Brooch is heavier than ring.
⇒ Brooch is cheaper than ring [statement IV].
'Brooch is the costliest' and 'Brooch is cheaper than ring' are contradicting each other. Hence, our assumption is wrong.
⇒ Ring is not the lightest.
⇒ Watch is the lightest.
Statement V:
If brooch is costlier than necklace, then it is lighter than watch. But we know that watch is the lightest.
⇒ Necklace is not lighter than watch.
⇒ Brooch is not costlier than necklace.
⇒ Necklace is costlier than brooch.
⇒ Necklace is the costliest [statement III].
⇒ Necklace is heavier than brooch.
⇒ Brooch is not the heaviest.
⇒ Ring is the heaviest [statement IV].
⇒ Ring is the cheapest [statement II].
Let us now rank the pieces of jewellery and show them in the following table.

Rank	Cost	Weight
1	Necklace	Ring
2	Brooch (or) Watch	Necklace
3	Brooch (or) Watch	Brooch
4	Ring	Watch

In the above table, 1st rank represents the costliest or the heaviest as the case may be.
Observation:
We do not know the rank of brooch and watch in terms of cost.

16. The logician's wife is talking about brooch and watch.

17. If we know the comparison between brooch and watch in terms of cost, we can determine the ranks list in both the comparisons.

18. Necklace is the costliest.

19. Watch is the lightest.

Solutions for questions 20 to 22: From the given information.
I round scores – 2, 3, 4, 5, 6, 7
II round scores – 1, 3, 4, 5, 6, 8.
From (5), if Delhi scored 4 goals in each round, then the team which scored one goal in 2nd round can never score more goals than Delhi, which contradicts (5).
∴ Delhi score of 3 goals in each of the first two rounds.
From (4), the only possibility of getting 10 goals in a match is 6 and 4 in 2nd round. Now, as the total number of goals of Maharashtra is not 10, in the first match the number of goals scored must be 7 and the number of goals conceded must be 3. [∵ 3 goals are scored by Delhi is known for us].

∴ In the first round, Delhi and Maharashtra played and Maharashtra won.

∴ Delhi won in the second round.

It must have conceded 1 goal in the second round.

∴ In the second round, as the total number of goals scored is 10 in the match which Maharashtra played, Maharashtra loses it by scoring 4 goals and conceding 6 goals.

From (3) and the above results, as Maharashtra conceded 6 goals in the second round; Karnataka cannot concede 6 goals each in the first two rounds. If Karnataka conceded 5 goals each in the first two matches, then in the first match it has scored either 5 + 2 or 5 – 2 goals, i.e., either 7 or 3 goals, which is not possible as Maharashtra and Delhi scored 7 and 3 goals, respectively in the first round matches.

For the same reason as above, it cannot concede 3 goals each in the first two matches.

∴ Karnataka conceded 4 goals each in the first two matches.

It scored one of (4 + 2) and (4 – 2) are the number of goals in the first round and in the second round.

In the second round, no team scored 2 goals.

∴ Karnataka scored 2 goals in the first round and 6 goals in the second round.

From (1) and the above results, as West Bengal scored a total of 9 goals, it must have scored 4 and 5 goals in the first two matches [∵ It cannot have scored 6 and 3 goals as Delhi scored 3 goals in each of the first two rounds]. In the second round as Maharashtra scored 4 goals, West Bengal have scored 4 goals and 5 goals in the first and the second rounds, respectively. As Karnataka conceded 4 goals in the first round, it played against West Bengal in the first round.

As West Bengal won in the first round, it should lose in the second round.

∴It should concede 8 goals in the second round.

From (2) and the above results, Tamil Nadu cannot score 6 goals in the first round, if so it has to concede 6 goals in the second round, which is not possible as Maharashtra conceded 6 goals in the second round.

∴ Tamil Nadu must have scored 5 goals in the first round and conceded 5 goals in the second round.

It had scored 8 goals in the second round. (West Bengal scored 5 goals and conceded 8 goals in the second round)

∴ Tamil Nadu has to lose in the first round. So, it must have conceded 6 goals in the second round.

AP scored 6 goals and Tamil Nadu conceded 6 goals in the first round and Andhra Pradesh scored 1 goal and conceded 3 goals in the second round.

∴ The final table for the 1st round will be as follows:

(1) AP – Tamil Nadu
  6   5

(2) West Bengal – Karnataka
  4   2

(3) Delhi – Maharashtra
  3   7

2nd round:

(1) AP - Delhi
  1   3

(2) Karnataka - Maharashtra
  6   4

(3) West Bengal – Tamil Nadu
  5   8

20. Maximum number of goals scored in any match is 5 + 8 = 13.

21. Andhra Pradesh scored the least number of goals (1) in the second round match.

22. Tamil Nadu scored 5 + 8 = 13 goals in the first two rounds, which is the maximum.

23. Against Andhra Pradesh, Delhi won the match in the first two rounds.

24. Maharashtra scored 7 goals in the match it won.

Solutions for questions 25 to 28: We can get the following table from the given information.

	Sing	Dance	Play music
Delhi	Divya	Harish	Seema
Mumbai			Rohini
Kolkata	~ Harish	Dilip	
Chennai	Harish	~ Divya	
Bangalore	Mahesh		
Hyderabad	Rohini		
	~ Mahesh		

In Delhi, as Harish is not performing with Divya or Seema they are not his partners. In Chennai, as Dilip is not performing with Harish, they are not partners.

If Rohini is Harish's partner then Harish can sing in Chennai and Hyderabad only, but Harish has to sing in three cities. So, Mahesh is Harish's partner and they will be singing in Chennai, Bangalore and Mumbai.

Now, Divya and Rohini or Divya and Dilip can be partners.

If Divya and Rohini are partners, then they will play music in at least three cities (Chennai, Bangalore, Mumbai). As Divya plays music in only two cities, Divya and Dilip have to be partners.

So, Seema and Rohini are partners.

We will get the final performances as below.

	Sing	Dance	Play music
Delhi	D, D	H, M	R, S
Mumbai	H, M	D, D	R, S

(Continued)

	Sing	Dance	Play music
Kolkata	D, D/R, S	H, M/D, D/R, S	H, M/D, D/R, S
Chennai	H, M	D, D	R, S
Bangalore	H, M	R, S	D, D
Hyderabad	R, S	D, D/H, M	D, D/H, M

D, D $\rightarrow$ Dileep and Divya
H, M $\rightarrow$ Harish and Mahesh
R, S $\rightarrow$ Rohini and Seema

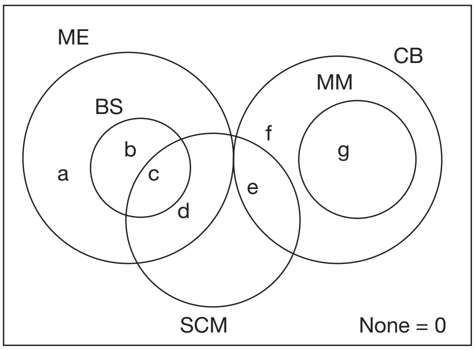
25. Dilip and Divya dance in Mumbai.

26. Dilip and Divya play music in Bangalore is true.

27. If Harish and Mahesh play music in Kolkata and since Divya plays music in two cities, then Dilip and Divya will play music in Hyderabad.

28. If Rohini and Seema dance in Kolkata then Dilip and Divya will Sing in Kolkata. So, Harish and Mahesh will play music in Kolkata. As Divya plays music in at least two cities she has to play music in Hyderabad. So Rohini and Seema will dance in Kolkata. This statement is sufficient to know the complete schedule. Choice (C)

Solutions for questions 29 to 33: From (1), (2) and (4), the information can be represented as follows:



From (3), the number of students who failed in four subjects is equal to the number of student who passed in only one subject = $a + f$.

The number of students who failed in Micro Economics $= e + f + g$

$a + f = e + f + g - 7 \Rightarrow a + 7 = e + g$ (1)

From (4), $c = 18$ (2)

From (6), $c + d + e = 46$

$\Rightarrow d + e = 28$ (3)

From (5), $a = 17$ and $g = 13$, $f = 10$

$a = 17$, $g = 13$

$\Rightarrow e = 11$ (from (1)) $\Rightarrow d = 17$ (From (3))

As $a + b + c + d + e + f + g = 108 \Rightarrow b = 22$

29. The number of students who passed in exactly two subjects $= b + d + e + g = 22 + 17 + 11 + 13 = 63$.

30. The required number of students $= b + c = 22 + 18 = 40$.

31. No student passed in all the subjects except ME and MM.

32. The required number of students $= d + e = 17 + 11 = 28$.

33. (I) $b + d + e + g + c = 22 + 17 + 11 + 13 + 18 = 81$ is true
(II) d = 17 is true
$\therefore$ Both (I) and (II) are true.

Solutions to questions 34 to 38: The table in the given data can be filled in completely, by filling in the data year wise using the given conditions.

From (i), N = 50 for Grade I across all years 2012 to 2016.

From (ii), for any grade either P = N + R or, N = 0, if R > P, thus for all grades except Grade I, N = 0 if P = 0 or P = 10; R is always a multiple of 10.

From (ix), all students of only Grade V of 2014 are promoted, therefore, R $\neq$ 0 except for Grade V in 2015.

Therefore, from (iv), value of R for all five grades in a year will be five different values among 10, 20, 30, 40 and 50. Thus the total number of students retained in any year will be 10 + 20 + 30 + 40 + 50 = 150 except for the year 2015, where Grade V has R = 0 and from (v), the lowest number of students retained in a year is 140. Thus, in the year 2015, the values of R are five different values among 0,20,30,40 and 50.

Further, from (vii), for any grade across each of the years 2013, 2014, 2015 and 2016, the value of R will be four different values among 10,20,30,40 and 50.

Across all years promotees in Grade I, i.e., 'P' will be zero, i.e., for Grade I, P = 0.

Enrolments in 2013:

	'12	2013				
	T	N	R	P	T	
I	50	**50**	**50**	**0**	100	N is 50; $\Rightarrow$ R = 100 – 50 = 50
II	50	**0**	**40**	**0**	40	P = 50 – 50 = 0; $\Rightarrow$ R > P $\Rightarrow$ N = 0
III	50	**0**	**20**	**10**	**30**	P = 50 – 40 = 10 $\Rightarrow$ N = 0; R = 20 as 30 from previous year are promoted.
IV	50	**20**	**10**	30	60	R = 10 (only possibility), N = 30 – 10
V	50	**10**	30	**40**	**80**	P = 50 – 10 = 40; N = 40 – 30 = 10; T = 80

Practice Set 2

Enrolments in 2014:

	'13	2014				
	T	**N**	**R**	**P**	**T**	
I	100	**50**	40	**0**	**90**	N is 50; $\Rightarrow T = 50 + 40 = 90$
II	40	**50**	**10**	**60**	**120**	$P = 100 - 40 = 60$; $\Rightarrow$ R cannot be 30(III), 40(I), 20(2015) or 50(2016); so, $R = 10 \Rightarrow N = 50$; $T = 120$
III	**30**	**0**	30	**30**	**60**	$P = 40 - 10 = 30 \Rightarrow N = 30 - 30 = 0$; $T = 30 + 30 = 60$
IV	60	**0**	**50**	**0**	**50**	$P = 30 - 30 = 0 \Rightarrow N = 0$; 10 promoted $\Rightarrow R = 50 \Rightarrow T = 50$
V	**80**	**0**	20	10	**30**	$R = 20$ (only possibility). $R > P \Rightarrow N = 0$; $T = 20 + 10 = 30$

Enrolments in 2015:

	'14	2015				
	T	**N**	**R**	**P**	**T**	
I	**90**	**50**	30	**0**	**80**	N is 50; $\Rightarrow T = 50 + 30 = 80$
II	**120**	**40**	20	**60**	120	$P = 90 - 30 = 60$; $\Rightarrow N = 60 - 20 = 40$
III	**60**	**50**	**50**	**100**	**200**	$R = 50$ (only possibility); $\Rightarrow N = 100 - 50 = 50$ and $T = 200$
IV	**50**	**0**	**40**	**10**	**50**	R cannot be 10(2013) or 50(2014); $R = 40$; $P = 60 - 50 = 10$; $R > P \Rightarrow N = 0$; $T = 50$
V	**30**	**10**	**0**	**10**	20	$R = 0$ in 2015 (given); $P = 50 - 40 = 10$; $\Rightarrow N = 20 - 10 = 10$

Enrolments in 2016:

	'14	2015				
	T	**N**	**R**	**P**	**T**	
I	**80**	**50**	**20**	**0**	**70**	N is 50; $\Rightarrow R = 70 - 50 = 20$
II	120	**10**	50	**60**	**120**	$P = 80 - 20 = 60$; $N = 60 - 50 = 10$; $T = 10 + 50 + 60 = 120$
III	**200**	**30**	**40**	**70**	**140**	$R = 40$ (only possibility). $P = 120 - 50 = 70$; $N = 70 - 40 = 30$; $T = 140$
IV	**50**	**130**	**30**	**160**	**320**	$R = 30$ (only possibility). $P = 200 - 40 = 160$; $N = 160 - 30 = 130 \Rightarrow T = 320$
V	20	**10**	**10**	**20**	40	$P = 50 - 30 = 20$; $N = 20 - 10 = 10$

Thus, the complete data will be as follows.

	2012	2013				2014				2015				2016			
Grade	**T**	**N**	**R**	**P**	**T**	**N**	**R**	**P**	**T**	**N**	**R**	**P**	**T**	**N**	**R**	**P**	**T**
I	50	50	50	0	100	50	40	0	90	50	30	0	80	50	20	0	70
II	50	0	40	0	40	50	10	60	120	40	20	60	120	10	50	60	120
III	50	0	20	10	30	0	30	30	60	50	50	100	200	30	40	70	140
IV	50	20	10	30	60	0	50	0	50	0	40	10	50	130	30	160	320
V	50	10	30	40	80	0	20	10	30	10	0	10	20	10	10	20	40
Certifications				20				60				30				10	

34. Total Grade V certifications issued = The total number of completed Grade V.
= (50 – 30) + (80 – 20) + (30 – 0) + (20-10) = 20 + 60 + 30 + 10 = 120.

35. 130 enrolments for Grade IV in 2016 is the highest.

36. In all total, 100 new enrolments made in 2014.

37. The total strength of students in the year 2015 = 80 + 120 + 200 + 50 + 20 = 470.

38. From the above total it can be seen that the highest new enrolment KBC Ltd. had seen is in 2012. Ans: (2012)

Solutions for questions 39 to 42: From the following given information.

Case (1):

Let us assume that A is the person who answered all the questions correctly.

Then the answers of the remaining person would be as follows.

	Question no. 1	Question no. 2	Question no. 3	Question no. 4	Question no. 5
B	✓	✗	✗	✗	✓
C	✗	✓	✗	✓	✗
D	✓	✓	✓	✗	✓
E	✓	✗	✓	✓	✓

From (iii), A is not the person who answered all questions correctly.

Case (2):

Let us assume that B is the person who answered all the questions correctly.

	Question no. 1	Question no. 2	Question no. 3	Question no. 4	Question no. 5
A	✓	✗	✗	✗	✓
C	✗	✗	✓	✗	✗
D	✓	✗	✗	✓	✓
E	✓	✗	✗	✗	✓

From (iii), B is not the person who answered all questions correctly.

Case (3):

Let us assume that C is the person who answered all the questions correctly.

	Question no. 1	Question no. 2	Question no. 3	Question no. 4	Question no. 5
A	✗	✓	✗	✓	✓
B	✗	✗	✓	✗	✗
D	✗	✓	✗	✗	✗
E	✗	✗	✗	✓	✗

From (iii), C also cannot be the person who answered all questions correctly.

Case (4):

Let D be the person who answered all questions correctly. Then the answers of the remaining people would be as follows.

	Question no. 1	Question no. 2	Question no. 3	Question no. 4	Question no. 5
A	✓	✓	✓	✗	✓
B	✓	✗	✗	✓	✓
C	✗	✓	✗	✗	✗
E	✓	✗	✓	✗	✓

From (ii) and (iii), D is the person who has got all the questions correct in the examination.

39. D is the person who has got all the questions correct.

40. The right choice for question 2 is ‘c’.

41. B answered 3 questions correctly.

42. D - 5 is the correct combination.

Solutions for questions 43 to 45:

43. When $n = 10$, we have to take one ball from the first box, two from the second and so on.
If the total weight is 1 kg more, then the required box is the first. If it is 2 kg more, then it is the second box and so on.
∴ Only one weighing is required.

44. Take one ball each from the first five boxes and two balls each from the remaining. If the weight is 1 kg more, then the required box is in the first five. If it is 2 kg more, then it is in remaining five. Now, number of these boxes is from 1 to 5. Take one ball each from first two and two balls each from the next two. If the weight is same, then the required box is the fifth one. If it is 1 kg more, then it is between the first two. If it is 2 kg more, then it is between the next two. Now, one more weighing is required to find the required box.
∴ A total of 3 weighing is required.

45. Take one ball each from the first three, two balls each from the next three and three balls each from the next three. If the weight is same, then the 10th box is the required one. If it is 1 kg more, then it is between the first three. Similarly, we can find for the other groups.
Now, number of these three boxes and one more weighing is required (similarly as described earlier).
∴ Two weighings are sufficient.

Solutions for questions 46 to 48: From the given information, it can be understood that each basket received at least one fruit or at most two fruits from each of the other baskets.

From (i), R has no guava in it, implies that the guavas which were there in R (eight in number) have been transferred to other baskets.

⇒ P, Q, S and T now contain 2 guavas each.

From (ii), P has at least three mangoes implies that P was originally having mangoes as every basket received at least one fruit from each of the other baskets. It is also mentioned that P now has ten fruits in all.

As there are only four apples, P could have received only one apple. Hence, the other six fruits in P consist two bananas, two oranges and 2 guavas.

∴ P = One apple + Three mangoes + Two bananas + Two oranges + Two guavas = 10 fruits.

From (iii), T has no apple in it but has six fruits in all. This implies that T was initially having apples in it. We know that T received two guavas.

As two out of five bananas are transferred to P, none of the other baskets can receive more than one banana.

Out of seven mangoes contained in P only four are transferred to other baskets. This implies that each basket received only one mango. Hence, the other two fruits in T are oranges.

∴ T = One banana + Two guavas + One mango + Two oranges = 6 fruits.

As there are six oranges in all and two oranges are transferred to each of T and P, R received only one orange.

∴ R = One mango + One apple + One banana + One orange = 4 fruits.

The basket which was initially containing oranges does not contain any orange now. Similarly, the one initially containing bananas now does not contain banana.

Hence, S initially contained bananas and Q does not contain any orange now.

Thus, we get the following arrangement.

Basket (Initially)		After changes					
		A	B	O	M	G	Total
P	7 Mangoes	1	2	2	3	2	10
Q	6 Oranges	1	1	0	1	2	5
R	8 Guavas	1	1	1	1	0	4
S	5 Bananas	1	0	1	1	2	5
T	4 Apples	0	1	2	1	2	6

46. S initially contained five fruits.

47. Mango is contained in all the baskets.

48. The fourth statement is true.

Solutions for questions 49 to 52: It is given that:

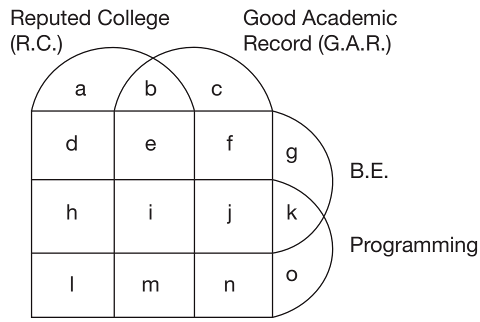
Each of the trainees are B.Es or M.Scs but not both.

⇒ B.Es + MSCs = 80

Similarly, the number of people with experience in Programming + Testing = 80.

Further, each person is either from a reputed college or with good academic background.

Let us represent the given data as follows.



Here, the features of trainees represented by 'a' represents neither B.Es nor programmers.

∴ They are MScs and testers.

a = (M.Sc., Testing, R.C.)

b = (M.Sc., Testing, R.C., G.A.R.)

c = (M.Sc., Testing, G.A.R.)

d = (Testing, R.C., B.E.)

e = (R.C., G.A.R., B.E., Testing)

f = (G.A.R., Testing, B.E.)

$g = 0$ as every person must have either G.A.R. or R.C.

Similarly, $k = 0$ and $o = 0$.

Similarly, the features of h, i, j, l, m and n can be written.

Now, it is given that:

$a + b + c + d + e + f = 40$
Testing = 40 and $e + f = 20$ } From statement (a)

$i = m = 0$
$h + j = 25$ } From statement (b)

$l + n = 15$
and $a + b + c + l + n = 30$ } From statement (c)

$b + e = 15$ From statement (d)

$b + c + e + f + j + n = 45$
$e + f + j = 30$
$j + n = 15$ } From statement (e)

As $e + f = 20$ and $e + f + j = 30$

$j = 10$

As $j + n = 15$, $n = 5$

As $h + j = 25$, $h = 15$

As $l + n = 15$, $l = 10$

As 50 trainees are B.Es:

$d + e + f + h + j = 50$,

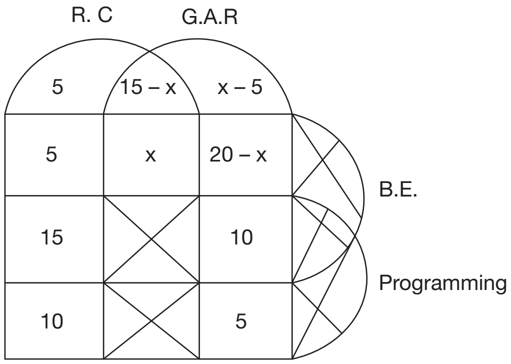
As $h + j = 25$

$d + e + j = 25$,

$\Rightarrow d = 5$

Similarly, $b + c = 10 \Rightarrow a = 5$

Let $e = x$



49. $j = 10$

50. Here, x cannot be determined.

51. $d = 5$

52. As $10 + 20 - x = 18$
$x = 12 \Rightarrow 15 - x = 3$

Solutions for questions 53 to 56: Given that no two people with consecutive room numbers bought the same type of product and the people with even numbered room numbers bought one type of products and others bought another type of products.

Given that B and C bought the same type of products. Also, A stays in odd numbered room and C stays in even numbered room.

$\Rightarrow$ A and C bought different types of products.

As A and D bought different types of products, B, C and D bought the same type of products, i.e., books as Bindu bought Maths book) and A, I and K bought mobiles.

As A stays in 3 and I, who bought LG cannot be in 5, K is in 5 and bought Nokia mobile. Hence, A bought Samsung and I is in 1.

As C stays in 6 and the person in 4 bought Chemistry book, B stays in 2 ($\because$ she bought maths book), C bought Physics and D stays in 4.

Room no.	Name	Item bought
1	Indu	LG
2	Bindu	Maths
3	Anand	Samsung
4	Dilip	Chemistry
5	Kiran	Nokia
6	Charu	Physics

53. Dilip bought the Chemistry book.

54. Bindu stays in room number 2.

55. The person in room number 1 bought LG mobile.

56. The person who bought Samsung mobile is adjacent to the person who bought Maths book.

Solutions for questions 57 to 59: Among Tarun and Uday, at least one statement must be false.

Let the first statement of Ram be true, then Shyam will be of No-No type.

Let Ram be Yes-Yes type.

So, Ram's second statement 'Uday is Yes-No' must be true.

$\Rightarrow$ 'Tarun is No-No' type. As Uday is 'yes – No' type.

But Shyam and Tarun both cannot be of the same type, so Ram's first statement must be false.

$\Rightarrow$ Now Ram, Tarun and Uday each have at least one false statement. Hence, Shyam must be Yes-Yes type and Tarun must be No-No type.

We know that Ram's first statement is false.

$\therefore$ Ram is No-Yes type and Uday is Yes-No type.

57. Uday belongs to Yes-No type.

58. Tarun belongs to No-No type.

59. Ram belongs to No-Yes type.

Solutions for questions 60 to 63: Given that England lost to Australia but won against Sri Lanka and New-Zealand. And South Africa lost to India but won against Sri Lanka and New-Zealand. These results can be written as:

Team \ Playing against	Australia	India	Sri Lanka (SL)	England (Eng)	South Africa (SA)	New Zealand (NZ)
Australia	×			W		
India		×			W	
Sri Lanka			×	L	L	
England	L		W	×		W
South Africa		L	W		×	W
New Zealand				L	L	×

Hence, Eng and SA won two matches each and lost one match each.
But from the Stage-1 observations:
Two teams lost all matches and therefore, they are SL and NZ.
India lost 1 match and Australia won all 3 matches. So, NZ did not play Australia.

Team \ Playing against	Australia	India	Sri Lanka	England	South Africa	New Zealand
Australia	×	W	W	W	NS	NS
India	L	×	NS	NS	W	W
Sri Lanka	L	NS	×	L	L	NS
England	L	NS	W	×	NS	W
South Africa	NS	L	W	NS	×	W
New Zealand	NS	L	NS	L	L	×

NS-next stage

Now in the second stage. Australia lost both the matches.
Australia played against NZ and SA where NZ and SA won.
Hence, SL lost the next 2 matches it played and NZ won both the matches in Stage-2.
∴ The other team that lost both the games in Stage-2 is England.
Now after Stage-2, the results are as follows:

Wins	Loses	Team \ Playing against	Australia	India	Sri Lanka	England	South Africa	New Zealand
3	2	Australia	X	W	W	W	L	L
4	1	India	L	×	W	W	W	W
0	5	Sri Lanka	L	L	×	L	L	L
2	3	England	L	L	W	×	L	W
4	1	South Africa	W	L	W	W	×	W
2	3	New Zealand	W	L	W	L	L	×

60. From the above table, Sri Lanka lost all the matches.

61. Australia lost 2 matches.

62. In Stage-1, New Zealand and Sri Lanka lost two each in Stage-2, Australia and England lost two each.

63. South Africa and New Zealand played in Stage-2 with Australia.

Solutions for questions 64 to 67: It is given that total 215 sprinters are selected.

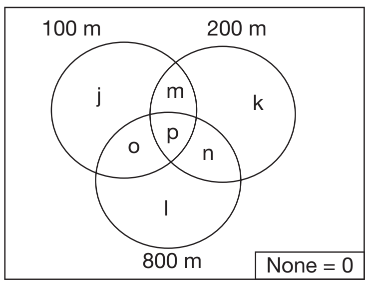
Sprinters in category A – 35
Sprinters in category B – 60
Sprinters in category C – 120

As each sprinter is trained in one or the other events, the number of sprinters who are not trained in any of the events is zero.

The sprinters of Category A are trained in at most three events, i.e., one event, two events or three events.

Similarly, the sprinters of Category B are trained in one of two events, the sprinters of Category C are trained in only one event.

The base Venn diagram is as follows.



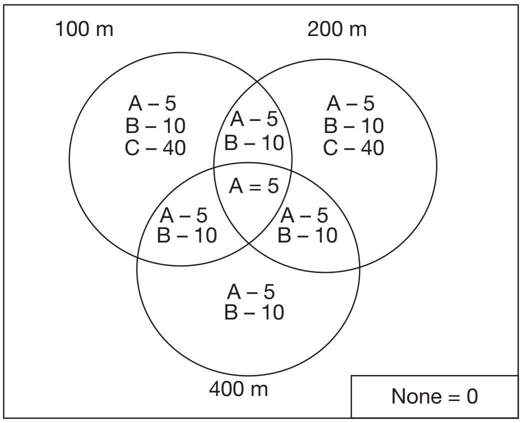
The diagram represents all the possible combination of events. As it is given that equal number of sprinters are trained in all the possible combination of events.

Practice Set 2

For sprinters in Category A = $j = k = l = m = n = o = p = \frac{35}{7} = 5$

For sprinters in Category B = $j = k = l = m = n = o = \frac{60}{6} = 10$

For sprinters in Category C = $j = k = l = \frac{120}{3} = 40$



64. The number of sprinters who are trained in exactly two events = 3(5 + 10) = 45.

65. The number of sprinters who are trained for only 400 m from Category A is 5. 60% of 5 is 3. Now the number of sprinters who are trained in only 100 m and 400 m is 10 + 5 + 3 = 18.

66. Here we have to select those 35 athletes who can participate in the maximum possible number of events.
Take the five athletes from Category A, who are trained in the three events. The names of these five athletes appear fifteen times. Take the fifteen athletes from Category A, who are trained in two events. The names of these fifteen athletes appear thirty times. Take the fifteen athletes from Category B, who are trained in two events. The names of these fifteen athletes appear thirty times.
Hence, the names of athletes appear 15 + 30 + 30 = 75 events.
This value holds good for other possible combination also.

67. The thirty athletes can be selected from the region common to 100 m and 400 m only, 200 m and 400 m only and the region common for all the three. In such case, the number of athletes who can participate in 200 m is 20.

Solutions for questions 68 to 70: Total expense = ₹610

They spent at least ₹100 per item.

∴ They have to buy at least 5 pens, 4 bracelets, 2 perfumes, 2 wristwatches and 1 teddy bear.

Then total cost = ₹560.

None of them purchased more than one of any items.

They cannot buy more than 5 pens.

∴ They bought 3 perfumes.

One of them bought each one of the five items.

Total cost to buy each one of the five items = (20 + 30 + 50 + 70 + 100) = 270.

So, it can be Arpit only.

Now one of them bought a pen, a bracelet, a perfume and a wristwatch.

Total cost of those = 20 + 30 + 50 + 70 = 170.

∴ Manas is that person.

Jemmy can buy one pen only.

Similarly, Tushar can buy a pen and a bracelet.

∴ Bhupen will buy a pen, a bracelet and a perfume.

The following distribution table represents the final result.

	Tushar	Bhupen	Jemmy	Arpit	Manas
Pen	✓	✓	✓	✓	✓
Bracelet	✓	✓	✗	✓	✓
Perfume	✗	✓	✗	✓	✓
Wristwatch	✗	✗	✗	✓	✓
Teddy bear	✗	✗	✗	✓	✗
Expenditure	50	100	20	270	170
Remaining	25	50	20	30	30

68. Bhupen is left with the maximum amount of money.

69. Jemmy did not buy bracelet.

70. The fourth statement is false.

Solutions for questions 71 to 74:

Stage I:

As P, Q, S and T won at least one match, R and U lost all the three matches.

As Q, S and T lost at least one match, P won all the three matches.

In Stage I, there are a total of 9 matches and so 9 wins. Q, S and T won two matches each.

As P (the top team in Stage-I) did not play against U, P played matches against Q and R.

∴ The ninth match was between Q and U.

So, the nine matches that have taken place are as follows.

Won	Lost	Won	Lost	Won	Lost
P	S	S	R	S	U
Q	T	T	R	T	U
P	Q	P	R	Q	U

Stage-II:

As each team played a total of five matches, in Stage-II, the matches that take place between the following pairs of teams.

P – T, P – U, Q – R, Q – S, T – S and R – U

Given that, in Stage-II, three teams lost all the two matches.

Given that P lost both the matches in Stage-II.

∴ Each of T and U won the two matches.

⇒ R and S lost the two matches.

∴ Q also won two matches.

71. T and U defeated P (the top team in Stage-I).

72. Only Q, T and U won both their matches in Stage-II.

73. S and U won exactly two matches in the event.

74. Q and T won exactly four matches each in the event.

Solutions for questions 75 to 78:

75. As Arjun sold gold on three consecutive days while Madhu sold gold on only one day, the only possibility is that the price of one gram at the end of day 1, day 2, day 3, day 4 and day 5 was ₹900, ₹1000, ₹1100, ₹1200 and ₹1100 respectively, i.e., on three consecutive days it increased and on the closing day it closed above ₹1100.

76. As the price at the beginning of the first day was ₹1000 and at the end of the fifth day was ₹1100, it means that the price increased by ₹100 on three days and decreased by ₹100 on two days. Therefore, effectively Arjun sold 10 grams of gold. If Arjun ended up with ₹13,000 more, it means that he sold 10 grams more than Madhu. Therefore, Madhu effectively sold zero grams of gold. As she could not both buy and sell 10 grams of gold in this 5-day period, she did not buy or sell any gold during this 5-day period. Therefore, on day 4, the price (per gram) of gold can only be ₹1000.

77. Since Arjun sold on three days and bought on two days, he would have ended up with 10 grams less than what he started with and since he ended up with 20 grams less than what Madhu had, Madhu would have bought 10 grams and it is possible only if the price of gold (per gram) on the five days are ₹900, ₹800, ₹900, ₹1000 and ₹1100, respectively.

78. If Madhu ended up with ₹1000 less than Arjun at the end of day 5, it can only be because the quantity of gold with them is equal as gold is only bought and sold in multiplies of 10 grams and if the difference in the quantity with them is 10 grams, the difference in amounts with them must be close to ₹10,000.

Solutions for questions 79 to 82:

79. Machine A will take 10 hours (4 hours + 2 hours break + 4 hours), Machine B will start after 4 hours from start when 50 kgs raw material is passed. B will work from fourth hour to 16th hour (4 cycles). Machine C will start from the 7th hour, work for 2 cycles taken a break, work for 3 cycles, break and continue till 20th hour. Machine D should start in 9th hour from start when 20 kgs of raw material reaches from C. Last cycle of D will be 20th to 21st hour. E should start when a minimum of 50 kgs of material resells from D, i.e., at 16th hour. Second cycle of E should start from 21st hour as only then all raw materials will reach C. Last cycle of E will be from 21st to 25th hour.

∴ It will take minimum 25 hours to process 100 kg raw material.

80. The processing rate of A, B, C, D and E including breaks is as follows.

A = 100/10 kg/hr, B =100/12 kg/hr, C =100/13 kg/hr, D = 100/8 kg/hr, E = 100/8 kg/hr

If capacity of A, B, D or E is doubled it wouldn't affect the time taken as the immediate next step is at most of equal speed, if not slower. If capacity of C is changed, there may be a decrease in processing time.

81. By doubling the capacity of C, the total time taken will be 23 hours. [Last cycle of D can complete in 18th to 19th hour and last cycle of E from 19th to 23rd hour].

The minimum time taken with old capacity was 25 hours.

Difference = 2 hours.

82. If the raw material was only 50 kgs. A would be fourth from start till 4th hour, B till 10th hour, C till 13th hour from start, D would work three cycles last been 13th hour to 14th hour. E will work only one cycle from 14th to 18th hour.

Solutions to questions 83 to 86: This given information can be tabulated as follows:

	Morning slot		Afternoon slot	
Day	**Name**	**Subject**	**Name**	**subject**
Monday				
Tuesday	D	Biology	X	X
Wednesday				
Thursday				
Friday	F	Maths		

Since there are 8 lecturers who teach only once during the week, two slots will be free. From (iii), (iv) and (vi), it can be concluded that one of those slots is Tuesday – afternoon slot.

From (vii), Maths should be taught on Wednesday. Since no subject is taught more than once in the same slot, Maths is taught on Wednesday afternoon. Also, from (vii), Physics is taught on both Monday and Wednesday.

∴ Physics is taught on Wednesday–morning slot and Monday–afternoon slot.

From (ii) and (iii), C and F teach the same subject. E and H teach the same subject. B teaches Chemistry and D teaches Biology and no one else teaches Biology or Chemistry.

∴ A and G teach the same subject and C and F teach Maths.

C is scheduled on Wednesday afternoon.

From (i), A is also scheduled on Wednesday.

∴ A teaches Physics on Wednesday – morning.

G also teaches Physics on Monday – afternoon.

From (i), H is also scheduled on Monday.

∴ H teaches English on Monday–morning.

From (ii) and (iii), E also teaches English and nobody else is scheduled to teach on the day E teaches.

∴ E teaches English on Thursday – afternoon and Thursday – morning slot is free.

Finally, B teaches Chemistry on Friday – afternoon.

The resulting table is as follows.

	Morning slot		Afternoon slot	
Day	**Name**	**Subject**	**Name**	**Subject**
Monday	H	English	G	Physics
Tuesday	D	Biology	X	X
Wednesday	A	Physics	C	Maths
Thursday	X	X	E	English
Friday	F	Maths	B	Chemistry

83. E teaches English.

84. Chemistry is not taught during the morning slot.

85. Thursday – morning slot is free.

86. English and Maths are not taught on the same day.

Solutions for questions 87 to 91:

87. If Laxman is included in the team, then the size of the team would be at least four. All others can be a member of a team of size 3.

88. A team of five must include Sanjay, Uday and Watson which means one of Mohan and Qureshi must be a member of the team along with Naveen.

89. The size of the largest possible team is five as exactly two of Pranav, Rahul and Sanjay, exactly one of Mohan and Qureshi and either Naveen and Uday or Kalyan and Laxman must not be selected.

90. The size of a team that includes Kalyan can only be four as Laxman, exactly one of Qureshi and Mohan and exactly one of Pranav and Rahul must be selected.

91. A team which includes Naveen can be constituted in six ways as given.

(1) Mohan, Rahul, Naveen
(2) Qureshi, Rahul, Naveen
(3) Mohan, Pranav, Naveen
(4) Qureshi, Pranav, Naveen
(5) Mohan, Sanjay, Uday, Watson, and Naveen
(6) Qureshi, Sanjay, Uday, Watson and Naveen

Solutions for questions 92 to 96: The following table can be made from the available information.

Games	Kerala	Karnataka	Tamil Nadu	Andhra Pradesh	Total
Athletes			1	1	6
Basketball	0	1	1	1	3
Football			2	1	6
Cricket			1	1	6
Total	4	8	5	4	21

92. Only the number of Cricket players from Kerala cannot be determined.

93. Since Tamil Nadu and Andhra Pradesh had a total of 3 Football players. Kerala and Karnataka should together have a total of 3 Football players.

94. Since Karnataka had a total of 8 sports people selected and if there is only Football player, there should be 3 Athletes and 3 Cricket players.

95. The number of Cricket players is one.

96. None of the given statements is true.

Solutions for questions 97 to 101: It is given that five students, namely Anil, Sunil, Pavan, Naveen and Sravan answered five questions. Each question has five answer choices A, B, C, D and E.

	Anil	Sunil	Pavan	Naveen	Sravan
1	B	C	D	B	C
2	A	D	B	A	C
3	C	A	D	A	D
4	A	B	C		A
5	E	B	B	E	C

As only one student answered the first question correctly, the correct choice for the first question cannot be B or C.

Hence, the correct choice for the first question is choice (D) and that is correctly marked by Pavan.

As only one student answered the second question correctly, the correct choice for the second question cannot be A. The correct choice is B or C or D.

As two students answered the third question correctly, the correct answer is either A or D.

As only one student answered the fourth question correctly, the correct answer is not A, it is either B or C.

As two students answered the fifth question correctly, the correct answer is either B or E.

From (iii), the correct choice for the third question is A.

As Naveen scored the maximum marks, the correct choice for the 5th question is E.

From (iv), the correct choice for the second and the fourth questions is B.

The above information can be represented as follows.

Q. No.	Correct choice	Correctly marked by
1	D	Pavan
2	B	Pavan
3	A	Sunil, Naveen
4	B	Sunil
5	E	Anil, Naveen

The marks scored by them is as follows.

	Questions attempted	Correct	Wrong	Net score
Anil	5	1	4	0
Sunil	5	2	3	5
Pavan	5	2	3	5
Naveen	4	2	2	6
Sravan	5	0	5	–5

97. Sravan scored the least.

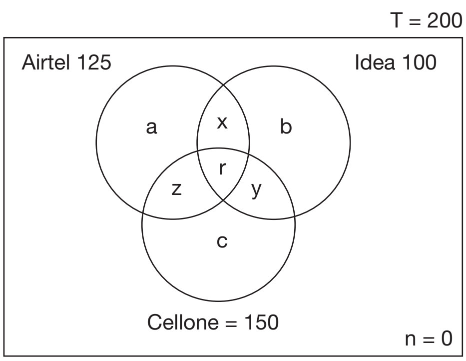
98. Naveen scored 6 marks.

99. Sunil and Pavan scored equal marks.

100. Highest score (Naveen) = 6
Lowest Score (Sravan) = –5
Diff = 6 – (–5)
= 6 + 5
= 11

101. The correct choice for question 4 is B. Naveen marked it as either C or D. In both the cases he get '– 1' score. Hence, his net score becomes 5.
Naveen, Pavan and Sunil are the highest scorers with 5 marks each.
Anil is the second highest with a score of zero.

Solutions for questions 102 to 104:



50 members do not subscribe to Cellone which implies that 150 subscribe to Cellone.

$$a + b + x = 50 \qquad (1)$$

75 do not subscribe to Airtel which implies that 125 subscribe to Airtel.

$$b + y + c = 75 \qquad (2)$$

100 do not subscribe to Idea which implies that 100 subscribe to Idea.

$$a + z + c = 100 \qquad (3)$$

125 subscribe to at least two companies.

$$x + y + z + r = 125 \qquad (7)$$

Airtel subscribers = $a + x + r + z = 125$ (4)

Idea subscribers = $b + x + r + y = 100$ (5)

Cellone subscribers = $c + z + r + y = 150$ (6)

(4) + (5) + (6)

$\Rightarrow$ $a + b + c + 2(x + y + z) + 3r = 375$ (8)

$a + b + c + x + y + z + r = 200$ (9)

Subtracting (9) from (8), we get:

$$x + y + z + 2r = 175 \qquad (10)$$

Subtracting (10) from (7), we have:

$$r = 50$$

Hence, $x + y + z = 75$.

(1) + (2) + (3) $\Rightarrow 2\ (a + b + c) + x + y + z = 225$

But $x + y + z = 75$

$\Rightarrow 2\ (a + b + c) = 150$

$\Rightarrow a + b + c = 75$

102. 75 customers subscribe to exactly one service.

103. 75 customers subscribe to exactly two services.

104. 50 customers subscribe to exactly three services.

Solutions for questions 105 to 109: Given that the number of notes they had are 2, 6, 8, 9 and 12, and the number of coins they had are 0, 3, 4, 8 and 12. Now, from (vi), we know that the number of coins Manjeet had is half of the number of coins somebody else had. This is possible only when if he had 4 coins. Now, condition (i) will satisfy only when Harish had 12 coins (thrice of 4). Thus, Manjeet had six notes. From (vii), we can say that Sohan had 8 coins. Also, only the pair 8 and 12 (notes) satisfy the condition (vii). Thus, Sohan had 12 notes and Harish had 8 notes.

Now, from (ii), (iii) and (vi) we can say that Dinesh, Manjeet and Sohan do not keep a wallet. Hence, Harish and Arpit had a wallet.

From (iv) and (v), we can say that Dinesh had two notes and no coins.

Thus, Arpit had 9 notes and 3 coins.

So, we get the following table:

Person	Notes	Coins	Wallet
Harish	8	12	✓
Dinesh	2	0	✗
Manjeet	6	4	✗
Arpit	9	3	✓
Sohan	12	8	✗

105. Sohan had 8 coins is true.

106. Arpit had 9 notes is true.

107. Manjeet having 6 notes, 4 coins but no wallet is true.

108. Arpit has three coins and nine notes, i.e., 1 : 3.

109. Only choice (D) is true.

Solutions to questions 110 to 113: Exactly one course should be picked from Group 1. Therefore, the number of credits that can be picked from Group 1 are 1 or 2 or 3.

Exactly 2 courses are to be picked from Group 2. Therefore, the numbers of credits that can be picked from Group 2 are

3 + 4/4 + 4/5 +3/5 + 4.

∴ 7/8/9 credits can be picked from Group 2. 8 credits can be picked in 2 ways. 7 credits can be picked in 2 ways. 9 credits can be picked in 2 ways. Exactly 2 courses are to be picked from Group 3. Therefore, the numbers of credits that can be picked from Group 3 are 3 + 3/3 + 5.

∴ 6/8 credits can be picked from Group 3.

6 credits can be picked in 3 ways.

8 credits can be picked in 3 ways.

110. Manju picked 17 credits but did not choose Hindi. Therefore, her possible credits from each Group are:

	Group 1	Group 2	Group 3
Case (i)	1	8	8
Case(ii)	3	8	6

Choice (A): If she picks French, then as per case (i), she has to pick 8 credits from Group 2. It is possible to pick Physics (along with Biology).

Choice (B): If she picks English, as per case (ii), she has to pick 8 credits from Group 2. She can choose Chemistry and Maths.

Choice (C): If she picks French, as per case (i), she needs 8 credits from Group 3. She can pick Sports and Music.

Choice (D) If she picks English, as per case (ii), she needs to pick 6 credits from Group 3. Therefore, she cannot pick Sports.

111. Ajay has already picked French and Chemistry. Group 1 is done. If he wants to maximize his credits, he has to pick Maths from Group 2 and Sports from Group 3 along with any one of the other three (Music/Dance/Arts and Crafts).

Choice (A), (B), and (D) are possible.

Choice (C) is not possible since Biology cannot be picked from Group 2.

112. Surabhi has chosen 18 credits. The possible combinations are:

	Group 1	Group 2	Group 3
Case (i)	1	9	8
Case (ii)	2	8	8
Case (iii)	3	7	8
Case (iv)	3	9	6

Choice (A): As per case (i), if she chooses French, she needs 9 credits from Group 2. It is not possible to get 9 credits by picking Chemistry.

Choice (B): As per case (ii), if she chooses Hindi, she needs 8 credits from Group 2. She can choose Biology and Physics.

Choice (B) is possible.

Choice (C): As per case (iii) or (iv), if she chooses English, she will need either 7 or 9 credits from Group 2. She can pick Maths and either Physics or Biology to get 9 credits.

Choice (C) is also possible.

∴ Choice (D) is the answer.

113. Madhuri needs 19 credits and she has already picked Maths and Music. The possible numbers of credits she can choose from each group are:

	Group 1	Group 2	Group 3
Case (i)	2	9	8
Case (ii)	3	8	8

Choice (A): As per case (i) if she picks Hindi, she needs 9 credits from Group 2. Since Madhuri has already picked Maths, choosing Chemistry will only give her 8 credits from Group 2. This choice is not possible.

Choice (B): As per case (ii), if she picks English, she needs 8 credits from Group 2. Since Maths is already chosen, she cannot choose Biology. This choice is also not possible.

Choice (C): As per case (ii), if she picks English, she needs 8 credits from Group 2. She has already chosen Maths. Therefore, Chemistry can be picked. This choice is possible.

Choice (D): French cannot be picked either in case (i) or in case (ii).

Solutions for questions 114 to 117: From the statements, we can make out that there are ten different amounts and each is the combined earnings of two people. Since there are five people and taking two at a time we can make ${}^5C_2$ or $(5 \times 4)/2$, i.e., 10 amounts. Since there are ten amounts given, it means they are all possible combinations of the five people taking two at a time. Also, since none of the ten amounts are equal, we can conclude that no two out of the five people had the same amount. Let us denote the five amounts with the five of them as a, b, c, d and e in descending order, i.e., a is the largest amount and e is the smallest amount. When we make combinations of two at a time out of five given items, each item occurs in four combinations. So, if we add up all the ten amounts, it should be equal to four times the sum of five amounts, i.e., $(a + b + c + d + e)$. If we add all the amounts given, it comes to ₹9600. Hence,

$$a + b + c + d + e = 2400 \qquad \text{(i)}$$

Since a, b, c, d and e are in descending order, among the amounts that we can make taking two at a time, $a + b$ will be the largest, $a + c$ will be the second largest, $d + e$ will be the smallest and $c + e$ will be the second smallest. From this data and looking at the amounts given, we have:

$$a + b = 1500 \qquad \text{(ii)}$$

$$a + c = 1400 \qquad \text{(iii)}$$

$$d + e = 400 \qquad \text{(iv)}$$

$$c + e = 600 \qquad \text{(v)}$$

From the five equations (i), (ii), (iii), (iv) and (v), we can very easily get the values of the five variables [For example: Add (ii) and (iv) together and subtract it from (i) to get c] as

$$a = 900;\ b = 600;\ c = 500;\ d = 300;\ e = 100$$

Once we know the five amounts, we proceed as below (by taking the statements given regarding the amounts earned) to identify the names, designations, etc. As we get all the required information one by one, we shall fill up the following table.

Amount	Name	Division	Designation	Dress
900 (a)	Saxena	Engineering	Production Manager	Safari
600 (b)	Mathur	Foods	General Manager	3-piece
500 (c)	Sastry	Oil Seeds	Deputy General Manager	3-piece
300 (d)	Verma	International Trading	Senior Vice President	Churidar kurta
100 (e)	Rao	Soaps	Vice President	2-piece

Since Mathur and the person from Engineering division together won 1500 and since we know $a + b = 1500$, we can then write:

$$\text{Mathur + Engineering division} = a + b \qquad (1)$$

Similarly, since Sastry and Saxena together earned 1400 and we know that $a + c = 1400$, we can write:

$$\text{Sastry + Saxena} = a + c \qquad (2)$$

By comparing the two equations (1) and (2), we find that 'a' is common on the RHS in both equations. Then looking at the LHS, because Mathur is not common, we can conclude that Mathur is 'b' and 'a' belongs to the person from Engineering division. Fill this information in the appropriate cells in the table.

Similarly, Saxena and the Senior Vice President earn 1200 together and we know $a + d = 1200$, we can write:

$$\text{Saxena + Senior Vice President} = a + d \qquad (3)$$

From equations (2) and (3), where 'a' is common on the RHS of both equations and Saxena common on the LHS of both equations, we can conclude that 'a' belongs to Saxena. Then from equation (2), 'c' belongs to Sastry. Form equation (3), 'd' belongs to the Senior Vice President. This information also should be filled in the table.

In this manner, we can write down the equations for all the other seven amounts also and make similar conclusions and fill in the table completely.

$$\text{Verma + Production Manager} = a + e \qquad (4)$$

$$\text{General Manager + International Trading division} = b + d \qquad (5)$$

$$\text{Foods division + Soaps division} = b + e \qquad (6)$$

$$\text{Vice President + Oil Seeds division} = c + e \qquad (7)$$

$$\text{Rao + Soaps division} = d + e \qquad (8)$$

$$\text{Deputy General Manager + Churidar} = c + d \qquad (9)$$

$$\text{3-piece suit + 3-piece suit} = b + c \qquad (10)$$

From (4), we get Production Manager is 'a' (since Verma cannot be 'a') and hence, Verma is 'e'. Thus Rao will be 'd' who is the only person left out. From (8), Soaps division is 'e'; from (6), Foods division is 'b'; from (7), VP is 'e' (because Oil Seeds division cannot be 'e') and Oil seeds division is 'c'; from (5), International Trading division is 'd' (because 'd' cannot be GM since he is already Senior Vice President) and so 'b' is General Manager; the only slot left for Deputy General Manager is 'c'. Hence, from (9) we get 'd' as Churidar-kurta; from (10) 'b' and 'c' are 3-piece suit.

The last condition in the problem states that Safari-suit person won more than the person in 2-piece suit; the only two people left are 'a' and 'e'; so 'a' is Safari-suit and 'e' is 2-piece suit. This completes the table.

Once we have the table filled, we can answer the questions easily.

114. Only I and III are false.

115. None of these.

116. All the three statements are not true.

117. Saxena won the maximum amount.

Solutions for questions 118 to 121: From the facts about M1 we can observe that M1 was the first to start. After then, M3 and M4 started. M2 and M7 were started before M1 and M5 were stopped. Also, we can say that M3 and M4 were stopped before M1 was stopped.

From the facts about M3, we can say that M3 was started after M4 and M5 were started.

From the facts known of M4, we can say that M3 was the first to be stopped and M4 was the next.

Now, from the facts of M2, we can say that M2 was started with M7 and at that time M3 and M4 had already stopped.

The information is tabulated as follows

Machines	Order of starting	Machines running with it	Order of stopping
M1	1	M2, M3, M4, M5, M7	
M2	5 (with M7)	M1, M5	
M3	3	M1, M4, M5	1
M4	2	M1, M3, M5	2
M5	4	M1, M3, M4	
M6			
M7	5(with M2)	M2, M7	

Now, from the information about M5, we can say that at the first time it was stopped, M2 and M7 were running. And when it was started again only M2 and M7 were running. Also, from the information known about M6 as it was operated with M5, we can say that no other machine was started or stopped when M5 was not running (between its two shifts of operations).

Now, from the information known about M6 we can say that M7 was stopped before M6 was started. After that M6 was stopped. Thus, M2 and M5 were the last to stop.

Thus, we get the following arrangement:

Machine	Order of starting	Machines running with it	Order of stopping
M1	1	M2, M3, M4, M5, M7	3 (with M5)
M2	5(with M7)	M1, M5, M6, M7	7/6
M3	3	M1, M4, M5	1
M4	2	M1, M3, M4	2
M5 (A)	4	M1, M2, M3, M4, M7	3 (with M1)
M5 (B)	6	M2, M6, M7	6/7
M6	7	M2, M5	5
M7	5(With M2)	M1, M5, M2)	4

118. M5 was started first.

119. M1 and M5 were running at the time M2 was started.

120. Three machines ___________ 00M1, M5 and M2 were operated with M7.

121. M2 and M5 were the last to stop.

Solutions for questions 122 to 125:

122. Only P, R and T who co-authored a paper with U had a change in their Einstein numbers. The other 5 participants did not have a change in their Einstein number.

123. It is given that on the third day, U co-authored a paper jointly with P and R. They would have reduced the Einstein number of P and R to $x + 1$ where x was U's Einstein number. As P had an infinite Einstein number at the beginning and as no co-authorship among any three other numbers would have made the average Einstein number to as low as 3, it means R initially did not have an Einstein number $x + 1$.
Now, since it is given that five of them had the same Einstein number at the end of day 3 and that the average of all the eight was 3 and as the three (other than the five) had Einstein number distinct from each other, the only possibility is that U had an Einstein number 1, five of them including P and R had an Einstein number 2 and the other two had an Einstein number which added up to $24 – 10 – 1$, i.e., 13.
Now as it is given, that T co-authored a paper with U, the average of the group reduced by 0×5, i.e., the total reduced by $0 \times 5 \times 8 = 4$ and that of T became 2 or it was initially $4 + 2 = 6$, which means the eighth person had an Einstein number $13 – 6$, i.e., 7.

124. Only three of the five participants who had an Einstein number of 2 at the end of day 3 had the same Einstein number at the beginning of the conference.

125. At the end of the third day each of P and R, have Einstein number $x + 1$, where x is the Einstein number of U. It is given that five scientists have the same number at the end of the third day it implies that three among Q, S, T, V and W also have $x + 1$, at the beginning of day one. But for these three, the rest of the five can have distinct Einstein numbers.

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Unit 4

OMET-based LR

1 Series

CHAPTER

Learning Objectives

In this chapter, you will:

- Learn about Number series and Letter series
- Learn about different types of series, like
  - Difference series
  - Product series
  - Squares/Cubes series
  - Miscellaneous series
  - Combination of two or more series types
- Learn different ways to identify patterns and find the missing term in a series
- Learn how to convert alphabets into their equivalent number and vice-versa

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❒ SERIES

Number and Letter Series form an important part of the 'Reasoning' section in various competitive examinations. There are two or three broad categories of questions that appear in various exams from this particular chapter.

In the first category of questions, a series of numbers/letters is given with one number/letter (or two numbers/letters) missing, represented by a blank or a question mark. The given series of numbers/letters will be such that each one follows its predecessor in a certain way, i.e., according to a definite pattern. Students are required to find out the way in which the series is formed and hence, work out the missing number/numbers or letter/letters to complete the series. Under these questions, there are a large variety of patterns that are possible and the student requires a proper understanding of various patterns to be able to do well in these types of questions.

Number Series

For better understanding, we will classify this into the following broad categories.

1. Difference series
2. Product series
3. Squares/Cubes series
4. Miscellaneous series
5. Combination series

Difference Series

The difference series can be further classified as follows.

1. Number series with a constant difference.
2. Number series with an increasing or decreasing difference.

In the number series with a **constant difference**, there is always a constant difference between two consecutive numbers. For example, the numbers of the series 1, 4, 7, 10, 13, … are such that any number is obtained by adding a constant figure of 3 to the preceding term of the series.

If we have to find the next number in the above series, we need to add 3 to the last term 13. Thus, 16 is the next term of the series.

Under the series with constant difference, we can have series of odd numbers or series of even numbers also.

In the series with **increasing/decreasing difference**, the difference between consecutive terms keeps increasing (or decreasing, as the case may be). For example, let us try to find out the next number in the series 2, 3, 5, 8, 12, 17, 23, …

Here, the difference between the first two terms of the series is 1; the difference between the second and third terms is 2; the difference between the third and the fourth terms is 3 and so on. That is, the difference between any pair of consecutive terms is one more than the difference between the first number of this pair and the number immediately preceding this number. Here, since the difference between 17 and 23 is 6, the next difference should be 7. So, the number that comes after 23 should be (23 + 7) = 30.

We can also have a number series where the difference is in decreasing order (unlike in the previous example where the difference is increasing). For example, let us find out the next term of the series 10, 15, 19, 22, 24, …

$$10 \xrightarrow{+5} 15 \xrightarrow{+4} 19 \xrightarrow{+3} 22 \xrightarrow{+2} 24$$

Here the differences between 1st and 2nd, 2nd and 3rd, 3rd and 4th numbers, etc., are 5, 4, 3, 2, and so on. Since the difference between 22 and 24 is 2, the next difference should be 1. So, the number that comes after 24 should be 25.

Product Series

A product series is usually a number series where the terms are obtained by a process of multiplication. Here also, there can be different types of series. We will look at these through examples.

Consider the series 2, 4, 8, 16, 32, 64, …

$$2 \xrightarrow{\times 2} 4 \xrightarrow{\times 2} 8 \xrightarrow{\times 2} 16 \xrightarrow{\times 2} 32 \xrightarrow{\times 2} 64$$

Here, each number in the series is multiplied by 2 to get the next term. So, the term that comes after 64 is 128. **So, each term is multiplied by a fixed number to get the next term.** Similarly, we can have a series where we have numbers obtained by **dividing** the previous term with a constant number. For example, in the series 64, 32, 16, 8, …, each number is obtained by dividing the previous number by 2 (or in other words, by multiplying the previous term by $^1/_2$). So, here, the next term will be 4 (obtained by dividing 8 with 2).

Consider the series 4, 20, 80, 240, …

$$4 \xrightarrow{\times 5} 20 \xrightarrow{\times 4} 80 \xrightarrow{\times 3} 240$$

Here, the first term is multiplied by 5 to get the second term; the second term is multiplied by 4 to get the third term; the third term is multiplied by 3 to get the fourth term. Hence, to get the fifth term, we have to multiply the fourth term by 2, i.e., the fifth term is 480. **So, each term is multiplied by a decreasing factor (or it could also be an increasing factor) to get the next term.** That is, with whatever number a particular term is multiplied to get the next term, this latest term is multiplied by a number different from the previous multiplying factor to get the next term of the series. All the multiplying factors follow a certain pattern (normally of increasing or decreasing order).

Consider the series 2, 6, 12, 20, 30, …

$$2 \xrightarrow{+4} 6 \xrightarrow{+6} 12 \xrightarrow{+8} 20 \xrightarrow{+10} 30$$

This can be looked at a series of increasing differences. The differences of consecutive pairs of terms are 4 (between 2 and 6), 6 (between 6 and 12), 8 (between 12 and 20), 10 (between 20 and 30) and so on. Hence, the difference between 30 and the next term should be 12 and so the next term will be 42. But this series can also be looked at as a product series.

$$\begin{array}{ccccc} 2, & 6, & 12, & 20, & 30 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 1 \times 2 & 2 \times 3 & 3 \times 4 & 4 \times 5 & 5 \times 6 \end{array}$$

The first term is the product of 1 and 2; the second term is the product of 2 and 3; the third term is the product of 3 and 4; the fourth term is the product of 4 and 5; the fifth term is the product of 5 and 6. Hence, the next term will be the product of 6 and 7, that is 42.

Squares/Cubes Series

There can be series where all the terms are related to the squares of numbers or cubes of numbers. With squares/cubes of numbers as the basis, there can be many variations in the pattern of the series. Let us look at the various possibilities of series based on squares/cubes.

Each term of the series may be the square of a natural number, such as 1, 4, 9, 16, …

$$\begin{array}{cccc} 1, & 4, & 9, & 16 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1^2 & 2^2 & 3^2 & 4^2 \end{array}$$

The numbers are squares of 1, 2, 3, 4 …, respectively. The number which follows 16 (which is the square of 4) will be 25 (which is the square of 5).

The terms of the series may be the squares of odd numbers (For example: 1, 9, 25, 49, …) or even numbers (For example: 4, 16, 36, 64, …).

The terms of the series could be such that a number and its square are both given one after the other and such pairs are given in some specific pattern. For example, take the series 2, 4, 3, 9, 4, 16, …

$$\begin{array}{cccccc} 2, & 4, & 3, & 9, & 4, & 16, \\ & +1 & & & +1 & \end{array}$$

Here, 2 is followed by its square 4; then comes the number 3 (which is one more than 2) followed by its square 9 and so on. Hence, the next number in the series is 5 and the one after that is its square, i.e., 25.

Similarly, each term could be the square root of its predecessor. For example, in the series 81, 9, 64, 8, 49, 7, 36, …, 81 is the square of 9, 64 the square of 8, and so on. Therefore, the next number which follows in the series should be the square root of 36, i.e., 6.

The terms of the series could be the squares of natural numbers increased or reduced by certain number. For example, in the series 3, 8, 15, 24, …

$$\begin{array}{cccc} 3, & 8, & 15, & 24 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 2^2 - 1 & 3^2 - 1 & 4^2 - 1 & 5^2 - 1 \end{array}$$

We have {Squares of natural numbers - 1} as the terms. The first term is $2^2 - 1$; the second term is $3^2 - 1$; the third term is $4^2 - 1$ and so on. Hence, the next term will be 6^2 - 1, i.e., 35 [Please note that the above series can also be looked at as a series with increasing differences. The differences between the 1st and 2nd terms, the 2nd and 3rd terms, and so on are 5, 7, 9, and so on. Hence, the next difference should be 11 giving us the next term as 35]. There could also be a series with (squares of natural numbers + some constant).

Like we have seen series with squares of numbers, we can have similar series with cubes of numbers. For example, take the series 1, 8, 27, 64, ...

$$\begin{array}{cccc} 1, & 8, & 27, & 64 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1^2 & 2^3 & 3^3 & 4^3 \end{array}$$

Here, all the terms are cubes of natural numbers. So, the next term will be 5^3, i.e., 125.

Consider the series 2, 9, 28, 65, …

$$\begin{array}{cccc} 2, & 9, & 28, & 65 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1^3 + 1 & 2^3 + 1 & 3^3 + 1 & 4^3 + 1 \end{array}$$

Here, the terms are (Cubes of natural numbers + 1). The first term is $1^3 + 1$; the second term is $2^3 + 1$; the third term is $3^3 + 1$ and so on. Hence, the next term will be $5^3 + 1$, i.e., 126.

Miscellaneous Series

There are series that do not come under the other patterns and are of general nature but are important and are fairly common. Even here, sometimes, there can be a specific pattern in some cases.

Take the series 3, 5, 7, 11, 13, … This is a series of consecutive PRIME NUMBERS. It is an important series and the student should look out for this as one of the patterns. The next term in this series is 17.

There can also be variations using prime numbers. Take the series 9, 25, 49, 121, …. In this series, the terms are squares of prime numbers. Hence, the next term is 13^2, i.e., 169.

Take the series 15, 35, 77, …. The first term is 3×5; the second term is 5×7; the third term is 7×11. Here, the terms are product of two consecutive prime numbers. So, the next term will be the product of 11 and 13, i.e., 143.

Take the series 8, 24, 48, 120, 168, …. Here, the 2nd term is 3 times the first term and the 3rd term is 2 times the 2nd term, but after that it does not follow this pattern any more. If you look at the terms carefully, you will find that the terms are (one less than squares of prime numbers). Hence, the next term will be $17^2 - 1$, i.e., 288.

Consider the series 1, 4, 9, 1, 6, 2, 5, 3, ….

At first sight there is nothing we can say about the series. This is actually a series formed by squares of natural numbers. However, if any of the squares is in two or more digits, each of the digits is written as a separate term of the series. Thus, the first terms are 1, 4 and 9, the squares of 1, 2 and 3, respectively. After this, we should get 16 (which is the square of 4). Since this has two digits 1 and 6, these two digits are written as two different terms 1 and 6 in the series. Similarly, the next square 25 is written as two different terms 2 and 5 in the series. So, the next square 36 should be written as two terms 3 and 6. Of these, 3 is already given. So, the next term of the series is 6.

Consider the series 1, 1, 2, 3, 5, 8, ….

$$\begin{array}{cccccc} 1, & 1, & 2, & 3, & 5, & 8 \\ & & \downarrow & \downarrow & \downarrow & \downarrow \\ & & 1+1 & 1+2 & 2+3 & 3+5 \end{array}$$

Here, each term, starting with the third number, is the sum of the two preceding terms. After taking the first two terms as given (1 and 1), then onwards, to get any term, we need to add the two terms that come immediately before that position. Hence, to get the next term of the series, we should take the two preceding terms 5 and 8 and add them up to get 13. So, the next term of the series is 13. The term after this will be 21 (= 8 + 13).

Combination Series

A number series which has more than one type of (arithmetic) operation performed or more than one series combined together is a combination series. The series that are combined can be two series of the same type or could be different types of series as described above. Let us look at some examples.

First let us look at those series which are formed by more than one arithmetic operation performed on the terms to get the subsequent terms.

Consider the series: 2, 6, 10, 3, 9, 13, 4, 12, …. Here, the first term 2 is multiplied by 3 to get the second term, and 4 is added to get the third term. The next term is 3 (one more than the first term 2) and it is multiplied by 3 to get 9 (which is the next term) and then 4 is added to get the next term 13. The next term 4 (which is one more than 3) which is multiplied with 3 to get 12. Then 4 is added to this to get the next number 16.

Consider the series: 1, 2, 6, 21, 88, …. Here, we can observe that 88 is close to 4 times 21. It is in fact $21 \times 4 + 4$. So, if we now look at the previous term 21, it is related to the previous term 6 as $6 \times 3 + 3$. Now we get the general pattern, to get any term, multiply the previous term with k and then add k where k is a natural number with values in increasing order from 1. So, to get the second term, the first term has to be multiplied with 1 and then 1 is added. To get the third term, the second term is multiplied with 2 and then 2 is added and so on. Hence, after 88, the next term is $88 \times 5 + 5$, i.e., 445.

Now, let us look at a series that is formed by combining two (or more) different series. The two (or more) series can be of the same type or of different types described above.

Consider the series: 8, 12, 9, 13, 10, 14, …. Here the 1st, 3rd, 5th, ... terms which are 8, 9, 10, …. form one series whereas the 2nd, 4th, 6th, etc., terms which are 12, 13, 14 form another series. Here, both series that are being combined are two simple constant difference series. Therefore, the missing number will be the next term of the first series 8, 9, 10, … which is equal to 11.

Consider the series: 0, 7, 2, 17, 6, 31, 12, 49, 20, …. Here, the series consisting of 1st, 3rd, 5th, … terms (i.e., the series consisting of the odd terms) which is 0, 2, 6, 12, 20, … is combined with another series consisting of 2nd, 4th, 6th, … terms (i.e., the series consisting of the even terms) which is 7, 17, 31, 49, …. The first series has the differences in increasing order 2, 4, 6, 8, 10 and so on. The second series also has the difference in increasing order 10, 14, 18, …. Since, the last term 20 belongs to the first series, a number from the second series should follow next. The next term of the second series will be obtained by adding 22 to 49, i.e., 71.

Consider the series: 1, 1, 4, 8, 9, 27, …. Here, the series of squares of natural numbers is combined with the series of cubes of natural numbers. The next term in the series will be $4^2 = 16$.

Consider the series: 2, 4, 5, 9, 9, 16, 14, ?, 20, …. Here, we have to find out the term that should come in place of the question mark. The odd terms form one series 2, 5, 9, 14, 20, … where the difference is increasing. The differences are 3, 4, 5, 6, …. This series is combined with the series of even terms 4, 9, 16, … where the terms are squares of numbers 2, 3, 4, …. Hence, the term that should come in place of the question mark is the next term of the second series which is 5^2, i.e., 25.

A General Approach to Number Series

The best way of approaching the number series questions is to first observe the difference between terms. If the difference is constant, it is a constant difference series. If the difference is increasing or decreasing by a constant number, then it is a series with a constant increasing or decreasing difference. If there is no constant increasing or decreasing difference, then try out the product series approach. For this, first divide the second term with the first term, third with the second and so on. If the numbers obtained are the same, then it is a product series. Alternatively, try writing each term of the series as a product of two factors and see if there is any pattern that can be observed. If still there is no inference, but the difference is increasing or decreasing in a rapid manner, then check out the square series. If the increase is very high, and it is not a square series, then try out the cube series.

If the difference is alternately decreasing and increasing (or increasing for some time and alternately decreasing), then it should most probably be a mixed series. Therefore, test out the series with alternate numbers. If still the series is not solved, then try out the general series.

Letter Series

The questions here are similar to the questions in Number Series Type I. Instead of numbers we have letters of the alphabet given here. We have to first identify the pattern that the series of letters follow. Then, we have to find the missing letter based on the pattern already identified. In Letter Series, in general, we have a series with constant or increasing or decreasing differences. The position of the letters in the English alphabet is considered to be the value of the alphabet in questions on Letter Series. Also, when we are counting, after we count from A to Z, we again start with A, i.e., we treat the letters as being cyclic in nature. Like in Number Series, in this type of Letter Series also, we can have a 'combination' of series, i.e., two series are combined and given. We need to identify the pattern in the two series to find out the missing letter. Sometimes, there will be some special types of series also. Let us look at a few examples to understand questions on Letter Series.

SOLVED EXAMPLES

1. Find the next letter in the series.
D, G, J, M, P, ____.

(A) Q (B) R
(C) S (D) T

Sol: Three letters are added to each letter to get the next letter in the series.
D^{+3}, G^{+3}, J^{+3}, M^{+3}, P^{+3}, $\underline{S}$
P + 3 and P = 16 and 16 + 3 = 19 and the 19th letter in the alphabet is S.

2. Find the next letter in the series.
A, B, D, H, ____.

(A) L (B) N
(C) R (D) P

Sol: Each letter in the given series is multiplied with 2 to get the next letter in the series.
A × 2 ⇒ 1 × 2 = 2 and the 2nd letter is B. B × 2 ⇒ 2 × 2 = 4 and the 4th letter is D.
Similarly, H × 2 ⇒ 8 × 2 = 16 and the 16th letter is P.

3. What is the next letter in the series?
B, D, G, K, P, ____

(A) S (B) V
(C) W (D) X

Sol: B^{+2}, D^{+3}, G^{+4}, K^{+5}, P^{+6}, ______
P + 6 = 16 + 6 = 22 and the 22nd letter is V.

4. I, X, J, W, K, V, L,____.

(A) M (B) U
(C) S (D) T

Sol: The given series is an alternate series.
I^{+1}, J^{+1}, K^{+1}, L is one series and X^{-1}, W^{-1}, V^{-1}, ____ is the other series.
X – 1 = W, W – 1 = V and V – 1 = 22 – 1 = 21 and the 21st letter is U.

EXERCISES

Directions for questions 1 to 5: Complete the following series.

1. 10, 130, 1430, 10010, 50050, _____
(A) 100100 (B) 110100
(C) 150150 (D) 150100

2. 361, 529, 841, 961, _____
(A) 1249 (B) 1269
(C) 1349 (D) 1369

3. 50, 51, 77.5, 156, 391, _____
(A) 1173 (B) 1174
(C) 1175 (D) 1369.5

4. 20, 35, 60, 105, 190, _____
(A) 365 (B) 360
(C) 355 (D) 350

5. 8, 4, 4, 6, 12, 30, _____
(A) 75 (B) 95
(C) 90 (D) 105

Directions for questions 6 to 10: Find the wrong number in the series.

6. 12, 33, 55, 84, 114, 147, 183
(A) 55 (B) 84
(C) 33 (D) 147

7. 1850, 1050, 650, 400, 350, 300, 275
(A) 650 (B) 350
(C) 400 (D) 275

8. 12, 31, 72, 95, 114, 131, 144
(A) 31 (B) 12
(C) 72 (D) 114

9. 24, 25, 29, 36, 54, 79, 115
(A) 24 (B) 29
(C) 79 (D) 36

10. 5, 10, 17, 33, 65, 129, 257
(A) 5 (B) 10
(C) 17 (D) 65

Directions for questions 11 to 15: In each of these questions a number series is given. After the series, a number is given along with (a), (b), (c), (d) and (e). You have to complete the series starting with the number given to find the values of (a), (b), (c), (d) and (e) applying the same pattern followed in the given series. Then answer the questions given below.

11. 4, 40, 89, 153, 234, 334
20 (a) (b) (c) (d) (e)
What is the value of (d) in the series?
(A) 169 (B) 250
(C) 196 (D) 269

12. 24, 12, 12, 18, 36, 90, 270
32 (a) (b) (c) (d) (e)
What is the value of (b) in the series?
(A) 16 (B) 24
(C) 75 (D) 48

13. 5, 6, 14, 45, 184, 925, 5556
10 (a) (b) (c) (d) (e)
What is the value of (a) in the series?
(A) 24 (B) 75
(C) 9 (D) 11

14. 0.75, 3, 12, 48, 192, 768, 3072
5 (a) (b) (c) (d) (e)
What is the value of (e) in the series?
(A) 1280 (B) 4680
(C) 5120 (D) 80

15. 16, 52, 84, 112, 136, 156, 72
30 (a) (b) (c) (d) (e)
What is the value of (c) in the series?
(A) 66 (B) 30
(C) 126 (D) 80

Directions for questions 16 to 20: In each of the following series two wrong numbers are given, out of which one differs by a margin of 1, i.e., + 1 or −1 and the other with a greater margin. From the given choices choose the number that differs by the greater margin. The first and the last numbers in the series are always correct.

16. 434, 629, 774, 874, 938, 972, 990
(A) 629 (B) 774
(C) 874 (D) 972

17. 5, 7, 18, 73, 499, 5487, 71314
(A) 7 (B) 18
(C) 73 (D) 499

18. 5300, 4300, 3571, 3061, 2716, 2501, 2375
(A) 4300 (B) 3571
(C) 3061 (D) 2716

19. 21491, 3071, 511, 102, 30, 9, 4
(A) 9 (B) 30
(C) 102 (D) 509

20. 201600, 100800, 33605, 8400, 1679, 28040
(A) 280 (B) 1679
(C) 8400 (D) 33605

Directions for questions 21 to 30: Complete the following series.

21. MTD, NSA, PVE, PVC, SXF, RYE, _____
(A) VXY (B) VZG
(C) UVW (D) UWV

22. ABFL, BDLX, CFRJ, DHXV, ____
(A) EICH (B) EJDI
(C) EJDH (D) FICH

23. DMP, IOM, ___, SSG, XUD
(A) MPI (B) MQI
(C) NPJ (D) NQJ

24. X, P, J, F, ___
(A) A (B) B
(C) C (D) D

25. IJLO, STVY, CDFI, MNPS, ___
(A) WYAB (B) WWAC
(C) WXZC (D) WXAC

26. EUILN, DWFPI, ___, CXEQH, GSKJP
(A) FTJKO (B) FUILM
(C) EULIM (D) ETJKO

27. C1B, G12E, K3H, O26K, ____
(A) T5O (B) U5P
(C) S5N (D) R5M

28. $\frac{C}{12}, \frac{E}{30}, \frac{G}{56}, \_\_\_, \frac{K}{132}, \frac{M}{182}$

(A) $\frac{J}{80}$ (B) $\frac{I}{80}$

(C) $\frac{J}{90}$ (D) $\frac{I}{90}$

29. XTV, UQS, RNP, OKM, ___
(A) HLJ (B) JHL
(C) HJL (D) LHJ

30. LMZ, NYO, XPQ, RSW, ___
(A) TUV (B) TVU
(C) VTU (D) SVQ

ANSWER KEYS

1. (C)	**6.** (A)	**11.** (B)	**16.** (D)	**21.** (B)	**26.** (A)
2. (D)	**7.** (C)	**12.** (A)	**17.** (B)	**22.** (C)	**27.** (C)
3. (B)	**8.** (A)	**13.** (D)	**18.** (C)	**23.** (D)	**28.** (D)
4. (C)	**9.** (D)	**14.** (C)	**19.** (B)	**24.** (D)	**29.** (D)
5. (C)	**10.** (B)	**15.** (C)	**20.** (D)	**25.** (C)	**30.** (B)

SOLUTIONS

Solutions for questions 1 to 5:

1. The given logic is as follows.

$10^{\times 13}, 130^{\times 11}, 1430^{\times 7}, 10010^{\times 5}, 50050^{\times 3}, \underline{150150}.$

2. The given logic is as follows.

361,	529,	841,	961,	$\underline{1369}$.
$(19)^2$	$(23)^2$	$(29)^2$	$(31)^2$	$(37)^2$

3. The given logic is as follows.

$50^{(\times 1+1)}, 51^{(\times 1.5+1)}, 77.5^{(\times 2+1)}, 156^{(\times 2.5+1)}, 391^{(\times 3+1)}, \underline{1174}.$

4. The given logic is as follows.

$20^{(\times 2\, -\, 5)}, 35^{(\times 2\, -\, 10)}, 60^{(\times 2\, -\, 15)}, 105^{(\times 2\, -\, 20)}, 190^{(\times 2\, -25)}, \underline{355}.$

5. The given logic is as follows.

$8^{\times 0.5}, 4^{\times 1}, 4^{\times 1.5}, 6^{\times 2}, 12^{\times 2.5}, 30^{\times 3}, \underline{90}.$

Solutions for questions 6 to 10:

6. The given logic is as follows.

$12^{+21}, 33^{+24}, \underline{57}^{+27}, 84^{+30}, 114^{+33}, 147^{+36}, 183$

Hence, the wrong number in the series is 55.

7. The given logic is as follows.

$1850^{-800}, 1050^{-400}, 650^{-200}, \underline{450}^{-100}, 350^{-50}, 300^{-25}, 275$

Hence, the wrong number in the series is 400.

8. The given logic is as follows.

$12^{+31}, \underline{43}^{+29}, 72^{+23}, 95^{+19}, 114^{+17}, 31^{+13}, 144$

Hence, the wrong number in the series is 31.

9. The given logic is as follows.

$$24^{(+1=1^2)}, 25^{(+4=2^2)}, 29^{(+9=3^2)}, \underline{38}^{(+16=4^2)}, 54^{(+25=5^2)}, 79^{(+36=6^2)}, 115$$

Hence wrong number in the series is 36.

10. The given logic is as follows.

$$5^{(+4=2^2)}, \underline{9}^{(+8=2^3)}, 17^{(+16=2^4)}, 33^{(+32=2^5)}, 65^{(+64=2^6)}, 129^{(+128=2^7)}, 257$$

Hence wrong number in the series is 10.

Solutions for questions 11 to 15:

The given logic is as follows.

$$4^{+6^2}, 40^{+7^2}, 89^{+8^2}, 153^{+9^2}, 234^{+10^2}, 334$$

11. The new series will be as follows.

20 (a) (b) (c) (d) (e)

$$20^{+6^2}, 56^{+7^2}, 105^{+8^2}, 169^{+9^2}, 250^{+10^2}, 350$$

$\therefore$ The value of (d) is 250.

12. The given logic is as follows.

$$24^{\times 0.5}, 2^{\times 1}, 12^{\times 1.5}, 18^{\times 2}, 36^{\times 2.5}, 90^{\times 3}, 270$$

The new series will be as follows.

32 (a) (b) (c) (d) (e)

$$32^{\times 0.5}, 16^{\times 1}, 16^{\times 1.5}, 24^{\times 2}, 48^{\times 2.5}, 120$$

$\therefore$ The value of (b) is 16.

13. The given logic is as follows.

$$5^{(\times 1+1)}, 6^{(\times 2+2)}, 14^{(\times 3+3)}, 45^{(\times 4+4)}, 184^{(\times 5+5)}, 925^{(\times 6+6)}, 5556$$

The new series will be as follows.

10 (a) (b) (c) (d) (e)

$$10^{(\times 1+1)}, 11^{(\times 2+2)}, 24^{(\times 3+3)}, 75(\times 4+4), 304^{(\times 5+5)}, 1525$$

$\therefore$ The value of (a) is 11.

14. The given logic is as follows.

$$0.75^{\times 4}, 3^{\times 4}, 12^{\times 4}, 48^{\times 4}, 192^{\times 4}, 768^{\times 4}, 3072$$

The new series will be as follows.

5 (a) (b) (c) (d) (e)

$$5^{\times 4}, 20^{\times 4}, 80^{\times 4}, 320^{\times 4}, 1280^{\times 4}, 5120$$

$\therefore$ The value of (e) is 5120.

15. The given logic is as follows.

$$16^{+36}, 52^{+32}, 84^{+28}, 112^{+24}, 136^{+20}, 156^{+16}, 172$$

The new series will be as follows.

30 (a) (b) (c) (d) (e)

$$30^{+36}, 66^{+32}, 98^{+28}, 126^{+24}, 150^{+20}, 170$$

$\therefore$ The value of (c) is 126.

Solutions for questions 16 to 20:

16. The given logic is as follows.

$434 + (14)^2 = \underline{630}$.

$630 + (12)^2 = 774$.

$774 + (10)^2 = 874$.

$874 + (8)^2 = 938$.

$938 + (6)^2 = \underline{974}$.

$974 + (4)^2 = 990$.

Hence, the wrong numbers are 629 and 972, where 972 has greater difference.

17. The given logic is as follows.

$$5^{\times 2-3}, 7^{\times 3-5}, \underline{16}^{\times 5-7}, 73^{\times 7-11}, \underline{500}^{\times 11-13}, 5487^{\times 13-17}, 71314$$

Hence, the wrong numbers are 18 and 499, where 18 has greater difference.

18. The given logic is as follows.

$$5300^{-10^3}, 4300^{-9^3}, 3571^{-8^3}, \underline{3059}^{-7^3}, 2716^{-6^3}, \underline{2500}^{-5^3}, 2375$$

Hence, the wrong numbers are 3061 and 2501, where 3061 has greater difference.

19. The given logic is as follows.

$$21491^{(+6 \div 7)}, 3071^{(-5 \div 6)}, 511^{(+4 \div 5)}, \underline{103}^{(-3 \div 4)}, \underline{25}^{(+2 \div 3)}, 9^{(-1 \div 2)}, 4$$

Hence, the wrong numbers are 102 and 30, where 30 has greater difference.

20. The given logic is as follows.

$$201600^{\div 2}, 100800^{\div 3}, \underline{33600}^{\div 4}, 8400^{\div 5}, \underline{1680}^{\div 6}, 280^{\div 7}, 40$$

Hence, the wrong numbers are 33605 and 1679, where 33605 has greater difference.

Solutions for questions 21 to 30:

21. The alternate groups are in different series.

MTD, PVE, SXF are in on series.

Pattern for the first letters:

$M^{+3}, P^{+3}, S^{+3}, \underline{V}$

Pattern for the second letters:

$T^{+2}, V^{+2}, X^{+2}, \underline{Z}$

Pattern for the third letters:

$D^{+1}, E^{+1}, F^{+1}, \underline{G}$

Hence, the next pair is VZG.

22. In this series the first letter in all the groups form a series of consecutive letters. Hence, the first letter in the next group is E. The other letters in each group is related as follows.

A^{x2}	B^{x3}	F^{x2}	L
B^{x2}	D^{x3}	L^{x2}	X
C^{x2}	F^{x3}	R^{x2}	J

D^{x2} H^{x3} X^{x2} V
Hence, the next group is obtained as follows.
$E^{x2}\, J^{x3}\, D^{x2}\, H$
Hence, the next group in the series is EJDH.

23. The given series is a mixed series.
Pattern for the first letters:
$D^{+5}, I^{+5}, \underline{N}^{+5}, S^{+5}, X$
Pattern for the second letters:
$M^{+2}, O^{+2}, \underline{Q}^{+2}, S^{+2}, U$
Pattern for the third letters:
$P^{-3}, M^{-3}, \underline{J}^{-3}, G^{-3}, D$
Hence, the missing pair is NQJ.

24. $X^{-8}, P^{-6}, J^{-4}, F^{-2}, \_\_$
The values that are subtracted are consecutive even numbers in decreasing order starting from 8. Hence, the next letter in the series is $F - 2 = D$.

25. The given logic is as follows.
$I^{+1}\, J^{+2}\, L^{+3}\, O^{+4}, S^{+1}\, T^{+2}\, V^{+3}\, Y^{+4}, C^{+1}\, D^{+2}\, F^{+3}\, I^{+4},$
$M^{+1}\, N^{+2}\, P^{+3}\, S^{+4}, \underline{W{+1}\ X{+2}\ Z{+3}\ C}$
Hence, WXZC is the next group in the series.

26. The given series is a mixed series.
Pattern for the first letters:
$E^{-1}, D^{+2}, \underline{F}^{-3}, C^{+4}, G$
Pattern for the second letters:
$U^{+2}, W^{-3}, \underline{T}^{+4}, X^{-5}, S$
Pattern for the third letters:
$I^{-3}, F^{+4}, \underline{J}^{-5}, E^{+6}, K$
Pattern for the fourth letters:
$L^{+4}, P^{-5}, \underline{K}^{+6}, Q^{-7}, J$
Pattern for the fifth letters:
$N^{-5}, I^{+6}, \underline{O}^{-7}, H^{+8}, P$
Hence, the missing group is FTJKO.

27.

3	2	7	5	11	8	15	11				
C	1	B,	G	12	E,	K	3	H,	O	26	K,
(3 – 2)	(7 + 5)	(11 – 8)	(15 + 11)								

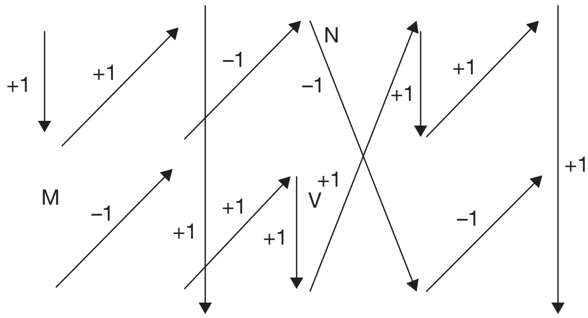
Pattern for the first letters:
$C^{+4}, G^{+4}, K^{+4}, O^{+4}, \underline{S}$

Pattern for the second letters:
$B^{+3}, E^{+3}, H^{+3}, K^{+3}, \underline{N}$
Hence, the required group is, $S(19 - 14)N \Rightarrow S5N$.

28. $\dfrac{C(3)}{3x(3+1)}, \dfrac{E(5)}{5x(5+1)}, \dfrac{G(7)}{7x(7+1)},$
$\dfrac{I(9)}{9x(9+1)}, \dfrac{K(11)}{11x(11+1)}, \dfrac{M(13)}{13x(13+1)}$
Pattern for the letters:
Letters in consecutive odd positions.
Hence, $\dfrac{I}{90}$ is the missing term in the series.

29. The given series is a mixed series.
Pattern for the first letters:
$X^{-3}, U^{-3}, R^{-3}, O^{-3}, \underline{L}$
Pattern for the second letters:
$T^{-3}, Q^{-3}, N^{-3}, K^{-3}, \underline{H}$
Pattern for the third letters:
$V^{-3}, S^{-3}, P^{-3}, M^{-3}, \underline{J}$
Hence, the next group in the series is LHJ.

30. In the given series each group is related to its previous group and the letters within the group are related to each other as follows.



Hence, the missing group in the series is TVU.

CHAPTER 2

Analogies

LEARNING OBJECTIVES

In this chapter, you will:
- Learn about analogy – a similar relationship between two or more entities
- Learn about number analogies and verbal analogies
- Learn about the different kinds of relationships between elements in number analogy questions like:
  - Multiples
  - Square/square roots
  - Cube/cube roots
  - Prime numbers

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❒ ANALOGIES

Analogy means 'similarity' or 'similar relationship'. In these type of questions regarding the number or letter or verbal analogies, a pair of entities that has a certain relationship is given. This number/letter/pair is followed by a third number/letter/word. The student is expected to identify the relationship between the pair given and find out a FOURTH element such that the relationship between the third and the fourth is similar to the relationship that exists between the first and the second (In some cases, it may not be the fourth one that has to be found out. The fourth one will be given and the student has to find out one of the other three, whichever is not given).

Number Analogies

Typical relationships between the numbers in a given pair can be any of the following:

- One number is a multiple of the other.
- One number is the square or square root of the other.
- One number is the cube or cube root of the other.
- The two numbers are squares of two other numbers which themselves are related. For example, the two numbers are squares of two consecutive integers or squares of two consecutive even integers or squares of two consecutive odd integers.
- The two numbers are such that they are obtained by subtracting a certain number from the squares or cubes of the two related numbers.
- The two numbers are such that they are obtained by adding a certain number to the squares or cubes of the two related numbers.
- The two numbers can be consecutive, even, odd or prime numbers.

There can be many more combinations that one can think of but the student has to note an important point in solving questions on Number Analogies. In Number Series related questions, since a series of numbers (more than two numbers) will be given, the relationship or pattern can be identified uniquely. In Number Analogies, since only two numbers are given, it may be possible to think of more than one relationship existing between the two numbers in the given pair. But, it should be kept in mind that generally, simple addition of one number or subtraction of one number is not what is given in Number Analogies. The questions try to test the insight that the student has got into the relationship between the numbers.

Let us take a few examples and understand the questions on Number Analogies.

SOLVED EXAMPLES

1. 25 is related to 36 in the same way as 49 is related to __________.

(A) 61 (B) 63
(C) 65 (D) 60

Sol: When the numbers in the question are considered the students tend to consider 25 and 36 as squares of two consecutive natural numbers. But the answer choices do not have any answer suitable to the above logic. Hence, it is important that the student keeps the answer choices in view in arriving at the logic.

$25 + 11 = 36$

Similarly, $49 + 11 = 60$.

2. 27 is related to 51 in the same way as 83 is related to ________.

(A) 102 (B) 117
(C) 123 (D) 138

Sol: The given analogy can be written as:

$5^2 + 2 : 7^2 + 2 : : 9^2 + 2 :$ ____.

5 and 7 are successive odd numbers.

Similarly, next odd number to 9 is 11 and $11^2 + 2 = 121 + 2 = 123$.

3. 11 is related to 25 in the same way as 17 is related to _______.

(A) 33 (B) 28
(C) 41 (D) 37

Sol: $11 \times 2 + 3 = 22 + 3 = 25$.

Similarly, $17 \times 2 + 3 = 34 + 3 = 37$.

Letter Analogies

The questions in this area are similar to Verbal Analogies. Here, the questions are based on the relationship between **two groups of letters** (instead of **two words** as in Verbal Analogies). Typically, three sets of letters are given followed by a question mark (where a fourth set of letters is supposed to be inserted). The student has to find the relation or order in which the letters have been grouped together in the first two sets of letters on the left hand side of the symbol ': :' and then find a set of letters to fit in place of the question mark so that the third and the fourth set of letters will also have the same relationship as the first and the second. The sequence or order in which the letters are grouped can be illustrated by the following examples.

SOLVED EXAMPLES

1. BDEG is related to DFGI in the same way as HKMO is related to _______.

(A) ILNP (B) JMOP
(C) JMOQ (D) JNOQ

Sol: Two letters are added to each letter to get the next letters in the analogy.

B D E G; Similarly, H K M O
+2 +2 +2 + 2 +2 +2 +2 +2
D F G I J M O Q

2. ACDF is related to CGJN in the same way as BEHI is related to ____.

(A) DJNQ (B) DINQ
(C) DINR (D) DHNQ

Sol: A C D F; Similarly, B E H I
+2 +4 +6 +8 +2 +4 +6 +8
C G J N D I N Q

3. SUWY is related to LPTX in the same way as PRTV is related to ______.

(A) INRU (B) INQU
(C) IMRU (D) IMQU

Sol: S U W Y; Similarly, P R T V
–7 –5 –3 –1 –7 –5 –3 –1
L P T X I M Q U

4. BCDE is related to DFHH in the same way as FGHI is related to ______.

(A) LJPL (B) LKPL
(C) JKPJ (D) IKPL

Sol: B C D E; Similarly, F G H I
x2 +3 x2 +3 x2 +3 x2 +3
D F H H L J P L

Verbal Analogies:

Here, the questions are based on relationship between two words. In these kind of questions three words are followed by a blank space, which the student has to fill up in such a way that the third and the fourth words have the same relationship between them as the first and the second words have. The following examples help in understanding the concepts.

SOLVED EXAMPLES

1. Gum is related to Stick in the same way as Needle is related to _____.
(A) Cloth (B) Prick
(C) Taylor (D) Stitch

Sol: Gum is used to Stick and Needle is used to Stitch.

2. Socks is related to Feet in the same way as Hands is related to _____.
(A) Arms (B) Shirt
(C) Gloves (D) Fingers

Sol: Socks are worn on Feet. Similarly, Gloves are worn on Hands.

3. Soft is related to Hard in the same way as Cold is related to _____.
(A) Hot (B) Ice
(C) Winter (D) Snow

Sol: Soft and Hard are antonyms. Similarly, the antonym of Cold is Hot.

EXERCISES

Directions for questions 1 to 25: Find the missing term.

1. 36 : 343 : : _____ : 1331
(A) 81 (B) 121
(C) 100 (D) 144

2. 24 : 576 : : 32 : _____
(A) 1024 (B) 992
(C) 1228 (D) 865

3. 13 : 2197 : : 16 : _____
(A) 256 (B) 2744
(C) 4096 (D) 3378

4. 81 : 729 : : 144 : _____
(A) 1728 (B) 1331
(C) 169 (D) 2197

5. 22 : 506 : : 27 : _____
(A) 675 (B) 756
(C) 702 (D) 783

6. 6 : 222 : : 9 : _____
(A) 738 (B) 767
(C) 729 (D) 744

7. 5 : 120 : : 8 : _____
(A) 520 (B) 504
(C) 448 (D) 512

8. 5 : 150 : : 8 : _____
(A) 520 (B) 516
(C) 512 (D) 576

9. 6 : 180 : : 9 : _____
(A) 729 (B) 738
(C) 632 (D) 648

10. 105 : 150 : : 39 : _____
(A) 68 (B) 64
(C) 60 (D) 72

11. 390 : 315 : : _____ : 564
(A) 663 (B) 689
(C) 653 (D) 674

12. 3864 : 5098 : : 4994 : _____
(A) 6228 (B) 6246
(C) 6194 (D) 6286

13. 1936 : 1360 : : _____ : 2142
(A) 2746 (B) 2718
(C) 2672 (D) 2466

14. 11 : 24 : : 37 : _____
(A) 68 (B) 92
(C) 74 (D) 78

15. 97 : 8 : : 43 : _____
(A) 4 (B) 2
(C) 3 (D) 7

16. PS : KH : : MT : _____
(A) NH (B) NG
(C) LG (D) LH

17. EOU : IUA : : AIU : _____
(A) EIO (B) IOE
(C) EOA (D) EAO

18. HRD : JSF : : XMP : _____
(A) ZNQ (B) ZOR
(C) YNR (E) YNQ

19. DATE : ECWI : : CHAIN : _____
(A) DJDMS (B) DJELR
(C) DIFMS (D) DIELS

20. MONTH : NMQPM : : PAPER : _____
(A) QYTBV (B) QXSBX
(C) QYTAV (D) QYSAW

21. 6P1 : 5Y2 : : 6J3 : _____
(A) 6L4 (B) 9K4
(C) 9W4 (D) 4L6

22. 3P2 : 2J0 : : 3R6 : _____
(A) 2M6 (B) 2N8
(C) 1H6 (D) 2L4

23. 12L : 24X : : 5E : _____
(A) 21U (B) 19S
(C) 10J (D) 20T

24. 2E3 : 4I5 : : 7O8 : _____
(A) 10U11 (B) 11W12
(C) 13A14 (D) 9U12

25. B6H : D10N : : K5P : _____
(A) M9V (B) T72
(C) R8J (D) B6D

Directions for questions 26 to 30: Select the correct alternative from the given choices.

26. Hand is related to Elbow in the same way as Leg is related to _________.

(A) Joint (B) Fingers
(C) Toes (D) Knee

27. Aeroplane is related to Pilot in the same way as Elephant is related to _________.
(A) Elephant Man (B) Saddle
(C) Mahout (D) Jockey

28. Bangladesh is related to Dhaka in the same way as Germany is related to _________.
(A) Paris (B) Berlin
(C) Baghdad (D) Rome

29. Spain is related to King in the same way as Brazil is related to _________.
(A) Chancellor (B) President
(C) Pope (D) Director

30. River is related to Bank in the same way as Sea is related to _________.
(A) Bank (B) Port
(C) Coast (D) Pebble

ANSWER KEYS

1. (C)	**6.** (A)	**11.** (A)	**16.** (B)	**21.** (C)	**26.** (D)
2. (A)	**7.** (B)	**12.** (A)	**17.** (C)	**22.** (D)	**27.** (C)
3. (C)	**8.** (D)	**13.** (B)	**18.** (D)	**23.** (C)	**28.** (B)
4. (A)	**9.** (D)	**14.** (D)	**19.** (A)	**24.** (A)	**29.** (B)
5. (B)	**10.** (A)	**15.** (B)	**20.** (D)	**25.** (A)	**30.** (C)

SOLUTIONS

Solutions for questions 1 to 25:

1. 36 : 343 : : ______ : 1331
$(6)^2 : (7)^3 : : (10)^2 : (11)^3$
Hence, $10^2 = 100$ is the missing number.

2. 24 : 576 : : 32 : ______
The given analogy is of the form $n : n^2$.
$24 : (24)^2 : : 32 : (32)^2$
$(32)^2 = 1024$ is the next number.

3. 13 : 2197 : : 16 : ______
The given analogy is of the form $n : n^3$.
$13 : (13)^3 : : 16 : \underline{(16)3}$
$(16)^3 = 4096$ is the missing number.

4. 81 : 729 : : 144 : ______
$(9)^2 : (9)^3 : : (12)^2 : \underline{(12)3}$
$(12)^3 = 1728$ is the next number.

5. 22 : 506 : : 27 : ______
$22 : (22)^2 + 22 : : 27 : \underline{(27)2 + 27}$
This is of the form $n : n^2 + n$.
$(27)^2 + 27 = 756$ is the next number.

6. 6 : 222 : 9 : ______
$6 : (6)^3 + 6 : : 9 : \underline{(9)3 + 9}$
$(9)^3 + 9 = 738$ is the next number.

7. 5 : 120: : 8 : ______
$5 : (5)^3 - 5 : : 8 : \underline{(8)3 - 8}$
$(8)^3 - 8 = 504$ is the next number.

8. 5 : 150 : : 8 : ______
$5 : 5^3 + 5^2 : : 8 : \underline{83 + 82}$
This is of the form $n : n^3 + n^2$
$8^3 + 8^2 = 576$ is the next number.

9. 6 : 180 : : 9 : ______
$6 : 6^3 - 6^2 : : 9 : \underline{93 - 92}$
This is of the form $n : n^3 - n^2$.
$9^3 - 9^2 = 648$.

10. 105 : 150 : : 39: ______

$$\underbrace{(10)^2 + \frac{10}{2} : (12)^2 + \frac{12}{2}}_{+2} : : \underbrace{(6)^2 + \frac{6}{2} : (8)^2 + \frac{8}{2}}_{+2}$$

This is of the form $n^2 + \dfrac{n}{2}$.

$(8)^2 + \dfrac{8}{2} = 68$ is the next number.

11. 390 : 315 : : ______ : 564

$$(20)^2 - \frac{20}{2} : (18)^2 - \frac{18}{2} :: (26)^2 - \frac{26}{2} : (24)^2 - \frac{24}{2}$$

+2 +2

This is of the form $n^2 - \frac{n}{2}$.

Hence, $(26)^2 - \frac{26}{2} = 663$ is the missing number.

12. 3864^{+1234} : 5068 : : 4994^{+1234} : ______

4994 + 1234 = 6228.

13. 1936^{-576} : 1360 : : ______ : 2142

2142 + 576 = 2718.

14. 11^{+13} : 24 : : 37 : ______

Next prime number to 11 is 13. 11 + 13 = 24.
Next prime to 37 is 41 and 37 + 41 = 78.

15. 97^{-89} : 8 : : 43 : ______

Previous prime number to 97 is 89 and 97 – 89 =8.
Similarly, the previous prime to 43 is 41 and 43 – 41 = 2.

16. PS : KH : : MT : ______

For P the opposite letter is K and for S the opposite letter is H.
Similarly, for M and T, the opposite letters are N and G, respectively.

17. EOU : IUA : : AIU : ______

For each of the vowels, its next vowel is given.
Similarly, for AIU the next vowels are EOA.

18. HRD : JSF : : XMP : ______

For each of these consonants its next consonant is given.
Similarly, for XMP their next constants are YNQ.

19. DATE : ECWI : : CHAIN : ______

				Similarly				
D	A	T	E		C	H	A	I	N
+1	+2	+3	+4		+1	+2	+3	+4	+5
E	C	W	I		D	J	D	M	S

20. MONTH : NMQPM : : PAPER : ______

					Similarly					
M	O	N	T	H		P	A	P	E	R
+1	+2	+3	+4	+5		+1	+2	+3	+4	+5
N	M	Q	P	M		Q	Y	S	A	W

21. 6P1 : 5Y2 : : 6J3 : ______

P is the 16th letter and 16 is $(Y)^2$.
The value is written in the reverse order.
Next number to 4 is 5 and $(5)^2 = 25$ Y.
The digits 25 are written in reverse order.
Similarly, $6^2 = 36 \Rightarrow$ 6J3and 9W4 is the next term.

22. 3P2 $\rightarrow$ P is the 16th letter and $16 \times 2 = 32$.
The digits in 32 are written on either side of P.
P^{-6} is J. J is the 10th letter and $10 \times 2 = 20. \Rightarrow$ 2J0.
R is the 18th letter and $18 \times 2 = 36 \Rightarrow$ 3 R6
R^{-6} is L. L is the 12th letter and $12 \times 2 = 24 \Rightarrow$ 2L4.

23. 12L : 24 X : : 5E : ______
L is the 12th letter and $12 \times 2 = 24$
The 24th letter is X.
Similarly, E is the 5th letter and $5 \times 2 = 10$
The 10th letter is J.

24. 2E3 : 4I5 : : 7O8 : ______
The letters are consecutive vowels and the sum of the numbers on either side of the letter is equal to the place value of the letter and the number are consecutive. Hence, 10U11 is the required term.

25. B6H : D8N : : K5P : ______
In each group, the difference and the place values of the letters is equal to the number written between then.

B	H
+2	+6
D	N

Similarly,

K	P
+2	+6
M	V

M is 13 and V is 22 and 22 – 13 = 9. Hence, M9V is the required group.

Solutions for questions 26 to 30:

26. Elbow is the joint of the Hand and Knee is the joint of the Leg.

27. Pilot, one who controls an Aeroplane and Mahout, one who controls an Elephant.

28. The capital of Bangladesh is Dhaka and the capital of Germany is Berlin.

29. The head of Spain is King and the head of Brazil is the President.

30. The land near the River is Bank. Similarly, land near the Ocean is Coast.

3 Odd Man Out

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:

- Learn how to pick the odd one out from the given items
- Learn about different ways of classifying elements like:
  - Number classification
  - Alphabet classification
  - Word classification

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❒ ODD MAN OUT

Finding the odd man out from the given alternatives is a very common type of questions that one comes across in different competitive examinations. In the questions on odd man out, all the items, except one follow a certain pattern (in their formation) or belong to a group. The item that does not follow the pattern or does not belong to the group has to be marked as the answer choice.

The problems of this variety often fall under the category of 'Classification'. When a given set of elements is classified under a single head, one of the items will not fall into that group to which the rest belong, i.e., it will not have the common property, which the others will have. Hence, it becomes the odd man out.

Questions on classification can be asked in any form. Some of the commonly asked ones are given below.

Number Classification

In this case, we need to choose the odd number from the given alternatives. The numbers may belong to a particular set, i.e., they may be odd, even, prime, rational, squares, cubes, etc., and only one of the choices will not follow the rule which others do and that is our answer. A few illustrations are given below.

1. Find the odd one among the following.

(a) (A) 17 (B) 27
(C) 37 (D) 47

(b) (A) 441 (B) 289
(C) 361 (D) 343

Sol: (a) All the given numbers except 27 are prime numbers whereas 27 is a composite number.

(b) The given numbers can be written as $(21)^2$, $(17)^2$, $(19)^2$, $(7)^3$, $(25)^2$. All except 343 are the squares whereas 343 is a cube.

Alphabet Classification

In this type, a group of jumbled letters typically consisting of three letters (but can be four or two or just a single letter) are put together. The pattern or order in which they are grouped is to be studied and we need to find out which groups have the same pattern or relationship between the letters. There will be one choice, which will have a pattern that is different from the rest and that is our answer.

2. Find the odd one among the following.
(A) ZW (B) TQ
(C) SP (D) NL

Sol: $Z^{-3}W$, $T^{-3}Q$, $S^{-3}P$, $N^{-2}L$, $P^{-3}M$
Hence, NL is the odd one.

3. Find the odd one among the following.
(A) CFD (B) GJH
(C) KNM (D) JMK

Sol: $C^{+3}F^{-2}D$, $G^{+3}J^{-2}H$, $K^{+3}N^{-1}M$, $J^{+3}M^{-2}K$, $V^{+3}Y^{-2}W$
Hence, KNM is the odd one.

Word Classification

Here, different items are classified based on common properties, like names, places, professions, parts of speech, etc. A few examples are illustrated below.

4. Find the odd one among the following.
(A) Mercury (B) Moon
(C) Jupiter (D) Saturn

Sol: All others except Moon are planets where as Moon is a satellite.

5. Find the odd one among the following.
(A) SORE (B) SOTLU
(C) NORGAE (D) MEJNIAS

Sol: The words are jumbled. The actual words are ROSE, LOTUS, ORANGE, JASMINE and LILLY. All, except ORANGE are flowers whereas ORANGE is a fruit.

EXERCISES

Directions for questions 1 to 30: Three of the following four are alike in a particular pattern and hence, form a group. Find the one which does not belong to that group.

1. (A) 16 (B) 9
(C) 49 (D) 121

2. (A) 248 (B) 224
(C) 236 (D) 268

3. (A) 1296 (B) 2304
(C) 2704 (D) 1764

4. (A) $\frac{312}{468}$ (B) $\frac{318}{477}$
(C) $\frac{230}{345}$ (D) $\frac{354}{472}$

5. (A) 2 (B) 5
(C) 10 (D) 54

6. (A) 144 (B) 169
(C) 225 (D) 196

7. (A) 543 (B) 435
(C) 345 (D) 354

8. (A) 39 (B) 636
(C) 24 (D) 37

9. (A) 346 (B) 469
(C) 742 (D) 427

10. (A) 4774 (B) 4174
(C) 363 (D) 666

11. (A) 744 (B) 852
(C) 654 (D) 473

12. (A) 5840 (B) 6530
(C) 7321 (D) 6422

13. (A) 18 (B) 32
(C) 72 (D) 88

14. (A) 29 (B) 341
(C) 129 (D) 67

15. (A) 43 (B) 59
(C) 13 (D) 21

16. (A) QDW (B) UHS
(C) BOL (D) SGT

17. (A) 2D (B) 3I
(C) 4P (D) 6A

18. (A) IED (B) OCL
(C) IDE (D) OHK

19. (A) PQUM (B) HIOE
(C) NOSK (D) RSWO

20. (A) MNPL (B) SUWR
(C) FHJE (D) JLNI

21. (A) ABB (B) KBV
(C) BCE (D) EDT

22. (A) EHFC (B) ILJG
(C) PSQN (D) ROSU

23. (A) AZ (B) CZYX
(C) DZYXW (D) EZYXW

24. (A) $\frac{A}{BB}$ (B) $\frac{CCC}{DDDD}$
(C) $\frac{BB}{CCC}$ (D) $\frac{DDDDD}{CCC}$

25. (A) 2W3 (B) 1Q7
(C) 1L2 (D) 2Z5

26. (A) Late (B) Mate
(C) Rate (D) Bite

27. (A) Spider (B) Housefly
(C) Mosquito (D) Bee

28. (A) Pistchios (B) Pecans
(C) Walnuts (D) Apple

29. (A) November (B) September
(C) June (D) February

30. (A) November (B) March
(C) August (D) December

ANSWER KEYS

1. (A)	**6.** (B)	**11.** (D)	**16.** (D)	**21.** (C)	**26.** (D)
2. (D)	**7.** (D)	**12.** (D)	**17.** (D)	**22.** (D)	**27.** (A)
3. (C)	**8.** (D)	**13.** (D)	**18.** (D)	**23.** (D)	**28.** (D)
4. (D)	**9.** (A)	**14.** (B)	**19.** (B)	**24.** (D)	**29.** (D)
5. (D)	**10.** (B)	**15.** (?)	**20.** (A)	**25.** (D)	**30.** (A)

SOLUTIONS

1. Except 16, all are odd numbers.

2. Except 268, in all the numbers, the last digit is the product of first two digits.

3. Except 2704, all the numbers are divisible by '6'.

4. Except $\frac{354}{472}$, all fractions become $\frac{2}{3}$ when simplified.

5. Except 54, all the numbers can be expressed in the form $n^2 + 1$.

6. Except 169, all the numbers are squares of composite numbers.

7. Except 354, all the numbers are odd numbers.

8. $6\,\underline{62} = 636$, $2\,\underline{22} = 24$, $8\,\underline{82} = 864$, $33^2 = 39$.

The above pattern is not followed in $\underline{37}$.

9. Except 346, all the numbers are divisible by 7.

10. Except 4174, all the numbers are palindromes.

11. Except 473, in all the numbers the sum of all the digits is 15.

12. Except in '6422', in all the numbers, the last two digits are product of the first digits.

13. $18 = 2 \times 3^2$, $32 = 2 \times 4^2$, $72 = 2 \times 6^2$, $98 = 2 \times 7^2$
Except 88, all the numbers follow similar pattern.

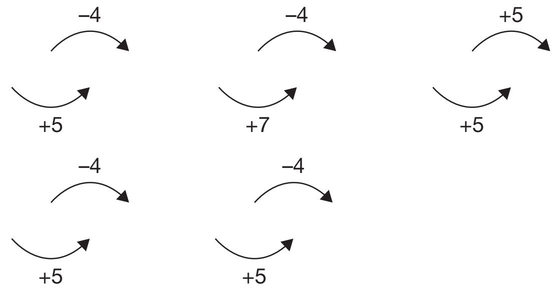
14. $29 = 3^3 + 2$, $129 = 5^3 + 4$, $67 = 4^3 + 3$, $221 = 6^3 + 5$,
Except 341, all the numbers follow similar pattern.

16. In all the groups, the first and last letters are corresponding and opposite letters of the 2nd letter, respectively except SGT.

17. In each of the group 2D, 3I, 4P, 5Y the number is the square root of the position value of the letter. This pattern is not followed by 6A.

18. In all the groups, the difference of the first letter and the last letter's place value is the place value of the second letter, except in 'OHK'.

19.



Except HIOE, all the other groups follow the similar pattern.

20. $M^{+1}N^{+2}P^{-4}L$, $S^{+2}U^{+2}W^{-5}R$, $F^{+2}H^{+2}J^{-5}E$, $J^{+2}L^{+2}N^{-5}I$, $C^{+2}E^{+2}G^{-5}B$. Except MNPL, all other groups follow the similar pattern.

21. In all the groups the product of place values of first two letters is the place value of last letter. Except in BCE.

22. $I^{+3}\,L^{-2}\,J^{-3}\,G$, $P^{+3}\,S^{-2}\,Q^{-3}\,N$, $M^{+3}\,P^{-2}\,N^{-3}\,K$,
$R^{-3}\,O^{+4}\,S^{+2}\,U$, $E^{+3}\,H^{-2}\,F^{-3}\,C$
Except ROSU, all follow the similar pattern.

23. In all the groups the number of letters of the place value of the first letter is taken from the back side, in reverse order, except in EZYXW.

24. In all the groups, the number of letters in the numerator are less than the number of letters in denominator. Except in $\frac{\text{DDDDD}}{\text{CCC}}$.

25. The digits on either side of the letter in each of the groups 2W3, 1Q7, 1L2, 1R8 indicate the place value of the letter in the alphabet. Except in 2Z5.

26. Except 'Bite', all the words are rhyming words.

27. Except 'Spider', all are flying insects.

28. Except 'Apple', all are types of nuts.

29. Except 'February', all the given months have 30 days.

30. Except 'November', all the given months have 31 days.

4 Coding and Decoding

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:

- Learn about coding (encrypting a message, according to a pattern or set of rules so that no one can understand the message without knowing the rule)
- Learn about decoding (decrypting a message and writing the original message so that anyone can understand it)
- Be exposed to various types of coding/decoding like:
  - Arranging in ascending/descending order
  - Adding/subtracting from each element to generate new elements
  - Coding letters as numbers and vice-versa

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❐ CODING AND DECODING

Before looking at the different types of questions and some of the codes that can be used with the help of examples, let us first understand what we mean by coding and decoding. When we say **coding**, a particular code or pattern is used to express a word in English language as a different word or in a different form. The coded word itself does not make any sense unless we know the pattern or code that has been followed.

Decoding refers to the process of arriving at the equivalent English word from the given code word.

In the questions, a particular code is given and on the basis of this given code, we have to find out how another word (in English language) can be coded. The correct code for the given word has to be selected from the answer choices on the basis of the code given in the question.

SOLVED EXAMPLES

1. In a certain code language, if the word 'PARTNER' is coded as OZQSMDQ, then what is the code for the word 'SEGMENT' in that language?

(A) TFHNFOU (B) RDFLDMS
(C) RDELDMS (D) RDFEDNS

Sol: Word: P A R T N E R
Logic: –1 –1 –1 –1 –1 –1 –1
Code: O Z Q S M D Q
Similarly, the code for SEGMENT is
Word: S E G M E N T
Logic: –1 –1 –1 –1 –1 –1 –1
Code: R D F L D M S

2. In a certain code language, if the word 'RECTANGLE' is coded as TGEVCPING, then how is the word 'RHOMBUS' coded in that language?
(A) TJOQDWV (B) TJQNDWU
(C) TJQODWU (D) TJQOEWU

Sol: Word: R E C T A N G L E
Logic: +2 +2 +2 +2 +2 +2 +2 +2 +2
Code: T G E V C P I N G
Similarly, the code for RHOMBUS is
Word: R H O M B U S
Logic: +2 +2 +2 +2 +2 +2 +2
Code: T J Q O D W U

3. In a certain code language, if the word 'SPHERE' is coded as EREHPS, then how is the word 'EXHIBITION' coded in that language?
(A) NOTITBIHXE (B) NOITIDIHXE
(C) NOITIBIHWE (D) NOITIBIHXE

Sol: Word: S P H E R E
Logic: The letters in the given word are reversed.
Code: E R E H P S
Similarly, the code for EXHIBITION is
Word: E X H I B I T I O N
Logic: The letters in the given word are reversed.
Code: N O I T I B I H X E

4. In a certain code language, if the word 'REJECTION' is coded as SGMIHZPWW, then how is the word 'MECHANIC' coded in that language?
(A) NGFLFTPK (B) NGPLFTPK
(C) NGFKFTPK (D) NGPTPKIL

Sol: Word: R E J E C T I O N
Logic: +1 +2 +3 +4 +5 +6 +7 +8 +9
Code: S G M I H Z P W W
Similarly, the code for MECHANIC is
Word: M E C H A N I C
Logic: +1 +2 +3 +4 +5 +6 +7 +8
Code: N G F L F T P K

5. In a certain code language, if the word 'PLAYER' is coded as AELPRY, then how is the word 'MANAGER' coded in that language?
(A) AEAGMNR (B) AAGEMNR
(C) AAEGMNR (D) AAEGNMR

Sol: Word: P L A Y E R
Logic: The letters in the word are arranged in the increasing order of their value as in the alphabet.
Code: A E L P R Y
Similarly, the code for MANAGER is AAEGMNR.

6. In a certain code language, if the number 1 is assigned to all the letters in odd numbered places in the alphabet and the remaining letters are assigned the number 2, then what is the code for the word 'INDIAN'?
(A) 121212 (B) 111222
(C) 112212 (D) 122112

Sol: The code for the word INDIAN is 122112.

7. In a certain code language, if CRICKET is coded as 3923564, ROCKET is coded as 913564 and KETTLE is coded as 564406, then how is LITTLE coded in that language?
(A) 244060 (B) 024406
(C) 020446 (D) 200446

Sol: As we observe that the letters and their corresponding codes are given in order, i.e., the code for C is 3, R is 9, I is 2 and so on. Hence, the code for LITTLE is 024406.

Directions for questions 8 to 11: In a certain code language, the codes for some words are as follows.

WORDS		CODES
NATION	-	agvnab
REMOTE	-	rzgrbe
STAIR	-	efgnv
FORMAL	-	bensyz
COMMON	-	zabzpb
FOR	-	ebs

Based on the above coding pattern answer the following questions.

8. What is the code for 'SCREEN'?
(A) fepcra (B) fpersa
(C) fpreba (D) fperra

9. What is the code for 'RATION'?
(A) ensvba (B) engvba
(C) engrba (D) engvca

10. What is the code for 'CREATOR'?
(A) prengbc (B) persbgc
(C) perngbe (D) pebrycn

11. What is the code for 'AMERICAN'?
(A) nzrevpna (B) nzrespna
(C) nzlespna (D) nzreqpna

Solutions for questions 8 to 11: The given words and their codes are as follows.

	WORDS		CODES
(1)	NATION	-	agvnab
(2)	REMOTE	-	rzgrbi
(3)	STAIR	-	efgnv
(4)	FORMAL	-	bensyz
(5)	COMMON	-	zabzpb
(6)	FOR	-	ebs

In the 1st word, the letter N is repeated and the code 'a' is repeated. Hence, for N, the code is 'a'. Similarly, from the 2nd word, the code for E is 'r'. In the 1st and 6th words, the letter O is common and so is the code b. Hence, the code for O is b. In the 5th word, the letter M is repeated and so is the code z. Hence, the code for M is Z. Similarly, the codes for the remaining letters can be determined.

The letters and their respective codes are as follows.

Letter	A	C	E	F	I	L	M	N	O	R	S	T
Code letter	n	p	R	s	v	y	z	a	b	e	f	G

8. The code for 'SCREEN' is fperra.

9. The code for 'RATION' is engvba.

10. The code for 'CREATOR' is perngbe.

11. The code for 'AMERICAN' is nzrevpna.

12. In a certain code if white is called as black, black as yellow, yellow as blue, blue as red, red as green, green as purple, then what is the colour of blood in that language?
(A) Red (B) Green
(C) Yellow (D) Purple

Sol: The colour of blood is Red and in this code, Red is called Green. Hence, blood is green in colour in that language.

EXERCISES

Directions for questions 1 to 20: Select the correct alternative from the given choices.

1. In a certain code language, if the word CLIMATE is coded as IAECLMT, then how is the word CALCULATE coded in that language?

(A) AUEACLCLT (B) AUACELCLT
(C) AUAECLCLT (D) AUAECLCCT

2. In a certain code language, if the word SOLUTIONS is coded as SNOITULOS, then how is the word ANSWER coded in that language?

(A) RENSWA (B) RENSAW
(C) REASWN (D) REWSNA

3. In a certain code language, if the word BASKET is coded as UFLTBC, then how is the word SIMPLE coded in that language?

(A) FMQNJG (B) FMQGNJ
(C) FMQNJT (D) MFNQTJ

4. In a certain code language, if the word SINGER is coded as XNSLJW, then how is the word DANCER coded in that language?

(A) IFSHJW (B) ISFHJW
(C) ISFJHW (D) IJWFSH

5. In a certain code language, if the word PLEASE is coded as GNRGUC, then how is the word CODING coded in that language?

(A) FQIPEK (B) FPKQEI
(C) FQKPIE (D) FQEIPK

6. In a certain code language, if the word KITE is coded as 4567 and the word RATE is coded as 8967, then how is the word TAKE coded in that language?

(A) 6974 (B) 6794
(C) 6947 (D) 6479

7. In a certain code language, if SP is coded as UR and LO is coded as NQ then TV is coded as ______.

(A) VW (B) WV
(C) VN (D) VX

8. In a certain code language, if the word ASIA is coded as 1431, the word AFRICA is coded as 125361 and the word FRANCE is coded as 251768, then how is the word ARIES coded in that language?

(A) 15348 (B) 15438
(C) 13584 (D) 15384

9. In a certain code language, if the word LOCAL is coded as MPDBM, then which word is coded as DBMMFS?

(A) CARROT (B) CODING
(C) CALLER (D) CARING

10. In a certain code language, if the word CREATE is coded as $\$\Omega\theta\#4\theta$ and the word INDIA is coded as 8768#, then how is the word ACCIDENT coded in that language?

(A) $\$\#\$86\theta74$ (B) $86\theta74\#\$\$$
(C) $\theta\#\#86\$74$ (D) $\#\$\$86\theta74$

11. In a certain code language, if the word PROGRAM is coded as RTQHTCO, then how is the word PLAYING coded in that language?

(A) RKPICZN (B) RCKPIZN
(C) RPICKZN (D) None of these

12. In a certain code language, if the word DOUBLE is coded as ODBUEL, then how is the word SINGLE coded in that language?

(A) ISNGEL (B) ISGNLE
(C) SINGEL (D) ISGNEL

13. In a certain code language, if the word SUMMER is coded as $\Omega\theta\#\#17$, and the word MOTION is coded as #$%2$6, then how can the word 'SECTOR' be coded in that language?

(A) Ω$©21% (B) $\Omega 21\#\theta\$$
(C) $\theta\$\Omega 861$ (D) Ω1©%$7

14. In a certain code language, if ADC is coded as 143 and BED is coded as 254 then how is DFG coded in that language?

(A) 456 (B) 465
(C) 467 (D) 645

15. In a certain code language, if sun means moon, moon means earth, earth means sky, sky mean sea, then on which of the following do we live according to that language?

(A) Sun (B) Moon
(C) Earth (D) Sea

16. In a certain code language, if the word STOP is coded as PWLS, then how is the word EXIT coded in that language?

(A) ABFW (B) BWFA
(C) WBFA (D) BAFW

17. In a certain code language, if the word LANGUAGE is coded as LNGGAUAE then how is the word FINANCE coded in that language?

(A) FNCNIAE (B) FNCNIEA
(C) FNNCIAE (D) FNNCAIE

18. In a certain code, if the word SOME is coded as MSEO and the word NAME is coded as MNEA, then what is the code for WARM in that language?

(A) RAMW (B) RMAW
(C) RWMA (D) RAWM

19. In a certain code language, if the word LETTER is coded as MUFFUS, then how is the word DECIDE coded in that language?

(A) EDFEEJ (B) EEFFJD
(C) FFEEDJ (D) EDEFJF

20. In a certain code language, if the word CORRECT is coded as ORCCRET, then what is the code for the word KINGDOM?

(A) IGOKNDM (B) HFNMJCL
(C) HFJNMCL (D) HFNCLMJ

Directions for questions 21 to 25: In a certain code language, the codes for the sentences in Column I are given in Column II. Each word has a unique code. Answer the questions based on these codes.

	Column I	Column II
1.	kite night right might	sap tap map cap
2.	might weight sight eight	zap cap wap yap
3.	night eight boat right	yup tap sap zap
4.	boat not weight night	wap sap yup lap

21. What is the code for the word 'not'?

(A) lap (B) map
(C) nap (D) zap

22. Which word is coded as 'cap'?

(A) weight (B) might
(C) night (D) not

23. What is the code for the word 'right'?

(A) yup (B) sap
(C) map (D) tap

24. What can be the code for 'might right boat correct'?

(A) cap fap tap yup (B) tap yap sap map
(C) wap yup lap cap (D) cap zap tap sap

25. What can be coded as 'sap tap map pop'?

(A) might right boat eight
(B) boat sight might weight
(C) right night site kite
(D) might not boat kite

Directions for questions 26 to 30: Given below are the codes for the digits/symbols. Study the conditions given below and answer the questions that follow.

Digit/ Symbol	5	2	©	#	4	@	8	Ω	7	$	3	9	®	*	1
Letter code	C	G	R	L	B	T	H	M	D	X	W	O	P	K	N

Conditions:

(i) If both the first and the last elements are odd digits, then code both of them as 'A'.
(ii) If both the first and the last elements are even digits, then code both of them as 'E'.
(iii) If the first element is an even digit and the last element is a symbol, then the codes for the first and the last elements get interchanged.
(iv) If both the first and the last elements are symbols, then reverse the code for the entire group.

What will be the codes for the following group of elements?

26. 25$391@4

(A) GCXWONTE (B) ECXWONTE
(C) ECWVOTNE (D) GWXCONTE

27. 1$Ω4925

(A) AXKPMGA (B) BPKXMGB
(C) BXKPMBA (D) None of these

28. $4@Ω2*5

(A) XTBMGKC (B) XBTMGKC
(C) XBTGMCK (D) CBTMGKX

29. *87$Ω2$

(A) KHDXMGX (B) HKXDGMO
(C) DXGKDHM (D) XGMXDHK

30. 872Ω@9®

(A) HDGMTOP (B) HGDMOTP
(C) PDGMTOH (D) PGMDOTH

ANSWER KEYS

1. (C)	6. (C)	11. (D)	16. (D)	21. (A)	26. (B)
2. (D)	7. (D)	12. (D)	17. (C)	22. (B)	27. (D)
3. (C)	8. (D)	13. (D)	18. (C)	23. (D)	28. (B)
4. (A)	9. (C)	14. (C)	19. (D)	24. (A)	29. (D)
5. (D)	10. (D)	15. (B)	20. (A)	25. (C)	30. (C)

SOLUTIONS

Solutions for questions 1 to 20:

1. Word: CLIMATE
Logic: Letters in the word are arranged as all vowels come together first and then consonants.
Code: IAECLMT
Similarly, the code for CALCULATE is AUAECLCLT.

2. Word: SOLUTIONS
Logic: The letters in the given word are reversed.
Code: SNOITULOS
Similarly, the code for ANSWER is REWSNA.

3.

Word:	B	A	S	K	E	T
Logic:	+1	+1	+1	+1	+1	+1
Code:	C	B	T	L	F	U

and then the code is reversed, i.e., UFLTBC.
Similarly,

Word:	S	I	M	P	L	E
Logic:	+1	+1	+1	+1	+1	+1
Code:	T	J	N	Q	M	F

and then the code is reversed, i.e., FMQNJT.

4.

Word:	S	I	N	G	E	R
Logic:	+5	+5	+5	+5	+5	+5
Code:	X	N	S	L	J	W

Similarly,

Word:	D	A	N	C	E	R
Logic:	+5	+5	+5	+5	+5	+5
Code:	I	F	S	H	J	W

∴ IFSHJW is the code for DANCER.

5.

Word:	P	L	E	A	S	E
Logic1:	+2	+2	+2	+2	+2	+2
Logic2:	R	N	G	C	U	G

Half coded and then reversed.
Code: GNRGUC.
Similarly,

Word:	C	O	D	I	N	G
Logic1:	+2	+2	+2	+2	+2	+2
Logic2:	E	Q	F	K	P	I

Half coded and then reversed.
Code: FQEIPK.
∴ FQEIPK is the code for CODING.

6.

WORD	CODE					
KITE	4567	→(I)				
RATE	8967	→(II)				

By comparing the positions of the letters T and E, in both the words with respective codes it can be identified that the code for each letter is in the corresponding position in the code.
Hence, the code for the word TAKE is 6947.

7.

Word1:	S	P	Word2:	L	O	
Logic:	+2	+2	Logic:	+2	+2	
Code:	U	R	Code:	N	Q	

Similarly,

Word:	T	V				
Logic:	+2	+2				
Code:	V	X				

∴ VX is the code for TV.

8. As we observe that the letters and their corresponding codes are given in order, i.e., the code for A is 1, S is 4 and so on. Hence, the code for ARIES is 15384.

9.

Word:	L	O	C	A	L	
Logic:	+1	+1	+1	+1	+1	
Code:	M	P	D	B	M	

Similarly,

Word:	C	A	L	L	E	R
Logic:	+1	+1	+1	+1	+1	+1
Code:	D	B	M	M	F	S

∴ CALLER is coded as DBMMFS.

10. As we observe that the letters and their corresponding codes are given in order, i.e., the code for C is \$, R is Ω and so on.
Hence, the code for ACCIDENT is '#\$\$86θ74'.

11.

Word:	P	R	O	G	R	A
M						
Logic:	+2	+2	+2	+1	+2	+2
+2						
Code:	R	T	Q	H	T	C
O						

Similarly,

Word:	P	L	A	Y	I	N
G						
Logic:	+2	+2	+2	+1	+2	+2
+2						
Code:	R	N	C	Z	K	P
I						

∴ RNCZKPI is the code for PLAYING.

12. Word: D O U B L E

Logic:

Code: O D B U E L

Similarly,

Word: S I N G L E

Logic:

Code: I S G N E L

∴ ISGNEL is the code for SINGLE.

13. As we observe that the letters and their corresponding codes are given in the following order, i.e., the code for S is Ω, U is θ and so on. Hence, the code for SECTOR is 'Ω1©%\$7'.

14. As we observe that the alphabets are coded with their place values according to the alphabetical order, i.e., the code for A is 1, B is 2 and so on. Hence, the code for DFG is 467.

15. We live on the earth and earth is called moon in the given code language.

16.

Word:	S	T	O	P		
Logic:	–3	+3	–3	+3		
Code:	P	W	L	S		

Similarly,

Word:	E	X	I	T		
Logic:	–3	+3	–3	+3		
Code:	B	A	F	W		

∴ BAFW is the code for EXIT.

17. As we observe that consonants come first, followed by vowels in the corresponding code. Hence, the code for the word FINANCE is FNNCIAE.

18. Word: S O M E

Logic:

Code: M S E O

Word: N A M E

Logic:

Code: M N E A

Similarly,

Word: W A R M

Logic:

Code: R W M A

∴ RWMA is coded for WARM.

19. First write the odd-positioned letters of the word and then even-positioned ones one after the other.

Word:	L	E	T	T	E	R
Logic:	L	T	E	E	T	R
+1	+1	+1	+1	+1	+1	
Code:	M	U	F	F	U	S

Similarly,

Word:	D	E	C	I	D	E
Logic:	D	C	D	E	I	E
+1	+1	+1	+1	+1	+1	
Code:	E	D	E	F	J	F

∴ EDEFJF is code for DECIDE.

20. Write the even-positioned letters of the word first and then the odd-positioned ones one after the other.

Word:	C	O	R	R	E	C
T						
Code:	O	R	C	C	R	E
T						

Similarly,

Word:	K	I	N	G	D	O
M						
Code:	I	G	O	K	N	D
M						

∴ IGOKNDM is the code for KINGDOM.

Solutions for questions 21 to 25:

From (1) and (2), the code for 'might' is 'cap'.
From (2) and (3), the code for 'eight' is 'zap'.
From (1) and (4), the code for 'night' is 'sap'.
From (3) and (4), the code for 'boat' is 'yup'.
From (3), the code for 'right' is 'tap' and from (1), the code for 'kite' is 'map'.
From (2) and (4), the code for 'weight' is 'wap' and from (2), the code for 'sight' is 'yap' and from (4), the code for 'not' is lap.
Hence, the codes are as follows:

Word	kite	night	right	might	weight	sight	Eight	boat	not
Code	map	sap	tap	cap	wap	yap	Zap	yup	lap

21. The code for 'not' is 'lap'.

22. 'might' is coded as 'cap'.

23. The code for 'right' is 'tap'.

24. The code for 'might right boat correct' can be 'cap tap yup fap'.

25. 'right night site kite' can be coded as 'sap tap map pop'.

Solutions for questions 26 to 30:

26. The given group of elements is 25$391@4.
This group follows condition (ii).
Hence, the code is ECXWONTE.

27. The given group of elements is 1$Ω4925.
This group follows condition (i).
Hence, the code is AXMBOGA.

28. The given group of elements is $4@Ω2*5.
This group does not follow any condition.
Hence, the code is XBTMGKC.

29. The given group of elements is *87$Ω2$.
This group follows condition (iv).
Hence, the code is XGMXDHK.

30. The given group of elements is 872Ω@9®.
This group follows condition (iii).
Hence, the code is PDGMTOH.

5 Symbols and Notations

CHAPTER

Learning Objectives

In this chapter, you will:

- Learn about symbols and notations as an extension of coding-decoding
- Learn how to draw conclusions, find the values or compare two or more quantities after understanding the notations/relations given

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❒ SYMBOLS AND NOTATIONS

The basic approach for the questions of this chapter is more or less similar to that of coding and decoding. As in the questions of coding and decoding, a basic word is coded in a particular way and the candidate is asked to code another word using the same logic.

Similarly, in 'Symbols and Notations', one has to study the symbols and their meanings carefully which are given against them. Then, the meanings given are to be used in place of those symbols in answering the questions. The word 'Notation' basically stands for the meaning which each symbol will be assigned.

SOLVED EXAMPLES

Directions for questions 1 to 5: These questions are based on the following information.

$a + b$ means a is greater than b.

$a - b$ means a is less than or equal to b.

$a \times b$ means a is greater than or equal to b.

$a \div b$ means a is less than b.

$a \# b$ means a is equal to b.

Each of the following questions consists of some statements followed by two conclusions.

Give your answer as:

(A) If only conclusion (I) follows.
(B) If only conclusion (II) follows.
(C) If both conclusions (I) and (II) follow.
(D) If neither (I) nor (II) follows.

1. Statements: m – n, n ÷ o, o # q.
Conclusions: I: q × m
II. q + m

Sol: The given statements are (i) $m \le n$, $n < o$ and $o = q$, i.e, $m \le n < o = q$.
Conclusion I: q × m ⇒ $q \ge m$ does not follow.
Hence, conclusion I does not follow.
Conclusion II: q + m ⇒ $q > m$ follows.
Hence, only conclusion II follows.

2. Statements: l ÷ m, o # q, m + o
Conclusions: I: m + q
II: l × o

Sol: The given statements are $l < m$, $o = q$ and $m > o$, i.e., $m > l$, $m > o = q$.
Conclusion I: m +q ⇒ $m > q$ follows.
Conclusion II: l × o ⇒ $l \ge o$ does not follow.
∴ Only conclusion I follows

3. Statements: $x \div y$, $y - z$, $z \div a$
Conclusions: I. $x - a$
II. $z - x$

Sol: By combining the statements, we get $x < y \le z < a$.
Conclusion I: $x - a \Rightarrow x \le a$ does not follow.
Conclusion I does not follow.
Conclusion II: $z - x \Rightarrow z \le x$ does not follow.
Conclusion II does not follow
∴ Neither I nor II follows.

4. Statements: e # f, f + g, g ÷ h
Conclusions: I. e + g
II. g + f

Sol: By combining all the statements, we get:
$e = f > g$; $g < h$
Conclusion I: e + g ⇒ $e > g$ follows.
Conclusion I follows.
Conclusion II: g + f ⇒ $g > f$ does not follow.
Conclusion II does not follow.
∴ Only I follows.

5. Statements: $a + b$, $c - d$, $d \div b$
Conclusions: I. $a + c$
II. $c \div b$

Sol: By combining all the statements, we get:
$a > b$, $c \le b$, $d < b$
$\Rightarrow a > b > d \ge c$
Conclusion I: $a + c \Rightarrow a > c$ follows.
Conclusion I follows.
Conclusion II: $c \div b \Rightarrow c < b$ follows.
Conclusions II follows.
∴ Both I and II follow.

6. If 'Δ' means 'is less than', '\$' means 'is greater than' and '£' means 'is equal to' and given that $a \,\Delta\, b$, $c \,£\, d$ and $c \,\$\, b$, then which of the following is true?
(A) $d \,\Delta\, a$ (B) $b \,\$\, d$
(C) $a \,£\, c$ (D) $a \,\Delta\, b \,\Delta\, c$

Sol: $a \,\Delta\, b \Rightarrow a < b$
$c \,\$\, b \Rightarrow c > b \Rightarrow b < c$
$c \,£\, d \Rightarrow c = d$
$\therefore a < b < c = d$
(A) d Δ a ⇒ $d < a \rightarrow$ does not follow
(B) $b \,\$\, d \Rightarrow b > d \rightarrow$ does not follow
(C) $a \,£\, c \Rightarrow a = c \rightarrow$ does not follow
(D) $a \,\Delta\, b \,\Delta\, c \Rightarrow a < b < c \rightarrow$ follows

EXERCISES

Directions for questions 1 to 5: In a certain code language, '+' means '×', '×' means '–', '–' means '÷' and '÷' means '+'.

Simplify the following expressions using the above directions, in which the mathematical operators are written according to the code language.

1. $9 + 4 - 6 \times 6 \div 8$
(A) 8.5 (B) 8
(C) 5.25 (D) 0

2. $10 + 10 \times 10 - 10 \div 10$
(A) 109 (B) 10
(C) 19 (D) 11

3. $16 \times 4 \div 4 + 14 - 2$
(A) 56 (B) 28
(C) 40 (D) 112

4. $16 - 2 + 4 \div 16 - 8 \times 2$
(A) –15 (B) 2
(C) 4 (D) 32

5. $2 \div 4 + 8 - 16 \times 32 \div 64 \times 128 \div 256$
(A) 4·5 (B) 164
(C) 4 (D) 163

Directions for questions 6 to 10: These questions are based on the following information.

In the following questions certain symbols are used to represent relations as given below.

A ϕ B means A is greater than B.
A δ B means A is greater than or equal to B.
A Ω B means A is equal to B.
A $\int$ B means A is not greater than B.
A λ B means is neither greater nor equal to B.

In each of the following questions four statements followed by four conclusions marked I, II, III, IV are given. Assuming that the statements to be true, find which of the four conclusions follows the given statements.

6. Statements: P Ω R, Q $\int$ S, R δ S, P λ T.
Conclusions:
I. P ϕ S,
II. P Ω Q
III. P Ω S
IV. Q λ P
(A) Either I or III follows.
(B) Either II or IV follows.
(C) Either I or III and either II or IV follow.
(D) All follow

7. Statements: J ϕ H, J λ G, I δ K, G $\int$ K.
Conclusions:
I. I ϕ G
II. J ϕ I
III. G Ω I
IV. H λ K
(A) Only IV follows. (B) Either I or III follows.
(C) Only II follows. (D) Both (A) and (B).

8. Statements: W Ω Z, Z δ X, W $\int$ V, Y ϕ V.
Conclusions:
I. V δ X
II. Y $\int$ X
III. Y ϕ Z
IV. V λ X
(A) Only I follows. (B) Only III follows.
(C) Only II follows. (D) Both (A) and (B).

9. Statements: P ϕ F, U λ W, R $\int$ F, R δ U.
Conclusions:
I. P δ W
II. W Ω F
III. W λ F
IV. P λ W
(A) Only I and III follow.
(B) Only II and IV follow.
(C) Either III or IV follows.
(D) None follows

10. Statements: A λ B, B $\int$ C, C δ E, E ϕ D.
Conclusions:
I. A λ C
II. B ϕ E
III. C ϕ D
IV. E Ω A
(A) Only I follows.
(B) Only III follows.
(C) Only II and III follows.
(D) Only I and III follows.

Directions for questions 11 to 15: Select the correct alternative from the given choices.

11. Which of the following symbols should replace the question marks in that order in the given expression, in order to make the expression '$A > J$' definitely true?
A? X? E? U? J? S
(A) =, ≥, =, =. > (B) >, ≥, =, <, >
(C) =, >, =, >, ≥ (D) <, =, ≤, <, >

12. Which of the following elements should replace the question marks in that order in the given expression, in order to make the expression '$L > P$' definitely true?
? > ? =? >? =? <?
(A) L, O, N, M, Q, P (B) L, M, N, O, P, Q
(C) P, O, N, Q, M, L (D) L, M, O, N, Q, P

13. Which of the following expressions is true if the given expressions 'B > K' as well as 'T < D' are true?
(A) $T < B \leq G < D = K$ (B) $B < T = G \leq D > K$
(C) $D < K > G = T \geq B$ (D) $B > D = G \geq K > T$

14. Which of the following expressions is not true if the given expression 'L < F' as well as 'H ≥ Q' are definitely true?
(A) $L < M \leq H = F \geq Q$ (B) $L < M \leq Q = F \leq H$
(C) $L < Q = M \geq H \leq F$ (D) $H \geq F = M \geq Q > L$

15. Which of the following symbols should replace the question mark in the given expression in order to make the expressions 'C < V' and 'Y < Q' definitely true?
C < J ≤ Q ? V ≥ W > Y
(A) = (B) <
(C) ≤ (D) >

Directions for questions 16 to 20: Study the following sequence carefully and answer the questions given below.

R K 5 9 # B 2 % * E ? A 8 L $ I 4 S V 7 ! C 6 N @ H 1 3 & D

16. Four of the following are alike. Find the odd one.
(A) RKB (B) ALI
(C) SVC (D) BLI

17. How many consonants are there, which are immediately followed by a digit but not immediately preceded by a consonant?
(A) 3 (B) 2
(C) 1 (D) 0

18. Find the next term in the following series.
5#2, *?8, $4V, _____
(A) LIS (B) !6@
(C) I4S (D) 13$

19. Which is the 9th element to the right of the 19th element from the left end?
(A) 7 (B) 3
(C) V (D) H

20. How many letters are there, each of which are immediately followed and immediately preceded by a symbol?
(A) One (B) Three
(C) Two (D) More than three

Directions for questions 21 to 25: Select the correct alternative from the given choices.

21. If $a \Delta b = a + b + ab$ and $a \$ b = a^2 + b^2$, then $(3 \Delta 4) \$ 5 =$
(A) 386 (B) 1625
(C) 336 (D) 436

22. If $p > q = p^2 + q^3$ and $p < q = p^3 - q^2$ then $(1 > 2) < 3 =$
(A) 503 (B) 720
(C) 648 (D) 960

23. If $x > y = x^3 + y^3$ and $x @ y = x^3 - y^3$ then $3 > (2 @ 1) =$
(A) 756 (B) 702
(C) 440 (D) 370

24. If $E \downarrow F = (E + F)^2 + (E - F)^2$ and $E \uparrow F = (E + F)^2 - (E - F)^2$, then $(2 \downarrow 5) \uparrow (4 \downarrow 3) =$
(A) 7808 (B) 15616
(C) 11600 (D) 23200

25. If $m \neq n = m^2 - mn + n^2$ and $m ? n = (m + n)^2 - mn$, then $(2 ? 3) \neq (5 \neq 3) =$
(A) 900 (B) 361
(C) 642 (D) 729

Directions for questions 26 to 30: In a certain instruction system the different computation processes are written as follows.

(a) 'A % B ! C' means 'A is added to the product of B and C'.
(b) 'A © B * C' means 'the product of B and C is subtracted from A'.
(c) 'A # B @ C' means 'the product of A and B is divided by C'.
(d) 'A • B $ C' means 'C is multiplied by the sum of A and B'.

You have to find out what will come in the place of question mark (?) in each question following the computation processes.

26. 100 © 20 * 3 = *a*
a % 40 ! 5 = ?
(A) 140 (B) 240
(C) 340 (D) 360

27. 16 • 14 $ 4 = t
t # 10 @ 12 = ?
(A) 700 (B) 300
(C) 400 (D) 100

28. 100 © 5 * 16 = q
q % 4! 12 = ?
(A) 140 (B) 68
(C) 98 (D) 102

29. 50 # 40 @ 200 = p
16 % 12 ! p = ?
(A) 112 (B) 136
(C) 126 (D) 226

30. 12 • 13 $ 5 = r
10 © 4 * r = ?
(A) 120 (B) –430
(C) –490 (D) 720

ANSWER KEYS

1. (B)	6. (C)	11. (C)	16. (D)	21. (A)	26. (B)
2. (A)	7. (D)	12. (B)	17. (A)	22. (B)	27. (D)
3. (C)	8. (D)	13. (D)	18. (B)	23. (D)	28. (B)
4. (D)	9. (D)	14. (C)	19. (B)	24. (C)	29. (B)
5. (B)	10. (D)	15. (A)	20. (A)	25. (B)	30. (C)

SOLUTIONS

Solutions for questions 1 to 5:

1. '+' means '×', '×' means '–', '–' means '÷' and '÷' means '+'
The given expression is $9 + 4 - 6 \times 6 \div 8$.
On converting this expression as per the directions, we get the expression $9 \times 4 \div 6 - 6 + 8$.
Let us simplify this expression using BODMAS rule, where
'B' stands for 'Brackets',
'O' stands for 'of',
'D' stands for 'Division',
'M' stands for 'Multiplication',
'A' stands for 'Addition' and
'S' stands for 'subtraction'.
$\therefore\ 9 \times 4 \div 6 - 6 + 8 = 36 \div 6 - 6 + 8 = 6 - 6 + 8 = 8$

2. The given expression is $10 + 10 \times 10 - 10 \div 10$.
As per the directions, it becomes $10 \times 10 - 10 \div 10 + 10$.
Using BODMAS rule, we get:
$10 \times 10 - 10 \div 10 + 10 = 100 - 1 + 10 = 109$.

3. The given expression $16 \times 4 \div 4 + 14 - 2$ becomes $16 - 4 + 4 \times 14 \div 2$ as per the given directions.
Using BODMAS rule, we get:
$16 - 4 + 4 \times 14 \div 2 = 12 + 56 \div 2 = 12 + 28 = 40$.

4. As per the given directions, the given expression
$16 - 2 + 4 \div 16 - 8 \times 2$ becomes
$16 \div 2 \times 4 + 16 \div 8 - 2$.
Using BODMAS rule, we get:
$16 \div 2 \times 4 + 16 \div 8 - 2 = 8 \times 4 + 2 - 2 = 32 + 0 = 32$.

5. $2 \div 4 + 8 - 16 \times 32 \div 64 \times 128 \div 256$ becomes
$2 + 4 \times 8 \div 16 - 32 + 64 - 128 + 256$.
Using BODMAS rule, we get:
$2 + 4 \times 8 \div 16 - 32 + 64 - 128 + 256$
$= 2 + 32 \div 16 + 32 + 128$
$= 2 + 2 + 160 = 164$

Solutions for questions 6 to 10:

A ϕ B means A is greater than B $\Rightarrow A > B$.
A δ B means A is greater than or equal to B $\Rightarrow A \geq B$.
A Ω B means A is equal to B $\Rightarrow A = B$.
A $\int$ B means A is not greater than B $\Rightarrow A \leq B$.
A λ B means A is neither greater nor equal to B $\Rightarrow A < B$.

6. Given statements are:
$P\ \Omega\ R \Rightarrow P = R$
$Q \int S \Rightarrow Q \leq S$
$R\ \delta S \Rightarrow R \geq S$
$P\ \lambda\ T \Rightarrow P < T$
By combining all the statements, we get:
$T > P = R \geq S \geq Q$
Conclusion I: $P\ \phi\ S \Rightarrow P > S$ does not follow.
Conclusion II: $P\ \Omega\ Q \Rightarrow P = Q$ does not follow.
Conclusion III: $P\ \Omega\ S \Rightarrow P = S$, does not follow.
Conclusion IV: $Q\ \lambda\ p \Rightarrow Q < P$ does not follow.
But either I or III and either II or IV follows.

7. Given statements are:
$J\ \phi H \Rightarrow J > H$
$J\ \lambda G \Rightarrow J < G$
$I\ \delta K \Rightarrow I \geq K$
$G \int K \Rightarrow G \leq K$
By combining all the statements, we get:
$H < J < G \leq K \leq I$
Conclusion I: $I\ \phi\ G \Rightarrow I > G$ does not follow.
Conclusion II: $J\ \phi\ I \Rightarrow J > I$ does not follow.
Conclusion III: $G\ \Omega\ I \Rightarrow G\ I$ does not follow.
Conclusion IV: $H\ \lambda K \Rightarrow H < K$ follows.
$\therefore$ Only IV and either I or III follow.

8. Given statements are:
$W\ \Omega\ Z \Rightarrow W = Z$; $Z\ \delta X \Rightarrow Z \geq X$
$W \int V \Rightarrow W \leq V$; $Y\ \phi V \Rightarrow Y > V$
By combining all the statements, we get:
$Y > V \geq W = Z \geq X$
Conclusion I: $V\ \delta X \Rightarrow V \geq X$ follows.
Conclusion II: $Y \int X \Rightarrow Y \leq X$ does not follow.
Conclusion III: $Y\ \phi Z \Rightarrow Y > Z$ follows.
Conclusion IV: $V\ \lambda X \Rightarrow V < X$ does not follow.
$\therefore$ Only I and III follow.

9. Given statements are:
$P\ \phi F \Rightarrow P > F; U\ \lambda W \Rightarrow U < W$
$R \int F \Rightarrow R \leq F; R\ \delta U \Rightarrow R \geq U$
By combining all the statements, we get:
$P > F \geq R \geq U < W$

Conclusion I: $P \,\delta\, W \Rightarrow P \geq W$ does not follow.
Conclusion II: $W \,\Omega\, F \Rightarrow W = F$ does not follow.
Conclusion III: $W \,\lambda\, F \Rightarrow W < F$ does not follow.
Conclusion IV: $P \,\lambda\, W \Rightarrow P < W$ does not follow.
$\therefore$ None follows.

10. Given statements are:
$A \,\lambda\, B \Rightarrow A < B$
$B \int C \Rightarrow B \leq C$; $C \,\delta\, E \Rightarrow C \geq E$; $E \,\phi\, D \Rightarrow E > D$
By combining all the statements, we get:
$A < B \leq C \geq E > D$
Conclusion I: $A \,\lambda C \Rightarrow A < C$ follows.
Conclusion II: $B \,\phi\, E \Rightarrow B > E$, does not follow.
Conclusion III: $C \,\phi\, D \Rightarrow C > D$, follows.
Conclusion IV: $E \,\Omega\, A \Rightarrow E = A$ does not follow.
$\therefore$ Only I and III follow.

Solutions for questions 11 to 15:

11. In order to make the given expression 'A > J' true, the symbols which are to be placed in the place of question mark are =, >, =, >, ≥. Then the expression becomes $A = X > E = U > J \geq S$.

12. In order to make the given expression 'L > P' true, the elements which are to be placed in the place of question mark are L, M, N, O, P and Q. Then the expression becomes $L > M = N > O = P < Q$.

13. In the expression $B > D = G \geq K > T$, the given expression 'B > K' as well as 'T < D' are true.

14. In all the other expressions except in $L < Q = M \geq H < F$ the expressions 'L < F' and 'H ≥ Q' are definitely true.

15. In the given expression the question mark should be replaced with '=' in order to make the expressions 'C < V' and 'Y < Q' definitely true. Then the expression becomes $C < J \leq Q = V \geq W > Y$.

Solutions for questions 16 to 20:

16. The given sequence is:
R K 5 9 # B 2 % * E ? A 8 L $ I 4 S V 7 ! C 6 N @ H 1 3 & D
Except BLI, in all others, the three letters are consecutive alphabets in the given sequence.

17. B, C and H are the three letters which are followed by a digit but not immediately preceded by a consonant.

18. The sequence is:
5 # 2, *?8, $4V _____
The logic is as follows:
$5^{+2}\ \#^{+2}\ 2^{+2}$, $*^{+2}\ ?^{+2}\ 8^{+2}$, $\$^{+2} 4^{+2}\ V^{+2}$, $!^{+2}\ 6^{+2}$@

19. The 9th element to the right of the 19th element form the left end is (9 + 19)th = 28th from the left end, that is 3.

20. There is only one letter, i.e., E, which is immediately followed and immediately preceded by symbol.

Solutions for questions 21 to 25:

21. $3 \,\Delta\, 4 = 3 + 4 + 3 \times 4 = 7 + 12 = 19$
$(3 \,\Delta\, 4) \,\$\, 5 = 19 \,\$\, 5 = 19^2 + 5^2$
$= 361 + 25 = 386$

22. $1 > 2 = 1^2 + 2^3 = 1 + 8 = 9$
$(1 > 2) < 3 = 9 < 3 = 9^3 - 3^2$
$= 729 - 9 = 720$

23. $2 \,@\, 1 \Rightarrow 2^3 - 1^3 = 8 - 1 = 7$
$3 > (2 \,@\, 1) = 3 > 7 = 3^3 + 7^3$
$\Rightarrow 27 + 343 = 370$

24. $E \downarrow F = (E + F)^2 + (E - F)^2 = 2\ (E^2 + F^2)$
$E \uparrow F = (E + F)^2 - (E - F)^2 = 4\,E\,F$
$2 \downarrow 5 = 2\ (2^2 + 5^2) = 2\ (4 + 25) = 58$
$4 \downarrow 3 = 2\ (4^2 + 3^2) = 2\ (16 + 9) = 50$
$\therefore\ (2 \downarrow 5) \uparrow (4 \downarrow 3) = 58 \uparrow 50$
$= 58 \uparrow 50$
$\Rightarrow 4 \times 58 \times 50 = 11600$

25. $2 \,?\, 3 = (2 + 3)^2 - 2 \times 3 = 25 - 6 = 19$
$5 \neq 3 = 5^2 - 5 \times 3 + 3^2 = 25 - 15 + 9 = 19$
$(2 \,?\, 3) \neq (5 \neq 3) = 19 \neq 19$
$= 19^2 - 19 \times 19 + 19^2 = 19^2 = 361$

Solutions for questions 26 to 30:

(a) 'A % B ! C' means 'A is added to the product of B and C'.
(b) 'A © B * C' means 'the product of B and C is subtracted from A'.
(c) 'A # B @ C' means 'the product of A and B is divided by C'.
(d) A • B $ C' means 'C is multiplied by the sum of A and B'.

26. Given, 100 © 20 * 3 = a
The value $a = 100 - (20 \times 3) = 40$
Then 40% 40 ! 5 = $40 + 40 \times 5 = 240$.

27. Given, 16 • 14 $ 4 = q
The value of $q = (16 + 14) \times 4 = 120$
Then 120 # 10 @ 12
$= \dfrac{120 \times 10}{12} = 10 \times 10 = 100.$

28. Given, 100 © 5 * 16 = t
$\therefore\ t = 100 - (5 \times 16) = 20$
Then 20 % 4 ! 12 = $20 + 4 \times 12 = 20 + 48 = 68$.

29. Given, 50 # 40 @ 200 = p
$\Rightarrow\ p = \dfrac{50 \times 40}{200} = 10$
Then 16% 12! 10 = $16 + 12 \times 10 = 16 + 120 = 136$.

30. Given, 12 • 13 $ 5 = r
$\Rightarrow r = (12 + 13) \times 5 = 125$
10 © 4 * 125 = $10 - (4 \times 125) = -490$.

Exercises

6 Blood Relations

CHAPTER

Learning Objectives

In this chapter, you will:

- Learn about different types of blood relations
- Learn how to draw a family tree
- Learn to differentiate between different stages/levels of the family tree
- Learn how to represent the gender of each person and the relationship between two persons

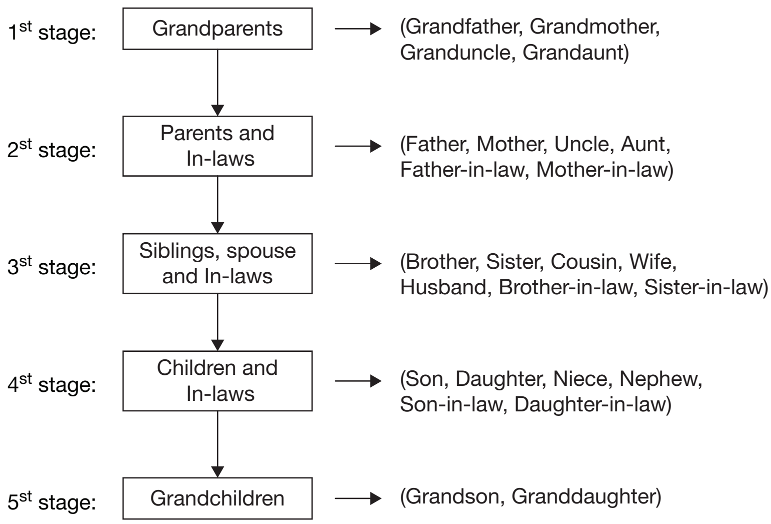
The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❒ BLOOD RELATIONS

There are two types of questions based on blood relations that are given in different competitive examinations. For the sake of convenience, we will refer to the two types of questions as Type I and Type II. (Please note that the questions on blood relationships are not categorized as above in the actual exam papers. It is being done purely from the point of better understanding).

In the exams, the success of a candidate in the questions on blood relations depends upon his knowledge about various blood relations. Some of the relationships given below help in solving the problems.

The easiest and non-confusing way to solve these types of problems would be to draw a family tree diagram and increase the levels in the hierarchy as shown in the next page:



Mother's or Father's son	:	Brother
Mother's or Father's daughter	:	Sister
Mother's or Father's brother	:	Uncle
Mother's or Father's sister	:	Aunt
Mother's or Father's mother	:	Grandmother
Mother's or Father's father	:	Grandfather
Grandmother's brother	:	Granduncle
Grandmother's sister	:	Grandaunt
Grandfather's brother	:	Granduncle
Grandfather's sister	:	Grandaunt
Sister's or Brother's son	:	Nephew
Sister's or Brother's daughter	:	Niece
Uncle or Aunt's (Son or Daughter)	:	Cousin
Son's wife	:	Daughter-in-law
Daughter's husband	:	Son-in-law
Husband's or Wife's sister	:	Sister-in-law
Husband's or Wife's brother	:	Brother-in-law
Sister's husband	:	Brother-in-law
Brother's wife	:	Sister-in-law
Children of same parents	:	Siblings (could be all brothers, all sisters or some brothers and some sisters)
Children	:	Son, Daughter
Children's Children	:	Grandchildren (Grandson, Granddaughter)

In addition, remember the word spouse which means either husband or wife.

Grandfather and grandmother will come in the first stage; mother, father, uncle and aunt will come in the second stage; sister, brother and cousin will come at the third stage; son, daughter, niece and nephew will come in the fourth stage and finally, granddaughters and grandsons will come. The above stages are made from the point of view of an individual.

In **Type – I questions**, the relationship between two people is given through a roundabout way of relating them through other people. We have to go through the series of relationships and finally determine the relationship between the two people given in the question. The relationship can be given as a simple statement or as a statement made by a person. In the first example given below, a person is involved in making a statement whereas in the **Type – II question**, there is no person involved in making a statement.

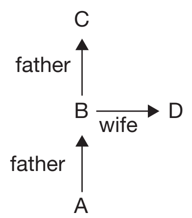
SOLVED EXAMPLES

1. A's father's mother-in-law's only daughter's son is B. How is A related to B?
(A) Brother
(B) Sister
(C) Nephew
(D) Cannot be determined

Sol: A's father's mother-in-law's only daughter is A's mother. A's mother's son is A's brother.
But A can be either brother or sister to B.

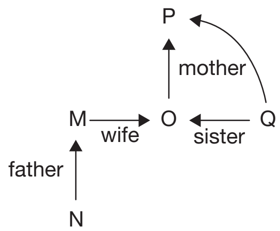
2. If A's father is B, C is the father of B and D is A's mother, then how is C related to D?
(A) Father
(B) Grandfather
(C) Father-in-law
(D) Uncle

Sol: A's father is B and mother is D. Therefore, D is B's wife and C is the father of B. Hence, C is D's father-in-law.



3. A + B means A is the son of B.
A – B means A is the daughter of B.
A × B means A is the father of B.
A ÷ B means A is the mother of B.
If M × N + O – P ÷ Q, then how is M related to Q?
(A) Husband
(B) Cousin
(C) Brother-in-law
(D) Uncle

Sol: M × N + O – P ÷ Q means M is the father of N, N is the son of O, O is the daughter of P, P is the mother of Q.
M is the father of N and N is the son of O means M is the husband of O.
O is the daughter of P and P is the mother of Q means O is the sister of Q.
M is the husband of O and O is the sister of Q means M is the brother-in-law of Q.



4. A + B means A is the son of B.
A – B means A is the daughter of B.
A × B means A is the father of B.
A ÷ B means A is the mother of B.
Which of the following means S is the son-in-law of P?
(A) P + Q ÷ R × S – T (B) P × Q ÷ R – S + T
(C) P + Q × R – S ÷ T (D) P × Q – R ÷ S × T

Sol: P + Q ÷ R × S – T means P is the son of Q. Q is the mother of R, R is the father of S and S is the daughter of T. Hence, S is the nephew of P.
P × Q ÷ R – S + T means P is the father of Q, Q is the mother of R, R is the daughter of S and S is the son of T. Hence, S is the son-in-law of P.
R + Q × R – S ÷ T means P is the son of Q, Q is the father of R, R is the daughter of S and S is the mother of T. Hence, S is the mother of P.
P × Q – R ÷ S × T means P is the father of Q, Q is the daughter of R, R is the mother of S and S is the father of T. Hence, S is the son of P.

5. Pointing to a person, Raju said, 'He is the only brother of my father's mother's daughter'. How is the person related to Raju?
(A) Brother (B) Father
(C) Uncle (D) Nephew

Sol: Raju's father's mother's daughter is Raju's father's sister. Raju's father's sister's only brother is Raju's father. Hence, the person is Raju's father.

6. A's mother's father is the husband of B's mother. How is A related to B, if A and B are males?
(A) Uncle (B) Father
(C) Nephew (D) Son

Sol: A's mother's father is the husband of B's mother.ThatmeansA'smotheristhesisterofB.Hence,Ais the nephew of B.

EXERCISES

Directions for questions 1 to 15: Select the correct alternate from the given choices.

1. How is my mother's daughter's only sister-in-law related to me if my father has only one child?
(A) Cousin (B) Sister
(C) Sister-in-law (D) Niece

2. A person who is my wife's mother's only son is my ________.
(A) Brother-in-law (B) Sister-in-law
(C) Cousin (D) Mother-in-law

3. How is my father's daughter's mother's only daughter-in-law related to my wife's son?
(A) Mother (B) Father
(C) Brother (D) Sister

4. How is my brother's son's sister related to my mother's husband?
(A) Son (B) Daughter
(C) Grandson (D) Granddaughter

5. A is the sister of B's wife's brother-in-law's father. If B's father-in-law has only one child, B is A's ________.
(A) Nephew (B) Niece
(C) Aunt (D) Daughter

6. Mr Rakesh's father-in-law's only child's sister-in-law's father is Rakesh's ________.
(A) Uncle (B) Father
(C) Brother (D) Grandfather

7. How is my mother's sister's only sibling's father related to me?
(A) Father (B) Uncle
(C) Father-in-law (D) Grandfather

8. How is my father's wife's son's daughter's brother related to me?
(A) Son (B) Nephew
(C) Niece (D) Cannot be determined

9. How is my son's daughter's sibling's mother related to my wife's only son?
(A) Sister (B) Sister-in-law
(C) Wife (D) Aunt

10. How is Mr Harry's son's paternal grandmother related to Harry's father-in-law's only daughter?
(A) Mother-in-law (B) Aunt
(C) Mother (D) Wife

11. How is my father's son's sister's mother related to me?
(A) Grandmother (B) Aunt
(C) Sister (D) Mother

12. C is the brother of D. A is the husband of B. B is the mother of C. E is the wife of D. How is B related to E?
(A) Mother (B) Father
(C) Mother-in-law (D) Father-in-law

13. How is my father's brother's wife's daughter related to me?
(A) Brother (B) Cousin
(C) Sister (D) Aunt

14. Ms Ritu's only brother's son's maternal grandmother is Sita. How is Sita's husband related to Ritu's brother's sister-in-law?
(A) Father (B) Uncle
(C) Grandfather (D) Brother

15. Ms Sneha introduced a person to Deepti and said, 'He is your sister's paternal grandfather and also my only brother's father'. How is Sneha related to Deepti?
(A) Mother (B) Sister
(C) Grandmother (D) Aunt

Directions for questions 16 to 20: Use the relationship given below and answer the following questions.

P @ Q means P is the father of Q.
P ⇒ Q means P is the mother of Q.
P * Q means P is the son of Q.
P # Q means P is the daughter of Q.
P $ Q means P is the brother of Q.
P % Q means P is the sister of Q.

16. Which of the following means A is the grandson of B?
(A) A ⇒ R * S * B (B) A $ R # S @ B
(C) A $ R # S *B (D) A $ R ⇒ S * B

17. Which of the following means W is the mother of Z?
(A) W * X * Y @ Z (B) W ⇒ X # Y @ Z
(C) W ⇒ X * Y @ Z (D) Both (B) and (C)

18. Which of the following means E is the nephew of G?
(A) E # F $ G (B) E * F $ G
(C) E * F % G (D) Both (B) and (C)

19. Which of the following means H is the father of L?
(A) H @ I % J # K ⇒ L (B) H ⇒ I % J # K @ L
(C) H @ I % J % K # L (D) H ⇒ I # J @ K $ L

20. Which of the following means M is the aunt of S?
(A) M $ N @ R $ S (B) M % N @ R $ S
(C) M % N ⇒ R @ S (D) M @ N ⇒ R % S

Directions for questions 21 to 25: These questions are based on the following information.

P, Q, R, S, T, U, V and W are eight members of a family. W is T's only brother. S's son is U and S is P's son. R is T's brother-in-law and his mother is Q. V is not Q's husband. R is not S's son. W has a niece.

21. Who is Q's daughter-in-law?
(A) S (B) T
(C) U (D) V

22. What is the ratio of males to females in the family?
(A) 1 : 1 (B) 3 : 5
(C) 5 : 3 (D) 3 : 1

23. How is R related to V?
(A) Brother (B) Father
(C) Daughter (D) Uncle

24. Which of the following statements is true?
(A) U is R's Son (B) V is W's daughter
(C) P is Q's brother (D) None of these

25. How is P's grandson related to T's brother?
(A) Nephew (B) Niece
(C) Son (D) Daughter

Directions for questions 26 to 30: These questions are based on the following information.

In a family of three generations there are six members A, B, C, D, E and F. All of them are from different professions. E is the son of a Teacher. The Architect is married to the Manager. C is an Engineer and his daughter is a Doctor. D is the wife of an Engineer. A is not C's father and C's mother is B. A's grandfather is a Manager. One of them is a Student, who is not F.

26. Which of the following pair represents a couple?
(A) FD (B) AB
(C) CD (D) AE

27. Who is a Student?
(A) A (B) B
(C) C (D) E

28. What is the profession of D's daughter?
(A) Doctor (B) Student
(C) Engineer (D) Manager

29. How is the Engineer related to the Student?
(A) Son (B) Daughter
(C) Father (D) Brother

30. Who is Teacher's father-in-law?
(A) A (B) B
(C) C (D) None of these

ANSWER KEYS

1. (C)	6. (B)	11. (D)	16. (C)	21. (B)	26. (C)
2. (A)	7. (D)	12. (C)	17. (D)	22. (C)	27. (D)
3. (A)	8. (D)	13. (B)	18. (D)	23. (D)	28. (A)
4. (D)	9. (C)	14. (A)	19. (A)	24. (D)	29. (C)
5. (A)	10. (A)	15. (D)	20. (B)	25. (A)	30. (D)

SOLUTIONS

Solutions for questions 1 to 15:

1. Note: For all the diagram, ▭ represent males and ○ represent females.
My mother's daughter is myself. As my father has only one child, my only sister-in-law is my sister-in-law.

2. My wife's mother's only son is my wife's brother. Hence, he is my brother-in-law.

3. My father's daughter's mother is my mother. My mother's only daughter-in-law is my wife. My wife is the mother of her son.

4. My brother's son's sister is my brother's daughter. My mother's husband is my father. therefore, he is my brother's father. My brother's daughter is my father's granddaughter.

5. B's wife's brother-in-law is B's brother as B's father-in-law has only one child. B's brother's father's sister is B's aunt. Thus, B is A's nephew.

6. Mr Rakesh's father-in-law's only child is Rakesh's wife. Rakesh's wife's sister-in-law is Rakesh's sister, whose father is Rakesh's father.

7. My mother's sister's only sibling is my mother. My mother's father is my grandfather.

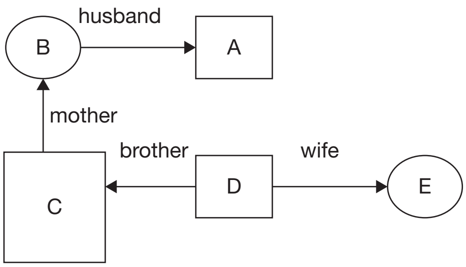
8. My father's wife's son can be me or my brother. Thus, the person here can be my son or my nephew. Hence, it cannot be determined.

9. My son's daughter's sibling's mother is my daughter-in-law. My wife's only son is my son. My daughter-in-law is my son's wife.

10. Harry's son's paternal grandmother is Harry's mother. Harry's father-in-law's only daughter is Harry's Wife. Harry's mother is Harry's wife's mother-in-law.

11. My father's son's sister's mother is my mother.

12. As A is the husband of B, B is the mother of C and C is the brother of D, we can say that C and D are children of A and B. Now, E is the wife of D. Thus, B is the mother-in-law of E.



13. My father's brother's wife's daughter is my cousin.

14. Ms. Ritu's only brother's son's maternal grandmother is Ritu's brother's mother-in-law, i.e., Sita.
Ritu's brother's sister-in-law is Sita's daughter. Sita's husband is the father of Sita's daughter.

15. He is your sister's paternal grandfather means he is Deepti's paternal grandfather. He is my only brother's father means he is Sneha's father. Thus, Sneha is the aunt of Deepti.

Solutions for questions 16 to 20:

16. Choice (A):
A ⇒ R * S * B means A is the mother of R, R is the son of S, S is the son of B. Hence, A is the daughter-in-law of B.
Choice (B):
A $ R # S @ B means A is the brother of R, R is the daughter of S, S is the father of B.
∴ A is the brother of B.
Choice (C):
A $ R # S * B means A is the brother of R, R is the daughter of S, S is the son of B.
∴ A is the grandson of B.

17. From (A), W is a male. Hence, he cannot be a mother.
Choice (B):
W ⇒ X # Y @ Z means W is the mother of X, X is the daughter of Y and Y is the father of Z. Hence, W is the mother of Z.
Choice (C):
W ⇒ X * Y @ Z means W is the mother of X, X is the son of Y and Y is the father of Z. Thus, W is the mother of Z.
Hence, (B) and (C) are correct.

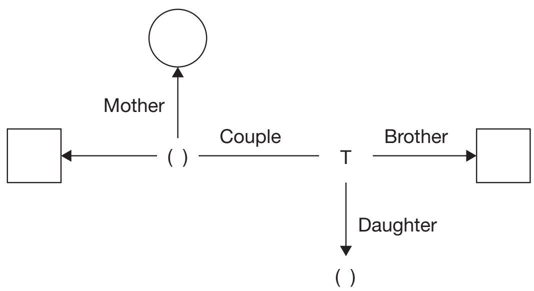
18. Choice (A):
E # F means, E is the daughter of F. Thus, E is a female and hence, she cannot be a nephew.
Choice (B):
E * F means, E is the son of F. F $ G means F is the brother of G. Hence, E is the nephew of G.
Choice (C):
E * F means, E is the son of F. F % G means F is the sister of G. Hence, E is the nephew of G.
∴ Both (B) and (C) are correct.

19. Choice (A):
H @ I means H is the father of I. I % J means I is the sister of J. J # K means J is the daughter of K and K ⇒ L means K is the mother of L. Hence, H is the father of L.

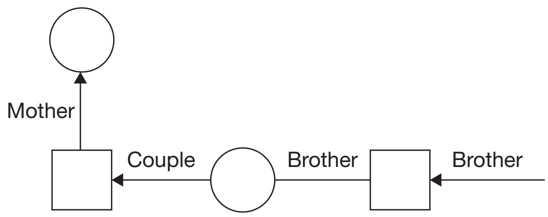
20. Choice (A):
M $ N means M is the brother of N. Thus, M is a male. Hence, he cannot be the aunt.
Choice (B):
M % N means M is the sister of N, N @ R means N is the father of R. R $ S means R is the brother of S. Thus, M is the aunt of S.

Solutions for questions 21 to 25: From the information, W is T's brother, R is T's brother-in-law and R's mother is Q and W has a niece. We get the following relationship diagram.

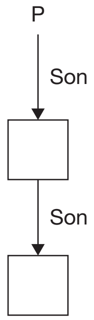
(i)



(ii)



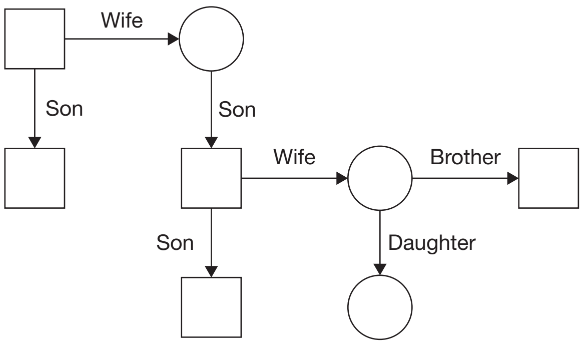
Now, S is P's son and U is S's son.



Now, Q's husband is not V (given) and not S (R is not S's son).

Thus, Q's husband is either P or U. But U can be Q's husband because then there is no way to find out R's sibling and T's daughter.

Thus, P is Q's husband and V is T's daughter. Case (ii) is not possible as S is a male. Thus, we get following relationship diagram.



21. Q's daughter-in-law is T.

22. P, R, S, W and U are males and Q, T and V are females. Hence, the ratio is 5 : 3.

23. R is V's uncle.

24. None of the statements is true.

25. P's grandson is U. T's brother is W. U is W's nephew.

Solutions for questions 26 to 30: It is given that C is an Engineer. Also, D is the wife of an Engineer, i.e., D is C's wife. Also, C's mother is B and C has a daughter. And from the information that F is not C's father. Thus, we get the following relationship diagram.

Now, we have got at least one person from each of the three generations of the family.

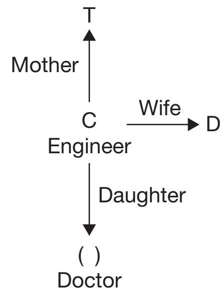
Now, E is the son of a Teacher and A's grandfather is a Manager. Also, Manager is married to an Architect.

Now, comparing the above three figures, there is only one way to combine them. C's father is a Manager. C's mother is an Architect. D is a Teacher. So, A and E are C's children. Now, as E is a male, E cannot be the Doctor.

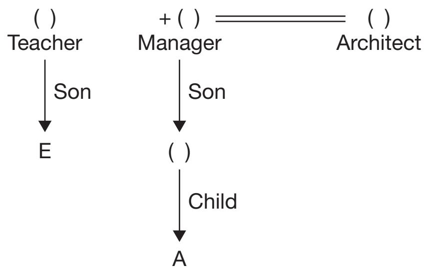
Hence, E is a Student. Thus, A is the Doctor. Finally, F is the Manager and is C's father.

∴ The family tree is represented as shown below.

It is give that C is an Engineer. Also D is the wife of an Engineer I.e., D is C's wife. Also, C's mother is B and C has a daughter. And from the information that F is not C's Father. Thus, we get the following relationship.



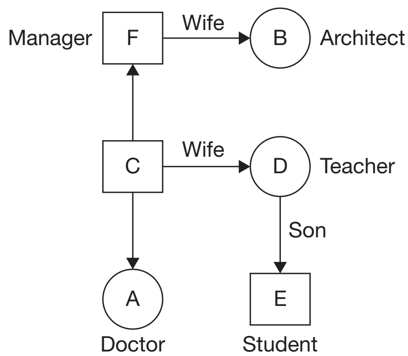
Now, we have got at least one person from each of the three generations of the family. Now E is the son of an Teacher. And, A's grandfather is a Manager. Also, Manager is married to an Architect.



Now, comparing the above three figures, there is only one way to combine them. C's father is a manager. C's Mother is an Architect. D is a Teacher. So, A and E are c's children. Now, as E is a Male, E cannot be the Doctor.

Hence, E is a Student. Thus, A is the Doctor. Finally, F is the Manager and is C's father.

∴ The family tree is represented as shown below.



26. CD represents a couple.

27. E is a Student.

28. D's daughter is A. A is a Doctor.

29. C is the Engineer and E is the Student, C is E's father.

30. D is the Teacher. D's father-in-law is F.

7 Direction Sense

CHAPTER

Learning Objectives

In this chapter, you will:

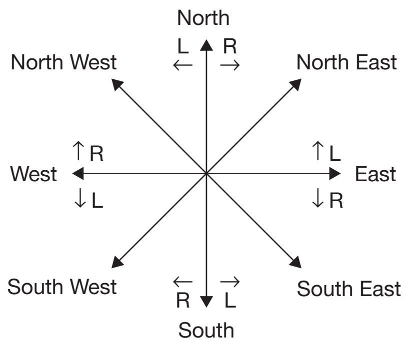
- Revisit the cardinal directions (North, South, East and West) and the intermediate directions (northeast, southeast, northwest and southwest)
- Become familiar with spatial visualization, and draw figures tracing the journey of a person/moving body
- Learn to apply Pythagoras theorem to find the distance between two points in a plane

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❒ DIRECTION SENSE

The questions on direction sense typically involve a person moving certain distances in specified directions. Then, the student is asked to find out the distance between the initial and the final points. The easiest way of solving these problems is to draw a diagram as you read the information given in the problem and ensure that the diagram reflects all the information given in the problem.

To solve these types of problems, the student should be aware of the directions. The student should also recognize the left and right of a person walking in a particular direction. The following diagram shows all the directions and left (L) and right (R) of a person walking in that direction and the student should memorize the diagram.



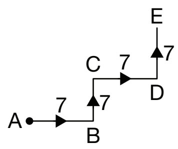
NOTE

The distance from a particular point after travelling a distance of x metres in the horizontal direction and a distance of y metres in the vertical direction is equal to $\sqrt{x^2 + y^2}$. (Please note that in common usage, North-South direction is referred to as 'vertical' direction and the East-West direction is referred to as the 'horizontal' direction).

SOLVED EXAMPLES

1. A person travels a distance of 7 km towards east from his house, then travels 7 km towards north and then a distance of 7 km towards east and finally 7 km towards north. What is the vertical distance travelled by him?

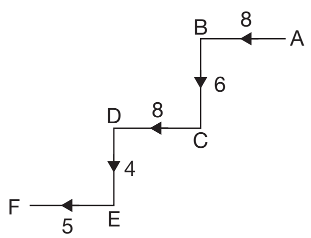
Sol:



Let A and E be the initial and the final positions. The vertical distance travelled = BC + ED = (7 + 7) km = 14 km.

2. A person starts from his house and travels 8 m towards west; then he travels 6 m towards his left, then 8 m towards west and then 4 m towards south. Finally, he turns right and travels 5 m. What is the horizontal distance travelled by him?

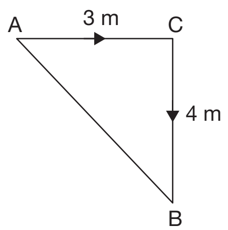
Sol:



Let A and F be the initial and the final positions.
∴ Horizontal distance travelled = FE + DC + BA = 5 + 8 + 8 = 21 m.

3. Surya travels 3 m towards east and then turns right and travels 4 m. What is the distance between the initial and the final positions of Surya?

Sol:

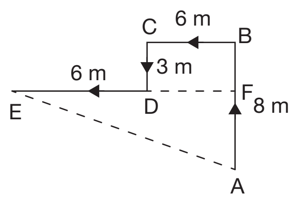


Let A and B be the initial and the final positions of Surya.

AB = $\sqrt{AC^2 + BC^2} = \sqrt{3^2 + 4^2} = 5$ m.

4. Starting from his house, Sachin walks a distance of 8 m towards north, then he turns left and walks 6 m, then he walks 3 m towards south and finally travels 6 m towards west to reach his office. What is the distance between his house and office and also find in which direction is his office situated with respect to his house?

Sol:



Let A be Sachin's house and E his office.
The distance between A and E

$= \sqrt{(EF)^2 + (AF)^2} = \sqrt{(ED + CB)^2 + (AB - FB)^2}$

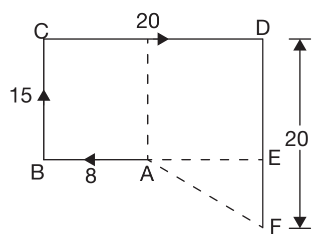
$= \sqrt{(ED + CB)^2 + (AB - FB)^2} = \sqrt{12^2 + 5^2}$

= 13 m.

His office is towards north-west of his house.

5. Starting from her house, Nisha travelled 8 m towards west, then turned right and travelled 15 m. She then travelled 20 m towards east, followed by another 20 m towards south to reach a hostel. How far is her house from the hostel and in which direction?

Sol:



Let A and F be the initial and final positions.

Now AF = $\sqrt{AE^2 + EF^2}$

AE = CD – BA
= 20 – 8 = 12 m
EF = DF – DE = DF – CB
= 20 – 15 = 5m

∴ AF = $\sqrt{144 + 25}$

= $\sqrt{169}$ = 13 m

Her house is towards north-west from the hostel.

EXERCISES

Directions for questions 1 to 20: Select the correct alternative from the given choices.

1. A boy walks 7 km towards the north, then turns to his right to travel 13 km then he turns to his left and travels 11 km. What is the vertical distance between the starting point and end point (in km)?

(A) 11 (B) 18
(C) 21 (D) 13

2. Shyam travels 8 km towards the east, then turns right and travels 4 km, then turns right and travels 18 km and then turns left and travels 17 km. What is the vertical distance between his starting and end points?

(A) 21 km (B) 26 km
(C) 47 km (D) 20 km

3. Akash travels 8 km towards the east then turns right and travels 9 km. After he travels 6 km towards east, he turns left and travels 5 km, then he travels 13 km towards the east and stops. How far is he from the starting point, in horizontal direction (in km)?

(A) 41 (B) 14
(C) 26 (D) 27

4. Mahesh runs 20 m towards the west, turns left and runs 10 m, again turns left and runs 8 m. Then he runs 25 m towards the north, turns left and runs 17 m and stops. In which direction is he from the starting point?

(A) South-west (B) South-east
(C) North-west (D) North-east

5. A person travels 2 km towards the south, turns left and travels 10 km, again turns left and travels 14 km and then turns right and travels 5 km. In which direction is he from the starting point?

(A) South-west (B) South
(C) North-west (D) North-east

6. A person travels 8 km, turns right and travels 5 km. Then, he turns right again and travels 8 km. Finally, he turns left and travels 17 km towards south. In which direction is his starting point with respect to his end point?

(A) North (B) South
(C) North-east (D) South-west

7. Sonal went for a walk from her house to the park. She walked for 100 m then turned right to walk 200 m towards south. Then she turned right and walked 50 m, then turned left and walked 20 m. Finally, she turned left and walked 250 m to reach the park. In which direction is her home with respect to the park?

(A) East (B) West
(C) North (D) North-west

8. Harilal travels 4 km towards east, turns left and travels 2 km and then travels another 6 km towards the north. He then turns right and travels 2 km to reach a temple. How far is the temple from his starting point (in km)?

(A) 6 (B) 8
(C) 10 (D) 12

9. Pooja starts from her house and travels 7 km towards the north. Then she turns left, travels 3 km, then turns left again and travels 4 km and then travels 2 km towards the west. Finally, she turns right and travels 9 km. How far is she from the starting point (in km)?

(A) 10 (B) 12
(C) 0 (D) 13

10. Kapil travelled 7 km towards the north-east followed by 12 km towards the south, then 7 km towards the south-west followed by 5 km towards the west. How far is he from the starting point (in km)?

(A) 13 (B) 12
(C) 5 (D) 14

11. Sunil travelled 15 km towards the east, then he took a left turn and travelled 4 km, then he turned right and travelled 9 km followed by 11 km after taking a right turn. How far is he from the starting point (in km)?

(A) 24 (B) 25
(C) 20 (D) 5

12. Pramod went to visit his friend Malik's home. He travelled 10 km towards the north, then turned right and travelled 6 km. Then he took a left turn and travelled 4 km, then took right and travelled 9 km, then again, he took right and travelled 6 km to reach Malik's home. Malik told him that there is a shortcut road which is of the shortest possible distance between their homes. How many kilometers less Pramod can travel if he follows the shortcut for returning back to his home?

(A) 17 (B) 35
(C) 15 (D) 18

13. Lokesh travelled 7 km towards the east. Then he turned left and travelled 4 km then he travelled 3 km towards the west, then he turned right and travelled 5 km. How far is he from the starting point?

(A) $13\sqrt{2}$ (B) $\sqrt{97}$
(C) $2\sqrt{13}$ (D) 13

14. Ramesh and Suresh started their journey from a common point. Ramesh travelled 7 km towards the north, then turned left and travelled 23 km, then he turned left again and travelled 15 km and Suresh travelled 2 km towards the south, then he turned left and travelled 12 km,

then he turned left again and travelled 6 km stopped. How far they are from each other (in km)?
(A) 35 (B) 37
(C) 12 (D) 38

15. Pallavi and Jessica decided to meet at a theatre. Pallavi travelled 9 km from her house towards the west. Then she took a right turn and travelled 10 km, then 7 km towards her right to reach the theatre. Jessica travelled 5 km from her house, then 7 km towards her right then 5 km towards her left. Then she took a left turn and travelled 12 km towards south and reached the theatre. How far is Pallavi's home from Jesscia's home (in km)?
(A) 14 (B) 15
(C) 16 (D) 17

16. A person walks 4 km towards the east, then turns to his left to travel 7 km, then turns towards the west and travels 10 km, finally he travels 15 km towards the south. How far and in which direction is he from the starting point?
(A) 10 km, South-west
(B) 12 km, South-east
(C) 12 km, North-east
(D) 10 km, North-east

17. A person started travelling from place A towards the east. After travelling 9 km he took a right turn and travelled 5 km, then he took a right turn and travelled 4 km, then again right and travelled for 6 km, then he travelled 10 km towards the east, then took a left turn and travelled 7 km to reach place B. How far and in which direction is B with respect to A?
(A) 17 km, South-west
(B) 15 km, North-east
(C) 17 km, North-east
(D) 25 km, North-west

18. A defective compass points towards the north when it should point towards the south-west. If a person is holding that compass facing north-east, then in which direction that defective compass will indicate?
(A) South (B) East
(C) South-west (D) North-west

19. Sujay walked 9 m facing towards the west, then took a left turn and walked a distance of 6 m. He then took a right turn and walked a distance of 9 m. Approximately, how far is he from the starting point?
(A) 18 m (B) 20 m
(C) 19 m (D) 21 m

20. Anil started walking towards the east. After walking 8 km, he took a left turn and walked for 4 km then again took a left turn and walked for 2 km. Further, he took a right turn and walked for 6 km then again took a right turn and walked for 2 km then took a left turn before taking another left turn and walked for 2 km and 8 km, respectively and stopped. In which direction is he from the starting point?
(A) North-east (B) South
(C) North-west (D) North

Directions for questions 21 to 23: These questions are based on the following information.

A person starts from place A and goes upto place J through B, C, D, E, F, G, H and I in that order. From A he travels 7 km towards the north to reach B, then he takes a right turn and travels 6 km to reach C, from C he travels 11 km towards the south to reach D, then he turned left and travelled 12 km to reach E, then he again turned left and travelled for 5 km to reach F. From F he travelled 9 km towards west to reach G, then he took a left turn and travelled 15 km to reach H, then he took a left turn, travelled 15 km to reach I, then again turned left and travelled 7 km to reach J.

21. How far and in which direction is A with respect to J?
(A) 29 km, North-west
(B) 25 km, North-west
(C) 29 km, South-west
(D) 25 km, South-east

22. H is in which direction from F?
(A) South-west (B) South-east
(C) South (D) North-east

23. How far is F from D (in km)?
(A) 10 (B) 17
(C) 15 (D) 13

Directions for questions 24 to 26: These questions are based on the following information.

Point A is 8 m towards the south of point E. Point C is 4 m towards the north of point B, which is 5 m towards the west of point D. Point C is 7 m towards the east of point E. Point I is 6 m towards the east of point F, which is 9 m towards the north of point H. Point D is 11 m towards the south of point G, which is 8 m towards the east of point H.

24. In which direction is point I with respect to point A?
(A) South-west (B) North-east
(C) South-east (D) North-west

25. If point K is 2 m towards the east of point I, then what is the distance between the points D and K?
(A) 19 m (B) 21 m
(C) 22 m (D) 20 m

26. Approximately how far and in which direction is point E with respect to point G?
(A) 14 m, South-west (B) 15 m, North-west
(C) 13 m, South-west (D) 14 m, South

Directions for questions 27 and 28: These questions are based on the following information.

Exercises

S, T, U, V, W, X, Y and Z are the eight class rooms in a college. Z is 10 km towards the south of X and is 2 km towards the east of U, which is 3 km towards the south of T. S is 5 km towards the west of T and is 6 km towards the south of V. W is 1 km towards the north of Y, which is 5 km towards the east of V.

27. How far and in which direction is X with respect to W?

(A) 2 km, West (B) 2 km, East
(C) 1 km, East (D) 2 km, North

28. If classroom R is constructed exactly between X and Z, then approximately how far is T with respect to R?

(A) 9 km (B) 8 km
(C) 3 km (D) 4 km

Directions for questions 29 and 30: These questions are based on the following information.

Eight cities T_1, T_2, T_3, T_4, T_5, T_6, T_7 and T_8 in a state are in different directions as given below.

T_6 is towards the south of T_4, which is towards the south-west of T_3. T_8 is towards the north of T_2 and is west of T_5. T_1 is towards the west of T_2. T_7 is towards the west of T_6 and is towards the south of T_5.

29. If the distance between the cities (T_8, T_2), (T_8, T_5) and (T_5, T_7) are equal, then in which direction is T_2 with respect to T_7?

(A) East (B) West
(C) North-east (D) South-west

30. In which direction is T_4 with respect to T_5?

(A) East (B) North-east
(C) South-east (D) North

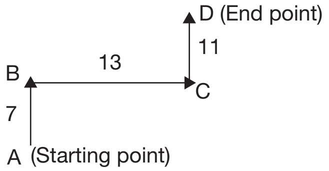
ANSWER KEYS

1. (B)	6. (A)	11. (B)	16. (A)	21. (B)	26. (A)
2. (A)	7. (D)	12. (D)	17. (C)	22. (A)	27. (B)
3. (D)	8. (C)	13. (B)	18. (A)	23. (D)	28. (C)
4. (C)	9. (D)	14. (B)	19. (C)	24. (B)	29. (B)
5. (D)	10. (A)	15. (D)	20. (D)	25. (D)	30. (A)

SOLUTIONS

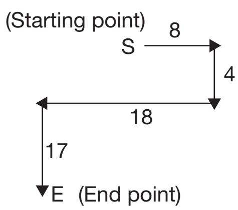
Solutions for questions 1 to 20:

1.



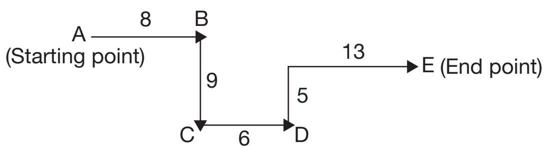
Vertical distance between A and D = 7 + 11 =18 km.

2.



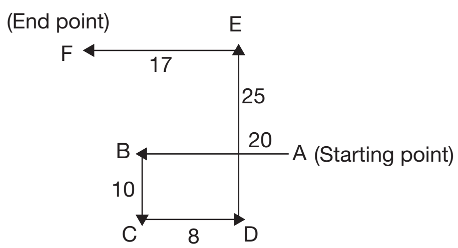
Vertical distance between S and E is 4 + 17 = 21 km.

3.



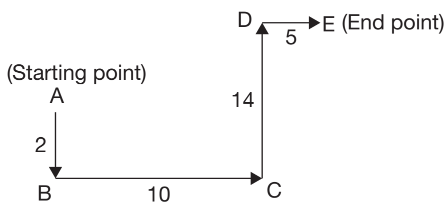
∴ Total distance in horizontal direction = AB + CD + DE = 8 + 6 + 13 = 27 km.

4.

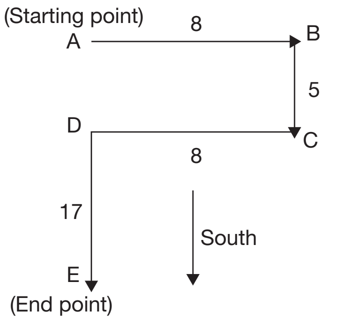


F is towards the north-west of A.

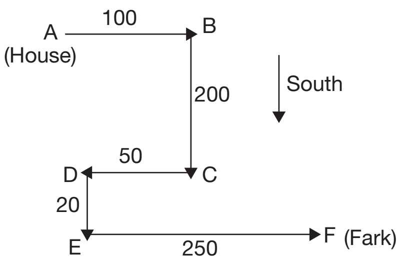
5.



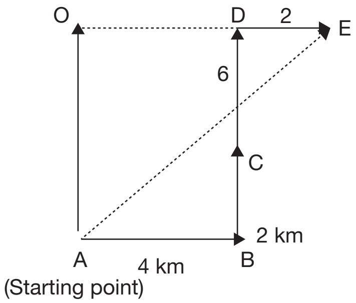
E is towards the north-east of A.

6. 

Hence, A (starting point) is towards the north of E (end point).

7. 

Hence, her home (A) is towards the north-west of park (F).

8. 

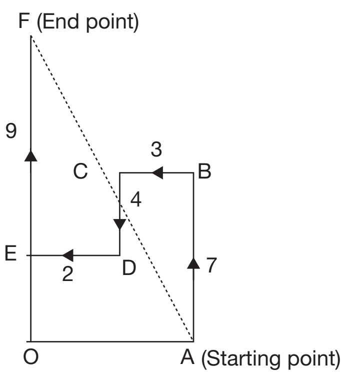
We have to find AE:

In ΔAOE, AE = $\sqrt{AO^2 + OE^2}$

AO = BC + CD = 8 km

OE = AB + DE = 6 km

∴ AE = $\sqrt{8^2 + 6^2}$ = 10 km

9. 

We have to find AF.

In ΔAOF,

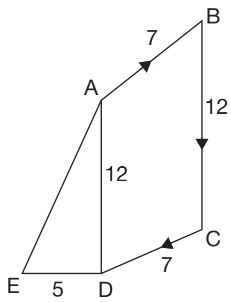
$AF^2 = \sqrt{AO^2 + OF^2}$

AO = BC + DE = 5 km.

OF = AB – CD + EF = 12 km.

AF = $\sqrt{25 + 144} = \sqrt{169}$.

AF = 13 km

10. 

We have to find AE.

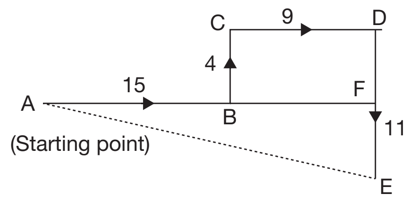
In ΔADE,

$AE^2 = AD^2 + DE^2$

AD = BC = 12

∴ AD = $\sqrt{12^2 + 5^2}$ = 13 km

11. We have to find AE.



In ΔAEF,

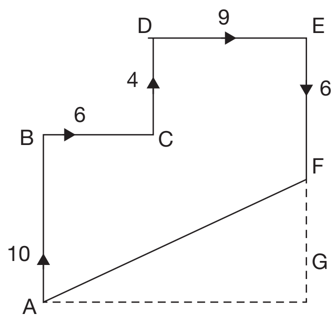
$AE^2 = AF^2 + EF^2$

AF = 15 + 9 = 24

EF = 11 – 4 = 7

∴ AE = $\sqrt{24^2 + 7^2}$

= 25 km

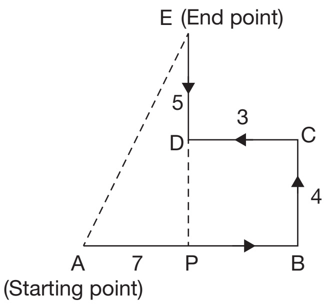
12. 

A is Pramod's home and F is Malik's home.

Exercises

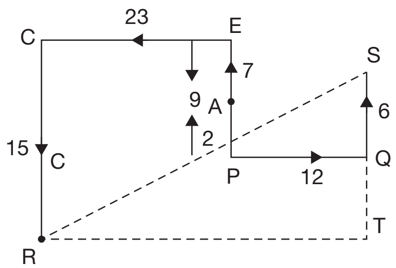
Total distance travelled by Pramod from A to F is
$= 10 + 6 + 4 + 9 + 6 = 35$ km
AF represents the shortcut road.
In ΔAGF,
$AF^2 = AG^2 + GF^2$
$AG = BC + DE = 15$.
$GF = AB + CD - EF = 10 + 4 - 6 = -8$
$\therefore AF^2 = \sqrt{15^2 + 8^2} = 17$ km
Hence, Pramod can travel $(35 - 17)$ km $= 18$ km less if he follows the shortcut for returning to his home.

13. We have to find AE.



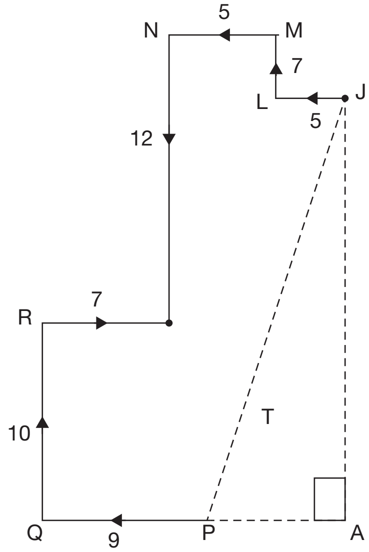
In ΔAPE,
$AE^2 = AP2 + EP^2$
$AP = AB - CD = 4$
$EP = DE + BC = 9$
$AE = \sqrt{4^2 + 9^2}$
$\Rightarrow AE = \sqrt{97}$ km

14.



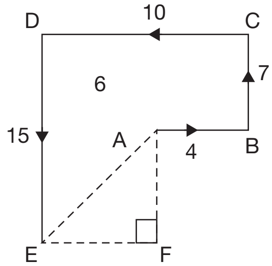
Here, A is the common point from where Ramesh and Suresh start. R is the final position of Ramesh and S is the final position of Suresh. We have to find RS.
Now, $RT = 23 + 12 = 35$
$QS = 6 + (15 - 9) = 12$
$\therefore$ In ΔRTS,
$RS^2 = RT^2 + TS^2$
$TS = QS + (RC - PE)$
$\therefore RS = \sqrt{35^2 + 12^2}$
$RS = 37$ km

15. Let P represents Pallavi's home. J represents Jessica's home and T represents the theatre where they meet.



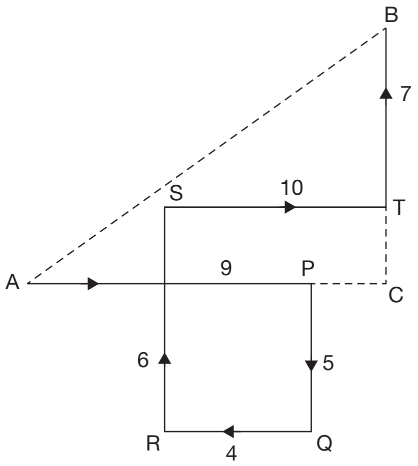
We have to find PJ.
In ΔPAJ,
$PJ^2 = PA^2 + JA^2$
$PA = 7 + 5 + 5 - 9 = 8$
$JA = 10 + 12 - 7 = 15$
$\therefore PJ = \sqrt{8^2 + 15^2}$
$PJ = 17$ km

16. In ΔAEF, $AF = 15 - 7 = 8$.



$EF = 10 - 4 = 6$
$\therefore AE = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100}$
$AE = 10$ km
E is towards the south-west of A.

17.



In ΔACB,

$AB^2 = BC^2 + AC^2$

$BC = BT + RS - PQ$

$= 6 + 7 - 5 = 8.$

$AC = AP + ST - RQ$

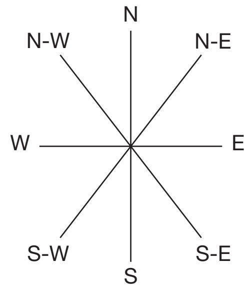
$= 9 + 10 - 4 = 15$

$\therefore AB = \sqrt{8^2 + 15^2}$

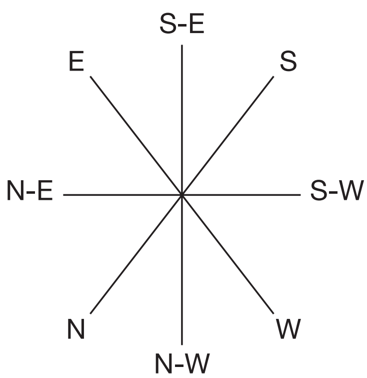
AB = 17 km.

B is towards the north-east of A.

18. Actual direction:

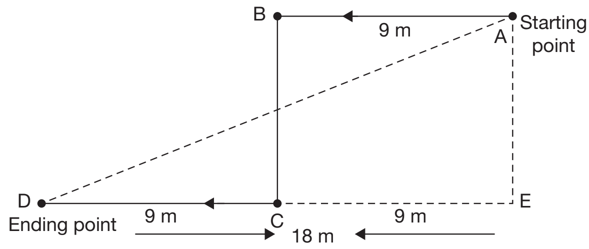


Damaged compass:



By comparing the two diagrams it is clear that, the pointer which was showing north-east is directed towards the south by damaged compass.

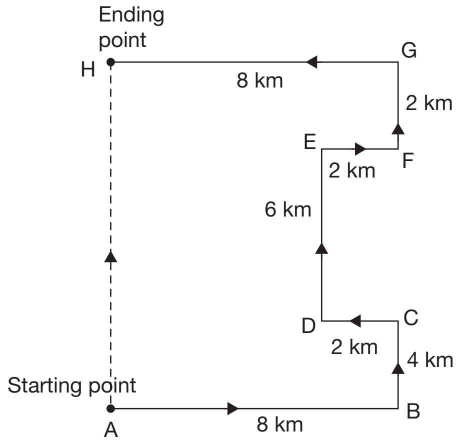
19. The path traced by Sujay is as shown below.



$AD^2 = DE^2 + AE^2$

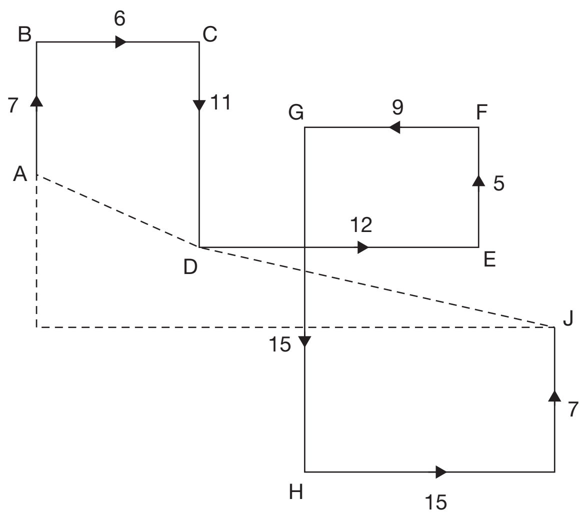
$AD = \sqrt{18^2 + 6^2} = \sqrt{360} \approx 19\ m$

20. The path traced by Anil is as shown below.



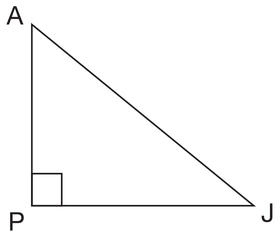
∴ He is towards the north of the starting point.

Solutions for questions 21 to 23:



21. In ΔAPJ, $AJ^2 = AP^2 + PJ^2$

$AP = (CD - AB) + (GH - FE - JI)$
$= (11 - 7) + (15 - 5 - 7)$
$= 4 + 3 = 7$
$PJ = BC + HI + (DE - GF)$
$= 6 + 15 + 12 - 9 = 24$



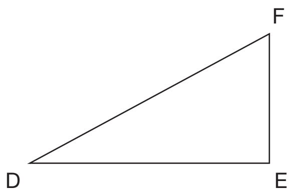
$AJ = \sqrt{7^2 + 24^2} = \sqrt{625}$

$AJ = 25$

A is towards the north-west of J.

22. H is towards the south-west of F.

23.



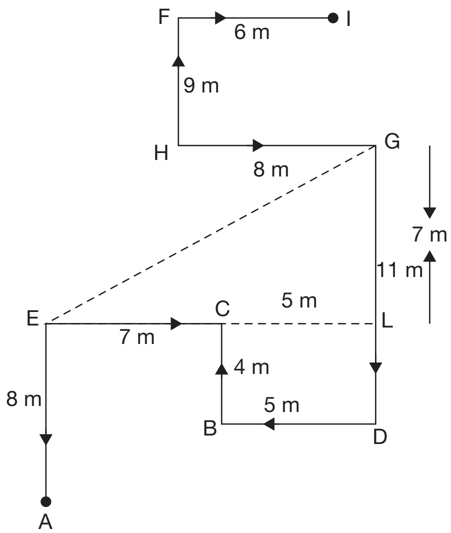
$FD^2 = EF^2 + DE^2$

$FD^2 = 5^2 + 12^2$

$FD = \sqrt{169} = 13$ km

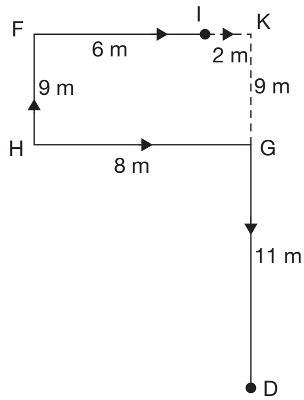
Solutions for questions 24 to 26:

24. The given data can be represented as shown below.



Point I is towards the north-east of point A.

25. If point K is 2 m towards the east of point I, then we shall draw the following route.



∴ The distance between D and K is
$DK = DG + GK = 11 + 9 = 20$ m.

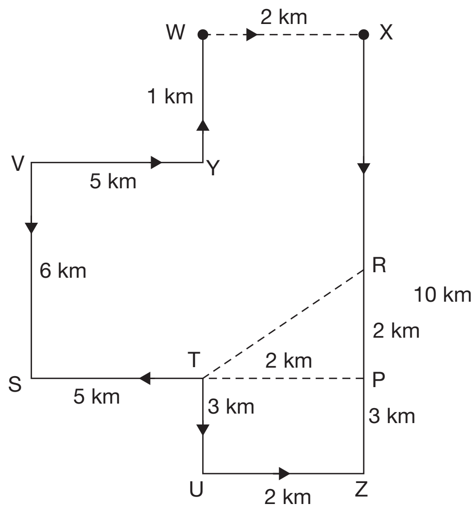
26. $GE = \sqrt{EL^2 + LG^2}$

$= \sqrt{(7+5)^2 + 7^2} = \sqrt{193} \approx 14m$

∴ Point E is 14 m towards the south-west of point G.

Solutions for questions 27 and 28:

27. The given data can be represented as shown below.

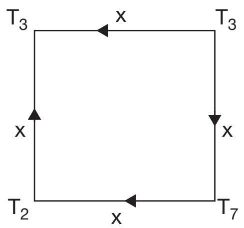


X is 2 km towards the east of W.

28. $TR = \sqrt{TP^2 + RP^2} = \sqrt{2^2 + 2^2} \approx 3\ km$

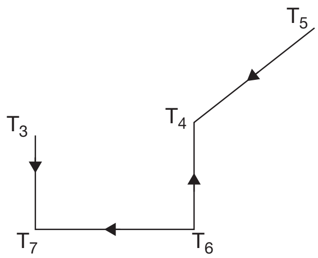
Solutions for questions 29 and 30:

29. From the given information, the path can be represented as shown below.

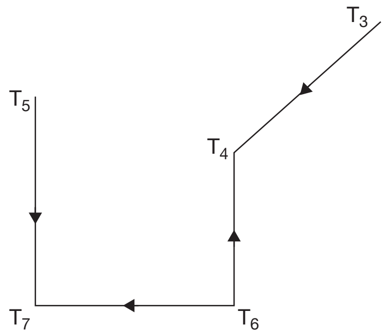


$\therefore$ T_2 is towards the west of T_7.

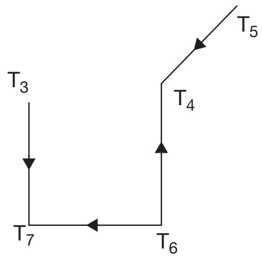
30. From the given information, we have the following possibilities.



Hence, T_4 is towards the north-east of T_5.



Hence, T_4 is towards the south-east of T_5.



Hence, T_4 is towards the east of T_5.

8 Clocks

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:
- Calculate the angular speed at which the minute hand and the hour hand of a clock move
- Learn to apply relative velocity concept to clocks
- Learn how many times in a day the hands of a clock coincide
- Learn how to find the angle between the hands of a clock at any time of the day

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❒ CLOCKS

The hour hand and the minute hand of a clock move in relation to each other continuously and at any given point of time, they make an angle between 0° and 180° with each other.

If the time shown by the clock is known, then the angle between the hands can be calculated. Similarly, if the angle between two hands is known, then the time shown by the clock can be found out.

When we say angle between the hands, we normally refer to the acute/obtuse angles (upto 180°) between the two hands and not the reflex angle (>180°).

For solving the problems on clocks, the following points will be helpful.

1. Minute hand covers 360° in 1 hour, i.e., in 60 minutes. Hence, MINUTE HAND COVERS 6° PER MINUTE.
2. Hour hand covers 360° in 12 hours. Hence, hour hand covers 30° per hour. Hence, HOUR HAND COVERS 1/2° PER MINUTE.

The following additional points also should be remembered. In a period of 12 hours, the hands make an angle as follows.

1. 0° with each other (i.e., they coincide with each other), 11 times.
2. 180° with each other (i.e., they lie on the same straight line), 11 times.
3. 90° or any other angle with each other, 22 times.

NOTE

We can also solve the problems on clocks using the method of 'Relative Velocity'.

In 1 minute, minute hand covers 6° and hour hand covers 1/2°.

Therefore, relative velocity = 6 – 1/2 = 51/2° per minute. Alternately, in 1 hour, the minute hand covers 60 minute divisions whereas the hour hand covers 5 minute divisions.

∴ Relative speed = 60 – 5 = 55 minutes per hour.

However, adopting the approach of actual angles covered is by far the simplest and does not create any confusion.

Important points to remember

1. Any angle is made 22 times in a period of 12 hours.
2. In a period of 12 hours, there are 11 coincidences of the two hands, when the two hands are in a straight line facing opposite directions.
3. The time gap between any two coincidences is 12/11 hours or $65^5/_{11}$ minutes.
4. If the hands of a clock (which do not show the correct time) coincide every p minutes, then

If $p > 65^5/_{11}$, then the watch is going slow or losing time. If $p < 65^5/_{11}$, then the watch is going fast or gaining time.

To calculate the angle 'θ' between the hands of a clock, we use the following formula (where m = minutes and h = hours).

1. $\theta = \frac{11}{2}m - 30h \left(\text{when} \frac{11}{2}m > 30h \right)$
2. $\theta = 30h - \frac{11}{2}m \left(\text{when } 30h > \frac{11}{2}m \right)$

SOLVED EXAMPLES

1. What is the angle between the minute hand and the hour hand of a clock at 3 hours 40 minutes?

(A) 20° (B) 70°
(C) 90° (D) 130°

Sol: The angle between the hands can be calculated by $\theta = \left| \frac{11}{2}m - 30h \right|$, where m is minutes and h is hours. Here, m = 40 and h = 3.

$\therefore \ \theta = \left| \frac{11}{2} \times 40 - 30 \times 3 \right| = |220 - 90| = 130°$

The angle between the two hands is 130°.

2. Find the time between 2 and 3 p.m. at which the minute hand and the hour hand

(i) make an angle of 60° with each other.
(ii) overlap
(iii) are perpendicular to each other.
(iv) are on the same straight line but are facing opposite directions.

Sol: (i) In the formula $\theta = \left| \frac{11}{2}m - 30h \right|$,

$\theta = 60°$ and h = 2

$\therefore \ 60 = \frac{11}{2}m - 30 \times 2$

$\frac{11}{2}m = 120$

$m = \frac{240}{11} = 21\frac{9}{11}$ minutes past 2

or

$60 = 30 \times 2 - \frac{11}{2}m$

$\therefore \ \frac{11}{2}m = 0$

m = 0

Therefore, the angle between the hour hand and the minute hand is 60° at 2 p.m. and at $21\frac{9}{11}$ minutes past 2 p.m..

(ii) When the two hands overlap, the angle between them is 0°.

$\theta = \left| \frac{11}{2}m - 30h \right|$

$\therefore \ \theta = 0°$ and h = 2

$\frac{11}{2}$ m = 30 × 2

$m = \frac{120}{11} = 10\frac{10}{11}$ minutes past 2.

(iii) When two hands are perpendicular: θ = 90° and h = 2

$\therefore \ \theta = \left(\frac{11}{2}m - 30h \right)$ or $\left(30h - \frac{11}{2}m \right)$

$90 = \frac{11}{2}$ m – 30 × 2

$\frac{11}{2}$ m = 150

$m = \frac{300}{11} = 27\frac{3}{11}$ minutes past 2.

or

$90 = 30 \times 2 - \frac{11}{2}m$

$\frac{11}{2}m = -30$

As m cannot be negative, this case is not possible.
So, the hands are perpendicular to each other only once, i.e., at $27\frac{3}{11}$ minutes past 2 p.m.

(iv) When two hands are pointing opposite directions and are on a straight line the angle between them would be 180°, i.e., $\theta = 180°$ and h = 2.

$180° = \frac{11}{2}m - 30h$

$\frac{11}{2}m = 180 + 60 = 240$

$m = \frac{480}{11} = 43\frac{7}{11}$

So, at $43\frac{7}{11}$ minutes past 2 p.m. the hands will be at 180°.

EXERCISES

Directions for questions 1 to 30: Select the correct alternative from the given choices.

1. What is the angle covered by the minute hand in 22 minutes?
   (A) 66° (B) 110°
   (C) 121° (D) 132°

2. By how many degrees does an hour hand move in one quarter of an hour?
   (A) 5° (B) 7.5°
   (C) 10° (D) 12.5°

3. By how many degrees will the minute hand move in the same time, in which the hour hand moves 6°?
   (A) 54° (B) 84°
   (C) 72° (D) 60°

4. What is the angle between the hands of the clock, when it shows 40 minutes past 6?
   (A) 40° (B) 70°
   (C) 80° (D) 90°

5. When the clock shows 3 hours 14 minutes, then what is the angle between the hands of the clock?
   (A) 10° (B) 12°
   (C) 13° (D) 14°

6. What is the angle between the two hands of a clock when the time is 25 minutes past 7 p.m.?
   (A) $62^1/_2°$ (B) $66^1/_2°$
   (C) $72^1/_2°$ (D) $69^1/_2°$

7. When the clock shows 20 minutes past 11 p.m., then what is the angle between the two hands of the clock?
   (A) 110° (B) 120°
   (C) 130° (D) 140°

8. At what time between 9 and 10 p.m., will both the two hands of the clock coincide?
   (A) $43\ ^3/_{11}$ minutes past 9 p.m.
   (B) $45\ ^6/_{11}$ minutes past 9 p.m.
   (C) $49\ ^1/_{11}$ minutes past 9 p.m.
   (D) $49\ ^6/_{11}$ minutes past 9 p.m.

9. At what time between 4 and 5 p.m. are the hands of a clock in the opposite directions?
   (A) $52^3/_{11}$ minutes past 4 p.m.
   (B) $54^6/_{11}$ minutes past 4 p.m.
   (C) $51^7/_{11}$ minutes past 4 p.m.
   (D) $53^9/_{11}$ minutes past 4 p.m.

10. The angle between the hands of a clock is 20° and the hour hand is in between 2 and 3. What is the time shown by the clock?
    (A) 2 hours $7^3/_{11}$ minutes
    (B) 2 hours $14^6/_{11}$ minutes
    (C) 2 hours $15^5/_{11}$ minutes
    (D) Both (A) and (B)

11. Which of the following can be the time shown by the clock, when the hour hand is in between 4 and 5 and the angle between the two hands of the clock is 60°?
    (A) $16^4/_{11}$ min past 4 (B) $18^9/_{11}$ min past 4
    (C) $32^8/_{11}$ min past 4 (D) $36^5/_{11}$ min past 4

12. Which of the following can be the time shown by the clock, when the hour hand is in between 2 and 3 and the angle between the two hands of the clock is 50°?
    (A) 25 min past 2 (B) $11\frac{9}{11}$ min past 2
    (C) $1\frac{9}{11}$ min past 2 (D) $2\frac{8}{11}$ min past 2

13. How many times the hands of a clock will be at 30° with each other in a day?
    (A) 36 (B) 40
    (C) 44 (D) 48

14. How many times the minute hand of a clock overlaps with the hour hand from 9 a.m. to 4 p.m. in a day?
    (A) 5 (B) 6
    (C) 7 (D) 8

15. A watch which gains uniformly was observed to be 1 minute slow at 8 a.m. on a day. At 6 p.m. on the same day it was 1 minute fast. At what time did the watch show the correct time?
    (A) 12 p.m. (B) 1 p.m.
    (C) 2 p.m. (D) 3 p.m.

16. A watch, which gains uniformly was observed to be 6 minutes slow at 9 a.m. on a Tuesday and 3 minutes fast at 12 p.m. on the subsequent Wednesday. When did the watch show the correct time?
    (A) 9 p.m. on Tuesday
    (B) 12 a.m. on Wednesday
    (C) 3 a.m. on Wednesday
    (D) 6 a.m. on Wednesday

17. A watch which gains uniformly was observed to be 3 minutes slow at 7 a.m. on a day. At 5 p.m. on the same day, it was 3 minutes fast. At what time did the watch show the correct time?

(A) 1 p.m. (B) 12 p.m.
(C) 11 a.m. (D) 2 p.m.

18. A watch showed 10 minutes past 6 a.m. on Thursday morning when the correct time was 6 a.m. It loses uniformly and was observed to be 15 minutes slow at 8 a.m. on Saturday morning. When did the watch show the correct time?
(A) 1 p.m. on Friday afternoon
(B) 12 p.m. noon on Friday
(C) 4 p.m. on Friday evening
(D) 2 a.m. on Friday morning

19. The minute hand of a clock overtakes the hour hand at intervals of 60 minutes of correct time. How much time does the clock gain or lose in one hour of correct time?
(A) Gains $5^5/_{11}$ minutes
(B) Loses $5^5/_{11}$ minutes
(C) Gains $^5/_{11}$ minutes
(D) Loses $^5/_{11}$ minutes

20. The minute hand of a clock overtakes the hour hand after every 70 minutes of correct time. How much time does the clock lose or gain in a day of normal time?
(A) $93^{39}/_{77}$ minutes
(B) $91^{31}/_{77}$ minutes
(C) $92^{24}/_{77}$ minutes
(D) $94^{56}/_{77}$ minutes

21. Two clocks are showing correct time at 4 p.m. One clock loses 3.5 minutes in an hour, while the other gains 2.5 minutes in one hour. At 10 p.m. on the same day, by how much time will the two clocks differ?
(A) 12 minutes (B) 36 minutes
(C) 24 minutes (D) 30 minutes

22. A watch which gains uniformly was observed to be three minutes slow at 8 a.m. on a particular day. At 8 p.m. on the same day it was three minutes fast. At what time did the watch show the correct time?
(A) 1 p.m. (B) 3 p.m.
(C) 2 p.m. (D) 6 p.m.

23. A watch which loses uniformly was observed to be two minutes fast at 6 p.m. on a Wednesday and four minutes slow at 3 p.m. on the next day, i.e., Thursday. When did the watch show the correct time?
(A) 2 a.m. on Thursday
(B) 12 p.m. on Thursday
(C) 1 a.m. on Wednesday
(D) 1 a.m. on Thursday

24. There are two clocks on a wall, both set to show the correct time at 5 p.m. The clocks lose 2 minutes and 3 minutes, respectively in an hour. If the clock which loses 2 minutes in one hour shows the time as 9.50 p.m. on the same day, then what time does the other clock show?
(A) 9.30 p.m. (B) 9.40 p.m.
(C) 9.45 p.m. (D) 10.15 p.m.

25. If the time in a clock is 10 hours 40 minutes, then what time does its mirror image show?
(A) 1 hour 25 minutes (B) 1 hour 15 minutes
(C) 1 hour 10 minutes (D) 1 hour 20 minutes

26. The reflection of a wall clock in a mirror shows the time as 3 hours 40 minutes. What is the actual time?
(A) 8 hours 20 minutes
(B) 8 hours 15 minutes
(C) 8 hours 45 minutes
(D) 8 hours 35 minutes

27. If the seconds hand moves by 240°, then by how many degrees does the minute hand move in the same time?
(A) 1° (B) 2°
(C) 3° (D) 4°

28. When the time is 10.30, if the minute hand points towards south, the hour hand will point towards
(A) North-east (B) North-west
(C) South-east (D) South-west

29. A clock strikes once at 1 p.m., twice at 2 p.m., three times at 3 p.m. and so on. If it takes 10 seconds to strike at 6 p.m., find the time taken by it to strike at 12 p.m.
(A) 18 seconds (B) 22 seconds
(C) 24 seconds (D) 26 seconds

30. At a particular point of time, the number of hours to 12 p.m. from that particular time is twice the number of hours to 12 p.m. after five hours from that particular time. Find the time.
(A) 2.30 p.m. (B) 4 p.m.
(C) 2 p.m. (D) None of these

ANSWER KEYS

1. (D)	**6.** (C)	**11.** (C)	**16.** (C)	**21.** (B)	**26.** (A)
2. (B)	**7.** (D)	**12.** (C)	**17.** (B)	**22.** (C)	**27.** (D)
3. (C)	**8.** (C)	**13.** (C)	**18.** (D)	**23.** (D)	**28.** (B)
4. (A)	**9.** (B)	**14.** (B)	**19.** (A)	**24.** (C)	**29.** (B)
5. (C)	**10.** (D)	**15.** (B)	**20.** (A)	**25.** (D)	**30.** (C)

SOLUTIONS

Solutions for questions 1 to 30:

1. The angle covered by the minute hand in 22 minutes is $22 \times 6 = 132°$.

2. The hour hand covers 360^0 in 12 hours.

$\therefore$ It covers $\frac{1}{2}^{o}$ in one minute. In quarter of an hour, i.e., in 15 minutes the hour hand will move 15 x $\frac{1}{2}^{\circ} = 7.5°$.

3. The hour hand will move by 6° in 12 minutes. So, minutes hand will move $12 \times 6° = 72°$ in 12 minutes, as the minute hand moves by 6° in one minute.

4. Angle will be $\theta = \left(\frac{11}{2}m - 30h\right)$

$\left(\frac{11}{2} \times 20 - 30 \times 6\right) = 40°$.

5. The angle between the hands will be:

$\theta = \left|\frac{11}{2}m - 30h\right|$

here, h = 3 and m = 14

$\theta = \frac{11}{2} \times 14 - 30 \times 3$

$\theta = |77 - 90| = 13°$.

6. Angle between two hands is given by:

$\theta = \left|\frac{11}{2}m - 30h\right|$

Here, h = 7 and m = 25

$\therefore\ \theta = \left|\frac{11}{2} \times 25 - 30 \times 7\right|$

$= \frac{275 - 210}{2} = \frac{145^{\circ}}{2} = 72\frac{1}{2}^{\circ}$

7. Angle between the two hands is given by:

$\theta = \left|\frac{11}{2}m - 30h\right|$, here m = 20 and h = 11

$\Rightarrow \theta = \left|\frac{11}{2} \times 20 - 30 \times 11\right| = 220$

As the angle is more than 180°, the angle must be 360° – 220° = 140°.

8. When the hands coincide with each other the angle between them is 0. Therefore, the angle between two hands is given by:

$\theta = 30h - \frac{11}{2}m \qquad (\because 30h > \frac{11}{2}m)$

Here, h = 9

$0 = 30 \times 9 - \frac{11}{2}m$

$270 \times \frac{2}{11} = m$

$\therefore$ m = $49\frac{1}{11}$ minutes

So, the hands coincide at $49\frac{1}{11}$ minutes past 9 hours.

9. When the hands of a clock are in opposite direction the angle between them is 180°.

Therefore, $\theta = \left|\frac{11}{2}m - 30h\right|$

Where $\theta = 180°$ and h = 4

$180 = \frac{11}{2}$ m 120

$\frac{11}{2}$ m = 300

m = $\frac{600}{11}$ m = $54\frac{6}{11}$ minutes

So, at $54\frac{6}{11}$ minutes past 4 hours, the hands are in opposite direction.

10. Given $\theta = 20°$ and h = 2

$\theta = \frac{11}{2}$ m – 30h or 30h – $\frac{11}{2}$ m

$20 = \frac{11}{2}$ m – 30 × 2

$\frac{11}{2}$ m = 80

m = $\frac{160}{11}$ = $14\frac{6}{11}$ minutes (or) 20 = 30 × 2 – $\frac{11}{2}$ m

$\frac{11}{2}$ m = 40

m = $\frac{80}{11}$ = $7\frac{3}{11}$ minutes

Therefore, the angle between the hands will be 20° at hours $14\frac{6}{11}$ minutes past 2 and $7\frac{3}{11}$ minutes past 2.

11. Given $\theta = 60°$ and h = 4

$\theta = \frac{11}{2}$ m – 30h or $\theta = 30$ h – $\frac{11}{2}m$

$60 = \frac{11}{2}$ m – 30 × 4 or 60 = 120 – $\frac{11}{2}$ m

$\frac{11}{2}$ m = 180 or

$\frac{11}{2}m = 60 \therefore m = \frac{2x60}{11} = 10\frac{10}{11}$ min *utes*

m = $\frac{360}{11}$ = $32\frac{8}{11}$ minutes

Hence, the angle between the hands will be 60° at $32\frac{8}{11}$ min past 4.

12. In the formula, $\theta = \left|\frac{11}{2}m - 30h\right|$

$\theta = 50°$, h = 2

$\therefore\ 50 = -\frac{11}{2}m - 30\times2$

$\frac{11}{2}m\ = 110$

m = 20 minutes past 2 p.m.

or

$50 =\ 30\times2 - \frac{11}{2}m \Rightarrow \frac{11}{2}m\ = 10$

m = $\frac{20}{11} = 1\frac{9}{11}$ minutes past 2 p.m.

13. In 12 hours the clock will be at 30° with each other for 22 times. So, they will be at 30° with each other for 44 times in a day.

14. The minute hand overlaps with the hour hand once between 9 and 10, 10 and 11, 1 and 2, 2 and 3, 3 and 4. But between 11 and 1, the overlap happens for only one time, i.e., a total of 6 times.

15. From 8 a.m. to 6 p.m., i.e., in 10 hours the clock gained 2 minutes.
So, it gains 1 minute in 5 hours.
So, it shows correct time at 1 p.m. on the same day.

16. The watch which was 6 minutes slow at 9 a.m. on a Tuesday and 3 minutes fast at 12 p.m. on Wednesday.
∴ The watch gained 9 minutes in 27 hours.

So, it gains 6 minutes in $\frac{6x27}{9}$ = 18 hours.

∴ It shows correct time after 18 hours, i.e., at 3 a.m. on Wednesday.

17. The duration from 7 a.m. to 5 p.m. on a day is 10 hours. The total number of minutes gained by the clock in these 10 hours is given as 6 (3 + 3) minutes. If the clock gains 3 minutes, then it shows the correct time. The time taken by the clock to gain 3 minutes is $\frac{3}{6}\times10$ =5 hours. 5 hours after 7 a.m., i.e., 12 p.m. Therefore, the clock shows the correct time.

18. The watch lost 25 minutes in 50 hours, i.e., 6 a.m. on Thursday to 8 a.m. on Saturday. So, it will lose 10 minutes in 20 hours. So, it will show correct time at 2 a.m. on Friday morning.

19. The minute hand gains $65\frac{5}{11} - 60 = 5\frac{5}{11}$ minutes in one hour.

20. In a normal clock, the minute hand overtakes the hour hand 11 times in 12 hours (i.e., 720 minutes). Hence, it takes $\frac{720}{11} = 65\frac{5}{11}$ minutes to overtake once. But in the given clock the minute hand overtakes the hour hand in 70 minutes. Minute hand loses 70 – 65 $\frac{5}{11}$, *i.e.*, $4\frac{6}{11}$ in 70 minutes.

So, it loses $\frac{50}{11}$ x $\frac{24x60}{70}$ minutes in 24 hours,

i.e., $\frac{7200}{77} = 93\frac{39}{77}$ minutes.

21. After 1 hour, the two clocks differ by 3.5 + 2.5 = 6 minutes. So, after 6 hours the two clocks differ by 36 minutes.

22. The duration from 8 a.m. to 8 p.m. is 12 hours. Total number of minutes gained by the clock in these 12 hours is given as 6 (3 + 3) minutes. If the clock gains three minutes, then it shows the correct time, the time taken by the clock to gain three $\frac{3}{6}$ × 12 = 6 hours.

6 hours after 8 a.m., i.e., at 2 p.m. the clock shows the correct time.

23. The duration from 6 pm on Wednesday to 3 p.m. on Thursday is 21 hours. The total number of minutes by the clock in this 21 hours is 6 minutes.
The clock shows correct time when it covers the initial 2 minutes.

The time taken by the clock to lose 2 minutes is $\frac{2}{6} \times 21$ = 7 hours.

$\therefore$ After 7 hours, i.e., at 1 a.m. on Thursday, the clock shows the correct time.

24. After 5 hours, i.e., at 10 p.m. the clock, which loses 2 minutes will lose 10 minutes and shows 9.50 p.m. So, the other clock will lose $3 \times 5 = 15$ minutes and show 9.45 p.m.

25. Mirror time = 12 – Actual time = 12 – 10.40 = 1.20.

26. When the time is 3 hours 40 minutes the hour hand will be between 3 and 4 and the minutes hand will be at 8. So, in the reflection, the hour hand will be between 8 and 9 and the minute hand will be at 4. So, the time is 8 hours 20 minutes.
Mirror time = 12 – Actual time = 12 – 3:40 = 8:20.

27. When the seconds hand moves by 360° (i.e., 1 minute) the minute hand moves by 6°. So when the seconds hand moves by 240°, the minute hand moves by $\frac{240 x 6}{360} = 4°$.

28. At 10.30 the angle between minute hand and hour hand will be $\theta = \left|\frac{11}{2} \times 30 - 30 \times 10\right|$

when the minute hand points towards south, the hour hand will be to the right of the minute hand which is north-west.

29. The clock strikes for 6 times at 6 p.m. Let the gap between two consecutive strikes be x.
Now, the total gap between 1st strike and 6th strike is $5x$.
Now $5x = 10 \Rightarrow x = 2$
Now, the gap between 1st strike and 12th strike at 12 p.m. is $11x$, i.e., 22 seconds.

30. Let the present time be x p.m.
$\therefore 2(12 - (x + 5)) = 12 - x$
$2(7 - x) = 12 - x$
$14 - 2x = 12 - x$
$x = 2$
$\therefore$ The present time is 2 p.m.

Exercises

9 Calendars

CHAPTER

Learning Objectives

In this chapter, you will:

- Learn concepts of calendars like:
  - Leap and non-leap years
  - Odd days
- Learn how to calculate the number of odd days when two dates are given
- Learn how to find the day of the week when a date is given
- Learn how to find the day of the week when two reference dates are given

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as SNAP, XAT, IIFT, MH-CET, MAT, etc.

❐ CALENDARS

Suppose you are asked to find the day of the week on 30 June 1974, it would be a tough job to find it if you do not know the method. The method of finding the day of the week lies in the number of 'odd days'.

NOTE

Every 7th day will be the same day count wise, i.e., if today is Monday, then the 7th day counting from Tuesday onwards will once again be Monday. Odd day is the days remaining after completion of an exact number of weeks. Odd day is the reminder obtained on dividing the total number of days with seven.

Example: 52 days ÷ 7 = 3 odd days.

Leap and Non-leap Year

A non-leap year has 365 days whereas a leap year has one extra day because of 29 days in the month of February. Every year which is divisible by 4 is called a leap year. Leap year consists of 366 days, (52 complete weeks + 2 days), the extra two days are the odd days. So, a leap year has two odd days.

A non-leap year consists of 365 days (52 complete weeks + 1 day). The extra one day is the odd day.

NOTE

For every century, the year which is a multiple of 400 is a leap year. A century year which is not divisible by 400 is a non-leap year.

Example: 400, 800, 1200, 1600 …. are leap years. 500, 700, 900, 1900 … are non-leap years.

Counting the Number of Odd Days

100 years consist of 24 leap years + 76 ordinary years. (100 years when divided by 4, we get 25. But at the

100th year is not a leap year, hence, only 24 leap years).

$= 2 \times 24$ odd days $+ 1 \times 76$ odd days

= 124 days

= 17 weeks + 5 days

The extra 5 days are the odd days.

So, 100 years contain 5 odd days.

Similarly, for 200 years we have 10 extra days (1 week + 3 days).

∴ 200 years contains 3 odd days.

Similarly, 300 years contain 1 odd day and 400 years contain 0 odd days.

Counting of Number of Odd Days, When Only One Date is Given

Here, we take January 1st 1 AD as the earlier date and we assume that this day is a Monday. We take its previous day, i.e., Sunday as the reference day. After this the above-mentioned method is applied to count the number of odd days and find the day of the week for the given date.

Counting Number of Odd Days, When Two Dates Are Given

Any month which has 31 days has 3 odd days.

(∴ 31 ÷ 7 leaves 3 as remainder) and any month which has 30 days has 2 odd days (30 ÷ 7 leaves 2 as remainder).

Then, the total number of odd days are calculated by adding the odd days for each month. The value so obtained is again divided by 7 to get the final number of odd days. The day of the week of the second date is obtained by adding the odd days to the day of the week of the earlier date.

SOLVED EXAMPLES

1. If you were born on 14th April 1992, which was a Sunday, then on which day of the week does your birthday fall in 1993?

(A) Monday (B) Tuesday
(C) Wednesday (D) Friday

Sol: 14th April 1992 to 14th April 1993 is a complete year, which has 365 days. Hence, the number of odd days from 14th April 1992 to 14th April 1993 is 1. Hence, 14th April 1993 is one day after Sunday, i.e., Monday.

2. If 1st January 1992 is a Tuesday then on which day of the week will 1st January 1993 fall?

(A) Wednesday (B) Thursday
(C) Friday (D) Saturday

Sol: Since 1992 is a leap year there are 2 odd days. Hence, 1st January 1992 is two days after Tuesday, i.e., Thursday.

3. If 1st April 2003 was Monday, then which day of the week will 25th December of the same year be?

(A) Tuesday (B) Wednesday
(C) Thursday (D) Friday

Sol: The number of days from 1st April to 25th December

(29 + 31 + 30 + 31 + 31 + 30 + 31 + 30 + 25) days

= 268 days

$$= \frac{268}{7} = 38 + 2 \text{ odd days}.$$

Hence, 25th December is two days after Monday, i.e., Wednesday.

4. On which day of the week does 4th June 2001 fall?

(A) Monday (B) Tuesday
(C) Wednesday (D) Thursday

Sol: 4th June 2001 ⇒ (2000) years + 1st January to 4th June 2001.

We know that 2000 years have zero odd days. The number of odd days from 1st January to 4th June 2001.

Month: Jan + Feb + Mar + Apr + May + June

Odd day: 3 + 0 + 3 + 2 + 3 + 4

$$\frac{15}{7} = 1 \text{ odd day}.$$

Hence, 4th June 2001 was a Monday.

5. Which year will have the same calendar as that of 2005?

(A) 2006 (B) 2007
(C) 2008 (D) 2011

Sol: Year: 2005+2006+2007+2008+2009+2010

Odd days: 1 + 1 + 1 + 2 + 1 + 1

Total number of odd days from 2005 to 2010 are $7 \cong 0$ odd days.
Hence, 2011 will have the same calendar as that of 2005.

6. What day of the week was 18th April 1901?
(A) Monday (B) Tuesday
(C) Wednesday (D) Thursday

Sol: 18th April 1901 $\Rightarrow$ (1600 + 300) years + 1st January to 18th April 1901.
1600 years have – 0 odd days
300 years have – 1 odd day
The number of days from 1st January 1901 to 18th April 1901 is (31 + 28 + 31 + 18) days
108 days $\cong$ 3 odd days
$\therefore$ Total number of odd days = 3 + 1 = 4
Hence, 18th April 1901 is Thursday.

EXERCISES

Directions for questions 1 to 30: Select the correct alternative from the given choices.

1. If 8th February 1995 was a Wednesday, then 8th February 1994 was on which day?
(A) Wednesday (B) Thursday
(C) Tuesday (D) Monday

2. If 17th September 1993 was a Friday, then which day of the week was 30th June 1989?
(A) Wednesday (B) Thursday
(C) Friday (D) Saturday

3. If 11th August 1985 was a Sunday, then which day of the week was 13th August 1986?
(A) Tuesday (B) Monday
(C) Thursday (D) Wednesday

4. How many odd days are there in 352 days?
(A) One (B) Two
(C) Three (D) zero

5. Which among the following years is a leap year?
(A) 3000 (B) 3100
(C) 3200 (D) 3300

6. If 1st January 2012 is a Sunday, then which day of the week will the new year be celebrated in 2016?
(A) Friday (B) Sunday
(C) Wednesday (D) Saturday

7. If 1st April 1963 was a Monday, then which day of the week will be 1st August 1959?
(A) Saturday (B) Monday
(C) Tuesday (D) Thursday

8. On which dates of October 1994 did Monday fall?
(A) 4, 11, 18, 25 (B) 2, 9, 16, 23
(C) 1, 8, 15, 22 (D) 3, 10, 17, 24, 31

9. Which year will have same calendar as 2002?
(A) 2008 (B) 2011
(C) 2009 (D) 2013

10. The calendar for year 2005 is the same as that for which of the following years?
(A) 2010 (B) 2012
(C) 2011 (D) 2009

11. Which of the following years will have the same calendar as that of 2020?
(A) 2050 (B) 2048
(C) 2046 (D) 2052

12. What will be the next leap year after 2096?
(A) 2100 (B) 2101
(C) 2104 (D) 2108

13. If in a calendar year, there are 541 days and 10 days a week, then how many odd days will be there in that year?
(A) One (B) Two
(C) Three (D) Four

14. The last day of a century cannot be
(A) Friday (B) Wednesday
(C) Monday (D) Tuesday

15. Which day of the week was 25th December 1995?
(A) Sunday (B) Monday
(C) Tuesday (D) Wednesday

16. Which day of the week was 23rd July 1776?
(A) Sunday (B) Wednesday
(C) Thursday (D) Tuesday

17. If holidays are declared only on Sundays and in a particular year 12th March is a Sunday, is 23rd September in that year a holiday?

(A) Yes
(B) No
(C) Yes, if it is a leap year.
(D) No, if it is a leap year.

18. Which day of the week was 15th January1601?
(A) Monday (B) Tuesday
(C) Wednesday (D) Thursday

19. The first Republic day was celebrated on 26th January 1950 and it was on which day?
(A) Thursday (B) Friday
(C) Monday (D) Tuesday

20. If 23rd April 2006 is a Sunday, then 23rd April 2106 will be a
(A) Wednesday (B) Thursday
(C) Friday (D) Saturday

21. If the first day of the years 2012 and 2023 are Mondays, then which day of the week will be the last days of years respectively?
(A) Tuesday, Tuesday (B) Tuesday, Monday
(C) Monday, Tuesday (D) Sunday, Monday

22. If 14th November 2006 is a Sunday, then 14th November 2706 is a
(A) Sunday (B) Friday
(C) Tuesday (D) Monday

23. In a year, if 23rd November is a Friday then 14th March in that year is on which day of the week?
(A) Monday
(B) Wednesday
(C) Sunday
(D) Tuesday

24. In a leap year, which month will have the same calendar as that of January in that year?
(A) April (B) July
(C) October (D) March

25. What is the next leap year after 2396?
(A) 2398 (B) 2408
(C) 2404 (D) 2400

26. Which day of the week is 21st April 2006?
(A) Tuesday (B) Wednesday
(C) Thursday (D) Friday

27. What was the day of the week on 5th May 1938?
(A) Friday (B) Sunday
(C) Thursday (D) Saturday

28. What day of the week was the Indian Independence Day in 2001?
(A) Monday (B) Wednesday
(C) Tuesday (D) Thursday

29. If a year starts on Saturday but does not end on Saturday, then what is the day of the week on 13th June in that year?
(A) Monday (B) Tuesday
(C) Sunday (D) Saturday

30. If a year starts on a Friday, then what is the maximum possible number of Tuesdays in that year?
(A) 58 (B) 52
(C) 51 (D) 49

ANSWER KEYS

1. (C)	6. (A)	11. (B)	16. (D)	21. (B)	26. (D)
2. (C)	7. (A)	12. (C)	17. (B)	22. (D)	27. (C)
3. (D)	8. (D)	13. (A)	18. (A)	23. (B)	28. (B)
4. (B)	9. (D)	14. (D)	19. (A)	24. (B)	29. (B)
5. (C)	10. (C)	15. (B)	20. (C)	25. (D)	30. (B)

SOLUTIONS

Solutions for questions 1 to 30:

1. 1994 is not a leap year.
∴ It has only 1 odd day.
8th February 1995 is one day before Wednesday.
Hence, 8th February 1994 is a Tuesday.

2. The number of odd days from 17th September 1993 to 30th June 1993 is as follows.
Month: Sep + Aug + Jul + Jun
Odd days: 2 + 3 + 3 + 1 = 2 odd days.
Hence, 30th June 1993 was two days back to Friday, i.e., Wednesday.
The number of years from 1993 to 1989 is 4 years out of which there is one leap year and 3 non-leap years.
∴ The number of odd days = 2 × 1 + 3 = 5 odd days.
Therefore, it is the same day, i.e., Friday.

3. It is given that 11th August 1985 was Sunday then 13th August 1985 is Tuesday.
1985 is not a leap year, hence, it has only one odd day.
So, 13th August 1986 is one day to Tuesday, i.e., Wednesday.

4. The number of odd days in 352 days

$$=\frac{352}{7}=50+2 \text{ odd days}$$

Hence, the total number of odd days is 2.

5. Century years which are divisible by 400 are leap years. As 3000, 3100 and 3300 are not divisible by 400, they are not leap years. But, 3200 is a leap year.

6. The total number of years from 2012 to 2016 is four out of which 2013, 2014 and 2015 are non-leap years. Hence, there is 1 odd day in each of these years, 2012 is a leap year, therefore, it has 2 odd days.
∴ The total number of odd days in these four years is 5.
Here, 1st Jan 2016 is five days to Sunday, i.e., Friday.

7. The total number of years from 1963 to 1959 is 4 years out of which 1959, 1961 and 1962 are ordinary years. Hence, they have 3 odd days and 1960 is leap year which has 2 odd days.
The total number of odd days in these 4 years is 5. Hence, 1st April 1959 is 5 days back to Monday, i.e., Wednesday.
Now, the number of odd days from 1st April to 1st August in 1959 is as follows.
Month: Apr + May + Jun + Jul + Aug
Odd days: 1 + 3 + 2 + 3 + 1
The total number of odd days is 3.
Hence, 1st August 1959 is 3 days to Wednesday, i.e., Saturday.

8. 1600 years contain zero odd days.
300 years contain 1 odd day.
93 years = (23 leap + 70 non-leap years)
Total number of odd days in 93 years = (23 × 2 + 70 × 1)
= 116 odd days ⇒ 4 odd days.
Number of odd days from 1st January to 1st October in 1994
Month: J + F + M + A + M + J + J + A + S + O
Odd days: 3 + 0 + 3 + 2 + 3 + 2 + 3 + 3 + 2 + 1
= 22 odd days ⇒ 1 odd day.
The total number of odd days = 1 + 4 + 1 = 6 odd days
∴ 1st October 1994 is Saturday.
Therefore, first Monday is on 3rd October.
So, 3, 10, 17, 24 and 31 are Mondays in October.

9. The number of odd days should be zero to have same calendar.

Years		Odd day
2002	–	1
2003	–	1
2004	–	2
2005	–	1
2006	–	1
2007	–	1
2008	–	2
2009	–	1
2010	–	1
2011	–	1
2012	–	2
2013	–	1

After the completion of 2013 we get 14 odd days.
Hence, the number of odd days is zero. So, 2013 will have the same calendar as that of 2002.

10. Year: 2005 2006 2007 2008 2009 2010
Odd days: 1 + 1 + 1 + 2 + 1 + 1
As the number of odd days from 2005 to 2010 is 7 ⇒ $\frac{7}{7}$ = 0 odd days.
Hence, 2011 had the same calendar as did 2005.

11. 2020 is a leap year. In the next 28 years we have 21 non-leap years and 7 leap years, which result in 35 odd days, i.e., effectively zero odd days.
Hence, 2020 + 28 = 2048 will have the same calendar as will 2020.

12. For a century year to be a leap year, it should be divisible by 400. As 2100 is not divided by 400 it is not a leap year. The next leap year is 2104.

13. To find the number of odd days, we have to find the remainder of 541/10. The remainder is one.
Hence, there is one odd day.

14. 100 years contain 5 odd days.
$\therefore$ The last day of the first century is Friday.
200 years contain 10 odd days, i.e., 3 odd days.
$\therefore$ The last day of the second century is Wednesday.
300 years contain 15 odd days, i.e., 1 odd day.
$\therefore$ The last day the third century is Monday.
400 years contain 20 odd days and the 400th year itself is a leap year. Hence, there is no odd day.
$\therefore$ The last day is Sunday. The last day of a century cannot be Tuesday, Thursday or Saturday.

15. The total number of odd days up to 25th December, 1995 is obtained as follows.
For 1600 years – zero odd days
For 300 years – 1 odd day
In 94 years there are 23 leap years and 71 non-leap years.
The total number of odd days in these 94 years is $(23 \times 2 + 71 \times 1) = (46 + 71) = 117$.
$\Rightarrow$ 5 odd days
The number of odd days from 1st January to 25th December 1995 is as follows.
Month: J + F + M + A + M + J + J + A + S + O + N + D
Odd days: $3 + 0 + 3 + 2 + 3 + 2 + 3 + 3 + 2 + 3 + 2 + 4$
$= 2$ odd days.
The total number of odd days $= 1 + 5 + 2 = 1$, i.e., Monday.
Hence, 25th December 1995 is Monday.

16. 1600 years – 0 odd days
100 years – 5 odd days
75 years = (18L + 57 NL)
$= 36 + 57 = 93$ odd days $\Rightarrow$ 2 odd days
Now 1st January – 24th July 1776
$= 205$ days $\Rightarrow$ 2 odd days
Total number of odd days $= 2$
$\therefore$ 23rd July 1776 was Tuesday.

17. The total number of odd days from 12th March to 23rd September is as follows.
Month: M + A + M + J + J + A + S
Odd days: $5 + 2 + 3 + 2 + 3 + 3 + 2 = 20$ days.
$\frac{20}{7} = 6$ odd days.
Hence, 23rd September is 6 days to Sunday, i.e., Saturday.
So, 23rd September is not a holiday.

18. The number of odd days upto 15th January 1601 is as follows.
1600 + (1st January to 15th January 1601)
1600 years have zero odd days and there is one odd day in 15 days.
Hence, 15th January 1601 is a Monday.

19. The number of odd days upto 26th January 1950.
1600 years = odd days
300 years = 1 odd day
In 49 years these are 12 leap years and 37 non-leap years.
Number of odd days in these 49 years is 61.
$= 61$ odd days $\Rightarrow$ 5 odd days.
26th January = 26 days = 5 odd days
Total odd days 11 odd days = 4 odd days
Therefore, the answer is Thursday.

20. Number of years from 2006 to 2106 is 100 years.
We know that 100 years have 5 odd days. Hence, 23rd April 2106 will be 5 days after Sunday, i.e., Friday.

21. 2012 is leap year, so it will have two odd days.
Hence, 1st January 2013 is two days after Monday, i.e., Wednesday.
So, 31st December 2012 is a Tuesday. 2023 is a non-leap year and have 1 odd day.
So, 1st January 2024 is Tuesday. Hence, 31st December 2023 is Monday.

22. Number of years from 2006 to 2706 is 700 years.
700 year (400 + 300) have 1 odd day.
Hence, 14th November 2706 is one day after Sunday, i.e., Monday.

23. The number of odd days from 23rd November to 14th March in that year.
Month: N + O + S + A + J + J + M + A + M
Odd days: $2 + 3 + 2 + 3 + 3 + 2 + 3 + 2 + 3$
23 odd days $\Rightarrow \frac{23}{7} = 2$ odd days.
Hence, it is two days before Friday, i.e., Wednesday.

24. In order to have the same calendar between these two months the number of odd days should be zero.
Month: Jan + Feb + Mar + Apr + May + Jun + Jul
Odd days: $3 + 1 + 3 + 2 + 3 + 2$
At the completion of June, the number of odd days is zero. Hence, January and July will have the same calendar.

25. A century year which is divisible by 400 is a leap year and a leap year comes for every 4 years.
Hence, $2396 + 4 = 2400$ is a leap year.

26. The number of odd days upto 21st April 2006 is as follows.
(200) years + 5 years + (1st January 2006 to 21st April 2006)
2000 years have 0 odd days.
In these 5 years there is a leap year and 4 non-leap years.
Odd days $= 1 \times 2 + 4 \times 1 = 6$ odd days.
The number of odd days from 1st January 2006 to 21st April 2006.
Month: Jan + Feb + Mar + Apr
Odd days: $3 + 0 + 3 + 0 = 6$ days
The total, number of odd days $= 6 + 6 = 12$
$\Rightarrow$ 5 odd days
Hence, 21st April 2006 is Friday.

27. 5th May 1938 = 1600 + 300 + 37 + (1st January 1937 to 5th May 1937)
1600 years have o odd days.
300 years have 1 odd day.

37 years contain 9 leap years + 28 non-leap year.
One leap year contains 2 odd days.
One non-leap year contains 1 odd day.
In 37 years number of odd days = $9 \times 2 + 28 = 46$ odd days.
Number of odd days from 1st January 1938 to 5th May 1938 is as follows.

Month	Jan	Feb	March	Apr	May	Total
Odd day	3	0	3	2	5	13

$\therefore$ The total number of odd days = 0 + 1 + 46 + 13 = 60 odd days.
4 odd days means thus = 4 odd days (60 ÷ 4)
$\therefore$ 5th May 1938 is Thursday.

28. 15th August 2001 = 2000 + (1st January 2001 to 15th August 2001)
Century year contains 'o' odd days.
The number of odd days from 1st January to 15th August is as follows.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Odd day	3	0	3	2	3	2	3	1

$\therefore$ The total number of odd days = $\dfrac{17}{7}$ = 3 odd days.
3 odd days means = Wednesday
$\therefore$ 15th August 2001 is Wednesday.

29. As the year is starting and ending on two different days, it is a leap year.
It is given that 1st January is Saturday.
The number of odd days from 1st of January to 13th June is
= 2 (Excluding 1st January) + 1 + 3 + 2 + 3 + 6
= 17 = 3 odd days.
$\therefore$ The 3rd day after Saturday is Tuesday.

30. If the year is a leap year, it will have 53 Fridays as well as 53 Saturdays. Each of the remaining days of week occurs only 52 times. Similarly, if it is a non-leap year, it will have 53 Fridays and all the other days of the week occur only 52 times.
$\therefore$ The number of Tuesday is 52.

CHAPTER

10 Decision Making

Learning Objectives

In this chapter, you will:

- Practise how to take decisions subject to the given constraints
- Learn to sift through conditions and data to arrive at an actionable conclusion
- Learn to identify sufficiency or inadequacy of data to take a decision

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as, XAT, IIFT, MH-CET, MAT, etc.

❒ INTRODUCTION

In Decision Making, a situation/ circumstance or a caselet is provided, based on it students are generally asked two types of questions, such as:

What would be the best course of action to take in such circumstances? And, What lead to such a conclusion/situation?

These kinds of questions require a logical approach and a clear understanding of the case. It often requires careful reading of complete paragraphs and then logically inferring the information presented.

EXERCISES

Directions for questions 1 to 5: These questions are based on the following information.

A device classifies numbers according to the following criteria. The number must have

(1) Seven digits
(2) At least two prime digits.
(3) Perfect squares as the first and the last digits.
(4) An even digit as the middle digit.
If the number satisfies criteria (1), (2), (3) and (4), then the number is classified as a superior number.
(5) If the number is a superior number and the digits are in increasing order from left to right according to the value, then the number is classified as an impressive number.
(6) If the number satisfies criteria (1), (2) and (4) but not (3), but the first and last digits are even, then the number is classified as a mutual number.
(7) If the number satisfies criteria (1) (2) and (3) but not (4) but all the digits are odd, then the number is classified as a classic number.

(8) If the number does not fall under any of the above-mentioned classification, then the number is classified as a garbage number.

Based on the above criteria, decide which number should be classified in which of the above-mentioned classification. Mark your answer choice as follows.

(A) If the number is classified as superior number.
(B) If the number is classified as impressive number.
(C) If the number is classified as mutual number.
(D) If the number is classified as classic number (or) garbage number.

Classify the following number as mentioned above.

1. 1778459

2. 4276891

3. 1367574

4. 9632578

5. 4556666

Directions for questions 6 to 11: Study the following information carefully and answer the questions given below.

Following are the criteria required to be fulfilled, to gain admission into any intermediate college in Andhra Pradesh (AP):

(a) The candidate should have scored at least 80% marks in SSC or 70% marks in CBSE.
(b) The candidate should have scored at least 60% marks in the entrance exam.
(c) The candidate should be a male and must be at least 16 years old, as on 1st June 2000.
(d) The candidate should be a resident of Andhra Pradesh (AP), i.e., should have been born and brought up in AP.
In case a condition given above is not satisfied, an alternate condition for the same is given as below:
(e) If condition (d) is violated, but at least one relative of the candidate is presently staying in Andhra Pradesh, for the last 5 years, then the candidate can be admitted.
(f) If condition (b) or (c) is violated, but not both and the student has a good sports record, then he/she should be advised to approach the Principal.
(g) If both conditions (a) and (b) are violated, but the candidate is ready to donate ₹1,00,000 to the college, then he is advised to approach the Secretary.

Now mark your answer as
(A) If admission can be granted.
(B) If admission cannot be granted.
(C) If the Principal (or) the Secretary of the school is to be approached.
(D) If data is inadequate.

The following cases are given to you as on 1st June 2000.

6. Jahangir has applied for a college in AP. He was born and brought up in Kerala. He was born in 1980 and has an excellent percentage of 95 in his SSC. He has done pretty well in the entrance test scoring 65%, and has a close relative working in AP for the past 10 years.

7. Rita was born in 1981 and has a very good academic record with 90% in her CBSE. She has done her entrance test well by scoring 80% and she is a resident of AP, where the college is also situated. She is a very good sports person and has represented the state for basketball.

8. Prakash is 17 years old and is a resident of AP where the college is located. He has scored 90% in the entrance test and 69% in SSC.

9. Ranga, a resident of state of AP is a gold medallist in swimming at National games in the junior's category. He has applied with a percentage of 75% in the SSC and a score of 55% in the entrance exam. The college is in the same locality as his residence is. He was born on 16th July 1981. He is ready to give a donation of ₹1,00,000.

10. Ram was born in 1980 and is a resident of AP. He is the present TT Champion of India. He was presented a gold medal by the President of India in the National Games. He scored below 40% of marks in both SSC and the entrance exam. He is ready to give a donation upto ₹2 lakhs.

11. Rahim, applying for a college in the same state as he is in has a bad academic record of 40% and 35% in CBSE and entrance exam, respectively. He is able to pay a donation of ₹1,00,000 to the college.

Directions for questions 12 to 17: Study the following information carefully. Answer the questions given below:

Following conditions are to be fulfilled in order to become the hero of a film:

(a) The person should have a good track record, i.e., the rate of success of that particular person as a hero should be at least 70%.
(b) The person should have been in the industry for at least 5 years.
(c) The person should ask for a remuneration of less than ₹10,00,000 per film.
(d) The person should be able to give dates continuously for at least one week.
In case a condition given above is not satisfied, the following alternate condition should be applied:
(e) If condition (a) is violated and the hero has a success rate of greater than 50% in films and recognized as a hit pair with the heroine he is going to work with in the film, then take the suggestion of the heroine; (OR)
If condition (a) is violated and the hero has acted in more than six films with that producer, then take the advice of the Producer.

(f) If condition (c) is violated, but the hero demands a remuneration of less than ₹50 lakhs, then he can be selected.

(g) If condition (d) is violated, but the hero can work for at least 15 days in a month, may not be continuous then the hero can be selected.

Now mark your answers as

(A) If the person is to be selected as a hero.

(B) If the person is not to be selected as a hero.

(C) If the heroine or the producer is to be consulted regarding the decision.

(D) If the data is insufficient.

12. Romeo has been in the film industry for the past 7 years and asks for a remuneration of ₹7,00,000. He is ready to give dates continuously for 10 days. As a hero he has 30 hits out of the 40 films that he has acted in for the last 7 years.

13. Chris has a very good reputation as an actor and has 80% hits in his career. He is famous for the kind of films the producer of this film is presently making and is considered as an ideal combination with the heroine of the film. However, he charges ₹40,00,000 per film.

14. Ritesh is a hero, who has 65% hits in his career and charges ₹8,00,000 per film. He has been in the film industry for the past 7 years and has acted in nine films with the producer of that movie. He is ready to give dates continuously for ten days.

15. Rajesh is a hero in the film industry for the past 10 years, having 85% hits and demands a remuneration of ₹9,00,000. He acted in 10 for the kind of films of that producer. He can give dates for 18 days in a month, but continuously for only 6 days at a stretch.

16. Hasmukh has been in the industry for the past 6 years and has 68% hits in his career. He demands a remuneration of ₹8,00,000 and is ready to give dates continuously for at least one week. The heroine and he are considered to be the hit pair of the industry.

17. Roopesh is a successful hero for the last 8 years in action films, delivering around 85% in that line. But the trend has changed for the past 5 years. Therefore, Roopesh had to act in romantic films, where he was a misfit. Because of his failure as a romantic hero, he reduced his remuneration to ₹5,00,000. The producer now wants to make an action movie.

Directions for questions 18 to 23: Study the following information carefully and answer the questions which follow:

The following are the conditions/criteria for a group of college students to watch a movie at a theatre.

(a) Every student in the group should have attended all their lectures, i.e., no student should miss any lecture for watching the movie.

(b) No student in the group should have watched the movie earlier.

(c) The movie should be screened at a theatre, which is not more than 3 kms away from their college.

(d) Every student of the group should agree to watch the same movie.

In case a condition given above is not satisfied, an alternate condition for the same is given below:

(e) If condition (a) alone is violated and less than half of the total number of students have to miss their lectures, then these students attend their lectures and others can proceed to watch the movie.

(f) If condition (b) alone is violated but the movie is a hit and more than half of the total number of students of the group have not watched the movie before, then all of them go to watch the movie.

(g) If condition (c) alone is violated but at least two seniors are accompanying them to watch the movie, then all of them go to watch the movie.

(h) If condition (d) alone is violated and more than one-fourth of the total number of students in the group are not willing to watch the movie, then all the members of the group should meet and take a decision.

Based on the above criteria and the information given in each of the following questions, you have to take decision with regard to each case. You should not assume anything beyond what is given in the question.

Now mark your answer as:

(A), If the group decides to watch the movie.

(B) If the decision is not to watch the movie.

(C) If the data is inadequate.

(D) If a meeting is to be held to take a decision. (or) If only a few attends the classes and remaining go to watch the movie.

18. A group of students, having attended all of their lectures plan to watch a particular movie. The movie has been released for the first time in the city and is being screened in a theatre, which is 2 kms away from their college. None of them has watched the movie earlier. All of them agree to watch the movie.

19. A group of 10 students plans to watch a particular movie in a theatre, which is 3 kms away from their college. None of them has watched the movie earlier. Everybody agrees to watch this movie. Only 3 students out of the group have not attended their classes.

20. A group of students has plans to watch a particular movie, along with three senior students. The movie is being screened in a theatre, which is 5 kms away from their college. None of them has seen this movie earlier. Everyone in the group has attended all their lectures. All of them agree to watch this movie.

Exercises

21. A group of students has planned to watch a particular movie. They attended all the lectures but only 8 out of 10 students in the group have not watched this movie earlier. The movie is being screened in a theatre, which is just 1 km away from their college. The movie is a hit. All of them agree to watch this movie.

22. A group of students plans to watch a particular movie, which has been watched before by less than half of the group. The movie is a hit. The movie is running in a theatre, which is 2 kms away from their college. All of the students in the group have attended all their lectures. All of them agree to watch this movie.

23. A group of students having attended all of their lectures, plan to watch a particular movie. The theatre in which the movie is being screened is just 2.5 km away from their college. Less than 20% of students of the group have already watched the movie and every one in the group wants to see the movie as it was a big hit.

Directions for questions 24 to 30: Study the following information and answer the questions that follow:

The following are the conditions for constructing a house.

(a) The cost of land should be less than ₹20,00,000.
(b) The estimated cost of construction should be less than ₹30,00,000.
(c) The land/house to be bought should be in a posh locality.
(d) A loan worth at least ₹20,00,000 should be available at an interest rate of not more than 18% per annum.
If a condition given above is not satisfied, then an alternate condition for the same is given below.
(e) If condition (d) is violated, but the interest rate is less than 12% per annum, and the amount of loan that can be availed is at least ₹10,00,000, then the matter should be discussed with the family members.
(f) If condition (b) is violated, then a decision is made to buy a flat on the ground floor in that location, subject to availability.
(g) If condition (c) is violated, but all the basic amenities are available in the vicinity nearby, then the matter should be discussed with the family members.

Now mark your answer as:

(A) If the decision is to buy a flat (or) to construct a house.
(B) If the data is inadequate.
(C) If a discussion is to be held with the family members.
(D) If the house cannot be constructed.

24. The cost of a piece of land is ₹12,00,000 and the cost of construction is ₹38,00,000. The land is located in a posh locality. A loan of ₹30,00,000 can be availed at an interest rate of 15% per annum. There are flats available in that locality on the ground floor.

25. The cost of a piece of land would be around ₹15,00,000 and the cost of construction of a house would be around ₹20,00,000. Loan can be availed at a rate of 15% per annum. The land is located in a posh locality where every facility is available.

26. Land is available for ₹30,00,000 and the cost of construction is ₹20,00,000. Land is available in a very posh locality and a loan worth ₹20,00,000 can be availed at an interest rate of 16% per annum.

27. Land is available in a posh locality at ₹20,00,000 and the cost of construction is around ₹25,00,000. A loan worth ₹15,00,000 can be availed at an interest rate of 10% per annum.

28. Cost of a land in a locality is ₹10,00,000 and the cost of construction is ₹20,00,000. A loan can be availed at an interest rate of 20%.

29. Land is available in a posh locality of the city at ₹19,00,000 and the cost of construction is around ₹28,00,000. A loan of ₹15,00,000 can be availed at the rate of 10% per annum.

30. Mr Ravi has seen a piece of land costing ₹14,00,000 and the cost of construction of house is ₹28,00,000. A loan of ₹20,00,000 can be availed at the annual interest rate of a maximum of 18% per annum. The land is not located in a posh locality but has all basic amenities available in the vicinity.

Directions for questions 31 to 36: Read the following passage and answer the questions that follow it.

In May 1993, the Swedish automobile major, Volvo AB (Volvo) announced the closure of its car manufacturing facility at Uddevalla, Sweden, barely five years since its launch in 1989. A year later, the company had to shutdown yet another world-famous facility, the car assembly plant at Kalmar, which is also in Sweden.

Reacting to the two closures within a year's gap, analysts said Volvo's human centric approach towards automobile manufacturing was no longer feasible in the fiercely competitive scenario of the 1990's with most companies striving hard to improve production efficiency. Volvo was well recognized in the industry for its employee-friendly policies ever since its inception.

Guided by the 'Volvo Way,' the company had made conscious efforts to implement job enrichment concepts such as job rotation, job enlargement and employee work groups in its manufacturing facilities. In the late 1960s and early 1970s, when the company faced the problem of increasing employee turnover and absenteeism, it introduced these concepts and obtained positive results.

Volvo was inspired to build a new facility keeping this work design as a basis. This reiterated the company's belief that the industry needed to adapt itself to the people's requirements and not vice versa. This concept was implemented successfully in other plants of the company too in the 1970s. The best practices in Human Relations (HR) tried and tested in these plants were passed on to new plants established in the 1980s. While investing heavily in developing new plants like Kalmar and Uddevalla, where new work design concepts were implemented, Volvo was conscious of the risks involved and the possible effect on the company's financial performance if the experiments failed.

Acknowledging this, Gyllenhammar, in Harvard Business Review wrote, 'Volvo's Kalmar plant, for example, is designed for a specific purpose: car assembly in working groups of about 20 people. If it didn't work, it would be a costly and visible failure, in both financial and social terms. We would lose credibility with our people and those who are watching from outside'.

Gyllenhammar's apprehensions proved correct when Volvo closed down the Kalmar plant in 1994. However, Volvo's efforts in bringing changes in work design offered valuable lessons to both the academic and corporate community.

Analysts appreciated Volvo for its constant emphasis on learning from experiences and implementing the lessons so learnt in its new initiatives. This contributed significantly to the development of human centric production systems. These systems brought to life several theories and concepts, which had earlier only been enunciated in textbooks but rarely practised with the kind of seriousness with which Volvo did.

31. Which of the following best captures Volvo's philosophy of work?

(A) Employees should update their skills according to the changing needs of the company.
(B) Industry needs to adapt itself to the employee's requirements.
(C) In order to maximize profits more emphasis should be laid on employee welfare.
(D) A company can gain recognition only through its pro-employee policies.
(E) Lavish financial incentives given to employees go a long way in keeping afloat employee morale.

32. Which of the following is perceived to be the main reason for the closure of Volvo's manufacturing facility?

(A) The so-called employee friendly policies which did more harm than good to the employees.
(B) The aggressively competitive scenario of the 90s.
(C) The incongruity between Volvo's human-centric work philosophy and the aggressively competitive atmosphere of the 90s.
(D) The mismatch between the number of workers and the work involved.
(E) Increasing absenteeism among the employees.

33. Which of the following can be inferred from the failure of the policies implemented by Volvo at Uddevalla and Kalmar?

(A) Policies should be designed according to the needs of the time.
(B) The policies which had positive results in the past need not necessarily have positive results in the present also.
(C) 'Employee-friendly' policies seldom improve production efficiency.
(D) Failure of policies implemented by the company will have a bearing on the company's financial performance.
(E) Policies introduced on an experimental basis will most often end in a fiasco.

(A) A + B + C (B) B + C + D
(C) C + D + E (D) A + B + D
(E) D + E

34. Which of the following is a positive outcome of Volvo's fiasco?

(A) It offered an opportunity to learn from experience.
(B) The lessons learnt from the mistakes were implemented in new ventures.
(C) It led to the development of human-centric production systems.
(D) The human-centric production system helped to put into practice theories and concepts which were enunciated only in text books.
(E) All of these

35. The closure of Volvo's new manufacturing facilities does not detract from the value of the example that was set in the area of

(A) Business strategy
(B) Cost-benefit analysis
(C) Business growth
(D) Human resource optimization
(E) Brand building

36. According to Gyllenhammar if the new concepts failed it would be a 'costly failure' because

(A) The credibility of the company was at stake.
(B) It would mean huge financial losses for the company.
(C) It may impact on company's image.
(D) It would be demoralizing for the employees.
(E) All of these

Direction for questions 37 to 40: Read the following situation and choose the best possible alternative.

37. A Programme Manager in a software outfit learned a valuable lesson firsthand. He had asked his team for some feedback on his leadership. While most of what he received was positive and supportive, he got some strong advice too, from his team, about how he could be more effective in helping a shared vision. 'We would benefit, as a team, when we walk with you while you create the goals and vision, so that we all get to the end vision together'.

What effect will the suggestion have on the team or the team leader?

(A) Team spirit will be fostered among the team members.
(B) It would help in fostering team spirit in the team leader.
(C) The team leader's morale may be adversely affected.
(D) The team would benefit because the leader would be able to take into account the abilities and hopes of people.
(E) The ego of the team leader may be hurt.

38. Who can help a junior executive at work? Who knows what's going on? Who gets around road blocks? Who are the critical links in the information chain? The boss can be a big help by identifying and introducing people, setting up meetings and so on. Others who can help are the office administrator (who always knows who's on the way up), those in the legal department (who have an idea of the major problems the organization is facing), those in liaison (who have an idea of impending changes, whether in the market or in government regulations).

What does this suggest about the working of an organization?

(A) Hierarchies are an inevitable part of every organization.
(B) Every individual has an important role to play in the smooth functioning of an organization.
(C) Networking is a skill every potential manager should develop.
(D) Networking is a binding force which is needed to ensure camaraderie among employees.
(E) Networking goes a long way in developing individual talent.

39. Many firms that have a system in place for employee suggestions are taking steps to improve the quality of ideas before they are submitted for review. They are encouraging employees to first discuss ideas with their colleagues to gain insights into their technical and market feasibility or how they fit with the company's objectives. Such steps are invaluable because

(A) They would either enhance the value of the ideas or lead to their early and appropriate demise.
(B) They would deter people from being forthcoming with their advice.
(C) There are too many suggestions being offered by employees.
(D) They would help in enhancing a sense of participation among employees.
(E) They would make the employees feel valued and important if their suggestions are taken into consideration.

40. Customers who are loyal to your competitors represent market share you don't have and will likely not get. Customers who are loyal to you represent market share you already have. Protecting your most loyal customers is an obvious priority in a downturn. However, if they are spending 25% less than in better times, most of that will come directly out of what they spend in your stores. As a smart retail manager, you would, in times of downturn

(A) Put in place ways of identifying customers who have needs but no established loyalties.
(B) Resort to aggressive marketing in order to attract more customers.
(C) Relax and rest on your laurels, i.e., rely on your brand name.
(D) Introduce attractive offers to all customers thus retaining old customers and attracting new ones.
(E) Offer your regular customers attractive deals to prompt them to spend more in your stores.

Directions for questions 41 to 43: Read the passages carefully and answer the questions that follow.

Udaan, with its headquarters in Delhi, India, began as an air taxi operator in 1991 and started its commercial operations a year later in 1992. It operated with just 24 flights across 10 destinations initially but showed exceptional growth and had more than 300 daily flights to about 60 domestic and international destinations in 2007. It was first listed in the National Stock Exchange (NSE) in the year 2004.

In January 2008, the Air Transport Association had predicted that, globally, the airline industry would lose about US $5.2 billion by the end of 2008 based on an average price of US $140 per barrel of oil as the rise in fuel prices would push the fuel bills of the industry for that year to US $186 billion. The case is about the retrenchment drama that unfolded in Udaan Limited in late 2008. After showing the door to more than 1000 employees in a bid to streamline its operations and reduce its losses, Udaan was faced with immense criticism and opposition by various organizations and political parties. Udaan chairman Brijesh Goswami reinstated the employees a day later saying that he was not aware of these sackings which were done by the senior business manager. The sudden decision not only took the employees by surprise but also caused alarm in the sector. Amidst great furore and opposition by various organizations and political parities, Brijesh Goswami Chairman of Udaan reinstated the employees a day later amidst great emotional drama. Mr Brijesh Goswami was quoted as saying he had been appalled by the retrenchments of his employees, which he claimed, he had come to know only through media reports. He added that he would 'not be able to live as long as he lives' with the tough decision his management had taken and clarified that he was taking back the employees as they were 'family to him and as the head of the family he would take care of them'.

41. Which of the following can be concluded from the passage?

(A) Brijesh Goswami had capitulated under pressure from external parties.

(B) All is not well with the organizational communication mechanisms at Udaan.

(C) The company had planned to retrench more employees had there been no protests.

(D) There were many loopholes in Udaan's organizational communication network.

(E) None of these

42. What is the flaw in the decision taken by the senior business manager from the business point of view?

(A) The company decided to layoff the employees without any prior notice.

(B) Such sudden lay offs make future recruitment difficult.

(C) If oil prices remain high for long time the company will have no other option but to close down.

(D) Such decision does not gain support from the industry.

(E) None of these

43. What can be a better way to cope with the losses?

(A) The company should have sold off a few of its airplanes.

(B) Instead of sacking the employees the management could have proposed a salary reduction and explained the circumstances behind it to all its employees.

(C) The company should reduce its fares below that of its competitors so as to increase the load factor.

(D) The company should sack all the experienced employees and start hiring new employees for lower salaries.

(E) All the above

Directions for question 44 and 15: Select the correct alternative from the given choices.

44. Shaheshah Law the Uttar Pradesh (UP) based law firm was founded in 1982. It has a number of in-house advocates operating in all the higher courts including the Supreme Court, High Court, Labour Courts and various Tribunals. It was also accredited by the law society for training purposes. They conduct training sessions for fresh law graduates, throughout the day starting from 5 a.m. to 8 p.m. They take up various legal cases on behalf of corporate companies and individuals. On 16 July 2007 the court gave a ruling that expanded the scope of law on disability discrimination to include those who were associated with or responsible for a disabled person. The case discusses in detail the events that led to this ruling which was considered a landmark and was expected to have huge implications on businesses. It all began when Sarla Damitri, a former legal secretary with Shaheshah Law, sued her former employer (and a partner in the firm) in August 2004 for constructive dismissal. Sarla, who had a disabled son, alleged that the firm had discriminated against her at the work place due to her association with a disabled person. She alleged that she was treated differently, subjected to criticism and insults, denied flexible working arrangements that would help her to take care of her badly disabled child and ultimately forced into accepting voluntary retirement. She claimed that other employees were allowed flexible working arrangements.

Which among the following weakens Sarla Damitri's argument that she was discriminated against because other employees were allowed flexible working arrangements while she was not.

(A) This is the first time there is such a request at Shaheshah Law.

(B) Sarla left her previous job for the same reason.

(C) Sarla Damitri has relatives at home to take care of her child.

(D) The firm had two legal secretaries and so her presence at all times was not critical.

(E) Only the trainees were allowed flexible working hours at the time.

45. Sumit Patil, a management trainee in the sales and marketing department of one of the largest hardware firms of the city, Sheeta Electronics, was greeted by Ganesh Singh, the zonal sales manager of that firm on the very first day of his joining. In addition to his normal responsibilities, Ganesh was entrusted with the job of training the sales executives of that city and the nearby areas on the outskirts of that city. The firm sold electronic instruments to industries, schools, colleges, banks, cinemas and other industries.

Ganesh gave Sumit the catalogues and pamphlets describing in detail the types of electronic equipment sold by the company and the company background and showed him to his assigned desk. Thereafter, Ganesh excused himself and did not return. He did not even respond to the calls made and the messages sent by Sumit to seek clarifications though he is free. Sumit spent the whole day scanning the material and late in the evening he picked up his things and went home.

Which of the following best describes Ganesh's training methodology?

(A) The trainees were expected to be fully prepared for the job before they join.

(B) The trainees were trained by letting them work on their own without the help of senior officers.

(C) Ganesh's training programme is not suited for the firm.
(D) Training aims at providing circumstantial pressure on the trainee.
(E) All the above

Directions for questions 46 and 47: Answer the questions based on the information given below.

Ambuja solutions, the logistics firm was growing steadily. The huge growth of the organization brought to the fore the need to recruit employees. The HR manager found that it is not difficult to recruit unskilled employees but recruiting people at the middle and top management level is going to be an uphill task. The HR team has formed a team to attract people who are already employed, busy in their jobs and are passive job seekers. The team has come up with their report indicating the reasons for looking for a change of job, their attachment to the present organization, the quantum of job satisfaction they are deriving now, their views regarding a conducive work atmosphere, etc. The team has found that these passive job seekers become active when a new employer approaches them through common friends.

46. Which of the following is a possible reason behind the HR manager's attempt to attract employees of other organizations?
(A) Fresh candidates lack experience.
(B) To weaken the other organizations.
(C) Amount spent on training new employees can be reduced.
(D) Experienced personnel are required for middle and top management.
(E) To infuse fresh competition among the middle and top managers.

47. Based on the findings of the report which of the following steps would help the organization to attract passive job seekers?
(A) Advertise the openings in the organization.
(B) Talk negative about other organizations.
(C) Encourage their employees to bring in potential candidates among friends and acquaintances.
(D) Increase the salary paid to the existing employees in the middle and top management
(E) Find out what keeps an employee loyal to an organization.

Directions for questions 48 to 50: Answer the questions based on the information given below.

Ravi's father is working as an agent for an insurance company. Ravi has developed interest in the field of insurance. He used to go through all insurance related laws and regulations. He appeared for several competitive exams conducted to recruit people into insurance companies. Finally, the day of joining his dream job had come. He was called for an interview for the post of sales manager in a major multinational insurance company. He prepared meticulously for the interview. He had gone through the fundamentals of his areas of interest and study. His father conducted several mock interviews.

Ravi arrived at the venue of the interview 15 minutes before time, but the interview started one hour late. During this time, he tried to complete the pre-interview formalities. The receptionist appeared to be ill-trained and lethargic. Ravi did not feel confident that the receptionist had gone through the formalities properly. The interview rooms were more intimidating. None of the six members on the panel retuned the pleasantries nor did they ask Ravi to sit. After standing for several minutes (which appeared to be several years for Ravi) one of the panel members asked Ravi to sit, but not in a pleasing manner. The chair offered to Ravi was quite uncomfortable. The attitude of the panel seemed like 'you need us, but we do not need you'. Most of the questions asked were not directly related to insurance, moreover several of them appeared sarcastic. Ravi kept his cool and answered most of the question carefully. At the end, when the panel asked him if Ravi had any questions, the exhausted Ravi, though he had prepared a list of questions, said no and left the room in disgust after thanking the panel.

48. Which among the following qualities is the interview panel trying to test in Ravi?
(i) Resistance to change
(ii) Aggressiveness
(iii) Patience
(iv) Behaviour while under pressure
(v) Persistence
(A) Only (i), (ii) and (iii)
(B) Only (iii) and (iv)
(C) Only (i), (iii) and (iv)
(D) Only (ii) and (v)
(E) Only (ii) and (iii)

49. Which of the following is a possible reason, why Ravi did not ask any question, even though he was given a chance to do so?
(A) Ravi had succumbed to the preemptive tactics played by the panel.
(B) Ravi thought that the panel was probably not interested in him.
(C) Ravi made an assessment of the work environment that he may have to face.
(D) Ravi had decided that insurance was not his cup of tea.
(E) None of the above

50. Which of the following denotes both the positive and negative traits of Ravi that were displayed during the interview?

(A) Punctuality; Resistance
(B) Enthusiasm; Pessimism
(C) Temperament; Hardworking nature
(D) Enthusiasm; Resistance
(E) Patience; Pessimism

Directions for questions 51 to 60: There is a passage given followed by certain directions. Read the directions given under the passage carefully before answering the questions that follow.

M/s. Anant Printers is a small printing press. It has three treadle machines, one modern offset machine and DTP equipment. It has nine workers for all the jobs in the press. The whole unit is just like a family and the proprietor has excellent relations with the workers. The press has a large number of orders. There is good demand and it can increase its operations, if the proprietor wishes to do so.

The proprietor has come across four offset machines in a medium-size press at Mumbai, which has been closed due to the death of its owner. If the proprietor purchases these machines, the total number of workers would increase to twenty. The trade union would come forward and build a union of the workers in M/s. Anant Printers. It is difficult to anticipate whether the wage rates could be maintained at the same level after union is formed.

One alternative is to import fully automatic machine from Germany, then the number of workers could be limited, but the capital cost is very high, besides skilled technicians will have to be appointed to run the modern automatic machines. The cost of production is expected to be very heavy.

Directions: The questions that follow relate to the preceding passage. Evaluate, in terms of the passage, each of the item given. Then select your answer from one of the following classifications.

(A) A Major Objective in making the decision: One of the goals sought by the decision.
(B) A Major Factor in making the decision: An aspect of the problem, specifically mentioned in the passage, that fundamentally affects and/or determines the decision.
(C) A Minor Factor in making the decision: A less important element bearing on/or affecting a major factor, rather than a major objective directly.
(D) A Major Assumption in making the decision: A projection or supposition arrived at by the decision maker before considering the factors and alternatives.
(E) An Unimportant issue in making the decision: An item lacking significant impact on, or relationship to, the decision.

Questions:

51. Funds are available for purchasing either of the machines.

52. Union may be formed and wage rate may not be at the present level.

53. Import of machinery would involve heavy capital cost, leading to rise in cost of the production.

54. Harmonious relations exist between the present set of workers and management.

55. Operations can be expanded.

56. Union may not accept the level of wages being paid now.

57. The owner of the offset printing press in Mumbai is dead.

58. If production capacity is not increased, then potential business could be lost to rivals.

59. Buying old machines may not keep the company in the forefront of technology.

60. New imported machinery will lead to better quality resulting in more business.

ANSWER KEYS

1.	(C)	11.	(B)	21.	(B)	31.	(B)	41.	(A)	51.	(C)
2.	(C)	12.	(C)	22.	(D)	32.	(B)	42.	(A)	52.	(B)
3.	(D)	13.	(A)	23.	(B)	33.	(D)	43.	(B)	53.	(B)
4.	(B)	14.	(D)	24.	(B)	34.	(E)	44.	(E)	54.	(B)
5.	(C)	15.	(B)	25.	(D)	35.	(C)	45.	(B)	55.	(D)
6.	(A)	16.	(D)	26.	(D)	36.	(E)	46.	(D)	56.	(D)
7.	(A)	17.	(B)	27.	(C)	37.	(B)	47.	(C)	57.	(E)
8.	(D)	18.	(A)	28.	(B)	38.	(C)	48.	(B)	58.	(C)
9.	(D)	19.	(A)	29.	(B)	39.	(A)	49.	(B)	59.	(E)
10.	(C)	20.	(C)	30.	(B)	40.	(D)	50.	(E)	60.	(C)

SOLUTIONS

Solutions for questions 1 to 5:

Q.No.	Number	(1) Seven digits	(2) At least 2 prime digits	(3) First and last digits are perfect squares. [6] First and last digits are even	(4) Middle digit even [7]. All digits are odd.	(5) Digits are in increasing order from left to right.
1	1778459	✓	✓	✓	✓	×
2	4276891	✓	✓	✓	✓	×
3	1367574	✓	✓	✓	× [×]	×
4	9632578	✓	✓	× [×]	✓	×
5	4556662	✓	✓	× [✓]	✓	×
6	4368579	✓	✓	✓	✓	×
7	1366789	✓	✓	✓	✓	✓
8	9517531	✓	✓	✓	× [✓]	×
9	1573934	✓	✓	✓	× [×]	×
10	964374	×	✓	✓	×	×

1. Satisfies all the basic conditions.
∴ Superior number.

2. Satisfies all the basic conditions.
∴ Superior number.

3. Does not satisfy condition (4) and also its alternate condition.
∴ Garbage number.

4. Does not satisfy condition (3) and also its alternate condition.
∴ Garbage number.

5. Does not satisfy condition (3) but satisfies the alternate condition.
∴ Mutual number.

Solutions for questions 6 to 11: The four basic conditions from (a) to (d), given in the selection criteria are as shown in the table below. In case a basic condition is violated, the case is verified for the respective alternate condition given. The alternate conditions are as given below:

(e) If condition (d) is violated, but at least one relative of the candidate is presently staying in Andhra Pradesh (AP) for the last 5 years, then the candidate can be admitted.

(f) If condition (b) or (c) is violated, but not both and the candidate has a good sports record, then the candidate should be advised to approach the Principal.

(g) If both the conditions (a) and (b) are violated, but the candidate is ready to donate ₹1,00,000 to the college, then the candidate should be advised to approach the Secretary.

Now let us scrutinize the applicants for the basic conditions, as given in the table below (a tick mark '✓'means that the condition is fulfilled, cross mark '×' means that the condition is violated).

Exercises

Question number	Name of the candidate	(a) SSC ≥ 80% or CBSE ≥ 70%	(b) Entrance exam ≥ 60%	(c) Age ≥ 16 years, male candidate	(d) Resident of AP	Remarks
6	Jahangir	✓	✓	✓	×	(d) violated
7	Rita	✓	✓	× (female)	✓	(c) violated
8	Prakash	×	✓	✓	✓	(a) violated
9	Ranga	×	×	✓	✓	(a) and (b) violated
10	Ram	×	×	✓	✓	(a) and (b) violated
11	Rahim	×	×	? (age)	?	(c) and (d) unknown, (a) and (b) violated

6. In this case, condition (d) is violated, i.e., Jahangir is not a resident of AP. Hence, his case is verified for the alternate condition (e). Since, a close relative of Jahangir has been staying in AP for the last 10 years (i.e., more than the required 5 years), Jahangir is selected.

7. In this case, condition (c) is violated, i.e., Rita is a female. Then the alternate condition (f) is applied. As Rita has a good sports record (represented state in basketball), she should approach the Principal.

8. In this case, condition (a) is violated. As there is no alternate condition given for violating condition (a) alone, admission cannot be granted to Prakash.

9. In this case, both the conditions (a) and (b) are violated. Then, Ranga's case is verified for the alternate condition (g). As Ranga is ready to pay a donation of ₹1,00,000, he should be advised to approach the Secretary of the school.

10. In this case, both the conditions (a) and (b) are violated, hence, the alternate condition (g) is applied. As Ram is ready to pay a donation of more than ₹one lakh, he should be advised to approach the Secretary of the school.

11. In this case both the conditions (a) and (b) are violated in lieu of which the alternate condition (g) is satisfied (i.e., Rahim can pay a donation of ₹1,00,000). But, no information is available to check conditions (c) and (d). Hence, we cannot take a decision as the data is inadequate.

Solutions for questions 12 to 17: The four basic conditions, from (a) to (d), given in this selection criteria are as shown in the table below.

In case a basic condition is violated, the respective alternate condition is applied in order to take a decision.

The alternate conditions are as given below:

(e) If condition (a) is violated and the hero has a success rate of greater than 50% in films but has a good success rate with the heroine he is going to work with in the film, then take the suggestion of the heroine. OR

If condition (a) is violated, but the hero is successful in the type of films the Producer is making, then take the advice of the Producer.

(f) If condition (c) is violated, but the hero demands a remuneration of less than ₹50 lakhs, then he can be selected.

(g) If condition (d) is violated, but the hero can work for at least 15 days in a month, may not be continuous, then the hero can be selected.

Now let us scrutinize all the applicants for the basic conditions (from (a) to (d)) as given in the table below (a tick mark '✓' means that the condition is fulfilled and a cross mark '×' means that the condition is violated):

Question number	Name of the hero	(a) Success rate as a hero ≥ 70%	(b) Time spent in the industry ≥ 5 yrs	(c) Remuneration < 10 lakhs	(d) Dates continuous ≥ one week	Remarks
12	Romeo	✓	✓	✓	✓	All conditions are satisfied
13	Chris	×	?	×	?	(b) and (d) unknown and (a) and (c) are not satisfied.

(Continued)

Exercises

Question number	Name of the hero	(a) Success rate as a hero ≥ 70%	(b) Time spent in the industry ≥ 5 yrs	(c) Remuneration < 10 lakhs	(d) Dates continuous ≥ one week	Remarks
14	Ritesh	×	✓	✓	✓	(a) violated
15	Rajesh	✓	✓	✓	×	(d) violated
16	Hasmukh	×	✓	✓	✓	(a) violated
17	Roopesh	?	✓	✓	?	(a) and (d) unknown

12. In this case, Romeo satisfies all the conditions. Hence, he is selected as a hero.

13. In this case, condition (a) is not satisfied, as the information given regarding the success rate is not that of his success as a hero. Similarly, (c) is not satisfied, as he charged more than 10 lakhs per film. Conditions (b) and (d) cannot be checked, as Chris's experience in the industry and whether he can give dates continuously for at least one week is not known. Hence, the data is inadequate for taking a decision.

14. In this case, condition (a) is violated, then the alternate condition (e) is tested. As Ritesh has 90% hits in the type of films the producer is making, Producer's advice should be taken.

15. In this case, condition (d) is violated. But as Rajesh can give dates for 18 days in a month, (i.e. more than the required 15 days in a month, as in condition (g)), hence Rajesh is to be selected as the hero.

16. In this case, condition (a) is violated. But as Hasmukh is very successful with the heroine of the film, he fulfils the alternate condition (e). Hence, the suggestion of the heroine should be taken.

17. In this case, both the conditions (a) and (d) cannot be checked, as Roopesh's track record for all the 8 years is unknown, and also his ability to give dates is not mentioned, hence data is insufficient to take a decision.

Solutions for questions 18 to 23:

Question number	(a) Every student attended all lectures	(b) No body watched the movie earlier	(c) Theatre ≤ 3 kms	(d) Every student agrees	Remarks
18	✓	✓	✓	✓	All conditions are satisfied.
19	×	✓	✓	✓	(a) violated
20	✓	✓	×	✓	(c) violated
21	✓	×	✓	✓	(b) violated
22	✓	×	✓	✓	(b) violated
23	✓	× [✓]	✓	✓	(b) violated

18. In this case, as all the conditions are satisfied. Hence, the group will watch the movie.

19. In this case, condition (a) is violated, as three students out of ten have not attended all their lectures. As three is less than half of ten (the total number of students in the group), these three students should attend their lectures and others should proceed to watch the movie, as given in the alternate condition (e).

20. In this case, the theatre is more than 3 km away. Hence, condition (c) is violated. But as at least two seniors are accompanying them to watch the movie, the alternate condition (g) is fulfilled. Hence, all of them can go to watch the movie.

21. Here, condition (b) is violated, i.e., 2 out of 10 students have already watched the movie earlier. But as the movie is a hit and more than half the total number of students (i.e., 8 out of 10) would be watching the movie for the first time, the alternate condition (f) is satisfied. Hence, all of them go to watch the movie.

22. In this case, condition (b) is violated as some students have seen the movie earlier. As the number of these students is less than half of the total number of students in the group, the alternate condition (f) is fulfilled. Hence, all of them go to watch the movie.

23. In this case, condition (b) is violated as some students have seen the movie earlier. As the number of these students is less than half of the total number of students in the group, the alternate condition (f) is fulfilled. Hence, all of them go to watch the movie.

Solutions for questions 24 to 30:

Question number	(a) Land cost < 20L	(b) Estimated cost of construction < 30L	(c) Posh locality	(d) Loan ≥ 20L @ maximum of 18% p.a.	Remarks
24	✓	×	✓	✓	(b) violated
25	✓	✓	✓	?	(d) unknown
26	×	✓	✓	✓	(a) violated
27	×	✓	✓	×	(a) and (d) violated
28	✓	✓	?	?	(c), (d) unknown
29	✓	✓	✓	×	(d) violated
30	✓	✓	×	✓	(c) violated

24. In this case, condition (b) is violated, as the estimated cost of construction is more than ₹30 lakhs. As there are flats available in that locality on the ground floor, the respective alternate condition (f) is fulfilled.

25. In this case, the loan amount is not specified, which is required to check condition (d). Hence, the data is inadequate to take a decision.

26. In this case, the cost of the land is more than ₹20 lakhs. Hence, condition (a) is violated. As there is no alternate condition for the same, the house cannot be constructed.

27. In this case, the land cost is ₹20 lakhs, whereas according to condition (a) the cost of land should be less than ₹20 lakhs. Hence, condition (a) is violated. As there is no alternate condition for the same, the house cannot be constructed.

28. As no information is available to check conditions (c) and (d), the data is inadequate to take any decision.

29. Here, condition (d) is violated, as the loan available is less than the stipulated ₹20 lakhs. But the alternate condition (e) is satisfied as the loan available is ₹15,00,000, i.e., more than the stipulated ₹12 lakhs and the interest rate is 10%, i.e., less than the stipulated 12%. Hence, the matter is to be discussed with the family members.

30. In this case, condition (c) is violated as the land is not located in a posh locality. However, as all the basic amenities are available nearby, the alternate condition (g) is satisfied. Hence, the matter should be discussed with the family members.

Solutions for questions 31 to 36:

31. It is implied in the second sentence of para 4, 'This reiterated the company's belief……' that the industry's policies should be employee friendly and the industry should adapt itself to the employees' requirements.

32. Refer to the 2nd para of the passage where it is stated that the human-centric work philosophy of Volvo did not match with the fiercely competitive work atmosphere of the 1990s.

33. It is implied in the passage that the closure of Volvo's plants at Uddevala and Kalmar convey that policies should be designed according to the needs of the time (choice A). It is implied in the second third and fourth paragraphs of the passage that choices (B) and (D) can be inferred from the failure of the policies implemented by Volvo at Uddevalla and Kalmar.

34. Refer to the last paragraph of the passage where it is stated that all the given options speak about the positive outcome of Volvo's fiasco.

35. It is implied in the passage (particularly in the last para) that valuable lessons were learnt in the area of employee relations and productivity.

36. Refer to para 5 of the passage according to which all the given options are applicable.

Solutions for questions 37 to 40:

37. The passage says that the Programme Manager sought feedback on his leadership this indicates that he is keen on doing his best. So, it is certain that he would act on the advice given to him by his team members and become more open to their suggestions. Choice (D) is incorrect as it states 'how' it would be effective but not about the effect it would have.

38. The passage speaks about the importance of networking in an organization. Hence, the skill of networking is a skill every potential manager should develop. Statement – A which talks about hierarchies is rather digressing. (B) does not capture the most essential aspect of the para. (D) and (E) are not relevant to the context.

39. Encouraging employees to discuss ideas with their colleagues in order to gain insights into their technical and market feasibility is essential because such a step would enhance the value of ideas or if they are not feasible it would be abandoned before implementation, thereby saving valuable time and resources. Hence, (A) is most logical and practical.

40. The loyalties of the old customers are already established. Hence, a retail manager need not make any special efforts to lure them. But during times of recession, when the spending is low, the best way of attracting new customers is by tempting them with attractive deals. This will serve the dual purpose of retaining old customers and attracting new ones.

Solutions for questions 41 to 43:

41. Brijesh Goswami was quoted as saying he had been appalled by the retrenchments and was not aware of the sackings of 1000 employees, but the decision to retrench 1000 employees cannot be taken without prior permission from the chairman. Hence, Brijesh Goswami had capitulated under pressure from external parties.

42. The company cannot lay off its employees without prior notice and hence,
Choice (A) is definitely a flaw in the decision of the senior business manager. The question of future recruitment comes into picture when the company survives. The present problem for the company is survival. Hence, (B) is not a flaw. Choice (C) is irrelevant in the context of the question. The decision taken by Udaan does not have any effect on other companies in the industry. Hence, (D) is not a flaw.

43. It is not known whether the company has excess planes or not and it is possible that as the situation improves the company would have a shortage of planes for its requirement. For a company in loss, it is a normal practice for the employees to take a pay out to save the job and also the company. So, his company could have considered this option after taking all the employees into confidence. Reducing fare below that of its competitors would force others to follow suit which would further bleed the industry and increase the loss. Employees with experience would have higher salaries than freshers, but as experience would count a lot in any situation, (D) is not a better way to cope with the losses.

Solution for question 44 and 45:

44. As only the trainees were allowed flexible working hours, the argument that there was a discrimination would be weakened. Hence, option (E) weakens her argument.

45. Ganesh gave Sumit all the details related to the company and its products and expected Sumit to learn the work on his own.

Solutions for questions 46 and 47:

46. Choice (A) indicates the drawback with fresh employees but does not talk about the advantage in recruiting employees of other organizations. From the passage it is clear that the organization is growing and there is a need to recruit new employees. Hence, (B) is not a reason. According to the HR manager the difficulty lies in finding the required people but not training them. Hence, (C) is not a reason. HR manager did not find a difficulty in recruiting unskilled people but the problem is with filling middle and top management posts. This implies that the HR manager is looking for experienced people. Hence, (D) is a reason.
(E) is out of context.

47. It is stated that the passive job seekers become active when a new employer approaches them through common friends. Hence,
choice (C) would help the HR manager to attract the target group.

Solutions for questions 48 to 50:

48. The panel is trying to reflect the 'you need me, but I do not need you' kind of attitude that some of the customers would show. The quality that is required in such cases is patience. The panel decided to make things unpleasant for Ravi to check how he would react when faced with a similar situation during the course of his work. The panel is trying to test his patience and his behaviour under pressure.

49. As Ravi faced a panel which was behaving in an unfriendly manner and was also asking him questions that were not directly related to insurance, he could have felt that the panel was not interested in him.

50. Ravi had kept his cool in the face of an unfriendly atmosphere. Thus, he is successful in showing good levels of patience. Ravi has seen the entire interview with pessimism. It is clear from the sentence. The attitude of the panel seemed like….' and the sentence 'several of them are sarcastic'. This is also evident from the fact that Ravi has not asked the panel any question, though he was prepared with some questions and he was given a chance to ask questions. From this it is clear that he was pessimistic throughout. He left the interview in disgust.

Solutions for questions 51 to 60: **Situation Analysis:**

The present status of Anant Printers is, it is comfortably placed in terms of orders and the other business basics like a dedicated work force and adequate number of machines. However, the important aspect/issue that is raised in the passage is evident from the statement, 'There is good demand and it can increase its operations, if the proprietor wishes to do so'. The issue, hence, is of growth. The important decisions that are under consideration are mentioned below:

(1) Whether to expand operations / volumes or to stay at the present level of volumes.

(2) If expanding of operations is chosen, then the choice between the 'four offset machines' and the 'fully automatic machine' is to be made.

The worker relations and the final cost of production are evidently, the major criteria that need consideration.

Also, the other criteria that is required to be considered would be the wage rates, availability of skilled technicians, required capital, costs, etc.

51. The availability of funds is not explicitly mentioned as a concern/problem in the passage. Hence, it can be concluded that funds are not a major factor in making the decision. At the same time, it cannot be said that the availability of funds is not important for the decision making involved. Hence, this is a minor factor in making the decision.

52. Since the present state is that '…the proprietor has excellent relations with the workers…' and 'The whole unit is like a family…', the formation of the union will have a significant impact on these pleasant relations (and also the wage rate). This is a major factor in making the decision.

53. The final cost of production is an important factor as it affects the basic competitiveness of the press. Hence, if imported machinery leads to high cost of production it will be a major factor in making the decision.

54. As already mentioned in the solution to Q. no. 22, the fact that harmonious relationships exist between the workers and the management is an important and major factor in making the decision.

55. The entire passage is regarding the issue of expanding operations, which is the major objective in making the decision. However, to suppose that it is possible to do so, (i.e., that there will be adequate demand) is in fact a major assumption in the decision-making process.

56. Since the basic information from the passage says 'It is difficult to anticipate whether the wage rates could be maintained at the same level after the union is formed', to suppose that the union may not accept the level of wages being paid now would be a major assumption.

57. The fact whether the owner of the offset printing press is dead or alive has practically no significance or impact on the decision-making process. It is an unimportant issue.

58. If there was a possibility of losing the "present business" to rivals, then it could be considered as a major factor in making the decision. However, the passage states 'There is good demand and it can increase its operation…' This implies that the 'present business' is not under any threat. 'Potential' business being lost to rival would not be a major factor but can be considered as a minor factor in the decision-making process.

59. The issue of being in the forefront of technology has neither been mentioned nor alluded to in the passage. Hence, this is a relatively unimportant issue in the present decision-making scenario.

60. Since it is mentioned in the passage that the present business is good and that the demand is adequate, with scope for expanding operations, improving quality is not a necessity or a major factor. However, since better quality will help win more business, it cannot be considered unimportant either. Hence, it is appropriate to consider this as a minor factor.

11 Non-verbal Reasoning

CHAPTER

LEARNING OBJECTIVES

In this chapter, you will:

- Learn about different models of non-verbal reasoning questions like:
  - Series
  - Analogies
  - Odd man out
- Learn about different patterns in which elements behave, like:
  - Shifting
  - Rotation
  - Image formation
  - Increasing/reducing the number of elements
  - Substitution

The questions in this section do not occur frequently in the CAT but carry a high weightage in the Other Management Entrance Tests (OMETs) such as, XAT, IIFT, MH-CET, MAT, etc.

INTRODUCTION

Under Non Verbal Section, you get combination of Reasoning questions but broadly asked in symbols/image format.

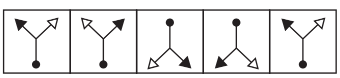
The questions could range from: Series, Analogies, to Odd man out etc. We can understand them better with the help of below exercise.

EXERCISES

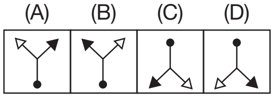
Directions for questions 1 to 5: In each of the following questions there are two sets of figures. One on the left side (problem figures) and the other on the right side marked (A), (B), (C) and (D) (answer figures). Select one figure from the answer set which will continue the same series as given in the problem set of figures.

Problem figures **Answer figures**

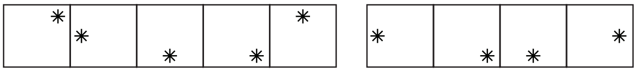
1.



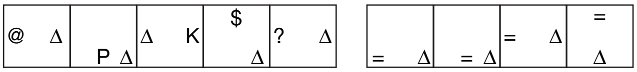
(A) (B) (C) (D)



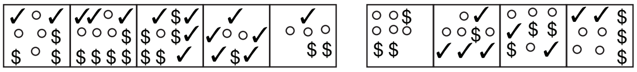
2. (A) (B) (C) (D)



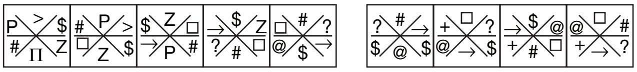
3. (A) (B) (C) (D)



4. (A) (B) (C) (D)



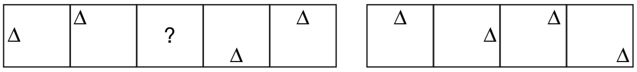
5. (A) (B) (C) (D)



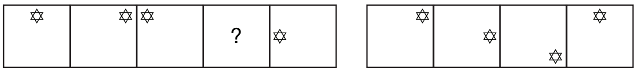
Directions for questions 6 to 10: In each of these questions there are two sets of figures. The figures on the left are 'Problem Figures' (Four figures and one question marked space) and those on the right are 'Answer Figures' indicated by A, B, C and D. A series is established if one of the four 'Answer Figures' is placed at the 'question marked space'. Figures form a series if they change from left to right according to the same rule. The number of the 'Answer Figure' which should be placed in the question marked space is the answer. All the five figures, i.e., four 'Problem Figures' and one 'Answer Figure' placed in the question marked space should be considered as forming the series.

Problem figures **Answer figures**

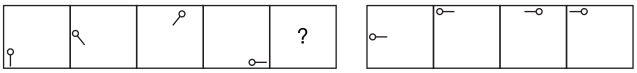
6. (A) (B) (C) (D)



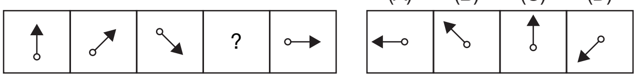
7. (A) (B) (C) (D)



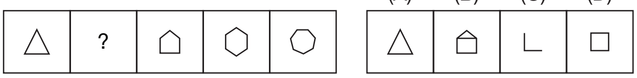
8. (A) (B) (C) (D)



9. (A) (B) (C) (D)



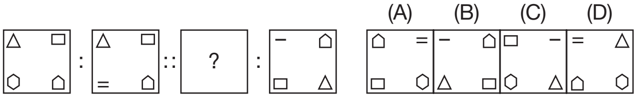
10. (A) (B) (C) (D)



Directions for questions 11 to 15: In each of the following questions, the first two figures are related to each other in a certain way. Find out which figure from the answer figures (A), (B), (C) and (D) should be placed at the '?' mark so that the second pair so formed will have a similar relationship and can be placed at 'question mark'.

Problem figures **Answer figures**

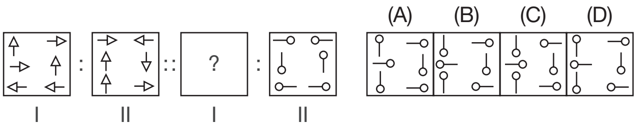
11. (A) (B) (C) (D)



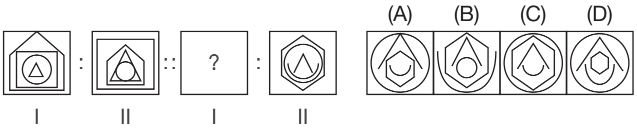
I II I II

Exercises

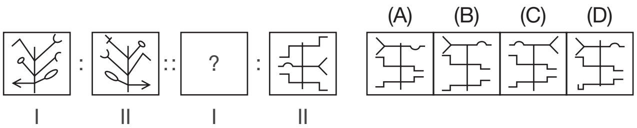
12.



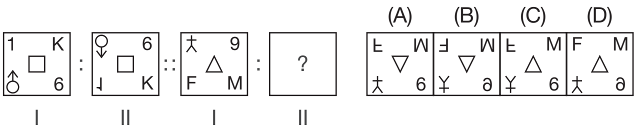
13.



14.



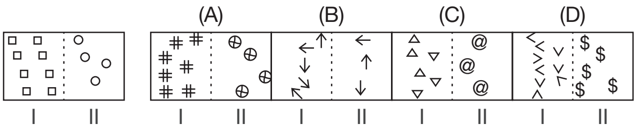
15.



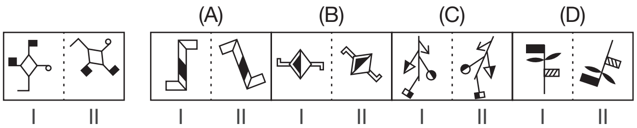
Directions for questions 16 to 20: In each of the following questions a pair of problem figures is given at the left extreme followed by four pairs of answer figures. The left figure in the problem figures bears a certain relationship with the right figure. Out of the four pairs given in the answer figures one is similar to that pair given in the problem figures. Find out the answer pair by comparison.

Problem figures **Answer figures**

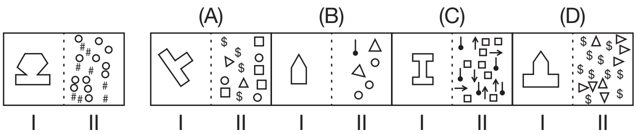
16.



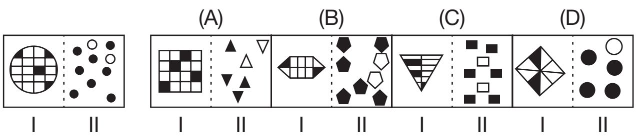
17.



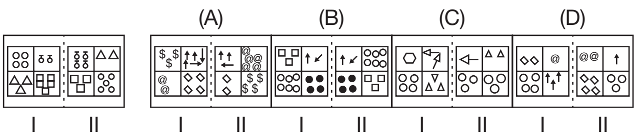
18.



19.

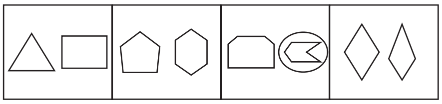


20.

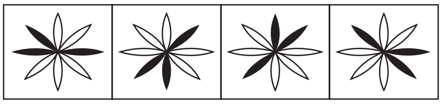


Directions for questions 21 to 25: In each of the following questions three out of the given four figures are similar in a certain way and hence, they form a group. Find the one which does not belong to that group.

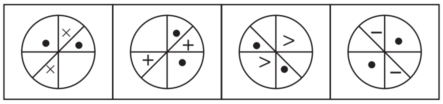
21. (A) (B) (C) (D)



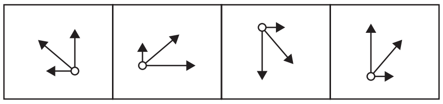
22. (A) (B) (C) (D)



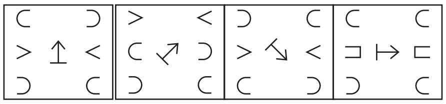
23. (A) (B) (C) (D)



24. (A) (B) (C) (D)

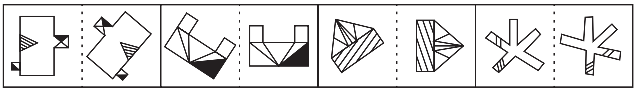


25. (A) (B) (C) (D)



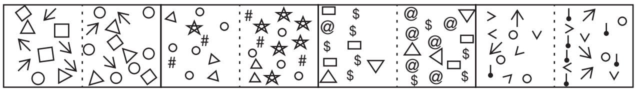
Directions for questions 26 to 35: In each of the following questions, in three out of the four pairs of figures, element II is related to element I in the same particular pattern. Find out the pair in which the element II is not so related to element I.

26. (A) (B) (C) (D)



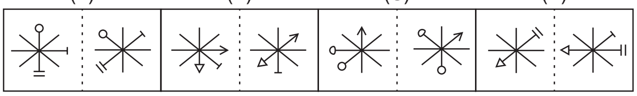
I II I II I II I II

27. (A) (B) (C) (D)



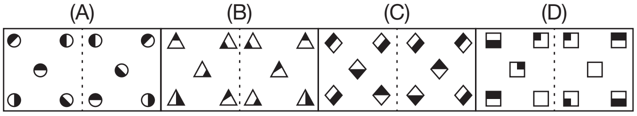
I II I II I II I II

28. (A) (B) (C) (D)



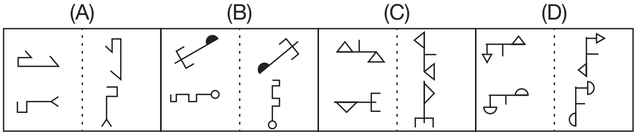
I II I II I II I II

29. (A) (B) (C) (D)



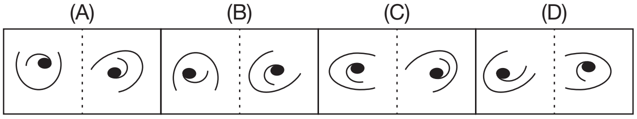
I II I II I II I II

30. (A) (B) (C) (D)



I II I II I II I II

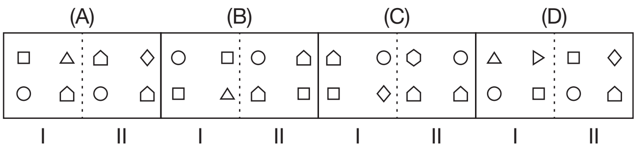
31. (A) (B) (C) (D)



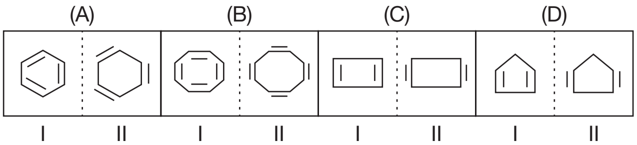
I II I II I II I II

Exercises

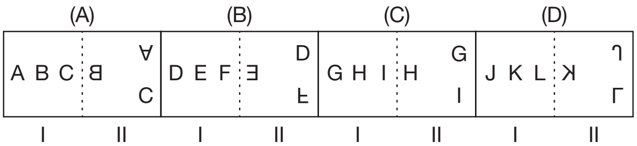
32.



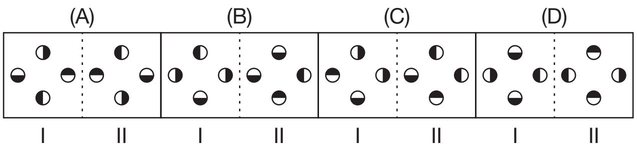
33.



34.



35.



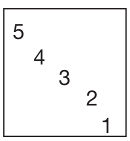
ANSWER KEYS

1. (A)	**7.** (B)	**13.** (C)	**19.** (C)	**25.** (D)	**31.** (C)
2. (C)	**8.** (D)	**14.** (B)	**20.** (A)	**26.** (B)	**32.** (A)
3. (B)	**9.** (A)	**15.** (C)	**21.** (D)	**27.** (A)	**33.** (D)
4. (D)	**10.** (D)	**16.** (C)	**22.** (A)	**28.** (A)	**34.** (C)
5. (D)	**11.** (A)	**17.** (A)	**23.** (D)	**29.** (D)	**35.** (B)
6. (C)	**12.** (D)	**18.** (D)	**24.** (D)	**30.** (C)	

SOLUTIONS

1. In each figure, the mirror image and the water image are taken alternately to get the appropriate answer figure (A).

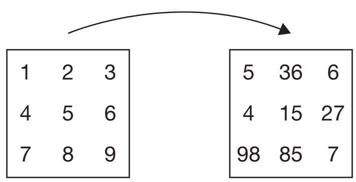
2. Let the elements in each figure be:



The elements are swapped. From the second figure to the third figure, the 2nd and the 3rd elements are swapped. Similarly, the 3rd and the 4th, the 4th and the 5th, and the 5th and the 1st elements are swapped cyclically in the remaining figures.

∴ The appropriate answer figure is (C).

3. The elements in the first figure are rearranged as shown below to get the next figure.



Previous figure

Next figure

Exercises

A similar logic is applied from the third to the fourth and from the fifth to get the answer figure.
∴ The appropriate answer figure is (B).

4. The element ✓ is appearing 2 times, 3 times, 4 times, 5 times, 1 time and 2 times cyclically.
The element o is appearing 4 times, 5 times, 1 time, 2 times, 3 times and 4 times cyclically.
The element $ is appearing 3 times, 4 times, 5 times, 1 time, 2 times and 3 times cyclically.
∴ The appropriate answer figure is (D).

5. The elements are shifted to adjacent blocks in clockwise direction and opposite blocks alternately and a new element is appearing in each figure in the place of bottom left element.
∴ The appropriate answer figure is (D).

6. The element is shifted by $\frac{1}{2}$ a side, 1 side, $1\frac{1}{2}$ sides, 2 sides in CW direction, respectively.
∴ The appropriate missing figure is (C).

7. The element is shifted by each time $\frac{1}{2}$ a side in CW, 1 side in ACW, $1\frac{1}{2}$ sides in CW and 2 sides in ACW directions, respectively.
∴ The appropriate missing figure is (B).

8. The element is shifted by each time $\frac{1}{2}$ a side, 1 side, $1\frac{1}{2}$ sides and 2 sides in CW direction and rotated by 45° in ACW, 90° in CW, 135° in ACW, 180° in CW, respectively.
∴ The appropriate missing figure is (D).

9. The element is rotated by 45°, 90°, 135° and 180° in CW direction, respectively.
∴ The appropriate missing figure is (A).

10. The number of sides of an element is increased each time by 1.
∴ The appropriate missing figure is (D).

11. In the first pair, from the second figure frame to the first figure frame, each element is increased by one side and shifts to the next position in clockwise direction. The similar pattern is followed in figure (A).

12. The elements in the first column are rotating by 90° and the elements in the second column are rotating by 180°. The top left and the bottom left elements are rotating in clockwise direction and the middle left element is rotating in anticlockwise direction.
Answer figure (D) is related to frame (II) of the problem figure in the same way.

13. In the first pair, from the second frame to the first frame the outer two figures are interchanged and the inner two figures are interchanged. Similar pattern is followed in figure (C).

14. The left hand side top element is interchanging with right side middle element. The right hand side top element is interchanging with the left hand side middle element. The bottom two elements are interchanged. Similar pattern is followed in figure (B).

15. The top left element and the bottom left element interchanged their positions and formed water images. Similar changes have taken place for the top right and the bottom right elements. The element at the centre rotated by 180°. This pattern can be established in the second pair by replacing the question mark with the answer figure (C).

16. In the question figure, the number of elements in second segment is half of the number of elements in the first segment. Figure (C) also posses the same relationship.

17. The water image of the element in the first segment of the question figure has been rotated by 45° in anticlockwise direction.
Figure (A) also posses the similar relationship.

18. In the question figure, the number of elements in the second segment of the figure is double the number of sides in the first segment of the figure.
Figure (D) also posses the similar relationship.

19. In the first segment of the question figure, the ratio of the number of shaded to that of unshaded elements is 1 : 4 and in the second segment it is 4 : 1. The ratio between the shaded to the unshaded is constant in the first segment and in the second segment. Figure (C) also posses the similar relationship.

20. Let us represent the figure as follows:

1	2
3	4

The number of elements in each segment will always remain the same and the elements are shifting in the second segment as follows:

4 → 3
3 → 2
1 → 4
2 → 1

Figure (A) posses the similar relationship.

21. In each figure, the left element is symmetric about the vertical axis and the second element is symmetric about the horizontal axis. Figure (D) does not follow this pattern.

22. Figure (B), (C) and (D) have the same figure in different rotated forms. Figure (A) cannot be obtained by rotating any of the other figures.

23. Two out of the four small elements are exactly at opposite places inside the circles except figure (D).

24. The 90° angle between the two arrows in the straight lines is in clockwise direction, but in (D) it is in anticlockwise direction.

25. All the elements are in opposite directions except in figure (D).

26. In each figure the second segment of the figure is the mirror image of the first figure, which has been rotated by 45° in clockwise directions.

Figure (B) does not follow such pattern.

27. Here, in each figure, except figure (1), the elements which are five in number are becoming four. Similarly, the elements which are 3, 2 and 1 in number are becoming 2, 3 and 6, respectively.

Figure (A) does not follow this pattern.

28. In each figure three lines are headed by three elements, out of three, one is shifting to the next line in anticlockwise direction and the other two are shifting to next line in clockwise direction. The figure (A) does not follow this pattern.

29. Each of the elements in the frame are shifting to new position without any rotation as shown below.

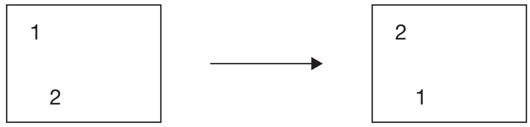


Figure (D) does not follow this pattern

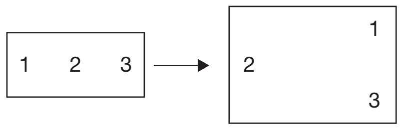
30. In each pair the mirror image of upper element is rotated by 90° in anticlockwise direction. The water image of lower element is rotated by 90° in clockwise direction. Fig. (C) does not follow the above pattern.

31. The outer element is rotated by 135° in an anticlockwise direction. The inner element is rotated by 135° in a clockwise direction. This is not true in figure (C).

32. One side is added to all the other elements except in the circle. This is not true in figure (A).

33. The number of elements connected to the main figure is equal to half of the total number of the number of sides of the main figure. This is not true in figure (D).

34. The elements are changing their position as follows:



Here, 1 is changing to its water image, 2 is changing to its mirror image and 3 is changing to its water image. This is not true in figure (C).

35. Every element is rotated by 180°. This is not followed in figure (B).

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Unit 5

Mock Tests

MOCK TEST 1

Directions for questions 1 to 4: Answer the questions on the basis of the information given below and select the correct alternative from the given choices.

Fifteen candidates from A to O were allotted seats in an examination hall. Some of the candidates are from the engineering stream and the others are from the medicine stream. There are 15 chairs in the examination hall, arranged uniformly in three columns and five rows.

The seats were allotted in such a way that no two students from the same stream are adjacent to each other either in a row or in a column. Further, the following information is also known about the students and their seating arrangements.

(i) Four students, L, M, N and O, not all of who were from the same stream, did not attend the exam and the seats allotted to them were left vacant.

(ii) In each row, at least one student from each stream attended the exam.

(iii) While the number of students who were allotted seats was more for one stream, the number of students who attended was more for the other stream.

(iv) A and B, who belong to the same stream are sitting in the same column and A is sitting in the second row of the second column.

(v) None of C, D and E are sitting in the corner chair of their respective rows and F, H and J are adjacent to C, D and E, not necessarily in the same order.

(vi) K and G are adjacent to A and I is to the left of B.

1. Who among the following belongs to a different stream than that of the others?

(A) K (B) G

(C) I (D) J

2. Which pair of candidates among the following pairs cannot sit in the same row?

(A) F and D (B) A and G

(C) J and I (D) B and I

3. If, in each column, there is at least one student from each of the streams, then what is the maximum possible number of students who attended in the right most column?

(A) 2 (B) 3

(C) 4 (D) 5

4. If L and K are from the same stream, then who among the following is sitting in the same row as the seat allotted to L is?

(A) B (B) A

(C) H (D) J

Directions for questions 5 to 8: A company has four departments, such as Finance, Marketing, Production and Administration. The following table provides information about the employees in the four departments. Select the correct alternative from the given choices.

	Finance		**Marketing**		**Production**		**Administration**	
Age (in years)	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**	**Male**	**Female**
Less than 30	18	11	22	5	15	7	7	14
30 to 50	15	8	17	3	18	2	5	8
Above 50	21	7	5	0	23	1	4	12
Total	54	26	44	8	56	10	16	34

5. What percentage of employees in the Production department are in the age group from 30 to 50 years?

(A) 25.2% (B) 30.3%
(C) 28.5% (D) 33.6%

6. What proportion of males in the Administration department are more than 50 years old?

(A) 33.3% (B) 21.2%
(C) 25% (D) 28.6%

7. What percentage of female employees in the organization are in the age group between 30-50 years?

(A) 5.2% (B) 7.18%
(C) 9.6% (D) None of the above

8. What is the difference (in percentage points) between the percentage of employees who are males and in the above 50 years age group, and the percentage of employees who are females and in the less than 30 years age group?

(A) 5.2 (B) 9.1
(C) 6.5 (D) 7.8

Directions for questions 9 to 12: Answer the questions on the basis of the information given below and select the correct alternative from the given choices.

A group of five people, namely Manish, Preetam, Akhil, Sunil and Varun visits at least one and at most two of the five cities, such as Bangalore, Chennai, Delhi, Kolkata and Mumbai. We have the following additional information.

(i) Each city is visited by at least one person and Kolkata is visited by three people.
(ii) Manish visits two cities and Bangalore is one of them.
(iii) The total number of visits (by the given five people) is 8.
(iv) Preetam visits more cities than Sunil, who visits Mumbai.
(v) Akhil visits Chennai but not Kolkata.
(vi) Varun visits the same cities that Manish visits.

9. Who visits Delhi?

(A) Akhil (B) Manish
(C) Preetam (D) Varun

10. Which is the city that is visited by Manish but not Preetam?

(A) Bangalore (B) Delhi
(C) Kolkata (D) Chennai

11. Who among the following visited Mumbai?

(A) Akhil (B) Manish
(C) Preetam (D) None of these

12. Who among the following did not visit any of the cities that Preetam visited?

(A) Manish (B) Varun
(C) Akhil (D) None of the above

Directions for questions 13 to 16: Answer the questions on the basis of the information given below and select the correct alternative from the given choices.

The following table gives details of the products manufactured by a company in 2016:

Product	Total Market (In crore)	Market Share of the Company (%)	Profitability of the Company (%)
A	680	12	14
B	1760	15	17
C	520	7	22
D	2350	13	8
E	980	18	11

The following table exhibits the projected increase in the total market size and the projected increase in the market share of the company for each of the given products for the year 2017 by subsequently comparing the corresponding values in 2016.

Product	Projected Increase in the Total Market Size (%)	Projected Increase in the Market Share of Company (percentage Points)
A	15	3
B	10	5
C	20	2
D	7	7
E	15	2

$$\text{Profitability } (\%) = \frac{\text{Profit}}{\text{Sales}} \times 100$$

13. If the company produces only the given five products, then what is the total sales (in crore) of the company in 2016?

(A) 864 (B) 852
(C) 817 (D) 788

14. The profit (in crore) made by the company in 2016 is approximately

(A) 99.8 (B) 102.5
(C) 108.2 (D) 112.7

15. If the profitability of the company for each of the five products in 2017 remains the same as that in the previous year, then the profit (in crore) made from the sales of the products in 2017 would be approximately

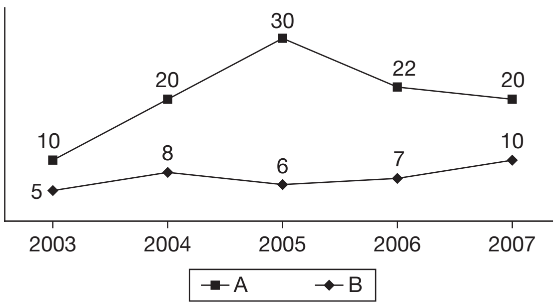
(A) 247 (B) 256
(C) 292 (D) None of the above

16. What is the approximate projected percentage increase in sales of the company in 2017 when compared to that in 2016?

(A) 82% (B) 56.7%
(C) 49.2% (D) 123%

Directions for questions 17 to 20: Answer the questions on the basis of the information given below and select the correct alternative from the given choices.

The following line graph depicts the percentage increase in the sales of two companies A and B, in four years from 2003 to 2007, when compared to their sales in the previous year:



17. In which year was the increase in sales for Company A the highest?

(A) 2004 (B) 2005
(C) 2006 (D) 2007

18. If the ratio of the sales of companies A and B in 2002 was 1 : 2, then in 2004, the sales of A forms approximately what percentage of the sales of B?

(A) 38% (B) 48%
(C) 58% (D) 68%

19. If the increase in sales of Company A was not more than that of Company B in any of the given years and it is known that the sales of Company A in 2003 were x% of that of Company B, then what is the maximum possible value of x?

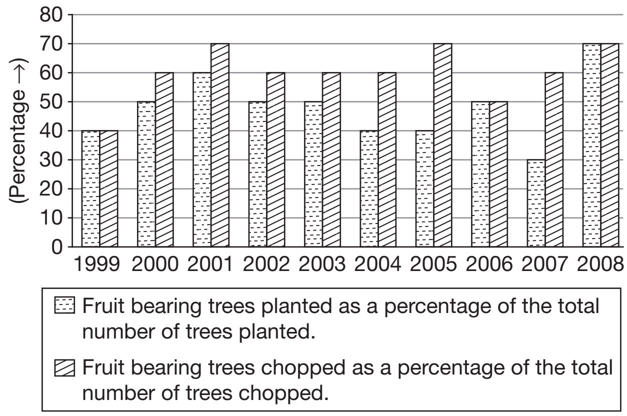
(A) 22 (B) 14
(C) 16 (D) 18

20. If the sales of Company B in 2002 was Rs 180 crore, then its sales in 2007 was approximately

(A) Rs 230 crore (B) Rs 242 crore
(C) Rs 255 crore (D) Rs 292 crore

Directions for questions 21 to 24: Answer the questions on the basis of the information given below and select the correct alternative from the given choices.

The bar graph show below provides information about the planting and chopping of fruit bearing trees for over a period of ten years.



The following changes occurred in the given time period:

- The total number of trees planted in a year increased over that of the previous year in every alternate year starting from 2000 and in the remaining years it decreased over that of the previous year.
- The total number of trees chopped in a year increased over that of the previous year for every year.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

21. If the total number of trees chopped increased by 10% every year from 1999 to 2003 and in 2001, the number of fruit bearing trees planted is equal to the number of fruit bearing trees chopped, then in at least how many years from 1999 to 2003 was the number of fruit bearing trees chopped less than the number of fruit bearing trees planted?

(A) 1 (B) 2
(C) 3 (D) 0

22. What is the minimum number of years in the given period in which the number of fruit bearing trees that are planted, increased over that of the previous year?__________

23. What is the maximum number of years from 2000 to 2008 in which the number of fruit bearing trees that are chopped can be equal to that of the previous year?________

24. The percentage change in the total number of trees planted with respect to the previous year is same as every year from 2000 to 2008, then the number of fruit bearing trees planted in the year 2002 can be equal to that of the year

(A) 2000 (B) 2004
(C) 2008 (D) Cannot be determined

Directions for questions 25 to 28: Answer the questions on the basis of the information given below and select the correct alternative from the given choices..

A group of six people A, B, C, D, E and F are compared with each other in terms of their heights. Further, no two people have the same height. The following information is also known:

(i) Neither F nor B is the shortest.
(ii) Neither A nor C is the tallest.
(iii) F is taller than not more than three people and B is taller than not less than three people.
(iv) A is taller than E, who is shorter than C, who is shorter than D, who is taller than B.
(v) F is taller than A but shorter than D.
(vi) C is taller than A but shorter than B.

25. Which of the following is definitely true?
(A) B is the second tallest.
(B) F is the third shortest.
(C) C is taller than F.
(D) A is the shortest.

26. Who among the following is ranked as the second shortest?
(A) A (B) E
(C) F (D) Cannot be determined

27. Which of the following can be the ascending order of their heights?
(A) DBFACE (B) DBCFEA
(C) DBFCAE (D) None of the above

28. If the number of people shorter than C is the same as the number of people taller than F, then who is the third tallest?
(A) D
(B) C
(C) F
(D) Cannot be determined

Directions for questions 29 to 32: Answer these questions on the basis of the information given below.

Ashok, Bhargav, Chetan, Dhruv, Eswar, Faizal and Govind were seven students in the same class. After when the marks for the class test in mathematics were announced, they gave the following replies as none of them was happy with their respective marks:

Ashok : Bhargav, Chetan and I, together scored 195.
Bhargav : Chetan, Dhruv and I, together scored 210.
Chetan : Dhruv, Eswar and I, together scored 196.
Dhruv : Eswar, Faizal and I, together scored 208.
Eswar : Faizal, Govind and I, together scored 213.
Govind : All seven of us together scored 491.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

29. How many marks were scored by Dhruv?______________

30. What are the marks scored by Govind?________________

31. If no student had a score less than 50, then who scored the highest marks?
(A) Bhargav
(B) Dhruv
(C) Govind
(D) Cannot be determined

32. If Eswar scored 65 marks, then how many marks did Bhargav score?________________

ANSWER KEYS

1. D	**6.** C	**11.** D	**16.** C	**21.** A	**25.** A	**29.** 83
2. C	**7.** D	**12.** C	**17.** D	**22.** 3	**26.** A	**30.** 88
3. B	**8.** C	**13.** A	**18.** C	**23.** 2	**27.** D	**31.** C
4. A	**9.** C	**14.** C	**19.** D	**24.** C	**28.** D	**32.** 79
5. B	**10.** A	**15.** D	**20.** C			

SOLUTIONS

Solutions for questions 1 to 4: As no candidates of the same stream are sitting adjacent to each other either in a row or in a column, so the candidates can be arranged in the following ways.

Case (i): Table (i)

E	M	E
M	E	M
E	M	E
M	E	M
E	M	E

Case (ii): Table (ii)

M	E	M
E	M	E
M	E	M
E	M	E
M	E	M

In the above arrangement, E denotes a candidate from the engineering stream and M denotes a candidate from the medical stream

Thus, there will be a maximum of 8 students and a minimum of 7 students from each of the stream who are allotted seats in the hall.

From condition (iii), the number of candidates who did not attend the exam is more for the stream in which more number of candidates are allotted to the hall.

From condition (i), there is at least one candidate from each of the stream who did not attend the exam. Thus, the number of candidates who did not attend is 3 for the stream in which 8 candidates are allotted to the hall and it is 1 for the other stream.

Thus, the number of candidates allotted, the number of people who did not attend the exam and the number of people who attended the exam can be tabulated as follows.

	Stream I	Stream II
Allotted	8	7
Did not attend	3	1
Attend	5	6

Using condition IV and V, we can determine that A, B, C, D and E are sitting in the middle column with A in the second row and B in the fourth and C, D and E in the remaining three rows in any order.

It is known that there is at least one candidate from each stream in each of the rows and as F, H, J belong to the same stream as B (as they are adjacent to C, D, E) and as K and G are adjacent to A so. I must be in the same row as B.

Based on the condition, the candidates are allotted seats as shown in the following table.

F/H/J/-	C/D/E	F/H/J/ -
k/4	A	k/4
F/H/J/-	C/D/E	F/H/J/-
I	B	–
F/H/J/-	C/D/E	F/H/J/-

Here, '–' indicates that the seat is / can be left vacant. From the above diagram it can be concluded that A, B, F, H and J belong to one stream whereas C, D, E, K, G and I belongs to the other stream.

1. K, G, I belong to one stream and J belongs to the other stream.
2. From Table (iii), we can see that only B and I are sitting in the fourth row and the third seat is vacant.
3. As K, G, and I belong to the same stream at least one among F, H, J should be in the left most column. The maximum possible number of students in the right most column is two among F, H, J and one among K, G, i.e., a total of 3 candidates.
4. Since L and K belong to the same stream, the seat allotted to L must be the seat to the right of B. From the given options, B is the only person who is sitting in the same row as the seat is allotted to L.

Solutions for questions 5 to 8:

5. Total employees in the Production department = 66
   Number of employees in the age group from 30 to 50 years = 20
   $\therefore$ Required value = $\frac{20}{65} \times 100 = 30.3\%$
6. Number of males in the Administration department = 16.
   Number of males of age greater than 50 years = 4.
   Required value = $\frac{4}{16} \times 100 = 25\%$.
7. Total number of employees in the organization = 248
   Number of females in the age group from 30 to 50 years = 21
   Required value = $\frac{21}{248} \times 100 = 8.46\%$

8. Number of males in the above 50 years age group as a percentage of total employees $= \frac{53}{248} \times 100 = 21.4\%$

Number of females less than 30 years age group as a percentage of total employees $= \frac{37}{248} = 14.9\%$

The required difference = 6.5 percentage points.

Solutions for questions 9 to 12: Given, Preetam visited more cities than Sunil.

- Preetam visited two cities and Sunil visited one city i.e., Mumbai. Also, given that Akhil did not visit Kolkata (but visited Chennai).
- Manish, Preetam and Varun visited Kolkata.

From the remaining three visits (among the 8 visits) it is given that Manish visited two cities and Bangalore is one of them. Also, Varun visited the same cities as Manish.

As Delhi must be visited by at least one, Preetam visited Delhi.

Akhil	–	Chennai
Manish	–	Bangalore, Kolkata
Preetam	–	Delhi, Kolkata
Sunil	–	Mumbai
Varun	–	Bangalore, Kolkata

9. Preetam visits Delhi.

10. Bangalore is the required city.

11. Only Sunil visited Mumbai.

12. Akhil did not visit any city that Preetam visited.

Solutions for questions 13 to 16:

13. The total sales of the company in 2016

$$= 680 \times \frac{12}{100} + 1760 \times \frac{15}{100} + 520 \times \frac{7}{100} + 2350 \times \frac{13}{100} + 980 \times \frac{18}{100}$$

$= 81.6 + 264 + 36.4 + 305.50 + 176.4 = 863.9$ crore

14. The profit made by the company is

$$= 81.6 \times \frac{14}{100} + 264 \times \frac{17}{100} + 36.4 \times \frac{22}{100} + 305.5 \times \frac{8}{100} + 17.4 \times \frac{11}{100}$$

$= 11.42 + 44.88 + 8.0 + 24.44 + 19.4$

$= 108.2$ crore (approximately).

15. The total market for all the products, the market share of the company, the sales of the company and the profit it makes in each product is given below.

Product	Market size	Market share of company (%)	Market share (in crore)	Profit (in crore)
A	782	15	117.3	16.42
B	1936	20	387.2	65.82
C	624	9	56.16	12.36
D	2514.50	20	503.0	40.24
E	1127	20	225.4	24.80

$\therefore$ The profit would be approximately = 159.6 crore.

16. The projected sales of the company in 2017 = 1289

The sales in 2016 = 864 crore

$\therefore$ Projected increase $= \frac{425}{864} \times 100 = 49.2\%$

Solutions for questions 17 to 20:

17. If the sales of Company A in 2004 is 100, then the sales in 2005 is 130, in 2006 it is 158.6 and in 2007 it is 190.32. The increase in the different years are given below.

Year	Increase
2005	30
2006	28.6
2007	31.72

$\therefore$ The increase is the highest in 2007.

18. If the sales of Company A was 100 in 2002, its sales in 2004 would be $100 \times 1.1 \times 1.2 = 132$.

The sales of Company D in 2002 would be 200 and the sales in 2004 would be $200 \times 1.05 \times 1.08 = 227$.

Required value $= \frac{132}{227} \times 100 = 58\%$.

19. Assume that the initial sales of A and B in 2002 were $1000k_1$ and $1000k_2$, respectively. The sales and the increase in sales in the subsequent years can be tabulated as shown below for both the companies A and B.

	A			B		
Year	Sales	Growth Rate	Increase in Sales	Sales	Growth Rate	Increase in Sales
2002	$1000k_1$	–	–	$1000k_2$	–	–
2003	$1100k_1$	10%	$100k_1$	$1050k_2$	5%	$50k_2$
2004	$1320k_1$	20%	$220k_1$	$1134k_2$	8%	$84k_2$
2005	$1716k_1$	30%	$396k_1$	$1202k_2$	6%	$68k_2$
2006	$2093k_1$	22%	$377k_1$	$1286k_2$	7%	$84k_2$
2007	$2512k_1$	20%	$419k_1$	$1415k_2$	10%	$129k_2$

Since A never had a higher increase in sales than B in any of the years, the following can be calculated.

$50k_2 \geq 100k_1$, $68k_2 \geq 396k_1$, $129k_2 \geq 419k_1$
$84k_2 \geq 220k_1$, $84k_2 \geq 377k_1$,

$$68k_2 > 396k_1 \Rightarrow \frac{k_2}{k_1} \geq \frac{396}{68} = 5.82$$

This takes care of all the other inequalities above. The ratio of sales of A to that of B in 2003 will be at most

$$= \frac{1100k_1}{1050k_2} \leq \frac{1100}{1050} \times \frac{1}{5.82} = 0.18 = 18\%$$

$\therefore$ Maximum value of x is 18.

20. Sales of B in 2005 would be $= 180 \times 1.05 \times 1.08 \times 1.06 \times 1.07 \times 1.1 = 255$

Solutions for questions 21 to 24:

21. Let the total number of trees planted in 2001 be x and the number of trees chopped in 1999 be 100.
$\therefore$The total number of fruit bearing trees chopped in 1999, 2000, 2001, 2002 and 2003 will be 100(0.4), 110(0.6), 121(0.7), 133.1(0.6) and 146.41(0.6), i.e., 40, 66, 84.7, 79.86 and 87.846, respectively.

$$x = \frac{84.7}{0.6} = 141.1$$

$\therefore$ The total number of trees planted in 2000 will be greater than 141.
$\Rightarrow$ The number of fruit bearing trees chopped in 2000 will be greater than 70.
As the number of trees planted in 1999 can be as low as 10, in this year the number of fruit bearing trees chopped can be more than the number of fruit bearing trees planted.
Similarly, if the number of trees planted in 2002 and 2003 are 150 and 10, respectively, then the number of fruit bearing trees chopped will be greater than the number of trees planted.
$\therefore$ In only 2000, the number of fruit bearing trees chopped was definitely less than the number of fruit bearing trees planted.

22. The years in which the percentage of fruit bearing trees and the total number of trees planted definitely increased, therefore, it would be the minimum number of years in which the number of fruit bearing trees increased.
This can be observed only in the year 2000, 2006 and 2008.

23. We cannot definitely say that the number of fruit bearing trees that are chopped are equal to that of the previous year but for years 2002 and 2005 as the total number of trees chopped has been increased in the next year, the percentage of trees chopped has been reduced in the next year. Hence, there is a possibility that they could be equal. Hence, the maximum number of years from 2000 to 2008 in which the number of fruit bearing trees that are chopped can be equal to that of the previous year and they are two.

24. When there is an equal percentage change, it increases and decreases alternately, starting with an increase in the final quantity which is always less than the initial quantity.
Let the number of trees planted in the year 2002 be x.
Let the number of fruit bearing trees planted in the years 2000, 2004 and 2008 be p, q and r.
$p > q > r$ and
$\therefore p > x > q > r$
Considering the given choices, we derive the following.
(A) $0.5p > 0.5x$
(B) $0.4q < 0.5x$
(D) $0.7s$ can be equal to $0.5x$.

Solutions for questions 25 to 27: The following information is available.

(i) F, B $\neq$ Shortest
(ii) A, C $\neq$ Tallest
(iii) F $\neq$ 1st or 2nd ($\Rightarrow$ F = 3, 4, or 5) B $\neq$ 4th, 5th or 6th ($\Rightarrow$ B = 1, 2, or 3)
(iv) A > E ; D > C > E ; D > B.
(v) D > F > A
(vi) B > C > A

Based on the above information, we get the following.

(A) From (iv) and (v), we conclude that D is the tallest.

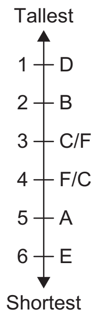
(B) From (i), (iii) and (iv), we conclude that E is the shortest.

(C) From (iv), (v) and (vi), we conclude that A is the 5th tallest or the second shortest.

(D) From (vi), B is taller than C, A and E (E is the shortest), but shorter than D. Hence, B must be the second tallest.

(e) No information available about comparing the heights of F and C. Hence, F, C = 3 and 4, in any order.

Hence, we get the following arrangement:



Based on the above data, we solve the given questions.

25. From conclusion (2), B is the second tallest.

26. From conclusion (c), A is ranked the second shortest.

27. As it can be observed, DBFCAE is the descending order of their heights and not the ascending order.

28. Even with the given information, the answer cannot be determined.

Solutions for questions 29 to 32: Let the marks of Ashok, Bhargav, Chetan, Dhruv, Eswar, Faizal and Govind be A, B, C, D, E, F and G, respectively.

$$A + B + C = 195 \rightarrow (1)$$
$$B + C + D = 210 \rightarrow (2)$$
$$C + D + E = 196 \rightarrow (3)$$
$$D + E + F = 208 \rightarrow (4)$$
$$E + F + G = 213 \rightarrow (5)$$
$$A + B + C + D + E + F + G = 491 \rightarrow (6)$$
$$(6) - ((1) + (4)) = 491 - (195 + 208) = 403$$
$$\Rightarrow G = 491 - 403 = 88$$
$$(6) - ((2) + (5)) = 491 - (210 + 213)$$
$$\Rightarrow A = 68$$
$$(2) - (1) \Rightarrow D - A = 15 \text{ or } D = 83$$

29. $D = 83$

30. $G = 88$

31. From (1), we get $B + C = 127$ and from (3), we get $C + E = 113$ and from (5), we get $E + F = 125$. Since no student scored less than 50, Govind should have scored the highest. Choice (C)

32. Given, $E = 65 \Rightarrow C = 196 - 65 - 83 = 48$.
$\therefore B = 210 - 48 - 83 = 79$

MOCK TEST 2

Directions for questions 1 to 4: Answer the questions based on the information given below and select the correct alternative from the given choices.

The following table provides some demographic details of the eight states from A through H in a particular year.

Name of the State	Total Population (in million)	Majors	Minor Males	Minor Females	Minors (in million)
		As a Percentage of Total Population			
A	29.7	61.6	20.3	18.1	11.40
B	45.7	67.2	17.0	15.8	15
C	21.3	70.1	15.3	14.6	6.36
D	30.2	56.6	23.8	19.6	13.1
E	26.1	64.1	16.7	19.2	9.37
F	10.3	55.7	24.3	20	4.56
G	11.7	56.2	23.2	20.6	5.12
H	38.5	67.2	15.4	17.4	12.63

1. Minors form approximately what percentage of the population of all the eight states put together?

(A) 28% (B) 36%
(C) 40% (D) 44%

2. Approximately what percentage of the total population of states A, D and H are majors?

(A) 57.2% (B) 59.1%
(C) 62.3% (D) 66.2%

3. What is the approximate number (in million) of minor females in states A, B and C together?

(A) 15.7 (B) 14.2
(C) 20.1 (D) 18.4

4. How many of the eight states have more than 6 million minor males?

(A) 3 (B) 4
(C) 2 (D) 5

Directions for questions 5 to 8: Answer the questions based on the information given below and select the correct alternative from the given choices.

A group of six friends P, Q, R, S, T and U attended a party. There are two housewives, one doctor, one editor, one manager and one engineer in the group.

There are two married couples in the group. The engineer is married to S, a housewife. T is not a housewife. No woman in the group is either an editor or a doctor. One of the two housewives is married to P. U, the manager is married to R, the doctor.

5. T is a/an

(A) Manager (B) Editor
(C) Doctor (D) Engineer

6. Who among the following is a married couple?

(A) P and S (B) Q and T
(C) S and T (D) P and Q

7. How many people in the group are males?
(A) 2
(B) 3
(C) 4
(D) Cannot be determined

8. Who among the following is married to Q?
(A) P
(B) S
(C) T
(D) None of the above

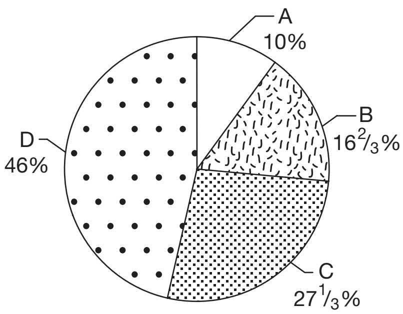
Directions for questions 9 to 12: Answer the questions based on the information given below and select the correct alternative from the given choices.

The citizens of Kya Kya island had to pay income tax on their income and for any individual, the rate of tax applicable increased with an increase in the income of the individual. Rupee was the currency of Kya Kya island and only the citizens who earned ₹12000 per year or less did not have to pay any income tax. For any individual who earned more than ₹ 12000 per year, the incident tax was calculated based on three income slabs, such as Slab 1, Slab 2 and Slab 3 as shown in the table given below.

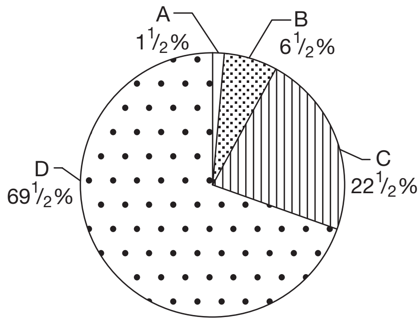
Income Slab	Range of Income (I) (in Rs.)	Total Tax Incident
Slab 1	$12,000 < I \le P$	10% of income above ₹12000
Slab 2	$P < I \le Q$	10% of (P – 12000) + 20% of income above P
Slab 3	$Q < I$	10% of (P – 12000) + 20% of (Q – P) + 40% of income above Q

The two pie-charts given pertains to the income and the tax paid by a family of four members A, B, C and D.

The first pie-chart shows the share of income of each member of the family, as a percentage of the total income of all the four put together and the second pie-chart shows the share of the tax paid by each member of the family as a percentage of the total tax paid by all the four put together.



Share of income



Share of tax paid

9. The value of P is
(A) ₹ 20,000 (B) ₹ 25,000
(C) ₹ 30,000 (D) Cannot be determined

10. What was the income of D?
(A) ₹ 46,000 (B) ₹ 69,000
(C) ₹ 92,000 (D) None of the above

11. What was the amount of tax paid by C?
(A) ₹ 9000 (B) ₹ 6750
(C) ₹ 4500 (D) ₹ 2250

12. Which of the following is the value of Q?
(A) ₹ 30,000 (B) ₹ 45,000
(C) ₹ 60,000 (D) None of the above

Directions for questions 13 to 16: Answer the questions on the basis of the information given below and select the correct alternative from the given choices.

A group of eight students A, B, C, D, E, F, G and H wrote an exam conducted by an Assistant Professor named Asim. Asim ranked the students in the decreasing order of their marks in the exam, such that two or more students who get the same mark will get the same rank (say N) and the student who has the next highest mark will get the next rank (i.e., N + 1) and so on.

Later, Asim submitted the same to his superior, Professor Ganesh. The Professor was not satisfied and asked Asim to give revised ranks from 1 to 8, to the eight students in the decreasing order of their marks. If two or more students get the same mark and compete for the same rank (say N), then the student with the least number of attempts among them should be ranked the best (i.e., N), the one with the next least number of attempts, the next best (i.e., N + 1) and so on. When Asim ranked the students accordingly, that the following information was observed.

(i) Of the students whose ranks were changed, the change in the rank of B was less than that of any other student and the change in the rank of G was more than that of any other student.
(ii) Only D's rank changed by 3.

(iii) The number of questions attempted by C was less than that of A, but his revised rank was not better than that of A, which, in turn was 6.
(iv) Initially, D and E had the same rank.
(v) Initially, there was no rank which was given to more than three students.
(vi) No two students had the same number of attempts.

NOTE

Between any two ranks, the numerically lower rank is considered to be the better rank.

13. What is the revised rank of D?
(A) 4 (B) 5
(C) 7 (D) Cannot be determined

14. Who are the students who initially got the 1st rank?
(A) H and F (B) B, D and E
(C) B, H and F (D) D, E and H

15. What is the change in the rank of C?
(A) 2 (B) 4
(C) 5 (D) Cannot be determined

16. Who got the revised rank of 1?
(A) B (B) F
(C) E (D) Cannot be determined

Directions for questions 17 to 20: Answer the questions based on the information given below and select the correct alternative from the given choices.

The following table gives the ranks of a few Indian companies based on Sales, Profit After Tax (PAT), Net Fixed Assets (NFA) and Market Capitalization (M. Cap) (all values in crore) for the year 2004.

Company	Sales (in crore)	Sales rank	PAT (in crore)	PAT rank	NFA (in crore)	NFA rank	M. Cap (in crore)	M. Cap rank
IOC	116775.50	1	7004.8	2	27333.2	4	52759	3
ONGC	32078.40	5	8664.4	1	30853.4	2	114089	1
Reliance	51801.50	2	5160.1	4	35146.0	1	75348	2
NTPC	19484.30	7	5281.30	3	28956.7	3	42877	7
HPCL	51517.60	3	1904.0	6	7074.3	11	11272.5	21
BPCL	47984.10	4	1694.6	10	7453.5	9	10884	23
TCS	7122.70	21	1612.40	11	674.7	76	51666	4
INFOSYS	4760.9	30	1243.50	13	970.3	56	45917	5
WIPRO	5132.70	27	914.90	16	794.7	69	44540	6
SAIL	21528.40	6	2512.10	5	13536.1	5	19949	12

17. If profit percentage is defined as $\frac{\text{PAT}}{\text{Sales}}$, which of the following companies had the highest profit percentage?
(A) NTPC (B) TCS
(C) INFOSYS (D) SAIL

18. How many of the companies given have a rank which is better (i.e., numerically smaller) than at least three other companies given in each of the four categories?
(A) 2 (B) 3
(C) 4 (D) 5

19. Which of the following companies has the highest ratio of M. Cap to sales?
(A) WIPRO (B) TCS
(C) NTPC (D) INFOSYS

20. If VFM of a company is defined as $\frac{\text{NFA}}{\text{M.Cap}}$, then which of the following is the correct order of the VFMs of the four companies, such as SAIL, BPCL, HPCL and NTPC?
(A) SAIL > BPCL > HPCL > NTPC
(B) SAIL > BPCL > NTPC > HPCL
(C) BPCL > SAIL > NTPC > HPCL
(D) SAIL > NTPC > BPCL > HPCL

Directions for questions 21 to 24: Answer the questions on the basis of the information given below.

Two robots Yasuku and Samurai guard Mr. Tigochenko's house. They are programmed in such a way that the movement of the puppet robot, Yasuku depends on the movement of the master robot, Samurai. Whenever the master robot takes a turn, the puppet robot will also take a similar turn, i.e., if the Samurai turns to the right by 45°, at the same time Yasuku also turns to the right by 45°. The speed of Yasuku is twice the speed of Samurai. Every night, both the robots will start moving from the same point and at the same time but in two opposite directions, i.e., if one goes to north, the other goes to south and they will never stop moving until the next day morning.

Directions for question 21: Select the correct alternative from the given choices.

21. One day Samurai started travelling and moved 48 metre west, then 14 metre south. How far is Yasuku from their starting point?
(A) 100 m, south-west
(B) 100 m, north-west
(C) 50 m, south-west
(D) 100 m, north-east

Directions for questions 22 to 24: Type in your answer in the space provided below the question.

22. One day Samurai started travelling and moved 100 metre towards south, 75 metre towards east and 200 metre towards north. What is the distance (in m) between the two robots?____________

23. The distance between the two robots is 30 metre and Yasuku is to the left of Samurai, who is facing north. If Samurai walks straight 40 metre and then 65 metre to his right, then the distance (in metre) between Yasuku and the starting point is ____________

24. One day Samurai started travelling and moved 54 metre towards east and 72 metre towards south. What is the minimum distance Yasuku must travel to meet Samurai? ____________

Directions for questions 25 to 28: Answer the questions on the basis of the information given below.

The following table gives the production capacity of the five refineries, namely P, Q, R, S and T of an oil company, SPCL, the demand at the company's four outlets, such as A, B, C and D and the transportation costs involved in transporting the oil from the different refineries to the outlets.

Transportation Cost (in Rupees/kilolitre)				
Refinery (production Capacity in Kilolitres/day)	**Outlet (Demand in kilolitres/day)**			
	A (40)	**B (30)**	**C (10)**	**D (20)**
P (30)	500	300	600	200
Q (10)	600	400	400	100
R (30)	300	400	700	400
S (20)	200	300	400	200
T (10)	400	300	600	100

In the above table, the number given in the brackets alongside each refinery gives the production capacity (in kilolitres per day) of that refinery and the number given in the brackets alongside each outlet gives the demand (in kilolitres per day) at that outlet. The number given in the cell corresponding to a refinery and an outlet gives the transportation cost (in rupees per kilolitre) incurred for transporting oil from that refinery to that outlet.

For example, the production capacity of refinery P is 30 kilolitres/day and the demand at outlet A is 40 kilolitres/day and the cost of transporting one kilolitre of oil from refinery P to outlet A is 500.

Assume that the company currently operates only the refineries and outlets mentioned above and the production at the refineries on any day is transported to the outlets such that the demand at all the outlets is met.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

25. The daily cost (in ₹) of transportation of oil to all the outlets is at least.___________

26. If due to a disruption in the supply lines to some of the outlets, refinery P can supply oil to only outlets A and C, then the daily cost (in ₹) of transportation of oil to all the outlets is at least. ___________

27. Consider that the daily cost of transportation of oil to all the outlets is the least possible. The daily cost of transportation to all the outlets would increase by the maximum amount, if there is a disruption (blockage) in the supply line connecting
(A) P and B (B) S and D
(C) Q and C (D) T and B

28. Consider that the daily cost of transportation of oil to all the outlets is the least possible. If the company plans to decrease the daily cost of transportation of oil to all the outlets by doubling the production capacity of exactly one of the refineries, then which refinery should it select so as to reduce the transportation cost by the maximum amount?
(A) S (B) Q
(C) R (D) P

Directions for questions 29 to 32: Answer the questions on the basis of the information given below.

As a part of selecting his team for the Indian Premier League, Shah Rukh Khan had shortlisted ten players, they are Ishant, Ponting, Akthar, Sourav, Warne, Lee, Dhoni, Symonds, Harbhajan and Sreesanth who were to be auctioned in the first stage. His idea was to select only players from among the shortlisted ten, adhering to all the following conditions.

(i) If the team includes one among Symonds, Sreesanth and Harbhajan, it must also include the other two.

(ii) The team must include either Akthar or Lee, but not both.
(iii) If the team includes Ishant, then it must also include Ponting and vice versa.
(iv) Ponting and Harbhajan cannot be selected together.
(v) The team must include exactly one among Warne, Dhoni and Symonds.
(vi) Ponting and Sourav cannot be selected together.

Directions: For the multiple choice questions, select the correct alternative from the given choices. For the non-multiple choice questions, write your answer in the space provided.

29. If exactly five of the ten players were selected, then who among the following must be selected?

(A) Ponting (B) Dhoni
(C) Warne (D) Sourav

30. If it is known that Ishant was one of the players selected, then how many of the other nine players were selected?

____

31. If only three of the ten players were selected, then who among the following is definitely not selected?

(A) Warne (B) Sourav
(C) Ponting (D) Lee

32. Among the ten players, at most how many can be selected together? __________

ANSWER KEYS

1. B	**7.** B	**13.** B	**19.** D	**25.** 26000	**30.** 3	**31.** B
2. C	**8.** D	**14.** C	**20.** C	**26.** 34000	**31.** C	**32.** 5
3. A	**9.** B	**15.** B	**21.** D	**27.** A	**32.** 5	**33.** C
4. A	**10.** B	**16.** D	**22.** 375	**28.** A	**29.** B	**34.** B
5. B	**11.** C	**17.** A	**23.** 170	**29.** D	**30.** B	
6. A	**12.** D	**18.** C	**24.** 180			

SOLUTIONS

1. Total population of minors of all the states put together (in million)

$= 11.4 + 15 + 6.36 + 13.1 + 9.37 + 4.56 + 5.12 + 12.63$
$= 77.54$

Total population of all the states put together (in million)

$= 29.7 + 45.7 + 21.3 + 30.2 + 26.1 + 10.3 + 11.7 + 38.5$
$= 213.5$

$$\therefore \text{Required } \% = \frac{77.54}{213.5} \times 100 \approx 36\%.$$

2. Total population of majors in states A, D and H (in million)

$$= 29.7 \times \frac{61.6}{100} + 30.2 \times \frac{56.6}{100} + 38.5 \times \frac{67.2}{100}$$

$= 18.3 + 17.1 + 25.87 = 61.27$

$$\therefore \text{Required}\% = \frac{61.27}{(29.7 + 30.2 + 38.5)} \times 100$$

$$= \frac{61.27}{98.4} \times 100 \approx 62.26\%$$

Alternative Solution:

As the population of A and D are nearly the same, the percentage in these two states put together is about 59 and if state H is also considered, then the percentage should be slightly above that as the value in state H is 67.2.

3. Total population of female minors (in million)

$$= 29.7 \times \frac{18.1}{100} + 45.7 \times \frac{15.8}{100} + 21.3 \times \frac{14.6}{100}$$

$\cong 5.4 + 7.2 + 3.1 = 15.7$

4. Number of minor males in A = $29.7 \times 0.203 = 6.0291$ million
Number of minor males in B = $45.7 \times 0.17 > 6$ million
Number of minor males in C = $21.3 \times 0.153 < 6$ million
Number of minor males in D = $30.2 \times 0.238 > 6$ million
Number of minor males in E = $26.1 \times 0.167 < 6$ million
Number of minor males in F and G are less than 6 million
Number of minor males in H = $38.5 \times 0.154 = 5.929 < 6$ million
Hence, in three states A, B and D, the given condition is satisfied.

Solutions for questions 5 to 8:

Given that:

Engineer (Male)	Housewife (S)
Manager (U)	Doctor (R)

As neither the editor nor the doctor is a female, in the second couple U is the female and R is the male.

As P is married to a housewife.

⇒ P must be the engineer and S is his wife.

Engineer ⇔ Housewife
(P) (S)

As T is not a housewife, he is the editor and Q is the housewife.

5. T is the editor.

6. P and S is a married couple.

7. There are three males.

8. Q is a housewife but is not married to any person among the six people in the group.
The answer is not given in the options.

Solutions for questions 9 to 12: The share of income and the share of taxes paid is as follows:

Share of income	Share of taxes paid
A – 10%	A – 1.5%
B – 16.67%	B – 6.5%
C – 27.33%	C – 22.5%
D – 46%	D – 69.5%

It is given that the tax rates are 0%, 10%, 20% and 40% depending upon the income. For the first 10% of the income, the tax paid is 1.5% of the total tax paid, for the next $6^2/_3$% ($16^2/_3 - 10$), it is 5% (6.5 – 1.5) and so on which is as given.

Let the total income be X and the total tax paid be Y.

Income (X)	Taxes paid (Y)
First 10% (for A) of X	1.5% of Y
Next $6^2/_3$ (for B) of X	5% of Y
Next $10^2/_3$ (for C) of X	16% of Y
Next $18^2/_3$ (for D) of X	47% of Y

Now as for the income up to ₹12,000 there is no tax and the income of A is more than ₹12,000. Some part of his income which exceeds ₹12000 is taxed at 10% and the remaining part at 20% or whole income which exceeds ₹12,000 is taxed at 10%.

Now from the last row of the table, $18^2/_3$% of X = 47% of Y and so at that rate of tax, X is approximately equal to 2.5 Y.

In the previous row, $10^2/_3$% of X = 16% of Y and so X = 1.5Y. From this we can say that all of this $10^2/_3$ is not taxed at the same rate as the last $18^2/_3$. From the second row, we get $6^2/_3$% of X = 5% of Y or X = 0.75Y which is exactly half of that in the previous case.

∴ The $10^2/_3$% was taxed at double the rate at which the $6^2/_3$% was taxed. As some part of the final $18^2/_3$% was taxed at a higher rate than what the $10^2/_3$% was taxed, the only possibility is that, some part of $18^2/_3$% was taxed at 40%, the whole of $10^2/_3$% was taxed at 20% and $6^2/_3$% was taxed at 10%. The maximum limit of Slab 1, i.e., P is the same as the income of B as the entire part of C's income which exceed B's income is taxed at 20%.

Now as all the income of B which exceed that of A, i.e., $6^2/_3$% of the total income is taxed at 10%, at that slab (i.e., 10%)

$$6^2/_3X = 5Y$$
$$\text{Or } X = 0.75Y$$

As A paid a tax of 1.5Y, his taxable income is 2X, or his income is Rs. 12,000 + 2X and the income of B (which is equal to P) is $12{,}000 + 2X + 6^2/_3X = 12{,}000 + 8^2/_3X$

$$\text{Now } 12{,}000 + 8^2/_3X = \frac{16\,^2/_3}{10}(12{,}000 + 2X)$$

$12{,}000 + 8^2/_3X = 20{,}000 + 3^1/_3X$

∴ 5.33X = 8000

Or X = 1500

∴ Income of A = 12000 + 3000 = ₹15,000 and the income of B, C and D are respectively ₹25,000, ₹41,000 and ₹69,000.

As the tax rate up to 25,000 is 10% (as P is the same as income of B, as found earlier), A would have paid a tax of ₹300 which is 1.5% of the total tax paid.

∴ The taxes paid by others are B – ₹1300, C – ₹4500 and D – ₹13,900.

As income of C is ₹41,000 and all of his income is taxed at 20%, the rate of 40% is definitely applicable only above ₹41,000.

The tax up to an income of ₹41,000 is ₹4,500 and for an income of ₹69,000 it is ₹13,900. At some amount Q, between 41,000 and 69,000, the tax rate doubles from 20% to 40%.

∴ $(Q - 41{,}000) \times 0.2 + (69{,}000 - Q) \times 0.4 = 13{,}900 - 4500$

∴ $0.2Q - 8200 + 27600 - 0.4\,Q = 9400$

$0.2Q = 10000$

$Q = 50{,}000$

9. The value of P is ₹25,000.

10. The income of D is ₹69,000.

11. The amount of tax paid by C is ₹4500.

12. The value of Q is ₹50,000.

Solutions for questions 13 to 16: Given that:

The change in G's rank is more than that of any other person. The person with maximum change should get a revised rank of 8.

∴ G's revised rank is 8.

Given that A got sixth rank.

The revised rank of C was not better than that of A.

∴ C must have got seventh rank.

Let us look at a particular case (Hypothetical) to understand the changing of ranks.

Rank	Students	
N	P	Q
N + 1	R	S
N + 2	T	U

One of the people among P and Q does not have any changes in the rank. Further, the change in the rank of the lesser ranked among P and Q = The change in the rank of higher ranked among R and S.

B's rank changed by the least value. Hence, it must have changed by one.

∴ There must be two other people who got the same rank as B. (∵ If there is only one person, then B cannot be the only person whose rank changed by 1)

Initial rank 1:	?	B	?
	Least attempts	Second least attempts	Most attempts

Similar is the case for G and there must be one more person who got the same rank as G.

G	

Further, C and G initially must have got the same rank. A must have got the next best rank to that of C and G.

Now, D's rank changed by 3.

The only possibility is that D must have got rank 2 initially and 5 in the final rankings. (As changing from 1 to 4 is ruled out)

E and A also must have got the same rank.

Further, B's rank changed by the least. Therefore, he must have got 1 initially and 2 in the final rankings. H and F are the remaining two people who got the first rank initially.

Similarly, we can sort out the ranks of all other people and the final table obtained will be as follows:

Initial ranks:

1	H/F	B	F/H
2	E	D	A
3	C	G	

Revised ranks:

1	F/H
2	B
3	H/F
4	E
5	D
6	A
7	C
8	G

13. Revised rank of D is 5.

14. B, H and F got 1st rank initially.

15. C's rank changed by four.

16. Either F or H would have got a revised rank of 1.

17. Only for NTPC, ONGC and Infosys PAT is more than 25% of sales.

For Infosys, profit percentage = $\frac{1243.5}{4760.9} \times 100$

$= 26.12\%$

For NTPC, profit percentage = $\frac{5281.3}{19484.3} \times 100$

$= 27.1\%$

18. Only four companies, i.e., IOC, ONGC, Reliance and NTPC satisfy the given condition.

19. By observation, the ratio of M. Cap to sales is the highest for Infosys.

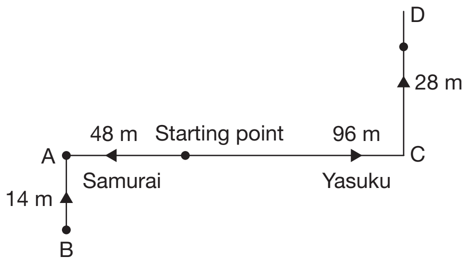
20. For BPCL it is $\frac{7453.5}{10884} = 68.5\%$

For SAIL it is $\frac{13536.1}{19949} = 67.85\%$

For HPCL it is $\frac{7074.3}{11272.5} = 62.75\%$

And for NTPC it is $\frac{28956.7}{42877} = 67.53\%$

21.



Now OD = $\sqrt{OC^2+CD^2} = \sqrt{96^2+28^2}$

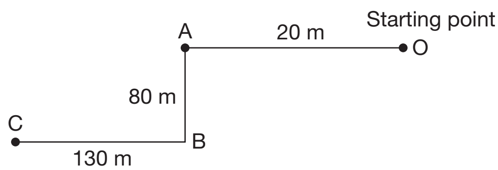
= 100 m towards north-east.

22. From starting point:
Horizontal distance of Samurai = 75 m
Vertical distance of Samurai = (200 – 100) = 100 m
The shortest distance of Samurai

$$= \sqrt{75^2+100^2} = 125 = 125$$

Now, as Yasuku moves with twice the speed of Samurai. The distance between Yasuku and Samurai is 125 × 3 = 375 m.

23. The distance between the robots is 30 m, i.e., Yasuku is 20 m away from the starting point. Now the path traced by Yasuku is as follows.



Now OC = $\sqrt{(CB+OA)^2+AB^2}$ = 170 m.

24. Since Samurai travelled 54 m east and 72 m south, Yasuku would have travelled 108 m west and 144 m north. Further, it is not possible for Yasuku and Samurai to meet at any point other than the starting point.
Hence, the minimum distance that Yasuku must travel will be along the line connecting the starting point to yasuku's location.
Minimum distance that Yasuku needs to travel

$$= \sqrt{108^2+144^2} = 180 \text{ m.}$$

25. The lowest cost of transportation occurs when the oil is transported along the cheapest routes available, which are shown below.
T – D → 10 kl → 10 × 100 = ₹1000
Q – D → 10 kl → 10 × 100 = ₹1000
P – B → 30 kl → 30 × 300 = ₹9000
S – A → 10 kl → 10 × 200 = ₹2000
R – A → 30 kl → 30 × 300 = ₹9000
S – C → 10 kl → 10 × 400 = ₹4000
Total = ₹26,000.

26. As P can be used for supplying only to A or C, the minimum cost of transportation in this case would be
P – C → 10 kl → ₹6000
P – A → 20 kl → ₹10,000
Q – D → 10 kl → ₹1000
T – D → 10 kl → ₹1000
R – B → 30 kl → ₹12,000
S – A → 20 kl → ₹4000
Total = ₹34,000

27. The cheapest supply lines were already determined in the first question of the set. Now we have to determine the supply line whose disruption would cause the transportation cost to increase by the maximum amount. Choices (B), (C) and (D) can be straight away ignored as they are not part of the cheapest route. Hence, the increase in transportation cost would be the highest, if the supply line P – B is disrupted.

28. As far as the cost of transporting at each of the outlets is considered, most of the outlets receive the oil at a low transportation cost from the refinery S Hence, if the capacity of refinery S is doubled, then the savings in the expenditure will be more.

29. Only one of Warne, Dhoni and Symonds can be selected and only one of Akthar and Lee can be selected. So also, Ponting and Harbhajan or Ponting and Sourav cannot be selected together.
∴ The team of five can be Symonds, Sreesanth, Harbhajan, Sourav and one among Akthar or Lee.

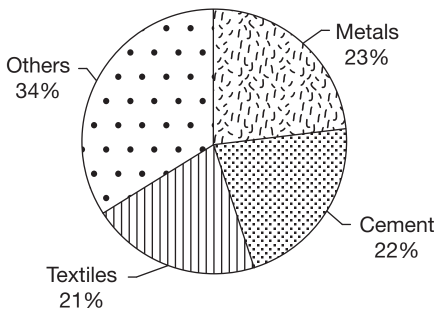
30. If Ishant was selected, then definitely Ponting has to be selected, which means that Sourav, Harbhajan, Symonds, Sreesanth, one of Akthar and Lee and one of Warne and Dhoni cannot be selected.
∴ A total of four players or three players other than Ishant can be selected.

31. One of Akthar and Lee and one of Warne and Dhoni must be definitely selected. Symonds cannot be selected as the players selected would exceed three in that case. Sourav can be selected while Ponting cannot be selected in which case Ishant also has to be selected and the number of players selected becomes four.

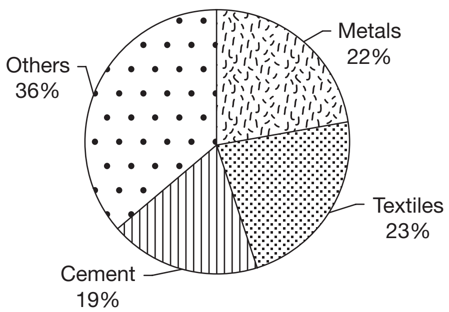
32. As all of Symonds, Sreesanth and Harbhajan must be selected or none of them must be selected, to have the largest number of players selected, all the three must be selected. In that case Warne, Dhoni, Ponting, Ishant and one of Akthar or Lee cannot be selected.
∴ At most five players can be selected.

MOCK TEST 3

Directions for questions 1 to 4: These questions are based on the following pie charts, which provides the breakup of the sales turnover of an industrial group from different sectors in 2013 and 2015. Select the correct alternative from the given choices.



Year 2013
Turnover = ₹ 450 crore



Year 2015
Turnover = ₹ 650 crore

1. Which sector recorded the maximum percentage growth in the two-year period?
(A) Metals (B) Cement
(C) Textiles (D) Others

2. If metals are expected to continue at the same percentage growth/decrease for the period from 2015 to 2017 as in from 2013 to 2015, then the turnover (in crore) from metals in 2017 would be
(A) 172.5 (B) 187.5
(C) 205.6 (D) 197.5

3. The annual growth rate of cement sales is
(A) 8.6% (B) 10.5%
(C) 12.4% (D) 13.6%

4. If 'Others' grew by 20% in 2016 and the total group turnover grew by 10%, both when compared to 2015, then 'Others' constituted what percentage of the total group turnover in 2016?
(A) 36.2% (B) 39.2%
(C) 38.4% (D) 40.8%

Directions for questions 5 to 8: Answer the questions based on the information given below and select the correct alternative from the given choices.

A software company is planning to conduct campus interviews in eight cities, namely in Delhi, Mumbai, Kolkata, Chennai, Hyderabad, Lucknow, Ahmedabad and Indore. While finalizing the order of the cities the company has to visit, it follows certain conditions as given below.

(i) Indore must be visited before Mumbai.
(ii) Kolkata must be visited before Delhi.
(iii) Chennai and Ahmedabad must be visited after Kolkata.
(iv) Ahmedabad must be visited after Hyderabad.
(v) Indore must be visited before Lucknow, which in turn must be visited before Kolkata.

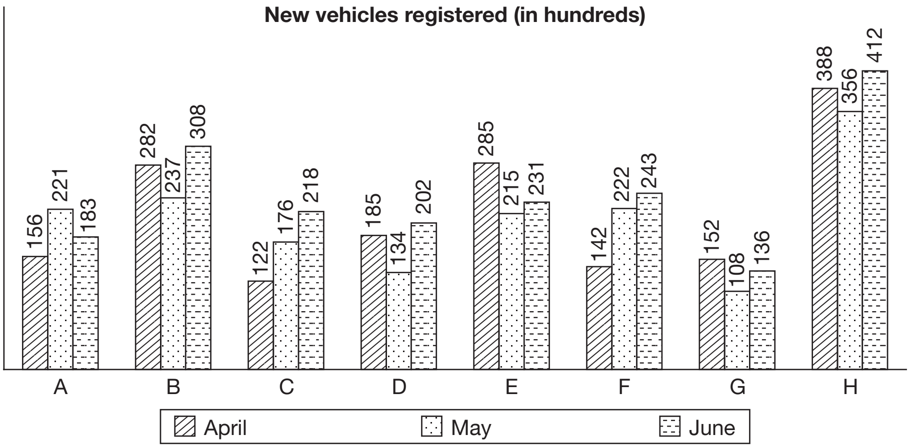
5. If Chennai is the fourth city to be visited by the company, then which of the following statements must be true?
(A) Delhi is to be visited before Mumbai.
(B) Ahmedabad is the seventh city to be visited by the company.
(C) Lucknow is the third city to be visited by the company.
(D) Hyderabad is not the third city to be visited by the company.

6. If Hyderabad is the seventh city to be visited by the company, then which of the following statements is definitely false?
(A) Lucknow is the second city to be visited by the company.
(B) Lucknow is the third city to be visited by the company.
(C) Kolkata is the fourth city to be visited by the company.
(D) Kolkata is the fifth city to be visited by the company.

7. If Hyderabad is to be visited after Chennai, then which of the following statements could be true?
(A) Lucknow is the fourth city to be visited by the company.
(B) Kolkata is the fifth city to be visited by the company.
(C) Delhi is the sixth city to be visited by the company.
(D) Chennai is the seventh city to be visited by the company.

8. If Kolkata is the fourth city to be visited by the company, then which of the following statements cannot be true?
(A) Indore is the second city to be visited by the company.
(B) Hyderabad and Ahmedabad are to be visited before visiting Chennai.
(C) Mumbai is the last city to be visited by the company.
(D) Hyderabad and Mumbai are to be visited after visiting Kolkata.

Directions for questions 9 to 12: These questions are based on the following bar chart, which shows the number of new vehicles registered in three different months in eight cities, such as A, B, C, D, E, F, G and H. Select the correct alternative from the given choices.



9. The average number of new vehicles registered per city in the month of April was
(A) 18,600 (B) 19,700
(C) 21,800 (D) None of these

10. From April to May, which city had the highest percentage growth in the number of new vehicles registered?
(A) A (B) C
(C) F (D) None of these

11. For how many of the given cities was the number of new vehicles registered in the month of May less than the average number of new vehicles registered (per month) in that city in the three given months?
(A) 7 (B) 4
(C) 5 (D) 6

12. Among the following cities, which city had the lowest number of new vehicles registered in the given period?
(A) A (B) C
(C) D (D) F

Directions for questions 13 to 16: Answer the questions based on the information given below and select the correct alternative from the given choices.

Five books are stacked up, one over the other, each belonging to a different subject among Botany, Anatomy, Zoology, Neurology and Dentistry, not necessarily in the same order. The following information is also known.

(i) The Neurology book is just below the Anatomy book.
(ii) There are four books above the Zoology book.
(iii) The Botany book is above the Dentistry book and there is at least one book between them.

13. How many books are there below the Anatomy book?
(A) 1 (B) 2
(C) 3 (D) 4

14. Which book has an equal number of books above it and below it?

(A) Anatomy (B) Neurology
(C) Dentistry (D) None of the above

15. The number of books between the Botany book and the Zoology book is

(A) 3 (B) 2
(C) 1 (D) None of the above

16. How many books are below the Dentistry book?

(A) 1 B) 2
(C) 3 (D) Cannot be determined

Directions for questions 17 to 20: Answer the questions based on the information given below.

Peoplereach.com is an internet company engaged in the business of collecting details of potential employees and selling these details to other organizations. The company collects data based on the details of people from different categories by evaluating on their job profile. For each person in each category, the company collects the details of one or more of the six features, such as Name, Age, Address, Experience, Phone Number and E-mail ID. The following table gives the information available in the database of the company about the number of people in each category, and the percentage of people in that category for whom the details of each feature are available.

		Percentage of people for whom the details of the feature are available					
Category	**Number of people**	**Name**	**Age**	**Address**	**Experience**	**Phone number**	**E-mail ID**
School Teachers	16000	100%	85%	70%	90%	65%	80%
Pharmacists	3000	100%	75%	95%	80%	70%	60%
Doctors	60000	100%	50%	60%	70%	65%	100%
Professors	10000	100%	70%	75%	60%	85%	90%
Civil Engineers	25000	100%	40%	60%	50%	65%	85%
MBAs	150000	100%	50%	55%	70%	65%	100%
CAs	2600	100%	80%	50%	40%	50%	90%
Mechanical Engineers	42000	100%	85%	70%	95%	60%	80%
Nurses	18000	100%	50%	40%	60%	75%	40%
Accountants	12000	100%	40%	70%	75%	90%	85%
Ex-Servicemen	15000	100%	65%	75%	40%	80%	60%
Electrical Engineers	22000	100%	70%	65%	60%	70%	90%
Computer Engineers	26000	100%	80%	60%	65%	50%	100%

In the above table, for example, the phone numbers of 65% of the school teachers in the database (i.e., 65% of 16,000) are available. Assume that no person belongs to more than one category.

17. The number of doctors, each of whose name, phone number and address is available, is at least.___________

18. The number of mechanical engineers, for whom the details of exactly four of the six features are available, is at least. ___________

19. The number of professors for whom at least two of the three features, like address, phone number and E-mail ID are available, is at least. ___________

20. For at most how many of the CAs are the details of exactly five of the six features available? ___________

Directions for questions 21 to 24: Answer the questions based on the information given below.

Moodys, a modelling agency was on the lookout for new models. It had called 150 candidates for the purpose of recruiting models who were tall, dark and handsome. The break-up of the candidates with different attributes in that group of 150 is as follows.

- Tall and handsome but not dark = 9
- Dark and handsome but not tall = 12
- Tall or dark but not handsome = 107

Each candidate had at least one of the three attributes that the agency was looking for.

The agency could find only one person who satisfied its criteria and so was considering relaxing the requirements a little. It was also found that, for any attribute, the number of candidates who had that attribute alone did not exceed one-third of the total number of candidates called.

21. What is the minimum number of candidates who had at least two of the three attributes? ___________

22. If the number of candidates who were dark is less than those who were tall, then at least how many candidates were dark and as well as tall? ___________

23. If exactly half of the candidates who were tall and were also dark and exactly half of the candidates who were dark were also tall, then how many candidates were only tall? ___________

24. If the number of candidates who were tall is twice that of those who had at least two attributes, then at least how many candidates were only dark? ___________

Directions for questions 25 to 28: Answer the questions based on the information given below and select the correct alternative from the given choices.

Seven politicians, namely P, Q, R, S, T, U and V are invited to a special meeting. They are seated in a row. As S and V had to leave early, they must be seated at the two extreme left seats in any order. T must be seated at the centre and P and R must have at least three people between them.

25. Which of the following people cannot be seated at either of the ends?

(A) P (B) R
(C) U (D) V

26. Which of the following pair of people cannot be seated together?

(A) P and U (B) T and R
(C) R and V (D) Q and S

27. Which of the following pair of people cannot be the two people adjacent to T?

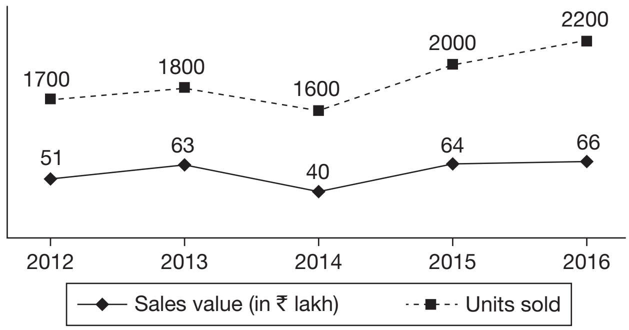
(A) R and U (B) Q and R
(C) Q and V (D) P and U

28. If there are exactly two people between Q and S, then who among the following is sitting four places to the left of U?

(A) T (B) R
(C) S (D) V

Directions for questions 29 to 32: Answer the questions based on the information given below and select the correct alternative from the given choices.

The following graph provides the details of the number of units sold and the total sales value of Company X, which produces bicycles for the period from 2012 to 2016.



29. What was the average price per bicycle sold in 2014?

(A) Rs. 2700 (B) Rs. 2200
(C) Rs. 2500 (D) Rs. 3000

30. In which of the following years was the percentage increase in the average price per bicycle the highest?

(A) 2013 (B) 2014
(C) 2015 (D) 2016

31. In how many of the given years was the sales (by volume) of bicycles less than that of the average sales (by volume) in the entire period?

(A) 1 (B) 2
(C) 4 (D) 3

32. If, in 2017, both the number of units sold increases by 25% and the total sales (value) by 50% when compared to 2016, then what is the average price per bicycle sold in 2017?

(A) Rs. 3200 (B) Rs. 3500
(C) Rs. 3900 D) Rs. 3600

ANSWER KEYS

1. C	6. D	11. C	16. A	21. 29	26. D	31. D
2. D	7. C	12. B	17. 15000	22. 12	27. C	32. D
3. C	8. D	13. C	18. 0	23. 34	28. C	
4. B	9. D	14. B	19. 7500	24. 41	29. C	
5. D	10. C	15. A	20. 1820	25. C	30. C	

SOLUTIONS

1. By observation we can say that though 'Others' and 'Textiles' have increased their share by the same percentage points, since 'Textiles' has a lower base, its growth rate should be maximum.

2. Turnover from metals in 2013 = $\frac{23}{100} \times 450 = 103.5$ crore

In 2015 = $\frac{22}{100} \times 650 = 143.0$

Growth rate = $\frac{143.0 - 103.5}{1035} \times 100 = 38.16$

$\therefore$ Turnover in 2017 = $143.0 \times \frac{138.16}{100} = 197.5$ crore

3. Turnover in 2013 = $\frac{22}{100} \times 450 = 99$ crore

In 2015 = $\frac{19}{100} \times 650 = 123.5$

% growth = $\frac{24.5}{99} = 24.75$

Annual growth = $\frac{24.75}{2} = 12.37\%$.

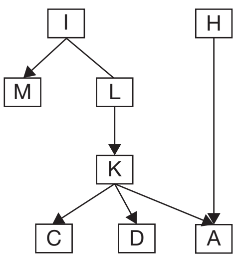
4. Let the total turnover in 2015 be 100 crore.
$\therefore$ 'Others' = 36 crore

'Others' in next year = $36 \times \frac{120}{100} = 43.2$ crore

Total = 100 crore

$\therefore$ Score of 'Others' = $\frac{43.2}{110} \times 100 = 39.2\%$

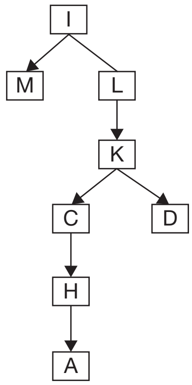
Solutions for questions 5 to 8: The given information can be represented as follows (Each city is represented by the first letter of its name).



5. Given that Chennai is the fourth city to be visited. As Indore, Lucknow and Kolkata are to be visited before Chennai, they are to be visited first, second and third, respectively. Hence, Hyderabad cannot be the third city to be visited.

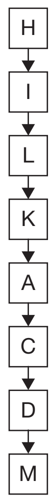
6. As Hyderabad is the seventh city to be visited, Ahmedabad is the eighth city to be visited.
Hence, the only cities that can be visited before Kolkata are Indore, Mumbai and Lucknow. Therefore, Kolkata cannot be visited fifth.

7. If Hyderabad is visited after Chennai, the representation is as follows.



In the above case, five cities are to be visited after Lucknow. Hence, Lucknow can be visited second and third. Therefore, choice (A) is false.
Similarly, choices (B) and (D) can be eliminated. Delhi can be the sixth city to be visited as in the following sequence.
IMLKCDHA.

8. If Kolkata is the fourth city to be visited, then the representation is as follows.



From the above representation, we can say that (A), (B) and (C) can be true. Choice (D) cannot be true.

9. The average number of new vehicles registered in the month of April

$$= \frac{156 + 282 + 122 + 185 + 285 + 142 + 152 + 388}{8}$$

$$= \frac{1712}{8} = 214 \text{ (in hundreds)}.$$

10. In May, only for city F there was more than 50% increase in the number of new vehicles registered.

11. The number of new vehicles registered in May was less than the average for the three months in cities B, D, E, G and H.

12. The number of new vehicles registered in different cities are
A – 156 + 221 +183 = 560
C – 122 + 176 + 218 = 516
D – 185 + 134 + 202 = 521
F – 142 + 222 + 243 = 607
It was the least for city C.

Solutions for questions 13 to 16: The 5 books are Botany, Anatomy, Zoology, Neurology and Dentistry. Let us represent these books by their starting letters, i.e., B, A, Z, N, D, respectively.

From (i), N is just below A. From (ii), Z is the bottom most book since there are a total of five books and Z has 4 above it. From (iii), Botany is above Dentistry and they are having at least one book between them, N and A must be between them.

∴ The arrangement is as follows.

B
A
N
D
Z

13. There are 3 books below Anatomy book.

14. Neurology book has 2 books above it and 2 below it.

15. There are 3 books between Botany and Zoology.

16. There is only one book below the Dentistry book.

17. As the name is available for all the people, i.e. 100%, the least number of doctors for whom the name, phone number and address are available happens when the number of doctors for whom the address and phone numbers are available is the minimum,
i.e., (60 + 65) – 100 = 25%, i.e., 15,000.

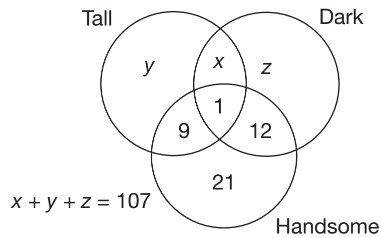
18. The number of mechanical engineers for whom exactly four of the six details were available can be zero, as the total data available is on an average of 4.9 items per person (100 + 85 + 70 + 95 + 60 + 80 = 490) and it can be due to a majority of them having exactly five of their details available and for the others on, two or three of the six details being available.

19. The number of professors for whom at least two of address, phone number and e-mail ID are available would be the least when for the maximum number of professors exactly one detail is available and for others all the three details are available.
Total = 75 + 85 +90 = 250
∴ The required value would be the least when for 25% of them exactly one detail is available and for the rest 75% of all the three details are available.

20. As one should have exactly five of six features, he should not have one of the features with least percentage values. Consider the three least percentages available, i.e., 50%, 40% and 50%.
Exactly two among these three must be available.

$$\therefore \frac{50 + 40 + 50}{2} = 70\%$$ and as other features have more than 70% availability, 70% of 2600 = 1820.

Solutions for questions 21 to 24: Drawing the Venn diagram with the given conditions:



Only handsome = 150 – (107 + 9 + 1 + 12) = 21

21. To find the minimum number of candidates who had at least two of the three attributes, we have to maximize the number of candidates with just one single attribute. But as it is mentioned that at most fifty candidates (i.e., one-third of 150) had any one single attribute, there must be at least seven people who were both tall and dark (i.e., $x \geq 7$). Required value = 7 + 9 + 1 + 12 = 29

22. Given that: Tall > Dark
Tall = 10 + x + y
Dark = 13 + x + z
$\Rightarrow 10 + x + y > 13 + x + z$
$\Rightarrow 10 + y > 13 + z$
$y > z + 3$
The maximum possible value of y is 50.
The maximum possible value of z is 46.
∴The minimum possible value of x is 107 – (50 + 46) = 11.
Hence, the number of people who are tall as well as dark is at least $(x + 1) = 12$.

23. Given that: 50% of tall are also dark and 50% of dark are also tall.

$\Rightarrow \quad 9 + y = 1 + x \quad$ (i)

and also,

$z + 12 = x + 1 \quad$ (ii)

Now adding (i) + (ii), we get:

$$21 + y + z = 2x + 2$$
$$21 + (107 - x) = 2x + 2$$
$$128 - 2 = 3x$$
$$x = \frac{126}{3} = 42$$

Only tall = $y = x - 8$ (from (i)) = 34

24. The number of people who were only dark (i.e., z) should be as minimum as possible.

$\therefore (x + y)$ should be the maximum.

The maximum possible value of y is 50.

Given that, tall = 2 (candidates with at least two attributes)

$(60 + x) = 2(22 + x)$

$x = 16$

$z = 107 - (x + y) = 107 - (16 + 50) = 41$

Solutions for questions 25 to 28: T is at the centre and S and V are at the extreme left. We have:

S/V V/S __ T ___ __

P and R are separated by at least three people, i.e., we have the following final arrangement.

S/V V/S P/R T Q/U U/Q R/P

25. U can never be at either of the ends.

26. Q and S can never be seated together.

27. V can never be adjacent to T.

28. From the given condition, V must be at the extreme left, S must be to the right of V. Q must be to the right of T and U must be to the right of Q. Hence, S is sitting four places to the left of U. Choice (C)

29. The average price per bicycle sold in the various years are as follows.

$$2012 = \frac{51}{1700} \text{lakhs} = ₹3000$$
$$2013 = \frac{63}{1800} \text{lakhs} = ₹3500$$
$$2014 = \frac{40}{1600} \text{lakhs} = ₹2500$$
$$2015 = \frac{64}{2000} \text{lakhs} = ₹3200$$
$$2016 = \frac{66}{2200} \text{lakhs} = ₹3000$$

30. The highest percentage increase occurred in the year 2015.

31. Average sales for the given period

$$= \frac{1700 + 1800 + 1600 + 2000 + 2200}{5}$$
$$= \frac{9300}{5} = 1860.$$

The sales was less than the average in 2012, 2013 and 2014.

32. Number of units sold in 2017 = $2200 \times 1.25 = 2750$

Total sales (value) = 66 lakhs × 1.5 = 99 lakhs

$$\text{Average price} = \frac{99}{2750} \text{lakhs} = 3600$$